Math Journal 2

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I'm finally starting a new pdf since the old one was getting so long. Unfortunately I won't be able to have hyperlinks directly to pages in the prior pdf. But oh well. If a hyperlink goes to this page, it means you should look at a previous journal.

Journal 1 contains pages 1-790.

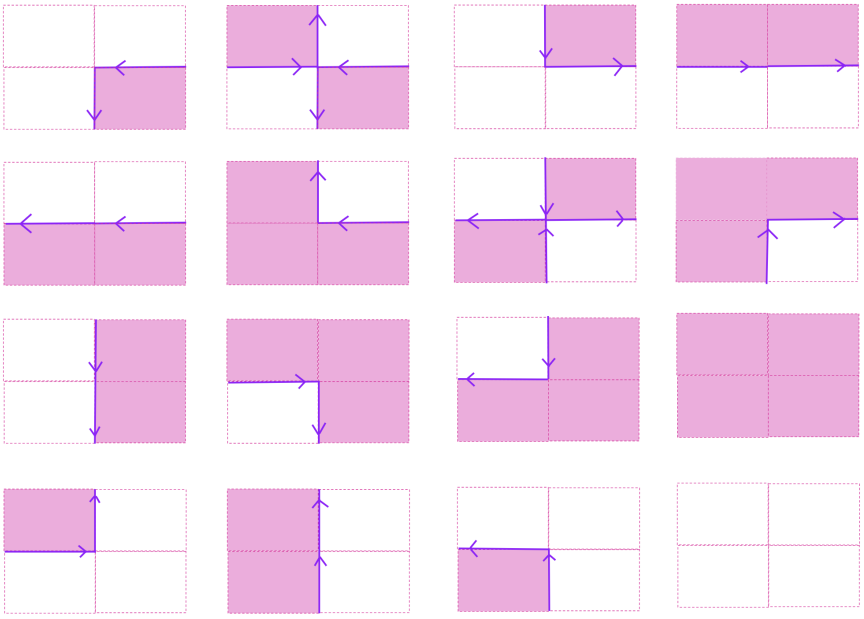
The first thing I want to do now that it is Spring break is finally give a formal justification to the assertion I wrote on *page 759*.

Recall the construction from *page 753*.

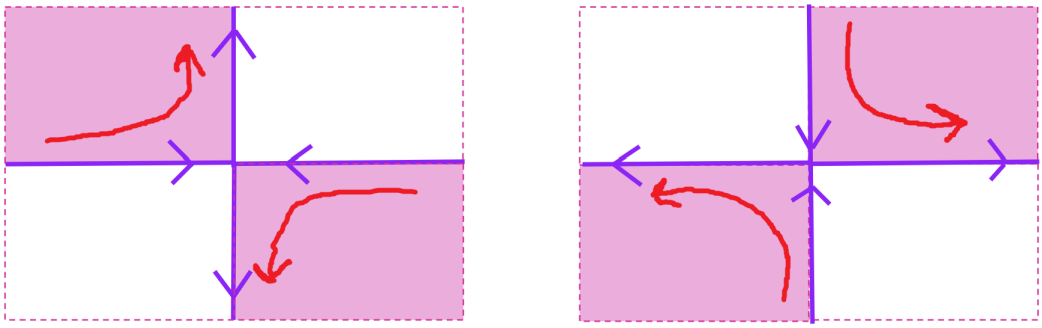
- Given a compact set $K \subseteq \mathbb{C}$, and $\delta > 0$, we overlay a grid of squares with side-length δ.
- Next, we consider the closed squares $Q_1, \ldots, Q_m$ that intersect K.
- Having oriented the boundary of each Q_i to be traveling counter-clockwise, we finally let $\sigma_1, \ldots, \sigma_n$ be the edges of $\partial Q_1, \ldots \partial Q_m$ which are not shared by distinct squares.

Claim: We can strategically concatenate the σ_j in order to get a bunch of continuous closed polygonal paths $\pi_1, \ldots, \pi_\ell$.

Proof:
To start off, note that an edge σ_j will move up, left, down, or right if and only if the square the edge borders to the left, below, right, or above (respectively) intersects K. By using this observation plus the fact that no σ_j are shared by two squares which both intersect K, we can thus eliminate many possible arrangements of lines. For example, here are all possible cases we can have of four adjacent squares and their shared edges:



Based on the prior image, we can assign each edge σ_j a successor edge $s(\sigma_j)$ and a predecessor edge $p(\sigma_j)$ such that $p(\sigma_j) + \sigma_j + s(\sigma_j)$ form a connected path. The only times we have any ambiguity on how we can choose $s(\sigma_j)$ and $p(\sigma_j)$ is in the two cases that we have squares diagonal to each other but not sharing any sides. Yet this isn't a big issue since for those two specific cases I can just declare the following convention.



With that, we now have that every edge has a unique predecessor and successor edge. Also, we can easily see that $p(s(\sigma_j)) = \sigma_j = s(p(\sigma_j))$. In other words, s and p are inverse functions.

Finally, consider the group action $\mathbb{Z} \curvearrowright \{\sigma_j : 1 \leq j \leq n\}$ given by $k \cdot \sigma_j = s^k(\sigma_j)$. Because there are only finitely many σ_j, we know every orbit of this group action has only finitely many elements. In particular, given any fixed σ_j we claim there exists $N \in \mathbb{N}$ such that $s^{k_1}(\sigma_j) = s^{k_2}(\sigma_j)$ iff $k_2 - k_1 \equiv 0 \pmod{N}$.

> Why?
> Since the orbit of σ_j is finite, we know that there exists $0 < k_1' < k_2'$ such that $s^{k_1'}(\sigma_j) = s^{k_2'}(\sigma_j)$. Then by composing on $p^{k_1} = s^{-k_1}$, we get that $\sigma_j = s^{k_2'-k_1'}(\sigma_j)$ where $k_2' - k_1' > 0$. This proves that the set of $k \in \mathbb{N}_{>0}$ such that $s^k(\sigma_j) = \sigma_j$ is nonempty. So, we can pick a minimal positive N such that $s^N(\sigma_j) = \sigma_j$. It's a basic algebra proof to go from here...

But now the orbit of σ_j consists of the collection $\{\sigma_j, s(\sigma_j), \ldots, s^{N-1}(\sigma_j)\}$. This orbit also does not intersect any distinct orbits. Also the concatenated path:

$$\sigma_j + s(\sigma_j) + \ldots + s^{N-1}(\sigma_j)$$

is connected, closed, and does not repeat any distinct edges.

Thus, we define $\pi_1, \ldots, \pi_\ell$ by considering the paths above defined from the orbits of our group action. ■

The next rabbit hole I want to descend is about $\text{weak-}L^p$. After all, this is the one topic from math 240 that even now I've not spent any time trying to learn now that the class is over and my life is somewhat solidified again.

Let $(X, \mathcal{M}, \mu)$ be a measure space. Then recall that given any measurable $f : X \to \mathbb{C}$, we define:

- the distribution function $\lambda_f : (0, \infty) \to [0, \infty]$ of f to be:
$$\lambda_f(\alpha) \coloneqq \mu(\{x \in X : |f(x)| > \alpha\});$$

- the weak-L^p quasinorm of f to be:
$$[f]_p \coloneqq \left(\sup_{\alpha>0} \alpha^p \lambda_f(\alpha)\right)^{1/p} = \sup_{\alpha>0} \alpha(\lambda_f(\alpha))^{1/p} \text{ (where } 0 < p < \infty\text{).}$$

Finally, weak-L^p (also denoted $L^{p,\infty}$) is the collection of all measurable functions f on X such that $[f]_p < \infty$.

Note that by rearranging terms, we can equivalently see that:
$$[f]_p = \min\{C \in [0, \infty] : \lambda_f(\alpha) \leq \tfrac{C^p}{\alpha^p} \text{ for all } \alpha > 0\}$$

Note that the latter set will in fact always have a minimum. The easiest way to see this is by noting that the latter set's complement (given by $\{C \geq 0 : \lambda_f(\alpha) > \frac{C^p}{\alpha^p}$ for some $\alpha > 0\}$) is an open set.

Thus, a function is in weak-L^p precisely iff there is some constant $D > 0$ such that $\mu(\{x : |f(x)| > \alpha\}) \leq \frac{D}{\alpha^p}$ for all $\alpha > 0$.

Unsurprisingly we have that $L^p \subseteq$ weak-L^p. After all, by Chebishev's inequality we know that $\lambda_f(\alpha) \leq \frac{\|f\|_p^p}{\alpha^p}$ for all $\alpha > 0$. Therefore, it follows that $[f]_p \leq \|f\|_p$. That said, the converse inclusion doesn't generally hold.

For example, consider the sequence $\{\frac{1}{n^{1/p}}\}_{n\in\mathbb{N}}$ as a function in $l^p(\mathbb{N})$. Its l^p norm is given by the harmonic series $\sum_{n\in\mathbb{N}} \frac{1}{n}$ which diverges. That said it is in weak-l^p. After all:
$$\lambda(\alpha) = \begin{cases} \lfloor \alpha^{-p} \rfloor & \text{if } \alpha^{-p} \notin \mathbb{N} \\ \alpha^{-p} - 1 & \text{if } \alpha^{-p} \in \mathbb{N} \end{cases}$$

Importantly, this means that $\lambda(\alpha) < \alpha^{-p}$ for all $\alpha > 0$ and we can make their difference arbitrarily small. In turn, we can show that $\sup_{\alpha>0} \alpha^p \lambda(\alpha) = 1$.

The big use case of weak-L^p is as follows. Recall that for any $0 < p < \infty$ and measurable function f on X we have that:
$$\int |f|^p \mathrm{d}\mu = p \int_0^\infty \alpha^{p-1} \lambda_f(\alpha) \mathrm{d}\alpha$$

But now if $f \in$ weak-L^p, we can say that:
$$p \int_0^\infty \alpha^{p-1} \lambda_f(\alpha) \mathrm{d}\alpha \leq p[f]_p^p \int_0^\infty \tfrac{1}{\alpha} \mathrm{d}\alpha$$

In some sense, the latter integral only just barely doesn't converge. So, given slightly more information bounding our function, we can say that f is in L^p. To further explore this, here is an exercise from Folland's *Real Analysis*.

(Folland) Exercise 6.36: If $f \in \text{weak-}L^p$ and $\mu(\{x : f(x) \neq 0\}) < \infty$, then $f \in L^q$ for all $q < p$.

Proof:
If $C := \mu(\{x : f(x) \neq 0\}) < \infty$, then we know that $\lambda_f(\alpha) \leq C$ for all $\alpha > 0$. In turn, for all $0 < q < p$ we have that:

$$\begin{aligned}\int |f|^q \mathrm{d}\mu = q\int_0^\infty \alpha^{q-1}\lambda_f(\alpha)\mathrm{d}\alpha &= q\int_0^1 \alpha^{q-1}\lambda_f(\alpha)\mathrm{d}\alpha + q\int_1^\infty \alpha^{q-1}\lambda_f(\alpha)\mathrm{d}\alpha \\ &\leq qC\int_0^1 \alpha^{q-1}\mathrm{d}\alpha + q\int_1^\infty \tfrac{\alpha^{q-1}}{\alpha^p}[f]_p^p \mathrm{d}\alpha \\ &= C + [f]_p^p \tfrac{q}{p-q}.\end{aligned}$$

On the other hand, if $f \in (\text{weak-}L^p) \cap L^\infty$, then $f \in L^q$ for all $q > p$.

Proof:
If $C := \|f\|_\infty < \infty$, then we know that $\lambda_f(\alpha) = 0$ for all $\alpha \geq C$. Hence, we can say for all $p < q < \infty$ that:

$$\begin{aligned}\int |f|^q \mathrm{d}\mu = q\int_0^\infty \alpha^{q-1}\lambda_f(\alpha)\mathrm{d}\alpha &= q\int_0^C \alpha^{q-1}\lambda_f(\alpha)\mathrm{d}\alpha \\ &\leq q\int_0^C [f]_p^p \tfrac{\alpha^{q-1}}{\alpha^p}\mathrm{d}\alpha = \tfrac{q}{q-p}C^{q-p}. \blacksquare\end{aligned}$$

One other thing on this topic I'd like to note is that $[\cdot]_p$ is in fact a quasinorm. After all:

- $[f]_p \geq 0$ with $[f]_p = 0$ if and only if $f = 0$ a.e.

  $[f]_p \geq 0$ because $\alpha^p\lambda_f(\alpha) \geq 0$ for all α. Also, because $\alpha^p > 0$ for all $\alpha > 0$, we have that $\sup_{\alpha>0} \alpha^p\lambda_f(\alpha) = 0$ if and only if $\lambda_f(\alpha) = 0$ for all $\alpha > 0$. Yet the latter will happen if and only if $\mu(\{x : f(x) \neq 0\}) = 0$.

- For any $c \in \mathbb{C}$ we have that $[cf]_p = |c|[f]_p$.

  Why?
  If $c = 0$ then this is trivial from the last bullet point. Meanwhile, if $c \neq 0$ then:

  $$\lambda_{cf}(\alpha) = \mu(\{x : |cf(x)| > \alpha\}) = \mu(\{x : |f(x)| > \tfrac{\alpha}{|c|}\}) = \lambda_f(\tfrac{\alpha}{|c|})$$

  Therefore:

  $$\sup_{\alpha>0} \alpha^p\lambda_{cf}(\alpha) = \sup_{\alpha>0} \alpha^p\lambda_f(\tfrac{\alpha}{|c|}) = |c|^p \sup_{\alpha>0}(\tfrac{\alpha}{|c|})^p\lambda_f(\tfrac{\alpha}{|c|}) = |c|^p \sup_{\beta>0} \beta^p\lambda_f(\beta)$$

- There exists $B \geq 1$ such that $[f+g]_p \leq B([f]_p + [g]_p)$ for all measurable functions f and g.

  Why?
  First note that:

  $$\begin{aligned}\lambda_{f+g}(\alpha) = \mu(\{x : |f(x)+g(x)| > \alpha\}) &\leq \mu(\{x : |f(x)| + |g(x)| > \alpha\}) \\ &\leq \mu(\{x : |f(x)| > \tfrac{\alpha}{2}\} \cup \{x : |g(x)| > \tfrac{\alpha}{2}\}) \\ &\leq \mu(\{x : |f(x)| > \tfrac{\alpha}{2}\}) + \mu(\{x : |g(x)| > \tfrac{\alpha}{2}\}) \\ &= \lambda_f(\tfrac{\alpha}{2}) + \lambda_g(\tfrac{\alpha}{2})\end{aligned}$$

  In turn $\sup_{\alpha>0} \alpha^p\lambda_{f+g}(\alpha) \leq \sup_{\alpha>0} \alpha^p(\lambda_f(\tfrac{\alpha}{2}) + \lambda_g(\tfrac{\alpha}{2})) \leq \sup_{\alpha>0} \alpha^p\lambda_f(\tfrac{\alpha}{2}) + \sup_{\beta>0} \beta^p\lambda_g(\tfrac{\beta}{2})$.

Next note that $\sup_{\alpha>0} \alpha^p \lambda_f(\frac{\alpha}{2}) = 2^p \sup_{\alpha>0} (\frac{\alpha}{2})^p \lambda_f(\frac{\alpha}{2}) = 2^p [f]_p^p$.

Similarly, we have that $\sup_{\beta>0} \beta^p \lambda_f(\frac{\beta}{2}) = 2^p [g]_p^p$. Thus, we can conclude:

$$[f+g]_p^p \leq 2^p([f]_p^p + [g]_p^p)$$

In other words, $[f+g]_p \leq 2([f]_p^p + [g]_p^p)^{1/p}$. To finish off, if $p \geq 1$ then this proves that $[f+g]_p \leq 2([f]_p + [g]_p)$.

> You can justify to yourself that $(a^p+b^p)^{1/p} \leq a+b$ for all $a, b \geq 0$ when $p \geq 1$ by considering that the p-norm on $\mathbb{R}^2$ satisfies the triangle inequality.

Meanwhile, if $0 < p < 1$, then we need to find a constant $A > 1$ such that $([f]_p^p + [g]_p^p)^{1/p} \leq A([f]_p + [g]_p)$.

Tangent: While I'm finding this constant A, I shall also prove that L^p is a quasinormed space when $0 < p < 1$.

Lemma 1: If $a, b \geq 0$ and $0 < p < 1$, then $(a+b)^p \leq a^p + b^p$.

Proof:
If $a = 0$ then this is obvious. Meanwhile, if $a \neq 0$ then we have that $(a+b)^p \leq a^p + b^p$ if and only if $(1+\frac{b}{a})^p = (\frac{a+b}{a})^p \leq \frac{a^p+b^p}{a^p} = 1 + (\frac{b}{a})^p$. Therefore, it suffices to show that $(1+t)^p \leq 1 + t^p$ for all $t \geq 0$.

Set $h(t) = 1 + t^p - (1+t)^p$. We want to show that $h(t) \geq 0$ for all $t \geq 0$. Fortunately, note that $h(0) = 0$. Also $h'(t) = pt^{p-1} - p(1+t)^{p-1}$. Since $p-1 < 0$ and $0 < t < 1+t$ implies that $(1+t)^{p-1} < t^{p-1}$, we can conclude that $h'(t) > 0$ for all $t > 0$. This proves that h is strictly increasing for $t > 0$. Hence, we can conclude that $h(t) \geq 0$ for all $t \geq 0$. ■

> As a side note, mirror reasoning shows that $h(t)$ is strictly decreasing for $t > 0$ if $p > 1$. So, the opposite inequality will hold if $p > 1$.

Lemma 2: If $a, b \geq 0$ and $q \geq 1$, then $(a+b)^q \leq 2^{q-1}(a^q + b^q)$.

Proof:
In order to make it clear how someone might have discovered this lemma without seeing the constant 2^{q-1} on a google search, I shall specifically write down the process for finding some $A \geq 1$ such that $(a+b)^q \leq A(a^q + b^q)$. Then, it will just so happen to work out that the minimal A that works is 2^{q-1}.

Firstly, note for any $A \geq 1$ that if $a = 0$ then we always have that $(a+b)^q \leq A(a^q + b^q)$. So like before, it suffices to assume $a \neq 0$. Then because $(a+b)^q \leq A(a^q + b^q)$ if and only if $(1+\frac{b}{a})^q \leq A(1 + (\frac{b}{a})^q)$, it suffices to show that $(1+t)^q \leq A(1+t^q)$ for all $t \geq 0$.

Let $h(t) = \frac{A(1+t^q)}{(1+t)^q}$. We want to fix $A \geq 1$ such that $h(t) \geq 1$ for all $t \geq 0$. Fortunately, note that $h(0) = A \geq 1$. Also, we can calculate for $t \geq 0$ that:

$$h'(t) = \frac{Aqt^{q-1}(1+t)^q - Aq(1+t^q)(1+t)^{q-1}}{(1+t)^{2q}} = \frac{Aq(t^{q-1}(1+t) - (1+t^q))}{(1+t)^{q+1}} = \frac{Aq(t^{q-1}-1)}{(1+t)^{q+1}}$$

It follows that h achieves an absolute minimum on $[0,\infty)$ at $t=1$. So, it suffices to pick A with $h(1)=\frac{2A}{2^q}\geq 1$. The smallest A satisfying this is $A=2^{q-1}$. ■

As a result of these two lemmas, we can now say for any measurable functions $f,g:X\to\mathbb{C}$ and $0<p<1$ that:

$$\|f+g\|_p\leq 2^{\frac{1}{p}-1}(\|f\|_p+\|g\|_p).$$

Proof:
As $t\mapsto t^p$ is a strictly increasing function, we can conclude from the triangle inequality that $|f(x)+g(x)|^p\leq(|f(x)|+|g(x)|)^p$ for all $x\in X$. In turn, by applying lemma 1 we get that:

$$\begin{aligned}\int|f(x)+g(x)|^p\mathrm{d}\mu&\leq\int(|f(x)|+|g(x)|)^p\mathrm{d}\mu\\&\leq\int|f(x)|^p+|g(x)|^p\mathrm{d}\mu=\int|f|^p\mathrm{d}\mu+\int|g|^p\mathrm{d}\mu\end{aligned}$$

Next, as $\frac{1}{p}>1$, we can conclude by lemma 2 plus the fact that $t\mapsto t^{1/p}$ is a strictly increasing function that:

$$\begin{aligned}(\int|f(x)+g(x)|^p\mathrm{d}\mu)^{1/p}&\leq(\int|f|^p\mathrm{d}\mu+\int|g|^p\mathrm{d}\mu)^{1/p}\\&\leq 2^{\frac{1}{p}-1}\left((\int|f|^p\mathrm{d}\mu)^{1/p}+(\int|g|^p\mathrm{d}\mu)^{1/p}\right).\end{aligned}$$ ■

Going back to the problem of finding A such that $([f]_p^p+[g]_p^p)^{1/p}\leq A([f]_p+[g]_p)$, note that lemma 2 from the prior tangent tells us that $A=2^{\frac{1}{p}-1}$ works. In turn this means that $[f+g]_p\leq 2^{1/p}([f]_p+[g]_p)$ if $0<p<1$.

As a side note, in general the $\text{weak-}L^p$ quasinorm will not be a true norm (i.e. it won't in general satisfy the triangle inequality). For an example of this, see my notes from math 240b.

Now it's significant that $[\cdot]_p$ is a quasinorm because any vector space $\mathcal{X}$ equipped with a quasinorm $[\cdot]$ can in turn be equipped with a topology turning that vector space into a topological vector space where convergence is equivalent to convergence in that quasi-norm.

Describing this topology will require another tangent though unfortunately.

3/30/2026

I'd like to continue the rabbit-hole from yesterday by taking notes on the book *Modern Methods in Topological Vector Spaces* by Albert Wilanksy (see *citation 2*).

To start off, here are some conventions that Wilanksy uses. Given a topological space $(X,\mathcal{T})$ as well as some $x\in X$, we write $\mathcal{N}_x$ to denote the collection of neighborhoods of x. Also, we say a local base of neighborhoods of x is a collection $\mathcal{B}\subseteq\mathcal{N}_x$ such that for all $N\in\mathcal{N}_x$ there exists $E\in\mathcal{B}$ with $E\subseteq N$.

Beware: This differs from the definition of a neighborhood base that math 240b and Folland uses in that the sets in $\mathscr{B}$ are *not* required to be open. That said, it is still the case that if $\mathscr{B}$ is a local base of neighborhoods of $x \in X$ then a net $\langle x_i \rangle_{i \in I}$ converges to x if and only if x_i is eventually in each $E \in \mathscr{B}$.

While nets have the advantage of looking and acting like sequences, it will also be helpful to introduce a different generalization of the theory of convergence.

Given a set X (and denoting its power set as $\mathcal{P}(X)$), we say $\mathscr{F} \subseteq \mathcal{P}(X)$ is a filter if:

- $A, B \in \mathscr{F} \implies A \cap B \in \mathscr{F}$,
- $A \in \mathscr{F}$ and $A \subseteq E \implies E \in \mathscr{F}$,
- $\emptyset \notin \mathscr{F}$ and $X \in \mathscr{F}$.

Also, we say $\mathscr{B} \subseteq \mathcal{P}(X)$ is a filterbase if it is nonempty, doesn't contain $\emptyset$, and is directed by inclusion (i.e if $A, B \in \mathscr{B}$ then there exists $C \in \mathscr{B}$ with $C \subseteq A \cap B$).

Note that if $\{\mathscr{F}_i\}_{i \in I}$ is a family of filters, then $\bigcap_{i \in I} \mathscr{F}_i$ is also a filter. Therefore, it makes sense to talk about the smallest filter containing some generating collection of sets (assuming any filters containing that collection even exist).

Given a filterbase $\mathscr{B}$, we claim the smallest filter containing $\mathscr{B}$ is the collection:

$$\mathscr{F} := \{A \in \mathcal{P}(X) : A \supseteq B \text{ for some } B \in \mathscr{B}\}.$$

Why?
It's clear that $\mathscr{F}$ must be contained in any filter containing $\mathscr{B}$. So, it suffices to justify why $\mathscr{F}$ is itself a filter. The only property not obvious from the definition of $\mathscr{F}$ is that if $A_1, A_2 \in \mathscr{F}$ then so is $A_1 \cap A_2 \in \mathscr{F}$. Fortunately, we can pick $B_1, B_2 \in \mathscr{B}$ with $A_1 \supseteq B_1$ and $A_2 \supseteq B_2$. Next, we can pick C in $\mathscr{B}$ with $C \subseteq B_1 \cap B_2 \subseteq A_1 \cap A_2$. So, $A_1 \cap A_2 \in \mathscr{F}$.

Given a topological space $(X, \mathcal{T})$ and some $x \in X$, notice that $\mathcal{N}_x$ is a filter. Also a local base of neighborhoods of x is merely a filterbase generating $\mathcal{N}_x$.

I mentioned before that filters give us an alternate description of convergence to that of nets. Here is what I mean:

(note for just this brief aside I'm going back to Folland...)

Given a topological space $(X, \mathcal{T})$ and a filter $\mathscr{F} \subseteq \mathcal{P}(X)$, we say $\mathscr{F}$ converges to x (written as $\mathscr{F} \to x$) iff $\mathcal{N}_x \subseteq \mathscr{F}$. This definition relates to the definition of convergence of nets as follows:

- Suppose $\langle x_i \rangle_{i \in I}$ is a net. Then its derived filter is the collection $\mathscr{F}$ of all $E \subseteq \mathcal{P}(X)$ such that x_i is eventually in E.

  (Hopefully it's obvious why $\mathscr{F}$ is a filter...)

  Lemma: The net $\langle x_i \rangle_{i \in I}$ converges to x if and only if its derived filter converges to x.

  This is clear from the fact that $\langle x_i \rangle_{i \in I}$ converges to x iff x_i is eventually in N for all $N \in \mathcal{N}_x$.

- Given any filter $\mathscr{F}$, we can view $\mathscr{F}$ as a set directed by (reverse) inclusion. Therefore, $\mathscr{F}$ can be used to index a net. We say a net $\langle x_F \rangle_{F \in \mathscr{F}}$ is associated with $\mathscr{F}$ iff $x_F \in F$ for all $F \in \mathscr{F}$.

  Lemma: A filter $\mathscr{F}$ converges to x if and only if all its associated nets converge to x.

  $(\Longrightarrow)$
  Suppose $\langle x_F \rangle_{F \in \mathscr{F}}$ is any net associated with $\mathscr{F}$. To prove that $x_F \to x$, it suffices to show that if $N \in \mathscr{N}_x$ then x_F is eventually in N. Fortunately, as $\mathscr{F} \to x$, we know that $N \in \mathscr{F}$. Then for all $F \subseteq N$ we have that $x_F \in N$.

  $(\Longleftarrow)$
  If $\mathscr{F} \not\to x$ then there must exist some neighborhood N of x such that $N \notin \mathscr{F}$. In turn, given any $F \in \mathscr{F}$ we must have that $F - N \neq \emptyset$. After all we'd have a contradiction if $F \subseteq N$.

  For each $F \in \mathscr{F}$ pick $x \in F$ with $x \notin N$. Then $\langle x_F \rangle_{F \in \mathscr{F}}$ is a net associated with $\mathscr{F}$ but it does not converge to x since x_F is not eventually in N. ■

One more definition is as follows. If V is a vector space, we say a collection $\mathscr{B} \subseteq \mathcal{P}(V)$ is additive if for each $B \in \mathscr{B}$ there exists $A \in \mathscr{B}$ with $A + A \subseteq B$.

Theorem 4-3-5: Let $\mathcal{X}$ be a real or complex vector space. Let $\mathscr{B}$ be an additive filterbase of balanced absorbing subsets of $\mathcal{X}$. Then there is a unique topology turning $\mathcal{X}$ into a topological vector space and satisfying that $\mathscr{B}$ is a local base of neighborhoods of 0.

(See *pages 230-232* for a refresher on the definitions of a balanced set and an absorbing set in a real or complex vector space. Importantly, even though I've priorly only used those terms when referring to vector spaces already equipped with some topology, the definitions of those terms are purely algebraic. So it still makes sense to talk about balanced sets and absorbing sets in any real or complex vector space.)

Proof (Existence):
We define our topology how one would reasonably expect. We shall say $U \subseteq \mathcal{X}$ is open if and only if for all $x \in U$ there exists $B \in \mathscr{B}$ with $x + B \subseteq U$.

Claim 1: This defines a topology on $\mathcal{X}$.

- It's trivial that $\emptyset$ and $\mathcal{X}$ are both open.
- Suppose U_1, U_2 are both open. Then for any $x \in U_1 \cap U_2$, we can pick $B_1, B_2 \in \mathscr{B}$ with $x + B_1 \subseteq U_1$ and $x + B_2 \subseteq U_2$. Since $\mathscr{B}$ is a filterbase, we know there exists $B \in \mathscr{B}$ with $B \subseteq B_1 \cap B_2$. In turn, $x + B \subseteq (x + B_1) \cap (x + B_2) \subseteq U_1 \cap U_2$. As x was arbitrary, this proves that $U_1 \cap U_2$ is open.

  By induction we can conclude that finite intersections of open sets are open.

- Suppose $\{U_\alpha\}_{\alpha \in A}$ is a family of open sets. Then if $x \in U := \bigcup_{\alpha \in A} U_\alpha$, we know that $x \in U_{\alpha_0}$ for some α_0. So, there exists $B \in \mathscr{B}$ with $x + B \subseteq U_{\alpha_0} \subseteq U$. Since x was arbitrary, this proves that $\bigcup_{\alpha \in A} U_\alpha$ is open.

Claim 2: $\mathscr{B}$ is a local base of neighborhoods of 0.

To start off, if N is a neighborhood of 0, then there exists an open set U such that $0 \in U \subseteq N$. In turn, by how we defined our topology we know there exists $B \in \mathscr{B}$ with $B = 0 + B \subseteq N$. This shows that $\mathscr{N}_0$ is contained in the filter generated by $\mathscr{B}$.

To finish showing that $\mathscr{N}_0$ is the filter generated by $\mathscr{B}$, we need to show that $\mathscr{B} \subseteq \mathscr{N}_0$ (i.e. all $B \in \mathscr{B}$ are neighborhoods of 0 in $\mathscr{X}$). So consider any $B \in \mathscr{B}$ and define $U := \{x \in \mathscr{X} : x + A \subseteq B \text{ for some } A \in \mathscr{B}\}$.

- Note that as each $A \in \mathscr{B}$ contains 0 (since each A is balanced), we know that $U \subseteq B$.
- Also $0 \in U$ as $0 + B = B$.
- Finally, consider any $x \in U$. Then by the definition of U there exists $A \in \mathscr{B}$ with $x + A \subseteq B$. And since $\mathscr{B}$ is additive, we can find a set $C \in \mathscr{B}$ with $C + C \subseteq A$. We claim that $x + C \subseteq U$. After all, suppose $y \in x + C$. Then $y + C \subseteq x + C + C \subseteq x + A \subseteq B$. This proves that $y \in U$, and since y was arbitrary, we conclude that $x + C \subseteq U$. Finally, by the definition of an open set plus the arbitrariness of x, this proves that U is open.

As an important side note, by slightly modifying the proof of the prior claim we can also show $x + B$ is a neighborhood of x for all $B \in \mathscr{B}$. Yet also since for any open set U containing x we have that there exists some $B \in \mathscr{B}$ with $x + B \subseteq U$, we can thus conclude that $x + \mathscr{B} := \{x + B : B \in \mathscr{B}\}$ is a local base of neighborhoods of x for any $x \in \mathscr{X}$. This will be used in the proof of the third and final claim.

Claim 3: The vector space operations are continuous with respect to our defined topology.

First we show vector addition is continuous. Consider any $x, y \in \mathscr{X}$ and let N be any neighborhood of $x + y$. Then it suffices to show that there are neighborhoods N_x and N_y of x and y respectively such that $x' + y' \in N$ for any $x' \in N_x$ and $y' \in N_y$. Fortunately, we know there exists $B \in \mathscr{B}$ with $x + y + B \subseteq N$. Next as $\mathscr{B}$ is additive, we know there exists $A \in \mathscr{B}$ with $A + A \subseteq B$. So, set $N_x = x + A$ and $N_y = y + A$. Then $N_x + N_y = x + y + A + A \subseteq x + y + B \subseteq N$.

Next, we show scalar multiplication is continuous. Consider any $c \in \mathbb{C}$ (or $\mathbb{R}$) and $x \in \mathscr{X}$. Then again let N be any neighborhood of cx. Then it suffices to find a neighborhood N_x of x in $\mathscr{X}$ and a disk Δ_c about c in $\mathbb{C}$ (or $\mathbb{R}$) such that $c'x' \in N$ for all $c' \in \Delta_c$ and $x' \in N_x$.

Let $B \in \mathscr{B}$ satisfy that $cx + B \subseteq N$. Then by the additivity of $\mathscr{B}$ we can pick $A \in \mathscr{B}$ such that $A + A + A + A \subseteq B$. Also, we can pick $D \in \mathscr{B}$ with $D \subseteq A$ and $cD \subseteq A$.

To see why, let $n \in \mathbb{N}_{>0}$ satisfy that $2^n \geq |c|$. Then choose D such that $\underbrace{D + \ldots + D}_{2^n \text{ terms}} \subseteq A$. Since D is balanced, we have that:

$$cD \subseteq 2^n D \subseteq \underbrace{D + \ldots + D}_{2^n \text{ terms}} \subseteq A$$

Since D is absorbing and contains 0, we can find some $r > 0$ such that $\alpha x \in D$ when $|\alpha| \leq r$. Let Δ_c be the disk of radius r centered at $c \in \mathbb{C}$ (or $\mathbb{R}$). Also let $N_x = x + D$. Then we claim that $\Delta_c N_x \subseteq N$.

Suppose $y \in N_x = x + D$ and $c' = c + \alpha \in \Delta_c$ where $|\alpha| < r$. Then:

$$c'y = (c + \alpha)(y - x + x) = c(y - x) + cx + \alpha(y - x) + \alpha x$$

But $y - x \in D$. Therefore, we can conclude that $c(y - x) \in cD \subseteq A$. Also as $|\alpha| < r$, we have that $\alpha x \in D \subseteq A$.

Finally, without loss of generality we can assume $r < 1$. Then as D is balanced and $|\alpha| < r$, we have that $\alpha(y - x) \in \alpha D \subseteq D \subseteq A$. So, we can conclude that $c'y \in cx + A + A + A \subseteq cx + B \subseteq N$.

Proof (Uniqueness):
To see that the topology we just defined is unique, first note that any topology satisfying our theorem must have the same collection $\mathcal{N}_0$ of neighborhoods of 0. In other words, $\mathcal{B}$ uniquely determines the neighborhoods of 0. Secondly, note that if our topology is specifically a vector space topology, then the convergence of any net is uniquely determined based on the neighborhoods of 0. Therefore, every topology satisfying our theorem will have the same nets converging. This proves that every topology satisfying our theorem is equal (see *page 229*). ■

As a special case of the prior theorem, we can prove that every quasinormed vector space (with quasinorm $[\cdot]$) has a unique vector space topology such that $\langle x_i \rangle_{i \in I}$ converges to x if and only if $[x_i - x] \to 0$.

Proof:
Let $\mathcal{B} := \{B_r = \{x \in \mathcal{X} : [x] < r\} : r > 0\}$. Checking that $\mathcal{B}$ is a filterbase consisting of balanced absorbing sets is trivial. The only nonobvious statement is that $\mathcal{B}$ is additive. Consider any $r > 0$ and pick $C \geq 1$ satisfying that $[x+y] \leq C([x]+[y])$ for all $x, y \in \mathcal{X}$. Then given any $\ell < \frac{r}{2C}$ we have for all $x, y \in B_\ell$ that $[x+y] \leq C([x]+[y]) \leq 2C\ell < r$. Therefore $B_\ell + B_\ell \subseteq B_r$.

By the prior theorem, there is thus a unique vector space topology with $\mathcal{B}$ as a local base of neighborhoods of 0. Finally, note that a net $\langle x_i \rangle_{i \in I}$ in this topology converges to 0 iff x_i is eventually in B_r for every $B_r \in \mathcal{B}$, and in turn that happens iff $[x_i] \to 0$. By applying a translation, we can thus see that the topology we constructed here is the topology of convergence with respect to this quasinorm.

As a side note, one particularly cursed observation we can make is that we shouldn't generally expect the balls $B_r := \{x : [x] < r\}$ to be open (even though they are neighborhoods of 0).

After all suppose $[x] < r$ and $[z + x] \leq C([z] + [x])$ for all $z \in \mathcal{X}$. Then if B_r were open we must be able to find a ball $B_\ell(x) := \{y : [y - x] < \ell\}$ such that $B_\ell(x) \subseteq B_r$. In other words, given that $[y - x] < \ell$ we'd want to know that

$[y] \leq C([y-x]+[x]) < r$. Yet assuming $C > 1$, we can pick x with $C[x] > r$. Therefore we'd need to have that $C\ell < r - C[x] < 0$. Yet as C and ℓ are positive, this is impossible.

On an optimistic note, you can interpret the prior reasoning as proving that $B_{r/C}$ is contained in the interior of B_r.

If we can get a sharper estimate for $[x+y]$ though, then it may be the case that all the balls are open. For example, the quasinorm topology on weak-L^p does contain all the balls $\{f \in \text{weak-}L^p : [f-g]_p < r\}$ (where $g \in \text{weak-}L^p$ and $r > 0$).

Claim: Suppose $g \in B_r := \{f \in \text{weak-}L^p : [f]_p < r\}$. Then there exists a ball $B_\ell(g) := \{f \in \text{weak-}L^p : [f-g] < \ell\}$ which is contained in B_r. Hence, B_r is open.

Proof:

Given any $t \in (0,1)$, we can conclude that $\lambda_f(\alpha) \leq \lambda_{f-g}((1-t)\alpha) + \lambda_g(t\alpha)$. Then in turn, like on *pages 794-795* we have that:

$$[f]_p \leq \left(\tfrac{1}{(1-t)^p}[f-g]_p^p + \tfrac{1}{t^p}[g]_p^p\right)^{1/p}$$

Now we want to find some ℓ such that if $[f-g]_p < \ell$ then $[f]_p < r$. Fortunately, for all $t \in (0,1)$ we have that:

$$\left(\tfrac{1}{(1-t)^p}[f-g]_p^p + \tfrac{1}{t^p}[g]_p^p\right)^{1/p} < r \text{ if and only if } [f-g]_p^p < (1-t)^p(r^p - \tfrac{1}{t^p}[g]_p^p)$$

As $[g]_p < r$, we can fix t close enough to 1 such that $r^p - \frac{1}{t^p}[g]_p^p$ is positive. Then as $(1-t)^p$ is also positive, we can conclude that $(1-t)^p(r^p - \frac{1}{t^p}[g]_p^p) > 0$. So, there exists some $\ell > 0$ such that if $\ell^p < (1-t)^p(r^p - \frac{1}{t^p}[g]_p^p)$. Then $[f-g]_p < \ell$ implies that $[f]_p < r$. ■

Corollary: The quasinorm topology on weak-L^p is the topology generated (in the typical topological sense) by the balls $B_\ell(g) := \{f \in \text{weak-}L^p : [f-g] < r\}$ (where $g \in \text{weak-}L^p$ and $r > 0$).

3/31/2026

Math 220C Lecture Notes:

A convention I will be using during spring quarter is that if $\Omega \subseteq \mathbb{R}^m$ is an open set and $u : \Omega \to \mathbb{R}$ is a function, then I may denote the jth partial derivative of u as u_{x_j}. Also u_{x_j,x_k} will refer to $(u_{x_j})_{x_k}$ as opposed to $(u_{x_k})_{x_j}$. Usually this distinction won't matter though since we'll typically assume we are working with functions that are C^2 if we need second order partial derivatives.

In specifically math 220c, we'll often identify $\mathbb{C}$ with $\mathbb{R}^2$ via the isometry $x+iy \mapsto (x,y)$. Thus if f is a complex variable function, we shall refer to $\partial_x f = f_x$ as the resulting partial derivative in the direction of the real axis. And similarly, $\partial_y f = f_y$ will be the partial derivative of f in the direction of the imaginary axis.

Recall from *page 332* that if $G \subseteq \mathbb{C}$ is a region, we say $u : G \to \mathbb{R}$ is a harmonic function if $u \in C^2(G)$ and $u_{x,x} + u_{y,y} = 0$.

For example, if $f \in O(G)$ then $u = \text{Re}(f)$ and $v = \text{Im}(f) = \text{Re}(-if)$ are both harmonic functions. In this case, we say u and v are harmonic conjugates.

Lemma: If G is simply connected, then any harmonic function $u : G \to \mathbb{R}$ admits a harmonic conjugate.

Proof:
Let $F = u_x - iu_y$ be a function from G to $\mathbb{C}$. Then note that $F \in C^1(G)$ and satisfies the Cauchy-Riemann equations. After all $(u_x)_x = (-u_y)_y$ by the Laplace equation (i.e. the PDE defining harmonic functions). Meanwhile $(u_x)_y = -(-u_y)_x$ is true since $u \in C^2(G)$. It follows that F is a holomorphic function. And since G is simply connected, we in turn know that $F = f'$ for some $f \in O(G)$.

Write $f(x + iy) = \alpha(x + iy) + i\beta(x + iy)$ where α, β are real-valued functions. Since f' exists (and equals F), we can get f' by differentiating in whatever direction we find convenient. In other words, we know that $u_x - iu_y = F = f' = \alpha_x + i\beta_x$. In turn, we know that $u_x = \alpha_x$ and $-u_y = \beta_x$. Since f is holomorphic and thus satisfies the Cauchy-Riemann equations, we also know $\beta_x = -\alpha_y$. So, this proves that $u_y = \alpha_y$.

Since u and α have the same partial derivatives, we can thus conclude that $u = \alpha + C$ where C is some constant. Yet without loss of generality we can assume $C = 0$. After all, if not then $f - C$ is as valid a primitive of F as f. Finally, we now have that a harmonic conjugate of u is given by the function β. ■

We can also prove the converse statement (as is shown by the following homework exercises). The key observation to make is just that even if an open set $U \subseteq \mathbb{C}$ isn't simply connected, every point in U still has a simply connected neighborhood contained in U (namely some disk of small enough radius). Therefore, any harmonic function *locally* admits a harmonic conjugate. And since satisfying a PDE is a local property, that is often enough for our purposes.

Set 1 Problem 1: If U, V are connected open sets in $\mathbb{C}$, $g : V \to U$ is holomorphic, and $u : U \to \mathbb{R}$ is harmonic, then $u \circ g : V \to \mathbb{R}$ is harmonic in V.

Proof:
Given any $z \in V$, by the continuity of g we can find disks $\Delta_z \subseteq V$ and $\Delta_{g(z)} \subseteq U$ centered at z and $g(z)$ respectively such that $g(\Delta_z) \subseteq \Delta_{g(z)}$. Then as $\Delta_{g(z)}$ is simply connected, we know u has a harmonic conjugate on $\Delta_{g(z)}$. So, we can consider a holomorphic function $f : \Delta_{g(z)} \to \mathbb{C}$ satisfying that $\text{Re}(f) = u$.

To finish off, by chain rule we know that $f \circ g$ is a holomorphic function on Δ_z. Also, we have that $\text{Re}(f \circ g) = u \circ g$. This proves that $u \circ g$ is a harmonic function on Δ_z. Since z was arbitrary, this proves that $(u \circ g)_{x,x} + (u \circ g)_{y,y} = 0$ everywhere on U. So, $u \circ g$ is harmonic in V. ■

Set 1 Problem 3: Let $G \subseteq \mathbb{C}$ be open and connected.

(i) Show that if $h : G \to \mathbb{C}$ is holomorphic and nowhere zero in G, then $u(z) = \log|h(z)|$ is harmonic in G.

Let $u(z) := \log|z|$. If we can show that u is harmonic in $\mathbb{C} - \{0\}$, then the statement we are trying to prove is just an application of problem 1.

Fortunately, note that $u(x+iy) = \log(\sqrt{x^2+y^2}) = \frac{1}{2}\log(x^2+y^2)$. Therefore, we can calculate that $u_x(x+iy) = \frac{x}{x^2+y^2}$ and in turn:

$$u_{x,x}(x+iy) = \frac{x^2+y^2-x(2x)}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}.$$

By symmetry, we thus also have that $u_{y,y}(x+iy) = \frac{x^2-y^2}{(x^2+y^2)^2}$. Hence, $u_{x,x} + u_{y,y} = 0$.

Also $u \in C^\infty(\mathbb{C} - \{0\})$ as $t \mapsto \log(t)$ is a smooth function from $(0,\infty)$ to $\mathbb{R}$ and $(x,y) \mapsto \sqrt{x^2+y^2}$ is a smooth function from $\mathbb{R}^2 - \{(0,0)\}$ to $(0,\infty)$. This finishes the proof that u is harmonic in $\mathbb{C} - \{0\}$.

(ii) Show that G is simply connected if every harmonic function in G admits a harmonic conjugate.

Let $h \in O(G)$ be any nowhere zero holomorphic function on G. To prove that G is simply connected, it suffices to proves that G is a logarithm domain, and to do that it suffices to prove that there exists a function $g \in O(G)$ with $h(z) = e^{g(z)}$.

Yet since every harmonic function in G admits a harmonic conjugate and we know from part (i) that $\log|h(z)|$ is harmonic in G, we know that there is a holomorphic function $g \in O(G)$ with $\mathrm{Re}(g(z)) = \log|h(z)|$. Next, we consider the function $f(z) := \frac{h(z)}{e^{g(z)}}$. Importantly, $f \in O(G)$ and $|f(z)| = \frac{|h(z)|}{e^{\mathrm{Re}(g(z))}} = \frac{|h(z)|}{e^{\log|h(z)|}} = 1$ for all $z \in G$. By the maximum modulus principle, we can thus conclude that f is a constant function (with magnitude 1). And this proves that there exists some $C \in \mathbb{C}$ with $|C| = 1$ and $Ce^{g(z)} = h(z)$. Finally, write $C = e^{is}$ for some $s \in [-\pi, \pi)$. Then $h(z) = e^{g(z)+is}$. This proves that h has a logarithm.

(iii) Show that $u(z) = \log|z|$ admits no harmonic conjugate in $G = \mathbb{C} - \{0\}$.

Suppose for the sake of contradiction that u has a harmonic conjugate $v : \mathbb{C} - \{0\} \to \mathbb{R}$. Then, we'd know the function $f(z) := u(z) + iv(z)$ is a holomorphic function in $O(\mathbb{C} - \{0\})$ satisfying that $|e^{f(z)}| = e^{u(z)} = |z|$ for all $z \in \mathbb{C} - \{0\}$. Thus by identical reasoning to that of part (ii), we can conclude that there exists some $C \in \mathbb{C}$ with $|C| = 1$ and $Ce^{f(z)} = z$ on $\mathbb{C} - \{0\}$. In turn, this would imply the existence of a branch of the logarithm function on $G = \mathbb{C} - \{0\}$.

Yet we know no branch of the logarithm exists on G.

If such a branch did exist, then we'd have to have that $\frac{1}{z}$ has a primitive on $\mathbb{C} - \{0\}$. In turn, we'd have to have that $\int_{|z|=1} \frac{1}{z}\mathrm{d}z = 0$. Yet we can calculate that $\int_{|z|=1} \frac{1}{z}\mathrm{d}z = \pm 2\pi i \neq 0$. ■

Corollary (Elliptic Regularity): If $U \subseteq \mathbb{C}$ is open and $u : U \to \mathbb{R}$ is a harmonic function then u is smooth.

Proof:
Given any $x \in U$ we can consider a disk $\Delta_x \subseteq U$ centered at x. Then $u|_{\Delta_x} = \mathrm{Re}(f)$ for some $f \in O(\Delta_x) \subseteq C^\infty(\Delta_x)$. Since f is smooth, we thus know that $u = \mathrm{Re}(f) = \frac{f+\overline{f}}{2}$ is also smooth on Δ_x (and in particular at x). ■

Given a continuous function $u : G \to \mathbb{R}$, we say u has the mean value property if for all $a \in G$ and $r > 0$ such that $\overline{\Delta}(a, r) \subseteq G$ we have that $u(a) = \frac{1}{2\pi} \int_0^{2\pi} u(a + re^{it})\mathrm{d}t$.

Theorem: If $u : G \to \mathbb{R}$ is harmonic then u satisfies the mean value property.

Proof:
Suppose $\overline{\Delta}(a, r) \subseteq G$. Then there exists $R > r$ such that $\Delta(a, R) \subseteq G$. And since $\Delta(a, R)$ is simply connected, there exists a holomorphic function $f \in O(a, R)$ such that $u = \mathrm{Re}(f)$.

By Cauchy's formula, we have that:

$$f(a) = \frac{1}{2\pi i} \int_{|z-a|=r} \frac{f(z)}{z-a}\mathrm{d}z = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(a+re^{it})}{re^{it}} ire^{it}\mathrm{d}t = \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{it})\mathrm{d}t.$$

Then by taking the real parts of both sides, we get that $u(a) = \frac{1}{2\pi} \int_0^{2\pi} u(a + re^{it})\mathrm{d}t$. ■

Theorem: Let $G \subseteq \mathbb{C}$ be open and connected. If $u : G \to \mathbb{R}$ is continuous and satisfies the mean value property, then u satisfies the maximal principle, meaning that if there exists $a \in G$ with $u(a) \geq u(z)$ for all $z \in G$ then u is a constant function.

Proof:
Suppose such an a exists and set $\Omega := \{z \in G : u(z) = u(a)\}$. Note that $a \in \Omega$ implies that $\Omega \neq \emptyset$. Also, Ω is closed as it is the preimage of a closed set under the continuous function u. Thus, if we can also show that Ω is open then we will know that $\Omega = G$ since G is connected.

Let $z_0 \in \Omega$. Then there exists a closed disk $\overline{\Delta}(z_0, r) \subseteq G$. We want to show $\Delta(z_0, r) \subseteq \Omega$. So let $w \in \Delta(z_0, r)$ and pick $\rho = |w - z_0|$. Then as $\overline{\Delta}(z_0, \rho) \subseteq G$, we know by the mean value property that $u(a) = u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + \rho e^{it})\mathrm{d}t$. In turn, we also have that $\frac{1}{2\pi} \int_0^{2\pi} u(a) - u(z_0 + \rho e^{it})\mathrm{d}t = 0$. Yet as $f(t) := u(a) - u(z_0 + \rho e^{it}) \geq 0$, this implies that $f(t) = 0$ a.e. Also because f is continuous, this implies $f = 0$ everwhere. So, $u(a) = u(z_0 + \rho e^{it})$ for all t. In particular, this proves that $u(w) = u(a)$. ■

Remark: This also shows that if there exists $a \in G$ with $u(a) \leq u(z)$ for all $z \in G$, then u is a constant function. After all, if u satisfies the mean value property then so does $-u$. Then u having a minimum implies $-u$ has a maximum, which in turn implies $-u$ is constant.

Remark 2: It will be relevent on *page 837* to note that the proof of this theorem still works if we assume the weaker statement that for all $z \in G$ there is a closed disk $\overline{\Delta}(z, r) \subseteq G$ such that $u(a) = \frac{1}{2\pi} \int_0^{2\pi} u(a + \rho e^{it})\mathrm{d}t$ for all $0 \leq \rho < r$.

Given an open connected set $G \subseteq \mathbb{C}$, we define its extended boundary:

$$\partial_\infty G \coloneqq \begin{cases} \partial G & \text{if } G \text{ is bounded} \\ \{\infty\} \cup \partial G & \text{if } G \text{ is not bounded.} \end{cases}$$

Another Maximum Modulus Theorem: Let $G \subseteq \mathbb{C}$ be a region and $f : G \to \mathbb{C}$ a holomorphic function. Assume $\exists M > 0$ with $\limsup_{z \to a} |f(z)| \leq M$ for all $a \in \partial_\infty G$. Then $|f(z)| \leq M$ for all $z \in G$.

Note, if $a \in \mathbb{C}$ then:

$$\limsup_{z \to a} |f(z)| \coloneqq \lim_{r \to 0^+} (\sup(\{|f(z)| : z \in G \text{ and } |z - a| < r\}))$$

Meanwhile, we define:

$$\limsup_{z \to \infty} |f(z)| \coloneqq \lim_{r \to \infty} (\sup(\{|f(z)| : z \in G \text{ and } |z| > r\}))$$

Proof:
Consider any $\varepsilon > 0$ and put $\Omega \coloneqq \{z \in G : |f(z)| > M + \varepsilon\}$. Our goal will be to prove that $\Omega = \emptyset$. After all, this will let us say that $|f(z)| \leq M + \varepsilon$ for all $z \in G$. And since ε was arbitrary, we can take $\varepsilon \to 0$ to get that $|f(z)| \leq M$ for all $z \in G$.

To start off, because $|f|$ is a continuous function we know that Ω is an open set. This is important because it means it makes sense to talk about a function being holomorphic on Ω. Next, note that if G is bounded then so is Ω automatically. Meanwhile, if G is unbounded then we still know from the fact that $\limsup_{z \to \infty} |f(z)| \leq M$ that Ω is bounded. This all proves that $\overline{\Omega}$ is a compact subset of $\mathbb{C}$. Thirdly, we can show that $\overline{\Omega} \subseteq G$. After all, as every $a \in \partial G$ has some disk Δ_a such that $\Delta_a \cap G \subseteq \Omega^C$, we know that no point on the boundary of G is an accumulation point of Ω. And since $\Omega \subseteq G$, this implies that $\overline{\Omega} \subseteq \overline{G} - \partial G = G$.

Now if Ω were nonempty, then we'd be able to conclude by the maximum modulus theorem that:

$$M + \varepsilon < \max_{z \in \overline{\Omega}} |f(z)| = \max_{z \in \partial\Omega} |f(z)|$$

Yet this is a contradiction. After all as Ω doesn't include its boundary (since Ω is open), we'd have to have that $|f(z)| \leq M + \varepsilon$ for all $z \in \partial\Omega$.

Thus, we conclude like we wanted that $\Omega = \emptyset$. ■

Theorem (Maximum Principle +): If $G \subseteq \mathbb{C}$ is a region and $u : G \to \mathbb{R}$ is continuous and has the mean value property, then the following holds. If $\forall a \in \partial_\infty G$ we have that $\limsup_{z \to a} u(z) \leq 0$, then either $u < 0$ on G or $u \equiv 0$ on G.

Proof:
It suffices to prove that $u \leq 0$ in G. After all, we can then conclude from the theorem on the bottom of the prior page that if there exists any $z \in G$ with $u(z) = 0$, then $u \equiv 0$ everywhere.

Suppose for the sake of contradiction that there exists $\alpha \in G$ such that $\varepsilon \coloneqq u(\alpha) > 0$. Then let $K \coloneqq \{z \in G : u(z) \geq \varepsilon\}$. We know $K \neq \emptyset$ as $\alpha \in K$. Also, suppose $\{z_n\}_{n \in \mathbb{N}} \subseteq K$. Since $\mathbb{C}_\infty$ is compact, we can pass to a subsequence to get that $z_n \to z$

for some $z \in \mathbb{C}_\infty$. Yet by our limsup assumption we can't have that $z \in \partial_\infty G$. So, we must have that $z \in G$. In turn, by the continuity of u we know that $u(z) \geq \varepsilon$ and thus $z \in K$. This proves that K is a nonempty sequentially compact set.

It follows that u achieves a maximum on K and in turn on G. Yet by the priorly mentioned theorem on page 804, this means u is a strictly positive constant on G. This contradicts that $\limsup_{z \to a} u(z) \leq 0$ for all $a \in \partial_\infty G$. ■

Corollary: If G is a bounded open set, $u : \overline{G} \to \mathbb{R}$ is continuous and has the mean value property on G, and $u \equiv 0$ on ∂G, then $u \equiv 0$ in G.

Proof:
By the prior theorem applied to u and $-u$ on each component of G, we have that:

- $u < 0$ or $u \equiv 0$ for each component,
- $-u < 0$ or $-u \equiv 0$ for each component.

The only way we don't have a contradiction is if $u \equiv 0$ everywhere on G. ■

Corollary 2: If $u, v : \overline{G} \to \mathbb{R}$ are continuous functions that are harmonic in G (where G is a bounded open set), then $u|_{\partial G} \equiv v|_{\partial G} \implies u \equiv v$.

Proof:
Apply the previous corollary to the function $u - v$. ■

As a side note, it is easily checked that harmonic functions form a vector space. Thus, $u - v$ does have the mean value property.

A major consequence of the corollary right above is that if $u : \overline{G} \to \mathbb{R}$ is continuous and also harmonic on G, then u is uniquely determined by its restriction to ∂G. This raises two questions (of which only the first will be partially answered during week 1). Suppose $G \subseteq \mathbb{C}$ is a bounded region.

1. Given a continuous function $u : \overline{G} \to \mathbb{R}$ that is harmonic on G, can we find a formula for u in terms of $u|_{\partial G}$?
2. (The Dirichlet Problem) Given a continuous function $f : \partial G \to \mathbb{R}$, can we find a continuous function $u : \overline{G} \to \mathbb{R}$ that is harmonic on G and satisfying that $u|_{\partial G} = f$?

For now, we will focus on the case that $G = \Delta := \Delta(0, 1)$.

Remark: If $u : \overline{\Delta} \to \mathbb{R}$ is continuous and also harmonic on Δ, then:

$$u(0) = \frac{1}{2\pi} \int_0^{2\pi} u(e^{it}) \mathrm{d}t.$$

Why?
Since u has the mean value property in Δ, we know that $u(0) = \frac{1}{2\pi} \int_0^{2\pi} u(re^{it}) \mathrm{d}t$ for all $r \in (0, 1)$. Yet also note that as u is uniformly continuous on $\overline{\Delta}$, we have that $u(re^{it}) \to u(e^{it})$ uniformly as $r \to 1$. Therefore, we can conclude that:

$$u(0) = \lim_{r \to 1^-} \frac{1}{2\pi} \int_0^{2\pi} u(re^{it}) \mathrm{d}t = \frac{1}{2\pi} \int_0^{2\pi} \lim_{r \to 1^-} u(re^{it}) \mathrm{d}t = \frac{1}{2\pi} \int_0^{2\pi} u(e^{it}) \mathrm{d}t.$$

As for any other $a = re^{i\theta} \in \Delta$, recall that $\varphi(z) := \frac{z+a}{1+\bar{a}z}$ is a biholomorphism of Δ which bijectively maps $\partial\Delta$ to $\partial\Delta$, sends 0 to a, and whose inverse is given by

$\psi(z) \coloneqq \frac{z-a}{1-\overline{a}z}$.

Let $\widetilde{u} \coloneqq u \circ \varphi$. Then $\widetilde{u} : \overline{\Delta} \to \mathbb{R}$ is continuous. Also by set 1 problem 1 (see *page 802*), we know that $\widetilde{u}$ is harmonic on Δ. Therefore, by the prior remark we know that $u(a) = \widetilde{u}(0) = \frac{1}{2\pi}\int_0^{2\pi} \widetilde{u}(e^{it})\mathrm{d}t = \frac{1}{2\pi}\int_0^{2\pi} u(\varphi(e^{it}))\mathrm{d}t$.

We can still do better with this formula however. After noting that $\psi(\partial\Delta) = \partial\Delta$, let's try to do a change of variables.

> In particular, note that there is a smooth bijective function $f : [\alpha, \beta] \to [0, 2\pi]$ satisfying that $\beta - \alpha = \pm 2\pi$ and $e^{if(s)} = \psi(e^{is})$ for all s.
>
> For example, if increasing s causes the argument of $\psi(e^{is})$ to increase then we can pick real numbers $s_0 < s_{\frac{\pi}{2}} < s_\pi < s_{\frac{3\pi}{2}} < s_{2\pi} = s_0 + 2\pi$ satisfying that $\psi(e^{is_{(*)}}) = e^{i(*)}$. Then after letting Log_α denote the obvious branch of the logarithm on the set $\{re^{i\xi} : r > 0 \text{ and } \xi \in (\alpha, \alpha + 2\pi)\}$, we define:
>
> $$f(s) = \begin{cases} -i\,\mathrm{Log}_{-\frac{\pi}{2}}(\psi(e^{is})) & \text{if } s \in [s_0, s_\pi) \\ -i\,\mathrm{Log}_0(\psi(e^{is})) & \text{if } s \in (s_{\frac{\pi}{2}}, s_{\frac{3\pi}{2}}) \\ -i\,\mathrm{Log}_{\frac{\pi}{2}}(\psi(e^{is})) & \text{if } s \in (s_\pi, s_0 + 2\pi] \end{cases}$$
>
> Every subcase of this definition gives a smooth increasing function, every point in $[s_0, s_0 + 2\pi]$ is contained in the interior of one of the domains of the subcases, and the subcases agree with each other on their shared domains due to how we chose the branches of the logarithm. Therefore, by the pasting lemma we can see that f is a well-defined smooth injective function. By the intermediate value theorem it is also then easily seen that f surjectively maps onto $[0, 2\pi]$.
>
> As for the case that increasing s causes the argument of $\psi(e^{is})$ to decrease, then we can do similar reasoning to get a smooth decreasing function f.

Now letting f be from the above note, we have that:

$$\frac{1}{2\pi}\int_0^{2\pi} u(\varphi(e^{it}))\mathrm{d}t = \frac{1}{2\pi}\int_{f^{-1}(0)}^{f^{-1}(2\pi)} u(\varphi(e^{if(s)}))f'(s)\mathrm{d}s = \frac{1}{2\pi}\int_{f^{-1}(0)}^{f^{-1}(2\pi)} u(e^{is})f'(s)\mathrm{d}s$$

Also, we can see that $f'(s) = \frac{-i\psi'(e^{is})}{\psi(e^{is})} \cdot ie^{is} = \frac{\psi'(z)}{\psi(z)}z$ (where $z = e^{is}$).

To quickly take the logarithmic derivative of $\psi(z) = \frac{z-a}{1-\overline{a}z}$, it suffices to sum the logarithmic derivatives of $z - a$ and $(1 - \overline{a}z)^{-1}$. Therefore, we get that:

$$\frac{\psi'(z)}{\psi(z)} = \frac{1}{z-a} + \frac{-(1-\overline{a}z)^{-2}\cdot -\overline{a}}{(1-\overline{a}z)^{-1}} = \frac{1}{z-a} + \frac{\overline{a}}{1-\overline{a}z}$$

In turn, $f'(s) = \frac{z}{z-a} + \frac{\overline{a}z}{1-\overline{a}z} = \frac{z}{z-a} + \frac{\overline{a}z}{1-\overline{a}z} \underbrace{-\frac{1}{2} + \frac{1}{2}}$.

Next note $\frac{z}{z-a} - \frac{1}{2} = \frac{z+a}{2(z-a)}$. Similarly $\frac{1}{2} + \frac{\overline{a}z}{1-\overline{a}z} = \frac{1+\overline{a}z}{2(1-\overline{a}z)} = \frac{\overline{z}z+\overline{a}z}{2(\overline{z}z-\overline{a}z)} = \frac{\overline{z}+\overline{a}}{2(\overline{z}-\overline{a})}$. (For one of those steps recall that $z\overline{z} = |z| = 1$).

Thus, we get that $f'(s) = \mathrm{Re}(\frac{z+a}{z-a}) = \mathrm{Re}(\frac{1+\frac{a}{z}}{1-\frac{a}{z}})$. And using the fact that $z = e^{is}$ and $a = re^{i\theta}$, we get that $f'(s) = \mathrm{Re}(\frac{1+re^{i(\theta-s)}}{1-re^{i(\theta-s)}})$.

To finish off, by rearranging things we get that $\mathrm{Re}(\frac{1+re^{i(\theta-s)}}{1-re^{i(\theta-s)}}) = \frac{1-r^2}{|1-re^{i(\theta-s)}|^2} > 0$ since $0 < r < 1$. This shows that f is in fact increasing. Hence $f^{-1}(2\pi) > f^{-1}(0)$. So, there are $\beta, \alpha \in \mathbb{R}$ with $\beta - \alpha = +2\pi$ satisfying that:

$$u(a) = \frac{1}{2\pi}\int_{\alpha}^{\beta} u(e^{is})\mathrm{Re}(\frac{1+re^{i(\theta-s)}}{1-re^{i(\theta-s)}})\mathrm{d}s$$

As the integrand of this expression is 2π-periodic, we can without loss of generality make $\alpha = 0$ and $\beta = 2\pi$.

Thus, our final simplified equation is that for any $a = re^{i\theta} \in \Delta$ we have that:

$$u(a) = \frac{1}{2\pi}\int_{0}^{2\pi} u(e^{is})\mathrm{Re}(\frac{1+re^{i(\theta-s)}}{1-re^{i(\theta-s)}})\mathrm{d}s. \blacksquare$$

We define the Poisson kernel as $P_r(\theta) = \sum_{n\in\mathbb{Z}} r^{|n|}e^{in\theta}$. Note that $P_r(\theta)$ converges absolutely on $\mathbb{R}$ for all $r \in [0, 1)$.

Lemma: If $z = re^{i\theta}$, then:

(a) $P_r(\theta) = \mathrm{Re}(\frac{1+z}{1-z})$

(b) $P_r(\theta) = \frac{1-|z|^2}{|1-z|^2}$

(c) $P_r(\theta) = \frac{1-r^2}{1-2r\cos(\theta)+r^2}$

Proof:
To show (a) note $\frac{1+z}{1-z} = 1 + \frac{2z}{1-z} = 1 + 2z\sum_{k=0}^{\infty} z^k$. (Since $z = re^{i\theta}$ where $0 \le r < 1$, we know $|z| < 1$ and thus the last equality holds.) It follows that:

$$\frac{1+z}{1-z} = 1 + 2\sum_{k=1}^{\infty} r^k e^{ik\theta}$$

Taking the real parts of both sides, we get that:

$$\mathrm{Re}(\frac{1+z}{1-z}) = 1 + 2\sum_{k=1}^{\infty} r^k\cos(k\theta) = 1 + \sum_{k=1}^{\infty} r^k(e^{ik\theta} + e^{-ik\theta}) = \sum_{k\in\mathbb{Z}} r^{|k|}e^{ik\theta} = P_r(\theta)$$

Next, to show (b) note that $\mathrm{Re}(\frac{1+z}{1-z}) = \mathrm{Re}(\frac{(1+z)(1-\bar{z})}{|1-z|^2}) = \mathrm{Re}(\frac{(1-|z|^2)+(z-\bar{z})}{|1-z|^2}) = \frac{1-|z|^2}{|1-z|^2}$.

Finally, to prove (c) just note that:
$|1-z|^2 = (1-r\cos(\theta))^2 + (r\sin(\theta))^2 = 1 - 2r\cos(\theta) + r^2(\cos^2(\theta) + \sin^2(\theta))$. $\blacksquare$

We can write our prior result about harmonic functions on the unit disk using the Poisson kernel. Namely:

Theorem: Let Δ be the unit disk. Given a continuous function $u : \overline{\Delta} \to \mathbb{R}$ that is harmonic on Δ, we have for all $a = re^{i\theta} \in \Delta$ that:

$$u(a) = \frac{1}{2\pi}\int_{0}^{2\pi} P_r(\theta - s)u(e^{is})\mathrm{d}s.$$

4/2/2026

Math 220c Homework:

Set 1 Problem 2: Let $u : G \to \mathbb{R}$ be a nonconstant harmonic function on a region $G \subseteq \mathbb{C}$. Show that u is an open map.

Lemma: Let $G \subseteq \mathbb{C}$ be an open region and $u : G \to \mathbb{R}$ be a harmonic function. If there is an open set $U \subseteq G$ such that $u|_U \equiv 0$, then $u \equiv 0$ on G.

Proof:
Let Ω be the interior of the set $u^{-1}(\{0\})$ in G. Then Ω is automatically an open set. Also, $U \subseteq \Omega$ implies that Ω is nonempty. So, if, we can show that Ω is closed (relative to G), then we will know from the connectedness of G that $\Omega = G$ and thus $u \equiv 0$ on G.

Suppose $a \in G$ is a limit point of Ω, and pick $r > 0$ such that $\Delta(a, r) \subseteq G$ Then there is a holomorphic function f such that $u = \mathrm{Re}(f)$ on $\Delta(a, r)$. Also, we know that there exists $b \in \Delta(a, r) \cap \Omega$. Hence, we can we find some $\rho > 0$ such that $\Delta(b, \rho) \subseteq \Delta(a, r)$ and $u(z) = \mathrm{Re}(f(z)) = 0$ if $|z - b| < \rho$.

However, as f is purely imaginary on the open set $\Delta(b, \rho)$, we must have by the Cauchy-Riemann equations that $f' = 0$ on that disk. Hence, $f|_{\Delta(b,\rho)} \equiv iC$ for some $C \in \mathbb{R}$. In turn, by the identity principle we know $f \equiv iC$ on $\Delta(a, r)$. And then this means that $u \equiv 0$ on $\Delta(a, r)$. This proves that $a \in \Omega$.

Since Ω contains all its limit points in G, we thus know that Ω is closed in G.

Corollary: Suppose $G \subseteq \mathbb{C}$ is an open region and $u : G \to \mathbb{R}$ is harmonic. If u is constant an open set $U \subseteq G$ then u is constant everywhere on G.

Proof:
Applying the prior lemma to the harmonic function $u - C$ where $C \in \mathbb{R}$ is a constant.

Now we can finally do the problem proper.

Given any open set $U \subseteq G$, consider any $a \in U$ and let $r > 0$ be such that $\Delta(a, r) \subseteq U$. Then $u = \mathrm{Re}(f)$ for some holomorphic function f on $\Delta(a, r)$. By the prior corollary, we know u is nonconstant on $\Delta(a, r)$. In turn, f is nonconstant on $\Delta(a, r)$. Hence, by the open mapping theorem we know that $f(\Delta(a, r))$ is an open subset of $\mathbb{C}$.

Finishing off, since $z \mapsto \mathrm{Re}(z)$ is an open map from $\mathbb{C} \to \mathbb{R}$, we can conclude that $u(\Delta(a, r)) = \mathrm{Re}(f(\Delta(a, r)))$ is an open subset of $\mathbb{R}$ containing $u(a)$. Also as $\Delta(a, r) \subseteq U$ we have that $u(\Delta(a, r)) \subseteq u(U)$. This proves that for any $a \in U$ there is an open neighborhood of $u(a)$ contained in $u(U)$. Hence, $u(U)$ is open. ■

Set 1 Problem 4: Let $G \subseteq \mathbb{C}$ be a region.

(i) Show that if $u : G \to \mathbb{R}$ is harmonic and $e^u + e^{2u}$ is also harmonic, then u is constant.

Since u is harmonic we know that $u_{x,x} + u_{y,y} = 0$. Yet also since $e^u + e^{2u}$ is harmonic, we can calculate that:

$$\begin{aligned} 0 &= (u_{x,x} + u_{y,y})e^u + (u_x^2 + u_y^2)e^u + 2(u_{x,x} + u_{y,y})e^{2u} + 4(u_x^2 + u_y^2)e^{2u} \\ &= 0 + (u_x^2 + u_y^2)e^u + 0 + 4(u_x^2 + u_y^2)e^{2u} \end{aligned}$$

Now as e^u, and e^{2u} are strictly positive and $(u_x^2 + u_y^2)$ is nonnegative, we must have that $u_x^2 + u_y^2 = 0$ on G. In turn, this implies that $u_x = 0 = u_y$ on G. And now as G is connected, this implies that u is constant on G.

(ii) Show that if $h : G \to \mathbb{C}$ is holomorphic and $|h| + |h|^2$ is harmonic, then h is constant.

Note that without loss of generality we can assume there exists $a \in G$ such that $h(a) \neq 0$. After all, if no such a existed then we'd already be done as h is trivially the constant zero function. Next, by the continuity of h as well as the openness of G, we can pick $r > 0$ such that $\Delta(a, r) \subseteq G$ and h is nowhere zero on $\Delta(a, r)$.

By set 1 problem 3(i) (see *page 803*), we then know that $u(z) \coloneqq \log |h(z)|$ is harmonic on $\Delta(a, r)$. Yet by the assumption of the problem we also know that $e^u + e^{2u} = |h| + |h|^2$ is harmonic on $\Delta(a, r)$. So by part (i) we can conclude that $u(z)$ is constant on $\Delta(a, r)$. As $\log$ is injective, this in turn implies that there exists some $R > 0$ such that $|h(z)| = R$ for all $z \in \Delta(a, r)$.

By the maximum modulus principle, this then implies that h is constant on G.

I had a different very overcomplicated proof of the prior problem that I wrote out before replacing it with the current solution shown. That said, I thought I should keep one of the results I showed during that proof.

Lemma: Suppose $G \subseteq \mathbb{C}$ is a connected open set and $F \subseteq G$ is a closed set with no limit points in G. Then $G - F$ is also connected and open.

It's clear that $G - F$ is open. So, we just need to show that it is connected.

Suppose for the sake of contradiction that there exists nonempty open disjoint sets U_1, U_2 whose union is $G - F$. Then consider any $a \in F$. By the openness of G plus the fact each point in F is isolated, we can find some $r > 0$ such that $\Delta(a, r) \subseteq G$ and $\Omega \coloneqq \Delta(a, r) - \{a\} \subseteq G - F$. Yet now as Ω is connected and $\Omega \cap U_1, \Omega \cap U_2$ are disjoint open subsets of Ω covering Ω, we must have that $\Omega \subseteq U_1$ or $\Omega \subseteq U_2$. It follows that $a \in \overline{U}_j$ for exactly one j. Also for the j such that $a \in \overline{U}_j$, we also have that $U_j \cup \{a\}$ is an open subset of G disjoint from $U_{j'} \cup (F - \{a\})$ where $j' \in \{1, 2\} - \{j\}$.

For both j let $A_j \coloneqq \{a \in F : a \in \overline{U}_i\}$ and $V_j \coloneqq \bigcup_{a \in A_i} (U_i \cup \{a\})$. Then:

- Each V_j is open as it is a union of sets which we proved are open in the prior paragraph.
- Also as $A_1 \cup A_2 = F$, we know that $V_1 \cup V_2 = G$.
- Thirdly, as $U_1 \cup \{a_1\}$ and $U_2 \cup \{a_2\}$ is disjoint for all $a_1 \in A_1$ and $a_2 \in A$, we have that $V_1 \cap V_2 = \emptyset$.
- Fourthly, as $\emptyset \neq U_j \subseteq V_j$ for both j, we have that neither V_1 nor V_2 is empty.

This contradicts that G is connected. So, we can conclude no such U_1, U_2 exist and thus $G - F$ is also connected. ■

The other way I tried going about this proof before I tried this approach was that you can show that any path γ in G going between two points $a, b \in G - F$ can be rerouted so as to avoid F. I stopped with that approach though because the exact expressions in that rerouting were symbolically messy.

Set 1 Problem 5:

(i) (Liouville's theorem) Show that if $u : \mathbb{R}^2 \to \mathbb{R}$ is harmonic and bounded either from below or from above, then u is constant.

To start off, it suffices to prove that u is constant if u is harmonic on $\mathbb{R}^2$ and $u \geq 0$. After all:

- u is harmonic and bounded above if and only if $-u$ is harmonic and bounded below. And similarly u is constant if and only if $-u$ is constant. So by replacing u with $-u$ if necessary we can assume without loss of generality that u is bounded below.
- Also u is harmonic with $u \geq C$ iff $u - C$ is harmonic with $u - C \geq 0$. Similarly, $u - C$ is constant if and only if u is constant. So, by replacing u with $u - C$ we can assume without loss of generality that u is bounded below by 0.

Now as $\mathbb{R}^2 \cong \mathbb{C}$ is simply connected, there exists an entire function $f \in O(\mathbb{C})$ such that $u(x, y) = \operatorname{Re}(f(x + iy))$. In turn, if we consider the entire function $g(z) := e^{-f(z)}$, we have that $|g(x + iy)| = |e^{-f(x+iy)}| = e^{-u(x+iy)} \leq 1$ for all $x, y \in \mathbb{R}$. By Liouville's theorem for holomorphic functions, we can thus conclude that $g = e^{-f} \equiv C$ for some constant C.

Since $\{z \in \mathbb{C} : e^z = C\}$ consists entirely of isolated points and $-f$ is continuous, this in turn implies that f must also be consant. So, $u = \operatorname{Re}(f)$ is also a constant function.

(ii) Let $G \subseteq \mathbb{C}$ be a simply connected region. Then show that $G = \mathbb{C}$ if and only if every positive harmonic function $h : G \to \mathbb{R}$ is constant.

$(\Longrightarrow)$
This is just a special case of part (i).

$(\Longleftarrow)$
Suppose $G \neq \mathbb{C}$. Then by composing the Cayley transform with a biholomorphism given by the Riemann mapping theorem, we can find a biholomorphism $\varphi : G \to \mathfrak{h}^+$ where $\mathfrak{h}^+$ is the upper half plane. In turn, $u := \operatorname{Im}(\varphi)$ is a strictly positive nonconstant harmonic function on G. ■

Set 1 Problem 6 (Hadamard's Three Circle Theorem): Let Δ be the unit disk and $f \in O(\Delta)$. Also assume $f(z) \neq 0$ for $z \neq 0$, and for $0 \leq r < 1$ let $M(r) := \max_{|z|=r} |f(z)|$.

(i) Show that the function $h_s(z) = s \log |z| + \log |f(z)|$ is harmonic in $\Delta - \{0\}$ for all $s \in \mathbb{R}$.

Firstly, real linear combinations of harmonic functions are also harmonic functions. Also we already showed in problem 3(i) (see *page 803*) that $\log|f(z)|$ and $\log|z|$ are harmonic in $\Delta - \{0\}$ since $f(z)$ and z are holomorphic functions that are nowhere zero on that set. Thus, we can conclude that h_s is harmonic on $\Delta - \{0\}$.

(ii) Show that there exists $s \in \mathbb{R}$ such that $\max_{|z|=\frac{1}{2}}(h_s(z)) = \max_{|z|=\frac{1}{8}}(h_s(z))$.

Note that $\max_{|z|=\frac{1}{2}}(h_s(z)) = -\log(2)s + C$ where $C = \max_{|z|=\frac{1}{2}}(\log|f(z)|)$ is constant with respect to s. Similarly, $\max_{|z|=\frac{1}{8}}(h_s(z)) = -3\log(2)s + C'$ where $C' = \max_{|z|=\frac{1}{8}}(\log|f(z)|)$ is constant with respect to s.

Next, since $\log$ is an increasing function, we can see that:

$$C = \max_{|z|=\frac{1}{2}}(\log|f(z)|) = \log(\max_{|z|=\frac{1}{2}}|f(z)|) = \log(M(\tfrac{1}{2})).$$

Similarly, we have that $C' = \log(M(\frac{1}{8}))$.

Finally, we now solve for s such that:

$$-\log(2)s + \log(M(\tfrac{1}{2})) = -3\log(2)s + \log(M(\tfrac{1}{8})).$$

The unique s that works is $s = \frac{\log(M(\frac{1}{8}))-\log(M(\frac{1}{2}))}{2\log(2)} = \frac{\log\left(\frac{M(\frac{1}{8})}{M(\frac{1}{2})}\right)}{2\log(2)}$.

(iii) Show that $M(\frac{1}{4})^2 \leq M(\frac{1}{2})M(\frac{1}{8})$.

Consider the annulus $G := \{z \in \Delta : \frac{1}{8} < z < \frac{1}{2}\}$. As h_s is a continuous function and $\overline{G}$ is a compact subset of Δ, we know that $\max_{z\in\overline{G}}(h_s(z))$ exists. Yet also note that as G is open and connected and h_s is harmonic on G, we know that h_s achieves a maximum on G iff h_s is constant on G. Yet if the latter happens, then h_s is also constant on $\overline{G}$ by the continuity of h_s. So no matter if h_s achieves a maximum on G or not, we can conclude that:

$$\max_{z\in\overline{G}}(h_s(z)) = \max_{z\in\partial G}(h_s(z))$$

In particular, if we let s be the real number we found in part (ii), then we can say that:

$$\begin{aligned}\max_{z\in\partial G}(h_s(z)) = \max_{|z|=\frac{1}{2}}(h_s(z)) &= -\left(\frac{\log\left(\frac{M(\frac{1}{8})}{M(\frac{1}{2})}\right)}{2\log(2)}\right)\log(2) + \log(M(\tfrac{1}{2})) \\ &= -\tfrac{1}{2}(\log(M(\tfrac{1}{8})) - \log(M(\tfrac{1}{2}))) + \log(M(\tfrac{1}{2})) \\ &= \tfrac{3}{2}\log(M(\tfrac{1}{2})) - \tfrac{1}{2}\log(M(\tfrac{1}{8}))\end{aligned}$$

To finish off, note that:

$$\begin{aligned}\max_{|z|=\frac{1}{4}}(h_s(z)) &= -2\left(\frac{\log\left(\frac{M(\frac{1}{8})}{M(\frac{1}{2})}\right)}{2\log(2)}\right)\log(2) + \log(M(\tfrac{1}{4})) \\ &= \log(M(\tfrac{1}{2})) - \log(M(\tfrac{1}{8})) + \log(M(\tfrac{1}{4})).\end{aligned}$$

Therefore, as $\max_{|z|=\frac{1}{4}}(h_s(z)) \leq \max_{z\in G}(h_s(z)) \leq \max_{z\in\overline{G}}(h_s(z)) = \max_{|z|=\frac{1}{2}}(h_s(z))$, we've proven that:

$$\log(M(\tfrac{1}{4})) \leq \tfrac{1}{2}\log(M(\tfrac{1}{2})) + \tfrac{1}{2}\log(M(\tfrac{1}{8})) = \log\left(\sqrt{M(\tfrac{1}{2})M(\tfrac{1}{8})}\right)$$

This proves that $M(\frac{1}{4})^2 \leq M(\frac{1}{2})M(\frac{1}{8})$.

(iv) Show that equality in (iii) holds if and only if $f(z) = az^n$ for some $a \in \mathbb{C}$ and integer $n \geq 0$.

$(\Longrightarrow)$

Let G be the annulus from part (iii) and let s be the real number from part (ii). If equality holds then we know h_s achieves a maximum on G and is therefore a constant function on G. By the lemma I wrote in my answer to problem 2 (see *page 809*), this also in turn lets us conclude that $h_s(z) = s\log|z| + \log|f(z)|$ is constant on $\Delta - \{0\}$. After some rearranging, we can thus conclude that there exist $C > 0$ such that $|f(z)| = C|z|^{-s}$ when $z \in \Delta - \{0\}$.

Next, as f is holomorphic on Δ, there exists $\{c_k\}_{k\in\mathbb{N}} \subseteq \mathbb{C}$ such that $f(z) = \sum_{k=0}^{\infty} c_k z^k$ (where the latter infinite sum converges absolutely when $|z| < 1$). Also let n be the least integer for which $c_k \neq 0$.

- By observing that $c_n = \lim_{z\to 0}|z^{-n}f(z)| = \lim_{z\to 0} C|z|^{-s-n}$, we can see that $-s-n \geq 0$. Or in other words, $-n \geq s$.
- Also by observing for any $r > n$ that $\infty = \lim_{z\to 0}|z^{-r}f(z)| = \lim_{z\to 0} C|z|^{-s-r}$, we can see that $-s-r < 0$ for all $r > n$. In particular, for all $\varepsilon > 0$ this shows that $-n-\varepsilon < s$. And by taking $\varepsilon \to 0$ this proves that $-n \leq s$.

This proves that $|f(z)| = C|z|^n = |Cz^n|$ for some integer $n \geq 0$ and $C > 0$.

To finish off, set $g(z) = \frac{f(z)}{z^n}$ when $z \in \Delta - \{0\}$. Then as $|g(z)| = C$ for all $z \in \Delta - \{0\}$, we know by the maximum modulus principle that $g(z) = a$ for some $a \in \mathbb{C}$. In turn, this proves that $f(z) = az^n$ on $\Delta - \{0\}$. By taking the limit as $z \to 0$, we can also then see that $f(z) = az^n$ holds when $z = 0$.

$(\Longleftarrow)$

Note that if $f(z) = az^n$ where $a \in \mathbb{C}$ and $n \geq 0$ is an integer, then we can calculate that $M(r) = \max_{|z|=r}|az^n| = |a|r^n$. In particular, this tells us that:

$$M(\tfrac{1}{4})^2 = |a|^2(\tfrac{1}{4})^{2n} = |a|^2(\tfrac{1}{16})^n = |a|^2(\tfrac{1}{2})^n(\tfrac{1}{8})^n = M(\tfrac{1}{2})M(\tfrac{1}{8})$$

4/3/2026

I'm choosing to audit math 240c this quarter just cause I'm not particularly happy with the amount of Fourier analysis I digested last Spring. And while I'm not planning on fully redoing all my notes for that class, I will be solving some things out which I feel motivated to do so. Also the professor this time for math 240c is the same person as my math 241a professor.

Given any positive integer n, we let $(\mathbb{R}^n, \mathcal{L}_n, m)$ denote the completion of the Lebesgue measure on the Borel σ-algebra $\mathcal{B}_{\mathbb{R}^n}$ of $\mathbb{R}^n$. As a reminder, we say a function $f : \mathbb{R}^m \to \mathbb{R}^n$ is:

- Borel measurable if f is $(\mathcal{B}_{\mathbb{R}^m}, \mathcal{B}_{\mathbb{R}^n})$-measurable;
- Lebesgue measurable if f is $(\mathcal{L}_m, \mathcal{B}_{\mathbb{R}^n})$-measurable.

An issue I didn't think to address last Spring was about proving that the measurability of $f * g(x) \coloneqq \int f(x-y)g(y)\mathrm{d}y$ is never an issue. So, I'm going to address that now.

Firstly define $s : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ by $s(x,y) \coloneqq x - y$. Since s is continuous, we always have that s is Borel-measurable. Hence, if $f, g : \mathbb{R}^n \to \mathbb{R}^n$ are also Borel measurable then we have that $f(s(x,y))g(y)$ is $(\mathcal{B}_{\mathbb{R}^n \times \mathbb{R}^n}, \mathcal{B}_{\mathbb{R}^n})$-measurable.

That said, if f and g are merely Lebesgue measurable then we need the following exercise in order to show that $f(s(x,y))g(y)$ is $(\mathcal{L}_{2n}, \mathcal{B}_{\mathbb{R}^n})$-measurable.

> As a side note $\mathcal{L}_{2n}$ is equivalently the completion of $\mathcal{L}_n \otimes \mathcal{L}_n$ and the completion of $\mathcal{B}_{\mathbb{R}^n} \otimes \mathcal{B}_{\mathbb{R}^n}$ (when we identify $\mathbb{R}^{2n}$ with $\mathbb{R}^n \times \mathbb{R}^n$).

(Folland) Exercise 8.5: Show that s is $(\mathcal{L}_{2n}, \mathcal{L}_n)$-measurable.

> Let $E \subseteq \mathbb{R}^n$ be a Lebesgue measurable set and consider the invertible linear map $T(x,y) = (x-y, x+y)$. It is easy to see that $T(s^{-1}(E)) = E \times \mathbb{R}^n$ and is thus in $\mathcal{L}_n \otimes \mathcal{L}_n \subseteq \mathcal{L}_{2n}$. In turn, as T is $(\mathcal{L}_{2n}, \mathcal{L}_{2n})$-measurable, we have that $s^{-1}(E) \in \mathcal{L}_{2n}$. ■

Now in either the case that f, g are Borel or Lebesgue measurable, we know that $y \mapsto f(x-y)g(y)$ is Lebesgue measurable for all fixed x. This is because affine maps of y are both $(\mathcal{B}_{\mathbb{R}^n}, \mathcal{B}_{\mathbb{R}^n})$-measurable and $(\mathcal{L}_n, \mathcal{L}_n)$-measurable.

Also let $E \subseteq \mathbb{R}^n$ be the set of all x such that $\int |f(x,y)g(y)|\mathrm{d}y < \infty$.

> As a side note, $K(x,y) \coloneqq f(x,y)g(y)$ being measurable implies $|K(x,y)|$ is as well. Then by either Tonelli's theorem for complete measures (see *pages 179-181*) or the Tonelli's theorem for product measures in my paper math 240a notes, we can conclude that $H(x) \coloneqq \int |K(x,y)|\mathrm{d}(m \times m)$ is a measurable function into $[0, \infty]$. Therefore, $E = H^{-1}([0,\infty))$ is guarenteed to be a measurable set.

Similarly to in that side note, we can also apply Tonelli's theorem to the positive and negative parts of the real and imaginary parts individually and then restrict to E in order to show that the integral of each part is measurable on E. In turn, their sum $x \mapsto \int f(x-y)g(y)\mathrm{d}y$ is a Lebesgue measurable function on E.

4/4/2026

Math 200c Lecture Notes:

I would like to start off with an exercise from the optional homework at the end of last quarter. As a side note, while professor Alireza used the convention for field extensions that $F \subseteq K \subseteq E$, professor Rogalski uses the convention that $F \subseteq E \subseteq K$.

(Math 200b) Set 9 Problem 10: Suppose K is a splitting field of $f \in F[x]$ over F.

(a) Prove that K/F is a normal extension.

Suppose $\alpha \in K$ and let $g(x) := f(x)m_{\alpha;F}(x)$. We need to show $m_{\alpha;F}(x)$ splits into a product of linear polynomials in $K[x]$. In order to do this let K' be a splitting field of $g(x)$ over F. Notably, we can embed the field K into K'. After all, $f \mid g$. So, K' must contain a splitting subfield of f which is thus isomorphic to K.

Consider any $\alpha' \in V_{m_{\alpha;F}}(K')$. Then there exists a unique isomorphism $\theta : F[\alpha] \to F[\alpha']$ fixing F and sending α to α' (see *pages 762-763*). Next, because $g = g^{\theta}$ and K' is a splitting field of g, we can thus conclude that there is a θ-isomorphism $\widetilde{\theta} \in \mathrm{Aut}_{\theta}(K')$.

Finally, note that $\widetilde{\theta}$ must permute the zeros of f. Hence, $\widetilde{\theta}(K) = K$. Consequently, as $\alpha \in K$ we also know that $\alpha' = \widetilde{\theta}(\alpha) \in K$. This proves that every zero of $m_{\alpha;F}$ is actually in K. And now we are done.

As a side note, this also shows K is a splitting field of $f(x)m_{\alpha;F}(x)$ for any $\alpha \in K$

(b) Prove that for every $\alpha \in K$, $\mathrm{Aut}_F(K)$ acts transitively on $V_{m_{\alpha;F}}(K)$.

In the proof of the prior part, $\widetilde{\theta}$ gives us an element of $\mathrm{Aut}_F(K)$ sending α to α' where α' was an arbitrary element of $V_{m_{\alpha;F}}(K)$. Thus if we consider $\mathrm{Aut}_F(K)$ as acting on $V_{m_{\alpha;F}}(K)$ via permuting the elements of that set, then this shows that there is only one orbit.

Let K/F be a finite extension. On *pages 775-778*, I wrote down a bunch of theorems concerning a correspondance between $\mathrm{Sub}(\mathrm{Aut}_F(K))$ and $\mathrm{Int}(K/F)$. There are two other theorems related to this correspondance to be aware of.

Theorem: Let K/F be a finite Galois extension and let $G = \mathrm{Gal}(K/F)$. Also let $E \in \mathrm{Int}(K/F)$.

1. The following are equivalent:
   - (i) E/F is a Galois extension;
   - (ii) E/F is a normal extension;
   - (iii) $\sigma(E) = E$ for all $\sigma \in G$
   - (iv) $\mathrm{Gal}(K/E) \lhd G$.

Proof:
$(i \iff ii)$
K/F being Galois means that K/F is separable. This in turn automatically means E/F is separable. And since E/F is also a finite extension, the only condition of E/F being Galois that is not immediate is that E/F is normal.

$(ii \iff iii)$
Suppose (ii) and given any $\sigma \in G$, let $a \in E$ and $f = m_{a;F}$. Since E/F is normal we know f splits in E. Also $\sigma(a)$ is another root of f. So, $\sigma(a) \in E$. And this proves that $\sigma(E) \subseteq E$ for all $\sigma \in G$. Since this reasoning also applies to σ^{-1}, we can in turn conclude $\sigma(E) = E$.

Conversely, suppose $\sigma(E) = E$ for all $\sigma \in G$. Then pick $a \in E$ with $f = m_{a;F}$. Since K/F is a normal extension, we know f has distinct roots $a_1, \ldots, a_n \in K$ (where $n = \deg(f)$). Yet for each i there also exists some $\sigma \in G$ such that $\sigma(a) = a_i$. (This is a consequence of the problem on the last page). So, every $a_i \in \sigma(E) = E$. And this proves that f splits over E.

$(iii \iff iv)$
Suppose (iii) and then consider any $\tau \in H = \mathrm{Gal}(K/E)$ and $\sigma \in G$. Then if $a \in E$, we have that $\sigma^{-1}\tau\sigma(a) = a$. Hence $\sigma^{-1}\tau\sigma \in H$ and we've shown that $H \lhd G$.

Conversely, if $H \lhd G$, $a \in E$, $\sigma \in G$, and $\tau \in H$, then;

$$\sigma^{-1}\tau\sigma(a) = a \implies \tau\sigma(a) = \sigma(a) \implies \sigma(a) \in \mathrm{Fix}(H).$$

Finally, as K/E is Galois, we have that $\mathrm{Fix}(H) = E$. So, $\sigma(a) \in E$ for all $a \in E$ and $\sigma \in G$. This proves that $\sigma(E) \subseteq E$ for all σ. By applying this to σ^{-1} we also get that $\sigma(E) = E$.

2. If E/F is a normal extension, then $\frac{\mathrm{Gal}(K/F)}{\mathrm{Gal}(K/E)} \cong \mathrm{Gal}(E/F)$.

Consider the map $\Phi : \mathrm{Gal}(K/F) \to \mathrm{Gal}(E/F)$ given by $\sigma \mapsto \sigma|_E$.

- This is well-defined because $\sigma(E) = E$ for all $\sigma \in G$.
- Also, this is a group homomorphism as $(\sigma \circ \tau)|_E = \sigma|_E \circ \tau|_E$.
- $\ker(\Phi) = \{\sigma \in \mathrm{Gal}(K/F) : \sigma|_E = \mathrm{Id}_E\} = \mathrm{Gal}(K/E)$.
- Φ is onto. After all, if $\tau \in \mathrm{Gal}(E/F)$ then since K is a splitting field of some $f(x)$ over E, there exists a lif $\widetilde{\tau} \in \mathrm{Gal}(K/E)$ such that $\widetilde{\tau}|_E = \tau$.

Now just apply the first isomorphism theorem. ■

Set 1 Problem 4: Let K/F be a finite Galois extension and let E, L be intermediate subfields of F and K. Then there's an isomorphism of fields $\theta : E \to L$ with $\theta : E \to L$ with $\theta|_F = \mathrm{Id}_F$ if and only if the subgroups $\mathrm{Gal}(K/L)$ and $\mathrm{Gal}(K/E)$ are conjugate subgroups of $G := \mathrm{Gal}(K/F)$.

$(\implies)$
Since K/F is a Galois extension, we know that K is a splitting field of some separable polynomial $f(x) \in F[x]$. In turn, we can also say K is a splitting field of f over both E

and L. So by the theorem on *page 764*, there is some automorphism $\widetilde{\theta} : K \to K$ such that $\widetilde{\theta}|_E = \theta$. Because $\theta|_F = \mathrm{Id}_F$, we know that $\widetilde{\theta} \in \mathrm{Gal}(K/F)$. And thus, we've proven that there exists some $\sigma \in \mathrm{Gal}(K/F)$ such that $L = \sigma(K)$.

To finish off, I establish the following result.

Lemma: Given a Galois extension K/F and an intermediate field $F \subseteq E \subseteq K$, we have that $\mathrm{Gal}(K/\sigma(E)) = \sigma\mathrm{Gal}(K/E)\sigma^{-1}$ for every $\sigma \in \mathrm{Gal}(K/F)$.

Proof:
Suppose $\tau \in \mathrm{Gal}(K/E)$ and consider any $a \in \sigma(E)$. Then there exists $b \in E$ such that $a = \sigma(b)$. Therefore $(\sigma \circ \tau \circ \sigma^{-1})(a) = (\sigma \circ \tau)(b) = \sigma(b) = a$, and this proves that $\sigma \circ \tau \circ \sigma^{-1} \in \mathrm{Gal}(K/\sigma(E))$ for all $\tau \in \mathrm{Gal}(K/E)$. Hence, $\sigma\mathrm{Gal}(K/E)\sigma^{-1} \subseteq \mathrm{Gal}(K/\sigma(E))$.

As for the other containment, just note that $\sigma^{-1}(\sigma(E)) = E$. So, by the above reasoning we also have that $\sigma^{-1}\mathrm{Gal}(K/\sigma(E))\sigma \subseteq \mathrm{Gal}(K/E)$. Equivalently, this gives us that $\sigma\mathrm{Gal}(K/E)\sigma^{-1} \supseteq \mathrm{Gal}(K/\sigma(E))$.

As a consequence of the above lemma, we have that $\mathrm{Gal}(K/L) = \sigma\mathrm{Gal}(K/E)\sigma^{-1}$ for some $\sigma \in \mathrm{Gal}(L/F)$. In other words, $\mathrm{Gal}(K/L)$ and $\mathrm{Gal}(K/E)$ are conjugate subgroups.

$(\Longleftarrow)$
If $\mathrm{Gal}(K/L)$ and $\mathrm{Gal}(K/E)$ are conjugate subgroups, then we know there exists $\sigma \in \mathrm{Gal}(K/F)$ such that $\mathrm{Gal}(K/L) = \sigma\mathrm{Gal}(K/E)\sigma^{-1}$. Yet by the prior lemma we also know that $\sigma\mathrm{Gal}(K/E)\sigma^{-1} = \mathrm{Gal}(K/\sigma(E))$. Therefore, we must have that:

$$L = \mathrm{Fix}(\mathrm{Gal}(K/L)) = \mathrm{Fix}(\mathrm{Gal}(K/\sigma(E))) = \sigma(E)$$

It follows that $\sigma|_E$ is an isomorphism of fields from E to L satisfying that $(\sigma|_E)|_F = \mathrm{Id}_F$. ■

As a side note which I hopefully will never say again:
The theorem on *page 774* says K/F is a Galois extension iff K is a splitting field of an irreducible-separable polynomial f. Yet by taking the square-free radical of f, we are then also able to say that K/F is a splitting field of a separable polynomial. So without loss of generality we can just assume f is a separable polynomial.

Given field extensions $F \subseteq E \subseteq K$, we often want to write $E = F[\alpha]$ for some α. (We call α a primitive element.) As we will show next, in most cases that we care about we can do this.

Lemma: If V is a finite dimension vector space over an infinite field F, then we cannot write $V = \bigcup_{i=1}^{m} W_i$ where each W_i is a proper F-subspace of V.

Proof:
We proceed by induction on $\dim_F(V)$. In the trivial case that $V = \{0\}$, then there are no proper F-subspaces. So the lemma is trivially true. Meanwhile if $\dim_F(V) = 1$, then the only proper subspace of V is $\{0\}$. So, we'd have that $\bigcup_{i=1}^{m} W_i = \{0\} \neq V$.

The actual non-trivial base case we need to prove is when $\dim_F(V) = 2$. As we will see, this is where we need F to be infinite.

Let v_1, v_2 be a basis of V and suppose for the sake of contradiction that $V = \bigcup_{i=1}^m W_i$ where each W_i is a proper subspace. Without loss of generality we can assume each W_i has dimension 1. After all, at least one W_i has dimension 1 (lest $\bigcup_{i=1}^m W_i = \{0\} \neq V$). Then, the dimension 1 subspaces each contain all of the dimension 0 subspaces.

Write $W_i = \text{span}(a_i v_1 + b_i v_2)$ for some $a_i, b_i \in F$. Then given any $av_1 + bv_2 \in V$ we must have that either $b = 0$ or $a/b = a_i/b_i$ for some i since $V = \bigcup_{i=1}^m W_i$. Yet as F is infinite, there exists some $c \in F$ such that $c \neq a_i/b_i$ for any i. And now we have a contradiction since for any nonzero $d \in F$ we know that $cdv_1 + dv_2 \in V$ but not in $\bigcup_{i=1}^m W_i$.

Finally, we now assume $\dim_F(V) = n \geq 3$ and that the lemma holds for all finite dimension F-vector spaces W where $\dim_F(W) < n$. If the lemma was not true for V, then there'd have to exist some minimum $m \in \mathbb{N}$ for which there exists proper subspaces $W_1, \ldots, W_m$ satisfying that $V = \bigcup_{i=1}^m W_i$.

Claim: For any fixed $j \in \{1, \ldots, m\}$, we have that $W_j \not\subseteq \bigcup_{i \neq j} W_i$.

To see why, note that if $W_j \subseteq \bigcup_{i \neq j} W_i$ then we'd also know $W_j = \bigcup_{i \neq j}(W_j \cap W_i)$. Yet now by our inductive hypothesis, the only way this would be possible is if $W_j \cap W_{i'} = W_j$ for some $i' \neq j$. So, we'd have that $W_j \subseteq W_{i'}$. And this contradicts the minimality assumption about m.

Because of the above claim, we can pick $w \in W_m - \bigcup_{i=1}^{m-1} W_i$ and $w' \in W_1 - \bigcup_{i=2}^m W_i$. Then, after setting $W = \text{span}(w, w')$, we know that W is a proper subspace of V since $\dim_F(V) > 2$. At the same time though, we know that $W \cap W_i \neq W$ for any i. Thus, we have a contradiction to our inductive hypothesis since:

$$W = W \cap V = W \cap (\bigcup_{i=1}^m W_i) = \bigcup_{i=1}^m (W \cap W_i). \blacksquare$$

Proposition: Let K/F be a finite extension of fields. Then the following are equivalent:

(a) $K = F[\alpha]$ for some $\alpha \in K$.

(b) There exists finitely many intermediate fields E with $F \subseteq E \subseteq K$.

$(b \implies a)$

If F is finite field, then so is K since K/F is a finite extension. In turn, we know that $K^\times$ is cyclic. And if $\alpha \in K$ is a generator for $K^\times$, we know $F[\alpha] = K$.

Meanwhile if F is infinite, then since there are finitely many intermediate fields we can index them to say that $E_1, \ldots, E_m$ are all the proper (i.e. $\neq K$) intermediate fields. By the prior lemma, we then know $K \neq \bigcup_{i=1}^m E_i$. So, if we pick $\alpha \in K - \bigcup_{i=1}^m E_i$ we have that $F[\alpha] = K$.

$(a \implies b)$

Assume $K = F[\alpha]$ and consider any intermediate field $F \subseteq E \subseteq K$. Also note that $K = F[\alpha] \subseteq E[\alpha] = K$.

Let $f(x) = m_{\alpha;E}(x) = x^m + a_{m-1}x^{m-1} + \ldots + a_1x + a_0 \in E[x]$. Then if we consider the field $E' \coloneqq F[a_0, a_1, \ldots, a_{m-1}] \subseteq E$, we have that $m_{\alpha;E'} = f$. Hence, it follows that $[K : E] = \deg(f) = [K : E']$, and in turn we know that $E = E'$.

Also note though that if $g = m_{\alpha;F} \in F[x]$, then we must have that $f \mid g$ in $E[x] \subseteq K[x]$. Therefore, every intermediate field E is given by the coefficients of some monic divisor of g in $K[x]$. But since $K[x]$ is a UFD, g only has finitely many monic divisors. So, there only are finitely many intermediate fields E. ■

This proposition is useful because we can show as an easy consequence of the following lemma that if K/F is a finite separable extension, then there are guarenteed to be only finitely many intermediate fields E with $F \subseteq E \subseteq K$. Hence, we only possibly have that $K \neq F[\alpha]$ for any $\alpha \in K$ if K/F is *not* a separable extension.

Lemma (Existence of Galois closures): If K/F is a finite separable extension, then there exists a finite extension L/K such that L/K and L/F are Galois extensions.

Since K/F is a finite extension, we can write $K = F[\alpha_1, \ldots, \alpha_n]$. Next, we let $f_i \coloneqq m_{\alpha_i;F}$. Then each f_i is separable since K/F is a separable extension. If we let g be the product of all distinct f_i, then g will be separable too.

Let L be a splitting field of g over K. Then L/K is automatically a Galois extension. Also if $\beta_1, \ldots, \beta_m$ are the roots of g then $L = K[\beta_1, \ldots, \beta_m]$. Yet also note that each of the α_i are included in the β_j. So, we have that $F[\beta_1, \ldots, \beta_m] = L$ as well. Hence, L is a splitting field of $g(x)$ over F. And this proves that L/F is a Galois extension. ■

Importantly, as $\mathrm{Gal}(L/F)$ has finitely many subgroups, we know there are only finitely many intermediate fields E with $F \subseteq E \subseteq L$. In turn, we also know there are only finitely many intermediate fields E with $F \subseteq E \subseteq K \subseteq L$.

Here's one more topic I want to cover before switching to the homework.

Let n be a positive integer and let K be the splitting field of $f(x) = x^n - 1$ over $\mathbb{Q}$. The roots of $f(x)$ are $1, \zeta, \ldots, \zeta^{n-1}$ where $\zeta \coloneqq e^{2\pi i/n}$. Hence, $K = \mathbb{Q}[\zeta]$.

We define the nth cyclotomic polynomial to be $\Phi_n(x) = \prod_{\substack{1 \leq i \leq n \\ \gcd(i,n)=1}} (x - \zeta^i)$.

In other words, $\Phi_n(x)$ has as roots all primitive nth roots of 1 (meaning the $\alpha \in \mathbb{C}$ such that $\alpha^n = 1$ and $\alpha^k \neq 1$ for all $1 \leq k < n$).

As an example, you can easily work out that $\Phi_1(x) = x - 1$, $\Phi_2(x) = x + 1$, $\Phi_3(x) = x^2 + x + 1$, and $\Phi_4(x) = x^2 + 1$.

Lemma: For all n we have that:

1. $x^n - 1 = \prod_{d|n} \Phi_d(x)$
2. $\Phi_n(x) \in \mathbb{Z}[x]$ and is monic.

Proof:
To start off, note that

$$x^n - 1 = \prod_{0 \le k \le n-1} (x - e^{2\pi i k/n}) = \prod_{d|n} \Big(\prod_{\substack{0 \le k \le n-1 \\ \gcd(k,n) = \frac{n}{d}}} (x - e^{2\pi i k/n}) \Big)$$

Next, note that if $0 \le k \le n-1$ and $\gcd(k,n) = \frac{n}{d}$, then we know that $\frac{kd}{n}$ is an integer in $\{0, 1, \ldots, d-1\}$ satisfying that $\gcd(\frac{kd}{n}, d) = 1$. Conversely, if $\gcd(j,d) = 1$ and $0 \le j \le d-1$ Then $j\frac{n}{d}$ is an integer in $\{0, 1, \ldots, n\}$ and $\gcd(j\frac{n}{d}, n) = \frac{n}{d}$. Therefore, we can write:

$$\prod_{\substack{0 \le k \le n-1 \\ \gcd(k,n) = \frac{n}{d}}} (x - e^{2\pi i k/n}) = \prod_{\substack{0 \le k \le n-1 \\ \gcd(k,n) = \frac{n}{d}}} (x - e^{\frac{2\pi i (kd/n)}{d}}) = \prod_{\substack{0 \le j \le d-1 \\ \gcd(j,d) = 1}} (x - e^{\frac{2\pi i j}{d}}) = \Phi_d(x)$$

And this proves part 1.

To show part 2 just do induction on n and use the long division theorem on $\frac{x^n - 1}{\prod_{\substack{d|n \\ d \ne n}} \Phi_d(x)}$. Our base case is clear since I wrote Φ_1 through Φ_4 on the last page. ■

Theorem: $\Phi_n(x)$ is irreducible over $\mathbb{Q}[x]$ for each n. Hence, $\Phi_n(x) = m_{\zeta_n;\mathbb{Q}}$ where $\zeta_n = e^{2\pi i/n}$. Also $[\mathbb{Q}[\zeta_n] : \mathbb{Q}] = \det(\Phi_n(x)) = \varphi(n)$ (where φ is Euler's totient function).

Proof:
Write $\Phi_n(x) = f(x)g(x)$ where f is an irreducible monic polynomial in $\mathbb{Q}[x]$ and $f(\zeta_n) = 0$. By Gauss' lemma, we know $f, g \in \mathbb{Z}[x]$ and are monic.

Given any root ζ of f, notice if p is a prime number with $\gcd(p,n) = 1$, then ζ^p is another root of $\Phi_n(x)$. We can show as follows that ζ^p is a root of f and not g.

Suppose for the sake of contradiction that ζ^p is a root of $g(x)$, Then as $g(\zeta^p) = 0$, we know that $g(x^p)$ has ζ as a root. Thus, it follows that $f(x) \mid g(x^p)$. So, let $g(x^p) = f(x)h(x)$ where $h \in \mathbb{Z}[x]$.

By passing to $(\mathbb{Z}/p\mathbb{Z})[x]$, we get that $\overline{g(x^p)} = \overline{f(x)} \cdot \overline{h(x)}$. Yet now since $\overline{a_i}^p = \overline{a_i}$ for all $\overline{a_i} \in \mathbb{Z}/p\mathbb{Z}$, we can conclude by the Frobenius homomorphism on $(\mathbb{Z}/p\mathbb{Z})[x]$ that $\overline{g(x^p)} = \overline{g(x)}^p$. Thus, $\overline{g}^p = \overline{f} \cdot \overline{h}$. And now we can conclude that any irreducible factor of $\overline{f}$ must also divide $\overline{g}$. (After all, $(\mathbb{Z}/p\mathbb{Z})[x]$ is a PID and thus a UFD. So all irreducible factors are prime factors.)

Since f and g must be monic, neither $\overline{f}$ nor $\overline{g}$ are constant polynomials. Hence, they must have a shared irreducible factor. But now that implies that $\overline{\Phi_n(x)} = \overline{f(x)} \cdot \overline{g(x)}$ is not separable, and in turn $\overline{x^n - 1}$ is not separable. This is a contradiction.

In $(\mathbb{Z}/p\mathbb{Z})[x]$ we have that $(\overline{x^n - 1})' = \overline{n}x^{n-1}$. Since $\overline{n} \ne \overline{p}$ since $\gcd(n,p) = 1$, we know that $\overline{n}x^{n-1}$ is not the zero polynomial. Hence, any nonconstant monic factor of $\overline{n}x^{n-1}$ will be of the form x^j for some $j \in \{1, \ldots, n-1\}$. At the same time, $x \not\mid \overline{x^n - 1}$ as $0^n - 1 \ne 0$ in $\mathbb{Z}/p\mathbb{Z}$. Thus, we know that $\gcd(\overline{x^n - 1}, \overline{n}x^{n-1}) = \overline{1}$. And this proves $\overline{x^n - 1}$ is separable in $(\mathbb{Z}/p\mathbb{Z})[x]$.

To finish off, suppose $f \neq \Phi_n(x)$ and let j be the minimal positive integer such that ζ_n^j is not a root of f and $\gcd(j,n) = 1$. Write $j = p_1^{e_1} \cdots p_\ell^{e_\ell}$ where each p_k is a prime number less than n and each e_k is a positive integer.

By our minimality assumption, we know ζ^s is a root of f where $s = p_1^{e_1} \cdots p_\ell^{e_\ell - 1}$. But then by the reasoning on the prior page, we have that $(\zeta^s)^{p_m} = \zeta^j$ is a root of f. This is a contradiction.

Hence, we conclude $f = \Phi_n$ is irreducible over $\mathbb{Q}[x]$. And from there all the other claims of the theorem are obvious from past results. ■

Theorem: Let K be a splitting field of $x^n - 1$ over $\mathbb{Q}$. I.e. $K = \mathbb{Q}[\zeta]$ where $\zeta = e^{2\pi i/n}$. Then $\mathrm{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$.

Proof:
If $\sigma \in G := \mathrm{Gal}(K/\mathbb{Q})$, then $\sigma(\zeta) = \zeta^i$ for some $1 \leq i \leq n-1$ such that $\gcd(i,n) = 1$. This is because σ permutes the roots of $\Phi_n(x) = m_{\zeta;\mathbb{Q}}$. Also as $|\mathrm{Gal}(K/\mathbb{Q})| = [K:\mathbb{Q}] = \varphi(n) = \#$ roots of Φ_n, we know all choices of i do actually occur for some $\sigma \in G$.

Define $\Psi : (\mathbb{Z}/n\mathbb{Z})^\times \to \mathrm{Gal}(K/\mathbb{Q})$ such that $\overline{j} \mapsto \sigma_j$ where σ_j is the unique automorphism sending ζ to ζ^j.

- To see that this is well-defined, just note that $\zeta^{j+kn} = \zeta^j$ for all $k \in \mathbb{Z}$. Also as mentioned before, through a simple cardinality argument we know there does exist an automorphism σ_j such that $\sigma(\zeta) = \zeta^j$ for every j such that $\gcd(j,n) = 1$.
- Surjectivity is obvious and injectivity then follows because
$$|(\mathbb{Z}/n\mathbb{Z})^\times| = \varphi(n) = |\mathrm{Gal}(K/\mathbb{Q})|.$$
- To see that Ψ is a group homomorphism, note that both $\Psi(\overline{jk})$ and $\Psi(\overline{j}) \circ \Psi(\overline{k})$ send ζ to ζ^{jk}. ■

Math 200c Homework:

Set 1 Problem 3: Let $\alpha = \sqrt{3+\sqrt{7}} \in \mathbb{C}$. Let K be a splitting field of $f(x) = m_{\alpha;F}(x)$ over $\mathbb{Q}$. Calculate $[K:\mathbb{Q}]$ and show that $G = \mathrm{Gal}(K/\mathbb{Q})$ is isomorphic to a familiar group.

Note that $(\alpha^2 - 3)^2 - 7 = 0$. Hence, α is a zero of the polynomial $(x^2-3)^2 - 7 = x^4 - 6x^2 + 2$. Furthermore, the latter polynomial is irreducible by Eisenstein's criterion (using $p = 2$). Hence, we know that $f(x) = x^4 - 6x^2 + 2$.

By the quadratic formula, we know that if $x^4 - 6x^2 + 2 = 0$ then $x^2 = 3 \pm \sqrt{7}$. Hence, the roots of f are $\pm\sqrt{3 \pm \sqrt{7}}$. In particular, if we let $\beta = \sqrt{3-\sqrt{7}}$, then this shows that $K = \mathbb{Q}[\alpha, \beta]$ $(= \mathbb{Q}[\pm\alpha, \pm\beta])$.

Also note that $\alpha\beta = \sqrt{9-7} = \sqrt{2}$. So, we can conclude that $K = \mathbb{Q}[\alpha, \beta] = \mathbb{Q}[\alpha, \sqrt{2}]$. And now it follows by the tower lemma that:

$$\begin{aligned}[K : \mathbb{Q}] = [\mathbb{Q}[\alpha, \sqrt{2}] : \mathbb{Q}[\alpha]] \cdot [\mathbb{Q}[\alpha], \mathbb{Q}] &= [\mathbb{Q}[\alpha, \sqrt{2}] : \mathbb{Q}[\alpha]] \cdot \deg(f) \\ &= [\mathbb{Q}[\alpha, \sqrt{2}] : \mathbb{Q}[\alpha]] \cdot 4\end{aligned}$$

We can also show that $\sqrt{2} \notin \mathbb{Q}[\alpha]$ and thus $[\mathbb{Q}[\alpha, \sqrt{2}] : \mathbb{Q}[\alpha]] = \deg(x^2 + 2) = 2$.

Note that $m_{\sqrt{2};\mathbb{Q}} = x^2 + 2$ is irreducible in $\mathbb{Q}[\alpha]$ if and only if $m_{\alpha;\mathbb{Q}} = x^4 - 6x^2 + 2$ is irreducible in $\mathbb{Q}[\sqrt{2}]$ as either statements hold iff $[\mathbb{Q}[\alpha, \sqrt{2}] : \mathbb{Q}] = 8$.

(This was also a homework result on *pages 764-765*.)

But now as $(\pm\sqrt{3 \pm \sqrt{7}})^2 = 3 \pm \sqrt{7}$, we know that if any root of $m_{\alpha;\mathbb{Q}}$ was in $\mathbb{Q}[\sqrt{2}]$ then that would imply $\mathbb{Q}[\sqrt{7}] \subseteq \mathbb{Q}[\sqrt{2}]$. Yet as that is not true, we can conclude that $m_{\alpha;\mathbb{Q}} = x^4 - 6x^2 + 2$ has no linear divisors in $\mathbb{Q}[\sqrt{2}]$.

To show that $m_{\alpha;\mathbb{Q}}$ has no quadratic divisors in $\mathbb{Q}[\sqrt{2}]$, it suffices to show that $(x - \alpha)(x + \alpha)$, $(x + \alpha)(x + \beta)$, and $(x + \alpha)(x - \beta)$ are not in $(\mathbb{Q}[\sqrt{2}])[x]$.

- $(x - \alpha)(x + \alpha) = x^2 - \alpha^2 = x^2 - 3 - \sqrt{7}$. Yet as $\sqrt{7} \notin \mathbb{Q}[\sqrt{2}]$, we thus have that $(x - \alpha)(x + \alpha) \notin (\mathbb{Q}[\sqrt{2}])[x]$.

- Note that $(x + \alpha)(x + \beta) = x^2 + (\alpha + \beta)x + \sqrt{2}$ and similarly that $(x + \alpha)(x - \beta) = x^2 + (\alpha - \beta)x - \sqrt{2}$. Thus, it suffices to show that $\alpha \pm \beta \notin \mathbb{Q}[\sqrt{2}]$ in order to show that our two polynomials are not in $(\mathbb{Q}[\sqrt{2}])[x]$.

  Fortunately, note that $g(x) := x^4 - 12x^2 + 28$ satisfies that $g(\alpha \pm \beta) = 0$. Also, we can see that g is irreducible in $\mathbb{Q}[x]$ via the $\pmod 3$ irreducibility criterion (see *page 523*).

  By passing to $(\mathbb{Z}/3\mathbb{Z})[x]$, we get that $\overline{g}(x) = x^4 + 1$. We can check that $\overline{g}(x)$ has no linear divisors since $g(0) = 1$, $g(1) = 2$, and $g(2) = 17 \equiv 2$. Also note that the only irreducible monic quadratics in $(\mathbb{Z}/3\mathbb{Z})[x]$ are $x^2 + 1$, $x^2 + x + 2$, and $x^2 + 2x + 2$. Yet by long division we can see that none of those divide $x^4 + 1$ either. So, $x^4 + 1$ is irreducible in $(\mathbb{Z}/3\mathbb{Z})[x]$.

  It follows that $[\mathbb{Q}[\alpha + \beta] : \mathbb{Q}] = 4 = [\mathbb{Q}[\alpha - \beta] : \mathbb{Q}]$. So, we can't have that $\alpha \pm \beta \in \mathbb{Q}[\sqrt{2}]$ since that would contradict that $[\mathbb{Q}[\sqrt{2}] : \mathbb{Q}] = 2$.

This finishes showing that $m_{\alpha;\mathbb{Q}}$ is irreducible in $\mathbb{Q}[\sqrt{2}]$. So, we can also conclude that $m_{\sqrt{2};\mathbb{Q}}$ is irreducible in $\mathbb{Q}[\alpha]$.

With that, we've proven that $[K : \mathbb{Q}] = 8$.

Now since $K = \mathbb{Q}[\alpha, \sqrt{2}]$, every $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$ is uniquely determined by where it sends α and $\sqrt{2}$. Yet we also must have that $\sigma(\alpha) \in \{\pm\alpha, \pm\beta\}$ and $\sigma(\sqrt{2}) = \pm\sqrt{2}$ for every $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$. Thus there are eight possible choices for where we could send α and $\sqrt{2}$.

And since $K/\mathbb{Q}$ is a Galois extension since K is the splitting field of the separable polynomial $x^4 - 6x^2 + 2$, we also know that $|\mathrm{Gal}(K/\mathbb{Q})| = [K : \mathbb{Q}] = 8$. Hence, all possible choices occur.

Let us fix $\tau, \sigma \in \mathrm{Gal}(K/\mathbb{Q})$ such that:

$$\tau(\sqrt{2}) = -\sqrt{2}, \tau(\alpha) = \alpha \text{ and } \sigma(\sqrt{2}) = -\sqrt{2}, \sigma(\alpha) = \beta.$$

Now it's immediate that $\tau^2 = \mathrm{Id}_K$. Meanwhile, we can show from the following scratch work that $o(\sigma) = 4$.

$$\sigma^2(\alpha) = \sigma(\beta) = \sigma(\tfrac{\sqrt{2}}{\alpha}) = \tfrac{-\sqrt{2}}{\beta} = \tfrac{-\sqrt{2}}{\frac{\sqrt{2}}{\alpha}} = -\alpha$$

$$\implies \sigma^3(\alpha) = -\beta \implies \sigma^4(\alpha) = \alpha.$$

$$\sigma^4(\sqrt{2}) = \sigma^3(-\sqrt{2}) = \sigma^2(\sqrt{2}) = \sigma(-\sqrt{2}) = \sqrt{2}.$$

Also note that:

$$\tau \circ \sigma \circ \tau(\alpha) = \tau \circ \sigma(\alpha) = \tau(\tfrac{\sqrt{2}}{\alpha}) = \tfrac{-\sqrt{2}}{\alpha} = -\beta \text{ and } \tau \circ \sigma \circ \tau(\sqrt{2}) = -\sqrt{2}.$$

Therefore we have that $\tau \circ \sigma \circ \tau = \sigma^3$. As for one last observation, note that $\tau \notin \langle\sigma\rangle$. After all, τ fixes α but $\sigma^k(\alpha) \neq \alpha$ whenever $k \not\equiv 0 \pmod 4$. It follows that $\tau\langle\sigma\rangle$ and $\langle\sigma\rangle$ are distinct cosets.

This lets us say that $\mathrm{Gal}(K/\mathbb{Q})$ is isomorphic to the dihedral group D_8.

I'm moving on even though I haven't constructed that isomorphism. Sorry.

Set 1 Problem 5: Let $\zeta \in \mathbb{C}$ be a primitive pth. root of 1 for some prime $p \geq 3$. Let $K = \mathbb{Q}[\zeta]$ be the splitting field of $x^p - 1$ inside $\mathbb{C}$.

1. Let $\alpha = \sum_{i=0}^{p-1} \zeta^{i^2}$. This is called a Gauss sum. Prove that $E = \mathbb{Q}[\alpha]$ is the unique subfield of K such that $[E : \mathbb{Q}] = 2$.

   Recall from *page 821* that $(\mathbb{Z}/p\mathbb{Z})^\times \cong \mathrm{Gal}(K/\mathbb{Q})$ by the isomorphism such that $\overline{j}$ maps to the unique automorphism sending ζ to ζ^j. In particular, as $\mathrm{Gal}(K/\mathbb{Q})$ is cyclic with order $p-1$, there exists a unique subgroup with order $\frac{p-1}{2}$. By the Galois correspondance, that then means that there is a unique intermediate field $\mathbb{Q} \subseteq E \subseteq K$ with $[K : E] = \frac{p-1}{2}$. Equivalently, by tower lemma that E is the unique intermediate field with $[E : \mathbb{Q}] = 2$.

   So now that we know there is a unique intermediate field E with $[E : \mathbb{Q}] = 2$, it suffices to prove that $E = \mathbb{Q}[\alpha]$. Since it's obvious that $\alpha \in K$, we in turn just need to show that $m_{\alpha;\mathbb{Q}}(x)$ is a quadratic. Also because $\mathbb{Q}$ has characteristic 0 and $m_{\alpha;\mathbb{Q}}(x)$ is irreducible, we know that $m_{\alpha;\mathbb{Q}}(x)$ is separable. Therefore, we merely need to show that $m_{\alpha;\mathbb{Q}}(x)$ has two roots. And finally, because $\mathrm{Gal}(K/\mathbb{Q})$ acts transitively on the roots of $m_{\alpha;\mathbb{Q}}(x)$ (see *page 815*), it suffices to show that the orbit of α under the action of $\mathrm{Gal}(K/\mathbb{Q})$ has two elements.

   Thanks @hchan for telling me to think in terms of the orbit of the group action of $\mathrm{Gal}(K/\mathbb{Q})$ on α. I was completely stuck before being told that idea.

   Now I establish some important facts.

(a) $\sum_{i=0}^{p-1} \zeta^i = 0$.

Why?

For a rigorous reason why this is true, notice that $-(\sum_{i=0}^{p-1} \zeta^i)$ is the coefficient of the x^{p-1} term of the polynomial $\prod_{i=0}^{p-1}(x - \zeta^i) = x^p - 1$. As for an intuitive reason why this is true, note that the ζ^i are equally spaced around the unit circle. So, we should expect their average point to be the center of the unit circle.

(b) We say $k \in \mathbb{Z}/p\mathbb{Z}$ is a square residue if $x^2 - k$ has a zero in $\mathbb{Z}/p\mathbb{Z}$. Otherwise, we say k is a square nonresidue. I shall denote the subset of square residues in $(\mathbb{Z}/p\mathbb{Z})^\times$ as S, and similarly I shall denote the subset of square nonresidues in $(\mathbb{Z}/p\mathbb{Z})^\times$ as N. Clearly, $\mathbb{Z}/p\mathbb{Z} = \{0\} \cup S \cup N$.

Observation: $H < (\mathbb{Z}/p\mathbb{Z})^\times$ with $[(\mathbb{Z}/p\mathbb{Z})^\times : H] = 2$.

Proof:

Consider the group homomorphism $\varphi : (\mathbb{Z}/p\mathbb{Z})^\times \to (\mathbb{Z}/p\mathbb{Z})^\times$ such that $\varphi(j) = j^2$. Then $H = \mathrm{im}(\varphi)$. Also $j \in \ker(\varphi)$ iff $j^2 = 1$. As p is an odd prime (so $1 \neq -1$), we thus know that $|\ker(\varphi)| = 2$. Now just apply Lagrange's theorem.

It follows that H and N are two cosets partitioning $(\mathbb{Z}/p\mathbb{Z})^\times$. In particular if $k \in N$ then the map $j \mapsto kj$ is a bijective map from H to N.

(c) Note that $(p - x)^2 = p^2 - 2px + x^2 = x^2 \pmod p$. It follows that the squares of $1, 2, \ldots, \frac{p-1}{2}$ are all distinct and correspond in a bijective fashion to the squares of $\frac{p-1}{2} + 1, \ldots, p - 1$. Hence, we may write:

$$\alpha = 1 + \sum_{i=1}^{p-1} \zeta^{i^2} = 1 + 2\sum_{j \in H} \zeta^j$$

Now we know for any $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$ that there exists k such that:

$$\sigma(\alpha) = 1 + 2\sum_{j \in H} \sigma(\zeta)^j = 1 + 2\sum_{j \in H} \zeta^{kj} = 1 + 2\sum_{j \in kH} \zeta^j$$

Thus, the orbit of α is precisely:

$$\alpha = 1 + 2\sum_{j \in H} \zeta^j \text{ and } \beta := 1 + 2\sum_{j \in N} \zeta^j$$

If we can prove that α and β are distinct then we will be done. So, suppose for the sake of contradiction that $\alpha = \beta$.

Since $\alpha + \beta = 2(\sum_{j=0}^{p-1} \zeta^j) = 2(0)$, we in turn must have that $\alpha = -\beta$. So, we can conclude that both α and β are equal to 0. This implies $\sum_{j \in H} \zeta^j = \frac{-1}{2} = \sum_{j \in N} \zeta^j$. Yet as $1, \zeta, \zeta^2, \ldots, \zeta^{p-2}$ are $\mathbb{Q}$-linearly independent (since $\deg(m_{\zeta;\mathbb{Q}}) = p - 1$), one of the sums over those cosets equaling $\frac{-1}{2}$ will give a contradiction to the linear independence of our $\mathbb{Q}$-basis.

And yay I managed to figure this all out without resorting to calculating α explicitly (which according to the internet is actually possible).

As a side note, since $0 = \sum_{i=0}^{p-1} \zeta^i$ we know $\zeta^{p-1} = -\sum_{i=0}^{p-2} \zeta^i$. This lets us then shown that $\zeta^1, \ldots, \zeta^{p-1}$ is a $\mathbb{Q}$-basis for K.

2. Show that $L = \mathbb{Q}[\zeta + \zeta^{-1}]$ is the unique subfield of K such that $[K : L] = 2$. Show that in fact $L = K \cap \mathbb{R}$.

We can conclude via identical reasoning as in part 1 there there is a unique intermediate field $\mathbb{Q} \subseteq L \subseteq K$ such that $[K : L] = 2$. So, it suffices to show that $L = \mathbb{Q}[\zeta + \zeta^{-1}]$ and $L = K \cap \mathbb{R}$.

First we establish that $L = \mathbb{Q}[K \cap \mathbb{R}]$. Let $\tau : \mathbb{C} \to \mathbb{C}$ be the map $z \mapsto \overline{z}$. Importantly, τ is a field automorphism fixing $\mathbb{R} \supseteq \mathbb{Q}$. Also note that $\tau(\zeta^j) = 1/\zeta^j$ for all integers j. Thus since every element of K is a $\mathbb{Q}$-linear combination of $\zeta, \ldots, \zeta^{p-1}$, we know that τ maps K to K. After all, τ fixes the coefficients of that linear combination while mapping each spanning element into K. This proves that τ restricted to K is an element of $\mathrm{Gal}(K/\mathbb{Q})$.

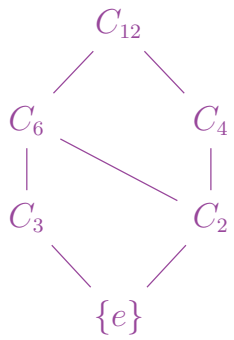
Now τ has order 2 in the Galois group. Thus by the Galois correspondance, it follows that $L = \mathrm{Fix}(\langle \tau \rangle) = K \cap \mathbb{R}$.

As for the claim that $L = \mathbb{Q}[\zeta + \zeta^{-1}]$, note that $\tau(\zeta + \zeta^{-1}) = \zeta^{-1} + \zeta$ and thus $\mathbb{Q}[\zeta + \zeta^{-1}] \subseteq L$. At the same time though, I claim that no automorphisms in $\mathrm{Gal}(K/\mathbb{Q})$ other than the identity and τ fix $\mathbb{Q}[\zeta + \zeta^{-1}]$. As a result, by the Galois correspondance we can conclude that $\zeta + \zeta^{-1}$ is not in any intermediate field properly contained in L. Thus, $\mathbb{Q}[\zeta + \zeta^{-1}] = L$.

To see why $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$ fixes $\zeta + \zeta^{-1}$ iff $\sigma \in \langle \tau \rangle$, recall that $\zeta, \ldots, \zeta^{p-1}$ is a $\mathbb{Q}$-basis for K. Thus fix k such that $\sigma(\zeta) = \zeta^k$. Then after writing $\zeta^{-1} = \zeta^j$ for some integer j (which we can do since ζ^{-1} is a pth roots of unity), we know by the linear independence of our basis that $\zeta + \zeta^j = \sigma(\zeta) + \sigma(\zeta^j) = \zeta^k + \zeta^{jk}$ if and only if $\{1, j\} = \{k, jk\}$ in $(\mathbb{Z}/p\mathbb{Z})^\times$. Equivalently, this means that either $\sigma(\zeta) = \tau(\zeta)$ or $\sigma(\zeta) = \zeta$. And since σ is uniquely determined by where it sends ζ, we are done. ■

Set 1 Problem 6: Let $\zeta = \zeta_{13}$ be a primitive 13th root of unity and let $K = \mathbb{Q}(\zeta) \subseteq \mathbb{C}$. Find all intermediate subfields E with $\mathbb{Q} \subseteq E \subseteq K$ and write each E explicitly as $E = \mathbb{Q}[\alpha]$ for some $\alpha \in K$.

Let C_k denote the cyclic group with k elements. Then from *page 821* we know that $G := \mathrm{Gal}(K/\mathbb{Q}) \cong C_{12}$. Also, the below diagram gives the subgroups of C_{12} (with the lines denoting containement as you go upwards).



In turn, by the Galois correspondance theorem there exists exactly four intermediate subfields E_2, E_3, E_4, E_6, each satisfying that $\dim_{\mathbb{Q}}(E_k) = k$. From problem 5, we know that $E_2 = \mathbb{Q}[\sum_{i=0}^{12} \zeta_{13}^{i^2}]$ and $E_6 = \mathbb{Q}[\zeta_{13} + \zeta_{13}^{-1}] = K \cap \mathbb{R}$. So, we just need to find

generators for E_3 and E_4 to finish this problem.

Observation: Let $H = (\mathbb{Z}/13\mathbb{Z})^\times$ and let $H^s := \{a^s : a \in H\}$. Since H^s is the image of the group homomorphism $\varphi_s(a) = a^s$, we know $H^s < H$. Also note that:

- When $s = 3$ then $|\ker(\varphi_s)| = 3$ since $x^3 - 1 = (x-1)(x-3)(x-9)$ in $(\mathbb{Z}/13\mathbb{Z})[x]$.
- When $s = 4$ then $|\ker(\varphi_s)| = 4$ since:
$$x^4 - 1 = (x-1)(x+1)(x-5)(x+5) \text{ in } (\mathbb{Z}/13\mathbb{Z})[x]$$

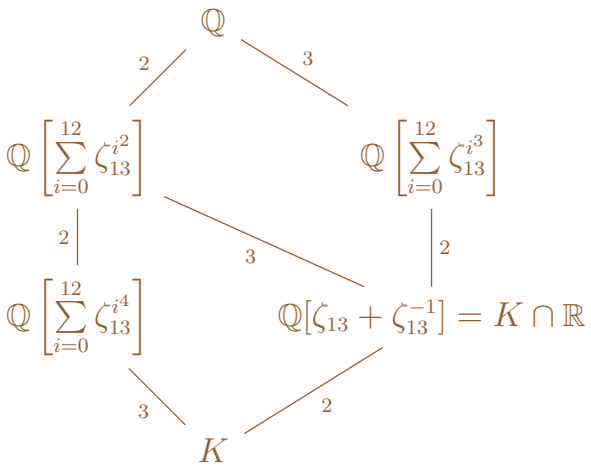
It follows that H^3 has 3 cosets and H^4 has 4 cosets in $(\mathbb{Z}/13\mathbb{Z})^\times$. One consequence of this is that:
$$\alpha := \sum_{i=0}^{12} \zeta_{13}^{i^3} = 1 + 3\sum_{j\in H^3} \zeta_{13}^j \text{ and } \beta := \sum_{i=0}^{12} \zeta_{13}^{i^4} = 1 + 4\sum_{j\in H^4} \zeta_{13}^j.$$

Another consequence of this is that the G-orbit of α has only three elements, and similarly the G action of β has only four elements.

Note that $\sum_{j\in k_1H^s} \zeta_{13}^j \neq \sum_{j\in k_2H^s} \zeta_{13}^j$ if $k_1H^s \neq k_2H^s$ since otherwise would contradict the $\mathbb{Q}$-linear independence of $\zeta_{13}, \zeta_{13}^2, \ldots, \zeta_{13}^{12}$.

By mirror reasoing to that in problem 5a, we thus know that $[\mathbb{Q}[\alpha] : \mathbb{Q}] = 3$ and $[\mathbb{Q}[\beta] : \mathbb{Q}] = 4$. Hence, we may write $E_3 = \mathbb{Q}[\alpha]$ and $E_4 = \mathbb{Q}[\beta]$. ■

As a side note, we've thus proven the below diagram of the intermediate fields of $K/\mathbb{Q}$ (with the lines denoting containment as you go downwards):



Interestingly, we can thus see that both $\sum_{i=0}^{12} \zeta_{13}^{i^2}$ and $\sum_{i=0}^{12} \zeta_{13}^{i^3}$ are real-valued.

Before doing problem 7, here is a result that will be helpful from lecture today (4/9/2026):

Theorem: Suppose p is prime. Then $x^{p^n} - x$ in $\mathbb{F}_p[x]$ is the product of all monic irreducible polynomials in $\mathbb{F}_p[x]$ of degree dividing n.

Proof:
Suppose $f(x)$ is irreducible of degree d over $\mathbb{F}_p$ where $d \mid n$. Then we have that $\mathbb{F}_p[x]/\langle f(x)\rangle \cong \mathbb{F}_{p^d}$. Yet also recall from page 780 that $\mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p) = \langle \sigma_p \rangle \cong C_n$ where σ_p is the Frobenius homomorphism. It follows that the intermediate fields of the

extension $\mathbb{F}_{p^n}/\mathbb{F}_p$ are precisely the fields $\mathbb{F}_{p^d}$ where $d \mid n$. So, we can say that f has a root $\alpha \in \mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^n}$. And since α is also a root of $x^{p^n} - x$, we can conclude that $f \mid x^{p^n} - x$ in $\mathbb{F}_p[x]$. (The last part works because $f(x) = m_{\alpha;\mathbb{F}_p}(x)$.)

Conversely, if $f \mid x^{p^n} - x$ in $\mathbb{F}_p[x]$ and f is irreducible over $\mathbb{F}_p$, then if α is a root of f in $\mathbb{F}_{p^n}$ we have that $\mathbb{F}_p \subseteq \mathbb{F}_p[\alpha] \subseteq \mathbb{F}_{p^n}$. Thus by our prior observation about the intermediate fields of $\mathbb{F}_{p^n}/\mathbb{F}$, we know that $\mathbb{F}_p[x]/\langle f \rangle \cong \mathbb{F}_p[\alpha] = \mathbb{F}_{p^d}$ for some integer d dividing n. It follows that $\deg(f) = d$.

Finally, each irreducible factor of $x^{p^n} - x$ only divides $x^{p^n} - x$ once since our polynomial is separable. Hence, we can conclude what we wanted. ■

Set 1 Problem 7: Let p be prime and let $\mathbb{F}_{p^n}$ be a field with p^n elements. Let S be the set of generators of the multiplicative group $(\mathbb{F}_{p^n})^\times$ (recall that $S \neq \emptyset$ since the multiplicative group of a finite field is cyclic).

(a) Let $f \in \mathbb{F}_p[x]$ be an irreducible polynomial of degree n. Show that f splits in $\mathbb{F}_{p^n}$ and that either all of its roots are in S or none of them are.

By the prior theorem we know that f divides $x^{p_n} - x$ where the latter is a separable polynomial splitting in $\mathbb{F}_{p^n}$. It follows that f splits in $\mathbb{F}_{p^n}$ as well.

Next suppose f has some zero $a \in S$. Then a has multiplictive order $p^n - 1$. Yet also note that $\mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)$ acts transitively on the set of zeros of f. Plus $\sigma(a)$ and a must have the same multiplicative order for all $\sigma \in \mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)$. It follows that all zeros of f have multiplicative order $p^n - 1$. Or in other words, all zeros of f are in S.

(b) Show that $n \mid \varphi(p^n - 1)$ for all primes p and all $n \geq 1$ (where φ is Euler's totient function).

To start off, we have that $|S| = \varphi(p^n - 1)$. Yet also note that since for every $\alpha \in S$ we have that $\mathbb{F}_p[\alpha] = \mathbb{F}_{p_n}$, we in turn must have that $\deg(m_{\alpha;\mathbb{F}_p}) = n$. Thus every element of S is the root of some irreducible polynomial of degree n in $\mathbb{F}_p[x]$.

By part (a), we thus know that S is a disjoint union of zero sets of certain irreducible degree n polynomials. Finally, as every irreducible polynomial in $\mathbb{F}_p$ is separable (or alternatively because our polynomials divide the separable polynomial $x^{p^n} - x$), we can conclude that each zero set has n elements. This proves that:

$$|S| = n \cdot (\#\text{polynomials with roots in } S).$$

(c) Consider the explicit case of the field $\mathbb{F}_{16}$. Find all irreducible polynomials of degree 4 over $\mathbb{F}_2$. Which ones have roots in S?

To start off, note by the chinese remainder theorem (see *page 593*) that we have an algebra isomorphism $\frac{\mathbb{Z}}{15\mathbb{Z}} \cong \frac{\mathbb{Z}}{5\mathbb{Z}} \oplus \frac{\mathbb{Z}}{3\mathbb{Z}}$. In particular, we know that an element is a unit in one $\mathbb{Z}$-algebra iff its image in the other $\mathbb{Z}$-algebra is a unit. Thus, we can restrict to a group isomorphism $(\frac{\mathbb{Z}}{15\mathbb{Z}})^\times \cong (\frac{\mathbb{Z}}{5\mathbb{Z}} \oplus \frac{\mathbb{Z}}{3\mathbb{Z}})^\times$. Then upon noting that $(\frac{\mathbb{Z}}{5\mathbb{Z}} \oplus \frac{\mathbb{Z}}{3\mathbb{Z}})^\times = (\frac{\mathbb{Z}}{5\mathbb{Z}})^\times \times (\frac{\mathbb{Z}}{3\mathbb{Z}})^\times$, we can conclude that $\varphi(15) = \varphi(5)\varphi(3) = 4 \cdot 2 = 8$.

All of this let's us say by part (b) that there are two irreducible polynomials of degree 4 over $\mathbb{F}_2$ with roots in $S \subseteq \mathbb{F}_{16}$.

(I did this reasoning first just so that I'd hopefully know when I can stop looking for polynomials with roots in S.)

Now for any polynomial in $\mathbb{F}_2[x]$ of degree > 1 to have no linear factors, it must have a constant coefficient of 1 (else 0 will be a root). In turn, we then also know it must have an even number of x^j factors with nonzero coefficients (else 1 will be a root). So, we can say that the only degree 4 polynomials in $\mathbb{F}_2[x]$ with no linear factors are:

$$x^4 + x + 1, \quad x^4 + x^2 + 1, \quad x^4 + x^3 + 1, \text{ and } x^4 + x^3 + x^2 + x + 1.$$

It could be the case though that these polynomials fail to be irreducible if they have quadratic factors. Yet fortunately, we can check that the only quadratic factor satisfying our criterion for not having linear factors is $x^2 + x + 1$. Thus, $x^2 + x + 1$ is the unique quadratic irreducible polynomial in $\mathbb{F}_2[x]$. In turn, it follows that our only candidate quartic polynomial that fails to be irreducible is $(x^2 + x + 1)^2 = x^4 + x^2 + 1$.

So, now we can conclude that the only degree 4 irreducible polynomials in $\mathbb{F}_2[x]$ are $x^4 + x + 1$, $x^4 + x^3 + 1$, and $x^4 + x^3 + x^2 + x + 1$. As discussed before, we know two of these have roots in S and the other doesn't. If we can find which of the three doesn't have roots in S, then we will be done.

Finally, note that if $f(x)$ is one of our three irreducible quartic polynomials then $\mathbb{F}_{16} \cong \frac{\mathbb{F}_2[x]}{\langle f \rangle}$ and that $x + \langle f \rangle$ is a zero of f in the latter field. So, it follows by the former reasoning plus part (a) that f has a root in S iff $k = 15$ is the first integer such that $x^{15} + \langle f \rangle = 1 + \langle f \rangle$. However, we want to find the f without any roots in S. Hence, we instead want to find an f such that $x^5 + \langle f \rangle = 1$.

Technically by Lagrange's theorem, if $o(x + \langle f \rangle) \neq 15$ then it must equal 1, 3, or 5. However, it is trivial to see that $o(x + \langle f \rangle) \neq 1, 3$ for any candidate f. So we only need to worry about finding the f such that $o(x + \langle f \rangle) = 5$.

Now we just plug and chug.

Note that $x^4 + \langle x^4 + x^3 + x^2 + x + 1 \rangle = x^3 + x^2 + x + 1 + \langle x^4 + x^3 + x^2 + x + 1 \rangle$. In turn we have that:

$$\begin{aligned} x^5 + \langle x^4 + x^3 + x^2 + x + 1 \rangle &= x^4 + x^3 + x^2 + x + \langle x^4 + x^3 + x^2 + x + 1 \rangle \\ &= 2x^3 + 2x^2 + 2x + 1 + \langle x^4 + x^3 + x^2 + x + 1 \rangle \\ &= 1 + \langle x^4 + x^3 + x^2 + x + 1 \rangle \end{aligned}$$

Thus $x^4 + x^3 + x^2 + x + 1$ is the irreducible polynomial without roots in S. ■

4/9/2026

I have not written up problems 1 and 2 yet for my math 200c homework. That said, I am desperately behind and need to move on to notes for a PDE class I decided to take at the

very last minute. However, I won't be including much of the homework for this class in my typed notes since the homework doesn't look to have as much note-taking value (it's an undergrad class).

Math 148 Lecture Notes:

A partial differential equation (PDE) is an equation $F(X, u, \mathrm{D}u, \mathrm{D}^2u, \ldots, \mathrm{D}^ku)$ where:

- $k \in \mathbb{N}$ is the order of our PDE,
- $\Omega \subseteq \mathbb{R}^N$ is an open set;
- $u : \Omega \to \mathbb{R}^M$ is our unknown function of $X \in \Omega$
  (here D^ju denotes the vector of all jth. order partial derivatives of u).
- $F : \Omega \times \mathbb{R}^{M(1+N+\ldots+N^k)} \to \mathbb{R}^L$ is the actual differential equation.

In theory N, M, L can be any positive integers. However in this class we will always have that $M = L = 1$. Also, in a lot of differential equations that arise in nature, it is useful to distinguish between spatial variables and time. In other words, we might want to write $X = (t, \vec{x})$.

As an example, the heat equation in N dimensions is defined on a domain $\Omega \subseteq \mathbb{R}^{N+1}$. Specifically, the heat equation is:

$$\frac{\partial u}{\partial t} - \kappa \triangle u = 0,$$

where κ is a real number and $\triangle u = \sum_{i=1}^{N} \frac{\partial^2}{\partial x_i^2}$ is the spatial Laplacian.

Example: Consider the PDE $u_{x,y} = 0$ on $\mathbb{R}^2$.

Solution:
$u_{x,y} = 0$ implies that $u_x(x, y) = f(x)$ for some $f : \mathbb{R} \to \mathbb{R}$. In turn, we have that $u(x, y) = F(x) + G(y)$ where $F, G : \mathbb{R} \to \mathbb{R}$ and F is differentiable with $F' = f$.

Conversely, one can also check that if $u(x, y) = F(x) + G(y)$ where F is a single time differentiable, then $u_{x,y} = 0$.

Note that similarly we have that $u_{y,x} = 0$ on $\mathbb{R}^2$ iff $u(x, y) = F(x) + G(y)$ where $F, G : \mathbb{R} \to \mathbb{R}$ and G is a differentiable function.

In this class we'll typically require solutions to be in $C^k(\Omega)$ where $k =$ the order of the PDE. Applying this condition to our example PDE, we can now see that $u_{x,y} = u_{y,x}$. And sure enough, both solutions are now the same. $u(x, y) = F(x) + G(y)$ where $F, G \in C^2(\mathbb{R})$.

Here are some derivative operators to be aware of:

- We write $\nabla u := (u_{x_1}, \ldots, u_{x_N})$ to be the (spatial) gradient of u.

- $\nabla \cdot (F_1, \ldots, F_N) = \sum_{n=1}^{N} (F_n)_{x_n}$ is the divergence of $(F_1, \ldots, F_N)$.
- The spatial Laplacian is $\triangle u = \nabla \cdot (\nabla u)$.
- Given a vector $\vec{a}$, we write $\frac{\partial u}{\partial \vec{a}} := \vec{a} \cdot \nabla u$ to be the directional derivative of u with respect to $\vec{a}$.

If a PDE models a physical phenonenon, then the science will also give initial/boundary condition which in turn hopefully let us have a unique solution.

An initial / boundary value problem is a PDE along with initial / boundary conditions. Its solution is any solution to the PDE in some class of functions attaining our initial / boundary conditions.

If an initial / boundary value problem satisfies that a solution exists, is unique, and depends continuously on the initial / boundary conditions (with respect to some topology), then we say the problem is well-posed.

Solutions to Linear PDEs in 2D:

We'll start by supposing our PDE is homogeneous with no 0th order term. In other words, we will first consider a PDE:

$$a(x,y)u_x + b(x,y)u_y = 0$$

If we write $\vec{\alpha}(x,y) := (a(x,y), b(x,y))$, then note that:

$$\frac{\partial u}{\partial \vec{\alpha}} = a(x,y)u_x + b(x,y)u_y$$

Also if $\gamma(t) = (x(t), y(t))$ is any curve in our domain with derivative vectors $\vec{\alpha}(\gamma(t))$, then we can see by chain rule that $\frac{\mathrm{d}}{\mathrm{d}t} u(\gamma(t)) = \frac{\partial u}{\partial \vec{\alpha}}(x(t), y(t))$. We call such a curve γ a characteristic of the PDE. And importantly our differential equation is stating that u must be constant on characteristics.

To find a characteristic passing through a point (x_0, y_0), we must solve the autonomous ODE system:

$$x' = a(x,y),\ y' = b(x,y) \text{ where } x(0) = x_0 \text{ and } y(0) = y_0.$$

Remark:
If a, b are locally Lipschitz, then we know from the tangent on *pages 289-294* that this ODE has a unique solution.

Importantly, note that the uniqueness theorem tells us that any two characteristics $\gamma_1(t), \gamma_2(t)$ can only intercept each other if there exists some t_0 such that $\widetilde{\gamma}_1(t) := \gamma_1(t - t_0)$ equals $\gamma_2(t)$ on their shared domain. In particular, any two "distinct" characteristics don't intercept.

Example: Consider the PDE $u_x - yu_y = 0$.

Solution:

Any characteristic must satisfy that $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 \\ -y \end{pmatrix}$. Thus every characteristic has the form $(x_0 + s, y_0 e^{-s})$ (where (x_0, y_0) is the point the characteristic passes through at $t = 0$).

In particular, $u(x_0 + s, y_0 e^{-s}) = u(x_0, y_0)$ can be said to be an implicit solution to our differential equation. After all, we can uniquely evaluate u if we are given as a boundary condition the value of u at a point in each characteristic.

As for getting an explicit solution, note that if $x = x_0 + s$ and $y = y_0 e^{-s}$, then we can rearrange to get that the characteristic is traced by the function $y = y_0 e^{x_0 - x}$. Or equivalently, we can say that the characteristics are precisely the level sets $ye^x = C$ $(= y_0 e^{x_0})$.

So, our explicit solution will be $u(x, y) = f(ye^x)$ where f is a C^1 function on $\mathbb{R}$.

Now we are equipped to handle a more general equation:

$$a(x, y)u_x + b(x, y)u_y + c(x, y)u = g(x, y)$$

Suppose $u \in C^1$ is a solution to this PDE and $(x(s), y(s))$ is a characteristic (which is still defined as being a curve $\gamma(s)$ satisfying that $\gamma'(s) = \alpha(\gamma(s))$ where $\alpha(x, y) = (a(x, y), b(x, y))$). If we set $\zeta(s) = u(x(s), y(s))$, then we have that:

$$\begin{aligned} \zeta'(s) &= u_x(x(s), y(s))x'(s) + u_y(x(s), y(s))y'(s) \\ &= u_x(x(s), y(s))a(x(s), y(s)) + u_y(x(s), y(s))b(x(s), y(s)) \\ &= g(x(s), y(s)) - c(x(s), y(s))\zeta(s) \end{aligned}$$

Hence, the value of u along a characteristic must satisfy a certain nonautonomous ODE. If we solve this ODE, then we will get an expression for u along the characteristic.

Example: Consider the PDE $u_x - yu_y = u$.

Solution:

We know from before that the characteristics of this PDE have the form $(x(s), y(s)) = (x_0 + s, y_0 e^{-s})$. Now supposing that $\zeta(s) := u(x(s), y(s))$ (where u is a solution to the PDE), then:

$$\zeta'(s) = \zeta(s) \implies \zeta(s) = \zeta(0)e^s = u(x_0, y_0)e^s$$

Hence, our implicit solution is:

$$u(x_0 + s, y_0 e^{-s}) = u(x_0, y_0)e^s \text{ for all } x_0, y_0, s \in \mathbb{R}$$

To get an exact solution, fix $x_0 = 0$. Then $(x, y) = (s, y_0 e^{-s})$ implies that $s = x$ and $y_0 = ye^x$. In turn, we have that:

$$u(x, y) = u(0, ye^x)e^x = \varphi(ye^x)e^x,$$

where $\varphi : \mathbb{R} \to \mathbb{R}$ is a differentiable or C^1 function. In particular note that $\varphi(y) = u(0, y)$.

On *pages 289-294* I proved that autonomous systems of ordinary differential equations combined with an initial condition have a unique solution in some interval J_0 provided that the function defining our ODE is locally Lipschitz continuous. Now, I want to explain why that theorem extends to nonautonomous systems of differential equations.

As a reminder on notation:
Let $\Omega \subseteq \mathbb{R}^{n+1}$ be an open set and suppose $V_1, \ldots, V_n$ are continuous real-valued functions defined on Ω. Let us write $V = (V_1, \ldots, V_n)$. Then given any initial condition $(t_0, \vec{c}) = (t_0, c_1, \ldots, c_n) \in \Omega$, we consider the problem of finding some function $y : J_0 \to \mathbb{R}^n$ such that:

- $y'(t) = V(t, y(t))$
- $y(t_0) = \vec{c}$

Note: from here on out I will assume $\Omega = J \times U$ where $J \subseteq \mathbb{R}$ is an open interval and $U \subseteq \mathbb{R}^n$ is an open set.

Observe that we can turn the above potentially nonautonomous ODE into an equivalent autonomous ODE as follows.

Define the functions:

- Let $W_0 : \mathbb{R} \times J \times U \to \mathbb{R}$ be given by $W_0 \equiv 1$.
- Also for $1 \leq i \leq n$ let $W_i : \mathbb{R} \times J \times U \to \mathbb{R}$ be given by $W_i(s, t, \vec{y}) = V_i(t, \vec{y})$.

Observation: If $V = (V_1, \ldots, V_n)$ is locally Lipschitz continuous then we also know $W = (W_0, W_1, \ldots, W_n)$ is. Similarly, if V is C^k then so will be W.

Next consider the initial value problem:

- $y_0'(t) = W_0(t, y_0(t), y_1(t), \ldots, y_n(t)) = 1$,
- $y_i'(t) = W_i(t, y_0(t), y_1(t), \ldots, y_n(t))$ for all $1 \leq i \leq n$,
- $y_0(t_0) = t_0$,
- $y_i(t_0) = c_i$ for all $1 \leq i \leq n$.

Observation: $(y_1(t), \ldots, y_n(t))$ is a solution to the original ODE (with V) if and only if $(y_0(t) = t, y_1(t), \ldots, y_n(t))$ is a solution to the new ODE (with W).

But now we can apply the theorems from pages 289-294 to the new autonomous ODE in order to prove the existence and uniqueness of solutions for our nonautonomous ODE (assuming V is locally Lipschitz in all arguments).

As a side note, similar reasoning also lets us apply the existence and uniqueness theorem from pages 289-294 to nth order ODEs (where $n > 1$) even though those theorems are only stated for 1st order ODEs.

There's a stronger result I want to prove specifically in the case of linear ODEs (which I will be using a bunch). That said, I don't have time to prove it now since I need to switch over to taking complex analysis notes. So I will continue this topic on *page 849*.

4/10/2026

Math 220c Lecture Notes:

Recall that if $\Delta \subseteq \mathbb{C}$ is the unit disk, then given a continuous function $u : \overline{\Delta} \to \mathbb{R}$ that is harmonic on Δ we gave a formula for u in terms of $u|_{\partial\Delta}$. This was on *page 808*. This formula is called the Poisson integral formula.

Since $u|_\Delta = \mathrm{Re}(f)$ for some $f \in O(\Delta)$, a natural next question is if we can write down f explicitly in terms of $u|_{\partial\Delta}$

Theorem (Schwarz Integral Formula): Suppose $u : \overline{\Delta} \to \mathbb{R}$ is continuous and that u is harmonic on Δ. Then for any $a \in \Delta$ set:

$$f(a) = \frac{1}{2\pi i}\int_{|z|=1} \frac{z+a}{z-a} \cdot \frac{u(z)}{z} \mathrm{d}z$$

We claim f is holomorphic in Δ and $\mathrm{Re}(f) = u$ in Δ.

Proof:
To show f is holomorphic, just apply *(Conway) exercise IV.2.2 on pages 380-381*. As for the other claim, note that:

$$\mathrm{Re}(f(a)) = \mathrm{Re}(\frac{1}{2\pi i}\int_0^{2\pi} \frac{e^{it}+a}{e^{it}-a} \cdot \frac{u(e^{it})}{e^{it}} \cdot ie^{it}\mathrm{d}t) = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{Re}(\frac{e^{it}+a}{e^{it}-a})u(e^{it})\mathrm{d}t$$

If we write $a = re^{i\theta}$, we get that:

$$\mathrm{Re}(\frac{e^{it}+a}{e^{it}-a}) = \mathrm{Re}(\frac{1+re^{i(\theta-t)}}{1-re^{i(\theta-t)}}) = P_r(\theta - t)$$

And thus, we can conclude that $\mathrm{Re}(f) = u$ by the Poisson integral formula. ■

The Schwarz integral formula will be a vital tool for proving the Dirichlet problem on $\overline{\Delta}$. That said, before attacking that problem we need to establish a few more facts about the Poisson kernel.

Lemma:

(i) $P_r(t)$ is an even, 2π-periodic, nonnegative function for all $r \in [0, 1)$

(ii) $\frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(t)\mathrm{d}t = 1$ for all $r \in [0, 1)$.

(iii) $\forall \delta > 0$, $P_r(t) \to 0$ on $[\delta, \pi] \cup [-\pi, -\delta]$ uniformly as $r \to 1$.

Proof:
(i) is obvious from the formulas we proved for P_r on *page 808*.

To show (ii), apply the Poisson integral formula to the function $u \equiv 1$ on $\overline{\Delta}$. By doing this we get $1 = u(re^{i\theta}) = \frac{1}{2\pi}\int_0^{2\pi} P_r(\theta - t)u(e^{it})\mathrm{d}t = \frac{1}{2\pi}\int_0^{2\pi} P_r(\theta - t)\mathrm{d}t$. Then by the 2π-periodicity and evenness of P_r, we in turn get by plugging in $\theta = 0$ that:

$$\int_0^{2\pi} P_r(-t)\mathrm{d}t = \int_0^{2\pi} P_r(t)\mathrm{d}t = \int_{-\pi}^{\pi} P_r(t)\mathrm{d}t$$

Finally to show (iii), fix any $\delta > 0$. By the evenness of P_r it suffices to show uniform convergence on the interval $[\delta, \pi]$. Fortunately, as $P_r(t) = \frac{1-r^2}{1-2r\cos(t)+r^2}$, we can see that $P_r(t)$ is decreasing for all $t \in [0, \pi]$. Hence:

$$\sup_{t\in[\delta,\pi]} |P_r(t)| = P_r(\delta) = \frac{1-r^2}{1-2r\cos(\delta)+r^2}.$$

And since $\cos(\delta) \neq 1$, we can conclude that the latter expression goes to 0 as $r \to 1$. ■

Intuitively, this can be thought of as saying that the Poisson kernel converges to the dirac delta function on $[-\pi, \pi]$ as $r \to 1$.

Now we are ready to solve the Dirichlet problem on $\overline{\Delta}$.

Theorem: Suppose $f : \partial\Delta \to \mathbb{R}$ is continuous. Then if we define:

$$u(re^{i\theta}) = \begin{cases} \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta - t)f(e^{it})\mathrm{d}t & \text{if } r < 1 \\ f(e^{i\theta}) & \text{if } r = 1 \end{cases}$$

we have that $u|_{\partial\Delta} = f$ and that u is harmonic in Δ.

Proof:

There are two things we need to accomplish. Firstly, we need to show that u is harmonic on Δ. Fortunately this is easy. After all, in the proof of the Schwarz integral formula we saw that the given formula for $u(a)$ at any $a \in \Delta$ is equal to $\mathrm{Re}(F(a))$ where:

$F(a) = \frac{1}{2\pi i}\int_{|z|=1} \frac{z+a}{z-a} \cdot \frac{f(z)}{z}\mathrm{d}z$ is holomorphic on Δ.

Thus, we know that u is harmonic on Δ.

As a side note, we can change the bounds from 0 and 2π to $-\pi$ and π in the definition of u since $P_r(\theta - t)f(e^{it})$ is 2π-periodic.

The other thing we need to do is show that u is continuous on the boundary. To accomplish this, we first make two simplifying observations.

1. If we want to show continuity at $e^{i\theta_0}$, we can let $\widetilde{f}(z) := f(e^{i\theta_0}z)$ and

$$\widetilde{u}(re^{i\theta}) = \begin{cases} \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta - t)\widetilde{f}(e^{it})\mathrm{d}t & \text{if } r < 1 \\ \widetilde{f}(e^{it}) & \text{if } r = 1. \end{cases}$$

   Then we can see that $\widetilde{u}(z) = u(e^{i\theta_0}z)$. Consequently, $\widetilde{u}$ is continuous at 1 iff u is continuous at $e^{i\theta_0}$. And hence, for our proof it suffices to prove why u is continuous at $1 \in \partial\Delta$.

2. We already know u is continuous when restricted to $\partial\Delta$. So, we just need to show that the limit from the interior of the disk approaches $f(1)$. I.e, we need to show that:

$$\lim_{\substack{r\to 1 \\ \theta\to 0}} \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta - t)f(e^{it})\mathrm{d}t = f(1).$$

Fix $\varepsilon > 0$. We will show there exists $\delta, \rho > 0$ such that:

(a) $|u(re^{i\theta}) - u(r)| < \varepsilon$ if $|\theta| < \delta$;

(b) $|u(r) - f(1)| < 2\varepsilon$ if $\rho < r < 1$.

To start off, as $t \mapsto f(e^{it})$ is a continuous function on a compact interval $[-3\pi, 3\pi]$, we can find some $0 < \delta < \pi$ such that for all $x, y \in [-2\pi, 2\pi]$ we have that:

$$|x - y| < \delta \Longrightarrow |f(e^{ix} - f(e^{iy}))| < \varepsilon$$

As a side note, the bounds of the interval $[-2\pi, 2\pi]$ aren't particularly important. I just want them big enough so that I can always apply the implication in the manipulations below.

Also by the extreme value theorem we can pick out $M \geq 0$ such that $|f| \leq M$ on $\partial\Delta$.

Proof (a):
Suppose $|\theta| < \delta$. Then:

$$\begin{aligned}
|u(re^{i\theta}) - u(r)| &= \tfrac{1}{2\pi} \left| \int_{-\pi}^{\pi} P_r(\theta - t) f(e^{it}) \mathrm{d}t - \int_{-\pi}^{\pi} P_r(-t) f(e^{it}) \mathrm{d}t \right| \\
&= \tfrac{1}{2\pi} \left| \int_{-\pi}^{\pi} P_r(-t) f(e^{i(t+\theta)}) \mathrm{d}t - \int_{-\pi}^{\pi} P_r(-t) f(e^{it}) \mathrm{d}t \right| \\
&\leq \tfrac{1}{2\pi} \int_{-\pi}^{\pi} P_r(-t) \left| f(e^{i(t+\theta)}) - f(e^{it}) \right| \mathrm{d}t \\
&\leq \tfrac{1}{2\pi} \int_{-\pi}^{\pi} P_r(-t) \varepsilon \mathrm{d}t = \varepsilon
\end{aligned}$$

Proof (b):

$$\begin{aligned}
|u(r) - f(1)| &= \left| \tfrac{1}{2\pi} \int_{-\pi}^{\pi} P_r(-t) f(e^{it}) \mathrm{d}t - \left(\tfrac{1}{2\pi} \int_{-\pi}^{\pi} P_r(-t) \mathrm{d}t \right) f(1) \right| \\
&= \tfrac{1}{2\pi} \left| \int_{-\pi}^{\pi} P_r(-t) (f(e^{it}) - f(1)) \mathrm{d}t \right| \\
&\leq \tfrac{1}{2\pi} \int_{-\pi}^{\pi} P_r(-t) \left| f(e^{it}) - f(1) \right| \mathrm{d}t \\
&= \tfrac{1}{2\pi} \int_{\delta \leq |t| \leq \pi} P_r(-t) \left| f(e^{it}) - f(1) \right| \mathrm{d}t + \tfrac{1}{2} \int_{-\delta}^{\delta} P_r(-t) \left| f(e^{it}) - f(1) \right| \mathrm{d}t.
\end{aligned}$$

By how we chose δ, we know that:

$$\tfrac{1}{2} \int_{\delta}^{\delta} P_r(-t) \left| f(e^{it}) - f(1) \right| \mathrm{d}t \leq \tfrac{1}{2} \int_{\delta}^{\delta} P_r(-t) \varepsilon \mathrm{d}t \leq \tfrac{1}{2} \int_{-\pi}^{\pi} P_r(-t) \varepsilon \mathrm{d}t = \varepsilon.$$

On the other hand, we know from part (iii) of the prior lemma that there exists $\rho < 1$ such that $|P_r(-t)| < \frac{\varepsilon}{2M}$ if $\rho < r < 1$. In turn:

$$\tfrac{1}{2\pi} \int_{\delta \leq |t| \leq \pi} P_r(-t) \left| f(e^{it}) - f(1) \right| \mathrm{d}t \leq \tfrac{1}{2\pi} \int_{\delta \leq |t| \leq \pi} \tfrac{\varepsilon}{2M} \cdot 2M \mathrm{d}t \leq \tfrac{1}{2\pi} \int_{-\pi}^{\pi} \varepsilon \mathrm{d}t = \varepsilon.$$

Finally by triangle inequality we know that $|u(re^{i\theta}) - u(1)| < 3\varepsilon$ whenever $|\theta| < \delta$ and $\rho < r < 1$. ■

As an interesting side note, if we consider the Poisson kernel $P_r(\theta - t) = \mathrm{Re}(\frac{e^{it}+z}{e^{it}-z})$ where $z = re^{i\theta}$ as being a function $P : \Delta \times S^1 \to \mathbb{R}$, then $f \mapsto \frac{1}{2\pi} \int_0^{2\pi} P(z, e^{it}) f(t) \mathrm{d}t$ will be a bounded integral operator from $L^\infty(S^1)$ to $L^\infty(\Delta)$. I won't go into further depth on this here. But it is good to see connections to stuff I've already done. Also, this proves that the Dirichlet problem on $\overline{\Delta}$ is well-posed.

Using the solution to the Dirichlet problem on $\overline{\Delta}$, we can also solve the Dirichlet problem for harmonic functions in simply connected regions whose boundary is a Jordan curve.

> Since G is simply connected and not equal to $\mathbb{C}$ since G has a nonempty boundary, we know there exists a biholomorphism $\varphi : G \to \Delta$. By Carathéodory's theorem (see *pages 744-749*), we know φ extends continuously to the boundary of G such that $\varphi|_{\partial G}$ is a homeomorphism from ∂G to $\partial\Delta$.
>
> Now given $f : \partial G \to \mathbb{R}$, we can solve the Dirichlet problem for $f \circ \varphi^{-1}$ on the unit disk to get a continuous function $h : \overline{\Delta} \to \mathbb{R}$ that is harmonic on Δ. In turn, the function $u := h \circ \varphi$ satisfies that $u|_{\partial G} = f$. Also, u is harmonic on G by *set 1 problem 1 on page 802*.

As a special case of this, we can solve for the Poisson integral Formula on $\Delta(a, R)$ (where $a \in \mathbb{C}$ and $R > 0$):

> Consider the biholomorphism $\varphi : \mathbb{C} \to \mathbb{C}$ given by $\varphi(z) = \frac{1}{R}(z - a)$ and $\varphi^{-1}(z) = a + Rz$. By restricting these functions we can observe that:
> $$\varphi(\Delta(a, R)) = \Delta \text{ and } \varphi(\partial\Delta(a, R)) = \partial\Delta.$$
>
> Now suppose $f : \partial\Delta(a, R)$ is a continuous function. We can solve the Dirichlet problem on $\overline{\Delta}$ to get that:
> $$h(re^{i\theta}) = \begin{cases} \frac{1}{2\pi}\int P_r(\theta - t) f(a + Re^{it})\mathrm{d}t & \text{if } r < 1 \\ f(a + Re^{i\theta}) & \text{if } r = 1 \end{cases}$$
>
> satisfies that h is continuous on $\overline{\Delta}$, harmonic on Δ, and equals $f \circ \varphi^{-1}$ on $\partial\Delta$. In turn $u := h \circ \varphi$ satisfies that u is continuous on $\overline{\Delta(a, R)}$, harmonic on $\Delta(a, R)$, and equals f on $\partial\Delta(a, R)$.
>
> As for explicitly calculating u, note for any $r < R$ that:
> $$\begin{aligned} u(a + re^{i\theta}) = h(\tfrac{r}{R}e^{i\theta}) &= \tfrac{1}{2\pi}\textstyle\int_0^{2\pi} P_{r/R}(\theta - t) f(a + Re^{it})\mathrm{d}t \\ &= \tfrac{1}{2\pi}\textstyle\int_0^{2\pi} \frac{1-(\frac{r}{R})^2}{1-2\frac{r}{R}\cos(\theta-t)+(\frac{r}{R})^2} f(a + Re^{it})\mathrm{d}t \\ &= \tfrac{1}{2\pi}\textstyle\int_0^{2\pi} \frac{R^2-r^2}{R^2-2rR\cos(\theta-t)+r^2} f(a + Re^{it})\mathrm{d}t \end{aligned}$$

Lemma: Suppose $u : G \to \mathbb{R}$ is harmonic, $u \geq 0$, and $\overline{\Delta}(a, R) \subseteq G$, and $r = |z - a| < R$. Then $\frac{R-r}{R+r}u(a) \leq u(z) \leq \frac{R+r}{R-r}u(a)$.

> Proof:
> Suppose $z = a + re^{i\theta}$. Then since $u \geq 0$ and:
> $$R^2 - 2Rr\cos(\theta) + r^2 \geq R^2 - 2rR + r^2 \geq (R - r)^2 > 0,$$
> we can conclude by the Poisson integral formula on $\Delta(a, R)$ that:
> $$u(z) \leq \tfrac{1}{2\pi}\textstyle\int_0^{2\pi} \frac{R^2-r^2}{R^2-2Rr+r^2} u(a + Re^{it})\mathrm{d}t = \frac{R+r}{R-r} \cdot \tfrac{1}{2\pi}\textstyle\int_0^{2\pi} u(a + Re^{it})\mathrm{d}t$$
>
> Also by the mean value property of harmonic functions (see *page 804*) we know that $\int_0^{2\pi} u(a + Re^{it})\mathrm{d}t = u(a)$. This proves one inequality. The other inequality is proven similarly. ■

Corollary (Harnack's inequality): If $u : G \to \mathbb{R}$ is harmonic and u is also nonnegative on $\overline{\Delta}(a, R) \subseteq G$, then for all $z \in \overline{\Delta}(a, \frac{R}{2})$ we have that $\frac{1}{3}u(a) \leq u(z) \leq 3u(a)$.

Proof:
$\frac{1}{3} \leq \frac{R-r}{R+r}$ and $\frac{R+r}{R-r} \leq 3$ when $r \leq \frac{R}{2}$. ■

Even though professor Oprea isn't strictly teaching from Conway, he is supposed to mirror Conway. As such, he brought up an oversight that Conway makes in his book *Functions of One Complex Variable*. Namely, while Conway defined the mean value property identically to how we did:

$$\forall a \in G, \forall r > 0, \ \ \overline{\Delta}(a, r) \subseteq G \implies u(a) = \frac{1}{2\pi} \int_0^{2\pi} u(a + re^{it}) \mathrm{d}t$$

the property Conway actually uses in proofs is:

$$\forall a \in G, \exists R > 0 \ s.t. \ \overline{\Delta}(a, R) \subseteq G \text{ and } \forall 0 < r < R, \ \ u(a) = \frac{1}{2\pi} \int_0^{2\pi} u(a + re^{it}) \mathrm{d}t.$$

We'll briefly call the second property the local mean value property. That said, this is not standard terminology. Also, it is a simple corollary of the following theorem that the local mean value property is equivalent to the mean value property.

Theorem (converse to the mean value property): If $u : G \to \mathbb{R}$ is continuous and has the local mean value property, then u is harmonic in G.

Proof:
Let $a \in G$. By the local mean value property of u, we can find $R > 0$ such that $\overline{\Delta}(a, R) \subseteq G$ and for all $0 < r < R$, $u(a) = \frac{1}{2\pi} \int_0^{2\pi} u(a + re^{it}) \mathrm{d}t$. Also, let us set $f = u|_{\partial\Delta(a,R)}$. Then f is continuous since u is.

We can solve the Dirichlet problem in $\Delta(a, R)$ to find a continuous function $h : \overline{\Delta}(a, R)$ that is harmonic on $\Delta(a, R)$ and satisfies that $h|_{\partial\Delta(a,R)} = f$. If we let $\Phi := u - h$ on $\overline{\Delta}(a, R)$ we then have that Φ is continuous, has the local mean value property on $\Delta(a, R)$, and satisfies that $\Phi|_{\partial\overline{\Delta}(a,R)} \equiv 0$.

Now Φ must achieve a max and min on $\overline{\Delta}(a, r)$ since it is a continuous function on a compact set. That said, by the theorem on the bottom half of *page 804*, we know that if Φ achieves a max or min in $\Delta(a, r)$ then Φ is constant. So, we can always conclude that Φ achieves its min or max in $\partial\Delta(a, r)$. Hence, $\Phi \equiv 0$. In turn, $u \equiv h$ is harmonic in $\Delta(a, r)$. ■

Corollary: If $u : G \to \mathbb{R}$ is continuous, the following are equivalent.

(a) u is harmonic;

(b) u has the mean value property;

(c) u has the local mean value property.

Lemma: If $\{u_n : G \to \mathbb{R}\}$ is a sequence of harmonic functions such that $u_n \xrightarrow{\ell.u.} u$, then u is harmonic.

Proof:
Since the u_n are continuous, so is u. So, we only need to show that u is harmonic. By the prior theorem, it equivalently suffices to show that u satisfies the mean value property. But this is incredibly easy.

Suppose $a \in G$. Then for any $R > 0$ such that $\overline{\Delta}(a, R) \subseteq G$, we have that $u_n(a) = \frac{1}{2\pi}\int_0^{2\pi} u_n(a + Re^{it})\mathrm{d}t$ for all n. Next, since $u_n \to u$ uniformly on $\partial\Delta(a, R)$, we can conclude that:

$$\begin{aligned} u(a) &= \lim_{n\to\infty} \tfrac{1}{2\pi}\int_0^{2\pi} u_n(a + Re^{it})\mathrm{d}t \\ &= \tfrac{1}{2\pi}\int_0^{2\pi} \lim_{n\to\infty} u_n(a + Re^{it})\mathrm{d}t = \tfrac{1}{2\pi}\int_0^{2\pi} u(a + Re^{it})\mathrm{d}t. \blacksquare \end{aligned}$$

Theorem (Harnack): Suppose $\{u_n : G \to \mathbb{R}\}_{n\in\mathbb{N}}$ is a sequence of harmonic functions on a region satisfying that $u_1 \leq u_2 \leq \cdots$. Then either $u_n \xrightarrow{\ell.u.} u$ for some function u or $u_n \xrightarrow{\ell.u.} \infty$.

Note: By $u_n \xrightarrow{\ell.u.} \infty$ I mean that for all compact $K \subseteq G$ and $M > 0$ we can find $N \in \mathbb{N}$ with $|u_n(z)| > M$ for all $n \geq N$.

Proof:
Without loss of generality we can assume $u_n \geq 0$ for all n. After all, if this is not true then we can just prove the theorem for the sequence $\{u_n - u_1\}_{n\in\mathbb{N}}$.

Let $A := \{z \in G : \lim_{n\to\infty} u_n(z) < \infty\}$ and $B := \{z \in G : \lim_{n\to\infty} u_n(z) = \infty\}$. Then $A \cup B = G$ and $A \cap B = \emptyset$. We'll also show below that A and B are both open. So by the connectivity of G we know that either $A = G$ or $B = G$.

- Suppose $a \in A$ and let $R > 0$ be such that $\overline{\Delta}(a, R) \subseteq G$. Then by Harnack's inequality we have that $u_n(z) \leq 3u_n(a)$ when $z \in \overline{\Delta}(a, \frac{R}{2})$. So, the sequence $\{u_n(z)\}_{n\in\mathbb{N}}$ is bounded for all $z \in \overline{\Delta}(a, \frac{R}{2})$ (since $\lim_{n\to\infty} u_n(a) < \infty$). And this proves that $\Delta(a, \frac{R}{2}) \subseteq A$.
- Similarly, if $b \in B$ and $R > 0$ satisfies that $\overline{\Delta}(b, R) \subseteq G$, then by Harnack's inequality we have that $u_n(z) \geq \frac{1}{3}u_n(b)$ for all $z \in \overline{\Delta}(b, \frac{R}{2})$. Hence, $\Delta(b, \frac{R}{2}) \subseteq B$.

This proves that either $\{u_n\}_{n\in\mathbb{N}}$ converges to a function on G or u converges pointwise to ∞. We still need to show that though that this convergence is locally uniform.

Case 1: Suppose $u_n(a) \to \infty$.
Then for all $M > 0$ there exists $N \in \mathbb{N}$ such that $u_n(a) \geq 3M$. In turn if $R > 0$ satisfies that $\overline{\Delta}(a, R) \subseteq G$ then we have that $u_n(z) \geq M$ for all $n \geq N$ and $z \in \overline{\Delta}(a, \frac{R}{2})$ by Harnack's inequality. This shows that $u_n \xrightarrow{u.} \infty$ in $\overline{\Delta}(a, \frac{R}{2})$.

Case 2: Suppose $u_n(a) \to u(a) < \infty$.
Since $\{u_n(a)\}_{n\in\mathbb{N}}$ converges, we know it is Cauchy. Hence, for any fixed $\varepsilon > 0$ and $R > 0$ such that $\overline{\Delta}(a, R) \subseteq G$, we have that there exists $N \in \mathbb{N}$ such that for all $n \geq m \geq N$ we have that $0 \leq u_n(a) - u_m(a) < \varepsilon$.

By Harnack's inequality applied to $u_n - u_m$ (which is nonnegative since $\{u_n\}_{n\in\mathbb{N}}$ is an increasing sequence), we in turn know that $0 \leq u_n(z) - u_m(z) < 3\varepsilon$ for all $z \in \overline{\Delta}(a, \frac{R}{2})$ and $n \geq m \geq N$. This proves that $\{u_n\}_{n\in\mathbb{N}}$ is uniformly Cauchy (and thus uniformly converges) in $\overline{\Delta}(a, \frac{R}{2})$. ■

If $\varphi : G \to \mathbb{R}$ is continuous and for all $a \in G$ there exists $\overline{\Delta}(a, R) \subseteq G$ such that:

$$\varphi(a) \leq \frac{1}{2\pi}\int_0^{2\pi} \varphi(a + re^{it})\mathrm{d}t \text{ for all } 0 \leq r \leq R,$$

then φ is called subharmonic.

Meanwhile if φ satisfied the opposite inequality for all $0 \leq r \leq R$, then we'd instead call φ superharmonic.

Remarks:

- φ is subharmonic if and only if $-\varphi$ is superharmonic.
- φ is harmonic if and only if φ is both subharmonic and superharmonic.

Set 2 Problem 0: Find the formula for the Laplacian $\triangle = \partial_{x,x} + \partial_{y,y}$ in polar coordinates.

Consider the change in variables from Cartesian coordinates to polar coordinates, $(r, \theta) \mapsto (r\cos(\theta) + x_0, r\sin(\theta) + y_0)$. Then for any function $u : G \to \mathbb{C}$ with all second derivatives existing, we have by chain rule that:

- $u_r = \cos(\theta)u_x + \sin(\theta)u_y$,
- $u_{r,r} = \cos^2(\theta)u_{x,x} + \sin(\theta)\cos(\theta)u_{x,y} + \sin(\theta)\cos(\theta)u_{y,x} + \sin^2(\theta)u_{y,y}$,
- $u_\theta = -r\sin(\theta)u_x + r\cos(\theta)u_y$,
- $u_{\theta,\theta} = -r\cos(\theta)u_x + r^2\sin^2(\theta)u_{x,x} - r^2\sin(\theta)\cos(\theta)u_{x,y}$
  $- r\sin(\theta)u_y - r^2\cos(\theta)\sin(\theta)u_{y,x} + r^2\cos^2(\theta)u_{y,y}$.

Now $u_{r,r} + \frac{u_{\theta,\theta}}{r^2} = u_{x,x} + u_{y,y} - \frac{\cos(\theta)u_x}{r} - \frac{\sin(\theta)u_y}{r} = \triangle u - \frac{u_r}{r}$. Therefore, we can conclude that:

$$\boxed{\partial_{x,x} + \partial_{y,y} = \Delta = \partial_{r,r} + \tfrac{1}{r}\partial_r + \tfrac{1}{r^2}\partial_{\theta,\theta}}.$$

Set 2 Problem 1: Assume that $\varphi : G \to \mathbb{R}$ is a C^2 function such that $\triangle\varphi = \varphi_{x,x} + \varphi_{y,y} \geq 0$. Let $a \in G$ and pick $R > 0$ such that $\overline{\Delta}(a, R) \subseteq G$. Then for $r \in [0, R)$ define the function:

$$h(r) = \frac{1}{2\pi}\int_0^{2\pi} \varphi(a + re^{it})\mathrm{d}t$$

(i) Show that h is non-decreasing.

Let $u(r, t) = \varphi(a + re^{it})$. I already showed how to calculate u_r and $u_{r,r}$ in problem 0. So I won't repeat that calculation. All that's important to know is that u_r and $u_{r,r}$ are continuous (since φ is C^2). Therefore, we can conclude by Leibniz' rule that:

$$h'(r) = \frac{1}{2\pi}\int_0^{2\pi} \frac{\partial}{\partial r}u\mathrm{d}t \implies rh'(r) = \frac{1}{2\pi}\int_0^{2\pi} r\frac{\partial}{\partial r}u\mathrm{d}t$$
$$\implies (rh')'(r) = \frac{1}{2\pi}\int_0^{2\pi} \frac{\partial}{\partial}(r\frac{\partial}{\partial r}u)\mathrm{d}t = \frac{1}{2\pi}\int_0^{2\pi} u_r + ru_{r,r}\mathrm{d}t$$
$$\implies r(rh')'(r) = \frac{1}{2\pi}\int_0^{2\pi} ru_r + r^2u_{r,r}\mathrm{d}t$$

But now by problem 0, we have that $ru_r + r^2 u_{r,r} = r^2 \triangle\varphi - u_{t,t}$. Therefore, we have that:

$$\begin{aligned} r(rh')'(r) &= \tfrac{1}{2\pi}\int_0^{2\pi} r^2 \triangle\varphi(a + re^{it})\mathrm{d}t - \tfrac{1}{2\pi}\int_0^{2\pi} u_{t,t}(r,t)\mathrm{d}t \\ &= \tfrac{1}{2\pi}\int_0^{2\pi} r^2 \triangle\varphi(a + re^{it})\mathrm{d}t - \tfrac{1}{2\pi}(u_t(r,2\pi) - u_t(r,0)) \end{aligned}$$

Yet u and thus u_t is 2π-periodic with respect to t. Thus $\frac{1}{2\pi}(u_t(r,2\pi) - u_t(r,0)) = 0$ and we are left with:

$$r(rh')'(r) = \tfrac{1}{2\pi}\int_0^{2\pi} r^2 \triangle\varphi(a + re^{it})\mathrm{d}t \geq 0$$

This proves that $(rh')' \geq 0$ when $r > 0$. Thus, we can conclude that rh' is a non-decreasing function on $[0, R)$. Then since $(0)h'(0) = 0$, this tells us that $rh'(r) \geq 0$ for all $r \in [0, r)$. The only way that is possible is if $h'(r) \geq 0$ when $r > 0$. So finally, we get that h is non-decreasing on $[0, r)$.

(ii) Using (i), show that φ is subharmonic if φ is C^2 and $\triangle\varphi \geq 0$ on G.

Let h be function from part (i). Then we know h is non-decreasing. Also, $h(0) = \frac{1}{2\pi}\int_0^{2\pi} \varphi(a)\mathrm{d}t = \varphi(a)$. Therefore, for all $r \in [0, R)$ we have that:

$$\varphi(a) \leq h(r) = \tfrac{1}{2\pi}\int_0^{2\pi} \varphi(a + re^{it})\mathrm{d}t$$

(iii) Now suppose $\psi : G - \{a\} \to \mathbb{R}$ is a harmonic function, and like before set $h(r) \coloneqq \frac{1}{2\pi}\int_0^{2\pi} \psi(a + re^{it})\mathrm{d}t$ for all $r \in (0, R)$. Show that $r(rh')' \equiv 0$ and conclude that $h(r) = \alpha\log(r) + \beta$ for some $\alpha, \beta \in \mathbb{R}$.

To start off, we can conclude by identical reasoning as that in part (i) that $r(rh')'(r) = \frac{1}{2\pi}\int_0^{2\pi} r^2 \triangle\psi(a + re^{it})\mathrm{d}t$.

Technically Leibniz's rule only applies to h when h is restricted to a compact subinterval $[\rho_1, \rho_2] \subseteq (0, R)$. That said, this isn't a limitation as we can just take $\rho_1 \to 0$ and $\rho_2 \to R$ to prove that $r(rh')'(r) = \frac{1}{2\pi}\int_0^{2\pi} r^2 \triangle u(r,t)\mathrm{d}t$ for all $r \in (0, R)$.

I realize now I should have also mentioned this when doing part (i). During that part of the problem, Leibniz' rule only technically applied to the compact subinterval $[0, \rho] \subseteq [0, R)$. That said, you could just take $\rho \to R$ to get the desired result.

But now since ψ is harmonic on $G - \{a\}$, this proves that $r(rh')'(r) \equiv 0$. Equivalently, we have that $rh'(r) = \alpha$ for some $\alpha \in \mathbb{R}$. Then by integrating $h'(r) = \frac{\alpha}{r}$, we get that $h(r) = \alpha\log(r) + \beta$ where $\beta \in \mathbb{R}$ is another constant.

(iv) Which of the following functions are subharmonic, harmonic, or neither?

(a) $f(x, y) = x^2 + y^2$

$\triangle f \equiv 4 > 0$. Hence while f is not harmonic, we know by part (ii) that f is subharmonic.

(b) $f(x, y) = x^2 - y^2$

$\triangle f \equiv 0$. Hence, this function is harmonic.

(c) $f(x, y) = x^2 + y$

$\triangle f \equiv 2$. Hence, this function is subharmonic by part (ii) but not harmonic. ■

Since the homework assignment for this class is due in three hours, I'm going to focus on doing the remaining homework problems on the 220c homework set before finishing my week 2 lecture notes.

Set 2 Problem 2 (Removable Singularities): Possibly using problem 1, show that if $u : G - \{a\} \to \mathbb{R}$ is a harmonic function and $\lim_{z\to a} u(z)$ exists and is finite, then u can be extended to a harmonic function on G.

If we extend u continuously to a, then it suffices to show u has the (local) mean value property at a in order to prove that the extension of u to all of G is harmonic. Thus, fix $R > 0$ such that $\overline{\Delta}(a, R) \subseteq G$. Then for all $r \in (0, R)$ let:

$$h(r) := \tfrac{1}{2\pi}\int_0^{2\pi} u(a + re^{it})\mathrm{d}t.$$

By part (iii) of the last problem, we know there are real constants α, β such that $h(r) = \alpha\log(r) + \beta$. At the same time though, by the continuity of u at a we know that $u(a + re^{it}) \to u(a)$ uniformly over all t as $r \to 0$. Hence, we can conclude that:

$$\lim_{r\to 0} h(r) = \tfrac{1}{2\pi}\lim_{r\to 0}\int_0^{2\pi} u(a + re^{it})\mathrm{d}t = \tfrac{1}{2\pi}\int_0^{2\pi}\lim_{r\to 0} u(a + re^{t})\mathrm{d}t = \tfrac{1}{2\pi}\int_0^{2\pi} u(a)\mathrm{d}t = u(a)$$

The only way the limit of $h(r)$ as $r \to 0$ can exist is if $\alpha = 0$. Hence, we know that $h(r) \equiv \beta$. Then since the limit of $h(r)$ as $r \to 0$ is specifically $u(a)$, this proves that $\beta = u(a)$. Hence, we've proven the (local) mean value property holds for u at a. ■

Set 2 Problem 3: Let $h : \Delta \to \mathbb{R}$ be a bounded harmonic function on the unit disk. Assume that $\limsup_{z\to a} h(z) \leq 0$ for all $a \in \partial\Delta - \{1\}$. Then show that $h \leq 0$ in Δ.

Let $\varepsilon > 0$ and define $h_\varepsilon(z) = h(z) + \varepsilon\log(\frac{|z-1|}{2})$. This will be a harmonic function on Δ (see *set 1 problem 3 on page 803*). Also note that if $a \in \partial\Delta - \{1\}$, then:

$$\limsup_{z\to a} h_\varepsilon(z) = \varepsilon\log(\tfrac{|a-1|}{2}) + \limsup_{z\to a} h(z) \leq \varepsilon\log(1) + \limsup_{z\to a} h(z) \leq 0.$$

Meanwhile, because h is a bounded function, we know that:

$$\limsup_{z\to 1} h_\varepsilon(z) = \lim_{z\to 1}\left(\varepsilon\log(\tfrac{|z-1|}{2})\right) = -\infty.$$

Hence, the theorem at the bottom of *page 805* tells us that $h_\varepsilon \leq 0$ on Δ. To finish off, note that $h_\varepsilon \to h$ pointwise on Δ as $\varepsilon \to 0$. Hence, we can conclude that $h \leq 0$ on Δ as well. ■

Set 2 Problem 4:

(i) Give an example of a harmonic function in the half plane, $u : \{z : \mathrm{Re}(z) > 0\} \to \mathbb{R}$ such that $\lim_{z\to iy} u(z) = 1$ for all $y > 0$ and $\lim_{z\to iy} u(z) = -1$ for all $y < 0$.

$u(z) = \frac{2}{\pi}\mathrm{Im}(\mathrm{Log}(z))$ works (where Log is the principal branch of the logarithm).

(ii) Using part (i), give an example of a harmonic function on the unit disc $v : \Delta \to \mathbb{R}$ such that $\lim_{r\to 1} v(re^{it}) = 1$ for all $t \in (0, \pi)$ and $\lim_{r\to 1} v(re^{it}) = -1$ for all $t \in (\pi, 2\pi)$.

$\varphi(z) = \frac{1-z}{1+z}$ is a biholomorphism mapping $\{z : \mathrm{Re}(z) > 0\}$ to Δ.

(I got this biholomorphism by plugging iz into the Cayley transform.)

We can solve that $\varphi^{-1}(z) = \frac{1-z}{1+z}$. Then if I set $v := -u \circ \varphi^{-1} = \frac{-2}{\pi}\mathrm{Im}(\mathrm{Log}(\frac{1-z}{1+z}))$, I shall have that v is a harmonic function on Δ.

Also note that for any $\theta \in (0,\pi) \cup (\pi, 2\pi)$, we have that:

$$\frac{1-e^{i\theta}}{1+e^{i\theta}} = \frac{(1-e^{i\theta})(1+e^{-i\theta})}{(1+e^{i\theta})(1+e^{-i\theta})} = \frac{-2i\sin(\theta)}{2-2\cos(\theta)} = -i\frac{\sin(\theta)}{1-\cos(\theta)}$$

Therefore, if $\sin(\theta) > 0$ (which happens when $\theta \in (0,\pi)$), we have that:

$$\lim_{r\to 1} v(re^{it}) = \frac{-2}{\pi}\mathrm{Im}(\mathrm{Log}(-i\frac{\sin(\theta)}{1-\cos(\theta)})) = \frac{-2}{\pi}\cdot\frac{-\pi}{2} = 1.$$

Meanwhile if $\sin(\theta) < 0$ (which happens when $\theta \in (\pi, 2\pi)$), we have that:

$$\lim_{r\to 1} v(re^{it}) = \frac{-2}{\pi}\mathrm{Im}(\mathrm{Log}(-i\frac{\sin(\theta)}{1-\cos(\theta)})) = \frac{-2}{\pi}\cdot\frac{\pi}{2} = -1. \blacksquare$$

Now I'm going back to taking lecture notes.

Remarks:

- I showed in set 2 problem 1 that if φ is C^2 and $\triangle\varphi \geq 0$, then φ is subharmonic. In turn, we can also conclude that if φ is C^2 and $\triangle\varphi \leq 0$ then φ is superharmonic. After all, $\triangle\varphi \leq 0 \implies \triangle(-\varphi) \geq 0 \implies -\varphi$ is subharmonic $\implies \varphi$ is superharmonic.
- If φ was a one variable function, then:
  - $\triangle\varphi = 0 \implies \varphi$ being linear.
  - $\triangle\varphi \geq 0 \implies \varphi$ being convex.
  - $\triangle\varphi \leq 0 \implies \varphi$ being concave.

  Thus subharmonicity can be viewed as a generalization of convexity.

Some prior theorems and proofs can be generalized for subharmonic functions.

Theorem (Maximum Principle): Suppose G is a region. If $\varphi : G \to \mathbb{R}$ is subharmonic and achieves a maximum at $a \in G$, then φ is a constant function.

Proof:
Repeat the steps of the proof on *page 804*. When proving that $\Omega := \{z \in G : \varphi(z) = \varphi(a)\}$ is open, you will get the expression that:

$$\varphi(a) = \varphi(z_0) \leq \frac{1}{2\pi}\int_0^{2\pi} \varphi(z_0 + \rho e^{it})\mathrm{d}t$$

Thus, we can conclude $\frac{1}{2\pi}\int_0^{2\pi} \varphi(a) - \varphi(z_0 + \rho e^{it})\mathrm{d}t \leq 0$. Yet since $\varphi(a) - \varphi(z_0 + \rho e^{it})$ is a nonnegative continuous function of t, the only way the integrand can be nonpositive is if $\varphi(a) - \varphi(z_0 + \rho e^{it}) \equiv 0$. And now the rest of the proof follows like on page 804. $\blacksquare$

Theorem (Maximum Principle +): Suppose G is a region, $\varphi : G \to \mathbb{R}$ is subharmonic, and $\forall a \in \partial_\infty G$ we have that $\limsup_{z\to a} \varphi(z) \leq 0$. Then $\varphi < 0$ or $\varphi \equiv 0$ on G.

Proof:
Do identical reasoning as on *pages 805-806*. Only, use the previous theorem instead of the theorem on page 804. $\blacksquare$

Corollary: If G is a bounded region, $\varphi : \overline{G} \to \mathbb{R}$ is a continuous function that is subharmonic on G, and $\varphi|_{\varphi G} \leq 0$, then $\varphi < 0$ or $\varphi \equiv 0$ on G.

We also have the following property:

Proposition: If φ_1, φ_2 is subharmonic then so is $\varphi := \max(\varphi_1, \varphi_2)$.

Proof:

Firstly note that φ is continuous. Also, if $a \in G$ then we can find R_1, R_2 such that:

- $\varphi_1(a) \leq \frac{1}{2\pi}\int_0^{2\pi} \varphi_1(a + re^{it})\mathrm{d}t \leq \frac{1}{2\pi}\int_0^{2\pi} \varphi(a + re^{it})\mathrm{d}t$ for all $r < R_1$,
- $\varphi_2(a) \leq \frac{1}{2\pi}\int_0^{2\pi} \varphi_2(a + re^{it})\mathrm{d}t \leq \frac{1}{2\pi}\int_0^{2\pi} \varphi(a + re^{it})\mathrm{d}t$ for all $r < R_2$.

Now set $R = \min(R_1, R_2)$. Then for $r < R$ we have that:

$$\varphi(a) = \max(\varphi_1(a), \varphi_2(a)) \leq \frac{1}{2\pi}\int_0^{2\pi} \varphi(a + re^{it})\mathrm{d}t$$

Thus φ is subharmonic. ■.

I'm going to switch to taking lecture notes on a different class now.

4/13/2026

Math 200c Lecture Notes:

The particular topic I'm about to take notes on is a branch of Galois theory called Kummer theory. Specifically, Kummer theory studies the case where Galois groups are cyclic.

Proposition: Let F be a field such that $x^n - 1$ splits in $F[x]$. Assume $\mathrm{char}(F) = 0$ or $p > 0$ and $p \nmid n$. Then let L/F be a field extension where L has a root α of $f(x) = x^n - a$ for some $a \in F$. If $K = F[\alpha]$ then K/F is a Galois extension and $\mathrm{Gal}(K/F)$ is cyclic with some order dividing n.

Proof:

The assumption on $\mathrm{char}(F)$ guarentees that $(x^n - 1)' = nx^{n-1} \neq 0$. Hence $\gcd(x^n - 1, nx^{n-1}) = 1$ and so $x^n - 1$ is separable with roots in F by assumption. If we let C be the nth roots of 1, then C is a subgroup of $F^\times$ with order n. In turn, C is cyclic by the following lemma.

Lemma: A finite subgroup of the multiplicative group of a field is always cyclic.

Proof:

Suppose $G < F^\times$ is a finite subgroup. For any positive integer n, we know that $|\{a \in G : a^n = 1\}| = |V_{x^n-1}(F)| \leq \deg(x^n - 1) = n$. Thus, we can apply set 1 problem 2 from math 200a (see *pages 259-260*) to see that G is a cyclic.

Thus $\mathbb{Z}/n\mathbb{Z} \cong C = \langle\zeta\rangle$ for some $\zeta \in F$. Also, we may write $C = \{1, \zeta, \ldots, \zeta^{n-1}\} \subseteq F$.

Now, the n distinct roots of $x^n - a$ are given by $\alpha, \alpha\zeta, \alpha\zeta^2, \ldots, \alpha\zeta^{n-1}$. Hence, we can say that $K = F[\alpha]$ is a splitting field of $x^n - a$ over F. Also $x^n - a$ is separable (for the same reason that $x^n - 1$ is). Thus, K/F is Galois.

To finish off, if $\sigma \in G := \mathrm{Gal}(K/F)$, then $\sigma(\alpha) = \zeta^i \alpha$ for some i. But σ is uniquely determined on K by where it sends α. So, we have a well-defined injective function $\Phi : G \to C$ such that $\sigma \mapsto \zeta^i$ where $\sigma(\alpha) = \zeta^i \alpha$.

Furthermore, it is easy to check that Φ is a group homomorphism. After all, if $\sigma_1(\alpha) = \zeta^{i_1}\alpha$ and $\sigma_2(\alpha) = \zeta^{i_2}\alpha$, then since both σ_k fix $\zeta \in F$, we know $(\sigma_1 \circ \sigma_2)(\alpha) = \zeta^{i_1+i_2}\alpha$.

Thus $G \cong \Phi(G)$ which is a subgroup of C. So, G is cyclic of some order dividing n. ■

Let G be a group. A character of G with values in a field F is a group homomorphism $\chi : G \to F^\times$. Importantly, any character lies in the F-vector space $F^G := \mathrm{Hom}_{\mathbf{Set}}(G, F)$ (equipped with pointwise addition and scalar multiplication).

Proposition: Let $\chi_1, \ldots, \chi_n$ be n distinct characters from G to $F^\times$. Then they are linearly independent in F^G.

Proof:
Suppose not. Then there must exist some minimal $d \leq n$ such that there exists constants $a_1, \ldots, a_d \in F^\times$ and distinct characteristics $\chi_{i_1}, \ldots, \chi_{i_d}$ in our collection of characteristics satisfying that $a_1\chi_{i_1} + \ldots + a_d\chi_{i_d} = 0$.

(For ease of notation, I'll henceforth relabel χ_{i_k} as χ_k.)

Firstly, note that $d \geq 2$ since $a_1\chi_1 \equiv 0$ and $a_1 \neq 0$ would imply $\chi_1 \equiv 0$ (which contradicts that χ_1 maps G into $F^\times$). Next, note that $a_1\chi_1(g) + \ldots + a_d\chi_d(g) = 0$ for all $g \in G$. By plugging in gh into this expression (where $h \in G$), we thus get that:

$$a_1\chi_1(g)\chi_1(h) + a_2\chi_2(g)\chi_2(h) + \ldots + a_d\chi_d(g)\chi_d(h) = 0 \text{ for all } g \in G$$

We can also multiply both sides of our original linear combination by $\chi_d(h)$ to get that:

$$a_1\chi_1(g)\chi_d(h) + a_2\chi_2(g)\chi_d(h) + \ldots + a_d\chi_d(g)\chi_d(h) = 0 \text{ for all } g \in G$$

Then by subtracting these expressions, we get for all $g \in G$ that:

$$a_1(\chi_1(h) - \chi_d(h))\chi_1(g) + \ldots + a_{d-1}(\chi_{d-1}(h) - \chi_d(h))\chi_{d-1}(g) + 0 = 0$$

This would contradict the minimality of d unless $a_i(\chi_i(h) - \chi_d(h)) = 0$ for all i. Or in other words, we must have that $\chi_i(h) = \chi_d(h)$ for all i. But since our characteristics are all distinct and $d \geq 2$, we know there exists some h such that $\chi_1(h) \neq \chi_d(h)$. By choosing that h, we can guarentee our contradiction happens. ■

Given a finite Galois extension K/F, we define the (field) norm of any $\alpha \in K$ as:

$$N_{K/F}(\alpha) = \prod_{\sigma \in \mathrm{Gal}(K/F)} \sigma(\alpha)$$

Note that $N_{K/F}$ maps K into F. After all, given any $\tau \in \mathrm{Gal}(K/F)$ we can easily check that $\tau(N_{K/F}(\alpha)) = N_{K/F}(\alpha)$. Hence, $N_{K/F}(\alpha) \in \mathrm{Fix}(\mathrm{Gal}(K/F)) = F$.

It's also easy to check that $N_{K/F}(\alpha\beta) = N_{K/F}(\alpha) \cdot N_{K/F}(\beta)$, and that $N_{K/F}(\alpha) = 0$ iff $\alpha = 0$. So, $N_{K/F}$ restricted to $K^\times$ is a characteristic into $F^\times$.

As an example of a field norm, note that if we consider the extension $\mathbb{Q}[\sqrt{d}]/\mathbb{Q}$ where $d \in \mathbb{Z}$ is square-free, then $\mathrm{Gal}(\mathbb{Q}[\sqrt{d}]/\mathbb{Q}) = \{\mathrm{Id}, \sigma\}$ where σ sends $\sqrt{d}$ to $\sqrt{-d}$. It follows that:

$$N_{\mathbb{Q}[\sqrt{d}]/\mathbb{Q}}(a + b\sqrt{d}) = (a + b\sqrt{d})(a - b\sqrt{d}) = a^2 - b^2 d$$

This norm should be familiar from math 100b.

Theorem (Hilbert Theorem 90): Let K/F be a Galois extension with $\mathrm{Gal}(K/F) = \langle\sigma\rangle$ for some element σ of order n. Suppose $\alpha \in K$ satisfies that $N_{K/F}(\alpha) = 1$. Then $\alpha = \sigma(\beta)\beta^{-1}$ for some $\beta \in K$.

Proof:

Pick any $\gamma \in K$. Then let:

$$\epsilon = \gamma + \alpha\sigma(\gamma) + \alpha\sigma(\alpha)\sigma^2(\gamma) + \ldots + \alpha\sigma(\alpha)\cdots\sigma^{n-2}(\alpha)\sigma^{n-1}(\gamma).$$

In turn, we have that:

$$\begin{aligned}\alpha\sigma(\epsilon) &= \alpha\sigma(\gamma) + \alpha\sigma(\alpha)\sigma^2(\gamma) + \ldots + \alpha\sigma(\alpha)\cdots\sigma^{n-2}(\alpha)\sigma^{n-1}(\gamma) + N_{K/F}(\alpha)\gamma. \\ &= \epsilon \text{ (since } N_{K/F}(\alpha) = 1).\end{aligned}$$

So assuming $\epsilon \neq 0$, then we can set $\beta := \epsilon^{-1}$ to get that $\alpha = \sigma(\beta)\beta^{-1}$.

What if $\epsilon = 0$ though?

Notice that $\sigma^i : K^\times \to K^\times$ is a character of $K^\times$. By the linear independence of characters we thus have that:

$$\mathrm{Id}_K + \alpha\sigma + \alpha\sigma(\alpha)\sigma^2 + \ldots + \alpha\sigma(\alpha)\cdots\sigma^{n-2}(\alpha)\sigma^{n-1} \not\equiv 0.$$

Hence, there must exist some $\gamma \in K^\times$ such that $\epsilon \neq 0$. ■

Theorem: Let K/F be a Galois extension where $x^n - 1$ splits in F. Suppose $\mathrm{char}(F) = 0$ or $\mathrm{char}(F) = p$ (with $p \nmid n$). Also suppose $\mathrm{Gal}(K/F) = \langle\sigma\rangle$ is cyclic of order n. Then $K = F[\alpha]$ for some $\alpha \in K$ such that $\alpha^n \in F$.

Proof:

The roots of $x^n - 1$ in F are a subgroup of $F^\times$ with order n (since our assumptions about $\mathrm{char}(F)$ guarentee that $x^n - 1$ is separable). Thus, by the lemma proven on page 843, we know that that group is cyclic. In other words, there exists a primitive root ζ such that $\{1, \zeta, \ldots, \zeta^{n-1}\}$ are the distinct nth roots of unity in F.

Next, note that because $\zeta \in F$ we know that $\sigma(\zeta) = \zeta$. As a result, we can calculate that:

$$N_{K/F}(\zeta) = \zeta\sigma(\zeta)\cdots\sigma^{n-1}(\zeta) = \zeta^n = 1$$

By Hilbert theorem 90 there thus exists some $\alpha \in K$ such that $\zeta = \sigma(\alpha)\alpha^{-1}$. In other words, $\sigma(\alpha) = \alpha\zeta$. Also note then that $\sigma(\alpha^n) = \zeta^n\alpha^n = \alpha^n$. Therefore, $\alpha^n \in \mathrm{Fix}(\langle\sigma\rangle) = F$.

Finally, write $F \subseteq F[\alpha] \subseteq K$. Then $\mathrm{Gal}(K/F[\alpha]) \leq \langle\sigma\rangle$. If this subgroup is nontrivial, there must exist some d such that $\sigma^d \neq \mathrm{Id}_K$ and σ^d fixes $F[\alpha]$. But $\sigma^d(\alpha) = \zeta^d\alpha \neq \alpha$ unless $n \mid d$. Hence, $\mathrm{Gal}(K/F[\alpha])$ is trivial and this proves that $F[\alpha] = K$. ■

There's one other theorem I want to cover from today. But, I don't think the professor covered it well in class today. Also, after looking at a different professor's proof, I think I need slightly more machinery. Thus, I'm going to hold off on taking more algebra notes for now.

Math 200c Homework:

Set 2 Problem 1: In this problem we'll prove the additive form of Hilbert Theorem 90. Let K/F be a Galois extension such that $\mathrm{Gal}(K/F) = \langle \sigma \rangle$ is a cyclic group of order n. Suppose that $\alpha \in K$ is an element such that:

$$\mathrm{Tr}_{K/F}(\alpha) \coloneqq \sum_{\sigma \in \mathrm{Gal}(K/F)} \sigma(\alpha) = 0$$

Then show there exists $\beta \in K$ such that $\alpha = \sigma(\beta) - \beta$.

Firstly, we'll need the following result.

Lemma: $\mathrm{Tr}_{K/F}$ is a nonzero surjective F-linear map from K to F.

$\mathrm{Tr}_{K/F}$ is F-linear because it is a sum of F-linear maps. Also note that for every $\alpha \in K$,

$$\tau(\mathrm{Tr}_{K/F}(\alpha)) = \sum_{\sigma \in \mathrm{Gal}(K/F)} (\tau \circ \sigma)(\alpha) = \sum_{\sigma' \in \mathrm{Gal}(K/F)} \sigma'(\alpha) = \mathrm{Tr}_{K/F}(\alpha)$$

Therefore $\mathrm{Tr}_{K/F}(K) \subseteq \mathrm{Fix}(\mathrm{Gal}(K/F)) = F$.

Finally, note that each σ is a distinct character from $K^\times$ to $K^\times$. As a result, they are linearly independent. And this implies that $\mathrm{Tr}_{K/F}(\alpha) \coloneqq \sum_{\sigma \in \mathrm{Gal}(K/F)} \sigma(\alpha) \not\equiv 0$ on $K^\times$.

Since F only has one F-dimension and the image of $\mathrm{Tr}_{K/F}$ is not a trivial subspace, we can thus conclude that $\mathrm{im}(\mathrm{Tr}_{K/F}) = F$.

Consider any $\gamma \in K$ such that $\mathrm{Tr}_{K/F}(\gamma) = -1$ and set:

$$\beta = \alpha\sigma(\gamma) + (\alpha + \sigma(\alpha))\sigma^2(\gamma) + \ldots + (\alpha + \sigma(\alpha) + \ldots + \sigma^{n-2}(\alpha))\sigma^{n-1}(\gamma)$$

Then as $\mathrm{Tr}_{K/F}(\alpha) = 0$ and $\mathrm{Tr}_{K/F} = -1$, we can evaluate that:

$$\begin{aligned}\sigma(\beta) &= (\alpha - \alpha + \sigma(\alpha))\sigma^2(\gamma) + (\alpha - \alpha + \sigma(\alpha) + \sigma^2(\alpha))\sigma^3(\gamma) \\ &\qquad + \ldots + (\alpha - \alpha + \sigma(\alpha) + \ldots + \sigma^{n-1}(\alpha))\gamma \\ &= -\alpha(\gamma + \sigma^2(\gamma) + \sigma^3(\gamma) + \ldots + \sigma^{n-1}(\gamma)) + (\alpha + \sigma(\alpha))\sigma^2(\gamma) \\ &\qquad + (\alpha + \sigma(\alpha) + \sigma^2(\alpha))\sigma^3(\gamma) \\ &\qquad + \ldots + (\alpha + \sigma(\alpha) + \ldots + \sigma^{n-2}(\alpha))\sigma^{n-1}(\gamma) \\ &\qquad + 0\gamma \\ &= \alpha + \alpha\sigma(\gamma) + (\alpha + \sigma(\alpha))\sigma^2(\gamma) + \ldots + (\alpha + \sigma(\alpha) + \ldots + \sigma^{n-2}(\alpha))\sigma^{n-1}(\gamma) \\ &= \alpha + \beta\end{aligned}$$

By rearranging we get that $\sigma(\beta) - \beta = \alpha$. ■

Set 2 Problem 2: Let F be a field of characteristic $p > 0$. Let K be the splitting field over F of the polynomial $f(x) = x^p - x - a \in F[x]$. K/F is called an Artin-Schreier extension. Assume that $K \neq F$ (i.e. $f(x)$ does not already split in F). Then prove that $K = F[\alpha]$ where α is any root of f. Also show that K/F is a Galois extension and $\mathrm{Gal}(K/F) \cong \mathbb{Z}/p\mathbb{Z}$.

Proof:
Suppose $f(\alpha) = \alpha^p - \alpha - a = 0$. Then we have for any $k \in K$ that:

$$(\alpha + k)^p - (\alpha + k) - a = \alpha^p + k^p - \alpha - k - a = 0 + k^p - k.$$

In particular, $k^p - k = 0$ iff $k \in \mathbb{F}_p \subseteq K$. Therefore, if α is a zero of f, then we know the p distinct zeros of f are $\alpha, \alpha + 1, \ldots, \alpha + p - 1$. This shows f is separable and in turn K/F is a Galois extension. Also, this shows that $F[\alpha]$ contains all the roots of f and thus $K = F[\alpha]$.

The next task I want to accomplish is showing that $[K : F] = p$. That way, as K/F is a Galois extension we will know that $|\mathrm{Gal}(K/F)| = p$. Then in turn, as any $\sigma \in \mathrm{Gal}(K/F)$ is uniquely determined by where it sends α and we know $\sigma(\alpha) = \alpha + k$ for some $k \in \mathbb{F}_p$, we can conclude that for every $k \in \mathbb{F}_p$ there exists some $\sigma \in \mathrm{Gal}(K/F)$ such that $\sigma(\alpha) = \alpha + k$.

Label the zeros of f as $\alpha_1, \ldots, \alpha_p$. Importantly, since $F[\alpha_i] = K = F[\alpha_j]$ for all i, j, we know that the degrees of the minimal polynomials $m_{\alpha_i;F}$ and $m_{\alpha_j;F}$ equal each other. Also the minimal polynomials of the α_i must also divide f. And thirdly, since f is separable, we know each of those minimal polynomials are separable. It follows that the zero sets $V_{m_{\alpha_i;F}}(K)$ of the $m_{\alpha_i;F}$ partition the zero set $V_f(K)$ of f into disjoint subsets whose size are all equal to the degree of $m_{\alpha_1;F}$.

It follows that $\deg(m_{\alpha_1;F}) \mid p$. And since p is prime, this proves that either $m_{\alpha_1;F}(x) = x^p - x - a$ or $\alpha_1 \in F$. Yet the latter case would imply $K = F[\alpha_1] = F$, which contradicts our initial assumption that $K \neq F$. Hence, we can conclude that $x^p - x - a$ is irreducible in $F[x]$ and thus $[K : F] = p$.

To finish off, we define an isomorphism from $\mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z}$ to $\mathrm{Gal}(K/F)$. Specifically, given any $k \in \mathbb{F}_p$ let $\Phi(k)$ equal the unique automorphism in $\mathrm{Gal}(K/F)$ sending α to $\alpha + k$. By our prior work, we know Φ is well-defined. Φ is also automatically injective. And since $|\mathbb{F}_p| = p = |\mathrm{Gal}(K/F)|$, we know Φ is surjective. Finally:

$$(\Phi(k_1 + k_2))(\alpha) = \alpha + k_1 + k_2 = (\Phi(k_1))(\alpha + k_2) = (\Phi(k_1) \circ \Phi(k_2))(\alpha).$$

Thus Φ is a group homomorphism from $(\mathbb{F}_p, +)$ to $\mathrm{Gal}(K/F)$. ■

Set 2 Problem 3: Let F be a field of characteristic $p > 0$. Let K/F be a Galois extension with $\mathrm{Gal}(K/F) \cong \mathbb{Z}/p\mathbb{Z}$.

Some commentary from professor Rogalski:
This is a situation not covered by the theorem on the bottom half of *page 845* since F cannot contain p many pth roots of 1. In fact, $x^p - 1 = (x - 1)^p$. So, the only pth root of 1 is 1 and thus we shouldn't expect to be able to write $K = F[\beta]$ where $\beta^n \in F$.

Instead, we can show as follows that K/F will always be an Artin-Schreier extension.

Show that there exists $\alpha \in K$ such that $K = F[\alpha]$ and α is a root of the polynomial $f(x) = x^p - x - a$ for some $a \in F$. (In turn, by the prior problem K is also the splitting field of f over F).

Let $\sigma \in \mathrm{Gal}(K/F)$ generate the Galois group. Then since $1 \in F$, we know that $\sigma(1) = 1$. It follows that $\mathrm{Tr}_{K/F}(1) = p(1) = 0$. And by the additive version of Hilbert theorem 90, we thus know that there exists $\alpha \in K$ satisfying that $\sigma(\alpha) - \alpha = 1$. Or in other words, $\sigma(\alpha) = \alpha + 1$.

Also note that $\sigma(\alpha^p - \alpha) = (\alpha+1)^p - (\alpha+1) = \alpha^p - \alpha + 1 - 1 = \alpha^p - \alpha$. Therefore $\alpha^p - \alpha \in \mathrm{Fix}(\langle\sigma\rangle) = F$. And if we let $a = \alpha^p - \alpha$, then α is a root of $f(x) = x^p - x - a \in F[x]$.

To finish off, write $F \subseteq F[\alpha] \subseteq K$. If $\mathrm{Gal}(K/F[\alpha])$ is nontrivial, then there must exist some integer $d \in \{1, \ldots, p-1\}$ such that $\sigma^d(\alpha) = \alpha + d = \alpha$. Yet $\alpha + d = \alpha$ iff $d = 0$ which happens iff $d = kp$ for some integer k. So we conclude that $\mathrm{Gal}(K/F[\alpha])$ is trivial. And this proves that $K = F[\alpha]$. ■

Set 2 Problem 4: Suppose that K/F is a Galois extension with $\mathrm{Gal}(K/F)$ being an abelian group of order n. Assume that $x^n - 1$ splits in $F[x]$ with distinct roots. Then show that K is a splitting field over F of some polynomial of the form:

$$f(x) = (x^{n_1} - a_1)(x^{n_2} - a_2)\cdots(x^{n_d} - a_d) \text{ where each } a_i \in F.$$

To start off, recall that an abelian group is equivalent to a $\mathbb{Z}$-module. And as $\mathbb{Z}$ is a PID, we can conclude by the fundamental theorem of finitely generated modules over PIDs that any finitely generated abelian group is isomorphic to some direct sum $\mathbb{Z}^r \oplus \bigoplus_{i=1}^d \mathbb{Z}/n_i\mathbb{Z}$ where $r, n_1, \ldots, n_d$ are unique integers (when all n_i are positive and $n_1 \mid n_2 \mid \cdots \mid n_d$).

Since $G \coloneqq \mathrm{Gal}(K/F)$ is a *finite* abelian group, we in particular can see that:

$$G \cong \bigoplus_{i=1}^d \mathbb{Z}/n_i\mathbb{Z}.$$

In turn, we must also have that $|G| = n = \prod_{i=1}^d n_i$. Hence, $n_i \mid n$ for all i in our expression above. And since $x^n - 1$ splits in F with distinct zeros, we thus know $x^{n_i} - 1$ splits in F with distinct zeros as well. (After all, every zero of $x^{n_i} - 1$ would be a zero of $x^n - 1$.)

One other observation I want to make is that $x^N - 1$ splitting in $F[x]$ and having distinct roots guarentees that $\mathrm{char}(F) = 0$ or $\mathrm{char}(F) \nmid N$. After all, if neither of these two cases hold, then there exists a prime number $p = \mathrm{char}(F)$ and an integer k such that $N = pk$. In turn, $x^N - 1 = x^{pk} - 1 = (x^k - 1)^p$ will not have distinct roots. I point this out just to say that we won't need to worry about the characteristic of F at all.

Let $H_i \coloneqq \bigoplus_{j \neq i}(\mathbb{Z}/n_i\mathbb{Z})$. As G is an abelian group, we automatically have that $H_i \lhd G$. Therefore, if we let $E_i \coloneqq \mathrm{Fix}(H_i)$, we know from the theorem on *pages 815-816* that E_i/F is a Galois extension and $\mathrm{Gal}(E_i/F) \cong G/H_i \cong \mathbb{Z}/n_i\mathbb{Z}$. Due to the observations in the prior paragraphs, we can then apply the theorem at the bottom of page 845 to get that $E_i = F[\alpha_i]$ for some $\alpha_i \in E_i$ satisfying that $\alpha_i^{n_i} = a_i \in F$.

Finally, we define $f(x) = (x^{n_1} - a_1)\cdots(x^{n_d} - a_d)$. A splitting field of f over F is given by $E \coloneqq F[\alpha_1, \ldots, \alpha_d] \subseteq K$. To prove that $E = K$, consider that the Galois group $\mathrm{Gal}(K/E)$ must fix α_i for all i. In particular, this means if $\sigma \in \mathrm{Gal}(K/E)$ then $\sigma \in \mathrm{Gal}(K/E_i) = H_i$ for all i. Yet $\bigcap H_i = \{\mathrm{Id}\}$. So, we've proven that $\mathrm{Gal}(K/E)$ is trivial and hence $E = K$. ■

4/15/2026

Before getting back into PDE notes, I want to go over some important results in ODE theory. To start off, I realized that the proofs of both theorem D.3 and D.4 on *pages 292-294* work identically if we assume the (*different*) hypothesis that:

- $U \subseteq \mathbb{R}^n$ is an open set and $J \subseteq \mathbb{R}$ is an open interval;
- $V : J \times U \to \mathbb{R}^n$ satisfies that $V(t, \vec{y})$ is continuous with respect t and Lipschitz (by some constant *independent* of t) with respect to $\vec{y}$;

in order to prove that the nonautonomous initial value problem $y'(t) = V(t, y(t))$ and $y(t_0) = \vec{c}_0$ (where $t_0 \in J$ and $\vec{c}_0 \in U$) has a solution on a small enough interval and also that any two solutions will agree on their common domain.

> There are literally no new ideas involved in proving this. In essence, all I'm stating is that the proofs I did before still work with the above nonautonomous ODEs so long as we slightly weaken the continuity requirements of V with respect to t and strengthen the continuity requirements of V with respect to $\vec{y}$.
>
> This fact is called the Picard-Lindelöf Theorem (if you want to look it up).

In the special case that $U = \mathbb{R}^n$, we can simplify the argument of the Picard-Lindelöf theorem and get a stronger result (which I will need in a moment). Hence, I'll explicitly write out this argument.

Theorem: Let $J \subseteq \mathbb{R}$ be an open interval and let $V : J \times \mathbb{R}^n \to \mathbb{R}^n$ be a function satisfying that $V(t, \vec{y})$ is continuous with respect to t and Lipschitz with respect to $\vec{y}$ by some constant C (which is independent of t).

1. Given any $t_0 \in J$ and $0 < \varepsilon < \frac{1}{C}$ such that $J_0 := [t_0 - \varepsilon, t_0 + \varepsilon] \subseteq J$, we can find a solution $y : J_0^\circ \to \mathbb{R}^n$ to the following initial value problem for any $\vec{c}_0 \in \mathbb{R}^n$:
$$y'(t) = V(t, y(t)) \text{ and } y(t_0) = \vec{c}_0. \ (*)$$

> Proof:
> Note that the the initial value problem above is equivalent to the integral equation $y(t) = \vec{c}_0 + \int_{t_0}^t V(s, y(s))\mathrm{d}s$. Thus if we consider the integral operator $(Iy)(t) := \vec{c}_0 + \int_{t_0}^t V(s, y(s))\mathrm{d}s$, a function will solve our initial value problem if it is a fixed point of the operator I.
>
> > As a side note, so long as $y(t)$ is a continuous function then the integral $\int_{t_0}^t V(s, y(s))\mathrm{d}s$ exists and is finite. To see why, first note that V is a continuous function (jointly with respect to both of its arguments).
> >
> > > To see this, consider any $(t_1, \vec{y}_1) \in J \times \mathbb{R}^n$ and let $\delta > 0$ be such that $\|V(t, \vec{y}_1) - V(t_1, \vec{y}_1)\|_2 < \varepsilon/2$ when $|t - t_1| < \delta$. Then if we have that $\|\vec{y} - \vec{y}_1\|_2 < \frac{\varepsilon}{2C}$ and $|t - t_1| < \delta$, we can guarentee by the triangle inequality that:
> > > $$\|V(t, \vec{y}) - V(t_1, \vec{y}_1)\| < \tfrac{\varepsilon}{2} + C\tfrac{\varepsilon}{2C}.$$

It follows that $V(s, y(s))$ is a continuous (and thus measurable) function of s (if y is a continuous function). And as the interval $[t_0, t]$ or $[t, t_0]$ is compact, we thus know the integral exists and is finite for all $t \in J_0$.

Let $\mathcal{M}_c := C(\overline{J_0}, \mathbb{R}^n)$. This is a Banach space when equipped with the uniform norm. Also note that I maps $\mathcal{M}_c$ into $\mathcal{M}_c$. After all, indefinite integrals of continuous functions are continuously differentiable. (This is just from my math 140b notes).

Finally, note that I is a contraction on $\mathcal{M}_c$. After all, given distinct $y, \widetilde{y} \in \mathcal{M}_c$, we have that:

$$\begin{aligned}\|Iy - I\widetilde{y}\|_u &= \sup_{t\in J_0} \left\| \int_{t_0}^t V(s, y(s)) - V(s, \widetilde{y}(s)) \mathrm{d}s \right\|_2 \\ &\leq \sup_{t\in J_0} \int_{t_0}^t \|V(s, y(s)) - V(s, \widetilde{y}(s))\|_2 \, \mathrm{d}s \\ &\leq \sup_{t\in J_0} \int_{t_0}^t C \, \|y(s) - \widetilde{y}(s)\|_2 \, \mathrm{d}s \leq C\varepsilon \|y - \widetilde{y}\|_u\end{aligned}$$

And since $\varepsilon < \frac{1}{C}$, we know $C\varepsilon < 1$.

By the Banach fixed point theorem, we know there is a unique function $y \in \mathcal{M}_c$ such that $(Iy)(t) = y(t)$. In turn, $y|_{J_0^\circ}$ satisfies our initial value problem.

As a side note, the significance of how I stated this result is that the same ε works for any $\vec{c} \in \mathbb{R}^n$ and t_0 farther than ε away from the boundary of J.

2. For any $t_0 \in \mathbb{R}$ and $\vec{c} \in \mathbb{R}^n$, any two solutions (with open intervals as domains) to the initial value problem $y'(t) = V(t, y(t))$ and $y(t_0) = \vec{c}$ $(**)$ are equal on their common domain.

   Suppose J_1, J_2 are both open subintervals of J containing t_0 and $y : J_1 \to \mathbb{R}^n$ and $y : J_2 \to \mathbb{R}^n$ both solve $(**)$. By setting $J_0 := J_1 \cap J_2$ we can now without loss of generality assume y and $\widetilde{y}$ have the same domain.

   Now set $u(t) = y(t) - \widetilde{y}(t)$ and note that $\|u'(t)\|_2 \leq C\|u(t)\|_2$. Therefore, if we apply the *ODE comparison theorem on pages 289-290* to the functions $u(t)$ as well $f(v) = Cv$ and $v(t) = 0$, we get that $\|u(t)\|_2 \leq v(|t - t_0|) = 0$ for all $t \in J_0$. This proves that $y(t) = \widetilde{y}(t)$ on J_0. ■

   As a consequence of this, we can take the union of any functions (with open intervals as domains) solving the initial value problem $(**)$ in order to get a well-defined solution function on a larger open interval. After all:

   - An arbitrary union of open sets is open. Also an arbitrary union of connected sets containing a common point is connected. So, if $\{y_\alpha : J_\alpha \to \mathbb{R}^n\}_{\alpha\in A}$ is a collection of solutions to our initial value problem, then $\bigcup_{\alpha\in A} J_\alpha$ is an open connected subset of $\mathbb{R}$. This means $\bigcup_{\alpha\in A} J_\alpha$ is an open interval.
   - Also, since the collection of functions y_α agree on their common domains, we know the union of those functions is a function. Plus, that union of functions clearly still satisfies the initial value problem as derivatives are a locally determined value.

Corollary: Let $J \subseteq \mathbb{R}$, $V : J \times \mathbb{R}^n \to \mathbb{R}^n$, and $C \geq 0$ be as in the prior theorem. Given any $t_0 \in J$ and $\vec{c}$, there exists a unique solution $y : J \to \mathbb{R}^n$ satisfying that $y'(t) = V(t, y(t))$ and $y(t_0) = \vec{c}_0$.

Proof:

By part 1 of the prior theorem, we can always find a solution on an open interval containing t_0. Therefore, if we consider the collection $\mathcal{F}$ of functions defined on an open interval containing t_0 and solving our initial value problem, we know that $\mathcal{F}$ is nonempty. Furthermore, by the observation I wrote after part 2, we know that any chain in $\mathcal{F}$ (ordered by inclusion) has an upper bound in $\mathcal{F}$.

Thus, we can apply Zorn's lemma to find a solution $y(t)$ to our initial value problem that is defined on a *maximal* open interval $J_0 \subseteq J$. To finish the proof, I claim that $J_0 = J$.

Suppose not and write $J_0 = (a, b)$. Then either $a \in J$ or $b \in J$. Since an analogous argument yields a contradiction when $b \in J$, I will just focus on the case where $a \in J$.

Since J is open and $a \in J$, we can find some $\varepsilon < \min(\frac{b-a}{4}, \frac{1}{C})$ such that $(a - 2\varepsilon, a + 4\varepsilon) \subseteq J$.

As a side note, I'm mandating $\varepsilon < \frac{b-a}{4}$ merely so that $(a, a + 4\varepsilon) \subseteq (a, b)$. If $b = \infty$, then $\frac{b-a}{4} = \infty$ and so we don't need to worry about this.

By part 1 of the prior theorem, we can find a solution $\widetilde{y} : (a - \varepsilon, a + 3\varepsilon) \to \mathbb{R}^n$ to the initial value problem:

$$\widetilde{y}'(t) = V(t, \widetilde{y}(t)) \text{ and } \widetilde{y}(a + \varepsilon) = y(a + \varepsilon).$$

But then by part 2 of the prior theorem, we know $\widetilde{y}$ and y agree on their common domains. So, we can take their union in order to get a solution to our original initial value problem on the interval $(a - \varepsilon, b)$. But this contradicts the maximality of the solution $y \in \mathcal{F}$. ■

Where the prior result is most relevant is in the theory of linear ODEs. An ordinary differential equation $y'(t) = V(t, y(t))$ is called linear if:

$$V(t, \vec{y}) = (A(t))\,\vec{y} + f(t),$$

where $A : J \to M_{n\times n}(\mathbb{R})$, $f : J \to \mathbb{R}^n$, and $J \subseteq \mathbb{R}$ is an open interval.

(I realize I never gave a proper definition for linear PDEs. That said, I feel like I can avoid doing that unless I'm trying to formulate a general theorem.)

We'll equip $M_{n\times n}(\mathbb{R})$ with the 2-operator norm. Note that if every (i, j)th component function of $A : \mathbb{R} \to M_{n\times n}(\mathbb{R})$ is continuous, then A will be a continuous function. This is simply because the 2-operator norm is bounded by the Frobenius norm.

Theorem: Let $J \subseteq \mathbb{R}$ be an open interval and suppose $A(t) : J \to M_{n\times n}(\mathbb{R})$, $f(t) : J \to \mathbb{R}^n$ are continuous functions. Then for any $t_0 \in J$ and $\vec{c} \in \mathbb{R}^n$ there exists a unique function $y : J \to \mathbb{R}^n$ satisfying that:

$$y'(t) = A(t)y(t) + f(t) \text{ and } y(t_0) = \vec{c}.$$

Proof:
The important observation to make is that if $V(t, \vec{y}) = A(t)\vec{y} + f(t)$, then V is automatically continuous with respect to t. Also, for any $\vec{y}_1, \vec{y}_2 \in \mathbb{R}^n$ and $t \in J$ we have that $\|V(t, \vec{y}_1) - V(t, \vec{y}_2)\|_2 \leq \|A(t)\|_{\text{op}} \|\vec{y}_1 - \vec{y}_2\|_2$. Therefore, if we can bound $\|A(t)\|_{\text{op}}$, then the prior corollary applies. The one complication we run into however is that we can only bound $\|A(t)\|_{\text{op}}$ for t in compact subsets of J.

To address this difficulty, let $\{J_n\}_{n\in\mathbb{N}}$ be an increasing sequence of bounded open intervals whose closures are contained in J and which satisfy that $J = \bigcup_{n\in\mathbb{N}} J_n$. By the prior corollary, there exists a unique solution $y_n : J_n \to \mathbb{R}^n$ to our initial value problem on the interval J_n. Also, we must have that $y_n|_{J_m} = y_m$ for all $m < n$. Therefore, we can conclude $y = \bigcup_{n\in\mathbb{N}} y_n$ is a well-defined function on all of J solving our initial value problem.

Also, if $\widetilde{J} \subseteq J$ is an open interval containing t_0 and $\widetilde{y} : \widetilde{J} \to \mathbb{R}^n$ is another solution to our initial value problem, then $y|_{\widetilde{J}} \equiv \widetilde{y}$.
To see why, note by the prior corollary that $y_n|_{J_n \cap \widetilde{J}} \equiv \widetilde{y}$ for all n. Also, $\widetilde{J} = J \cap \widetilde{J} = \bigcup_{n\in\mathbb{N}}(J_n \cap \widetilde{J})$. Therefore, we can conclude that:
$$y|_{\widetilde{J}} = \bigcup_{n\in\mathbb{N}} y_n|_{J_n\cap\widetilde{J}} = \bigcup_{n\in\mathbb{N}} \widetilde{y}|_{J_n\cap\widetilde{J}} = \widetilde{y}. \blacksquare$$

I'm now going to take notes on the PDE class again.

Math 148 Lecture Notes:

Here is the general theorem for solving linear first order PDEs with arbitrary coefficients.

Theorem: Let $\Omega \subseteq \mathbb{R}^N$ be an open set. Also suppose $a = (a_1, \ldots, a_n) : \Omega \to \mathbb{R}$ is a locally Lipschitz function, and $c, g : \Omega \to \mathbb{R}$ are continuous functions.

A characteristic of the PDE $a(x) \cdot \nabla u + c(x)u = g(x)$ is any curve $x : (s_0, s_1) \to \Omega$ such that $x' = a(x)$ on (s_0, s_1).
Note that as a is locally Lipschitz and $x' = a(x)$ is an autonomous ODE, we know from *pages 289-294* that every point in Ω has a characteristic passing through it, and that no two "distinct" characteristics pass through that point.

Now $u \in C^1(\Omega)$ solves the PDE iff for each characteristic $x : (s_0, s_1) \to \Omega$ the function $\zeta(s) := u(x(s))$ satisfies that $\zeta'(s) + c(x(s))\zeta(s) = g(x(s))$.
$(\Longrightarrow)$:
If $x(s)$ is a characteristic and $\zeta(x) = u(x(s))$, then:
$$\zeta'(s) = \nabla u(x(s)) \cdot x'(s) = a(x(s)) \cdot \nabla u(x(s)) = g(x(s)) - c(x(s))\zeta(s)$$

$(\Longleftarrow)$
Pick any $x_0 \in \Omega$ and let $x(s)$ be a characteristic of the PDE such that $x(0) = x_0$. Then after letting $\zeta(s) := u(x(s))$, note that:
$$a(x_0) \cdot \nabla u(x_0) = \zeta'(0) = -c(x(0))\zeta(0) + g(x(0)) = g(x_0) - c(x_0)u(x_0)$$

Therefore, the PDE holds at x_0 for u. $\blacksquare$

This theorem also shows any solution to our first order linear PDE is unique when given an initial condition at a single point on every characteristic. After all, the continuity assumptions on $c : \Omega \to \mathbb{R}$ and $g : \Omega \to \mathbb{R}$ guarentee the linear ODE involving $\zeta(s)$ has a unique solution when provided with an initial condition.

Unfortunately, in order to get an existence theorem, I need a stronger smoothness result about the solutions of linear and arbitrary ODEs as we change their initial conditions. These results do exist and are even in John Lee's book that I was following on *pages 289-294*. But I do not have time to show them now.

I'll go into more depth on the smoothness of solutions to ODEs on *page* ____.

The next subject covered in lecture is the wave equation on $\mathbb{R}$. Specifically, given any $c > 0$ we consider C^2 solutions of the PDE: $u_{t,t} = c^2 u_{x,x}$.

Theorem: If $c > 0$ and $\Omega = \mathbb{R} \times \mathbb{R}$ or $\mathbb{R} \times \mathbb{R}_{>0}$, then all solutions to $u_{t,t} = c^2 u_{x,x}$ from $C^2(\Omega)$ are precisely given by:

$$u(x,t) = f(x+ct) + g(x-ct) \text{ where } f, g \in C^2(\mathbb{R}).$$

($\Longleftarrow$)
You can look at my physics notes from last summer for the scratch work on checking that $f(x+ct)$ and $g(x-ct)$ (and thus their sum) are solutions of $u_{t,t} = c^2 u_{x,x}$.

($\Longrightarrow$)
Suppose $u \in C^2(\Omega)$ solves the PDE and write $y = x + ct$ and $z = x - ct$. By solving for x and t in terms of y and z, we get that:

$$x = \tfrac{y+z}{2} \text{ and } t = \tfrac{y-z}{2c}$$

Now define $v(y,z) := u(\frac{y+z}{2}, \frac{y-z}{2c})$. Depending on whether $\Omega = \mathbb{R} \times \mathbb{R}$ or $\Omega = \mathbb{R} \times \mathbb{R}_{>0}$, we will either have that v is well-defined on $\mathbb{R}^2$ or that v is well-defined on the region $U := \{(y,z) : y > z\} \subseteq \mathbb{R}^2$. Also, we can calculate that:

$$v_{y,z} = (\tfrac{1}{2}u_x + \tfrac{1}{2c}u_t)_z = \tfrac{1}{4}u_{x,x} - \tfrac{1}{4c}u_{x,t} + \tfrac{1}{4c}u_{t,x} - \tfrac{1}{4c^2}u_{t,t} = -\tfrac{1}{4c^2}(u_{t,t} - c^2 u_{x,x}) = 0$$

Thus by repeated reasoning to the example PDE on *page 829*, we know $v(y,z) = f(y) + g(z)$ for some $f, g \in C^2(\mathbb{R})$. And to finish off, we just substitute back in $y = x + ct$, $z = x - ct$, and notice that $v(y,z) = u(x,t)$. Therefore, $u(x,y) = f(x+ct) + g(x-ct)$. ■

Theorem: Let $c > 0$ and $\Omega = \mathbb{R} \times \mathbb{R}$ or $\mathbb{R} \times \mathbb{R}_{>0}$, and consider the initial value problem:

- $u_{t,t} = c^2 u_{x,x}$ on Ω,
- $u(x,0) = \varphi(x)$ on $\mathbb{R}$,
- $u_t(x,0) = \psi(x)$ on $\mathbb{R}$.

If $\varphi \in C^2(\mathbb{R})$ and $\psi \in C^1(\mathbb{R})$, then the initial value problem has a unique $C^2(\Omega) \cap C(\overline{\Omega})$ solution with $u_t \in C(\Omega)$.

What I mean here is that u_t can be extended continuously to being a well-defined function to the line $\mathbb{R} \times \{0\}$ even if that's not contained in Ω.

Also, u is given by the d'Alembert Formula:

$$u(x,t) = \tfrac{1}{2}(\varphi(x+ct) + \varphi(x-ct)) + \tfrac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)\mathrm{d}s$$

Proof (uniqueness + the d'Alembert Formula):
Suppose u is a solution. Then recall from the last theorem that:

$$u(x,t) = f(x+ct) + g(x-ct) \text{ for some } f, g \in C^2(\mathbb{R}).$$

But now we can see by plugging in $t = 0$ to u and u_t that $f(x) + g(x) = \varphi(x)$ and $cf'(x) - cg'(x) = \psi(x)$. Hence, we get the following system of equations:

$$f' + g' = \varphi' \text{ and } f' - g' = \tfrac{1}{c}\psi.$$

By solving for f' and g' with respect to φ' and ψ, we get that:

$$f' = \tfrac{1}{2}\varphi' + \tfrac{1}{2c}\psi \text{ and } g' = \tfrac{1}{2}\varphi' - \tfrac{1}{2c}\psi$$

Therefore, we can solve that $f = \tfrac{1}{2}\varphi + \tfrac{1}{2c}\Psi_1$ and $g = \tfrac{1}{2}\varphi - \tfrac{1}{2c}\Psi_2$ where Ψ_1 and Ψ_2 are antiderivatives of ψ. In particular, for any fixed x we can say that there is a constant B_x such that:

$$\begin{aligned} &f(x+ct) + g(x-ct) \\ &\quad = \tfrac{1}{2}(\varphi(x+ct) + \varphi(x-ct)) + \tfrac{1}{2c}\left(\int_x^{x+ct} \psi(s)\mathrm{d}s - \int_x^{x-ct} \psi(s)\mathrm{d}s\right) + B_x \\ &\quad = \tfrac{1}{2}(\varphi(x+ct) + \varphi(x-ct)) + \tfrac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)\mathrm{d}s + B_x \end{aligned}$$

Yet in particular, if we plug in $t = 0$ then:

$$\varphi(x) = f(x + c(0)) + g(x - c(0)) = \varphi(x) + 0 + B_x$$

Thus $B_x = 0$. And so, we've proven the D'Alembert formula holds for all (x,t). Uniqueness is an obvious consequence.

Proof (existence):
Given φ and ψ as in the formula, define:

$$u(x,t) = \tfrac{1}{2}(\varphi(x+ct) + \varphi(x-ct)) + \tfrac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)\mathrm{d}s$$

Then $u(x,0) = \varphi(x)$. Also we can calculate that:

$$u_t = \tfrac{c}{2}(\varphi'(x+ct) - \varphi'(x-ct)) + \tfrac{1}{2c}(c\psi(x+ct) + c\psi(x-ct))$$

Therefore $u_t(x,0) = \psi(x)$.

Finally, you do the rest of the work to prove that u is C^2 and $u_{t,t} = c^2 u_{x,x}$. :) ■

Remark 1: This formula makes sense even when $\varphi \notin C^2$ and $\psi \notin C^1$. We sometimes treat its output as a "generalized solution".

Remark 2: The value of u at any (x,t) is only affected by the value of φ at $x+ct$ and $x-ct$, and by the value of ψ on the interval $[x-ct, x+ct]$. This is called the principle of causality.

I'm getting slightly bored with differential equations. So I'm going to switch back over to studying algebra. Hopefully, I'll finish my field theory notes in the next section.

4/16/2026

Math 200c Notes:

A field F is algebraically closed if whenever $F \subseteq K$ is an algebraic extension, then $L = K$.

Lemma: The following are equivalent:

(a) F is algebraically closed,

(b) every $f \in F[x]$ has a root in F,

(c) every $f \in F[x]$ splits over F.

Proof:
That (c) implies (b) is obvious. Meanwhile, that (b) implies (c) is a consequence of the long division theorem. So, we just need to show that (a) holds if and only if (b) or (c) holds.

If F is algebraically closed and $f \in F[x]$, then there exists a field extension K/F such that f has a root α in K. Also in particular, we can consider the intermediate field $F \subseteq F[\alpha] \subseteq K$. Yet $[F[\alpha] : F]$ is a finite and thus algebraic extension. Thus as F is algebraically closed, we know $F = F[\alpha]$. Hence, we've proven that f has a root in F.

Conversely, if every $f \in F[x]$ splits over f and K/F is an algebraic extension, then given any $\alpha \in K$ we can consider the polynomial $m_{\alpha;F} \in F[x]$. Yet as f splits in F, we thus know all roots of f (including α) are in F. So $F = K$. ■

Lemma: Let $F \subseteq K \subseteq L$ be field extensions such that K/F and L/K are algebraic extensions. Then L/F is also an algebraic extension.

Proof:
Suppose $\alpha \in L$. Then as L/K is algebraic, we can consider the minimal polynomial $m_{\alpha;K} \in K[x]$. In particular, we can write $m_{\alpha;K} = x^n + a_{n-1}x^{n-1} + \ldots + a_0$ where all $a_i \in K$.

Next, as each a_i is algebraic over F (since K/F is an algebraic extension), we can consider a finite extension $F \subseteq E \subseteq K$ where $E = F[a_0, a_1, \ldots, a_{n-1}]$. Then since $m_{\alpha;K} \in E[x]$, we know that α is algebraic over E and hence $[E[\alpha] : E] < \infty$.

By setting $E' = F[a_0, \ldots, a_{n-1}, \alpha] = E[\alpha]$, we thus know by the tower lemma that $[E' : F] < \infty$. Hence, it follows that E'/F is an algebraic extension. Since $\alpha \in E'$, this proves that α is algebraic in $E' \subseteq L$ over F. Also as α was arbitrary, this finishes showing L/F is an algebraic extension. ■

Given a field F, an algebraic closure of F is an algebraic extension $\overline{F}/F$ such that $\overline{F}$ is algebraically closed.

Here are two specfic examples of algebraic closures.

1. The algebraic closure of $\mathbb{R}$ is $\mathbb{C}$. After all, $[\mathbb{C} : \mathbb{R}] = 2 < \infty$. So, $\mathbb{C}/\mathbb{R}$ is an algebraic extension. Also, the fundamental theorem of algebra plus the first lemma on the prior page tells us that $\mathbb{C}$ is algebraically closed.

2. Let $\mathbb{A} \coloneqq \{\alpha \in \mathbb{C} : \alpha \text{ is algebraic over } \mathbb{Q}\}$. Then $\mathbb{A}$ is a field.

   Why?
   Suppose $\alpha, \beta \in \mathbb{A}$. Then we know that $\mathbb{Q}[\alpha, \beta]/\mathbb{Q}$ is a finite and thus algebraic field extension. Also, $\alpha \pm \beta$ and $\alpha\beta^{\pm 1}$ are in $\mathbb{Q}[\alpha, \beta]$. Hence, all of these are algebraic over $\mathbb{Q}$.

   Furthermore, $\mathbb{A}$ is algebraically closed.

   Why?
   Given $f \in \mathbb{A}[x]$, we know that f splits in $\mathbb{C}[x]$. In other words, we can write $f = (x - \alpha_1) \cdots (x - \alpha_n)$ where each $\alpha_i \in \mathbb{C}$.

   Now each α_i is algebraic over $\mathbb{A}$. By the prior lemma, we can conclude that $\mathbb{A}[\alpha_i]/\mathbb{Q}$ is an algebraic extension. Yet in turn this implies that each $\alpha_i \in \mathbb{A}$. So, we've shown that f splits in $\mathbb{A}[x]$. This proves by the first lemma from today that $\mathbb{A}$ is algebraically closed.

   It's also worth noting that $\mathbb{A}$ is a proper subset of $\mathbb{C}$.

   Why?
   $\mathbb{A}$ is equal to the union of all zero sets of polynomials in $\mathbb{Q}[x]$. As each zero set is a finite set and $\mathbb{Q}[x]$ is a countable set, we can thus conclude that $\mathbb{A}$ is also countable. As a result, if we consider $\mathbb{C} \cong \mathbb{R}^2$ and equip $\mathbb{C}$ with the lebesgue measure, then $\mathbb{A}$ will be a null set.

An important result which I didn't give a fully correct proof to in my math 100b notes is that every field has an algebraic closure. (It's ok though because I didn't attempt to hide that my proof wasn't fully correct in my math 100b notes.) I'll give an actually correct proof now.

Theorem: Let F be a field. Then there exists an algebraic closure $\overline{F}/F$.

Choose a set S with cardinality greater $\mathbb{N} \times F[x]$. (We can always do this by taking a power set). Also without loss of generality assume $F \subseteq S$.

The point of taking a really big set S is just so that every field extension we consider lives in a single big set. Otherwise, we wouldn't be able to "take a union" of a bunch of field extensions like I was stating in the proof outline I gave in my math 100b notes.

Once we have that big set S, the reasoning in my math 100b notes should work with only minor alterations. That said, we can shorten the proof by applying Zorn's lemma instead of transfinite induction.

Define the collection:
$\mathcal{F} \coloneqq \{(E, +_E, \cdot_E) : F \subseteq E \subseteq S,\ (E, +_E, \cdot_E)$ is a field,
$+_E|_F = +_F,\ \cdot_E|_F = \cdot_F$, and E/F is an algebraic field extension$\}$.

Note that $\mathcal{F}$ is a poset when ordered by inclusion. Also, $\mathcal{F}$ is nonempty because $(F, +_F, \cdot_F) \in \mathcal{F}$. Furthermore, if $\{(E_\beta, +_\beta, \cdot_\beta)\}_{\beta \in I}$ is a chain in $\mathcal{F}$, then $E \coloneqq \bigcup E_\beta$ is an upper bound to our chain when eqipped with the operations $+_E = \bigcup +_\beta$ and $\cdot_E = \bigcup \cdot_\beta$.

Note that E is a well-defined field because given any $a, b, c \in E$, we can find some E_β in our chain containing each of those three elements. And now $+_E$ and $\cdot_E$ satisfy the field axioms on a, b, c since $+_\beta$ and $\cdot_\beta$ do.

Also, to see that E/F is algebraic, note that for any $\alpha \in E$ that $\alpha \in E_\beta$ for some β. And since E_β/F is an algebraic extension, we know that α is algebraic over F.

By Zorn's lemma, $\mathcal{F}$ has a maximal element $(\overline{F}, +_{\overline{F}}, \cdot_{\overline{F}})$. It's clear from the definition of $\mathcal{F}$ that $\overline{F}/F$ is a maximal extension. Also, we can show as follows that $\overline{F}$ is algebraically closed.

Lemma: If K/F is an algebraic extension, then we must have that K has cardinality less than $\mathbb{N} \times F[x]$.

To see why, note that every $\alpha \in K$ must be in the zero set of some polynomial in $F[x]$. Since each of those zero sets are finite, we can pick a finite indexing function for each zero set (using axiom of choice) and then define an injection from K to $\mathbb{N} \times F[x]$.

Now suppose for the sake of contradiction that $L/\overline{F}$ is an algebraic extension where $L \neq \overline{F}$. Then from a prior result we know that L/F is also an algebraic extension. Hence, by the above lemma both L and $\overline{F}$ have cardinality at most equal to that of $\mathbb{N} \times F[x]$.

Since $S - \overline{F}$ has strictly larger cardinality than $L - \overline{F}$, we can find an injection from $L - \overline{F}$ to $S - \overline{F}$. Yet this then lets us define a field $(K, +_K, \cdot_K)$ in S containing $\overline{F}$ as a subfield and satisfying that there exists an isomorphism from K to L and fixing $\overline{F}$. In particular, this isomorphism between K and L will guarentee that $K/\overline{F}$ and in turn K/F is an algebraic extension. Since K is defined on a larger subset of S than $\overline{F}$, this contradicts the maximality of $\overline{F}$. ■

A less important result is that the algebraic closure of a field is unique up to isomorphism.

Lemma 1: Suppose F, F' are fields, $\theta : F \to F'$ is a field isomorphism, K/F is an algebraic extension, and L/F' is another extension such that L is algebraically closed. Then there exists a field homomorphism $\phi \in \mathrm{Embed}_\theta(K, L)$.

Proof:
Let $\mathcal{F} \coloneqq \{\varphi \in \mathrm{Embed}_\theta(E, L) : E \in \mathrm{Int}(K/F)\}$. Also note that $\mathcal{F}$ is a poset when ordered by inclusion. Furthermore, $\mathcal{F} \neq \emptyset$ since if j is the embedding $F' \hookrightarrow L$ then $j \circ \theta \in \mathcal{F}$. Finally, if we have a chain of embeddings $\{\varphi_\beta : E_\beta \to L\}_{\beta \in I}$ then we can take their union to get an upper bound in $\mathcal{F}$.

By Zorn's lemma, $\mathcal{F}$ has a maximal element $\phi : E \to L$. If we can show that $E = K$, then we will be done. So suppose for the sake of contradiction that $E \neq K$.

Since K/F is an algebraic extension, we know $\alpha \in K - E$ is algebraic over F (and in turn over E). Thus, we can consider some irreducible polynomial $f := m_{\alpha;E} \in E[x]$ with α as a zero. And then, there is a field extension $E \subseteq E[\alpha] \subseteq K$ such that $2 \leq [E[\alpha] : E] < \infty$.

Next consider that f^ϕ is a polynomial in $(\phi(E))[x] \subseteq L[x]$. Since L is algebraically closed, we know that f^ϕ has a root $\alpha' \in L$. In turn, we can consider the finite extension $(\phi(E))[\alpha']/\phi(E)$.

But now by the theorem at the bottom of *page 762*, we know there is a unique $\psi \in \text{Isom}_\phi(E[\alpha], (\phi(E))[\alpha'])$ mapping α to α'. This contradicts the maximality of the field homomorphism ϕ. ■

Lemma 2: If K/F is an algebraic extension and $\phi : K \to K$ is a field homomorphism fixing F, then ϕ is an isomorphism.

Proof:
Since ϕ is not the zero map, we necessarily know ϕ is injective. As for showing surjectivity, consider any $\alpha \in K$ and let $f := m_{\alpha;F} \in F[x]$ be a monic irreducible polynomial with α as a zero. If we restrict ϕ to just the zero set of f in K, we then must have that ϕ is an injective function on this zero set. Yet an injective function from a finite set to itself is always surjective. Hence, there must exist some $\beta \in K$ such that $f(\beta) = 0$ and $\phi(\beta) = \alpha$. This proves $\alpha \in \phi(K)$. Hence, $K = \phi(K)$. ■

Corollary: Suppose F is a field and K/F and L/F are two algebraic closures of F. Then there exists an isomorphism $\phi \in \text{Isom}_{\text{Id}_F}(K, L)$.

Proof:
By lemma 1, we can find injective field isomorphism $\phi : K \to L$ and $\psi : L \to K$ which both fix F. Then if ϕ is not a surjection as well, we must have that $\psi \circ \phi$ is an injective field homomorphism fixing F that is also not surjective. Yet this contradicts lemma 2. Hence, we must have that ϕ is surjective and thus a field isomorphism. ■

The main use of an algebraic closure is just that it gives us a large set for any field extensions to be contained in. Plus, some applications (like taking Jordan forms of matrices), require us to know that polynomials split in our field.

Set 2 Problem 6: Suppose that $\text{char}(F) = p > 0$. An algebraic extension K/F is called purely inseparable if for all $\alpha \in K/F$, $m_{\alpha;F}$ is an inseparable polynomial.

(a) Prove that K/F is purely inseparable if and only if every $\alpha \in K$ satisfies that $\alpha^{p^k} \in F$ for some $k \geq 0$.

($\Longrightarrow$)
Consider any $\alpha \in K$ and recall from *page 770* that $m_{\alpha;F}(x) = f_{\text{sep}}(x^{p^n})$ for some unique nonnegative integer n and irreducible, separable polynomial $f_{\text{sep}} \in F[x]$. Yet since K/F is a purely inseparable extension and f_{sep} is a separable irreducible polynomial, we must have that every zero of f_{sep} is in K. This proves that $\alpha^{p^n} \in F$.

($\Longleftarrow$)
Suppose $\alpha \in K - F$ and let n be a nonnegative integer such that $\alpha^{p^n} = a \in F$. Then we know that $m_{\alpha;F} \mid x^{p^n} - a$ in $F[x]$. Yet also note that:

$$(x^{p^n} - a) = (x^{p^n} - \alpha^{p^n}) = (x - \alpha)^{p^n} \text{ since } \operatorname{char}(F) = p.$$

Therefore, $m_{\alpha;F}$ has only one zero in K. Also, as $\alpha \notin F$, we know $\deg(m_{\alpha;F}) \geq 2$. This proves that $m_{\alpha;F}$ is not a separable polynomial.

(b) Let $K := \mathbb{F}_p(x, y)$ be the field of fractions of $\mathbb{F}_p[x, y]$, and define $F := \mathbb{F}_p(x^p, y^p)$ as the field of fractions of $\mathbb{F}_p[x^p, y^p]$. Then K/F is a purely inseparable extension.

Consider any $\frac{g(x,y)}{h(x,y)} \in \mathbb{F}_p(x, y)$. Importantly, as $\operatorname{char}(\mathbb{F}_p[x, y]) = p$, we know the Frobenius map is a ring homomorphism on this polynomial ring. Also, the Frobenius map fixes any coefficients in $\mathbb{F}_p$. Hence, we can conclude that $(g(x, y))^p = g(x^p, y^p)$ and $(h(x, y))^p = h(x^p, y^p)$.

In turn, this let's us conclude $\left(\frac{g(x,y)}{h(x,y)}\right)^p = \frac{g(x^p,y^p)}{h(x^p,y^p)} \in F$. By part (a), this proves that K/F is a purely inseparable extension.

(c) The extension K/F in part (b) has $[K : F] = p^2$. In turn, there can be no primitive element α such that $K = F[\alpha]$.

To start off, note that $K = F[x, y]$. After all, it is clear how every polynomial in $\mathbb{F}_p[x, y]$ can be written as a sum of $x^i y^j f_{i,j}(x, y)$ where $0 \leq i, j < p$ and $f_{i,j} \in \mathbb{F}_p[x^p, y^p]$ for all i, j. By applying this process to both the numerator and denominator of any $\frac{g(x,y)}{h(x,y)} \in K$, we can thus conclude that $K = F[x, y]$ like we wanted.

Next, I claim that $m_{x;F}(z) \in F[z]$ is precisely the polynomial $z^p - x^p$.

To see why, again note that there is a unique integer n and irreducible, separable polynomial $f_{\text{sep}}(z) \in F[z]$ with $m_{x;F}(z) = f_{\text{sep}}(z^{p^n})$. Yet since K/F is a purely inseparable extension and $x \notin F$, we must have that n is strictly positive. Also note that x is a zero of the polynomial $z^p - x^p \in F[z]$. The only way this is possible is if $f_{\text{sep}}(z) = z - x^p$ and $m_{x;F}(z) = z^p - x^p$.

As a result, we know that $[F[x] : F] = p$. I also claim that $m_{y;F[x]}(z) \in (F[x])[z]$ is precisely the polynomial $z^p - y^p$.

To see why, first note that if K/F is a purely inseparable extension and $E \in \operatorname{Int}(K/F)$, then we know from part (a) that K/E is also a purely inseparable extension. After all, if $\alpha^{p^k} \in F$ for some k then we also know that $\alpha^{p^k} \subseteq E$. But now we can just repeat all the reasoning we used for showing $m_{x;F}(z) = z^p - x^p$.

This proves that $[F[x, y] : F[x]] = p$. And by tower lemma we can now conclude that $[K : F] = p^2$.

To finish off, it is now clear that K has no primitive element α. After all, in part (b) I showed that every $\alpha \in K$ satisfies that $\alpha^p \in F$. Therefore, it follows that $\deg(m_{\alpha;F}) \leq p$ and thus $[F[\alpha] : F] \leq p$ for all $\alpha \in K$. ■

Note that by the proposition on *page 818*, this then also shows that there are infinitely many intermediate fields E with $F \subseteq E \subseteq K$.

Set 2 Problem 7: Let K/F be an algebraic extension, not necessarily of finite degree.

(a) Let S be the set of elements $\alpha \in K$ such that $m_{\alpha;F}$ is separable over F. Show that S is a subfield of K containing F. S is called the separable closure of F (inside K).

To start off, it's obvious that $F \subseteq S$ since all the minimal polynomials of elements of F are linear and thus trivially separable.

Meanwhile, consider any $\alpha, \beta \in S$ and let $\overline{K}$ be the algebraic closure of K.

As a side note, $\overline{K}$ is thus also the algebraic closure of F since $\overline{K}/K$ and K/F being algebraic extensions implies that $\overline{K}/F$ is an algebraic extension. The purpose of considering the algebraic closure of K is just so that the intersection of K and any splitting field of a polynomial in $F[x]$ is well-defined.

If we consider the polynomial $f(x) = m_{\alpha;F}(x)m_{\beta;F}$, we know that f splits in $\overline{K}[x]$. Hence, it follows that $\overline{K}$ contains a splitting field L of f. Also note that $f(x)$ is an irreducible-separable polynomial (i.e. a function whose irreducible factors are separable). Hence, we know that L/F is a finite Galois (and thus separable) extension.

With that, we can now conclude that the minimal polynomial in $F[x]$ for any $\gamma \in K \cap L \subseteq L$ must have a separable minimal polynomial in $F[x]$. Since $\alpha \pm \beta$ and $\alpha\beta^{\pm 1} \in K \cap L$, we thus can conclude $\alpha \pm \beta$ and $\alpha\beta^{\pm 1}$ are in S.

As a side note, this also proves that if $\alpha_1, \ldots, \alpha_n \subseteq S$, then $F[\alpha_1, \ldots, \alpha_n] \subseteq S$. Remembering this is useful.

(b) With the same notation as in (a), show that K/S is purely inseparable. Thus, any algebraic extension decomposes into a separable extension followed by a purely inseparable extension.

To start off, note that if $\mathrm{char}(F) = 0$, then we automatically have that $K = S$ and thus K/S is trivially purely inseparable. After all, if $\mathrm{char}(F) = 0$ then all irreducible polynomials (and hence minimal polynomials of $\alpha \in K$) are separable. Thus the only case we need to prove is when $\mathrm{char}(F) = p > 0$. From now on, I'll assume that p is some fixed prime number.

Suppose $\beta \in K$. Then we know that $m_{\beta;F}(x) = f_{\mathrm{sep}}(x^{p^n})$ where n is a unique nonnegative integer and f_{sep} is a unique irreducible and separable polynomial in $F[x]$. But now all the zeros of f_{sep} are in S. Hence, we must have that $\beta^{p^n} \in S$. By part (a) of problem 6, this implies that K/S is purely inseparable.

(c) Suppose that $F \subseteq L \subseteq K$ where K/F is an algebraic extension. Show that K/F is separable if and only if L/F and K/L are separable.

$(\Longrightarrow)$
This direction is trivial.

($\Longleftarrow$)
Let S be as in part (a) and (b), and note that since L/F is separable, we must have that $L \subseteq S$. To prove that K/F is a separable extension, it suffices to prove that $S = K$. Fortunately, we know from assumption that K/L is a separable extension, and from part (b) that K/S is a purely inseparable extension. Since $L[x] \subseteq S[x]$, we have for any $\alpha \in K$ that $m_{\alpha;S} \mid m_{\alpha;L}$ where the latter polynomial is separable. It follows $m_{\alpha;S}$ is separable as well. Yet the only way this is possible when K/S is a purely inseparable extension is if $\alpha \in S$. Hence, we've proven $K = S$. ■

4/18/2026

Math 220c Lecture Notes:

Suppose G is a region, $\varphi : G \to \mathbb{R}$ is subharmonic, and $\overline{\Delta} \subseteq G$ is any closed disk. If we set $f = \varphi|_{\partial\Delta}$, then we can solve the Dirichlet problem on $\overline{\Delta}$ to get a continuous function $h : \overline{\Delta} \to \mathbb{R}$ that is harmonic on Δ and satisfies that $h|_{\partial\Delta} = f$. Then we set:

$$\widetilde{\varphi} = \begin{cases} \varphi & \text{in } G - \overline{\Delta} \\ h & \text{in } \overline{\Delta}. \end{cases}$$

$\widetilde{\varphi}$ is called the Poisson modification / bumping of φ along Δ.

Set 3 Problem 2: Show that $\widetilde{\varphi}$ is subharmonic on G.
We know by the pasting lemma that f is continuous on all of G. Also, it's automatic that $\widetilde{\varphi}$ is subharmonic on $G - \partial\Delta$. So, we just need to show $\widetilde{\varphi}$ is also subharmonic on $\partial\Delta$.

Lemma: $\varphi \leq \widetilde{\varphi}$ on G with equality everywhere iff φ was already harmonic on Δ.
Proof:
If $z \notin \Delta$, we then we know that $\varphi(z) = \widetilde{\varphi}(z)$. Meanwhile, note that since $\widetilde{\varphi}(z)$ is harmonic on Δ, we know $\varphi - \widetilde{\varphi}$ is a subharmonic function on Δ. Also $\varphi - \widetilde{\varphi} \equiv 0$ on $\partial\Delta$. It follows by the maximum principle that either $\varphi - \widetilde{\varphi} < 0$ on Δ or $\varphi - \widetilde{\varphi} \equiv 0$ on Δ.

Now fix any $a \in \partial\Delta$. Since φ is subharmonic at a, we can find some $R > 0$ such that $\overline{\Delta}(a, R) \subseteq G$ and $\varphi(a) \leq \frac{1}{2\pi}\int_0^{2\pi} \varphi(a + re^{it})\mathrm{d}t$ for all $0 \leq r \leq R$. In turn, as $\varphi \leq \widetilde{\varphi}$ on G and $\widetilde{\varphi}(a) = \varphi(a)$, we know that:

$$\widetilde{\varphi}(a) \leq \tfrac{1}{2\pi}\int_0^{2\pi} \widetilde{\varphi}(a + re^{it})\mathrm{d}t \text{ for all } 0 \leq r \leq R.$$

So, $\widetilde{\varphi}$ is subharmonic at a. ■

Here's one other result we'll need.

Proposition: If φ_1, φ_2 are subharmonic functions on a region G, $\overline{\Delta}$ is any closed disk, and $\varphi_1 \leq \varphi_2$, then the Poisson modifications of φ_1 and φ_2 along Δ satisfy that $\widetilde{\varphi}_1 \leq \widetilde{\varphi}_2$.

Proof:
Let $f_1 = \varphi_1|_{\partial\Delta}$ and $f_2 = \varphi_2|_{\partial\Delta}$. Then let $h_1, h_2 : \overline{\Delta} \to \mathbb{R}$ solve the Dirichlet problem on Δ with boundary values f_1 and f_2 respectively. To show that $\widetilde{\varphi}_1 \leq \widetilde{\varphi}_2$, it suffices to show that $h_1 \leq h_2$ in $\overline{\Delta}$.

Fortunately, note that $h_1 - h_2$ is harmonic on Δ, continuous in $\overline{\Delta}$ and satisfies that $(h_1 - h_2)|_{\partial\Delta} = f_1 - f_2 = \varphi_1|_{\partial\Delta} - \varphi_2|_{\partial\Delta} \leq 0$. By the maximum principle, we thus know $h_1 - h_2 \leq 0$ in $\overline{\Delta}$. ■

We can use subharmonic functions in order to prove a more general existence and uniqueness theorem for solutions to the Dirichlet problem on an arbitrary region G. Although, unfortunately our more general solution will not be constructive.

Given a bounded region $G \subseteq \mathbb{C}$ and a continuous function $f : \partial G \to \mathbb{R}$, we define the Perron family:

$$\mathcal{P}(G, f) = \{(\varphi : G \to \mathbb{R}) : \varphi \text{ is subharmonic and } \limsup_{z \to a} \varphi(z) \leq f(a) \text{ for all } a \in \partial G\}.$$

Also, we define the Perron function $u : G \to \mathbb{R}$ by:

$$u(z) \coloneqq \sup\{\varphi(z) : \varphi \in \mathcal{P}(G, f)\}.$$

Some remarks:

- $\mathcal{P}(G, f) \neq \emptyset$. After all since G is bounded we know ∂G is compact. Then as f is continuous, we know that there exists $m, M \in \mathbb{R}$ with $m \leq f \leq M$ in ∂G. And now, $\varphi \equiv m$ is in $\mathcal{P}(G, f)$.
- We also thus know u is a well-defined function. After all, the set we are taking a sup over is nonempty. Also, as $f \leq M$, we can conclude by the maximal principle that $\varphi \leq M$ for all $\varphi \in \mathcal{P}(G, f)$. Therefore $u(z) = \sup\{\varphi(z) : \varphi \in \mathcal{P}(G, f)\} \leq M$.

A hopefully obvious fact is that given $\varphi, \psi \in \mathcal{P}(G, f)$, we also know $\max(\varphi, \psi) \in \mathcal{P}(G, f)$.
Remember on *page 843* that I showed $\max(\varphi, \psi)$ is subharmonic.

Also, if we take a Poisson modification of $\varphi \in \mathcal{P}(G, f)$ along $\overline{\Delta}$, then we have that $\widetilde{\varphi} \in \mathcal{P}(G, f)$.

Theorem: The Perron function u is harmonic on G.
Proof:
Fix any $x \in G$ and let $\overline{\Delta} \subseteq G$ be a closed disk centered at x. Also for the sake of convenience, write $\mathcal{P} = \mathcal{P}(G, f)$. We can show that u is harmonic in Δ.

Since $u(x) = \sup\{\varphi(x) : \varphi \in \mathcal{P}\}$, we can find a sequence $\{\varphi_n\}_{n \in \mathbb{N}} \subseteq \mathcal{P}$ such that $\varphi_n(x) \to u(x)$. Without loss of generality, we can also assume $\varphi_1 \leq \varphi_2 \leq \cdots$.
After all, if this is not the case then we can define $\varphi_n^{\text{new}} = \max(\varphi_1, \ldots, \varphi_n)$. Then $\varphi_n^{\text{new}} \in \mathcal{P}$ for all n. Also $\varphi_n(x) \leq \varphi_n^{\text{new}}(x) \leq u(x)$ for all n. So, $\varphi_n^{\text{new}}(x) \to u(x)$.

Next, we consider the Poisson bumping $\widetilde{\varphi}_n$ of φ_n along Δ. Imporantly, we still have that $\widetilde{\varphi}_n \in \mathcal{P}$ and that $\widetilde{\varphi}_1 \leq \widetilde{\varphi}_2 \leq \cdots$. By identical reasoning to before, we also know that $\widetilde{\varphi}_n(x) \to u(x)$. By Harnack's theorem, we thus know that $\widetilde{\varphi}_n$ converges locally uniformly to some harmonic function U in Δ.

Clearly $U(x) = u(x)$. And if we can show more generally that $U \equiv u$ in Δ, then we will have proven that u is harmonic in Δ. So, let us consider any $y \in \Delta - \{x\}$ and then prove that $U(y) = u(y)$.

Note that there exists a sequence $\{\psi_n\}_{n\in\mathbb{N}} \subseteq \mathcal{P}$ such that $\psi_n(y) \to u(y)$. Without loss of generality, we may assume $\varphi_n \leq \psi_n$ for all n.

> After all, otherwise we can just define $\psi_n^{\text{new}} = \max(\varphi_n, \psi_n)$. Then $\psi_n^{\text{new}} \in \mathcal{P}$ and also $\psi_n^{\text{new}}(y) \to u(y)$ since $\psi_n(y) \leq \psi_n^{\text{new}}(y) \leq u(y)$ for all n.

Yet we now know $\psi_n(x) \to u(x)$ since $\varphi_n(x) \leq \psi_n(x) \leq u(x)$ for all n. Furthermore, by identical reasoning that we did to the φ_n, we can assume without loss of generality that $\psi_1 \leq \psi_2 \leq \cdots$. And by considering the Poisson bumping of the ψ_n along Δ, we get an increasing sequence of functions $\{\widetilde{\psi}_n\}_{n\in\mathbb{N}}$ that are harmonic on Δ and satisfy that $\widetilde{\psi}_n(x) \to u(x)$ and $\widetilde{\psi}_n(y) \to u(y)$. By Harnack's theorem, there thus exists a harmonic function V on Δ which the $\widetilde{\psi}_n$ converge locally uniformly to.

Now observe that:

- $V(x) = u(x) = U(x)$;
- $V(y) = u(y)$;
- $\varphi_n \leq \psi_n$ for all n implies $\widetilde{\varphi}_n \leq \widetilde{\psi}_n$ for all n. Hence, $U \leq V$.

Let $h = V - U$. This is a harmonic function in Δ. Also $h \geq 0$, and h acheieves a minimum of 0 at x. By the minimal principle, $h \equiv 0$. Hence, we've proven that $U \equiv V$. In particular, $u(y) = V(y) = U(y)$. ■

It's now clear that if a function v solves the Dirchlet problem on the region G with the boundary condition $f : \partial G \to \mathbb{R}$, then $v|_G \equiv u$ where u is the Perron function associated with $\mathcal{P}(G, f)$.

> After all, by the definition of u we know that $v \leq u$ on G. At the same time though, if $\varphi \in \mathcal{P}(G, f)$ then we know that:
> $$\limsup_{z\to a}(\varphi(z) - v(z)) = \limsup_{z\to a}(\varphi(z)) - \lim_{z\to a} v(z) = \limsup_{z\to a}(\varphi(z)) - f(a) \leq 0.$$

> Therefore $\varphi \leq v$ for all $\varphi \in \mathcal{P}(G, f)$, and this proves that $u \leq v$. So, $u \equiv v$ on G.

Conversely, we can ask if in the case we are given no information about whether a solution exists to the Dirichlet problem on G with the boundary condition $f : \partial G \to \mathbb{R}$, then does the Perron function solve the Dirichlet problem? In general, the answer is *no* and the next problem gives an explicit counter example.

Set 3 Problem 4: Let $G = \Delta(0,1) - \{0\}$, and then let $f : \partial G \to \mathbb{R}$ be the function defined by $f \equiv 0$ on $\partial\Delta(0,1)$ and $f(0) = 1$. Then any continuous function $u : \overline{G} \to \mathbb{R}$ that is harmonic on G (including the Perron function) cannot satisfy that $u|_{\partial G} = f$.

Proof:
Recall from *set 2 problem 2 on page 841* that since $u|_G$ has a removable discontinuity at 0, we can actually conclude that the continuous extension of u to $G \cup \{0\}$ is harmonic. Hence, we have that u is a continuous function on $\overline{\Delta}(0,1)$ that is harmonic on $\Delta(0,1)$.

But now $u|_{\partial\Delta(0,1)} \equiv 0$ implies that $u \equiv 0$ on $\overline{\Delta}(0,1)$. Therefore, we can't have that $u(0) = 1$ and $u|_{\partial\Delta(0,1)} \equiv 0$. ■

That all said, often times the Perron function does give a solution to the Dirichlet problem on G with the boundary condition $f : \partial G \to \mathbb{R}$. In fact, we can give a necessary and sufficient condition for when the Perron function is guarenteed to solve the Dirichlet problem. Also, we can show this condition holds for many "reasonable" domains.

Given a bounded region $G \subseteq \mathbb{C}$ and $\zeta_0 \in \partial G$, we say G admits a barrier at ζ_0 if there is a continuous function $w : \overline{G} \to \mathbb{R}$ that is harmonic on G and satisfies that $w(\zeta_0) = 0$ and $w > 0$ on $\partial G - \{\zeta_0\}$.

Two remarks:

- In the above definition, notice by the minimal principle that $w > 0$ on $\overline{G} - \{\zeta_0\}$.
- We sometime refer to the function w as the barrier at ζ_0. Also, the terminology of "barriers" is specifically due to Lebesgue, who observed the fact below.

Set 3 Problem 5(b): Let $G \subseteq \mathbb{C}$ be a bounded region, let $\zeta_0 \in \partial G$, and suppose ℓ is a line segment with endpoints ζ_0 and ζ_1 satisfying that $\ell \cap \overline{G} = \{\zeta_0\}$. Then G admits a barrier at ζ_0.

Proof:
Let $h(z) := \frac{z-\zeta_0}{z-\zeta_1}$. Importantly, h is holomorphic on G and continuous on $\overline{G}$ (since $\zeta_1 \notin \overline{G}$). Furthermore, the only zero of h is at ζ_0, and also $h(z) \in \mathbb{R}_{\leq 0}$ if and only if $z \in \ell - \{\zeta_1\}$.

For an algebraic way of seeing this, note that $\frac{z-\zeta_0}{z-\zeta_1} \leq 0$ if and only if there exists $\beta > 0$ such that $z - \zeta_0 = \beta(\zeta_1 - z)$. But then in turn we have that $z = \frac{\beta}{1+\beta}(\zeta_1 - \zeta_0)$. Since $\frac{\beta}{1+\beta} \in [0,1)$ for all $\beta > 0$, this proves that $z \in \ell$.

As for a geometric way of seeing this, note that h is a Möbius transformation satisfying in the extended complex plane that $\zeta_0 \mapsto 0$, $\zeta_1 \mapsto \infty$, and $\infty \mapsto 1$. Therefore, h will send the line passing through ζ_0 and ζ_1 to the line passing through 0 and 1 (i.e. to $\mathbb{R}_\infty$). In particular, h will either map ℓ to $(-\infty, 0] \cup \{\infty\}$ or to $[0,\infty) \cup \{\infty\}$. Yet the latter can't happen since $\infty \notin \ell$ and $h(\infty) \in (0,\infty)$.

Let us denote Log as the principal branch of the logarithm. A useful observation is that $g(z) := e^{\frac{1}{2}\operatorname{Log}(z)}$ is a biholomorphic function mapping $\mathbb{C} - \mathbb{R}_{\leq 0}$ to the region $\{z \in \mathbb{C} : \operatorname{Re}(z) > 0\}$. And since $h(\overline{G} - \{\zeta_0\}) \subseteq \mathbb{C} - \mathbb{R}_{\leq 0}$ by our prior reasoning, we can get a well-defined function $f \in C(\overline{G} - \{\zeta_0\}) \cap O(G)$ satisfying that $\operatorname{Re}(f)(z) > 0$ on $\overline{G} - \{\zeta_0\}$ by setting:

$$f(z) := (g \circ h)(z) = \exp\left(\tfrac{1}{2}\operatorname{Log}\left(\tfrac{z-\zeta_0}{z-\zeta_1}\right)\right).$$

To finish off, we let $w = \mathrm{Re}(f)$. Then it's clear that w is a continuous strictly positive function on $\overline{G} - \{\zeta_0\}$ that is also harmonic on G. The only issue we still need to address is that w is not currently defined at ζ_0. But once we have shown that $\lim_{z\to\zeta_0} w(z) = 0$, we will have proven that G admits a barrier at ζ_0.

Fortunately, note that since $\lim_{z\to\zeta_0} h(z) = 0$, we can also conclude that:

$$\lim_{z\to\zeta} |f(z)| = \lim_{z\to\zeta_0} e^{\frac{1}{2}\log|h(z)|} = 0.$$

Thus $f(z) \to 0$ as $z \to \zeta_0$. In turn, $w(z) = \mathrm{Re}(f(z)) \to 0$ as $z \to \zeta_0$. ■

As a side note, it is apparently possible for a region G to admit a barrier at some $\zeta_0 \in \partial G$ even if no line segment like in the above problem exists. However, the professor didn't give any explicit constructions of this. Also, most "reasonable" regions we would care about will satisfy the hypothesis of this problem.

We'll call a bounded region G a Dirichlet region if for every continuous function $f : \partial G \to \mathbb{R}$ there is a function $u \in C(\overline{G}, \mathbb{R})$ that is harmonic on G and satisfies that $u|_{\partial G} = f$.

Remark:
By my prior observation that the Perron function will be a solution to the Dirichlet problem with boundary condition f if a solution actually exists, we can equivalently say that G is a Dirichlet region if and only if for every $f \in C(\partial G, \mathbb{R})$, the Perron function u associated with $\mathcal{P}(G, f)$ solves the Dirichlet problem on the region G with boundary condition f.

Theorem: A bounded region G is a Dirichlet region if and only if G admits barriers at all $a \in \partial G$.

($\Longrightarrow$) (This direction is also set 3 problem 5(a)):
Given any $a \in G$, define $f : \partial G \to \mathbb{R}$ by $f(z) = |z - a|$. Then f is a continuous nonnegative function from the boundary of G which is zero only at a. If we let w be the solution to the Dirichlet problem on G with boundary condition f, we will thus have gotten our desired barrier.

($\Longleftarrow$)
Suppose $f \in C(\partial G, \mathbb{R})$ and let $u : G \to \mathbb{R}$ be the Perron function associated with $\mathcal{P}(G, f)$. Since we already know that u is harmonic, all we need to do to prove that u solves the Dirichlet problem associated to f is show that $\lim_{z\to a} u(z) = f(a)$ for all $a \in \partial G$.

Fix any $a \in \partial G$. Without loss of generality, we can assume $f(a) = 0$.

Why?
If $f(a) \neq 0$, then we can consider the function $f^{\mathrm{new}} = f - f(a)$. I claim the Perron function u^{new} associated with $\mathcal{P}(G, f^{\mathrm{new}})$ is $u - u(a)$. This is because we have that $\varphi \in \mathcal{P}(G, f)$ iff $\varphi - f(a) \in \mathcal{P}(G, f^{\mathrm{new}})$.

If we can now show $\lim_{z\to a} u^{\mathrm{new}}(z) = f^{\mathrm{new}}(a) = 0$, then we will have proven that $\lim_{z\to a} u(z) = f(a)$.

Next let us fix any $\varepsilon > 0$. By the continuity of f, we can pick a disk Δ centered at a such that $-\varepsilon < f < \varepsilon$ in $\partial G \cap \Delta$. Also by the extreme value theorem, we can find some $M \geq 0$ such that $-M \leq f \leq M$ on ∂G. Thirdly, since G admits a barrier at all $a \in \partial G$, we can consider a barrier w at a. Since $w > 0$ on $\overline{G} - \{a\}$ and $\overline{G} - \Delta$ is a closed bounded set, we can conclude by the extreme value theorem that $w_0 := \min_{z \in \overline{G} - \Delta} w(z) > 0$.

Claim A: $\liminf_{z \to a} u(z) \geq -\varepsilon$.

Why?
Let $V(z) := -\varepsilon - \frac{w(z)}{w_0} M$. Then note that $V \in \mathcal{P}(G, f)$. After all, V is a continuous function on $\overline{G}$ that is harmonic function on G. Furthermore, if $z \notin \Delta$ then $V(z) \leq -M$ and hence $\lim_{z \to b} V(z) \leq f(b)$ for all $b \in \partial G - \Delta$. Meanwhile, if $z \in \Delta$ then $V(z) \leq -\varepsilon$ and hence $\lim_{z \to b} V(z) \leq f(b)$ for all $b \in \partial G \cap \Delta$.

It follows that $V \leq u$. But then as a consequence, we have that:

$$-\varepsilon = \lim_{z \to a} V(z) \leq \liminf_{z \to a} u(z)$$

Claim B: $\limsup_{z \to a} u(z) \leq \varepsilon$.

Why?
Let $W := \varepsilon + \frac{w}{w_0} M$. Then W is continuous in $\overline{G}$ and harmonic in G. I also claim that $\varphi \leq W$ for all $\varphi \in \mathcal{P}(G, f)$. After all, by analogous reasoning as in claim A, we know that $\lim_{z \to a} W(z) \geq f(a)$. In turn, we can conclude that:

$$\limsup_{z \to a} (\varphi(z) - w(z)) \leq \limsup_{z \to a} \varphi(z) - \lim_{z \to a} W(z) \leq 0$$

Since $\varphi - W$ is a subharmonic function on G, we can conclude by the maximum principle that $\varphi - W \leq 0$ in G. In other words $\varphi \leq W$.

Now since $\varphi \leq W$ for all $\varphi \in \mathcal{P}(G, f)$, we can conclude by the definition of u that $u \leq W$. In turn, we can conclude that:

$$\limsup_{z \to a} u(z) \leq \lim_{z \to a} W(z) = \varepsilon$$

By taking $\varepsilon \to 0$, this proves that $\lim_{z \to a} u(z) = 0 = f(a)$. ■

Some remarks:

- Note that in the proof of the $(\Longleftarrow)$ direction, we didn't actually need w to be fully harmonic. Instead, all the reasoning would have worked if w was merely superharmonic. Because of this, some authors merely require barriers to be superharmonic as opposed to being harmonic. Because of the $(\Longrightarrow)$ direction of the proof, it is clear that:

  Proposition: G admits a superharmonic barrier at all $a \in \partial G$ if and only if G admits a harmonic barrier at all $a \in \partial G$.

- Given any $a \in \partial G$, let $G(a) = G \cap \Delta(a, R)$ be a neighborhood of a in G. We say $w : \overline{G(a)} \to \mathbb{R}$ is a local barrier at a if:
  - w is continuous on $\overline{G(a)}$ and superharmonic on $G(a)$,
  - $w(a) = 0$ and $w > 0$ on $\overline{G(a)} - \{a\}$.

Some authors use local barriers instead of (global) superharmonic barriers. That said, we can show a local barrier exists at $a \in \partial G$ if and only if a global superharmonic barrier exists at a.

$(\Longleftarrow)$
This direction is obvious.

$(\Longrightarrow)$
Let $w : \overline{G}(a) \to \mathbb{R}$ be a local barrier at a, and write $G(a) = G \cap \Delta(a, R)$. Next let $\Delta' \coloneqq \overline{\Delta}(a, r) \subseteq \Delta(a, R)$, and set $w_0 \coloneqq \min_{z \in \overline{G(a)} - \Delta'} w(z) > \varepsilon$. Finally, define:

$$\widehat{w}(z) = \begin{cases} \min(w(z), w_0) & \text{if } z \in \overline{G} \cap \Delta' \\ w_0 & \text{if } z \in \overline{G} - \Delta' \end{cases}$$

Then $\widehat{w}$ is a global superharmonic barrier.

Note that $\widehat{w}$ is continuous by the pasting lemma. After all, we know from the definition of w_0 that $\min(w(z) < w_0) = w_0$ if $z \in \partial\Delta'$. Plus, both cases in the definition of $\widehat{w}$ are continuous functions.

As for checknig that $\widehat{w}$ is superharmonic, note that similarly to how the maximum of two subharmonic functions is subharmonic, we can show the minimum of two superharmonic functions is superharmonic. Thus, it's clear that $\widehat{w}$ is superharmonic everywhere except $\partial\Delta'$. Meanwhile, if $z \in \partial\Delta'$, then we can use the superharmonicity of $\min(w, w_0)$ on $\Delta(a, R)$ in order to conclude for all $\rho \in [0, \frac{R}{4}]$ that:

$$\begin{aligned} \tfrac{1}{2\pi} \int_0^{2\pi} \widehat{w}(z + \rho e^{it}) \mathrm{d}t = \tfrac{1}{2\pi} \int_0^{2\pi} \min(w(z + \rho e^{it}), w_0) \mathrm{d}t &\leq \min(w(z), w_0) \\ &= w_0 = \widehat{w}(z) \end{aligned}$$

Thus, we are lead to the following result:

Proposition: G admits barriers at all $a \in \partial G$ if and only if G admits local barriers at all $a \in \partial G$.

Math 220c Homework Problems:

Set 3 Problem 3:

(i) Let $\varphi : \mathbb{R}^2 \to \mathbb{R}$ be a C^2 function with $\varphi \geq 0$ and $\triangle\varphi \geq 0$. Assume that $\int \varphi < \infty$. Then show that $\varphi \equiv 0$.

Suppose for the sake of contradiction that $a \in \mathbb{R}^2$ satisfies that $\varphi(a) > 0$. Then recall from *set 2 problem 1 on pages 839-840* that:

$$h(r) \coloneqq \tfrac{1}{2\pi} \int_0^{2\pi} \varphi(a + r(\cos(t), \sin(t))) \mathrm{d}t$$

is a non-decreasing function on $[0, \infty)$ satisfying that $h(0) = \varphi(a)$. In turn, we get a contradiction as:

$$\int \varphi = \int_0^\infty r h(r) \mathrm{d}r \geq \varphi(a) \int_0^\infty r \mathrm{d}r = \varphi(a) \cdot \infty.$$

(ii) Show that if $f : G \to \mathbb{R}$ is holomorphic, then $|f|^2$ is subharmonic.

Write $f(x+iy) = u(x,y) + iv(x,y)$. Then we have that $|f|^2 = u^2 + v^2$. In turn, we can calculate that:

$$\partial_x |f|^2 = 2uu_x + 2vv_x \text{ and } \partial_x^2 |f|^2 = 2u_x^2 + 2uu_{x,x} + 2v_x^2 + 2vv_{x,x}$$

By symmetry we also know what $\partial_y^2 |f|^2$ is. Hence, we can conclude that:

$$\triangle |f|^2 = 2(u_x^2 + uu_{x,x} + v_x^2 + vv_{x,x} + u_y^2 + uu_{y,y} + v_y^2 + vv_{y,y})$$

Yet since u and v are harmonic, we know that $u(u_{x,x} + u_{y,y}) \equiv 0$ and $v(v_{x,x} + v_{y,y}) \equiv 0$. Hence, our expression simplifies to:

$$\triangle |f|^2 = 2(u_x^2 + v_x^2 + u_y^2 + v_y^2).$$

One final simplication I can make (that is completely unnecessary at this point) is that since $f' = u_x + iv_x = v_y - iu_y$, we know $|f'|^2 = u_x^2 + v_x^2 = v_y^2 + u_y^2$. Therefore, we've proven that $\triangle |f|^2 = 4|f'|^2$.

This proves that $\triangle |f|^2 \geq 0$. Hence, as $|f|^2$ is a smooth function, we can conclude it is subharmonic.

(iii) Conclude that if $f : \mathbb{C} \to \mathbb{C}$ is entire and satisfies that $\int_{\mathbb{C}} |f|^2 \mathrm{d}x\mathrm{d}y < \infty$, then $f \equiv 0$.

It's automatic that $|f|^2 \geq 0$ and C^∞ (since f is C^∞). Also in part (b) I showed that $|f|^2$ is subharmonic by proving that $\triangle |f|^2 \geq 0$. Therefore, part (i) applies and we conclude:

$$\int |f|^2 < \infty \implies |f|^2 \equiv 0 \implies f \equiv 0. \blacksquare$$

Set 1 Problem 1: Let $u : G \to \mathbb{R}$ be harmonic, and suppose $\overline{\Delta}(a,r) \subseteq G$.

(i) Show that $u(a) = \frac{1}{\pi r^2} \int_{\overline{\Delta}(a,r)} u(x,y)$

To start off, by switching to polar coordinates centered at a we get that:

$$\begin{aligned} \tfrac{1}{\pi r^2} \int_{\overline{\Delta}(a,r)} u(x+iy) &= \tfrac{1}{\pi r^2} \int_0^r \int_0^{2\pi} u(a + \rho e^{i\theta}) \cdot \rho \mathrm{d}\theta \mathrm{d}\rho \\ &= \tfrac{2}{r^2} \int_0^r \rho \left(\tfrac{1}{2\pi} \int_0^{2\pi} u(a + \rho e^{i\theta}) \mathrm{d}\theta \right) \mathrm{d}\rho \end{aligned}$$

In turn, by the mean value property of u we know that the last double integral simplifies to:

$$\tfrac{2}{r^2} \int_0^r \rho u(a) \mathrm{d}\rho = \tfrac{2u(a)}{r^2} \cdot (\tfrac{r^2}{2} - 0) = u(a)$$

(ii) Show that the derivatives u_x and u_y are also harmonic. Hence, part (i) can be applied to u_x and u_y.

Recall from the elliptic regularity result for harmonic functions (on *page 804*) that any harmonic function is automatically C^∞. In particular, since u is C^3 (meaning we can swap the order of differentiation as much as we want for all third-order derivatives), we can conclude that:

$$\triangle u_x = u_{x,x,x} + u_{x,y,y} = (u_{x,x} + u_{y,y})_x = (0)_x = 0$$

Similarly, we can conclude that $\triangle u_y \equiv 0$.

This was not the end of problem 1. But, I didn't finish the rest before the deadline to turn my homework in. Also, part (iii) would have required Green's theorem which I've not proved yet. Since I have other things to be taking notes on now, I'm going to move on.

4/19/2026

Math 200c Notes:

On *page 846*, I mentioned there being a theorem I still wanted to prove. Now that I've gone over algebraic closures, I think I am ready to prove this theorem finally.

> Btw, because I'm taking these notes from Alireza and not Rogalski, I'll be returning *briefly* to the convention that $F \subseteq K \subseteq E$ instead of that $F \subseteq E \subseteq K$.

A finite degree extension E/F is a root extension if there exists intermediate fields $F = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_m = E$ such that for all i we have that $K_{i+1} = K_i[\alpha_{i+1}]$ and $\alpha_{i+1}^{n_{i+1}} \in K_i$ for some positive integer n_{i+1}.

Next, we say $f(x) \in F[x]$ is solvable by radicals if $f(x)$ splits in $E[x]$ for some root extension E/F.

> If it were possible to express all the roots of $f(x)$ by mathematical expressions of numbers in F using only $+$, $-$, $\times$, $\div$, and $\sqrt[n]{}$, then we'd have to have that f is solvable by radicals. A historically significant theorem by Galois tells us exactly when f is solvable by radicals. And although this theorem is apparently not that important in practice today, I'll still be covering it just so that I can be a cultured mathematician.

Lemma: Suppose $\mathrm{char}(F) = 0$ or that $\mathrm{char}(F) = p$ where $p \not| n$. Also suppose E is a splitting field of $x^n - 1$ over F. Then there is an injection $\mathrm{Gal}(E/F) \hookrightarrow (\mathbb{Z}/n\mathbb{Z})^\times$.

> Proof:
> By our assumption on $\mathrm{char}(F)$, we know that $x^n - 1$ is separable. Hence, $|V_{x^n-1}(E)| = n$. Also, we can easily check that $V_{x^n-1}(E)$ is a subgroup of $E^\times$. Hence, by a prior lemma on *page 843* we know that $V_{x^n-1}(E)$ is a cyclic group. And now, this proves the existence of a primitive nth root of unity ζ. In other words, the zero set of $x^n - 1$ is given by the cyclic group $\langle \zeta \rangle \cong \mathbb{Z}/n\mathbb{Z}$ where ζ has multiplicative order n in $E^\times$.
>
> Now no matter what the group structure of $\mathrm{Gal}(E/F)$ is exactly, we know that any $\sigma \in \mathrm{Gal}(E/F)$ is uniquely determined by where it sends ζ since $E = F[\zeta]$. Also, we know that ζ must be mapped to a different nth root of unity (which is not the trivial nth root 1). Hence, there is a well-defined injective map $\mathrm{Gal}(E/F) \to (\mathbb{Z}/n\mathbb{Z})^\times$ given by $\sigma \mapsto j + n\mathbb{Z}$ where j satisfies that $\sigma(\zeta) = \zeta^j$. Also, this map is easily checked to be a homomorphism. ■
>
> > In the case that $F = \mathbb{Q}$, then we proved on *page 821* that the above injection is also a surjection. That said, if F were a different field then this might not be the case. For example, a splitting field of $x^5 - 1$ over $\mathbb{R}$ would be given by $\mathbb{C}$. In turn, the Galois group would only have an order of 2 instead of $4 = |(\mathbb{Z}/5\mathbb{Z})^\times|$.

Another random fact I want to point out is as follows.

Lemma: Suppose we have a field isomorphism $\theta : F \to F'$. If E/F and E'/F' are arbitrary field extensions and if there exists $\widehat{\theta} \in \mathrm{Isom}_\theta(E, E')$, then we have an isomorphism $\mathrm{Aut}_F(E) \to \mathrm{Aut}_{F'}(E')$ given by $\sigma \mapsto \widehat{\theta} \circ \sigma \circ \widehat{\theta}^{-1}$.

The significance of this lemma is that when considering a splitting field of a polynomial, we can always assume that splitting field is embedded in whichever larger field we want and it won't change the Galois group.

I realize I've implicitely used this fact a few times.

Moving on, suppose E/F is a field extension and $K_1, \ldots, K_n \in \mathrm{Int}(E/F)$. Then the composite of the K_i (denoted $K_1 \cdots K_n$) is the smallest subfield of E containing $\bigcup_{i=1}^n K_i$.

Note that if $K, F \in \mathrm{Int}(E/F)$ satisfy that $K = F[\alpha_1, \ldots, \alpha_n]$ and $L = F[\beta_1, \ldots, \beta_m]$, then $KL = F[\alpha_1, \ldots, \alpha_n\beta_1, \ldots, \beta_m]$.

Also, given $K_1, K_2, K_3 \in \mathrm{Int}(E/F)$, we have that $K_1K_2K_3 = (K_1K_2)K_3 = K_1(K_2K_3)$ and $K_1K_2 = K_2K_1$.

Lemma: If E/F is an algebraic field extension, $K, L \in \mathrm{Int}(E/F)$, and K/F and L/F are finite Galois extensions, then KL/F is also a Galois extension.

By the theorem of the primitive element, we can write $K = F[\alpha]$ and $L = F[\beta]$ for some $\alpha, \beta \in E$. Thus, $KL = F[\alpha, \beta]$. Also, if $\overline{F}$ is an algebraic closure of F (containing E) then given any $\phi \in \mathrm{Embed}_{\mathrm{Id}_F}(E, \overline{F})$ we know:

$$\phi(KL) = F[\phi(\alpha), \phi(\beta)] = F[\phi(\alpha)]F[\phi(\beta)] = \phi(F[\alpha])\phi(F[\beta]) = \phi(K)\phi(L)$$

Yet simultaneously, since K/F and L/F are normal extensions and ϕ must act as a bijective permutation on any zero set of a polynomial in $F[x]$, we can conclude that $\phi(K) = K$ and $\phi(L) = L$. This proves that $\phi(KL) = KL$ for all $\phi \in \mathrm{Embed}_{\mathrm{Id}_F}(E, \overline{F})$. In turn, we know KL/F is a normal extension.

Why?
Consider any $\alpha \in KL$ and let $f := m_{\alpha;F}$. Then for any $\alpha' \in V_f(\overline{F})$ we know that there is some $\phi \in \mathrm{Embed}_{\mathrm{Id}_F}(E, \overline{F})$ sending α to α'. To see why, recall from *page 762* that there is a unique isomorphism $\theta : F[\alpha] \to F[\alpha']$ fixing F and mapping α to α'. Then by lemma 1 on *pages 857-858* we can extend θ to an embedding $\phi : E \to \overline{F}$ such that $\theta|_{F[\alpha]} = \theta$.

But by our prior reasoning, we know that $\phi(KL) = KL$. Hence, as $\alpha \in KL$, this proves that $\alpha' = \phi(a) \in KL$.

As for showing that KL is a separable extension, just note the comment on set 2 problem 7(a) on *page 860*. This proves that KL is a Galois extension.

One more lemma I need to cover is as follows.

(Also I'm getting this lemma's proof from Rogalski.)

Lemma: Suppose $\text{char}(F) = 0$, and E/F is a root extension. Then there is a field extension K/E such that K/F and K/E are Galois extensions, and also K/F is a root extension.

Proof:

To start off, note that E/F is a separable extension.

To see why, write $F = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_m = E$ where for all i we have that $K_{i+1} = K_i[\beta_{i+1}]$ and $\beta_{i+1}^{n_{i+1}} \in K_i$ for some positive integer n_{i+1}. As $\text{char}(F) = 0$, we know that the polynomial $x^{n_{i+1}} - \beta_{i+1}^{n_{i+1}}$ is separable in $K_i[x]$ for all i. It follows that K_{i+1}/K_i is a separable extension for all i. And as can be proven using set 2 problem 7(c) on page 860 as well as induction, a tower of separable extensions gives another separable extension.

By the theorem of the primitive element, we can thus find some $\alpha \in E$ such that $E = F[\alpha]$. So, we let $f := m_{\alpha;F}$ and then we let K be the splitting field of f over E. Importantly, if $\alpha_1, \ldots, \alpha_n$ are the roots of $f(x)$ in K, then since $\alpha_i = \alpha$ for some i, we know $E[\alpha_1, \ldots, \alpha_n] = F[\alpha_1, \ldots, \alpha_n]$. Hence K is also the splitting field of f over F. Then because $f = m_{\alpha;F}$ is separable, we know K/F and K/E are Galois extensions.

What's left to do is prove that K/F is a root extension. Fortunately, we know $\text{Gal}(K/F)$ acts transitively on $\{\alpha_1, \ldots, \alpha_n\}$. Also, we must have that $|\text{Gal}(K/F)| = [K : F] = n$. Thus, for each α_i there is a unique $\sigma_i \in \text{Gal}(K/F)$ such that $\sigma_i(\alpha) = \alpha_i$. And so, we define $E_i := F[\alpha_i] = F[\sigma_i(\alpha)] = \sigma_i(F[\alpha]) = \sigma_i(E)$. Then, we will have that $K = E_1 \cdots E_n$, and also we can easily justify that E_i/F is a root extension for each i.

Why?
Write $F = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_m = E$ where for all j we have that $K_{j+1} = K_j[\beta_{j+1}]$ and $\beta_{j+1}^{n_{j+1}} \in K_j$ for some positive integer n_{j+1}. Then set $K_j' := \sigma_i(K_j)$.

For any $\ell \in \{0, \ldots, \ell\}$ we can write $K_\ell = F[\beta_1, \ldots, \beta_\ell]$. And then in turn, we get that $K_\ell' = F[\sigma(\beta_1), \ldots, \sigma(\beta_\ell)]$. By induction, this let's us conclude that $K_{j+1}' = K_j[\sigma_i(\beta_{j+1})]$ for all j. Also, $\beta_{j+1}^{n_{j+1}} \in K_j$ implies that:

$$\sigma_i(\beta_{j+1})^{n_{j+1}} = \sigma_i(\beta_{j+1}^{n_{j+1}}) \in \sigma_i(K_j) = K_j'.$$

So, the sequence $F = K_0' \subseteq K_1' \subseteq \cdots \subseteq K_m' = E_i$ proves that E_i/F is a root extension.

To finish off, we need to show that $E_1 \cdots E_n$ is a root extension of f. To do this, it suffices due to a simple induction argument to just prove that a composite of two root extensions in a larger field is also a root extension. Fortunately, we have the following result.

Proposition: Suppose K/F is a field extension and $E_1, E_2 \in \text{Int}(K/F)$ both satisfy that E_1/F and E_2/F are root extensions. Then E_1E_2/F is also a root extension.

Why?
Write $F = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_m = E_1$ where $K_{i+1} = K_i[\alpha_{i+1}]$ and $\alpha_{i+1}^{n_{i+1}} \in K_i$. Similarly write $F = L_0 \subseteq L_1 \subseteq \cdots \subseteq L_p = E_2$ where $L_{i+1} = L_i[\beta_{i+1}]$ and $\beta_{i+1}^{s_{i+1}} \in L_i$. Then consider the sequence:

$$F = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_m = E_1 = E_1L_0 \subseteq E_1L_1 \subseteq \cdots \subseteq E_1L_p = E_1E_2$$

This sequence works. ■

At last, we are ready to prove a theorem which the undergrad abstract algebra class was very weirdly centered around.

Galois' Theorem (Part 1): Suppose $\mathrm{char}(F) = 0$, $f \in F[x]$, and E is a splitting field of f over F. If f is solvable by radicals then $\mathrm{Gal}(E/F)$ is a solvable group.

Proof:
Suppose f is solvable by radicals in a field K and and write:
$F = K_0 \subseteq \cdots \subseteq K_m = K$ where $K_{i+1} = K_i[\alpha_{i+1}]$ and $\alpha_{i+1}^{n_{i+1}} \in K_i$ for some $n_{i+1} \in \mathbb{N}$.

By our prior lemma, we can assume without loss of generality that K/F is a Galois extension. Also without loss of generality, we can consider E to be the splitting field of f contained in K. Plus, we can consider the algebraic closure $\overline{F}$ of K. Then as K/F and $\overline{F}/K$ are algebraic extensions, we also have that $\overline{F}$ is the algebraic closure of F.

Let $E_0 \subseteq \overline{F}$ be the splitting field of $x^{\prod_{i=1}^m n_i} - 1$ over F. Also let $E_{i+1} \subseteq \overline{F}$ be the splitting field of $x^{n_{i+1}} - \alpha_{i+1}^{n_{i+1}}$ over E_i for all i. By induction, we can conclude that $K_i \subseteq E_i$ for all i. In particular, this lets us conclude $E \subseteq F_m \subseteq E_m$.

Now if we let $n = \prod_{i=1}^m n_i$, then we know there is an injective homomorphism $\mathrm{Gal}(E_0/F) \hookrightarrow (\mathbb{Z}/n\mathbb{Z})^\times$. Thus, $\mathrm{Gal}(E_0/F)$ is an abelian group. Also, because $x^{n_{i+1}} - 1$ splits over E_i for all i, we can apply our first proposition from Kummer theory (see *pages 843-844*) to get an injective homomorphism $\mathrm{Gal}(E_{i+1}/E_i) \hookrightarrow (\mathbb{Z}/n_{i+1}\mathbb{Z})$ for each i.

Finally, note that if ζ is a primitive $(\prod_{i=1}^m n_i)$th root of unity, then:
$$E_m = F[\zeta, \alpha_1, \ldots, \alpha_n] = F[\zeta]F[\alpha_1, \ldots, \alpha_n] = E_0 K.$$

Since K/F and E_0/F are Galois extensions, we can conclude by a prior lemma that E_m/F is a Galois extension. It follows that E_m/E_i is also a Galois extension for all i. And since E_{i+1}/E_i is a Galois extension, we know from *pages 815-816* that:
$$\mathrm{Gal}(E_m/E_{i+1}) \lhd \mathrm{Gal}(E_m/E_i) \text{ and } \frac{\mathrm{Gal}(E_m/E_i)}{\mathrm{Gal}(E_m/E_{i+1})} \cong \mathrm{Gal}(E_{i+1}/E_i) \text{ for all } i.$$

(Similarly, we know $\mathrm{Gal}(E_m/E_0) \lhd \mathrm{Gal}(E_m/F)$ and $\frac{\mathrm{Gal}(E_m/F)}{\mathrm{Gal}(E_m/E_0)} \cong \mathrm{Gal}(E_0/F)$.)

With that we've proven a normal series:
$$\mathrm{Gal}(E_m/F) \rhd \mathrm{Gal}(E_m/E_0) \rhd \cdots \rhd \mathrm{Gal}(E_m/E_{m-1}) \rhd \{\mathrm{Id}\}$$
where each quotient of adjacent subgroups in this sequence is an abelian group. Hence, $\mathrm{Gal}(E_m/F)$ is a solvable group.

Since E/F is a Galois extension and $E \subseteq E_m$, we know $\mathrm{Gal}(E/F) \cong \frac{\mathrm{Gal}(E_m/F)}{\mathrm{Gal}(E_m/E)}$ is solvable as well. After all, if G is solvable and $N \lhd G$, then G/N is solvable as well. ■

We can also prove the converse statement. (That's part 2 of the theorem.) However, I'm short on time and so I won't prove it. The more important fact is part 1 anyways.

Notably, if F is a field of characteristic 0, $f \in F[x]$ has degree n, and E is a splitting field of f over F, then we can embed $\mathrm{Gal}(E/F)$ into S_n by observing that any automorphism $\sigma \in \mathrm{Gal}(E/F)$ is uniquely determined by how it permutes the roots f (of which there are at most n). But S_n is a solvable group so long as $n \leq 4$. And since subgroups of solvable groups are also solvable, we can thus conclude by Galois' theorem that every polynomial of degree at most 4 in $F[x]$ is solvable by radicals.

That said, S_n is not solvable when $n \geq 5$. And since we can generally find polynomials of degree ≥ 5 whose splitting field extension has a Galois group congruent to all of S_n, this proves that polynomials of degree at least 5 are not in general solvable by radicals.

> Caution: Having a polynomial be solvable by radicals is merely a ncessary condition for there to exist some equation for its zeros in terms of its coefficients. That said, the fact that quintics don't all satisfy this necessary condition proves that there cannot be any general quintic formula (or at least one only using the operations $+$, $-$, $\times$, $\div$, and $\sqrt[n]{}$).

Math 148 Lecture Notes:

To start off, I want to give a slightly more generalized way of proving the form of all solutions to the wave equation on $\mathbb{R}$ than what I gave on *page 853*. Namely, note that the 1-dimension wave equation can be factorized as:

$$0 = (\partial_t^2 - c^2\partial_x^2)u = (\partial_t - c\partial_x)(\partial_t + c\partial_x)u.$$

Using standardizable arguements, we can solve many PDEs of the form:

$$(\partial_t + a_1\partial_x)(\partial_t + a_2\partial_x)\cdots(\partial_t + a_n\partial_x)u = 0.$$

Lemma 1: Any solution to a PDE of the form $au_x + bu_t = 0$ where $a, b \in \mathbb{R}$ and $b \neq 0$ has the form $u(x,t) = f(x - \frac{a}{b}t)$ where f is a C^1 function such that $f(x) = u(x,0)$.

Proof:
First we solve that the characteristic passing through any (x_0, t_0) is given by $(x(s), t(s)) = (as + x_0, bs + t_0)$. Supposing $b \neq 0$, then if we set $t_0 = 0$ we get that $v(x_0, 0) = v(as + x_0, bs)$ for all x_0 and s. Yet we can then solve that $x_0 = x - \frac{a}{b}t$. Hence, we get that:

$$v(x,t) = v(as + x_0, bs) = v(x_0, 0) = f(x_0) = f(x - \tfrac{a}{b}t)$$

where $f \in C^1$ satisfies that $u(x,0) = f(x)$ for all x. ■

Lemma 2: Given any PDE of the form $au_x + bu_t = f(x - ct)$ where $a - bc \neq 0$ and f is continuous, we can find a particular solution of the form $g(x - ct)$ where g is a C^1 function.

Proof:
If a function $u(x,t) = g(x - ct)$ where g is any C^1 function is to satisfy our lemma, then we need to have that:

$$(a - bc)g'(x - ct) = f(x - ct)$$

In other words, we need $g' = \frac{1}{a-bc} f$. But this is doable since f is continuous and thus has an antiderivative. ■

Theorem: Suppose we are given the PDE $(\partial_t + a_1\partial_x)\cdots(\partial_t + a_n\partial_x)u = 0$ where all a_i are distinct real numbers. Then any solution will have the form $u(x,t) = \sum_{i=1}^{n} f_i(x - a_i t)$ where each f_i is a C^n function.

Proof:
Write $v_1 = (\partial_t + a_2\partial_x)\cdots(\partial_t + a_n\partial_x)u$. Then we know that $(\partial_t + a_1\partial_x)v_1 = 0$. Hence, by lemma 1 we can solve that $v_1(x,t) = f(x - a_1 t)$ for some C_1 function f.

Next, we set $v_k = (\partial_t + a_{k+1}\partial_x)\cdots(\partial_t + a_n\partial_x)$ where $2 \leq k \leq n-1$. Then suppose by induction that $v_{k-1}(x,t) = \sum_{i=1}^{k-1} g_i(x - a_i t)$. Then we know that:

$$\sum_{i=1}^{k-1} g_i(x - a_i t) = (\partial_t + a_k\partial_x)v_k$$

To solve for v_k is suffices to write v_k as a sum of inhomogenous solutions plus one homogenous solution. Fortunately, by lemma 1 we know any homogenous solution is given by $f_k(x - a_k t)$ for some C^1 function f. Meanwhile, as $a_k \neq a_i$ for any $i < k$, we can conclude by lemma 2 that there exists a particular solution for the PDE $g_i(x - a_i t) = (\partial_t + a_k\partial_x)v$ of the form $f_i(x - a_i t)$ for all i. Hence, we've proven that $v_k = \sum_{i=1}^{k} f_i(x - a_i t)$.

As a side note, the difference between two different particular solutions to the PDE $g_i(x - a_i t) = (\partial_t + a_k\partial_x)v$ will be a solution to the homogeneous PDE $0 = (\partial_t + a_k\partial_x)v$. Hence, so long as we know the form of one particular solution and all homogenous solutions, we know the form of all particular solutions.

Now once we reach $k = n - 1$, we will have that $v_k = u$. Hence, we will be done. ■
(As a side note, we force all the f_i to be C^n functions at the end just because we want u to be C^n.)

In the specific case of the wave equation on the real number line, we have that $0 = u_{t,t} - c^2 u_{x,x} = (\partial_t + c\partial_x)(\partial_t - c\partial_x)u$. Thus, this gives another explanation of why all solutions to this PDE have the form $f(x - ct) + g(x + ct)$.

One particular fruitful way of studying the wave equation comes from physics. Namely, suppose u is any solution to $u_{t,t} - c^2 u_{x,x} = 0$ such that $u_x(\cdot,t), u_t(\cdot,t)$ are in $L^2(\mathbb{R})$ for all t. Then define:

- $E_k(t) \coloneqq \int u_t(x,t)^2 \mathrm{d}x$ (this is kinetic energy),
- $E_p(t) \coloneqq c^2 \int u_x(x,t)^2 \mathrm{d}x$ (this is potential energy),
- $E(t) = E_k(t) + E_p(t)$ (this is total energy).

Assuming the first and second derivatives of u go to zero fast enough so that on some neighborhood of t_0 we have that $|2u_t u_{t,t} + 2c^2 u_x u_{x,t}|$ is bounded by a nonnegative L^1 function, then we can conclude by Leibniz's rule plus the fact that $u_{t,t} = c^2 u_{x,x}$ that:

$$E'(t_0) = \int 2u_t u_{t,t} + 2c^2 u_x u_{x,t} \mathrm{d}x = 2c^2 \int u_t u_{x,x} + u_x u_{x,t} \mathrm{d}x.$$

Next, assuming $u_t(x,t), u_x(x,t) \to 0$ as $x \to \pm\infty$ for all t, then we can show via integration by parts that:

$$\int u_t u_{x,x} \mathrm{d}x = u_t u_x \Big|_{-\infty}^{\infty} - \int u_{t,x} u_x \mathrm{d}x = 0 - \int u_{t,x} u_x \mathrm{d}x$$

Hence, we can conclude that $E'(t_0) = 2c^2 \int -u_{t,x} u_x + u_x u_{x,t} \mathrm{d}x = 2c^2 \int 0 \mathrm{d}x = 0.$

In turn, we've shown that $E(t)$ is constant.

I think for now I'm going to skip taking notes in this pdf at least on the wave equation in higher dimensions. My reasoning for this is that the proofs use the divergence theorem, something I've not proven and don't have a good grasp on. In fact, in general I've kinda been realizing today that I do not know enough vector calculus and am not comfortable enough with manifolds, differential forms, and etc. Hence, the next topic for me to cram is the diffusion equation in $\mathbb{R}^N$.

We shall consider solutions to the equation $u_t = \kappa \triangle u$ (where $\kappa > 0$) in $C_b^{2,1}(\Omega) \cap C(\overline{\Omega})$ where $\Omega \subseteq \mathbb{R}^N \times \mathbb{R}_{>0}$

> Note that $C_b^{2,1}(\Omega)$ denotes the space of bounded continuous functions on Ω which are twice continuously partial differentiable in space and once continuously partial diffferentiable in time.
>
> (If we want to remove the bounded condition, we just write $C^{2,1}$).

Theorem (Maximum Principle): Let $\Omega = \Omega' \times (a,b)$ where $\Omega' \subseteq \mathbb{R}^N$ is an open bounded set. Also let $\kappa > 0$. If $u \in C^{2,1}(\Omega) \cap C(\overline{\Omega})$ and satisfies $u_t = \kappa \triangle u$ on Ω, then:

$$\max_{(x,t) \in \overline{\Omega}} u(x,t) = \max_{(x,t) \in \Gamma} u(x,t)$$

where $\Gamma = (\Omega' \times \{a\}) \cup (\partial\Omega' \times [a,b])$.

> Caution: Γ does not include the entire boundary set of Ω. Also, this theorem does not prove anything about when the local maximum is attained inside Ω (though it is possible to show if Ω' is connected then that happens iff u is a constant function).

> Proof:
> To start off, since $\overline{\Omega}$ is compact in $\mathbb{R}^{N+1}$, we know $M := \max_{(x,t) \in \overline{\Omega}} u(x,t)$ exists.
>
> Similarly, Γ is a compact set (since we can write Γ as a union of two compact sets $\overline{\Omega'} \times \{a\}$ and $\partial\Omega' \times [a,b]$). Therefore, we know $m := \max_{(x,t) \in \Gamma} u(x,t)$ exists.
>
> Clearly $m \leq M$. But assume $m < M$. Then we know u acheieves its maximum on either Ω or $\Omega \times \{b\}$. Hence, there must exist some $\delta > 0$ such that $\Omega_\delta := \Omega' \times (a, b-\delta)$ and $M_\delta := \max_{(x,t) \in \overline{\Omega_\delta}} u(x,t) > m$.
>
> > This works because u is continuous on $\overline{\Omega}$. So, if u acheieves a global maximum in $\Omega' \times (a,b)$, we just set δ small enough so that M_δ includes that global maximum.

Meanwhile, if u acheieves its global maximum in $\Omega \times \{b\}$, then so long as Ω_δ contains points sufficiently close to that maximum, then M_δ will be larger than m.

Let $v(x,t) = u(x,t) + \varepsilon \|x\|_2^2$ for any $\varepsilon > 0$. Then in Ω we have that:
$(\partial_t - \kappa\triangle)v = (\partial_t - \kappa\triangle)u + \varepsilon(\partial_t - \kappa\triangle)\|x\|_2^2 = 0 + \varepsilon(0 - 2\kappa N) = -2\varepsilon\kappa N < 0.$

Meanwhile if we let $R > 0$ be such that $\Omega' \subseteq B_R(0)$ and we fix $\varepsilon \in (0, \frac{M_\delta - m}{R^2})$, then we can conclude that $\max_{(x,t)\in\Gamma} v(x,t) \leq m + \varepsilon R^2 < M_\delta$.

Yet also note that $v \geq u$ on Ω. So, if $(x_0, t_0) \in \overline{\Omega_\delta}$ is such that $v(x_0,t_0) = \max_{(x,t)\in\overline{\Omega_\delta}} v(x,t)$, then we know that $v(x_0,t_0) \geq M_\delta > \max_{(x,t)\in\Gamma} v(x,t)$. Hence:
$$(x_0, t_0) \in \overline{\Omega_\delta} - \Gamma \subseteq \Omega.$$

But now we must have $v_t(x_0,t_0) \geq 0$ and $v_{x_i}(x_0,t_0) = 0$ for all i.
The reason why we only know $v_t(x_0,t_0) \geq 0$ is that it could be that $(x_0,t_0) \in \Omega \times \{b - \delta\}$ That said, if the first partial derivatives don't satisfy these relations then we can follow the gradiant of the function v (staying inside Ω_δ) to get another point contradicting the maximality of $v(x_0,t_0)$.

Also, we must have that $v_{x_i,x_i}(x_0,t_0) = 0$ for all i.
This is because the function $f(s) = v(x_0 + se_i, t_0)$ must have a maximum at $s = 0$. Yet by chain rule we can calculate that $f'(s) = v_{x_i}(x_0 + se_i, t_0)$ and $f''(s) = v_{x_i,x_i}(x_0 + se_i, t_0)$. So $f''(0) = v_{x_i,x_i}(x_0,t_0) \leq 0$.

And now we have a contradiction because we've proven that $(\partial_t - \kappa\triangle)v \geq 0$ and $(\partial_t - \kappa\triangle)v < 0$. ■

Some remarks:

- If u satisfies the PDE $u_t - \kappa\triangle u = 0$, then any scalar multiple of u also satisfies the PDE. Hence, by applying the maximum principle to $-u$, we get an analogous minimum principle for u.
- A physics description for the maximum and minimum principles would be that diffusion processes do not lead to clumping.

Corollary (Comparison Principle): Let $\Omega = \Omega' \times (a,b)$ where $\Omega' \subseteq \mathbb{R}^N$ is an open bounded set. Also let $\kappa > 0$. If $u, \widetilde{u} \in C^{2,1}(\Omega) \cap C(\overline{\Omega})$ solve the PDE $u_t = \kappa\triangle u$ on Ω, then:
$$u \leq \widetilde{u} \text{ on } \Gamma \text{ implies } u \leq \widetilde{u} \text{ on } \overline{\Omega},$$
where $\Gamma = (\Omega' \times \{a\}) \cup (\partial\Omega' \times [a,b])$.
Proof:
Apply the maximum principle to $u - \widetilde{u}$. ■

This also leads to the following uniqueness result on bounded domains in space.

Theorem (Uniqueness): Suppose Ω' is an open bounded subset of $\mathbb{R}^N$ and $\kappa > 0$. Then after setting $\Omega = \Omega' \times (0, \infty)$, given the initial value problem:

- $u_t - \kappa \triangle u = g(x,t)$ on Ω,
- $u(x,0) = \varphi(x)$ for all $x \in \overline{\Omega'}$,
- $u(x,t) = h(x,t)$ for all $(x,t) \in \partial\Omega' \times (0,\infty)$,

(where φ, h are continuous functions...)

there is at most one solution in $C^{2,1}(\Omega) \cap C(\overline{\Omega})$.

Proof:
Suppose $u, \widetilde{u}$ are two solutions to this initial value problem. Then note that both $u - \widetilde{u}$ and 0 solve the same initial value problem except with g, φ, h replaced by the zero function. If we now apply the comparison principle on the set $\Omega' \times (0,b)$ to the solutions $u - \widetilde{u}$ and 0, we get that $u - \widetilde{u} \equiv 0$ on $\Omega' \times (0,b)$. Then to finish, take $b \to \infty$. ■

4/21/2026

Math 220c Lecture Notes:

Suppose $f : G \to \mathbb{C}$ is a holomorphic function, f is nowhere zero on G, and $\overline{\Delta}(a,r) \subseteq G$. By set 1 problem 3(i) on *page 803*, we thus know $\log|f(z)|$ is a harmonic function in G. And by the mean value property, we can conclude that:

$$\log|f(a)| = \tfrac{1}{2\pi}\int_0^{2\pi} \log|f(a+re^{it})|\mathrm{d}t$$

What if f has zeros around a though?

Theorem: Suppose $f : G \to \mathbb{C}$ is a holomorphic function, $\overline{\Delta}(a,r) \subseteq G$, and $f(a) \neq 0$. Let $b_1, \ldots, b_n$ be all the zeros of f in $\Delta(a,r)$ repeated according to their multiplicity (there can only be finitely many). Then:

$$\log|f(a)| + \sum_{j=1}^{n} \log(\tfrac{r}{|b_j - a|}) = \tfrac{1}{2\pi}\int_0^{2\pi} \log|f(a+re^{it})|\mathrm{d}t$$

Proof:
Without loss of generality we can assume $a = 0$. Otherwise just translate everything. Next, by restricting G we can assume without loss of generality that $G = \Delta(0,R)$ where $R > r$. Additionally, we can assume without loss of generality that $r = 1$. After all, if not then we can just consider the function $f^{\text{new}}(z) := f(rz)$ defined on $G^{\text{new}} = \Delta(0, \frac{R}{r})$. This leaves us the job of proving that:

$$\log|f(0)| + \sum_{j=1}^{n} \log(\tfrac{1}{|b_j|}) = \log|f(0)| - \sum_{j=1}^{n} \log|b_j| = \tfrac{1}{2\pi}\int_0^{2\pi} \log|f(e^{it})|\mathrm{d}t$$

In addition to letting $b_1, \ldots, b_n$ be the zeros of f in $\Delta(0,1)$, let $c_1, \ldots, c_m$ be the zeros of $f \in \partial\Delta(0,1)$ (repeated according to multiplicity).

Recall that $\varphi_b(z) := \frac{z-b}{1-\bar{b}z}$ is a biholomorphism mapping $\Delta(0,1)$ to $\Delta(0,1)$, $\partial\Delta(0,1)$ to $\partial\Delta(0,1)$, b to 0, and 0 to $-b$. Now set:

$$F(z) := f(z) \cdot \frac{1}{\prod_{j=1}^{n} \varphi_{b_j}(z)} \cdot \prod_{k=1}^{m} \frac{c_k}{c_k - z}$$

Then F has no zeroes in $\overline{\Delta}(0,1)$, and so by the observation at the start of lecture, we know:

$$\log|F(0)| = \tfrac{1}{2\pi}\int_0^{2\pi} \log|F(e^{it})|\mathrm{d}t$$

Yet $F(0) = \frac{f(0)}{\prod_{j=1}^{n}(-b_j)} \cdot 1$. Hence, we know that:

$$\log|F(0)| = \log|f(0)| - \textstyle\sum_{j=1}^{n} \log|b_j|.$$

Similarly, we can see that:

$$\int_0^{2\pi} \log|F(e^{it})|\mathrm{d}t = \int_0^{2\pi} \log|f(e^{it})|\mathrm{d} - \sum_{j=1}^{n} \int_0^{2\pi} \log|\varphi_{b_j}(e^{it})|\mathrm{d}t + \sum_{k=1}^{m} \int_0^{2\pi} \log\left|\tfrac{c_k}{c_k - e^{it}}\right|\mathrm{d}z$$

Claim 1: If $c \in \mathbb{C}$ satisfies that $|c| = 1$ then $\int_0^{2\pi} \log\left|\frac{c}{c-e^{it}}\right|\mathrm{d}t = 0$.

Let $c = e^{i\alpha}$. Then $\int_0^{2\pi} \log\left|\frac{c}{c-e^{it}}\right|\mathrm{d}t = \int_0^{2\pi} \log\left|\frac{e^{i\alpha}}{e^{i\alpha}-e^{it}}\right|\mathrm{d}t = \int_0^{2\pi} \log\left|\frac{1}{1-e^{i(t-\alpha)}}\right|\mathrm{d}t$. The last integral can be simplified to $\int_0^{2\pi} \log\left|\frac{1}{1-e^{it}}\right|\mathrm{d}t$ since $\log\left|\frac{1}{1-e^{i(t-\alpha)}}\right|$ is 2π-periodic. And thus, it suffices to prove that $-\int_0^{2\pi} \log|1-e^{it}|\mathrm{d}t = 0$.

Now's a good time to mention that $\log|1-e^{it}|$ is definitely integrable on $[0,2\pi]$ since it's positive part is bounded by $2\pi\log(2)$. Thus if this integral converges to a finite value, it will converge abolutely. Next note that:

$$\begin{aligned}|1-e^{it}|^2 &= (1-\cos(t))^2 + \sin^2(t)\\ &= 2 - 2\cos(t) = 2(\cos^2(\tfrac{t}{2}) + \sin^2(\tfrac{t}{2})) - 2(\cos^2(\tfrac{t}{2}) - \sin^2(\tfrac{t}{2})) = 4\sin^2(\tfrac{t}{2})\end{aligned}$$

Therefore $\int_0^{2\pi} \log|1-e^{it}|\mathrm{d}t = \int_0^{2\pi} \log|2\sin(\frac{t}{2})|\mathrm{d}t = \frac{1}{2}\int_0^{\pi} \log|2\sin(u)|\mathrm{d}u$. In turn, it suffices to show $0 = \int_0^{\pi} \log|2\sin(u)|\mathrm{d}u = \pi\log(2) + \int_0^{\pi} \log|\sin(u)|\mathrm{d}u$. Or in other words, we need to show that:

$$\int_0^{\pi} \log|\sin(u)|\mathrm{d}u = -\pi\log(2)$$

But $\sin(u) \in [0,1]$ for all $u \in [0,\pi]$. Thus $\int_0^{\pi} \log|\sin(u)|\mathrm{d}u = \int_0^{\pi} \log(\sin(u))\mathrm{d}u$ is integrable and there exists $I \in [-\infty, 0]$ such that $I = \int_0^{\pi} \log(\sin(u))\mathrm{d}u$. By a change of variables as well as some further manipulations, we then get:

$$\begin{aligned}I &= 2\int_0^{\pi/2} \log(\sin(2v))\mathrm{d}v\\ &= 2\int_0^{\pi/2} \log(2\sin(v)\cos(v))\mathrm{d}v\\ &= 2\int_0^{\pi/2} \log(2)\mathrm{d}v + 2\int_0^{\pi/2} \log(\sin(v))\mathrm{d}v + 2\int_0^{\pi/2} \log(\cos(v))\mathrm{d}v\\ &= \pi\log(2) + 2\int_0^{\pi/2} \log(\sin(v))\mathrm{d}v + 2\int_0^{\pi/2} \log(\sin(\tfrac{\pi}{2}+v))\mathrm{d}v\\ &= \pi\log(2) + 2\int_0^{\pi} \log(\sin(v))\mathrm{d}v = \pi\log(2) + 2I\end{aligned}$$

So, either $I = -\infty$ or $I = -\pi\log(2)$. To disprove the former case, recall that $\sin(u) \geq \frac{2}{\pi}u$ for all $u \in [0, \frac{\pi}{2}]$. And since $\sin(\frac{\pi}{2} - u) = \sin(\frac{\pi}{2} + u)$, we can thus conclude that:

$$I = 2\int_0^{\pi/2} \log(\sin(u))\mathrm{d}u \geq 2\int_0^{\pi/2} \log(\tfrac{2}{\pi}u)\mathrm{d}u$$
$$= \pi \int_0^1 \log(v)\mathrm{d}v$$
$$= \pi(v\log(v) - v)\,|_0^1 = -\pi - \pi \lim_{v\to 0^+}(v\log(v) - v)$$

By a single application of L'Hôpital's rule applied $v\log(v) = \frac{\log(v)}{v^{-1}}$ we can show the latter limit equals 0. Hence $I \geq -\pi$ and this proves that $I \neq -\infty$.

Claim 2: $\int_0^{2\pi} \log|\varphi_{b_j}(e^{it})|\mathrm{d}t = 0$ for all b_j.
This is because $|\varphi_{b_j}(e^{it})| = 1$ for all t and hence $\log|\varphi_{b_j}(e^{it})| = 0$ for all t.

Therefore, we've proven that $\int_0^{2\pi} \log|F(e^{it})|\mathrm{d}t = \int_0^{2\pi} \log|f(e^{it})|\mathrm{d}t$. ■

The formula proved in the prior theorem is called Jensen's Formula. The most important case for our purposes will be when $a = 0$. So from now on I will just assume our closed disk of radius r is centered at 0. A natural next question for us to ask is if we can find an equation for $\log(f(z_0))$ when $z_0 \neq 0$.

Theorem (The Poisson-Jensen Formula): Let $f : G \to \mathbb{C}$ be holomorphic, $\overline{\Delta}(0,r) \subseteq G$, $z_0 \in \Delta(0,r)$, and $f(z_0) \neq 0$. Also let $b_1, \ldots, b_n$ be the zeros (repeated according to multiplicity) of f in $\Delta(0,r)$. Then:

$$\log|f(z_0)| + \sum_{j=1}^{n} \log\left|\frac{r^2 - \overline{b_j}z_0}{r(z_0 - b_j)}\right| = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{Re}\left(\frac{re^{it}+z_0}{re^{it}-z_0}\right)\log|f(re^{it})|\mathrm{d}t$$

Some Remarks:

- If $z_0 = \rho e^{i\theta}$, then:
$$\mathrm{Re}\left(\frac{re^{it}+z_0}{re^{it}-z_0}\right) = \mathrm{Re}\left(\frac{1+\frac{\rho}{r}e^{i(\theta-t)}}{1-\frac{\rho}{r}e^{i(\theta-t)}}\right) = P_{\rho//r}(\theta - t)$$
Thus, if there are no zeros of f in $\Delta(0,r)$, then this formula simplifies to just being the Poisson integral formula (on $\overline{\Delta}(0,r)$) applied to $\log|f|$.

- Meanwhile, if $z_0 = 0$ then this formula simplifies to just being Jensen's formula.

Proof:
Like before, by restricting G we can assume without loss of generality that $G = \Delta(0,R)$ where $R > r$. Then, we can assume without loss of generality that $r = 1$. After all, if not then we can just consider the function $f^{\text{new}}(z) := f(rz)$ defined on the region $G^{\text{new}} = \Delta(0, \frac{R}{r})$. But now the proposed formula we want to prove becomes:

$$\log|f(z_0)| + \sum_{j=1}^{n} \log\left|\frac{1 - \overline{b_j}z_0}{z_0 - b_j}\right| = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{Re}\left(\frac{e^{it}+z_0}{e^{it}-z_0}\right)\log|f(e^{it})|\mathrm{d}t$$

Our solution path will now be to recenter our function using a biholomorphism. Namely, consider the functions $L(w) = \frac{w+z_0}{1+\overline{z_0}w}$ and $M(w) = \frac{w-z_0}{1-\overline{z_0}w}$. Then let $\widetilde{f} := f \circ L$. By

noting that L and M are inverse biholomorphisms on $\Delta(0,1)$, we can see the zeros of $\widetilde{f}$ in $\Delta(0,1)$ are given by $M(b_1), \ldots M(b_n)$. In turn, by Jensen's formula applied to $\widetilde{f}$, we have that:

$$\log|\widetilde{f}(0)| - \sum_{j=1}^{n} \log\left|\tfrac{1}{M(b_j)}\right| = \tfrac{1}{2\pi}\int_0^{2\pi} \log|\widetilde{f}(e^{it})|\mathrm{d}t$$

Or to put in other words, we have that:

$$\log|f(z_0)| - \sum_{j=1}^{n} \log\left|\tfrac{1-\overline{z_0}b_j}{b_j - z_0}\right| = \tfrac{1}{2\pi}\int_0^{2\pi} \log|(f\circ L)(e^{it})|\mathrm{d}t$$

But now by the same change of variables as on *pages 807-808*, we can conclude that $\frac{1}{2\pi}\int_0^{2\pi} \log|(f\circ L)(e^{it})|\mathrm{d}t = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{Re}\left(\frac{e^{is}+z_0}{e^{is}-z_0}\right)\log|f(e^{is})\mathrm{d}s$

(where $\mathrm{Re}\left(\frac{e^{is}+z_0}{e^{is}-z_0}\right) = P_\rho(\theta - s)$ and $z_0 = \rho e^{i\theta}$).

Also $|1-\overline{z_0}b_j| = |1 - z_0\overline{b_j}|$ and $|b_j - z_0| = |z_0 - b_j|$. And thus the proof follows. ■

For a quick application of Jensen's formula, suppose $f : \mathbb{C} \to \mathbb{C}$ is an entire function and $f(0) = 1$. Then if:

- $M(R) = \sup_{|z|=R}|f(z)|$,
- $N(R) = \#$ of zeros of f in $\overline{\Delta}(0,R)$ (counted according to multiplicities),

by Jensen's formula applied to $\overline{\Delta}(0,3R)$, we have that:

$$0 + \sum_{j=1}^{N(3R)} \log\left|\tfrac{3R}{a_j}\right| = \log|f(0)| + \sum_{j=1}^{N(3R)} \log\left|\tfrac{3R}{a_j}\right| = \tfrac{1}{2\pi}\int_0^{2\pi}\log|f(3Re^{it})|\mathrm{d}t \leq \log(M(3R))$$

But now in turn we have $N(R) \leq N(R)\log(3) \leq \sum_{j=1}^{N(R)} \log\left|\frac{3R}{a_j}\right| \leq \log(M(3R))$.

The quick application that the prior lecture ended with shows that the growth of holomorphic functions and the distribution of their zeros is positively correlated.

For another illustration of this correlation, suppose f is a polynomial of degree d. Then if $M(R)$ and $N(R)$ are defined as before, $N(R) = d$ and $M(R) \approx |\mathrm{lt}(f)|R^d$ when R is large. Therefore,

$$\lim_{R\to\infty}\frac{\log(M(R))}{\log(R)} = \lim_{R\to\infty}\frac{\log|\mathrm{lt}(f)| + d\log(R)}{\log(R)} = d$$

Conversely, suppose f is an entire function and $\lim_{R\to\infty}\frac{\log(M(R))}{\log(R)} = d$. Then for any $\varepsilon > 0$,

$\log(M(R)) < (1+\varepsilon)d\log(R)$ when R is large enough.

Hence, we can conclude that $|f(z)| < e^{1+\varepsilon}|z|^d$ for $|z| \gg 0$. By (Conway) exercise IV.3.1 on *page 397*, this lets us conclude f is a polynomial of degree at most d. By our prior calculation though, we know that if f is a polynomial then the limit is exactly the degree of f. So, we can conclude f is a polynomial of exactly degree d.

Since our end-goal is to vastly strengthen the Weierstraß factorization theorem (see *pages 575-581*) and recalling the definition of the Weierstraß elementary factors, it would be especially helpful to get a measure of the growth of functions of the form $e^{p(z)}$ where p is a polynomial. But note that by taking one log of $M(R)$, we can reduce our problem to the polynomial case at the end of the last page. This motivates the following definition.

If $f : \mathbb{C} \to \mathbb{C}$ is entire and $f \not\equiv c$ where $|c| \leq 1$, then after letting $M(R) = \sup_{|z|=R} |f(z)|$ we define the order of f to be:

$$\lambda(f) := \limsup_{R\to\infty} \frac{\log(\log(M(R)))}{\log(R)}.$$ (Note $\lambda(f)$ can be infinite.)

To show that this is well-defined, it suffices to show that if $f \not\equiv c$ with $|c| \leq 1$ then $M(R) > 1$ for all R large enough. If $f \equiv C$ with $|C| > 1$, then this is trivial. So from now on, we can assume f is not a constant function.

Without loss of generality, suppose there exists a sequence $\{R_n\}_{n\in\mathbb{N}} \subseteq \mathbb{R}_{>0}$ with $R_n \to \infty$ and $M(R_n) \leq 1$ for all n. Then by Cauchy's estimate, we know that:

$$|f^{(1)}(0)| \leq \frac{k!}{R_n^k} \text{ for all } R_n.$$

But now this would imply $f^k(0) = 0$ for all k. In turn, we'd know that f is a constant function, contradicting our assumption about f.

Note: To prove that the function f has order λ that is finite (when we can't easily calculate a limit of $\frac{\log(\log(M(R)))}{\log(R)}$ as $R \to \infty$), it suffices to show the following:

1. $\forall \varepsilon > 0, \exists r > 0$ such that $|f(z)| < e^{|z|^{\lambda+\varepsilon}}$ for all $|z| > r$.

   This equivalently means $M(R) < e^{R^{\lambda+\varepsilon}}$ for all $R > r$. Thus in turn, we have that:

   $$\lambda(f) = \limsup_{R\to\infty} \frac{\log(\log(M(R)))}{\log(R)} \leq \lambda + \varepsilon$$

   And by taking $\varepsilon \to 0$ we get that $\lambda(f) \leq \lambda$.

2. $\forall \varepsilon > 0, \exists \{z_n\}_{n\in\mathbb{N}} \subseteq \mathbb{C}$ such that $z_n \to \infty$ and $|f(z_n)| > e^{|z_n|^{\lambda-\varepsilon}}$.

   This proves that:

   $$\lambda(f) = \limsup_{R\to\infty} \frac{\log(\log(M(R)))}{\log(R)} \geq \limsup_{n\to\infty} \frac{\log(\log|f(z_n)|)}{\log|z_n|} \geq \lambda - \varepsilon.$$

   And by taking $\varepsilon \to 0$, we get that $\lambda(f) \geq \lambda$.

For example, $\lambda(z^m) = 0$ for all $m \geq 0$. After all, $M(R) = R^m$. Therefore:

$$\frac{\log(\log(M(R)))}{\log(R)} = \frac{\log(m)+\log(\log(R))}{\log(R)} \to 0 \text{ as } R \to \infty.$$

More generally, given any polynomial $p(z) \in \mathbb{C}[z]$ not equal to a constant in $\overline{\Delta}(0,1)$, we have that $\lambda(p) = 0$.

To see why, first note $\frac{\log(\log(M(R)))}{\log(R)} \to 0$ trivially when the numerator is constant. So, without loss of generality we can assume $d := \deg(p) > 0$.

Next note that $\lim_{z\to\infty} \frac{p(z)}{z^d} = \mathrm{lt}(f)$. Thus if $|z| = R$ is large and $\alpha := |\mathrm{lt}(f)|$, we know $e < |P(z)| < (\alpha+1)R^d$. In turn, for any $\varepsilon > 0$ we have for $R \gg 0$ that $e^{R^{-\varepsilon}} \leq e \leq M(R) \leq e^{R^{\varepsilon}}$. This verifies the two conditions needed for proving that $\lambda(p) = 0$.

4/25/2026

Before taking more complex analysis notes, I'm going to do some homework problems. Given a function $f : G \to \mathbb{C}$ with $\overline{\Delta}(0,r) \subseteq G$, $M(R)$ and $N(R)$ will be defined as in the prior section.

Set 4 Problem 1: Let $f : G \to \mathbb{C}$ be holomorphic and let $\overline{\Delta}(0,r) \subseteq G$. Assume that $f(0) \neq 0$ and $z_1, \ldots, z_k$ are some of the zeros of f in $\Delta(0,r)$ (which are allowed to repeated up to their corresponding multiplicities).

(i) Using Jensen's formula, show that $|f(0)| \leq |z_1 \cdots z_k| \cdot \frac{M(r)}{r^k}$.

Let $z_{k+1}, \ldots, z_n$ be the remaining zeros in $\Delta(0,r)$. Then by Jensen's formula we have that:

$$\log|f(0)| + \sum_{j=1}^{n} \log(\tfrac{r}{|z_j|}) = \tfrac{1}{2\pi}\int_0^{2\pi} \log|f(re^{it})|\mathrm{d}t \leq \log(M(r)).$$

But now since $\frac{r}{|z_j|} > 1$ for all j, we know that:

$$\log|f(0)| \leq \log(M(r)) - \sum_{j=1}^{n} \log(\tfrac{r}{|z_j|}) \leq \log(M(r)) - \sum_{j=1}^{k} \log(\tfrac{r}{|z_j|})$$
$$= \log(M(r)^{\frac{|z_1\cdots z_k|}{r^k}})$$

By exponentiating both sides, this proves that $|f(0)| \leq M(r)^{\frac{|z_1\cdots z_k|}{r^k}}$.

(ii) Assume $f : G \to \mathbb{C}$ is holomorphic, $\overline{\Delta}(0,1) \subseteq G$, and $|f(z)| \leq 1$ for all $z \in G$. Assume $f(\frac{1}{2}) = f(\frac{i}{2}) = 0$. Then $|f(0)| \leq \frac{1}{4}$.

If $f(0) = 0$, we have nothing to prove. Otherwise, we know by part 1 that $|f(0)| \leq 1 \cdot \frac{|\frac{1}{2}\cdot\frac{i}{2}|}{1^2} = \frac{1}{4}$.

Set 4 Problem 2: Assume f is a bounded holomorphic function $f : \Delta(0,1) \to \mathbb{C}$ with zeros $a_1, a_2, \ldots$ listed with multiplicity. Show that $\sum_n (1 - |a_n|) < \infty$.

To start off, the problem is trivial if there are only finitely many zeros. So, we can assume without loss of generality now that f has countably infinitely many zeros. Also let $M > 0$ satisfy that $|f| \leq M$ on $\Delta(0,1)$.

First we'll assume the case that $f(0) \neq 0$. Then notice that since $f^{-1}(\{0\})$ can't have any limit points in $\Delta(0,1)$, we know for any $r \in (0,1)$ that $S_r \coloneqq \{n \in \mathbb{N} : a_n \in \Delta(0,r)\}$ is a finite set. Plus, by Jensen's formula we have that:

$$|f(0)| + \sum_{n \in S_r} \log(\tfrac{r}{|a_n|}) = \tfrac{1}{2\pi} \int_0^{2\pi} \log|f(re^{it})| \leq M$$

In other words, $\sum_{n \in S_r} \log(\frac{r}{|a_n|}) \leq M - |f(0)|$. But now since the left side is positive and strictly increasing as $r \to 1$, we know $\lim_{r \to 1^-} \sum_{n \in S_r} \log(\frac{r}{|a_n|})$ exists and is at most $M - |f(0)|$. Also by the monotone convergence theorem, we know:

$$\lim_{r \to 1^-} \sum_{n \in S_r} \log(\tfrac{r}{|a_n|}) = \sum_{n \in \mathbb{N}} \log(\tfrac{1}{|a_n|}) = \sum_{n \in \mathbb{N}} -\log|a_n|.$$

So, $\sum_{n \in \mathbb{N}} -\log|a_n|$ converges absolutely.

To finish off the argument for when $f(0) \neq 0$, note that when $|a_n| \in (0,1]$ we have that $1 - |a_n| \leq -\log|a_n|$.

To prove this inequality, set $f(x) = -\log(x) - 1 + x$ then we can see that $f'(x) = \frac{-1}{x} + 1 \leq 0$ for $x \leq 1$ and that $f(1) = 0$. So, $f \geq 0$ for $x \leq 1$.

Thus by the comparison test (and using the fact that $1 - |a_n| \geq 0$ for all n), we can conclude $\sum_{n \in \mathbb{N}} (1 - |a_n|)$ converges.

As for the case that $f(0) = 0$, note that since f is not identically zero (since f was assumed to have countably many zeros), we can write $f(z) = z^d g(z)$ where $g : \Delta(0,1) \to \mathbb{C}$ is another holomorphic function.

By the continuity of g on the compact set $\overline{\Delta(0, \frac{1}{2})}$, we know g acheieves a maximum on $\overline{\Delta(0, \frac{1}{2})}$. Meanwhile $g(z) = z^{-d} f(z) \leq \frac{M}{2^d}$ on $\Delta(0,1) - \overline{\Delta(0, \frac{1}{2})}$. Hence, we can conclude that g is also bounded on the unit disk.

By the prior case I proved, we thus know if $S \coloneqq \{n \in \mathbb{N} : a_n = 0\}$ then:

$$\sum_{n \in \mathbb{N} - S} (1 - |a_n|) < \infty$$

And now we just add $|S| = d$ to our series to prove that $\sum_{n \in \mathbb{N}} (1 - |a_n|) < \infty$.

In particular, show that there is no bounded holomorphic function $f : \Delta(0,1) \to \mathbb{C}$ with zeros only at $a_n = 1 - \frac{1}{n}$ for all $n \geq 1$.

If a function f had zeros as described here, then we'd have that:

$$\textstyle\sum_{n \in \mathbb{N}} (1 - |a_n|) = \sum_{n \in \mathbb{N}} \frac{1}{n} = \infty.$$

But now if f is also bounded, then that would contradict the prior part of the problem. ■

Set 4 Problem 3: Let $f : \mathbb{C} \to \mathbb{C}$ be an entire function with $f(0) = 1$.

(i) Show that $N(r)\log(2) \leq \log(M(2r))$.

Let $a_1, \ldots, a_{N(r)}$ denote the zeros of f in $\overline{\Delta}(0,r)$, and let $a_{N(r)+1}, \ldots, a_{N(2r)}$ denote the zeros of f in $\overline{\Delta}(0,2r) - \overline{\Delta}(0,r)$. Since $\log|f(0)| = 0$, we can conclude by Jensen's formula that:

$$\sum_{j=1}^{N(r)} \log(\tfrac{2r}{|a_j|}) \leq \sum_{j=1}^{N(2r)} \log(\tfrac{2r}{|a_j|}) = \tfrac{1}{2\pi}\int_0^{2\pi} \log|f(2re^{it})|\mathrm{d}t \leq \log(M(2r)).$$

But now $\sum_{j=1}^{N(r)} \log(\tfrac{2r}{|a_j|}) \geq \sum_{j=1}^{N(r)} \log(2) = N(r)\log(2)$. This proves what we wanted.

(ii) Assume that $|f(z)| \leq \exp(A|z|^k)$ where $A > 0$ and k is a natural number. Then show that:

$$\limsup_{r\to\infty} \tfrac{\log(N(r))}{\log(r)} \leq k$$

Note: This problem has a slight error in that we need to explicitly assume $|f(z)|$ has a zero (lest $\log(N(r))$ be just not defined for any r). After all, if $f(z) = e^z$, then we'd know f is an entire function with $f(0) = 1$. Also $|f(z)| = e^{\mathrm{Re}(z)} \leq e^{|z|}$. But f has no zeros and hence the aforementioned problem occurs.

By part (i), we know that:

$$\begin{aligned}\limsup_{r\to\infty} \tfrac{\log(N(r))}{\log(r)} \leq \limsup_{r\to\infty} \tfrac{\log(\frac{\log(M(2r))}{\log(2)})}{\log(r)} &= \limsup_{r\to\infty}\left(\tfrac{\log(\log(M(2r)))}{\log(r)} - \tfrac{\log(\log(2))}{\log(r)}\right) \\ &= \limsup_{r\to\infty} \tfrac{\log(\log(M(2r)))}{\log(r)} - 0\end{aligned}$$

Next, given any $\varepsilon > 0$, note that $2^k A r^k < r^{k+\varepsilon}$ when $r \gg 0$. In turn, for $r \gg 0$ we can say that $M(2r) \leq e^{A(2r)^k} < e^{r^{k+\varepsilon}}$ and thus $\limsup_{r\to\infty} \tfrac{\log(\log(M(2r)))}{\log(r)} \leq k + \varepsilon$.

By taking $\varepsilon \to 0$, this proves what we wanted. ■

Set 4 Problem 4: Let $f : \mathbb{C} \to \mathbb{C}$ be an entire function with finite order λ.

(i) Show that for all $\varepsilon > 0$, there exists $r > 0$ such that $M(R) < e^{R^{\lambda+\varepsilon}}$ for all $R > r$. (This is the converse of the first claim on *page 881*.)

We know for any $\varepsilon > 0$ that there exists $r > 0$ such that for all $R > r$ we have that $\tfrac{\log(\log(M(R)))}{\log(R)} < \lambda + \varepsilon$. In turn $\log(\log(M(R))) < \log(R^{\lambda+\varepsilon})$. And by exponentiating twice we get that $M(R) < e^{R^{\lambda+\varepsilon}}$.

(ii) Using (i), show that if f, g are entire functions of orders λ_1, λ_2, then fg has order at most $\max(\lambda_1, \lambda_2)$ (assuming $\lambda(fg)$ is well-defined).

To start off, the theorem is trivial if λ_1 or λ_2 is ∞. So, we can assume both are finite. Then let $M_f(R), M_g(R), M_{fg}(R)$ be the maximum of the respective functions on the cirle $\partial\Delta(0, r)$. Also consider any $\varepsilon > 0$. Now $M_{fg}(R) \leq M_f(R)M_g(R)$, and by part (i) we know that there is some $r > 0$ such that for all $R > r$ we have that:

$$M_f(R)M_g(R) < e^{R^{\lambda_1+\varepsilon/2}+R^{\lambda_2+\varepsilon/2}}$$

Now set $\lambda := \max(\lambda_1, \lambda_2)$. Then for some $\widetilde{r} > 0$ we know that:

$$R^{\lambda+\varepsilon} > R^{\lambda_1+\varepsilon/2} + R^{\lambda_2+\varepsilon/2} \text{ for all } R > \widetilde{r}.$$

And now we can conclude for all $M_{fg}(R) < e^{R^{\lambda+\varepsilon}}$ when $R > \widetilde{r}$. Hence the degree of fg is at most $\lambda = \max(\lambda_1, \lambda_2)$. ■

Set 4 Problem 5: Let $|a_1| \leq |a_2| \leq \cdots$ be a sequence of non-zero complex numbers converging to ∞. (Repetitions are allowed but repeated values appear only finitely many times.) Let:

$$\alpha := \inf\left\{t > 0 : \sum_{n\in\mathbb{N}} \frac{1}{|a_n|^t} < \infty\right\}$$

(assuming such a t exists). The exponent of convergence α is a measure of the growth of the sequence $\{a_n\}$.

(i) Show that for any $\varepsilon > 0$, we have that $\sum_{n\in\mathbb{N}} \frac{1}{|a_n|^{\alpha+\varepsilon}} < \infty$ and $\sum_{n\in\mathbb{N}} \frac{1}{|a_n|^{\alpha-\varepsilon}} = \infty$.

Firstly, by the infimum definition of α we know for any $\varepsilon > 0$ that $\sum_{n\in\mathbb{N}} \frac{1}{|a_n|^{\alpha-\varepsilon}} = \infty$. After all, otherwise α would be smaller.

Also note from the infimum definition of α that for any $\varepsilon > 0$ there must exist $\beta \in (\alpha, \alpha+\varepsilon)$ with $\sum_{n\in\mathbb{N}} \frac{1}{|a_n|^{\beta}} < \infty$. But now we must have that $\frac{1}{|a_n|^{\beta}} \to 0$ and in turn there exists $N \in \mathbb{N}$ with $|a_n| > 1$ for all $n \geq N$. Then for all $n \geq N$ we also know $\frac{1}{|a_n|^{\beta}} \geq \frac{1}{|a_n|^{\alpha+\varepsilon}}$. So, $\sum_{n\geq N} \frac{1}{|a_n|^{\alpha+\varepsilon}} < \infty$. Finally, by adding the remaining N terms we get what we wanted.

(ii) Next assume f is an entire function with zeros only at $a_1, a_2, \ldots$ (with corresponding multiplicities if repeated values occur). Let λ be the order of f and show that $\alpha \leq \lambda$. This is an application of problem 3 and establishes a connection between the growth of zeros (measured by α) and the growth of f (measured by λ).

If $\lambda = \infty$ there is nothing to prove. So assume $\lambda < \infty$ and consider any $\varepsilon > 0$. Then by problem 3(i) we know that $n \leq N(|a_n|) \leq \frac{\log(M(2|a_n|))}{\log(2)}$.

Also note by problem 4(i) that $\log(M(r)) \leq r^{\lambda+\varepsilon}$ when $r \gg 0$. Therefore, for n sufficiently large so that $|a_n|$ is sufficently large, we can guarentee that:

$$n \leq \frac{(2|a_n|)^{\lambda+\varepsilon}}{\log(2)}$$

After some rearranging, this shows there exists some $N \in \mathbb{N}$ such that:

$$\frac{1}{|a_n|^{\lambda+2\varepsilon}} \leq \left(\frac{n\log(2)}{2^{\lambda+\varepsilon}}\right)^{\frac{-(\lambda+2\varepsilon)}{\lambda+\varepsilon}} \text{ for all } n \geq N.$$

And since $\frac{-(\lambda+2\varepsilon)}{\lambda+\varepsilon} < -1$, we can conclude that:

$$\sum_{n\geq N} \frac{1}{|a_n|^{\lambda+2\varepsilon}} \leq \left(\frac{\log(2)}{2^{\lambda+\varepsilon}}\right)^{\frac{-(\lambda+2\varepsilon)}{\lambda+\varepsilon}} \cdot \sum_{n\geq N} n^{\frac{-(\lambda+2\varepsilon)}{\lambda+\varepsilon}} < \infty$$

This proves that $\alpha \leq \lambda + 2\varepsilon$. And by taking $\varepsilon \to 0$ we get that $\alpha \leq \lambda$. ■

4/27/2026

I haven't studied math 200c in a while so I'm going to do that right now.

Math 200c Lecture Notes:

We're now going to move on to talking about ring theory again. A different convention of Rogalski versus Alireza is that Rogalski says that rings are unital and commutative unless explicitly mentioned otherwise. Also, ring homomorphisms are assumed to be unital unless explicitly mentioned otherwise.

Consider a ring homomorphism $\phi : A \to B$. If $I \lhd A$ then I defined its extension to B (denoted I^e) on *page 728*. Meanwhile, if $J \lhd B$ then its contraction to A is $J^c := \phi^{-1}(J)$.

Lemma: If $P \lhd B$ is a prime ideal, then $P^c \lhd A$ is also a prime ideal.

Proof:

Consider the map $A/P^c \hookrightarrow B/P$ given by $a + P^c \mapsto \phi(a) + P$. This map is well-defined because ϕ maps P^c into P. It's a clearly a ring homomorphism. And thirdly, it's injective. After all if $\phi(a-b) \in P$ then $a - b \in \phi^{-1}(P) = P^c$. Therefore:

$$\phi(a) + P = \phi(b) + P \implies a + P^c = b + P^c.$$

As a result, we know A/P^c is isomorphic to a subgroup of B/P. And since the latter is an integral domain, this also proves A/P^c is an integral domain. So, P^c is a prime ideal of A. ■

As an immediate corollary of the prior lemma, we can say that ϕ induces a map $\phi^* : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ given by $P \mapsto P^c$.

Proposition: ϕ^* is a continuous function with respect to the Zarisiki topologies on $\operatorname{Spec}(A)$ and $\operatorname{Spec}(B)$.

Proof:

It suffices to prove that for any closed set $V(I) = \{P \in \operatorname{Spec}(A) : I \subseteq P\}$ that $(\phi^*)^{-1}(V(I))$ is also closed in $\operatorname{Spec}(B)$. Fortunately, note that:

$$\begin{aligned}\{Q \in \operatorname{Spec}(B) : \phi^*(Q) \in V(I)\} &= \{Q \in \operatorname{Spec}(B) : \phi^{-1}(Q) \in V(I)\} \\ &= \{Q \in \operatorname{Spec}(B) : \phi(I) \subseteq Q\} = V(I^e)\end{aligned}$$

So $(\phi^*)^{-1}(V(I)) = V(I^e)$. ■

A question we can also ask about ϕ^* is if the function is closed. Sometimes it is. For example, suppose $I \lhd A$ and $\phi : A \to A/I$ is the quotient map.

By the ideal correspondence theorem, any ideal of A/I can be written J/I where $I \subseteq J \lhd A$. Also, $P/I \in \operatorname{Spec}(A/I)$ is prime iff $P \in V(I) \subseteq \operatorname{Spec}(A)$. (I technically showed this on *page 462* but the below proof is better...)

$(\implies)$

To show this direction just apply the first lemma on this page.

$(\impliedby)$

Recall that $(A/I)/(P/I) \cong A/P$. Thus if P is prime we know that $(A/I)/(P/I)$ is an integral domain, and this proves that P/I is a prime ideal in A/I.

Therefore if $V(J/I)$ is a closed set in $\mathrm{Spec}(A/I)$, and $P/I \in V(J/I)$ we know that $\phi^*(P/I) = P \supseteq J$. Hence $\phi^*(V(J/I)) \subseteq V(J)$. On the other hand, suppose $P \in V(J)$. Then $P/I \in \mathrm{Spec}(A/I)$ and $P/I \supseteq J/I$. Hence, $P/I \in V(J/I)$. Also, $\phi^*(P/I) = P$. This proves that $V(J) \subseteq \phi^*(V(J/I))$.

Since $\phi^*(V(J/I)) = V(J)$ this finishes the proof that ϕ^* is a closed map.

It's also worth noting that $\phi^* : \mathrm{Spec}(A/I) \to \mathrm{Spec}(A)$ is injective with an image of $V(I)$. Therefore, $\mathrm{Spec}(A/I)$ is homeomorphic to $V(I)$ equipped with the subspace topology.

Another interesting case is when $S \subseteq A$ is a multiplicatively closed set and $\phi : A \to S^{-1}A$ is the map $a \mapsto \frac{a}{1}$. Then we can recall from *pages 624-625* that:

- There is a bijection $\mathcal{C}_S := \{P \in \mathrm{Spec}(A) : P \cap S = \emptyset\} \longleftrightarrow \mathrm{Spec}(S^{-1}A)$ such that $P \in \mathcal{C}_S \mapsto P^e$ and $Q \in \mathrm{Spec}(S^{-1}A) \mapsto Q^c$.
- If J is a proper ideal of $S^{-1}A$ then $J = S^{-1}I = I^e$ where $I = J^c$ and $I \cap S = \emptyset$.

  But then $J = I^e = J^{ce} = I^{ece}$. This proves for any $I \lhd A$ with $I \cap S = \emptyset$ that $I^e = I^{ece}$ (at least according to this particular ϕ).

Let B/A be an extension of rings. In other words, we have an injective unital ring homomorphism $\varphi : A \to B$. Usually, we'll just view A as a subring of B.

We say $b \in B$ is integral over A if $f(b) = 0$ for some monic polynomial $f(x) \in A[x]$.

We B/A is a module-finite (or just finite) extension if B is a finitely generated A-module.

Theorem: Let B/A be an extension of rings. The following are equivalent:

1. $b \in B$ is integral over A;
2. The ring generated by A and b inside B (denoted $A[b]$) is a finitely generated A-module;
3. There is a subring $A \subseteq C \subseteq B$ such that C/A is a module finite extension and $b \in C$.

$(1 \implies 2)$
If $f(b) = 0$ for $f(x) = x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0$ where each $a_i \in A[x]$, then $b^n = -a_{n-1}b^{n-1} - \cdots - a_1 b - a_0$. By induction, we can then show $b^{n-k} \in Ab^{n-1} + \ldots + Ab + A$ for all $k \in \mathbb{N}$. So, $A[b] = \sum Ab^i = Ab^{n-1} + \ldots + Ab + A$.

$(2 \implies 3)$ is obvious.

$(3 \implies 1)$
Write $C = \sum_{i=1}^n Ac_i$. Then multiplication by b is a homomorphism $C \to C$. And if we write $bc_i = \sum_{j=1}^n d_{i,j}c_j$ and let D be the matrix $[d_{i,j}]$, then we have that $(bI - D)|(c_1, \ldots, c_n)\rangle = 0$. From there we know that $\det(bI - D) = 0$. Hence, the characteristic polynomial of D is a monic polynomial in $A[x]$ with $f(b) = 0$. ■

Note from 5/18/2026: If $(bI - D)\vec{c} = 0$ then $\det(bI - D)\vec{c}$ $= \text{adj}(bI - D)(bI - D)\vec{c} = 0$. But now $\det(bI - D)c_i = 0$ for all i. And since $1 \in \langle c_1, \ldots, c_n \rangle$, this implies $\det(bI - D) = 0$.

As a side note, we say a ring extension B/A is integral if every $b \in B$ is integral over A. The implication $(3 \implies 1)$ tells us that any module-finite extension B/A is an integral extension.

Corollary: If B/A is a ring extension and $b_1, \ldots, b_n \in B$ are integral over A, then $A[b_1, \ldots, b_n]$ is a finitely generated A-module.

Proof:
We do induction. The base case of when $n = 1$ is covered by the last theorem. Meanwhile, if we've proven the corollary true for $n - 1$ many b's, then set $B' = A[b_1]$. We know $A[b_1, \ldots, b_n] = B'[b_2, \ldots, b_n]$ where the latter is a finitely generated B'-module by induction. Also B' is a finitely generated A-module by our base case. It then follows that $A[b_1, \ldots, b_n] = B'[b_2, \ldots, b_n]$ is a finitely generated A-module.

The proof of the last assertion follows just like in the first part of the proof of the tower lemma on *page 762*.

Corollary: If $A \subseteq B \subseteq B'$ are ring extensions, B is integral over A, and B' is integral over B, then B' is integral over A.

Proof:
Suppose $b' \in B'$ and let $f(x) = x^n + b_{n-1}x^{n-1} + \ldots + b_1 x + b_0$ satisfy that all $b_i \in B$ and $f(b') = 0$. Then we know b' is integral over $A[b_0, \ldots, b_{n-1}]$. Hence, $A[b_0, \ldots, b_{n-1}][b']$ is a finitely generated $A[b_0, \ldots, b_{n-1}]$-module.

Simultaneously though, because each b_i is integral over A, we know from the prior corollary that $A[b_0, \ldots, b_{n-1}]$ is a finitely generated A-module. Therefore, we can conclude that $C \coloneqq A[b_0, b_1, \ldots, b_{n-1}, b']$ is a finitely generated A-module. Yet since $A \subseteq C \subseteq B'$, $b' \in C$, and C/A is a finite extension, we can conclude that b' is integral over A. ■

Corollary: Given a ring extension B/A, define $C \coloneqq \{b \in B : b \text{ is integral over } A\}$. Then C is a subring of B and if $b \in B$ is integral over C, then $b \in C$.

Proof:
Given $b_1, b_2 \in C$, we know that $A[b_1, b_2]$ is a finitely generated A-module. And since $b_1 - b_2$ and $b_1 b_2$ are in this ring, we thus know $b_1 - b_2$ and $b_1 b_2$ are integral over A. It follows that $b_1 - b_2$ and $b_1 b_2$ are in C. It's also trivial that $A \subseteq C$ and in particular $1_A = 1_B \in C$. Thus, we've proven that $A \subseteq C \subseteq B$ are ring extensions.

To finish off, if $b \in B$ is integral over C, then $C[b]$ is a finitely generated C-module. Hence, $C[b]$ is integral over C. Also, C is integral over A. Hence by our prior lemma, we know $C[b]$ is integral over A. In turn, b is integral over A and this proves that $b \in C$. ■

We call the ring C the integral closure of A inside B. Also, because of the property that $b \in B$ being integral over C implies $b \in C$, we say C is integrally closed in B.

Be aware that there is a more specific meaning of the phrase integrally closed that gets used. Specifically, if D is an integral domain then we say D is integrally closed if D is integrally closed (in the prior sense) inside the field of fractions K of D.

Lemma: Let D be a UFD. Then D is integrally closed.
(This was set 1 problem 3(b) in math 200b. See *pages 534-535* for a proof.)

Here is an important construction. Let $K/\mathbb{Q}$ be a finite degree field extension. Then while we know $\mathbb{Z}$ is integrally closed inside $\mathbb{Q}$, we don't know that $\mathbb{Z}$ is integrally closed inside K. Hence, it is not pointless to define $\mathcal{O}_K$ to be the integral closure of $\mathbb{Z}$ inside K.

We often call K a number field and $\mathcal{O}_K$ the ring of algebraic integers inside K.

Set 3 Problem 6:

(a) Let $\alpha \in K$. Prove that $\alpha \in \mathcal{O}_K$ if and only if $m_{\alpha;\mathbb{Q}} \in \mathbb{Z}[x]$.

$(\Longrightarrow)$
If $\alpha \in \mathcal{O}_K$ then we know there is a monic polynomial $f \in \mathbb{Z}[x]$ such that $f(\alpha) = 0$. In turn, we must have that $m_{\alpha;\mathbb{Q}}$ divides f in $\mathbb{Q}[x]$. Hence, there exists a polynomial $g \in \mathbb{Q}[x]$ such that $f(x) = g(x)m_{\alpha;\mathbb{Q}}(x)$. Also, because both f and $m_{\alpha;\mathbb{Q}}$ are monic, we must have that g is monic as well.

But now by Gauss' lemma, we know there are constants $c_1, c_2 \in \mathbb{Q}^\times$ such that $c_1c_2 = 1$ and $c_1 g$ and $c_2 m_{\alpha;\mathbb{Q}}$ are in $\mathbb{Z}[x]$. Yet since $g, m_{\alpha;\mathbb{Q}}$ are monic, the only way we don't have a contradiction is if $c_1, c_2 \in \mathbb{Z}^\times = \{-1, 1\}$. This proves that g and $m_{\alpha;\mathbb{Q}}$ are in $\mathbb{Z}[x]$.

$(\Longleftarrow)$
This direction is trivial.

(b) Consider the case $K = \mathbb{Q}[\sqrt{d}]$ where d is a squarefree integer (i.e. no prime integer divides d twice). Then show that $\mathcal{O}_K = \mathbb{Z} + \mathbb{Z}\omega$ where:

$$\omega = \begin{cases} \sqrt{d} & \text{if } d \not\equiv 1 \pmod 4 \\ \frac{1+\sqrt{d}}{2} & \text{if } d \equiv 1 \pmod 4 \end{cases}$$

Since $[K : \mathbb{Q}] = 2$, we know for any $\alpha \in K - \mathbb{Q}$ that $\deg(m_{\alpha;\mathbb{Q}}) = 2$. Hence, we can solve for any $\alpha = a + b\sqrt{d} \in K$ where $a, b \in \mathbb{Q}$ and $b \neq 0$ that:

$$m_{\alpha;\mathbb{Q}}(x) = x^2 - 2ax + (a^2 - db^2).$$

By part (a), we can then conclude $a + b\sqrt{d} \in \mathcal{O}_K$ if and only if $-2a \in \mathbb{Z}$ and $a^2 - db^2 \in \mathbb{Z}$. Yet as $-2a \in \mathbb{Z}$, we must have that $a = \frac{n}{2}$ where $n \in \mathbb{Z}$.

Next, because $a^2 - db^2 = \frac{n^2 - 4db^2}{4} \in \mathbb{Z}$, we must have that $n^2 - 4db^2 \in 4\mathbb{Z}$. One consequence of this is that $4db^2 \in \mathbb{Z}$. So if we write $b = r/s$, we must have that $\frac{4d}{s^2} \in \mathbb{Z}$. Yet as d is square free, this is only possible if $s \mid 4$. After all, d can only cancel out each prime factor of s at most one time. So, it's then up to 4 to cancel out all the of the prime factors of s a second time. Next, we can manually check that $s = 4$ doesn't work. After all, $\frac{4d}{4^2} \in \mathbb{Z}$ iff $4 \mid d$. But that violates that d is square-free. Thus in conclusion, we can write $b = \frac{m}{2}$ where $m \in \mathbb{Z}$.

To summarize my work so far, we've shown that if $a+b\sqrt{d} \in \mathcal{O}_K$ where $b \neq 0$, then $a = \frac{n}{2}$ and $b = \frac{m}{2}$ where $n, m \in \mathbb{Z}$, and also that $n^2 - 4d(\frac{m}{2})^2 = n^2 - dm^2 \in 4\mathbb{Z}$.

Case 1: Suppose $d \not\equiv 1 \pmod 4$. Then since d is square free, we know $d \equiv 2 \pmod 4$ or $d \equiv 3 \pmod 4$.

Note that $n^2 \equiv 1 \pmod 4$ if n is odd and $n^2 \equiv 0 \pmod 4$ if n is even. Thus, we can check that $n^2 - dm^2 \equiv 0 \pmod 4$ iff both n and m are even. Yet this then guarentees that $a, b \in \mathbb{Z}$. So, $a+b\sqrt{d} \in \mathcal{O}_K$ (where $b \neq 0$) iff $a, b \in \mathbb{Z}$ and $b \neq 0$.

Case 2: Suppose $d \equiv 1 \pmod 4$.

Then by similar manual checking, we can see that $n^2 - dm^2 \equiv 0 \pmod 4$ iff either both n and m are even or both n and m are odd. Or to put in other words, if we defined $\omega = \frac{1+\sqrt{d}}{2}$, then every $a+b\sqrt{d} \in \mathcal{O}_K$ (where $b \neq 0$) can be written as $k + \ell\omega$ where $k, \ell \in \mathbb{Z}$ and $\ell \neq 0$. Also, for every such k, ℓ we have that $k + \ell\omega \in \mathcal{O}_K$.

To finish off, if $b = 0$ then the minimal polynomial of $a + 0\sqrt{d}$ is just $x - a$. Hence by part (a), $a + 0\sqrt{d} \in \mathcal{O}_K$ iff $a \in \mathbb{Z}$. ■

Here are some more facts about integral extensions.

Lemma: Suppose B/A is an integral extension of rings.

(1) A is a field iff B is a field (assuming B is an integral domain).

(2) If $J \subseteq B$ is an ideal, then B/J is integral over $A/J^c = A/(J \cap A)$.

(3) If $S \subseteq A$ is multiplicatively closed then $S^{-1}B$ is integral over $S^{-1}A$.

Proof:
Claim 2 is easy. The embedding $A \hookrightarrow B$ uniquely quotients to an embedding $A/J^c \hookrightarrow B/J$. Also, if $b \in B$ is a zero of a monic polynomial $f \in A[x]$, we can just consider $f \pmod J$ to get a monic polynomial with $b + J$ as a zero.

Similarly, claim 3 is also easy since if $\phi : A \to B$ is our embedding, then we can extend ϕ to being an embedding $S^{-1}A \to S^{-1}B$ (technically equal to $\phi(S)^{-1}B$) by defining $\phi(\frac{a}{s}) = \frac{\phi(a)}{\phi(s)}$.

It is trivial to check that $\phi(S)$ is multiplicatively closed, and that ϕ is a unital ring homomorphism.

Then if $\frac{b}{s} \in S^{-1}B$, we can consider a monic polynomial:
$f(x) = x^n + a_{n-1}x^{n-1} + \ldots + a_1x + a_0 \in A[x]$ such that $f(b) = 0$.

If we replace each a_{n-k} with $\frac{a_{n-k}}{s^k}$, we get another monic polynomial in $S^{-1}A$ with $\frac{b}{s}$ as a zero. This finishes showing claim 2 and leaves us with claim 1.

($\Longrightarrow$)
If A is a field, then $A[x]$ is a PID. In turn for any $b \in B - \{0\}$ there exists a minimal monic polynomial $f \in A[x]$ such that $f(b) = b^n + a_{n-1}b^{n-1} + \ldots a_1 b + a_0 = 0$. By minimality we know $a_0 \neq 0$ (since B is an integral domain). So, $1 = b \cdot -a_0^{-1}(b^{n-1} + a_{n-1}b^{n-2} + \ldots + a_1)$. This proves every nonzero element of B has an inverse. So, B is a field.

($\Longleftarrow$)
Suppose B is a field and $a \in A - \{0\}$. We know a^{-1} exists in B and is thus integral over A. Thus, we can find constants $a_0, \ldots, a_{n-1} \in A$ such that:

$$(a^{-1})^n + a_{n-1}(a^{-1})^{n-1} + \ldots + a_1 a^{-1} + a_0 = 0$$

By rearranging, we thus get that $a^{-1} = -a_{n-1} - a_{n-2}a - \ldots - a_1 a^{n-2} - a_0 a^{n-1} \in A$. So, $a^{-1} \in A$ and we've proven that A is a field. ■

Corollary: Let B/A be an integral extension and let $Q \in \mathrm{Spec}(B)$. Then $Q^c = Q \cap A$ is maximal in A if and only if Q is maximal in B.

Proof:
Q is maximal in B iff B/Q is a field. But note that since B/Q is an integral domain (since $Q \in \mathrm{Spec}(B)$), B/Q is a field iff A/Q^c is a field, which in turn happens iff Q^c is maximal in A. ■

Proposition: If B/A is an integral extension:

(1) (Lying over): If $P \in \mathrm{Spec}(A)$ there exists $Q \in \mathrm{Spec}(B)$ such that $Q^c = P$.

(2) (Incomparability): If $Q_1 \subseteq Q_2$ are primes in B and $Q_1^c = Q_2^c$, then $Q_1 = Q_2$.

(3) (Going up): If $P_1 \subseteq P_2$ are prime ideals in A and $Q_1 \in \mathrm{Spec}(B)$ satisfies that $Q_1^c = P_1$, then there exists $Q_2 \in \mathrm{Spec}(B)$ with $Q_1 \subseteq Q_2$ such that $Q_2^c = P_2$.

Proof:
We start with (2). Let $P := Q_1^c = Q_2^c$, and then let $S := A - P$. Then by considering the localizations $A_P = S^{-1}A$ as well as $S^{-1}B$, we get an integral extension $S^{-1}B/A_P$. Also importantly, A_P is a local ring whose only maximal ideal is $S^{-1}P$.

But now note that $Q_i \cap S = Q_i \cap (A \cap S) = (Q_i \cap A) \cap S = P \cap S = \emptyset$ for both i. It follows that $S^{-1}Q_1$ and $S^{-1}Q_2$ are in $\mathrm{Spec}(S^{-1}B)$. Also, we can show for both i that $S^{-1}Q_i \cap S^{-1}A = S^{-1}(Q_i \cap A) = S^{-1}P$.

It's obvious that $S^{-1}(Q_i \cap A) \subseteq S^{-1}A \cap S^{-1}Q_i$. Meanwhile, suppose $\frac{a}{s_1} = \frac{q}{s_2}$ where $a \in A$, $q \in Q_i$, and $s_1, s_2 \in S$. Then there additionally exists $t \in S$ such that $as_2t = qs_1t$. But now we know $as_2t = qs_1t \in Q_i$ since Q_i is an ideal. And since Q_i is prime and $s_2, t \notin Q_i$, the only way for as_2t to be in Q_i is if a is in Q. So, we've proven that $\frac{a}{s_1} \in S^{-1}(A \cap Q_i)$. And this proves that $S^{-1}A \cap S^{-1}Q_i \subseteq S^{-1}(A \cap Q_i)$.

More generally, this argument shows for any ring extension B/A, any multiplicatively closed set $S \subseteq A$, and any prime ideal $Q \in \mathrm{Spec}(B)$ such that $Q \cap S = \emptyset$, we have that $S^{-1}Q \cap S^{-1}A = S^{-1}(Q \cap A)$. As

for the case that $Q \cap S \neq \emptyset$, then we trivally have that:

$$S^{-1}(Q \cap A) = S^{-1}A = S^{-1}Q \cap S^{-1}A.$$

By the prior corollary, we can thus conclude that $S^{-1}Q_1$ and $S^{-1}Q_2$ are both maximal ideals in $S^{-1}B$. And since $S^{-1}Q_1 \subseteq S^{-1}Q_2$, this implies $S^{-1}Q_1 = S^{-1}Q_2$. Finally, as localization induces a bijection $\text{Spec}(S^{-1}B) \to \{Q \in \text{Spec}(B) : Q \cap S = \emptyset\}$, this proves that $Q_1 = Q_2$.

Next we prove (1). Let $S \coloneqq A - P$ and consider the integral extension $S^{-1}B/A_P$. Then let M be any maximal ideal of the ring $S^{-1}B$. By our prior corollary, we know $M \cap A_p$ is a maximal ideal in A_P. Hence, $M \cap A_p = S^{-1}P$. Yet we also know that $M = S^{-1}Q$ for some $Q \in \text{Spec}(B)$ such that $Q \cap S = \emptyset$. Hence, we claim that $P = Q \cap A$.

We know $S^{-1}Q \cap S^{-1}A = S^{-1}P$. At the same time, we can conclude through identical reasoning to before that $S^{-1}(Q \cap A) = S^{-1}Q \cap S^{-1}A$. Hence, we know that $S^{-1}(Q \cap A) = S^{-1}P$. And by using the fact that localization induces a bijection from the proper ideal of $S^{-1}A$ to the ideals of A not intercepting S, we know that $Q \cap A = P$.

Finally, we prove (3). Note that B/Q_1 is integral over A/P_1. Also by the ideal correspondance theorem, we know $P_2/P_1 \in \text{Spec}(A/P_1)$. In turn, by (lying over) we know that there exists $Q_2/Q_1 \in \text{Spec}(B/Q_1)$ (which has this form by the ideal correspondance theorem) such that $(Q_2/Q_1)^c = P_2/P_1$.

Yet now $Q_2 \in \text{Spec}(B)$ and $Q_1 \subseteq Q_2$ by the ideal correspondance theorem. Also, note that $(Q_2/Q_1)^c = \{a + P_1 \in A/P_1 : a \in Q_2 \cap A\} = (Q_2 \cap A)/P_1$. Therefore $(Q_2 \cap A)/P_1 = P_2/P_1$. And by the ideal correspondance theorem this proves $Q_2 \cap A = P_2$. ■

Corollary: Let B/A be an integral extension. Then $\dim(A) = \dim(B)$.

Here $\dim(\cdot)$ denotes the Krull dimension of a ring (see page 487).

Proof:

Suppose $Q_0 \subsetneq Q_1 \subsetneq \cdots \subsetneq Q_n$ is a chain of prime ideals in B. Then if we set $P_i = Q_i^c = Q_i \cap A$, we know that $P_0 \subseteq P_1 \subseteq \cdots \subseteq P_n$. Yet also note that by incomparability, $P_k = P_{k+1}$ for some k would imply $Q_k = Q_{k+1}$. Hence, we can actually conclude the P_k are strictly increasing. And this proves that $\dim(A) \geq \dim(B)$.

Conversely, if $P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_n$ is a chain of prime ideals in A, we can construct another strictly increasing chain of prime ideals in B as follows.

Choose $Q_0 \in \text{Spec}(B)$ lying over P_0. Then by inductively applying (going up), we

can choose primes $Q_1, \ldots, Q_n$ such that $Q_0 \subseteq Q_1 \subseteq \cdots \subseteq Q_n$ and $Q_k \cap A = P_k$ for all k. Yet because $Q_k \cap A$ is strictly increasing with k, we must have that the Q_k are strictly increasing.

This proves that $\dim(B) \geq \dim(A)$. ■

Returning to the topic of rings of algebraic integers, given a number field K containing $\mathbb{Q}$, we now can say due to the prior corollary that $\dim(\mathcal{O}_K) = \dim(\mathbb{Z}) = 1$. Also note that the field of fractions can be embedded as a subfield F of K. And since $\mathcal{O}_K$ is integrally closed in K, we know $\mathcal{O}_K$ is also integrally closed in F.

This finishes two parts of proving that $\mathcal{O}_K$ is a Dedekind ring, meaning that $\mathcal{O}_K$ is an integral domain that is Noetherian, has Krull dimension 1, and is integrally closed.

We aren't going to give a proof yet that $\mathcal{O}_K$ is Noetherian. That said, in the specific case that $K = \mathbb{Q}[\sqrt{d}]$ where d is a squarefree integer, we can already see that $\mathcal{O}_K$ is finitely generated and thus Noetherian.

Dedekind rings have an incredibly versatile theory of factorization of ideals (which works independently of being a UFD) and are thus very important to number theory.

Lemma (Prime Avoidance): Let A be a ring, let $P_1, \ldots, P_n \in \operatorname{Spec}(A)$, and let I be an ideal of A such that $I \subseteq \bigcup_{i=1}^{n} P_i$. Then $I \subseteq P_i$ for some i.

Proof:

We shall proceed by induction in order to prove the contrapositive of the lemma. In other words, we want to show that if $I \not\subseteq P_i$ for all i, then $I \not\subseteq \bigcup_{i=1}^{n} P_i$.

If $n = 1$, this is trivial. Meanwhile suppose $n > 1$ and that $I \not\subseteq P_i$ for any i. By our induction hypothesis, we know that:

$$I \not\subseteq \bigcup_{\substack{i=1 \\ i \neq j}}^{n} P_i \text{ for every } j \in \{1, \ldots, n\}.$$

Thus for every j, we can pick $x_j \in I - \bigcup_{\substack{i=1 \\ i \neq j}}^{n} P_i$.

Now if $x_j \notin P_j$ for some j, then we automatically know that $x_j \notin \bigcup_{i=1}^{n} P_i$ and thus $I \not\subseteq \bigcup_{i=1}^{n} P_i$. Therefore, we can now assume $x_j \in P_j$ for all j.

Finally, let $y = \sum_{k=1}^{n} (\prod_{k \neq j} x_j) \in I$. Then we have that $y \in I$. That said, we can't have that $y \in P_j$ for any j. After all, if $y \in P_j$ then we must have that $\prod_{i \neq j} x_i \in P_j$. Yet as each $x_i \notin P_j$ unless $i = j$, this would contradict that P_j is a prime ideal.

This proves that $I \not\subseteq \bigcup_{i=1}^{n} P_i$. ■

4/30/2026

Before anything else, I want to do my remaining 200c homework.

Set 3 Problem 1: Let A be a nonzero ring.

(a) Let Σ be the set of all multiplicatively closed subsets S in A such that $0 \notin S$. Show that Σ has maximal elements, and that $S \in \Sigma$ is maximal iff $A - S = P$ for a minimal prime ideal P of A.

Note: We say P is a minimal prime ideal of A if for every $Q \in \operatorname{Spec}(A)$ we have that $Q \subseteq P \implies Q = P$.

We prove Σ has maximal elements by invoking Zorn's lemma. To start off, Σ is nonempty because $\{1\} \in \Sigma$. Next, suppose $\{S_\alpha\}_{\alpha \in A}$ is chain in Σ. Then I claim $\bigcup_{\alpha \in A} S_\alpha$ is in Σ.

After all, $1 \in S_\alpha$ for each α. Hence, $1 \in \bigcup_{\alpha \in A} S_\alpha$. Furthermore, since $0 \notin S_\alpha$ for any α, we know that $0 \notin \bigcup_{\alpha \in A} S_\alpha$. Finally, given any $s_1, s_2 \in \bigcup_{\alpha \in A} S_\alpha$, we know there exists some α' such that $s_1, s_2 \in S_{\alpha'}$. (This works because $\{S_\alpha\}_{\alpha \in A}$ is a chain.) But now $s_1 s_2 \in S_{\alpha'} \subseteq \bigcup_{\alpha \in A} S_\alpha$.

With that we've shown Zorn's lemma applies, and hence it makes sense to talk about a maximal element $S \in \Sigma$. To finish, I shall show $S \in \Sigma$ is maximal if and only if $A - S = P$ for a minimal prime ideal P.

$(\implies)$
Suppose S is maximal in Σ and let $P \coloneqq A - S$. I first claim that P is an ideal.

To see why, first suppose $a \in A$ and $p \in P$. Then note that $S' \coloneqq \{sp^k : k \in \mathbb{N}_{\geq 0} \text{ and } s \in S\}$ is a multiplicatively closed set containing S as well as p which is not in S. Thus by the maximality of S, we must have that S' contains 0, and hence there exists $s \in S$ and $k \geq 0$ such that $sp^k = 0$. In particular, we can guarentee that $k \neq 0$ since $0 \notin S$.

But now $s(ap)^k = a^k \cdot (sp^k) = a^k \cdot 0 = 0$. So if $ap \in A - P = S$, that would contradict that S is multiplicatively closed and doesn't contain zero. Hence, we've proven that P is closed under multiplication by scalars in A.

Next, suppose $p_1, p_2 \in P$. By repeating the reasoning two paragraphs ago, we can find $s_i \in S$ and $k_i \in \mathbb{N}_{>0}$ such that $s_i p_i^{k_i} = 0$. Then by taking $s = s_1 s_2$ and $k = k_1 + k_2$, we get that $sp_1^k = sp_2^k = 0$. Thus in turn, we have that:

$$s(p_1 + p_2)^{2k} = sp_2^k \left(\sum_{i=0}^{k} \binom{2k}{i} p_1^i p_2^{2k-i} \right) + sp_1^k \left(\sum_{i=k+1}^{2k} p_1^{i-k} p_2^{2k-i} \right) = 0$$

But now this shows that $p_1 + p_2 \notin A - P = S$ since otherwise would contradict that S is multplicatively closed and doesn't contain zero. Hence, P is also closed under addition.

Now since P doesn't contain 1 (since $1 \in S$), we know P is a proper ideal. Also, $P \in \operatorname{Spec}(A)$ since if $s_1, s_2 \notin P$ then $s_1 s_2 \notin P$ as well. Finally, suppose

P is not a minimal prime ideal. Then there exists $Q \in \operatorname{Spec}(A)$ such that $Q \subsetneq P$. Yet now $S' = A - Q$ would be a multiplicatively closed set not containing 0 but also properly containing S. This contradicts the maximality of S. Hence, we must have that P is a minimal prime ideal.

$(\Longleftarrow)$
Suppose P is a minimal prime ideal in A and let $S = A - P$. Then $S \in \Sigma$. Hence by using Zorn's lemma, we can in particular find a maximal element $S' \in \Sigma$ such that $S \subseteq S'$.

If the TA is unhappy about me using this slightly strengthened version of Zorn's lemma, I can foward them a proof on *pages 20-21* of this journal.

Yet now by the forward direction of the proof, we know that $P' = A - S'$ is a prime ideal in A. Also $P' \subseteq P$. Thus by the minimality of P we know that $P' = P$. Hence, $S = S'$ and we've proven that S is a maximal element of Σ.

As a side note, the fact that Σ has maximal elements guarentees that any nonzero ring A has a minimal prime ideal. More strongly, by using the slightly strengthened Zorn's lemma we can show that any prime ideal contains a minimal prime ideal.

After all, if $P \in \operatorname{Spec}(A)$ and $S := A - P$, then we can find a set $S' \supseteq S$ such that $P' := A - S'$ is a minimal prime ideal. Yet we also have that $P' \subseteq P$.

(b) A multiplicatively closed subset $S \subseteq A$ is called saturated if $xy \in S$ implies that $x \in S$ and $y \in S$. Prove that S is saturated if and only if $A - S$ is a union of prime ideals.

$(\Longrightarrow)$
Suppose $x \in A - S$. Then because S is saturated we know that $\langle x \rangle \cap S = \emptyset$. In turn, by a theorem from math 200a (see *pages 456-457*), we know there is some ideal $P_x \in \operatorname{Spec}(A)$ such that $\langle x \rangle \subseteq P_x$ and $P_x \cap S = \emptyset$. But now $A - S = \bigcup_{x \in A-S} P_x$. This proves the right hand side.

$(\Longleftarrow)$
Suppose $A - S = \bigcup_{\beta \in B} I_\beta$ where $\{I_\beta\}_{\beta \in B}$ is a collection of ideals. (Our argument won't actually require the I_β to be prime). Then to show that S is saturated, it suffices to note that for any $x \notin S$, we know $\langle x \rangle \cap S = \emptyset$.

For a proof this just note that $x \in I_\beta$ for some β. Hence, $\langle x \rangle \subseteq I_\beta \subseteq A - S$.

Consequently, if $x \notin S$ then $xy \notin S$ for any $y \in A$. ■.

Set 3 Problem 2: The set S_0 of all nonzero-divisors in A is a saturated multiplicatively closed set. Thus by problem 1(b) we know the set D of zero-divisors of A is a union of prime ideals.

(a) Show that every minimal prime ideal of A is contained in D.

Suppose P is a minimal prime ideal in A. Then we know that $S := A - P$ is a maximal multiplicatively closed set. Also note that $S_0 S := \{xy : x \in S_0, y \in S\}$ is a multiplicatively closed set as well which contains both S and S_0. Yet $S_0 S$ doesn't contain 0 because if $xy = 0$ then that would require $y = 0$ by the definition of S_0. Thus $0 \notin S$ and so we'll never have that $y = 0$.

By the maximality of S, we can conclude $S_0 S = S$. Hence, $S_0 \subseteq S$ and this proves that $P \subseteq A - S_0 = D$.

(b) The ring $S_0^{-1}A$ is called the total ring of fractions of A. Prove that $S = S_0$ is the unique largest multiplicatively closed subset of A such that the map $A \to S^{-1}A$ given by $a \mapsto \frac{a}{1}$ is injective.

For the adjective "largest" to even make sense, we would want to prove that the map $A \to S_0^{-1}A$ is injective, and that if S is any other multiplicatively closed set with the map $A \to S^{-1}A$ being injective, then we know $S \subseteq S_0$.

- To show the first claim, suppose $\frac{a_1}{1} = \frac{a_2}{1}$ in $S_0^{-1}A$. Then there exists $t \in S_0$ such that $t(a_1 - a_2) = 0$. Yet since t is not a zero-divisor, the only way this is possible is if $a_1 - a_2 = 0$. So, $a_1 = a_2$.

- As for the second claim, suppose S is a multiplicatively closed set containing any element $t \in A - S_0$. Then there exists $a \in A - \{0\}$ such that $t(a - 0) = 0$. In turn, the map $A \to S^{-1}A$ is not injective as $\frac{a}{1} = \frac{0}{1}$.

(c) Prove that every element of $S_0^{-1}A$ is either a zero divisor or a unit.

Consider any $\frac{a}{s} \in S_0^{-1}A$. Then we have two cases. If $a \in S_0$, we know that $\frac{s}{a} \in S_0^{-1}A$ and $\frac{a}{s} \cdot \frac{s}{a} = 1$. Hence $\frac{a}{s}$ is a unit. Meanwhile, if $a \notin S_0$ then we know $ab = 0$ for some $b \neq 0$. Since the map $b \mapsto \frac{b}{1}$ is injective, we thus know $\frac{b}{1} \neq 0$. Yet $\frac{a}{s} \cdot \frac{b}{1} = \frac{0}{s} = 0$. Hence $\frac{a}{s}$ is a zero-divisor. ■

Set 3 Problem 3: Let A be a ring and let S be a multiplicatively closed subset. Consider the ring homomorphism $\phi : A \to S^{-1}A$ given by $a \mapsto \frac{a}{1}$ and let ϕ^* be the induced spectra map from $\operatorname{Spec}(S^{-1}A) \to \operatorname{Spec}(A)$.

(a) Let $\mathcal{O}_S := \{P \in \operatorname{Spec}(A) : P \cap S = \emptyset\}$ be the image of ϕ^* considered as a subspace of $\operatorname{Spec}(A)$. Show that ϕ^* is a homeomorphism from $\operatorname{Spec}(S^{-1}A)$ to $\mathcal{O}_S$.

I've already shown in math 200b that ϕ^* is a bijection from $\operatorname{Spec}(S^{-1}A)$ to $\mathcal{O}_S$ (see *pages 624-625*). Also, we automatically have that ϕ^* is continuous. Hence, all we need to do now is show $(\phi^*)^{-1}$ is also continuous.

Fortunately, if $V(S^{-1}I)$ is a nonempty closed set in $\operatorname{Spec}(S^{-1}A)$, then $S^{-1}I \neq S^{-1}A$ and so $I \cap S = \emptyset$. Also $S^{-1}P \in V(S^{-1}I)$ iff $S^{-1}I \subseteq S^{-1}P$. Since P is prime with $P \cap S = \emptyset$, we know $I \subseteq P$ by a comment on *page 625*. So, $\phi^*(V(S^{-1}I)) \subseteq V(I) \cap \mathcal{O}_S$ and the other inclusion is obvious. Hence, $\phi^*(V(S^{-1}I))$ is closed in the subspace topology of $\mathcal{O}_S$.

(b) In the case that $S = A - P$ for some prime ideal P, show that $\mathcal{O}_S$ is also equal to the intersection of all open subsets U of $\operatorname{Spec}(A)$ such that $P \in U$.

In the case that $S = A - P$, we know that $\mathcal{O}_S = \{Q \in \operatorname{Spec}(A) : Q \subseteq P\}$. Meanwhile, any open subset U of $\operatorname{Spec}(A)$ containing P has the form $\operatorname{Spec}(A) - V(I)$ where $P \notin V(I)$, which equivalently means that $I \not\subseteq P$. If we let $\{I_\beta\}_{\beta \in B}$ be the collection of all ideals of A such that $I_\beta - P \neq \emptyset$, our goal according to the problem is thus to show that:

$$\{Q \in \operatorname{Spec}(A) : Q \subseteq P\} = \bigcap_{\beta \in B} (\operatorname{Spec}(A) - V(I_\beta))$$

- Suppose $Q \in \text{Spec}(A)$ satisfies that $Q \subseteq P$. Then for any $I_\beta \lhd A$ satisfying that $I_\beta - P \neq \emptyset$, we also know $I_\beta - Q \neq \emptyset$. Hence, $Q \notin V(I_\beta)$ and, this proves that:
$$\{Q \in \text{Spec}(A) : Q \subseteq P\} \subseteq \bigcap_{\beta \in B} (\text{Spec}(A) - V(I_\beta))$$

- Meanwhile, suppose $Q \in \text{Spec}(A)$ satisfies that $Q \not\subseteq P$. Then we can find $x \in Q - P$. And since $\langle x \rangle \not\subseteq P$, we know that $\langle x \rangle = I_{\beta_0}$ for some β_0. That said, $\langle x \rangle \subseteq Q$ and thus $Q \in V(I_{\beta_0})$. It follows that:
$$\{Q \in \text{Spec}(A) : Q \subseteq P\}^{\mathsf{C}} \subseteq \left(\textstyle\bigcap_{\beta \in B} (\text{Spec}(A) - V(I_\beta))\right)^{\mathsf{C}}.$$

Set 3 Problem 4: Let A be a ring and put the Zariski topology on $X = \text{Spec}(A)$. For any $f \in A$ define $X_f = X - V(\langle f \rangle)$. The sets X_f are called the principal open subsets of X.

(a) The sets $\{X_f : F \in A\}$ form a basis for the Zariski topology.

To prove we have a basis, we need to show that given any open set $U \subseteq X$ and $P \in U$, there exists f such that $P \in X_f \subseteq U$. Equivalently, it suffices to show for any closed set $V(I)$ not containing P that there exists $f \in A$ such that $V(I) \subseteq V(\langle f \rangle)$ and $P \notin V(\langle f \rangle)$. But this is easy (and in fact I already showed this in the prior problem).

Suppose $P \in \text{Spec}(A)$ and consider any $I \lhd A$ such that $I - P \neq \emptyset$. Then after picking $f \in I - P$, we have that $\langle f \rangle - P \neq \emptyset$ and hence $P \notin V(\langle f \rangle)$. That said, $\langle f \rangle \subseteq I$ and hence $V(I) \subseteq V(\langle f \rangle)$.

(b) $X_f \cap X_g = X_{fg}$.

By DeMorgan's laws, we know that $V(\langle f \rangle)^{\mathsf{C}} \cap V(\langle g \rangle)^{\mathsf{C}} = (V(\langle f \rangle) \cup V(\langle g \rangle))^{\mathsf{C}}$. Yet also note that $V(\langle f \rangle) \cup V(\langle g \rangle) = V(\langle f \rangle \cdot \langle g \rangle) = V(\langle fg \rangle)$. Thus, the identity we wanted falls out.

(c) $X_f = \emptyset$ if and only if f is nilpotent.

Recall that $\text{Nil}(A) = \bigcap_{P \in \text{Spec}(A)} P$. Therefore:
$$\begin{aligned} X_f = \emptyset &\iff \langle f \rangle \subseteq P \text{ for all } P \in \text{Spec}(A) \\ &\iff f \in P \text{ for all } P \in \text{Spec}(A) \\ &\iff f \in \textstyle\bigcap_{P \in \text{Spec}(A)} P = \text{Nil}(A). \end{aligned}$$

(d) $X_f = X_g$ if and only if $\sqrt{\langle f \rangle} = \sqrt{\langle g \rangle}$.

As a reminder, if $I \lhd A$ then $\sqrt{I} := \{x \in A : \exists n \in \mathbb{N}_{>0} \; s.t. \; x^n \in I\}$. On *pages 461-462* we found that $\sqrt{I} = \bigcap_{P \in V(I)} P$.

$(\Longrightarrow)$
Now $X_f = X_g$ if and only if $V(\langle f \rangle) = X_f^{\mathsf{C}} = X_g^{\mathsf{C}} = V(\langle g \rangle)$. It then follows that:
$$\sqrt{\langle f \rangle} = \textstyle\bigcap_{P \in V(\langle f \rangle)} P = \bigcap_{P \in V(\langle g \rangle)} P = \sqrt{\langle g \rangle}$$

$(\Longleftarrow)$
Conversely, if $\sqrt{\langle f \rangle} = \sqrt{\langle g \rangle}$ then we know that $\bigcap_{P \in V(\langle f \rangle)} P = \bigcap_{P \in V(\langle g \rangle)} P$. What we want to show however is that $V(\langle f \rangle) = V(\langle g \rangle)$ because then it will follow that

$X_f = X_g$.

Suppose for the sake of contradiction that $V(\langle f \rangle) \neq V(\langle g \rangle)$. Then without loss of generality we can assume there exists $P \in V(\langle f \rangle) - V(\langle g \rangle)$. Yet now since $\langle g \rangle \not\subseteq P$ we know $g \notin P$. Hence, $g \notin \bigcap_{P \in V(\langle f \rangle)} P$. This is a contradiction because we know $g \in \bigcap_{P \in V(\langle g \rangle)} P$ and the latter set equals $\bigcap_{P \in V(\langle f \rangle)} P$ by assumption.

(e) X is compact.

Let $\mathcal{U}$ be an arbitrary open cover of X. Then because of part (a), we can define an open cover $\mathcal{V}$ of X consisting entirely of principal open subsets of X and which is a refinement of $\mathcal{U}$.

Specifically, for each $P \in X$ let $U \in \mathcal{U}$ be an open set containing P. Then there is a principal open subset X_{f_P} satisfying that $P \in X_{f_P} \subseteq U$. Hence, if we let $\mathcal{V} := \{X_{f_P}\}_{P \in \mathrm{Spec}(A)}$, we have that $\mathcal{V}$ is also an open cover of X and that every $V \in \mathcal{V}$ is contained in some $U \in \mathcal{U}$.

Yet now in order to find a finite subcover over X using sets in $\mathcal{U}$, it suffices to instead find a finite subcover over X using sets in $\mathcal{V}$. And thus, we've proven that we can without loss of generality assume our open cover of X is precisely of the form $\{X_{f_\beta}\}_{\beta \in B}$ where each X_{f_β} is some principal open set.

Next note that $X = \bigcup_{\beta \in B} X_{f_\beta}$ iff $\emptyset = \bigcap_{\beta \in B} V(\langle f_\beta \rangle)$. Yet also note:

$$\bigcap_{\beta \in B} V(\langle f_\beta \rangle) = V(\textstyle\sum_{\beta \in B} \langle f_\beta \rangle) = V(\langle f_\beta : \beta \in B \rangle).$$

Since every proper ideal is contained in a maximal (and thus prime) ideal, the only way for $V(\langle f_\beta : \beta \in B \rangle) = \emptyset$ is if $A = \langle f_\beta : \beta \in B \rangle$. Equivalently, we know $1 \in \langle f_\beta : \beta \in B \rangle$.

But now to finish off, there must exist $\beta_1, \ldots, \beta_m \in B$ as well as $a_1, \ldots, a_m \in A$ such that $a_1 f_{\beta_1} + \ldots + a_m f_{\beta_m} = 1$. Hence, we can conclude $A = \langle f_{\beta_1}, \ldots, f_{\beta_m} \rangle$. In turn, $\emptyset = V(\langle f_{\beta_1}, \ldots, f_{\beta_m} \rangle) = V(\sum_{i=1}^m \langle f_{\beta_i} \rangle) = \bigcap_{i=1}^m V(\langle f_{\beta_i} \rangle)$. Or in other words:

$$X = \bigcup_{i=1}^m X_{f_{\beta_i}}$$

Set 3 Problem 5: Let B/A be an integral extension, let $\phi : A \to B$ be the inclusion map, and let $\phi^* : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ be the induced spectra map. Prove the following properties:

(a) ϕ^* is surjective.

Suppose $P \in \mathrm{Spec}(A)$. Then by lying over we know there exists $Q \in \mathrm{Spec}(B)$ such that $Q^c = P$. Hence, $P \in \phi^*(\mathrm{Spec}(B))$.

(b) ϕ^* is a closed map.

Consider any closed set $V(I) \subseteq \mathrm{Spec}(B)$. Then $P \in \phi^*(V(I))$ iff $P = Q^c$ for some $Q \in V(I)$. But then as $I \subseteq Q$, we know that $I^c \subseteq Q^c = P$. So, we've proven that $P \in V(I^c)$. Hence, $\phi^*(V(I)) \subseteq V(I^c)$.

Conversely, suppose $P \in V(I^c)$. Then note that B/I is integral over A/I^c (where the extension is given by the map $\psi : a + I^c \mapsto \phi(a) + I$). Also, $P/I^c \in \mathrm{Spec}(A/I^c)$.

Hence, by lying over plus the correspondance theorem, we know there exists $Q \in V(I)$ such that $\psi^{-1}(Q/I) = P/I^c$.

Yet $a + I^c \in \psi^{-1}(Q/I)$ iff $\phi(a) \in Q$. Thus, $\psi^{-1}(Q/I) = \phi^{-1}(Q)/I^c = Q^c/I^c$. Hence, we've proven that $Q^c = P$. In turn, $P \in \phi^*(V(I))$ and so $V(I^c) \subseteq \phi^*(V(I))$.

Since we've proven that $\phi^*(V(I)) = V(I^c)$, this proves that ϕ^* is a closed map.

After going to office hours, I was made aware that there was a theorem from a class I skipped that I didn't take notes on that I really need here. So I shall deal with that now.

Theorem: Let A be a nontrivial Noetherian ring.

1. We can write $0 = P_1 \cdots P_n$ as a product of prime ideals (where not all P_i are necessarily distinct).

   Proof:
   For the sake of contradiction suppose not and let S be the set of ideals of A not containing a product of prime ideals. Due to our supposing we know that $S \neq \emptyset$. Then since A is Noetherian we know that S contains a maximal element I. And clearly $I \neq A$ since A contains a prime ideal.

   Now I can't be a prime ideal since then I would trivially be a product of primes, thus contradicting that $I \in S$. Yet we can also show that I must be prime. After all, if I were not prime, we'd be able to find two ideal J_1, J_2 of A such that $I \subsetneqq J_1$, $I \subsetneqq J_2$, but $J_1 J_2 \subseteq I$.

   The professor claims this is the definition we received on *page 451*. However, the professor is mistaken. So, I want to actually prove we can do this.

   Lemma: $P \lhd A$ is prime if and only if $P \neq A$ and there doesn't exist ideals $I, J \lhd A$ satisfying that $P \subsetneqq I$, $P \subsetneqq J$, but $IJ \subseteq P$.

   ($\Longrightarrow$)
   If P is prime, then we know from the definition given on *page 451* that $P \neq A$ and $IJ \subseteq P \Longrightarrow I \subseteq P$ or $J \subseteq P$. Hence, the right side of the implication holds.

   ($\Longleftarrow$)
   Suppose P is not prime. If $P \neq A$ then we know there exists $\widetilde{I}, \widetilde{J} \lhd A$ such that $\widetilde{I} \not\subseteq P$, $\widetilde{J} \not\subseteq P$, and $\widetilde{I}\widetilde{J} \subseteq P$. Now let $x \in \widetilde{I} - P$ and $y \in \widetilde{J} - P$. Next, set $I := \langle x, P \rangle$ and $J := \langle y, P \rangle$. Then $P \subsetneqq I$ and $P \subsetneqq J$. That said, since $xy \in P$, we have that $IJ \subseteq P$.

   By the maximality of I in S, we know that $J_1, J_2 \notin S$. Hence, we can write $J_1 = P_1 \cdots P_k$ and $J_2 = Q_1 \cdots Q_\ell$ as products of prime ideals. In turn, we know that $(P_1 \cdots P_k)(Q_1 \cdots Q_\ell) = J_1 J_2 \subseteq I$. And this contradicts I doesn't contain a product of primes.

Remark 2: Something I want to quickly mention is that given ideals $P_1, \dots, P_k$, we define the product $P_1 \cdots P_k$ to be the ideal:

$$\{\textstyle\sum_{i=1}^{n}(\prod_{j=1}^{k} p_{j,i}) : n \in \mathbb{N}_{>0} \text{ and } p_{j,i} \in P_j \text{ for all } i, j\}.$$

The proof that this is an ideal is equivalent to that on *page 452* when we proved this works for $k = 2$. Also importantly, we can easily see that $(P_1 \cdots P_k)P_{k+1} = P_1 \cdots P_{k+1}$ and $P_0(P_1 \cdots P_k) = P_0 \cdots P_k$. Through an inductive argument we can then show the placement of parentheses in our product of primes doesn't matter.

2. A has only finitely many minimal primes.

Write $0 = P_1 \cdots P_k$ where each P_j is a prime ideal in A. If Q is a minimal prime ideal, then we automatically know from the fact Q is an ideal that $\{0\} \subseteq Q$. In other words, $Q \mid P_1 \cdots P_k$. Next, because Q is prime we can specifically conclude that $Q \mid P_j$ for some j. Yet now as $P_j \subseteq Q$, Q is a minimal prime, and P_j is another prime, we must have that $Q = P_j$. This proves that the set of all minimal prime ideals in A is contained in $\{P_1, \dots, P_k\}$. ■

As a side note, apparently in the lecture I missed Rogalski also proved that every prime ideal contains a minimal prime ideal. To summarize his proof, essentially:

The intersection over any *de*scending chain of prime ideals in a ring will be another prime ideal. Hence, given any prime ideal P we can apply Zorn's lemma on the poset of prime ideals contained in P ordered by reverse inclusion. This poset is nonempty since it contains P. Also, every chain has an upper bound due to the comment about intersections. I am stretched for time so I won't prove this.

Another side note is that if $J \lhd A$ is a proper ideal, then we can apply this lemma to show that A/J has only finitely many minimal primes. In other words, there are only finitely many minimal primes over J.

With that out of the way, we can now finish the prior homework problem.

Set 3 Problem 5(c): If the ring B is Noetherian, then ϕ^* has finite fibers (i.e. for any $P \in \operatorname{Spec}(A)$ the preimage $(\phi^*)^{-1}(P)$ is a finite set).

Quicker Proof (hinted in office hours):
Let $I = P^e = \phi(P)B$. Then we still have that the quotient ring B/I is Noetherian.
Also, if $Q \in \operatorname{Spec}(B)$ lies over P, we must have that $I \subseteq Q$. After all Q must contain $\phi(P)$ and I is the smallest ideal containing $\phi(P)$. Hence, we will have that $Q/I \in \operatorname{Spec}(B/I)$.

Now note that if $I \subseteq Q_1 \subseteq Q_2$ where $Q_1, Q_2 \in \operatorname{Spec}(B)$ and Q_2 is lying over P, we can conclude that Q_1 also lying over P.

To see why, first note $\phi^{-1}(I) = P$. After all, because ϕ is injective we know that $P = \phi^{-1}(\phi(P)) \subseteq \phi^{-1}(\phi(P)B) = \phi^{-1}(I)$. Also, by lying over we know that there exists $Q \in \operatorname{Spec}(B)$ such that $\phi^{-1}(Q) = P$. Yet as mentioned before, $I \subseteq Q$. Hence, $P \subseteq \phi^{-1}(I) \subseteq \phi^{-1}(Q) = P$ implies that $\phi^{-1}(I) = P$.

As a side note, this reasoning proves for any integral ring extension B/A that $P^{ec} = P$ for all $P \in \operatorname{Spec}(A)$.

Now since $I \subseteq Q_1 \subseteq Q_2$, we know that $P = \phi^{-1}(I) \subseteq \phi^{-1}(Q_1) \subseteq \phi^{-1}(Q_2) = P$. Hence Q_1 also lies over P.

By incomparability, we thus must have that $Q_1 = Q_2$. And hence, we've proven by the correspondance theorem that if $Q \in \mathrm{Spec}(B)$ lies over P then we must have that Q/I is a *minimal* prime ideal in B/I. But since B/I is Noetherian, we know only finitely many $Q/I \in \mathrm{Spec}(B/I)$ are minimal. Hence, there can only be finitely many $Q \in \mathrm{Spec}(B)$ lying over P.

Longer Proof (following my idea before office hours):
Consider any $P \in \mathrm{Spec}(A)$ and let $S = A - P$. Then S is a multiplicatively closed subset of A. Also $S' := \phi(S)$ is a multiplicatively closed subset of B. By localizing, we get that $(S')^{-1}B/S^{-1}A$ is an integral extension via the embedding $\psi(\frac{a}{s}) = \frac{\phi(a)}{\phi(s)}$. Also, observe the following:

- Localizations of Noetherian rings are always still Noetherian and hence $(S')^{-1}B$ is Noetherian.

  To see why, note that a ring C is Noetherian if and only if every ideal $I \lhd C$ is finitely generated. Yet given any multiplicatively closed set T, we have that $T^{-1}\langle x_1, \ldots, x_n\rangle = \langle \frac{x_1}{1}, \ldots, \frac{x_n}{1}\rangle$. Therefore, if every ideal of C is finitely generated then every ideal of $T^{-1}C$ is also finitely generated.

- $\mathrm{Max}(S^{-1}A) = \{S^{-1}P\}$.

- The map $Q \mapsto (S')^{-1}Q$ gives a bijection from the prime ideals $Q \in \mathrm{Spec}(B)$ lying over P to the prime ideals of $(S')^{-1}B$ lying over $(S')^{-1}P$.

  We already proved in class (see *pages 891-892*) that if $Q \cap S' = \emptyset$ then $\psi^{-1}((S')^{-1}Q) = S^{-1}(\phi^{-1}(Q))$. Also, any $Q \in \mathrm{Spec}(B)$ lying over P would necessarily not intersect S', and thus $Q \mapsto (S')^{-1}Q$ is a bijection from a subset containing all the Q we care about to the prime spectrum of $(S')^{-1}B$. The observation then follows.

Thus to keep my notation clean, I'll henceforth assume without loss of generality that A is a local ring and P is the unique maximal ideal of A. This gives us the advantage that since B/A is an integral extension, any $Q \in \mathrm{Spec}(B)$ that contracts to P must be maximal, and conversely any $Q \in \mathrm{Max}(B)$ must contract to a maximal ideal in A (of which P is the only option). Hence, we know $(\phi^*)^{-1}(P) = \mathrm{Max}(B)$.

Also note that because P is a maximal ideal, we know that $\{P\} = V(P)$ is a closed set in $\mathrm{Spec}(A)$. Hence, by the continuty of ϕ^*, we know that $\mathrm{Max}(B)$ must be a closed set in $\mathrm{Spec}(B)$. Or in other words, there must exist some ideal $I \lhd B$ such that $\mathrm{Max}(B) = V(I)$.

To finish off, since every prime ideal containing I is maximal, we can conclude that B/I is a ring with Krull dimension 0. In other words, every prime ideal of B/I is both a maximal ideal and a minimal prime ideal. Also note that B/I is Noetherian because B is. Hence, by the theorem on the prior page, B/I only has finitely many prime ideals. In turn, this also proves that $\mathrm{Max}(B)$ is finite, which is what we wanted. ■

As a side note, a commutative unital Noetherian ring with Krull dimension 0 is called an Artinian ring. And now I actually know how one would come across one that isn't just a field.

I've already done problems 7(a) and 7(b) on *pages 857-858*. So, all that's left to do is problem 7(c).

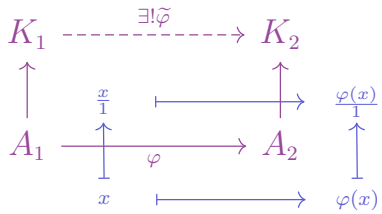
Set 3 Problem 7(c): Consider an integral ring extension B/A and let $\phi : A \to F$ be a ring homomorphism where F is an algebraically closed field. Then show that ϕ can be extended to a homomorphism $\psi : B \to F$ such that $\psi|_A = \phi$.

To start off, because Rogalski assumes by default that ring homomorphisms are unital, we know φ is not the zero map. In turn, because F is an integral domain we know that $P := \ker(\phi) \in \operatorname{Spec}(A)$. Then as B/A is an integral extension, we know that there exists $Q \in \operatorname{Spec}(B)$ lying over P. Therefore, we can conclude B/Q is integral over A/P by the homomorphism $a + P \mapsto a + Q$. Also importantly, we know ϕ quotients to an injective homomorphism:

$$\overline{\phi} : A/P \to F \text{ given by } \overline{\phi}(a + P) = \phi(a).$$

Next, since A/P and B/Q are both integral domains, we can consider their fields of fractions (which I will call E and K respectively).

Lemma 1: If $\varphi : A_1 \to A_2$ is an injective ring homomorphism and A_1, A_2 are integral domains with fields of fractions K_1, K_2 respectively, then there exists a unique field homomorphism $\widetilde{\varphi} : K_1 \to K_2$ such that the below diagram commutes:



Proof:
Clearly if $\widetilde{\varphi}$ is to be a homomorphism such that $\widetilde{\varphi}(\frac{x}{1}) = \frac{\varphi(x)}{1}$ for all $x \in A_1$, then we are forced to have that $\widetilde{\varphi}(\frac{x}{y}) = \frac{\varphi(x)}{\varphi(y)}$ for all $\frac{x}{y} \in K_1$. This handles the uniqueness part of the claim. So, now we just need to prove that $\widetilde{\varphi}$ is actually a well-defined homomorphism.

- To start off, because φ is injective we know that $y \neq 0$ implies that $\varphi(y) \neq 0$. Hence, for any $\frac{x}{y} \in K_1$ we know that the fraction $\frac{\varphi(x)}{\varphi(y)}$ is well-defined in K_2.
- Next suppose $\frac{x_1}{y_1} = \frac{x_2}{y_2}$ in K_1. Then $x_1y_2 - x_2y_1 = 0$, which in turn means that $\varphi(x_1)\varphi(y_2) - \varphi(x_2)\varphi(y_1) = 0$. And this proves that $\frac{\varphi(x_1)}{\varphi(y_1)} = \frac{\varphi(x_2)}{\varphi(y_2)}$. Hence, we've shown that $\widetilde{\varphi}$ is a well-defined function on K_1.
- Finally, we need to show that $\widetilde{\varphi}$ is a homomorphism. Fortunately, it's clear that $\widetilde{\varphi}$ respects multiplication and sends $\frac{1}{1}$ to $\frac{1}{1}$. As for addition, suppose $\frac{x_1}{y_1}, \frac{x_2}{y_2} \in K_1$. Then:
$$\widetilde{\varphi}(\tfrac{x_1}{y_1} + \tfrac{x_2}{y_2}) = \tfrac{\varphi(x_1y_2 + x_2y_1)}{\varphi(y_1y_2)} = \tfrac{\varphi(x_1)\varphi(y_2) + \varphi(x_2)\varphi(y_1)}{\varphi(y_1)\varphi(y_2)} = \tfrac{\varphi(x_1)}{\varphi(y_1)} + \tfrac{\varphi(x_2)}{\varphi(y_2)} = \widetilde{\varphi}(\tfrac{x_1}{y_1}) + \widetilde{\varphi}(\tfrac{x_2}{y_2})$$

Using the prior lemma, we can in particular conclude that $\overline{\phi}$ uniquely extends to a field homomorphism given by:

$$\widetilde{\overline{\phi}} : E \to F \text{ where } \widetilde{\overline{\phi}}(\tfrac{a_1+P}{a_2+P}) = \tfrac{\varphi(a_1)}{\varphi(a_2)}.$$

Also by the prior lemma, we can conclude that we have a field extension K/E given by $\frac{a_1+P}{a_2+P} \mapsto \frac{a_1+Q}{a_2+Q}$. And since B/Q is integral over A/P, we can conclude that K/E is an algebraic extension.

To explain why, I shall prove the following:

Lemma 2: If A, B are integral domains and B/A is an integral extension, then after letting E and K be the fields of fractions of A and B respectively we have that K/E is an algebraic extension (when given the embedding from lemma 1).

Proof:
Note that the algebraic elements of K over E form a subfield of K. After all, if $\alpha, \beta \in K$, then $E[\alpha, \beta]$ will be a finite (and thus algebraic) extension from E. And since $\alpha \pm \beta$ and $\alpha\beta^{\pm 1}$ are in $E[\alpha, \beta]$, this proves that those four elements are also algebraic.

Due to this, it suffices to show that $\frac{b}{1} \in K$ is algebraic over E for all $b \in B$. Yet this is automatic from the fact that B is integral over A.

Thus, we can now apply problem 7(a) (i.e. lemma 1 on *pages 857-858*) to the algebraic field extensions K/E as well as the homomorphism $\widetilde{\overline{\phi}} : E \to F$ in order to say that there exists some embedding $\widetilde{\overline{\psi}} : K \to F$ such that $\widetilde{\overline{\psi}}|_E = \widetilde{\overline{\phi}}$.

By restricting $\widetilde{\overline{\psi}}$ to the subring B/Q of K, we get a well-defined ring homomorphism $\overline{\psi} : B/Q \to F$. Also, it is easy to see that $\overline{\psi}|_{A/P} = \overline{\phi}$.

Finally, define $\psi : B \to F$ by $\psi(b) = \overline{\psi}(b+Q)$. Then ψ is a well-defined ring homomorphism since it is the composition of two other ring homomorphisms. Also suppose $a \in A$. Then $\psi(a) = \overline{\psi}(a+Q) = \overline{\phi}(a+P) = \phi(a)$. This shows that $\psi|_A = \phi$. ■

5/2/2026

Math 220c Lecture Notes:

Firstly, I want to make more observations about the order $\lambda(f)$ of an entire function $f : \mathbb{C} \to \mathbb{C}$. Like before, for any $R > 0$ we'll denote $M(R) := \sup\{|f(z)| : |z| = R\}$ and $N(R) :=$ the number of zeros of f in $\overline{\Delta}(0, R)$ (counted with multiplicity). To make it clear which function M or N is referring to, I'll also sometimes write M_f or N_f.

- Let f be an entire function with a well-defined order λ. Then $\lambda \geq 0$.

  Why?
  In *set 4 problem 4 on page 884* we proved that for all $\varepsilon > 0$ there exists $r > 0$ such that $M(R) < e^{R^{\lambda+\varepsilon}}$ for all $R > r$. In other words, $|f(z)| \leq e^{|z|^{\lambda+\varepsilon}}$ if $|z| > r$. But now if $\lambda < 0$, we'd be able take ε to be small enough so that $\lambda + \varepsilon < 0$. Therefore, $|f(z)| \leq e^{|z|^{\lambda+\varepsilon}} \leq e^{|z|^0} = e$ for $|z| > \max(r, 1)$.

Yet by the extreme value theorem we also know f is bounded on $\overline{\Delta}(0,r)$. Hence, we've proven that f is bounded. By Liouville's theorem this proves that f is a constant function, and we showed on *page 882* that constant functions have order 0. This contradicts that $\lambda(f) < 0$.

- Given any entire function f with order $\lambda \coloneqq \lambda(f)$, if $\alpha \in \mathbb{C} - \{0\}$ then either $\lambda(\alpha f) = \lambda(f)$ or $f \equiv c$ with $|c| \leq 1/|\alpha|$.

  Why?
  If f is a constant function then this claim is obvious from the fact that αf is also a constant function and has order 0 (assuming its order even exists). Hence, we'll now assume f is nonconstant. This guarentees that the order of αf exists for any $\alpha \neq 0$

  Suppose $\lambda(f) < \infty$. Then given any $\varepsilon > 0$, we can find some $r_1 > 0$ such that $|f(z)| < e^{|z|^{\lambda+\varepsilon/2}}$ when $|z| > r_1$. In other words, $|\alpha f(z)| < |\alpha| e^{|z|^{\lambda+\varepsilon/2}}$. But now we can find some $r_2 > r_1$ such that $|\alpha| e^{|z|^{\lambda+\varepsilon/2}} < e^{|z|^{\lambda+\varepsilon}}$ when $|z| > r_2$. Hence, we've proven there exists $r_2 > 0$ such that $|\alpha f(z)| < e^{|z|^{\lambda+\varepsilon}}$.

  This proves that $\lambda(\alpha f) \leq \lambda(f)$. To get the other inequality, just note that we can repeat all this reasoning but with the function αf and the constant α^{-1} in order to get that $\lambda(f) \leq \lambda(\alpha f)$.

  Meanwhile, suppose $\lambda(f) = \infty$. Then for all $N > 0$ we can find some $r_1 > 0$ such that $\frac{\log(\log(M_f(R)))}{\log(R)} > N$ when $R > r_1$. Equivalently, this means that $M_f(R) > e^{R^N}$ when $R > r_1$. Yet note that $|\alpha| M_f(R) = M_{\alpha f}(R)$. Hence, we've shown that $M_{\alpha f}(R) > |\alpha| e^{R^N}$ when $R > r_1$. And in turn there exists $r_2 > r_1$ such that when $R > r_2$ then $M_{\alpha f}(R) > e^{R^{N-1}}$. This proves $\lambda(\alpha f) = \infty = \lambda(f)$.

Because of these two observations, we'll make an ad hoc definition that if $f \equiv c$ where $0 < |c| \leq 1$, then $\lambda(f) \coloneqq 0$. Then:

- For any entire $f \not\equiv 0$ and $\alpha \neq 0$ we have that $\lambda(f) = \lambda(\alpha f)$.
- Also by *set 4 problem 4(ii) on page 884* plus my prior work, we know for any entire functions f, g with $f \not\equiv 0 \not\equiv g$ that $\lambda(fg)$ is well-defined and $\lambda(fg) \leq \max(\lambda(f), \lambda(g))$.
- Also $\lambda(P(z)) = 0$ for any nonzero polynomial $P(z)$.

  As a side note, in class the professor was saying that the limsup definition of $\lambda(f)$ was valid for any nonzero function. This was bugging me because the limsup doesn't exist when $f \equiv c$ where $0 < |c| \leq 1$. Also, part 2 of the procedure for calculating $\lambda(f)$ on *page 881* does not work when $f \equiv c$ with $|c| \leq 1$. This is because $e^{R^t} > e^0 = 1$ for all $R > 0$ and $t \in \mathbb{R}$.

  Interestingly, this same reasoning also shows that part 1 of the procedure on *page 881* will never fail at "proving" that $\lambda(f) \leq 0$ if $f \equiv c$ where $0 < |c| < 1$. Similarly *set 4 problem 4(i) on page 884* won't fail either for such f.

Hence, it is still the case that if $0 \leq \lambda < \infty$ then the order of f is at most λ iff for all $\varepsilon > 0$ there exists $r > 9$ such that $M(R) < e^{R^{\lambda+\varepsilon}}$ when $R > r$.

Lemma: Given any two entire functions $f, g \in O(\mathbb{C}) - \{0\}$ such that $f \neq -g$, we have that $\lambda(f+g) \leq \max(\lambda(f), \lambda(g))$.

Proof:

Let λ_1 and λ_2 denote the orders of f and g respectively. There is nothing to show if either λ_1 or λ_2 is ∞. So, we can assume they are finite. Then by the triangle inequality, we can conclude that $M_{f+g}(R) \leq M_f(R) + M_g(R)$ for all R. Also, for any $\varepsilon > 0$ we know there exists some $r > 0$ such that $M_f(R) < e^{R^{\lambda_1+\varepsilon}}$ and $M_g(R) < e^{R^{\lambda_2+\varepsilon}}$ when $R > r$.

But now $M_{f+g}(R) < e^{R^{\lambda_1+\varepsilon}} + e^{R^{\lambda_2+\varepsilon}}$ when $R > r$. In turn, if $\lambda = \max(\lambda_1, \lambda_2)$, we know there exists $\widetilde{r} > 0$ with $e^{R^{\lambda_1+\varepsilon}} + e^{R^{\lambda_2+\varepsilon}} < e^{R^{\lambda+2\varepsilon}}$ for all $R > \widetilde{r}$. This proves that $\lambda(f+g) < \lambda + 2\varepsilon$. ■

As a side note, if $f \in O(\mathbb{C}) - \{0\}$, $\alpha \in \mathbb{C}$, and $f \not\equiv -\alpha$, then we know $\lambda(f+\alpha) = \lambda(f)$.

If $\alpha = 0$ this is trivial. Meanwhile, if $\alpha \neq 0$ then we can use the prior lemma to prove that $\lambda(f+\alpha) \leq \lambda(f)$ and $\lambda(f) \leq \lambda(f+\alpha)$.

Lemma: If f is an entire function with $f \not\equiv 0$ and P is a nonzero polynomial, then $\lambda(f) = \lambda(Pf)$.

Proof:

We already proved the case when P is a constant polynomial. So from now on assume P is not a constant polynomial. Then $|f| \leq |Pf|$ if $|z| \gg 0$. hence $\lambda(f) \leq \lambda(Pf)$. On the other hand, $\lambda(Pf) \leq \max(\lambda(P), \lambda(f)) = \lambda(f)$. Hence, $\lambda(f) = \lambda(Pf)$.

Set 5 Problem 2: If f is a nonconstant entire function of finite order λ, then f' also has order λ.

Proof:

First we show $\lambda(f') \leq \lambda(f)$. Note that for any $\varepsilon > 0$ we can find some $r > 0$ such that $|f(z)| \leq e^{|z|^{\lambda+\varepsilon}}$ when $|z| > r$. Without loss of generality we'll take $r > 1$. Next consider any $a \in \mathbb{C}$ with $|a| > 2r$. Then by considering the disk $\Delta(a, r)$, we can conclude by Cauchy's estimate (see *page 368*) that:

$$|f'(a)| \leq \frac{\max_{z \in \overline{\Delta}(a,r)} |f(z)|}{r} \leq \frac{1}{r} \max_{z \in \overline{\Delta}(a,r)} e^{|z|^{\lambda+\varepsilon}} \leq e^{(|a|+r)^{\lambda+\varepsilon}}$$

Yet now we can find some $R > 0$ such that $e^{(x+r)^{\lambda+\varepsilon}} < e^{x^{\lambda+2\varepsilon}}$ when $x > R$. Therefore, when $|a| > \max(2r, R)$ we have that $|f'(a)| \leq e^{|a|^{\lambda+2\varepsilon}}$. By bringing $\varepsilon \to 0$ this proves $\lambda(f') \leq \lambda$.

Next we show $\lambda(f) \leq \lambda(f')$. To prove this, start by noting that if f' is a constant function then we trivially have that f and f' are polynomials and have the same order. Therefore, we may assume now that f' is not constant. Then by the maximum modulus principle, we must have that $M_{f'}(R)$ is increasing as $R \to \infty$.

As a side note, if I had made this connection earlier, I could have written a *way* shorter proof at the top of *page 881*.

By the fundamental theorem of calculus we know $f(z) - f(0) = z\int_0^1 f'(tz)\mathrm{d}t$. Hence:

$$\begin{aligned}|f(z)| = \left|z\int_0^1 f'(tz)\mathrm{d}t + f(0)\right| &\leq |z|\int_0^1 |f'(tz)|\mathrm{d}t + |f(0)| \\ &\leq |z|M_{f'}(|z|) + |f(0)|\end{aligned}$$

Let μ be the order of f' (which we know is finite since it is less than λ). Then for any $\varepsilon > 0$ we know there exists $r > 0$ such that $M_{f'}(R) < e^{R^{\mu+\varepsilon}}$ for all $R > r$. In turn:

$$|f(z)| < |z|e^{|z|^{\mu+\varepsilon}} + |f(0)|.$$

Finally, we can find some $\widetilde{r} > r$ such that $Re^{R^{\mu+\varepsilon}} + |f(0)| < e^{R^{\mu+2\varepsilon}}$ when $R > \widetilde{r}$. Hence, we've proven that $|f(z)| < e^{|z|^{\mu+2\varepsilon}}$ when $|z| > \widetilde{r}$. In turn, the order of f is at most μ. ■

As a side note we can also conclude that if f has order ∞ then so does f'. After all, if f' had finite order then the reasoning that proved that $\lambda(f) \leq \lambda(f')$ could be applied to show that f doesn't have infinite order. But that's a contradiction. So, we always have that $\lambda(f) = \lambda(f')$.

Set 5 Problem 6: Show that if P is a polynomial of degree d, then $f = e^P$ has order exactly d.

To start off, if $d = 0$ then f is a constant function and the problem is trivial. Hence, we can assume without loss of generality that $d > 1$. Now, I will first show that $\lambda(f) \geq d$.

Consider any $\varepsilon > 0$ and write $P(z) = a_d z^d + \ldots + a_1 z + a_0$ where $a_d \neq 0$. Since $|a_d| > 0$, we crucially can find some $\delta \in (0, |a_d|)$. Next, because $\frac{P(z)}{z^d} \to a_d$ as $|z| \to \infty$, we can find some $r > 0$ such that $|\frac{P(z)}{z^d} - a_d| < \delta$ whenever $|z| > r$. Equivalently, this means that $|P(z) - a_d z^d| = |a_{d-1}z^{d-1} + \ldots + a_1 z + a_0| < \delta|z|^d$ when $|z| > r$.

Next, note that $|f(z)| = e^{\mathrm{Re}(P(z))} = e^{\mathrm{Re}(a_d z^d) + \mathrm{Re}(P(z) - a_d z^d)}$. Importantly, if $a_d = |a_d|e^{i\alpha}$ and $z = |z|e^{i\theta}$ where $\alpha, \theta \in \mathbb{R}$, we have that:

$$\mathrm{Re}(a_d z^d) = |a_d| \cdot |z|^d \cdot \cos(\alpha + d\theta).$$

Thus by strategicially picking the angle θ of z we can guarentee $\cos(\alpha + d\theta) = 1$ and hence $\mathrm{Re}(a_d z^d) = |a_d| \cdot |z|^d$. At the same time, if $|z| > r$ then we can say that $\mathrm{Re}(P(z) - a_d z^d) \geq -|P(z) - a_d z^d| > -\delta|z|^d$. And thus, we've proven for all $R > r$ that there exists some $z \in \mathbb{C}$ with $|z| = R$ and:

$$|f(z)| = e^{\mathrm{Re}(a_d z^d) + \mathrm{Re}(P(z) - a_d z^d)} \geq e^{(|a_d| - \delta)|z|^d}$$

To finish off, because $|a_d| - \delta > 0$, we can find some $\widetilde{r} > 0$ such that:

$$R^{d-\varepsilon} < (|a_d| - \delta)R^d \text{ whenever } R > \widetilde{r}.$$

Thus, we've proven that $M(R) > e^{R^{d-\varepsilon}}$ whenever $R > \widetilde{r}$. This finishes showing that $\lambda(f) \geq d$.

The other inequality is easy. After all, if we write $P(z)$ as we did before, then we can see when $|z| > 1$ that:

$$\mathrm{Re}(P(z)) \leq |a_d||z|^d + \ldots + |a_1||z| + |a_0| \leq |z|^d \cdot (|a_d| + |a_{d-1}| + \ldots + |a_1| + |a_0|).$$

Therefore, we know there exists some $B > 0$ (with the inequality strict since $|a_d| > 0$) such that $|f(z)| \leq e^{B|z|^d}$ when $|z| > 1$. Then to finish off, given any $\varepsilon > 0$ we can find some $r > 0$ such that $e^{BR^d} < e^{R^{d+\varepsilon}}$ when $R > r$. In turn, when $|z| > r$ we've shown that $|f(z)| < e^{R^{d+\varepsilon}}$. And this finishes showing that $\lambda(f) \leq d$. ■

Assume f has zeros $|a_1| \leq |a_2| \leq \cdots \leq |a_n| \leq \cdots$ where $|a_n| \to \infty$ as $n \to \infty$ and no a_n is equal to 0. We define the following quantities to measure the growth of zeros:

- We say p is the rank of f if p is the smallest integer such that $\sum_{n=1}^{\infty} \frac{1}{|a_n|^{p+1}} < \infty$. If such a p doesn't exist we we say the rank of f is ∞.
- We say $\alpha := \inf\left\{t > 0 : \sum_{n=1}^{\infty} \frac{1}{|a_n|^t} < \infty\right\}$ is the critical exponent of f. If no such t exists we set $\alpha = \infty$.

  See *set 4 problem 5 on page 885* for more information on this. Importantly, we know $p \leq \alpha \leq p + 1$. Thus, if $\alpha \notin \mathbb{Z}$ we know α uniquely determines p.

- Also we still have the quantity $N(R)$ for any $R > 0$.

  Interestingly enough, while we won't prove or use this equation in this class, $\alpha = \limsup_{R\to\infty} \frac{\log(N(R))}{\log(R)}$. Hence, $N(R)$ uniquely determines α which often uniquely determines p.

Finally, if we now suppose f has m many zeros at 0 (in addition to the zeros at $a_1, a_2, \ldots$), then recall from the Weierstraß factorization theorem and the note on *page 581* that if $f(z)z^{-m}$ has rank p then we can write $f(z) = z^m e^{g(z)} \prod_{n=1}^{\infty} E_p(\frac{z}{a_n})$ where g is an entire function and

$$E_p(z) = \begin{cases} (1-z) & \text{if } p = 0 \\ (1-z)\exp(z + \frac{z^2}{2} + \ldots + \frac{z^p}{p}) & \text{if } p > 1. \end{cases}$$

Now we define $h = \operatorname{genus}(f) := \begin{cases} \max(p, q) & \text{if } g(z) \text{ is a polynomial of degree } q \\ \infty & \text{if } g(z) \text{ is not a polynomial.} \end{cases}$

Meanwhile if $f(z)z^{-m}$ has infinite rank, we say $\operatorname{genus}(f) := \infty$.

Hadamard's Theorem: If λ is the order of f then $h \leq \lambda \leq h + 1$.

Proof part 1: $\lambda \leq h + 1$
Our key lemma is as follows.

Lemma: $\log|E_p(w)| \leq C_p|w|^{p+1}$ for some $C_p > 0$.

Proof:
We do induction on p. When $p = 0$, $\log|1 - w| \leq \log(1 + |w|) \leq |w|$. So, we can take $c_0 = 1$.

Meanwhile suppose $p \geq 1$. Then $E_p(w) = E_{p-1}(w)\exp(\frac{w^p}{p})$. Hence when $|w| \neq 0$ we have that:

$$\begin{aligned}\log|E_p(w)| &= \log|E_{p-1}(w)| + \log|\exp(\tfrac{w^p}{p})| \\ &\leq C_{p-1}|w|^p + \operatorname{Re}(\tfrac{w^p}{p}) \leq (C_{p-1} + \tfrac{1}{p})|w|^p\end{aligned}$$

In particular, if $|w| \geq \frac{1}{2}$ then we can say that $2(C_{p-1} + \frac{1}{p})|w|^{p+1} \geq (C_{p-1} + \frac{1}{p})|w|^p$.

Meanwhile, when $|w| < 1$ then recall the taylor expenasion (where Log denote the principal branch): $\operatorname{Log}(1-w) = -w - \frac{w^2}{2} - \cdots - \frac{w^k}{k} - \cdots$. Hence, when $|w| < 1$ we may write:

$$E_p(w) = (1-w)\exp(-\operatorname{Log}(1-w) - \textstyle\sum_{k=p+1}^{\infty}\frac{w^k}{k}) = \exp(-\sum_{k=p+1}^{\infty}\frac{w^k}{k})$$

In particular, if $|w| \leq \frac{1}{2}$ we can write:

$$\log|E_p(w)| = \operatorname{Re}(-\sum_{k=p+1}^{\infty}\frac{w^k}{k}) \leq \sum_{k\geq p+1}\left|\frac{w^k}{k}\right| \leq |p|^{p+1}\sum_{k\geq 0}\frac{|w|^k}{1} \leq 2|w|^{p+1}.$$

By taking $C_p \coloneqq \max(2, 2(C_{p-1} + \frac{1}{p}))$, we thus get that $\log|E_p(w)| \leq C_p|w|^{p+1}$. ■

Now we can use this to prove $\lambda \leq h+1$. To start off, without loss of generality we can assume h is finite. Otherwise, there is nothing to prove. But then in turn, when we write $f(z) = z^m e^{g(z)}\prod_{n=1}^{\infty}E_p(\frac{z}{a_n})$ we must have that g is a polynomial. By set 5 problem 6 on *pages 906-907*, we know $\lambda(e^{g(z)}) = \deg(g(z)) \leq h$. Hence, all that we need to do to prove the order of f is at most $h+1$ is show that the order of $\prod_{n=1}^{\infty}E_p(\frac{z}{a_n})$ is at most $p+1 \leq h+1$.

Fortunately, $\log|\prod_{n\in\mathbb{N}}E_p(\frac{z}{a_n})| = \sum_{n\in\mathbb{N}}\log|E_p(\frac{z}{a_n})|$. Then by our prior lemma we have that $\sum_{n\in\mathbb{N}}\log|E_p(\frac{z}{a_n})| \leq C_p\sum_{n\in\mathbb{N}}|\frac{z}{a_n}|^{p+1} = K|z|^{p+1}$ where $K = C_p\sum_{n\in\mathbb{N}}\frac{1}{|a_n|^{p+1}}$ is finite.

This proves that the order of $\prod_{n\in\mathbb{N}}E_p(\frac{z}{a_n})$ is at most $p+1$.

Proof part 2: $h \leq \lambda$

To start off, without loss of generality we can assume λ is finite. Otherwise, there is nothing to prove. Furthermore, we can assume $f(0) = 1$. After all, if not then we know there exists $f^{\text{new}}(z)$ such that $f(z) = cz^m f^{\text{new}}(z)$ and $f^{\text{new}}(0) = 1$. Also, $\lambda(f^{\text{new}}) = \lambda(f)$ and $\operatorname{genus}(f^{\text{new}}) = \operatorname{genus}(f)$.

Now if α is the critical exponent of f, recall from set 4 problem 5 on *page 885* that $\alpha \leq \lambda$. In turn, this shows that f is a function with finite rank $p \leq \alpha \leq \lambda$.

After we write $f(z) = e^{g(z)}\prod_{n=1}^{\infty}E_p(\frac{z}{a_n})$ (where as a side note we know $g(0) = 0$ since $f(0) = 1$), it now suffices to prove that $g(z)$ is a polynomial of degree at most λ.

Let m be an integer such that $0 \leq m \leq \lambda < m+1$. Our goal will be to show that $\mathrm{D}^{m+1} g \equiv 0$ in $\mathbb{C} - \{a_1, a_2, \ldots\}$ (where D is the differentiation operator). By the identity principle, this will then let us conclude $D^{m+1} g \equiv 0$ everywhere. Hence, g is a polynomial of degree at most $m \leq \lambda$.

Write $P(z) = \prod_{n=1}^{\infty} E_p(\frac{z}{a_n})$ so that $f = e^g P$. By logarithmically diferentiating both sides we get that $\frac{f'}{f} = g' + \frac{P'}{P} = g'$ on the set of points where $P(z) \neq 0$. Next, by taking m usual derivatives we get that:

$$\mathrm{D}^m \tfrac{f'}{f} = \mathrm{D}^{m+1} g + \mathrm{D}^m \tfrac{P'}{P}$$

Lemma A: $\mathrm{D}^m \frac{P'}{P} = -m! \sum_{n \in \mathbb{N}} \frac{1}{(a_n - z)^{m+1}}$.

Proof:
Note that $\frac{P'}{P} = \sum_{n \in \mathbb{N}} \frac{E_p'(\frac{z}{a_n})}{a_n E_p(\frac{z}{a_n})}$ where the latter sum converges locally uniformly on $\mathbb{C} - \{a_1, a_2 \ldots\}$. (This is due to a theorem on page 549.)

By applying the Weierstraß convergence theorem m times, we can thus conclude that $\mathrm{D}^m \frac{P'}{P} = \sum_{n \in \mathbb{N}} \mathrm{D}^m \left(\frac{E_p'(\frac{z}{a_n})}{a_n E_p(\frac{z}{a_n})} \right)$.

Next since $E_p(z) = (1-z)\exp(z + \frac{z^2}{2} + \ldots + \frac{z^p}{p})$, we know that:
$\frac{E_p'(z)}{E_p(z)} = \frac{-1}{1-z} + 1 + z + \cdots + z^{p-1} \Longrightarrow \frac{E_p'(\frac{z}{a_n})}{a_n E_p(\frac{z}{a_n})} = \frac{-1}{a_n - z} + \frac{1}{a_n} + \frac{z}{a_n^2} + \cdots + \frac{z^{p-1}}{a_n^p}$.

But since m is the largest integer $\leq \lambda$ and we know p is another integer $\leq \lambda$, we can now conclude that:

$$\mathrm{D}^m \left(\frac{E_p'(\frac{z}{a_n})}{a_n E_p(\frac{z}{a_n})} \right) = -\mathrm{D}^m \left(\frac{1}{a_n - z} \right) + 0 = \frac{-m!}{(a_n - z)^{m+1}}$$

This gives us the result we wanted.

One other other lemma we need (although I will state and prove it in a stronger form than is strictly required by this theorem) is as follows:

Lemma B: Suppose f is an entire function, $f \not\equiv 0$, and $m + 1 > \lambda := \lambda(f)$. If $\{a_n\}_{n \in S}$ is the set of all zeros of f (where S is either $\mathbb{N}$ or $\{1, \ldots, N\}$ for some $N \in \mathbb{N}$) with elements repeated according to multiplicity, then:

$$\mathrm{D}^m \tfrac{f'}{f} = -m! \sum_{n \in S} \tfrac{1}{(a_n - z)^{m+1}}$$

As a side note, if f is a polynomial then this lemma is obvious. This lemma is especially useful for letting us calculate derivatives of functions while only knowing the locations of zeros and growth rate of f.

Proof:
Without loss of generality we can assume $f(0) = 1$. After all, otherwise we could write $f(z) = cz^m F(z)$ where $F(z)$ is an entire function with $F(0) = 1$. Then f and F will have the same order. Also:
$\frac{f'}{f} = \frac{k}{z} + \frac{F'}{F} \Longrightarrow \mathrm{D}^m \frac{f'}{f} = k \cdot \frac{m!(-1)^m}{z^{m+1}} + \mathrm{D}^m \frac{F'}{F} = -k \frac{m!}{(-z)^{m+1}} + \mathrm{D}^m \frac{F'}{F}$.

So if we can prove the lemma for F then we will have proven the lemma for f.

Without loss of generality suppose $|a_k|$ is nondecreasing as k increases. If we recall the Poisson-Jensen formula on *page 879*, we get for any $z \in \Delta(0,R)$ such that $z \neq a_k$ for any $k \in S$ that:

$$\log|f(z)| + \sum_{k=1}^{N(R)} \log\left|\frac{R^2-\overline{a_k}z}{R(z-a_k)}\right| = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{Re}\left(\frac{Re^{it}+z}{Re^{it}-z}\right)\cdot \log|f(Re^{it})|\mathrm{d}t$$

As a side note, if $|z| = R$ then:

$$|R^2 - \overline{a_k}z| = |R^2 - a_k\overline{z}| = R^2|1-\tfrac{a_k}{z}| = \tfrac{R^2}{|z|}|z-a_k| = |R(z-a_k)|$$

As a result, if $a_k \in \partial\Delta(0,R)$ then $\log\left|\frac{R^2-\overline{a_k}z}{R(z-a_k)}\right| = \log(1) = 0$. In turn, although the equation written on page 879 is only summing over the zeros of $\Delta(0,R)$, adding in the terms associated with the zeros in $\partial\Delta(0,R)$ doesn't change the validity of the formula.

Tangent on Wirtinger derivatives:

Given a function $f : \mathbb{C} \to X$, instead of thinking of f as being a multivariable function $(x,y) \in \mathbb{R}^2 \mapsto f(x+iy)$, it is natural to perform a change of variables and write f as a function of $z = x+iy$ and $\overline{z} = x-iy$.

To complete our formulation of this change of variables, we want to define what it means to partially differentiate a function $f : \mathbb{C} \to \mathbb{C}$ with respect to z and $\overline{z}$ as variables. So, note that $f(x,y) = f(\frac{z+\overline{z}}{2}, -i\frac{z-\overline{z}}{2})$. Then by chain rule we "should" have that:

$$\frac{\partial f}{\partial z} = \frac{1}{2}\left(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\right) \text{ and } \frac{\partial f}{\partial \overline{z}} = \frac{1}{2}\left(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\right)$$

I say "should" with quotation marks because I've never actually defined what a partial derivative with respect to a complex variable is. Hence, right now me invoking chain rule is just abstract symbol manipulation.

Based on that, given a C^1 function $f : \mathbb{R}^2 \to \mathbb{C}$ we define the derivative operators:

$$\partial_z = \frac{\partial}{\partial z} := \frac{1}{2}(\partial_x - i\partial_y) \text{ and } \partial_{\overline{z}} = \frac{\partial}{\partial \overline{z}} := \frac{1}{2}(\partial_x + i\partial_y)$$

These operators are called Wirtinger derivatives and are easily checked to have the following algebraic properties:

- ∂_z and $\partial_{\overline{z}}$ are $\mathbb{C}$-linear.
- $\partial_z(fg) = (\partial_z f)g + f(\partial_z g)$ and $\partial_{\overline{z}}(fg) = (\partial_{\overline{z}} f)g + f(\partial_{\overline{z}} g)$.
- $\frac{\partial}{\partial z}z = 1$, $\frac{\partial}{\partial \overline{z}}z = 0$, $\frac{\partial}{\partial z}\overline{z} = 0$, and $\frac{\partial}{\partial \overline{z}}\overline{z} = 1$.
- $\overline{(\frac{\partial}{\partial z}f)} = \frac{\partial}{\partial \overline{z}}\overline{f}$ and $\overline{(\frac{\partial}{\partial \overline{z}}f)} = \frac{\partial}{\partial z}\overline{f}$.

There is also a chain rule in terms of Wirtinger derivatives but it is not perfectly analogous to the usual expression for chain rule unforunately. The details are not relevant enough for me to bring up here.

Note that if we write $f(x,y) = u(x,y) + iv(x,y)$ then:

- $\frac{\partial}{\partial z} f = \frac{1}{2}((u_x + iv_x) - i(u_y + iv_y)) = \frac{1}{2}((u_x + v_y) + i(v_x - u_y))$,
- $\frac{\partial}{\partial \overline{z}} f = \frac{1}{2}((u_x + iv_x) + i(u_y + iv_y)) = \frac{1}{2}((u_x - v_y) + i(v_x + u_y))$.

Now $\frac{\partial}{\partial \overline{z}} f \equiv 0$ iff $u_x - v_y \equiv 0$ and $v_x + u_y \equiv 0$. As a result, we can see that a function $f : \mathbb{C} \cong \mathbb{R}^2 \to \mathbb{C}$ is holomorphic iff $\frac{\partial}{\partial \overline{z}} f \equiv 0$. Furthermore, note that if f is holomorphic then $\frac{1}{2}((u_x + v_y) + i(v_x - u_y)) = u_x + iv_x = f'$. Thus $\frac{\partial}{\partial z} f = f'$ when f is holomorphic.

One last observation I want to to make is that if $f(x,y) = u(x,y) + iv(x,y)$ is holomorphic, then $\frac{\partial}{\partial z} u = \frac{1}{2}(u_x - iu_y) = \frac{1}{2}(u_x + iv_x) = \frac{1}{2} f'$.

Returning to the lemma I was proving before, note that the principal branch of the logarithm satisfies $\operatorname{Log}(re^{-i\theta}) = \overline{\operatorname{Log}(re^{i\theta})}$ for all $r > 0$ and $-\pi < \theta < \pi$. Hence if $f(z) \notin \mathbb{R}_{\leq 0}$, we get that:

$$\begin{aligned} 2\tfrac{\partial}{\partial z} \log|f(z)| = \tfrac{\partial}{\partial z} \log(|f(z)|^2) &= \tfrac{\partial}{\partial z} \log(f(z)\overline{f(z)}) \\ &= \tfrac{\partial}{\partial z} \operatorname{Log}(f(z)) + \tfrac{\partial}{\partial z} \operatorname{Log}(\overline{f(z)}) \\ &= \tfrac{\partial}{\partial z} \operatorname{Log}(f(z)) + \overline{\tfrac{\partial}{\partial \overline{z}} \operatorname{Log}(f(z))} = \tfrac{f'(z)}{f(z)} + \overline{0} \end{aligned}$$

Similarly, if $f(z) \in \mathbb{R}_{<0}$ then we consider the branch of the logarithm Log_1 and Log_2 defined for angles in $(0, 2\pi)$ and $(-2\pi, 0)$ respectively. Then:

$$\begin{aligned} 2\tfrac{\partial}{\partial z} \log|f(z)| = \tfrac{\partial}{\partial z} \log(f(z)\overline{f(z)}) &= \tfrac{\partial}{\partial z} \operatorname{Log}_1(f(z)) + \tfrac{\partial}{\partial z} \operatorname{Log}_2(\overline{f(z)}) \\ &= \tfrac{f'}{f} + \tfrac{\partial}{\partial z} \operatorname{Log}_2(\overline{f(z)}) \end{aligned}$$

But now for any $\theta \in (0, 2\pi)$ and $r > 0$ that we have that:

$$\operatorname{Log}_2(re^{-i\theta}) = \log(r) - i\theta = \overline{\log(r) + i\theta} = \overline{\operatorname{Log}_1(re^{i\theta})}.$$

Thus $\frac{\partial}{\partial z} \operatorname{Log}_2(\overline{f(z)}) = \frac{\partial}{\partial z} \overline{\operatorname{Log}_1(f(z))} = \overline{\frac{\partial}{\partial \overline{z}} \operatorname{Log}_1(f(z))} = 0.$

And in all cases we've thus proven that $2\frac{\partial}{\partial z} \log|f(z)| = \frac{f'(z)}{f(z)}$ whenever $z \neq a_k$ for any $k \in S$.

By applying this formula to the special case that $f(z) = \frac{R^2 - \overline{a_k} z}{R(z - a_k)}$, we can in turn see that if $z \neq a_k$ then:

$$2\tfrac{\partial}{\partial z} \log\left|\tfrac{R^2 - \overline{a_k} z}{R(z - a_k)}\right| = -\tfrac{\overline{a_k}}{R^2 - \overline{a_k} z} - \tfrac{1}{z - a_k}$$

Hence by applying $2\frac{\partial}{\partial z}$ to the Poisson-Jensen formula expression at the top of the prior page, we can conclude for $z \in \Delta(0, R) - f^{-1}(\{0\})$ that:

$$\tfrac{f'}{f} = \sum_{k=1}^{N(R)} \tfrac{\overline{a_k}}{R^2 - \overline{a_k} z} - \sum_{k=1}^{N(R)} \tfrac{1}{a_k - z} + \tfrac{1}{\pi}\left(\tfrac{\partial}{\partial z} \int_0^{2\pi} \operatorname{Re}\left(\tfrac{Re^{it} + z}{Re^{it} - z}\right) \cdot \log|f(Re^{it})| \mathrm{d}t\right)$$

We want to bring the derivative operator into the integral. To do this, note that:

$$\left(\frac{Re^{it}+z}{Re^{it}-z}\right)' = \left(-1+\frac{2Re^{it}}{Re^{it}-z}\right)' = \frac{2Re^{it}}{(Re^{it}-z)^2}.$$

Also because $\frac{Re^{it}+z}{Re^{it}-z}$ is holomorphic, we know that:

$$\frac{\partial}{\partial x}\left(\frac{Re^{it}+z}{Re^{it}-z}\right) = \left(\frac{Re^{it}+z}{Re^{it}-z}\right)' = -i\frac{\partial}{\partial y}\left(\frac{Re^{it}+z}{Re^{it}-z}\right).$$

Now if $|z| < \rho$, we have that:

$$\left|\frac{\partial}{\partial x}\mathrm{Re}\left(\frac{Re^{it}+z}{Re^{it}-z}\right)\right| = \left|\mathrm{Re}\left(\left(\frac{Re^{it}+z}{Re^{it}-z}\right)'\right)\right| = \left|\mathrm{Re}\left(\frac{2Re^{it}}{(Re^{it}-z)^2}\right)\right| \leq \left|\frac{2Re^{it}}{(Re^{it}-z)^2}\right| \leq \frac{2R}{(R-\rho)^2}$$

Similarly, if $|z| < \rho$ then:

$$\begin{aligned}\left|\frac{\partial}{\partial y}\mathrm{Re}\left(\frac{Re^{it}+z}{Re^{it}-z}\right)\right| &= \left|\mathrm{Re}\left(i\cdot -i\frac{\partial}{\partial y}\left(\frac{Re^{it}+z}{Re^{it}-z}\right)\right)\right| \\ &= \left|\mathrm{Re}\left(i\cdot\left(\frac{Re^{it}+z}{Re^{it}-z}\right)'\right)\right| = \left|\mathrm{Re}\left(i\cdot\frac{2Re^{it}}{(Re^{it}-z)^2}\right)\right| \leq \left|\frac{2Re^{it}}{(Re^{it}-z)^2}\right| \leq \frac{2R}{(R-\rho)^2}\end{aligned}$$

Since $\frac{2R}{(R-\rho)^2}\log|f(Re^{it})|$ is an integrable function on $[0,2\pi]$, we can apply Leibniz's rule (see *pages 189-190*) to get that on $\Delta(0,\rho)$ that:

$$\begin{aligned}&\tfrac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)\int_0^{2\pi}\mathrm{Re}\left(\frac{Re^{it}+z}{Re^{it}-z}\right)\cdot\log|f(Re^{it})|\mathrm{d}t \\ &\qquad = \int_0^{2\pi}\left(\tfrac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)\mathrm{Re}\left(\frac{Re^{it}+z}{Re^{it}-z}\right)\right)\cdot\log|f(Re^{it})|\mathrm{d}t\end{aligned}$$

By taking $\rho \to R$, we can then conclude for all $z \in \Delta(0,R)$ that:

$$\begin{aligned}\frac{\partial}{\partial z}\int_0^{2\pi}\mathrm{Re}\left(\frac{Re^{it}+z}{Re^{it}-z}\right)\cdot\log|f(Re^{it})|\mathrm{d}t &= \int_0^{2\pi}\frac{\partial}{\partial z}\mathrm{Re}\left(\frac{Re^{it}+z}{Re^{it}-z}\right)\cdot\log|f(Re^{it})|\mathrm{d}t \\ &= \int_0^{2\pi}\tfrac{1}{2}\left(\frac{Re^{it}+z}{Re^{it}-z}\right)'\cdot\log|f(Re^{it})|\mathrm{d}t \\ &= \int_0^{2\pi}\frac{Re^{it}}{(Re^{it}-z)^2}\cdot\log|f(Re^{it})|\mathrm{d}t\end{aligned}$$

As a side note, at this point in the calculation the professor has the integrand as being $\mathrm{Re}\left(\frac{2Re^{it}}{(Re^{it}-z)^2}\right)\cdot\log|f(Re^{it})|$. I think the cause of the difference in our integrands is that the professor got excited and forgot which derivative operator he was trying to move through the integral sign.

Anyways a little further down in Oprea's notes he abruptly drops the real part, and once he does that his integrand is a scalar multiple of mine. Also it will go to 0 at the end of this proof so that scalar discrepancy won't be impactful.

To summarize the calculation so far, we have proven that we have for all $z \in \Delta(0,R) - f^{-1}(\{0\})$ that:

$$\frac{f'}{f} = \sum_{k=1}^{N(R)}\frac{\overline{a_k}}{R^2-\overline{a_k}z} - \sum_{k=1}^{N(R)}\frac{1}{a_k-z} + \frac{1}{\pi}\int_0^{2\pi}\frac{Re^{it}}{(Re^{it}-z)^2}\cdot\log|f(Re^{it})|\mathrm{d}t$$

Now take m (typical complex) derivatives of both sides to get that:

$$\mathrm{D}^m\frac{f'}{f} = m!\sum_{k=1}^{N(R)}\frac{\overline{a_k}^{m+1}}{(R^2-\overline{a_k}z)^{m+1}} - m!\sum_{k=1}^{N(R)}\frac{1}{(a_k-z)^{m+1}} + \frac{1}{\pi}\mathrm{D}^m\int_0^{2\pi}\frac{Re^{it}}{(Re^{it}-z)^2}\cdot\log|f(Re^{it})|\mathrm{d}t$$

So long as $\left(\frac{Re^{it}}{(Re^{it}-z)^2}\right)^{(k)} \cdot \log|f(Re^{it})|$ is a continuous function jointly in t and z (where $1 \leq k \leq m$), we can apply Leibniz's rule for path integrals (see *pages 380-381*) to bring a derivative operator with respect to z through the integral. Luckily, this can be guarenteed if R satisfies that $f(Re^{it}) \neq 0$ for any t.

Since f has at mostly countably infinite many zeros, we can choose some increasing sequence $\{R_j\}_{j\in\mathbb{N}} \subseteq \mathbb{R}_{>0}$ such that $R_j \to \infty$ and $\partial\Delta(0,R_j) \cap f^{-1}(\{0\}) = \emptyset$ for all j. Then we can conclude for all $j \in \mathbb{N}$ and $z \in \Delta(0,R_j) - f^{-1}(\{0\})$ that:

$$\mathrm{D}^m \frac{f'}{f} = m! \sum_{k=1}^{N(R_j)} \frac{\overline{a_k}^{m+1}}{(R_j^2 - \overline{a_k}z)^{m+1}} - m! \sum_{k=1}^{N(R_j)} \frac{1}{(a_k - z)^{m+1}} + \frac{1}{\pi}\int_0^{2\pi} \frac{(m+1)!R_j e^{it}}{(R_j e^{it} - z)^{m+2}} \cdot \log|f(R_j e^{it})|\mathrm{d}t$$

To finish off, we need to show that:

$$\sum_{k=1}^{N(R_j)} \frac{\overline{a_k}^{m+1}}{(R_j^2 - \overline{a_k}z)^{m+1}} \to 0 \text{ and } \int_0^{2\pi} \frac{R_j e^{it}}{(R_j e^{it} - z)^{m+2}} \cdot \log|f(R_j e^{it})|\mathrm{d}t \to 0$$

with convergence pointwise in z as $R_j \to \infty$. Once we prove that, then we will be able to take $R_j \to \infty$ to finish proving that $\mathrm{D}^m \frac{f'}{f} = -m! \sum_{n\in S} \frac{1}{(a_n - z)^{m+1}}$ for all $z \in \mathbb{C} - f^{-1}(\{0\})$.

Claim 1: $\sum_{k=1}^{N(R)} \frac{\overline{a_k}^{m+1}}{(R^2 - \overline{a_k}z)^{m+1}} \to 0$ (pointwise in z) as $R \to \infty$.

Pick any fixed $z \in \mathbb{C} - f^{-1}(\{0\})$ and then consider any $R > 2|z|$. We have for any $a_k \in \Delta(0,R)$ that:

$$|R^2 - \overline{a_k}z| \geq R^2 - |a_k||z| > R^2 - \frac{R^2}{2} = \frac{R^2}{2}$$

As a result, $\left|\frac{\overline{a_k}}{R^2 - \overline{a_k}z}\right|^{m+1} \leq \frac{R^{m+1}}{(R^2/2)^{m+1}} = \frac{2^{m+1}}{R^{m+1}}$.

Then in turn we can say:

$$\left|\sum_{k=1}^{N(R)} \frac{\overline{a_k}^{m+1}}{(R^2 - \overline{a_k}z)^{m+1}}\right| \leq \sum_{k=1}^{N(R)} \left|\frac{\overline{a_k}}{R^2 - \overline{a_k}z}\right|^{m+1} \leq N(R) \cdot \frac{2^{m+1}}{R^{m+1}}$$

Now since $m+1 > \lambda$, we can pick some $\varepsilon > 0$ with $m+1 > \lambda + \varepsilon$. In particular, by recalling the quick application of Jensen's formula on *page 880*, we can thus say that:

$$N(R) \leq \log(M(3R)) < \log(e^{(3R)^{\lambda+\varepsilon}}) = (3R)^{\lambda+\varepsilon} \text{ when } R \gg 0.$$

But now when $R \gg 0$, we have that $\left|\sum_{k=1}^{N(R)} \frac{\overline{a_k}^{m+1}}{(R^2 - \overline{a_k}z)^{m+1}}\right| \leq 3^{\lambda+\varepsilon} \cdot 2^{m+1} \cdot \frac{R^{\lambda+\varepsilon}}{R^{m+1}}$.

Since $m+1 > \lambda + \varepsilon$ the last expression goes to zero as $R \to \infty$.

Claim 2: $\int_0^{2\pi} \frac{Re^{it}}{(Re^{it}-z)^{m+2}} \cdot \log|f(Re^{it})|\mathrm{d}t \to 0$ (pointwise in z) as $R \to \infty$.

To start off, note that if $z \in \Delta(0,R)$ and $\gamma(t) = Re^{it}$ then:

$$\int_0^{2\pi} \frac{Re^{it}}{(Re^{it}-z)^{m+2}}\mathrm{d}t = \int_\gamma \frac{-i}{(w-z)^{m+2}}\mathrm{d}w = 0.$$

The prior equation works because $m+2 \geq 2$ so the residue of $\frac{-i}{(w-z)^{m+2}}$ at $w = z$ is 0. As a consequence of this relation, we can write:

$$\int_0^{2\pi} \frac{Re^{it}}{(Re^{it}-z)^{m+2}} \cdot \log|f(Re^{it})|\mathrm{d}t = \int_0^{2\pi} \frac{Re^{it}}{(Re^{it}-z)^{m+2}} \cdot (\log|f(Re^{it})| - \log(M(R)))\,\mathrm{d}t$$

In turn, we get that:

$$\left|\int_0^{2\pi} \frac{Re^{it}}{(Re^{it}-z)^{m+2}} \cdot \log|f(Re^{it})|\mathrm{d}t\right| \leq \int_0^{2\pi} \frac{R}{|Re^{it}-z|^{m+2}} (\log(M(R)) - \log|f(Re^{it})|)\,\mathrm{d}t$$

Now suppose $R > 2|z|$. Then $|Re^{it} - z|^{m+2} \geq (R/2)^{m+2}$. In turn, we have that:

$$\frac{R}{|Re^{it}-z|^{m+2}} (\log(M(R)) - \log|f(Re^{it})|) \leq \frac{2^{m+2}}{R^{m+1}} (\log(M(R)) - \log|f(Re^{it})|).$$

Hence, we have that:

$$\int_0^{2\pi} \frac{R}{|Re^{it}-z|^{m+2}} (\log(M(R)) - \log|f(Re^{it})|)\,\mathrm{d}t = \frac{2^{m+3}\pi}{R^{m+1}} \log(M(R)) - \frac{2^{m+2}}{R^{m+1}} \int_0^{2\pi} \log|f(Re^{it})|\mathrm{d}t$$

By Jensen's formula, we can write:

$$\int_0^{2\pi} \log|f(Re^{it})|\mathrm{d}t = 2\pi \left(\sum_{k=1}^{N(R)} \log(\tfrac{R}{|a_k|}) + \log|f(0)| \right) = 2\pi \sum_{k=1}^{N(R)} \log(\tfrac{R}{|a_k|}) + 0$$

Meanwhile, since f has order $\lambda < m+1$, we can find some $\varepsilon > 0$ such that $\lambda + \varepsilon < m+1$. Then we also know for $R \gg 0$ that $\log(M(R)) < R^{\lambda+\varepsilon}$. In turn:

$$\frac{2^{m+3}\pi}{R^{m+1}} \log(M(R)) - \frac{2^{m+2}}{R^{m+1}} \int_0^{2\pi} \log|f(Re^{it})|\mathrm{d}t = \frac{2^{m+3}\pi}{R^{m+1}} \left(\log(M(R)) - \sum_{k=1}^{N(R)} \log(\tfrac{R}{|a_k|}) \right)$$
$$\leq \frac{2^{m+3}\pi}{R^{m+1}} \log(M(R)) \leq 2^{m+3}\pi \frac{R^{\lambda+\varepsilon}}{R^{m+1}} \to 0 \text{ as } R \to \infty. \blacksquare$$

By combining lemmas A and B, we can conclude $\mathrm{D}^{m+1} g \equiv 0$. And now we are done. ■

Considering how much effort I put into proving lemma B, I'd like to use it to prove another random result. This wasn't part of lecture though so I'm just going on a random tangent.

An easy to see consequence of Hadamard is that if $f \not\equiv 0$ is any function of finite order that has an infinite set of zeros, then $f(z)$ can be written in the form $\frac{1}{z^m} e^{P(z)} \prod_{k=1}^{\infty} E_p(\frac{z}{a_k})$ where P is a polynomial and p and $\deg(P)$ are at most the order of f. But what about the the case where f has only finitely many zeros?

Proposition: Suppose f is an entire function with finite order λ as well as only finitely many zeros $a_1, \ldots, a_n$ (repeated according to multiplicity). Then λ is an integer and $f(z) = e^{P(z)} \cdot \prod_{k=1}^{n} (z - a_k)$ where $P(z)$ is a polynomial of degree λ.

Proof:
To start off, we know there exists an entire function F with no zeros such that $f(z) = F(z) \cdot \prod_{k=1}^{n}(z - a_k)$. Also because F is nonzero everywhere on $\mathbb{C}$ and $\mathbb{C}$ is simply connected, we know that F has some logarithm P. Therefore, we've proven that we can write $f(z) = e^{P(z)} \cdot \prod_{k=1}^{n}(z - a_k)$ where P is an entire function.

Now importantly f and e^P must both have order λ. Also $e^P \neq 0$ anywhere. Thus if $m \in \mathbb{N}$ satisfies that $m + 1 > \lambda$ then we know from lemma B that:

$$P^{(m+1)} = \mathrm{D}^m P' = \mathrm{D}^m \frac{(e^P)'}{e^P} = -m! \cdot 0 = 0.$$

This proves that $P(z)$ is a polynomial of degree at most m. Then to finish off, recall from *set 5 problem 6 on page 906* that $e^{P(z)}$ has order exactly equal to $\deg(P(z))$. So, the order λ of f is an integer and $\deg(P(z)) = \lambda$. ■

One corollary of this proposition is that if f is any entire function of finite order $\lambda \notin \mathbb{Z}$, then f must assume each of the values in its range infinitely many times. After all, if $\alpha \in f(\mathbb{C})$ is assumed only finitely many times then we'd know from prior proposition that $f(z) - \alpha$ has an integer order. Yet that would contradict that f doesn't have an integer order.

As a consequence of this, if f has only finitely many zeros $a_1, \ldots, a_n$ then it is natural to define $\mathrm{genus}(f) := \deg(P)$ if there exists a polynomial P such that $f(z) = e^{P(z)} \prod_{k=1}^{n}(z - a_k)$. Otherwise, we define $\mathrm{genus}(f) := \infty$. Then Hadamard's theorem still trivially holds in this case.

There are two more results from lecture that are fair game for the qualifying exam.

Proposition: Suppose f is entire, not constant, and of finite order. Then the image of f in $\mathbb{C}$ omits at most one value.

Proof:
Assume f omits distinct values α and β. By defining $f^{\text{new}} := \frac{f(z)-\alpha}{\beta-\alpha}$ we can assume without loss of generality that $\{\alpha, \beta\} = \{0, 1\}$.

Next, since f omits 0 we know that $f = e^g$ for some entire function g. Also, since f omits 1 we know that g omits 0. Yet since the order of f is finite, we know that $\mathrm{genus}(f) < \infty$ and thus g is a polynomial. The only way a polynomial g can have no zeros is if $g \equiv C$ for some constant C. Yet now we've proven that f is constant, which contradicts our initial assumption. ■

Remark: Picard's little theorem (which we won't prove in this class) strengthens this result by removing the assumption that f is of finite order.

One other result to be aware of is that the order of $\cos(z)$ and $\sin(z)$ is exactly 1.

To see why, note that $\cos(z) = \mathrm{Re}(e^{iz}) \leq |e^{iz}| = e^{\mathrm{Re}(iz)} \leq e^{|z|}$. A similar story holds for $\sin(z)$. Hence both functions have order at most 1.

Meanwhile, let $z_n = ni$. Then for any $\varepsilon > 0$ we can see that:

$$\cos(z_n) = \tfrac{1}{2}(e^{-n} + e^{+n}) > e^{n^{1-\varepsilon}} = e^{|z_n|^{1-\varepsilon}} \text{ when } |z_n| \gg 0.$$

A similar proof works for $\sin$ since $|\sin(z_n)| = \tfrac{1}{2}|e^{-n} - e^{n}|$.

I did not finish all the homework I was supposed to have. But here are two exercises I did do.

Set 5 Problem 3: Let f be an entire function satisfying that $|f'(z)| \leq e^{|z|}$ and $f(\sqrt{n}) = 0$ for all $n \in \mathbb{N}_{>0}$. Show that $f \equiv 0$.

Suppose for the sake of contradiction that $f \not\equiv 0$. Since f has zeros we know f is not a constant function. Hence $f' \not\equiv 0$ either. Also recall from *set 5 problem 2 on pages 905-906* that the order of f is equal to the order of f'. Thus since $|f'(z)| \leq e^{|z|}$, we know f has order 1. In turn, by Hadamard's theorem we know f has rank 0 or 1.

Yet simultaneously note that for any $t \leq 2$ we have that $\sum_{n=1}^{\infty} \frac{1}{(\sqrt{n})^t} = \infty$ Therefore, that would imply f has order at least 2, which is a contradiction. ■

Set 5 Problem 1:

(i) Let f, g be two entire functions of finite order $\leq \lambda$. Let $\varepsilon > 0$. Assume $f(a_n) = g(a_n)$ for a sequence $\{a_n\}_{n\in\mathbb{N}}$ of non-zero complex numbers with $\sum_{n=1}^{\infty} \frac{1}{|a_n|^{\lambda+\varepsilon}} = \infty$. Then show that $f = g$.

Consider the function $h = f - g$, and suppose for the sake of contradiction that $h \not\equiv 0$. Then by our assumption that $\sum_{n=1}^{\infty} \frac{1}{|a_n|^{\lambda+\varepsilon}} = \infty$ plus a prior homework problem (*set 4 problem 5 on page 885*) about the critical exponent α of h, we know that $\lambda + \varepsilon \leq \alpha \leq \lambda(h)$. In other words, h has order at least $\lambda + \varepsilon$.

Yet $h \not\equiv 0$ and is the difference of two functions with order at most λ. Thus by a prior lemma from my notes today, we know the order of h is at most λ. This is a contradiction.

(ii) Find all entire functions of f of finite order such that $f(\log(n)) = n^2$ for all integers $n \geq 1$.

Firstly, fix $f(z) = e^{2z}$. Then f is an entire function with order exactly equal to 1 (see *set 5 problem 6 on page 906* which I did prior to this problem).

Next, I claim that f is the only such function satisfying our claim. After all, if g were a distinct finite order entire function also satisfying our problem, then $h := f - g$ would have finite order as well as zeros at $\log(n)$ for every $n \geq 1$.

Yet note that for every integer k we have that $\sum_{n=2}^{\infty} \frac{1}{(\log(n))^k} = \infty$.

To see why, first evaluate by L'Hôpital's rule that:

$$\lim_{n\to\infty} \frac{n}{(\log(n))^k} = \lim_{x\to\infty} \frac{x}{(\log(x))^k} = \lim_{x\to\infty} \frac{x}{k(\log(x))^{k-1}} = \cdots = \lim_{x\to\infty} \frac{x}{k!} = \infty$$

Therefore, we know for large enough $N > 0$ that $\frac{1}{n} < \frac{1}{(\log(n))^k}$. In turn by the comparison test, we know $\sum_{n=2}^{\infty} \frac{1}{(\log(n))^k} = \infty$ diverges.

As a result, h must have infinite genus, and in turn by Hadarmard's theorem h must have infinite order. This is a contradiction. ■

5/5/2026

Math 148 Lecture Notes:

I am really far behind in this class so I'm probably going to spend several days catching up. The faster I catch up in this class the sooner I can copy down my notes on distributions from math 240c.

Firstly, here is a theorem from an earlier homework. Also for the sake of convenience, given any $x \in \mathbb{R}^N$ I will write $|x|$ to mean $\|x\|_2$.

Theorem (Maximum Principle on $\mathbb{R}^N \times \mathbb{R}_{>0}$): Let $\Omega = \mathbb{R}^N \times \mathbb{R}_{>0}$ and suppose u is a $C_b^{2,1}(\Omega) \cap C(\overline{\Omega})$ solution to the heat equation $u_t = \kappa \triangle u$ on Ω (where $\kappa > 0$). If $u(x,0) \leq M$ for all $x \in \mathbb{R}^N$ then $u(x,t) \leq M$ for all $(x,t) \in \Omega$.

Proof:
First consider any $T > 0$ and $\varepsilon > 0$, and then let:

$$v_{T,\varepsilon}(x,t) := u(x,t) - \frac{\varepsilon}{(2T-t)^{N/2}} e^{\frac{|x|^2}{4\kappa(2T-t)}}.$$

Then $v_{T,\varepsilon}$ satisfies the heat equation on $\mathbb{R}^N \times (0, 2T)$.

It's clear that $v_{T,\varepsilon}$ is well defined and $C^{2,1}$ on $(0,2T)$. The rest is just an annoying computation:

- $\partial_t v_{T,\varepsilon}(x,t) = u_t(x,t) - \frac{N\varepsilon}{2(2T-t)^{(N/2)+1}} e^{\frac{|x|_2^2}{4\kappa(2T-t)}} - \frac{\varepsilon|x|^2}{4\kappa(2T-t)^{(N/2)+2}} e^{\frac{|x|^2}{4\kappa(2T-t)}}$
- $\partial_{x_i} v_{T,\varepsilon}(x,t) = u_{x_i}(x,t) - \frac{\varepsilon x_i}{2\kappa(2T-t)^{(N/2)+1}} e^{\frac{|x|^2}{4\kappa(2T-t)}}$
- $\partial_{x_i,x_i} v_{T,\varepsilon}(x,t) = u_{x_i,x_i}(x,t) - \frac{\varepsilon}{2\kappa(2T-t)^{(N/2)+1}} e^{\frac{|x|^2}{4\kappa(2T-t)}} - \frac{\varepsilon x_i^2}{4\kappa^2(2T-t)^{(N/2)+2}} e^{\frac{|x|^2}{4\kappa(2T-t)}}$
- $\triangle v_{T,\varepsilon}(x,t) = \triangle u(x,t) - \frac{N\varepsilon}{2\kappa(2T-t)^{(N/2)+1}} e^{\frac{|x|^2}{4\kappa(2T-t)}} - \frac{\varepsilon|x|^2}{4\kappa^2(2T-t)^{(N/2)+2}} e^{\frac{|x|^2}{4\kappa(2T-t)}}.$

Thus $\partial_t v_{T,\varepsilon} - \kappa \triangle v_{T,\varepsilon} = u_t - \kappa \triangle u + 0 = 0.$

(The motivation for us considering the function $v_{T,\varepsilon}$ is that the term we are subtracting from u is always positive and in fact rapidly increases as $|x| \to \infty$. Hence, $v_{T,\varepsilon} < u$ for all $(x,t) \in \mathbb{R}^N \times [0,2T)$ and it will hopefully be way easier to prove $v_{T,\varepsilon}$ is bounded by M. Also importantly, we can see that $v_{T,\varepsilon} \to u$ pointwise on $\mathbb{R}^N \times [0,2T)$ as $\varepsilon \to 0$. So by taking $\varepsilon \to 0$ we can translate any boundedness result on $v_{T,\varepsilon}$ to a boundedness result on u.)

Since u is bounded above on $\partial\Omega$ by the constant M, and also because we assumed in the theorem statement that u is bounded on Ω, we can overall say that there exists some $K > 0$ such that $u(x,t) \leq K$ for all $(x,t) \in \overline{\Omega}$. Without loss of generality we can specifically pick $K > M$. Then I claim that there exists $R > 0$ such that $v_{T,\varepsilon}(x,t) \leq M$ for all $(x,t) \in (\mathbb{R}^N - B_R(0)) \times [0,T]$.

To see why, notice that it suffices for $u - v_{T,\varepsilon} \geq K - M$ if we want to have that $v_{T,\varepsilon}(x,t) \leq M$. Therefore, we just need to find some R such that:

$$u - v_{T,\varepsilon} = \frac{\varepsilon}{(2T-t)^{N/2}} e^{\frac{|x|^2}{4\kappa(2T-t)}} \geq K - M \text{ when } |x| > R \text{ and } t \in [0,T].$$

Fortunately, note that if $t \in [0,T]$ then $u - v_{T,\varepsilon} \geq \frac{\varepsilon}{(2T)^{N/2}} e^{\frac{|x|^2}{8\kappa T}}$.

Next, note that:

$$\frac{\varepsilon}{(2T)^{N/2}} e^{\frac{|x|^2}{8\kappa T}} \geq K - M \iff |x|^2 \geq 8\kappa T \cdot \log\left(\frac{(2T)^{N/2}}{\varepsilon}(K-M)\right) =: \alpha.$$

Therefore if $\alpha \leq 0$ we can set R to anything. Otherwise, we set $R \geq \sqrt{\alpha}$.

Yet now by the maximum principle (see *pages 875-876*) applied to $v_{T,\varepsilon}$ on the domain $\overline{B_R(0)} \times [0,T]$, we can conclude that:

$$\max_{\substack{x \in \overline{B_R(0)} \\ t \in [0,t]}} (v_{T,\varepsilon}(x,t)) = \max_{(x,t) \in \Gamma} (v_{T,\varepsilon}(x,t))$$

where $\Gamma := (\overline{B_R(0)} \times \{0\}) \cup (\partial B_R(0) \times [0,T])$.

Furthermore, on the first set in the above union we know $v_{T,\varepsilon} \leq u \leq M$. Meanwhile, we know from our prior reasoning that $v_{T,\varepsilon} \leq M$ on the second set in the above union. This proves that $v_{T,\varepsilon} \leq M$ on $\overline{B_R(0)} \times [0,T]$.

With that, we've shown that $v_{T,\varepsilon} \leq M$ on $\mathbb{R}^N \times [0,T]$. By taking $\varepsilon \to 0$ this then proves that $u \leq M$ on $\mathbb{R}^N \times [0,T]$ as well. And finally as T was arbitrary, we can take $T \to \infty$ to get that $u \leq M$ on $\mathbb{R}^N \times \mathbb{R}_{\geq 0}$. ■

The significance of this theorem is that we can conclude an analogous comparison principle and uniqueness theorem as on *pages 876-877*.

Corollary (Comparison Principle): Let $\Omega = \mathbb{R}^N \times \mathbb{R}_{>0}$. Also let $\kappa > 0$. If $u, \widetilde{u} \in C_b^{2,1}(\Omega) \cap C(\overline{\omega})$ solve the PDE $u_t = \kappa \triangle u$ on Ω, then $u \leq \widetilde{u}$ on $\mathbb{R}^N \times \{0\}$ implies that $u \leq \widetilde{u}$ on Ω.

Proof:
Apply the maximum principle to $u - \widetilde{u}$. ■

Theorem (Uniqueness of bounded solutions): Let $\Omega = \mathbb{R}^N \times \mathbb{R}_{>0}$. Then given any $\kappa > 0$, $\varphi \in C_b(\mathbb{R}^N)$, and $h \in C(\Omega)$, there is at most one solution in $C_b^{2,1}(\Omega) \cap C(\overline{\Omega})$ to the initial value problem:

$$u_t = \kappa \triangle u + h \text{ on } \Omega \text{ and } u(x,0) = \varphi(x) \text{ for all } x \in \mathbb{R}^N.$$

Proof:
If u and $\widetilde{u}$ are two solutions, we can apply the comparison principle to $u - \widetilde{u}$ and 0 as well as $\widetilde{u} - u$ and 0 to get that $u = \widetilde{u}$ on Ω. ■

Next, I want to follow Folland Real Analysis chapter 8.7 in order to give motivation for the solution to the heat equation.

The Time-Independent Case:
A linear differential operator L is a linear map on some space of functions $\mathbb{R}^N \to \mathbb{R}$ of the form $Lf(x) := \sum_{|\alpha| \leq m} a_\alpha(x) \partial^\alpha f(x)$ where each $a_\alpha \in C^\infty(\mathbb{R}^n)$.

If all the a_α are constant functions, we say L is a constant-coefficient operator.

Here $\alpha = (\alpha_1, \ldots, \alpha_N) \in (\mathbb{Z}_{\geq 0})^N$ is a multi-index. As a reminder, we say $|\alpha| := \sum_{k=1}^N \alpha_k$, $\alpha! := \prod_{k=1}^N \alpha_k!$, and $(x_1, \ldots, x_N)^\alpha := \prod_{k=1}^N x_k^{\alpha_k}$. Also if $f \in C^{|\alpha|}(U)$ then $\partial^\alpha f$ is defined by differentiating f (α_k)-many times with respect to x_k for each k.

In the special case of a constant-coefficient operator, note that by applying the Fourier transform on L^1 (see my math 240c notes), we have for well enough behaved f that:

$$(Lf)^\wedge(\xi) = \sum_{|\alpha| \leq m} a_\alpha (\partial^\alpha f)^\wedge(\xi) = \sum_{|\alpha| \leq m} a_\alpha (2\pi i \xi)^\alpha \widehat{f}(\xi)$$

As a reminder since different people have different conventions, if $f \in L^1(\mathbb{R}^n)$ then $\widehat{f}(\xi) := \int_{\mathbb{R}^N} f(x) e^{-2\pi i \xi \cdot x} \mathrm{d}x$. Also, if f is Schwartz then f will definitely be well enough behaved for the above reasoning.

In particular, define the constants $b_\alpha = (2\pi i)^{|\alpha|} a_\alpha$ and the operators $\mathrm{D}^\alpha := (2\pi i)^{-|\alpha|} \partial^\alpha$. Then $L = \sum_{|\alpha| \leq m} b_\alpha D^\alpha$ and $(Lf)^\wedge = \sum_{|\alpha| \leq m} b_\alpha \xi^\alpha \widehat{f}$.

If we write $P(\xi) = \sum_{|\alpha| \leq m} b_\alpha \xi^\alpha$, we call the polynomial P the symbol of the operator L (which we also write as $P(D)$). Then $(P(D)f)^\wedge = P\widehat{f}$.

This is useful for solving PDEs because if $P(D)u = f$ is a partial differential equation, then by applying the Fourier transform to both sides we get $\widehat{u} = \frac{1}{P}\widehat{f}$ and in turn $u = (\frac{1}{P}\widehat{f})^\vee$. In particular, in the special case that $\frac{1}{P}$ is the Fourier transform of some function ϕ we get that $u = \phi * f$.

Of course this method is heavily limited by the fact though that this reasoning only works if u is well enough behaved and if it is well-defined to take the Fourier transform of f and $\frac{1}{P}\widehat{f}$. Hence, none of this reasoning is very concrete yet.

Before moving on, another important observation is that:

$$\triangle = \sum_{k=1}^n \frac{\partial^2}{\partial x_k^2} = -4\pi^2 \sum_{k=1}^n D_k^2 = P(D).$$

Hence, the symbol of the Laplacian is $P(\xi) = -4\pi^2 |\xi|^2$.

A Time-Dependent Case:
We next consider differential operators L on a space of functions $f(x,t) : \mathbb{R}^N \times \mathbb{R} \to \mathbb{R}$ (or $\mathbb{R}^N \times \mathbb{R}_{>0} \to \mathbb{R}$). In particular, we consider the case where L has no mixed spatial and temporal partial derivatives, and every coefficient is constant.

Thus for any multi-index $\alpha \in (\mathbb{Z}_{\geq 0})^N$, we define ∂^α and D^α purely by taking partial derivatives with respect to $x = (x_1, \ldots, x_N)$. Then we want to write $L = P(D) + Q(\partial_t)$ where $P \in \mathbb{C}[\xi_1, \ldots, \xi_N]$ and $Q \in \mathbb{C}[t]$.

A general strategy we can then employ for solving the PDE $(Q(D)+P(\partial_t))u(x,t) = f(x,t)$ is that we shall take the Fourier transformation only with respect to the spatial argument.

For the sake of notation I will write $\widehat{u}(\xi,t) = \int_{\mathbb{R}^N} u(x,t)e^{-2\pi i \xi \cdot x}\mathrm{d}x$. Assuming u is well-behaved enough to apply Leibniz's rule, we can show that $\widehat{u}$ is (jointly) continuous and that $\partial_t \widehat{u}(\xi,t) = (\partial_t u)\widehat{\ }(\xi,t)$.

By applying the Fourier transformation in this way, we will get the equation that:

$$Q(\xi)\widehat{u}(\xi,t) + P(\partial_t)\widehat{u}(\xi,t) = \widehat{f}(\xi,t).$$

But now for each fixed ξ this is a linear ODE which will ideally let us solve for $\widehat{u}(\xi,t)$. Then hopefully by applying the (spatial) inverse Fourier transformation to $\widehat{u}$ we can solve for $u(x,t)$.

Here is how the above ideas were used to originally "solve" the heat equation $(\partial_t - \kappa\triangle)u = 0$ on $\mathbb{R}^N \times [0,\infty)$ where $\kappa > 0$ and we are given an initial condition $u(x,0) = f(x)$. (Also I'm intentionally being vague about which assumptions are needed because all I'm trying to do right now is give intuition for the formulas which will be proven rigorously in the actual 148 lecture.)

To start off, by applying the Fourier transformation to the above PDE, we get that:

$$(\partial_t + 4\pi^2\kappa|\xi|^2)\widehat{u}(\xi,t) = 0.$$

Also note that $\widehat{u}(\xi,0) = \widehat{f}(\xi)$.

Thus for any fixed ξ we consider the function $v(t) = \widehat{u}(\xi,t)$. Then we know v satisfies the initial value problem $v'(t) + 4\pi^2\kappa|\xi|^2 v(t) = 0$ and $v(0) = \widehat{f}(\xi)$. By solving for v we get that $v(t) = \widehat{f}(\xi)e^{-4\pi^2\kappa|\xi|^2 t}$. Hence, we've "shown" that:

$$\widehat{u}(\xi,t) = \widehat{f}(\xi)e^{-4\pi^2\kappa t|\xi|^2}$$

Yet now recalling that $\mathcal{F}(e^{-a\pi|x|^2}) = a^{-N/2}e^{-\pi|\xi|^2/a}$, we can solve that:

$$S^{(N)}(x,t) := (4\pi\kappa t)^{-N/2}e^{\frac{-|x|^2}{4\kappa t}} \text{ satisfies that } (S^{(N)})\widehat{\ }(\xi,t) = e^{-4\pi^2\kappa t|\xi|^2}$$

Therefore $\widehat{u}(\xi,t) = (f * S^{(N)})\widehat{\ }(\xi,t)$.

(Here convolution is with respect to the spatial argument of our functions.)

In turn, we should have that:

$$u(x,t) = (f * S^{(N)})(x,t) = \int_{\mathbb{R}^N} f(y) \cdot (4\pi\kappa t)^{-N/2} e^{\frac{-|x-y|^2}{4\kappa t}} \mathrm{d}y.$$

Going back to the actual lecture notes, we call $S^{(N)}(x,t) := (4\pi\kappa t)^{-N/2} e^{\frac{-|x|^2}{4\kappa t}}$ the fundamental solution of the heat equation $(\partial_t - \kappa\triangle)u = 0$ on $\mathbb{R}^N \times \mathbb{R}_{>0}$.

One can manually check that $S^{(N)}(x,t)$ does satisfy the heat equation. I don't feel like doing that though because it's just a bunch of symbol pushing.

> A physicist would have just accepted the prior two pages of napkin math and then concluded that $S^{(N)}$ satisfies the heat equation where we let the initial condition f be the Dirac delta function. :P

Theorem: Let $\kappa > 0$, $\Omega = \mathbb{R}^N \times \mathbb{R}_{>0}$, and $\varphi \in C_b(\mathbb{R}^N)$. Then the initial value problem $u_t = \kappa\triangle u$ on Ω and $u(x,0) = \varphi(x)$ on $\mathbb{R}^N$ is solved by:

$$u(x,t) = \int_{\mathbb{R}^N} \varphi(y) \cdot (4\pi\kappa t)^{-N/2} e^{\frac{-|x-y|^2}{4\kappa t}} \mathrm{d}y.$$

> As a side note, this integral is easily seen to be well-defined when $t > 0$. That said, the integrand isn't defined when $t = 0$. Thus, when we say $u(x,0) = \varphi(x)$ what we really mean is that $\lim\limits_{\substack{x \to x_0 \\ t \to 0}} u(x,t) = \varphi(x_0)$ for all $x_0 \in \mathbb{R}^N$.

Proof:

To start off, $u(x,t)$ is bounded by $\|\varphi\|_u \cdot 1$ due to Hölder's inequality. Also by Leibniz's rule on $\mathbb{R}^N \times \mathbb{R}_{>0}$ plus the fact that $S^{(n)}$ satisfies the equation on $\mathbb{R}^N \times \mathbb{R}_{>0}$, we can easily conclude that:

$$\begin{aligned} u_t(x,t) &= \int_{\mathbb{R}^N} S_t^{(N)}(x-y,t) \cdot \varphi(y) \mathrm{d}y \\ &= \int_{\mathbb{R}^N} -\kappa\triangle S^{(N)}(x-y,t) \cdot \varphi(y) \mathrm{d}y = -\kappa\triangle u(x,t). \end{aligned}$$

> For tips on getting the bounding functions to apply Leibniz's rule, note that:
>
> - $|S^{(N)}(x-y,t)| \le (4\pi\kappa t_1)^{-N/2} e^{\frac{-|x-y|^2}{2\kappa t_2}} \in L^1(y)$ if $t_1 < t < t_2$,
> - $|S^{(N)}(x'-y,t)| \le (4\pi\kappa t)^{-N/2} e^{\frac{-|x-y|^2 + r|x-y|}{2\kappa t_2}} \in L^1(y)$ if $x' \in B_r(x)$.

Meanwhile, to prove the needed limit first note that since φ is continuous we can find for any $\varepsilon > 0$ and $x_0 \in \mathbb{R}^N$ some $\delta > 0$ such that $|\varphi(x) - \varphi(x_0)| < \varepsilon/4$ when $x \in B_{2\delta}(x_0)$. Thus, it just suffices now to prove that there exists some $T > 0$ such that:

$$|u(x,t) - \varphi(x)| < 3\varepsilon/4 \text{ for all } x \in B_\delta(x_0) \text{ and } t \in (0,T).$$

Note that $\int_{\mathbb{R}^N} S^{(N)}(x,t)\mathrm{d}x = 1$ for all $t > 0$.

> To see why, first note by Tonelli's theorem that:
>
> $$\begin{aligned} \int_{\mathbb{R}^N} (4\pi\kappa t)^{-N/2} e^{\frac{-|x|^2}{4\kappa t}} \mathrm{d}x &= \prod_{k=1}^N \int_{\mathbb{R}} (4\pi\kappa t)^{-1/2} e^{\frac{-x_k^2}{4\kappa t}} \mathrm{d}x_k \\ &= \left(\int_{\mathbb{R}} (4\pi\kappa t)^{-1/2} e^{\frac{-y^2}{4\kappa t}} \mathrm{d}y \right)^N \end{aligned}$$
>
> Therefore, it suffices to prove that $\int_{\mathbb{R}} S^{(1)}(x,t)\mathrm{d}x = 1$. Fortunately, by an easy u-substitution we get that:
>
> $$\int_{\mathbb{R}} S^{(1)}(x,t)\mathrm{d}x = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-u^2} \mathrm{d}u = 1$$

Therefore, for any $x \in B_\delta(x_0)$ and $t \in \mathbb{R}_{>0}$ we have that:

$$\begin{aligned} |u(x,t) - \varphi(x)| &= \left| \int_{\mathbb{R}^N} S^{(N)}(x-y,t)\varphi(y)\mathrm{d}y - \left(\int_{\mathbb{R}^N} S^{(N)}(x-y,t)\mathrm{d}y \right) \varphi(x) \right| \\ &\leq \int_{\mathbb{R}^N} S^{(N)}(x-y,t) \cdot |\varphi(y) - \varphi(x)| \mathrm{d}y \end{aligned}$$

If we let $C = \|\varphi\|_u$, then we can in turn say that:

$$\begin{aligned} |u(x,t) - \varphi(x)| &\leq \int_{B_{2\delta}(x_0)} S^{(N)}(x-y,t) \cdot |\varphi(y) - \varphi(x)| \mathrm{d}y + 2C \int_{B_{2\delta}(x_0)^\mathsf{c}} S^{(N)}(x-y,t)\mathrm{d}y \\ &\leq \sup_{y \in B_{2\delta}(x_0)} |\varphi(y) - \varphi(x)| \cdot \int_{\mathbb{R}^N} S^{(N)}(y,t)\mathrm{d}y + 2C \int_{B_{2\delta}(x_0)^\mathsf{c}} S^{(N)}(x-y,t)\mathrm{d}y \\ &\leq \tfrac{2\varepsilon}{4}(1) + 2C \int_{B_{2\delta}(x_0)^\mathsf{c}} S^{(N)}(x-y,t)\mathrm{d}y. \end{aligned}$$

Now to finish off, we just need to pick $T > 0$ such that $2C \int_{B_{2\delta}(x_0)^\mathsf{c}} S^{(N)}(x-y,t)\mathrm{d}y < \frac{\varepsilon}{4}$ when $x \in B_\delta(x_0)$ and $t \in (0,T)$.

Fortunately, we know that if $|y - x_0| > 2\delta$ and $|x - x_0| < \delta$, then $|y - x| > \delta$. Hence, we can guarentee for all $x \in B_\delta(x_0)$ that:

$$2C \int_{B_{2\delta}(x_0)^\mathsf{c}} S^{(N)}(x-y,t)\mathrm{d}y \leq 2C \int_{B_\delta(x)^\mathsf{c}} S^{(N)}(x-y,t)\mathrm{d}y = 2C \int_{B_\delta(0)^\mathsf{c}} S^{(N)}(y,t)\mathrm{d}y$$

Thus, we've reduced our problem to finding some $T > 0$ such that:

$$\int_{B_\delta(0)^\mathsf{c}} S^{(N)}(y,t)\mathrm{d}y < \tfrac{\varepsilon}{4(2C+1)} \text{ for all } t \in (0,T).$$

Fortunately, note that there exists some positive constant A such that:

$$\int_{B_\delta(0)^\mathsf{c}} S^{(N)}(y,t)\mathrm{d}y = A \cdot \int_\delta^\infty \tfrac{r^{N-1}}{t^{N/2}} e^{\frac{-r^2}{4\kappa t}} \mathrm{d}r$$

Next, let $\psi(r,t) := \frac{r^{N-1}}{t^{N/2}} e^{\frac{-r^2}{4\kappa t}}$ and note that:

$$\psi_t(r,t) = \tfrac{r^{N-1} t^{-N/2}}{4\kappa t^2}(r^2 - 2N\kappa t) e^{\frac{-r^2}{4\kappa t}}.$$

Thus once $t < \frac{\delta^2}{2N\kappa}$, then we know for any fixed $r > \delta$ that $\psi(r,t)$ is strictly decreasing as t decreases. Yet also note that $\psi(r,t) > 0$ for all r and t. Hence, by the dominated convergence theorem (using $\int_\delta^\infty \psi(r, \frac{\delta^2}{2N\kappa})\mathrm{d}r$ as an upper bound), we can conclude that:

$$\lim_{t \to 0^+} \int_\delta^\infty \psi(r,t)\mathrm{d}r = \int_\delta^\infty (\lim_{t \to 0^+} \psi(r,t))\mathrm{d}r.$$

Finally, it is not difficult to prove for all $r > 0$ that $\psi(r,t) \to 0$ as $t \to 0^+$. Hence, $\int_\delta^\infty (\lim_{t \to 0^+} \psi(r,t))\mathrm{d}r = \int_\delta^\infty 0 \mathrm{d}r = 0$.

Since we've proven $\int_{B_\delta(0)^\mathsf{c}} S^{(N)}(y,t)\mathrm{d}y \to 0$ as $t \to 0^+$, it thus follows that there exists some $T > 0$ such that $0 < \int_{B_\delta(0)^\mathsf{c}} S^{(N)}(y,t)\mathrm{d}y < \frac{\varepsilon}{4(2C+1)}$ when $t < T$. ■

One notable remark is that the above equation for u is well-defined on $\mathbb{R}^N \times (0,\infty)$ even if φ is not continuous. Additionally, by inductively doing Leibniz's rule and using the fact that $e^{-a|x|^2}$ is Schwartz for all $a > 0$, we can show that u will be smooth.

Unfortunately some hackers just took down Canvas and so I don't have access to the professor's course notes anymore. Thus, unfortunately I have little choice but to switch over to studying the textbook on the course website (which sucks compared to Zlatos' notes). See *item 3* in the bibliography for a citation.

What about the diffusion equation $u_t - \kappa \triangle u = 0$ on the half line (with the homogeneous Dirichlet condition at the end point $x = 0$)? In other words, upon letting $\Omega = \mathbb{R}_{>0} \times \mathbb{R}_{>0}$ we want to solve the initial value problem:

$$u_t - \kappa u_{x,x} = 0 \text{ in } \Omega, \; u(x,0) = \varphi(x) \text{ for } x \in \mathbb{R}_{>0}, \text{ and } u(0,t) = 0 \text{ for } t \in \mathbb{R}_{>0},$$

where $\varphi \in C_b(\mathbb{R}_{>0})$ satisfies that $\lim_{x \to 0^+} \varphi(x) = 0$.

By slightly modified reasoning to that on *pages 917-919*, we can show that any solution to this initial value problem is unique. Hence, we shall only worry about finding a solution when given φ

To start off, let us define $\widetilde{\varphi}$ to be the *odd extension* of φ to all of $\mathbb{R}$. In other words:

$$\widetilde{\varphi}(x) = \begin{cases} \varphi(x) & \text{if } x > 0 \\ -\varphi(-x) & \text{if } x < 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Then $\widetilde{\varphi}$ is a continuous function on all of $\mathbb{R}$. Hence by our prior work, we know $\widetilde{u}(x,t) \coloneqq \int_{\mathbb{R}} S^{(1)}(x-y,t)\widetilde{\varphi}(y)\mathrm{d}y$ is the unique solution in $C_b^{2,1}(\mathbb{R} \times \mathbb{R}_{>0}) \cap C(\mathbb{R} \times \mathbb{R}_{\geq 0})$ to the initial value problem $\widetilde{u}_t - \kappa \widetilde{u}_{x,x} = 0$ in $\mathbb{R} \times \mathbb{R}_{>0}$ and $\widetilde{u}(x,0) = \widetilde{\varphi}(x)$ for all $x \in \mathbb{R}$.

Claim: $\widetilde{u}(x,t) = -\widetilde{u}(-x,t)$.

Proof:

Not only is $\widetilde{\varphi}$ an odd function but we also know $S^{(1)}(+x,t) = S^{(-1)}(-x,t)$. Hence:

$$\begin{aligned} -\widetilde{u}(-x,t) = \int_{\mathbb{R}} S^{(1)}(-x-y,t) \cdot -\widetilde{\varphi}(y)\mathrm{d}y &= \int_{\mathbb{R}} S^{(1)}(x-(-y),t) \cdot \widetilde{\varphi}(-y)\mathrm{d}y \\ &= \int_{\mathbb{R}} S^{(1)}(x-y,t) \cdot \widetilde{\varphi}(y)\mathrm{d}y = \widetilde{u}(x,t). \end{aligned}$$

In particular, by putting $x = 0$ we can see that $\widetilde{u}(0,t) = 0$ for all $t > 0$.

By letting $u = \widetilde{u}|_{\overline{\Omega}}$ we thus get that u satsfies the initial value problem for the half line.

> Remark: If we were to instead take the *even extension* of φ to all of $\mathbb{R}$ then that will similarly guarentee $\widetilde{u}$ is an even function (in x). But note that if f is even then f' is odd. Hence, we must have that $u_x(0,t) = 0$ for all $t > 0$. In turn, this shows that the even extension of φ yields a solution to the homogeneous Neumann condition at the end point $x = 0$.

A different question to ask about the heat equation is how do we deal wth the inhomogeneous diffusion equation $u_t - k u_{x,x} = f(x,t)$? Once again, to motivate the next results I'm first going to look at what the Fourier transform would predict.

Suppose u satisfies that $u_t - \kappa\triangle u = f(x,t)$ for all $(x,t) \in \mathbb{R}^N \times \mathbb{R}_{>0}$ and that $u(x,0) = \phi(x)$ for all $x \in \mathbb{R}^N$.

At all following steps we'll assume u, ϕ , and f are as well-behaved as necessary.

Firstly, we apply the Fourier transformation to our PDE with respect to the spatial variable to get that:

$$(\partial_t + 4\pi^2\kappa|\xi|^2)\widehat{u}(\xi,t) = \widehat{f}(\xi,t) \text{ and } \widehat{u}(\xi,0) = \widehat{\varphi}(\xi).$$

In particular, for any fixed ξ we consider the functions $v(t) = \widehat{u}(\xi,t)$ and $g(t) = \widehat{f}(\xi,t)$. Then we know v satisfies the initial value problem:

$$v'(t) + 4\pi^2\kappa|\xi|^2 v(t) = g(t) \text{ and } v(0) = \widehat{\varphi}(\xi).$$

To solve for $v(t)$, we multiply on an integrating factor $\mu(t) = e^{4\pi^2\kappa|\xi|^2 t}$. Then by product rule we can write:

$$(v(t)e^{4\pi^2\kappa|\xi|^2 t})' = e^{4\pi^2\kappa|\xi|^2 t} g(t).$$

By taking an antiderivative of both sides, we can then conclude there is some constant $C \in \mathbb{R}$ such that:

$$v(t) = e^{-4\pi^2\kappa|\xi|^2 t}C + e^{-4\pi^2\kappa|\xi|^2 t}\int_0^t e^{4\pi^2\kappa|\xi|^2 s} g(s)\mathrm{d}s.$$

By plugging in $t = 0$ we can solve that $C = \widehat{\varphi}(\xi)$. And with that, we've shown that:

$$\begin{aligned}\widehat{u}(\xi,t) &= e^{-4\pi^2\kappa|\xi|^2 t}\widehat{\varphi}(\xi) + \int_0^t e^{4\pi^2\kappa|\xi|^2(s-t)}\widehat{f}(\xi,s)\mathrm{d}s \\ &= (S^{(N)}(\cdot,t) * \varphi)^\wedge(\xi) + \int_0^t (S^{(N)}(\cdot,t-s) * f(\cdot,s))^\wedge(\xi)\mathrm{d}s\end{aligned}$$

To finish off, by applying Fubini's theorem we get that:

$$\int_0^t (S^{(N)}(\cdot,t-s) * f(\cdot,s))^\wedge(\xi)\mathrm{d}s = \left(\int_0^t (S^{(N)}(\cdot,t-s) * f(\cdot,s))\mathrm{d}s\right)^\wedge(\xi)$$

By inverting the Fourier transform, we thus should have that:

$$u(x,t) = (S^{(N)}(\cdot,t) * \varphi)(x) + \int_0^t (S^{(N)}(\cdot,t-s) * f(\cdot,s))(x)\mathrm{d}s$$

Gee, isn't math so easy when you don't need to justify anything?

Hopefully that is enough motivation to justify how someone would guess the following theorem.

Theorem: Let $\kappa > 0$, $\Omega = \mathbb{R}^N \times \mathbb{R}_{>0}$, and $\varphi \in C_b(\mathbb{R}^N)$. Also suppose $f \in C_b^{2,1}(\Omega)$ with bounded first and second derivatives. Then the initial value problem $u_t - \kappa\triangle u = f$ on Ω and $u(x,0) = \varphi(x)$ on $\mathbb{R}^N$ is solved by:

$$u(x,t) = \int_{\mathbb{R}^N} \varphi(y)\cdot S^{(N)}(x-y,t)\mathrm{d}y + \int_0^t\int_{\mathbb{R}^N} f(y,s)S^{(N)}(x-y,t-s)\mathrm{d}y\mathrm{d}s.$$

Proof:
To start off, note that:

$$\left|\int_0^t\int_{\mathbb{R}^N} f(y,s)S^{(N)}(x-y,t-s)\mathrm{d}y\mathrm{d}s\right| \le \int_0^t (\|f\|_u \cdot 1)\mathrm{d}s = t\|f\|_u.$$

As a result, we know that $\int_0^t\int_{\mathbb{R}^N} f(y,s)S^{(N)}(x-y,t-s)\mathrm{d}y\mathrm{d}s \to 0$ uniformly over x as $t \to 0$. Consequently, we can now assume without loss of generality $\varphi \equiv 0$ on $\mathbb{R}^N$ and therefore that the first term in the definition of u cancels. After all, if we were to prove that

$$v(x,t) \coloneqq \int_0^t \int_{\mathbb{R}^N} f(y,s) S^{(N)}(x-y, t-s) \mathrm{d}y\mathrm{d}s$$

is a particular solution to $v_t - \kappa \triangle v = f$, then we would know by the super-position principle that our original function $u(x,t)$ solves for the initial condition we actually want.

So now our task is just to show that $u(x,t) = \int_0^t \int_{\mathbb{R}^N} f(y,s) S^{(N)}(x-y,t-s)\mathrm{d}y\mathrm{d}s$ satisfies that $u_t - \kappa \triangle u = f$ for all $(x,t) \in \Omega$.

Unfortunately differentiating u is highly nontrivial and I have no fucking idea how to do this at least right now. After all, the integral expressions which we'd want to prove the partial derivatives equal are *not* absolutely convergent. Hence, all of our expressions for the derivatives of u will involve improper integrals. And right now I'm way too behind in this class to spend a day trying to figure this out, especially when the professor skipped going into detail on this as well.

Maybe I'll come back to this on *page 1224*. I somehow didn't notice until just now that f is assumed to be differentiable in the theorem statement. So right now my best educated guess for the proof of the derivative with respect to t (for example) is that for some $\varepsilon > 0$ we write the expression for $u_t(x,t)$ as:

$$\begin{aligned}\lim_{h\to 0+} \Bigg(&\int_0^{t-\varepsilon} \int_{\mathbb{R}^N} \frac{S^{(N)}(x-y,t+h-s) - S^{(N)}(x-y,t-s)}{h} f(y,s)\mathrm{d}y\mathrm{d}s \\ &+ \int_{t-\varepsilon}^t \int_{\mathbb{R}^N} \frac{S^{(N)}(x-y,t+h-s) - S^{(N)}(x-y,t-s)}{h} f(y,s)\mathrm{d}y\mathrm{d}s \\ &+ \tfrac{1}{h} \int_t^{t+h} \int_{\mathbb{R}^N} S^{(N)}(x-y,t+h-s) f(y,s)\mathrm{d}y\mathrm{d}s \Bigg)\end{aligned}$$

Then we can probably apply reasoning like in the proof of Leibniz' rule (see *pages 189-190*) to the first term. The second term meanwhile can probably be approximated as $\varepsilon f_t(x,t)$, and by taking $\varepsilon \to 0$ it will vanish. And finally, the third term will probably approach being $f(x,t)$.

Note from 10/22/2026: Yeah I was going about this the wrong way…

5/10/2026

Not having access to vector calculus theorems has been getting increasingly debilitating. So, I'm going to try and study that. The textbook I will be looking at for inspiration and guidance for this is Guillemin and Haine's *Differential Forms*. Also, instead of working with generalized manifolds as on *pages 130-151*, I'll be working with manifolds (and their surface measures) as developed on *pages 77-99*.

To motivate the first result, recall that the zero set of a function (satisfying certain assumptions) is a manifold without boundary. The next result will show a partial converse, that any C^r manifold without boundary is locally the zero set of a function.

Theorem B.18: Suppose $U \subseteq \mathbb{R}^k$ is an open set containing a point p and $\phi : U \to \mathbb{R}^N$ is a C^r map. Also suppose $\mathrm{D}\phi(p)$ has full rank.

(Guillen and Haine state the last requirement as $\mathrm{D}\phi(p) : \mathbb{R}^k \to \mathbb{R}^N$) is an injective map.

Then there exists a neighborhood U_p of p in U, a neighborhood V of $\phi(p)$ in $\mathbb{R}^N$, and a C^r diffeomorphism $\psi : V \to U_p \times \mathbb{R}^{N-k}$ such that $\psi \circ \phi : U_p \to U_p \times \mathbb{R}^{N-k}$ is equal to the embedding ι given by $\iota(x_1, \ldots, x_k) = (x_1, \ldots, x_k, \underbrace{0, \ldots, 0}_{N-k \text{ entries}})$.

Proof:
Write $\phi(x) = (\phi_1(x), \ldots, \phi_N(x))$ where each $\phi_i : \mathbb{R}^k \to \mathbb{R}$. Since $\mathrm{D}\phi(p)$ has full rank, we know $\mathrm{D}\phi(p)$ has k-linearly independent rows. By a change of basis in the codomain, we can assume without loss of generality that it is $\nabla\phi_1, \ldots, \nabla\phi_k$ which are linearly independent.

Next, let $\pi : \mathbb{R}^N \to \mathbb{R}^k$ be given by $(x_1, \ldots, x_n) \mapsto (x_1, \ldots, x_k)$. Then we can define $f := \pi \circ \phi$ and say that f is a C^r function with $\mathrm{D}f(p)$ being an invertible matrix. By the inverse function theorem, there is a neighborhood $U_p \subseteq U$ containing p and an open set $V' \subseteq \mathbb{R}^k$ such that f is a C^r diffeomorphism from U_p to V'. Now let $g := (f|_{U_p})^{-1}$ and $V := V' \times \mathbb{R}^{N-k}$ and define $G : V \to U_p \times \mathbb{R}^{N-k}$ by the map:

$$G(x_1, \ldots, x_N) = (g(x_1, \ldots, x_k), x_{k+1}, \ldots, x_N).$$

Now note that for any $x \in U_p$ we have that:

$$G \circ \phi(x) = (g(\pi \circ \phi(x)), \phi_{k+1}(x), \ldots, \phi_N(x)) = (x_1, \ldots, x_k, \phi_{k+1}(x), \ldots, \phi_N(x)).$$

Remark 1: G is itself an important function because it shows $\phi(U)$ is locally diffeomorphic to the graph of a function $\mathbb{R}^k \to \mathbb{R}^{N-k}$.

Meanwhile, consider the function $H : U_p \times \mathbb{R}^{N-k} \to U_p \times \mathbb{R}^{N-k}$ defined by:

$$H(x_1, \ldots, x_N) = (x_1, \ldots, x_k, x_{k+1} - \phi_{k+1}(x_1, \ldots, x_k), \ldots, x_N - \phi_N(x_1, \ldots, x_k))$$

Note that H is a C^r function with a C^r inverse given by:

$$(x_1, \ldots, x_N) \mapsto (x_1, \ldots, x_k, x_{k+1} + \phi_{k+1}(x_1, \ldots, x_k), \ldots, x_N + \phi_N(x_1, \ldots, x_k))$$

Finally, define $\psi : V \to U_p \times \mathbb{R}^{N-k}$ by $\psi := H \circ G$. Then ψ is a C^r diffeomorphism and $\psi \circ \phi(x_1, \ldots, x_k) = (x_1, \ldots, x_k, 0, \ldots, 0)$. ■.

Remark 2: If we project onto the final $N - k$ coordinates, we get a function $F : V \to \mathbb{R}^{N-k}$ such that $\phi(U_p) = F^{-1}(\{(0, \ldots, 0)\})$. Furthermore, $\mathrm{D}F(x)$ has rank $N - k$ for all $x \in V$.

Next, given a C^r k-dimensional manifold $M \subseteq \mathbb{R}^N$ without boundary as well as a point $p \in M$, we have two definitions of the tangent plane at p:

1. Given a chart $\phi : U \to V'$ where $p \in V' \subseteq M$, let $q = \phi^{-1}(p)$ and then define:
$$T_pM := \{(p, \vec{y}) : \vec{y} \in \mathrm{im}(\mathrm{D}\phi(q) : \mathbb{R}^k \to \mathbb{R}^N)\}.$$

2. Let $V \subseteq \mathbb{R}^N$ be an open set containing p and suppose $f : V \to \mathbb{R}^{N-k}$ is a C^r function such that $\mathrm{D}f(x)$ has rank $N - k$ for all $x \in V$ and $V \cap M = f^{-1}(\{0\})$.
Such a function f and open set V exists due to the prior theorem. Namely, let

$\psi : U' \to V' \subseteq M$ be any coordinate chart containing p and define via the prior theorem + remark the function $F : W \to \mathbb{R}^{N-k}$ (where $W \subseteq \mathbb{R}^N$ is some open set containing p satisfying that $W \cap V' = F^{-1}(\{0\})$). To finish off, note that since $(W \cap M) \cap V'$ is an open subset of M, there exists some open set $A \subseteq \mathbb{R}^N$ such that $A \cap M = (W \cap M) \cap V'$. And by setting $V = A \cap W$ and $f = F|_V$ we thus have that $V \cap M = f^{-1}(\{0\})$ like we wanted.

Now we define $T_pM \coloneqq \{(p, \vec{y}) : \vec{y} \in \ker(\mathrm{D}f(p) : \mathbb{R}^N \to \mathbb{R}^{N-k})\}$.

Lemma 4.2.8: Both of the above definitions are equivalent.

Proof:
Let $\phi : U \to V'$ and $f : V \to \mathbb{R}^{N-k}$ be functions as described in the prior two definitions, and also let $\phi^{-1}(p) = q$. By restricting U and V', we can assume without loss of generality that $V' \subseteq V$. In turn, the function $f \circ \phi : U \to \mathbb{R}^{N-k}$ is a well-defined function. Yet since $V' \subseteq M$, we also must have that $f \circ \phi$ is the constant zero function. Therefore, we can conclude that $\mathrm{D}f(p)\mathrm{D}\phi(q) = \mathbf{0}$.

This proves that $\mathrm{im}(\mathrm{D}\phi(q)) \subseteq \ker(\mathrm{D}f(p))$. And since both vector spaces are k-dimensional, the only way this is possible is if $\mathrm{im}(\mathrm{D}\phi(q)) = \ker(\mathrm{D}f(p))$. ■

Remark:
Since the first definition doesn't depend on the particular f and the second definition doesn't depend on the particular ϕ, this lemma also proves the tangent plane is defined independently of whichever ϕ and f is chosen.

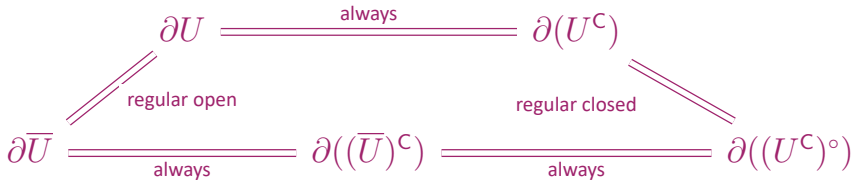
One particular note to be aware of is that T_pM is an $\mathbb{R}$-vector space when equipped with the operations $a(p, \vec{y}) = (p, a\vec{y})$ and $(p, \vec{y}_1) + (p, \vec{y}_2) = (p, \vec{y}_1 + \vec{y}_2)$. In particular, we can embed T_pM as a vector subspace of $T_p\mathbb{R}^N \coloneqq \{(p, \vec{y}) : \vec{y} \in \mathbb{R}^N\}$ in an obvious way. That said, I don't have time right now to get into the theory of differential forms and how they are defined as tensors on these spaces.

Given a topological space X, we say an open set $U \subseteq X$ is regular if $\partial U = \partial\overline{U}$. Meanwhile, we say a closed set $F \subseteq X$ is regular if $\partial F = \partial(F^\circ)$.

Lemma: A set $U \subseteq X$ is regular open if and only if U^C is regular closed.

Proof:
Note that for any $E \subseteq X$ we have that $(E^\mathsf{C})^\circ = (\overline{E})^\mathsf{C}$. Using this fact, we can then conclude for any open set $U \subseteq X$ that $\partial(U^\mathsf{C}) = \partial U$. After all, if $x \in \partial(U^\mathsf{C})$ then $x \notin U$ and $x \notin (U^\mathsf{C})^\circ = (\overline{U})^\mathsf{C}$. Hence $x \in \overline{U} - U = \partial U$, and this proves that $\partial(U^\mathsf{C}) \subseteq \partial U$. Meanwhile, if $x \in \partial U$ we know $x \notin U$ and $x \notin (\overline{U})^\mathsf{C} = (U^\mathsf{C})^\circ$. It follows that $x \in U^\mathsf{C} - (U^\mathsf{C})^\circ = \partial(U^\mathsf{C})$, and this proves that $\partial U \subseteq \partial(U^\mathsf{C})$.

Note that the above reasoning also lets us conclude that $\partial F = \partial F^\mathsf{C}$ for all closed sets $F \subseteq X$. Hence, we at last get the following equivalences:



■

Remark: The reason we care about if U is a regular open set is that if U is regular then for any $x \in \partial U$ and neighborhood N of x we have that N has a nonempty intersection with both U and $(U^C)^\circ = (\overline{U})^C$.

The result I shall prove next is that if a manifold (without boundary) is an open subset of the topological boundary of a regular open set (in the subspace topology), then we can strengthen theorem B.18 from before.

Lemma: Suppose $U \subseteq \mathbb{R}^N$ is a regular open set, M is a C^r $(N-1)$-dimensional manifold without boundary that is also an open subset of ∂U, and $p \in M$. Then there is an open set $V \subseteq \mathbb{R}^N$ containing p as well as a C^r function $f : V \to \mathbb{R}$ with full rank such that $V - M$ has two nonempty connected components, $V \cap U$ and $V - \overline{U}$, and we also have that $U \cap V = f^{-1}((-\infty, 0))$ and $M \cap V = f^{-1}(\{0\})$.

Proof:
Let $\widetilde{V} \subseteq \mathbb{R}^N$ be an open set and $\widetilde{\psi} : \widetilde{V} \to \mathbb{R}^N$ be an injective C^r function of full rank such that $\widetilde{V} \cap M = \widetilde{\psi}^{-1}(\mathbb{R}^{N-1} \times \{0\})$. (The fact such a function exists is theorem B.18 plus the comment between pages 926 and 927). By shrinking $\widetilde{V}$, we can further assume $\widetilde{V} \cap \partial U = \widetilde{V} \cap M$ since M is an open subset of ∂U. Then note $\widetilde{\psi}(\widetilde{V})$ is an open subset of $\mathbb{R}^N$ containing $\widetilde{\psi}(p)$. In turn, we can find an open ball $B_r(\widetilde{\psi}(p))$ contained in $\widetilde{\psi}(\widetilde{V})$. Hence, we define $V \coloneqq \widetilde{\psi}^{-1}(B_r(\widetilde{\psi}(p)))$ and $\psi \coloneqq \widetilde{\psi}|_V$.

Now we have that $\psi : V \to B_r(\psi(p))$ is a diffeomorphism (which importantly for us implies ψ is also a homeomorphism). Also note that:

$$B_r(\psi(p)) - \psi(M \cap V) = (B_r(\psi(p)) \cap \mathbb{H}^+) \cup (B_r(\psi(p)) \cap \mathbb{H}^-)$$

where $\mathbb{H}^+$ and $\mathbb{H}^-$ denote the upper and lower half spaces respectiely. The latter expression is a union of two convex disjoint open sets. Hence $B_r(\psi(p)) - \psi(M \cap V)$ has exactly two connected components. And since ψ is a homeomorphism, this also implies that $V - M$ has exactly two connected components, each of which maps homeomorphically to a distinct component of $B_r(\psi(p)) - \psi(M \cap V)$.

To finish off, note that because U is a regular open set and V is a neighborhood of $p \in \partial U$, we know that $(V \cap U)$ and $(V \cap (U^C)^\circ)$ are disjoint nonempty open sets partitioning $V - M$. It follows that the connected components of $V - M$ are precisely $V \cap U$ and $V - \overline{U}$. If we now define $f : V \to \mathbb{R}$ by projecting $\psi(x)$ onto its Nth coordinate, we will get that f is a C^r map with rank 1 on V and the sign of f will be positive on one of those components of $V - M$ and negative on the other.

Just negate f if necessary to force $f(U \cap V) \subseteq \mathbb{R}_{<0}$. ■

The significance of the prior result is that we now can talk about the unit normal vector to M at any point $p \in M$ that points away from U.

To see why, let all the variables refer to what they did in the prior lemma. Then note that the orthogonal complement of $\ker(\mathrm{D}f(p))$ in $\mathbb{R}^N$ is given by the span of $\nabla f(p)$ (where the latter is a nonzero vector because $\ker(\mathrm{D}f(p))$ is only $(N-1)$-dimensional). We can get a unit vector out of this by just defining $\widehat{n} := \frac{\nabla f(p)}{|\nabla f(p)|}$. Also by the way we constructed f, we know that by moving in a direction that increases f, we will leave $\overline{U}$. Similarly, by moving in a direction that decreases f we will enter U. Hence, $\widehat{n}$ points outside of $\overline{U}$ and $-\widehat{n}$ points inside of U.

5/11/2026

The more I look into it, proving divergence theorem is still too far out of reach for me to do, at least in the generality that I'll need it. Because of that and my stupid insistence on proving everything here instead of just accepting black boxes, I'm going to move my PDE notes to being entirely on paper for the time being. I'll hopefully be taking the grad PDE class anyways next year, and by then hopefully I'll have proven Divergence theorem for real.

5/15/2026

I feel like studying something fun today since I had my PDE midterm yesterday. So I'm going to study geometric measure theory from chapter 11 of Folland.

Let (X, ρ) be a metric space. An outer measure μ^* on X is called a metric outer measure when $\rho(A,B) > 0 \implies \mu^*(A \cup B) = \mu^*(A) + \mu^*(B)$.

> For a reminder of the definition of an outer measure, see my 240a notes. Also, given two nonempty sets A, B we define $\rho(A,B) := \inf\{\rho(a,b) : a \in A, b \in B\}$. If either A or B is empty, we define $\rho(A,B) = \emptyset$.

Theorem 11.16: If μ^* is a metric outer measure on X then every Borel subset of X is μ^*-measurable.

> Proof: It suffices to show that every closed set is μ^*-measurable since $\mathcal{B}_X$ is generated by the closed sets. So, let $F \subseteq X$ be closed and let $A \subseteq X$ be an arbitrary set with $\mu^*(A) < \infty$. To prove F is μ^*-measurable we need to show that:
>
> $$\mu^*(A \cap F) + \mu^*(A - F) \leq \mu^*(F).$$
>
> Let $B_n := \{x \in A - F : \rho(x, F) \geq n^{-1}\}$. Then $\{B_n\}_{n\in\mathbb{N}}$ is an increasing sequence of sets satisfying that $\bigcup_{n=1}^{\infty} B_n = A - F$ since F is a closed set. Also clearly $\rho(B_n, F) \geq n^{-1}$. Hence:
>
> $$\mu^*(A) \geq \mu^*((A \cap F) \cup B_n) = \mu^*(A \cap F) + \mu^*(B_n).$$

If we can show that $\mu^*(B_n) \to \mu^*(A-F)$ as $n \to \infty$ we will be done. Fortunately, as $\mu^*(B_n) \leq \mu^*(A-F)$ for all n we already know that $\limsup_{n\to\infty} \mu^*(B_n) \leq \mu^*(A-F)$. Therefore we can focus entirely on the liminf. Let $C_n := B_{n+1} - B_n$. By subadditivity we know $\mu^*(A-F) \leq \mu^*(B_n) + \sum_{j=n}^{\infty} \mu^*(C_j)$ for all n. Thus, if we can show that $\sum_{j=n}^{\infty} \mu^*(C_j) \to 0$ as $n \to \infty$, we will have proven $\liminf_{n\to\infty} \mu^*(B_n) \geq \mu^*(A-F)$.

Note that if $x \in C_{n+1}$ then $\rho(x,F) < \frac{1}{n+1}$. In turn, if $\rho(x,y) < \frac{1}{n(n+1)}$ we have that:

$$\rho(y,F) \leq \rho(x,y) + \rho(x,F) < \frac{1}{n(n+1)} + \frac{1}{n+1} = \frac{1}{n}.$$

It follows that $y \notin B_n$. Hence, we've proven that $\rho(C_{n+1}, B_n) \geq \frac{1}{n(n+1)}$.

Yet now it follows when $n \geq 2$ that:

$$\mu^*(B_n) \geq \mu^*(C_{n-1} \cup B_{n-2}) = \mu^*(C_{n-1}) + \mu^*(B_{n-2})$$

By induction, we can conclude for all n that:

- $\sum_{j=1}^{n} \mu^*(C_{2j}) \leq \mu^*(B_{2n+1}) \leq \mu^*(A-F)$,
- $\sum_{j=1}^{n} \mu^*(C_{2j-1}) \leq \mu^*(B_{2n}) \leq \mu^*(A-F)$.

Hence, $\sum_{j=1}^{\infty} \mu^*(C_j)$ is a convergent series with a total sum at most $2\mu^*(A-F)$. Consequently, $\sum_{j=n}^{\infty} \mu^*(C_j) \to 0$ as $n \to \infty$. ■

As for constructing metric outer-measures, I want to use slightly more general terminology than what Folland uses.

Given a collection of sets $\mathcal{F} \subseteq \mathcal{P}(X)$ such that $\emptyset \in \mathcal{F}$, we say $p : \mathcal{F} \to [0,\infty]$ is a gauge if $p(\emptyset) = 0$. Also, given any $\delta > 0$ we say $(E_i)_{i\in I} \subseteq \mathcal{P}(X)$ is a δ-covering of $E \subseteq X$ if $E \subseteq \bigcup_{i\in I} E_i$ and $\operatorname{diam}(E_i) \leq \delta$ for all i. (Here $\operatorname{diam}(\emptyset) := 0$.)

Now given a gauge $p : \mathcal{F} \to [0,\infty]$ and any $\delta > 0$, let us define:

$$\psi_\delta(E) := \inf(\{\textstyle\sum_{n\in\mathbb{N}} p(E_n) : \{E_n\}_{n\in\mathbb{N}} \subseteq \mathcal{F} \text{ is a } \delta \text{ covering of } E\} \cup \{\infty\})$$

Clearly ψ_δ is increasing as δ decreases. Hence, we can also define:

$$\psi(E) := \sup_{\delta>0} \psi_\delta(E) = \lim_{\delta\to 0} \psi_\delta(E).$$

I'm getting this terminology from some lecture notes by Marco Inversi (see *item 5* in the bibliography).

Theorem: ψ is a metric outer measure.

Why?

We can conclude from page 21 of my math 240a notes that ψ_δ is an outer measure for every δ, and from there it is easy to see ψ is also an outer measure.

Also, we can easily see that if $\rho(A,B) > \delta$ then $\psi_\delta(A \cup B) = \psi_\delta(A) + \psi_\delta(B)$. In turn, by taking $\delta \to 0$ we get that $\psi(A \cup B) = \psi(A) + \psi(B)$ and this proves that ψ is a metric outer measure.

We get the p-dimensional Hausdorff outer measure (denoted $\mathcal{H}_p$) by applying the prior construction to the gauge defined by $E \mapsto \operatorname{diam}(E)^p$ for all $E \in \mathcal{P}(X)$.

5/16/2026

I ended up falling asleep and I have other things I need to pivot to working on for the time being.

Math 200c Lecture Notes:

Proposition: Let A be an integrally closed domain, F its field of fractions, and L/F a finite degree normal extension. Let B be the integral closure of A in L. Also let $P \in \operatorname{Spec}(A)$ and $\mathcal{T} \coloneqq \{Q \in \operatorname{Spec}(B) : Q^c = P\}$. Then $G = \operatorname{Aut}(L/F)$ restricts to a group of automorphisms of B fixing A and G acts transitively on $\mathcal{T}$. In particular, this shows that $\mathcal{T}$ is finite (with at most $|\operatorname{Aut}_F(L)|$ elements).

Proof:
Suppose $\sigma \in G$. Then σ fixes F pointwise and in turn fixes A pointwise. Also as $A \subseteq B$ is an integral extension so is $A = \sigma(A) \subseteq \sigma(B)$. Yet now by the definition of the integral closure of A in K, we know that $\sigma(B) \subseteq B$. Since the same reasoning also shows that $\sigma^{-1}(B) \subseteq B$, we can conclude $\sigma(B) = B$. Hence, each automorphism of L fixing F restricts to an automorphism of B fixing A.

Next consider any fixed $Q \in \mathcal{T}$. Then:

$$\sigma(Q)^c = \sigma(Q) \cap A = \sigma(Q) \cap \sigma(A) = \sigma(Q \cap A) = \sigma(P) = P.$$

As a result, $\{\sigma(Q) : \sigma \in G\} = \{Q_1, \ldots, Q_n\}$ (where $Q_1 = Q$) is a subset of $\mathcal{T}$. To finish our proof, we need to show this subset is not proper.

Let $K \coloneqq \operatorname{Fix}(\operatorname{Aut}_F(L))$. By a corollary on *page 777*, we know L/K is a Galois extension. Meanwhile, I also will make the following claim about K/F.

Lemma 1: If L/F is a finite degree normal extension of fields and $K = \operatorname{Fix}(\operatorname{Aut}_F(L))$, then K/F is a purely inseparable extension.

Proof:
Suppose $\alpha \in K - F$. Then $\deg(m_{\alpha;F}) > 1$. Yet also note that since L/F is a normal extension, $m_{\alpha;F}$ splits in L. Plus, $\operatorname{Aut}_F(L)$ acts transitively on the zeros of $m_{\alpha;F}$ in L.

To see this, just note that since L/F is a finite degree extension, we know that there is some F-basis for L given by $\beta_1, \ldots, \beta_m$. In turn as L/F is a normal extension we can say that L is the splitting field of $f \coloneqq \prod_{i=1}^{m} m_{\beta_i;F}$ over F. And now we can conclude $\operatorname{Aut}_F(L)$ acts transitively on the zeros of $m_{\alpha;F}$ due to the problem on *page 815*.

Yet since $\operatorname{Aut}_F(L)$ fixes K, we can thus conclude that $m_{\alpha;F}(x) = (x - \alpha)^{\deg(m_{\alpha;F})}$. This proves that $m_{\alpha;F}$ is not separable.

Now let C be the integral closure of A in K and note that $A \subseteq C \subseteq B$. I shall make the following claim:

Lemma 2: Let A be an integrally closed domain, F its field of fractions, K/F a finite purely inseparable extension, and C the integral closure of A in K. Then for any $P \in \mathrm{Spec}(A)$ there is a unique $\widetilde{P} \in \mathrm{Spec}(C)$ with $\widetilde{P} \cap A = P$.

Proof:
To start off, if $K = F$ then this theorem is trivial. Meanwhile if K/F is purely inseparable and not a trivial extension, then we must have that $\mathrm{char}(F) = p > 0$. Yet in turn we know from set 2 problem 6 on *page 858* that for all $\alpha \in K$ there exists $m \geq 0$ such that $\alpha^{p^m} \in F$.

Consider any fixed $P' \in \mathrm{Spec}(C)$ satisfying that $P' \cap A = P$. (We know such an ideal exists by the going up theorem.) Then if $\alpha \in P'$ and m is such that $\alpha^{p^m} \in F$, we know that $\alpha^{p^m} \in F \cap P'$. Since A is integrally closed in F, we also know $C \cap F = A$. Hence $F \cap P' = F \cap C \cap P' = A \cap P' = P$.

It follows that if $\widetilde{P} := \{\alpha \in C : \exists m \geq 0 \text{ with } \alpha^{p^m} \in P\}$, then $P' \subseteq \widetilde{P}$. To finish off the proof of this lemma, we need to show that $\widetilde{P} \in \mathrm{Spec}(C)$ and that $\widetilde{P} \cap A = P$. That way, by incomparability we must have that $P' = \widetilde{P}$. And since the latter was defined independently of our choice of P', this will prove what we wanted.

Firstly I'll show $\widetilde{P} \lhd C$. Suppose $\alpha_1, \alpha_2 \in \widetilde{P}$ and let $c_1, c_2 \in C$. Then let $m_i \geq 0$ be such that $\alpha_i^{p^{m_i}} \in P$. Also let $n_i \geq 0$ be such that $c_i^{p^{n_i}} \in A$. Then if $N = \max(m_1, m_2, n_1, n_2)$, we have that:

$$(c_1\alpha_1 + c_2\alpha_2)^{p^N} = c_1^{p^N}\alpha_1^{p^N} + c_2^{p^N}\alpha_2^{p^N} \in P.$$

Hence $c_1\alpha_1 + c_2\alpha_2 \in \widetilde{P}$.

Next I show that $\widetilde{P} \in \mathrm{Spec}(C)$. Let $\alpha_1, \alpha_2 \in C$ satisfy that $\alpha_1\alpha_2 \in \widetilde{P}$. Then let $m \geq 0$ be such that $(\alpha_1\alpha_2)^{p^m} = \alpha_1^{p^m} \cdot \alpha_2^{p^m} \in P$. By increasing m if necessary we can also assume $\alpha_i^{p^m} \in A$ for both i. But now as P is prime, we must have that either $\alpha_1^{p^m} \in P$ or $\alpha_2^{p^m} \in P$. Hence, $\alpha_1 \in \widetilde{P}$ or $\alpha_2 \in \widetilde{P}$.

Finally, suppose $\alpha \in \widetilde{P} \cap A$, and let $m \geq 0$ be such that $\alpha^{p^m} \in P$. Since $P \in \mathrm{Spec}(A)$ and $\alpha \in A$, the only way this is possible is if $\alpha \in P$. So, $A \cap \widetilde{P} \subseteq P$. The other inclusion is trivial.

Now given any $Q \in \mathcal{T}$, we must have that $Q \cap C$ is unique prime ideal $\widetilde{P}$ in C whose contraction to A is P. Conversely, suppose $Q \in \mathrm{Spec}(B)$ satisfies that $Q \cap C = \widetilde{P}$. Then $Q \cap A = \widetilde{P} \cap A = P$. Hence, $\mathcal{T}$ is equivalently the set of all prime ideals of B lying over $\widetilde{P}$ in C. And if we can show that the set $\{\sigma(Q) : \sigma \in G\} = \{Q_1, \ldots, Q_n\}$ from before contains all such ideals lying over $\widetilde{P}$, we will be done. Fortunately now we will be aided by the fact that L/K is a Galois extension with $\mathrm{Gal}(L/K) = \mathrm{Aut}_F(L)$ and $\mathrm{Fix}(\mathrm{Gal}(L/K)) = K$.

Suppose $Q' \in \mathcal{T}$. Then for any $b \in Q'$ we have that $N_{L/K}(b) = \prod_{\sigma \in G} \sigma(b) \in K$. Yet also note $b \in B$ and $\sigma(B) = B$ for all $\sigma \in G$ implies that $N_{L/K}(b) \in B$. Hence, $N_{L/K}(b) \in B \cap K = C$ (since C is integrally closed in K). In turn:

$$N_{L/K}(b) \in Q' \cap C = \widetilde{P} \subseteq Q_1.$$

Since $\prod_{\sigma \in G} \sigma(b) \in Q_1$ and Q_1 is a prime ideal of B, we must have that $\sigma_0(b) \in Q_1$ for some $\sigma_0 \in G$. In turn $b \in \sigma_0^{-1}(Q_1) = Q_j$ for some j.

With that, we've proven that $b \in \bigcup_{j=1}^n Q_j$ for all $b \in Q'$. Or in other words, $Q' \subseteq \bigcup_{j=1}^n Q_j$. By the prime avoidence lemma on *page 893*, we can conclude that $Q' \subseteq Q_j$ for some j. Then by incomparability plus the fact that both Q' and Q_j contract to $\widetilde{P}$, we can conclude that $Q' = Q_j$. This shows $\mathcal{T} \subseteq \{Q_1, \ldots, Q_n\}$. ■

Theorem (Going Down): Let A be an integrally closed domain, F its field of fractions, K/F a finite extension, and B the integral closure of A in K. Given primes $P_1 \subseteq P_2$ in $\mathrm{Spec}(A)$, if $Q_2 \in \mathrm{Spec}(B)$ has $Q_2^c = P_2$ then there exists $Q_1 \subseteq Q_2$ with $Q_1 \in \mathrm{Spec}(B)$ such that $Q_1^c = P_1$.

Proof:
Let $\beta_1, \ldots, \beta_m$ span K as an F-vector space, and define $f := \prod_{i=1}^m m_{\beta_i;F}$. Then let L be the splitting field of f over K. Importantly, we can easily see that L is also the splitting field of f over F. Hence, L/F and L/K are finite normal extensions.

Next, let C be the integral closure of A in L. Thus $A \subseteq B \subseteq C$ with C/A and C/B both being integral extensions. Using lying over and going up, we can find $Q_1' \subseteq Q_2'$ in $\mathrm{Spec}(C)$ such that $Q_i' \cap A = P_i$. Similarly, we can find $Q_2'' \in \mathrm{Spec}(C)$ such that $Q_2'' \cap B = Q_2$. Yet now note that $Q_2'' \cap A = P_2 = Q_2' \cap A$. Thus by our prior theorem, there exists $\sigma \in \mathrm{Aut}(L/F)$ such that $\sigma(Q_2') = Q_2''$.

Let $Q_1'' := \sigma(Q_1')$ and then let $Q_1 := Q_1'' \cap B$. Then $Q_1 \subseteq Q_2'' \cap B = Q_2$. Also $Q_1 \cap A = Q_1'' \cap A = \sigma(Q_1') \cap A = P_1$. ■

An affine k-algebra is another name for a f.g. unital commutative k-algebra (where k is a field).

Note that if A is an affine k-algebra generated by $a_1, \ldots, a_n$, then equivalently that means that the natural map $\phi : k[x_1, \ldots, x_n] \to A$ sending k to k and x_i to a_i is surjective. In turn, by the first isomorphism theorem we have that $A \cong \frac{k[x_1, \ldots, x_n]}{\ker(\phi)}$. Also because $k[x_1, \ldots, x_n]$ is Noetherian, we know $\ker(\phi)$ is generated as an ideal by finitely many polynomials $f_1, \ldots, f_m$.

Let A be a k-algebra and $a_1, \ldots, a_n \in A$. We say $a_1, \ldots, a_n$ are algebraically independent over k if there does not exist $0 \neq f \in k[x_1, \ldots, x_n]$ with $f(a_1, \ldots, a_n) = 0$. Equivalently, $a_1, \ldots, a_n$ are algebraically independent over K iff the natural map $\phi : k[x_1, \ldots, x_n] \to A$ sending x_i to a_i and k to k is injective.

Lemma: Let $k[x_1, \ldots, x_n]$ be a polynomial ring over a field k. Suppose $0 \neq f \in k[x_1, \ldots, x_n]$. Then there are elements $x_1', \ldots, x_{n-1}', x_n$ in $k[x_1, \ldots, x_n]$ generating the entire ring and satisfying that $f = cx_n^d + g_{d-1}x_n^{d-1} + \ldots + g_1 x_n + g_0$ where each $g_i \in k[x_1', \ldots, x_{n-1}']$ and $c \in k^\times$. In particular, we let $x_i' = x_i - x_n^{d_i}$ for some integers $d_1, \ldots, d_{n-1}$.

Proof:

Write $0 \neq f = \sum_{(i_1,\ldots,i_n)\in(\mathbb{Z}_{\geq 0})^n} a_{(i_1,\ldots,i_n)} x_1^{i_1} \cdots x_n^{i_n}$ where all but finitely many summands are 0.

Next, for $1 \leq i \leq n-1$ we introduce the polynomial $x_i' = x_i - x_n^{d_i}$ where $d_i \geq 1$ is some integer. Note that $x_1', \ldots, x_{n-1}', x_n$ is still an algebraically independent generating set for $A \coloneqq k[x_1, \ldots, x_n]$.

To see why, note that the natural map $\phi : A \to A$ given by $x_i \mapsto x_i - x_n^{d_i}$ for $1 \leq i \leq n-1$ and $x_n \mapsto x_n$ has an inverse ϕ^{-1} which is the natural map given by $x_i \mapsto x_i + x_n^{d_i}$ for $1 \leq i \leq n-1$ and $x_n \mapsto x_n$. Hence, ϕ is injective and surjective.

Next, we rearrange to get that $x_i = x_i' + x_n^{d_i}$ for all $1 \leq i \leq n-1$. In turn by substituting this into a monomial $x_1^{i_1} \cdots x_n^{i_n}$ we get the expression:

$$(x_1' + x_n^{d_1})^{i_1} \cdots (x_{n-1}' + x_n^{d_{n-1}})^{i_{n-1}} x_n^{i_n}$$
$$= x_n^{d_1 i_1 + \ldots + d_{n-1} i_{n-1} + i_n} + \text{ lower degree terms in } x_n \text{ with coefficients in } k[x_1', \ldots, x_{n-1}'].$$

To finish off, let $S \coloneqq \{(i_1, \ldots, i_n) \in (\mathbb{Z}_{\geq 0})^n : a_{(i_1,\ldots,i_n)\neq 0}\}$. As mentioned before, S is finite. Thus, I claim that we can choose $d_1, \ldots, d_{n-1}$ such that the map $\psi : S \to \mathbb{Z}$ given by:

$$(i_1, \ldots, i_n) \mapsto d_1 i_1 + \ldots + d_{n-1} i_{n-1} + i_n$$

is injective.

There are many ways of doing this. One such way is to let:

$$N = \max\{\max_{1\leq k\leq n} i_k : (i_1, \ldots, i_n) \in S\} \text{ and } d_i = N^i.$$

Given a permutation $\sigma \in S_{n-1}$, it would also be valid to let $d_{\sigma(i)} = N^i$. Either way, our function ψ would then just be outputting some integer with digits i_j in base N.

Hopefully it's then clear how f takes on the form we want. ■

As a side note, if all we care about is the ideal generated by f or a particular zero of f, then we can rescale f by c^{-1} to force f to be monic.

Corollary: $\dim(k[x_1, \ldots, x_n]) = n$.

To start off, $\dim(k[x_1, \ldots, x_n]) \geq n$ since:

$$\{0\} \subsetneq \langle x_1 \rangle \subsetneq \langle x_1, x_2 \rangle \subsetneq \cdots \subsetneq \langle x_1, \ldots, x_n \rangle$$

is a chain of prime ideals in $k[x_1, \ldots, x_n]$. As for the other inequality, suppose by induction that the corollary holds for integers less than n. (Our base case is covered by the fact that if $n = 1$ then $k[x_1]$ is a PID and not a field, and thus $\dim(k[x_1]) = 1$).

Let $\{0\} \subsetneq P_1 \subsetneq \cdots \subsetneq P_m$ be a chain of prime ideals in $k[x_1, \ldots, x_n]$ of maximal length. Then choose $f \in P_1 - \{0\}$ so that $\{0\} \subsetneq \langle f \rangle \subseteq P_1$. We then shall consider the ring $A \coloneqq k[x_1, \ldots, x_n]/\langle f \rangle = k[\overline{x_1}, \ldots, \overline{x_n}]$.

By the prior lemma, we can find $x'_1, \ldots, x'_{n-1} \in k[x_1, \ldots, x_n]$ such that $k[x'_1, \ldots, x'_{n-1}, x_n]$ and f is a polynomial in x_n over the ring $k[x'_1, \ldots, x'_{n-1}]$ whose leading term is in k. By rescaling f, we can without loss of generality assume f is monic.

But then $B := k[\overline{x'_1}, \ldots, \overline{x'_{n-1}}]$ is a subring of A. Furthermore $\overline{x_n} \in A$ is a zero of the monic polynomial $\overline{f}(x) \in B[x]$. It follows that $B \subseteq B[\overline{x_n}] = A$ is an integral extension, and in turn $\dim(B) = \dim(A)$.

Next note that $\overline{x'_1}, \ldots, \overline{x'_{n-1}}$ are algebraically independent in A. Therefore, by our induction hypothesis we can conclude that $\dim(B) = \dim(\overline{x'_1}, \ldots, \overline{x'_{n-1}}) = n - 1$.
Yet we also can easily see that $\dim(A) \geq m - 1$. Thus, we've proven:

$$n - 1 \geq \dim(k[x_1, \ldots, x_n]) - 1.$$

Or in other words, $n \geq \dim(k[x_1, \ldots, x_n])$. ■

Noether Normalization Theorem: Let A be an affine k-algebra with generators $a_1, \ldots, a_n$. Then there is a subring $k[b_1, \ldots, b_r]$ where $b_1, \ldots, b_r$ are algebraically independent over k and A is a finitely generated $k[b_1, \ldots, b_r]$-module.

Remark: In particular, this says that A is integral over $k[b_1, \ldots, b_r]$ since module finite extensions are integral. Also if $k[b_1, \ldots, b_r] \subseteq A$ is an integral extension then $\dim(A) = \dim(k[b_1, \ldots, b_r]) = \dim(k[x_1, \ldots, x_r]) = r$.

Also, it is valid in this theorem for $\dim(A) = 0$. In that case, $r = 0$ and A is a finitely generated k-vector space.

Proof:
We shall induct on n. If $a_1, \ldots, a_n$ are already algebraically independent, then just take $b_i = a_i$ for all i. Then A is trivially a finitely generated $k[b_1, \ldots, b_n]$-module. Alternatively, in the trivial case that $n = 1$ and a_1 is not algebraically independent by itself (meaning $\exists f \in k[x]$ with $f(a_1) = 0$), then $A = k[a_1]$ is a finitely generated k-vector space with $\deg(f)$ many generators.

Now suppose $n > 1$ and there exists $0 \neq f \in k[x_1, \ldots, x_n]$ such that $f(a_1, \ldots, a_n) = 0$. Then let $d_1, \ldots, d_{n-1}$ be integers as in the lemma on the prior page, and set $x'_i = x_i - x_n^{d_i}$ for $1 \leq i \leq n - 1$. By rescaling f if necessary, we can assume that:
$f = x_n^d + g_{d-1}x_n^{d-1} + \ldots + g_1 x_n + g_0$ for some polynomials $g_j \in k[x'_1, \ldots, x'_{n-1}]$.

Yet now if we set $b_i = a_j - a_n^{d_i}$, we have that $a_n^d + \sum_{j=0}^{d-1} g_j(b_1, \ldots, b_{n-1})a_n^j = 0$. This proves that a_n is integral over $k[b_1, \ldots, b_{n-1}]$, and in turn $A = k[b_1, \ldots, b_{n-1}][a_n]$ is a finitely generated $k[b_1, \ldots, b_{n-1}]$-module.

By our inductive hypothesis, $k[b_1, \ldots, b_{n-1}]$ is a finitely generated module over some polynomial ring $k[c_1, \ldots, c_r] \subseteq A$ (here r may perhaps be 0). And by the same reasoning as was used to prove the tower lemma on *page 762*, we can conclude A is a finitely generated $k[c_1, \ldots, c_r]$-module. ■

Here are a few important consequences of Noether normalization:

Proposition: Let A be an affine k-algebra which is also a field. Then A/k is a finite degree field extension.

Proof:
By Noether normalization, $\dim(A) = 0$ and A being a finitely generated k-algebra implies that A is a finitely generated k-vector space. ■

Corollary: Let P be a maximal ideal of a polynomial ring $A = k[x_1, \ldots, x_n]$. Then A/P is a field and $k \subseteq K \coloneqq A/P$ is a finite degree field extension. If $k = \overline{k}$ (i.e. k is algebraically closed), then $P = \langle x_1 - a_1, x_2 - a_2, \ldots, x_n - a_n \rangle$ for some $a_1, \ldots, a_n \in k$.

Proof:
$K \coloneqq A/P$ is obviously a field. Yet it is also a finitely generate k-algebra still since $K = k[\overline{x_1}, \ldots, \overline{x_n}]$ where $\overline{x_i} = x_i + P$. So by our prior proposition, we know $k \subseteq K$ is a finite degree field extension.

Next, suppose $k = \overline{k}$. Then $P \in \operatorname{Spec}(k[x_1, \ldots, x_n])$ implies $k[x_i] \cap P \in \operatorname{Spec}(k[x_i])$ for all i. Furthermore, as $\frac{k[x_1,\ldots,x_n]}{P}$ is a finite field extension of k, we know there is some nonzero polynomial $g(x) \in k[x]$ with $g(x_i + P) = 0 + P$. Hence, $g(x_i) \in P \cap k[x_i]$ and this proves that $P \cap k[x_i] \neq \{0\}$. Because $k[x_i]$ is a PID, we can then conclude there is an irreducible polynomial $f \in k[x_i]$ such that $P \cap k[x_i] = \langle f \rangle$. Yet since k is algebraically closed, the only way f can be irreducible is if $f(x_i) = x_i - a_i$ for some $a_i \in k$.

Hence, for all i there exists $a_i \in k$ such that $P \cap k[x_i] = \langle x_i - a_i \rangle$. In turn, we can conclude that $I \coloneqq \langle x_1 - a_1, x_2 - a_2, \ldots, x_n - a_n \rangle \subseteq P$.

Yet note that I is the kernel of the evaluation map $e : k[x_1, \ldots, x_n] \to k$ given by $f(x_1, \ldots, x_n) \mapsto f(a_1, \ldots, a_n)$.

To see this, first note that $x_i - a_i \in I$ for all i implies $x_i \equiv a_i \pmod{I}$. As a result, we can conclude that $f(x_1, \ldots, x_n) \equiv f(a_1, \ldots, a_n) = e(f) \pmod{I}$ for all $f \in k[x_1, \ldots, x_n]$. Consequently, $f(x_1, \ldots, x_n) \equiv 0 \pmod{I}$ iff $e(f) \in I$. And since $e(f) \in k$ and $k \cap I = \{0\}$, this happens iff $f \in \ker(e)$. Hence, we've proven that $I = \ker(e)$.

Since the evaluation map e is surjective (which we know by considering constant polynomials), we can conclude by the first isomorphism theorem that $\frac{k[x_1,\ldots,x_n]}{I} \cong k$. In particular, this shows that $\frac{k[x_1,\ldots,x_n]}{I}$ is a field and thus I is a maximal ideal. And since $I \subseteq P$ with P being a proper ideal, we can conclude $I = P$. ■

This last result will be important when studying algebraic geometry.

Let A be a ring. The Jacobson radical of A is $J(A) = \bigcap_{P \in \operatorname{Max}(A)} P$. Clearly $\operatorname{Nil}(A) = \bigcap_{P \in \operatorname{Spec}(A)} P \subseteq \bigcap_{P \in \operatorname{Max}(A)} P = J(A)$. Usually this containment will be proper but in a special class of rings we have that $J(A) = \operatorname{Nil}(A)$.

A ring A is Jacobson if every prime ideal $P \lhd A$ is an intersection of maximal ideals.

Proposition (Algebraic Weak Nullstellensatz): Let A be any affine k-algebra. Then A is a Jacobson ring.

Proof:

Suppose not. Then there is some prime P that is not an intersection of maximal ideals. Since A is Noetherian, we can in turn pick P to be a maximal such ideal that is not an intersection of maximal ideals. In turn, if we let $B = A/P$ then we have that $\{0\}$ is not an intersection of maximal ideals in B and yet every nonzero prime $Q \in \text{Spec}(B)$ is an intersection of maximal ideals in B.

Since $\{0\}$ is not an intersection of maximal ideals in B, we know $J(B) \neq \{0\}$. Hence, we can pick $f \neq 0$ in $J(B)$. Yet now consider the localized ring $B_f := S^{-1}B$ where $S = \{1, f, f^2, \ldots\}$. Any prime ideal of B_f corresponds to a prime ideal Q of B such that $f \notin Q$. Since $f \in J(B)$ and every nonzero prime of B contains $J(B)$, we can conclude that the only prime ideal of B_f is the zero ideal. This implies the zero ideal of B_f is a maximal ideal, and in turn we know B_f is a field.

Yet now note that B_f is still a finitely generated k-algebra.

After all, we can write that B is a f.g. k-algebra generated by $b_1, \ldots, b_n$ over k. In turn, B_f is finitely generated (as a k-algebra) by $b_1, \ldots, b_n, f^{-1}$.

By the proposition on the prior page, we have that $k \subseteq B_f$ is a finite degree field extension. Yet as B is a domain and thus S has no zero-divisors, we know the natural embedding $B \to B_f$ given by $b \mapsto \frac{b}{1}$ is an injection. Hence, we have a ring extension $B \subseteq B_f$, and in turn we know B is a k-module of finite rank.

Lemma: If B is an integral domain and a k-vector space of finite dimension, then B is a field.

Proof:

Let $b \in B - \{0\}$. Since B is finite dimensioned, we know that $\{1, b, b^2, \ldots\}$ are not k-linearly independent. Hence, we can find a nonzero polynomial $g(x) \in k[x]$ such that $g(b) = 0$.

Without loss of generality, let us now assume g is a minimal monic polynomial of $k[x]$ with $g(b) = 0$. (Such a polynomial exists because $k[x]$ is a PID.) Then let us write $g(x) = c_n x^n + \ldots + c_1 x + c_0$ where $c_n \neq 0$. By the minimality of g we can also conclude $c_0 \neq 0$. Otherwise, we'd be able to write $g(b) = b \cdot h(b)$ for some $h(x) \in k[x]$ of lesser degree. And since $b \neq 0$ and B is a domain, we'd have that $g(b) = 0$ iff $h(b) = 0$.

Now $0 = c_n b^n + \ldots + c_1 b + c_0$ where $c_n \neq 0 \neq c_0$. Hence, after rearranging we get that $1 = b \cdot \frac{-1}{c_0}(c_n b^{n-1} + \ldots + c_1)$. This proves that b is a unit. And since b was arbitrary, we've shown that B is a field.

As a result of the above lemma, we know B is a field and thus $\{0\}$ is a maximal ideal of B. Yet this contradicts our prior conclusion that $\{0\}$ is not an intersection of maximal ideals of B. ■

For this next section, we'll assume k is an algebraically closed field. Then we call $\mathbb{A}^n := k^n$ the affine n-space. (The reason for the new symbol is purely psychological since we will be less interested in the vector space properties of $\mathbb{A}^n$ and moreso interested in the geometric properties).

We'll call $B := k[x_1, \ldots, x_n]$ be the coordinate ring of $\mathbb{A}^n$.

Also, given any collection of polynomials $S \subseteq B$ we define:

$$Z(S) := \{(a_1, \ldots, a_n) \in \mathbb{A}^n : f(a_1, \ldots, a_n) = 0 \text{ for all } f \in S\}.$$

> If $S = \{f\}$ then we abbreviate $Z(S)$ as $Z(f)$. Also, note that this is just slightly modified notation from the example of a functor on *pages 635-636*. (The difference in notations is entirely just the personal whims of Alireza versus Rogalski.)

Note that $Z(S) = Z(BS)$ where BS is the ideal generated by S in B. Hence, we can restrict ourselves to just considering $S = I$ where I is an ideal of B (if we wanted).

Conversely, let $X \subseteq \mathbb{A}^n$ and define the ideal:

$$I(X) = \{f \in B : f(a_1, \ldots, a_n) = 0 \text{ for all } (a_1, \ldots, a_n) \in X\}.$$

> Note that $I(X)$ is an ideal because it is the intersection of the kernels of the evaluation maps $f \mapsto f(a_1, \ldots, a_n)$ for all $(a_1, \ldots, a_n) \in X$.

Our next goal will be to establish some facts about the functions:

$$\mathcal{P}(\mathbb{A}^n) \underset{Z(\cdot)}{\overset{I(\cdot)}{\rightleftarrows}} \mathcal{P}(B)$$

Some hopefully obvious observations are as follows:

1. $S \subseteq T \subseteq B \implies Z(T) \subseteq Z(S)$,
2. $X \subseteq Y \subseteq \mathbb{A}^n \implies I(Y) \subseteq I(X)$,
3. $X \subseteq Z(I(X))$ for all $X \subseteq \mathbb{A}^n$.
4. $S \subseteq I(Z(S))$ for all $S \subseteq B$.

> You may notice these obvious relations are incredibly similar to the obvious relations of the Galois correspondance (see *page 776*). In fact, some people call two maps satisfying these observations a Galois connection.

Lemma:

- $I(Z(I(X))) = I(X)$ for all $X \subseteq \mathbb{A}^n$.

  Why?
  We know $X \subseteq Z(I(X))$. Hence $I(Z(I(X))) \subseteq I(X)$ by observation 2. Yet we also know from observation 4 that $I(X) \subseteq I(Z(I(X)))$.

- $Z(I(Z(S))) = Z(S)$ for all $S \subseteq B$.

  Why?
  The same idea as above works but using observations 1 and 3. ■

Theorem: There is a bijective correspondance given by:

$$\left\{ X \subseteq \mathbb{A}^n : X = Z(S) \text{ for some } S \subseteq B \right\} \underset{Z(\cdot)}{\overset{I(\cdot)}{\rightleftarrows}} \left\{ J \lhd B : J = I(X) \text{ for some } X \subseteq \mathbb{A}^n \right\}$$

Proof:
This should be obvious from the prior lemma. ■

Now it'd be nice to have an actual description of which sets and ideals are in the above domains for which $I(\cdot)$ and $Z(\cdot)$ are bijections. To this end, we define a topology on $\mathbb{A}^n$.

We'll say $X \subseteq \mathbb{A}^n$ is a *closed* set if $X = Z(S)$ for some $S \subseteq B$.

To check that this defines a topology on $\mathbb{A}^n$, note that:

- $\emptyset = Z(c)$ for any constant nonzero polynomial $c \in B$. Hence, $\emptyset$ is a closed set.
- $\mathbb{A}^n = Z(0)$ where 0 is the constant zero polynomial in B. Hence, $\mathbb{A}^n$ is a closed set.
- Let $X_1 = Z(S_1), \ldots, X_n = Z(S_n)$ be closed subsets of $\mathbb{A}^n$. Then let:
  $$S := \{\textstyle\prod_{i=1}^n f_i : f_i \in S_i \text{ for all } i\}.$$
  Note that if $f = \prod_{i=1}^n f_i$ is a polynomial in S then $f(a_1, \ldots, a_n) = 0$ if and only if $f_i(a_1, \ldots, a_n) = 0$ for some i. As a result, we can see that $\bigcup_{i=1}^n X_i \subseteq Z(S)$. To get the other inclusion, suppose $\vec{a} \notin \bigcup_{i=1}^n X_i$. Then for each i we can find some $\widetilde{f_i} \in S_i$ with $f_i(\vec{a}) \neq 0$. In turn $\widetilde{f} = \prod_{i=1}^n \widetilde{f_i}$ is an element of S with $\widetilde{f}(\vec{a}) \neq 0$. So, $\vec{a} \notin Z(S)$. Hence, we've proven that $\bigcup_{i=1}^n X_i$ is closed.
- Let $\{X_\alpha\}_{\alpha \in A}$ be an arbitrary collection of closed subsets of $\mathbb{A}^n$, and for each α let $S_\alpha \subseteq B$ be such that $X_\alpha = Z(S_\alpha)$. Then let $S := \bigcup_{\alpha \in A} S_\alpha$. It is easy to see that $\bigcap_{\alpha \in A} X_\alpha = Z(S)$.

The topology that this defines on $\mathbb{A}^n$ is called the Zariski topology. Clearly from the name of this topology, there is some connection to the Zariski topology on $\operatorname{Spec}(B)$.

Consider $Z(J)$ where $J \lhd B$. Then $(a_1, \ldots, a_n) \in Z(J)$ implies $f(a_1, \ldots, a_n) = 0$ for all $f \in J$. Equivalently, this means that f is contained in the maximal ideal $\langle x_1 - a_1, \ldots, x_n - a_n \rangle$ for all $f \in J$. Or in other words, $J \subseteq \langle x_1 - a_1, \ldots, x_n - a_n \rangle$.

As a result, we can write that:

$$Z(J) = \{(a_1, \ldots, a_n) \in \mathbb{A}^n : J \subseteq \langle x_1 - a_1, \ldots, x_n - a_n \rangle\}$$

But now by the proposition on *page 937*, we know there is a bijective correspondance from $\mathbb{A}^n$ to $\mathrm{Max}(B)$ given by $(a_1, \ldots, a_n) \mapsto \langle x_1 - a_1, \ldots, x_n - a_n \rangle$.

(Here we are finally using that k is an algebraically closed field.)

By identifying any $\vec{a} \in \mathbb{A}^n$ with it's corresponding ideal in $\mathrm{Max}(B)$, we can thus say that:

$$Z(J) = \{P \in \mathrm{Max}(B) : J \subseteq P\} = V(J) \cap \mathrm{Max}(B)$$

As a result, the map $(a_1, \ldots, a_n) \mapsto \langle x_1 - a_1, \ldots, x_n - a_n \rangle$ is a homeomorphism from $\mathbb{A}^n$ equipped with the Zariski topology to $\mathrm{Max}(B)$ equipped with the Zariski subspace topology (relative to $\mathrm{Spec}(B)$).

As a side note, this proves that the Zariski topology on $\mathbb{A}^n$ is T_1.

Proposition: Any prime ideal $P \lhd k[x, y]$ (where $k = \overline{k}$) is of the form:

- $P = \langle x - a, y - b \rangle$,
- $P = \langle f \rangle$ (where f is irreducible in $k[x, y]$),
- $P = \{0\}$.

Proof:
$k[x, y]$ has a Krull dimension of 2. Therefore, the height of any prime ideal P of $k[x, y]$, (which is defined as $\max\{n \in \mathbb{Z}_{\geq 0} : \exists Q_0 \subsetneq \cdots \subsetneq Q_n = P \text{ in } \mathrm{Spec}(k[x, y])\}$), is either 0, 1, or 2.

Since $k[x, y]$ is a domain, we know the only prime ideal with height 0 is $\{0\}$. Meanwhile, consider the following lemma.

Lemma: If D is a UFD and $P \lhd D$ is a prime ideal with height 1, then P is a principal ideal.

Proof:
Since P has height 1 we know $P \neq \{0\}$. Hence, we can find $x \in P$ such that $x \notin \{0\} \cup D^\times$. Then as D is a UFD, we can factor x as a product of irreducible elements $p_1 \cdots p_n$. In particular, as P is a prime ideal, we then must have that $p_i \in P$ for some i. Yet also, because D is a UFD we know $\langle p_i \rangle$ is a nonzero prime ideal that is contained in P. Since P has height 1, the only way we don't have a contradiction is if $P = \langle p_i \rangle$.

This lemma shows every height 1 ideal in $k[x, y]$ is a principal ideal. To finish off, note that if a prime ideal has height 2 than it must be maximal. But by the proposition on *page 937* we know what every maximal ideal looks like. ■

As a consequence of this proposition, we can say that any proper closed subset $F \subseteq \mathbb{A}^2$ is a finite union of irreducible curves in $\mathbb{A}^2$ of the form:

$$\{(a_1, a_2) : f(a_1, a_2) = 0 \text{ where } f \in k[x, y] \text{ is irreducible}\}$$

as well as individual points $\{(a_1, \ldots, a_n)\}$.

Why?
If $F = \emptyset$ then it is trivially a finite union of zero irreducible curves. So, we can now assume that $F \neq \emptyset$.

Next, note that $F = V(J) \cap \text{Max}(k[x,y])$ for some $J \lhd k[x,y]$. And importantly since $F \neq \emptyset$, we know that $J \neq k[x,y]$. In other words, J is a proper ideal of $k[x,y]$. But now since $k[x,y]$ is a Noetherian ring, we know that there are only finitely many minimal primes $Q_1, \ldots, Q_n$ over J (see *page 900*). And then, we can see that $V(J) = \bigcup_{i=1}^{n} V(Q_i)$.

Since $F \neq \mathbb{A}^2$, we know none of the minimal primes over J can be the zero ideal. Hence, we now can just apply the prior prosition in order to describe what $V(Q_i) \cap \text{Max}(k[x,y])$ looks like for each i.

Question: What are the ideals of B of the form $I(X)$ where $X \subseteq \mathbb{A}^n$ is a closed set?

Proposition: Given $J \lhd B$, we have that:
$I(Z(J)) = \sqrt{J}$ $(= \bigcap_{P \in V(J)} P = \{f \in B : f^n \in J \text{ for some } n \in \mathbb{Z}_{\geq 0}\})$.

Proof:
Note that
$$I(Z(J)) = \{f \in B : f(\vec{a}) = 0 \text{ for all } \vec{a} \in Z(J)\} = \bigcap_{P \in V(J) \cap \text{Max}(B)} P.$$

Then since B is a Jacobson ring, every prime ideal P is an intersection of maximal ideals. In particular, if $P \in V(J)$ then P must be an intersection of ideals in $V(J) \cap \text{Max}(B)$. Hence:
$$\bigcap_{P \in V(J) \cap \text{Max}(B)} P = \bigcap_{P \in V(J)} P = \sqrt{J}$$

Corollary: There are inverse inclusion-reversing bijections:

$$\{X \subseteq \mathbb{A}^n : X \text{ is Zariski closed}\} \underset{Z(\cdot)}{\overset{I(\cdot)}{\rightleftarrows}} \left\{J \lhd B : J = \sqrt{J'} \text{ for some } J' \lhd B\right\}$$

This next section will be on a completely different topic. For this section, we'll mostly assume A is Noetherian throughout.

Let M be an A-module. Then recall that $\text{Ann}_A(M) \lhd A$. Also note that M is naturally an $A/\text{Ann}_A(M)$-module (where we define $(a + \text{Ann}_A(M))m = am$).
After all, if $a_1 - a_2 \in \text{Ann}_A(M)$ then $(a_1 - a_2)m = 0$. Hence $a_1 m = a_2 m$.

We define the Krull dimension of M to be $\dim(M) = \dim(A/\text{Ann}_A(M))$.
Note: If M is finitely generated then $\dim(M)$ is equal to the maximal chain length of $\text{supp}(M) \subseteq \text{Spec}(A)$. (See *pages 672-673*).

Let M be an A-module. An associated prime of M is a prime ideal P of A such that $P = \text{Ann}_A(m)$ for some $m \in M$ with $m \neq 0$. The set of all associated primes is $\text{Ass}(M)$.

Remark: If $P \in \text{Spec}(A)$ and we view A/P as an A-module, then the fact that A/P is an integral domain means that $\text{Ass}(A/P) = \{P\}$.

Proposition: Let M be an A-module. Then $P \in \text{Ass}(M)$ iff M has as a submodule $N \cong A/P$.

$(\Longrightarrow)$

Suppose $P \in \text{Ass}(M)$ with $P = \text{Ann}_A(m)$ for some $m \neq 0$, then $N := Am$ is a submodule of M such that $N \cong A/P$. After all, consider the map $\phi : A \to N$ given by $\phi(a) = am$. Then $\ker(\phi) = P$. Hence, by the first isomorphism theorem we get that $A/P \cong N$.

$(\Longleftarrow)$

Let $\phi : A/P \to N$ be the module isomorphism and then let $m := \phi(1 + P)$. Now $am = a\phi(1 + P) = \phi(a + P)$ for all $a \in A$. Since ϕ is injective, we thus have that $am = 0$ iff $a + P = 0 + P$. Hence, $\text{Ann}_A(m) = P$. ■

Lemma: Let $\{0\} \longrightarrow M_1 \xhookrightarrow{\phi_1} M_2 \xrightarrow{\phi_2} M_3 \longrightarrow \{0\}$ be a short exact sequence of A-modules. Then:

1. $\text{Ass}(M_1) \subseteq \text{Ass}(M_2) \subseteq \text{Ass}(M_1) \cup \text{Ass}(M_3)$.
2. If the SES splits then $\text{Ass}(M_2) = \text{Ass}(M_1) \cup \text{Ass}(M_3)$.

Proof:
To start off $\phi_1(M_1)$ is a submodule of M_2. Hence any submodules of M_1 are isomorphic to a submodule of M_2. In particular, if M_1 has a submodule isomorphic to A/P for some $P \in \text{Spec}(A)$, then so does M_2. Hence, $\text{Ass}(M_1) \subseteq \text{Ass}(M_2)$.

Next, let $0 \neq m \in M_2$ be such that $\text{Ann}_A(m) = P \in \text{Spec}(A)$. If $m \in \phi_1(M_1) = \ker(\phi_2)$ then we know $P \in \text{Ass}(M_1)$ by analogous reasoning as in the first paragraph. Hence, we may assume $\phi_2(m) \neq 0$. But then also note by the first isomorphism theorem that $M_2/\phi(M_1) \cong M_3$ by the map $m' + \phi_1(M_1) \mapsto \phi_2(m')$. So, we shall now ask what $\text{Ann}_A(m + \phi_1(M_1))$ is.

Clearly $P = \text{Ann}_A(m) \subseteq \text{Ann}_A(m + \phi_1(M_1))$. On the other hand, suppose that $a(m + \phi_1(M_1)) = 0$ for some $a \in A - P$. Then $am \in \phi_1(M_1)$ but $am \neq 0$. Also note that since P is prime and $a \notin P$, we have that $\text{Ann}_A(am) = P$. Therefore, we have once again shown that $P \in \text{Ass}(M_1)$.

Hence, we now know that either $P = \text{Ann}_A(m + \phi_1(M_1))$ or $P \in \text{Ass}(M_1)$. And in the former case, we then equivalently know that $P = \text{Ann}_A(\phi_2(m))$. So, $P \in \text{Ass}(M_3)$. This finishes the proof of claim 1.

As for claim 2, if our SES splits then M_3 is a isomorphic to a submodule of M_2. Hence by identical reasoning to the claim $\text{Ass}(M_1) \subseteq \text{Ass}(M_2)$, we can conclude $\text{Ass}(M_3) \subseteq \text{Ass}(M_2)$. It follows that $\text{Ass}(M_1) \cup \text{Ass}(M_3) \subseteq \text{Ass}(M_2)$. ■

Proposition: Let M be a nontrivial finitely generated module over a nontrivial Noetherian ring A.

1. $\text{Ass}(M) \neq \emptyset$. (This claim doesn't require M to be finitely generated.)
2. M has a prime filtration. In other words, there exists a sequence of modules $\{0\} = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_n = M$ with $M_{i+1}/M_i \cong A/P_i$ for some $P_i \in \text{Spec}(A)$.
3. Every prime $P \in \text{Ass}(M)$ is one of the P_i in claim 2 (which in turn means $\text{Ass}(M)$ is finite).

Proof:
To show claim 1, consider the set of ideals $\{\text{Ann}_A(m') : m' \in M - \{0\}\}$. Because A is Noetherian, there is a maximal ideal $P = \text{Ann}_A(m)$ in this set. Then we claim P is prime (and thus in $\text{Ass}(M)$).

After all, firstly note that since $1m = m \neq 0$, we know P is a proper ideal. Also, suppose $ab \in P$ and $a \notin P$. Then $am \neq 0$, meaning $\text{Ann}_A(am)$ is in our original set of ideals. Yet also note that $P = \text{Ann}_A(m) \subseteq \text{Ann}_A(am)$. So by the maximality of P in our original set, we must have that $P = \text{Ann}_A(am)$. Since $b \in \text{Ann}_A(am)$, this proves that $b \in P$.

To show claim 2, note that since M has at least one associated prime P_1, we know M has a submodule $M_1 = Am_1 \cong A/P_1$. Similarly M/M_1 has an associated prime P_2. Thus, M/M_1 has a submodule $M_2/M_1 \cong A/P_2$. And now we keep repeating the same reasoning so on and so forth with M/M_2, M/M_3, etc. until we get to the situation where $M_n = M$ and can ascend no further.

Note that because M is a finitely generated module over a Noetherian ring, we know M is a Noetherian module and thus there will in fact be some $n \in \mathbb{N}$ with $M_n = M$.

If not then we'd have constructed an infinite ascending chain of submodules of M, violating that M is Noetherian.

Hence, this shows claim 2.

Finally to show claim 3, consider any prime filtration

$$\{0\} = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_n = M$$

as described in claim 2. Also let $P_1, \ldots, P_n \in \text{Spec}(A)$ such that $M_{i+1}/M_i \cong A/P_i$ for each i. Then consider the short exact sequence below:

$$\{0\} \longrightarrow M_{n-1} \xhookrightarrow{i} M_n \xrightarrow{\pi} \frac{M_n}{M_{n-1}} \longrightarrow \{0\}$$

From the prior lemma we know $\text{Ass}(M) \subseteq \text{Ass}(M_{n-1}) \cup \text{Ass}(M_n/M_{n-1})$. Yet, we also know $\text{Ass}(M_n/M_{n-1}) = \text{Ass}(A/P_n) = \{P_n\}$. Thus, we can conclude that $\text{Ass}(M) \subseteq \text{Ass}(M_{n-1}) \cup \{P_n\}$. By repeating this reasoning but with M_{n-1}, M_{n-2}, etc., we can eventually conclude that $\text{Ass}(M) \subseteq \{P_1, \ldots, P_n\}$. ■

A question we can ask about the prior theorem is if every prime from our prime filtration is an associated prime of M. The answer is unfortunately not necessarily yes. For example, let k be a field and $A \coloneqq k[x,y]$. Then let $M = \langle x, y\rangle$.

For any torsion free module (over a domain), we have that the only associated prime ideal is the zero ideal. Hence, $\operatorname{Ass}(M) = \{\{0\}\}$.

On other hand, consider the ascending chain of submodules:

$$0 \subseteq \langle x\rangle \subseteq \langle x, y\rangle = M$$

Since $\langle x\rangle$ is a principal ideal, we know $\langle x\rangle \cong A \cong A/\{0\}$ as A-modules. Meanwhile, $\langle x, y\rangle/\langle x\rangle \cong A/\langle x\rangle$.

To prove this, define $\psi : A \to \frac{\langle x,y\rangle}{\langle x\rangle}$ by $\psi(f(x,y)) = yf(x,y) + \langle x\rangle$. Then ψ is a well-defined module homomorphism. Furthermore, I claim ψ is surjective. After all, any $g(x,y) \in \langle x, y\rangle$ can be written in the form $g(x,y) = h_1(x,y)x + yh_2(y)$. In turn, $g(x,y) + \langle x\rangle = yh_2(y) + \langle x\rangle = \psi(h_2(y))$. Thirdly, note that $\ker(\psi) = \langle x\rangle$. Hence, we can use the first isomorphism theorem.

More generally, if M is a finitely generated torsion-free module that is not a free-module, then any prime filtration will contain non-associated primes.

After all, if not then we'd have that $\{0\} = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_n = M$ satisfies that $M_{i+1}/M_i \cong A$ for all i. But now as A is free (and thus projective), and also because we have an SES:

$$\{0\} \longrightarrow M_i \hookrightarrow M_{i+1} \twoheadrightarrow \tfrac{M_{i+1}}{M_i} \longrightarrow \{0\}\ ,$$

we can conclude $M_{i+1} \cong M_i \oplus A$ for all i. Hence $M \cong A^n$ and this contradicts that M is not a free-module.

For another example calculation, suppose A is a PID and M is a finitely generated A-module. By the fundamental theorem of PIDs we can write $M \cong A^r \oplus \bigoplus_{k=1}^{m} \frac{A}{\langle a_k\rangle}$ where $a_1 \mid a_2 \mid \cdots \mid a_m$. Then after considering the prime decompositions of each a_k, we can use the Chinese remainder theorem to get that $M \cong A^r \oplus \bigoplus_{i=1}^{\ell} \frac{A}{\langle p_i^{e_i}\rangle}$ where each p_i is a prime element of A and each e_i is a positive integer.

By applying the short exact sequence lemma from two pages ago, we can see that:

$$\operatorname{Ass}(M) = \underbrace{\operatorname{Ass}(A)}_{\text{if } r>0} \cup \bigcup_{i=1}^{\ell} \operatorname{Ass}(\tfrac{A}{\langle p_i^{e_i}\rangle}).$$

In turn, we can see that $\operatorname{Ass}(M) = \{\langle p_i\rangle : 1 \le i \le \ell\} \cup \underbrace{\{\{0\}\}}_{\text{if } r>0}$.

Proposition: Let A be a nontrivial Noetherian ring and let D be the set of zero divisors in A (including 0). Then $D = \bigcup_{P \in \mathrm{Ass}(A)} P$.

Proof:
It's obvious that $\bigcup_{P \in \mathrm{Ass}(A)} \subseteq D$. Conversely, if $a \in D$ then we know there exists $b \in A - \{0\}$ such that $ab = 0$. Or in other words, $a \in \mathrm{Ann}_A(b)$.

But now consider the set $\{\mathrm{Ann}_A(x'b) : x'b \neq 0 \text{ and } x' \in A\}$. This set is nonempty since $\mathrm{Ann}_A(1b) = \mathrm{Ann}_A(b)$ is in this set. So because A is Noetherian, we can consider an ideal $P = \mathrm{Ann}_A(xb)$ that is maximal among the elements of this set. Then, we can see that P is a prime ideal.

P is a proper ideal because $1xb = xb \neq 0$. Hence $1 \notin P$. Also suppose $cd \in P$ but $c \notin P$. Then as P is maximal in our original set and $c \notin P$, we must have that $\mathrm{Ann}_A(cxb) = \mathrm{Ann}_A(xb)$. In turn, $dc \in P \implies d \in \mathrm{Ann}_A(cxb) = P$.

Now note that $\mathrm{Ann}_A(b) \subseteq \mathrm{Ann}_A(xb) = P$. Hence, we have found an associated prime containing $\mathrm{Ann}_A(b)$ (which contains a). This proves that $D \subseteq \bigcup_{P \in \mathrm{Ass}(A)} P$. ■

One application of the above proposition is as follows. Suppose A is a nontrivial reduced ring (meaning $\mathrm{Nil}(A) = \{0\}$). Then after letting $\mathrm{Min}(A)$ denote the set of minimal primes of A, we have that $\mathrm{Ass}(A) \subseteq \mathrm{Min}(A)$.

Why?
Suppose $P \in \mathrm{Ass}(A)$ and let $0 \neq a \in A$ be such that $P = \mathrm{Ann}_A(a)$. Then suppose $Q \subsetneq P$ is another prime ideal properly contained in P. Then we can find $b \in P - Q$. By the definition of P, we know $ba = 0 \in Q$. In turn, since Q is prime we know $a \in Q \subseteq P$. Hence, $a^2 = 0$. This contradicts that $\mathrm{Nil}(A) = \{0\}$.

So, we can conclude that $\mathrm{Ass}(A)$ is contained in the set of minimal primes of A.

If A is also Noetherian then we can also say $\mathrm{Min}(A) \subseteq \mathrm{Ass}(A)$.

Why?
If A is a domain, then trivially the only minimal prime (which is also the only associated prime) is $\{0\}$. Hence, we'll now suppose A is not a domain.

By the theorem on *pages 899-900*, we can find a bunch of primes $P_1, \ldots, P_n$ with $0 = P_1 \cdots P_n$. Without loss of generality, we can assume n is minimal (and since A is not an integral domain, we know $n \geq 2$).

But now by the minimality of n we know $\prod_{\substack{i=1 \\ i \neq k}}^{n} P_i \neq \{0\}$ for any $k \in \{1, \ldots, n\}$.

Let P be a minimal prime. Then we know $P = P_k$ for some k, and we can also pick some $0 \neq a \in \prod_{\substack{i=1 \\ i \neq k}}^{n} P_i$ with $P \subseteq \mathrm{Ann}_A(a) \subsetneq A$.

Finally, by identical reasoning as in the proof of the above proposition, we can

find some $x \in A$ such that $xa \neq 0$ and also $\text{Ann}_A(xa)$ is prime. Yet now $P \subseteq \text{Ann}_A(a) \subseteq \text{Ann}_A(xa)$, and we also know $\text{Ann}_A(xa) \in \text{Ass}(A) \subseteq \text{Min}(A)$. By the minimality of $\text{Ann}_A(xa)$ we thus know $P = \text{Ann}_A(xa) \in \text{Ass}(A)$. Since P was arbitrary, this proves that $\text{Min}(A) \subseteq \text{Ass}(A)$.

I'd like to briefly rant that the professor proved neither inclusion in class.

By our prior proposition, we can thus conclude that if A is a nontrival Noetherian reduced ring, then the set of zero divisors of A is precisely given by $\bigcup_{P \in \text{Min}(A)} P$.

If it is ambiguous whether M could be viewed as an A-module or a B-module, we'll annotate $\text{Ass}(M)$ with a subscript to indicate which ring we are considering the associated primes of M with respect to.

Proposition: Let A be a Noetherian ring and M an A-module. Also let $S \subseteq A$ be a multiplicatively closed set. Then:

$$\text{Ass}_{S^{-1}A}(S^{-1}M) = \{S^{-1}P : P \in \text{Ass}_A(M) \text{ and } P \cap S = \emptyset\}$$

Proof:
First we note the following lemma.

Lemma: For any $m \in M$, we have that $S^{-1}(\text{Ann}_A(m)) = \text{Ann}_{S^{-1}A}(\frac{m}{1})$.

Proof:
Suppose $a \in \text{Ann}_A(m)$ and let $s \in S$. Then $\frac{a}{s} \cdot \frac{m}{1} = \frac{am}{1} = \frac{0}{1}$. So, $\frac{a}{s} \in \text{Ann}_{S^{-1}A}(\frac{m}{1})$ and this proves one inclusion.

Meanwhile, suppose $\frac{a}{s} \in \text{Ann}_{S^{-1}A}(\frac{m}{1})$. Then we know $\frac{am}{s} = \frac{0}{1}$. Hence, there exists $t \in S$ with $tam = 0$. But now $ta \in \text{Ann}_A(m)$. In turn, $\frac{a}{s} = \frac{ta}{ts} \in S^{-1}(\text{Ann}_A(M))$. So, $P \in \text{Ass}_A(M)$. ■

Now suppose $P \in \text{Ass}(M)$ with $P \cap S = \emptyset$. Then $S^{-1}P$ is prime in $S^{-1}A$. Also, let us write $P = \text{Ann}_A(m)$ where $m \neq 0$. By the prior lemma we know $S^{-1}P = \text{Ann}_{S^{-1}A}(\frac{m}{1})$. Hence, $S^{-1}P \in \text{Ass}_{S^{-1}A}(S^{-1}M)$. This proves one inclusion we wanted.

Conversely, suppose $Q \in \text{Ass}_{S^{-1}A}(S^{-1}M)$. By the prime correspondance I proved on *page 625*, we know $Q = S^{-1}P$ for some $P \in \text{Spec}(A)$ with $P \cap S = \emptyset$. Also let us write $Q = \text{Ann}_{S^{-1}A}(\frac{m}{s})$. Since s is a unit, we can equivalently say that $Q = \text{Ann}_{S^{-1}A}(\frac{m}{1})$. But now by our prior lemma, we have that $S^{-1}P = Q = S^{-1}(\text{Ann}_A(m))$.

By a comment on *page 625*, we know $\text{Ann}_A(m) \subseteq P$. Hence, it makes sense to consider the localization $S^{-1}(\frac{P}{\text{Ann}_A(M)})$ which we know is isomorphic to $\frac{S^{-1}P}{S^{-1}(\text{Ann}_A(m))} = \frac{Q}{Q} \cong \{0\}$ as a module. It follows that for all $a \in P$ there exists $t \in S$ with $ta \in \text{Ann}_A(m)$. Yet since A is Noetherian, we know $P = \langle a_1, \ldots, a_n \rangle$ for some elements $a_1, \ldots, a_n \in A$. Let $t_i \in S$ satisfy that $t_i a_i = 0$ for all i, and then set $t := \prod_{i=1}^n t_i$. Then $ta \in \text{Ann}_A(m)$ for all $a \in P$.

To finish off, I now show that $P \in \text{Ass}(A)$ since $P = \text{Ann}_A(tm)$. To start off, from our

construction of t we know $P \subseteq \text{Ann}_A(tm)$. On the other hand, if $b \in \text{Ann}_A(tm)$ then $bt \in \text{Ann}_A(m) \subseteq P$. Since $t \in S \subseteq A - P$, we thus must have that $b \in P$. So, $\text{Ann}_A(tm) \subseteq P$. ■

There is one more lecture to cover before I am caught up with everything (except the homework) for this class.

A proper ideal ideal Q of a ring A is called primary if $xy \in Q \implies x \in Q$ or $y \in \sqrt{Q}$.

Remark 1:
If Q is primary then $P := \sqrt{Q}$ is prime. After all, if $xy \in P$ then $(xy)^n = x^n y^n \in Q$. So either $x^n \in Q$ or $y^n \in P$ (which in turn means that $(y^n)^\ell \in Q$ for another power ℓ). Either way, this proves that either x to some power is in Q or y to some power is in Q. Hence, $x \in P$ or $y \in P$.

I'll also note that if Q is a proper ideal then trivially $\sqrt{Q}$ is not a proper ideal as $1 \notin \sqrt{Q}$. Hence, this finishes showing that $P \in \text{Spec}(A)$.

Remark 2:
Based on the prior remark, if Q is primary and $P = \sqrt{Q}$ then we say Q is P-primary.

Lemma: The following are equivalent for a proper ideal Q of a Noetherian ring A.

1. Q is primary,
2. The set of zero divisors of $\frac{A}{Q}$ is precisely $\frac{\sqrt{Q}}{Q}$,
3. $\text{Ass}_{A/Q}(A/Q) = \{\frac{\sqrt{Q}}{Q}\}$,
4. $\text{Ass}_A(A/Q) = \{\sqrt{Q}\}$.

$(1 \implies 2)$
Suppose $xy \in Q$ but $x \notin Q$. Then as Q is primary we must have that $y^n \in Q$ for some n. Hence, the zero divisors of A/Q are contained in $\sqrt{Q}/Q$. On the other hand, suppose $y \in \sqrt{Q}$. Then we know $y^n \in Q$ for some integer n, and in turn $y + Q$ is a zero divisor in A/Q. After all, if we take the minimal positive integer n for which $y^n \in Q$, then $y \cdot y^{n-1} \in Q$ but $y^{n-1} \notin Q$. This proves that $\sqrt{Q}/Q$ is a subset of the set of zero divisors of A/Q.

$(2 \implies 1)$
Suppose $xy \in Q$ but $x \notin Q$. Then $y + Q$ is a zero divisor of $\frac{A}{Q}$, which in turn means $y \in \sqrt{Q}$. So, $y^n \in Q$ for some n. And this proves Q is primary.

$(2 \implies 3)$
We know from page 945 that $\frac{\sqrt{Q}}{Q} = \bigcup_{P/Q \in \text{Ass}_{A/Q}(A/Q)} P/Q$. Hence, every $P/Q \in \text{Ass}_{A/Q}(A/Q)$ is contained in $\sqrt{Q}/Q$. Yet we also know $\sqrt{Q}/Q$ is contained in every prime ideal of A/Q. Hence, the only possible prime ideal in $\text{Ass}_{A/Q}(A/Q)$ is $\sqrt{Q}/Q$. Also, we know $\text{Ass}_{A/Q}(A/Q)$ is nonempty or else $\frac{\sqrt{Q}}{Q} \neq \bigcup_{P/Q \in \text{Ass}_{A/Q}(A/Q)} P/Q$.

$(3 \implies 2)$
If $\mathrm{Ass}_{A/Q}(A/Q) = \{\sqrt{Q}/Q\}$ then we know from page 945 that the set of zero divisors of $\frac{A}{Q}$ is exactly $\frac{\sqrt{Q}}{Q}$.

$(3 \iff 4)$
In general, if $I = \mathrm{Ann}_A(M)$ then $\mathrm{Ass}_{A/I}(M) = \{P/I : P \in \mathrm{Ass}_A(M)\}$.

Why?
Note that every $P \in \mathrm{Ass}_A(M)$ contains I. Also suppose $P \in \mathrm{Ass}_A(M)$ and let $P = \mathrm{Ann}_A(m)$ for some $m \in M - \{0\}$. Then $P/I = \mathrm{Ann}_{A/I}(m)$ since $(a+I)m = am = 0$ if and only if $a \in P$. Also by the prime correspondance theorem, we know P/I is a prime ideal.

Conversely, suppose $Q/I \in \mathrm{Ass}_{A/i}(M)$ and let $Q/I = \mathrm{Ann}_{A/I}(m)$ for some $m \in M - \{0\}$. Then $(a+I)m = am = 0$ iff $a \in Q$. Hence, $\mathrm{Ann}_A(m) = Q$. Also by the prime correspondance theorem, we know $Q \in \mathrm{Spec}(A)$.

Now $\mathrm{Ann}_A(A/Q) = Q$. Hence, we can just apply the above claim to prove that (3) and (4) are equivalent. ■

For an example of a primary ideal that isn't necessarily prime, suppose $P \in \mathrm{Max}(A)$ and $Q \lhd A$ with $\sqrt{Q} = P$. Then Q is P-primary.
A/Q only has primes P'/Q where $P' \in V(Q)$. Yet since $P = \sqrt{Q} = \bigcap_{P' \in V(Q)} P'$, we know every prime containing Q also contains P. Then since P is a maximal ideal, that implies the only prime ideal containing Q is P. Also, since A is Noetherian and A/Q is a finitely generated A-module, we know $\mathrm{Ass}_A(A/Q) \neq \emptyset$. It follows that $\mathrm{Ass}_A(A/Q) = \{P\} = \{\sqrt{Q}\}$. Hence by the prior lemma, Q is primary.

Let I be an ideal of A. A primary decomposition of I is an expression $I = Q_1 \cap \cdots \cap Q_n$ where each Q_i is P_i-primary for some prime P_i.

A primary decomposition for I is called irredundant if each Q_j does not contain the intersection of the other Q_i and all the primes P_i are distinct.

Example (Set 5 Problem 2(a)): If A is a UFD and $f \in A$, then we can write $f = up_1^{e_1} \cdots p_n^{e_n}$ where the p_i are pairwise nonassociate primes and each e_i is a positive integer. In turn, we have that $I = \bigcap_{i=1}^n \langle p_i^{e_i} \rangle$ is an irredundant primary decomposition of I.
Why?
Firstly, $\langle p_i^{e_i} \rangle$ is $\langle p_i \rangle$-primary. After all, suppose $g, h \in A$ satisfy that $gh \in \langle p_i^{e_i} \rangle$ but $g \notin \langle p_i^{e_i} \rangle$. Equivalently, this means $p_i^{e_i} \mid gh$ but $p_i^{e_i} \nmid g$. Hence, $p_i \mid h$, which in turn means $p_i^{e_i} \mid h^{e_i}$. This proves that $h^{e_i} \in \langle p_i^{e_i} \rangle$, thus showing that $\langle p_i^{e_i} \rangle$ is primary. Similar reasoning also shows $g^n \in \langle p_i^{e_i} \rangle$ for some integer n if and only if $p_i \mid g$ (meaning $g \in \langle p_i \rangle$). So $\langle p_i^{e_i} \rangle$ is $\langle p_i \rangle$-primary for each i.

Now to finish off, just note that $\langle f \rangle = \langle p_1^{e_1} \cdots p_n^{e_n} \rangle = \bigcap_{i=1}^n \langle p_i^{e_i} \rangle$. Also:
$$p_j^{e_j} \in \langle p_j^{e_j} \rangle - \bigcap_{i \neq j} \langle p_i^{e_i} \rangle \text{ for all } j.$$

I'm going to move onto doing homework now.

Set 4 Problem 1: Let k be an infinite field.

(a) Show for any nonzero polynomial $f \in k[x_1, \ldots, x_n]$ that there is $(a_1, \ldots, a_n) \in k^n$ such that $f(a_1, \ldots, a_n) \neq 0$.

We proceed by induction on n. If $n = 1$, then the set $Z(f)$ of zeros of f in k has at most $\deg(f)$ ($< \infty$) elements in k (since f is a nonzero polynomial). Since k is infinite, we thus know $k - Z(f) \neq \emptyset$. Hence, ther exists $a \in k$ with $f(a) \neq 0$.

Meanwhile, suppose $n > 1$ and that we have proven the problem for integers less than n. Then note that we can write $f(x_1, \ldots, x_n)$ as:

$$g_d(x_1, \ldots, x_{n-1})x_n^d + \ldots + g_1(x_1, \ldots, x_{n-1})x_n + g_0(x_1, \ldots, x_{n-1})$$

where $d \geq 0$ and $g_d(x_1, \ldots, x_{n-1}) \neq 0$.

By induction, we can find $a_1, \ldots, a_{n-1} \in k$ such that $g_d(a_1, \ldots, a_{n-1}) \neq 0$. In turn, if we set $c_i = g_i(a - 1, \ldots, a_{n-1})$ and $h(x) = c_d x^d + \ldots + c_1 x + c_0 \in k[x]$ we then have that h is a nonzero polynomial and $f(a_1, \ldots, a_{n-1}, x_n) = h(x_n)$. By our base case, we know there exists $a_n \in k$ with $h(a_n) \neq 0$. And now we are done since $f(a_1, \ldots, a_n) \neq 0$.

A notable consequence of this result is that it proves that distinct polynomials in an infinite field k always define distinct functions from k^n to k. I.e., equality in the polynomial ring $k[x]$ is equivalent to equality in the set of functions $k^{(k^n)}$.

(b) Over an infinite field, the lemma on *page 934* can be simplified by using a nicer change of variables. This was Noether's original result:

Suppose $f \in k[x_1, \ldots, x_n]$. Then there is a change of variables $x_i' = x_i - a_i x_n$ for some $a_i \in k$ such that for some $d \geq 0$, f obtains the form $f = \lambda x_n^d + g_{n-1}x_n^{d-1} + \ldots + g_1 x_n + g_0$ where each $g_i \in k[x_1', \ldots, x_{n-1}']$ for all $1 \leq i \leq n-1$ and $0 \neq \lambda \in k$.

Proof:
Consider a monomial $x_1^{i_1} \cdots x_n^{i_n}$. For now we'll just let a_i be any element of k and then set $x_i' = x_i - a_i x_n$. In turn, we know that $x_i = x_i' + a_i x_n$. Hence our original monomial becomes:

$$(x_1' + a_1 x_n)^{i_1} \cdots (x_{n-1}' + a_{n-1}x_n)^{i_{n-1}} x_n^{i_n}$$
$$= (\textstyle\prod_{j=1}^{n-1} a_j^{i_j}) x_n^{\sum_{j=1}^n i_j} + \text{ lower degree terms in } x_n$$

Now if we consider the entire polynomial f, let us write $f = f_{\max} + f_{\text{low}}$ where $f_{\max}$ consists of only the max total degree monomials in the polynomial (in other words the $cx_1^{i_1} \cdots x_n^{i_n}$ terms for which $\sum_{j=1}^n i_j$ is maximized) and f_{low} consists only of the lower total degree terms. I'll also let d denote the maximal value for $\sum_{j=1}^n i_j$. Then based on the prior observation about how our change of variables affects monomials, we can see that:

$$f(x_1, \ldots, x_n) = f_{\max}(a_1, \ldots, a_{n-1}, 1)x_n^d + \text{ lower degree terms in } x_n$$
$$\text{with coefficients in } k[x_1', \ldots, x_{n-1}'].$$

But now $g(y_1, \ldots, y_{n-1}) = f_{\max}(y_1, \ldots, y_{n-1}, 1)$ is a nonzero polynomial. By part (a), we can set $a_1, \ldots, a_{n-1}$ such that $g(a_1, \ldots, a_{n-1}) \neq 0$. ■

Set 4 Problem 2: The point of this problem is to give an alternate proof that if F is a field and an affine k-algebra then $[F : k] < \infty$. This proof won't use Noether normalization but it will also depend on k being uncountable.

(a) Let $k(x)$ be the field of rational functions in one variable over k. Thinking of $k(x)$ as a k-vector space, prove that the set of elements $\{(x-a)^{-1} : a \in k\}$ is linearly independent over k.

For any $n \in \mathbb{N}$ let $a_1, \ldots, a_n \in k$ be distinct and then suppose:

$$\tfrac{c_1}{x-a_1} + \ldots + \tfrac{c_n}{x-a_n} = 0$$

We need to show that $c_i = 0$ for all i. To do this consider, consider fixing an i and multiplying both sides of our equation by $x - a_i$. By doing this, we get that:

$$c_i + \sum_{j \neq i} c_j \tfrac{x-a_i}{x-a_j} = 0.$$

Next, we consider evaluating this expression when $x = a_i$. By doing so, most stuff cancels out and we are left with $c_i = 0$. This proves the linear independence we wanted.

Note that since we can use Zorn's lemma to complete a linearly independent set (i.e. make it span all of a vector space), this proves that we can find a k-basis for $k(x)$ with cardinality at least that of k.

(b) let k be any field. Prove that if A is a f.g. k-algebra, then $\dim_k(A)$ is countable.

Suppose A is generated by $a_1, \ldots, a_n$, and then consider any k-basis: $\{v_\beta\}_{\beta \in B}$, for A. For every $\beta \in B$ we can pick some $f_\beta \in k[x_1, \ldots, x_n]$ with $v_\beta = f_\beta(a_1, \ldots, a_n)$. And now, we let B_m be the set of $\beta \in B$ such that the max degree of f_β with respect to x_1, x_2, ..., or x_n individually is at most m.

Now $B = \bigcup_{m=1}^{\infty} B_m$. Also note that the linear independence of the v_β forces each B_m to be finite with cardinality at most m^n. If this weren't the case we could associate a nontrivial linear combination of v_β that is associated to the zero polynomial.

It follows that B is a countable set. Hence, every basis of $\dim_k(A)$ is countable.

(c) In the case that k is uncountable, prove that if F is a field and a f.g. k-algebra then $[F : k] < \infty$.

First we show an easier result, that F/k must be an algebraic extension.

Suppose to the contrary that there exists $\alpha \in F$ that is transcendental over k (meaning there does not exist any nonzero polynomial $f \in k[x]$ with $f(\alpha) = 0$). In that case, evaluation at α yields a well-defined injective map from $k(x)$ to F. So by part (a) plus the fact that k is an uncountable field, we can find an uncountable k-basis for F. Yet this contradicts part (b) which says that every k-basis for F must be countable. Hence, we conclude no such α can exist.

Now that we know that F/k is an algebraic extension, let $a_1, \ldots, a_n \in F$ be

generators of F. $F = k[a_1, \ldots, a_n]$ is a finite k-vector space with dimension at most $\prod_{i=1}^{n} \deg(m_{a_i;k})$. ■

Set 4 Problem 3: Let k be any field (not necessarily algebraically closed). Let $A \coloneqq k[x_1, \ldots, x_n]$. In this case, the maximal ideals of A do not necessarily have the form $\langle x_1 - a_1, \ldots, x_n - a_n \rangle$ so the maximal ideals of A are harder to describe exactly.

(a) Show that if P is maximal in A, then for each i there is an irreducible polynomial $f_i \in k[x_i]$ such that $f_i \in k[x_i] \cap P$. Thus, the ideal $I = \langle f_1(x_1), \ldots, f_n(x_n) \rangle$ is contained in P.

Note that if $P \in \operatorname{Spec}(k[x_1, \ldots, x_n])$ then $P \cap k[x_i] \in \operatorname{Spec}(k[x_i])$. Furthermore, as $k[x_1, \ldots, x_n]/P$ is a field with finite k-dimension, there is some nonzero polynomial $g \in k[x]$ with $g(x_i + P) = 0 + P$. Hence, $g(x_i) \in P \cap k[x_i]$, and this proves that $P \cap k[x_i] \neq \{0\}$.

Since $k[x_i]$ is a PID, we can thus conclude there is an irreducible polynomial $f_i \in k[x_i]$ with $P \cap k[x_i] = \langle f_i(x_i) \rangle$. In particular, as $f_i(x_i) \in P$ for all $1 \leq i \leq n$, we can conclude that $I = \langle f_1(x_1), \ldots, f_n(x_n) \rangle$ is contained in P.

(b) Show that in the context of part (a) we have $A/I \cong K_1 \otimes_k K_2 \otimes_k \cdots \otimes_k K_n$ as rings where $K_i \coloneqq k[x_i]/\langle f_i(x_i) \rangle$ is a field.

It suffices to prove for any $n \geq 2$ that $\frac{k[x_1,\ldots,x_n]}{\langle f_1(x_1),\ldots,f_n(x_n)\rangle} \cong \frac{k[x_1,\ldots,x_{n-1}]}{\langle f_1(x_1),\ldots,f_{n-1}(x_{n-1})\rangle} \otimes_k \frac{k[x_n]}{\langle f_n(x_n)\rangle}$.

After all, we can then conclude the claim of part (b) by induction plus the fact that the tensor functor always sends isomorphisms to isomorphisms.

But now recall from *page 728* that:

$$\left(\frac{k[x_1,\ldots,x_{n-1}]}{\langle f_1(x_1),\ldots,f_{n-1}(x_{n-1})\rangle}\right) \otimes_k \frac{k[x_n]}{\langle f_n\rangle} \cong \frac{\left(\frac{k[x_1,\ldots,x_{n-1}]}{\langle f_1(x_1),\ldots,f_{n-1}(x_{n-1})\rangle}\right)[x_n]}{f_n(x_n)\left(\frac{k[x_1,\ldots,x_{n-1}]}{\langle f_1(x_1),\ldots,f_{n-1}(x_{n-1})\rangle}\right)[x_n]} \text{ (as } k\text{-algebras).}$$

Then on *pages 459-460* I showed that (as rings):

$$\left(\frac{k[x_1,\ldots,x_{n-1}]}{\langle f_1(x_1),\ldots,f_{n-1}(x_{n-1})\rangle}\right)[x_n] \cong \frac{k[x_1,\ldots,x_{n-1}][x_n]}{\langle f_1(x_1),\ldots,f_{n-1}(x_{n-1})\rangle[x_n]} \cong \frac{k[x_1,\ldots,x_n]}{\langle f_1(x_1),\ldots,f_{n-1}(x_{n-1})\rangle[x_n]}$$

Hence we know:

$$\frac{\left(\frac{k[x_1,\ldots,x_{n-1}]}{\langle f_1(x_1),\ldots,f_{n-1}(x_{n-1})\rangle}\right)[x_n]}{f_n(x_n)\left(\frac{k[x_1,\ldots,x_{n-1}]}{\langle f_1(x_1),\ldots,f_{n-1}(x_{n-1})\rangle}\right)[x_n]} \cong \frac{\left(\frac{k[x_1,\ldots,x_n]}{\langle f_1(x_1),\ldots,f_{n-1}(x_{n-1})\rangle[x_n]}\right)}{f_n(x_n)\left(\frac{k[x_1,\ldots,x_n]}{\langle f_1(x_1),\ldots,f_{n-1}(x_{n-1})\rangle[x_n]}\right)} \text{ (as rings).}$$

To finish off, if we can calculate the ideal J generated by $\{f_n(x_n)\} \cup \langle f_1(x_1), \ldots, f_{n-1}(x_{n-1})\rangle[x_n]$, then we will be able to conclude that:

$$\frac{\left(\frac{k[x_1,\ldots,x_n]}{\langle f_1(x_1),\ldots,f_{n-1}(x_{n-1})\rangle[x_n]}\right)}{f_n(x_n)\left(\frac{k[x_1,\ldots,x_n]}{\langle f_1(x_1),\ldots,f_{n-1}(x_{n-1})\rangle[x_n]}\right)} \cong \frac{\left(\frac{k[x_1,\ldots,x_n]}{\langle f_1(x_1),\ldots,f_{n-1}(x_{n-1})\rangle[x_n]}\right)}{\left(\frac{J}{\langle f_1(x_1),\ldots,f_{n-1}(x_{n-1})\rangle[x_n]}\right)} \cong \frac{k[x_1,\ldots,x_n]}{J} \text{ (as rings).}$$

Fortunately, it is easy to see that $f_i(x_i) \in J$ for all i and hence $\langle f_1(x_1), \ldots, f_n(x_n)\rangle \subseteq J$. Meanwhile:

$$J = \langle f_n(x_n)\rangle + \langle f_1(x_1), \ldots, f_{n-1}(x_{n-1})\rangle[x_n].$$

But then any element of J can be written as:

$$\begin{aligned} &p_n(x_1, \ldots, x_n)f_n(x_n) + \sum_{k=0}^{d}\left(\sum_{j=1}^{n-1} p_{k,j}(x_1, \ldots, x_{n-1})f_j(x_j)\right)x_n^k \\ &\qquad = p_n(x_1, \ldots, x_n)f_n(x_n) + \sum_{j=1}^{n-1}\left(\sum_{k=0}^{d} p_{k,j}(x_1, \ldots, x_{n-1})x_n^k\right)f_j(x_j) \end{aligned}$$

Hence, $J \subseteq \langle f_1(x_1), \ldots, f_n(x_n)\rangle$.

As a side note, the ring isomorphism described above is given by $\overline{g} \otimes \overline{h} \mapsto \overline{gh}$. Clearly this preserves scalar multiplication by elements of k. So, this isomorphism is a k-algebra isomorphism.

(c) Describe explicitly all the maximal ideals of $A \coloneqq \mathbb{R}[x, y]$.

Given a maximal ideal $P \lhd \mathbb{R}[x, y]$, let I be the ideal described in parts (a) and (b). Then since every irreducible nonzero polynomial in $\mathbb{R}[x]$ is either degree one with a real root or degree two with conjugate complex roots, we have exactly three cases to deal with.

Firstly, suppose $I = \langle x - a, y - b\rangle$. Then we can see that $A/I \cong \mathbb{R} \otimes_{\mathbb{R}} \mathbb{R} \cong \mathbb{R}$ as rings. It follows that A/I is a field and thus I is a maximal ideal. And since $I \subseteq P$, we can conclude that $P = \langle x - a, y - b\rangle$.

The maximal ideal $P = \langle x - a, y - b\rangle$ can be thought of as corresponding to the point $(a, b) \in \mathbb{R}^2$.

Secondly, suppose $I = \langle x^2 + ax + b, y - c\rangle$. Then we have that $A/I \cong \mathbb{C} \otimes_{\mathbb{R}} \mathbb{R} \cong \mathbb{C}$ as rings. In turn, by identical reasoning as before we can conclude that:

$$P = I = \langle x^2 + ax + b, y - c\rangle.$$

(Symmetric reasoning works if $I = \langle x - a, y^2 + cy + d\rangle$.)

Thirdly, suppose $I = \langle x^2 + ax + b, y^2 + cy + d\rangle$. Then we have that $A/I \cong \mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}^2$ as rings. (See *page 729*.) But the latter is no longer a field and hence we don't have that I is a maximal ideal. At the very least though, the only two nonzero proper ideals of $\mathbb{C}^2$ are $\mathbb{C} \times \{0\}$ and $\{0\} \times \mathbb{C}$. Thus, there are exactly two proper ideals in A containing I and P could be either of the two.

I am not done calculating what P is but I have other things I need to get done and it looks like it will be time-consuming to work backwards to calculate P. If I come back to this, it will be on *page 956*. I can see though that P will be generated by three polynomials.

Here is a short proposition about the Jacobson radical of a ring A.

Lemma: $a \in J(A)$ if and only if $1 - ab$ is a unit in A for all $b \in A$.

$(\Longrightarrow)$

Suppose $1 - ab$ is not a unit for some $b \in A$. In that case, $\langle 1 - ab \rangle$ is a proper ideal of A and so there exists a maximal ideal $M \in \operatorname{Max}(A)$ with $1 - ab \in M$. But now it follows that $1 + M = ab + M$. Since $1 \notin M$, we thus know $ab \notin M$ and in turn that $a \notin M$. This proves that $a \notin J(A)$.

$(\Longleftarrow)$

Suppose $a \notin J(A)$. Then there exists $M \in \operatorname{Max}(A)$ such that $a \notin M$. Yet then since $\langle a, M \rangle$ is an ideal properly containing the maximal ideal M, we must have that $A = \langle a, M \rangle$. Hence there exists $b \in A$ and $m \in M$ with $ab + m = 1$. Or in other words, $1 - ab = m \in M$. So, $1 - ab$ is not a unit in A. ■

Set 4 Problem 4: For any ring A, show that $J(A[x]) = \operatorname{Nil}(A[x])$.

Proof:

Recall from *pages 460-461* that:

- $\operatorname{Nil}(A[x]) = \operatorname{Nil}(A)[x]$,
- $(A[x])^\times = \{a_0 + xf(x) : a_0 \in A^\times \text{ and } f \in \operatorname{Nil}(A)[x]\}$.

Now we always have that $\operatorname{Nil}(A) \subseteq J(A)$. Hence, we just need to show the other containment. Fortunately, note by the prior lemma that if $g(x) \in J(A)$ then $1 - xg(x) \in (A[x])^\times$. In turn, we must have that $-g \in \operatorname{Nil}(A)[x]$, which is equivalent to what we needed to show. ■

Set 4 Problem 5: Let X be a topological space. X is called Noetherian if X satisfies the descending chain condition on closed subsets (i.e. every descending chain of closed sets $\{F_i\}_{i \in I}$ has a minimal element).

Remark 1: Note that the following are equivalent:

1. X is Noetherian
2. Every nonempty family of closed sets in X has a minimal element.
3. Given any descending sequence of closed sets $F_1 \supseteq F_2 \supseteq \cdots$ there exists some $n \in \mathbb{N}$ with $F_n = F_{n+k}$ for all $k \in \mathbb{N}$.

Proof:

$(1 \implies 2)$

This is just an application of Zorn's lemma.

$(2 \implies 3)$

Let $\{F_n\}_{n \in \mathbb{N}}$ be a descending sequence of closed sets. By (2) we know there is a minimal element F_N. But now since $\{F_n\}_{n \in \mathbb{N}}$ is descending we must have that $F_N = F_{N+1} = \cdots$.

$(3 \implies 1)$

Suppose the chain $\{F_i\}_{i \in I}$ had no minimal element. Then we could construct a sequence of closed sets $\{F_{i_n}\}$ violating claim 3.

Remark 2: All three of the above properties are equivalent to properties concerning an ascending chain condition on open sets.

A closed set $Z \subseteq X$ is called irreducible if one cannot write $Z = Z_1 \cup Z_2$ where Z_1 and Z_2 are both *proper* closed subsets of Z (which are *not* necessarily disjoint).

(a) Show that any closd set $Z \subseteq X$ (where X is a Noetherian topological space) is a union of finitely many irreducible closed sets.

Let $\mathcal{S}$ be the set of closed subsets of X which aren't a union of finitely many irreducible closed sets. Then suppose for the sake of contradiction that $\mathcal{S}$ is nonempty. Since X is Noetherian, we then know that $\mathcal{S}$ has a minimal element $Z \subseteq \mathcal{S}$. Yet since $Z \in \mathcal{S}$, we must have that Z is not irreducible. Hence, there exists proper closed subsets of Z: F_1, F_2, such that $F_1 \cup F_2 = Z$. Then since $F_1, F_2 \notin \mathcal{S}$ (since Z was minimal), we know that we can write each F_i as a finite union $Z_1^{(i)} \cup \cdots \cup Z_{n_i}^{(i)}$ of closed irreducible sets. And now we have a contradiction because $Z = \bigcup_{i=1}^{2}(Z_1^{(i)} \cup \cdots \cup Z_{n_i}^{(i)})$.

Furthermore, explain why we can choose an expression $Z = Z_1 \cup \cdots \cup Z_n$ where the Z_i are irreducible and closed and no Z_i is contained in Z_j for any $i \neq j$.

Let $Z \subseteq X$ be a closed set and let n be the minimal integer such that we can write $Z = Z_1 \cup \cdots \cup Z_n$ with all Z_i being irreducible. If $Z_i \subseteq Z_j$ for any distinct i, j that would contradict the minimality of n. So, we know that must not occur.

Finally, show that such an expression $Z = Z_1 \cup \cdots \cup Z_n$ as described above is unique up to permutation of the Z_i.

Let n be the minimal integer such that we can write $Z = Z_1 \cup \ldots \cup Z_n$ with all Z_i being irreducible and closed. Then suppose we can also write $Z = F_1 \cup \ldots \cup F_m$ where each F_i is irreducible and closed, and $F_i \not\subseteq F_j$ for any distinct i, j.

Firstly, I shall show that every F_i is equal to some Z_j. To do this, first note that without generality we can just let $i = 1$. Next note that as $F_1 \subseteq Z = \bigcup_{i=1}^{n} Z_i$ we have that $F_1 \cap Z_1$ and $F_1 \cap (\bigcup_{i=2}^{n} Z_i)$ are both closed subsets of F_1 whose union is all of F_1. By the irreducibility of F_1 we thus know that:

$$\text{either } F_1 \cap Z_1 = F_1 \text{ or } F_1 \cap (\textstyle\bigcup_{i=2}^{n} Z_i) = F_1.$$

Now if the former statement holds true, then we have already successfully shown that $F_1 \cap Z_j = F_1$ for some j. Otherwise, we can just repeat the same reasoning to say that either $F_1 \cap Z_2 = F_1$ or $F_1 \cap (\bigcup_{i=3}^{n} Z_i) = F_1$.

And if we keep going without ever having the first case happening, we will eventually get that either $F_1 \cap Z_{n-1} = F_1$ or $F_1 \cap Z_n = F_1$. Hence, we are guarenteed to have that $F_1 \cap Z_j = F_1$ for some Z_j.

Next, since $Z_j \subseteq Z = \bigcup_{i=1}^{n} F_i$, we know that $Z_j \cap F_1$ and $Z_j \cap (\bigcup_{i=2}^{m} F_i)$ are both closed subsets of Z_j whose union is all of Z_j By the irreducibility of Z_j, we must have that:

$$\text{either } Z_j \cap F_1 = Z_j \text{ or } Z_j \cap (\textstyle\bigcup_{i=2}^{m} F_i) = Z_j.$$

In the former case, we are done since $F_1 = Z_j \cap F_1 = Z_j$. So suppose the latter case that $Z_j \cap (\bigcup_{i=2}^{m} F_i) = Z_j$. Then like before we can conclude either $Z_j \cap F_2 = Z_j$ or $Z_j \cap (\bigcup_{i=3}^{m} F_i) = Z_j$. And by repeating this over and over, we will eventually be forced to conclude that $Z_j \cap F_k = Z_j$ for some $1 \leq k \leq m$.

Yet now $F_1 = F_1 \cap Z_j = F_1 \cap Z_j \cap F_k \subseteq F_k$. If $k \neq 1$ then this contradicts that no F_i is contained in another F_k. So we can conclude the former case must hold.

Now that we've shown that every F_i is equal to some Z_j, the requirement that no F_{i_1} is contained in another F_{i_2} forces the assignment of F_i to their corresponding Z_j to be injective. And since $m \geq n$, we also must have by the pigeonhole principle that this assignment is surjective. This finishes the proof. ■

The Z_i are called the irreducible components of Z.

(b) Show that $\mathrm{Spec}(A)$ (equipped with the Zariski topology) is a Noetherian topological space for any Noetherian ring A.

Let $\{V(I_k)\}_{k\in\mathbb{N}}$ be a descending chain of closed sets in $\mathrm{Spec}(A)$.

Note that for any ideal $I \lhd A$ we have that $V(I) = V(\sqrt{I})$. After all, since $I \subseteq \sqrt{I}$ we know that $V(\sqrt{I}) \subseteq V(I)$. Meanwhile, if $P \in V(I)$ then $P \supseteq \bigcap_{Q\in V(I)} Q = \sqrt{I}$. So $P \in V(\sqrt{I})$, thus proving that $V(I) \subseteq V(\sqrt{I})$. Due to this observation, we can replace each ideal I_n defining our closed sets with $\sqrt{I_n}$.

Next note that if $V(\sqrt{I}) \subseteq V(\sqrt{J})$ then $\sqrt{J} = \bigcap_{P\in V(J)} P \subseteq \bigcap_{P\in V(I)} P = \sqrt{I}$.

But now we have shown that we have an ascending chain $\{\sqrt{I_n}\}_{n\in\mathbb{N}}$ of ideals in A. Since A is a Noetherian ring, there must be some fixed n such that $\sqrt{I_n} = \sqrt{I_{n+k}}$ for all $k \in \mathbb{N}$. In turn, $V(\sqrt{I_n}) = V(\sqrt{I_{n+k}})$ for all $k \in \mathbb{N}$.

(c) Given a closed subset $Z \coloneqq V(I)$ of $\mathrm{Spec}(A)$ where A is a Noetherian ring and I is an ideal of A, show that the irreducible components of Z are precisely the subsets $V(P_i)$ for the finitely many primes P_i where $\frac{P_i}{I}$ is a minimal prime of A/I.

Since A/I is a Noetherian ring, we know that it has finitely many minimal primes $P_1/I, \ldots, P_n/I$. My first goal is to show that each set $V(P_i)$ is irreducible.

Suppose J_1, J_2 are two ideals such that $V(P_i) = V(J_1) \cup V(J_2)$. Then since $P_i \in V(P_i)$ we know that $P_i \in V(J_1)$ or $P_i \in V(J_2)$. Without loss of generality, suppose the former happens. Then $J_1 \subseteq P_i$, which in turn means that $V(P_1) \subseteq V(J_1)$. Yet we also know from our initial expression that $V(J_1) \subseteq V(P_1)$. Hence, $V(P_1) = V(J_1)$ and this proves that $V(P_1)$ is irreducible.

More generally, this shows that for any $P \in \mathrm{Spec}(A)$ we have that $V(P)$ is an irreducible closed set.

Now for every prime P containing I there is a minimal prime P_i/I with $P_i/I \subseteq P/I$. Hence, we can conclude that $I \subseteq P_i \subseteq P$. Or in other words $V(I) \subseteq \bigcup_{i=1}^{n} V(P_i)$. The other containment is obvious.

Next note that $P_i \notin V(P_j)$ unless $P_i = P_j$. This is because since P_i/I is a minimal prime in A/I we have that $I \subseteq P_j \subseteq P_i \implies P_i = P_j$. As a result, we can see that

$V(P_i) \not\subseteq V(P_j)$ if $i \neq j$. This finishes showing what we wanted.

(d) Give an example of a ring A such that $\mathrm{Spec}(A)$ is a Noetherian space but A is not a Noetherian ring.

Given a monoid $M := \bigoplus_{i=1}^{\infty} \mathbb{Z}_{\geq 0}$, consider the monoid ring $\mathbb{Q}[M]$ which we express as the polynomial ring of countably infinitely many variables: $\mathbb{Q}[x_1, x_2, \ldots]$. Next, let $I \lhd \mathbb{Q}[x_1, x_2, \ldots]$ be the ideal generated by $\{x_n^2 : n \in \mathbb{N}\}$. Then, we let:

$$A := \tfrac{\mathbb{Q}[x_1, x_2, \ldots]}{I}.$$

Note that I contains every polynomial in $\mathbb{Q}[x_1, x_2, \ldots]$ such that every monomial in said polynomial has at least one variable raised to a power of 2. As a result, we can see that $x_n \notin I$ for all n.

Meanwhile, let $J \lhd \mathbb{Q}[x_1, x_2, \ldots]$ be the ideal generated by $\{x_n : n \in \mathbb{N}\}$. Then J contains every polynomial in $\mathbb{Q}[x_1, x_2 \ldots]$ without a constant term.

Now notably, every $\overline{x_n}$ in A is nilpotent. Hence $J/I \subseteq P/I$ for all $P \in \mathrm{Spec}(A)$. At the same time though, we can easily see that $J \in \max(\mathbb{Q}[x_1, x_2, \ldots])$. After all, given any $f(x_1, x_2, \ldots) \in \mathbb{Q}[x_1, x_2, \ldots] - J$ we'll be able to subtract off any higher degree terms and be left with an element of $\mathbb{Q}$ (which is a unit). Hence, we have proven that A has exactly one prime ideal which is given by $\frac{\langle x_1, x_2, \ldots \rangle}{\langle x_1^2, x_2^2, \ldots \rangle}$. Because of that, $\mathrm{Spec}(A)$ is trivially a Noetherian space.

On the other hand though, let us define $I_N \lhd \mathbb{Q}[x_1, x_2, \ldots]$ to be the ideal generated by $\{x_1, \ldots, x_N\} \cup \{x_n^2 : n \in \mathbb{N}\}$. Then $I_N \subsetneq I_{N+1}$ for all N, which we can verify by noting that $x_{N+1} \in I_{N+1} - I_N$.

But now $\frac{I_1}{I} \subsetneq \frac{I_2}{I} \subsetneq \cdots$ is an asscending chain of ideals not satisfying the ascending chain property for ideals. Hence, A is not a Noetherian ring. ■

I'm going to spend 30 more minutes doing problem 3 before I turn in what I have and call it a night. Go back to *page 952* for a refresher on what I was proving.

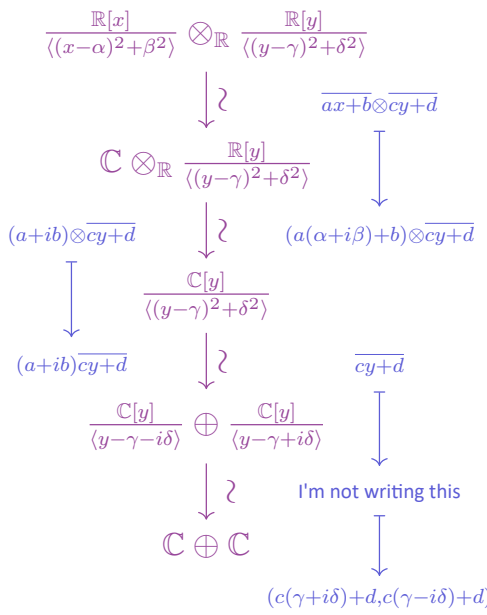
Set 3 Problem 3 (continued):

To start off, since $x^2 + ax + b$ and $y^2 + cx + d$ are irreducible polynomials in $\mathbb{R}[x]$, we can rewrite them as $(x - \alpha)^2 + \beta^2$ and $(y - \gamma)^2 + \delta^2$ so that their respective zeros are given by $\alpha \pm i\beta$ and $\gamma \pm i\delta$ (where $\beta, \delta \neq 0$).

Next, we need to write out and then walk backwards through the isomorphism going from $\frac{\mathbb{R}[x]}{\langle (x-\alpha)^2 + \beta^2 \rangle} \otimes_{\mathbb{R}} \frac{\mathbb{R}[x]}{\langle (y-\gamma)^2 + \delta^2 \rangle}$ to $\mathbb{C}^2$.

My diagram chase / scratch work is on the next page since it's too large to fit on this page. That said, there are roughly three steps in the isomorphism to invert in order to find an element corresponding to $(1, 0)$ in $\mathbb{C} \oplus \mathbb{C}$. (Also at this point I'm working after the deadline…)

Firstly, we need to find a complex linear polynomial $p(y) = (a + ib)y + (c + id)$ such that $p(\gamma + i\delta) = 1$ and $p(\gamma - i\delta) = 0$. We can do this by solving a certain system of linear equations to get that the only linear polynomial that works is $p(y) = \frac{-i}{2\delta} y + (\frac{1}{2} + i\frac{\gamma}{2\delta})$.



As for reversing the second part we need to, unfortunately there is no single complex number $a+ib$ such that $\frac{p(z)}{a+ib} \in \mathbb{R}[y]$. So, we instead need to settle for taking a sum of two pure tensors. Namely:

$$p(y) = -i(\tfrac{1}{2\delta}y - \tfrac{\gamma}{2\delta}) + (0y + \tfrac{1}{2}).$$

Hence, we now consider the element:

$$-i \otimes \overline{\tfrac{1}{2\delta}y - \tfrac{\gamma}{2\delta}} + 1 \otimes \overline{0y + \tfrac{1}{2}}.$$

To finish off, $a(\alpha+i\beta)+b = -i$ would imply that $a = \frac{-1}{\beta}$ and $b = \frac{\alpha}{\beta}$

Meanwhile, $a(\alpha+i\beta)+b = 1$ would imply that $a = 0$ and $b = 1$.

Hence, the element in $\frac{\mathbb{R}[x]}{\langle(x-\alpha)^2+\beta^2\rangle} \otimes_{\mathbb{R}} \frac{\mathbb{R}[x]}{\langle(y-\gamma)^2+\delta^2\rangle}$ corresponding to $(1,0)$ in $\mathbb{C} \oplus \mathbb{C}$ can be written as:

$$\tfrac{-1}{2}\left(\overline{\tfrac{1}{\beta}x - \tfrac{\alpha}{\beta}} \otimes \overline{\tfrac{1}{\delta}y - \tfrac{\gamma}{\delta}}\right) + \tfrac{1}{2}(\overline{1} \otimes \overline{1})$$

Now that we know this element, its also easy to get an element corresponding to $(0,1)$. After all note that $(0,1) = (1,1) - (1,0)$. Also, we can easily see $1 \otimes 1$ corresponds to $(1,1)$ since both are the respective multiplicative identies of their rings. Hence, the element corresponding to $(0,1)$ is given by:

$$(1 \otimes 1) - \left(\tfrac{-1}{2}\left(\overline{\tfrac{1}{\beta}x - \tfrac{\alpha}{\beta}} \otimes \overline{\tfrac{1}{\delta}y - \tfrac{\gamma}{\delta}}\right) + \tfrac{1}{2}(\overline{1} \otimes \overline{1})\right)$$

In other words, the element corresponding to $(0,1)$ can be written as:

$$\tfrac{+1}{2}\left(\overline{\tfrac{1}{\beta}x - \tfrac{\alpha}{\beta}} \otimes \overline{\tfrac{1}{\delta}y - \tfrac{\gamma}{\delta}}\right) + \tfrac{1}{2}(\overline{1} \otimes \overline{1})$$

Now to finish off, recall that the isomorphism from $\frac{\mathbb{R}[x]}{\langle(x-\alpha)^2+\beta^2\rangle} \otimes_{\mathbb{R}} \frac{\mathbb{R}[x]}{\langle(y-\gamma)^2+\delta^2\rangle}$ to $\frac{A}{\langle(x-\alpha)^2+\beta^2,(y-\gamma)^2+\delta^2\rangle}$ is given merely by $\overline{g} \otimes \overline{h} \mapsto \overline{gh}$. Therefore, the two maximal ideals of A/I are principal and generated respectively by the polynomials:

$$\pm\tfrac{1}{2}(\tfrac{1}{\beta}x - \tfrac{\alpha}{\beta})(\tfrac{1}{\delta}y - \tfrac{\gamma}{\delta}) + \tfrac{1}{2}$$

In other words, we've shown the only two maximal ideals containing I are:

- $\langle (x-\alpha)^2+\beta^2,\ \ (y-\gamma)^2+\delta^2,\ \ (x-\alpha)(y-\gamma)+\beta\delta\rangle$
- $\langle (x-\alpha)^2+\beta^2,\ \ (y-\gamma)^2+\delta^2,\ \ (x-\alpha)(y-\gamma)-\beta\delta\rangle$. ■

6/8/2026

It's been a while since I studied math while not doing homework. Today I'm going to study distributions on $\mathbb{R}^n$. Also if I don't specify what measure I'm integrating aginst, assume I'm integrating against the Lebesgue measure m. Finally, see *page 229* for where I left off.

We associate with any $f \in L^1_{\text{loc}}$ the linear functional $(f, \phi) \coloneqq \int f\phi$ where $\phi \in C_c^\infty(\mathbb{R}^n)$.

> Note that $(f, \cdot)$ is a continuous linear functional. After all, suppose $K \subseteq \mathbb{R}^n$ is compact and let $\{\phi_j\}_{j\in\mathbb{N}}$ be a sequence converging to ϕ in $C_c^\infty(K)$. In other words, $\partial^\alpha \phi_j \to \partial^\alpha \phi$ uniformly as $j \to \infty$ for all multi-indices α. But now by Hölder's inequality:
> $$\int f(\phi_j - \phi) \le \|f\|_{L^1(K)} \cdot \|\phi_j - \phi\|_u \to 0 \text{ as } j \to \infty.$$
>
> Hence, $(f, \cdot)$ meets the requirements for being a continuous linear functional on $C_c^\infty(\mathbb{R}^n)$ (i.e. a distribution).

Furthermore, note that $f, g \in L^1_{\text{loc}}$ define the same linear functional if and only if $f = g$ a.e.

> To see why, note that if f and g define the same linear functional then $\int (f-g)\phi = 0$ for all $\phi \in C_c^\infty(\mathbb{R}^n)$. But now I claim if $h \in L^1_{\text{loc}}$ and $\int h\phi = 0$ for all $\phi \in C_c^\infty(\mathbb{R}^n)$ then $h = 0$ a.e.
>
> This is because for any $x \in \mathbb{R}^n$ and $r, \varepsilon > 0$, we can find a smooth function $\phi_{r,\varepsilon} : \mathbb{R}^n \to [0,1]$ that satisfies that $\phi_{r,\varepsilon} \equiv 1$ on $B_r(x)$ and $\phi_n \equiv 0$ on $(B_{r+\varepsilon}(0))^{\mathsf{C}}$. In turn, we have that:
> $$0 = (h, \ \tfrac{\phi_{r,\varepsilon}}{m(B_r(x))}) = \tfrac{1}{m(B_r(x))}\int_{B_r(x)} h(y)\mathrm{d}y + \tfrac{1}{m(B_r(x))}\int_{\{y : r \le \|y-x\|_2 < r+\varepsilon\}} h(y)\phi_{r,\varepsilon}(y)\mathrm{d}y$$
>
> But now by the Lebesgue differentiation theorem, $\frac{1}{m(B_r(x))}\int_{B_r(x)} h(y)\mathrm{d}y \to h(x)$ as $r \to 0$ for a.e. $x \in \mathbb{R}^n$. Meanwhile by Holder's inequality we have that:
> $$\begin{aligned} 0 \le \tfrac{1}{m(B_r(x))}\left|\int_{\{y : r \le \|y-x\|_2 < r+\varepsilon\}} h(y)\phi_{r,\varepsilon}(y)\mathrm{d}y\right| &\le \tfrac{1}{m(B_r(x))}\int_{\{y : r \le \|y-x\|_2 < r+\varepsilon\}} |h(y)|\mathrm{d}y \\ &\le \tfrac{m(B_{r+\varepsilon}(x)) - m(B_r(x))}{m(B_r(x))}\|h\|_{L^1(B_1(0))}. \end{aligned}$$
>
> For every fixed $r > 0$ we know that as $\varepsilon \to 0$ the latter bound goes to 0. Hence, by strategically picking a sequence of r_n and ε_n, we can conclude for a.e. $x \in \mathbb{R}^n$ that there is some sequence of C_c^∞ test-functions ϕ_n such that $0 = (h, \phi_n) \to h(x)$. And this proves that $h = 0$ a.e.

It follows that L^1_{loc} is a subspace of the space of distributions $\mathcal{D}'$.

Another set that can be embedded into $\mathcal{D}'$ is the space of complex Radon measures $M(\mathbb{R}^n)$. Namely, we identify any $\mu \in M(\mathbb{R}^n)$ with the linear functional $(\mu, \phi) \coloneqq \int \phi \mathrm{d}\mu$ where $\phi \in C_c^\infty(\mathbb{R}^n)$.

> By nearly identical reasoning as with $f \in L^1_{\text{loc}}$, we can conclude that the linear functional associated with μ is continuous in the distribution sense. As for seeing that if μ_1 and μ_2 in $M(\mathbb{R}^n)$ both define the same distribution then $\mu_1 = \mu_2$, note that any Radon measure $\nu \in M(\mathbb{R}^n)$ is uniquely determined by the linear functional $f \mapsto \int f \mathrm{d}\nu$ defined for $f \in C_c(\mathbb{R}^n)$. Yet since $C_c^\infty(\mathbb{R}^n)$ is uniformly dense in $C_c(\mathbb{R}^n)$ and $f \mapsto \int f\mathrm{d}\nu$ is a continuous function on $C_c(\mathbb{R}^n)$, we can conclude that the value of the linear functional defined by ν in $C_c^\infty(\mathbb{R}^n)$ uniquely determines ν.

To see that $C_c^\infty(\mathbb{R}^n)$ is uniformly dense in $C_c(\mathbb{R}^n)$, just suppose $f \in C_c(\mathbb{R}^n)$. Then f is bounded and uniformly continuous. Next, consider any $\varphi \in L^1(\mathbb{R}^n) \cap C_c^\infty(\mathbb{R}^n)$ with $\int \varphi(x)\mathrm{d}x = 1$, and put $\varphi_t(x) := \frac{1}{t^n}\varphi(\frac{x}{t})$. Then we can guarentee that $f * \varphi_t \in C_c^\infty(\mathbb{R}^n)$ for all $t > 0$. Also, $f * \varphi_t \to f$ uniformly as $t \to 0$. (See pages 27-30 of my Spring 2025 math 240c notes...)

For the sake of consistency, given a general distribution $F \in \mathcal{D}'$ and $\phi \in C_c^\infty(\mathbb{R}^n)$ we shall often use the notation (F, ϕ) to mean $F(\phi)$.

One might ask why we defined distributions to specifically be linear functions on $C_c^\infty(\mathbb{R}^n)$. The reason why we want our test functions to be compactly supported is so that any locally integrable function defines a distribution. After all, a lot of important familes of functions that I have now seen in my life are not globally integrable. Meanwhile, the full power of requiring our test functions to be smooth is demonstrated by the following reasoning.

Suppose $F \in \mathcal{D}'$ and let α be a multi-index. Then we define the αth distributional derivative of F to be: $\partial^\alpha F := (-1)^{|\alpha|}F(\partial^\alpha \cdot)$.

To see that $\partial^\alpha F$ is in fact a distribution, notice that if $\phi_j \to \phi$ in $C_c^\infty(\mathbb{R}^n)$ then we also have that $\partial^\alpha \phi_j \to \partial^\alpha \phi$ in $C_c^\infty(\mathbb{R}^n)$ due to how we defined convergence in that space. In turn, as F is continuous we have that $F(\partial^\alpha \phi_j) \to F(\partial^\alpha \phi)$ as $j \to \infty$. And multiplication by a constant doesn't change that convergence.

Notably, all distributions have all distributional derivatives defined automatically. Thus, when working with distributions we never need to worry about how smooth they are. Two other cool observations are as follows.

- If α and β are two different multi-indices and $F \in \mathcal{D}'$, then $\partial^\alpha(\partial^\beta F) = \partial^{\alpha+\beta}F$.

  Why?
  Since our test functions ϕ are smooth, we can swap the order of differentiation of ϕ without any worry. Thus:

$$\begin{aligned}\partial^\alpha(\partial^\beta F)(\phi) = (-1)^{|\alpha|}\partial^\beta F(\partial^\alpha \phi) &= (-1)^{|\alpha|} \cdot (-1)^{|\beta|}F(\partial^\beta(\partial^\alpha \phi)) \\ &= (-1)^{|\alpha+\beta|}F(\partial^{\alpha+\beta}\phi) \\ &= \partial^{\alpha+\beta}F(\phi)\end{aligned}$$

- $f \in C^1(\mathbb{R}^n)$, then for any $j \in \{1, \dots, n\}$ we have that the x_jth distributional derivative of f is precisely the distribution corresponding to the classical partial derivative $\partial_{x_j} f$. Hence, distributional derivatives generalize the standard definition of a derivative.

  Why?
  Let $\phi \in C_c^\infty(\mathbb{R}^n)$. By using Fubini's theorem, we can say that:

$$\int(\partial_{x_j} f)\phi = \iint(\partial_{x_j} f)\phi \mathrm{d}x_j \mathrm{d}(x_1, \dots, \widehat{x}_j, \dots x_n).$$

  But now by integration by parts plus the fact that $f\phi \to 0$ as x_j gets large, we can say that $\int(\partial_{x_j} f)\phi \mathrm{d}x_j = -\int f(\partial_{x_j}\phi)\mathrm{d}x_j$. In turn, after undoing Fubinis' theorem we are left with $\int(\partial_{x_j} f)\phi = -\int f(\partial_{x_j}\phi)$.

One more definition is that if $f \in L^1_{\text{loc}}$, α is a multi-index, and the αth distributional derivative of f is equivalent to the distribution given by another function $g \in L^1_{\text{loc}}$, we say that $g = \partial^\alpha f$ is the αth partial weak derivative of f.

One other note is that we could have replaced $\mathbb{R}^n$ by any open subset $U \subseteq \mathbb{R}^n$ and none of the prior results would have changed much.

Hopefully it's now clear that the reason distributions were defined the way they are is because they allow for PDEs to be solvable without having to make restrictive assumptions about what domains we are working on, or what our initial and boundary conditions are allowed to be.

Whenever I prove a theorem or exercise, in this section it will be from Folland Real Analysis. Also, we'll let $\widetilde{\phi}(x) := \phi(-x)$ for all $\phi \in C^\infty_c(\mathbb{R})$.

Proposition 9.1: If $f \in L^1(\mathbb{R}^n)$ and $\int f = a$, then $f_t \to a\delta_0$ in $\mathcal{D}'$ as $t \to 0$.

Note: Here δ_0 denotes the dirac delta function at 0. Also, $f_t(x) := \frac{1}{t^n} f(\frac{x}{t})$.

Proof:
For all $\phi \in C^\infty_c(\mathbb{R}^n)$ we have that $(f_t, \phi) = \int f_t(y)\widetilde{\phi}(-y)\mathrm{d}y = (f_t * \widetilde{\phi})(0)$. But the latter then converges to $a\widetilde{\phi}(0) = (a\delta_0, \phi)$ as $t \to 0$. ■

See page 30 of my math 240c spring 2025 notes and note that ϕ is uniformly continuous and bounded to understand why $(f_t * \widetilde{\phi})(0) \to a\delta_0$. Technically, I only proved the result for when $a = 1$. That sad, the proof of that result in 240c notes doesn't change much if $a \neq 1$.

If $F, G \in \mathcal{D}'(U)$, $V \subseteq U$ is an open set, and $\forall \phi \in C^\infty_c(V)$ we have that $(F, \phi) = (G, \phi)$ (in other words $F|_{C^\infty_c(V)} = G|_{C^\infty_c(V)}$), we say that $F = G$ on V.

Proposition 9.2: If $F, G \in \mathcal{D}'(U)$ and $F = G$ on V_α for all $\alpha \in A$ (with V_α an open subset of U), then $F = G$ on $\bigcup_{\alpha \in A} V_\alpha$.

Proof:
Let $\phi \in C^\infty_c(\bigcup_{\alpha \in A} V_\alpha)$. Since $\text{supp}(\phi)$ is compact, we can pick $\alpha_1, \ldots, \alpha_m \in A$ such that $\text{supp}(\phi) \subseteq \bigcup_{i=1}^m V_{\alpha_i}$. Then, we can let $\psi_1, \ldots, \psi_m \in C^\infty_c(\mathbb{R}^n)$ be a partition of unity on $\text{supp}(\phi)$ that is subordnate to $\{V_{\alpha_1}, \ldots, V_{\alpha_m}\}$ In turn:

$$(F, \phi) = \sum_{j=1}^m (F, \psi_j\phi) = \sum_{j=1}^m (G, \psi_j\phi) = (G, \phi). \blacksquare.$$

Based on the last proposition, we define the support of $F \in \mathcal{D}'(U)$ to be the set:

$$\text{supp}(F) := U - \bigcup\{V \subseteq U \text{ open } : F = 0 \text{ on V}\}.$$

There's a general procedure for extending linear operators from acting on functions to acting on distributions. Suppose U, V are open subsets of $\mathbb{R}^n$, $\mathcal{X}$ is a vector subspace of $L^1_{\text{loc}}(U)$, and $T : \mathcal{X} \to L^1_{\text{loc}}(V)$ is a linear operator. Then we try to solve for a continuous

linear operator $T' : C_c^\infty(V) \to C_c^\infty(U)$ such that $\int_V (Tf)\phi = \int_U f(T'\phi)$ for all $f \in \mathcal{X}$ and $\phi \in C_c^\infty(V)$. Finally, we define $T : \mathcal{D}'(U) \to \mathcal{D}'(V)$ by $(TF, \phi) \coloneqq (F, T'\phi)$.

Firstly, I need to show that T is a well-defined map. So, suppose $F \in \mathcal{D}'(U)$. Fortunately, it's obvious that TF is a linear map on $C_c^\infty(V)$. As for proving continuity, let $\{\phi_j\}_{j\in\mathbb{N}}$ be a sequence of functions converging to another function ϕ in $C_c^\infty(V)$. Since T' is a continuous map from $C_c^\infty(V)$ to $C_c^\infty(U)$, we know $T'\phi_j \to T'\phi$ in $C_c^\infty(U)$. And finally, by the continuity of F we thus know that $(F, T'\phi_j) \to (F, T'\phi)$. This finishes showing that TF is a distribution.

Next it's trivial to show that $T : \mathcal{D}'(U) \to \mathcal{D}'(V)$ is a linear map. However, I also claim that T is a continuous map with respect to the topology on the space of distributions. To see this, suppose $\{F_j\}_{j\in\mathbb{N}}$ is a sequence of distributions converging weak*-ly to F in $\mathcal{D}'(U)$. Then given any fixed $\phi \in C_c^\infty(V)$ we have that $(F_j, T'\phi) \to (F, T'\phi)$ as $j \to \infty$.

Some important continuous linear operators on distributions defined by the above procedure are as follows:

- (Differentiation): If $U \subseteq \mathbb{R}^n$ is open and α is a multi-index, then $\partial^\alpha : \mathcal{D}'(U) \to \mathcal{D}'(U)$ is a continuous map.

  Why?
  If you squint you can see that differentiation of distributions is defined via the procedure mentioned priorly where $\mathcal{X} = C^{|\alpha|}(U)$ and $T' : C_c^\infty(U) \to C_c^\infty(U)$ is defined by $\phi \mapsto (-1)^{|\alpha|}\partial^\alpha\phi$. Also, T' is easily checked to be a continuous linear map.

  This is because T' continuously maps $C_c^\infty(K)$ into $C_c^\infty(K)$ for all compact sets $K \subseteq U$.

  Note that ∂^α being continuous is very useful to the study of PDEs because it means that if $\{F_j\}_{j\in\mathbb{N}}$ is a sequence of distributions converging to F and satisfying some linear PDE, then F automatically satisfies that same linear PDE (as a distribution).

- (Multiplication by smooth functions): If $U \subseteq \mathbb{R}^n$ is open and $\psi \in C^\infty(U)$, then multiplication by ψ (defined as $(\psi F, \phi) \coloneqq (F, \psi\phi)$) is a continuous linear map from $\mathcal{D}'(U)$ to $\mathcal{D}'(U)$.

  Why?
  Let $T' : C_c^\infty(U) \to C_c^\infty(U)$ be defined by $T'\phi \coloneqq \psi\phi$. This is a linear map which I also claim is continuous. After all, suppose $\phi_j \to \phi$ in $C_c^\infty(U)$ and let $K \subseteq U$ be any compact set satisfying that $\operatorname{supp}(\phi_j) \subseteq K$ for all j. Then clearly T' maps $C_c^\infty(K)$ into $C_c^\infty(K)$. Furthermore, by product rule we have for any multi-index α that:

$$\begin{aligned}\|\partial^\alpha(\psi(\phi_j - \phi))\|_u &= \left\| \sum_{\beta+\gamma=\alpha} \tfrac{\alpha!}{\beta!\gamma!} (\partial^\beta\psi)(\partial^\gamma(\phi_j - \phi)) \right\|_u \\ &\leq \sum_{\beta+\gamma=\alpha} \tfrac{\alpha!}{\beta!\gamma!} \|\partial^\beta\psi\|_u \|\partial^\gamma(\phi_j - \phi)\|_u\end{aligned}$$

Since $\|\partial^\beta \psi\|_u$ is fixed for all mult-indices β while $\|\partial^\gamma(\phi_j - \phi)\|_u \to 0$ as $j \to \infty$ for all multi-indicies γ, we can thus conclude the above finite sum goes to 0 as $j \to \infty$.

Furthermore, if $\psi \in C_c^\infty(U)$ then multiplication by ψ is a well-defined continuous linear map from $\mathcal{D}'(U)$ to $\mathcal{D}'(\mathbb{R}^n)$.

Why?
If $K \subseteq \mathbb{R}^n$ is any compact set and we let T' be defined as before, then T' is a linear map from $C_c^\infty(K)$ into $C_c^\infty(K \cap \text{supp}(\psi))$. Also, the same reasoning still shows that T' is continuous.

- (Translation): Suppose $U \subseteq \mathbb{R}^n$ is open and $y \in \mathbb{R}^n$. Let $V = U + y$. Then we define a continuous map $\tau_y : \mathcal{D}'(U) \to \mathcal{D}'(V)$ by $(\tau_y F, \phi) := (F, \tau_{-y}\phi)$

  Explanation:
  Recall for any $f \in L^1_{\text{loc}}(U)$ that $\tau_y f(x) := f(x - y)$ is a function defined on $U + y = V$. Importantly, note that for any $\phi \in C_c^\infty(V)$ that:
  $$\int (\tau_y f)\phi = \int f(x - y)\phi(x)\mathrm{d}x = \int f(x)\phi(x + y)\mathrm{d}x = \int f \cdot \tau_{-y}\phi$$

  So, we define $T' : C_c^\infty(V) \to C_c^\infty(U)$ by $\phi \mapsto \tau_{-y}\phi$. This map is obviously linear and continuous.

- (Composition with Linear Maps): Suppose $U \subseteq \mathbb{R}^n$ is open, $S \in \text{GL}_n(\mathbb{R})$, and $V := S^{-1}(U)$. Then "composition with S" (which is defined and denoted as $(F \circ S, \phi) := |\det(S)|^{-1}(F, \phi \circ S^{-1})$) is a continuous linear map from $\mathcal{D}'(U)$ to $\mathcal{D}'(V)$.

  Annoyingly, the distribution definition of $F \circ S$ does *not* correspond to the set theoretic definition of function composition which would instead be the map $\phi \mapsto (F, S \circ \phi)$. However, as we will see the distribution definition is what generalizes the definition of $f \circ S$ when $f \in L^1_{\text{loc}}(U)$.

  Explanation:
  Given any $f \in L^1_{\text{loc}}$ and $\phi \in C_c^\infty(V)$ we have that:
  $$\int (f \circ S) \cdot \phi = |\det(S)|^{-1} \int f \cdot (\phi \circ S^{-1}).$$

  Hence, we define $T' : C_c^\infty(V) \to C_c^\infty(U)$ by $\phi \mapsto |\det(S)|^{-1}(\phi \circ S^{-1})$. It's easy to see that T' is linear, and that for any compact set $K \subseteq V$ we have that T' maps $C_c^\infty(K)$ into $C_c^\infty(S(K))$. Meanwhile, suppose $\phi_j \to \phi$ in $C_c^\infty(V)$. Then $\|T'(\phi_j - \phi)\|_u = |\det(S)|^{-1}\|\phi_j - \phi\|_u \to 0$ as $j \to \infty$. And by substituting ϕ with $\partial^\alpha \phi$ for any arbitrary multi-index α, this finishes proving that T' is a continuous map.

I want to finish my notes for today with a short calculation. This should be reminiscent of *page 874*. I claim that any locally integrable function of the form $u(x, t) = f(x - ct)$ gives a distribution solution to the PDE $\partial_t + c\partial_x = 0$.

Let $\phi \in C_c^\infty(\mathbb{R}^n)$. Then set $v = x - ct$ and $w = x + ct$, and let $\alpha\ (= 2c)$ be the Jacobian of the map from the variables (x,t) to the variables (v,w). In turn, we know that $x = \frac{v+w}{2}$ and $t = \frac{w-v}{2c}$. Also:

$$(\partial_t u, \phi) = \int_{\mathbb{R}^2} -f(x-ct)\phi_t(x,t)\mathrm{d}(x,t) = \int_{\mathbb{R}^2} -\alpha^{-1} f(v)\phi_t(\tfrac{v+w}{2}, \tfrac{w-v}{2c})\mathrm{d}(v,w)$$

Yet now consider by chain rule that $\phi_w(v,w) = \frac{1}{2}\phi_x(\frac{v+w}{2}, \frac{w-v}{2c}) + \frac{1}{2c}\phi_t(\frac{v+w}{2}, \frac{w-v}{2c})$.

Therefore:

$$\begin{aligned}&\int_{\mathbb{R}^2} -\alpha^{-1} f(v)\phi_t(\tfrac{v+w}{2}, \tfrac{w-v}{2c})\mathrm{d}(v,w)\\ &\qquad = \int_{\mathbb{R}^2} c\alpha^{-1} f(v)\phi_x(\tfrac{v+w}{2}, \tfrac{w-v}{2c})\mathrm{d}(v,w) - \int_{\mathbb{R}^2} 2c\alpha^{-1} f(v)\phi_w(v,w)\mathrm{d}(v,w).\end{aligned}$$

If we reverse our change of variables, the first of the two integrals becomes:

$$\int_{\mathbb{R}^2} cf(x-ct)\phi_x(x,t)\mathrm{d}(v,w) = -c(\partial_x u, \phi).$$

Meanwhile, we can use Fubini's theorem to evaluate that the second integral cancels. This is because:

$$\int_{\mathbb{R}^2} 2c\alpha^{-1} f(v)\phi_w(v,w)\mathrm{d}(v,w) = \int_{-\infty}^{\infty} 2c\alpha^{-1} f(v) \int_{-\infty}^{\infty} \phi_w(v,w)\mathrm{d}w\mathrm{d}v$$

But since ϕ is compactly supported (even in the variables v and w):

$$\int_{-\infty}^{\infty} \phi_w(v,w)\mathrm{d}w = \lim_{\substack{b\to\infty \\ a\to-\infty}} [\phi(v,b) - \phi(v,a)] = 0 \text{ for all } v.$$

So, we've finished showing that $\partial_t u = -c\partial_x u$ as distributions. ■

6/11/2026

Starting off today, here are some relavant exercises from Folland to the prior notes.

Exercise 9.1: Let $U \subseteq \mathbb{R}^n$ be open and suppose that $f_1, f_2, \ldots,$ and f are in $L^1_{\text{loc}}(U)$. Then the following implies that $f_j \to f$ in $\mathcal{D}'(U)$:

(a) Each $f_j \in L^p(U)$ (where $1 \le p \le \infty$) and $f_j \to f$ weakly or in the L^p operator norm.

Let ϕ be a fixed test function in $C_c^\infty(U)$, and also let q be the conjugate exponent of p. If $f_j \to f$ in the L^p operator norm, we have that:

$$|\int (f_j - f)\phi| \le \|f_j - f\|_p \cdot \|\phi\|_q \to 0 \text{ as } j \to \infty.$$

Meanwhile, if $f_j \to f$ weakly in L^p then note that because $\phi \in L^q$, $\int(\cdot)\phi$ defines a continuous linear functional on L^p. Hence $\int f_j\phi \to \int f\phi$ as $j \to \infty$. In either case, this proves $f_j \to f$ in $\mathcal{D}'(U)$.

(b) There exists $g \in L^1_{\text{loc}}(U)$ with $|f_j| \le g$ for all j, and $f_j \to f$ a.e.

Again fix some test function $\phi \in C_c^\infty(U)$. Then $|f_j\phi| \le g\|\phi\|_u \chi_{\text{supp}(\phi)} \in L^1$ for all $j \in \mathbb{N}$ Also, $f_j \to f$ pointwise a.e. implies that $f_j\phi \to f\phi$ pointwise a.e. as well. So, now just apply the dominated convergence theorem to get what we want.

Meanwhile $f_j \to f$ pointwise on U does not imply that $f_j \to f$ in $\mathcal{D}'(U)$.

For example, consider any $f \in C_c^\infty(\mathbb{R})$ satisfying that $\int f = 1$ and $f(0) = 0$. Then for any $t > 0$ consider the map $f_t(x) \coloneqq \frac{1}{t} f(\frac{x}{t})$. As $t \to 0$ we have that $f_t \to 0$ pointwise on $\mathbb{R}$. After all, $f_t(0) = 0$ for all t. Meanwhile, we can find some $R > 0$ such that $\text{supp}(f) \subseteq S\{x : \|x\|_2 \leq R\}$. And now for any $y \neq 0$, we know for small enough t that $\frac{\|y\|_2}{t} > R$. So $f_t(y) \to 0$ as $t \to 0$.

But now recalling *proposition 9.1 on page 960*, we know that $f_t \to \delta_0$ in $\mathcal{D}'(\mathbb{R})$ as $t \to 0$. Additionally, δ_0 is a different distribution from the zero distribution. So, $f_j \to 0$ pointwise, $f_j \not\to 0$ in $\mathcal{D}'(\mathbb{R})$, and instead $f_j \to \delta_0$ in $\mathcal{D}'(\mathbb{R})$. ■

Exercise 9.2: The product rule for derivatives is valid for products of smooth functions and distributions.

Let $U \subseteq \mathbb{R}^n$ be an open set, $F \in \mathcal{D}'(U)$, and $\psi \in C^\infty(U)$. Then consider any fixed test function $\phi \in C_c^\infty(U)$. Importantly, by product for functions we have that $\partial_{x_j}(\psi\phi) = \psi_{x_j}\phi + \psi\phi_{x_j}$. In turn, we can say that:

$$\begin{aligned}(\partial_{x_j}(\psi F), \phi) = -(\psi F, \phi_{x_j}) &= (F, -\psi\phi_{x_j}) \\ &= (F, -\partial_{x_j}(\psi\phi) + \psi_{x_j}\phi) = (\partial_{x_j}F, \psi\phi) + (\psi_{x_j}F, \phi)\end{aligned}$$

Or in other words, $\partial_{x_j}(\psi F) = (\partial_{x_j}\psi)F + \psi(\partial_{x_j}F)$. ■

Exercise 9.5: Suppose that f is continuously differentiable on $\mathbb{R}$ except at $x_1, \ldots, x_m$ where f has jump discontinuities, and that its pointwise derivative $\mathrm{d}f/\mathrm{d}x$ (defined except at the x_j's) is in $L^1_{\text{loc}}(\mathbb{R})$. Then the distribution derivative f' of f is given by:

$$f' = (\mathrm{d}f/\mathrm{d}x) + \textstyle\sum_{j=1}^m [f(x_j+) - f(x_j-)]\delta_{x_j}.$$

Here δ_{x_j} denotes the Dirac delta function at x_j. Also $f(x_j+) \coloneqq \lim_{x \to x_j^+} f(x)$ and $f(x_j-) \coloneqq \lim_{x \to x_j^-} f(x)$.

Let $\phi \in C_c^\infty(\mathbb{R}^n)$ be a fixed test function. Also without loss of generality suppose $x_1 < x_2 < \ldots < x_m$. Then we can write that:

$$(f', \phi) = -\textstyle\int_{-\infty}^{x_1} f(x)\phi'(x)\mathrm{d}x - \int_{x_m}^{\infty} f(x)\phi'(x)\mathrm{d}x - \sum_{j=2}^m \int_{x_{j-1}}^{x_j} f(x)\phi'(x)\mathrm{d}x.$$

But now by integration by parts, we have for $2 \leq j \leq m$ that:

$$\textstyle\int_{x_{j-1}}^{x_j} f(x)\phi'(x)\mathrm{d}x = f(x_j-)\phi(x_j) - f(x_{j-1}+)\phi(x_{j-1}) - \int_{x_{j-1}}^{x_j} \frac{\mathrm{d}f}{\mathrm{d}x}(x)\phi(x)\mathrm{d}x$$

Similarly, as ϕ is compactly supported we know that:

- $\int_{-\infty}^{x_1} f(x)\phi'(x)\mathrm{d}x = f(x_1-)\phi(x_1) - 0 - \int_{-\infty}^{x_1} \frac{\mathrm{d}f}{\mathrm{d}x}(x)\phi(x)\mathrm{d}x$,
- $\int_{x_m}^{\infty} f(x)\phi'(x)\mathrm{d}x = 0 - f(x_m+)\phi(x_m) - \int_{x_m}^{\infty} \frac{\mathrm{d}f}{\mathrm{d}x}(x)\phi(x)\mathrm{d}x$.

Therefore, we can conclude that:

$$\begin{aligned}(f', \phi) = &+ \textstyle\int_{-\infty}^{x_1} \frac{\mathrm{d}f}{\mathrm{d}x}(x)\phi(x)\mathrm{d}x + \int_{x_m}^{\infty} \frac{\mathrm{d}f}{\mathrm{d}x}(x)\phi(x)\mathrm{d}x + \sum_{j=2}^m \int_{x_{j-1}}^{x_j} \frac{\mathrm{d}f}{\mathrm{d}x}(x)\phi(x)\mathrm{d}x \\ &+ \textstyle\sum_{j=1}^m (f(x_j+) - f(x_j-))\phi(x_j) \\ = &\textstyle\int_{-\infty}^{\infty} \frac{\mathrm{d}f}{\mathrm{d}x}(x)\phi(x)\mathrm{d}x + \sum_{j=1}^m (f(x_j+) - f(x_j-))\phi(x_j) \\ = &(\tfrac{\mathrm{d}f}{\mathrm{d}x}, \phi) + \textstyle\sum_{j=1}^m (f(x_j+) - f(x_j-))(\delta_{x_j}, \phi).\end{aligned}$$ ■

An important operation for distributions I need to establish is convolutions. We have two ways of doing this (which I will eventually show are equivalent).

Note from 6/17/2026: I also just realized that since this point in my journal I've been short-handing $\|x\|_2$ as $|x|$ when $x \in \mathbb{R}^n$.

1. Suppose $U \subseteq \mathbb{R}^n$ is open. Then for a given $\psi \in C_c^\infty$, let
$$V \coloneqq \{x \in \mathbb{R}^n : x - y \in U \text{ for all } y \in \operatorname{supp}(\psi)\}.$$

Note that V is open (although it may be empty). After all, if $x \in V$ then we know $x + (-\operatorname{supp}(\psi))$ is a compact subset of U. In turn, we can find some $r > 0$ with $|a - b| \geq r$ for all $a \in x + (-\operatorname{supp}(\psi))$ and $b \in U^{\complement}$.

Now suppose $x' \in B_r(x)$. Then for any $y \in \operatorname{supp}(\psi)$ we have that $x' - y \in U$. After all, $b \coloneqq x' - y = x' - x + x - y$. So if we set $a \coloneqq x - y \in x + (-\operatorname{supp}(\psi))$, we have that $|a - b| = |x' - x| < r$. Hence, we know $b \notin U^{\complement}$. And this proves that $B_r(x) \subseteq V$.

If $f \in L^1_{\text{loc}}(U)$ then the integral
$$f * \psi(x) = \int f(x-y)\psi(y)\mathrm{d}y = \int f(y)\psi(x-y)\mathrm{d}y = \int f\tau_x\widetilde{\psi} = (f, \tau_x\widetilde{\psi})$$
is well-defined for all $x \in V$. In turn, to extend convolutions to distributions we define for any $F \in \mathcal{D}'(U)$ the *function* $F * \psi : V \to \mathbb{C}$ given by $F * \psi(x) \coloneqq (F, \tau_x\widetilde{\psi})$.

Proposition 9.3 (Part 1): $F * \psi \in C^\infty(V)$ and $\partial^\alpha(F * \psi) = \partial^\alpha F * \psi = F * \partial^\alpha\psi$ for all multi-indices α.

Proof:
Let $e_1, \ldots, e_n \in \mathbb{R}^n$ denote the standard basis. Then given any $x \in \mathbb{R}^n$, we can pick $t_0 > 0$ such that $x + te_j \in V$ whenever $|t| \leq t_0$.

Claim: $\frac{1}{t}(\tau_{x+te_j}\widetilde{\psi} - \tau_x\widetilde{\psi}) \to \tau_x\widetilde{\partial_{x_j}\psi}$ in $C_c^\infty(U)$ as $t \to 0$.

Why?
$\frac{1}{t}(\tau_{x+te_j}\widetilde{\psi} - \tau_x\widetilde{\psi})(y) = \frac{\psi(x+te_j-y)-\psi(x-y)}{t}$ while $\tau_x\widetilde{\partial_{x_j}\psi}(y) = \psi_{x_j}(x-y)$. We need to show that as functions in y, the former expression goes to the latter in $C_c^\infty(U)$ as $t \to 0$.

To start off, note that $\operatorname{supp}_y(\psi(x + te_j - y)) = x + te_j - \operatorname{supp}(\psi)$. Thus, we need to find a compact set $K \subseteq U$ which (perhaps after shrinking t_0) contains $x + te_j - \operatorname{supp}(\psi)$ for all $t \in [-t_0, t_0]$. Fortunately, as U^c is a closed set which is disjoint from the compact set $x - \operatorname{supp}(\psi)$, we know there is some $r > 0$ such that $|x - y - z| > r$ for all $y \in \operatorname{supp}(\psi)$ and $z \notin U$. Now shrink t_0 if necessary so that $0 < t_0 \leq r$. Then set:
$$K \coloneqq \{z \in \mathbb{R}^n : \inf_{y \in \operatorname{supp}(\psi)} |x - y - z| \leq r\}.$$

- Clearly $K \subseteq U$.
- I also claim that K is closed in $\mathbb{R}^n$. After all, $\rho(z) \coloneqq \inf_{y \in \operatorname{supp}(\psi)} |y - z|$ is easily

checked to be a continuous function. In turn, $\rho(x-z)$ is also a continuous function of z. And thus, the preimage of $[0,r]$ with respect to it is closed.

- Thirdly $\operatorname{diam}(K) \leq 2r + \operatorname{diam}(x - \operatorname{supp}(\psi)) < \infty$. So, K is a bounded set. Since K is both closed and bounded, this proves that K is compact.
- Finally, $x + te_j - \operatorname{supp}(\psi) \subseteq K$ for all $t \in [-t_0, t_0]$.

With that, it now easily follows that $\frac{1}{t}(\tau_{x+te_j}\widetilde{\psi} - \tau_x\widetilde{\psi})$ are all supported in a single compact subset of U once t is small enough. Hence, we can move on to proving the other requirement for convergence in $C_c^\infty(U)$, which is that the derivatives of $\frac{\psi(x+te_j-y)-\psi(x-y)}{t}$ (with respect to y) converge uniformly to the derivatives of $\psi_{x_j}(x-y)$.

Note that for any multi-index β, we have that:

$$\partial^\beta\left(\tfrac{\psi(x+te_j-y)-\psi(x-y)}{t}\right) = (-1)^{|\beta|}\tfrac{\partial^\beta\psi(x+te_j-y)-\partial^\beta\psi(x-y)}{t}.$$

Similarly:

$$\partial^\beta_{(y)}(\psi_{x_j}(x-y)) = (-1)^{|\beta|}\cdot\partial^\beta(\partial_{x_j}\psi)(x-y) = (-1)^{|\beta|}\cdot(\partial^\beta\psi)_{x_j}(x-y)$$

So by cancelling out the $(-1)^{|\beta|}$ if necessary, it actually suffices to only do the proof that $\frac{\psi(x+te_j-y)-\psi(x-y)}{t} \to \psi_{x_j}(x-y)$ uniformly over all y (since the same proof will also work when ψ is replaced by $\partial^\beta\psi$).

By the mean value theorem though, we know for every $t \in [-t_0, t_0]$ and $y \in U$ that there exists some $s_{t,y}$ (dependent on both t and y) that is between 0 and t and satisfying that:

$$\psi(x+te_j-y) - \psi(x-y) = t\psi_{x_j}(x+s_{t,y}e_j-y).$$

In turn, for all $t \in [-t_0,t_0]-\{0\}$ and $y \in U$ we have that:

$$\left|\tfrac{\psi(x+te_j-y)-\psi(x-y)}{t} - \psi_{x_j}(x-y)\right| = |\psi_{x_j}(x+s_{t,y}e_j-y) - \psi_{x_j}(x-y)|$$

And since ψ_{x_j} is uniformly continuous and $s_{t,y} \to 0$ as $t \to 0$ uniformly over all y, this finishes proving what we wanted.

Now that we've shown the prior claim, it now is trivial that $\partial_{x_j}(F*\psi)(x)$ is defined (in the traditional non-distribution sense) and equal to:

$$(F, \tau_x\widetilde{\partial_{x_j}\psi}) = F*\psi_{x_j}(x).$$

By induction, we can then conclude for any multi-index α that $\partial^\alpha(F*\psi) = F*\partial^\alpha\psi$.

As for showing that these partial derivatives are continuous, note that $\tau_x\widetilde{\psi} \to \tau_{x_0}\widetilde{\psi}$ in $C_c^\infty(U)$ as $x \to x_0$ for any $x_0 \in U$. This is simply because every partial derivative of ψ is uniformly continuous (since it is smooth and compactly supported). In turn,

we know that $F * \psi$ is continuous. And by replacing ψ with $\partial^\alpha \psi$ for any multi-index α, we can also conclude that $F * \partial^\alpha \psi$ is continuous.

Finally, note that:

$$\begin{aligned}(F * \partial^\alpha \psi)(x) = (F, \tau_x \widetilde{\partial^\alpha \psi}) &= (F, (-1)^{|\alpha|} \partial^\alpha (\tau_x \widetilde{\psi})) \\ &= (\partial^\alpha, \tau_x \widetilde{\psi}) = ((\partial^\alpha F) * \psi)(x).\end{aligned}$$

This shows that $F * \partial^\alpha \psi = \partial^\alpha F * \psi$. ■

2. Letting ψ and V be from before, we can alternatively extend the definition of a convolution by defining $F * \psi$ as a distribution.

If $f \in L^1_{\text{loc}}(U)$ then $f * \psi$ is a bounded continuous function. Hence, it is locally integrable and defines a distribution. Next, let ϕ be a test function in $C_c^\infty(V)$. Then:

$$\begin{aligned}\int (f * \psi)\phi &= \int (\int f(y)\psi(x-y)\mathrm{d}y)\phi(x)\mathrm{d}x \\ &= \int f(y)(\int \psi(x-y)\phi(x)\mathrm{d}x)\mathrm{d}y = \int f \cdot (\phi * \widetilde{\psi})\end{aligned}$$

In other words, if $T : L^1_{\text{loc}}(U) \to L^1_{\text{loc}}(V)$ is the linear map $Tf = f * \psi$, then $T' : C_c^\infty(V) \to C_c^\infty(U)$ given by $T'\phi = \phi * \widetilde{\psi}$ is a linear map satisfying that:

$$\int (Tf)\phi = \int f(T'\phi).$$

The linearity of T' is obvious. Also note that $\phi * \widetilde{\psi} = \int \phi(x-y)\widetilde{\psi}(y)$. Hence, $\phi * \widetilde{\psi}(x) \neq 0$ implies that $x - y \in \text{supp}(\phi) \subseteq V$ and $-y \in \text{supp}(\psi)$ for some y. But now let $r = \min\{|x-y| : x \in \text{supp}(\phi), y \in V^C\}$. By the definition of V, we know that $B_r(x-y) + y = B_r(x) \subseteq U$. Hence, $\text{supp}(\phi * \widetilde{\psi}) \subseteq U$.

It's also trivial from other theorems that $T'\phi$ is smooth for any $\phi \in C_c^\infty(V)$. Going a step further though, I claim T' is continuous as a map from $C_c^\infty(V)$ to $C_c^\infty(U)$. After all, let $\{\phi_j\}_{j \in \mathbb{N}}$ be a sequence converging in $C_c^\infty(V)$ to ϕ. Then we have that $\partial^\alpha(T'\phi_j) = (\partial^\alpha \phi_j) * \widetilde{\psi}$ and similarly that $\partial^\alpha(T'\phi) = (\partial^\alpha \phi) * \widetilde{\psi}$. But now $\|(\partial^\alpha \phi - \partial^\alpha \phi_j) * \widetilde{\psi}\|_u \leq \|\widetilde{\psi}\|_1 \cdot \|\partial^\alpha \phi - \partial^\alpha \phi_j\|_u \to 0$ as $j \to \infty$ for all multi-indices α.

It follows that for any distribution $F \in \mathcal{D}'(U)$, we can define a distribution on V given by $(F * \psi, \phi) \coloneqq (F, \phi * \widetilde{\psi})$. Also, the map $F \mapsto F * \psi$ is continuous from $\mathcal{D}'(U)$ to $\mathcal{D}'(V)$.

Proposition 9.3 (Part 2): The prior two definitions of convolutions yield identical distributions. In other words, if $F * \psi$ is the function from part 1 of this proposition and $\phi \in C_c^\infty(V)$, then $\int (F * \psi)\phi = (F, \phi * \widetilde{\psi})$.

Proof:
Note that $\phi * \widetilde{\psi}(x) = \int \phi(y)\psi(y-x)\mathrm{d}y = \int \phi(y)\tau_y \widetilde{\psi}(x)\mathrm{d}y$. We shall proceed by approximating the latter integral with Riemann sums.

Namely, for any $m \in \mathbb{N}$ we can lay the grid $(2^{-m}\mathbb{Z})^n$ over $\mathbb{R}^n$. Then we consider sampling a point y_j^m from each cube in the grid. Because $\text{supp}(\phi)$ is compact in V, we know that $\phi(y_j^m) = 0$ for all but finitely many cubes. Hence, without loss of generality we can say that $y_1^m, \ldots, y_{k(m)}^m$ are the only sampled points where ϕ is nonzero. And then, we set:

$$S_m(x) \coloneqq 2^{-nm} \sum_j \phi(y_j^m) \tau_{y_j^m} \widetilde{\psi}(x)$$

Claim: $S_m(x) \to \phi * \widetilde{\psi}$ uniformly on U.

Note that this trick specifically works because $\text{supp}(\phi)$ is bounded and also because both ϕ and $\widetilde{\psi}$ are uniformly continuous and bounded.

Let Q_j^m denote the cube that y_j^m was sampled from. Then let α equal the lebesgue measure of $\text{supp}(\phi) + \overline{B_{\sqrt{n}}(0)}$. Importantly, each Q_j^m has volume 2^{-nm}. Furthermore, each Q_j has a null interception with every other $Q_{j'}$, and the union of all the Q_j must be contained in $\text{supp}(\phi) + \overline{B_{\sqrt{n}}(0)}$. It follows that $\sum_j \int \chi_{Q_j^m} \leq \alpha$ for all m.

Now note that:

$$\begin{aligned}
|\phi * \widetilde{\psi}(x) - S_m(x)| &= \left| \int \phi(y) \tau_y \widetilde{\psi}(x) - \sum_j \phi(y_j^m) \tau_{y_j^m} \widetilde{\psi}(x) \chi_{Q_j^m} \mathrm{d}y \right| \\
&\leq \sum_j \int_{Q_j} |\phi(y) \tau_y \widetilde{\psi}(x) - \phi(y_j^m) \tau_{y_j^m} \widetilde{\psi}(x)| \mathrm{d}y \\
&\leq \sum_j \int_{Q_j} |\phi(y) \tau_y \widetilde{\psi}(x) - \phi(y_j^m) \tau_y \widetilde{\psi}(x)| \mathrm{d}y \\
&\qquad + \sum_j \int_{Q_j} |\phi(y_j^m) \tau_y \widetilde{\psi}(x) - \phi(y_j^m) \tau_{y_j^m} \widetilde{\psi}(x)| \mathrm{d}y
\end{aligned}$$

But now as $|y - y_j^m| < 2^{-m}\sqrt{n}$ for all $y \in Q_j^m$, we can guarentee for any $\delta > 0$ that there exists $M \in \mathbb{N}$ with $|y - y_j^m| < \delta$ for all $m \geq M$, $j \in \{1, \ldots, k(m)\}$, and $y \in Q_j^m$. Also because ϕ and $\widetilde{\psi}$ are uniformly continuous, we know that there exists some $\delta > 0$ such that:

$|\phi(y') - \phi(y)| < \frac{\varepsilon}{2\alpha \|\widetilde{\psi}\|_u}$ and $|\widetilde{\psi}(y') - \widetilde{\psi}(y)| < \frac{\varepsilon}{2\alpha \|\phi\|_u}$ when $|y' - y| < \delta$.

And now after putting these inequalities together, we get that when m is made large enough, then $|\phi * \widetilde{\psi}(x) - S_m(x)| < \varepsilon$ for all x.

Also note that $\partial^\alpha S_m = \sum_j \phi(y_j^m) \tau_{y_j^m} (\partial^\alpha \widetilde{\psi})(x)$ and $\partial^\alpha (\phi * \widetilde{\psi})(x) = (\phi * \partial^\alpha \widetilde{\psi})(x)$.

Thus by repeating the proof of our prior claim, we can conclude $\partial^\alpha S_m \to \phi * \widetilde{\psi}$ uniformly for all multi-indices α. And now, this lets us conclude that:

$$\begin{aligned}
(F, \phi * \widetilde{\psi}) = \lim_{m \to \infty} (F, S_m) &= \lim_{m \to \infty} 2^{-nm} \sum_j \phi(y_j^m) \cdot (F, \tau_{y_j^m} \widetilde{\psi}) \\
&= \lim_{m \to \infty} 2^{-nm} \sum_j \phi(y_j^m) \cdot (F * \psi)(y_j^m)
\end{aligned}$$

$$= \lim_{m\to\infty} \int \sum_j (F * \psi)(y_j^m)\phi(y_j^m)\chi_{Q_j^m}.$$

Finally, by the uniform continuity of $(F * \psi)\phi$ we know that:

$$\sum_j (F * \psi)(y_j^m)\phi(y_j^m)\chi_{Q_j} \to (F * \psi)\phi \text{ pointwise.}$$

By the dominated convergence theorem (using $\|\phi\|_u \|F * \psi\|_u \chi_{\text{supp}(\phi)+\overline{B_{\sqrt{n}}(0)}}$ as a bounding function), we can conclude that:

$$(F, \phi * \widetilde{\psi}) = \lim_{m\to\infty} \int \sum_j (F * \psi)(y_j^m)\phi(y_j^m)\chi_{Q_j} = \int (F * \psi)\phi. \blacksquare$$

A big use of convolutions is that they let us approximate distributions with functions in a restricted nice-to-work-with subspace.

Lemma 9.4: Suppose that $\phi, \psi \in C_c^\infty(\mathbb{R}^n)$ and $\int \psi = 1$. Then let $\psi_t(x) = \frac{1}{t^n}\psi(\frac{x}{t})$.

(a) Given any neighborhood U of $\text{supp}(\phi)$, we have that $\text{supp}(\phi * \psi_t) \subseteq U$ for t sufficiently small.

(b) $\phi * \psi_t \to \phi$ in $C_c^\infty(U)$ as $t \to 0$.

Proof:
Pick $R > 0$ such that $\text{supp}(\psi) \subseteq \overline{B_R(0)}$. Then $\text{supp}(\phi * \psi_t)$ is contained in the set $\text{supp}(\phi) + \overline{B_{tR}(0)}$. Furthermore, because $\text{supp}(\phi)$ is compact and disjoint from the closed set U^C, we know there is some $\varepsilon > 0$ with $|x - y| > \varepsilon$ for all $x \in \text{supp}(\phi)$ and $y \in U^C$. In turn, once t is small enough so that $tR < \varepsilon$, we know that $\text{supp}(\phi * \psi_t) \subseteq U$.

As for showing convergence in $C_c^\infty(U)$, first note that we can get a fixed compact set K which is contained in U and contains the support of $\phi * \psi_t$ by taking the closure of $\text{supp}(\phi) + \overline{B_{\varepsilon/2}(0)}$ and then only considering $t < \frac{\varepsilon}{2R}$.

Finally, for all multi-indices α we know $\partial^\alpha(\phi * \psi_t) = (\partial^\alpha\phi) * \psi_t \to \partial^\alpha\phi$ uniformly as $t \to 0$. (Just check pages 27-28 and 30 on my math 240c spring 2025 notes.) $\blacksquare$

Proposition 9.5: For any open $U \subseteq \mathbb{R}^n$, $C_c^\infty(U)$ is (sequentially) dense in $\mathcal{D}'(U)$ in the topology of $\mathcal{D}'(U)$.

Proof:
Suppose $F \in \mathcal{D}'(U)$. Our first goal will be to approximate F with a distribution supported in a compact subset of U.

Let $\{V_j\}_{j\in\mathbb{N}}$ be an increasing sequence of precompact open subsets of U whose union is U. (In particular assume $V_j \subseteq \overline{V_j} \subseteq V_{j+1}$ for all j.) By the C^∞ Urysohn lemma, we can pick a function $\omega_j \in C_c^\infty(U)$ with $\omega_j = 1$ on $\overline{V_j}$ and $\omega_j = 0$ on V_{j+1}^C. But now given any test function $\phi \in C_c^\infty(U)$, we know for j sufficiently large that $\text{supp}(\phi) \subseteq V_j$ and hence $(F, \phi) = (F, \omega_j\phi) = (\omega_j F, \phi)$.

Something else that is convenient to note is that because ω_j is compactly supported, we know $\omega_j F$ gives a well-defined distribution in $\mathcal{D}'(\mathbb{R}^n)$. (See *pages 961-962*...) This is nice because it means that we won't need to worry about the domains of any convolutions of $\omega_j F$.

Next, we approximate the distribution $\omega_j F$ with functions in $C^\infty(U)$.

For every j we can pick some $r_j > 0$ with $|x - y| > r_j$ for every $x \in \overline{V_j}$ and $y \in V_{j+1}^{\complement}$. Next, suppose ψ and ψ_t are as in lemma 9.4. Then for each j, we can follow the reasoning of lemma 9.4 all over again to pick some $T_j > 0$ with $\operatorname{supp}(\widetilde{\psi}_t) \subseteq \overline{B_{r_j}(0)}$ for all $t < T_j$. Using these bounds, we can also pick a decreasing sequence $\{t_j\}_{j\geq 2}$ in $\mathbb{R}_{>0}$ converging to 0 and satisfying that $t_j < T_{j-1}$ and $t_j < T_{j+1}$ for all j.

But now if $\phi \in C_c^\infty(U)$ with $\operatorname{supp}(\phi) \subseteq V_{j-1}$, we have that $\operatorname{supp}(\phi * \widetilde{\psi}_{t_j}) \subseteq \overline{V_j}$. In turn:

$$\Big((\omega_j F) * \psi_{t_j}, \phi\Big) = \Big(\omega_j F, \phi * \widetilde{\psi}_{t_j}\Big) = \Big(F, (\phi * \widetilde{\psi}_{t_j})\omega_j\Big) = \Big(F, \phi * \widetilde{\psi}_{t_j}\Big)$$

Combining the above fact with lemma 9.4(b) which tells us that $\phi * \widetilde{\psi}_{t_j} \to \phi$ in $C_c^\infty(U)$ as $t_j \to 0$, we can conclude after restricting the domain of $(\omega_j F) * \psi_{t_j}$ that:

$$(\omega_j F) * \psi_{t_j} \to F \text{ in } \mathcal{D}'(U) \text{ as } j \to \infty.$$

Finally, we already know from earlier that $(\omega_j F) * \psi_{t_j}$ will be a smooth function. Hence, we just need to show that this function is also compactly supported. To do this, note for any x that:

$$((\omega_j F) * \psi_{t_j})(x) = (\omega_j F, \tau_x \widetilde{\psi}_{t_j}) = (F, \omega_j \cdot \tau_x \widetilde{\psi}_{t_j}).$$

For the latter to not equal zero, we must have that $\omega_j \cdot \tau_x \widetilde{\psi}_{t_j} \not\equiv 0$. Hence, there must exist some $y \in V_{j+1}$ with $\psi_{t_j}(x - y) \neq 0$. Yet then, $\operatorname{supp}(\psi_{t_j}) \subseteq \overline{B_{r_{j+1}}(0)}$ implies that $x = (x - y) + y \in V_{j+2}$. So, $\operatorname{supp}((\omega_j F) * \psi_{t_j}) \subseteq \overline{V_{j+2}} \subseteq U$. ■

(As a side note: Folland proved denseness but not sequential denseness. So the result I showed is slightly stronger.)

6/15/2026

Before moving on, there are some more exercises I want to do. In the first exercise I want to do, I will specifically build off of the proof of the prior proposition to prove a stronger statement than what Folland actually originally asked.

Exercise 9.13: Let $U \subseteq \mathbb{R}^n$ be a nonempty, open, and connected set. Then suppose $F \in \mathcal{D}'(U)$ satisfies that $\partial_{x_k} F \equiv 0$ for $k = 1, \ldots, n$. Prove that F is given by a constant function.

To start off, note that when we picked our increasing sequnce of precompact open subsets $\{V_j\}_{j\in\mathbb{N}}$ while proving the prior lemma, it was possible to modify that sequence so that each V_j is connected.

One way to do this would be by picking a base point $x_0 \in V_1$ and then letting V_j^{new} be just the connected component of V_j containing x_0.

- It's then trivial that V_j^{new} is connected for all j. Plus, since open subsets of $\mathbb{R}^n$ are locally connected, we know that each V_j^{new} is open. And thirdly, each V_j^{new} is automatically precompact since it is a subset of a precompact set.

- Note that $\overline{V_j^{\text{new}}}$ is a connected subset of $\overline{V_j}$ containing x_0. In turn, it's also a connected subset of V_{j+1} containing x_0. Hence by the definition of V_{j+1}^{new}, we know $\overline{V_j^{\text{new}}} \subseteq V_{j+1}^{\text{new}}$. And with that, we've shown that the increasing condition on our modified sequence still holds. In other words:
$$V_j^{\text{new}} \subseteq \overline{V_j^{\text{new}}} \subseteq V_{j+1}^{\text{new}} \text{ for all } j.$$

- Finally, we need to show that the V_j^{new} still exhaust all of U (i.e. their union is all of U). To do this, note for any $x \in U$ that we can find a compact a compact subset K of U containing both x and x_0.
  After all, U is path connected. So just consider a path going from x_0 to x.

  But now K must be contained in V_j when j is sufficiently large. In turn, this proves that $x \in V_j^{\text{new}}$ when j is sufficiently large. And since x was arbitrary, we can conclude that $U = \bigcup_{j\in\mathbb{N}} V_j^{\text{new}}$.

Now define the sequence of functions $\{(\omega_j F) * \psi_{t_j}\}_{j\geq 2}$ in $C_c^\infty(U)$ identically as in the prior proposition. Then from exercise 9.2 on *page 964* as well as proposition 9.3 part 1 on *pages 965-967*, we know that:
$$\partial_{x_k}((\omega_j F) * \psi_{t_j}) = (\partial_{x_k}(\omega_j F)) * \psi_{t_j} = ((\partial_{x_k}\omega_j)F) * \psi_{t_j} + (\omega_j \partial_{x_k} F) * \psi_{t_j}$$

But now $((\omega_j \partial_{x_k} F) * \psi_{t_j})(x) = \left(\partial_{x_k} F, \omega_j \cdot \tau_x \widetilde{\psi}_{t_j}\right) = 0$ for all x by assumption. Hence:
$$\partial_{x_k}((\omega_j F) * \psi_{t_j}) = ((\partial_{x_k}\omega_j)F) * \psi_{t_j}.$$

Also, note that since $\omega_j \equiv 1$ on V_j, we know $\partial_{x_k}\omega_j \equiv 0$ on V_j. Then as $\partial_{x_k}\omega_j$ is continuous and the zero set of a continuous function is closed, we can also conclude that $\partial_{x_k}\omega_j \equiv 0$ on $\overline{V_j}$. But now for any $\phi \in C_c^\infty(V_{j-1})$ we have that $\operatorname{supp}(\phi * \widetilde{\psi}_{t_j}) \subseteq \overline{V_j}$ and hence:
$$\left(((\partial_{x_k}\omega_j)F) * \psi_{t_j}, \phi\right) = (F, \partial_{x_k}\omega_j \cdot (\phi * \widetilde{\psi}_{t_j})) = (F, 0) = 0.$$

It follows that $((\partial_{x_k}\omega_j)F) * \psi_{t_j}$ and the constant zero function both define identical distributions in $\mathcal{D}'(V_{j-1})$. So, they equal each other almost everywhere on V_{j-1}. And since both are continuous, we can further conclude they are equal on V_{j-1}. Thus by the connectedness of V_{j-1} plus the fact that $\partial_{x_k}((\omega_j F) * \psi_{t_j}) \equiv 0$ on V_{j-1} for every k, we have shown that $(\omega_j F) * \psi_{t_j}$ is constant on V_{j-1}.

For the sake of brevity and since the convolution in the definition won't matter hereafter, I'm going to name $f_j := (\omega_j F) * \psi_{t_j}$. All that's important to remember is that:

- each f_j is a measurable function.
- there exists $c_j \in \mathbb{C}$ with $f_j \equiv c_j$ on V_{j-1},
- $f_j \to F \in \mathcal{D}'(U)$.

To finish off, note that if $\phi \in C_c^\infty(V_1)$ is a test function satisfying that $\int \phi = 1$, then $c_j = \int f_j \phi \to (F, \phi)$ as $j \to \infty$. It follows that the sequence $\{c_j\}_{j\geq 2}$ converges to some constant $C \in \mathbb{C}$.

Next note that the f_j converge uniformly on compact subsets of U to the constant C. That is enough to conclude for any test function $\phi \in C_c^\infty(U)$ that $\int f_j \phi \to C \int \phi$ as $j \to \infty$. And since $f_j \to F \in \mathcal{D}'(U)$, we can also conclude that $(F, \phi) = C \int \phi$ for all $\phi \in C_c^\infty(U)$. This proves that F is given by a constant function. ■

On a related note, the next exercise describes all the distributions F satisfying the condition that $\operatorname{supp}(F) = \{0\}$.

Exercise 9.11: Let F be a distribution on $\mathbb{R}^n$ such that $\operatorname{supp}(F) = \{0\}$.

(a) There exists $N \in \mathbb{N}$ and $C > 0$ such that for all $\phi \in C_c^\infty(\mathbb{R}^n)$:
$$|(F, \phi)| \leq C \sum_{|\alpha| \leq N} \sup_{|x| \leq 1} |\partial^\alpha \phi(x)|$$

By the C^∞-Urysohn lemma, there is a function $\omega \in C_c^\infty(\mathbb{R}^n, [0,1])$ satisfying that $\omega \equiv 1$ on $\overline{B_{1/2}(0)}$ and $\omega \equiv 0$ on $B_1(0)^{\mathsf{C}}$. Then note for any test function $\phi \in C_c^\infty(\mathbb{R}^n)$ and $\varepsilon > 0$ that:
$$(F, \phi) = (F, \omega\phi) + (F, (1-\omega)\phi) = (F, \omega\phi) + 0.$$

The second term above cancels since $\operatorname{supp}(F) = \{0\}$.

Yet now since F is a continuous linear functional on the Fréchet space $C_c^\infty(\overline{B_1(0)})$, we know there must exist some $C > 0$ and $N \in \mathbb{N}$ such that for all $\psi \in C_c^\infty(\overline{B_1(0)})$:
$$|(F, \psi)| \leq C \sum_{|\alpha| \leq N} \sup_{|x| \leq 1} |\partial^\alpha \psi(x)|.$$

In particular, we can thus conclude that:
$$|(F, \phi)| = |(F, \omega\phi)| \leq C \sum_{|\alpha| \leq N} \sup_{|x| \leq 1} |\partial^\alpha (\omega\phi)(x)|$$

Then by product rule, we have that $\partial^\alpha(\omega\phi) = \sum_{\beta+\gamma=\alpha} \frac{\alpha!}{\beta!\gamma!} (\partial^\beta \omega)(\partial^\gamma \phi)$. This lets us conclude for each multi-index α that:
$$\sup_{|x| \leq 1} |\partial^\alpha(\omega\phi)(x)| \leq \sum_{\beta+\gamma=\alpha} \left(\frac{\alpha!}{\beta!\gamma!} \sup_{|x| \leq 1} |\partial^\beta \omega(x)| \right) \cdot \sup_{|x| \leq 1} |\partial^\gamma \phi(x)|$$

Hence, it is now clear how to find a larger constant C^{new} such that for all test functions $\phi \in C_c^\infty(\mathbb{R}^n)$:
$$|(F, \phi)| = |(F, \omega\phi)| \leq C^{\text{new}} \sum_{|\alpha| \leq N} \sup_{|x| \leq 1} |\partial^\alpha \phi(x)|.$$

(b) Fix $\psi \in C_c^\infty(\mathbb{R}^n)$ with $\psi(x) = 1$ for $|x| \leq 1$ and $\psi(x) = 0$ for $|x| \geq 2$. If $\phi \in C_c^\infty(\mathbb{R}^n)$, let $\phi_k(x) = \phi(x)[1 - \psi(kx)]$. If $\partial^\alpha \phi(0) = 0$ for $|\alpha| \leq N$, then $\partial^\alpha \phi_k \to \partial^\alpha \phi$ uniformly as $k \to \infty$ for $|\alpha| \leq n$.

As a reminder, if $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ and $\alpha = (\alpha_1, \ldots, \alpha_n)$ is a multi-index, we denote $x^\alpha := \prod_{i=1}^n x_i^{\alpha_i}$.

Suppose α is a multi-index with $|\alpha| \leq N$. Then we know $\partial^\alpha \phi \in C^{N+1-|\alpha|}(\mathbb{R}^n)$. Hence, by Taylor's theorem we can write that:

$$\partial^\alpha \phi(x) = \sum_{|\beta| \leq N+1-|\alpha|} (\partial^{\beta+\alpha}\phi)(0) \cdot \tfrac{x^\beta}{\beta!} + E(x)$$

where $E(x)$ is an error term satisfying that $\frac{E(x)}{|x|^{N+1-|\alpha|}} \to 0$ as $x \to 0$. Another fact which will be relevant is that E must be continuous since every other term in the above equation is continuous.

Now note that if $|\beta| \leq N - |\alpha|$, then $|\beta + \alpha| \leq N$ and so $(\partial^{\beta+\alpha}\phi)(0) = 0$. Meanwhile, at the same we can fairly easily show that $|x^\beta| \leq |x|^{|\beta|}$. Hence, we get that:

$$\begin{aligned} |\partial^\alpha \phi(x)| &= \left| \sum_{|\beta|=N+1-|\alpha|} (\partial^{\beta+\alpha}\phi)(0) \cdot \tfrac{x^\beta}{\beta!} + E(x) \right| \\ &\leq \sum_{|\beta|=N+1-|\alpha|} \tfrac{|(\partial^{\beta+\alpha}\phi)(0)|}{\beta!} \cdot |x|^{N+1-|\alpha|} + |E(x)| \end{aligned}$$

Next note that by the extreme value theorem applied to the function:

$$f(x) := \begin{cases} \frac{E(x)}{|x|^{N+1-|\alpha|}} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0, \end{cases}$$

we can find some constant $C' \geq 0$ such that $|E(x)| \leq C'|x|^{N+1-|\alpha|}$.

Then upon letting $C_\alpha := C' + \sum_{|\beta|=N+1-|\alpha|} \frac{|(\partial^{\beta+\alpha}\phi)(0)|}{\beta!}$, we know $|\partial^\alpha \phi(x)| \leq C_\alpha |x|^{N+1-|\alpha|}$.

To finish off, note that when $|x| > \frac{2}{k}$ we trivially have that $\partial^\alpha \phi = \partial^\alpha \phi_k$ for any multi-index α. Therefore, it suffices to show that:

$$\sup_{|x| \leq \frac{2}{k}} |\partial^\alpha \phi - \partial^\alpha \phi_k| \to 0 \text{ as } k \to \infty \text{ when } |\alpha| \leq N.$$

Fortunately, note for every k that:

$$\partial^\alpha \phi - \partial^\alpha \phi_k = \partial^\alpha(\phi(x)\psi(kx)) = \sum_{\beta+\gamma=\alpha} \tfrac{\alpha!}{\beta!\gamma!} k^{|\gamma|} \partial^\beta \phi(x) \partial^\gamma \psi(kx)$$

In turn, we get that:

$$\begin{aligned} |\partial^\alpha \phi(x) - \partial^\alpha \phi_k(x)| &\leq \sum_{\beta+\gamma=\alpha} \tfrac{\alpha!}{\beta!\gamma!} k^{|\gamma|} \|\partial^\gamma \psi\|_u \cdot \partial^\beta \phi(x) \\ &\leq \sum_{\beta+\gamma=\alpha} \tfrac{\alpha!}{\beta!\gamma!} k^{|\gamma|} \|\partial^\gamma \psi\|_u \cdot C_\beta |x|^{N+1-|\alpha|+|\gamma|} \end{aligned}$$

And in particular when $|x| \leq \frac{2}{k}$, we have that:

$$|\partial^\alpha \phi(x) - \partial^\alpha \phi_k(x)| \leq \sum_{\beta+\gamma=\alpha} \tfrac{\alpha!}{\beta!\gamma!} 2^{|\gamma|} C_\beta \|\partial^\gamma \psi\|_u \cdot |x|^{N+1-|\alpha|}$$

This last bound is independent of k and also crucially goes to 0 as x goes to 0. (This specifically works because $|\alpha| \leq N$.) Hence, it is now easy to conclude that:

$$\sup_{|x| \leq \frac{2}{k}} |\partial^\alpha \phi - \partial^\alpha \phi_k| \to 0 \text{ as } k \to \infty.$$

(c) If $\phi \in C_c^\infty(\mathbb{R}^n)$ and $\partial^\alpha \phi(0) = 0$ for $|\alpha| \leq N$, then $(F, \phi) = 0$.

Let ϕ_k be defined as in part (b). Then because $\text{supp}(\phi_k) \subseteq \mathbb{R}^n - \{0\}$, we trivially know that $(F, \phi_k) = 0$ for all k. That said, by part (a) and (b) put together, we can also see that:

$$|(F, \phi)| = |(F, \phi) - 0| = |(F, \phi) - (F, \phi_k)| = |(F, \phi - \phi_k)| \to 0 \text{ as } k \to \infty.$$

Therefore, we must also have that $(F, \phi) = 0$.

(d) There exists constants c_α (for $|\alpha| \leq N$) such that $F = \sum_{|\alpha| \leq N} c_\alpha \partial^\alpha \delta_0$.

Let ϕ be any test function in $C_c^\infty(\mathbb{R}^n)$. Also let $\omega \in C_c^\infty(\mathbb{R}^n)$ be a function satisfying that $\omega \equiv 1$ on $\overline{B_1(0)}$. That way, we have that $g(x) := \omega(x) \sum_{|\alpha| \leq N} \frac{(\partial^\alpha \phi)(0)}{\alpha!} x^\alpha$ is compactly supported, smooth, and equals the polynomial $P(x) := \sum_{|\alpha| \leq N} \frac{(\partial^\alpha \phi)(0)}{\alpha!} x^\alpha$ on a neighborhood of the origin.

But now note that $\partial^\alpha(\phi - g)(0) = \partial^\alpha(\phi - P)(0) = 0$ for all multi-indices α with $|\alpha| \leq N$. Hence, by part (c) we can conclude that:

$$(F, \phi) = (F, \phi - g) + (F, g) = 0 + (F, g)$$

Furthermore, $(F, g) = \sum_{|\alpha| \leq N} \frac{(\partial^\alpha \phi)(0)}{\alpha!} (F, \omega(x) x^\alpha) = \sum_{|\alpha| \leq N} \frac{(-1)^{|\alpha|} (F, \omega(x) x^\alpha)}{\alpha!} (\partial^\alpha \delta_0, \phi)$.

So, we just need to set $c_\alpha := \frac{(-1)^{|\alpha|} (F, \omega(x) x^\alpha)}{\alpha!}$. Then as ϕ was arbitrary, we've proven that $F = \sum_{|\alpha| \leq N} c_\alpha \partial^\alpha \delta_0$. ■

6/17/2026

My goal for maybe today and tomorrow is to type all my remaining math 200c notes. This will mean finishing going over primary decompositions before then taking notes on Artinian rings, discrete valuation rings (DVRs), and Dedekind rings.

Math 200c Lecture Notes:

Lemma: Let A be a Noetherian ring and $J \lhd A$ a proper ideal. If A/J has only one associated prime then $\text{Ass}_A(A/J) = \{\sqrt{J}\}$. In turn, by the lemma on *page 947* we also know J is primary.

Proof:
We know $\sqrt{J} \subseteq P$ for all $P \in \text{Ass}_A(A/J)$ since every prime ideal containing J contains $\sqrt{J}$. Meanwhile, suppose for the sake of contradiction that $P \in \text{Ass}_A(A/J)$ and that $P - \sqrt{J} \neq \emptyset$. Then we can fix some $x \in P - \sqrt{J}$ and let $S := \{1, x, x^2, \ldots\}$.

But now by the proposition on *page 946*, we know that:

$$\text{Ass}_{S^{-1}A}(S^{-1}(\tfrac{A}{J})) = \{S^{-1}Q : Q \in \text{Ass}_A(\tfrac{A}{J}) \text{ and } Q \cap S = \emptyset\}$$

Since $\text{Ass}_A(\frac{A}{J}) = \{P\}$ and $P \cap S \neq \emptyset$, we thus know that $\text{Ass}_{S^{-1}A}(S^{-1}(\frac{A}{J})) = \emptyset$.

Yet simultaneously since $S^{-1}A$ is a Noetherian ring, we know from the proposition on *page 943* that if $S^{-1}A$ and $S^{-1}(\frac{A}{J})$ are nontrivial then $\text{Ass}_{S^{-1}A}(S^{-1}(\frac{A}{J})) \neq \emptyset$. So, we can only avoid a contradiction if $S^{-1}A$ and $S^{-1}(\frac{A}{J})$ are trivial.

Yet $S^{-1}(\frac{A}{J})$ would be trivial if and only if for all $a \in A$ there exists $n \in \mathbb{N}$ with $x^n(a+J) = 0+J$. In particular, we'd have to have that $x^n(1+J) = x^n + J = 0 + J$. Yet that contradicts that $x \notin \sqrt{J}$. Hence, we know $S^{-1}(\frac{A}{J})$ is not a trivial module. ■

Lemma: Let A be a Noetherian ring. If Q_1, Q_2 are both P-primary ideals, then $Q_1 \cap Q_2$ is also P-primary.

Proof:
Consider the map $A \to \frac{A}{Q_1} \oplus \frac{A}{Q_2}$ given by $a \mapsto (\overline{a}, \overline{a})$. The kernel of this map is $Q_1 \cap Q_2$. Hence, $\frac{A}{Q_1 \cap Q_2}$ embeds into $\frac{A}{Q_1} \oplus \frac{A}{Q_2}$.

But now note from the lemma at the bottom of *page 942* that:
$\text{Ass}_A(\frac{A}{Q_1 \cap Q_2}) \subseteq \text{Ass}_A(\frac{A}{Q_1} \oplus \frac{A}{Q_2})$ and $\text{Ass}_A(\frac{A}{Q_1} \oplus \frac{A}{Q_2}) = \text{Ass}_A(\frac{A}{Q_1}) \cup \text{Ass}_A(\frac{A}{Q_2}) = \{P\}$.

Also note from the proposition on *page 943* that $\text{Ass}_A(\frac{A}{Q_1 \cap Q_2}) \neq \emptyset$. It follows that $\text{Ass}_A(\frac{A}{Q_1 \cap Q_2}) = \{P\}$. And by the prior lemma we can conclude that $\sqrt{Q_1 \cap Q_2} = P$ and that $Q_1 \cap Q_2$ is P-primary. ■

Theorem: Any proper ideal I of a Noetherian ring A has an irredundant primary decomposition $I = Q_1 \cap \cdots \cap Q_n$ where each Q_i is P_i-primary. Moreover, $\text{Ass}_A(A/I) = \{P_1, \ldots, P_n\}$.

Proof:
We'll say a proper ideal J is irreducible if there doesn't exist ideals $K_1 \supsetneq J$ and $K_2 \supsetneq J$ with $J = K_1 \cap K_2$.

Claim 1: If $J \lhd A$ is irreducible then it is primary.

Why?
Suppose A/J has two distinct associated primes P_1, P_2. Then A/J has submodules $N_1 \cong A/P_1$ and $N_2 \cong A/P_2$. And since $P_1 \neq P_2$, we know $N_1 \cap N_2 = \{0\}$.

Otherwise, we'd have that $\text{Ass}_A(N_1 \cap N_2) \neq \emptyset$ since $N_1 \cap N_2$ is a nontrivial module over a Noetherian ring (see *page 943*). In turn, there'd exist $a \in N_1 \cap N_2$ with $\text{Ann}_A(a+J) \in \text{Spec}(A)$. Yet since $N_1 \cap N_2$ embeds into both A/P_1 and A/P_2, we also must have that:

- $\text{Ann}_A(a+J) \in \text{Ass}_A(A/P_1) = \{P_1\}$,
- $\text{Ann}_A(a+J) \in \text{Ass}_A(A/P_2) = \{P_2\}$.

So, $\text{Ann}_A(a+J)$ must equal both P_1 and P_2, which is a contradiction.

But now upon writing $N_1 = K_1/J$ and $N_2 = K_2/J$, we have that $N_1 \cap N_2 = \{0\}$ implies that $K_1 \cap K_2 = J$. This contradicts that J is irreducible. So, A/J has at most one associated prime.

Since A/J is nontrivial and A is nontrivial and Noetherian, we also know A/J has at least one associated prime (see *page 943*). Hence, A/J has exactly one associated prime. And by two lemmas ago, that proves that J is primary.

Claim 2: Any proper ideal is a finite intersection of irreducible ideals.

Why?

If not, since A is Noetherian we could choose a maximal proper ideal J which is not a finite intersection of irreducible ideals. Then J is not irreducible. So, $J = K_1 \cap K_2$ where $K_1 \supsetneq J$ and $K_2 \supsetneq J$. Yet now by the maximality of J, we must have that each K_i is eiter a finite intersection of irreducible ideals or the entire ring A. Although, we can't have that both $K_i = A$ since that would imply $J = A$ which is not true. So after ignorning potentially intersecting J with A, we've written J has a finite intersection of irreducible ideals. This is a contradiction.

Putting the two prior claims together, we've thus shown every ideal I has a primary decomposition. So, let us write $I = Q_1 \cap \cdots \cap Q_n$ where each Q_i is P_i-primary.

By the lemma before this theorem, we can without loss of generality intercept pairs of our Q_i together until no P_i is repeated. Next, if $Q_I \subseteq \bigcap_{j \neq i} Q_j$ for any i, then we can without loss of generality just remove Q_i from our decomposition. After finitely many steps of this, we will get an irredundant primary decomposition.

As for the other claim of the theorem, suppose $I = Q_1 \cap \cdots \cap Q_n$ is an irredundant primary decomposition where each Q_i is P_i-primary. Then $A/I = A/(Q_1 \cap \cdots \cap Q_n)$ is isomorphic to a submodule of $\bigoplus_{j=1}^n A/Q_j$.

(Consider the homomorphism $a \mapsto (a + Q_1, \ldots, a + Q_n)$. The kernel of this map is $Q_1 \cap \cdots \cap Q_n$.)

But now from the lemma at the bottom of *page 942*, we know that:

$$\operatorname{Ass}_A(\tfrac{A}{I}) \subseteq \operatorname{Ass}_A(\textstyle\bigoplus_{j=1}^n \frac{A}{Q_j}) = \bigcup_{j=1}^n \operatorname{Ass}_A(\frac{A}{Q_j}) = \bigcup_{j=1}^n \{P_j\}.$$

Hence, $\operatorname{Ass}_A(A/I) \subseteq \{P_1, \ldots, P_n\}$. As for the other inclusion, consider any fixed P_i and look at $J_i = \bigcap_{j \neq i} Q_j$. Note that $\frac{J_i}{I} = \frac{J_i}{J_i \cap Q_i} \cong \frac{J_i + Q_i}{Q_i}$ is a nonzero submodule of A/Q_i. So, we know that $\emptyset \subsetneq \operatorname{Ass}_A(J_i/I) \subseteq \operatorname{Ass}_A(A/Q_i) = \{P_i\}$. It follows that J_i/I has P_i as an associate prime. So, there is a submodule of $J_i/I \subseteq A/I$ isomorphic to A/P_i. And this proves that $P_i \in \operatorname{Ass}_A(A/I)$. ■

Some relevant homework problems are as follows:

Set 5 Problem 1: In the polynomial ring $A = k[x, y, z]$ where k is a field, let $P_1 = \langle x, y \rangle$, $P_2 = \langle x, z \rangle$, and let $I = P_1 P_2 = \langle x^2, xy, xz, yz \rangle$.

(a) Find a prime filtration of the module A/I.

To start off, $\langle x, y, z \rangle$ is a maximal ideal of A. We know this because $\frac{k[x,y,z]}{\langle x,y,z \rangle} \cong k$ is a field. Yet also note that $\langle x, y, z \rangle \subseteq \operatorname{Ann}_A(x + I)$. By maximality plus the fact that $1(x + I) \neq 0 + I$, we can thus conclude that $\langle x, y, z \rangle = \operatorname{Ann}_A(x + I) \in \operatorname{Ass}(A/I)$. Also, if we let M_1 be the submodule $\langle x + I \rangle$, we have that $M_1 \cong \frac{A}{\langle x,y,z \rangle}$.

Next, consider that $M_1 = \frac{\langle x, x^2, xy, xz, yz\rangle}{\langle x^2, xy, xz, yz\rangle} = \frac{\langle x, yz\rangle}{I}$. Furthermore, $\frac{A/I}{\langle x,yz\rangle/I} \cong \frac{A}{\langle x,yz\rangle}$ by the isomorphism $\overline{f+I} \mapsto \overline{f}$. So, we need to find an associated prime of $A/\langle x, yz\rangle$.

Fortunately, consider that $P_1 = \langle x, y\rangle$ is a prime ideal of A. After all $\frac{k[x,y,z]}{\langle x,y\rangle} \cong k[z]$ is a domain. Furthermore, it's obvious that $P_1 \subseteq \mathrm{Ann}_A(z)$, while the other inclusion comes from the fact that if $f(x, y, z)$ contains any term not divisible by x or y, then $zf(x, y, z)$ will also contain a term which is not divisible by xz (which is divisible by x) or yz. Therefore, $P_1 = \mathrm{Ann}_A(z + \langle x, yz\rangle)$.

$\widetilde{M_2}$ be the submodule of $\frac{A}{\langle x,yz\rangle}$ generated by $z + \langle x, yz\rangle$. Then notice that $\widetilde{M_2} = \frac{\langle z,x,yz\rangle}{\langle x,yz\rangle} = \frac{P_2}{\langle x,yz\rangle}$. Hence, if we let $M_2 := \frac{P_2}{I}$, we have that our original isomorphism from $\frac{A/I}{\langle x,yz\rangle/I}$ to $\frac{A}{\langle x,yz\rangle}$ sends M_2/M_1 to $\widetilde{M_2}$. Furthermore, $M_2/M_1 \cong \widetilde{M_2} \cong A/P_1$.

Finally, note that $P_2 \in \mathrm{Spec}(A)$ for identical reasons that $P_1 \in \mathrm{Spec}(A)$ is. Furthermore, $\frac{A/I}{P_2/I} \cong \frac{A}{P_2}$. Hence, we can conclude that a prime filtration of A/I is given by:

$$\{0\} \subseteq \frac{\langle x,yz\rangle}{I} \subseteq \frac{\langle x,z\rangle}{I} \subseteq \frac{A}{I}.$$

Also, the primes in this filtration are $\langle x, y, z\rangle$ followed by P_1 and then P_2.

(b) Find two different primary decompositions of I.

Here is a way to get an infinite family of primary decompositions. Let n, m be positive integers and define $Q = \langle x^2, y^n, z^m, xy, xz, xy\rangle$. Then we have that $x, y, z \in \sqrt{Q}$. And since $\langle x, y, z\rangle$ is a maximal ideal and $\sqrt{Q}$ is a proper ideal, we can conclude that $\sqrt{Q} = \langle x, y, z\rangle$. By the worked example on *page 948*, we can thus conclude that Q is $\langle x, y, z\rangle$-primary.

Next, note that $I = \langle x^2, xy, xz, yz\rangle$ contains precisely the polynomials for which every nonzero monomial term is divisible by x^2, xy, xz, or yz. Meanwhile f is contained in $P_1 \cap P_2 \cap Q$ if and only if each of the three following things are true for every nonzero monomial term of f:

1. the term is divisible by x or y,
2. the term is divisible by x or z,
3. the term is divisible by x^2, y^n, z^m, xy, xz, or yz.

By breaking into cases, we can thus see that $P_1 \cap P_2 \cap Q = I$. Since P_1 and P_2 are prime ideals and thus automatically primary, this proves that $P_1 \cap P_2 \cap Q$ is a primary decomposition of I.

Finally, I want to show this decomposition is irredundant. To start off P_1, P_2, and $\langle x, y, z\rangle$ are all distinct primes. Hence we have no issue there. Meanwhile:

- $P_1 \cap P_2$ contains the element x which is not in Q. Hence, $P_1 \cap P_2 \nsubseteq Q$.
- $P_1 \cap Q$ contains the element y^n which is in P_2. Hence, $P_1 \cap Q \nsubseteq P_2$.

- $P_2 \cap Q$ contains the element z^m which is not in P_1. Hence, $P_2 \cap Q \not\subseteq P_1$. ■

Set 5 Problem 2(b): Suppose A is a Noetherian domain. Then show that A is a UFD if and only if for every principal ideal $\langle f \rangle$ in A, every prime ideal minimal over $\langle f \rangle$ is also principal.

$(\Longrightarrow)$

Suppose A is a UFD. Then we can write $f = up_1^{e_1} \cdots p_n^{e_n}$ where each p_i is a pairwise nonassociate prime, each e_i is a positive integer, and u is a unit. But then in turn, we know $\langle f \rangle = \prod_{i=1}^{n} \langle p_i^{e_i} \rangle = \prod_{i=1}^{n} (\prod_{j=1}^{e_i} \langle p_i \rangle)$ (see *pages 594-595*). So, we've written $\langle f \rangle$ as a product of prime ideals $P_1 \cdots P_m$ which are all principal.

Now to finish off, just use identical reasoning as on *page 900* to show that any minimal prime Q over $\langle f \rangle$ is one of those P_j.

As a side note, none of the reasoning in this direction required A to be Noetherian.

$(\Longleftarrow)$

Note that because A is a Noetherian domain, it suffices to prove that every irreducible element of A is prime in order to prove that A is a UFD. (See the theorem on *page 486* as well as the lemma on *pages 488-489*).

Fortunately, note that p is an irreducible element of A if and only if $\langle p \rangle \neq A$, $\langle p \rangle \neq \{0\}$, and $\langle p \rangle$ is maximal among principal proper ideals. But then since every prime ideal minimal over $\langle p \rangle$ is principal, and also because we know such minimal primes exist (since $A/\langle p \rangle$ is nontrivial), we in turn must have that $\langle p \rangle$ is the minimal prime ideal over itself. And now we've proven that p is prime. ■

On *page 901* I briefly defined Artinian rings. That said, that definition is not where Rogalski wants to start covering them. So instead we'll define them differently here and later prove the two definitions are equivalent.

We say a ring A is Artinian if it has the descending chain condition on ideals. (See set 4 problem 5 and especially remark 1 on *page 953*).

More generally, we say an A-module M is Artinian if it has the descending chain condition on submodules. Note that A is an Artinian A-module iff A is an Artinian ring.

Proposition: If $\{0\} \longrightarrow M' \hookrightarrow M \twoheadrightarrow M'' \longrightarrow \{0\}$ is an S.E.S. of A-modules, then M is Artinian if and only if M' and M'' are Artinian.

Proof:

To start off, by applying an isomorphism of short exact sequences we can without loss of generality assume M' is a submodule N of M and $M'' = M/N$. Then you may note that I already showed on *pages 570-571* that this proposition holds if we replace the word "Artinian" with "Noetherian". As it happens, we can basically readapt the proof from then almost verbatim. I don't care to write the details here. ■

Lemma: An Artinian domain A is a field.

Proof:
Consider any $x \in A - \{0\}$ and note that $\langle x \rangle \supseteq \langle x^2 \rangle \supseteq \cdots$ is a descending chain of ideals. So, there must exist n such that $\langle x^n \rangle = \langle x^{n+1} \rangle$. But then $x^n = ux^{n+1}$ where u is a unit. Since we are in a domain, we can equivalently say that $1 = ux$. So, x is a unit. ■

Proposition: Let A be an Artinian ring.

1. Every prime ideal of A is maximal. Hence, $\dim(A) = 0$.
2. There are finitely many maximal ideals.

Proof:
Firstly, note that if P is prime then A/P is an artinian domain. So by the prior lemma, we have that A/P is a field. And that implies that P is a maximal ideal.

Secondly, suppose to the contrary that $P_1, P_2, \ldots$ was an infinite list of distinct max ideals of A. Then $P_1 \supseteq P_1 \cap P_2 \subseteq P_1 \cap P_2 \cap P_3 \supseteq \cdots$ would be a descending chain of ideals. Hence, we must have that $P_1 \cap \cdots \cap P_{n+1} = P_1 \cap \cdots \cap P_n$ for some n. Yet, this would imply that $P_1 \cdots P_n \subseteq P_1 \cap \cdots \cap P_n \subseteq P_{n+1}$. Thus by the definition of a prime ideal, we must have that $P_i \subseteq P_{n+1}$ for some $i \leq n$. Since P_i is a maximal ideal though, this in turn implies $P_i = P_{n+1}$. And this contradicts that each prime in our list was distinct. ■

One consequence of the last proposition is that the Jacobson radical $J(A)$ and nilradical $\mathrm{Nil}(A)$ of an Artinian ring A agree with each other.

Also let $J := J(A) = P_1 \cap \cdots \cap P_n$ where the P_i are the max ideals of A. Then since the P_i are pairwise coprime (i.e. $P_i + P_j = A$ for all $i \neq j$), we can conclude by CRT that $P_1 \cdots P_n = P_1 \cap \cdots \cap P_n$ and $\frac{A}{J} = \frac{A}{P_1 \cap \cdots \cap P_n} \cong \frac{A}{P_1} \oplus \cdots \oplus \frac{A}{P_n} \cong K_1 \times \cdots \times K_n$ where each K_i is a field.

In turn, we have that every $\frac{A}{J}$-module has the form $V_1 \oplus \cdots \oplus V_n$ where each V_i is a K_i-vector space.

Why?
For each i let $e_i \in K_1 \times \cdots \times K_n$ be the element $(0, \ldots, 0, \underbrace{1}_{i\text{th. element}}, 0, \ldots, 0)$.

Next, suppose M is an arbitrary $\frac{A}{J}$-module and let $V_i = e_i M = \{e_i x : x \in M\}$ for each i. Note that $\sum_{i=1}^n e_i = 1_{A/J}$. Hence, every $x \in M$ can be written as $\sum_{i=1}^n e_i x \in \sum_{i=1}^n V_i$. Going a step further, note that $e_i^2 = e_i$ for all i and that $e_i e_j = 0_{A/J}$ for all $i \neq j$. Therefore, suppose $0 = \sum_{i=1}^n e_i y_i$ where $y_1, \ldots, y_n \in M$. Then we know that $e_i y_i = 0$ for each i. After all, $e_i y_i = -\sum_{j \neq i} e_j y_j$. In turn:

$$e_i y_i = e_i^2 y_i = -\sum_{j \neq i} e_i e_j y_j = 0$$

This finishes showing that $M = V_1 \oplus \cdots \oplus V_n$ as an internal direct sum. And as is also hopefully clear, each V_i can be viewed as K_i-vector space where scalar multiplication is defined by $c \cdot x = (ce_i)x$.

A simple A-module M is a nonzero module whose only submodules are $\{0\}$ and M itself. Meanwhile, a semisimple A-module is a direct sum of simple A-modules. I have some examples below:

- If A is an Artinian ring and $J \coloneqq J(A)$, then the prior reasoning shows that any $\frac{A}{J}$-module is semisimple. After all, we can express the module as $V_1 \oplus \cdots \oplus V_n$ where each V_i is a K_i-vector space for some field K_i. Then, we just pick a K_i-basis for each V_i.

- Given any nontrivial unital commutative ring A, an A-module M is simple if and only if it is isomorphic to A/P for some $P \in \operatorname{Max}(A)$.

  The $\Longleftarrow$ implication is obvious. Meanwhile, suppose M is a simple A-module, and let $x \in M - \{0\}$. Then since M is simple, we must have that $Ax = M$. In turn, by considering the map $a \mapsto ax$, we can find some ideal $J \lhd A$ with $\frac{A}{J} \cong M$. Finally, J must be maximal as otherwise $\frac{A}{J}$ wouldn't be a simple module.

- As a side note, the prior bullet point also shows that if M is a semisimple A-module then $J(A) \subseteq \operatorname{Ann}_A(M)$. Meanwhile, if A is Artinian, then the first bullet point says that $J(A) \subseteq \operatorname{Ann}_A(M)$ implies that M is semisimple. Thus, for specifically Artinian rings we have that:

  An A-module M is semisimple if and only if $J(A) \subseteq \operatorname{Ann}_A(M)$.

Remark: Given any commutative unital ring A, we trivially have that every simple A-module is Artinian. Also, by using the proposition on *page 978* we can conclude that any finite direct sum of simple A-modules is Artinian.

Proposition: Let A be an Artinian ring. Then $J \coloneqq J(A)$ is nilpotent (i.e. $J^n = \{0\}$ for some $n \geq 1$).

Proof:
$J \supseteq J^2 \supseteq \cdots$ is a descending chain of ideals. Since A is Artinian, this chain must stabilize. So take n with $J^n = J^{n+1} = \cdots$.

Next, we look at the socle series of A. For our purposes, we define:

$$\operatorname{soc}^k(A) \coloneqq \{x \in A : J^k x = \{0\}\}.$$

This is easily checked to be an ideal of A. Also note that $\{0\} \subseteq \operatorname{soc}^1(A) \subseteq \operatorname{soc}^2(A) \subseteq \cdots$. And due to how we picked n before, we know $\operatorname{soc}^n(A) = \operatorname{soc}^{n+1}(A) = \cdots$.

For some intuition on why we are doing this, note that the socle of a module M (denoted $\operatorname{soc}(M)$) is defined to be the internal sum of all simple submodules of M. Note that this sum is a direct sum.

To see why, suppose $\{N_i\}_{i \in I}$ is an indexing of all simple submodules of M (where $N_{i_1} \neq N_{i_2}$ if $i_1 \neq i_2$). Then for any fixed $i' \in I$, as $N_{i'}$ is simple we

know that either:

$$N_{i'} \cap \left(\sum_{i \neq i'} N_i\right) = \{0\} \text{ or } N_{i'} \cap \left(\sum_{i \neq i'} N_i\right) = N_{i'}.$$

The latter being true though would imply $N_{i'}$ contains N_i for some $i \neq i'$. And since each of those N_i are nonzero and distinct from $N_{i'}$, this would contradict that $N_{i'}$ is simple. So, we've proven that $N_{i'} \cap \left(\sum_{i \neq i'} N_i\right) = \{0\}$ for all i'.

It follows that the socle of a module M is the "largest" semisimple submodule of M (i.e. the semisimple submodule that contains all other semisimple submodules). This also hints at the reason for the name "socle" (which is French for "foundation" or "base"). [Thus socle is not a proper noun nor from a language that capitalizes all nouns.]

In the special case of an Artinian ring A where $I \lhd A$ is semisimple iff $JI = \{0\}$, our first definition for $\mathrm{soc}^1(A)$ is merely just using another way of getting the largest semisimple submodule of A. Furthermore:

$$\begin{aligned} J^{k+1}x = \{0\} &\iff J^k(Jx) = \{0\} \iff Jx \subseteq \mathrm{soc}^k(A) \\ &\iff J(x + \mathrm{soc}^k(A)) = \{\mathrm{soc}^k(A)\}. \end{aligned}$$

Hence, $\frac{\mathrm{soc}^{k+1}(A)}{\mathrm{soc}^k(A)}$ is the socle of $\frac{A}{\mathrm{soc}^k(A)}$. The claim we are about to make (and in turn the proposition as a whole) is thus akin to saying that A has a filtration where each factor is a semisimple module.

We claim that $\mathrm{soc}^n(A) = A$. After all, suppose not. Then $\frac{A}{\mathrm{soc}^n(A)} \neq \{0\}$. Yet we also know that $\frac{A}{\mathrm{soc}^n(A)}$ is Artinian. Hence, the set of nonzero ideals must have a minimal element $I/\mathrm{soc}^n(A)$. Yet now $I/\mathrm{soc}^n(A)$ being simple implies that $J(I/\mathrm{soc}^n(A)) = \{0\}$. So, I properly contains $\mathrm{soc}^n(A)$ and is contained in $\mathrm{soc}^{n+1}(A)$. Yet this contradicts our earlier observation that $\mathrm{soc}^n(A) = \mathrm{soc}^{n+1}(A)$.

Finally, note that $\mathrm{soc}^n(A) = A$ iff $J^n A = \{0\}$ iff $J^n \cdot 1 = J^n = \{0\}$. ■

Theorem: If A is Artinian then A is also Noetherian.

Let $J := J(A)$. We know that $J^n = \{0\}$ for some positive integer n. Therefore, we now consider the filtration:

$$\{0\} = J^n \subseteq J^{n-1} \subseteq \cdots \subseteq J \subseteq A =: J^0.$$

By considering short exact sequences, it thus sufices to show that $\frac{J^k}{J^{k+1}}$ is a Noetherian module for all $0 \leq k < n$ in order to prove that A is Noetherian.

Fortunately, for any k we can observe that $\frac{J^k}{J^{k+1}}$ is semisimple since it is annihilated by J. Furthermore, we know $\frac{J^k}{J^{k+1}}$ is an Artinian module. Therefore, we must have that $\frac{J^k}{J^{k+1}}$ is a *finite* direct sum of simple A-modules.

(We'd be able to violate that $\frac{J^k}{J^{k+1}}$ is Artinian if $\frac{J^k}{J^{k+1}}$ were an infinite direct sum.)

But now we can write $\frac{J^k}{J^{k+1}} = S_1 \oplus S_2 \oplus \cdots \oplus S_m$ where each S_i is a simple module (and thus Noetherian). Also, by doing induction with short exact sequences (see *page 571*), we know finite direct sums of Noetherian modules are Noetherian. Hence, we've proven that $\frac{J^k}{J^{k+1}}$ is Noetherian. ■

Corollary: A is an Artinian ring if and only if A is Noetherian with $\dim(A) = 0$.

$(\Longrightarrow)$

This is just the prior theorem as well as the proposition on *page 979*.

$(\Longleftarrow)$

Let A be a Noetherian ring with $\dim(A) = 0$. It suffices to assume $A \neq \{0\}$. Then since Noetherian rings have only finitely many minimal prime ideals and $\dim(A) = 0$ implies that each minimal prime is a maximal ideal, we can conclude $N \coloneqq \text{Nil}(A) = P_1 \cap \cdots \cap P_n$ for some distinct maximal ideals P_i.

In turn, by the same reasoning as on *page 979* we can conclude that A/N is isomorphic to a product of fields $K_1 \times \cdots \times K_n$ where $K_i = A/P_i$ for every i. Since finite direct sums of Artinian modules are Artinian and each K_i is trivially Artinian, this proves that A/N is an Artinian ring.

Next note that in a Noetherian ring, $N \coloneqq \text{Nil}(A)$ is nilpotent (i.e. $N^m = \{0\}$ for some $m \geq 1$).

To see why, note that if A is trivial then this fact is trivial. Meanwhile, if A is not trivial then we can write $\{0\} = P_1 \cdots P_m$ where each $P_i \in \text{Spec}(A)$. (See *page 899*.) In turn, as $N \subseteq P_i$ for each i, we can conclude that:

$$\{0\} \subseteq N^m \subseteq P_1 \cdots P_m = \{0\} \Longrightarrow N^m = \{0\}.$$

Therefore, we can consider the filtration $0 = N^m \subseteq N^{m-1} \subseteq \cdots \subseteq N \subseteq A$. Since we already know that A/N is Artinian, it now suffices to show that $\frac{N^k}{N^{k+1}}$ is an Artinian module for every $1 \leq k < m$ in order to prove that A is Artinian.

Fortunately, note that $\frac{N^k}{N^{k+1}}$ is Noetherian and an A/N-module. Since $A/N \cong K_1 \times \cdots \times K_n$ where each K_i is a field, we can thus conclude like on *page 979* that $\frac{N^k}{N^{k+1}} \cong V_1 \oplus \cdots \oplus V_n$ where each V_i is a K_i-vector space. Also each V_i must be finite-dimensioned or else we'd able to contradict the Noetherian-ness of $\frac{N^k}{N^{k+1}}$. So, we can conclude that $\frac{N^k}{N^{k+1}}$ is a finitely generated semisimple A-module. Hence $\frac{N^k}{N^{k+1}}$ is Artinian. ■

With that we have successfully unified Rogalski's definition to the one I mentioned off-handedly on *page 901*. I have one more theorem from lecture on the topic of Artinian rings as well as a homework exercise.

Theorem: Let A be an Artinian ring. Then $A \cong A_1 \times \cdots \times A_n$ where each A_i is a local and Artinian ring.

Proof:

Let $P_1, \ldots, P_n$ be the distinct max ideals of A. Then:

$$J \coloneqq J(A) = P_1 \cap \cdots \cap P_n = P_1 \cdots P_n \text{ by CRT.}$$

Also $J^m = \{0\}$ for some m. Hence, $P_1^m P_2^m \cdots P_n^m = \{0\}$.

But now note that the P_i^m are pairwise coprime. To see why, note by the definition of a prime ideal that if $P \in \operatorname{Spec}(A)$ then $P_i^m \subseteq P \iff P_i \subseteq P$. But since P_i is maximal, we in turn know that the only prime ideal containing P_i^m is P_i itself. Consequently, suppose $i \neq j$. Then $P_i^m + P_j^m$ is not contained in any maximal ideal. So, $P_i^m + P_j^m$ must be the entire ring.

By CRT we can conclude that $\{0\} = P_1^m \cap \cdots \cap P_n^m$ and $A \cong \frac{A}{\{0\}} \cong \frac{A}{P_1^m} \times \cdots \times \frac{A}{P_n^m}$. Also, each A/P_i^m is local with a unique max ideal P_i/P_i^m. ■

Set 5 Problem 3: Let A be an affine k-algebra for some field k. Prove that the following are equivalent.

(i) $\dim_k(A) < \infty$

(ii) A is an Artinian ring.

Firstly, A is automatically Noetherian. Hence, A is an Artinian ring if and only if the Krull dimension of A is 0. Yet also by the Noether normalization theorem, we have that the Krull dimension of A being 0 would imply that A is finitely generated k-vector space. This proves the direction (ii $\implies$ i). Conversely, if the Krull dimension of A is greater than 0, then Noether normalization tells us that A contains a polynomial ring which has infinite k-dimension. This proves the contrapositive of (i $\implies$ ii). ■

The next major topic of the class is discrete valuation rings. A lot of what there is to cover can be easily concluded from the following two exercises. So I shall do those first.

Set 5 Problem 4: An integral domain A with field of fractions K is a valuation ring if for all $x \in K^\times$, either $x \in A$ or $x^{-1} \in A$.

An ordered abelian group Γ is an abelian group equipped with with a total order $\leq$ that is compatible with the addition operation of the group (i.e. $\alpha \leq \beta \implies \alpha + \gamma \leq \beta + \gamma$ for all $\alpha, \beta, \gamma \in \Gamma$).

Note that the compatibility also guarentees that:

$$0 \leq \alpha \iff -\alpha \leq 0 \text{ for all } \alpha \in \Gamma.$$

Since $-\alpha = 0 \iff \alpha = 0$, we in turn know for every $\alpha \neq 0$ in Γ that exactly one of the following cases is true:

- $\alpha > 0$ and $-\alpha < 0$,
- $\alpha < 0$ and $-\alpha > 0$.

Consequently, in an ordered abelian group we have that $\alpha = -\alpha$ if and only if $\alpha = 0$. After all, if α weren't zero and $\alpha = -\alpha$ then we'd have that $0 < \alpha < 0$. But that violates the law of trichotemy. Meanwhile, if $\alpha = 0$ then $\alpha = -\alpha$ trivially.

Let Γ be an ordered abelian group. By convention we adjoin an element ∞ to Γ such that:

- $\alpha < \infty$ for all $\alpha \in \Gamma$,
- $\alpha + \infty = \infty$ for all $\alpha \in \Gamma$,
- $\min(\alpha, \infty) = \alpha$ for all $\alpha \in \Gamma \cup \{\infty\}$.

A valuation of a field K is a surjective map $v : K \to \Gamma \cup \{\infty\}$ satisfying that:

- $v(0) = \infty$ and $v(a) \in \Gamma$ for all $a \neq 0$,
- $v(ab) = v(a) + v(b)$ for all $a, b \in K$,
- $v(a + b) \geq \min(v(a), v(b))$ for all $a, b \in K$.

(Here Γ is called the value group.)

The valuation is called discrete in the case $\Gamma = \mathbb{Z}$ with the usual ordering.

Recalling *pages 274, 491, and 518*, hopefully it's clear how valuations here are a generalization of our previous p-adic valuations.

(a) Show that if $v : K \to \Gamma \cup \{\infty\}$ is a valuation, then $A = \{x \in K : v(x) \geq 0\}$ is a valuation ring (called the associated valuation ring of v).

To start off, it's clear that for all $x \in K^\times$ we have that $v(x^{-1}) = -v(x)$.

After all, $v(1) = v(1 \cdot 1) = v(1) + v(1) \implies v(1) = 0$. Then in turn, if $x \in K^\times$ we have that $0 = v(1) = v(x \cdot x^{-1}) = v(x) + v(x^{-1})$. Hence, $-v(x) = v(x^{-1})$.

Now if $v(x) < 0$ then $0 < -v(x) = v(x^{-1})$. Hence, $x \notin A \implies x^{-1} \in A$.

It follows that once we show that A is a ring then we will know that A is a valuation ring (since we can embed the field of fractions of A into K). Fortunately, showing that A is a ring isn't that bad.

- $v(0) = \infty \geq 0$. Hence $0 \in A$.
- $0 = v(1)$. Hence, $1 \in A$.
- $0 = v(1) = v(-1 \cdot -1) = v(-1) + v(-1) \implies v(-1) = -v(-1)$. Since Γ is an ordered abelian group and $-1 \neq 0$ in K implies $v(-1) \neq \infty$, we thus know from an earlier observation that $v(-1) = 0$.
- Suppose $a, b \in A$. Then $a - b \in A$ since:
$$\begin{aligned} v(a - b) \geq \min(v(a), v(-b)) &= \min(v(a), v(-1) + v(b)) \\ &= \min(v(a), v(b)) \geq 0. \end{aligned}$$
- Suppose $a, b \in A$. Then $v(ab) = v(a) + v(b)$. Since $v(a), v(b) \geq 0$, we can conclude that $v(a) + v(b) \geq 0 + v(b) = v(b) \geq 0$. Hence, $ab \in A$.

In fact, we can actually say that K is isomorphic to the field of fractions of A since the embedding of the latter into the former will be surjective. After all, if $c \in K - A$ then $c \neq 0$ and $c^{-1} \in A$. So, $c = \frac{1}{c^{-1}}$ is in the field of fractions of A.

(b) If A is a valuation ring, let $\Gamma := K^\times / U$ where K is the field of fractions of A and U is the multiplicative group of units of A.

Note that as $K^\times$ is an abelian group, all subgroups of $K^\times$ (including U) are normal subgroups. Hence, $K^\times / U$ is a well-defined quotient group. Plus $K^\times / U$ is an abelian group since $K^\times$ is as well.

Show that Γ is an ordered abelian group when we define $xU \leq yU$ if $yx^{-1} \in A$.

Firstly, I'll show that $\leq$ is a well-defined ordering.

- Firstly, suppose $xU = yU$. Then $xy^{-1}, yx^{-1} \in U \subseteq A$. In turn, $yU \leq xU$ and $xU \leq yU$.

  Conversely, suppose $xU \leq yU$ and $yU \leq xU$. Then yx^{-1} and xy^{-1} are in A. Yet also note $1 = yx^{-1} \cdot xy^{-1}$. Hence, $xy^{-1} \in U$ and in turn $xU = yU$. This finishes proving that:
  $$xU = yU \iff xU \leq yU \text{ and } yU \leq xU.$$

- Next suppose $xU \leq yU \leq zU$. Then $yx^{-1} \in A$ and $zy^{-1} \in A$. In turn, $zx^{-1} = zy^{-1} \cdot yx^{-1} \in A$ as well. Hence, $xU \leq zU$. This proves transitivity.

- Finally, suppose $xU \not\leq yU$. Then $yx^{-1} \notin A$. Yet since A is a valuation ring, we must in turn have that $(yx^{-1})^{-1} = xy^{-1} \in A$. So, $yU \leq xU$.

I also still need to show $\leq$ is compatible with the group operation of Γ. (Note that it will be convenient to express this operation as multiplication instead of addition...)

Let $xU, yU, zU \in \Gamma$ satisfy that $xU \leq yU$. Then:
$$yx^{-1} = yz \cdot z^{-1}x^{-1} = (yz) \cdot (xz)^{-1} \in A,$$
which in turn tells us that $xU \cdot zU = xzU \leq yzU = yU \cdot zU$.

Also show that the map $v : K \to \Gamma \cup \{\infty\}$ given by $v(x) = xU$ for $x \in K^\times$ and $v(0) = \infty$ is a valuation.

As I did prior, I shall use multplicative notation for the group operation on $\Gamma = K^\times/U$.

Consider any $a, b \in K$. Since K is a field (and thus a domain), we know that $ab = 0$ iff $a = 0$ or $b = 0$, and this guarentees that if $ab = 0$ then trivially $v(ab) = v(a) \cdot v(b)$. Meanwhile, if $ab \neq 0$ then $a \neq 0$ and $b \neq 0$ as well. Hence, $v(ab) = abU = aU \cdot bU = v(a) \cdot v(b)$. And this finishes proving that:
$$v(ab) = v(a) \cdot v(b).$$

Next, suppose without loss of generality that $v(a) \leq v(b)$. Then it suffices to prove that $v(a) \leq v(a+b)$. Yet if $a + b = 0$ this is trivial. Similarly, if $a = 0$ that $\infty = v(a) \leq v(b) \implies v(b) = \infty \implies b = 0$ as well. But then $a + b = 0$ and $v(a) = \infty = v(a + b)$.

Hence, we can now assume a and $a + b$ are nonzero. Yet now $v(a) \not\leq v(a+b) \implies (a+b)a^{-1} = 1 + ba^{-1} \notin A$. Also, the only way the latter expression can hold is if $ba^{-1} \notin A$. But that would contradict that $v(a) = aU \leq bU = v(b)$. Hence, we can conclude that $v(a) \leq v(a + b)$.

The final thing that needs pointing out is that v is surjective since it is the natural projection map $K^\times \to K^\times/U$ with one additional assignment appended on.

(c) Show that if A is a valuation ring and we define $v : K \to \Gamma \cup \{\infty\}$ as in part (b), then $A = \{x \in K : v(x) \geq 1_\Gamma\}$.

Note that $0 \in A$ and $v(0) = \infty \geq 1_\Gamma$. Meanwhile, suppose $x \in K^\times$. Then $v(x) = xU \geq 1U = 1_\Gamma$ iff $x = x \cdot 1^{-1} \in A$.

Conversely, if v is a valuation from some field K to the value group Γ and we use $A := \{x \in K : v(x) \geq 0_\Gamma\}$ to define a valuation $w : K \to \Gamma' \cup \{\infty\}$ as in part (b), then there is an order-preserving group isomorphism $\phi : \Gamma \to \Gamma'$ such that $\phi \circ v = w$.

Let us define the map $\phi : \Gamma \to \Gamma'$ by the assignment $\phi(\alpha) = yU$ where y is some element in $K^\times$ with $v(y) = \alpha$.

To see that this map is well-defined, suppose $v(y_1) = v(y_2)$. Then $v(y_2 y_1^{-1}) = v(y_2) - v(y_1) = 0$, which in turn tells us that $y_2 y_1^{-1} \in A$. Similarly, $v(y_1 y_2^{-1}) = v(y_1) - v(y_2) = 0$, which in turn tells us that $y_1 y_2^{-1} \in A$. Yet now this proves that $y_1 U \leq y_2 U$ and $y_2 U \leq y_1 U$ in Γ' and from before this implies that $y_1 U = y_2 U$. So, $\phi(v(y_1)) = \phi(v(y_2))$.

Next, I shall show that this map is a group homomorphism. Let $\alpha, \beta \in \Gamma$ and let $x, y \in K^\times$ satisfy that $v(x) = \alpha$ and $v(y) = \beta$. Then $v(xy) = \alpha + \beta$. In turn, $\phi(\alpha + \beta) = xyU = xU \cdot yU = \phi(\alpha)\phi(\beta)$.

Thirdly, I show ϕ is order preserving. Suppose $\alpha \leq \beta$ in Γ and let $x, y \in K^\times$ satisfy that $v(x) = \alpha$ and $v(y) = \beta$. Since $\alpha \leq \beta$, we can in turn say that $0 \leq \beta - \alpha$. Also $v(yx^{-1}) = v(y) - v(x) = \beta - \alpha$. Hence, $yx^{-1} \in A$, which in turn implies that $xU \leq yU$ in Γ'. So, $\alpha \leq \beta \implies \phi(\alpha) \leq \phi(\beta)$.

To finish off, note that ϕ is surjective since for all $xU \in \Gamma'$, $\phi(v(x)) = xU$. Meanwhile, to see that ϕ is injective, suppose $\phi(\alpha) = 1U$ in Γ'. Then we know $\alpha = v(1) = 0_\Gamma$. Hence, $\ker(\phi) = \{0_\Gamma\}$.

The rest of the problem is now obvious. ■

Set 5 Problem 5: Let $v : K \to \Gamma \cup \{\infty\}$ be a valuation for some ordered group Γ, and let A be the associated valuation ring. Assume $A \neq K$.

(a) Prove that the principal ideals of A are exactly the ideals $I_g = \{a \in A : v(a) \geq g\}$ as g ranges over elements of $\Gamma \cup \{\infty\}$ with $g \geq 0$.

Firstly, I want to establish that I_g is an ideal for every $g \in \Gamma \cup \{\infty\}$.

I need to show that if $a_1, a_2 \in I_g$ and $c_1, c_2 \in A$, then $c_1 a_1 + c_2 a_2 \in I_g$. Fortunately:

$$v(c_1 a_1 + c_2 a_2) \geq \min(v(c_1 a_1), v(c_2 a_2)) = \min(v(c_1) + v(a_1), v(c_2) + v(a_2)).$$

Now as $c_i \in A$, we know $v(c_i) \geq 0$. In turn, $v(c_i) + v(a_i) \geq v(a_i) \geq g$. Since this works for both i, we've proven that $v(c_1 a_1 + c_2 a_2) \geq g$.

Now suppose $I = \langle b \rangle$ is a principal ideal of A. Then let $g = v(b) \geq 0$. I claim that $I = I_g$.

If $a \in A$ then $v(ab) = v(a) + v(b)$. And since $v(a) \geq 0$, we know that $v(a) + v(b) \geq v(b) = g$. Hence, $ab \in I_g$ for all a. This shows $I \subseteq I_g$.

Conversely, suppose $d \in I_g$. If $b = 0$ then $g = \infty$, then $v(d) \geq g$ iff $d = 0$. Hence, $I_g = \{0\} = I$. So without loss of generality we can assume $b \neq 0$. Next, if $d = 0$ then trivially $d \in I$. Meanwhile, if $d \neq 0$ then because K is a field, we know there exists $c \in K^\times$ with $cd = b$. In turn:

$$g = v(b) = v(cd) = v(c) + v(d) \geq v(c) + g \implies 0 \geq v(c).$$

But now we know $v(c^{-1}) \geq 0$. Hence, $c^{-1} \in A$ and $d = c^{-1}b \in I$. This proves that $I_g \subseteq I$.

Conversely, I claim for any $g \in \Gamma \cup \{\infty\}$ with $g \geq 0$ that there exists $b \in A$ with with $I_g = \langle b \rangle$.

Since v is by definition surjective, we can pick $b \in A$ with $v(b) = g$. And now from our prior reasoning, we know $\langle b \rangle = I_g$.

As a side note since the TA didn't seem to understand this: an easy consequence of this part of the problem is that the principal ideals of A are simply ordered by inclusion. After all, suppose I_{g_1} and I_{g_2} are principal ideals. Then $g_1 \leq g_2$ implies that $I_{g_2} \subseteq I_{g_1}$. And since $\Gamma \cup \{\infty\}$ is simply ordered, we know either $g_1 \leq g_2$ or $g_2 \leq g_1$.

(b) Prove that A is a Bezout domain. (See *page 532* for a reminder of what that is...)

Let $a, b \in A$ and consider the ideal $I = \langle a, b \rangle$. By the surjectivity of v, we know there exists $d \in A$ with $v(d) = \min(v(a), v(b))$. Also from part (a), we know for all $c \in A$ that:

$$d \mid c \text{ in } A \iff c \in \langle d \rangle \iff v(d) \leq v(c).$$

So, $v(d) \leq v(a)$ and $v(d) \leq v(b)$ tells us that $a, b \in \langle d \rangle$, which in turn means that $\langle a, b \rangle \subseteq \langle d \rangle$. At the same, since either $v(d) \geq v(a)$ or $v(d) \geq v(b)$, we know that either $d \in \langle a \rangle$ or $d \in \langle b \rangle$. Either way, this proves that $\langle d \rangle \subseteq \langle a, b \rangle$.

(c) Prove that if A is Noetherian, then A is a discrete valuation ring, that is $\Gamma \cong \mathbb{Z}$ by an order-preserving group isomorphism.

To start off, note that A being Noetherian is equivalent to $\Gamma^{\geq} := \{\alpha \in \Gamma : \alpha \geq 0_\Gamma\}$ being a well-ordered set.

$(\implies)$

Suppose $S \subseteq \Gamma^{\geq}$ is a nonempty set. Since A is Noetherian, we know the set $\{I_g : g \in S\}$ has a maximal ideal I_{g_0}. Yet since the principal ideals of A are simply ordered by inclusion, we know $I_g \subseteq I_{g_0}$ for all $g \in S$. Hence, $g_0 \leq g$ for all $g \in S$, and this proves that S has a minimal element g_0.

$(\impliedby)$

It suffices to prove that all ideals of A are finitely generated. So suppose for the sake of contradiction that $I \lhd A$ is some ideal that isn't finitely generated. Then we can pick a sequence $\{a_n\}_{n \in \mathbb{N}} \subseteq I$ such that $a_m \notin \langle a_1, \ldots, a_{m-1} \rangle$ for all $m \in \mathbb{N}$.

But now since A is a Bezout domain, we know that for every $n \in \mathbb{N}$ there exists $b_n \in A$ with $\langle b_n \rangle = \langle a_1, \ldots, a_n \rangle$. And since $\{\langle b_n \rangle\}_{n \in \mathbb{N}}$ is a strictly increasing sequence of principal ideals in A, by part (a) we can find a strictly decreasing sequence $\{g_n\}_{n \in \mathbb{N}} \subseteq \Gamma^{\geq}$ with $I_{g_n} = \langle b_n \rangle$ for all n. This contradicts that $\Gamma^{\geq}$ is well-ordered.

Next, note that $A \neq K$ iff $\Gamma^{\geq} \neq \{0_\Gamma\}$.

($\Longrightarrow$)
$A \neq K \iff \exists c \in K$ with $v(c) < 0_\Gamma$. Yet as $c \neq 0$ since otherwise $v(c) = \infty \geq 0_\Gamma$, we can guarentee that c^{-1} exists in K and $v(c^{-1}) > 0_\Gamma$. So, $0_\Gamma \neq v(c^{-1}) \in \Gamma^{\geq}$.

($\Longleftarrow$)
Since v is by definition surjective, if $\Gamma^{\geq} \neq \{0_\Gamma\}$ then we can find $a \in A - \{0\}$ satisfying that $v(a) > 0_\Gamma$. But then $v(a^{-1}) < 0_\Gamma$. So, $a^{-1} \in K - A$.

It follows that $\{\alpha \in \Gamma : \alpha > 0_\Gamma\}$ is a nonempty subset of $\Gamma^{\geq}$. Thus by the well-ordering of $\Gamma^{\geq}$, it must have some minimal element which I will call 1_Γ.

Now define $\phi : \mathbb{Z} \to \Gamma$ by the map $\phi(n) = n1_\Gamma$. (Here $n1_\Gamma$ is defined in the usual way involving repeated addition). It's obvious that ϕ is a group homomorphism. Furthermore, if $n \leq m$ then $m - n \geq 0$. Since we can easily prove that $k1_\Gamma \geq 0_\Gamma$ if $k \geq 0$, it follows that $m1_\Gamma - n1_\Gamma = (m-n)1_\Gamma \geq 0_\Gamma$.

After all, $1_\Gamma \geq 0_\Gamma \implies 21_\Gamma \geq 1_\Gamma \geq 0_\Gamma$. And now we can continue on from there.

But now $m1_\Gamma \geq n1_\Gamma$. This finishes proving that ϕ is order-preserving as well.

To finish off, I just need to show ϕ is injective and surjective.

(Injectivity):
Since ϕ is a group homomorphism, it suffices to show that $\phi(n) = n1_\Gamma = 0_\Gamma$ only if $n = 0$. So suppose for the sake of contradiction that $n \in \mathbb{Z} - \{0\}$ satisfies that $n1_\Gamma = 0_\Gamma$. Note that $0_\Gamma = -0_\Gamma = -(n1_\Gamma) = (-n)1_\Gamma$. Therefore, without loss of generality we can assume that n is positive.

But now by similar reasoning to how we proved $k1_\Gamma \geq 0_\Gamma$ for all $k \geq 0$, we can also show $k1_\Gamma \geq 1_\Gamma$ for all $k \geq 1$. Hence, we know $1_\Gamma \leq n1_\Gamma = 0_\Gamma$. Yet we also know from how we picked out 1_Γ that $0_\Gamma < 1_\Gamma$ This is a contradiction.

(Surjectivity):
Suppose for the sake of contradiction that there exists $g \in \Gamma - \phi(\mathbb{Z})$. By negating g if necessary, we can assume without loss of generality that $g \in \Gamma^{\geq}$. And now we consider the sequence $\{g - n1_\Gamma\}_{n \in \mathbb{Z}_{\geq 0}}$.

Note that as ϕ is an order-preserving injective map, we know that if $n < m$ then $-m1_\Gamma < -n1_\Gamma$ and in turn $g - m1_\Gamma < g - n1_\Gamma$. It follows that our sequence is strictly decreasing and thus has no minimal element. In turn, since $\Gamma^{\geq}$ is well-ordered, we can only avoid a contradiction if there exists some minimal $N \in \mathbb{Z}_{\geq 0}$ with $g - N1_\Gamma < 0_\Gamma$. By the minimality of the N we chose though, we also know $g - (N-1)1_\Gamma \geq 0_\Gamma$. Hence, $-1_\Gamma \leq g - N1_\Gamma < 0_\Gamma$.

Equivalently, we have that $0_\Gamma < N1_\Gamma - g \leq 1_\Gamma$. And since 1_Γ is by definition the minimal element of Γ greater than 0_Γ, we can conclude that $N1_\Gamma - g = 1_\Gamma$. This is a contradiction as it implies that $(N-1)1_\Gamma = g$. ■

Remark 1: A fun thing about how I proved part (c) is that I abstracted the problem to showing that up to isomorphism, the only well-ordered abelian groups (whose order respects the group operation) are $(\mathbb{Z}, +)$ and the trivial group. There is a much more direct way of proving 5(c) though that doesn't abstract the problem like I did.

Remark 2: Once we have established that A is a Noetherian valuation ring, the prior problem lets us conclude that A is a local PID.

Why?
If A is a field then A is trivially local and a PID. So, we can assume without loss of generality that A is not a field (so that part (c) of the prior problem applies).

Then being a PID is equivalent to being a Noetherian Bezout domain. Hence A is a PID. Also, since every ideal is principal and thus has the form I_n for some $n \in \mathbb{Z}_{\geq 0}$, and also since $n < m \implies I_m \subseteq I_n$, we can conclude that every proper ideal of A is contained in I_1. After all, $v(1) = 0$ implies that $A = I_0$ since $1 \in I_0$ and $A \neq I_1$ since $1 \notin I_1$. Then every other candidate ideal is contained in I_1 by the prior observation.

Remark 3: Conversely, suppose $v : K \to \mathbb{Z} \cup \{\infty\}$ is a discrete valuation whose associated valuation ring is A. Then A is Noetherian.

This is because $\mathbb{Z}_{\geq 0}$ is a well-ordered set and I did unnecessary extra work in part (c) to show that the nonnegative elements of $\mathbb{Z}$ being well-ordered implies A is Noetherian.

Furthermore, if A is a local PID that is not a field and x generates the unique maximal ideal P of A, then the unique factorization of any $a \in A - \{0\}$ into a product of irreducibles has the form $a = ux^n$ where u is a unit and $n \in \mathbb{Z}_{\geq 0}$.

Why?
Since A is a PID, every irreducibe element generates an ideal which is maximal over all proper principal ideals (and hence all proper ideals) of A. In turn, every irreducible element generates the same unique maximal ideal. So up to companion class, x is the unique irreducible element of A.

Consequently, if K is the field of fractions of the local PID A we were working with before, then every element of $K^\times$ can be uniquely written as ux^n where $u \in A^\times$ and $n \in \mathbb{Z}$.

Why?
Any element of $K^\times$ has the form $\frac{u_1 x^s}{u_2 x^\ell} = u_1 u_2^{-1} x^{s-\ell} = ux^n$ where $n = s - \ell$ and $u = u_1 u_2^{-1}$. Next suppose $ux^n = u'x^N$. Without loss of generality, we can assume $N \geq n$ (meaning $N - n \geq 0$). Hence, $1 = u'u^{-1}x^{N-n}$ in A. And by the uniqueness we just proved, this tells us that $N = n$ and $u' = u$.

Hence, we can define the discrete valuation $v : K \to \mathbb{Z} \cup \{\infty\}$ where $v(0) = \infty$ and $v(ux^n) = n$ for all $u \in A^\times$ and $n \in \mathbb{Z}$.

This is just a p-adic valuation so it should be clear that this meets all the criteria that a valuation must fulfill.

Also, it's then trivial to see from the prior two uniqueness claims that $A \subseteq K$ is precisely the set of elements with $v(a) \geq 0$.

To summarize the prior remarks as well as problems 4 and 5, we have three equivalent definitions for when a ring A is a discrete valuation ring (DVR):

(1.) A is the associated valuation ring of a discrete valuation $v : K \to \mathbb{Z} \cup \{\infty\}$,

(2.) A is a Noetherian valuation ring that is not a field,

(3.) A is a local PID that is not a field.

Remark 4: Let A be a local PID and also let x be a generator of the unique maximal ideal of A. Then the distinct ideals of A are precisely A, $\{0\}$, and $\langle x^n \rangle$ where $n \in \mathbb{N}$.

Why?
Let v be the discrete valuation constructed in the prior remark. Then we can see that $I_n = \langle x^n \rangle$ for all $n \in \mathbb{N}$.

Remark 5: The prime spectrum of a DVR A consists of only the unique maximal ideal and the zero ideal. This captures the idea that a DVR is the simplest type of PID that is not a field.

Returning to actual lecture notes, here are two examples of DVRs:

- Let $K = \mathbb{Q}$. Then given a prime number p, consider the p-adic valuation $v_p : \mathbb{Q} \to \mathbb{Z} \cup \{\infty\}$ defined on *page 491*. (i.e., if $n \in \mathbb{Z} - \{0\}$ then:
$$v_p(n) \coloneqq \max\{k \in \mathbb{Z}_{\geq 0} : p^k \mid n\}.$$
(Also, $v_p(0) \coloneqq \infty$). And more generally, if $n/m \in \mathbb{Q}$ we define:
$$v_p(n/m) \coloneqq v_p(n) - v_p(m).)$$

  On page 491 I never showed that $v_p(\frac{n_1}{m_1} + \frac{n_2}{m_2}) \geq \min(v_p(\frac{n_1}{m_1}), v_p(\frac{n_2}{m_2}))$. That said, note that:
$$v_p(\tfrac{n_1}{m_1} + \tfrac{n_2}{m_2}) = v_p(n_1 m_2 + m_1 n_2) - v_p(m_1) - v_p(m_2)$$

  Next, we can easily show for two integers r, k that $v_p(r+k) \geq \min(v_p(r), v_p(k))$. After all, if $p^e \mid r$ and $p^e \mid k$ then $p^e \mid r + k$. It follows that:
$$\begin{aligned} v_p(n_1 m_2 + m_1 n_2) &\geq \min(v_p(n_1 m_2), v_p(m_1 n_2)) \\ &= \min(v_p(n_1) + v_p(m_2), v_p(m_1) + v_p(n_2)) \end{aligned}$$

  So, $v_p(\frac{n_1}{m_1} + \frac{n_2}{m_2}) \geq \min(v_p(n_1) - v_p(m_1), v_p(n_2) - v_p(m_2)) = \min(v_p(\frac{n_1}{m_1}), v_p(\frac{n_2}{m_2}))$.

  The associated valuation ring of v_p is $A = \{\frac{a}{b} : v_p(\frac{a}{b}) \geq 0\} = \{\frac{a}{b} : b \notin \langle p \rangle\}$. In other words, the valuation ring of v_p is the localized ring $\mathbb{Z}_{\langle p \rangle}$.

- Let $K = k(x)$ be the field of rational functions in some field k. Then define $v(\frac{p(x)}{q(x)}) = i - j$ where i and j are the least degrees of nonzero terms in p and q respectively.

  Notice that v is equivalent to the prime valuation on $k(x)$ associated with the irreducible polynomial $x \in k[x]$. Hence, the proof that v is a valuation is identical to the proof that v_p from the first bullet point is a valuation.

  The associated valuation ring of v is the localized ring $A = (k[x])_{\langle x \rangle}$.

I have one more characterization of DVRs to go over from lecture.

Let A be a Noetherian local ring with a maximal ideal M. We denote the pair as (A, M). A is called regular-local if $M = \langle a_1, \ldots, a_n \rangle$ for some $a_1, \ldots, a_n \in A$ where $n = \dim(A)$.

Also we take the convention that $\langle \emptyset \rangle = \{0\}$. So, if A is such that $\dim(A) = 0$ then we A is regular if $M = \{0\}$. In other words, a Noetherian dimension zero local ring is regular-local iff it is a field.

More generally for a nonlocal Noetherian ring A, we say A is regular if A_P is a regular-local ring for all $P \in \mathrm{Spec}(A)$.

Note: I will not be showing or using in this class that A being regular-local implies that A is regular.

Theorem: Let (A, M) be a local Noetherian domain with $\dim(A) = 1$. Then the following are equivalent:

i. A is a DVR,

ii. A is regular-local (i.e. M is principal),

iii. A is integrally closed in its field of fractions.

(i $\implies$ iii):
Assume A is a DVR and K is the field of fractions of A. Then supposing $c \in K$ is integral over A, let us write $c^n + b_{n-1}c^{n-1} + \ldots + b_1 c + b_0 = 0$ where all $b_i \in A$. If $c \notin A$ then we know that c^{-1} exists and is in A. Hence, after multiplying both sides of our polynomial expression by c^{-n+1} we get that:

$$c + b_{n-1} + b_{n-2}c^{-1} + \ldots + b_1 c^{-n+2} + b_0 c^{-n+1} = 0$$

Yet now $c = -(b_{n-1} + b_{n-2}c^{-1} + \ldots + b_1 c^{-n+2} + b_0 c^{-n+1}) \in A$. This contradicts the premise that $c \notin A$.

(iii $\implies$ ii):
Pick some $a \in M - \{0\}$. Then note that $M^n \subseteq \langle a \rangle$ for some $n \geq 1$.

To see why, first note that A being local with $\dim(A) = 1$ implies that the only prime ideals of A are $\{0\}$ and M. Hence, since $\langle a \rangle \neq \{0\}$, it follows that $\sqrt{\langle a \rangle} = M$.

Next note that since A is a Noetherian ring, we know that M is finitely generated. So let $M = \langle y_1, \ldots, y_d \rangle$. Importantly, since $\sqrt{\langle a \rangle} = M$ there exists ℓ with $y_i^\ell \in \langle a \rangle$ for all i. But now $M^{d\ell}$ is generated by elements which are all in $\langle a \rangle$. So, $M^{d\ell} \subseteq \langle a \rangle$.

If $M^1 \subseteq \langle a \rangle$ then we are already finished showing ii. After all, we knew from how we chose a that $\langle a \rangle \subseteq M^1$. So, we'd have already proven that M is principal.

Now suppose n is the least integer such that $M^n \subseteq \langle a \rangle$, and also that $n > 1$. Then choose $b \in M^{n-1} - \langle a \rangle$. Note that $bM \subseteq M^n \subseteq \langle a \rangle$. So, let K be the field of fractions of A and consider $x := ab^{-1} \in K$. We have that $x^{-1}M = a^{-1}bM \subseteq \langle a \rangle a^{-1} \subseteq A$. Hence $x^{-1}M$ is a submodule (and thus ideal) of A.

For the sake of contradiction, assume $x^{-1}M \neq A$. Then we'd have that $x^{-1}M \subseteq M$. In turn, $x^{-k}M = x^{-k+1}(x^{-1}M) \subseteq x^{-k+1}M \subseteq \cdots \subseteq M$. So by taking linear combinations of negative powers of x, we can further say that $A[x^{-1}]M \subseteq M$.

In particular, this means for any $c \in M - \{0\}$ that $A[x^{-1}]c$ is a proper nonzero ideal of A. Yet now note that $A[x^{-1}]c$ being an ideal of A means it's a Noetherian module. Also $A[x^{-1}]c \cong A[x^{-1}]$ as A-modules. Hence, we've proven that $A[x^{-1}]$ is a Noetherian A-submodule of K. Consequently, we know that $A[x^{-1}]$ is finitely generated as an A-module. In turn, this also means that x^{-1} is intgral over A.

> (Suppose d is large enough so that $1, x^{-1}, \ldots, x^{-d}$ generate $A[x^{-1}]$. We can always find such a d by taking any finite generating set of polynomials in $A[x^{-1}]$ and then letting $-d$ be the most negative power order of any of those polynimials. Then we can express x^{-d-1} as an A-linear combination of $1, x^{-1}, \ldots, x^{-d}$. And this gives us our monic polynomial.)

Since A is integrally closed in K though, we've proven that $x^{-1} \in A$. In turn, $ba^{-1} \in A \implies b \in aA = \langle a \rangle$. But this contradicts that we specifically chose b to not be in $\langle a \rangle$.

With that, we've now proven that $x^{-1}M = A$. Equivalently, we can conclude that $M = xA$. Since $x(1) \in M \subseteq A$, we know $x \in A$. And now we've shown that M is a principal ideal with $M = \langle x \rangle$.

(ii $\implies$ i):
Since A is regular-local, we know that M is principal. So, we can pick $x \in A$ generating M. Also note that $\dim(A) = 1 \implies A$ is not a field. So, to prove that A is a DVR it now suffices to prove that A is a PID.

Let $I \lhd A$. If $I = \langle 0 \rangle$ then I is trivially principal. Meanwhile suppose $I \neq \{0\}$. Then by identical reasoning as in the proof of (iii $\implies$ ii), we can conclude that $\sqrt{I} = M = \langle x \rangle$. So, there exists some largest $n \geq 1$ such that $I \subseteq \langle x^n \rangle = M^n$.

Since $I \not\subseteq \langle x^{n+1} \rangle$, we can find $y \in I - \langle x^{n+1} \rangle$. Then $y \in I \subseteq \langle x^n \rangle$ implies that $y = x^n z$ where $z \notin M = \langle x \rangle$. But now we know that z is a unit. Therefore, $\langle x^n \rangle = \langle y \rangle \subseteq I$. So, we've proven that $I = \langle x^n \rangle$ is a principal ideal. And this finishes showing that A is a PID. ■

6/20/2026

Instead of studying Dedekind rings next, I'm going to take a break from 200c notes to study point set topology. In particular, I want to learn about uniform spaces. So, I shall be reading *Topological and Uniform Spaces* by Ioan James (see *citation 6* in my bibliography).

Here are some conventions for relations on X (i.e. subsets of $X \times X$).

- If $R \subseteq X \times X$ then we shorthand $(a, b) \in R$ as aRb.
- We write $\Delta X \coloneqq \{(a, a) : a \in X\}$. Note that $\Delta X \subseteq R$ if and only if R is a reflexive relation.
- Given $R \subseteq X \times X$ we define $R^{-1} \coloneqq \{(a, b) : (b, a) \in R\}$. Note that $R^{-1} \subseteq R$ if and only if R is a symmetric relation.
- Given $R, S \subseteq X \times X$, we define $R \circ S \subseteq X \times X$ where $aR \circ Sb$ iff there exists $c \in X$ such that aRc and cSb. Note that $R \circ R \subseteq R$ if and only if R is a transitive relation.

Remark 1: If $R_1, R_2, R_3 \subseteq X \times X$, then $(R_1 \circ R_2) \circ R_3 = R_1 \circ (R_2 \circ R_3)$.
Proof:
If $a(R_1 \circ R_2) \circ R_3 b$ then there exists c with $a(R_1 \circ R_2)c$ and cR_3b. Next, there exists d with aR_1d and dR_2c. But now $dR_2 \circ R_3b$. So, $aR_1 \circ (R_2 \circ R_3)b$. This proves that $(R_1 \circ R_2) \circ R_3 \subseteq R_1 \circ (R_2 \circ R_3)$. Mirror reasoning shows the other inclusion. ■

Remark 2: If $R, S \subseteq X \times X$ then $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$.
Proof: $a(R \circ S)^{-1}b$ iff $bR \circ Sa$. But that happens iff there exists c such that bRc and cSa, and that happens iff there exists c such that $aS^{-1}c$ and $cR^{-1}b$. That is equivalent to saying that $aS^{-1} \circ R^{-1}b$. ■

Remark 3: If $S, R \subseteq X \times X$ with $\Delta X \subseteq S$ then $R \subseteq R \circ S$ and $R \subseteq S \circ R$.
Why?
If aRb then aSa implies that $aS \circ Rb$. Similarly bSb implies that $aR \circ Sb$.

Remark 4: If $R, R', S \subseteq X \times X$ and $R \subseteq R'$ then $S \circ R \subseteq S \circ R'$ and $R \circ S \subseteq R' \circ S$.
Hopefully this is obvious.

For our purposes here, the relations on X that we will care about will be relations giving a way of describing when any two elements a and b in X are "close" to each other.

A uniform structure on a set X is a filter $\mathcal{U}$ on $X \times X$ satisfying that:

- $\Delta X \subseteq D$ for all $D \in \mathcal{U}$
- $D \in \mathcal{U} \implies D^{-1} \in \mathcal{U}$
- If $D \in \mathcal{U}$ then there exsts $D' \in \mathcal{U}$ with $D' \circ D' \subseteq D$.

A uniform space $(X, \mathcal{U})$ is a set X equipped with a uniform structure $\mathcal{U}$.

Two trivial uniform structures are as follows:

1. The discrete uniform structure on a set X is the filter consisting of all sets containing ΔX.

2. The indiscrete uniform structure on a set X is the filter whose only element is $X \times X$.

In order to get more interesting uniform structures, we consider uniform structures generated by collections of sets. Unfortunately, the third bullet point in the definition of a uniform structure makes it so that an intersection of many uniform structures may not be a uniform structure. Hence, we can't always define a "smallest" uniform structure generated by some collection of sets in the same way that we can for topologies. Something we can do though is consider a filterbase for a uniform structure. Then a smallest uniform structure containing that filterbase is guarenteed to exist as we will see.

We say a uniformity base on a set X is a filterbase $\mathscr{B}$ on $X \times X$ satisfying that:

- $\Delta X \subseteq E$ for all $E \in \mathscr{B}$
- $E \in \mathscr{B} \implies E^{-1} \in \mathscr{B}$
- If $E \in \mathscr{B}$ then there exsts $E' \in \mathscr{B}$ with $E' \circ E' \subseteq E$.

Claim: The filter $\mathscr{U}$ generated by a uniformity base $\mathscr{B}$ is a uniform structure.

Proof:
Let $D \in \mathscr{U}$. Then there exists $E \in \mathscr{B}$ with $E \subseteq D$. Since $\Delta X \subseteq E$, we can conclude $\Delta X \subseteq D$ as well. Also, note that $E \subseteq D \implies E^{-1} \subseteq D^{-1}$. And since $E^{-1} \in \mathscr{B}$ we know $D^{-1} \in \mathscr{U}$. Finally, let $E' \in \mathscr{B}$ satisfy that $E' \circ E' \subseteq E$. Then we also have that $E' \in \mathscr{U}$ and $E' \circ E' \subseteq D$. ■

Here are two* important uniform structures defined using uniformity bases.

1. Suppose (X, ρ) is a metric space. Then we define $\mathscr{B} := \{D_\varepsilon : \varepsilon > 0\}$ where $D_\varepsilon := \{(x, y) \in X \times X : \rho(x, y) < \varepsilon\}$. Next, we let the metric uniform structure $\mathscr{U}$ be generated $\mathscr{B}$.

   $\mathscr{B}$ is trivially directed by inclusion since we can make a stronger statment that $\{D_\varepsilon\}_{\varepsilon>0}$ is ordered by inclusion. Thus, $\mathscr{B}$ is a filterbase. Furthermore, $\rho(x, x) = 0$ for all $x \in X$ guarentees that $\Delta X \subseteq D_\varepsilon$ for all ε. Similarly, $\rho(x, y) = \rho(y, x)$ guarentees that $D_\varepsilon = D_\varepsilon^{-1}$ for all ε.

   Finally, let $C \geq 1$ satisfy that $\rho(x, z) \leq C(\rho(x, y) + \rho(y, z))$ for all $x, y, z \in X$. Then we can see that $D_{\frac{\varepsilon}{2C}} \circ D_{\frac{\varepsilon}{2C}} \subseteq D_\varepsilon$.

   I used an arbitrary constant C instead of just setting $C = 1$ cause I wanted to show that this construction also works for certain semimetrics (such as quasinorms).

2. Suppose $(G, \mathcal{T})$ is a topological group. We define the left uniform structure $\mathscr{U}_\ell$ to be generated by the collection $\mathscr{B}$ of sets of the form $L_N := \{(x, y) \in G \times G : x^{-1}y \in N\}$ where N is a neighborhood of the identity element e of G.

   In other words, xL_Ny iff $y \in xN$.

   To see why $\mathscr{B}$ is a filterbase, first note $G \times G \in \mathscr{B}$ always. Hence, $\mathscr{B}$ is nonempty. Also, since N is a neighborhood of the identity, we know that $x = xe \in xN$ for

all $x \in G$. Hence, $\Delta X \subseteq L_N$ for all neighborhoods N of the identity. This also means $\emptyset \notin \mathscr{B}$. Finally, consider L_{N_1} and L_{N_2} in $\mathscr{B}$. Then $N := N_1 \cap N_2$ is still a neighborhood of e. Also, $y \in xN \implies y \in xN_1$ and $y \in xN_2$. So, $L_N \subseteq L_{N_1} \cap L_{N_2}$. And this proves that $\mathscr{B}$ is directed by inclusion.

As for additionally showing that $\mathscr{B}$ is a uniformity base, note that if N is an open neighborhood of e then so is N^{-1}. Also $y \in xN$ iff $x^{-1}y \in N$ iff $y^{-1}x \in N^{-1}$ iff $x \in yN^{-1}$. So $L_N^{-1} = L_{N^{-1}} \in \mathscr{B}$.

Finally, suppose N is a neighborhood of e. Then recall from *page 333* that we can find a neighborhood N' of e with $N'N' \subseteq N$. But now if $y \in xN'$ and $z \in yN'$, we have that $z \in xN'N' \subseteq xN$. So $L_{N'} \circ L_{N'} \subseteq L_N$.

Similarly, we define the right uniform structure $\mathscr{U}_r$ to be generated by the collection $\mathscr{B}$ of sets of the form $R_N := \{(x, y) : yx^{-1} \in N\}$ where N is a neighborhood of the identity element e.

In other words, xR_Ny iff $y \in Nx$.

Let $(X, \mathscr{U})$ be a uniform space. Then we define a relation $R := \bigcap_{D \in \mathscr{U}} D$.

- Note that R is reflexive since $\Delta X \subseteq D$ for all $D \in \mathscr{U}$.
- Also note that if xRy then xDy for all $D \in \mathscr{U}$. Yet we also know $D \in \mathscr{U}$ implies that $D^{-1} \in \mathscr{U}$ So, $xRy \in R \implies xD^{-1}y$ for all $D \in \mathscr{U}$. So, yDx for all $D \in \mathscr{U}$. In other words, xRy implies yRx. So R is symmetric.
- Finally, suppose xRy and yRz. Then for any fixed $D \in \mathscr{U}$ we can pick $D' \in \mathscr{U}$ with $D' \circ D' \subseteq D$. Yet now note that $xD'y$ and $yD'z$. So, xDz. And since D was arbitrary, we've shown that xRz. So, R is transitive.

As a result of the prior reasoning, we know R is an equivalence relation on X. We say $(X, \mathscr{U})$ is a separated uniform space if $\Delta X = R$.

Intuitively, $(X, \mathscr{U})$ is separated iff for all distinct $x, y \in X$ we can find some measure of "closeness" in $\mathscr{U}$ for which x is not close to y. As we will see once we connect uniform structures to topology, a uniform space being separated is equivalent to it being Hausdorff as a topological space.

Let $\phi : X \to Y$ be a map between two uniform spaces $(X, \mathscr{U}_X)$ and $(Y, \mathscr{U}_Y)$. We denote $\phi \times \phi : X \times X \to Y \times Y$ to be the map $(x_1, x_2) \mapsto (\phi(x_1), \phi(x_2))$. Then, we say ϕ is uniformly continuous if $(\phi \times \phi)^{-1}(E) \in \mathscr{U}_X$ for all $E \in \mathscr{U}_Y$.

Remark 1: Let $\phi : X \to Y$ and $\psi : Y \to Z$ be uniformly continuous maps between uniform spaces. Then $\psi \circ \phi$ is a uniformly continuous map from X to Z.

Remark 2: Any constant function on a uniform space X is uniformly continuous. Similary, $\mathrm{Id}_X : X \to X$ is always uniformly continuous.

Both remarks are trivial to verify.

Proposition: If $\phi : (X, \mathcal{U}_X) \to (Y, \mathcal{U}_Y)$ is a map between uniform spaces and $\mathcal{B}_Y$ is a uniformity base for $\mathcal{U}_Y$, then $\phi : X \to Y$ is uniformly continuous iff $(\phi \times \phi)^{-1}(E) \in \mathcal{U}_X$ for all $E \in \mathcal{B}_Y$.

$(\implies)$

This direction is trivial from the fact that $\mathcal{B}_Y \subseteq \mathcal{U}_Y$.

$(\impliedby)$

Consider any $D \in \mathcal{U}_Y$ and then pick $E \in \mathcal{B}_Y$ with $E \subseteq D$. Then $(\phi \times \phi)^{-1}(D) \supseteq (\phi \times \phi)^{-1}(E) \in \mathcal{U}_X$. So, $(\phi \times \phi)^{-1}(D) \in \mathcal{U}_X$ since $\mathcal{U}_X$ is a filter. ■

Corollary: If (X, ρ_X) and (Y, ρ_Y) are metric spaces equipped with their metric uniform structures, then our new definition of uniform continuity agrees with our old definition.

Why?

Let $\mathcal{U}_X$ and $\mathcal{U}_Y$ denote the uniform structures on X and Y respectively, and suppose $\phi : X \to Y$ is a map. Now for any E_ε in the standard uniformity base for $\mathcal{U}_Y$ we have that $(\phi \times \phi)^{-1}(E_\varepsilon) \in \mathcal{U}_X$ iff there exists D_δ in the standard uniformity base for $\mathcal{U}_X$ such that $D_\delta \subseteq (\phi \times \phi)^{-1}(E_\varepsilon)$. Equivalently this means that ϕ is uniformly continuous according to our new definition iff for all $\varepsilon > 0$ there exists $\delta > 0$ with $\rho_X(x, y) < \delta \implies \rho_Y(\phi(x), \phi(y)) < \varepsilon$.

One related headache to mention is as follows:

Corollary 2: Let G be a topological group equipped with the left uniform structure $\mathcal{U}_L$, and also let $\mathbb{R}$ be equipped with the (Euclidean) metric uniform structure. Then left uniform continuity by our new definition is equivalent to Folland's notion of right uniform continuity (see *page 334*).

$(\implies)$

Let $\phi : G \to \mathbb{R}$ be a map. Then for any $\varepsilon > 0$ we know that $E_\varepsilon := \{(x, y) : |x - y| < \varepsilon\}$ satisfies that $(\phi \times \phi)^{-1}(E_\varepsilon) \in \mathcal{U}_\ell$ Hence, there exists a filter $L_N \in \mathcal{U}_\ell$ satisfying that $L_N \subseteq (\phi \times \phi)^{-1}(E_\varepsilon)$. In other words, if $z \in xN$ then $|f(x) - f(z)| < \varepsilon$. Yet this is equivalent to saying that for all $x \in G$ and $y \in N$, we have that $|f(xy) - f(x)| < \varepsilon$.

$(\impliedby)$

Just reverse all the steps in the prior direction. ■

Similarly, right uniform continuity by our new definition is equivalent to Folland's notion of left uniform continuity.

The proof is almost entirely identical to the one before except for an additional step where you shrink your neighborhood N to a symmetric neighborhood.

Unfortunately, both James and Folland's notions of left vs right make sense in the context of what they were thinking about. Thus for the time being, I'm going to stick to James's convention.

A uniform equivalence $\phi : X \to Y$ (where both X and Y are uniform spaces) is a bijective function such that both ϕ and ϕ^{-1} are uniformly continuous.

Example: Suppose $\phi : G \to H$ is a continuous group homomorphism between topological groups. Then ϕ is uniformly continuous if G and H are both equipped with the left uniform structure, and also if G and H are both equipped with the right uniform structures.

> I'll only show this for the left uniform structure. Suppose N is any neighorhood of the identity e_H of H. Then let $N' = \phi^{-1}(N)$. Because ϕ is continuous, we also know that N' is a neighborhood of the identity e_G of G. But now if $y \in xN'$, we know that $\phi(y) \in \phi(x)N$ since ϕ is a group homomorphism. Hence, $L_{N'}$ in the uniform structure on G satisfies that $L_{N'} \subseteq (\phi \times \phi)^{-1}(L_N)$. And since the L_N generate the left uniform structure on H, we've proven that ϕ is uniformly continuous. ■

As a consequence, we can also say that if $\phi : G \to H$ is a topological group isomorphism (i.e. a homeomorphism and group isomorphism), then ϕ is a left and right uniform equivalence.

One reason to care about uniform equivalences is that we can study properties that are invariant under uniform equivalences.

Proposition: If $(X, \mathcal{U}_X)$ and $(Y, \mathcal{U}_Y)$ are uniform spaces, X is separated, and $\phi : X \to Y$ is a bijective map such that ϕ^{-1} is uniformly continuous, then Y is separated as well.

> Proof:
> Let $R_Y \coloneqq \bigcap_{E \in \mathcal{U}_Y} E$. We want to prove $R_Y = \Delta Y$. So consider any $(y_1, y_2) \in Y \times Y$ such that $y_1 \neq y_2$. Since ϕ is bijective, we know that there exists unique distinct elements $x_1, x_2 \in X$ such that $\phi(x_1) = y_1$ and $\phi(x_2) = y_2$. And now since X is separated, we can find some $D \in \mathcal{U}_X$ such that $(x_1, x_2) \notin D$.
>
> But now set $E = (\phi \times \phi)(D)$. Since ϕ^{-1} is uniformly continuous, we know that $(\phi \times \phi)(D) \in \mathcal{U}_Y$. Furthermore, $(\phi \times \phi)^{-1}(\{(y_1, y_2)\}) = \{(x_1, x_2)\} \not\subseteq D$. So $(y_1, y_2) \notin E$. And with that, we've shown that $(y_1, y_2) \notin \Delta Y \implies (y_1, y_2) \notin R_Y$. ■

Corollary: If $(X, \mathcal{U}_X)$ and $(Y, \mathcal{U}_Y)$ are uniform spaces and $\phi : X \to Y$ is a uniform equivalence, then X is separated iff Y is separated.

6/22/2026

I have more to go over from *Topological and Uniform Spaces* today. For the time being I'll still be working with uniform structures as a thing defined separately from topological spaces.

Firstly, here's another convention of James for relations on a set X. Suppose $R \subseteq X \times X$. If aRb we say that a is R-related to b. Then, we define $R[a] \coloneqq \{b \in X : aRb\}$. In other words, $R[a]$ denotes the set of elements that a is R-related to.

Similarly, if $S \subseteq X$ then we define $R[S] \coloneqq \bigcup_{a \in S} R[a]$.

Suppose $(X, \mathcal{U})$ is a uniform space. We say that X is totally bounded if for all $D \in \mathcal{U}$ there exists a finite set $S \subseteq X$ such that $X = D[S]$.

Remark: If $\mathcal{B}$ is a filterbase for $\mathcal{U}$ then X is totally bounded iff for all $D \in \mathcal{B}$ we can find a finite set $S \subseteq X$ such that $X = D[S]$. From this observation, hopefully it is clear how this definition of total boundedness agrees with the metric space definition when we equip our metric space with the metric uniform structure.

Proposition: Suppose $(X, \mathcal{U}_X)$ and $(Y, \mathcal{U}_Y)$ are uniform spaces and $\phi : X \to Y$ is a uniformly continuous surjection. If X is totally bounded then so is Y.

Proof:
Consider any $E \in \mathcal{U}_Y$ and then let $D := (\phi \times \phi)^{-1}(E) \in \mathcal{U}_X$. Since X is totally bounded, we can find a finite set $S \subseteq X$ with $D[S] = X$. In turn, we have that $Y = E[\phi(S)]$. After all, suppose $y \in Y$. Since ϕ is surjective, there exists $x \in X$ such that $\phi(x) = y$. Yet now pick $s \in S$ such that sDx. Then $\phi(s)Ey$. ■

Corollary: If X and Y are uniform spaces, and $\phi : X \to Y$ is a uniform equivalence, then X is totally bounded iff Y is totally bounded.

One of the main reasons for caring about uniform spaces is that they let us define what it means for a sequence to be Cauchy.

Suppose $(X, \mathcal{U})$ is a uniform space and $\{x_n\}_{n \in \mathbb{N}}$ is a sequence in X. Then we say $\{x_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence if:

$$\forall D \in \mathcal{U},\ \exists N \in \mathbb{N}\ s.t.\ \forall n, m \geq N,\ x_n D x_m$$

Remark: Like with total boundedness, if $\mathcal{B}$ is a filterbase for $\mathcal{U}$ then it suffices to check the condition for a sequence being Cauchy only for the elements in $\mathcal{B}$. With this in mind, hopefully it is clear how this definition of a sequence being Cauchy agrees with the metric space definition when we equip out metric space with the metric uniform structure.

Similarly, we say a net $\langle x_\alpha \rangle_{\alpha \in A}$ is Cauchy if:

$$\forall D \in \mathcal{U},\ \exists \alpha \in A\ s.t.\ \forall \beta, \gamma \geq \alpha,\ \beta D \gamma$$

Once we define a topology associated with a uniform structure, we'll be able to discuss whether a uniform space is complete or not (i.e. if all Cauchy sequences and nets converge or not). Furthermore, we'll be able to discuss taking completions of uniform spaces.

Proposition: Suppose $(X, \mathcal{U}_X)$ and $(Y, \mathcal{U}_Y)$ are uniform spaces, $\langle x_\alpha \rangle_{\alpha \in A}$ is a Cauchy net in X, and $\phi : X \to Y$ is a uniformly continuous map. Then $\langle \phi(x_\alpha) \rangle_{\alpha \in A}$ is a Cauchy net in Y.

Proof:
Let E be a relation in $\mathcal{U}_Y$.

By the way, something I've not mentioned before is that a relation in $\mathcal{U}_Y$ would normally be called an entourage. The reason I wasn't using that terminology before

is that the word "entourage" makes my mind immediately think of a neighborhood in Y as opposed to a subset of $Y \times Y$. That said, it would be helpful to not have to keep explicitely naming the uniform structure of X and Y. So, I'm going to start calling relations in uniform structures "entourages".

Since ϕ is uniformly continuous, we know that $D := (\phi \times \phi)^{-1}(E)$ is an entourage on X. In turn, as $\langle x_\alpha \rangle_{\alpha \in A}$ is a Cauchy net, we know there exists $\alpha_0 \in A$ with $x_\beta D x_\gamma$ for all $\beta, \gamma \geq \alpha_0$. And now $\phi(x_\beta) E \phi(x_\gamma)$ for all $\beta, \gamma \geq \alpha_0$. Since E was arbitrary, we can conclude that $\langle \phi(x_\alpha) \rangle_{\alpha \in A}$ is a Cauchy net in Y. ■

I would mostly rather work with nets if possible since they along with sequences are much more natural to think about then filters are (at least when it comes to things like convergence). That said, I think it is unavoidable that I need to introduce some facts about them here.

Suppose $\mathscr{F}$ is a filter. We say another filter $\mathscr{F}'$ refines $\mathscr{F}$ if $\mathscr{F} \subseteq \mathscr{F}'$.

Note that we can let $\mathcal{F}$ be the collection of all filters $\mathscr{F}'$ refining $\mathscr{F}$. Importantly, $\mathcal{F}$ is nonempty since $\mathscr{F} \in \mathcal{F}$. Next, we can partially order $\mathcal{F}$ via inclusion. Then suppose we are given a chain of filters $\{\mathscr{F}'_\alpha\}_{\alpha \in A}$ contained in $\mathcal{F}$. By defining $\mathscr{F}' := \bigcup_{\alpha \in A} \mathscr{F}'_\alpha$, we get another filter in $\mathcal{F}$ which is an upper bound to our chain. With that, we can conclude by Zorn's lemma that $\mathcal{F}$ has a maximal element.

So, we've shown for any filter that there is a maximal filter refining our original filter.

If $\mathscr{F}$ is a maximal filter (meaning that for any filter $\mathscr{G}$ refining $\mathscr{F}$ we have that $\mathscr{F} = \mathscr{G}$), then we call $\mathscr{F}$ an ultrafilter.

Proposition: Let $\mathscr{F}$ be a filter on a set X. Then $\mathscr{F}$ is an ultrafilter if and only if whenever $A \cup B \in \mathscr{F}$ where $A, B \subseteq X$ then $A \in \mathscr{F}$ or $B \in \mathscr{F}$.

$(\Longrightarrow)$

Suppose that A and B are subsets of X such that $A \cup B \in \mathscr{F}$ but $A \notin \mathscr{F}$ and $B \notin \mathscr{F}$. Then we can define another filter $\mathscr{G} := \{B' \subseteq X : A \cup B' \in \mathscr{F}\}$.

To see that $\mathscr{G}$ is a filter, first note that $\emptyset \notin \mathscr{G}$ since $A = A \cup \emptyset \notin \mathscr{F}$. Meanwhile, $X \in \mathscr{G}$ since $X = A \cup X \in \mathscr{F}$. Next, suppose $B' \in \mathscr{G}$ and $B' \subseteq C' \subseteq X$. Then note that $A \cup C' \supseteq A \cup B' \in \mathscr{F}$. So, $A \cup C' \in \mathscr{F}$ and this proves that $C' \in \mathscr{G}$. Finally, suppose $B'_1, B'_2 \in \mathscr{G}$. Then $(A \cup B'_1) \cap (A \cup B'_2) = A \cup (B'_1 \cap B'_2) \in \mathscr{F}$. So, $B'_1 \cap B'_2 \in \mathscr{G}$.

Yet now note that $\mathscr{G}$ refines $\mathscr{F}$. After all, if $B' \in \mathscr{F}$ then:

$$A \cup B' \subseteq B' \Longrightarrow A \cup B' \in \mathscr{F} \Longrightarrow B' \in \mathscr{G}.$$

At the same time, $B \in \mathscr{G} - \mathscr{F}$. This proves that $\mathscr{F}$ is not an ultrafilter since $\mathscr{G}$ properly contains $\mathscr{F}$.

$(\Longleftarrow)$

Suppose $\mathscr{F}$ is not an ultrafilter and let $\mathscr{G}$ be a filter properly containing $\mathscr{F}$. In turn, we know that there exists a set $A \in \mathscr{G} - \mathscr{F}$. But now consider that $X - A \notin \mathscr{G}$ since otherwise

$\emptyset = (X - A) \cap A$ would be in $\mathcal{G}$. Since $\mathcal{F} \subseteq \mathcal{G}$, we can also thus conclude that $X - A \notin \mathcal{F}$. Simultanously though, we know $X \in \mathcal{F}$. So $X = A \cup (X - A) \in \mathcal{F}$ but A and $X - A$ are both not in $\mathcal{F}$. ■

Remark 1:, we can more strongly conclude from the reasoning of the prior proof that $\mathcal{F}$ is an ultrafilter on X if and only if for all $A \subseteq X$ we have that $A \in \mathcal{F}$ or $X - A \in \mathcal{F}$.

Remark 2: If $\mathcal{F}$ is an ultrafilter and $\bigcup_{i=1}^{n} A_i \in \mathcal{F}$, then we can easily now conclude by induction that at least one A_i is in $\mathcal{F}$.

A useful application of ultrafilters is as follows. We say a net $\langle x_\alpha \rangle_{\alpha \in A}$ on a set X is an ultra-net if and only if for all $E \subseteq X$ we have that x_α is eventually in E or x_α is eventually in $X - E$. Importantly, every net $\langle x_\alpha \rangle_{\alpha \in A}$ has an ultra-subnet.

Proof:
For all $\alpha \in A$ we let $T_\alpha := \{x_\beta : \beta \in A \text{ with } \beta \geq \alpha\}$. Importantly, the collection of all T_α constitutes a filterbase. In turn, we can find some ultrafilter $\mathcal{F}$ containing each of the T_α.

Next, let $B \subseteq A \times \mathcal{F}$ be defined to be the set of all pairs $(\alpha, F) \in A \times \mathcal{F}$ with $x_\alpha \in F$. Importantly, B is a directed set when we define $(\alpha_1, F_1) \lesssim (\alpha_2, F_2)$ if $\alpha_1 \leq \alpha_2$ and $F_1 \supseteq F_2$.

Why?
The only nonobvious thing to prove is that if $(\alpha_1, F_1), (\alpha_2, F_2) \in B$ then there exists $(\alpha', F') \in B$ such that $(\alpha', F') \gtrsim (\alpha_1, F_1)$ and $(\alpha', F') \gtrsim (\alpha_2, F_2)$. Fortunately, note that A being a directed set means that there exists $\beta \in A$ with $\beta \gtrsim \alpha_1$ and $\beta \gtrsim \alpha_2$. Next, as $\mathcal{F}$ is a filter containing F_1, F_2, and T_β, we know that $F' := F_1 \cap F_2 \in \mathcal{F}$ and $F' \cap T_\beta \in \mathcal{F}$. In particular, this means that $F' \cap T_\beta \neq \emptyset$. So, there exists $\alpha' \in A$ such that $x_{\alpha'} \in F'$ and $\alpha' \geq \beta$. And with that, we've now show that $(\alpha', F') \in B$. Also, this element is clearly an upper bound to our original two elements of B.

Now for any $(\alpha, F) \in B$ we define $y_{(\alpha,F)} = x_\alpha$. Then we consider the net $\langle y_{(\alpha,F)} \rangle_{(\alpha,F) \in B}$.
To see that this is a subnet of $\langle x_\alpha \rangle_{\alpha \in A}$, suppose that $\alpha_0 \in A$ and then pick any $F \in \mathcal{F}$. Since $T_{\alpha_0} \in \mathcal{F}$ we know that $F \cap T_{\alpha_0} \in \mathcal{F}$. In particular, this means that $F \cap T_{\alpha_0} \neq \emptyset$. So, we can find $\alpha_1 \in A$ with $(\alpha_1, F) \in B$ and $\alpha_1 \geq \alpha_0$. Now it is easy to see that if $(\alpha', F') \in B$ satisfies that $(\alpha', F') \gtrsim (\alpha_1, F)$ then $\alpha' \geq \alpha_0$.

Finally, I claim that $\langle y_{(\alpha,F)} \rangle_{(\alpha,F) \in B}$ is an ultra-net.
After all, suppose $E \subseteq X$ is any set. Since $\mathcal{F}$ is an ultrafilter, we know that either $E \in \mathcal{F}$ or $X - E \in \mathcal{F}$.

- Suppose $E \in \mathcal{F}$. Then we can pick some $\alpha_0 \in A$ with $x_{\alpha_0} \in E$. (This is because $E \cap T_\alpha \in \mathcal{F}$ and thus $E \cap T_\alpha \neq \emptyset$ for all $\alpha \in A$.) But now if $(\alpha, F) \in B$ satisfies that $(\alpha, F) \gtrsim (\alpha_0, E)$, we know $y_{(\alpha,F)} = x_\alpha \in E$. So, our subnet is eventually in E.

- Meanwhile, the case where $X - E \in \mathcal{F}$ can be dealt with using analogous reasoning. (Just replace E with $X - E$.)

One other note I want to make is that if $\langle x_\alpha \rangle_{\alpha \in A}$ is an ultra-net and $X = \bigcup_{i=1}^n E_i$, then we can conclude that there is at least one E_i such that x_α is eventually in E_i.

Proof:
We proceed by induction on n. For our base case, suppose $X = E_1 \cup E_2$. Then if x_α is not eventually in E_1, we know that x_α is eventually in $X - E_1$. And since $X - E_1 \subseteq E_2$, we can conclude that x_α is eventually in E_2.

Now suppose $n > 2$ and that $X = \bigcup_{i=1}^n E_i$. Then by writing:

$$X = (E_1 \cup E_2) \cup E_3 \cup \cdots \cup E_n,$$

we can conclude by induction that either there exists $i \geq 3$ with x_α eventually in E_i, or we have that x_α is eventually in $E_1 \cup E_2$. We're trivially done in the former case. So assume the latter case is what happens.

Next, write $X = E_1 \cup (E_2 \cup E_3) \cup E_4 \cup \cdots \cup E_n$. Then we know from our induction hypothesis that either there exists $i \neq 2, 3$ with x_α eventually in E_i or we have that x_α is eventually in $E_2 \cup E_3$. We assume the former case doesn't happen since otherwise we'd be trivially done. In particular, this means that x_α is not eventually in E_1.

But now we have that x_α is eventually in $(E_1 \cup E_2)$, and yet at the same time since $\langle x_\alpha \rangle_{\alpha \in A}$ is an ultrafilter, we can conclude that x_α is eventually in $X - E_1$. In turn, we can conclude that x_α is eventually in $(E_1 \cup E_2) \cap (X - E_1) = E_2 - E_1 \subseteq E_2$. ■

Proposition: Suppose X is a uniform space. Then X is totally bounded if and only if every net $\langle x_\alpha \rangle_{\alpha \in A}$ in X admits a Cauchy subnet.

$(\Longrightarrow)$
Let $\langle x_{\alpha_\beta} \rangle_{\beta \in B}$ be an ultra-subnet of $\langle x_\alpha \rangle_{\alpha \in A}$. Then I claim that $\langle x_{\alpha_\beta} \rangle_{\beta \in B}$ is Cauchy since more generally, any ultra-net $\langle y_\gamma \rangle_{\gamma \in C}$ is Cauchy in a totally bounded space.

To see why, let $\{y_\gamma\}_{\gamma \in C}$ be an ultra-net. Then suppose E is any entourage of X. Firstly, we find an entourage E' such that $E' = (E')^{-1}$ and $E' \circ E' \subseteq E$. Next, note that since X is totally bounded, there exists $x_1, \ldots, x_n \in X$ such that $X = \bigcup_{i=1}^n E'[x_i]$. Since $\{y_\gamma\}_{\gamma \in C}$ is an ultra-net, we know there must exist some x_i such that y_γ is eventually in $E'[x_i]$.

But now let $\gamma_0 \in C$ be such that $y_\gamma \in E'[x_i]$ for all $\gamma \geq \gamma_0$. Then for all $\gamma_1, \gamma_2 \geq \gamma_0$ in C we have that $x_i E' y_{\gamma_1}$ and $x_i E' y_{\gamma_2}$. This implies that $y_{\gamma_1} E y_{\gamma_2}$.

$(\Longleftarrow)$
Suppose that X is not a totally bounded space. Thus, we know there exists an entourage E such that $E[S]$ is a proper subset of X for all finite subsets S of X.

Let $\mathcal{S}$ be the collection of all finite subsets of X. This becomes a directed set when partially ordered by inclusion. Next, for every $S \in \mathcal{S}$ let x_S be an element of $X - E[S]$. Then, we can consider the net $\langle x_S \rangle_{S \in \mathcal{S}}$.

I claim that $\langle x_S \rangle_{S \in \mathcal{S}}$ has no Cauchy subnets. To see why, suppose that $\langle x_{S_\beta} \rangle_{\beta \in B}$ is any subnet of $\langle x_S \rangle_{S \in \mathcal{S}}$. Then consider any fixed $\beta_0 \in B$. It suffices to show that there exists $\beta_1, \beta_2 \geq \beta_0$ in B such that $(x_{S_{\beta_1}}, x_{S_{\beta_2}}) \notin E$. Fortunately, just let $\beta_1 = \beta_0$ and then let $\beta_2 \in B$ be such that $\beta_2 \geq \beta_0$ and $S_{\beta_2} \supseteq S_{\beta_0} \cup \{x_{S_{\beta_0}}\}$. Then we know $(x_{S_{\beta_1}}, x_{S_{\beta_2}}) \notin E$. ■

Before wrapping up for the day, I also want to translate the prior result into a result about Cauchy sequences. I think this is especially worth doing since the constructions in this simpler case are less set-theoretic.

Proposition: Suppose X is a uniform space whose uniform structure is generated by a countable filterbase $\{E_k\}_{k \in \mathbb{N}}$. Then X being totally bounded implies every sequence $\{x_n\}_{n \in \mathbb{N}}$ in X admits a Cauchy subsequence.

Proof:

For each k we can pick E'_k such that $E'_k = (E'_k)^{-1}$ and $E'_k \circ E'_k \subseteq E_k$. Then because X is totally bounded, we know for each k that there exists $y_1^{(k)}, \ldots, y_{m(k)}^{(k)} \in X$ such that $X = \bigcup_{j=1}^{m(k)} E'_k[y_j^{(k)}]$.

But now suppose $\{x_n\}_{n \in \mathbb{N}}$ is any sequence in X. Then there exists some $y_j^{(1)}$ as well as some subsequence $\{x_{n_i^{(1)}}\}_{i \in \mathbb{N}}$ such that $x_{n_i^{(1)}} \in E'_1[y_j^{(1)}]$ for all i. Next, there exists a subsequence $\{x_{n_i^{(2)}}\}_{i \in \mathbb{N}}$ of $\{x_{n_i^{(1)}}\}_{i \in \mathbb{N}}$ as well as some $y_j^{(2)}$ with $x_{n_i^{(2)}} \in E'_2[y_j^{(2)}]$ for all i. Continuing on in this manner, for every $k \in \mathbb{N}$ we can find a sequence $\{x_{n_i^{(k)}}\}_{i \in \mathbb{N}}$ which is a subsequence of $\{x_{n_i^{(k')}}\}_{i \in \mathbb{N}}$ for all $k' < k$ and which also satisfies that there exists $y^{(k)} \in X$ with $x_{n_i^{(k)}} \in E'_k[y^{(k)}]$ for all i.

To finish off, we now do a diagonal argument on all of the previous sequences we constructed. Namely, consider the sequence $\{z_k\}_{k \in \mathbb{N}}$ where $z_k := x_{n_k^{(k)}}$. This is a subsequence of $\{x_n\}_{n \in \mathbb{N}}$. Also I claim it is Cauchy.

After all, suppose D is an entourage of X. Then we can find an entourage E_k such that $E_k \subseteq D$. Next, note for all $k_1, k_2 \geq k$ that $y^{(k)} E'_k z_{k_1}$ and $y^{(k)} E'_k z_{k_2}$. Hence, we know that $z_{k_1} E_k z_{k_2}$. So, $(z_{k_1}, z_{k_2}) \in E_k \subseteq D$ for all $k_1, k_2 \geq k$. ■

Proposition: Suppose X is a uniform space. If every sequence $\{x_n\}_{n \in \mathbb{N}}$ in X admits a Cauchy subsequence then X is totally bounded.

Proof:

Suppose X is not totally bounded and let E be an entourage such that $E[S] \neq X$ for all finite subsets S of X. By shrinking E, we can without loss of generality assume that E is symmetric (i.e. $E = E^{-1}$). Now pick any $x_1 \in X$. We can find $x_2 \in X - E[x_1]$. Next, we can find $x_3 \in X - E[x_1, x_2]$. And continuing on in this fashion, we can find $x_n \in X - \bigcup_{i=1}^{n-1} E[x_i]$. Then we consider the sequence $\{x_n\}_{n \in \mathbb{N}}$.

Trivially, no subsequence (or even any subnet I now realize) of $\{x_n\}_{n \in \mathbb{N}}$ will be Cauchy since $(x_{n_1}, x_{n_2}) \notin E$ whenever $n_1 \neq n_2$. ■

6/23/2026

Today I want to start tying the theory of uniform spaces to the theory of topology. So, I'll continue loosely following the book I was following before.

Let $(X, \mathcal{U})$ be a uniform space. Then taking inspiration from *pages 798-800* as well as from how the metric space topology is defined, we'll define that a set $U \subseteq X$ is open in the uniform topology if for all $x \in U$ there exists an entourage $E \in \mathcal{U}$ with $E[x] \subseteq U$.

Proposition: The uniform topology is a well-defined topology.

Proof:
To start off, we trivially have that $\emptyset$ and X are open sets in the uniform topology.

Next suppose U_1 and U_2 are open sets in the uniform topology, and consider any $x \in U_1 \cap U_2$. Since U_1 and U_2 are open, we can find entourages E_1 and E_2 such that $E_1[x] \subseteq U_1$ and $E_2[x] \subseteq U_2$. So, we then consider the entourage $E := E_1 \cap E_2$. Now $E[x] \subseteq U_1 \cap U_2$. And since x was arbitrary, we've proven that $U_1 \cap U_2$ is an open set in the uniform topology.

(By induction we can in turn conclude that any finite intersection of open sets is open in the uniform topology.)

Finally, suppose $\{U_\alpha\}_{\alpha \in A}$ is an arbitrary collection of open sets in the uniform topology. Then for any $x \in \bigcup_{\alpha \in A} U_\alpha$ we know that there exists $\alpha_0 \in A$ with $x \in U_{\alpha_0}$. Next, since U_{α_0} is open we know that there exist an entourage E such that $E[x] \subseteq U_{\alpha_0}$. And now $E[x] \subseteq \bigcup_{\alpha \in A} U_\alpha$. This proves that $\bigcup_{\alpha \in A} U_\alpha$ is open. ■

Proposition: Suppose $\mathcal{B}$ is a filterbase for $\mathcal{U}$. Then for any $x \in X$ we have that $\{E[x] : E \in \mathcal{B}\}$ is a local base of neighborhoods of x. (See *page 796* for the definition of a local base of neighborhoods.)

Proof:
To start off, suppose N is any neighborhood of x. Then we know there is an open set U such that $x \in U \subseteq N$. But now since U is open, we know there exists some entourage $D \in \mathcal{U}$ such that $D[x] \subseteq U$. And finally, since $\mathcal{B}$ is a filterbase for $\mathcal{U}$ we know that there exists another entourage $E \in \mathcal{B}$ with $E \subseteq D$. In turn, $E[x] \subseteq D[x]$. And with that, we've proven that for all neighborhoods N of x there exists $E \in \mathcal{B}$ with $x \in E[x] \subseteq N$.

Meanwhile, we also need to show that $E[x]$ is a neighborhood of x for all $E \in \mathcal{B}$. So, having fixed our entourage E, let $U := \{y \in X : \exists D \in \mathcal{U} \text{ s.t. } D[y] \subseteq E[x]\}$. Note that:

- $x \in U$ since $E \in \mathcal{U}$ and $E[x] \subseteq E[x]$.
- Also since $y \in D[y]$ for all entourages $D \in \mathcal{U}$ and $y \in X$, we know that $U \subseteq E[x]$.
- Finally, U is open. After all, consider any $y \in U$ and let $D \in \mathcal{B}$ be such that $D[y] \subseteq E[x]$. But now since $\mathcal{U}$ is a uniform structure, we can find some $D' \in \mathcal{U}$ such that $D' \circ D' \subseteq D$. And now I claim that $D'[y] \subseteq U$.

  After all, suppose $z \in D'[y]$. Then for any $w \in D'[z]$ we have that $yD'z$ and $zD'w$. It follows that $yD' \circ D'w$. So, $w \in D[y] \subseteq E[x]$. And this proves that

$D'[z] \subseteq E[x]$. In turn, by the definition of U we know that $z \in U$. And since z was arbitrary, this proves that $D'[y] \subseteq U$.

With that, we can conclude that $E[x]$ is a neighborhood of x for every entourage $E \in \mathscr{B}$. ■

Corollary: If (X, ρ) is a metric space, then the uniform topology associated with the metric uniform structure is precisely the standard metric topology.

Proof:
Let $\{D_\varepsilon\}_{\varepsilon>0}$ be the standard uniformity base for the metric uniform structure. Then we can see for all $x \in X$ that $D_\varepsilon[x]$ is just the ball of radius ε centered at x. Hence, for any $x \in X$ we have that $\{D_\varepsilon[x], \varepsilon > 0\}$ is a local base of neighborhoods of x in both the standard metric topology and in the uniform topology. It follows that convergence in both topologies are equivalent, and in turn the topologies are equal to each other. ■

Corollary 2: Let G be a topological group. Then the uniform topologies associated with the left uniform structure and the right uniform structure are both equivalent to the original topology on G.

Proof:
Just show like in the prior corollary that either of the uniform topologies on G admits a local base of neighborhoods at any $x \in G$ which is also a local base of neighborhoods of x in the original topology. After all given any neighborhood N of the identity element $e \in G$ we have that $L_N[x] = xN$ and $R_N[x] = Nx$.

As a side note, since the left and right uniform structures on a topological group are generally not the same, this corollary also demonstrates that while we can speak of a single canonical topology associated with a uniform space, we generally can't speak of a single canonical uniform structure whose associated topology is some desired topology.

There are two propertes of the uniform topology which had better be true. Namely, uniform continuity should be stronger than continuity, and also every convergent net should be Cauchy. I'll prove these facts below.

Proposition: Suppose X and Y are uniform spaces and $\phi : X \to Y$ is uniformly continuous. Then ϕ is continuous when X and Y are equipped with the uniform topologies.

Proof:
Consider any $x \in X$ and then consider any neighborhood N of $\phi(x)$ in Y. We know there exists an entourage E of Y with $E[\phi(x)] \subseteq N$. Yet also note that $D := (\phi \times \phi)^{-1}(E)$ is an entourage of X since ϕ is uniformly continuous. And now I claim that $D[x]$ is a neighborhood of $x \in X$ satisfying that $\phi(D[x]) \subseteq N$. After all, if $y \in D[x]$ then $\phi(y) \in E[\phi(x)] \subseteq N$. This proves that ϕ is continuous at x. And since x was arbitrary, we can conclude that ϕ is continuous everywhere. ■

Proposition: Suppose X is a uniform space and that the net $\langle x_\alpha \rangle_{\alpha \in A}$ converges with respect to the uniform topology to a point $x \in X$. Then $\langle x_\alpha \rangle_{\alpha \in A}$ is a Cauchy net.

Proof:
Let E be any entourage of X and then pick a symmetric entourage E' of X such that $E' \circ E' \subseteq E$. Since $x_\alpha \to x$ in the uniform topology, we know that the x_α are eventually in $E'[x]$. But now let $\alpha_0 \in A$ satisfy that $x_\alpha \in E'[x]$ for all $\alpha \geq \alpha_0$. Then for all $\alpha, \beta \geq \alpha_0$ we have that $xE'x_\alpha$ and $xE'x_\beta$ implies that $x_\alpha E' \circ E' x_\beta$. So, $x_\alpha E x_\beta$ for all $\alpha, \beta \geq \alpha_0$. And this proves that our net is Cauchy. ■

Next I want to prove another useful fact about uniform the uniform topology.

Proposition: Suppose X is a uniform space. Then the following are equivalent:

(a) X is a separated uniform space,

(b) the uniform topology on X is T_0,

(c) the uniform topology on X is T_1,

(d) the uniform topology on X is Hausdorff.

$(a \implies d)$
Let x and y be two distinct points in X. Since X is separated, we know that there exists an entourage E such that $y \notin E[x]$. Next, we pick a symmetric entourage E' such that $E' \circ E' \subseteq E$. And I claim that $E'[x]$ and $E'[y]$ are disjoint neighborhoods of x and y respectively.

To see why they are disjoint, suppose $z \in E'[x]$. Then we know that $z \notin E'[y]$ since otherwise $xE'z$ and $yE'z$ would imply that xEy (a contradiction). So, $E'[x] \cap E'[y] = \emptyset$. This proves that X is Hausdorff.

$(d \implies c)$ and $(c \implies b)$
This is obvious.

$(b \implies a)$
Suppose X is T_0 and let x and y be distinct elements of X. Without loss of generality we can assume there exists an open set U such that $x \in U$ and $y \notin U$. In turn, we can find an entourage $E[x]$ such that $x \in E[x] \subseteq U$. And now we know $(x, y) \notin E$. This proves that X is separated. ■

A particularly important thing we ought to be able to do is consider uniform structures on product spaces. After all, multi-variable functions ought to be able to be jointly continuous.

Suppose $\{X_\alpha, \mathcal{U}_\alpha\}_{\alpha \in A}$ is a collection of uniform spaces and let us write $X = \prod_{\alpha \in A} X_\alpha$. We can identify $X \times X$ with $\prod_{\alpha \in A} X_\alpha \times X_\alpha$ via the obvious map. And now, we let the collection of $\prod_{\alpha \in A} D_\alpha$ form a uniformity base for X (where each $D_\alpha \in \mathcal{U}_\alpha$ and for all but finitely many α we have that $D_\alpha = X_\alpha \times X_\alpha$).

First, I want to justify that this is in fact a uniformity base for X.

- To start off, note that $X \times X$ is always in this base. Hence our base is nonempty.
- Also note that $\Delta X = \prod_{\alpha \in A} \Delta X_\alpha$ is a subset of very entourage in our base.
- Our base is a directed collection of sets. After all, suppose $\prod_{\alpha \in A} D_\alpha$ and $\prod_{\alpha \in A} E_\alpha$ are both entourages in our base. Then we have that $\prod_{\alpha \in A}(D_\alpha \cap E_\alpha)$ is also an entourage in our base contained in both of the original entourages.

  You check if you aren't convinced!!
- Suppose $\prod_{\alpha \in A} D_\alpha$ is an entourage in our base. Then we can see that $(\prod_{\alpha \in A} D_\alpha)^{-1} = \prod_{\alpha \in A}(D_\alpha^{-1})$ is also in our base.

  Again, you check if you aren't convinced!!
- Finally, suppose $\prod_{\alpha \in A} D_\alpha$ is an entourage in our base. Then for each α such that $D_\alpha \neq X_\alpha \times X_\alpha$, we can pick an entourage D'_α of X_α such that $D'_\alpha \circ D'_\alpha \subseteq D_\alpha$. Meanwhile, if $D_\alpha = X_\alpha \times X_\alpha$ then we trivially set $D'_\alpha = D_\alpha$ and observe that $D'_\alpha \circ D'_\alpha \subseteq D_\alpha$. But now:

  $$(\prod_{\alpha \in A} D'_\alpha) \circ (\prod_{\alpha \in A} D'_\alpha) = \prod_{\alpha \in A}(D'_\alpha \circ D'_\alpha) \subseteq \prod_{\alpha \in A} D_\alpha.$$

  I'll show that $(\prod_{\alpha \in A} D'_\alpha) \circ (\prod_{\alpha \in A} D'_\alpha) = \prod_{\alpha \in A}(D'_\alpha \circ D'_\alpha)$ just to prove I know how to.

  Let $f, g : A \to \bigcup_{\alpha \in A} X_\alpha$ be elements of $\prod_{\alpha \in A} X_\alpha$. Then $(f, g) \in (\prod_{\alpha \in A} D'_\alpha) \circ (\prod_{\alpha \in A} D'_\alpha)$ iff there exists $h : A \to \bigcup_{\alpha \in A} X_\alpha$ such that $(f, h) \in \prod_{\alpha \in A} D'_\alpha$ and $(h, g) \in \prod_{\alpha \in A} D'_\alpha$. Yet now note that the latter is possible iff for every α there exsts $h(\alpha) \in X_\alpha$ such that $f(\alpha) D'_\alpha h(\alpha)$ and $h(\alpha) D'_\alpha g(\alpha)$ for all α, and that's equivalent to saying $f(\alpha) D'_\alpha \circ D'_\alpha g(\alpha)$ for all α. Or in other words, $(f, g) \in \prod_{\alpha \in A}(D'_\alpha \circ D'_\alpha)$.

  Also note that $\prod_{\alpha \in A} D'_\alpha$ is in our base.

Next, I want to justify why this is the natural definition for a uniformity base on X. For each α let $\pi_\alpha : X \to X_\alpha$ be the projection map. Then for any entourage D_α of X_α we have that:

$$(\pi_\alpha \times \pi_\alpha)^{-1}(D_\alpha) = D_\alpha \times \left(\prod_{\alpha' \in A - \{\alpha\}} X_{\alpha'} \times X_{\alpha'}\right).$$

It follows that π_α is uniformly continuous for every $\alpha \in A$ when X is equipped with the product uniform structure (i.e. the structure generated by the uniformity base described priorly). More generally, by taking finite intersections we can also see that every uniform structure on X satisfying that $\pi_\alpha : X \to X_\alpha$ is uniformly continuous for all α will contain our uniformity base. So, the product uniform structure on X is the smallest uniform strucure such that all the π_α are uniformly continuous.

Finally, if all the X_α are equipped with their corresponding uniform topologies, then the uniform topology associated with X (given the product uniform structure) is precisely the same as the product topology on $X = \prod_{\alpha\in A} X_\alpha$.

Why?

Note that $(\prod_{\alpha\in A} D_\alpha)[(x_\alpha)_{\alpha\in A}] = \prod_{\alpha\in A} D_\alpha[x_\alpha]$. Hence, we can see that both topologies have identical local bases of neighborhoods at any $(x_\alpha)_{\alpha\in A}$ in X.

Proposition: Let $\{X_\alpha\}_{\alpha\in A}$ be a collection of uniform spaces and then let $X := \prod_{\alpha\in A} X_\alpha$ be equipped with the product uniform structure. Next suppose A is another uniform space and $\phi : A \to X$ is a map. Then ϕ is uniformly continuous if and only if for all $\alpha \in A$ we have that $\phi_\alpha := \pi_\alpha \circ \phi : A \to X_\alpha$ is uniformly continuous.

$(\Longrightarrow)$

This direction is clear from the fact that compositions of uniformly continuous functions are uniformly continuous and π_α is uniformly continuous for all α.

$(\Longleftarrow)$

It suffices to show that $(\phi \times \phi)^{-1}(\prod_{\alpha\in A} D_\alpha)$ is an entourage of A for any collection of entourages $\{D_\alpha\}_{\alpha\in A}$ with $D_\alpha = X_\alpha \times X_\alpha$ for all but finitely many α. (After all, see the proposition at the top of *page 996*).

Fortunately, upon letting $B \subseteq A$ be the set of all $\alpha \in A$ with $D_\alpha \neq X_\alpha \times X_\alpha$, we can say that:

$$\begin{aligned}(\phi \times \phi)^{-1}(\textstyle\prod_{\alpha\in A} D_\alpha) &= (\phi \times \phi)^{-1}(\bigcap_{\alpha\in B}(\pi_\alpha \times \pi_\alpha)^{-1}(D_\alpha)) \\ &= \bigcap_{\alpha\in B}(\phi \times \phi)^{-1}((\pi_\alpha \times \pi_\alpha)^{-1}(D_\alpha)) \\ &= \bigcap_{\alpha\in B}((\pi_\alpha \circ \phi) \times (\pi_\alpha \circ \phi))^{-1}(D_\alpha).\end{aligned}$$

Since the latter is a finite intersection of entourages of A (since $\pi_\alpha \circ \phi$ is uniformly continuous), we can conclude that $(\phi \times \phi)^{-1}(\prod_{\alpha\in A} D_\alpha)$ is an entourage of A. So, ϕ is uniformly continuous. ■

6/25/2026

I'm feeling burnt out talking about uniform spaces. So I'm going to take a break from that and instead study manifolds. I'll return to talking about uniform spaces on *page* ____.

My first goal for today is to give some proofs for some important facts about topological manifolds. An algebraic purist might not like the proofs I'm following to get these results. That said, my attempts to study manifolds prior to this point have been incredibly hampered by not having these results.

First is a proof that I'm getting from Terence Tao's blog (see *citation 7*).

Theorem (Invariance of Domain): Suppose $U \subseteq \mathbb{R}^n$ is open and $f : U \to \mathbb{R}^n$ is an injective continuous function. Then $f(U)$ is an open set.

To start off, for any $x \in U$ we can find a closed ball $\overline{B_r(x)}$ centered at x and contained in U. And now to prove that $f(U)$ is an open set, it suffices to prove that $f(x)$ is an interior point of $f(\overline{B_r(x)})$. But now by scaling and translated our domain, we can without loss of generality assume that $r = 1$ and $x = 0$. So having denoted $B^n := \overline{B_1(0)}$, we are left with the task of showing that:

> If $f : B^n \to \mathbb{R}^n$ is a continuous injective function then $f(0)$ lies in the interior of $f(B^n)$.

Fortunately, note that $f : B^n \to f(B^n)$ is a continuous bijection from a compact set to a Hausdorff space. So, the inverse map $f^{-1} : f(B^n) \to B^n$ must also be continuous. Furthermore, since $f(B^n)$ is a compact (and thus closed subset) of $\mathbb{R}^n$, we can apply a corollary of the Tietze Extension theorem to find a continuous function $G : \mathbb{R}^n \to \mathbb{R}^n$ with $G|_{F(B^n)} = f^{-1}$.

See Folland corollary 4.17 in my math 240b notes.

Now $G(f(x)) = x$ for all $x \in B^n$. In particular this means that $G(f(0)) = 0$. So, G has a zero on $f(B^n)$. As it turns out, this remains true when we perturb G.

Lemma: Let $\widetilde{G} : f(B^n) \to \mathbb{R}^n$ be a continuous function with $|G(y) - \widetilde{G}(y)| \leq 1$ for all $y \in f(B^n)$. Then $\widetilde{G}$ has at least one zero.

Why?

Consider the function $h : B^n \to \mathbb{R}^n$ given by:

$$h(x) = x - \widetilde{G}(f(x)) = G(f(x)) - \widetilde{G}(f(x)).$$

By our assumpion that $|G(y) - \widetilde{G}(y)| \leq 1$ for all $y \in f(B^n)$ we know that h maps B^n into B^n. So by Brouwer's fixed point theorem (see *pages 324-326*), there must exist $x \in B^n$ with $x = h(x) = x - \widetilde{G}(f(x))$. But now we have that $\widetilde{G}(f(x)) = 0$.

The rest of the proof will now be spent showing that if $f(0)$ is not an interior point of $f(B^n)$ then we can find a perturbed $\widetilde{G}$ contradicting our lemma.

Fix $\varepsilon > 0$ with $|G(y)| < \frac{1}{3}$ if $|y - f(0)| \leq 2\varepsilon$. (We can do this since G is continuous and $G(f(0)) = 0$.) Then supposing for the sake of contradiction that $f(0)$ is not an interior point of $f(B^n)$, we can find some $c \in \mathbb{R}^n - f(B^n)$ with $|c - f(0)| < \varepsilon$. By translating our codomain, we can assume without loss of generality that $c = 0$. So, we have that:

$$0 \notin f(B^n), |f(0)| < \varepsilon, \text{ and } |G(y)| < \tfrac{1}{3} \text{ if } |y| < \varepsilon.$$

Next let $\Sigma := \Sigma_1 \cup \Sigma_2$ where:

$$\Sigma_1 := \{y \in f(B^n) : |y| \geq \varepsilon\} \text{ and } \Sigma_2 := \{y \in \mathbb{R}^n : |y| = \varepsilon\}.$$

Since $0 \notin f(B^n)$, we can define a continuous function $\Phi : f(B^n) \to \Sigma$ given by $\Phi(y) := \max(\frac{\varepsilon}{|y|}, 1)y$. Also note that since Σ_1 is compact and G is nonzero on Σ_1, we can find some $0 < \delta < \frac{1}{3}$ with $|G(y)| \geq \delta$ for all $y \in \Sigma_1$. Also because Σ is compact, we can conclude by the Stone Weierstrass theorem that there is a smooth function (even a polynomial) $P : \mathbb{R}^n \to \mathbb{R}^n$ with $|P(y) - G(y)| < \frac{\delta}{2}$ for all $y \in \Sigma$.

In particular, since $|P(y) - G(y)| < \frac{\delta}{2}$ and $|G(y)| \geq \delta$ for all $y \in \Sigma_1$, we can conclude that P is nonzero on Σ_1. That said, it may be the case that P has a zero on Σ_2. So, we want to modify P such that P is nowhere zero on Σ_2.

Fortunately, note that Σ_2 has measure zero. It then follows that $P(\Sigma_2)$ also has measure zero. After all, consider the following lemma:

Lemma: Let $U \subseteq \mathbb{R}^n$ be an open set. If $g : U \to \mathbb{R}^n$ is a Lipschitz continuous function and $N \subseteq U$ has Lebesgue measure zero, then $g(N)$ is Lebesgue measurable and has Lebsgue measure zero.

Proof:
Let $C > 0$ be such that $|g(y_1) - g(y_2)| \leq C|y_1 - y_1|$ for all $y_1, y_2 \in U$.
Then importantly, for any closed ball $\overline{B_r(y)}$ contained in U we have that $g(\overline{B_r(y)})$ is measurable (since it is a compact set) and also $g(\overline{B_r(y)}) \subseteq \overline{B_{rC}(g(y))}$.

Hence, we can say that $m(g(\overline{B_r(y)})) \leq m(\overline{B_{rC}(g(y))}) = C^n m(\overline{B_r(y)})$.

As a side note, this is the exact sentence of the lemma where we use that g is mapping U into $\mathbb{R}^n$ as opposed to $\mathbb{R}^\ell$ where $\ell \neq n$. After all, let V_n be the volume of the unit ball in $\mathbb{R}^n$ and let V_ℓ be the unit ball in $\mathbb{R}^\ell$. Then $m(B_{rC}(g(y))) = r^\ell C^\ell V_\ell$. Meanwhile $m(\overline{B_r(y)}) = r^n V_n$. We want to say that $r^\ell C^\ell V_\ell = C^n r^n V_n$ for all $r > 0$. Yet this is equivalent to asking that $r^{\ell - n} = C^{n-\ell} V_n / V_\ell$ for all $r > 0$. And if $\ell - n \neq 0$ this doesn't work.

But now since N is a null set, for any $\varepsilon > 0$, we can find a cover of closed balls $\{B_k\}_{k \in \mathbb{N}}$ contained in U, covering N, and satisfying that $\sum_{k=1}^{\infty} m(B_k) < \varepsilon$. In turn, we have that:

$$m(\bigcup_{k \in \mathbb{N}} g(B_k)) \leq \sum_{k \in \mathbb{N}} m(g(B_k)) \leq C^n \sum_{k=1}^{\infty} m(B_k) < C^n \varepsilon$$

Then by taking $\varepsilon \to 0$, we can show that that $g(N)$ is contained in a countable intersection of measurable sets and that that intersection has measure 0. Therefore since the σ-algebra of Lebesgue measurable sets is complete, we know that $g(N)$ is also measurable with a measure of 0.

Since Σ_2 is compact, we know it is bounded and can find $0 < r < R$ such that $\Sigma_2 \subseteq B_r(0)$. Then as P is smooth on $B_{R+1}(0)$ we know that P is Lipschitz continuous on $\overline{B_r(0)}$ (see *page 291*). In particular, we can find a Lipschitz constant for P on $B_r(0)$ and apply the prior lemma to the null set Σ_2 contained in $B_r(0)$ to get that $P(\Sigma_2)$ is a null set.

Since $B_{\delta/2}(0)$ doesn't have measure zero, we know that $B_{\delta/2}(0) - P(\Sigma_2) \neq \emptyset$. Hence, there must exist a constant $a \in B_{\delta/2}(0) - P(\Sigma_2)$. And by defining $Q(y) := P(y) - a$, we have that $Q(y)$ is nowhere zero on Σ_2. Plus, $|Q(y) - G(y)| < \delta$ for all $y \in \Sigma$. Hence, we still don't have that $P = 0$ anywhere on Σ_1.

To finish off, let $\widetilde{G} : f(B^n) \to \mathbb{R}^n$ be defined as $\widetilde{G}(y) := Q(\Phi(y))$. Then Q is continuous and nowhere zero. Also note that if $|y| \geq \varepsilon$ then $|\widetilde{G}(y) - G(y)| < \delta < 1$. Meanwhile, if $|y| < \varepsilon$ then $|\Phi(y) - f(0)| \leq |\Phi(y)| + |f(0)| < 2\varepsilon$ implies that $|G(\Phi(y))| < \frac{1}{3}$. Since $|y| < \varepsilon$ also implies $|G(y)| < \varepsilon$, we can conclude that:

$$\begin{aligned}|\widetilde{G}(y) - G(y)| &\leq |Q(\Phi(y)) - G(\Phi(y))| + |G(\Phi(y)) - 0| + |G(y) - 0| \\ &< \delta + \tfrac{2}{3} < 1.\end{aligned}$$

This contradicts the lemma on the prior page. ■

The invariance of domain theorem has far reaching consequences for the study of manifolds. Now that I've received my bazooka from Terrance Tao, these next results will be easy to show on my own.

In the statements of the below proposiions, $H^n := \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_n \geq 0\}$.

Proposition: If $m < n$ and $U \subseteq \mathbb{R}^n$ is an open set, then there is no continuous injective function $f : U \to \mathbb{R}^m$.

Let $i : \mathbb{R}^m \hookrightarrow \mathbb{R}^n$ be the obvious embedding. Then if such a function f did exist, we'd have that $i \circ f$ is an injective continuous function from $\mathbb{R}^n$ into $\mathbb{R}^n$. Yet we'd also have that $i \circ f(U)$ is contained in an m-dimensional hyperplane and thus has an empty interior. This contradicts the invariance of domain theorem. ■

Corollary: Suppose that n and m are positive integers, that M is an n-dimensional topological manifold, and that for some nonempty open set $U \subseteq M$ and some nonempty open set $V \subseteq \mathbb{R}^m$ or H^m there is a homeomorphism $\varphi : U \to V$. Then, we must have that $m = n$.

Proof:

To start off, by shrinking U and V if necessary, we can without loss of generality assume V is an open subset of $\mathbb{R}^m$. Next, pick a point $p \in U$ and then pick a chart (W, ψ) such that $p \in W$ and ψ maps W homeomorphically to an open set $\psi(W)$ of H^n or $\mathbb{R}^n$. Again, by shrinking W and maybe shifting p, we can assume with loss of generality that $W \subseteq U$ and $\psi(W)$ is open in $\mathbb{R}^n$.

But now we have homeomorphisms $\varphi|_W : W \to \varphi(W)$ and $\psi : W \to \psi(W)$ where $\varphi(W)$ is an open subset of $\mathbb{R}^m$ and $\psi(W)$ is an open subset of $\mathbb{R}^n$.

- If $n < m$ then $\psi \circ \varphi^{-1}$ contradicts our prior proposition.
- Meanwhile if $m < n$ then $\varphi \circ \psi^{-1}$ contradicts our prior proposition.

It follows that $m = n$. ■

Based on the prior corollary, we can now easily conclude that the dimension of a topological manifold is uniquely defined. In fact, we can make an even stronger statement that if M is an n-dimensional topological manifold and N is an m-dimensional topological manifold, then M being homeomorphic to N implies that $m = n$. Thus, the dimension of a topological manifold is invariant under homeomorphisms.

Proposition: Suppose M is an n-dimensional topological manifold. Then x is a boundary point of M if and only if there is a chart (U, φ) on M such that $x \in U$, $\varphi(U)$ is open in H^n, and $\varphi(x) \in \partial H^n$.

($\Longrightarrow$)
The definition of a boundary point that I gave on *page 146* is that there are no charts (V, ψ) on M with $x \in V$ and $\psi(V)$ open in $\mathbb{R}^n$. So suppose x is a boundary point. By the definition of a topological manifold there must exist a chart (U, φ) on M with $x \in U$. But since x is a boundary point, we know that $\varphi(U)$ is not open in $\mathbb{R}^n$. Hence, we know that $\varphi(U)$ is only open in H^n. Furthermore, by intersecting $\varphi(U)$ with balls of decreasing radius around $\varphi(x)$, we can't be able to get a restricted chart whose image is open in $\mathbb{R}^n$. Hence, it follows that $\varphi(x) \in \partial H^n$.

($\Longleftarrow$)
Suppose $x \in M$ and (U, φ) is a chart on M such that $x \in U$, $\varphi(U)$ is open in H^n, and $\varphi(x) \in \partial H^n$. Then suppose for the sake of contradiction that (V, ψ) is another chart on M such that $x \in V$ and $\psi(V)$ is open in $\mathbb{R}^n$.

Let $W \coloneqq U \cap V$. Then we must have that $\varphi \circ \psi^{-1}$ is a continuous function mapping the open subset $\psi(W) \subseteq \mathbb{R}^n$ into $\mathbb{R}^n$. So by the invariance of domain theorem, we must have that $\varphi(W)$ is an open subset $\mathbb{R}^n$. Yet simultaneously note that $\varphi(W)$ is a subset of H^n containing $\varphi(x) \in \partial H^n$. Thus, we can't have that $\varphi(W)$ is open in $\mathbb{R}^n$ since x is not an interior point of $\varphi(W)$ (considered as a subset of $\mathbb{R}^n$ instead of H^n). This is a contradiction. ■

Anyways, hopefully the prior results were enlightening.

6/26/2026

I have notes from math 220c that I can use to supplement an intro to Riemann surfaces. So I shall follow that.

Math 220c Notes:

Let M be a 2-dimensional topological manifold (without boundary). We identify $\mathbb{R}^2$ with $\mathbb{C}$.

Next, suppose we have an atlas $\mathcal{A}$ on M such that for all charts (U, φ) and (V, ψ) satisfying that $U \cap V \neq \emptyset$ we have that $\psi \circ \varphi^{-1}$ is a biholomorphic function from $\varphi(U \cap V)$ to $\psi(U \cap V)$. We call such an atlas a complex atlas. By identical reasoning as on *pages 142-143*, we can find a maximal complex atlas containing an initial complex atlas. We call this maximal atlas a complex structure on M.

Now a Riemann surface is defined as a 2-dimensional topological manifold M without boundary and equipped with a complex structure. I will refer to all the charts in the complex structure on M as complex charts.

For some standard examples:

- Any open subset U of $\mathbb{C}$ equipped with the standard structure (i.e. the one generated by the chart (U, Id_U)) is a Riemann surface.

- The Riemann sphere (i.e. $\mathbb{C}_\infty$) is a Riemann surface.

  We specifically equip it with the structure generated by the charts $(\mathbb{C}, z \mapsto z)$ and $(\mathbb{C}_\infty - \{0\}, z \mapsto \frac{1}{z})$. Proving these charts are homeomorphisms into $\mathbb{C}$ is easy enough. Also, both of our transition maps are given by $z \mapsto \frac{1}{z}$ (defined on $\mathbb{C} - \{0\}$).

- Let $\omega_1, \omega_2 \in \mathbb{C} - \{0\}$ satisfy that $\omega_1/\omega_2 \notin \mathbb{R}$. Then let:
$$\Lambda := \{n\omega_1 + m\omega_2 : n, m \in \mathbb{Z}\} \text{ be a lattice in } \mathbb{C}.$$

  We know that Λ is a subgroup of $\mathbb{C}$ (when considered a group equipped with addition). Therefore, we now consider the topological group $X := \mathbb{C}/\Lambda$ equipped with the quotient topology. Also, we let $\pi : \mathbb{C} \to X$ be the projection map. All that I want to say about X as a Riemann surface is long enough that I'm going to go on a separate tangent first.

Let X and Y be topological spaces. We say a map $p : X \to Y$ is locally injective if every $x \in X$ has a neighborhood N such that $p|_N : N \to Y$ is an injective function.

For example, let $\pi : \mathbb{C} \to \mathbb{C}/\Lambda$ be the quotient map from before. Then note that $\pi(z) = \pi(w)$ if and only if $z = w + n\omega_1 + m\omega_2$ for some $n, m \in \mathbb{Z}$. But now note that if $n \neq 0 \neq m$ (which happens iff $z \neq w$), then $z - w \in \Lambda - \{0\}$. At the same time, recall from *pages 582-583* that there exists $c > 0$ with $|n\omega_1 + m\omega_2| \geq c(|n| + |m|)$ for all $n, m \in \mathbb{Z}$. In turn, we can see that $\pi(z) = \pi(w)$ where $z \neq w$ implies that $|z - w| \geq c$. Or alternatively, we can see that if $U \subseteq \mathbb{C}$ satisfies that $\mathrm{diam}(U) < c$, then π is injective on U.

Also recall from *page 121* the definition of a covering map.

We can go a step further and say that the map $\pi : \mathbb{C} \to \mathbb{C}/\Lambda$ from before is a covering map. To see why, let c be the constant from the prior paragraph such that $|n\omega_1 + m\omega_2| \geq c(|n| + |m|)$. Next, consider any $z + \Lambda \in \mathbb{C}/\Lambda$. Then $B_{c/3}(z) + \Lambda$ is a union of the collection of disjoint open sets: $\{B_{c/3}(z) + n\omega_1 + m\omega_2 : n, m \in \mathbb{Z}\}$.

As for why these sets are disjoint, note that if they weren't all disjoint then we could find $n_1, n_2, m_1, m_2 \in \mathbb{Z}$ with $|n_1 - n_2| + |m_1 - m_2| \geq 1$ as well as $w_1, w_2 \in \mathbb{C}$ with $|w_i| < c/3$ for both i, all satisfying that:
$$z + w_1 + n_1\omega_1 + m_1\omega_2 = z + w_2 + n_2\omega_1 + m_2\omega_2$$

By moving terms, we can see that $w_1 - w_2 = (n_2 - n_1)\omega_1 + (m_2 - m_1)\omega_2 \in \Lambda - \{0\}$. So, $|w_1 - w_2| \geq c$. But this contradicts that $|w_1| < c/3$ and $|w_2| < c/3$ implies that $|w_1 - w_2| < c$.

Now note that $\pi^{-1}(\pi(B_{c/3}(z) + \Lambda)) = B_{c/3}(z) + \Lambda$. Since π is a quotient map, it follows that $\frac{B_{c/3}(z)+\Lambda}{\Lambda}$ is an open subset of $\mathbb{C}/\Lambda$. Plus, it contains $z + \Lambda$.

To finish off, I need to show that $B_{c/3}(z)+n\omega_1+m\omega_2$ is homeomorphically mapped by π to $\frac{B_{c/3}(z)+\Lambda}{\Lambda}$ for all integers n and m. Fortunately since translation on $\mathbb{C}$ is a homeomoprhism, it already suffices to assume that $n = m = 0$. So, we just need to show that $B_{c/3}(z)$ is homeomorphically mapped by π to $\frac{B_{c/3}(z)+\Lambda}{\Lambda}$.

Yet since $\operatorname{diam}(B_{c/3}(z)) < c$, we know that π is injective on $B_{c/3}(z)$. Also note that, π maps $B_{c/3}(z)$ surjectively onto $\frac{B_{c/3}(z)+\Lambda}{\Lambda}$.

We trivially know that π is continuous. And on *page 350* I proved that if G is a topological group and $H \lhd G$, then the quotient map $G \mapsto G/H$ is an open map. This shows that π is a homeomorphism from $B_{c/3}(z)$ to $\frac{B_{c/3}(z)+\Lambda}{\Lambda}$.

One last important observation is that if X and Y are topological spaces, $p : X \to Y$ is an open, surjective, continuous map, and X is second countable, then Y is also second countable.

To see why, let $\{U_n\}_{n\in\mathbb{N}}$ be a countable base for X. Then suppose $y \in Y$ and $V \subseteq Y$ is an open set containing y. Since p is surjective, we can find $x \in X$ with $p(x) = y$. Next, as p is continuous we know that $p^{-1}(V)$ is an open subset of X containing x. Hence, we can find an open set U_n in our base such that $x \in U_n \subseteq p^{-1}(V)$. And finally since p is an open map, we know that $p(U_n)$ is an open subset of Y containing $y = p(x)$ and which also satisfies that $p(U_n) \subseteq V$. This proves that $\{p(U_n)\}_{n\in\mathbb{N}}$ is also a countable base for Y.

(And since $\pi : \mathbb{C} \to \mathbb{C}/\Lambda$ is an open, surjective, and continuous map, the prior result applies.)

Based on all the prior reasoning, we can now easily see that $\mathbb{C}/\Lambda$ is a topological manifold. We define a complex structure generated by the maps $\pi|_V : V \to \mathbb{C}/\Lambda$ where $V \subseteq \mathbb{C}$ is an open set such that π is an injection from V into $\mathbb{C}/\Lambda$. The transition maps between the charts corresponding to these maps will all have the form $z \mapsto z+n\omega_1+m\omega_2$ for some $n, m \in \mathbb{Z}$. So this is a well-defined complex structure.

One other note I'd like to make is that by the universal property of quotient maps, we have that any function on $\mathbb{C}/\Lambda$ is associated with a doubly periodic function on $\mathbb{C}$ (i.e. $f(z) = f(z+\omega_1) = f(z+\omega_2)$). Also, any continuous function on $\mathbb{C}/\Lambda$ is associated with a continuous doubly periodic function on $\mathbb{C}$.

Among the two example complex manifolds I gave before, I already know that $\mathbb{C}_\infty$ is compact. I can also show that $\mathbb{C}/\Lambda$ is compact.

Let $\Lambda = \{n\omega_1 + m\omega_2 : n, m \in \mathbb{Z}\}$ where $\omega_1, \omega_2 \in \mathbb{C} - \{0\}$ with $\omega_1/\omega_2 \notin \mathbb{R}$. Next, let K be the fundamental parallelogram $\{x\omega_1 + y\omega_2 : x, y \in [0, 1]\}$. Note that K is a compact subset of $\mathbb{C}$. Also if $\pi : \mathbb{C} \to \mathbb{C}/\Lambda$ is the quotient map, then we can see that $\pi(K) = \mathbb{C}/\Lambda$.

Next let $\{U_\alpha\}_{\alpha\in A}$ be any open cover of $\mathbb{C}/\Lambda$. Then we have that $\{\pi^{-1}(U_\alpha)\}_{\alpha\in A}$ is an open cover of $\mathbb{C}$ and in turn also of K. Since K is compact, we can find $\alpha_1, \ldots, \alpha_n$ with

$K \subseteq \bigcup_{j=1}^{n} \pi^{-1}(U_{\alpha_j}) = \pi^{-1}(\bigcup_{j=1}^{n} U_{\alpha_j})$. And finally since π is surjective, we have that $\pi(\pi^{-1}(\bigcup_{j=1}^{n} U_{\alpha_j})) = \bigcup_{j=1}^{n} U_{\alpha_j}$. So, $\mathbb{C}/\Lambda = \pi(K) \subseteq \bigcup_{j=1}^{n} U_{\alpha_j}$.

Knowing a Riemann surface is compact will be relevant once I define holomorphic functions on and between Riemann surfaces.

Recalling the definition of a differentiable map between manifolds on *page 147*, we make the following definition:

> Suppose M and N are Riemann surfaces. Then a map $F : M \to N$ is holomorphic if for all $p \in M$ there is a complex chart (U, φ) on M and a complex chart (V, ψ) on N such that $p \in U$, $F(U) \subseteq V$, and $\psi \circ F \circ \varphi^{-1}$ is a holomorphic function (in the complex analytic sense) from $\varphi(U)$.

Note by identical reasoning as on *pages 147-148*, we can see that a holomorphic function between Riemann surfaces M and N is continuous.

Proposition: If M and N are Riemann surfaces and $F : M \to N$ is a holomorphic function, then for any complex charts (U, φ) on M and (V, ψ) on M we have that $\psi \circ F \circ \varphi^{-1}$ is a holomorphic function (in the complex analytic sense) on $\varphi(U \cap F^{-1}(V))$

> Proof:
> Because F is continuous, we know that $U \cap F^{-1}(V)$ is an open subset of M. Now we have nothing to show if $U \cap F^{-1}(V) = \emptyset$. So, we can assume $U \cap F^{-1}(V)$ is nonempty. Then by shrinking U, we can without loss of generality just say that $U \subseteq F^{-1}(V)$. That way, our problem becomes showing that $\psi \circ f \circ \varphi^{-1}$ is a holomorphic function (in the complex analytic sense) from $\varphi(U)$ into $\psi(V) \subseteq \mathbb{C}$.
>
> Fortunately let $x = \varphi(p) \in \varphi(U)$. Then we know there exists complex charts (U', φ') on M and (V', ψ') on N with $p \in U'$, $F(U') \subseteq V'$, and $\psi' \circ F \circ (\varphi')^{-1}$ a holomorphic function (in the complex analytic sense) from $\varphi'(U')$.
>
> But now $\psi \circ F \circ \varphi^{-1} = (\psi \circ (\psi')^{-1}) \circ (\psi' \circ F \circ (\varphi')^{-1}) \circ (\varphi' \circ \varphi^{-1})$ is a holomorphic function from $\varphi(U \cap U')$ to $\psi(V \cap V')$. Since $\varphi(U \cap U')$ is an open set containing x (since $U \cap U'$ is an open set containing p), we've proven that $\psi \circ F \circ \varphi^{-1}$ is holomorphic at $x \in \varphi(U)$. And since x was arbitrary, we can conclude that $\psi \circ F \circ \varphi^{-1}$ is a holomorphic function from $\varphi(U)$. ■

Having shown the prior proposition, we can now conclude two other results really easily.

Firstly, if $U, V \subseteq \mathbb{C}$ are open sets, then a function $f : U \to V$ is holomorphic in the complex analytic sense if and only if it is holomorphic in the Riemann surface sense.

> $(\Longrightarrow)$
> Just consider the complex charts (U, Id_U) and (V, Id_V) on U and V respectively. Then any $p \in U$ is trivially contained in our chart on U, $f(U)$ is contained in V, and $f = \mathrm{Id}_V \circ f \circ \mathrm{Id}_V^{-1}$ is a holomorphic function (in the complex analytic sense). So, f is holomorphic in the Riemann surface sense.

($\Longleftarrow$)
If f is holomorphic in the Riemann surface sense, then we know the complex charts (U, Id_U) and (V, Id_V) satisfy that $f = \mathrm{Id}_V \circ f \circ \mathrm{Id}_U^{-1}$ is holomorphic by our prior proposition. ■

Secondly, suppose M is any Riemann surface. Also suppose $V \subseteq \mathbb{C}$ is an open set. Then the following are equivalent:

- $f : M \to V$ is holomorphic in the Riemann surface sense
- For any complex chart (U, φ) on M we have that $f \circ \varphi^{-1}$ is a holomorphic function (in the complex analytic sense) from $\varphi(U)$.
- For all $p \in M$ there exists a complex chart (U, φ) of M with $p \in U$ and $f \circ \varphi^{-1}$ a holomorphic function (in the complex analytic sense) from $\varphi(U)$.

The proof of this fact is mostly identical to the prior proof.

Given any Riemann surfaces M and N we let $O(M, N)$ denote the set of holomorphic functions from M to N. In the special case that $N = \mathbb{C}$, we can now easily see from the second result prior that $O(M, \mathbb{C})$ is a complex algebra.

Theorem (Identity Principle): Suppose M and N are Riemann surfaces with M connected. Also suppose $F : M \to N$ and $G : M \to N$ are two holomorphic functions. If $S := \{x \in M : F(x) = G(x)\}$ has an accumulation point in M then $S = M$.

Proof:
Let $A = \mathrm{acc}(S) \subseteq M$ be the set of accumulation points of S in M. By assumption we know that A is nonempty. I also immediately claim that A is a closed set.

To see why, note that $\overline{A} = A \cup \mathrm{acc}(A)$. So A is closed iff $\overline{A} = A$ iff $\mathrm{acc}(A) \subseteq A$. To prove the last statement does in fact hold, suppose $x \in \mathrm{acc}(A)$ and that $U \subseteq M$ is any open set containing x. We have that $(U - \{x\}) \cap A \neq \emptyset$. Hence, we can find a point $y \in U \cap A$ such that $y \neq x$. Next, since M is Hausdorff (and thus T_1), we know that $U - \{x\}$ is an open set containing y. Hence since $y \in \mathrm{acc}(S)$, we can find $z \in (U - \{x, y\}) \cap S$. But this proves that $S \cap (U - \{x\}) \neq \emptyset$. Since U was arbitrary, we've proven that $x \in \mathrm{acc}(S) = A$. And since x was arbitrary, we can conclude $\mathrm{acc}(A) \subseteq A$.

But now by the identity principle for holomorphic functions (in the complex analytic sense), we can also conclude that A is also an open set.

To see why, first note that F and G being continuous implies that $A \subseteq S$. After all, if $x \in A = \mathrm{acc}(S)$ then we know that there is a net $\langle x_\beta \rangle_{\beta \in B}$ in $S - \{x\}$ converging to x in the topology of M.

Technically as M is by definition second countable, we can work with sequences instead of nets. However, it doesn't really matter.

Since F and G are continuous, we thus know $F(x_\beta) \to F(x)$ and $G(x_\beta) \to G(x)$ in the topology on N. Yet also note that since $x_\beta \in S$ for all β, we know that $F(x_\beta) = G(x_\beta)$ for all β. Therefore, we can actually conclude that $F(x_\beta)$ converges to both $F(x)$ and $G(x)$. Finally, since N is Hausdorff by definition, we must have that $F(x) = G(x)$. So, $x \in S$. And since x was arbitrary, we've proven that $A \subseteq S$.

Next, we prove that every $x \in A$ is an interior point of A. To do this, first note that since F is holomorphic, we can find a complex chart (U, φ) on M and a complex chart (V, ψ) on N such that $x \in U$, $F(U) \subseteq V$ and $\psi \circ F \circ \varphi^{-1}$ is a holomorphic function from $\varphi(U)$. Next, since $x \in A \subseteq S$, we know that $G(x) = F(x) \in V$. So, $G^{-1}(V) \cap U$ is a nonempty set containing x. Also as G is holomorphic (and thus continuous), we know that $U \cap G^{-1}(V)$ is an open set of M.

In turn, we can shrink U in order to without loss of generality have that $x \in U$, $U \subseteq M$ is open, $F(U) \subseteq V$, and $G(U) \subseteq V$. Additionally, note that since $\varphi(U)$ is locally connected (since its an open subset of $\mathbb{C}$), we can shrink U one more time in order to guarentee without loss of generality that $\varphi(U)$ is connected.

So to summarize, we've shown there exists complex charts (U, φ) on M and (V, ψ) on N such that:

- $x \in U$,
- $\varphi(U)$ is connected,
- $F(U) \subseteq V$ and $G(U) \subseteq V$,
- $f \coloneqq \psi \circ F \circ \varphi^{-1} \in O(\varphi(U), \mathbb{C})$ and $g \coloneqq \psi \circ G \circ \varphi^{-1} \in O(\varphi(U), \mathbb{C})$.

But now $S \cap U$ has x as a limit point inside U. In turn, we also know that $\varphi(S \cap U)$ has $\varphi(x)$ as a limit point inside $\varphi(U)$. And since $\varphi(U)$ is an open connected set and $f = g$ on $\varphi(S \cap U)$, we can conclude by the identity principle (from math 220a) that $f = g$ everywhere on $\varphi(U)$. In turn $F = \psi^{-1} \circ f \circ \varphi$ and $G = \psi^{-1} \circ g \circ \varphi$ must equal each other everywhere on U. In other words, $U \subseteq S$.

To finish off, its easy to see that every point of $\varphi(U)$ is a limit point of $\varphi(U)$ in the topology on $\mathbb{C}$. (This is just a property of open subsets in the Euclidean topology on $\mathbb{R}^n$.) In turn, we can also see that every point of U is a limit point of $U \subseteq S$. So, we have shown that $U \subseteq A$. And this proves that $x \in A$ is an interior point. Also, since x was arbitrary we can conclude that A is an open set.

Since M is connected, the only way for A to be a nonempty clopen set is if $A = M$. Then since $A \subseteq S \subseteq M$, we can also conclude that $S = M$. So, $F = G$ on M. ■

6/27/2026

Today I'm going to continue introducing Riemann surfaces. Although, the first result I want to introduce actually applies to more general topological manifolds and is an application of the invariance of domain theorem.

Theorem: Suppose M and N are both n-dimensional topological manifolds (without boundary) and suppose $U \subseteq M$ is an open set. Then U is also an n-dimensional topological manifold when equipped with the subspace topology. If $F : U \to N$ is a locally injective continuous function then $F(U)$ is an open subset of N.

Proof:
Consider any $p \in U$ and let $W_1 \subseteq U$ be an open neighborhood of p on which F is injective. Next let (W_2, φ) be a chart containing p. Thirdly, let (V, ψ) be a chart on N containing $F(p)$. Since F is continuous, we know that $W_3 = F^{-1}(V)$ is an open set containing p. So now by setting $W = W_1 \cap W_2 \cap W_3$, we get that $p \in W$, W is an open subset of U, φ is a homeomorphism between W and some open set $\varphi(W) \subseteq \mathbb{R}^n$, $F(W) \subseteq V$, and ψ is a homeomorphism between V and some open set $\psi(V) \subseteq \mathbb{R}^n$.

Now to show that $F(U)$ is an open set, it suffices to show that $F(p)$ is in the interior of $F(U)$. Yet also note that $W \subseteq U \implies F(W) \subseteq F(U)$. Plus, we know that $F(p) \in F(W)$. So if we can conclude that $F(W)$ is an open subset of N, then we will have successfully shown that $F(p)$ is an interior point of $F(U)$.

Fortunately, note that $f := \psi \circ F \circ \varphi^{-1}$ is a continuous injective function from the open set $\varphi(W) \subseteq \mathbb{R}^n$ into $\psi(V) \subseteq \mathbb{R}^n$. By the invariance of domain theorem, we can conclude that $(\psi \circ F \circ \varphi^{-1})(\varphi(W))$ is an open subset of $\mathbb{R}^n$. Then since ψ is continuous, we in turn know that $\psi^{-1}((\psi \circ F \circ \varphi^{-1})(\varphi(W))) = F(W)$ is an open subset of $V \subseteq N$. ■

Corollary: Any bijective continuous function between topological manifolds is automatically a homeomorphism.

Returning to the topic of Riemann surfaces, we have the following result for holomorphic functions:

Open Mapping Theorem: Suppose M and N are Riemann surfaces with M connected, and also that $F : M \to N$ is a nonconstant holomorphic function. Then F is an open map.

Proof:
Let $U \subseteq M$ be any open set. To show that $F(U)$ is an open set, it suffices to prove that if $p \in U$ then $F(p)$ is an interior point of $F(U)$. Fortunately, as F is holomorphic we know that there exists complex charts (W, φ) on M and (V, ψ) on N such that $p \in W$, $F(W) \subseteq V$, and $\psi \circ F \circ \varphi^{-1}$ is a holomorphic function from $\varphi(W)$. By shrinking W if necessary, we can assume without loss of generality that $W \subseteq U$. Also because $\mathbb{C}$ is locally connected, we can additionally shrink W and $\varphi(W)$ in order to guarentee without loss of generality that $\varphi(W)$ is a connected open subset of $\mathbb{C}$.

By the identity principle proven on the prior two pages, we know that F cannot be constant on W or else we'd be able to also show that F is contant on all of M. So to avoid any contradictions, we must have that F is nonconstant on W. In turn, $f := \psi \circ F \circ \varphi^{-1}$ must also be nonconstant on the set $\varphi(W)$. So by the open mapping theorem proven in math 220a (see *page 429*), we can conclude that f is an open map.

With that we've shown $f(\varphi(W))$ is an open subset of $\psi(V) \subseteq \mathbb{C}$. Then since ψ is continuous, we can conclude that $F(W) = (\psi^{-1} \circ f \circ \varphi)(W)$ is also an open subset of $V \subseteq N$. Finally, since $p \in W$ and $W \subseteq U$, we know $F(p) \in F(W)$ and $F(W) \subseteq F(U)$. This proves that $F(p)$ is an interior point of $F(U)$. ■

Here's one more pair of results to justify why we will care about meromorphic functions a lot once we define them in a second.

Maximum Modulus Theorem: Suppose M is a connected Riemann surface and $F : M \to \mathbb{C}$ is a holomorphic function. If $|F|$ attains a local maximum on M then F is a constant function.

Proof:
Suppose $|F|$ has a local maximum at $p \in M$. Then let (U, φ) be a complex chart containing p. By shrinking U if necessary, we can also without loss of generality assume $\varphi(U)$ is a connected set. But now $F \circ \varphi^{-1}$ is a holomorphic function on the region $\varphi(U)$ of $\mathbb{C}$ which also satisfies that $|F \circ \varphi^{-1}|$ attains a local maximum at $\varphi(p)$. It follows by the maximum modulus principle from complex analysis (see *page 384*) that $F \circ \varphi^{-1}$ is a constant function on $\varphi(U)$. In turn $F \circ \varphi^{-1} \circ \varphi = F$ is also a constant function on U. And by the identity principle, we can then conclude that F is constant on M. ■

As a side note since I realize that I never showed this, any constant function between Riemann surfaces is automatically holomorphic. To see why, suppose M and N are Riemann surfaces, and $F : M \to N$ satisfies that $F(p) = q \in N$ for all $p \in M$. Then having picked a complex chart (V, ψ) in N containing Q, we have for every complex chart (U, φ) in M that $F(U) \subseteq V$ and $\psi \circ F \circ \varphi^{-1}$ is a constant function from $\varphi(U) \subseteq \mathbb{C}$ into $\mathbb{C}$. So, $\psi \circ F \circ \varphi^{-1}$ is trivially holomorphic.

Corollary: If M is a compact Riemann surface then $O(M, \mathbb{C})$ consists entirely of locally constant functions. In particular, if M is connected then $O(M, \mathbb{C})$ consists only of constant functions.

Proof:
Note that every connected component of M is closed. And since closed subsets of compact sets are compact, we can thus conclude that every connected component of M is compact.

Also note that since M is locally connected, we know that every connected component of M is open. Therefore every connected component of M (equipped with the subspace topology) can sill still be viewed as a Riemann surface where F restricted to each component is holomorphic.

Finally, suppose $F \in O(M, \mathbb{C})$ and let N be any connected component of M. Then F being continuous on N plus N being compact implies that $|F|$ attains a local maximum on N. Yet also note that F being holomorphic on N as a Riemann surface means that the maximum modulus theorem applies. So, we must have that F is constant on N. ■.

Since a lot of the Riemann surfaces we actually care about are compact and connected, we thus can see that $O(M, \mathbb{C})$ is often not the collection of functions we want to study on a Riemann surface. Hence, we will instead often consider $O(M, \mathbb{C}_\infty)$.

To motivate the next bit, observe that if $U \subseteq \mathbb{C}$ is an open set then any meromorphic function $f : U \to \mathbb{C}$ extends to a holomorphic function from U into $\mathbb{C}_\infty$ (in the Riemann surface sense).

If you squint you can see that this is just my comment from *page 467*. After all, $z \mapsto f(z)$ is a holomorphic function from $U - f^{-1}(\{\infty\})$ into $\mathbb{C}$ and $z \mapsto 1/f(z)$ is a holomorphic function from $U - f^{-1}(\{0\})$ into $\mathbb{C}$.

Conversely, suppose $G \subseteq \mathbb{C}$ is a region and $f : G \to \mathbb{C}_\infty$ is a holomorphic function (in the Riemann surface sense) satisfying that there exists $z \in G$ with $f(z) \neq \infty$. Since f is not the constant ∞-valued function, we know by the identity principle that $f^{-1}(\{\infty\})$ can't have any limit points. This also means every $a \in f^{-1}(\{\infty\})$ must be an isolated point of G.

Let $\Omega \coloneqq G - f^{-1}(\{\infty\})$. Since f is continuous, we can conclude that Ω is an open subset of G. So, we consider the chart $(\Omega, \mathrm{Id}_\Omega)$. Next, we consider the chart $(\mathbb{C}, \mathrm{Id}_\mathbb{C})$ in the standard structure on $\mathbb{C}_\infty$. So, $\mathrm{Id}_\mathbb{C} \circ f \circ \mathrm{Id}_\Omega = f$ is a holomorphic function from Ω into $\mathbb{C}$ (in the complex analytic sense).

But now $f|_\Omega$ has singularities at every $a \in f^{-1}(\{\infty\})$. Also by the continuity of f on G, we know that $|f(z)| \to \infty$ as $z \to a$. Therefore, we can conclude that $f|_\Omega$ has poles at all $a \in G - \Omega$. Or in other words, f is a meromorphic function on G (in the complex analytic sense).

All in all, we can conclude that if $U \subseteq \mathbb{C}$ is an open set, then the set of meromorphic functions on U is precisely equal to $O(U, \mathbb{C}_\infty) - \mathcal{F}$ where $\mathcal{F}$ is the collection of holomorphic functions that are constantly ∞ on any neighborhood. For the time being, I'm going to denote the set of meromorphic functions on U as $\mathcal{M}(U)$.

I realize that $\mathcal{M}$ and M look similar. So, I'll probably start denoting Riemann surfaces by X and Y instead.

6/29/2026

Proposition: Let X be a connected Riemann surface. And suppose $f : X \to \mathbb{C}_\infty$ is a function such that $f \not\equiv \infty$ on X. Then the following are equivalent.

(a) $f : X \to \mathbb{C}_\infty$ is a holomorphic function.

(b) For all complex charts (U, φ) on X we have that $f \circ \varphi^{-1}$ is a meromorphic function on $\varphi(U)$.

(c) For every $p \in X$ there exists a complex chart (U, φ) on X with $p \in U$ and $f \circ \varphi^{-1}$ a meromorphic function on $\varphi(U)$.

$(a \implies b)$
Let (U, φ) be any complex chart on X. Since f is continuous and $f \not\equiv \infty$ on X, we know that $f^{-1}(\{\infty\}) \cap U$ must be a closed set (relative to U) with no limit points in U. In other words, $\Omega \coloneqq U - f^{-1}(\{\infty\})$ is an open set with $f(\Omega) \subseteq \mathbb{C}_\infty - \{\infty\}$. If we now consider

the charts $(\Omega, \varphi|_\Omega)$ and $(\mathbb{C}, \mathrm{Id})$, we know that $\mathrm{Id} \circ f \circ \varphi^{-1} = f \circ \varphi^{-1}$ is a holomorphic map from $\varphi(\Omega) = \varphi(U) - \varphi(f^{-1}(\{\infty\}))$ into $\mathbb{C}$ since f is holomorphic in the Riemann surface sense. Furthermore, we know every $a \in \varphi(U - \Omega)$ has a neigborhood N where $f \circ \varphi^{-1}(z) \neq \infty$ for any $z \in N - \{a\}$. Plus by the continuity of $f \circ \varphi^{-1}$, we know that $f \circ \varphi^{-1}(z) \to \infty$ as $z \to a$. So, every $a \in \varphi(U - \Omega)$ is a pole. And this finishes showing that $f \circ \varphi^{-1}$ is meromorphic on $\varphi(U)$.

$(b \implies c)$
This is obvious.

$(c \implies a)$
Suppose $p \in X$ and let (U, φ) be a complex chart on X such that $p \in U$ and $f \circ \varphi^{-1}$ a meromorphic function on $\varphi(U)$. If $f(p) \neq \infty$ then by shrinking U and $\varphi(U)$ to a coordinate ball around p, we can assume $f \neq \infty$ anywhere on B. And now $f(U) \subseteq \mathbb{C}$ and $\mathrm{Id} \circ f \circ \varphi^{-1} = f \circ \varphi^{-1}$ is a holomomorphic function on $\varphi(U)$.

Meanwhile, if $f(p) = \infty$ then we know $\varphi(p)$ is a pole of $f \circ \varphi^{-1}$. So, we can shrink U and $\varphi(U)$ to a coordinate ball so that $f \neq 0$ anywhere on U. Next, $f(U) \subseteq \mathbb{C}_\infty - \{0\}$ and $z \mapsto 1/f(\varphi^{-1}(z))$ is a holomorphic function on $\varphi(U)$. ■

Given a Riemann surface X, we say $f : X \to \mathbb{C}_\infty$ is meromorphic if $f \in O(X, \mathbb{C}_\infty)$ and $f \not\equiv \infty$ on any neighborhood in X. We denote the set of meromorphic functions on X as $\mathcal{M}(X)$.

Example: What are the meromorphic functions on $\mathbb{C}_\infty$?

Suppose $f : \mathbb{C}_\infty \to \mathbb{C}_\infty$ is meromorphic. Since $\mathbb{C}_\infty$ is compact, we know that f has only finitely many poles (i.e. points in the set $f^{-1}(\{\infty\})$). So, we can let $p_1, \ldots, p_n$ be the poles of f in $\mathbb{C}$ (repeated according to multiplicity). Then $g(z) := f(z) \prod_{i=1}^n (z - p_i)$ is a well-defined entire function on $\mathbb{C}$.

Next, let $g(z) = \sum_{k=0}^\infty c_k z^k$ be the Taylor expansion of g about 0. Since g is an entire function, we know this power series has an infinite radius of convergence. In turn, we also know $g(\frac{1}{z}) = \sum_{k=0}^\infty c_k z^{-k}$ is absolutely convergent for all $z \in \mathbb{C} - \{0\}$. Yet also note $g(\frac{1}{z}) = f(\frac{1}{z}) \prod_{i=1}^n (\frac{1}{z} - p_i)$. And since f is meromorphic, we know that $h(z) := f(\frac{1}{z})$ has either a removable singularity or pole at $z = 0$. It follows that $z \mapsto g(\frac{1}{z})$ has a pole or removable singularity at $z = 0$ as well.

Yet now we can conclude there exists $K \in \mathbb{N}$ with $c_k = 0$ for all $k \geq K$. (If not then $z \mapsto g(\frac{1}{z})$ would have an essential singularity.) So, we've proven there exists a polynomial $P(z) \in \mathbb{C}[z]$ with $P(z) = f(z) \prod_{i=1}^n (z - p_i)$. Finally, by rearranging our expression and factoring $P(z)$, we can conclude that there exists $\lambda \in \mathbb{C}$ as well as $a_1, \ldots, a_m \in \mathbb{C}$ such that:

$$f(z) = \lambda \frac{(z-a_1)\cdots(z-a_m)}{(z-p_1)\cdots(z-p_n)}$$

So, $\mathcal{M}(\mathbb{C}_\infty) \subseteq \mathbb{C}(z)$ (where $\mathbb{C}(z)$ is the field of rational functions in $\mathbb{C}$). Conversely, it's easy to show every rational function is a holomorphic map from $\mathbb{C}_\infty$ into $\mathbb{C}_\infty$. ■

Here's another interesting observation that the prior example brought to mind.

Lemma: Suppose X and Y are connected Riemann surface with X also compact. Then any nonconstant holomorphic function $F \in O(X,Y)$ must be surjective.

Proof:
Since F is continuous and X is compact, we know that $F(X)$ must be a compact subset of Y. In turn, since Y is Hausdorff we also know that $F(X)$ is a closed subset of Y. Meanwhile, since F is a nonconstant holomorphic function and X is connected, we know F is an open map. In turn, as X is open we must have that $F(X)$ is an open subset of Y. Finally, we trivially know that $F(X) \neq \emptyset$. So, we can conclude from the connectedness of Y that $F(X) = Y$. ■

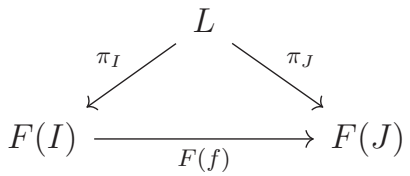
As a result of this lemma, any nonconstant rational function in $\mathbb{C}(z)$ must define a surjective map from $\mathbb{C}_\infty$ into $\mathbb{C}_\infty$. In particular, this means that every nonconstant meromorphic function on $\mathbb{C}_\infty$ will have both poles and zeros.

6/30/2026

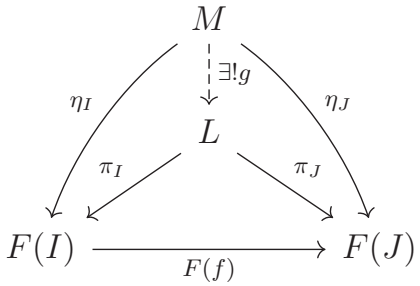
I moved to the room next door today. I was hoping to cover some category theory but all I managed to cover were the following definitions. I was not very productive today. Firstly, here are two definitions I got from *Algebra: Chapter 0* by Paulo Aluffi. (see *citation 8*...)

Suppose **I** is category (which we can think of as an indexing category). Next, we consider a functor $F : \mathbf{I} \to \mathbf{C}$. The limit of F (if it exists) is defined to be an object L in **C** along with a class of morphisms $\pi_I \in \mathrm{Hom}_{\mathbf{C}}(L, F(I))$ such that:

1. the following diagram commutes for all $I, J \in \mathrm{Obj}(\mathbf{I})$ and $f \in \mathrm{End}_{\mathbf{I}}(I,J)$



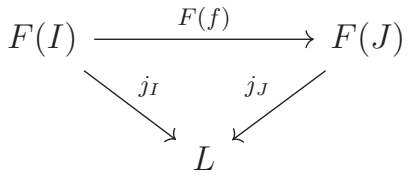
2. If $M \in \mathrm{Obj}(\mathbf{C})$ and $\eta_I \in \mathrm{Hom}_{\mathbf{C}}(M, F(I))$ is any other object and collection of morphisms satisfying the prior commuting diagram, then there exists a unique morphism $g : M \to L$ such that:



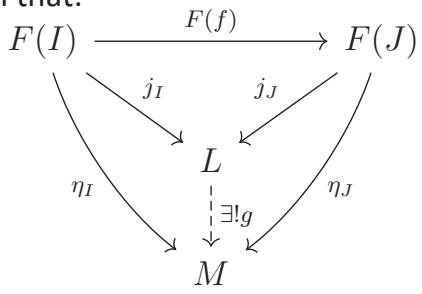
Remark: It's easy to see any two limits of a functor $F : \mathbf{I} \to \mathbf{C}$ are isomorphic. Thus, we can refer to there being a single limit (denoted as $\varprojlim(F)$). Also, we call this limit a projective limit.

Similarly, the colimit of F (if it exists) is defined to be an object C in $\mathbf{C}$ along with a class of morphisms $j_I \in \mathrm{Hom}_{\mathbf{C}}(F(I), C)$ such that:

1. the following diagram commutes for all $I, J \in \mathrm{Obj}(\mathbf{I})$ and $f \in \mathrm{End}_{\mathbf{I}}(I, J)$



2. If $M \in \mathrm{Obj}(\mathbf{C})$ and $\eta_I \in \mathrm{Hom}_{\mathbf{C}}(M, F(I))$ is any other object and collection of morphisms satisfying the prior commuting diagram, then there exists a unique morphism $g : C \to M$ such that:



Remark: It's easy to see any two colimits of a functor $F : \mathbf{I} \to \mathbf{C}$ are isomorphic. Thus, we can refer to there being a single colimit (denoted as $\varinjlim(F)$). Also, we call this limit an injective/direct limit.

Suppose $\mathbf{I}$ is a small category and let $F : \mathbf{I} \to A$-**mod** be a functor (where A is some unital commutative ring). I'll also denote $M_I := F(I)$ for all $I \in \mathrm{Obj}(\mathbf{I})$.

Firstly, here's how to construct $\varprojlim(F)$.

Let $L \subseteq \prod_{I \in \mathrm{Obj}(\mathbf{I})} M_I$ be given by the set:

$$L = \{(x_I)_{I \in \mathrm{Obj}(\mathbf{I})} : \forall I, J \in \mathrm{Obj}(\mathbf{I}) \text{ and } \forall f \in \mathrm{Hom}_{\mathbf{I}}(I, J),\ \ (F(f))(x_I) = x_J\}.$$

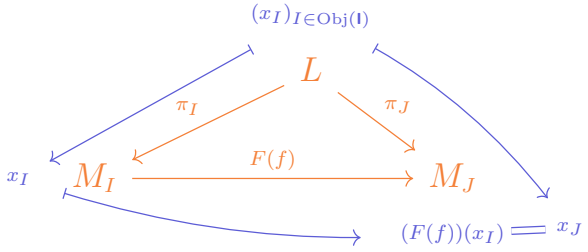
I claim that L is a submodule of $\prod_{I \in \mathrm{Obj}(\mathbf{I})} M_I$. After all, $(0_{M_I})_{I \in \mathrm{Obj}(\mathbf{I})} \in N$ since all module homomorphisms send zero elements to zero elements. Also suppose $(x_I)_{I \in \mathrm{Obj}(\mathbf{I})}$, $(y_I)_{I \in \mathrm{Obj}(\mathbf{I})} \in N$ and $a, a' \in A$. Then:

$$(F(f))(ax_I + a'y_I) = a(F(f))(x_I) + a'(F(f))(y_I) = ax_J + a'y_J$$

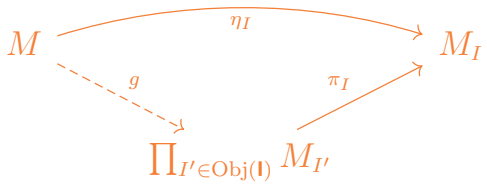
for all $I, J \in \mathrm{Obj}(\mathbf{I})$ and $f \in \mathrm{Hom}_{\mathbf{I}}(I, J)$. Hence $a(x_I)_{I \in \mathrm{Obj}(\mathbf{I})} + a'(y_I)_{I \in \mathrm{Obj}(\mathbf{I})} \in N$.

Next, for every $I \in \mathbf{I}$ let π_I by the projection from $\prod_{I' \in \mathrm{Obj}(\mathbf{I})} M_{I'}$ to M_I. I shall prove that L along with all the π_I (with their domains restricted to L) forms the limit of F.

1. Suppose $I, J \in \mathrm{Obj}(\mathbf{I})$ and $f \in \mathrm{Hom}_{\mathbf{I}}(I, J)$. Then the following diagram holds:



2. Now suppose another A-module M as well as the homomorphisms $\eta_I : M \to M_I$ satisfy the diagram that L and the π_I satisfy. By the universal property of products of modules, we know there is a unique module homomorphism $g : M \to \prod_{I\in\mathrm{Obj}(\mathbf{I})} M_I$ satisfying for all $I \in \mathrm{Obj}(\mathbf{I})$ that:



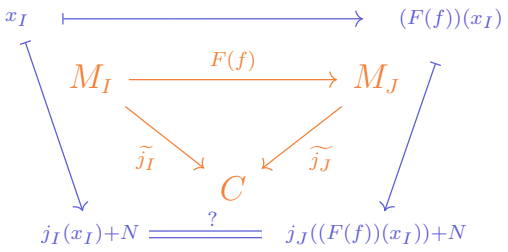
Since our desired homomorphism must satisfy the above commuting diagram, we can conclude that our desired morphism is unique. Also note that since M and the η_I satisfy the same commuting diagram that L and the π_I satisfy, we have that $\mathrm{im}(g) \subseteq L$. So, g is our desired morphism.

Secondly, here's how to construct $\varinjlim(F)$.

For all $I \in \mathrm{Obj}(\mathbf{I})$ let $j_I : M_I \to \bigoplus_{I'\in\mathrm{Obj}(\mathbf{I})} M_{I'}$ be the embedding homomorphism. Then, define $N \subseteq \bigoplus_{I'\in\mathrm{Obj}(\mathbf{I})} M_{I'}$ to be the submodule generated by all elements of the form $j_I(x_I) - j_J((F(f))(x_I))$ where $I, J \in \mathrm{Obj}(\mathbf{I})$, $x_I \in M_I$, and $f \in \mathrm{Hom}_{\mathbf{I}}(I, J)$.

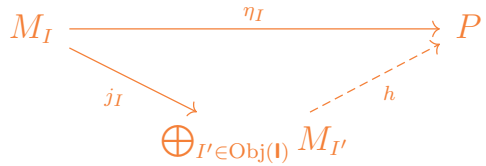
We let $C := (\bigoplus_{I\in\mathrm{Obj}(\mathbf{I})} M_I)/N$. After letting $\pi : \bigoplus_{I\in\mathrm{Obj}(\mathbf{I})} M_I \to (\bigoplus_{I\in\mathrm{Obj}(\mathbf{I})} M_I)/N$ be the quotient map, we then define the map $\widetilde{j_I} := \pi \circ j_I$ for all $I \in \mathrm{Obj}(\mathbf{I})$ to be. I claim that C along with the $\widetilde{j_I}$ is the colimit of F.

1. Suppose $I, J \in \mathrm{Obj}(\mathbf{I})$ and $f \in \mathrm{Hom}_{\mathbf{I}}(I, J)$. Then we need to verify that the following diagram holds for all $X_I \in M_I$:

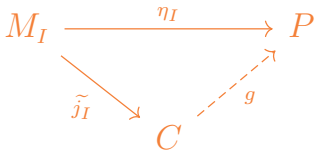


Fortunately, note that $j_I(x_I) - j_J((F(f))(x_I)) \in N$ because we defined N to have that element as a generator. So, the equality we needed does hold.

2. Suppose another A-module M and a bunch of homomorphisms $\eta_I : M_I \to M$ also satisfy the same commuting diagram that C and the $\widetilde{j_I}$ satisfy. Then by the universal property of direct sums of modules, there is a unique module homomorphism $h : \bigoplus_{I \in \text{Obj}(\mathbf{I})} M_I \to M$ satisfying that for all $I \in \text{Obj}(\mathbf{I})$ the below diagram holds:



Yet note that $\eta_I(x_I) - \eta_J((F(f))(x_I)) = 0_P$ for all $I, J \in \text{Obj}(\mathbf{I})$, $x_I \in M_I$, and $f \in \text{Hom}_{\mathbf{I}}(I, J)$. As a result, $N \subseteq \ker(h)$ and in turn we can define a module homomorphism $g : C \to P$ by $g((x_I)_{I \in \text{Obj}(\mathbf{I})} + N) = h((x_I)_{I \in \text{Obj}(\mathbf{I})})$. This will still be the unique homomorphism satisfying that:



Since A-**mod** has a limit and colimit respectively for all functors F going into A-**mod** from a small category, we say A-**mod** is complete and cocomplete respectively. Since A-**mod** is *both* complete *and* cocomplete, we also call A-**mod** a bicomplete category.

Note from 7/28/2026: Before today I had priorly written a false claim here that the prior construction generalizes to limits and colimits of functors to the category $\mathbf{CAlg}_A$ (the category of commutative A-algebras). However, I failed to realize that a "direct sum" of commutative A-algebras does not have an obvious well-defined multiplication operation turning that direct sum into a categorical coproduct. So, while the limit construction from before generalizes to functors to $\mathbf{CAlg}_A$, the colimit construction doesn't. I'll fix this oversight of mine on *pages 1067-1096*.

A point I do want to get across though is that A-**mod** and $\mathbf{CAlg}_A$ are both bicomplete. Furthermore, since $\mathbf{Ab} = \mathbb{Z}$-**mod** and $\mathbf{CRing} = \mathbf{CAlg}_{\mathbb{Z}}$, we can also construct limits and colimits of abelian groups and commutative rings.

One more colimit I want to construct is as follows. Suppose $(I, \lesssim)$ is a directed set (see my math 240b notes to recall what that means). We can turn I into a small category $\mathbf{I}$ whose objects are just the elements of I and whose morphisms are just the relations $i \lesssim j$. Now suppose $F : \mathbf{I} \to \mathbf{Set}$ is a functor. Then we can construct $\varinjlim(F)$ as follows.

Let $S_i := F(i)$ for all $i \in I$. Also, for reasons that will become apparent later, denote $F(i \lesssim j)$ as r^i_j. To construct the colimit of F, first let $S := \bigcup_{i \in I}(\{i\} \times S_i)$. Next, we use the fact that I is a directed set to define an equivalence relation. Namely, we say that $(i, x) \sim (j, y)$ if there is an element $k \in I$ with $i \lesssim k$ and $j \lesssim k$, and there exists $z \in S_k$ with $r^i_k(x) = z = r^j_k(y)$.

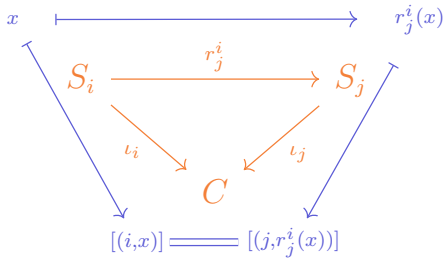
I claim that $\sim$ is an equivalence relation. After all, since $i \lesssim i$ and $r_i^i = \mathrm{Id}_{S_i}$ for all $i \in I$, we know that $(i,x) \sim (i,x)$ for all $i \in I$ and $x \in S_i$. Also, its obvious from our definition that $(i,x) \sim (j,y)$ iff $(j,y) \sim (i,x)$. Finally, suppose $(i_1,x_1) \sim (i_2,x_2) \sim (i_3,x_3)$. Then we know there exists (j,y) such that $r_j^{i_1}(x_1) = y = r_j^{i_2}(x_2)$. Similarly, we know there exists (j',y') such that $r_{j'}^{i_1}(x_2) = y' = r_{j'}^{i_2}(x_3)$. And finally, we let $k \in I$ be any element with $j \lesssim k$ and $j' \lesssim k$. Note that $r_k^j \circ r_j^{i_2} = r_k^{i_2} = r_k^{j'} \circ r_{j'}^{i_2}$, and also that $r_k^{i_1} = r_k^j \circ r_j^{i_1}$ and $r_k^{i_3} = r_k^{j'} \circ r_{j'}^{i_3}$. Thus:

$$r_k^{i_1}(x_1) = r_k^j \circ r_j^{i_1}(x_1) = r_k^j \circ r_j^{i_2}(x_2) = r_k^{j'} \circ r_{j'}^{i_2}(x_2) = r_k^{j'} \circ r_{j'}^{i_3}(x_3) = r_k^{i_3}(x_3)$$

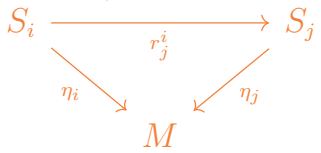
And this proves that $(i_1,x_1) \sim (i_3,x_3)$.

We now let $C := S/_\sim$. Also for every $i \in I$ we define $\iota_i : S_i \to C$ by $x \mapsto [(i,x_i)]$. I claim that C along with the ι_i forms a colimit.

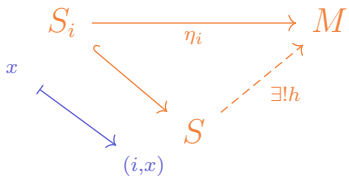
- Note that $(i,x) \sim (j, r_j^i(x))$ for all $j \in I$ with $i \lesssim j$. Hence, the following diagram holds for all $i,j \in I$ with $i \lesssim j$:



- Next, suppose a set M and functions η_i also satisfy for all $i \lesssim j$ that:



Then by the universal property of disjoint unions of sets (i.e. that they are categorical coproducts of sets), we know that there is a unique function $h : S \to M$ such that the following diagram holds for all $i \in I$:



In other words, $h(i,x) = \eta_i(x)$ for all $(i,x) \in S$. Yet now suppose $(i,x) \sim (j,y)$. Thus, there exists $k \in I$ with $i \lesssim k$, $j \lesssim k$, and $r_k^i(x) = r_k^j(y)$. Since we also know that $\eta_k \circ r_k^i = \eta_i$ and $\eta_k \circ r_k^j = \eta_j$, we can say that:

$$h(i,x) = \eta_i(x) = \eta_k \circ r_k^i(x) = \eta_k \circ r_k^j(y) = \eta_j(y) = h(j,y)$$

It follows that we can define a function $g : C \to M$ by $g([(i,x)]) \coloneqq h(i,x)$. And since h was uniquely determined, we also know g is uniquely determined. This finishes showing that C is a colimit.

7/3/2026

Today I want to try and take some notes on presheaves, stalks, and germs. I will be loosely following *Differential Analysis on Complex Manifolds* by Raymond Wells. (See *citation 9*...)

To start off, suppose **C** is a category. Then a presheaf is just a functor $F : \mathbf{C}^{\mathrm{op}} \to \mathbf{Set}$. If we replace the category **Set** with a different category **D** (say A-**mod**), we just say our presheaf is **D**-valued.

A morphism between presheaves F_1 and F_2 is just a natural transformation between the two functors. In the special case that the natural transformation $\eta : F_1 \to F_2$ consists of only injective functions, then we say F_1 is a subpresheaf of F_2.

Given a topological X, a relevant category for us to define is the category $\mathbf{Op}(X)$ of nonempty open subsets of X. The objects of this category are open subsets, and the morphisms are given by set inclusions. Clearly this is a small category. Using this category, we now define a presheaf $\mathcal{F}$ *over a topological space* X to be:

- An assignment to every nonempty open set $U \subseteq X$ some other set $\mathcal{F}(U)$,
- A collection of homomorphisms $r_V^U : \mathcal{F}(U) \to \mathcal{F}(V)$ (called restriction homomorphisms) for each pair of nonempty open sets $V \subseteq U \subseteq X$ such that:
  - $r_U^U = \mathrm{Id}_{\mathcal{F}(U)}$,
  - If $W \subseteq V \subseteq U$ then $r_W^U = r_W^V \circ r_V^U$.

    As a side note, we call the objects $\mathcal{F}(U)$ sections of the presheaf $\mathcal{F}$.

The subpresheafs in this case that we care about are when we restrict the domain of $\mathcal{F}$ to only open sets containing some arbitrary point $x \in X$.

Let **D** denote either **Set**, **Ab**, **CRing**, or A-**mod** or $\mathbf{CAlg}_A$ (where A is some unital commutative ring). Then let $\mathcal{F}$ be a **D**-valued presheaf over a topological space X. Given any $x \in X$ we define the stalk of $\mathcal{F}$ at x to be the direct limit (i.e. colimit) of $\mathcal{F}$ restricted to the open sets containing x:

$$\mathcal{F}_x \coloneqq \varinjlim_{x \in U \in \mathrm{Obj}(\mathbf{Op}(X))} \mathcal{F}(U)$$

Then, an element of $\mathcal{F}_x$ is called a germ.

> The intuition for this naming scheme is that in agriculture, a sheaf refers to a bundle of wheat, then each stalk is a single wheat strand, and finally the germs are the seeds at end of the stalk. Similarly in the math sense, a sheaf (which will be a presheaf equipped with a few more assumptions) is a structure encoding local data on an entire topological space, each stalk is a structure encoding local data at a specific point, and the germs are all the possible configurations of local data at a given point.

Here are three examples of presheaves over a topological space X:

- Suppose Y is another topological space and then for every $U \in \mathrm{Obj}(\mathbf{Op}(X))$ define:
$$\mathcal{F}(U) = C(U,Y) \coloneqq \{(f : U \to Y) : f \text{ is continuous}\}.$$
Also if $V \subseteq U \subseteq X$ let $r_V^U(f) \coloneqq f|_V$. This is easily checked to be a presheaf which we denote $\mathscr{C}_{X,Y}$. Also if $Y = k$ (where $k = \mathbb{R}$ or $\mathbb{C}$) then this presheaf is $(\mathbf{CAlg}_k)$-valued.

- Suppose that X in particular is a topological manifold equipped with a C^r structure (where $r \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$). Then for every $U \in \mathrm{Obj}(\mathbf{Op}(X))$ let $\mathscr{C}_X^r(U)$ denote the space of real-valued C^r functions from U to $\mathbb{R}$. Also for $V \subseteq U \subseteq X$ let $r_V^U(f) \coloneqq f|_V$. This is also easily checked to be a $(\mathbf{CAlg}_{\mathbb{R}})$-valued presheaf.
- Meanwhile, suppose that X is a topological manifold equipped with a complex structure. Then for every $U \in \mathrm{Obj}(\mathbf{Op}(X))$ let $\mathcal{O}_X$ denote the space of holomorphic functions from U. Also for $V \subseteq U \subseteq X$ let $r_V^U(f) \coloneqq f|_V$. This is also easily checked to be a $(\mathbf{CAlg}_{\mathbb{C}})$-valued presheaf.

Now a presheaf $\mathcal{F}$ over a topological space X is called a sheaf if for every collection of open sets $\{U_\alpha\}_{\alpha \in A}$ with $U \coloneqq \bigcup_{\alpha \in A} U_\alpha$ the following two axioms are satisfied:

1. If $s, t \in \mathcal{F}(U)$ satisfy that $r_{U_\alpha}^U(s) = r_{U_\alpha}^U(t)$ for all $\alpha \in A$ then $s = t$.
2. Suppose there exists $s_\alpha \in \mathcal{F}(U_\alpha)$ for all $\alpha \in A$ satisfying that if $\beta \in A$ and $U_\alpha \cap U_\beta \neq \emptyset$ then $r_{U_\alpha \cap U_\beta}^{U_\alpha}(s_\alpha) = r_{U_\alpha \cap U_\beta}^{U_\beta}(s_\beta)$. Then there exists an $s \in \mathcal{F}(U)$ such that $r_{U_\alpha}^U(s) = s_\alpha$ for all $\alpha \in A$.

Hopefully it is clear why all three examples of presheaves I just gave are also sheaves. The first bullet point is satisfied because because functions are uniquely determined by their values at each point. Meanwhile, the second bullet point is satisfied because we can glue compatible continuous, smooth, or holomorphic functions together.

7/6/2026

Augh I wanted to read the book by Wells but I've been having immense difficulty. I think my two gaps in knowledge are that I still don't have enough knowledge about manifolds, and also that I don't know any complex analysis on multivariable complex functions. Since I don't have a textbook covering the latter yet, I'll just read John Lee's *Introduction to Smooth Manifolds*. As a reminder, $H^n := \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_n \geq 0\}$.

Firstly, I want to generalize the notes about tangent planes on *pages 926 and 927* to abstract differentiable manifolds.

Let M be a C^∞ manifold with or without boundary and consider any $p \in M$. A linear map $v : C^\infty(M, \mathbb{R}) \to \mathbb{R}$ is a derivation at p if

$$v(fg) = f(p)v(g) + g(p)v(f) \text{ for all } f, g \in C^\infty(M, \mathbb{R}).$$

It will be clear later why we require our functions to be C^∞ as opposed to just C^1.

Also, we let T_pM be the set of all derivations at p. Importantly, we can see that T_pM is a real vector space. So, we say an element of T_pM is a tangent vector at p.

Lemma: Suppose $v \in T_pM$.

- If $f \equiv C$ where $C \in \mathbb{R}$ then $v(f) = 0$.

  Why?
  $v(f) = \frac{1}{c}v(cf) = \frac{1}{c}v(f^2) = \frac{1}{c}(2f(p)v(f)) = 2v(f)$. By subtracting $v(f)$ from both sides we then get that $0 = v(f)$.

- If $f, g \in C^\infty(M, \mathbb{R})$, $f(p) = 0$, and $g(p) = 0$ then $v(fg) = 0$.

  Hopefully this is obvious.

I'd like to give some intuition on why derivations should be called "tangent vectors". For now, suppose we have a k-dimensional C^r manifold $M \subseteq \mathbb{R}^n$ (where $r \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$), and for the sake of convenience assume M has no boundary. (Here I'm taking M to be a manifold as was studied on *pages 77-99* as opposed to a general abstract manifold as covered by John Lee. As a side note, M would also be an abstract manifold and its C^r structure would be generated by all full rank bijective C^r maps from open subsets of $\mathbb{R}^k$ to open subsets of M which also have continuous inverses.)

Next, consider any $p \in M$ and let ϕ be a C^r diffeomorphism from an open set U of $\mathbb{R}^k$ to an open set V' of M containing p. Also let $q = \phi^{-1}(p)$. Then recalling *pages 926 and 927*, we define the "geometric tangent plane" based at p as:

$$\widetilde{T_pM} := \{(p, \vec{y}) : \vec{y} \in \text{im}(\text{D}\phi(q) : \mathbb{R}^k \to \mathbb{R}^n)\}$$

Importantly, $\widetilde{T_pM}$ is an k-dimensional vector space. Also suppose that $\vec{v}_p = (p, \vec{v}) \in \widetilde{T_pM}$.

I realize I have two diffent definitions for what it means for a function $f : M \to \mathbb{R}$ to be a C^s function (where $s \leq r$). So I want to quickly rectify that.

- Definition 1: $f : M \to \mathbb{R}$ is a C^s map if there is an open set $W \subseteq \mathbb{R}^n$ containing M and a C^s function $g : W \to \mathbb{R}$ with $g|_M = f$.
- Definition 2: f is a C^s function in the manifold sense. Equivalently, this means for any open sets $U \subseteq \mathbb{R}^k$ and $V \subseteq M$ as well as any bijective C^∞ function $\varphi : U \to V$ such that $\mathrm{D}\varphi(x)$ has full rank for all $x \in U$ and φ^{-1} is continuous, we have that $f \circ \varphi$ is a C^s map. (Note: this is equivalent to the manifold definition because the φ form a C^r subatlas of the C^r structure of M).

  $(1 \implies 2)$
  This is hopefully obvious.

  $(2 \implies 1)$
  Recall from lemma 23.1 on *page 83* that by passing to a partition of unity, it suffices to show definition 1 only locally. In other words, we must show for any $p \in M$ that there exists a C^s function $g : W \to \mathbb{R}$ for some open set $W \subseteq \mathbb{R}^n$ containing p such that $g|_{M \cap V} = f|_{M \cap V}$.

  Fortunately, we can find open sets $U \subseteq \mathbb{R}^k$ and $V \subseteq M$ and a homeomorphism $\varphi : U \to V$ such that $p \in V$, φ is a C^r function, and $\mathrm{D}\varphi(x)$ has full rank for all $x \in U$. Next, note from theorem 24.1 on *page 87* that φ^{-1} can be extended to a C^r function $\psi : W \to \mathbb{R}^k$ such that $W \subseteq \mathbb{R}^n$ is open and $\psi|_V = \varphi^{-1}$. But now as ψ is continuous and $\psi(V) = \varphi^{-1}(V) = U$, we know that $\psi^{-1}(U)$ is an open subset of W containing V. Plus, $p \in V \subseteq \psi^{-1}(U)$. In other words, we can without loss of generality shrink W so that $\psi(W) \subseteq U$ and still have that $p \in W$. And finally, $(f \circ \varphi) \circ \psi = f$ on M and $(f \circ \varphi) \circ \psi$ is a composition of a C^s map and a C^r map on W. Hence, we've proven that definition 1 holds locally for f.

Next, suppose $f \in C^1(M, \mathbb{R})$. We define its directional derivative $\mathrm{D}_{\vec{v}}|_p f$ in the direction of $\vec{v}$ as follows:

Let $\gamma : [-1, 1] \to M$ be any C^1-path such that $\gamma(0) = p$ and $\gamma'(0) = \vec{v}$.
As for how we know such a path γ exists, recall that $\vec{v} \in \mathrm{D}\phi(q)$. Hence, there exists $w \in \mathbb{R}^k$ such that $(\mathrm{D}\phi(q))\vec{w} = \vec{v}$. Let $\widetilde{\gamma} : [-1, 1] \to U$ be a C^1-path such that $\widetilde{\gamma}(0) = q$ and $\widetilde{\gamma}'(0) = \vec{w}$. Then let $\gamma := \phi \circ \widetilde{\gamma}$. Clearly $\gamma(0) = p$ and $\gamma([-1, 1])$ is contained in M. Also by chain rule we have that:

$$\gamma'(0) = (\mathrm{D}\phi(q))\widetilde{\gamma}'(0) = (\mathrm{D}\phi(q))\vec{w} = \vec{v}.$$

Now we define $\mathrm{D}_{\vec{v}}|_p f = (f \circ \gamma)'(0)$. I claim this definition is independent of the specific path γ chosen. After all:
Suppose γ_1 and γ_2 are two distinct C^1-paths such that $\gamma_i(0) = p$ and $\gamma_i'(0) = \vec{v}$ for both i. Then after extending f to a C^r function on an open set $V \subseteq \mathbb{R}^n$ with $V' = V \cap M$ (which we can do since $f \in C^r(M, \mathbb{R})$), we know that:

$$(f \circ \gamma_1)'(0) = \nabla f(p) \cdot \gamma_1'(0) = \nabla f(p) \cdot \vec{v} = \nabla f(p) \cdot \gamma_2'(0) = (f \circ \gamma_2)'(0).$$

It's also easy to see when M is a C^∞ manifold that the directional derivative $\mathrm{D}_{\vec{v}}|_p$ restricted to C^∞ functions is a derivation at p. After all, we can just apply the product rule to the gradiant operator and also note that taking a dot product against a fixed vector $\vec{v}$ defines

a linear functional.

Now supposing that M is a C^∞ manifold, we can consider the function $\widetilde{T_pM}$ to T_pM given by $(p, \vec{v}) \mapsto \mathrm{D}_{\vec{v}}|_p$. We can see from the fact that taking a dot product against a fixed vector $\nabla f(p)$ defines a linear functional that our mapping $(p, \vec{v}) \mapsto \mathrm{D}_{\vec{v}}|_p$ is actually a linear map between vector spaces.

Furthermore, I claim that our linear map is injective. After all, suppose $(p, \vec{v})$ maps to the zero derivation (i.e. the derivation mapping every C^∞ function to 0). Then let $(p, u_1), \ldots, (p, u_k)$ be an orthonormal $\mathbb{R}$-basis for the vector space $\widetilde{T_pM}$. We can uniquely express $\vec{v} = c_1u_1 + \ldots + c_ku_k$ where $c_1, \ldots, c_k \in \mathbb{R}$. And now for any $f \in C^\infty(M, \mathbb{R})$ we have that:

$$\mathrm{D}_{\vec{v}}|_p f = \sum_{j=1}^k c_j \mathrm{D}_{u_j}|_p f$$

To finish off, we can define $f_i \in C^\infty(M, \mathbb{R})$ by $f_i(x) = x \cdot u_i$ for all $x \in M \subseteq \mathbb{R}^n$. This is a C^∞ map because it more strongly extends to a smooth map on all of $\mathbb{R}^n$. Furthermore, note that $\nabla f_i(p) = u_i$. So, $\mathrm{D}_{u_j}|_p f = u_i \cdot u_j$. And this equals 1 if $i = j$ and 0 otherwise. So, we can conclude for all i that:

$$0 = \mathrm{D}_{\vec{v}}|_p f_i = \sum_{j=1}^k c_j \mathrm{D}_{u_j}|_p f_i = c_i$$

In other words $\vec{v} = 0u_1 + \ldots + 0u_k = 0$. And with that we've proven that $(p, \vec{v}) \mapsto \mathrm{D}_{\vec{v}}|_p$ has a trivial kernel.

Now to finish off, note that $\widetilde{T_pM}$ is k-dimensional. We'll also soon show for a general abstract C^∞ k-dimensional manifold that T_pM is k-dimensional. Hence, our injective linear map must also be surjective. And thus, we've proven that our linear map $\widetilde{T_pM}$ to T_pM given by $(p, \vec{v}) \mapsto \mathrm{D}_{\vec{v}}|_p$ is a bijection.

The T.L.D.R. to all of this is that you should think of a tangent plane as being isomorphic to a vector space of directional derivative operators. Also, derivations are a generalization of directional derivatives.

Suppose M and N are C^∞ manifolds with or without boundary, and $F : M \to N$ is a C^∞ map. We define a map $\mathrm{d}F_p : T_pM \to F_{F(p)}N$ called the differential of F at p such that $\mathrm{d}F_p(v)f \coloneqq v(f \circ F)$ for all $v \in T_pM$ and $f \in C^\infty(N, \mathbb{R})$.

To see that this map is well-defined, suppose $v \in T_pM$ and let $f, g \in C^\infty(N, \mathbb{R})$. Also suppose $c \in \mathbb{R}$. Then $\mathrm{d}F_p(v)(cf) = v(cf \circ F) = cv(f \circ F) = c\mathrm{d}F_p(v)f$. Similarly,

$$\begin{aligned}\mathrm{d}F_p(v)(f+g) &= v((f+g) \circ F) \\ &= v((f \circ F) + (g \circ F)) = v((f \circ F)) + v((g \circ F)) \\ &= \mathrm{d}F_p(v)f + \mathrm{d}F_p(v)g\end{aligned}$$

Thus, we've shown that $\mathrm{d}F_p(v)$ is a linear map on $C^\infty(N, \mathbb{R})$. As for additionally showing that $\mathrm{d}F_p(v)$ is a derivation at $F(p)$, note that:

$$\begin{aligned} \mathrm{d}F_p(v)(fg) &= v((fg)\circ F) \\ &= v((f\circ F)(g\circ F)) = (f\circ F)(p)v(g\circ F) + (g\circ F)(p)v(f\circ F) \\ &= f(F(p))\mathrm{d}F_p(v)g + g(F(p))\mathrm{d}F_p(v)f \end{aligned}$$

Proposition 3.6: Let M, N, and P be C^∞ manifolds with or without boundary. Also let $F : M \to N$ and $G : N \to P$ be C^∞ maps, and let $p \in M$.

(a) $\mathrm{d}F_p : T_pM \to T_{F(p)}N$ is a linear map.

Proof:
Suppose $a, b \in \mathbb{R}$ and $v, w \in T_pM$. Then for any $f \in C^1(N)$ we have that:

$$\begin{aligned} \mathrm{d}F_p(av+bw)f &= (av+bw)(f\circ F) \\ &= av(f\circ F) + bw(f\circ F) = a\mathrm{d}F_p(v)f + b\mathrm{d}F_p(w)f \end{aligned}$$

(b) $\mathrm{d}(G\circ F)_p = \mathrm{d}G_{F(p)}\circ \mathrm{d}F_p$.

Proof:
Let $v \in T_pM$ and $f \in C^1(P, \mathbb{R})$. Then:

$$\mathrm{d}(G\circ F)_p(v)f = v(f\circ G\circ F) = (\mathrm{d}F_p(v))(f\circ G) = (\mathrm{d}G_{F(p)}(\mathrm{d}F_p(v)))f.$$

Thus $\mathrm{d}(G\circ F)_p(v) = \mathrm{d}G_{F(p)}(\mathrm{d}F_p(v))$.

(c) $\mathrm{d}(\mathrm{Id}_M)_p = \mathrm{Id}_{T_pM}$.

Proof:
Let $v \in T_pM$ and $f \in C^1(M, \mathbb{R})$. Then:

$$(\mathrm{d}(\mathrm{Id}_M)_p(v))f = v(f\circ \mathrm{Id}_M) = v(f).$$

As a result, $\mathrm{d}(\mathrm{Id}_M)_p v = v$. In other words, $\mathrm{d}(\mathrm{Id}_M)_p = \mathrm{Id}_{T_pM}$.

(d) If F is a C^∞-diffeomorphism from M to N (i.e. a bijection such that F and F^{-1} are C^∞ maps) then $\mathrm{d}F_p : T_pM \to T_{F(p)}N$ is an isomorphism and $(\mathrm{d}F_p)^{-1} = \mathrm{d}(F^{-1})_{F(p)}$.

Why?
From parts (b) and (c) we know that:

- $\mathrm{Id}_{T_pM} = \mathrm{d}(\mathrm{Id}_M)_p = \mathrm{d}(F^{-1}\circ F)_p = \mathrm{d}(F^{-1})_{F(P)}\circ \mathrm{d}F_p$,
- $\mathrm{Id}_{T_{F(p)}N} = \mathrm{d}(\mathrm{Id}_N)_{F(p)} = \mathrm{d}(F\circ F^{-1})_{F(p)} = \mathrm{d}F_p \circ \mathrm{d}(F^{-1})_{F(p)}$. ■

7/13/2026

In order to get anywhere studying differentials of C^∞ maps, we will want to be able study differentials as being a local property as opposed to a global property. Hence, we shall first prove the following results.

Lemma (Existence of Bump Functions): Suppose (U, φ) is a chart on a C^r manifold M with or without boundary (where $r \in \mathbb{N}\cup\{\infty\}$). Then for any compact set $K \subseteq U$ we can find a compactly supported function $\phi \in C^r(M, [0,1])$ with $\phi \equiv 1$ on K and $\mathrm{supp}(\phi) \subseteq K$.

Proof:
The proposition doesn't specify if $\varphi(U)$ is open in $\mathbb{R}^n$ or in H^n. That said, it will still be the case no matter what that $K' := \varphi(K)$ is compact in $\mathbb{R}^n$. Now if $\varphi(U)$ is open in H^n but not $\mathbb{R}^n$ then let V' be an open set in $\mathbb{R}^n$ with $V' \cap H^n = \varphi(U)$. Otherwise just set $V' = U$.

By the C^∞-Urysohn lemma for functions on $\mathbb{R}^n$, we can find a compactly supported smooth function $f : \mathbb{R}^n \to [0,1]$ such that $\text{supp}(f) \subseteq V'$ and $f \equiv 1$ on K'. And then if we define $\widetilde{f} : U \to \mathbb{R}$ by $f \circ \varphi$, we have that $\widetilde{f}$ is a C^r function, $\widetilde{f} \equiv 1$ on K, and $\text{supp}(\widetilde{f})$ is a compact subset of U.

Next define $\widetilde{g} : M - \text{supp}(f)$ to be the constant zero function. This will be a C^r map.

Finally, by the gluing lemma (on *page 150*) (since $\widetilde{f}$ and $\widetilde{g}$ agree on their shared domains), we know there is a unique C^r map $\phi : M \to [0,1]$ such that $\phi|_U = \widetilde{f}$ (and thus $\widetilde{f} \equiv 1$ on K), and also $\text{supp}(\phi) = \text{supp}(\widetilde{f}) \subseteq U$. ■

Proposition (Existence of Partitions of Unity): Suppose M is a C^r manifold with or without boundary. Then for any open cover $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ of M there is partition of unity $\{\phi_\alpha\}_{\alpha \in A}$ specifically satisfying that $\text{supp}(\phi_\alpha) \subseteq U_\alpha$ and $\phi_\alpha \in C^r(M, [0,1])$ for all α.

Proof:
For every $p \in M$ let $(V_p, \varphi_p)_{p \in M}$ be a smooth chart containing p such that $V_p \subseteq U_\alpha$ for some α. Then $\mathcal{V} := \{V_p\}_{p \in M}$ is a refinement of $\mathcal{U}$. Next, by the exercise on *page 138*, we can find two locally finite refinements $\mathcal{W} := \{W_m\}_{m \in \mathbb{N}}$ and $\mathcal{X} := \{X_m\}_{m \in \mathbb{N}}$ of $\mathcal{V}$ such that $\overline{W_m}$ is compact and $\overline{X_m} \subseteq W_m$ for all m. And then by the prior lemma, we can pick a compactly supported function $f_m \in C^r(M, [0,1])$ with $f_m \equiv 1$ on $\overline{X_m}$ and $\text{supp}(f_m) \subseteq W_m \subseteq V_p$ for some $p \in M$.

If we set $f = \sum_{m=1}^\infty f_m$, we have that f is finite and a C^r map on all of M because each $p \in M$ has a neighborhood which intercepts only finitely many W_m (and hence locally f is only a sum of finitely many C^r functions). Furthermore, we know $f(p) \geq 1$ for all $p \in M$ since every p is contained in some X_m. Therefore, we can set $\psi_m = f_m / f$. Again, it is easily checked that since both f_m and f are C^r maps and f is nowhere zero, ψ_m is also a C^r map. Also, $\psi_m(M) \subseteq [0,1]$ and $\text{supp}(\psi_m)$ is contained in W_m which in turn is contained in some V_p which is contained in some U_α. Also $\sum_{m \in \mathbb{N}} \psi_m = 1$. And for identical reasons as with the f_m, we know that every $p \in M$ has a neighborhood that intersects only finitely many of the supports of the ψ_m.

To finish off, let $h : \mathbb{N} \to A$ be a function such that $W_m \subseteq U_{h(m)}$ for all $m \in \mathbb{N}$. Then upon taking the convention that $\sum_{m \in \emptyset} \phi_m \equiv 0$, for every $\alpha \in A$ let $\phi_\alpha := \sum_{m \in h^{-1}(\alpha)} \psi_m$. Each ϕ_α will be a C^r map for the same reason f was a C^r map. Furthermore, we will still have that $\sum_{\alpha \in A} \phi_\alpha \equiv 1$ (where every p has a neighborhood on which only finitely many ϕ_α are nonzero). And since all the ϕ_α are nonnegative, we can conclude that $\phi_\alpha(M) \subseteq [0,1]$.

Finally, for every m let $K_m := \text{supp}(\psi_m)$ and note that each K_m is a closed subset of W_m. Also, $\{K_m\}_{m \in \mathbb{N}}$ is a locally finite collection of sets since $\mathcal{W}$ is. Hence by an exercise on *pages 139-140*:

$$\text{supp}(\phi_\alpha) \subseteq \overline{\bigcup_{m \in h^{-1}(\alpha)} K_m} = \bigcup_{m \in h^{-1}(\alpha)} K_m \subseteq \bigcup_{m \in h^{-1}(\alpha)} W_m \subseteq U_\alpha. \blacksquare$$

Corollary (Urysohn's Lemma for C^r manifolds): Let M be a C^r manifold with or without boundary. Then for any disjoint closed sets $A, B \subseteq M$ there exists a function $\phi \in C^r(M, [0,1])$ such that $\phi \equiv 1$ on A and $\phi \equiv 0$ on B.

Proof:
Let $U_0 = M - A$ and $U_1 = M - B$. Because A and B are disjoint closed sets, $\{U_0, U_1\}$ is an open cover of M. So by the prior proposition, we can find two C^r maps ϕ_0, ϕ_1 in $C^r(M, [0,1])$ such that $\phi_0 + \phi_1 \equiv 1$, $\text{supp}(\phi_0) \subseteq U_0$, and $\text{supp}(\phi_1) \subseteq U_1$. But now since $\phi_0 \equiv 0$ on $A = M - U_0$ and $\phi_0 + \phi_1 \equiv 1$, we must have that $\phi_1 \equiv 1$ on A. Also, $\phi_1 \equiv 0$ on $B = M - U_1$. So, ϕ_1 is our desired function. ■

Another way of stating this lemma is that if $A \subseteq M$ is a closed set and $U \subseteq M$ is an open set such that $A \subseteq U$, then there exists $\phi \in C^r(M, [0,1])$ such that $\phi \equiv 1$ on A and $\text{supp}(\phi) \subseteq U$.

Now that we have the ability to construct all the bump functions we need, we can go back to studying the differential of a C^∞ map between C^∞ manifolds.

Proposition 3.8: Let M be a C^∞ manifold with or without boundary, $p \in M$, and $v \in T_pM$. If $f, g \in C^\infty(M, \mathbb{R})$ agree on some neighborhood of p, then $v(f) = v(g)$.

Proof:
Let $h = f - g$ and note that $h \in C^\infty(M, \mathbb{R})$ with $h \equiv 0$ on some neighborhood U of p. Next, by the prior corollary let $\phi : M \to [0,1]$ be a C^∞ map such that $\phi \equiv 1$ on $\text{supp}(h)$ and $\text{supp}(\phi) \subseteq M - \{p\}$. Since $\phi \equiv 1$ everywhere that $h \neq 0$, we know that $h = h\phi$. Hence:

$$v(f) - v(g) = v(h) = v(h\phi) = v(h)\phi(p) + v(\phi)h(p) = 0 + 0 = 0 \text{ ■}$$

Proposition 3.9: Let M be a C^∞ manifold with or without boundary, let $U \subseteq M$ be an open subset, and let $\iota : U \hookrightarrow M$ be the inclusion map. For every $p \in U$ the differential $\mathrm{d}\iota_p : T_pU \to T_pM$ is a vector space isomorphism.

Proof:
First we show injectivity. Suppose $v \in T_pU$ satisfies that $\mathrm{d}\iota_p(v) = 0 \in T_pM$. Then let V be a precompact open set such that $p \in V \subseteq \overline{V} \subseteq U$. By Urysohn's lemma for C^∞ manifolds, we know there is a C^∞ function $\phi : M \to [0,1]$ such that $\phi \equiv 1$ on $\overline{V}$ and $\text{supp}(\phi) \subseteq U$.

Now suppose $f \in C^\infty(U, \mathbb{R})$. We can define a function $\widetilde{f} \in C^\infty(M, \mathbb{R})$ by setting $\widetilde{f} := \phi f$ on U and $\widetilde{f} = 0$ outside of U. Note that $\widetilde{f}$ is a C^∞ map by the gluing lemma (see *page 150*). But also by our prior proposition we know that:

$$v(f) = v(\widetilde{f}|_U) = v(\widetilde{f} \circ \iota) = \mathrm{d}\iota_p(v)\widetilde{f} = 0.$$

So, we've proven that $v(f) = 0$. And since f was arbitrary, we've proven that v is the zero derivation at p. This finishes proves that $\mathrm{d}\iota_p$ has a trivial kernel and is thus injective.

Next we show surjectivity. Suppose $w \in T_pM$ is arbitrary. Then we define $v \in T_pU$ by $v(f) = w(\widetilde{f})$ where $\widetilde{f} \in C^\infty(M, \mathbb{R})$ is any function such that $f = \widetilde{f}$ on the neighborhood V of p from before.

Note that such a function $\widetilde{f}$ exists by the reasoning given before. Also, v is a well-defined function from $C^\infty(U,\mathbb{R})$ to $C^\infty(M,\mathbb{R})$ simply due to proposition 3.8. Also if $\widetilde{f}$ and $\widetilde{g}$ are C^∞ maps on M agreeing with f and g respectively on V, then $\widetilde{f}\widetilde{g}$ agrees with fg on V and $\widetilde{f}+\widetilde{g}$ agrees with $f+g$ on V. Thus, it's straightforward to also see that v is a derivation.

Now I claim that $\mathrm{d}\iota_p(v) = w$. After all, let f be any function in $C^\infty(M,\mathbb{R})$. Then since f agrees with $f|_U = f\circ\iota$ on V, we have that $\mathrm{d}\iota_p(v)f = v(f\circ\iota) = w(f)$. ■

We now have two different cases where we know the differential of a function is a vector space isomorphism. Those two cases are when we are working with diffeomorphisms or with inclusion maps. Yet even just these two cases are useful. After all, suppose M is a C^∞ manifold, $p\in M$ and (U,φ) is a smooth chart containing p.

Lemma: Suppose M is a C^r manifold and (U,φ) is a smooth chart on M. Let $\widehat{U} = \varphi(U)$ and equip $\widehat{U}$ with the standard C^r structure. Then $\varphi: U\to\widehat{U}$ is a C^r-diffeomorphism.

Why?

Obviously φ is a bijection. Also note that φ is a C^r map since the chart (U,φ) trivially covers all of U and also $\varphi\circ\varphi^{-1} = \mathrm{Id}_{\widehat{U}}$ is trivially a smooth function. Similarly, φ^{-1} is a C^r function since $\varphi\circ\varphi^{-1}\circ\mathrm{Id}_{\widehat{U}} = \mathrm{Id}_{\widehat{U}}$ is trivially a smooth map. ■

As a result if $\iota: U\hookrightarrow M$ is the inclusion map, we have the following vector space isomorphisms:

$$T_pM \xleftarrow[\mathrm{d}\iota_p]{\sim} T_pU \xleftarrow[\mathrm{d}\varphi_p]{\sim} T_{\varphi(p)}\widehat{U}$$

This is especially nice because $T_{\varphi(p)}\widehat{U}$ is easy to describe.

First we describe the tangent space of $\mathbb{R}^n$. Note that $\mathbb{R}^n$ is a parametrized C^∞ manifold via the chart $(\mathbb{R}^n,\mathrm{Id})$. Therefore, we can apply the observations on *pages 1028-1030*. In particular, note that $\mathrm{D}(\mathrm{Id})(a):\mathbb{R}^n\to\mathbb{R}^n$ has full rank and hence the "geometric tangent plane" based at any $a\in\mathbb{R}^n$ is just $\widetilde{T_a\mathbb{R}^n} := \{(a,\vec{y}) : \vec{y}\in\mathbb{R}^n\}$. Furthermore, we have an injective linear map $\widetilde{T_a\mathbb{R}^n}$ to $T_a\mathbb{R}^n$ given by $(a,\vec{y})\mapsto \mathrm{D}_{\vec{y}}|_p$.

Lemma 3.2: The map described above is surjective.

Proof:

Let $w\in T_a\mathbb{R}^n$ be any derivation. Then having let $e_1,\ldots,e_n$ be the standard basis, for all $1\le i\le n$ let $c_i := w(x\mapsto x\cdot e_i)$. I claim $w = \mathrm{D}_v|_a$ where $v = c_1e_1+\ldots+c_ne_n$.

To see why, let f be any C^∞ function on $\mathbb{R}^n$. Next for any $x \in \mathbb{R}^n$ let $h(t) \coloneqq f(a + t(x-a))$. If we say $a = (a_1, \ldots, a_n)$ and $x = (x_1, \ldots, x_n)$, then note that:

$$\begin{aligned} f(x) - f(a) = h(1) - h(0) &= \int_0^1 \frac{\mathrm{d}h}{\mathrm{d}t}(s)\mathrm{d}s \\ &= \int_0^1 (x-a) \cdot \nabla f(a + s(x-a))\mathrm{d}s \\ &= \sum_{i=1}^n (x_i - a_i) \int_0^1 \frac{\partial}{\partial x_i} f(a + s(x-a))\mathrm{d}s. \end{aligned}$$

In particular, we set $h_i(x) \coloneqq \int_0^1 \frac{\partial}{\partial x_i} f(a + s(x-a))\mathrm{d}s$. By Leibniz's rule applied h_i (which we can apply to x in bounded subsets of $\mathbb{R}^n$ since f is a C^∞ function), we know h_i is also a C^∞ function on $\mathbb{R}^n$.

As a side note, if f were only finitely many times differentiable then this paragraph would not work. This hints at how we need an alternate definition of tangent vectors for when we are working on C^r manifolds where $r < \infty$.

And now we have that $w(f) = w\left(f(a) + \sum_{i=1}^n (x_i - a_i) g_i(x)\right)$. Yet $w(f(a)) = 0$ since $f(a)$ is a constant. Meanwhile:

$$\begin{aligned} w((x_i - a_i)g_i(x)) &= (a_i - a_i)w(g_i) + g_i(a)w(x_i - a_i) \\ &= 0 + (w(x \mapsto x \cdot e_i) - w(a_i)) \int_0^1 \frac{\partial}{\partial x_i} f(a)\mathrm{d}s = (c_i - 0)\frac{\partial}{\partial x_i} f(a). \end{aligned}$$

So, $w(f) = \sum_{i=1}^n c_i \frac{\partial}{\partial x_i} f(a) = D_v|_a f$. ■

As a side note, if we consider "derivations" whose domain is the vector space $C^k(M, \mathbb{R})$ where M is a C^r manifold and $k < \min(r+1, \infty)$, then we can ask about "exotic derivations" which aren't described by directional derivatives.

Corollary 3.3: A vector space basis for $T_a\mathbb{R}^n$ is given by the partial derivative operators $\frac{\partial}{\partial x_i}$ at a.

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Lemma 3.11: Suppose $\iota : H^n \hookrightarrow \mathbb{R}^n$ is the inclusion map. (Hopefully it is obvious that this is a C^∞-map.) For any $a \in \partial H^n$ the differential $\mathrm{d}\iota_a : T_a H^n \to T_a \mathbb{R}^n$ is an isomorphism.

Proof:

First I shall show injectivity. Suppose $v \in T_a H^n$ and $\mathrm{d}\iota_a(v) = 0 \in T_a\mathbb{R}^n$. Then consider any smooth function $f : H^n \to \mathbb{R}$. By definition we know that f extends to a smooth function $g : U \to \mathbb{R}$ where $U \subseteq \mathbb{R}^n$ is open and contains H^n. Furthermore, H^n is a closed subset of U. Thus by identical reasoning as in the proof of proposition 3.9, we can find a smooth function $\widetilde{f} : \mathbb{R}^n \to \mathbb{R}$ such that $\widetilde{f}|_{H^n} = g|_{H^n} = f$.

But now $v(f) = v(\widetilde{f}|_{H^n}) = v(\widetilde{f} \circ \iota) = \mathrm{d}\iota_a(v)\widetilde{f} = 0$. And since f was arbitrary, we've proven that $v = 0$. So, $\mathrm{d}\iota_a$ has a trivial kernel and is thus injective.

Next we show surjectivity. Suppose $w \in T_a\mathbb{R}^n$ and then define $v \in T_a H^n$ by $v(f) \coloneqq w(\widetilde{f})$ where $\widetilde{f}$ is any smooth extension of f to all of $\mathbb{R}^n$.

To see that v is well-defined, note by the prior corollary that $w = \sum_{i=1}^{n} c_i \frac{\partial}{\partial x_i}|_a$ for some $c_1, \ldots, c_n \in \mathbb{R}$. Yet since $\widetilde{f}$ has a continuous derivative, $\frac{\partial}{\partial x_i}\widetilde{f}(a)$ is uniquely determined by $\widetilde{f}$'s restriction to H^n.

Now its easy to check that v is a derivation and that $\mathrm{d}\iota_a(v) = w$. ■

Proposition 3.12: Given any n-dimensional C^∞ manifold M with or without boundary, for all $p \in M$ we have that T_pM is an n-dimensional vector space.

Proof:
We already know from *page 1034* that T_pM is isomorphic to $T_a\widehat{U}$ where $\widehat{U}$ is some open subset of H^n or $\mathbb{R}^n$ and $a \in \mathbb{R}^n$. But now it's straightforward to prove that $T_a\widehat{U}$ is isomorphic to $T_a\mathbb{R}^n$ by just chaining together the isomorphisms induced by varying inclusion functions. And finally, *lemma 3.2* tells us that $T_a\mathbb{R}^n$ is n-dimensional. Thus, T_pM is also n-dimensional. ■

I want to switch to covering some category theory related notes about manifolds.

Firstly, we let **Man** be the category whose objects are topological manifolds without boundaries and whose morphisms are continuous maps. If we want to allow our manifolds to have a boundary, we instead consider the category $\mathbf{Man}_b$.

Proposition: Suppose $M_1, \ldots, M_{k-1}$ are topological manifolds without boundary and M_k is a topological manifold with boundary. Also suppose each M_i has dimension n_i. Then the topological product space $M := M_1 \times \cdots \times M_k$ is an $n_1 + \ldots + n_k$ dimensional topological manifold with boundary.

Proof:
Countable products of second countable topological spaces are still second countable. Hence M is second countable.

This is easy to verify.

Also arbitrary products of Hausdorff spaces are still Hausdorff. Hence, M is Haudorff.

Finally, to show M is locally Euclidean, let $(p_1, \ldots, p_k) \in M$. Then for each $1 \leq i \leq k$ we can pick a chart (U_i, φ_i) on M_i containing p_i. Now we identify $\mathbb{R}^{n_1} \times \cdots \times \mathbb{R}^{n_k}$ with $\mathbb{R}^{n_1+\ldots+n_k}$ in the obvious way. Then we consider the chart $(U_1 \times \ldots \times U_k, \varphi_1 \times \cdots \times \varphi_k)$. This will be a homeomorphism to an open subset of $\mathbb{R}^{n_1+\ldots+n_k}$ or $H^{n_1+\ldots+n_k}$. ■

Remark: By the proposition on *pages 1010-1011*, we know that $(p_1, \ldots, p_k)$ is a boundary point of M if and only if p_k is a boundary point of M_k. Hence if M_k has an empty boundary then our constructed chart from before will always be an interior chart instead of a boundary chart.

Since $M = M_1 \times \cdots \times M_k$ is also a product in the category of topologies, we know M interacts with morphisms (i.e. continuous maps) in the way that a category theoretic product should. And by the above remark, we've thus proven that **Man** admits finite products.

As for whether $\mathbf{Man}_b$ admits finite products, we can actually assume in the prior proposition that $M_1, \dots, M_{k-1}$ have boundaries. To see why, let:

$$\overline{\mathbb{R}_+}^k := \{(x_1, \dots, x_k) : x_i \geq 0 \text{ for all } i\}.$$

Proposition: $\overline{\mathbb{R}_+}^k$ is homeomorphic to H^k.

Proof:

Firsty, note that $\overline{\mathbb{R}_+}^1 = H^1$. Also $\overline{\mathbb{R}_+}^2$ is homeomorphic to H^2 via the map:

$$\varphi(x, y) := (x^2 - y^2, 2xy).$$

Why?

It's clear that φ is continuous and also that it maps $\overline{\mathbb{R}_+}^2$ into H^2. After all, $x \geq 0$ and $y \geq 0$ implies that $2xy \geq 0$. As for showing the other requirements for φ to be a homeomorphism, I want to explicitely solve for the inverse function φ^{-1}.

Suppose $\varphi(x, y) = (z, w)$.

Suppose $y \neq 0$. Then $w = 2xy \implies x^2 = \frac{w^2}{4y^2}$. In turn, we get that $z = x^2 - y^2 = \frac{w^2}{4y^2} - y^2$. Or in other words, $4y^4 + 4zy^2 - w^2 = 0$. By using the quadratic formula, we can then solve that:

$$y^2 = \frac{-4z \pm \sqrt{16z^2 + 16w^2}}{8} = \frac{-z \pm \sqrt{z^2 + w^2}}{2}$$

We know $y^2 \geq 0$ though. So, we can actually further solve that:

$$y^2 = \frac{-z + \sqrt{z^2 + w^2}}{2}$$

And then since y is nonnegative, we can finally say that $y = \sqrt{\frac{-z + \sqrt{z^2 + w^2}}{2}}$. And this is a well-defined real number because $\sqrt{z^2 + w^2} \geq \max(|-z|, 0)$. Also, since $y \neq 0$ by assumption we can say that:

$$x = \frac{w}{2y} = \frac{w}{2\sqrt{\frac{-z + \sqrt{z^2 + w^2}}{2}}}.$$

So, we define: $\psi : H^2 - \{(z, 0) : z \geq 0\} \to \mathbb{R}^2_+ - \{(x, 0) : x \geq 0\}$ by:

$$\psi(z, w) = \left(\frac{w}{2\sqrt{\frac{-z + \sqrt{z^2 + w^2}}{2}}}, \sqrt{\frac{-z + \sqrt{z^2 + w^2}}{2}} \right)$$

Clearly ψ is a continuous function. And by how we solved for ψ we know that $\psi(\varphi(x, y)) = (x, y)$. Similarly, we can solve that $\varphi(\psi(z, w)) = (z, w)$.

Finally, note that if $y = 0$ then $w = 0$ and $z = x^2$. Therefore, we want to define $\psi(z, 0) := (\sqrt{z}, 0)$ when $z \geq 0$. And once we define this, we know that $\psi = \varphi^{-1}$. That said, we need to show our extension of ψ to the ray $\{(z, 0) : z \geq 0\}$ is still continuous.

To do this, consider any $z \geq 0$ and pick $R > \sqrt{z}$. Then $N := \overline{B_R(0)} \cap \overline{\mathbb{R}_+}^2$ is a compact neighborhood of $\varphi^{-1}(z, 0)$ relative to $\overline{\mathbb{R}_+}^2$. Also, we can see that:

$$\varphi(N) = \varphi(\overline{B_R(0)} \cap \mathbb{R}^2_+) = \overline{B_{R^2}(0)} \cap H^2 =: N'.$$

This is because $|\varphi(x,y)| = |(x,y)|^2$ and we already know from earlier that φ surjectively maps $\overline{\mathbb{R}_+}^2$ into H^2. So, φ is a bijective continuous map from the compact neighborhood N of $\varphi^{-1}(z,0)$ onto a compact neighborhood N' of $(z,0)$. It follows from the compactness of N and the Hausdorffness of N' that φ^{-1} is a continuous map from N' to N. In other words, φ^{-1} is continuous at $(z,0)$.

But now its easy to conclude by induction that $\overline{\mathbb{R}_+}^k$ is homeomorphic to H^k. After all, $\overline{\mathbb{R}_+}^k = \overline{\mathbb{R}_+}^2 \times \overline{\mathbb{R}_+}^{k-2}$ is homeomorphic to $H^2 \times \overline{\mathbb{R}_+}^{k-2}$ by the prior reasoning, and then that is trivially homeomorphic to $\mathbb{R} \times \overline{\mathbb{R}_+}^{k-1}$. And now we just conclude by induction that $\overline{\mathbb{R}_+}^{k-1} \cong H^{k-1}$. So, $\mathbb{R} \times \overline{\mathbb{R}_+}^{k-1} \cong \mathbb{R} \times H^{k-1} \cong H^k$. ■

Corollary: Suppose $M_1, \ldots, M_k$ are topological manifolds with boundary. Also suppose each M_i has dimension n_i. Then the topological product space $M := M_1 \times \cdots \times M_k$ is an $n_1 + \ldots + n_k$ dimensional topological manifold with boundary.

Proof:
Second countability and Hausdorffness is obvious. Hence, we only need to worry about if M is locally Euclidean.

Let $(p_1, \ldots, p_k)$ be any point in M and let (U_i, φ_i) be a chart in M_i containing p_i. By shrinking each U_i, we can assume without loss of generality that $\widehat{U}_i := \varphi(U_i)$ is a rectangle in H^{n_i} whose ends are all open intervals in $\mathbb{R}$ or $[0,\infty]$. Then, we consider the function $\varphi_1 \times \cdots \times \varphi_n : U_1 \times \cdots \times U_k \longrightarrow \widehat{U}_1 \times \cdots \times \widehat{U}_k$. By composing this function with some permutation of our coordinates, we get that there are open sets $V \subseteq \mathbb{R}^{n_1+\ldots+n_k-k}$ and $W \subseteq \overline{\mathbb{R}_+}^k$, as well a homeomorphism $\psi : U \to V \times W$. Finally, transform W into an open set of H^k via a homeomorphism we know exists from the prior proposition. ■

Next, given any $r \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$ we let **Diff**^{r} denote the category whose objects are manifolds without boundaries and that are equipped a C^r-structure. The morphisms of **Diff**^{r} are C^r maps between manifolds. Also if we want to allow our manifolds to have boundaries then we consider the category **Diff**^r_b.

Proposition: Suppose $M_1, \ldots, M_{k-1}$ are C^r manifolds without boundary and M_k is a C^r manifold with boundary. Then there is a canonical product C^r-structure we can equip $M := M_1 \times \cdots \times M_k$ with to turn M into a C^r manifold.

Proof:
Given any smooth charts (U_i, φ_i) on M_i for $1 \leq i \leq k$ let $(U_1 \times \cdots \times U_k, \varphi_1 \times \cdots \times \varphi_k)$ be a smooth chart on M. Hopefully it's clear that these charts cover all of M. Also, suppose $(V_1 \times \cdots \times V_k, \psi_1 \times \cdots \times \psi_k)$ is another smooth chart on M of this form. Then the transition function between our charts would be given by:

$$(\psi_1 \circ \varphi_1^{-1}) \times \cdots \times (\psi_k \circ \varphi_k^{-1}).$$

And since ψ_i and φ_i are C^r-compatible for each i, we know our transition function is a C^r-map. So, $(U_1 \times \cdots \times U_k, \varphi_1 \times \cdots \times \varphi_k)$ and $(V_1 \times \cdots \times V_k, \psi_1 \times \cdots \times \psi_k)$ are C^r-compatible.

Now to finish, just take the maximal atlas containing all the charts we just named. ■

Proposition: Suppose $M_1, \ldots, M_k$ are C^r manifolds and at most one of the M_i has a nonempty boundary. Then let $M := M_1 \times \cdots \times M_k$ and equip M with the product structure.

- The projection map $\pi_i : M \to M_i$ is a C^r map.

  Why?
  Given any $(p_1, \ldots, p_k) \in M$, we can for each j pick smooth charts (U_j, φ_j) on M_j containing p_j. Then upon setting $U := U_1 \times \cdots \times U_k$ and $\varphi := \varphi_1 \times \cdots \times \varphi_k$, we have that (U, φ) is a smooth chart on M and (U_i, φ_i) is a smooth chart on M_i such that $(p_1, \ldots, p_k) \in U$ and $\pi(U) \subseteq U_i$. Finally, let $V_j := \varphi_j(U_j)$ for all j. Then $\varphi_i \circ \pi_i \circ \varphi^{-1}$ the projection map from $V_1 \times \cdots \times V_k$ onto V_i (which is C^∞).

- Suppose N is a C^r-manifold with or without boundary. Then given any C^s maps $F_i : N \to M_i$ (where $s \leq r$), we have that $F : N \to M$ defined by $F(p) = (F_1(p), \ldots, F_k(p))$ is a C^s-map from $N \to M$.

  Why?
  Consider any $p \in N$. Since each F_i is a C^s map, we know for each i that there is a smooth chart $(U^{(i)}, \varphi^{(i)})$ on N containing p as well as a smooth chart (V_i, ψ_i) on M_i such that $F_i(U^{(i)}) \subseteq V_i$ and $\psi_i \circ F \circ (\varphi^{(i)})^{-1}$ is a C^s function. By letting $U := U^{(1)} \cap \cdots \cap U^{(k)}$ and letting φ be any of the $\varphi^{(i)}$ priorly chosen, we thus get a single smooth chart (U, φ) on N such that $F_i(U) \subseteq V_i$ and $\psi_i \circ F_i \circ \varphi^{-1}$ is a C^s map for all i.

  Now let $\psi := \psi_1 \times \cdots \times \psi_k$ and $V := V_1 \times \cdots \times V_k$. Then (V, ψ) is a smooth chart on M such that $F(U) \subseteq V$. Plus $\psi \circ F \circ \varphi^{-1}$ is precisely the map $x \mapsto (\psi_1 \circ F_1 \circ \varphi^{-1}(x), \ldots, \psi_k \circ F_k \circ \varphi^{-1}(x))$. And since all the components of this function are C^s functions, we know the entire thing is a C^s function. So, F is a C^s map from N to M.

  Remark: This also proves that M along with the π_i is a category theoretic product. After all, $F : M \to N$ is the unique function with $\pi_i \circ F = F_i$ for each i when we forget the manifold structures on everything and just look at M, N, and the M_i as sets.

- Conversely, suppose $F : N \to M$ is a C^s map (where $s \leq r$). Then let $F_i := \pi_i \circ F$ for each i. Since each π_i is a C^r map, we know that each F_i is a C^s map. Also $F(p) = (F_1(p), \ldots, F_k(p))$. ■

Corollary: let $(p_1, \ldots, p_k)$ be a fixed point in M. The inclusion map $\iota_i : M_i \to M$ given by $\iota(q) := (p_1, \ldots, p_{i-1}, q, p_{i+1}, \ldots, p_k)$ is a C^r map.
This is because constant maps from M_i to M_j as well as the identity map $\mathrm{Id}_{M_i} : M_i \to M_i$ are all C^r. ■

In particular, the prior two propositions show that $\mathbf{Diff}^r$ admits products over any finite collection of objects, and also that $\mathbf{Diff}^r_b$ admits products over finite collections of objects satisfying special conditions.

In general though, we'd have to consider C^r manifolds with corners in order to take products over arbitrary finite collections of objects in $\mathbf{Diff}_b^r$.

An n-dimensional topological manifold with corners M is defined to be a second countable Hausdorff space such that every $p \in M$ has an open neighborhood U that is homeomorphic to an open subset V of $\mathbb{R}^n$, H^n, or $\overline{\mathbb{R}_+}^n$.

> Since $\overline{\mathbb{R}_+}^n$ is homeomorphic to H^n, we can say that M is an n-dimensional topological manifold with corners if and only if M is an n-dimensional topological manifold with a boundary.

Now suppose $r \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$, U is an open subset of $\overline{\mathbb{R}_+}^n$, and $f \in C^r(U)$. Then we know there is an open set $V \subseteq \mathbb{R}^n$ containing U and a C^r map $g \in C^r(V)$ such that $g|_U = f$. Yet by identical reasoning as on *pages 83-84*, we can then say that $\partial^\alpha g(x)$ is uniquely determined by f for all $x \in \partial\overline{\mathbb{R}_+}^n$ and multi-indices α with $|\alpha| \leq r$. Hence, we can uniquely refer to f as having derivatives on $U \cap \partial\overline{\mathbb{R}_+}^n$.

Consequently from the prior paragraph, we can now talk about a corner chart on M (i.e. a chart (U, φ) with U open in $\overline{\mathbb{R}_+}^n$ but not H^n or $\mathbb{R}^n$) being C^r compatible with another chart on M. And then, we can consider a maximal atlas on M containing some atlas of C^r-compatible charts (some of which potentially may be corner charts). As a side note, the proposition on *pages 142-143* still holds by identical reasoning.

We say a topological manifold with corners M equipped with a maximal C^r atlas (a.k.a a C^r structure) is a C^r-manifold with corners. Also we shall still call the charts in that structure "smooth charts".

Supposing M is a C^r-manifold with corners, we call $p \in M$ a corner point if there exists a smooth corner chart (U, φ) on M containing $p \in U$ such that at least two coordinates of $\varphi(p)$ in U are zero.

7/17/2026

Before doing anything else today, I want to go further in depth into the claim that $\overline{\mathbb{R}_+}^n$ is homeomorphic to H^n.

> Recall from *pages 1037-1038* that $\varphi(x, y) := (x^2 - y^2, 2xy)$ is a homeomorphism between $\overline{\mathbb{R}_+}^2$ and H^2. But then importantly, we also have that $(x, y) \mapsto (x, \frac{1}{2}y)$ is a homeomorphism between H^2 and H^2. Therefore, $\widetilde{\varphi}(x, y) := (x^2 - y^2, xy)$ is also a homeomorphism from $\overline{\mathbb{R}_+}^2$ and H^2. And now by induction we can explicitely conclude that the map ψ defined by:
>
> $$(x_1, \ldots, x_k) \mapsto (x_1^2 - x_2^2, \quad x_1^2 x_2^2 - x_3^2, \quad \ldots, \quad (\textstyle\prod_{i=1}^{n-1} x_i^2) - x_n^2, \quad \prod_{i=1}^{n} x_i)$$
>
> is a homeomorphism from $\overline{\mathbb{R}_+}^n$ to H^n.
>
> One particular thing I want to point out is that $\psi(\partial\overline{\mathbb{R}_+}^n) = \partial H^n$.

One may notice that the homeomorphism I specifically wrote before is also a smooth map from $\overline{\mathbb{R}_+}^n$ into H^n. A simple corollary of the following theorem will let us conclude that the function's inverse won't be differentiable from H^n into $\overline{\mathbb{R}_+}^n$ however.

Theorem: Suppose M is a n-dimensional C^r-manifold with corners and that $p \in M$ is a corner point due to the smooth chart (U, φ). Then for any smooth chart (V, ψ) on M containing p, we have that (V, ψ) is also a corner chart and at least two coordinates of $\psi(p)$ in V are zero.

Proof:

To start off, we can't have that $\psi(V)$ is open in $\mathbb{R}^n$. After all, if not then the function $\varphi \circ \psi^{-1}$ would violate the invariance of domain theorem (see *page 1008*). In a similar vein, we can't have that $\psi(V)$ is open in H^n or $\overline{\mathbb{R}_+}^n$ and also have that $\psi(p)$ is not on the boundary of those sets. After all, we'd otherwise be able to shrink V to an open subset of $\mathbb{R}^n$. So there are exactly two possibilities. Either:

- $\psi(V)$ is open in H^n but not $\mathbb{R}^n$ and $\psi(p) \in \partial H^n$,
- $\psi(V)$ is open in $\overline{\mathbb{R}_+}^n$ but not H^n or $\mathbb{R}^n$ and $\psi(p) \in \partial \overline{\mathbb{R}_+}^n$.

We'll first focus on disproving the first possibility. So, suppose for the sake of contradiction that $\psi(V)$ is open in H^n but not $\mathbb{R}^n$ and that $\psi(p) \in \partial H^n$.

Let $\widehat{U} = \varphi(U)$ and without loss of generality suppose that:

$$\varphi(p) = (0, 0, x_3, \ldots, x_n).$$

Note that we can always permute our coordinates to move two of the zero coordinates of $\varphi(p)$ to the first indices. After all, a coordinate permutation function P is a smooth diffeomorphism of $\overline{\mathbb{R}_+}^n$ to itself. Hence, the chart $(U, P \circ \varphi)$ will be C^r compatible with any other chart (V, ψ) since $P \circ (\varphi \circ \psi^{-1})$ and $(\psi \circ \varphi^{-1}) \circ P^{-1}$ are both C^r maps.

Next let $e_1, \ldots, e_n$ be the standard basis for $\mathbb{R}^n$ and then consider the hyperplane $S \coloneqq \{y \in \mathbb{R}^n : y \cdot e_n = 0\}$. Importantly, we know that $S' \coloneqq S \cap \psi(U \cap V)$ is an open subset of S containing $\psi(p)$. Also let $\alpha = \varphi \circ \psi^{-1}$. Then since α is a C^r function from $\psi(U \cap V)$ to $\varphi(U \cap V)$ with a C^r inverse function, we know that $\mathrm{D}\alpha$ must be an invertible linear map at all $y \in \psi(U \cap V)$. In particular, this means that $T \coloneqq (\mathrm{D}\alpha(\psi(p)))\,(S)$ is an $(n-1)$-dimensional subspace of $\mathbb{R}^n$.

But now consider that there must exist $v = (v_1, \ldots, v_n) \in T$ such that either $v_1 \neq 0$ or $v_2 \neq 0$. After all, if not then T would be contained in an $(n-2)$-dimensional subspace of $\mathbb{R}^n$. By replacing v with $-v$ if needed, we can assume that either $v_1 < 0$ or $v_2 < 0$.

Finally, let $\gamma : (-\varepsilon, \varepsilon) \to S'$ be any C^1 curve such that $\gamma(0) = \psi(p)$ and $(\alpha \circ \gamma)'(0) = (\mathrm{D}\alpha(\psi(p)))\gamma'(0) = v$. But now for small enough $t \in (0, \varepsilon)$, we must have that $(\alpha \circ \gamma)(t) \cdot e_1 < 0$ or $(\alpha \circ \gamma)(t) \cdot e_2 < 0$. And this contradicts that α maps $S' \subseteq \psi(U \cap V)$ into $\varphi(U \cap V) \subseteq \overline{\mathbb{R}_+}^n$.

With that, we can now conclude that $\psi(V)$ is open in $\overline{\mathbb{R}_+}^n$ but not H^n or $\mathbb{R}^n$, and also we must have that $\psi(p) \in \partial \overline{\mathbb{R}_+}^n$. In particular, if $\psi(p) = (y_1, \ldots, y_n)$ that at least one y_i

must equal 0. To finish off, we need to prove another coordinate y_j equals 0 (where $j \neq i$).

Suppose otherwise. Then after permuting the coordinate of ψ we can assume without loss of generality that $\psi(p) = (y_1, \ldots, y_{n-1}, 0)$ where $y_j \neq 0$ for all $1 \leq j \leq n-1$. Yet now we can shrink V such that $p \in V$ still and also $\psi(V)$ is open in H^n.

For example, restrict V to the set $V \cap \psi^{-1}(((0,\infty)^{n-1} \times [0,\infty)) \cap \psi(V))$.

And now we've contradicted the earlier conclusion in our proof. ■

We can define what it means for a map F from a manifold with corners M to another manifold N to be a C^r-map identically to how we defined C^r maps from manifolds without corners. And everything on *pages 146-151* still works via identical reasoning. Furthermore, the proof of the lemma on *pages 1031-1032* still works for manifolds with corners when we just replace every instance of H^n with $\overline{\mathbb{R}_+}^n$. And once we have that lemma, we can construct partitions of unity and prove Urysohn's lemma without even adknowledging that M has corners.

One more related note is that if M is a C^∞ manifold with corners, then for any $p \in M$ we can define derivations at p, the vector space T_pM, and the differental $\mathrm{d}F_p$ of a C^∞ function $F : M \to N$ at p identically to how we priorly did so when M was a manifold without corners.

By analogous reasoning as in lemma 3.11 on *pages 1035-1036*, we can show that if $\iota : \overline{\mathbb{R}_+}^n \hookrightarrow \mathbb{R}^n$ is the inclusion map then $\mathrm{d}\iota_a : T_a\overline{\mathbb{R}_+}^n \to T_a\mathbb{R}^n$ is an isomorphism for all $a \in \partial\overline{\mathbb{R}_+}^n$. And it thus follows that every fact about C^∞ manifolds without corners proven since page 1028 still holds for C^∞ manifolds with corners.

7/18/2026

Firstly, I realized that $(\mathbb{R}_{\geq 0})^n$ is way better notation than $\overline{\mathbb{R}_+}^n$. So I'm going to write the corner space in my way instead of John Lee's way. Anyways, before I get back to progressing through chapter 3 of Lee's book, I want to define one more standard manifold structure.

Suppose V is a finite dimensioned real vector space. Then there is a unique Hausdorff vector space topology on V (see *pages 442-443*). Furthermore, if $f : \mathbb{R}^n \to V$ is any invertible linear map and V is equipped with the unique Hausdorff topology, then f will be a homeomorphism. Therefore, can trivially say that any finite dimensioned real vector space is a topological manifold (without a boundary).

The standard smooth structure we equip V is the one generated by all the charts (V, f) where $f : V \to \mathbb{R}^n$ is a bijective linear map. Importantly, given any bijective linear maps $f, g : V \to \mathbb{R}^n$ we have that $g \circ f^{-1}$ is a bijective linear map from $\mathbb{R}^n$ to $\mathbb{R}^n$ (which is thus smooth). So, these charts do define a C^∞-structure.

Suppose V is an n-dimensional real vector space equipped with the standard smooth structure. Then let $p \in V$. Given any $v \in V$ we define $\mathrm{D}_v|_p : C^\infty(V, \mathbb{R}) \to \mathbb{R}$ by $\mathrm{D}_v|_p f \coloneqq \frac{\mathrm{d}}{\mathrm{d}t}\big|_{t=0} f(p + tv)$.

First I want to show $D_v|p$ is even well-defined. So suppose $f \in C^\infty(V)$ and let $S : \mathbb{R}^n \to V$ be any invertible linear map. Then we must have that $f \circ S \in C^\infty(\mathbb{R}^n)$. Yet also let $a, y \in \mathbb{R}^n$ be such that $S(a) = p$ and $S(y) = v$. Then because:

$$S(a + ty) = S(a) + tS(y) = p + tv,$$

we can say that $f(p + tv) = (f \circ S)(a + ty)$.

It follows that $\mathrm{D}_v|_p f = \nabla(f \circ S)(a) \cdot y = \nabla(f \circ S)(S^{-1}(p)) \cdot S^{-1}(v)$.

And based on the formula I just derived, it's also now clear that $D_v|_p$ is a derivation at p and that the map $V \to T_pV$ given by $v \mapsto D_v|_p$ is a linear map.

As a side note, if $V = \mathbb{R}^n$ then $\mathrm{D}_v|_p$ is just the directional derivative at p in the direction of v.

Lemma: Suppose V is an n-dimensional real vector space and W is an m-dimensional real vector space. Also equip V and W with their standard smooth structure. Then any linear map $T : V \to W$ is a C^∞ map.

Why?

Let $S_V : \mathbb{R}^n \to V$ and $S_W : \mathbb{R}^m \to W$ be bijective linear maps. Then (V, S_V^{-1}) is a smooth chart on V covering all of V, (W, S_W^{-1}) is a smooth chart on W covering all of W, and $S_W^{-1} \circ T \circ S_V$ is a linear map (and thus a C^∞-map) from $\mathbb{R}^n$ to $\mathbb{R}^m$. ■

Corollary: Let V be an n-dimensional real vector space equipped with the standard smooth structure, and also let V^* denote the "algebraic dual space" of V (i.e. the vector space of all linear functionals $\phi : V \to \mathbb{R}$ that are continuous or not continuous. [Although since V is finite-dimensioned, all linear functionals will be continuous...]). Then $V^* \subseteq C^\infty(V, \mathbb{R})$.

Why?

The standard vector space smooth structure on $\mathbb{R}$ as well as the standard smooth structure on $\mathbb{R}$ are both generated by the chart $(\mathbb{R}, \mathrm{Id}_\mathbb{R})$. Hence, both smooth structures are the same and we can in turn apply the prior lemma. ■

Proposition 3.13: Let V be an n-dimensional real vector spaces equipped with the standard smooth structure and suppose $p \in V$. The map $v \mapsto \mathrm{D}_v|_p$ is a vector space isomorphism from V to T_pV.

Proof:

To start off, since V is an n-dimensional manifold, we know that T_pV is also an n-dimensional vector space. Hence, so long as we can prove that the linear map $v \mapsto D_v|_p$ is injective then we must have that $v \mapsto D_v|_p$ is a vector space isomorphism.

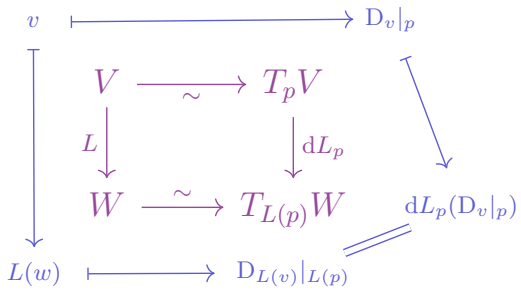
Fortunately, suppose for the sake of contradiction that there exists $v \in V - \{0\}$ such that $\mathrm{D}_v|_p f = 0$ for all $f \in C^\infty(V)$. Then we can pick vectors $v_2, \ldots, v_n \in V$ such that $\mathrm{span}(v, v_2, \ldots, v_n) = V$. Next, there's a well-defined linear functional $f : V \to \mathbb{R}$ given by $f(c_1 v + \sum_{i=2}^n c_i v_i) \coloneqq c_1$. Importantly, $f \in C^\infty(V, \mathbb{R})$ by the prior corollary. Also since $C^\infty(V, \mathbb{R})$ is a real vector space containing all constant functions, we can additionally say that $g \coloneqq f - f(p)$ is a C^∞ map from V to $\mathbb{R}$.

With that, we now have that:

$$0 = \mathrm{D}_v|_p g = \tfrac{\mathrm{d}}{\mathrm{d}t}\big|_{t=0} g(p+tv) = \tfrac{\mathrm{d}}{\mathrm{d}t}\big|_{t=0} f(tv) = \tfrac{\mathrm{d}}{\mathrm{d}t}\big|_{t=0} t = 1.$$

This is a contradiction. So, we can conclude that $\ker(v \mapsto \mathrm{D}_v|_p) = \{0\}$.

Furthermore, suppose W is an m-dimensional real vector space equipped with the standard smooth structure and $L : V \to W$ is a linear map. Then for any $p \in V$ the below diagram commutes:



Proof:
It suffices to show that $\mathrm{d}L_p(\mathrm{D}_v|_p) = \mathrm{D}_{L(v)}|_{L(p)}$. Fortunately, for any $f \in C^\infty(W, \mathbb{R})$ we have that:

$$\begin{aligned}\mathrm{d}L_p(\mathrm{D}_v|_p) f = \mathrm{D}_v|_p (f \circ L) &= \tfrac{\mathrm{d}}{\mathrm{d}t}\big|_{t=0} (f \circ L)(p+tv) \\ &= \tfrac{\mathrm{d}}{\mathrm{d}t}\big|_{t=0} f(L(p) + tL(v)) = \mathrm{D}_{L(v)}|_{L(p)} f. \blacksquare\end{aligned}$$

One may notice that the definition of a derivation on a C^∞ manifold M which was given prior specifically required us to work with maps in $C^\infty(M, \mathbb{R})$ maps. Also, we only defined differentials of C^∞ maps between maps. But what if we want to work with less smooth manifolds?

To start off, I want to recall a comment I made on *page 1035*. Namely, suppose we are working with C^r manifold M (where $r \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$). Then given any $p \in M$ and $k < \min(\infty, r+1)$, what is the space of "derivations at p" mapping $C^k(M, \mathbb{R})$ to $\mathbb{R}$?

I'm only slightly modifying a proof that I read from math stack exchange (see *citation 11* in my bibliography).

Recall that $C^k(M, \mathbb{R})$ is a unital real algebra. So, let us consider the set:

$$\mathfrak{m} := \{f \in C^k(M, \mathbb{R}) : f(p) = 0\}.$$

I claim this is a maximal ideal of $C^k(M, \mathbb{R})$.

To see that this is an ideal, just note that if $f \in \mathfrak{m}$ and $g \in C^k(M, \mathbb{R})$ then $fg \in C^k(M, \mathbb{R})$ with $(fg)(p) = 0 \cdot g(p) = 0$. So, $fg \in \mathfrak{m}$. Similarly, if $f, g \in \mathfrak{m}$ then $f + g \in C^k(M, \mathbb{R})$ with $(f+g)(p) = f(p) + g(p) = 0 + 0 = 0$. So, $f + g \in \mathfrak{m}$.

As for showing that $\mathfrak{m}$ is a maximal ideal, suppose $g \in C^k(M, \mathbb{R}) - \mathfrak{m}$. Then

consider any $f \in C^k(M,\mathbb{R})$. Since $g(p) \neq 0$, we can say that $f - \frac{f(p)}{g(p)}g \in \mathfrak{m}$. In turn, $f \in \mathbb{R}g + \mathfrak{m}$. And since f was arbitrary, we've proven that:

$$C^k(M,\mathbb{R}) = \mathbb{R}g + \mathfrak{m} \subseteq \langle \{g\} \cup \mathfrak{m} \rangle \subseteq C^k(M,\mathbb{R}).$$

Also since g was arbitrary, we've shown that $\mathfrak{m}$ is a maximal ideal (and even a maximal proper vector subspace) of $C^k(M,\mathbb{R})$.

Now $\mathfrak{m}^2$ will be an ideal (and thus also a vector subspace) of $C^k(M,\mathbb{R})$ contained in $\mathfrak{m}$. Hence, we can consider the quotient space $V := \mathfrak{m}/\mathfrak{m}^2$. I claim that the space of "derivations at p" mapping $C^k(M,\mathbb{R})$ to $\mathbb{R}$ is isomorphic to the algebraic dual space V^*.

Proof:
Firstly suppose $\varphi \in V^* = (\mathfrak{m}/\mathfrak{m}^2)^*$. Equivalently, we can view φ as linear functional on $\mathfrak{m}$ such that $\varphi(\mathfrak{m}^2) = \{0\}$. Next, we know from before that $C^k(M,\mathbb{R}) = \{\text{constant functions}\} \oplus \mathfrak{m}$. So, we can uniquely extend φ to being a linear functional on $C^k(M,\mathbb{R})$ by just setting $\varphi(\text{constant function}) := 0$. For convenience, I'll give this extended linear functional the name D_φ.

I claim D_φ is a derivation at p. After all, we already know its a linear map from $C^k(M,\mathbb{R})$ to $\mathbb{R}$. Furthermore, suppose $f, g \in C^k(M,\mathbb{R})$. Then we know that $\widetilde{f} := f - f(p)$ and $\widetilde{g} := g - g(p)$ are both in $\mathfrak{m}$. Also:

$$\begin{aligned}\mathrm{D}_\varphi(fg) &= \mathrm{D}_\varphi((\widetilde{f} + f(p))(\widetilde{g} + g(p))) \\ &= \mathrm{D}_\varphi(\widetilde{f}\widetilde{g}) + f(p)\mathrm{D}_\varphi(g - g(p)) + g(p)\mathrm{D}_\varphi(f - f(p)) + \mathrm{D}_\varphi(f(p)g(p)).\end{aligned}$$

But now as $\mathrm{D}_\varphi(\mathfrak{m}^2) = \{0\}$, we know that $\mathrm{D}_\varphi(\widetilde{f}\widetilde{g}) = 0$. Also, $\mathrm{D}_\varphi(f(p)g(p)) = 0$, $\mathrm{D}_\varphi(g - g(p)) = \mathrm{D}_\varphi(g)$, and $\mathrm{D}_\varphi(f - f(p)) = \mathrm{D}_\varphi(f)$ since D_φ is a linear map sending constant functions to zero. Hence, we can conclude that:

$$\mathrm{D}_\varphi(fg) = f(p)\mathrm{D}_\varphi(g) + g(p)\mathrm{D}_\varphi(f).$$

Now after letting $\mathcal{D}_pM$ denote the space of derivations at p mapping $C^k(M,\mathbb{R})$ to $\mathbb{R}$ define the map $\Phi : V^* \to \mathcal{D}_pM$ by $\varphi \mapsto \mathrm{D}_\varphi$. It is obvious from the construction of D_φ that Φ is a linear map. Meanwhile, to show that Φ is bijective we shall note that Φ has an incredibly simple inverse.

Suppose $w \in \mathcal{D}_pM$ and consider $w|_{\mathfrak{m}}$. Since w is a derivation, we know from the lemma on *page 1028* that $w(\mathfrak{m}^2) = \{0\}$. [Technically that lemma assumes all the functions being input into the derivation are C^∞. That said, nothing in the proof of that lemma actually requires any smoothness conditions.] It follows that $w|_{\mathfrak{m}}$ quotients to a well defined map $\Psi(w) \in (\mathfrak{m}^2/\mathfrak{m})^*$. Also, we can easily see that $\Psi = \Phi^{-1}$. ■

You may note that none of our reasoning in this section so far has required that k is finite. However, a fact one can prove is that if k is finite then V will be infinite dimensioned. In turn, we'll also have that V^* is infinite dimensioned. I'm tired, its almost 1am, and proving this fact isn't the most insanely interesting thing ever. So, I'm going to skip showing this. But the T.L.D.R. is that if $k < \infty$ then there are way more derivations from $C^k(M,\mathbb{R})$ into $\mathbb{R}$ then there should be if derivations are purely an abstraction of directional derivatives. And we don't want this.

7/19/2026

If our goal is to have the tangent space to a C^r-manifold M (where r is potentially finite) only contain directional derivatives, then we can try to formalize a tangent space in the following different way from how we were doing so before. (Also for this section I'll be assuming by default that our manifolds can have corners and a boundary.)

Pick a point $p \in M$. Next, we consider any C^1 path $\gamma : J \to M$ where $J \subseteq \mathbb{R}$ is a closed interval containing 0 and having a nonempty interior, and $\gamma(0) = p$.

As a side note, I'm allowing 0 to be either endpoint of J.

Furthermore, as J is a 1-dimensional manifold (with a boundary) it does make sense to talk about γ as being C^1 or not. In fact, given any smooth chart (U, φ) on M, we can say that γ is a C^1 map if and only if γ is continuous and $\varphi \circ \gamma$ is a C^1 map from $\gamma^{-1}(U) \subseteq \mathbb{R}$.

We shall consider the set $\mathcal{C}(M, p)$ of all such C^1-paths γ described priorly. And then, we define an equivalence relation on $\mathcal{C}(M, p)$. Namely, we say $\gamma_1 \sim \gamma_2$ iff:

$$(f \circ \gamma_1)'(0) = (f \circ \gamma_2)'(0) \text{ for all } f \in C^1(M, \mathbb{R}).$$

The fact that this is an equivalence relation is trivial to show. Hence, we can consider the space $\mathcal{V}_p M$ of equivalence classes of $\mathcal{C}(M, p)$. Plus, if $\gamma \in \mathcal{C}(M, p)$ then we consider its equivalence class in $\mathcal{V}_p M$ to be $[\gamma]$.

Lemma: Suppose $F : M \to N$ is a C^1 map and $\gamma_1 : J_1 \to M$ and $\gamma_2 : J_2 \to M$ are paths in $\mathcal{C}(M, p)$ such that $[\gamma_1] = [\gamma_2]$.

(a) We know $(F \circ \gamma_i) : J_i \to N$ is a path in $\mathcal{C}(N, F(p))$.

Hopefully this is obvious. After all, $F \circ \gamma_i$ is a composition of C^1 maps and is thus a C^1 map itself. Also $(F \circ \gamma_i)(0) = F(p)$.

(b) Furthermore, $[F \circ \gamma_1] = [F \circ \gamma_2]$ in $\mathcal{V}_{F(p)} N$.

Why?

Suppose $g \in C^1(N, \mathbb{R})$. Then $g \circ F \in C^1(M, \mathbb{R})$. Hence, we know that:

$$((g \circ F) \circ \gamma_1)'(0) = ((g \circ F) \circ \gamma_2)'(0). \blacksquare$$

Based on this, we'll define the differential at p of a C^1 map $F : M \to N$ to be the function $\mathrm{d}F_p : \mathcal{V}_p M \to \mathcal{V}_{F(p)} N$ given by $\mathrm{d}F_p[\gamma] := [F \circ \gamma]$.

(Ignore the conflict in notation now. I'll deal with it later...)

Proposition: Let M, N, P be C^r manifolds, let $F : M \to N$ and $G : N \to P$ be C^1 maps, and let $p \in M$.

(a) $\mathrm{d}(G \circ F)_p = \mathrm{d}G_{F(p)} \circ \mathrm{d}F_p$.

Why?

$\mathrm{d}(G \circ F)_p([\gamma]) = [G \circ F \circ \gamma] = \mathrm{d}G_{F(p)}([F \circ \gamma]) = \mathrm{d}G_{F(p)}(\mathrm{d}F_p([\gamma]))$.

(b) $\mathrm{d}(\mathrm{Id}_M)_p = \mathrm{Id}_{\mathcal{V}_p M}$.

Hopefully this is obvious from the fact $[\mathrm{Id}_M \circ \gamma] = [\gamma]$.

(c) If $F : M \to N$ is a C^1-diffeomorphism then $\mathrm{d}F_p$ is a bijection and $(\mathrm{d}F_p)^{-1} = \mathrm{d}(F^{-1})_{F(p)}$.

To see why, just combine parts (a) and (b). ■.

Now a downside of viewing the tangent space $\mathcal{V}_p M$ as a collection of equivalence classes of curves is that the vector space structure on $\mathcal{V}_p M$ is not at all obvious. So my next goal will be to make that structure clear.

Lemma: Let M be an n-dimensional C^r manifold, let $p \in M$, and let (U, φ) be any smooth chart on M containing p. Then $\gamma_1, \gamma_2 \in \mathcal{C}(M, p)$ satisfy that $[\gamma_1] = [\gamma_2]$ if and only if $(\varphi \circ \gamma_1)'(0) = (\varphi \circ \gamma_2)'(0)$.

Proof:

Note for any $f \in C^1(M, \mathbb{R})$ that:

$$(f \circ \gamma_i)'(0) = \nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot (\varphi \circ \gamma_i)'(0)$$

Since $\nabla(f \circ \varphi^{-1})(\varphi(p))$ depends only on f and not on γ_i, it's now clear that if $(\varphi \circ \gamma_1)'(0) = (\varphi \circ \gamma_2)'(0)$ then we'll also have that $(f \circ \gamma_1)'(0) = (f \circ \gamma_2)'(0)$.

Conversely, suppose $v_1 := (\varphi \circ \gamma_1)'(0) \neq (\varphi \circ \gamma_2)'(0) =: v_2$. Then it suffices to find some $f \in C^1(M, \mathbb{R})$ such that:

$$\nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot v_1 \neq \nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot v_2$$

Fortunately, because $v_1 \neq v_2$ in $\mathbb{R}^n$, we know there is some $w \in \mathbb{R}^n$ with $w \cdot v_1 \neq w \cdot v_2$. Next we can define $g : \mathbb{R}^n \to \mathbb{R}$ by $g(x) = x \cdot w$. Clearly g is a C^∞ function on $\varphi(U)$ with $\nabla g(\varphi(p)) = w$. If we then further set $\widetilde{g} := g \circ \varphi$, we have that $\widetilde{g} \in C^r(M, \mathbb{R})$.

By standard arguments on LCH spaces as well as with bump functions on C^r manifolds, we can find a precompact open set $V \subseteq M$ with $p \in V \subseteq \overline{V} \subseteq U$ as well as a function $f \in C^r(M, \mathbb{R}) \subseteq C^1(M, \mathbb{R})$ such that $f = \widetilde{g}$ on $\overline{V}$. And at last, we have on a neighborhood of $\varphi(p)$ that:

$$\begin{aligned}\nabla(f \circ \varphi^{-1})(\varphi(p)) &= \nabla(\widetilde{g} \circ \varphi^{-1})(\varphi(p)) \\ &= \nabla(g \circ \varphi \circ \varphi^{-1})(\varphi(p)) = \nabla g(\varphi(p)) = w.\end{aligned}$$ ■

One consequence of the prior lemma is that for any smooth chart (U, φ) on an n-dimensional manifold M containing a point p, there is a well-defined injective map $T_\varphi : \mathcal{V}_p M \to \mathbb{R}^n$ given by $T_\varphi([\gamma]) := (\varphi \circ \gamma)'(0)$.

Proposition: Suppose p is *not* a corner point of M. Then the map T_φ defined above is also surjective.

Proof:

Consider any $w \in \mathbb{R}^n$ and note that since $\varphi(U)$ must be an open neighborhood of $\varphi(p)$ in either H^n or $\mathbb{R}^n$, we know for a small enough $R > 0$ that either $\varphi(p) + tw \in \varphi(U)$ or $\varphi(p) - tw \in \varphi(U)$ for all $t \in [0, R]$.

Now we can define a map γ going from either $[0, R]$ or $[-R, 0]$ to M by setting $\gamma(t) \coloneqq \varphi^{-1}(\varphi(p) + tw)$. In either case, we have that $\gamma \in \mathcal{C}(M, p)$ and:

$$(\varphi \circ \gamma)'(0) = \tfrac{\mathrm{d}}{\mathrm{d}t}\big|_{t=0} (\varphi(p) + tw) = w. \blacksquare$$

Unfortunately, this proof doesn't work if p is a corner point of M. In fact, if p is a corner point of M then we'll not even have that $T_\varphi(\mathcal{V}_p M)$ is a vector subspace of $\mathbb{R}^n$. So from here on out in this section I'll be assuming my C^r manifolds do *not* have corners.

Now that we have a bijection $T_\varphi : \mathcal{V}_p M \to \mathbb{R}^n$, we can let this bijection induce a vector space structure on $\mathcal{V}_p M$. In other words, given any $[\gamma_1], [\gamma_2] \in \mathcal{V}_p M$ as well as any $c_1, c_2 \in \mathbb{R}$, we define:

$$c_1[\gamma_1] + c_2[\gamma_2] \coloneqq T_\varphi^{-1}(c_1 T_\varphi([\gamma_1]) + c_2 T_\varphi([\gamma_2])).$$

Proposition: The vector space structure on $\mathcal{V}_p M$ is defined independently of any particular smooth chart (U, φ) on M containing p. In particular, suppose (V, ψ) is another smooth chart on M containing p. Then for any $[\gamma_1], [\gamma_2] \in \mathcal{V}_p M$ as well as any $c_1, c_2 \in \mathbb{R}$ we have that:

$$T_\varphi^{-1}(c_1 T_\varphi([\gamma_1]) + c_2 T_\varphi([\gamma_2])) = T_\psi^{-1}(c_1 T_\psi([\gamma_1]) + c_2 T_\psi([\gamma_2]))$$

Proof:
Define the vectors:

- $v \coloneqq c_1 T_\varphi([\gamma_1]) + c_2 T_\varphi([\gamma_2]) = c_1(\varphi \circ \gamma_1)'(0) + c_2(\varphi \circ \gamma_2)'(0)$,
- $w \coloneqq c_1 T_\psi([\gamma_1]) + c_2 T_\psi([\gamma_2]) = c_1(\psi \circ \gamma_1)'(0) + c_2(\psi \circ \gamma_2)'(0)$.

Next let $[\widetilde{\gamma}] \coloneqq T_\varphi^{-1}(v)$. Thus $\widetilde{\gamma}$ is a C^1 path on M with $\widetilde{\gamma}(0) = p$ and $T_\varphi([\widetilde{\gamma}]) = (\varphi \circ \widetilde{\gamma})'(0) = v$. To prove the proposition, we need to show that $[\widetilde{\gamma}] = T_\psi^{-1}(w)$. Or in other words, we need to show that $T_\psi([\widetilde{\gamma}]) = (\psi \circ \widetilde{\gamma})'(0) = w$.

Fortunately, $\widetilde{\gamma}$ will map some open neighborhood of 0 in the interval $\widetilde{\gamma}$ is defined on into the set $U \cap V$. Therefore, we can say by chain rule that:

$$(\psi \circ \widetilde{\gamma})'(0) = (\psi \circ \varphi^{-1} \circ \varphi \circ \widetilde{\gamma})'(0) = \left[(\mathrm{D}(\psi \circ \varphi^{-1}))(\varphi(p))\right](\varphi \circ \widetilde{\gamma})'(0)$$

Next note that the matrix $\left[(\mathrm{D}(\psi \circ \varphi^{-1}))(\varphi(p))\right]$ defines a linear map. Therefore as $(\varphi \circ \widetilde{\gamma})'(0) = v = c_1(\varphi \circ \gamma_1)'(0) + c_2(\varphi \circ \gamma_2)'(0)$, we can say that:

$$\begin{aligned}&\left[(\mathrm{D}(\psi \circ \varphi^{-1}))(\varphi(p))\right](\varphi \circ \widetilde{\gamma})'(0)\\ &\quad = c_1\left[(\mathrm{D}(\psi \circ \varphi^{-1}))(\varphi(p))\right](\varphi \circ \gamma_1)'(0) + c_2\left[(\mathrm{D}(\psi \circ \varphi^{-1}))(\varphi(p))\right](\varphi \circ \gamma_2)'(0)\end{aligned}$$

Finally, by one more application of chain rule (which we can use on γ_1 and γ_2 for identical reasons as why we could use it on $\widetilde{\gamma}$) we get that:o

$$\left[(\mathrm{D}(\psi \circ \varphi^{-1}))(\varphi(p))\right](\varphi \circ \gamma_i)'(0) = (\psi \circ \gamma_i)'(0).$$

Thus, we've proven that $(\psi \circ \widetilde{\gamma})'(0) = c_1(\psi \circ \gamma_1)'(0) + c_2(\psi \circ \gamma_2)'(0) = w$. $\blacksquare$

Now that $\mathcal{V}_p M$ has been equipped with a standard vector space structure, I've now almost shown everything I want to on this tangent. What's left to do is show how $\mathcal{V}_p M$ corresponds to the space of derivations at p on M.

Lemma: Suppose M and N are C^r manifolds and $F : M \to N$ is a C^1 map. Then for any $p \in M$ we have that $\mathrm{d}F_p : \mathcal{V}_pM \to \mathcal{V}_{F(p)}N$ is a linear map.

Proof:

Consider any $c_1, c_2 \in \mathbb{R}$ as well as $\gamma_1, \gamma_2 \in \mathcal{C}(M, p)$. Next let $\widetilde{\gamma} \in \mathcal{C}(M, p)$ be such that $[\widetilde{\gamma}] = c_1[\gamma_1] + c_2[\gamma_2]$ in $\mathcal{V}_pM$. To prove this lemma it suffices to show that:

$$[F \circ \widetilde{\gamma}] = \mathrm{d}F_p[\widetilde{\gamma}] = c_1\mathrm{d}F_p[\gamma_1] + c_2\mathrm{d}F_p[\gamma_2] = c_1[F \circ \gamma_1] + c_2[F \circ \gamma_2]$$

in $\mathcal{V}_{F(p)}N$.

But now because $F \in C^1(M, N)$ we know there is a smooth chart (U, φ) on M containing p as well as a smooth chart (V, ψ) on N containing $F(p)$ such that $F(U) \subseteq V$ and $\psi \circ F \circ \varphi^{-1}$ is a C^1 map. Based on how we defined the vector space structure for $\mathcal{V}_{F(p)}N$, we just need to show that:

$$(\psi \circ F \circ \widetilde{\gamma})'(0) = c_1(\psi \circ F \circ \gamma_1)'(0) + c_2(\psi \circ F \circ \gamma_2)'(0).$$

Fortunately, we also know since $[\widetilde{\gamma}] = c_1[\gamma_1] + c_2[\gamma_2]$ that:

$$(\varphi \circ \widetilde{\gamma})'(0) = c_1(\varphi \circ \gamma_1)'(0) + c_2(\varphi \circ \gamma_2)'(0).$$

And now if we let $\mathbf{A}$ denote the matrix $(\mathrm{D}(\psi \circ F \circ \varphi^{-1}))(\varphi(p))$, we have by chain rule that:

$$\begin{aligned}(\psi \circ F \circ \widetilde{\gamma})'(0) = \mathbf{A}(\varphi \circ \widetilde{\gamma})'(0) &= c_1\mathbf{A}(\varphi \circ \gamma_1)'(0) + c_2\mathbf{A}(\varphi \circ \gamma_2)'(0) \\ &= c_1(\psi \circ F \circ \gamma_1)'(0) + c_2(\psi \circ F \circ \gamma_1)'(0). \blacksquare\end{aligned}$$

Lemma: Let M be a C^r manifold and let $\mathcal{D}_pM$ denote the space of derivations at $p \in M$ mapping $C^k(M, \mathbb{R})$ to $\mathbb{R}$ (where $k \leq r$). Then there is a well-defined map $\Psi : \mathcal{V}_pM \to \mathcal{D}_pM$ given by $\Psi([\gamma])f \coloneqq (f \circ \gamma)'(0)$.

Why?

The fact that $\Psi([\gamma])$ is a well-defined injective function on $C^1(M, \mathbb{R})$ is obvious from the definition of the equivalence class of γ. As for showing that $\Psi([\gamma])$ is a derivation, just apply normal derivative rules to $((c_1f + c_2g) \circ \gamma)'$ and $((fg) \circ \gamma)'$.

Furthermore, Ψ is a linear map.

Why?

Fix any $f \in C^1(M, \mathbb{R})$ and smooth chart (U, φ) on M containing p, and recall from the proof of the lemma on *page 1047* that:

$$(f \circ \gamma)'(0) = \nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot (\varphi \circ \gamma)'(0) \text{ for all } \gamma \in \mathcal{C}(M, p).$$

But now if $[\widetilde{\gamma}] = c_1[\gamma_1] + c_2[\gamma_2]$, we know that:

$$(\varphi \circ \widetilde{\gamma})'(0) = c_1(\varphi \circ \gamma_1)'(0) + c_2(\varphi \circ \gamma_2)'(0).$$

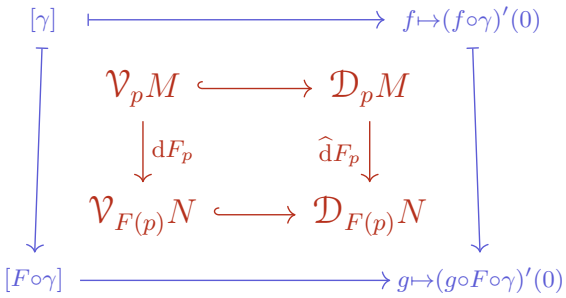
Since $x \mapsto \nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot x$ is a linear map, we can now easily conclude that if $[\widetilde{\gamma}] = c_1[\gamma_1] + c_2[\gamma_2]$ then $(f \circ \gamma)'(0) = c_1(f \circ \gamma_1)'(0) + c_2(f \circ \gamma_2)'(0)$.

Plus I claim no matter what k is, Ψ is still injective.

When $k = 1$ then this is just a result of the definition of the equivalence classses in $\mathcal{V}_pM$. After all, $[\gamma_1] \neq [\gamma_2]$ if and only if there exists some function $f \in C^1(M, \mathbb{R})$ with $(f \circ \gamma_1)'(0) \neq (f \circ \gamma_2)'(0)$. Meanwhile, if $1 < k \leq r$ then the thing we need to specifically show is that if $[\gamma_1] \neq [\gamma_2]$ then we can more strongly take f to be a

C^k map (as opposed to just a C^1 map) such that $(f \circ \gamma_1)'(0) \neq (f \circ \gamma_2)'(0)$. Since $C^r(M, \mathbb{R}) \subseteq C^k(M, \mathbb{R})$ for all $k \leq r$, I might as well by default just look for some $f \in C^r(M, \mathbb{R})$. Fortunately, I already showed how to construct such a function $f \in C^r(M, \mathbb{R})$ in my proof of the lemma on *page 1047*. ■

Proposition: Let M, N be C^r manifolds and suppose $F : M \to N$ is a C^k map (where $k \leq r$). Given any $p \in M$ we let $\mathcal{D}_p M$ denote the space of derivations at p from $C^k(M, \mathbb{R})$ to $\mathbb{R}$. Similarly, we let $\mathcal{D}_{F(p)} N$ denote the space of derivations at $F(p)$ from $C^k(N, \mathbb{R})$ to $\mathbb{R}$. And finally, we let $\widehat{\mathrm{d}}F_p : \mathcal{D}_p M \to \mathcal{D}_{F(p)}$ be the linear map $\widehat{\mathrm{d}}F_p(v)f = v(f \circ F)$. Then the following diagram commutes:



Hopefully this is obvious. Also as a side note, if $k = \infty$ then the horizontal arrows must be vector space isomorphisms. After all, $\mathcal{V}_p M$ and $\mathcal{D}_p M$ will both be n-dimensional vector spaces (assuming M is an n-dimensional manifold), and we just showed the arrow $\mathcal{V}_p M \to \mathcal{D}_p M$ is given by an injective linear map.

7/21/2026

I actually do have two more observations to mention before I'm fully done talking about $\mathcal{V}_p M$.

1. Suppose M is a C^r-manifold without corners, $p \in M$, and $U \subseteq M$ is an open set containing p. Then if we consider the inclusion map $\iota : U \hookrightarrow M$, we have that $\mathrm{d}\iota_p : \mathcal{V}_p U \to \mathcal{V}_p M$ is a vector space isomorphism.

   Why?
   Firstly note that both $\mathcal{V}_p U$ and $\mathcal{V}_p M$ will be n-dimensional vector spaces (where n is the number of dimensions M has). Also $\mathrm{d}\iota_p$ is a linear map. So all we need to do is show that $\mathrm{d}\iota_p$ is an injective map.

   Fortunately, we can just consider a smooth chart (V, φ) on M such that $p \in V \subseteq U$. Then (V, φ) is also a smooth chart on U containing p. So for $\gamma_1, \gamma_2 \in \mathcal{C}(M, p)$:
   $$[\gamma_1] = [\gamma_2] \text{ in } \mathcal{V}_p U \text{ iff } (\varphi \circ \gamma_1)'(0) = (\varphi \circ \gamma_2)'(0) \text{ iff } [\gamma_1] = [\gamma_2] \text{ in } \mathcal{V}_p M.$$

2. Consider the inclusion map $\iota : H^n \hookrightarrow \mathbb{R}^n$. Then for any $a \in \partial H^n$ we have that $\mathrm{d}\iota_a : \mathcal{V}_a H^n \to \mathcal{V}_a \mathbb{R}^n$ is a vector space isomorphism.

Why?
For any $\gamma_1, \gamma_2 \in \mathcal{C}(H^n, a)$ we have that $[\gamma_1] = [\gamma_2]$ in $\mathcal{V}_a H^n$ iff $\gamma_1'(0) = \gamma_2'(0)$ iff $[\gamma_1] = [\gamma_2]$ in $\mathcal{V}_a \mathbb{R}^n$.

Anyways, derivations in my opinion are much more tangible and thus nicer to work with than equivalence classes of curves. That said, I want to be able to work with C^r manifolds and C^k functions where $k \leq r < \infty$. So my solution is that if M is a C^r manifold without corners, then I'll let $\mathcal{D}_p M$ denote the space of derivations at p from $C^1(M, \mathbb{R})$ to $\mathbb{R}$. Then I'll let $\Psi : \mathcal{V}_p M \to \mathcal{D}_p M$ be the map defined two pages ago. And finally, I'll define $T_p M \coloneqq \Psi(\mathcal{V}_p M)$.

Importantly, this definition is identical to our prior definition when we work with C^∞ manifolds without corners and we restrict the domains of our derivations to just functions in $C^\infty(M, \mathbb{R})$. But this definition also lets us work with C^r manifolds without corners (where $r < \infty$).

Also, given a C^1 map $F : M \to N$, I'll let $\mathrm{d}F_p$ refer to the map $T_p M \to T_{F(p)} N$ given by $\mathrm{d}F_p(v) f = v(f \circ F)$. Due to the commuting diagram on the prior page, we know this map is well-defined. And also from all the work I've done in this section, we know every theorem about $\mathrm{d}F_p$ proved in the case of our old definition still works for our new definition.

I believe this deals with all notation inconsistencies I introduced in this section.

I'm tired of writing in red as opposed to black. Fortunately, the next stuff will for a while be actually in Lee's book instead of just an aside from me.

Let M be a C^r-manifold and $p \in M$. Also suppose $T_p M$ is well-defined. Now we can consider any smooth chart (U, φ) on M containing p and we let $\widehat{U} \coloneqq \varphi(U)$. Also we identify any $q \in U$ with $\varphi(q) = (x_1, \ldots, x_n) \in \widehat{U}$.

Note that a basis for $T_{\varphi(p)} \widehat{U}$ is given by the operators $\left.\frac{\partial}{\partial x_1}\right|_{\varphi(p)}, \ldots, \left.\frac{\partial}{\partial x_n}\right|_{\varphi(p)}$.

In the case of a C^∞-manifold (potentially with corners), we already showed that a basis for $T_{\varphi(p)} \mathbb{R}^n$ is given by the partial derivative operators. Then, we can simply trace backwards from the differential of $\iota : \widehat{U} \hookrightarrow \mathbb{R}^n$ at $\varphi(p)$ in order to show $\left.\frac{\partial}{\partial x_1}\right|_{\varphi(p)}, \ldots, \left.\frac{\partial}{\partial x_n}\right|_{\varphi(p)}$ forms a basis for $T_{\varphi(p)} \widehat{U}$.

After all, $\mathrm{d}\iota_{\varphi(p)}$ being a vector space isomorphism means that:

$$(\mathrm{d}\iota_{\varphi(p)})^{-1}\left(\left.\frac{\partial}{\partial x_1}\right|_{\varphi(p)}\right), \ldots, (\mathrm{d}\iota_{\varphi(p)})^{-1}\left(\left.\frac{\partial}{\partial x_n}\right|_{\varphi(p)}\right) \text{ forms a basis for } T_{\varphi(p)} \widehat{U}.$$

Yet also because $\mathrm{d}\iota_{\varphi(p)}$ is injective and maps $\left.\frac{\partial}{\partial x_i}\right|_{\varphi(p)}$ (as an operator on $C^\infty(\widehat{U}, \mathbb{R})$) to the operator $\left.\frac{\partial}{\partial x_i}\right|_{\varphi(p)}$ on $C^\infty(\mathbb{R}^n, \mathbb{R})$, we must have that:

$$(\mathrm{d}\iota_{\varphi(p)})^{-1}\left(\frac{\partial}{\partial x_i}\Big|_{\varphi(p)}\right) = \frac{\partial}{\partial x_i}\Big|_{\varphi(p)}.$$

Meanwhile, if we instead go by our new definition, then just note that the operator $\frac{\partial}{\partial x_i}\Big|_{\varphi(p)}$ corresponds to the path $\gamma_i : [0, R] \to \mathbb{R}^n$ defined by:

$$\gamma_i(t) = \varphi(p) + te_i \text{ (and } R \text{ is some small positive real number).}$$

[Here $e_1, \ldots, e_n$ is the standard basis on $\mathbb{R}^n$.] And since $\gamma_1'(0) = e_1, \ldots, \gamma_n'(0) = e_n$ forms a basis for $\mathbb{R}^n$, we know by the definition of the vector space structure on $\mathcal{V}_{\varphi(p)}\widehat{U}$ in the prior section that $[\gamma_1], \ldots, [\gamma_n]$ forms a basis for $\mathcal{V}_{\varphi(p)}\widehat{U}$. In turn, $\frac{\partial}{\partial x_1}\Big|_{\varphi(p)}, \ldots, \frac{\partial}{\partial x_n}\Big|_{\varphi(p)}$ forms a basis for $T_{\varphi(p)}\widehat{U}$.

In turn, the preimages of these operators under the map $\mathrm{d}\varphi_p$ gives a basis for T_pU (which in turn gives a basis for T_pM under the identification given by $\mathrm{d}\iota : T_pU \xrightarrow{\sim} T_pM$.) We write these preimage tangent vectors as $\frac{\partial}{\partial x_i}\Big|_p$. In other words:

$$\frac{\partial}{\partial x_i}\Big|_p := (\mathrm{d}\varphi_p)^{-1}\left(\frac{\partial}{\partial x_i}\Big|_{\varphi(p)}\right) = \mathrm{d}(\varphi^{-1})_{\varphi(p)}\left(\frac{\partial}{\partial x_i}\Big|_{\varphi(p)}\right).$$

Thus in particular we have that $\frac{\partial}{\partial x_i}\Big|_p f = \frac{\partial}{\partial x_i}\Big|_{\varphi(p)} f \circ \varphi^{-1} = \frac{\partial \widehat{f}}{\partial x_i}(\widehat{p})$ where $\widehat{f} := f \circ \varphi^{-1}$ and $\widehat{p} := (p_1, \ldots, p_n) = \varphi(p)$.

We call $\frac{\partial}{\partial x_1}\Big|_p, \ldots, \frac{\partial}{\partial x_n}\Big|_p$ the coordinate basis vectors of T_pM associated with the local coordinates on M given by the chart (U, φ). Because they form a basis for T_pM, any $v \in T_pM$ can be written as an $\mathbb{R}$-linear combination $\sum_{i=1}^n v_i \frac{\partial}{\partial x_i}\Big|_p$. The real numbers $v_1, \ldots, v_n$ are called the components of v with respect to our local coordinates.

Given an initial tangent vector $v \in T_pU \ (= T_pM)$, here is how to calculate the components of v. Define $f_j \in C^r(U, \mathbb{R})$ given by $f_j(q) = e_j \cdot \varphi(q)$. (In other words, if $q = (x_1, \ldots, x_n)$ via the chart (U, φ) then $f_j(q) = x_j$.) But then:

$$v(f_j) = \textstyle\sum_{i=1}^n v_i \frac{\partial}{\partial x_i}\Big|_p f_i = v_j \frac{\partial}{\partial x_j}\Big|_{\varphi(p)} (x \mapsto e_j \cdot x) = 1v_j.$$

One more topic I want to bring up before switching to looking at the next chapters in John Lee's book (because I'm bored with this chapter) is what the differential of a C^k map looks like in coordinates.

Suppose $F : M \to N$ is a C^k map. Then let (U, φ) be a smooth chart on M containing p and (V, ψ) be a smooth chart on N with $F(U) \subseteq V$. We associate to any $q \in U$ the coordinates $\varphi(q) = (x_1, \ldots, x_n)$ and to any $\overline{q} \in V$ the coordinates $\psi(\overline{q}) = (y_1, \ldots, y_m)$. Also we let $\widehat{F} = (\widehat{F}_1, \ldots, \widehat{F}_m) := \psi \circ F \circ \varphi^{-1}$, and we set $\widehat{p} := \varphi(p)$.

Because $\psi^{-1} \circ \widehat{F} = F \circ \varphi^{-1}$, we can calculate that:

$$\begin{aligned}
\mathrm{d}F_p\left(\left.\frac{\partial}{\partial x_i}\right|_p\right) &= \mathrm{d}F_p\left(\mathrm{d}(\varphi^{-1})_{\varphi(p)}\left(\left.\frac{\partial}{\partial x_i}\right|_{\varphi(p)}\right)\right) \\
&= \mathrm{d}(F\circ\varphi^{-1})_{\varphi(p)}\left(\left.\frac{\partial}{\partial x_i}\right|_{\varphi(p)}\right) = \mathrm{d}(\psi^{-1}\circ\widehat{F})_{\varphi(p)}\left(\left.\frac{\partial}{\partial x_i}\right|_{\varphi(p)}\right) \\
&= \mathrm{d}(\psi^{-1})_{\widehat{F}(\widehat{p})}\left(\mathrm{d}\widehat{F}_{\widehat{p}}\left(\left.\frac{\partial}{\partial x_i}\right|_{\widehat{p}}\right)\right).
\end{aligned}$$

Next, note for any $f\in C^1(\psi(V),\mathbb{R})$ that by chain rule:

$$\mathrm{d}\widehat{F}_{\widehat{p}}\left(\left.\frac{\partial}{\partial x_i}\right|_{\widehat{p}}\right)f = \frac{\partial(f\circ\widehat{F})}{\partial x_i}(\widehat{P}) = \sum_{j=1}^m \frac{\partial f}{\partial y_j}(\widehat{F}(\widehat{p}))\frac{\partial\widehat{F}_j}{\partial x_i}(\widehat{p}) = \left(\sum_{j=1}^m\left(\frac{\partial\widehat{F}_j}{\partial x_i}(\widehat{p})\right)\left.\frac{\partial}{\partial y_j}\right|_{\widehat{F}(\widehat{p})}\right)f.$$

Therefore, $\mathrm{d}\widehat{F}_{\widehat{p}}\left(\left.\frac{\partial}{\partial x_i}\right|_{\widehat{p}}\right) = \sum_{j=1}^m\left(\frac{\partial\widehat{F}_j}{\partial x_i}(\widehat{p})\right)\left.\frac{\partial}{\partial y_j}\right|_{\widehat{F}(\widehat{p})}$ and in turn we can say that:

$$\begin{aligned}
\mathrm{d}(\psi^{-1})_{\widehat{F}(\widehat{p})}\left(\mathrm{d}\widehat{F}_{\widehat{p}}\left(\left.\frac{\partial}{\partial x_i}\right|_{\widehat{p}}\right)\right) &= \mathrm{d}(\psi^{-1})_{\widehat{F}(\widehat{p})}\left(\sum_{j=1}^m\left(\frac{\partial\widehat{F}_j}{\partial x_i}(\widehat{p})\right)\left.\frac{\partial}{\partial y_j}\right|_{\widehat{F}(\widehat{p})}\right) \\
&= \sum_{j=1}^m\left(\frac{\partial\widehat{F}_j}{\partial x_i}(\widehat{p})\right)\mathrm{d}(\psi^{-1})_{\psi(F(p))}\left(\left.\frac{\partial}{\partial y_j}\right|_{\psi(F(p))}\right) \\
&= \sum_{j=1}^m\left(\frac{\partial\widehat{F}_j}{\partial x_i}(\widehat{p})\right)\left.\frac{\partial}{\partial y_j}\right|_{F(p)}
\end{aligned}$$

In other words, we've proven that:

$$\mathrm{d}F_p\left(\left.\frac{\partial}{\partial x_i}\right|_p\right) = \sum_{j=1}^m\left(\frac{\partial\widehat{F}_j}{\partial x_i}(\widehat{p})\right)\left.\frac{\partial}{\partial y_j}\right|_{F(p)}$$

Thus if we write the matrix representation of $\mathrm{d}F_p : T_pM \to T_{F(p)}N$ where T_pM is equipped with the coordinate basis vectors associated with (U,φ) and $T_{F(p)}N$ is equipped with the coordinate basis vectors associated with (V,ψ), then that matrix will be precisely the Jacobian matrix:

$$\mathrm{D}\widehat{F}(\widehat{p}) = \begin{bmatrix} \frac{\partial\widehat{F}_1}{\partial x_1}(\widehat{p}) & \cdots & \frac{\partial\widehat{F}_1}{\partial x_n}(\widehat{p}) \\ \vdots & \ddots & \vdots \\ \frac{\partial\widehat{F}_m}{\partial x_1}(\widehat{p}) & \cdots & \frac{\partial\widehat{F}_m}{\partial x_n}(\widehat{p}) \end{bmatrix} \text{ (where } \widehat{F} = \psi\circ F\circ\varphi^{-1} \text{ and } \widehat{p} = \varphi(p)\text{)}$$

7/22/2026

For the time being I think it will be wise to assume we are working with manifolds *without* corners (unless otherwise mentioned). After all, a bunch of the results I'm going to show next don't even work unless we further limit ourselves to working with manifolds without boundaries.

Suppose M and N are C^r manifolds. Given any $F \in C^1(M,N)$ and $p \in M$ we define the rank of F at p to be the rank of the linear map $\mathrm{d}F_p : T_pM \to T_{F(p)}N$. If F has the same rank R at every point, we say F has constant rank R and write $\mathrm{rank}(F) = R$.

Note that $\mathrm{rank}(\mathrm{d}F_p) \leq \min(\dim(T_pM), \dim(T_{F(p)}N)) = \min(\dim(M), \dim(N))$. If $\mathrm{rank}(\mathrm{d}F_p)$ equals this upper bound, we say that F has full rank at p. And if F has full rank everywhere, we say F has full rank.

Also, we call F a submersion if $\mathrm{rank}(F) = \dim(N)$, and we call F an immersion if $\mathrm{rank}(F) = \dim(M)$.

Proposition 4.1: Suppose $F : M \to N$ is a C^1 map and $p \in M$. If $\mathrm{d}F_p$ is surjective then p has an open neighborhood U such that $F|_U$ is a submersion. Meanwhile, if $\mathrm{d}F_p$ is injective, then p has an open neighborhood U such that $F|_U$ is an immersion.

Proof:
Let (U, φ) be a smooth chart on M containing p and (V, ψ) be a smooth chart on N such that $F(U) \subseteq V$. It suffices to show that F has full rank on some open set $W \subseteq U$ containing p. Fortunately, if we let $\widehat{F} := \psi \circ F \circ \varphi^{-1}$, then we have for any $q \in U$ that $\mathrm{d}F$ has full rank if and only if the Jacobian matrix $\mathrm{D}\widehat{F}(\varphi(q))$ has full rank.

Yet now if $n = \dim(M)$ and $m = \dim(N)$, let $M(m \times n, \mathbb{R})$ denote the set of all $m \times n$ real matrices. There is an obvious identification of $M(m \times n, \mathbb{R})$ with $\mathbb{R}^{nm}$. Next let $k \leq \min(m,n)$ and let $M_k(m \times n, \mathbb{R})$ be the subset of $m \times n$ matrices of rank k.

What follows is a bunch of linear algebra results I've used before without proof and I thought I'd finally prove them now.

Lemma A: Let F be any field and let $\mathbf{A} \in M(m \times n, F)$ be any matirx. Then:

$$\mathrm{rank}(\mathbf{A}) = \mathrm{rank}(\mathbf{A}^T).$$

Proof:
Since $(\mathbf{A}^T)^T = \mathbf{A}$, it suffices to prove that $\mathrm{rank}(\mathbf{A}^T) \geq \mathrm{rank}(\mathbf{A})$. So, let $e_1, \ldots, e_n$ be the standard basis for F^n and $u_1, \ldots, u_m$ be the standard basis for F^m. Next let e_i^* and u_j^* for each i and j be the corresponding dual vectors in $(F^n)^*$ and $(F^m)^*$ respectively. Then recall that the transpose matrix $\mathbf{A}^T$ describes the map $(F^m)^* \to (F^n)^*$ given by $f \mapsto f \circ A$ (where $(F^m)^*$ and $(F^n)^*$ are equipped with the dual basises $u_1^*, \ldots, u_m^*$ and $e_1^*, \ldots, e_m^*$ respectively). [See *pages 101-102* for a reminder on what I'm talking about...]

Now let $k := \mathrm{rank}(\mathbf{A}) = \dim(\mathrm{im}(\mathbf{A}))$, and note that the vectors $\mathbf{A}e_1, \ldots, \mathbf{A}e_n$ span $\mathrm{im}(\mathbf{A})$. So we can remove vectors until we have k many numbers $1 \leq i_1 < i_2 < \cdots < i_k \leq n$ such that $\mathbf{A}e_{i_1}, \ldots, \mathbf{A}e_{i_k}$ are F-linearly independent in $\mathrm{im}(\mathbf{A}) \subseteq F^m$.

Next, let us set $v_j := \mathbf{A}e_{i_j} \in F^m$ for each $1 \leq j \leq k$. We can additionally find vectors $v_{k+1}, \ldots, v_m \in F^m$ such that $v_1, \ldots, v_m$ forms a basis for F^m. And then, we let $v_1^*, \ldots, v_m^*$ be the corresponding dual vectors in $(F^m)^*$.

I claim $\mathbf{A}^T v_1^*, \ldots, \mathbf{A}^T v_k^*$ are linearly independent in $(F^n)^*$. To see why, suppose $\sum_{j=1}^k c_j(v_j^* \circ A)$ is the zero map. Then:

$$0 = \sum_{j=1}^k c_j(v_j^* \circ A)e_{i_{j'}} = \sum_{j=1}^k c_j v_j^*(v_{j'}) = \sum_{j=1}^k c_j \delta_{j,j'} = c_{j'}$$

In other words, $0 = \sum_{j=1}^k c_j(v_j^* \circ A) \implies c_j = 0$ for all j. And with that, I've proven that $\text{rank}(\mathbf{A}^T) = \dim(\text{im}(\mathbf{A}^T)) \geq k$. ■

Lemma B: Let F be any field and let $\mathbf{A} \in M_k(m \times n, F)$. Then $\mathbf{A}$ contains an invertible $k \times k$-submatrix $\mathbf{A}_0$. In other words, if we write $\mathbf{A} = [a_{i,j}]$, then there exists $1 \leq i_1 < i_2 < \cdots < i_k \leq n$ and $1 \leq j_1 < j_2 < \cdots < j_k \leq m$ so that $\mathbf{A}_0 = [a_{i_r,j_s}]$ is an invertible matrix.

Proof:
By the prior lemma, we know that $\text{rank}(\mathbf{A}^T) = \text{rank}(\mathbf{A})$. So, can pick k linearly independent columns of the matrix $\mathbf{A}^T$, and then we let the matrix $\mathbf{B}$ consist of just those columns. Importantly, we will still have that $\text{rank}(\mathbf{B}) = \text{rank}(\mathbf{A}^T)$, and by the prior lemma, we in turn know that $\text{rank}(\mathbf{B}^T) = \text{rank}(\mathbf{B})$. So, we can pick k linearly independent columns of $\mathbf{B}^T$, and we let $\mathbf{A}_0$ consist of just those columns.

Now $\mathbf{A}_0$ is a kxk submatrix of $\mathbf{A}$ such that all the columns of $\mathbf{A}_0$ are linearly independent in F^k. So, $\mathbf{A}_0$ is an invertible matrix. ■

Lemma C: Conversely, suppose $\mathbf{A} \in M(m \times n, F)$ has an invertible kxk submatrix $\mathbf{A}_0$. Then $\text{rank}(\mathbf{A}) \geq k$.

Why?
The existence of this matrix $\mathbf{A}_0$ trivially guarentees the existence of k linearly independent columns of $\mathbf{A}$. ■

Lemmas B and C in particular have the following consequence.

Lemma D: Let $k = \min(m, n)$. Then having identified $M(m \times n, \mathbb{R})$ with $\mathbb{R}^{mn}$, we have that $M_k(m \times n, \mathbb{R})$ is an open subset of $\mathbb{R}^{mn}$.

Proof:
Let $\mathbf{A} = [a_{i,j}]$ be any matrix in $M_k(m \times n, \mathbb{R})$. Then by lemma B we know there exists $1 \leq i_1 < i_2 < \cdots < i_k \leq n$ and $1 \leq j_1 < j_2 < \cdots < j_k \leq m$ so that $\mathbf{A}_0 = [a_{i_r,j_s}]$ is an invertible matrix. In particular, this means that $\det(\mathbf{A}_0) \neq 0$.

Yet now note that $\det : \mathbb{R}^{k^2} \to \mathbb{R}$ is a continuous function (as its even a polynomial). Hence, $\text{GL}(k, \mathbb{R}) = \det^{-1}(\mathbb{R} - \{0\})$ is an open subset of $\mathbb{R}^{k^2}$. In particular, this means we can find some $\varepsilon > 0$ such that

$$\|\mathbf{B_0} - \mathbf{A_0}\|_F := \left(\sum_{r,s}(b_{i_r,j_s} - a_{i_r,j_s})^2\right)^{1/2} < \varepsilon \implies \mathbf{B}_0 \text{ is invertible.}$$

But now suppose $\mathbf{B} \in M(m \times n, \mathbb{R})$ with $\|\mathbf{B} - \mathbf{A}\|_F < \varepsilon$. Then after making $\mathbf{B}_0$ be the submatrix of $\mathbf{B}$ consisting of the same indices as the submatrix $\mathbf{A}_0$ of $\mathbf{A}$, we have that $\|\mathbf{B}_0 - \mathbf{A}_0\|_F \leq \|\mathbf{B} - \mathbf{A}\|_F < \varepsilon \implies \mathbf{B}_0$ is invertible. By lemma C, we can conclude that $\|\mathbf{B} - \mathbf{A}\|_F < \varepsilon$ implies that $\text{rank}(\mathbf{B}) \geq k$. Yet since we must also have that $\text{rank}(\mathbf{B}) \leq k$, we have thus shown that for any $\mathbf{A} \in M_k(m \times n, \mathbb{R})$ there exists a ball $B_\varepsilon(\mathbf{A}) \subseteq M_k(m \times n, \mathbb{R})$. ■

Finally, this is relevant because we know that $\mathrm{D}\widehat{F}(\varphi(p))$ has full rank. So, we can pick some $\varepsilon > 0$ such that if $\mathbf{B} \in M(m \times n, \mathbb{R})$ satisfies that $\|\mathbf{B} - \mathrm{D}\widehat{F}(\varphi(p))\|_F < \varepsilon$ then $\mathbf{B}$ has full rank. Next, because $\mathrm{D}\widehat{F}$ is a C^1 map we know that the matrix components of $\mathrm{D}\widehat{F}$ vary continuously. Hence, we can pick $\delta > 0$ such that $B_\delta(\varphi(p)) \subseteq \varphi(U)$ and $\widehat{q} \in B_\delta(\varphi(p)) \implies \|\mathrm{D}\widehat{F}(\widehat{q}) - \mathrm{D}\widehat{F}(\varphi(p))\|_F < \varepsilon \implies \mathrm{D}\widehat{F}(\widehat{q})$ has full rank. So, we set $W = \varphi^{-1}(B_\delta(\varphi(p)))$. And now W is an open set containing p and F has full rank on W. ■

Suppose M, N are C^r manifolds and $F : M \to N$ is a C^k map (where $k \leq r$). We say F is a local C^k-diffeomorphism if every $p \in M$ has an open neighborhood $U \subseteq M$ such that $F(U)$ is open in N and $F|_U : U \to F(U)$ is a C^k-diffeomorphism.

Theorem 4.5 (Inverse Function Theorem for Manifolds): Suppose M and N are C^r manifolds *without boundaries* and $F : M \to N$ is a C^k map (where $k \leq r$). If $p \in M$ is a point where $\mathrm{d}F_p$ is invertible then there are connected open neighborhoods $U_0 \subseteq M$ of p and $V_0 \subseteq N$ of $F(p)$ such that $F|_{U_0} : U_0 \to V_0$ is a C^k-diffeomorphism.

Proof:
Since $\mathrm{d}F_p$ is invertible, we must have that $\dim(M) = \dim(N)$. So, let us call their common number of dimensions n. Next, let us pick smooth charts (U, φ) on M containing p and (V, ψ) on N such that $F(U) \subseteq V$. Then $\widehat{F} := \psi \circ F \circ \varphi^{-1}$ is a C^k map from the open set $\widehat{U} := \varphi(U) \subseteq \mathbb{R}^n$ into the open set $\widehat{V} := \psi(V) \subseteq \mathbb{R}^n$.

But now after setting $\widehat{p} := \varphi(p)$, we know that the Jacobian matrix $\mathrm{D}\widehat{F}(\widehat{p})$ is invertible. Hence by the inverse function theorem from real analysis, we know there are open subsets $\widehat{U}_0 \subseteq \widehat{U}$ containing $\widehat{p}$ and $\widehat{V}_0 \subseteq \widehat{V}$ containing $\widehat{F}(\widehat{p})$ such that $\widehat{F}|_{\widehat{U}_0} : \widehat{U}_0 \to \widehat{V}_0$ is a C^k-diffeomorphism.

You can see in the proof in my math 140c notes that $\widehat{U}_0$ as constructed in that proof happens to already be connected. That said, since homeomorphisms bijectively match the connected components of $\widehat{U}_0$ to the connected components of $\widehat{V}_0$, and also because our sets being locally connected means that the connected components are open, we could also just restrict $\widehat{U}_0$ and $\widehat{V}_0$ to being the connected components containing $\widehat{p}$ and $\widehat{F}(\widehat{p})$ respectively.

Also see *page 86* in this journal for why the inverse map is C^k.

Now just let $U_0 := \varphi^{-1}(\widehat{U}_0)$ and $V_0 := \psi^{-1}(\widehat{V}_0)$. Then U_0 is a connected open set in M containing p and V_0 is a connected open set in N containing $F(p)$. Also, we can easily see that $F|_{U_0} : U_0 \to V_0$ is a C^k-diffeomorphism. ■

As a side note, the prior theorem can fail when we allow M and N to have nonempty boundaries. For an obvious example of this, let $M = H^n$ and $N = \mathbb{R}^n$. Then let $\iota : H^n \hookrightarrow \mathbb{R}^n$ be the inclusion map and consider any $p \in \partial H^n$. We already know from earlier that $\mathrm{d}\iota_p$ is a bijection. That said, the prior theorem holding in this situation would

contradict the invariance of domain theorem on *page 1008*. So, no such open sets U_0 and V_0 can exist in this case.

Exercise 4-2: Suppose M is a C^r manifold (without a boundary) and N is a C^r manifold with a boundary. Also let $\text{Int}(N)$ denote the set of interior / non-boundary points of N. Then if $F : N \to M$ is a C^k map (where $k \leq r$), $p \in M$, and $\text{d}F_p$ is bijective, we must have that $F(p) \in \text{Int}(N)$.

Proof:
Once again, the fact that $\text{d}F_p$ is injective means that there exists an integer n with $\dim(M) = n = \dim(N)$. Now let (U, φ) be a smooth chart on M containing p and (V, ψ) be a smooth chart on N with $F(U) \subseteq V$. Since M has no boundary, we know that $\widehat{U} := \varphi(U)$ is open in $\mathbb{R}^n$.

Yet now note that $\widehat{F} := \psi \circ F \circ \varphi^{-1}$ is a C^k map from $\widehat{U}$ into $\mathbb{R}$ such that $\text{D}\widehat{F}(\varphi(p))$ is invertible. It follows by the inverse function theorem in my math 140c notes that there are open sets $\widehat{U}_0$ and $\widehat{V}_0$ of $\mathbb{R}^n$ such that $\varphi(p) \in \widehat{U}_0 \subseteq \widehat{U}$ and $\widehat{F}$ is a bijection from $\widehat{U}_0$ to $\widehat{V}_0$. So $(\psi|_{\psi^{-1}(\widehat{V}_0)}, \psi^{-1}(\widehat{V}_0))$ is an interior chart on N containing $F(p)$. ■

7/23/2026

To start off, it's been bugging me for several days that I don't have a way of talking about tangent spaces to corner points of C^r manifolds (where $r < \infty$). So I'm going to bodge a solution. I am not getting this construction from any resources though so I'm sorry if my construction disagrees with a better solution.

Let M be a C^r manifold with a corner point p. Then we define $\mathcal{C}(M, p)$ and $\mathcal{V}_pM$ as on *page 1046*. Also, we let $\mathcal{D}_pM$ denote the space of derivations at p from $C^1(M, \mathbb{R})$ to $\mathbb{R}$. Now for any $\gamma \in \mathcal{C}(M, p)$ we know $D_\gamma : C^1(M, \mathbb{R}) \to \mathbb{R}$ given by $D_\gamma(f) := (f \circ \gamma)'(0)$ is a well-defined derivation at p. Also, we know by the definition of the equivalence classes in $\mathcal{V}_p(M)$ that $[\gamma_1] = [\gamma_2]$ in $\mathcal{V}_p$ if and only if $D_{\gamma_1} = D_{\gamma_2}$.

Lemma: Let $S := \{D_\gamma : [\gamma] \in \mathcal{V}_pM\}$. Then let T_pM be the subspace of $\mathcal{D}_pM$ spanned by S.

(a) If M is an n-dimensional manifold then T_pM is also n-dimensional.

Proof:
Let (U, φ) be a smooth chart on M containing p, and then recall for any $f \in C^1(M, \mathbb{R})$ and $[\gamma] \in \mathcal{V}_pM$ that:

$$\text{D}_\gamma(f) = (f \circ \gamma)'(0) = \nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot (\varphi \circ \gamma)'(0)$$

Now suppose $\gamma_1, \ldots, \gamma_{n+1}$ are all in $\mathcal{C}(M, p)$. Then for any $c_1, \ldots, c_{n+1}$ we have that:

$$\left(\sum_{i=1}^{n+1} c_i D_{\gamma_i}\right)(f) = \nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot \left(\sum_{i=1}^{n+1} c_i (\varphi \circ \gamma_i)'(0)\right)$$

Yet because $\mathbb{R}^n$ is an n-dimensional vector space, we know there exists $c_1, \ldots, c_{n+1}$ which aren't all zero such that $\sum_{i=1}^{n+1} c_i (\varphi \circ \gamma_i)'(0) = 0 \in \mathbb{R}^n$.

And then we'll have $\left(\sum_{i=1}^{n+1} c_i D_{\gamma_i}\right)(f) = \nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot 0 = 0$ for all $f \in C^1(M, \mathbb{R})$. This proves that $\dim(T_pM) \leq n$.

As for the other inequality, note that because $\varphi(U)$ is open in $(\mathbb{R}_{\geq 0})^n$, we can guarentee if $e_1, \ldots, e_n$ is the standard basis for $\mathbb{R}^n$ then there exists $R > 0$ with $\varphi(p) + te_i \in \varphi(U)$ for all $t \in [0, R]$ and each i. So, I define $\gamma_i : [0, R] \to M$ by $\gamma(t) \coloneqq \varphi^{-1}(\varphi(p) + te_i)$. Then for any $c_1, \ldots, c_n$ we have that:

$$\left(\sum_{i=1}^{n} c_i D_{\gamma_i}\right)(f) = \nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot (c_1, \ldots, c_n)$$

At last, we just need to show that if $\vec{c} \coloneqq (c_1, \ldots, c_n) \neq 0$ then there exists some $f \in C^1(M, \mathbb{R})$ with $\nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot \vec{c} \neq 0$. Fortunately, we know from the proof of the lemma on *page 1047* how to go a step further and find $f \in C^r(M, \mathbb{R})$ with $\nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot \vec{c} \neq 0$. Therefore:

- $D_{\gamma_1}, \ldots, D_{\gamma_n}$ is a collection of linearly independent vectors in $S \subseteq \mathcal{D}_pM$.
- If we restrict the domains of all the derivations in T_pM to just $C^r(M, \mathbb{R})$ then we still have that $D_{\gamma_1}, \ldots, D_{\gamma_n}$ are linearly independent vectors.

(b) Let $F : M \to N$ be a C^1 map. Then let $S' \coloneqq \{D_\gamma : [\gamma] \in \mathcal{V}_{F(p)}N\}$ and let $T_{F(p)}N$ be the subspace of $\mathcal{D}_{F(p)}N$ spanned by S'.

[Note that if $F(p)$ is not a corner point of N, then this definition agress with our definition from *page 1051* because $T_{F(p)}N$ would equal S'].

Now define $\mathrm{d}F_p : \mathcal{D}_pM \to \mathcal{D}_{F(p)}N$ by $\mathrm{d}F_p(v)f \coloneqq v(f \circ F)$. Then we have that $\mathrm{d}F_p(T_pM) \subseteq T_{F(p)}N$.

Proof:
It suffices to show that $\mathrm{d}F_p(S) \subseteq S'$. Fortunately, given any $D_\gamma \in S$ we have that $\mathrm{d}F_p(D_\gamma)f = D_\gamma(f \circ F) = (f \circ F \circ \gamma)'(0) = D_{F\circ\gamma}f$. Therefore $\mathrm{d}F_p(D_\gamma) = D_{F\circ\gamma} \in S'$. ■

Based on the prior lemma, we now have a way of defining T_pM when p is a corner point. Also by part (b), we know that the differential of $F : M \to N$ at p gives a well defined linear map $T_pM \to T_{F(p)}N$.

The proofs for propositions 3.6, 3.8, and 3.9 on pages *1031* and *1033-1034* can now be repeated verbatim for our modified definition of T_pM for when p is a corner chart and M is a C^r manifold with $r \neq \infty$ necessarily. Similarly, nothing on *pages 1051-1053* breaks down with our definition for this special case either. Finally, proposition 4.1 on *pages 1054-1056* as well as exercise 4-2 on the prior page still work by identical reasoning.

And with that I'll now allow my manifolds to have corners again by default.

Next, I want to talk about product manifolds. Let $M_1, \ldots, M_k$ be C^r-manifolds of dimensions $n_1, \ldots, n_k$ respectively. Then let $M \coloneqq M_1 \times \cdots \times M_k$ be their product manifold, and for each i let $\pi_i : M \to M_i$ be the projection map $(p_1, \ldots, p_k) \mapsto p_i$.

Remark: Given any finite collection of C^r manifolds $M_1, \ldots, M_k$ with corners, the standard product C^r-structure on $M := M_1 \times \cdots \times M_k$ defined on *page 1038* is still well-defined. Furthermore, the proof of the proposition on *page 1039* still works verbatim. Thus the category of C^r manifolds with corners (which I'll denote $\mathbf{Diff}_c^r$) does admit products over any finite collection of objects. In particular, we can see that $(p_1, \ldots, p_k)$ is a corner point of M if at least one p_i is a corner point of M_i or there are two components p_i and p_j which are boundary points of M_i and M_j respectively.

Proposition 3.14: Let $p = (p_1, \ldots, p_k) \in M$. Then the map $\alpha : T_pM \to \bigoplus_{i=1}^k T_{p_i}M_i$ defined by $\alpha(v) := (\mathrm{d}(\pi_1)_p(v), \ldots, \mathrm{d}(\pi_k)_p(v))$ is an isomorphism.

Proof:
For each i let us define $\iota_i : M_i \to M$ by $\iota_i(q) := (p_1, \ldots, p_{i-1}, q, p_{i+1}, \ldots, p_k)$. Then, we define $\beta : \bigoplus_{i=1}^k T_{p_i}M_i \to T_pM$ by $\beta(v_1, \ldots, v_k) := \sum_{i=1}^k \mathrm{d}(\iota_i)_{p_i}(v_i)$. I claim that $\beta = \alpha^{-1}$. To see why, first note that:

$$\begin{aligned}(\alpha \circ \beta)(v_1, \ldots, v_k) &= \alpha\left(\sum_{i=1}^k \mathrm{d}(\iota_i)_{p_i}(v_i)\right) \\ &= \left(\mathrm{d}(\pi_1)_p\left(\sum_{i=1}^k \mathrm{d}(\iota_i)_{p_i}(v_i)\right), \ldots, \mathrm{d}(\pi_k)_p\left(\sum_{i=1}^k \mathrm{d}(\iota_i)_{p_i}(v_i)\right)\right) \\ &= \left(\sum_{i=1}^k \mathrm{d}(\pi_1)_p(\mathrm{d}(\iota_i)_{p_i}(v_i)), \ldots, \sum_{i=1}^k \mathrm{d}(\pi_k)_p(\mathrm{d}(\iota_i)_{p_i}(v_i))\right) \\ &= \left(\sum_{i=1}^k \mathrm{d}(\pi_1 \circ \iota_i)_{p_i}(v_i), \ldots, \sum_{i=1}^k \mathrm{d}(\pi_k \circ \iota_i)_{p_i}(v_i)\right)\end{aligned}$$

But now $\pi_j \circ \iota_i$ is a constant map if $j \neq i$. As a result, if $v \in T_{p_i}M_i$ and $f \in C^1(M_j, \mathbb{R})$ then $f \circ \pi_j \circ \iota_i$ is a constant function from M_i to $\mathbb{R}$ and in turn:

$$\mathrm{d}(\pi_j \circ \iota_i)_{p_i}(v_i)f = v_i(f \circ \pi_j \circ \iota_i) = 0.$$

Since f was arbitrary, this shows that if $j \neq i$ then $\mathrm{d}(\pi_j \circ \iota_i)_{p_i}(v)$ is the zero map. And since v was arbitary, we can conclude $\mathrm{d}(\pi_j \circ \iota_i)_{p_i}$ is also the zero map. Meanwhile, $\pi_i \circ \iota_i = \mathrm{Id}_{M_i}$. Hence, we can conclude that:

$$\begin{aligned}\left(\sum_{i=1}^k \mathrm{d}(\pi_1 \circ \iota_i)_{p_i}(v_i), \ldots, \sum_{i=1}^k \mathrm{d}(\pi_k \circ \iota_i)_{p_i}(v_i)\right) &= (\mathrm{d}(\mathrm{Id}_{M_1})_{p_1}(v_1), \ldots, \mathrm{d}(\mathrm{Id}_{M_k})_{p_k}(v_k)) \\ &= (v_1, \ldots, v_k).\end{aligned}$$

With that, we've proven that β is a right inverse of α. Based on the fact that α has a right inverse, we know that α is a surjective linear map. And since T_pM and $\bigoplus_{i=1}^k T_{p_i}M_i$ are both $(n_1 + \ldots + n_k)$-dimensional vector spaces and α is a linear map, we can conclude more strongly that α must be a bijection. And finally, any right inverse of a bijective function must also be a left inverse. Therefore, $\beta = \alpha^{-1}$. ■

The map α in the prior proposition is taken to define a standard identification between T_pM and $\bigoplus_{i=1}^k T_{p_i}M_i$.

7/24/2026

Proposition 4.6 (Properties of local C^k-diffeomorphisms):

(a) Let M, N, P be C^r manifolds and suppose $k \leq r$. If $F : M \to N$ and $G : N \to P$ are local C^k-diffeomorphisms, then so is $G \circ F : M \to P$.

Why?
Consider any $p \in M$. Since F is a local C^k-diffeomorphism, we know there exists an open set $U \subseteq M$ containing p and an open set $V \subseteq N$ such that $F|_U$ is a C^k-diffeomorphism from U to V. Next, since G is a local C^k-diffeomorphism, we know that there is an open set $V' \subseteq N$ containing $F(p)$ and an open set $W' \subseteq P$ such that $G|_{V'}$ is a C^k-diffeomorphism from V' to W'. And now we set $V_0 := V \cap V'$, $W_0 := G(V_0)$, and $U_0 := F^{-1}(V_0) \cap U$. We know that U_0 and W_0 are open sets in M and P respectively, that $p \in U_0$, and that $(G \circ F)$ is a C^k diffeomorphism from U_0 to W_0.

(b) Suppose $M_1, \ldots, M_k$ are C^r manifolds with $M := M_1 \times \cdots \times M_k$, $N_1, \ldots, N_k$ are also C^r manifolds with $N := N_1 \times \cdots \times N_k$, and $F_i : M_i \to N_i$ is a local C^s diffeomorphism for all $1 \leq i \leq k$ (where $s \leq r$). Then $F_1 \times \cdots \times F_k : M \to N$ is a local C^s diffeomorphism.

Why?
Let $(p_1, \ldots, p_k)$ be any point in M. For each i we can pick open sets $U_i \subseteq M_i$ containing p and $V_i \subseteq N_i$ such that $F|_{U_i} : U_i \to V_i$ is a C^s diffeomorphism. And now $U := U_1 \times \cdots \times U_k$ is an open subset of M containing $(p_1, \ldots, p_k)$ and $V := V_1 \times \cdots V_k$ is an open subset of N. I claim $F := F_1 \times \cdots \times F_k$ is a C^s diffeomorphism from U to V. This is because of the following lemma.

Lemma: Suppose $M_1, \ldots, M_k$ are C^r manifolds with $M := M_1 \times \cdots \times M_k$, $N_1, \ldots, N_k$ are also C^r manifolds with $N := N_1 \times \cdots \times N_k$, and $F_i : M_i \to N_i$ is a C^s diffeomorphism for all $1 \leq i \leq k$. Then $F_1 \times \cdots \times F_k : M \to N$ is a C^s diffeomorphism.

Proof:
It's incredibly straightforward to show that $F_1 \times \cdots \times F_k$ is a C^s map. Also, for each i let G_i be the inverse map of F_i. Then as each G_i is C^s, we can conclude by identical reasoning that $G_1 \times \cdots \times G_k : N \to M$ is a C^s map. And also it is easy to check that

$$G_1 \times \cdots \times G_k = (F_1 \times \cdots \times F_k)^{-1}.$$

(c) Let M and N be C^r manifolds, and suppose $F : M \to N$ is a local C^k diffeomorphism (where $k \leq r$). Then F is a local homeomorphism and an open map.

Proof:
Since diffeomorphisms are homeomorphisms, the first claim should be obvious. Next suppose $U \subseteq M$ is any open set. Then for any $p \in U$ we need to prove that $F(p)$ is in the interior of $F(U)$. Fortunately, as F is a local diffeomorphism we know there exists an open set $U_p \subseteq M$ containing p and an open set $V_p \subseteq N$ such that $F|_{U_p} : U_p \to V_p$

is a diffeomorphism. Next, as $U \cap U_p$ is an open subset of U_p, we know $F|_{U_p}(U \cap U_p)$ must be an open subset of V_p. So, $V \coloneqq V_p \cap F(U \cap U_p)$ is an open set in N which is contained in $F(U)$. Also, $F(p) \in V$. ■

I'm bored and don't want to prove anymore parts of the prior proposition. So I'm moving on.

Proposition 4.8: Suppose M is C^r manifold without a boundary, N is any other C^r manifold, and $F : M \to N$ is a C^k function (where $k \leq r$).

(a) F is a local C^k-diffeomorphism if and only if it is both an immersion and submersion.

$(\Longrightarrow)$
Given any $p \in M$, there is an open set $U \subseteq M$ containing p such that $F(U)$ is an open set and F is a C^k-diffeomorphism from U to $F(U)$. But then in turn we must have that $\mathrm{d}(F|_U)_p : T_pU \to T_{F(p)}(F(U))$ is an invertible map. Equivalently, this means that $\mathrm{d}F_p : T_pM \to T_pN$ is invertible. And since was p was arbitrary, we can conclude that $\mathrm{rank}(F) = \dim(M) = \dim(N)$. This shows that F is an immersion and a submersion.

$(\Longleftarrow)$
Conversely, suppose F is an immersion and a submersion. Then we must have that $\mathrm{d}F_p$ is invertible for all $p \in M$. By the inverse function theorem for manifolds as well as exercise 4-2 on *pages 1056-1057*, we thus know that $F(M) \subseteq \mathrm{Int}(N)$ and F is a local C^k-diffeomorphism.

(b) If $\dim(M) = \dim(N)$ and F is either an immersion or submersion, then F is a local C^k-diffeomorphism.

Why?
If $\dim(M) = \dim(N)$ then F is an immersion if and only if F is a submersion. ■

Exercise 4.10: Suppose M, N, and P are C^r manifolds and $F : M \to N$ is a local C^k diffeomorphism.

(a) If $G : P \to M$ is a continuous map then G is C^k if and only if $F \circ G$ is C^k.

$(\Longrightarrow)$
This is obvious.

$(\Longleftarrow)$
Let $p \in P$. Since $F \circ G$ is C^k, we know that there are smooth charts (U, φ) on P containing p and (V, ψ) on N with $(F \circ G)(U) \subseteq V$ and $\psi \circ F \circ G \circ \varphi^{-1}$ a C^k map. Next, because F is a local C^k diffeomorphism we know that there are open sets $W' \subseteq M$ containing $G(p)$ and $V' \subseteq N$ such that F is a C^k diffeomorphism from W' to V'. By shrinking W' and V', we can assume without loss of generality that $V' \subseteq V$ and that there is a smooth chart (W', η) on M. Finally, because G is continuous, we know that $U' \coloneqq G^{-1}(W')$ is an open set of P. So, we set $U_0 \coloneqq U \cap U'$.

Let $H = (F|_{W'})^{-1}$. Then (U_0, φ) is a smooth chart on P containing p, (W', η) is a smooth chart on M with $G(U_0) \subseteq W'$, and:

$$\eta \circ G \circ \varphi^{-1} = (\eta \circ H \circ \psi^{-1}) \circ (\psi \circ F \circ G \circ \varphi^{-1}) \text{ is a } C^k \text{ map.}$$

(b) If F is surjective and $G : N \to P$ is any map, then G is C^k if and only if $G \circ F$ is C^k.

$(\Longrightarrow)$
Again this is obvious.

$(\Longleftarrow)$
Consider any $p \in N$. Since F is surjective, we know there exists $q \in M$ such that $F(q) = p$. Next, since $G \circ F$ is a C^k map, we know there is a smooth chart (U, φ) on M containing q and a smooth chart (V, ψ) on P such that $(G \circ F)(U) \subseteq V$ and $\psi \circ G \circ F \circ \varphi^{-1}$ is a C^k map. But now as F is a local C^k diffomorphism, we can find an open set $U' \subseteq M$ containing q and an open set $W' \subseteq N$ such that F is a C^k diffeomorphism from U' to W'. Finally, by shrinking U' and W', we can assume that $U' \subseteq U$ and that there is a smooth chart (W', η) on N.

Now W' is an open set containing $p = F(q)$ with $G(W') = (G \circ F)(U')$ $\subseteq (G \circ F)(U) \subseteq V$. Also let $H = (F|_{U'})^{-1}$. Then:

$$\psi \circ G \circ \eta^{-1} = (\psi \circ G \circ F \circ \varphi^{-1}) \circ (\varphi \circ H \circ \eta^{-1}) \text{ is a } C^k \text{ map. } \blacksquare$$

7/26/2026

Welp, I've been fearing this day for a long time but I finally need to actually understand the rank theorem.

Theorem 4.12 (The Rank Theorem): Suppose M and N are C^k manifolds *without boundaries* of dimension m and n respectively, and also suppose $F : M \to N$ is a C^k map of constant rank r. Then for each $p \in M$ there exists a smooth chart (U, φ) on M with $p \in U$ as well as a smooth chart (V, ψ) on N with $F(U) \subseteq V$ such that $\varphi(p) = 0$ and $\widehat{F} := \psi \circ F \circ \varphi^{-1}$ has the form $\widehat{F}(x_1, \ldots, x_r, x_{r+1}, \ldots, x_m) = (x_1, \ldots, x_r, \underbrace{0, \ldots, 0}_{n-r \text{ many zeros}})$.

As a side note, if $k_1 > k_2$ in $\mathbb{Z}_{\geq 1} \cup \{\infty\}$ then every C^{k_1} structure on a manifold M is contained in a unique C^{k_2} structure. Therefore, it's not too too restrictive to assume F is as smooth a map as the structure on the manifold is smooth. That said, if I didn't make this restriction clear than one might falsely think that this theorem proves every constant rank C^1 map between C^k manifolds without boundaries is actually a C^k map.

Proof:
(Step 0: Simplying our notation)
Firstly, by choosing local coordinates around p it suffices to assume M is an open set $U \subseteq \mathbb{R}^m$ and N is an open set $V \subseteq \mathbb{R}^n$.

To see why, consider picking smooth charts $(\widetilde{U}, \widetilde{\varphi})$ on M containing p and $(\widetilde{V}, \widetilde{\psi})$ on N such that $F(\widetilde{U}) \subseteq \widetilde{V}$. Then set $U \coloneqq \widetilde{\varphi}(\widetilde{U})$ and $V \coloneqq \widetilde{\psi}(\widetilde{V})$ and note that U and V are open in $\mathbb{R}^m$ and $\mathbb{R}^n$ respectively. Once we've shown that the theorem holds for $\widetilde{\psi} \circ F \circ (\widetilde{\varphi})^{-1} : U \to V$, then we can just find two C^k diffeomorphisms $\varphi : U_0 \to \varphi(U_0)$ and $\psi : V_0 \to \psi(V_0)$ such that $\widetilde{\varphi}(p) \subseteq U_0 \subseteq U$, $V_0 \subseteq V$, $\varphi(\widetilde{\varphi}(p)) = 0$, and $\widetilde{\psi} \circ F \circ (\widetilde{\varphi})^{-1}(U_0) \subseteq V_0$, and $(\psi \circ \widetilde{\psi}) \circ F \circ ((\widetilde{\varphi})^{-1} \circ \varphi^{-1})$ has the desired form. And since $(\widetilde{\varphi}^{-1}(U_0), \varphi \circ \widetilde{\varphi})$ is a smooth chart on M, $(\widetilde{\psi}^{-1}(V_0), \psi \circ \widetilde{\psi})$ is a smooth chart on N, we'll have shown our theorem holds at p for the map F.

Next, note that since $\mathrm{D}F(p)$ has rank r, we know the Jacobian matrix has some $r \times r$ invertible submatrix. By reordering our coordinates, we can assume without loss of generality that the $r \times r$ invertible submatrix makes up the top left portion of $\mathrm{D}F(p)$. So, let us rename the standard coordinates on $\mathbb{R}^m$ and $\mathbb{R}^n$ as:

$$(x, y) \coloneqq (x_1, \ldots, x_r, y_{r+1}, \ldots, y_m) \text{ and } (v, w) \coloneqq (v_1, \ldots, v_r, w_{r+1}, \ldots, w_n).$$

By applying a translation to the open sets U and V, we can without loss of generality assume $p = (0, 0)$ and $F(p) = (0, 0)$. Also, we write $F(x, y) \coloneqq (Q(x, y), R(x, y))$ where $Q : U \to \mathbb{R}^r$ and $R : U \to \mathbb{R}^{n-r}$ are C^k functions. Therefore:

$$\mathrm{D}F(0,0) = \begin{bmatrix} \left[\frac{\partial Q_i}{\partial x_j}(0,0)\right]_{\substack{1 \le i \le r \\ 1 \le j \le r}} & \left[\frac{\partial Q_i}{\partial y_j}(0,0)\right]_{\substack{1 \le i \le r \\ r+1 \le j \le m}} \\ \left[\frac{\partial R_i}{\partial x_j}(0,0)\right]_{\substack{1 \le i \le n-r \\ 1 \le j \le r}} & \left[\frac{\partial R_i}{\partial y_j}(0,0)\right]_{\substack{1 \le i \le n-r \\ r+1 \le j \le m}} \end{bmatrix}$$

where $\left[\frac{\partial Q_i}{\partial x_j}(0,0)\right]_{\substack{1 \le i \le r \\ 1 \le j \le r}}$ is an invertible matrix.

(Step 1: Getting φ)

Now define $\varphi : U \to \mathbb{R}^m$ by $\varphi(x, y) = (Q(x, y), y)$. This will be C^k map whose total derivative is given by:

$$\mathrm{D}\varphi(x,y) = \begin{bmatrix} \left[\frac{\partial Q_i}{\partial x_j}(x,y)\right]_{\substack{1 \le i \le r \\ 1 \le j \le r}} & \left[\frac{\partial Q_i}{\partial y_j}(x,y)\right]_{\substack{1 \le i \le r \\ r+1 \le j \le m}} \\ \mathbf{0} & \mathbf{I}_{m-r} \end{bmatrix}$$

where $\mathbf{I}_{m-r}$ is the $(m-r) \times (m-r)$ identity matrix. Importantly, we can specifically see that all the columns of $\mathrm{D}\varphi(0, 0)$ are linearly independent. Hence, $\mathrm{D}\varphi(0, 0)$ is invertible. And by the inverse function theorem we know there are connected open sets $U_0 \subseteq U$ and $\widehat{U}_0 \subseteq \mathbb{R}^m$ satisfying that $(0, 0) \in U_0$, $\varphi(0, 0) = (0, 0) \in \widehat{U}_0$, and $\varphi : U_0 \to \widehat{U}_0$ is a C^k-diffeomorphism. By shrinking U_0 and $\widehat{U}_0$ we can without loss of generality assume that $\widehat{U}_0 = (-\varepsilon, \varepsilon)^m$ for some $\varepsilon > 0$.

Now we write the inverse map $\varphi^{-1} : \widehat{U}_0 \to U_0$ as $\varphi^{-1}(x, y) = (A(x, y), B(x, y))$ where $A : \widehat{U}_0 \to \mathbb{R}^r$ and $B : \widehat{U}_0 \to \mathbb{R}^{m-r}$ are C^k maps. Importantly, we know that:

$$(x, y) = \varphi(A(x, y), B(x, y)) = (Q(A(x, y), B(x, y)),\ B(x, y))$$

By comparing y-components we can see that $B(x, y) = y$. Hence, φ^{-1} has the form

$\varphi^{-1}(x,y) = (A(x,y),\ y)$. Meanwhile, by comparing x-components we can see that $x = Q(A(x,y),\ y)$. Therefore, upon letting $\widetilde{R}(x,y) \coloneqq R(A(x,y),y)$ then we have that:

$$(F \circ \varphi^{-1})(x,y) = (x,\ \widetilde{R}(x,y))$$

(Step 2: Simplifying $\widetilde{R}$)

Consider that the Jacobian matrix of $F \circ \varphi^{-1}$ at any $(x,y) \in \widehat{U}_0 = (-\varepsilon,\varepsilon)^m$ is given by:

$$\mathrm{D}(F \circ \varphi^{-1})(x,y) = \begin{bmatrix} \mathbf{I}_r & \mathbf{0} \\ \left[\frac{\partial \widetilde{R}_i}{\partial x_j}(x,y)\right]_{\substack{1 \le i \le n-r \\ 1 \le j \le r}} & \left[\frac{\partial \widetilde{R}_i}{\partial y_j}(x,y)\right]_{\substack{1 \le i \le n-r \\ r+1 \le j \le m}} \end{bmatrix}$$

Yet we also have that $\mathrm{D}(F \circ \varphi^{-1})(x,y) = \mathrm{D}F(\varphi^{-1}(x,y))\mathrm{D}(\varphi^{-1})(x,y)$. By assumption, we know that $\mathrm{rank}(\mathrm{D}F(\varphi^{-1}(x,y))) = r$ for all $(x,y) \in \widehat{U}_0$. Meanwhile, since φ^{-1} is a diffeomorphism, we know that $\mathrm{D}(\varphi^{-1})(x,y)$ defines a bijective linear map for all $(x,y) \in \widehat{U}_0$. It follows that $\mathrm{rank}(\mathrm{D}(F \circ \varphi^{-1})(x,y))$ will still equal r for all $(x,y) \in \widehat{U}_0$.

Yet the only way this matrix:

$$\begin{bmatrix} \mathbf{I}_r & \mathbf{0} \\ \left[\frac{\partial \widetilde{R}_i}{\partial x_j}(x,y)\right]_{\substack{1 \le i \le n-r \\ 1 \le j \le r}} & \left[\frac{\partial \widetilde{R}_i}{\partial y_j}(x,y)\right]_{\substack{1 \le i \le n-r \\ r+1 \le j \le m}} \end{bmatrix},$$

can ever have rank r is if $\left[\frac{\partial \widetilde{R}_i}{\partial y_j}(x,y)\right]_{\substack{1 \le i \le n-r \\ r+1 \le j \le m}} = \mathbf{0}$.

> This is because the first r columns are already linearly independent, and the intersection of their span with the span of the last $m-r$ columns is just the trivial subspace $\{0\}$.

Consequently, we know that $\widetilde{R}$ must be constant with respect to y. In other words, there exists some C^k map $S : (-\varepsilon,\varepsilon)^r \to \mathbb{R}^{n-r}$ such that $\widetilde{R}(x,y) = S(x)$. And with that, we can write:

$$(F \circ \varphi^{-1})(x,y) = (x, S(x))$$

(Step 3: Getting ψ)

Let $V_0 \coloneqq \{(v,w) \in V : (v,0) \in \widehat{U}_0\}$. Then V_0 will be an open subset of V containing $F(0,0) = (0,0)$. Also because $\widehat{U}_0$ is a cube and $(F \circ \varphi^{-1})(x,y) = (x,S(x))$, we can see that $(F \circ \varphi^{-1})(\widehat{U}_0) \subseteq V_0$. (And this also implies $F(U_0) \subseteq V_0$).

To finish off, define $\psi : V_0 \to \mathbb{R}^n$ by $\psi(v,w) = (v, w - S(v))$. Then ψ is a C^k map which is also a diffeomorphism onto its image. After all, by the invariance of domain theorem we know its image is an open subset of $\mathbb{R}^n$. Also, its inverse map is precisely given by $\psi(v,w) = (v, w + S(v))$.

And now just note that $(\psi \circ F \circ \varphi^{-1})(x,y) = \psi(x,S(x)) = (x, S(x) - S(x)) = (x,0)$.

■

Corollary 4.13: Let M and N be C^k manifolds without boundaries and let $F : M \to N$ be a C^k map. Also suppose M is connected. The following are equivalent.

(a) For each $p \in M$ there exists smooth charts (U, φ) on M and (V, ψ) on N with $p \in U$, $F(U) \subseteq V$, and $\widehat{F} := \psi \circ F \circ \varphi$ a linear map.

(b) F has constant rank.

Proof:
Firstly suppose (a) holds and for every $r \leq N := \min(\dim(M), \dim(N))$ let $A_r \subseteq M$ be the set of all $p \in M$ at which F has rank r. Trivially each A_r is disjoint with $\bigcup_{r=0}^{N} A_r = M$. By (a) we also know each A_r is an open set. By the connectedness of M we can thus conclude there exists a single r with $A_r = M$. So, F has constant rank on M.

Conversely, if (b) is true than the rank theorem tells us how to find the smooth charts described in (a). ■

Theorem 4.14: Let M, N be C^k manifolds without boundaries and let $F : M \to N$ be a C^1 map of constant rank.

(a) If F is surjective then F is a submersion.

Proof:
Let $m = \dim(M)$ and $n = \dim(N)$, and suppose F has constant rank r. We'll suppose to the contrary that F is *not* a submersion. Thus $r < n$.

Now without loss of generality we can assume $k = 1$ (i.e. that M and N are C^1 manifolds).

This is because the C^k structures $\mathcal{A}_M$ and $\mathcal{A}_N$ that M and N are equipped with are both contained in unique C^1 structures $\mathcal{B}_M$ and $\mathcal{B}_N$ respectively. Furthemore, a map $f : M \to \mathbb{R}$ is C^1 with respect to $\mathcal{A}_M$ if and only if it is C^1 with respect to $\mathcal{B}_M$. And a similar story holds for any $f : N \to \mathbb{R}$. It follows that the tangents spaces to M and N stay identical when each is equipped with either structure. In turn $F : (M, \mathcal{A}_M) \to (N, \mathcal{A}_N)$ is a submersion if and only if $F : (M, \mathcal{B}_M) \to (N, \mathcal{B}_N)$ is a submersion.

Due to the prior reasoning, the rank theorem now applies. So, for all $p \in M$ there exists a smooth chart (U_p, φ_p) on M containing p and a smooth chart (V_p, ψ_p) on N with $F(U_p) \subseteq V_p$ and $\widehat{F}_p : \psi \circ F \circ \varphi^{-1}$ having the form:

$$\widehat{F}_p(x_1, \ldots, x_r, x_{r+1}, \ldots, x_m) = (x_1, \ldots, x_r, \underbrace{0, \ldots, 0}_{\text{nonzero \# of zeros}}).$$

Since M is an LCH space, we can for each p pick a precompact open set $W_p \subseteq M$ such that $p \in W_p \subseteq \overline{W_p} \subseteq U_p$. Then as $\overline{W_p}$ is compact and N is Hausdorff, we know that $F(\overline{W_p})$ is a closed subset of N containing $F(W_p)$. Moreover, $F(\overline{W_p})$ has an empty interior. After all, let $\widehat{U}_p := \varphi_p(U_p)$ and $\widehat{V}_p := \psi_p(V_p)$. Then we have that $(\psi \circ F)(\overline{W_p}) \subseteq \widehat{F}_p(\widehat{U}_p) \subseteq$ (a hyperplane) $\cap\, \widehat{V}_p$. Since the latter set has an empty interior relative to $\widehat{V}_p$, we can conclude that $(\psi \circ F)(\overline{W_p})$ has an empty interior

relative to $\widehat{V}_p$ as well. Then since ψ is a homeomorphism from V_p to $\widehat{V}_p$, we can conclude that $F(\overline{W_p})$ has an empty interior relative to V_p. And since $F(\overline{W_p})$ is a closed subset of N containing $F(W_p)$ and contained in V_p, we can finally conclude that $F(W_p)$ is nowhere dense.

But now to get our contradiction, note that because M is second countable there must be a sequence $\{p_\ell\}_{\ell\in\mathbb{N}}$ such that $M = \bigcup_{\ell\in\mathbb{N}} W_{p_\ell}$. Since F is surjective, we in turn know that $N = F(M) = \bigcup_{\ell\in\mathbb{N}} F(W_{p_\ell})$. So, we've proven that N is a countable union of nowhere dense sets. In other words N is meager. Yet N is an LCH space and so this contradicts the Baire category theorem. [See *pages 540-541*.]

(b) If F is injective then F is an immersion.

Proof:
Let m, n, and r be as in the prior proof, and this time suppose to the contrary that F is *not* an immersion. For identical reasons as in the last proof, we can assume without loss of generality that $k = 1$ (i.e. that M and N are C^1 manifolds). And this is important because it lets us apply the rank theorem.

Consider any $p \in M$ and let (U, φ) and (V, ψ) be smooth charts on M such that $p \in U$, $F(U) \subseteq V$, and $\widehat{F} \coloneqq \psi \circ F \circ \varphi^{-1}$ has the form:

$$\widehat{F}(x_1, \ldots, x_r, \underbrace{x_{r+1}, \ldots, x_m}_{\text{nonzero \# of coordinates}}) = (x_1, \ldots, x_r, \underbrace{0, \ldots, 0}_{n-r \text{ many zeros}}).$$

But now F being injective implies that $\widehat{F}$ is also injective. Simultaneously, write $\varphi(p) = (p_1, \ldots, p_r, p_{r+1}, \ldots, p_m)$. Since $\varphi(U)$ is open in $\mathbb{R}^n$ we know for some small enough $\varepsilon > 0$ that $\widehat{q} \coloneqq (p_1, \ldots, p_r, p_{r+1} + \varepsilon, \ldots, p_m) \in \varphi(U)$. But now if we let $q = \varphi^{-1}(q)$ we have that $(\psi \circ F)(p) = \widehat{F}(\varphi(p)) = \widehat{F}(\varphi(q)) = (\psi \circ F)(q)$. Since ψ is injective, we in turn have that $F(p) = F(q)$. Yet $p \neq q$ since φ^{-1} is injective. So, this all contradicts that F is injective. ■

Corollary: Let M, N be C^r manifolds without boundaries and let $F : M \to N$ be a C^k map of constant rank (where $k \leq r$). If F is a bijection then F is a C^k diffeomorphism.

Proof:
By the prior theorem, F being a bijection means that F is both a submersion and an immersion. It then follows by proposition 4.8 on *page 1061* that F is a local C^k diffeomorphism. Yet finally, we can easily see by the pasting lemma that any local C^k diffeomorphism that is also bijective is a (global) C^k diffeomorphism. So, F is a C^k diffeomorphism. ■

7/27/2026

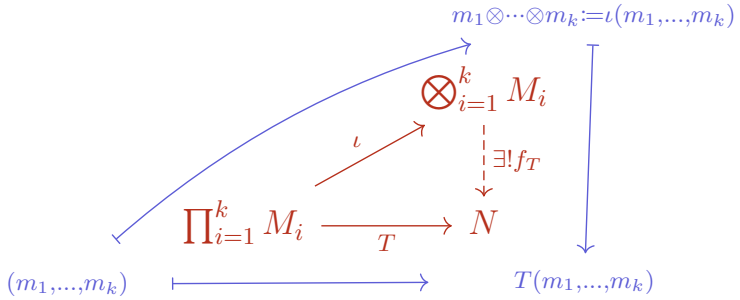
I mentioned a few days ago that my construction of the tangent space to a corner point might not be optimal. After talking to a colleague, I can now say that while nothing I've

written is wrong, there is a cleaner equivalent way of defining the tangent space. This improved way will also be the way that generalizes to analytic and complex manifolds. So, it wouldn't hurt to go over it now.

Firstly, I'm going to fix an oversight I priorly made on *page 1024*. I already deleted the false statement on that page but I did leave a note explaining where my deleted remark went wrong.

Firstly, since it will make my life easier and serve as a good review, I want to generalize the universal property of tensor products of module. Note that this will only use knowledge from math 200B.

Given a collection of A-modules $M_1, \dots, M_k$, we say their tensor product $\bigotimes_{i=1}^k M_i$ is an A-module equipped with a k-linear map $\iota : \prod_{i=1}^k M_i \to \bigotimes_{i=1}^k M_i$ such that for any k-linear map $T : \prod_{i=1}^k M_i \to N$ the following diagram holds:



In math 200b I learned how to construct tensor products of 2 modules and I also showed that $M_1 \otimes (M_2 \otimes M_3) \cong (M_1 \otimes M_2) \otimes M_3$. But for the sake of better notation and easier thinking, I want to be able to talk about tensor products of any finite number of modules. Fortunately, by standard reasoning we can see that any two A-modules satisfying the universal property in the above diagram are isomorphic.

Lemma A: Let $k \geq 2$ and suppose $M_1, \dots, M_k$ are all A-modules. Then there exists a module $\bigotimes_{i=1}^k M_i$ and a k-linear map $\iota : \prod_{i=1}^k M_i \to \bigotimes_{i=1}^k M_i$ fulfilling the above universal property. Furthermore, $\iota(\prod_{i=1}^k M_i)$ is a generating / spanning set for $\bigotimes_{i=1}^k M_i$.

Proof:

Our base case of $k = 2$ is already covered. So let us now proceed by induction and assume that we've already defined $\bigotimes_{i=2}^k M_i$ and $\bar{\iota} : \prod_{i=2}^k M_i \to \bigotimes_{i=2}^k M_i$. Then let's define $M := M_1 \otimes (\bigotimes_{i=2}^k M_i)$ and $\iota : \prod_{i=1}^k M_i \to M$ by:

$$\iota(m_1, \dots, m_k) := m_1 \otimes \bar{\iota}(m_2, \dots, m_k).$$

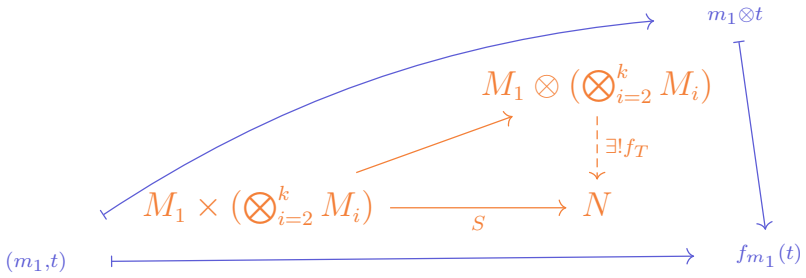
It's easy to see that ι is a k-linear map. Furthemore, suppose $T : \prod_{i=1}^k M_i \to N$ is any k-linear map. Then for any fixed $m_1 \in M_1$, we can see that $T_{m_1} : \prod_{i=2}^k M_i \to N$ defined by $T_{m_1}(m_2, \dots, m_k) := T(m_1, \dots, m_k)$ is a well-defined $(k-1)$-linear map. So by the universal property of $\bigotimes_{i=2}^k M_i$ and $\bar{\iota} : \prod_{i=2}^k M_i \to \bigotimes_{i=2}^k M_i$, we know that there exists a unique A-module homomorphism $f_{m_1} : \bigotimes_{i=2}^k M_i \to N$ such that $f_{m_1} \circ \bar{\iota} = T_{m_1}$.

Next, we define $S : M_1 \times (\bigotimes_{i=2}^k M_i) \to N$ by $S(m_1, t) \coloneqq f_{m_1}(t)$. Since each f_{m_1} is A-linear, we know that S is A-linear in its second variable. Meanwhile, note that:

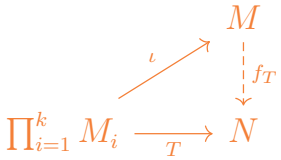
$$\begin{aligned} f_{am_1+a'm_1'}(\bar{\iota}(m_2, \ldots, m_k)) &= T(am_1 + a'm_1', m_2, \ldots, m_k) \\ &= aT(m_1, m_2, \ldots, m_k) + a'T(m_1', m_2, \ldots, m_k) \\ &= af_{m_1}(\bar{\iota}(m_2, \ldots, m_k)) + a'f_{m_1'}(\bar{\iota}(m_2, \ldots, m_k)). \end{aligned}$$

It follows S is A-linear in its first variable if its second variable is contained in $\bar{\iota}(\prod_{i=2}^k M_i)$. And since S is a A linear in its second variable and $\bar{\iota}(\prod_{i=2}^k M_i)$ spans all of $\bigotimes_{i=2}^k M_i$, we can in turn conclude that S is A-linear in its first variable regardless of the value of its second variable. And with that we've shown S is a bilinear map.

By the universal property of tensor products, we know there is a unique A-module homomorphism $f_T : M_1 \otimes (\bigotimes_{i=2}^k M_i) \to N$ such that the below diagram commutes:



I further claim that f_T is the unique A-module homomorphism making this diagram commute:



After all, the fact that this diagram commutes is simply because:

$$\begin{aligned} f_T(\iota(m_1, \ldots, m_k)) &= f_T(m_1 \otimes \bar{\iota}(m_2, \ldots, m_k)) \\ &= S(m_1, \bar{\iota}(m_2, \ldots, m_k)) = f_{m_1}(\bar{\iota}(m_2, \ldots, m_k)) \\ &= T(m_1, m_2, \ldots, m_k). \end{aligned}$$

Meanwhile, if g was another A-module homomorphism making the diagram commute, we'd similarly be able to say that:

$$\begin{aligned} g(m_1 \otimes \bar{\iota}(m_2, \ldots, m_k)) &= g((\iota(m_1, \ldots, m_k))) \\ &= T(m_1, m_2, \ldots, m_k) = f_{m_1}(\bar{\iota}(m_2, \ldots, m_k)) \\ &= S(m_1, \bar{\iota}(m_2, \ldots, m_k)). \end{aligned}$$

In turn since g is A-linear and $\bar{\iota}(\prod_{i=2}^k M_i)$ spans all of $\bigotimes_{i=2}^k M_i$, we can conclude that $g(m_1 \otimes t) = S(m_1, t)$ for all $(m_1, t) \in M_1 \times (\bigotimes_{i=2}^k M_i)$. By the uniqueness of f_T, we thus must have that $g = f_T$. And with that, we've proven that M equipped with ι satisfies the universal property of a tensor product of k many modules.

To finish off the proof, we already know our construction of $M_1 \otimes (\bigotimes_{i=2}^k M_i)$ in math 200b that that space is spanned by pure tensors of the form $m_1 \otimes t$ where $t \in \bigotimes_{i=2}^k M_i$. So, we can write any arbitrary element in M as a sum of pure tensors $\sum_{j=1}^{\ell} m_1^{(j)} \otimes t^{(j)}$.

(And via A-balance I'm letting $m_1^{(j)}$ absorb any scalars in this sum.)

But by induction we can then write $t^{(j)} = \sum_{j'=1}^{r_j} \bar{\iota}(m_2^{(j,j')}, \ldots, m_k^{(j,j')})$ (where again I'm letting the $m_i^{(j,j')}$ absorb any of the scalars). So:

$$\begin{aligned}\sum_{j=1}^{\ell} m_1^{(j)} \otimes t^{(j)} &= \sum_{j=1}^{\ell} \sum_{j'=1}^{r_j} m_1^{(j)} \otimes \bar{\iota}(m_2^{(j,j')}, \ldots, m_k^{(j,j')}) \\ &= \sum_{j=1}^{\ell} \sum_{j'=1}^{r_j} \iota(m_1^{(j)}, m_2^{(j,j')}, \ldots, m_k^{(j,j')}). \blacksquare\end{aligned}$$

As a side note: The prior proof can also be interpretted as saying that $M_1 \otimes (M_2 \otimes (\cdots \otimes (M_{k-1} \otimes M_k) \cdots))$ along with the k-linear map $\iota(m_1, \ldots, m_k) = m_1 \otimes (m_2 \otimes (\cdots \otimes (m_{k-1} \otimes m_k) \cdots))$ satisfies the universal property of a tensor product of k many A-modules.

Note that we can by identical reasoning then prove an analogous proposition as on *pages 726-728* if $M_1, \ldots, M_k$ are A-algebras (albeit with more cumbersome notation).

In other words, if $M_1, \ldots, M_k$ are A-algebras then $\bigotimes_{i=1}^k M_i$ is also an A-algebra when equipped with the multiplication operation:

$$(m_1 \otimes \cdots \otimes m_k)(m_1' \otimes \cdots \otimes m_k') = m_1 m_1' \otimes \ldots \otimes m_k m_k'$$

Similarly, given A-module homomorphisms $f_i : M_i \to M_i'$ we can show identically as on *page 700* that there is a unique A-module homomorphism $\bigotimes_{i=1}^k M_i \to \bigotimes_{i=1}^k M_i'$ such that $f(m_1 \otimes \cdots \otimes m_k) = f(m_1) \otimes \cdots \otimes f(m_k)$.

Lemma B: Suppose $N, M_1, \ldots, M_k$ are commutative A-algebras and $f_i : M_i \to N$ is an A-algebra homomorphism for $1 \leq i \leq k$. Then there is a unique A-module homomorphism $f : \bigotimes_{i=1}^k M_i \to N$ such that $f(m_1 \otimes \cdots \otimes m_k) = \prod_{i=1}^k f_i(m_i)$.

Proof:

To start off, if we define $T : \prod_{i=1}^k M_i \to N$ by $T(m_1, \ldots, m_k) = \prod_{i=1}^k f_i(m_i)$ then we can easily see that T is k-linear. So by the universal property of $\bigotimes_{i=1}^k M_i$, we know there is a unique A-module homomorphism $f : \bigotimes_{i=1}^k M_i \to N$ such that:

$$f(m_1 \otimes \cdots \otimes m_k) = \prod_{i=1}^k f_i(m_i).$$

What we need to do now though is show that f also happens to be algebra homomorphism. This is not difficult but involves a lot of tedious symbol pushing which I don't want to type out. $\blacksquare$

Another observation to make is that if each $f_i : M_i \to N$ is unital then f is also unital.

7/31/2026

This tangent has actually become significantly longer and deeper than I originally thought it would be. I just don't know enough about algebras. So at this point I'm going to start following the textbook *Multilinear Algebra* by Douglas Northcott. See *citation 12* for more information. Also I'm going to switch back to the normal color scheme.

For the time being, I'll let R denote a fixed commutative unital ring. I'm letting R as opposed to A denote the ring both because the ring I will care about eventually is just $\mathbb{R}$ and also because this frees up A to refer to an R-algebra.

Let A be an R-algebra (which doesn't have to be commutative). We say a family $\{A_n\}_{n\in\mathbb{Z}}$ of R-submodules of A is a grading on A if $A = \bigoplus_{n\in\mathbb{Z}} A_n$ as an internal direct sum of R-modules and $a_r \in A_r$ and $a_s \in A_s$ implies that $a_r a_s \in A_{r+s}$.

- An R-algebra with a grading $\{A_n\}_{n\in\mathbb{Z}}$ is said to be a graded R-algebra. If all $A_n = \{0\}$ for $n < 0$ then we say our grading is nonnegative.
- Also any elements of A_n are called homogeneous of degree n.
- By definition, any $a \in A$ can be written uniquely as $\sum_{n\in\mathbb{Z}} a_n$ where each $a_n \in A$ and all but finitely many a_n are zero. We call the a_n in this sum the homogeneous coefficients of a.

Theorem 3.5: Let A be a unital graded R-algebra with the grading $\{A_n\}_{n\in\mathbb{N}}$. Then 1_A is homogeneous of degree 0.

Let us write $1_A = \sum_{n\in\mathbb{Z}} a_n$ as a sum of the homogeneous coefficients of 1_A. Then for any $m \in \mathbb{Z}$ we consider any $b_m \in A_m$. But now note that:

$$\sum_{n\in\mathbb{Z}} a_n b_m = 1_A b_m = b_m = b_m 1_A = \sum_{n\in\mathbb{Z}} b_m a_n$$

By comparing homogeneous coefficients, we get that $a_0 b_m = b_m = b_m a_0$ (and also that $a_n b_m = 0 = b_m a_n$ if $n \neq 0$). And thus since every $b \in A$ is a sum of $b_m \in A_m$ as m ranges over some finite subset of $\mathbb{Z}$, we can conclude for all $b \in A$ that $a_0 b = b = b a_0$. This shows that a_0 is an identity element of A. And thus $1_A = 1_A a_0 = a_0$. ■

Given a unital R-algebra A, I'll say $B \subseteq A$ is a subalgebra if it is also an R-algebra under the algebra operations of A and $1_B = 1_A$.

Corollary: Let A be a graded unital R-algebra with the grading $\{A_n\}_{n\in\mathbb{N}}$. Then A_0 is a subalgebra of A.

Why?

A_0 is an R-module and we know from the definition of a grading that if $a, b \in A_0$ then so is $ab \in A_0$. Finally, $1_A \in A_0$ lets us conclude A_0 is a subalgebra. ■

Let A be an R-algebra. Note that if $\{B_i\}_{i\in I}$ is a collection of subalgebras of A then so is $B := \bigcap_{i\in I} B_i$. As a result, given a subset S of A we let the R-subalgebra generated by S (denoted $\langle S\rangle$) to be the smallest subalgebra of A containing S. We can construct $\langle S\rangle$ as follows:

Let B be the collection of all finite linear combinations of finite products of elements of S (where a product of zero-many elements is interpretted as equalling 1_A). It is easy to justify that $B \subseteq \langle S \rangle$ and that B is itself a subalgebra of A which contains S. So, $\langle S \rangle \subseteq B$.

Lemma 3.1: Let $\{A_n\}_{n\in\mathbb{Z}}$ be a grading on a unital R-algebra A, and suppose that $A = \langle A_1 \rangle$. Then the grading on A is nonnegative and for all $n \geq 0$ we have that any $a \in A_n$ is a finite linear combinations of products of n many elements of A_1. In particular, this means that A_0 is generated as an R-module by 1_A. So, A_0 is contained in the center of A.

Proof:
A is generated as an R-module by all finite products of elements of A_1. Therefore, let $B_0 = R1_A$, let $B_n = \{0\}$ for $n < 0$, and finally for $n > 0$ let B_n be the R-module generated by all finite products of n many elements of A_1. We know $A = \sum_{n\in\mathbb{Z}} B_n$, and we can also easily see that $B_n \subseteq A_n$ for all n. The only way both statements can be true is if $A_n = B_n$ for all n. ■

Let A and B be unital graded R-algebras with gradings $\{A_n\}_{n\in\mathbb{Z}}$ and $\{B_n\}_{n\in\mathbb{Z}}$ respectively. We say $f : A \to B$ is a homomorphism of graded R-algebras if it is a unital R-algebra homomorphism and $f(A_n) \subseteq B_n$ for all n.

Lemma: If $f : A \to B$ is a homomorphism of graded R-algebras and a bijection, then $f(A_n) = B_n$ for all n.

Proof:
Consider any $n \in \mathbb{Z}$ and $b_n \in B_n$. Then write $f^{-1}(b_n) = \sum_{m\in\mathbb{Z}} a_m$ where $a_m \in A_m$ for each m. In turn, $b_n = f(f^{-1}(b_n)) = f(\sum_{m\in\mathbb{Z}} a_m) = \sum_{m\in\mathbb{Z}} f(a_m)$. So if $m \neq 0$ then we can compare homogeneous coefficients of b_n to get that $f(a_m) = 0$. Since f is injective, it follows that $a_m = 0$ for all $m \neq 0$. And so, we've shown that $f^{-1}(b_n) \in A_n$. In other words, $f^{-1}(B_n) \subseteq A_n$. And since $f(A_n) \subseteq B_n$, we can now conclude that $f(A_n) = B_n$. ■

Let $I \subseteq A$ be a two-sided ideal of a graded unital R-algebra A. We say I is homogeneous if $a \in I$ implies that all the homogeneous components of A are also in I.

Lemma 3.2: Let $I \subseteq A$ be a two-sided ideal of a graded unital R-algebra A (with a grading $\{A_n\}_{n\in\mathbb{Z}}$). Then I is homogeneous if and only if it can be generated by homogeneous elements.

(Note: to distinguish between the subalgebra generated by a set $S \subseteq A$ and an ideal generated by a set $S \subseteq A$, I'll denote the objects $\langle S \rangle$ and (S) respectively.)

$(\Longrightarrow)$
Let S be a generating set for I. Then if I is homogeneous, we know the set S' of all the homogeneous components of S is also contained in I. Therefore, $I = (S) \subseteq (S') \subseteq I$.

$(\Longleftarrow)$
Suppose S is a set of homogeneous elements such that $(S) = I$. Then note that any element $y \in I$ is a finite sum of elements of the form $a_j x_j b_j$ where $a_j, b_j \in A$ and $x_j \in S$ for all j. By writing each a_j and b_j as finite sums of homogeneous coefficients, we can in turn say without loss generality that any element $y \in I$ is a (larger) finite sum of

elements $a_j x_j b_j$ where a_j and b_j are homogeneous in A and $x_j \in S$. But now each term $a_j x_j b_j$ is homogeneous and in I. So by collecting together homogeneous coefficients, we can say that each the homogeneous coefficients of y are all in I. ■

Now if A is an R-algebra and $I \subseteq A$ is a two-sided ideal of A, then we know that $B \coloneqq A/I$ is a well-defined ring and R-module when equipped with the operations $c(a+I) \coloneqq (ca)+I$, $(a+I)+(b+I) = (a+b)+I$, and $(a+I)(b+I) \coloneqq (ab)+I$ for all $c \in R$ and $a, b \in A$. Then proving that B is also an R-algebra is easy. And if A is unital then so is B.

Moreover, suppose A is graded by $\{A_n\}_{n\in\mathbb{Z}}$ and that I is homogeneous. Then there is a standard grading on $B = A/I$.

Put $I_n = I \cap A_n$ for all n. Then we can see that $I = \bigoplus_{n\in\mathbb{Z}} I_n$ as an internal direct sum of R-modules. Also put $B_n = (A_n + I)/I$. Then the B_n will be R-submodules of B such that $B = \sum_{n\in\mathbb{Z}} B_n$.

Claim: $\sum_{n\in\mathbb{Z}} B_n$ is a direct sum of R-modules.

Why?

Suppose $\sum_{n\in\mathbb{Z}} b_n = 0$ where each $b_n \in B_n$ and all but finitely many b_n are zero. Then for each n we can pick $a_n \in A_n$ such that $b_n = a_n + I$ (and $a_n = 0$ if $b_n = 0$). Yet $0 + I = (\sum_{n\in\mathbb{Z}} a_n) + I$ implies that $\sum_{n\in\mathbb{Z}} a_n \in I$. Since I is homogeneous, we can thus conclude $a_n \in I$ for each i. And thus, $b_n = a_n + I = 0 + I$ for all n.

Since it's obvious that if $b_r \in B_r$ and $b_s \in B_s$ then $b_r b_s \in B_{r+s}$, we can now conclude that $B = A/I$ is graded by the submodules $\{B_n = (A_n + I)/I\}_{n\in\mathbb{Z}}$.

Furthermore, consider the quotient map $\pi : A \twoheadrightarrow A/I$. When A/I is equipped with the grading described above, we have that π is a homomorphism of graded algebras. By applying the first isomorphism theorem for modules, we can see that $(A_n+I)/I \cong A_n/I_n$ as R-modules. In turn, $A/I \cong \bigoplus_{n\in\mathbb{Z}} A_n/I_n$ as R-modules.

8/1/2026

First and foremost, today I want to go over some variations on tensor powers.

Observation: Let $M_1, \ldots, M_k$ be free R-modules and in particular suppose $M_i = \bigoplus_{b_i \in B_i} Rb_i$ for each i. Then for any R-module N, any map $\phi : \prod_{i=1}^k B_i \to N$ has a unique extension to a k-linear map $\widetilde{\phi} : \prod_{i=1}^k M_i \to N$.

This is because the only thing the extension $\widetilde{\phi}$ of ϕ can possibly be is:

$$\widetilde{\phi}(m)\left(\sum_{b_1\in B_1} c_{b_1} b_1, \ldots, \sum_{b_k \in B_k} c_{b_k} b_k\right) = \sum_{b_1\in B_1,\ldots,b_k\in B_k} c_{b_1}\cdots c_{b_k}\phi(b_1,\ldots,b_k)$$

Theorem 1.3: Let $M_1, \ldots, M_k$ be free R-modules and in particular suppose $M_i = \bigoplus_{b_i \in B_i} Rb_i$ for each i. Then $\bigotimes_{i=1}^k M_i = \bigoplus_{b_1\in B_1,\ldots,b_k\in B_k} R(b_1 \otimes \cdots \otimes b_k)$ where the latter is an internal direct sum. This also shows that $\bigotimes_{i=1}^k M_i$ is a free module (although I already knew this via a less specific result in math 200b).

Proof:
Let $M := \bigoplus_{b_1 \in B_1, \ldots, b_k \in B_k} R(b_1, \ldots, b_k)$ be the free R-module associated to the set $\prod_{i=1}^k B_i$. Then there is a unique k-linear map $\iota : \prod_{i=1}^k M_i \to M$ satisfying that $\iota(b_1, \ldots, b_k) = (b_1, \ldots, b_k) \in M$.

Furthermore, suppose N is any other R-module and $T : \prod_{i=1}^k M_i \to N$ is a k-linear map. If we want the following commuting diagram to hold:

$$\begin{array}{ccc} & & M \\ & \overset{\iota}{\nearrow} & \big\downarrow f \\ \prod_{i=1}^k M_i & \xrightarrow[T]{} & N \end{array}$$

then we need $f(b_1, \ldots, b_n) = T(b_1, \ldots, b_n)$ for all $(b_1, \ldots, b_n) \in \prod_{i=1}^k B_i$. Fortunately, by the universal property of free modules we know a unique such $f : M \to N$ exists. And now since $f \circ \iota$ is a k-linear map agreeing with T on the set $\prod_{i=1}^k B_i$, we can conclude by the prior observation that $f \circ \iota = T$.

With that, we've proven that M and ι satisfy the universal property of a tensor product of $M_1, \ldots, M_k$. And now it is easy to see that $\bigotimes_{i=1}^k M_i$ will be isomorphic to M via an isomorphism sending $b_1 \otimes \cdots \otimes b_k$ to $(b_1, \ldots, b_k) \in M$. ■

Now given any R-module M and $p \in \mathbb{N}$, we let $T_p(M) := \bigotimes_{i=1}^p M$. This R-module is called the pth tensor power of M. By the prior theorem, if M is a free module with a linearly independent basis $B \subseteq M$ then $T_p(M)$ is also a free module with a linearly independent basis given by $\{b_1 \otimes \cdots \otimes b_p : b_i \in B \text{ for all } 1 \leq i \leq p\}$.

We also want to consider liftings of alternating or symmetric p-linear maps on $\prod_{i=1}^p M$. So, we consider the following constructions.

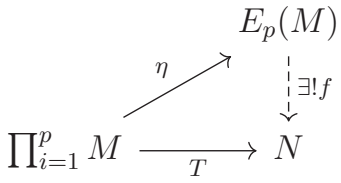
If N is an R-module and $p > 1$ then we say that a p-linear map $T : \prod_{i=1}^p M \to N$ is alternating if $T(m_1, \ldots, m_p) = 0$ if $m_i = m_j$ for any distinct integers i and j. Meanwhile, we say all R-module homomorphisms (i.e. all 1-linear maps) are alternating.

Note that the definition of an alternating k-linear map given on *page 104* implies this new definition provided that $1_R + 1_R \neq 0_R$. Conversely, note that:

$$\begin{aligned} &T(m_1, \ldots, m_i, \ldots, m_j, \ldots m_p) \\ &\quad = T(m_1, \ldots, m_i + m_j, \ldots, m_j, \ldots, m_p) - T(m_1, \ldots, m_j, \ldots, m_j, \ldots, m_p) \\ &\quad = T(m_1, \ldots, m_i + m_j, \ldots, m_j, \ldots, m_p) - 0 \\ &\quad = T(m_1, \ldots, m_i + m_j, \ldots, m_i + m_j, \ldots, m_p) \\ &\qquad\qquad - T(m_1, \ldots, m_i + m_j, \ldots, m_i, \ldots, m_p) \\ &\quad = 0 - T(m_1, \ldots, m_i + m_j, \ldots, m_i, \ldots, m_p) \\ &\quad = -T(m_1, \ldots, m_j, \ldots, m_i, \ldots, m_p) - T(m_1, \ldots, m_i, \ldots, m_i, \ldots, m_p) \\ &\quad = -T(m_1, \ldots, m_j, \ldots, m_i, \ldots, m_p) - 0. \end{aligned}$$

Therefore, by composing transpositions we can conclude that our new definition always implies the old definition when M is a vector space over any field.

We say an R-module $E_p(M)$ along with an alternating p-linear map $\eta : \prod_{i=1}^p M \to E_p(M)$ is the pth exterior power of M if for any R-module N and any alternating p-linear map $T : \prod_{i=1}^p M \to N$ there is a unique R-module homomorphism $f : E_p(M) \to N$ such that $f \circ \eta = T$.



Proposition: Such an R-module $E_p(M)$ and alternating p-linear map η exists, and the elements $m_1 \wedge \cdots \wedge m_p := \eta(m_1, \ldots, m_p)$ generate $E_p(M)$ as an R-module. Also if $(E_p(M)^{(1)}, \eta^{(1)})$ and $(E_p(M)^{(2)}, \eta^{(2)})$ both satisfy the above universal property then there exists inverse isomorphism $f : E_p(M)^{(1)} \to E_p(M)^{(2)}$ and $g : E_p(M)^{(2)} \to E_p(M)^{(1)}$ such that $f \circ \eta^{(1)} = \eta^{(2)}$ and $f \circ \eta^{(2)} = \eta^{(1)}$.

Proof:
The uniqueness of $E_p(M)$ upto isomorphism claim can be verified through the same standard arguements that were used on page 686. So, it's only the first sentence of the propositon that needs verifying.

Consider the pth tensor power of M and let $J_p(M) \subseteq T_p(M)$ be the submodule generated by all pure tensors $m_1 \otimes \cdots \otimes m_p$ where $(m_1, \ldots, m_p)$ contains at least one repeated element.

Note that $J_1(M)$ is generated by an empty set of elements and so we just take by convention that $J_1(M) = \{0\}$.

Next, we let $E_p(M) = T_p(M)/J_p(M)$. And assuming $\pi : T_p(M) \twoheadrightarrow E_p(M)$ is the quotient map and $\iota : \prod_{i=1}^p M \to T_p(M)$ is the multilinear map sending $(m_1, \ldots, m_p)$ to $m_1 \otimes \cdots \otimes m_p$, we define $\eta := \pi \circ \iota$. Thus η is a multilinear map as well. And by the definition of $J_p(M)$ we know that η is alternating.

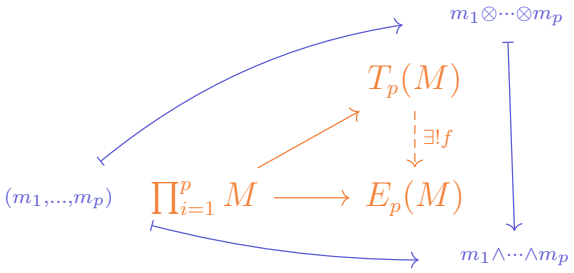
I next claim that $E_p(M)$ and η satisfy the universal property we want. To prove this, suppose N is any R-module and $\phi : \prod_{i=1}^p M \to N$ is an alternating p-linear map. By the universal property of $T_p(M)$ we know there is a unique R-module homomorphism $f : T_p(M) \to N$ such that $f(m_1 \otimes \cdots \otimes m_p) = \phi(m_1, \ldots, m_p)$ for all $(m_1, \ldots, m_p)$ in $\prod_{i=1}^p M$. Yet we can also see that each of the generating elements of $J_p(M)$ are in $\ker(f)$. Hence, $J_p(M) \subseteq \ker(f)$ and we can quotient them out to get a well-defined R-module homomorphism $\widetilde{f} : E_p(M) \to N$.

To show that $\widetilde{f}$ is unique first we note that since $\pi : T_p(M) \to E_p(M)$ is a surjective map and the pure tensors $m_1 \otimes \cdots \otimes m_p$ generate $T_p(M)$ as an R-module, we know that the elements of the form $m_1 \wedge \cdots \wedge m_p = \pi(m_1 \otimes \cdots \otimes m_p)$ also generate $E_p(M)$ as an R-module. Next note that if $g : E_p(M) \to N$ were another R-module homomorphism such that $\phi = g \circ \eta$, then we'd have to have that $g = \widetilde{f}$ on all elements of the form $m_1 \wedge \cdots \wedge m_p$. Since those elements generate all of $E_p(M)$, we in turn know that $g = \widetilde{f}$ everywhere on $E_p(M)$. ■

In general it will be convenient to view the structure of $E_p(M)$ as a blackbox and just rely on its universal property (similarly to how we were already treating $T_p(M)$). And with that mentioned, I now define the canonical homomorphism from $T_p(M)$ to $E_p(M)$:

Note that the map $\eta(m_1, \ldots, m_p) \coloneqq m_1 \wedge \cdots \wedge m_p$ is a p-linear map from $\prod_{i=1}^{p} M$ into $E_p(M)$. As a result, we know by the universal property of $T_p(M)$ that there is a unique R-module homomorphism $f : T_p(M) \to E_p(M)$ such that:

$$f(m_1 \otimes \cdots \otimes m_p) = m_1 \wedge \cdots \wedge m_p.$$



In my explicit construction of a model of $E_p(M)$ on the prior page, f would just be the quotient map π and the next theorem would be obvious. However, again I'm trying to treat $E_p(M)$ and $T_p(M)$ as black boxes.

Theorem 1.5: Let $f : T_p(M) \to E_p(M)$ be the canonical homomorphism and let $J_p(M)$ be the submodule of $T_p(M)$ defined on the prior page. Then f is surjective and $\ker(f) = J_p(M)$. Moreover, $J_p(M)$ is generated as an R-module by all pure tensors $m_1 \otimes \cdots \otimes m_p$ where $m_i = m_{i+1}$ for some i.

Proof:

The fact that f is surjective is simply a result of the fact that f maps a generating set for $T_p(M)$ (i.e. the set of pure tensors) to a generating set for $E_p(M)$. As for showing the rest of the theorem's claim, let $J \coloneqq J_p(M)$, let J' be the R-submodule of $T_p(M)$ generated by just the pure tensors $m_1 \otimes \cdots \otimes m_p$ where $m_i = m_{i+1}$ for some i, and finally let $J'' \coloneqq \ker(f)$. It's clear that $J' \subseteq J \subseteq J''$. And we just need to show all these submodules are equal.

Lemma 1.2: Let $\phi : \prod_{i=1}^{p} M \to N$ be a p-linear map. Then ϕ is alternating iff $\phi(m_1, \ldots, m_p) = 0$ whenever there exists i such that $m_i = m_{i+1}$.

Why?

The forwards implication is trivial. Conversely, note by identical reasoning as on two pages, if the latter claim holds then for any i we have that:

$$\phi(m_1, \ldots, m_i, m_{i+1}, \ldots, m_p) = -\phi(m_1, \ldots, m_{i+1}, m_i, \ldots, m_p).$$

But now suppose $m_i = m_j$ for some indices $i < j$. Then:

$$\begin{aligned}\phi(m_1, \ldots, m_i, m_{i+1}, m_{i+2}, \ldots, m_j, \ldots m_p) \\ &= -\phi(m_1, \ldots, m_{i+1}, m_i, m_{i+2} \ldots, m_j, \ldots m_p) \\ &= +\phi(m_1, \ldots, m_{i+1}, m_{i+2}, m_i \ldots, m_j, \ldots m_p) \\ &= \cdots = \pm\phi(m_1, \ldots, m_{j-2}, m_i, m_j, \ldots, m_p) = \pm 0 = 0.\end{aligned}$$

So, ϕ is alternating.

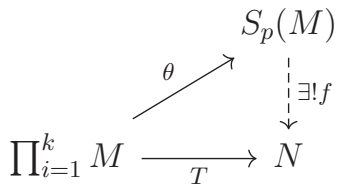
Based on the prior lemma, we can now say that if $\pi : T_p(M) \to T_p(M)/J'$ is the quotient map and $\iota : \prod_{i=1}^p M \to T_p(M)$ is the multilinear map $\iota(m_1, \ldots, m_p) = m_1 \otimes \cdots \otimes m_p$, then $\pi \circ \iota$ is an alternating multilinear map. It follows by the universal property of $E_p(M)$ that there is a unique R-module homomorphism $\lambda : E_p(M) \to T_p(M)/J'$ such that $\lambda(m_1 \wedge \cdots \wedge m_p) = (m_1 \otimes \cdots \otimes m_p) + J'$.

Yet also note that $\lambda \circ f = \pi$ After all, by the universal property of $T_p(M)$ we know that $h = \pi$ is the unique R-module homomorphism $T_p(M) \to T_p(M)/J'$ with $h \circ \iota = \pi \circ \iota$. And since $\lambda \circ f \circ \iota = \pi \circ \iota$, we must have that $\lambda \circ f = \pi$.

Consequently, $J'' = \ker(f) \subseteq \ker(\pi) = J'$. And since $J' \subseteq J \subseteq J'' \subseteq J'$, we can conclude that $J' = J = J''$. ■

A different type of p-linear map is a symmetric multilinear map. Let G denote the pth symmetric group. We say a p-linear map $T : \prod_{i=1}^p M \to N$ is symmetric if $T(m_1, \ldots, m_p) = T(m_{\sigma^{-1}(1)}, \ldots, m_{\sigma^{-1}(p)})$ for all $\sigma \in G$. Note that trivially all 1-linear maps are symmetric.

We say an R-module $S_p(M)$ along with a symmetric p-linear map $\theta : \prod_{i=1}^p M \to S_p(M)$ is the pth symmetric power of M if for any R-module N and any symmetric p-linear map $T : \prod_{i=1}^p M \to N$ there is a unique R-module homomorphism $f : S_p(M) \to N$ such that $f \circ \theta = T$.



Proposition: Such an R-module $S_p(M)$ and symmetric p-linear map θ exists, and the elements $m_1 m_2 \cdots m_p := \theta(m_1, m_2 \ldots, m_p)$ generate $S_p(M)$ as an R-module. Also if $(S_p(M)^{(1)}, \theta^{(1)})$ and $(S_p(M)^{(2)}, \theta^{(2)})$ both satisfy the above universal property then there exists inverse isomorphism $f : S_p(M)^{(1)} \to S_p(M)^{(2)}$ and $g : S_p(M)^{(2)} \to S_p(M)^{(1)}$ such that $f \circ \theta^{(1)} = \theta^{(2)}$ and $f \circ \theta^{(2)} = \theta^{(1)}$.

Proof:
The second sentence of this proposition can be proven through identical arguments as it was for $T_p(M)$ and $E_p(M)$. So like before, we only need to focus on the existence of $S_p(M)$ and θ.

Let G denote the pth symmetric group. Then consider the pth tensor power of M and let $H_p(M) \subseteq T_p(M)$ be the submodule generated by all differences:

$$(m_1 \otimes \cdots \otimes m_p) - (m_{\sigma^{-1}(1)} \otimes \cdots \otimes m_{\sigma^{-1}(p)})$$

where $\sigma \in G$ and $(m_1, \ldots, m_p) \in \prod_{i=1}^p M$.

Note that $H_1(M) = \{0\}$.

Let $S_p(M) = T_p(M)/H_p(M)$. Next let $\pi : T_p(M) \to S_p(M)$ be the quotient map and $\iota : \prod_{i=1}^p M \to T_p(M)$ be the p-linear map sending $(m_1, \ldots, m_p)$ to $m_1 \otimes \cdots \otimes m_p$. Thus $\theta := \pi \circ \iota$ is a p-linear map which is clearly symmetric.

By identical reasoning as in the proof that the pth exterior power of M exists, we can conclude that $S_p(M)$ and θ satisfy the universal property we want. ■

As a side note, θ being a symmetric p-linear map means that $m_1 m_2 \cdots m_p = m_{\sigma^{-1}(1)} m_{\sigma^{-1}(2)} \cdots m_{\sigma^{-1}(p)}$ for any $\sigma \in G$ and $m_1, \ldots, m_p \in M$.

Like with $E_p(M)$, it will be more convenient to view the structure of $S_p(M)$ as a blackbox and just rely on its universal property. Therefore, like before we shall define the canonical homomorphism from $T_p(M)$ to $S_p(M)$ using only universal properties.

To start off, the map $\theta(m_1, m_2 \ldots, m_p) := m_1 m_2 \cdots m_p$ is a p-linear map from $\prod_{i=1}^{p} M$ into $S_p(M)$. Thus by the universal property of $T_p(M)$ there is a unique R-module homomorphism $f : T_p(M) \to S_p(M)$ such that:

$$f(m_1 \otimes m_2 \otimes \cdots \otimes m_p) = m_1 m_2 \cdots m_p.$$

$f : T_p(M) \to S_p(M)$ is called the canonical homomorphism from $T_p(M)$ to $S_p(M)$.

Theorem 1.8: Let $f : T_p(M) \to S_p(M)$ be the canonical homomorphism and let $H_p(M)$ be the submodule of $T_p(M)$ defined on the prior page. Then f is surjective and $\ker(f) = H_p(M)$. Moreover, $H_p(M)$ is generated as an R-module by all differences:

$$(m_1 \otimes \cdots \otimes m_i \otimes m_{i+1} \otimes \cdots \otimes m_p) - (m_1 \otimes \cdots \otimes m_{i+1} \otimes m_i \otimes \cdots \otimes m_p)$$

where $1 \leq i < p$ and $m_1, \ldots, m_p \in M$.

Proof:

f is surjective because f maps a generating set for $T_p(M)$ (i.e. the set of pure tensors) to a generating set for $S_p(M)$. As for showing the rest of the theorem, let $H := H_p(M)$, let H' be the R-submodule of $T_p(M)$ generated by just the differences described above, and let $H'' := \ker(f)$. Clearly $H' \subseteq H \subseteq H''$.

Like in the proof of theorem 1.5 two pages ago, we shall show that $H'' \subseteq H'$. So let $\pi : T_p(M) \to T_p(M)/H'$ be the quotient map and let $\iota : \prod_{i=1}^{p} M \to T_p(M)$ be the multilinear map $\iota(m_1, \ldots, m_p) = m_1 \otimes \cdots \otimes m_p$. We can see that $\pi \circ \iota$ is a symmetric multilinear map.

This is because in order to prove a p-linear map ϕ is symmetric, it suffices to show for any $m_1, \ldots, m_p \in M$ and $1 \leq i < p$ that:

$$\phi(m_1, \ldots, m_i, m_{i+1}, \ldots, m_p) = \phi(m_1, \ldots, m_{i+1}, m_i, \ldots, m_p)$$

After all, any permutation of indices can be decomposed into a composition of interchanges of adjacent indices.

By the universal property of $S_p(M)$ we thus know there is a unique R-module homomorphism $\lambda : S_p(M) \to T_p(M)/H'$ such that:

$$\lambda(m_1 m_2 \cdots m_p) = (m_1 \otimes m_2 \otimes \cdots \otimes m_p) + H'.$$

Yet also by the universal property of $T_p(M)$ we know that $\lambda \circ f = \pi$. In turn, $H'' = \ker(f) \subseteq \ker(\pi) = H'$. And since $H' \subseteq H \subseteq H'' \subseteq H'$, we can conclude that $H' = H = H''$. ■

Today's notes took longer than I was hoping. So to finish off I want to restate theorem 1.3 on *pages 1072-1073* but in a way that applies to $E_p(M)$ and $S_p(M)$.

Let M be a free R-module and in particular let $B \subseteq M$ be a linearly independent basis for M. Also let $p \in \mathbb{Z}_{\geq 1}$ and let G denote the pth symmetric group.

- Consider the set of all sequences of p many distinct elements of B. We can define an equivalence relation on this set by saying that $(b_1, \ldots, b_p) \sim (b_{\sigma^{-1}(1)}, \ldots, b_{\sigma^{-1}(p)})$ for all $\sigma \in G$. And then we let $I_p(B)$ be a set containing one representative $(b_1, \ldots, b_p)$ from each equivalence class.

Theorem 1.6: $E_p(M)$ is a free R-module with a linearly independent basis given by the $b_1 \wedge \cdots \wedge b_p$ as $(b_1, \ldots, b_p)$ ranges over $I_p(B)$.

Proof:
Let $N := \bigoplus_{(b_1,\ldots,b_p) \in I_p(B)} R(b_1, \ldots, b_p)$ be the free R-module associated to $I_p(B)$. We can define a map $\phi : \prod_{i=1}^p B \to N$ given by:

- $\phi(b_1, \ldots, b_p) := 0$ if $(b_1, \ldots, b_p)$ contains any repeated elements of B.
- If $b_1, \ldots, b_p$ are all distinct then there exists a unique $\sigma \in G$ such that $(b_{\sigma^{-1}(1)}, \ldots, b_{\sigma^{-1}(p)}) \in I_p(B)$. So, we define $\phi(b_1, \ldots, b_p) := \operatorname{sgn}(\sigma) \cdot (b_1, \ldots, b_p)$.

By the observation on *page 1072*, ϕ has a unique extension $\eta : \prod_{i=1}^p M \to N$ such that η is a p-linear map. Also, I claim that η is alternating.

Why?
Consider any $1 \leq i < p$ as well as any elements $b_1, \ldots, b_{i-1}, b_{i+2}, \ldots, b_p \in B$. Then for any $m \in M$ let us pick $b^{(1)}, \ldots, b^{(n)} \in B$ as well as $c^{(1)}, \ldots, c^{(n)} \in R$ such that $m = \sum_{j=1}^n c^{(j)} b^{(j)}$. I claim that:

$$\eta(b_1, \ldots, b_{i-1}, m, m, b_{i+2}, \ldots, b_p) = 0.$$

This is because:

$$\begin{aligned}
&\eta(b_1, \ldots, b_{i-1}, m, m, b_{i+2}, \ldots, b_p) \\
&\quad = \sum_{j=1}^n \sum_{j'=1}^n c^{(j)} c^{(j')} \phi(b_1, \ldots, b_{i-1}, b^{(j)}, b^{(j')}, b_{i+2}, \ldots b_p) \\
&\quad = \left(\sum_{j=1}^n (c^{(j)})^2 \phi(b_1, \ldots, b_{i-1}, b^{(j)}, b^{(j)}, b_{i+2}, \ldots, b_p) \right) \\
&\qquad + \left(\sum_{j=1}^{n-1} \sum_{j'=j+1}^n c^{(j)} c^{(j')} \left(\phi(b_1, \ldots, b_{i-1}, b^{(j)}, b^{(j')}, b_{i+2}, \ldots, b_p) \right. \right. \\
&\qquad\qquad \left. \left. + \phi(b_1, \ldots, b_{i-1}, b^{(j')}, b^{(j)}, b_{i+2}, \ldots, b_p) \right) \right)
\end{aligned}$$

Yet now $\phi(b_1, \ldots, b_{i-1}, b^{(j)}, b^{(j)}, b_{i+2}, \ldots, b_p) = 0$ for all j. Meanwhile, $\phi(b_1, \ldots, b_{i-1}, b^{(j)}, b^{(j')}, b_{i+2}, \ldots, b_p) = -\phi(b_1, \ldots, b_{i-1}, b^{(j')}, b^{(j)}, b_{i+2}, \ldots, b_p)$ for all $j < j'$. So all, the summands in the above manipulation cancel and we are left with the equation $\eta(b_1, \ldots, b_{i-1}, m, m, b_{i_2}, \ldots, b_p) = 0$.

In turn, it's now clear that η is alternating because for any elements $m_1, \ldots, m_{i-1}, m_{i+2}, \ldots, m_p$ in M we can write:

$$\eta(m_1, \ldots, m_{i-1}, m, m, m_{i+2}, \ldots, m_p)$$

as a linear combination of elements $\eta(b_1, \ldots, b_{i-1}, m, m, b_{i+2}, \ldots, b_p)$.

Now importantly, N and η satisfy the universal property of the pth exterior power of M. After all, suppose K is another R-module and $\psi : \prod_{i=1}^{p} M \to k$ is an alternating p-linear map. By the universal property of free modules we know there is a unique R-module homomorphism $f : N \to K$ such that $f(b_1, \ldots, b_p) = \psi(b_1, \ldots, b_p)$ for all $(b_1, \ldots, b_p) \in I_p(B)$. Yet then as $f \circ \eta$ is an alternating p-linear map, we can conclude that $f \circ \eta = \psi$ on $\prod_{i=1}^{p} B$. In turn, as any p-linear map from $\prod_{i=1}^{p} M$ is uniquely determined by its restriction to $\prod_{i=1}^{p} B$, we can conclude that $f \circ \eta = \psi$ on all of $\prod_{i=1}^{p} M$.

Furthermore, note that if $g : N \to K$ also were to satisfy that $g \circ \eta = \psi$, we'd have to have that $g(b_1, \ldots, b_p) = \psi(b_1, \ldots, b_p)$ for all $(b_1, \ldots, b_p) \in I_p(B)$. This would in turn imply that $f = g$ by the universal property of free R-modules. So, we have shown that N along with η is a model for the pth exterior power of M.

With that all proven, we can now conclude that that $E_p(M)$ is isomophic to M via an R-module homomorphism sending $b_1 \wedge \cdots \wedge b_p \in E_p(M)$ to $(b_1, \ldots, b_p) \in N$ for all $(b_1, \ldots, b_p) \in I_p(B)$. ■

- Next consider the set of all sequences of p many elements of B (where this time we allow repetitions). We define an equivalence relation on this set by saying that $(b_1, \ldots, b_p) \sim (b_{\sigma^{-1}(1)}, \ldots, b_{\sigma^{-1}(p)})$ for all $\sigma \in G$. And then we let $I_p^*(B)$ be a set containing one representative $(b_1, \ldots, b_p)$ from each equivalence class.

Theorem 1.9: $S_p(M)$ is a free R-module with a linearly independent basis given by the elements $b_1 b_2 \cdots b_p$ as $(b_1, b_2, \ldots, b_p)$ ranges over $I_p^*(B)$.

Proof:
Let N be the free R-module associated with the set $I_p^*(B)$ and then let $\theta : \prod_{i=1}^{p} M \to N$ be the unique p-linear map satisfying that for all $(b_1, \ldots, b_p) \in \prod_{i=1}^{p} B$ we have that $\theta(b_1, \ldots, b_p) = (b_{\sigma^{-1}(1)}, \ldots, b_{\sigma^{-1}(p)})$ where $\sigma \in G$ is the unique permutation with $(b_{\sigma^{-1}(1)}, \ldots, b_{\sigma^{-1}(p)}) \in I_p^*(B)$.

Once we show that θ is symmetric, the rest of the proof of this theorem follows identically as in the proof of the prior theorem.

Fortunately, note for any sequence of p many elements $(b_1, \ldots, b_p)$ in B and any $\sigma \in G$ that $\theta(b_1, \ldots, b_p) = \theta(b_{\sigma^{-1}(1)}, \ldots, B_{\sigma^{-1}(p)})$. Therefore, for any $m_1, \ldots, m_p \in M$ we just need to decompose $\theta(m_{\sigma^{-1}}, \ldots, m_{\sigma^{-1}(p)})$ into a linear combination of $\theta(b_{\sigma^{-1}(1)}, \ldots, B_{\sigma^{-1}(p)})$, apply the prior observation, and then reassemble the linear combination into $\theta(m_1, \ldots, m_p)$. ■

Unfortunately, there is still a little more infrastructure that needs to be built up before I can get to constructing interesting R-algebras satisfying certain universal properties.

Theorem 2.1: Let $M_1, \ldots, M_p$ be R-modules (where $p > 2$) and suppose $1 \leq q < p$. Then there is an R-module isomorphism from $(\bigotimes_{i=1}^{q} M_i) \otimes (\bigotimes_{i=q+1}^{p} M_i)$ to $\bigotimes_{i=1}^{p} M_i$ such that $(m_1 \otimes \cdots \otimes m_q) \otimes (m_{q+1} \otimes \cdots \otimes m_p) \mapsto m_1 \otimes \cdots \otimes m_p$ for all $m_i \in M_i$.

Proof:

I shall proceed by induction on p. For our base case of $p = 3$, recall from *pages 1067-1069* that my original construction of $M_1 \otimes M_2 \otimes M_3$ was the set $M_1 \otimes (M_2 \otimes M_3)$ where $m_1 \otimes m_2 \otimes m_3$ is equal to $m_1 \otimes (m_2 \otimes m_3)$. Meanwhile, I showed on *pages 730-731* that $M_1 \otimes (M_2 \otimes M_3)$ is isomorphic to $(M_1 \otimes M_2) \otimes M_3$ via an isomorphism mapping $m_1 \otimes (m_2 \otimes m_3)$ to $(m_1 \otimes m_2) \otimes m_3$. Hence, our base case is entirely dealt with via prior results.

Next suppose $p > 3$.

- If $q = 1$ then recall that $\bigotimes_{i=2}^{p} M_i$ is isomorphic to $M_2 \otimes (\cdots \otimes (M_{p-1} \otimes M_p) \cdots)$ via an isomorphism ϕ such that:
$$m_2 \otimes \cdots \otimes m_{p-1} \otimes m_p \mapsto m_2 \otimes (\cdots \otimes (m_{p-1} \otimes m_p) \cdots).$$
In turn, $\mathrm{Id}_{M_1} \otimes \phi$ is an isomorphism from $M_1 \otimes (\bigotimes_{i=2}^{p} M_i)$ to $M_1 \otimes (M_2 \otimes (\cdots \otimes (M_{p-1} \otimes M_p) \cdots))$ such that:
$$m_1 \otimes (m_2 \otimes \cdots \otimes m_p) \mapsto m_1 \otimes (m_2 \otimes (\cdots \otimes (m_{p-1} \otimes m_p) \cdots)).$$
And via the reasoning on *pages 1067-1069*, we can compose that map with an isomorphism $\psi : M_1 \otimes (M_2 \otimes (\cdots \otimes (M_{p-1} \otimes M_p) \cdots)) \to \bigotimes_{i=1}^{p} M_i$ sending $m_1 \otimes (m_2 \otimes (\cdots \otimes (m_{p-1} \otimes m_p) \cdots))$ to $m_1 \otimes \cdots \otimes m_p$.

- Meanwhile, if $q > 1$ then we can conclude by our induction hypothesis that $\bigotimes_{i=1}^{q} M_i$ is isomorphic to $M_1 \otimes \bigotimes_{i=2}^{q} M_i$ via an isomorphism ϕ mapping $m_1 \otimes \cdots \otimes m_q$ to $m_1 \otimes (m_2 \otimes \cdots \otimes m_q)$.
In turn, $\phi \otimes \mathrm{Id}_{\bigotimes_{i=q+1}^{p} M_i}$ is an isomorphism from $(\bigotimes_{i=1}^{q} M_i) \otimes (\bigotimes_{i=q+1}^{p} M_i)$ to $(M_1 \otimes (\bigotimes_{i=2}^{q} M_i)) \otimes (\bigotimes_{i=q+1}^{p} M_i)$ such that $(m_1 \otimes \cdots \otimes m_q) \otimes (m_{q+1} \otimes \cdots \otimes m_p)$ is mapped to $(m_1 \otimes (m_2 \otimes \cdots \otimes m_q)) \otimes (m_{q+1} \otimes \cdots \otimes m_p)$.
Next, by the exercise on *pages 730-731* we have an isomorphism from $(M_1 \otimes (\bigotimes_{i=2}^{q} M_i)) \otimes (\bigotimes_{i=q+1}^{p} M_i)$ to $M_1 \otimes ((\bigotimes_{i=2}^{q} M_i) \otimes (\bigotimes_{i=q+1}^{p} M_i))$ sending $(m_1 \otimes (m_2 \otimes \cdots \otimes m_q)) \otimes (m_{q+1} \otimes \cdots \otimes m_p)$ to $m_1 \otimes ((m_2 \otimes \cdots \otimes m_q) \otimes (m_{q+1} \otimes \cdots \otimes m_p))$.
Then by our induction hypothesis, $(\bigotimes_{i=2}^{q} M_i) \otimes (\bigotimes_{i=q+1}^{p} M_i)$ is isomorphic to $\bigotimes_{i=2}^{p} M_i$ via an isomorphism ψ mapping $(m_2 \otimes \cdots \otimes m_q) \otimes (m_{q+1} \otimes \cdots \otimes m_p)$ to $m_2 \otimes \cdots \otimes m_p$.
In turn, $\mathrm{Id}_{M_1} \otimes \psi$ is an isomorphism from $M_1 \otimes ((\bigotimes_{i=2}^{q} M_i) \otimes (\bigotimes_{i=q+1}^{p} M_i))$ to $M_1 \otimes (\bigotimes_{i=2}^{p} M_i)$ sending $m_1 \otimes ((m_2 \otimes \cdots \otimes m_q) \otimes (m_{q+1} \otimes \cdots \otimes m_p))$ to $m_1 \otimes (m_2 \otimes \cdots \otimes m_p)$.

Finally, we can now use our reasoning for the case when $q = 1$ to finish up. ■

Next let $A_1, \ldots, A_p$ be R-algebras. I already mentioned how $\bigotimes_{i=1}^p A_i$ can naturally be turned into an R-algebra itself. However, here is another interpretation of this natural product operation on $\bigotimes_{i=1}^p A_i$.

Consider the $2p$-linear map $T : A_1 \times \cdots \times A_p \times A_1 \times \cdots \times A_p \to \bigotimes_{i=1}^p A_i$ given by:
$$T(a_1, \ldots, a_p, a'_1, \ldots, a'_p) \coloneqq (a_1 a'_1, \ldots, a_p a'_p).$$

By the universal property of tensor products we thus know there is a unique R-module homomorphism $\phi : A_1 \otimes \cdots \otimes A_p \otimes A_1 \otimes \cdots \otimes A_p \to \bigotimes_{i=1}^p A_i$ such that:
$$\phi(a_1 \otimes \cdots \otimes a_p \otimes a'_1 \otimes \cdots \otimes a'_p) \coloneqq (a_1 a'_1, \ldots, a_p a'_p).$$

Then by the prior theorem, we can also say that there is a unique R-module homomorphism $\widetilde{\phi} : (\bigotimes_{i=1}^p A_i) \otimes (\bigotimes_{i=1}^p A_i) \to \bigotimes_{i=1}^p A_i$ such that:
$$\widetilde{\phi}((a_1 \otimes \cdots \otimes a_p) \otimes (a'_1 \otimes \cdots \otimes a'_p)) \coloneqq (a_1 a'_1, \ldots, a_p a'_p).$$

And finally, we can conclude that there is a unique bilinear map $\mu : (\bigotimes_{i=1}^p A_i) \times (\bigotimes_{i=1}^p A_i) \to \bigotimes_{i=1}^p A_i$ such that:
$$\mu((a_1 \otimes \cdots \otimes a_p), (a'_1 \otimes \cdots \otimes a'_p)) \coloneqq (a_1 a'_1, \ldots, a_p a'_p).$$

Lemma: Suppose M is an R-module generated by a set B, and $\mu : M \times M \to M$ is a bilinear map such that $\mu(\mu(b_1, b_2), b_3) = \mu(b_1, \mu(b_2, b_3))$ for all $b_1, b_2, b_3 \in B$. Then $\mu(\mu(m_1, m_2), m_3) = \mu(m_1, \mu(m_2, m_3))$ for all $m_1, m_2, m_3 \in M$.

Proof:

Write $m_1 = \sum_{i=1}^n c_i b_i$, $m_2 = \sum_{j=1}^m c'_j b'_j$, and $m_3 = \sum_{k=1}^\ell c''_k b''_k$. Then:

- $\mu(\mu(m_1, m_2), m_3) = \sum_{i=1}^n \sum_{j=1}^m \sum_{k=1}^\ell c_i c'_j c''_k \mu(\mu(b_i, b'_j), b''_k)$,
- $\mu(m_1, \mu(m_2, m_3)) = \sum_{i=1}^n \sum_{j=1}^m \sum_{k=1}^\ell c_i c'_j c''_k \mu(b_i, \mu(b'_j, b''_k))$.

And now we just invoke that $\mu(\mu(b_i, b'_j), b''_k) = \mu(b_i, \mu(b'_j, b''_k))$. ■

Based on that lemma, its now clear that μ defines an associative operation on M. Clearly, μ is the standard multiplication map on $\bigotimes_{i=1}^p A_i$.

Lemma: Suppose A_1, A_2 are R-algebras, A_1 is generated as an R-module by a subset B, $f : A_1 \to A_2$ is an R-module homomorphism, and $f(bb') = f(b)f(b')$ for all $b, b' \in B$. Then f is an R-algebra homomorphism.

Proof:

We need to show for any $a, a' \in A_1$ that $f(aa') = f(a)f(a')$. Fortunately, we can write $a = \sum_{i=1}^n c_i b_i$ and $a' = \sum_{j=1}^m c'_j b'_j$. Then:

$$\begin{aligned} f(aa') = f((\textstyle\sum_{i=1}^n c_i b_i)(\sum_{j=1}^m c'_j b'_j)) &= \textstyle\sum_{i=1}^n \sum_{j=1}^m c_i c'_j f(b_i b'_j) \\ &= \textstyle\sum_{i=1}^n \sum_{j=1}^m c_i c'_j f(b_i) f(b'_j) \\ &= \left(\textstyle\sum_{i=1}^n c_i f(b_i)\right)\left(\sum_{j=1}^m c'_j f(b'_j)\right) \\ &= f(\textstyle\sum_{i=1}^n c_i b_i) f(\sum_{j=1}^m c'_j b'_j) = f(a)f(a'). \blacksquare \end{aligned}$$

Theorem 3.2: Let $A_1, \ldots, A_p$ be R-algebras (where $p > 2$) and suppose $1 \leq q < p$. Then the R-module isomorphism $f : (\bigotimes_{i=1}^{q} A_i) \otimes (\bigotimes_{i=q+1}^{p} A_i) \to \bigotimes_{i=1}^{p} A_i$ given by theorem 2.1 is also an R-algebra homomorphism.

Why?
Just apply the prior lemma where:

$$B = \{(a_1 \otimes \cdots \otimes a_q) \otimes (a_{q+1} \otimes \cdots \otimes a_p) : a_1, \ldots, a_p \in A\}.$$

Theorem 3.4: Let A be an R-algebra and note that R itself is trivially an R-algebra. Thus, we can consider the standard R-algebra structure on $R \otimes A$. I claim that the R-module isomorphism $\phi : R \otimes A \to A$ satisfying that $\phi(c \otimes a) = ca$ is also an R-algebra homomorphism.

Proof:
Let $B = \{(c \otimes a) : c \in R \text{ and } a \in A\}$. Then:

$$\phi((c \otimes a) \cdot (c' \otimes a')) = \phi(cc' \otimes aa') = cc'aa' = cac'a' = \phi(c \otimes a)\phi(c' \otimes a').$$

So, we can apply the lemma at the bottom of the prior page. ■

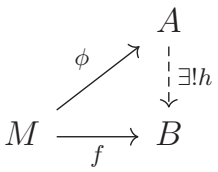
Obviously a similar thing holds if we replace $R \otimes A$ with $A \otimes R$.

8/3/2026

Today I can look forward to constructing another important object which I'll mostly try to treat as a black box satisfying a universal property!!

Let M be an R-module. Then suppose A is an R-algebra and $\phi : M \to A$ is an R-module homomorphism. If B is a second R-algebra and $h : A \to B$ is a unital R-algebra homomorphism, we know that $h \circ \phi$ is an R-module homomorphism from M to B. This brings up the following question:

> Does there exist a unital R-algebra A and an R-module homomorphism $\phi : M \to A$ such that for any unital R-algebra B and any R-module homomorphism $f : M \to B$ there exists a unique unital R-algebra homomorphism $h : A \to B$ with $h \circ \phi = f$?



> If such an R-algebra A and R-module homomorphism ϕ exists then we say A along with ϕ is the tensor algebra of M. Furthermore, if (A_1, ϕ_1) and (A_2, ϕ_2) both satisfy the above universal property then it is easy to see that there exists inverse R-algebra isomorphisms $f : A_1 \to A_2$ and $g : A_2 \to A_1$ such that $f \circ \phi_1 = \phi_2$ and $g \circ \phi_2 = \phi_1$. So, assuming a model of the tensor algebra of M exists, it is unique up to isomorphism.

Theorem 4.1: Such a unital R-algebra A and R-module homomorphism $\phi : M \to A$ exists. Furthermore:

(a) $\phi : M \to A$ is an injection.

(b) A is generated as an R-algebra by $\phi(M)$.

(c) There is a grading on $\{A_n\}_{n\in\mathbb{Z}}$ on A such that $A_1 = \phi(M)$.

(d) The grading $\{A_n\}_{n\in\mathbb{Z}}$ is nonnegative and $A_0 = R1_A$.

(e) For $p \geq 1$, $T_p(M)$ is isomorphic to A_p via the unique R-module homomorphism sending $m_1 \otimes \cdots \otimes m_p$ to $\phi(m_1) \cdots \phi(m_p)$.

Proof:
I'll first focus on just constructing A and ϕ. So let $A_p := T_p(M)$ for all $p \in \mathbb{Z}_{\geq 1}$ and let $A_0 = R$. Then let us set $A := \bigoplus_{p=0}^{\infty} A_p$. Clearly A has an R-module structure. But we still need to define a multiplication operation on A such that A becomes an algebra.

Fortunately for any $p, q \in \mathbb{Z}_{\geq 1}$ we know by theorem 2.1 that there is a (unique) R-module homomorphism $\phi_{p,q} : T_p(M) \otimes T_q(M) \to T_{p+q}(M)$ such that:
$$\phi_{p,q}((m_1 \otimes \cdots \otimes m_p) \otimes (m'_1 \otimes \cdots \otimes m'_q)) = m_1 \otimes \cdots \otimes m_p \otimes m'_1 \otimes \cdots \otimes m'_q$$

In turn, we know there is a (unique) bilinear map $\mu_{p,q} : T_p(M) \times T_q(M) \to T_{p+q}(M)$ such that:
$$\mu_{p,q}((m_1 \otimes \cdots \otimes m_p), (m'_1 \otimes \cdots \otimes m'_q)) = m_1 \otimes \cdots \otimes m_p \otimes m'_1 \otimes \cdots \otimes m'_q$$

Similarly for any $p \in \mathbb{Z}_{\geq 1}$ we know there is a (unique) R-module homomorphism $\phi_{0,q} : R \otimes T_p(M) \to T_p(M)$ such that:
$$\phi_{0,q}(c \otimes (m_1 \otimes \cdots \otimes m_p)) = c(m_1 \otimes \cdots \otimes m_p).$$

In turn, we get a (unique) bilinar map $\mu_{0,p} : R \times T_p(M) \to T_p(M)$ such that $\mu(c, m_1 \otimes \cdots \otimes m_p) = c(m_1 \otimes \cdots \otimes m_p)$. And we can similarly construct (unique) maps $\mu_{0,0} : R \times R \to R$ and $\mu_{p,0} : T_p(M) \times R \to T_p(M)$ for all $p \geq 1$.

I now define multiplication on A as follows. Note that any element $x \in A$ can be written uniquely as $\sum_{p=0}^{\infty} x_p$ where $x_p \in A_p$ for all p and all but finitely many x_p are zero. So, having written $x = \sum_{p=0}^{\infty} x_p$ and $y = \sum_{p=0}^{\infty} y_p$ then I define:
$$\mu(x, y) := \sum_{p=0}^{\infty} \sum_{q=0}^{\infty} \mu_{p,q}(x_p, y_q).$$

- I claim $\mu : A \times A \to A$ is bilinear. After all, let $c_1, c_2 \in \mathbb{R}$ and let $x = \sum_{p=0}^{\infty} x_p$, $y = \sum_{p=0}^{\infty} y_p$, and $z = \sum_{p=0}^{\infty} z_p$ be elements of A. Then:
$$\begin{aligned}\mu(x, c_1y + c_2z) = \sum_{p,q\geq 0} \mu_{p,q}(x_p, c_1y_q + c_2z_q) &= \sum_{p,q\geq 0} (c_1\mu_{p,q}(x_p, y_q) + c_2(x_p, z_q)) \\ &= c_1 \sum_{p,q\geq 0} \mu_{p,q}(x_p, y_q) + c_2 \sum_{p,q\geq 0} \mu_{p,q}(x_p, z_q) \\ &= c_1\mu(x, y) + c_2\mu(x, z)\end{aligned}$$

And analogous reasoning shows that $\mu(c_1x + c_2y, z) = c_1\mu(x, z) + c_2\mu(y, z)$.

- I also claim $\mu(\mu(x,y),z) = \mu(x,\mu(y,z))$ for all $x,y,z \in A$. After all, let $B = \{1_R\} \cup \left(\bigcup_{p=1}^{\infty}\{m_1 \otimes \cdots \otimes m_p \in A_p : m_1,\ldots,m_p \in M\}\right)$. Then B generates A as an R-module and its easy to see that:
$$\mu(\mu(b_1,b_2),b_3) = \mu(b_1,\mu(b_2,b_3)) \text{ for all } b_1,b_2,b_3 \in B.$$
So by the lemma on the middle of *page 1081*, we know this claim holds.

- Also note that $1_R \in A_0$ satisfies that $\mu(1_R,x) = x = \mu(x,1_R)$. Therefore, upon letting μ be the multiplication operation that A is equipped with, then A is unital with $1_A = 1_R$.

Thus A equipped with μ becomes a unital A-algebra. If we set $A_n = \{0\}$ for all $n < 0$ then it is clear that A is graded by the nonnegative grading $\{A_n\}_{n\in\mathbb{Z}}$. Also $A_0 = R1_A$. And since $M = T_1(m)$, if we let $\phi : M \to A = R \oplus \bigoplus_{p=1}^{\infty} T_p(M)$ be the embedding map then we have that ϕ is injective and $A_1 = \phi(M)$. Plus, it is easy to see that A is generated as an R-algebra by A_1 and that each A_p is isomorphic to $T_p(M)$ in the way wanted by property (e).

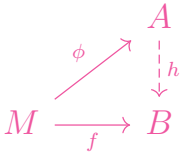
To finish off the main part of the proof, I claim that A and ϕ satisfy the universal property of the tensor algebra of M.

After all, let B an R-algebra and suppose $f : M \to B$ is an R-module homomorphism. Then note for all $p \geq 1$ that $f^{(p)} : M \times \cdots \times M \to B$ given by $f(m_1,\ldots,m_p) = f(m_1)f(m_2)\cdots f(m_p)$ is also a p-linear map. So, by the universal property of $T_p(M)$ there is a unique R-module homomorphism $\psi^{(p)} : T_p(M) \to B$ such that:
$$\psi^{(p)}(m_1 \otimes m_2 \otimes \cdots \otimes m_p) = f(m_1)f(m_2)\cdots f(m_p).$$

We also define the R-module homomorphism $\psi^{(0)} : R \to B$ by $\psi^{(0)}(c) = c1_B$.

By the universal property of coproducts of R-modules, there is a unique R-module homomorphism h from $A = \bigoplus_{p=0}^{\infty} A_p$ to B such that $h(x_p) = \psi^{(p)}(x_p)$ for all $x_p \in A_p$ (where $p \geq 0$). Also, it's clear that $h(1_A) = \psi^{(0)}(1_R) = 1_B$. And we can further verify h is an R-algebra homomorphism via the second lemma on *page 1081*. After all if $p,q \geq 1$ then:
$$\begin{aligned} h((m_1 \otimes \cdots \otimes m_p)\cdot(m_1' \otimes \cdots \otimes m_q')) &= h(m_1 \otimes \cdots \otimes m_p \otimes m_1' \otimes \cdots \otimes m_q') \\ &= f(m_1)\cdots f(m_p)f(m_1')\cdots f(m_q') \\ &= h(m_1 \otimes \cdots \otimes m_p)h(m_1' \otimes \cdots \otimes m_q') \end{aligned}$$

Finally, I claim h is the unique unital R-algebra homomorphism such that:



After all, if $h' : A \to B$ was another unital R-algebra homomorphism that worked then we'd have to have that $h = h'$ on A_1. Yet since A_1 generates A as a unital R-algebra, we can conclude $h = h'$ everywhere on A.

This finishes showing that there exists a model (A, ϕ) of the tensor algebra of M.

One more note I want to give is that if (A', ϕ') is another model of the tensor algebra of M, then properties (a) through (e) will still hold for A' and ϕ'.

To see why, note that there are unique inverse R-algebra isomorphisms $f : A \to A'$ and $g : A' \to A$ satisfying that $f \circ \phi = \phi'$ and $g \circ \phi' = \phi$. And this already guarentees (a) and (b) will hold.

Futhermore, let $A'_n = f(A_n)$ for all $n \in \mathbb{Z}$. Then we can easily see the grading $\{A'_n\}_{n\in\mathbb{Z}}$ on A' will satisfy (c), (d), and (e). ■

For hopefully obvious reasons, we denote the tensor algebra of M as $T(M)$. And if $T(M)$ is equipped with the grading $\{A_n\}_{n\in\mathbb{Z}}$ described in the last theorem then we identify the pth tensor power of M, $T_p(M)$ with A_p for all $p \geq 0$ (where $T_0(M)$ is understood to just be R). In particular, this means we'll identify $M = T_1(M)$ as a submodule of $T(M)$ and $R = T_0(M)$ as a subring of $T(M)$ sharing the same identity element.

Also we'll typically use $\otimes$ as the multiplication sign in $T(M)$.

One notable consequence of having a natural way of viewing M as a submodule of $T(M)$ is as follows:

Suppose M and N are R-modules and $f : M \to N$ is an R-algebra homomorphism. By viewing N as an R-submodule of $T(N)$, we can then say by the universal property of $T(M)$ that there is a unique unital R-algebra homomorphism $T(f) : T(M) \to T(N)$ such that $T(f)|_M = f$.

Furthermore, note that $x \in T(M)$ is homogeneous of degree 1 iff $x \in M$. Yet then $(T(f))(x) = f(x) \in N$. So, $(T(f))(x) \in T(N)$ is homogeneous of degree 1 as well and so we can conclude $(T(f))(T_1(M)) \subseteq T_1(N)$.

By lemma 3.1 on *page 1071* it then follows that $T(f)$ maps a generating set for $T_p(M)$ (as an R-submodule) into $T_p(N)$ for all $p \geq 2$. Plus, it's obvious that:

$$(T(f))(T_0(M)) = R(T(f))(1_{T(M)}) = R1_{T(N)} = T_0(N).$$

So, $T(f) : T(M) \to T(N)$ is a graded R-algebra homomorphism.

Another way of describing the prior consequence is as follows. Let $\mathbf{GrAlg}_R$ denote the category whose objects are graded unital R-algebras and whose morphisms are (unital) graded algebra homomorphisms.

We can define a functor $T : R\textbf{-mod} \to \textbf{GrAlg}_R$ mapping any R-module M to its tensor algebra and mapping any R-module homomorphism $f : M \to N$ to the graded R-algebra homomorphism $T(f) : T(M) \to T(N)$ described earlier.

Note that $T(\mathrm{Id}_M) = \mathrm{Id}_{T(M)}$ for all R-modules M. After all, $\mathrm{Id}_{T(M)} : T(M) \to T(M)$ is a graded R-algebra homomorphism whose restriction to M is just the inclusion map $M \hookrightarrow T(M)$. Yet by the univeral property of tensor algebras there is only one unital R-algebra homomorphism $T(\mathrm{Id}_M)$ with that property.

In a similar vein, if $f : M \to N$ and $g : N \to K$ are R-module homomorphisms, then $T(g \circ f) = T(g) \circ T(f)$. After all, the restriction of either map to M is just the map $g \circ f$. So, both maps must be the same unique map satisfying that. And this proves that $T : R\textbf{-mod} \to \textbf{GrAlg}_R$ is a functor.

One other note is that every element of $T_1(M) = M$ is homogeneous in $T(M)$. Therefore, if $K \subseteq M$ is a submodule then we know from lemma 3.2 on *page 1071* that the two-sided ideal $I(K) \subseteq T(M)$ generated by K is homogeneous.

Theorem 4.3: Let $f : M \to N$ be a surjective R-module homomorphism and let $K = \ker(f)$. Then $T(f) : T(M) \to T(N)$ is surjective and $\ker(T(f)) = I(K)$ where $I(K)$ is the two-sided ideal in $T(M)$ generated by K.

Proof:
Firstly, f being surjective means that $T_1(N) \subseteq (T(f))(T(M))$. And since $T_1(N)$ is a generating set for $T(N)$ as an R-algebra and we can easily verify that the image of a unital R-algebra homomorphism is a subalgebra, we in turn must have that $(T(f))(T(M)) = T(N)$. So, $T(f)$ is surjective.

Next note that it's obvious that $I(K) \subseteq \ker(T(f))$ since $K \subseteq \ker(T(f))$ and we can easily verify that the kernel of an R-algebra homomorphism is an ideal and R-module.

As for the other inclusion, consider the quotient map $\pi : T(M) \to T(M)/I(K)$. We know from *page 1072* that $T(M)/I(K)$ has a natural grading $\{A_n\}_{n\in\mathbb{N}}$ satisfying that M/K is isomorphic to A_1 as an R-module via the map $\phi(m+K) \coloneqq m+I(K)$. Also note by the first isomorphism theorem that $f : M \to N$ induces an R-module isomorphism $\widetilde{f} : M/K \to N$.

So, $g \coloneqq \phi \circ \widetilde{f}^{-1}$ is an R-module homorphism from N to $A_1 \subseteq T(M)/I(K)$ satisfying that $g(n) = m + I(K)$ where $m \in M$ is any element satisfying that $f(m) = n$. By the universal property of $T(N)$, we can then extend g to a unique R-algebra homomorphism $T(g) : T(N) \to T(M)/I(K)$.

But now note that $T(g) \circ T(f) = \pi$. After all, we can easily check for pure tensors that $(T(f))(m_1 \otimes \cdots \otimes m_p) = f(m_1) \otimes \cdots \otimes f(m_p)$. And then if $n_i = f(m_i)$, we have that:

$$\begin{aligned}(T(g))(n_1 \otimes \cdots \otimes n_p) &= g(n_1) \cdots g(n_p) \\ &= (m_1 + I(K)) \cdots (m_p + I(K)) = (m_1 \otimes \cdots \otimes m_p) + I(K).\end{aligned}$$

So, $T(g) \circ T(f)$ agrees with π on a generating set for $T(M)$ as an R-module and it follows that $T(g) \circ T(f) = \pi$ everywhere.

Consequently, $\ker(T(f)) \subseteq \ker(\pi) = I(K)$ and thus we've shown the other inclusion we wanted. ■

As a side note, $T(f)$ restricted to $T_p(M)$ is the map $f^{\otimes p} := \underbrace{f \otimes \cdots \otimes f}_{p \text{ times}}$.
This is because both maps satisfy that:

$$m_1 \otimes \cdots \otimes m_p \mapsto f(m_1) \otimes \cdots \otimes f(m_p).$$

But also note that we can now conclude that:

$$\ker(f^{\otimes p}) = I(K) \cap T_p(M) \subseteq T(M)$$

where $K = \ker(f)$ and $I(K)$ is the two-sided ideal in $T(M)$ generated by K.

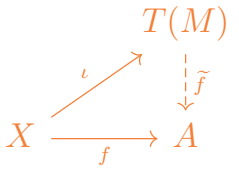
8/4/2026

Welp I realized my tangent for the last 5 days might have not been necssary after all. I'd still like to think it was informative though since I don't have much experience right now working with R-algebras. I'll start off today though going over two more results about tensor algebras from Northcott's book.

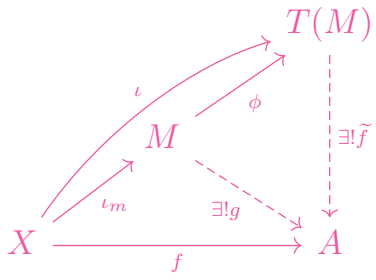
To start off, given a set X, here's how to construct the free R-algebra of X.

Let M be the free R-module of X (equipped with the embedding $\iota_m : X \to M$). Next, consider the tensor algebra $T(M)$ of M equipped with the R-module homomorphism $\phi : M \to T(M)$. And finally let $\iota : X \to T(M)$ be defined by $\iota := \phi \circ \iota_m$. Note that ι will be injective since both ϕ and ι_m are. Also, $\iota(X)$ will generate $T(M)$ as an R-algebra.

Theorem 4.4: If A is a unital R-algebra and $f : X \to A$ is any function, then there is a unique unital R-algebra homomorphism $\widetilde{f} : T(M) \to A$ such that the following diagram holds:



Proof:
Firstly such an $\widetilde{f}$ exists. After all, A is also an R-module, we can conclude by the universal property of M and ι_M that there is a unique R-module homomorphism $g : M \to A$ such that $g \circ \iota_m = f$. Next, we can use the universal property of $T(M)$ and ϕ to say that there is a unique R-algebra homomorphism $\widetilde{f} : T(M) \to A$ such that $\widetilde{f} \circ \phi = g$. And thus $\widetilde{f} \circ \iota = (\widetilde{f} \circ \phi) \circ \iota_m = g \circ \iota_m = f$.



Next suppose $h : T(M) \to A$ is another unital R-algebra homomorphism satisfying that $h \circ \iota = f$. Then as $\iota(X)$ generates all of $T(M)$ as an R-algebra, we can say that h is uniquely determind by its restriction to $\iota(X)$. Yet the commuting diagram for $\widetilde{f}$ says that $h(\iota(x)) = f(x) = \widetilde{f}(\iota(x))$ for all $x \in X$. So, h must equal $\widetilde{f}$ everywhere on $T(M)$. ■

As a side note we can also show through standard arguments that if (A, ι) and (A', ι') both satisfy the universal property of a free R-algebra of a set X, then there are inverse R-algebra isomorphisms $f : A \to A'$ and $g : A' \to A$ such that $f \circ \iota = \iota'$ and $g \circ \iota' = \iota$.

On a different note, it's easy to see that if M is a free R-module with a linearly independent basis B then $T(M)$ is also a free R-module with a linearly independent basis consisting of all pure tensors whose components are in B.

A related result we can prove is that M is a projective R-module if and only if the tensor algebra $T(M)$ of M is a projective R-module.

Lemma 4.1: Let $u : N \to M$ and $v : M \to N$ be R-module homomorphisms such that $v \circ u = \mathrm{Id}_N$. Then u is an injection, v is a surjection, and $M = u(N) \oplus \ker(v)$ as an internal direct sum.

Proof:
That u is an injection and v a surjection is obvious. Also note for any $m \in M$ that $m - u(v(m)) \in \ker(v)$ since:

$$v(m - u(v(m))) = v(m) - v(u(v(m))) = v(m) - v(m) = 0.$$

Similarly $u(v(m)) \in u(N)$. And since $m = u(v(m)) + (m - u(v(m)))$ we can conclude that $u(N) + \ker(v) = M$. Finally, suppose $m \in u(N) + \ker(v)$. Then there exists $n \in N$ such that $u(n) = m$. Yet also note that:

$$n = \mathrm{Id}_N(n) = v(u(n)) = v(m) = 0. \text{ So } m = u(0) = 0.$$

And ths proves that $u(N) + \ker(v)$ is actually an internal direct sum. ■

Theorem 4.5: Let P be an R-module and let $T(P)$ be the tensor algebra of P. Then P is a projective R-module if and only if $T(P)$ is a projective R-module.

$(\Longleftarrow)$
Note that $T(P) \cong P \oplus (\bigoplus_{\substack{p \geq 0 \\ p \neq 1}} T_p(P))$ as R-modules.

But now note that $T(P)$ is projective if and only if there is a free R-module $\mathcal{F}$ and some R-module K such that $T(P) \oplus K \cong \mathcal{F}$. In turn:

$$\mathcal{F} \cong (P \oplus (\bigoplus_{\substack{p \geq 0 \\ p \neq 1}} T_p(P))) \oplus K \cong P \oplus ((\bigoplus_{\substack{p \geq 0 \\ p \neq 1}} T_p(P)) \oplus K)$$

So, P is also a direct summand of a free R-module and is thus projective.

As a side note, analogous reasoning lets us further say that if $T(P)$ is a projective R-module then $T_p(P)$ is a projective R-module for all $p \in \mathbb{Z}_{\geq 0}$.

$(\Longrightarrow)$
Since P is projective we can choose an R-module Q such that $P \oplus Q \cong \mathcal{F}$ where $\mathcal{F}$

is a free R-module. Then we let $\iota : P \hookrightarrow \mathcal{F}$ be the direct sum embedding and we let $\pi : \mathcal{F} \twoheadrightarrow P$ be the direct sum projection. By the universal property of $T(P)$ and $T(\mathcal{F})$, we know that ι and π uniquely extend to R-algebra homomorphisms:

$$T(\iota) : T(P) \to T(\mathcal{F}) \text{ and } T(\pi) : T(\mathcal{F}) \to T(P).$$

But now as $\pi \circ \iota = \mathrm{Id}_P$, we can conclude that $T(\pi) \circ T(\iota) = T(\mathrm{Id}_P) = \mathrm{Id}_{T(P)}$. So by the prior lemma, we can conclude that $T(\mathcal{F}) \cong T(P) \oplus \ker(T(\pi))$. Yet also note that since $\mathcal{F}$ is a free R-module, we know $T(\mathcal{F})$ is also a free R-module. So, $T(P)$ is a direct summand of a free R-module and is thus projective. ■

I'm now going to return to what I was trying and failing to construct on 7/27/2026. Also for the sake of continuity I'll go back to the red and orange text tangent color scheme.

My endgoal is to show that if R is a unital commutative ring then the category $\mathbf{CAlg}_R$ (whose objects are unital commutative R-algebras and whose morphisms are unital R-algebra homomorphisms) is cocomplete. There will be three steps to showing this.

1. I need to show $\mathbf{CAlg}_R$ admits *filtered* colimits. In other words, if $(I, \lesssim)$ is a directed set (which we turn into a category $\mathbf{I}$ as described on *page 1024*), then any functor $F : \mathbf{I} \to \mathbf{CAlg}_R$ has a colimit.
2. Then I need to use the filtered colimit construction to construct coproducts / "tensor products" over arbitrary collections of commutative unital R-algebras.
3. Finally, I can adapt the construction on *pages 1023-1024* for constructing colimits of R-modules in order to construct general colimits in the category $\mathbf{CAlg}_R$.

Suppose $(I, \lesssim)$ is a directed set, that every $i \in I$ is associated with a commutative unital R-algebra A_i, and that for all $i, j \in I$ with $i \lesssim j$ there exists a unital R-algebra homomorphism $r^i_j : A_i \to A_j$ such that:

- $i \lesssim j \lesssim k$ in I implies that $r^i_k = r^j_k \circ r^i_j$,
- $r^i_i = \mathrm{Id}_{A_i}$ for all $i \in I$.

In other words, we have a functor $F : I \to \mathbf{CAlg}_R$. Now here's how to construct $\varinjlim(F)$.

Let $S \coloneqq \bigcup_{i \in I}(\{i\} \times A_i)$ be the disjoint union of the algebras A_i. Then equip S with the equivalence relation where $(i, x) \sim (j, y)$ if there exists $k \in I$ with $i \lesssim k$ and $j \lesssim k$ such that $r^i_k(x) = r^j_k(y)$. Finally, let $A \coloneqq S/_\sim$ and let $\iota_i : A_i \to A$ be the map $x \mapsto [(i, x)]$ for all $i \in I$.

If $\mathcal{G} : \mathbf{CAlg}_R \to \mathbf{Set}$ is the forgetful functor than we know from the construction on *pages 1024-1026* that A equipped with all the ι_i is the colimit $\varinjlim(\mathcal{G} \circ F)$.

Thus if we can equip A with a unital R-algebra structure and then show that $\iota_i : A_i \to A$ is a unital R-algebra homomorphism for all i, then this will be enough to conclude that A equipped with all the ι_i is the colimit $\varinjlim(F)$.

Here's how we define are algebra operations on A.

- (Scaling): Given any $c \in R$ and $[(i,x)] \in A$, we let $c[(i,x)] \coloneqq [(i,cx)]$.
  To see that this is well-defined, suppose $[(i,x)] = [(j,y)]$. Then there exists $k \in I$ with $i \lesssim k$, $j \lesssim k$, and $r^i_k(x) = r^j_k(y)$. Then as r^i_k and r^j_k are R-algebra homomorphisms, we have that $r^i_k(c \cdot x) = c \cdot r^i_k(x) = c \cdot r^j_k(y) = r^j_k(c \cdot y)$. As a result, $[(i,cx)] = [(j,cy)]$.

- (Addition): Suppose $[(i,x)], [(j,y)] \in A$. Then given any $k \in I$ with $i \lesssim k$ and $j \lesssim k$ we let $[(i,x)] + [(j,y)] \coloneqq [(k, r^i_k(x) + r^j_k(y))]$.
  To see that this is well-defined, suppose $[(i_1,x_1)] = [(i_2,x_2)]$ and $[(j_1,y_1)] = [(j_2,y_2)]$, and then consider any $k_1, k_2 \in I$ such that $i_1 \lesssim k_1$, $j_1 \lesssim k_1$, $i_2 \lesssim k_2$, and $j_2 \lesssim k_2$. We need to show that:
  $$[(k_1, r^{i_1}_{k_1}(x_1) + r^{j_1}_{k_1}(y_1))] = [(k_2, r^{i_2}_{k_2}(x_2) + r^{j_2}_{k_2}(y_2))].$$

  Fortunately, note that as $[(i_1,x_1)] = [(i_2,x_2)]$ there exists $\ell \in I$ such that $r^{i_1}_\ell(x_1) = r^{i_2}_\ell(x_2)$. Also without loss of generality we can assume $k_1 \lesssim \ell$ and $k_2 \lesssim \ell$. Similarly, as $[(j_1,y_1)] = [(j_2,y_2)]$ we can find $\ell' \in I$ such that $r^{j_1}_{\ell'}(y_1) = r^{j_2}_{\ell'}(y_2)$. And now I claim without loss of generality that we can just take $\ell = \ell'$ and also assume that $k_1, k_2 \lesssim \ell$.

  This is because since I is a directed set, we know there exists $\ell^{\text{new}} \in I$ with $\ell, \ell', k_1, k_2 \lesssim \ell^{\text{new}}$. And then:
  $$r^{i_1}_{\ell^{\text{new}}}(x_1) = (r^\ell_{\ell^{\text{new}}} \circ r^{i_1}_\ell)(x_1) = (r^\ell_{\ell^{\text{new}}} \circ r^{i_2}_\ell)(x_2) = r^{i_2}_{\ell^{\text{new}}}(x_2)$$

  Similarly, we can show $r^{j_1}_{\ell^{\text{new}}}(y_1) = r^{j_2}_{\ell^{\text{new}}}(y_2)$.

  So, there exists $\ell \in I$ such that $i_1, i_2, j_1, j_2, k_1, k_2 \lesssim \ell$, $r^{i_1}_\ell(x_1) = r^{i_2}_\ell(x_2)$, and $r^{j_1}_\ell(y_1) = r^{j_2}_\ell(y_2)$. And finally, we can now conclude that:
  $$\begin{aligned} r^{k_1}_\ell(r^{i_1}_{k_1}(x_1) + r^{j_1}_{k_1}(y_1)) &= r^{i_1}_\ell(x_1) + r^{j_1}_\ell(y_1) \\ &= r^{i_2}_\ell(x_2) + r^{j_2}_\ell(y_2) = r^{k_2}_\ell(r^{i_2}_{k_2}(x_2) + r^{j_2}_{k_2}(y_2)) \end{aligned}$$

- (Multiplication): Suppose $[(i,x)], [(j,y)] \in A$. Then given any $k \in I$ with $i \lesssim k$ and $j \lesssim k$ we let $[(i,x)][(j,y)] = [(k, r^i_k(x) r^j_k(y))]$.
  We can use basically identical reasoning to prove multiplication is well-defined as we used to prove addition is well-defined. Just replace all the addition operations with multiplication operations in the prior reasoning.

Now one major reason why I showed all the algebra operations are well-defined *before* I showed they satisfy the scaling/addition/multiplication axioms for a commutative R-algebra is that we can now always insist on working with whichever elements of an equivalence class are most convenient to work with.

As a general observation, it's easy to see that if $i \lesssim j$ then $[(i,x)] = [(j, r^i_j(x))]$. Also, we can see that $[(i,x)] + [(i,y)] = [(i, x+y)]$ and that $[(i,x)][(i,y)] = [(i,xy)]$. Thus, here's how we can "quickly" prove A is a commutative R-algebra.

Let $[(i,x)],[(j,y)],[(k,z)] \in A$. By picking a common upper bound to i, j, and k in I we can assume without loss of generality that $i = j = k$. And then since A_i is a commutative algebra:

- $([(i,x)][(i,y)])[(i,z)] = [(i,xyz)] = [(i,x)]([(i,y)][(i,z)])$,
- $[(i,x)]([(i,y)] + [(i,z)]) = [(i,xy+xz)] = [(i,x)][(i,y)] + [(i,x)][(i,z)]$,
- $([(i,x)] + [(i,y)])[(i,z)] = [(i,xz+xy)] = [(i,x)][(i,z)] + [(i,y)][(i,z)]$,
- $[(i,x)][(i,y)] = [(i,xy)] = [(i,yx)] = [(i,y)][(i,x)]$,
- $[(i,x)] + [(i,y)] = [(i,x+y)] = [(i,y+x)] = [(i,y)] + [(i,x)]$.

Additionally suppose $c_1, c_2 \in R$. Then:

- $c_1(c_2[(i,x)]) = [(i,c_1c_2x)] = c_2c_2[(i,x)]$
- $(c_1+c_2)[(i,x)] = [(i,(c_1+c_2)x)] = [(i,c_1x+c_2x)] = c_1[(i,x)]+c_2[(i,x)]$.

Also if $c \in R$ then:

- $(c[(i,x)])[(i,y)] = [(i,(cx)y)] = [(i,x(cy))] = [(i,x)](c[(i,y)])$,
- $(c[(i,x)])[(i,y)] = [(i,(cx)y)] = [(i,c(xy))] = c([(i,x)][(i,y)])$,
- $c([(i,x)] + [(i,y)]) = [(i,c(x+y))] = [(i,cx+cy)] = c[(i,x)] + c[(i,y)]$.

Finally, we just need to show A has an element 1_A and an element 0_A. So, I define $0_A \coloneqq [(i,0_{A_i})]$ and $1_A \coloneqq [(i,1_{A_i})]$ where i is any element of I.

> To see that this is well-defined, note that $[(i,0_{A_i})] = [(j,0_{A_j})]$ and $[(i,1_{A_i})] = [(j,1_{A_j})]$ for all $i,j \in I$. This is because if we let $k \in I$ be such that $i \lesssim k$ and $j \lesssim k$ then $r^i_k(0_{A_i}) = 0_{A_k}$, $r^i_k(1_{A_i}) = 1_{A_k}$, $r^j_k(0_{A_j}) = 0_{A_k}$, and $r^j_k(1_{A_j}) = 1_{A_k}$.

And now we can easily see that $0_A + [(i,x)] = [(i,x)]$ and $1_A[(i,x)] = [(i,x)]$. Also $[(i,x)] + [(i,-x)] = [(i,0_{A_i})] = 0_A$. So every element of A has an additive inverse.

It's also now obvious that $\iota_i : A_i \to A$ is a unital R-algebra homomorphism for all $i \in I$. ■

With that I've finished showing how to construct a *filtered* colimit.

Remark 1: This is actually all we need in order to study stalks and germs on topological spaces. However, I made the claim earlier that $\mathbf{CAlg}_R$ is a cocomplete category. And if I want to actually justify that assertion, I need to do more work.

Remark 2: The prior construction also works for taking filtered colimits of R-modules. I mention that because it's much easier to interpret what any element of this construction looks like compared to what any element of the general construction on *pages 1023-1024* looks like.

Remark 3: One really important detail I also want to note about this construction is that $A = \bigcup_{i \in I} \iota_i(A_i)$

8/5/2026

I have two more constructions to make before I can conclude that $\mathbf{CAlg}_R$ is cocomplete. Firstly, given an arbitrary collection $\{A_\beta\}_{\beta\in B}$ of unital commutative algebras I need to construct their coproduct.

Lemma A: Let $A_1,\ldots,A_n$ be unital R-algebras and let $1\leq k\leq n$ be a positive integer. Then there is a unique unital R-algebra homomorphism $j:\bigotimes_{i=1}^k A_i\to\bigotimes_{i=1}^n A_i$ satisfying that:

$$j(a_1\otimes\cdots\otimes a_k)=a_1\otimes\cdots\otimes a_k\otimes 1_{A_{k+1}}\otimes\cdots\otimes 1_{A_n}.$$

Proof:
Note that $\phi:\prod_{i=1}^k A_i\to\bigotimes_{i=1}^n A_i$ given by

$$\phi(a_1,\ldots,a_k)=a_1\otimes\cdots\otimes a_k\otimes 1_{A_{k+1}}\otimes\cdots\otimes 1_{A_n}.$$

is a k-linear map. Thus by the universal property of tensor products, we know there is a unique R-module homomorphism $j:\bigotimes_{i=1}^k A_i\to\bigotimes_{i=1}^n A_i$ such that:

$$j(a_1\otimes\cdots\otimes a_k)=a_1\otimes\cdots\otimes a_k\otimes 1_{A_{k+1}}\otimes\cdots\otimes 1_{A_n}.$$

Also note that $j(a_1\otimes\cdots\otimes a_k)j(a_1'\otimes\cdots\otimes a_k')=j((a_1\otimes\cdots\otimes a_k)(a_1'\otimes\cdots\otimes a_k'))$ for every pure tensor $a_1\otimes\cdots\otimes a_k$ and $a_1'\otimes\cdots\otimes a_k'$. Thus by the lemma at the bottom of *page 1081*, we can conclude that j is an R-algebra homomorphism.

Finally, note that $j(1_{\bigotimes_{i=1}^k A_i})=j(1_{A_1}\otimes\cdots\otimes 1_{A_k})=1_{A_1}\otimes\cdots\otimes 1_{A_n}=1_{\bigotimes_{i=1}^n A_i}$. So, j is unital. ■

Lemma B: Let $A_1,\ldots,A_n$ be unital R-algebras and let $\sigma:\{1,\ldots,n\}\to\{1,\ldots,n\}$ be a bijective function (i.e. a permutation). Then there is a unique unital R-algebra isomorphism $\phi:\bigotimes_{i=1}^n A_i\to\bigotimes_{i=1}^n A_{\sigma^{-1}(i)}$ such that:

$$\phi(a_1\otimes\cdots\otimes a_n)=a_{\sigma^{-1}(1)}\otimes\cdots\otimes a_{\sigma^{-1}(n)}.$$

Proof:
Showing that such a unique unital R-algebra homomorphism ϕ exists uses the same standard argument used in the proof of the last lemma. As for why ϕ is an isomorphism, note that we can by identical reasoning conclude that there is a unique unital R-algebra homomorphism $\psi:\bigotimes_{i=1}^n A_{\sigma^{-1}(i)}\to\bigotimes_{i=1}^n A_i$ such that:

$$\psi(a_1\otimes\cdots\otimes a_n)=a_{\sigma(1)}\otimes\cdots\otimes a_{\sigma(n)}.$$

But now $\psi\circ\phi=\mathrm{Id}_{\bigotimes_{i=1}^n A_i}$ since both agree on all pure tensors. Similarly, $\phi\circ\psi=\mathrm{Id}_{\bigotimes_{i=1}^n A_{\sigma^{-1}(i)}}$ since both agree on all pure tensors. ■

Using the prior lemmas, we're now ready to construct a functor whose (filtered) colimit will be the categorical coproduct of the collection $\{A_\beta\}_{\beta\in B}$ of unital commutative R-algebras.

Let I_f be the collection of finite subsets of B, and let I_f be directed by increasing inclusion. Then after viewing I_f as a category, we'll define a functor $F:I_f\to\mathbf{CAlg}_R$ as follows:

For each $J \in I_f$ we choose an indexing $J = \{\beta_1^{(J)}, \ldots, \beta_n^{(j)}\}$. Importantly, if $J_1 \subseteq J_2$ then we know there is a unique injection:

$$\sigma_{J_1 \subseteq J_2} : \{1, \ldots, |J_1|\} \to \{1, \ldots, |J_2|\}$$

such that $\beta_i^{(J_1)} = \beta_{\sigma_{J_1 \subseteq J_2}(i)}^{(J_2)}$ for all $i \in \{1, \ldots, |J_1|\}$. Also, it's easy to verify that if $J_1 \subseteq J_2 \subseteq J_3$ then $\sigma_{J_1 \subseteq J_3} = \sigma_{J_2 \subseteq J_3} \circ \sigma_{J_1 \subseteq J_2}$. And also note that $\sigma_{J \subseteq J} = \mathrm{Id}_{\{1,\ldots,|J|\}}$.

Next, for any $J \in I_f$ let $F(J) := \bigotimes_{i=1}^{|J|} A_{\beta_i^{(J)}}$. Note that as multiplication is commutative on pure tensors (since each $A_{\beta_i^{(J)}}$ is commutative), we can further conclude that $F(J)$ is a unital commutative R-algebra for all J.

Also, if $J_1 \subseteq J_2$ then we define $r_{J_2}^{J_1} = F(J_1 \subseteq J_2) : \bigotimes_{i=1}^{|J_1|} A_{\beta_i^{(J_1)}} \to \bigotimes_{i=1}^{|J_2|} A_{\beta_i^{(J_2)}}$ to be the unique unital R-algebra homomorphism such that:

$$r_{J_2}^{J_1}(a_1 \otimes \cdots a_{|J_1|}) = \widetilde{a}_1 \otimes \cdots \otimes \widetilde{a}_{|J_2|}$$

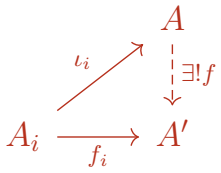
where $\widetilde{a}_i = a_j$ if there exists $1 \leq j \leq |J_1|$ with $\sigma_{J_1 \subseteq J_2}(j) = i$ and $\widetilde{a}_i = 1_{A_{\beta_i^{(J_2)}}}$ otherwise.

Note that we can show $r_{J_2}^{J_1}$ is unique and well-defined by combining lemmas A and B on the prior page. Also, we can see that $r_J^J = \mathrm{Id}_{F(J)}$ for all pure tensors and similarly that $J_1 \subseteq J_2 \subseteq J_3 \implies \sigma_{J_1 \subseteq J_3} = \sigma_{J_2 \subseteq J_3} \circ \sigma_{J_1 \subseteq J_2} \implies R_{J_3}^{J_1} = r_{J_3}^{J_2} \circ r_{J_2}^{J_1}$ for all pure tensors. So by R-linearity we can conclude that:

$$r_J^J = \mathrm{Id}_{F(J)} \text{ and } R_{J_3}^{J_1} = r_{J_3}^{J_2} \circ r_{J_2}^{J_1} \text{ for all } J, j_1, J_2, J_3 \in I_f.$$

With that we now have a well-defined functor $F : I_f \to \mathbf{CAlg}_R$. By yesterday's work we know how to construct a unital R-algebra C as well a bunch of unital R-algebra homomorphisms $\iota_J : \bigotimes_{i=1}^{|J|} A_{\beta_i^{(J)}} \to C$ such that C along with all the ι_J is a colimit of F.

Lemma C: Let $A_1, \ldots, A_n$ be unital R-algebras and let $A = \bigotimes_{i=1}^n A_i$. Also let $\iota_i : A_i \to A$ be the unital R-algebra homomorphism $\iota_i(a) = 1_{A_1} \otimes \cdots \otimes 1_{A_{i-1}} \otimes a \otimes 1_{A_{i+1}} \otimes \cdots \otimes 1_{A_n}$. If A' is another unital R-algebra and for each i we have a unital R-algebra homomorphism $f_i : A_i \to A'$, then there is a unique unital R-algebra homomorphism $f : A \to A'$ such that following diagram holds for all i:



Proof:
Note that the unique R-algebra homomorphism $f : \bigotimes_{i=1}^n A_i \to A'$ such that $f(a_1 \otimes \cdots \otimes a_n) = \prod_{i=1}^n f_i(a_i)$ satisfies the diagram we want. Furthermore, if g were any other R-algebra homomorphism also satisfying our diagrams, we'd have to have that $g(a_1 \otimes \cdots \otimes a_n) = g(\prod_{i=1}^n \iota_i(a_i)) = \prod_{i=1}^n g(\iota_i(a_i)) = \prod_{i=1}^n f_i(a_i) = f(a_1 \otimes \cdots \otimes a_n)$. So g and f will agree on all pure tensors and in turn on all of $A = \bigotimes_{i=1}^n A_i$. ■

As a side note, lemma C also shows us that a tensor product is a categorical coproduct over a finite collection of commutative unital R-algebras. This fact will be used when showing that the colimit from earlier is a categorical coproduct over an arbitrary collection of commutative unital R-algebras.

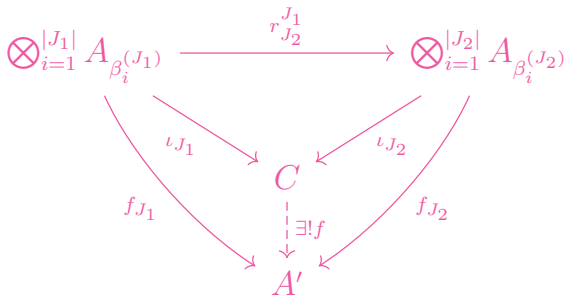
Let's now return to talking about the functor $F : I_f \to \mathbf{CAlg}_R$ defined on the prior two pages, as well as it's colimit C equipped with the unital R-algebra homomorphisms $\iota_J : F(J) \to C$. In particular, if $J \in I_f$ is the singleton set $\{\beta\}$ then I'll denote ι_J as ι_β.

I claim now that C equipped with the homomorphisms ι_β is the coproduct of the collection $\{A_\beta\}_{\beta \in B}$ of commutative unital R-algebras.

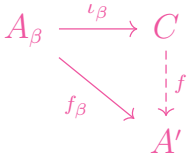
Suppose A' is a unital commutative R-algebra and $\{f_\beta : A_\beta \to A'\}_{\beta \in B}$ is a collection of unital R-algebra homomorphisms.

If $J = \{\beta_1^{(J)}, \ldots, \beta_n^{(J)}\}$ (where $n = |J|$) then we can find a unique unital R-algebra homomorphism $f_J : F(J) \to A'$ such that $f_J(a_1 \otimes \cdots \otimes a_n) = \prod_{i=1}^n f_{\beta_i^{(J)}}(a_i)$.

Yet also note that if $J_1 \subseteq J_2$ then $F_{J_2} \circ r_{J_2}^{J_1}$ agrees with F_{J_1} for all pure tensors (and thus for everything). So by the universal property of the colimit R-algebra C, there exists a unique unital R-algebra homomorphism $f : C \to A'$ such that if $J_1 \subseteq J_2$ then the following commuting diagram holds:



In particular, if we let $J_1 = \{\beta\}$ for any $\beta \in B$ and then focus on just part of the above diagram, we can conclude that:



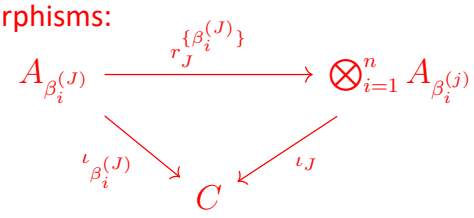
Next I claim that if $g : C \to A'$ is any other unital R-algebra homomorphism such that $g \circ \iota_\beta = f_\beta$ for all β, then $g = f$. This is because of the following observation:

Lemma: $\bigcup_{\beta \in B} \iota_\beta(A_\beta)$ generates C as an R-algebra.

To see why, recall from my remark yesterday that $C = \bigcup_{J \in I_f} \iota_J(F(J))$.

Then consider any $J = \{\beta_1^{(J)}, \ldots, \beta_n^{(J)}\} \in I_f$ and then pick any pure tensor: $a_1 \otimes \cdots \otimes a_n \in \bigotimes_{i=1}^n A_{\beta_i^{(J)}} = F(J)$.

Importantly, we can observe from the following commuting diagram of R-algebra homomorphisms:



that:

$$\begin{aligned}\iota_J(a_1 \otimes \cdots \otimes a_n) &= \iota_J(\textstyle\prod_{i=1}^n r_J^{\{\beta_i^{(J)}\}}(a_i)) \\ &= \textstyle\prod_{i=1}^n (\iota_J \circ r_J^{\{\beta_i^{(J)}\}})(a_i) = \prod_{i=1}^n \iota_{\{\beta_i^{(J)}\}}(a_i)\end{aligned}$$

But now since $F(J)$ is generated as an R-module by the pure tensors $a_1 \otimes \cdots \otimes a_n$, we know that $\iota_J(F(J))$ is also generated as an R-module by the elements $\iota_J(a_1 \otimes \cdots \otimes a_n)$ which I just showed are products of elements in $\bigcup_{\beta \in B} \iota_\beta(A_\beta)$.

So, C is a union of R-submodules, each of which are generated by elements in the subalgebra of C generated by $\bigcup_{\beta \in B} \iota_\beta(A_\beta)$. It follows that $\bigcup_{\beta \in B} \iota_\beta(A_\beta)$ is a generating set for C as a unital R-algebra.

If $g \circ \iota_\beta = f \circ \iota_\beta$ for all $\beta \in B$ then that implies $f = g$ on $\bigcup_{\beta \in B} \iota_\beta(A_\beta)$. As a result of the prior lemma, we in turn must have that $f = g$ on C.

Before continuing on, I'd like to further unpack the prior construction a little bit. Usually, the set C constructed priorly is written as $\bigotimes_{\beta \in B} A_\beta$ and called a "tensor product". In order to justify this nomenclature, we consider the following observations.

Let $\prod'_{\beta \in B} A_\beta \coloneqq \{(a_\beta)_{\beta \in B} \in \prod_{\beta \in B} A_\beta : a_\beta = 1_{A_\beta} \text{ for all but finitely many } \beta \in B\}$.

Given a commutative unital R-algebra A', we can say a function $T : \prod'_{\beta \in B} A_\beta \to A'$ is "multilinear" if it is R-linear with respect to each component of $\prod'_{\beta \in B} A_\beta$ individually.

For example, if we have a collection of unital R-algebra homomorphisms $\{f_\beta : A_\beta \to A'\}_{\beta \in B}$ then we can get a well-defined multilinear map $T : \prod'_{\beta \in B} A_\beta \to A'$ given by:

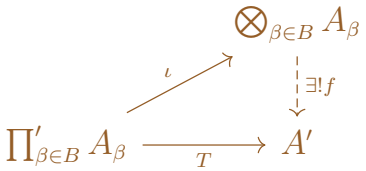
$$T((a_\beta)_{\beta \in B}) \coloneqq \textstyle\prod_{\beta \in B} f_\beta(a_\beta)$$

T is well-defined because $f_\beta(a_\beta) = 1_{A'}$ for all but finitely many $\beta \in B$. Ths is because each f_β is unital and $a_\beta = 1_{A_\beta}$ for all but finitely many β. Seeing that T is multilinear is also easy.

In particular, consider $\bigotimes_{\beta \in B} A_\beta$ equipped with all the ι_β. Then set $\iota : \prod'_{\beta \in B} A_\beta \to \bigotimes_{\beta \in B} A_\beta$ to be the map $\iota((a_\beta)_{\beta \in B}) \coloneqq \prod_{\beta \in B} \iota_\beta(a_\beta)$. From the prior reasoning we know that ι is well-defined and multilinear.

Furthermore, suppose A' is a commutative R-algebra and we are given any collection of unital

R-algebra homomorphisms $\{f_\beta : A_\beta \to A'\}_{\beta\in B}$, and we let $T((a_\beta)_{\beta\in B}) \coloneqq \prod_{\beta\in B} f_\beta(a_\beta)$. Then there is a unique unital R-algebra homomorphism $f : \bigotimes_{\beta\in B} A_\beta \to A'$ such that:



The uniqueness is because $\bigcup_{\beta\in B} \iota_\beta(A_\beta) \subseteq \iota(\prod'_{\beta\in B} A_\beta)$ where the former set generates $\bigotimes_{\beta\in B} A_\beta$ as an R-algebra. So any two unital R-algebra homomorphisms completing this diagram must equal each other everywhere.

As for showing that f exists, just use the fact that $\bigotimes_{\beta\in B} A_\beta$ is a coproduct in order to conclude there is a unique unital R-algebra homomorphism $f : \bigotimes_{\beta\in B} A_\beta \to A'$ such that $f \circ \iota_\beta = f_\beta$ for all β. Then as multiplication passes through f, we can conclude:

$$\begin{aligned}(f\circ\iota)((a_\beta)_{\beta\in B}) &= f(\textstyle\prod_{\beta\in B} \iota_\beta(a_\beta)) \\ &= \textstyle\prod_{\beta\in B}(f\circ\iota_\beta)(a_\beta) = \prod_{\beta\in B} f_\beta(a_\beta) = T((a_\beta)_{\beta\in B}).\end{aligned}$$

Anyways, this connection isn't the most important thing ever.

To finish off for today, I'll just mention that now that we know how to take coproducts in $\mathbf{CAlg}_R$, the construction on *pages 1023-1024* can be easily adapted to construct general colimits of functors $F : \mathbf{I} \to \mathbf{CAlg}_R$.

We just need to change the direct sum to a tensor product and then we need to make N a two sided ideal instead of a submodule.

I don't think it is worth my time going further in depth on this. So I'm going to finish my tangent on algebras and colimits and etc now.

8/6/2026

I'm finally going to talk about manifolds again. If you recall from *page 1067*, I was originally going to go over a cleaner equivalent way of defining a tangent space which importantly generalizes to analytic and complex manifolds. I'm going to finally go over that now that I've described how to take a filtered colimits of commutative unital $\mathbb{R}$-algebras.

To give some quick motivation for what comes next, if M is a C^r manifold (perhaps with corners) and $U \subseteq M$ is an open set containing a point p, then our prior ability to identify T_pM with T_pU relied on us being able to construct C^r bump functions. In other words, given any closed set $K \subseteq U$ also containing p we needed to be able to find a function $\phi \in C^r(M, [0,1])$ such that $\phi \equiv 1$ on K and $\operatorname{supp}(\phi) \subseteq U$. Unfortunately, the identity principle (see *page 1015* for one case of it) guarentees that there won't exist analytic or holomorphic bump functions.

To address this short coming we instead consider derivations acting on a stalk. Recall the definition of sheaves, stalks, and germs given on *pages 1026-1027*. Then given any C^r manifold M (with $r \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$) and any $0 \leq k \leq r$, we consider the sheaf $\mathscr{C}^k_M : \mathbf{Op}(M) \to \mathbf{CAlg}_{\mathbb{R}}$ such that $\mathcal{F}(U) = C^k(U, \mathbb{R})$ and for any open sets $V \subseteq U \subseteq M$ we have that $r^U_V(f) = f|_V$.

Here $C^0(U, \mathbb{R}) \coloneqq C(U, \mathbb{R})$.

For any $p \in M$ we denote the stalk of $\mathscr{C}^k_M$ at p to be the set $C^k_p(M)$. Also, given any open set U and C^k map $f : U \to \mathbb{R}$, we typically denote their germ as either $[(U, f)]$ or $[f]$.

Given any germ $[f] \in C^k_p(M)$, it is well-defined to evaluate $f(p)$.

This is because if $[(U, f)] = [(V, g)]$ then there is some open set $W \subseteq U \cap V$ containing p such that $f|_W = g|_W$. But in turn we know that $f(p) = g(p)$.

Now, we say a derivation on $C^1_p(M)$ is an $\mathbb{R}$-linear map $v : C^1_p(M) \to \mathbb{R}$ which also satisfies that if $[f]$ and $[g]$ are germs then $v([f][g]) = f(p)v([g]) + g(p)v([f])$. The set of all derivations on $C^1_p(M)$ is easily checked to be a real vector space and we denote that vector space by $\mathcal{D}_p(M)$.

Let M be an n-dimensional C^r manifold (where $r \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$). Then let $U \subseteq M$ be an open set, let $p \in M$, and consider any smooth chart (V, φ) such that $p \in V \subseteq U$. Note that φ defines local coordinates $(x_1, \ldots, x_n)$ on V. I'll also write $\varphi(p) = (x_1(p), \ldots, x_n(p))$.

Lemma 1: If $f \in C^k(U, \mathbb{R})$ (where $k \geq 1$) and $f(p) = 0$ then there is some open subset $W \subseteq V$ containing p such that:

$$f|_W = \sum_{i=1}^{n} (x_i - x_i(p))h_i,$$

where $h_i \in C^{k-1}(W, \mathbb{R})$ for all i.

Proof:

To start off, if $\varphi(V)$ is not open in $\mathbb{R}^n$, then by definition $f \circ \varphi^{-1}$ can be extended to being a C^k function $\widetilde{f}$ from an open set $\widetilde{V} \subseteq \mathbb{R}^n$.

Next, let $0 < r$ be a real number such that $\overline{B_r(\varphi(p))} \subseteq \widetilde{V}$. Then for any $x = (x_1, \ldots, x_n) \in \overline{B_r(\varphi(p))}$ we can write:

$$\widetilde{f}(x) = \sum_{i=1}^{n} (x_i - x_i(p)) \int_0^1 \frac{\partial}{\partial x_i} \widetilde{f}(\varphi(p) + s(x - \varphi(p)))\mathrm{d}s$$

By Leibniz's rule, we can conclude that $\widetilde{h}_i : B_r(\varphi(p)) \to \mathbb{R}$ given by:

$$\widetilde{h}_i(x) = \int_0^1 \frac{\partial}{\partial x_i} \widetilde{f}(\varphi(p) + s(x - \varphi(p)))\mathrm{d}s$$

is a C^{k-1} map. And now we let $W = \varphi^{-1}(\varphi(V) \cap B_r(\varphi(p)))$ and $h_i : W \to \mathbb{R}$ be defined by $h_i = \widetilde{h}_i \circ \varphi$. And then each h_i will be a C^{k-1} map and $f|_W = \sum_{i=1}^{n} (x_i - x_i(p))h_i$. ■

Now let (V, φ) be the smooth chart from before and suppose $f \in C^1(U, \mathbb{R})$ with $f(p) = 0$. If there exists an open set $W \subseteq U \cap V$ such that $f|_W = \sum_{i=1}^n (x_i - x_i(p))h_i$ where each $h_i \in C^0(W, \mathbb{R})$ and $h_i(\varphi(p)) = 0$ for each i, we say f vanishes to first order at p.

Lemma 2: Let $W, W' \subseteq U \cap V$ both be open sets containing p and then suppose $h_1, \ldots, h_n \in C^0(W, \mathbb{R})$ and $\overline{h_1}, \ldots, \overline{h_n} \in C^0(W', \mathbb{R})$ satisfy that:

$$f|_W = \sum_{i=1}^n (x_i - x_i(p))h_i \text{ and } f|_{W'} = \sum_{i=1}^n (x_i - x_i(p))\overline{h_i}.$$

Then $h_i(p) = \overline{h_i}(p)$ for all i.

Proof:

By considering the set $W \cap W'$ we can assume without loss of generality that $W = W'$. Also, we can assume without loss of generality that W is an open set in $\mathbb{R}^n_{\geq 0}$, H^n, or $\mathbb{R}^n$ since $\overline{h_i}(p) = h_i(p) = 0$ iff $(\overline{h_i} \circ \varphi^{-1})(\varphi(p)) = (h_i \circ \varphi^{-1})(\varphi(p)) = 0$. Finally, note that $f = \sum_{i=1}^n (x_i - x_i(p))h_i = \sum_{i=1}^n (x_i - x_i(p))\overline{h_i}$ means that:

$$0 \equiv \sum_{i=1}^n (x_i - x_i(p)) \cdot (h_i - \overline{h_i}).$$

Thus, we will be done once we show that if $0 \equiv \sum_{i=1}^n (x_i - x_i(p))g_i$ where $g_i \in C^0(W, \mathbb{R})$ then $g_i(x(p)) = 0$ for all i (where $x(p) = (x_1(p), \ldots, x_n(p))$).

Fortunately, consider any $s > 0$ such that $x(p) + se_i \in W$. (Here $e_1, \ldots, e_n$ is the standard basis for $\mathbb{R}^n$). Then $0 = \int_0^s 0 \mathrm{d}t = \int_0^s t g_i(x(p) + te_i)\mathrm{d}t$. But now if $g_i(x(p)) \neq 0$ then by continuity we know there is some $s > 0$ and $\varepsilon > 0$ such that $|g_i(x(0) + te_i)| > \varepsilon$ when $0 \leq t \leq s$. In turn, we have a contradiction as $|\int_0^s t g_i(x(p) + te_i)\mathrm{d}t| \geq s\varepsilon$ means that $0 = \int_0^s t g_i(x(p) + te_i)\mathrm{d}t \neq 0$. ■

More specifically, what the prior lemma shows is that if $f \in C^1(U, \mathbb{R})$ with $f(p) = 0$ and $f|_W = \sum_{i=1}^n (x_i - x_i(p))h_i$ for some open set $W \subseteq U \cap V$ containing p and some C^0 maps $h_1, \ldots, h_n$ then:

$$h_i(p) = \frac{\partial}{\partial x_i}\Big|_p f \text{ for each } i.$$

After all, the above equation holds for the functions $h_1, \ldots, h_n$ constructed in lemma 1, and lemma 2 then says that all other possible $h_1, \ldots, h_n$ must agree with the functions constructed in lemma 1. As a result:

$$f \text{ vanishes to first order iff } \frac{\partial}{\partial x_i}\Big|_p f = 0 \text{ for all } i \text{ iff } v(f) = 0 \text{ for all } v \in T_pU$$

This leads naturally to another result:

Lemma 3: Now suppose (V', ψ) is another smooth chart containing p. We'll let ψ define the local coordinates $(y_1, \ldots, y_n)$ on V'. Also I'll write $\psi(p) = y(p) = (y_1(p), \ldots, y_n(p))$. If $W \subseteq U \cap V$ is an open set containing p and there exists functions $h_i \in C^0(W, \mathbb{R})$ such that $f|_W = \sum_{i=1}^n (x_i - x_i(p))h_i$ and $h_i(p) = 0$ for each i, then for any open set $\Omega \subseteq U \cap V'$ containing p as well as functions $g_i \in C^0(\Omega, \mathbb{R})$ such that $f|_\Omega = \sum_{i=1}^n (y_i - y_i(p))g_i$, we know that $g_i(p) = 0$ for each i.

Proof:
We can conclude via lemmas 1 and 2 that $g_i(p) = \frac{\partial}{\partial y_i}\Big|_p f$ for each i. But now note from page 1052 that each linear operator $\frac{\partial}{\partial y_i}\Big|_p$ is a linear combination of the operators $\frac{\partial}{\partial x_j}\Big|_p$. And since $\frac{\partial}{\partial x_j}\Big|_p f = h_j(p) = 0$ for each j by assumption, we can conclude $g_i(p) = 0$ for each i. ■

As a result of the prior lemmas, we can say that f vanishing to first order at p is independent of our chosen local coordinates or the particular functions in our expression:

$$f|_W = \sum_{i=1}^{n}(x_i - x_i(p))h_i.$$

Also as is hopefully obvious, if $[(U, f)] = [(\widetilde{U}, \widetilde{f})]$ in $C^1_p(M)$ and f vanishes to first order at p then so does $\widetilde{f}$. So, we can talk about a germ in $C^1_p(M)$ vanishing to first order.

Now here's the new and improved definition of the tangent plane of p. For now I'll denote my old construction as T_pM. Meanwhile, I'll define a new construction by:

$$\mathcal{T}_pM \coloneqq \{v \in \mathcal{D}_p(M) : v([f]) = 0 \text{ if } [f] \text{ vanishes to first order}\}.$$

Clearly $\mathcal{T}_pM$ is a vector subspace of $\mathcal{D}_p(M)$.

Theorem: There is a bijective linear map $\Phi : T_pM \to \mathcal{T}_pM$ such that $(\Phi(v))([f]) \coloneqq v(f)$ for all $f \in C^k(M, \mathbb{R})$.

Proof:
Suppose $[(U, f)]$ is any germ and note that by multiplying on a C^r bump function, we can find a function $g \in C^1(M, \mathbb{R})$ such that $[(M, g)] = [(U, f)]$. So, for any $v \in T_pM$ we let $(\Phi(v))([f]) \coloneqq v(g)$.

To see that Φ is well-defined, note that if $[(M, g)] = [(U, f)] = [(M, \widetilde{g})]$ then $g = f = \widetilde{g}$ on a neighborhood of p. So by proposition 3.8 on page 1033 we know that $v(g) = v(\widetilde{g})$.

Showing that $\Phi(v) \in \mathcal{D}_p(M)$ is then easy. Furthermore, if $[(U, f)]$ vanishes to first order then we know for any $g \in C^k(M, \mathbb{R})$ with $[(M, g)] = [(U, f)]$ that $v(g) = 0$. So, $\Phi(v) \in \mathcal{T}_pM$. Finally, it is easy to verify that $\Phi(c_1v_1 + c_2v_2) = c_1\Phi(v_1) + c_2\Phi(v_2)$ for all $c_1, c_2 \in \mathbb{R}$ and $v_1, v_2 \in T_pM$. So, Φ is a linear map from T_pM to $\mathcal{T}_pM$.

To show that Φ is an isomorphism, let (U, φ) be a smooth chart containing p and defining local coordiantes $\varphi(m) = x(m) = (x_1(m), \ldots, x_n(m))$. Then recall $\frac{\partial}{\partial x_1}\Big|_p, \ldots, \frac{\partial}{\partial x_n}\Big|_p$ is a basis for T_pM. Meanwhile, we can define a basis for $\mathcal{T}_pM$ as follows:

Let $[(V, f)] \in C^1_p(M)$. By lemma 1 from a few pages ago can find an open subset $W \subseteq V \cap U$ as well as functions $h_i \in C^0(W, \mathbb{R})$ such that:

$$f|_W = f(p) + (x_i - x_i(p))h_i.$$

And now we define $\omega_i([f]) \coloneqq h_i(p)$. This is well-defined because if $[(V,f)] = [(\widetilde{V},\widetilde{f})]$ then we know there is some open set $\Omega \subseteq V \cap \widetilde{V}$ such that $f|_\Omega = \widetilde{f}|_\Omega$. And now upon writing that $f|_W = f(p) + \sum_{i=1}^n (x_i - x_i(p))h_i$ and $\widetilde{f}|_{\widetilde{W}} = f(p) + \sum_{i=1}^n (x_i - x_i(p))\widetilde{h}_i$, we know that:
$(f - f(p))|_{\Omega\cap\widetilde{W}} = \sum_{i=1}^n (x_i - x_i(p))\widetilde{h}_i$ and $(f - f(p))|_W = \sum_{i=1}^n (x_i - x_i(p))h_i$.

So by lemma 2 from before, we know that $h_i(p) = \widetilde{h}_i(p)$ for each i. And this proves that $\omega_i([f]) = \omega_i([\widetilde{f}])$.

Showing that $\omega_i \in \mathcal{T}_pM$ is now easy.

To show one necessary algebraic property of ω_i suppose $[f],[g] \in C_p^k(M)$ and write:
$f|_W = f(p) + \sum_{i=1}^n (x_i - x_i(p))\widetilde{h}_i$ and $g|_{\widetilde{W}} = g(p) + \sum_{i=1}^n (x_i - x_i(p))\widetilde{h}_i$.

Then after regrouping terms we can see that:
$$(fg)|_{W\cap\widetilde{W}} = (fg)(p) + \sum_{i=1}^n (x_i - x_i(p)) \cdot \left(h_i g + \widetilde{h}_i f - (x_i - x_i(p))h_i\widetilde{h}_i\right)$$

So, $\omega_i([f][g]) = \omega_i([f])g(p) + \omega_i([f])f(p) - 0$.

I also claim $\omega_1,\ldots,\omega_n$ span $\mathcal{T}_pM$. After all, let $\nu \in \mathcal{T}_pM$ be an arbitrary derivation.
We can consider the germ $[(U,f_i)]$ such that $f_i(m) = x_i(m) - x_i(p)$.
And then we let $c_i = \nu([f_i])$. Now for any germ $[f] \in C_p^k$, upon writing $f = f(p) + \sum_{i=1}^n (x_i - x_i(p))h_i$ we know that:
$f - f(p) - \sum_{i=1}^n (x_i - x_i(p))h_i(p)$ vanishes to first order at p.

So:
$$\begin{aligned} 0 &= \nu\left([f - f(p) - \textstyle\sum_{i=1}^n (x_i - x_i(p))h_i(p)]\right) \\ &= \nu([f]) - f(p)\nu([0]) - \textstyle\sum_{i=1}^n \nu([x_i - x_i(p)])h_i(p) \\ &= \nu([f]) - 0 - \textstyle\sum_{i=1}^n c_i\omega_i([f]). \end{aligned}$$

Or in other words, $\nu = \sum_{i=1}^n c_i\omega_i$.

And finally, I claim that $\omega_1,\ldots,\omega_n$ are all linearly independent. After all, suppose $c_1,\ldots,c_n \in \mathbb{R}$ satisfy that $\sum_{i=1}^n c_i\omega_i([f]) = 0$ for every germ $[f]$. Then in particular we know that $c_j = \sum_{i=1}^n c_i\omega_i([x_j - x_j(p)]) = 0$ for every j.

And now to finish the proof, just note for any germ $[(U,f)]$ that:
$$\Phi\left(\left.\frac{\partial}{\partial x_i}\right|_p\right)([f]) = \left.\frac{\partial}{\partial x_i}\right|_p f = \omega_i([f]) \text{ for each } i. \blacksquare$$

For the sake of convenience, I'll identify ω_i from the prior theorem with $\left.\frac{\partial}{\partial x_i}\right|_p$.

One last thing to sort out is how to define the differential of a C^1 map $F : M \to N$ with respect to the vector space $\mathcal{T}_pM$. To do this, we merely say:
$(\mathrm{d}F_p(\nu))([(U,f)]) \coloneqq \nu([(F^{-1}(U), f\circ F)])$ for every $\nu \in \mathcal{T}_pM$ and $[(U,f)] \in C^1_{F(p)}(N)$.

Checking that $\mathrm{d}F_p(\nu)$ is a well-defined operator in $\mathcal{D}_{F(p)}(N)$ for every $\nu \in \mathcal{T}_pM$ and that $\mathrm{d}F_p : \mathcal{T}_pM \to \mathcal{D}_{F(p)}(N)$ is a well-defined linear map is easy.

As for proving that $\mathrm{d}F_p(\nu)$ annihilates all germs vanishing to first order, it suffices to prove that if $[f] \in C^1_{F(p)}(N)$ vanishes to first order then so does $[f \circ F] \in C^1_p(M)$.

To do this, let $[(V, f)]$ be a germ and let $\widetilde{\mathrm{d}}F_p : T_p(F^{-1}(V)) \to T_{F(p)}V$ be the old definition of the differential of F. Then note that $f \circ F$ vanishes to first order iff $v(f \circ F) = 0$ for all $v \in T_p(F^{-1}(V))$. Yet simultaneously $v(f \circ F) = (\mathrm{d}F_p(v))(f)$. And since f vanishes to first order and $\mathrm{d}F_p(v) \in T_{F(p)}V$, we know that:

$$(\mathrm{d}F_p(v))(f) = 0.$$

Also if $\Phi_M : T_pM \to \mathcal{T}_pM$ and $\Phi_N : T_{F(p)}N \to \mathcal{T}_{F(p)}N$ are the maps constructed in the prior theorem, then we can see by looking at a representive for each germ whose domain is all of M or N that:

$$\begin{array}{ccc} & f \mapsto v(f) \qquad\qquad g \mapsto v(g \circ F) & \\ T_pM & \xrightarrow{\widetilde{\mathrm{d}}F_p} & T_{F(p)}N \\ \wr \downarrow \Phi_M & & \wr \downarrow \Phi_N \\ \mathcal{T}_pM & \xrightarrow{\mathrm{d}F_p} & \mathcal{T}_{F(p)}N \\ & [f] \mapsto v(f) \qquad\qquad [g] \mapsto v(g \circ F) & \end{array}$$

Anyways this new definition of the tangent plane is the most general way the tangent plane of a manifold is defined. It is also nice because as mentioned before, it generalizes to types of manifolds (i.e. real-analytic and complex) where germs don't have representatives in any arbitrary neighborhood of a point.

8/17/2026

I'm going to keep being distracted due to the fact I've been in and out of being homeless for two weeks. But right now I kinda want to study again. I'll be studying Lee's book again where I left off.

Local Immersion Theorem For Manifolds with Boundaries: Let M be an m-dimensional C^r manifold with a boundary (but no corners), and let N be an n-dimensional C^r manifold without a boundary. Also suppose $F : M \to N$ is a C^r immersion. For any $p \in \partial M$ there exists a smooth boundary chart (U, φ) on M containing p as well as a smooth chart (V, ψ) on N such that $\varphi(p) = 0$, $F(U) \subseteq V$, and $\widehat{F} := \psi \circ F \circ \varphi^{-1}$ is given by:

$$\widehat{F}(x_1, \ldots, x_m) = (x_1, \ldots, x_m, \underbrace{0, \ldots, 0}_{n-m \text{ zeros}})$$

Proof:
By choosing local coordinates around p it suffices to assume M is an open subset of H^m and N is an open set $\mathbb{R}^n$. This follows by identical reasoning as in the rank theorem. Also by identical reasoning as in the rank theorem, we can assume $p = 0$ and $F(p) = 0$.

Next we know that F extends to a C^r function $\widetilde{F} : W \to \mathbb{R}^n$ such that $M \subseteq W$ and $\widetilde{F}|_M = F$. Because $\mathrm{d}\widetilde{F}_0 = \mathrm{d}F_0$ is an injection (here we recall that we can identify T_pM with T_pW), we can shrink W and M such that $\widetilde{F}$ is a C^r immersion. Also, we can shrink W (without shrinking M) such that $\widetilde{F}(W) \subseteq N$. And now let us write the coordinates on W as $(x_1, \ldots, x_m)$ and the coordinates on N as $((v_1, \ldots, v_m), (w_1, \ldots, w_{n-m}))$.

By the rank theorem (see *page 1062*), there exists smooth charts (U_0, φ_0) and (V_0, ψ_0) such that $0 \in U_0 \subseteq W$, $F(U_0) \subseteq V_0 \subseteq N$, $\varphi_0(0) = 0$, and $\widehat{F} \coloneqq \psi_0 \circ \widetilde{F} \circ \varphi_0^{-1}$ is given by:

$$\widehat{F}(x_1, \ldots, x_m) = (x_1, \ldots, x_m, \underbrace{0, \ldots, 0}_{n-m \text{ zeros}})$$

The one issue that needs to be worked out is that $\varphi_0(U_0 \cap M)$ may not be contained in H^m. To address this, we make the following modification:

φ_0 is a C^r diffeomorphism from $U_0 \subseteq \mathbb{R}^m$ to an open set $\widehat{U}_0 \subseteq \mathbb{R}^m$ such that $\varphi_0(0) = 0$. In turn, $\varphi_0^{-1} \times \mathrm{Id}_{\mathbb{R}^{n-m}} : \widehat{U}_0 \times \mathbb{R}^{n-m} \to U_0 \times \mathbb{R}^{n-m}$ is a C^r diffeomorphism defined on an open neighborhood of 0.

Next let $\psi \coloneqq (\varphi_0^{-1} \times \mathrm{Id}_{\mathbb{R}^{n-m}}) \circ \psi_0$. This is well-defined C^r diffeomorphism from an open set $V \coloneqq V_0 \cap \psi_0^{-1}(\widehat{U}_0 \times \mathbb{R}_{n-m})$ of N containing $F(0) = 0$.

Finally $U \coloneqq U_0 \cap F^{-1}(V) \cap M$ is an open subset of M containing 0 such that for all $x \in U$:

$$\begin{aligned}\psi \circ F(x_1, \ldots, x_m) &= (\varphi_0^{-1} \times \mathrm{Id}_{\mathbb{R}^{n-m}}) \circ \psi_0 \circ F \circ \varphi_0^{-1} \circ \varphi_0(x_1, \ldots, x_m) \\ &= (\varphi_0^{-1} \times \mathrm{Id}_{\mathbb{R}^{n-m}})(\varphi(x_1, \ldots, x_m), 0) = (x_1, \ldots, x_m, 0, \ldots, 0)\end{aligned}$$

Therefore the original coordinates for M restricted to U as well as the smooth chart (V, ψ) works. ■

Given C^r manifolds M and N, we say a map $F : M \to N$ is a C^k embedding (where $k \leq r$) if F is a C^k immersion and F is a homeomorphism from M to $F(M)$.

Exercise 4.16: If M, N, P are C^r manifolds, $k \leq r$, and $F : M \to N$ and $G : N \to P$ are C^k embeddings then $(G \circ F) : M \to P$ is a C^k embedding as well.

Proof:
$G \circ F$ is obviously C^k. Also a composition of immersions will be an immersion. After all, for all $p \in M$ we have that $\mathrm{d}(G \circ F)_p = \mathrm{d}G_{F(p)} \circ \mathrm{d}F_p$ is injective since $\mathrm{d}F_p$ and $\mathrm{d}G_{F(p)}$ are both injective. Finally, let $h_F : F(M) \to M$ and $h_G : G(N) \to N$ be the continuous inverse functions to F and G respectively. Then $(h_F \circ h_G)|_{G(F(M))}$ is a continuous inverse function to $G \circ F$. ■

Some examples of C^r embeddings are:

1. If M is a C^r manifold and $U \subseteq M$ is an open set, then the inclusion map $\iota : U \hookrightarrow M$ is a C^r embedding.

   It's C^r, at every point it's differential is just the identity linear map, and also it's injective with an inverse map from $\iota(U) \to U$ just being the identity function (which is obviously continuous).

2. Suppose $M_1, \ldots, M_k$ are C^r manifolds and $p_i \in M_i$ is an arbitrarily chosen point for each i. Then for every $1 \leq j \leq k$ we have that the map $\iota_j : M_j \to M_1 \times \cdots \times M_k$ given by $\iota_j(m_j) = (p_1, \ldots, p_{j-1}, m_j, p_{j+1}, \ldots, p_k)$ is a C^r embedding.

   That ι_j is C^r is immediate. Also we know that it has a continuous C^r inverse $\pi_j : M_1 \times \cdots \times M_k \to M_j$ given by $(m_1, \ldots, m_{j-1}, m_j, m_{j+1}, \ldots, m_k) \mapsto m_j$. Finally, for any $q \in M_j$ we have that$\mathrm{d}(\mathrm{Id}_{M_j})_q = \mathrm{d}\pi_{\iota_j(q)} \circ \mathrm{d}(\iota_j)_q$ implies that $\mathrm{d}\pi_{\iota_j(q)}$ is surjective and $\mathrm{d}(\iota_j)_q$ is injective. From this we can conclude that ι_j is an immersion and π_j is a submersion.

Next, here are two nonexamples of smooth embeddings:

1. Consider the map $\gamma : \mathbb{R} \to \mathbb{R}^2$ given by $\gamma(t) = (t^3, 0)$. This is a C^∞ map with a continuous inverse function given by $(t, 0) \mapsto (\sqrt[3]{t}, 0)$. That said, $\mathrm{d}\gamma_0$ is a trivial linear map since $\gamma'(0) = 0$. So, γ fails to be an immersion.

2. Consider the lemniscate $\beta : (-\pi, \pi) \to \mathbb{R}^2$ defined by $\beta(t) = (\sin(2t), \sin(t))$.

   Proving that β is injective just involves considering a bunch of subcases. So suppose $s, t \in (-\pi, \pi)$ satisfy that $\sin(2t) = \sin(2s)$ and $\sin(t) = \sin(s)$.

   - If $t \in (0, \pi/2)$ and $s \neq t$ then $\sin(t) = \sin(s) \implies s = \pi - t$. Yet then $\sin(2s) < 0$ while $\sin(2t) > 0$. So if $t \in (0, \pi/2)$ then $s = t$.
   - If $t \in (\pi/2, \pi)$ and $s \neq t$ then $\sin(t) = \sin(s) \implies s = \pi - t$ again. Yet now we have that $\sin(2t) < 0$ while $\sin(2s) > 0$. So if $t \in (\pi/2, \pi)$ then $s = t$.
   - Given that $t \in (-\pi, \pi)$, we have that $\sin(t) = 0$ iff $t = 0$ and $\sin(t) = 1$ iff $t = \pi/2$. So, β is injective on $[0, \pi)$.
   - By the oddness of $\sin(2t)$ and $\sin(t)$ we can also conclude that β is injective on $(-\pi, 0]$. And since $\sin(t) < 0$ for all $t < 0$ and $\sin(t) > 0$ for all $t > 0$, we can conclude that β is injective on $(-\pi, \pi)$.

   We can also see that β is a C^∞ map. In particular, $\beta'(t) = (2\cos(2t), \cos(t))$. But now $\cos(t) = 0$ iff $t = \pm\pi/2$. Yet $\cos(2(\pi/2)) = -1 = \cos(2(-\pi/2))$. So $\beta'(t) \neq 0$ for any $t \in (-\pi, \pi)$ and we can conclude that β is an immersion.

   That all said, β does not have a continuous inverse function. After all, $\beta(0) = 0$ but also $\beta(t) \to 0$ as $t \to \pi$ or as $t \to -\pi$. Therefore, if $f : \beta((-\pi, \pi)) \to (-\pi, \pi)$ was the inverse function to β then we'd have to have that f is not continuous at 0.

Proposition 4.22: Suppose M and N are C^r manifolds with or without boundary, and $F : M \to N$ is an injective C^k immersion (where $k \leq r$). Then F is a C^k embedding if any of the following conditions hold:

(a) F is an open or closed map into N.

Why?
F being an open map into N trivially means F is an open map into the subspace topology of $F(M) \subseteq N$.

Meanwhile, if F is a closed map then let $U \subseteq M$ be an open set. Then $M - U$ is a closed set so $F(M - U)$ is a closed subset of N. It follows that $N - F(M - U)$ is an open subset of N. And so $F(M) \cap (N - F(M - U)) = F(U)$ (which holds because F is injective) is an open set in the subspace topology of $F(M) \subseteq N$.

More generally, this reasoning shows that if X and Y are topological spaces then any injective continuous map $f : X \to Y$ that is also open or closed is a topological embedding (i.e. a homeomorphism from X to $f(X) \subseteq Y$).

(b) F is a proper map.

Why?
Let E be any closed subset of M and then suppose q is any accumulation point of $F(E)$ in N. Since N is an LCH space, we can pick a precompact open set $V \subseteq N$ containing q. And now $\overline{V} \cap F(E)$ also has q as an accumulation point. Yet because F is proper, we know that $F^{-1}(\overline{V})$ is compact. Then since closed subsets of compact sets are also compact, we know that $E \cap F^{-1}(\overline{V})$ is compact. In turn, as F is continuous we have that $F(E \cap F^{-1}(\overline{V})) = F(E) \cap \overline{V}$ is compact in N (and thus closed since N is Hausdorff). So $F(E) \cap \overline{V}$ must contain all its accumulation points and so $q \in F(E)$.

Since q was arbitrary, we've proven that $F(E)$ is a closed subset of N. So, F is a closed map into N and we can finish by citing part (a).

More generally, this reasoning shows that if X is a topological space and Y is an LCH space then any continuous proper map $f : X \to Y$ is also a closed map.

(c) M is a compact manifold.

Why?
$F : M \to F(N)$ is a continuous bijection, M is compact, and $F(N)$ is Hausdorff. Thus F is a homeomorphism.

(d) M has no boundary and $\dim(M) = \dim(N)$.

Why?
By proposition 4.8(b) on *page 1061*, we know F is a local C^k-diffeomorphism. It then follows by proposition 4.6(c) on the page before proposition 4.8(b) that F is an open map. So, we finish by invoking part (a) of this theorem. ■

8/20/2026

I was realizing yesterday that I've conflated open maps with closed maps an unhealthy amount of times in my brain and am not sure if that's made some of my prior notes incorrect. I've double checked a bunch of pages and not found any errors but still.

Anyways a thing I want to get across first and foremost today is that a closed surjective continuous map is a quotient map for different reasons then an open surjective continuous map is.

Proposition: Suppose X and Y are topological spaces and $f : X \to Y$ is a continuous surjective map. Then f is a quotient map if either of the following holds:

(a) f is an open map.

Why?
Suppose $U \subseteq Y$ satisfies that $f^{-1}(U)$ is an open subset of X. Then because f is surjective, we know that $f(f^{-1}(U)) = U$. And because f is open, we know that U is open.

(b) f is a closed map.

Why?
Suppose $U \subseteq Y$ again satisfies that $f^{-1}(U)$ is an open subset of X. Then we know that $X - f^{-1}(U)$ is a closed subset of X. So because f is a closed map, we know that $f(X - f^{-1}(U)) = f(X) - U$ is a closed subset of Y. Since f is surjective, we in other words know that $Y - U$ is a closed set. Equivalently, U is an open set. ■

Another thing I want to mention is that I should think of a C^k embedding as the differential-topology equivalent of a topological embedding. Similarly, a (surjective) C^k submersion is the differential-topology equivalent of a quotient map as I will soon show.

One final observation I want to make before getting back into taking notes on Lee's book is that we can precisely restate the definition of a coordinate patch α for a k-dimensional C^r manifold in $\mathbb{R}^n$ given on *page 84* as being that α is a C^r-embedding. I will come back to this soon when talking about embedded submanifolds.

Theorem 4.25 (Local Embedding Theorem): Suppose M and N are C^r manifolds where M has no corners and $F : M \to N$ is a C^k map. Then F is a C^k immersion if and only if for all $p \in M$ there exists an open set $U \subseteq M$ such that $F|_U : U \to N$ is a C^k embedding.

$(\Longleftarrow)$
This direction is immediate.

$(\Longrightarrow)$
Suppose F is a C^k immersion. Without loss of generality we can assume M and N are equipped with the unique C^k structures containing their original C^r structures since that would have no effect on whether F is a C^k immersion or $F|_U$ is a C^k embedding. And this assumption is useful cause it let's us apply the rank theorem.

Firstly we show that any $p \in M$ has a neighborhood U on which F is injective.

First suppose $F(p) \notin \partial N$. Then as the set of interior points $\text{Int}(N)$ is an open set and F is continuous, we know that there is an open set $\Omega \subseteq M$ containing p with $F(\Omega) \subseteq \text{Int}(N)$. Then by either the rank theorem on *page 1062* or the local immersion theorem on *page 1101*, we know there is an open set $U_1 \subseteq M$ of p where F has the coordinate representation:

$$(x_1, \ldots, x_m) \mapsto (x_1, \ldots, x_m, 0, \ldots, 0).$$

So F is injective on U_1.

Meanwhile, suppose $F(p) \in \partial N$ and let (W, ψ) be a smooth chart on N containing $F(p)$. Then $U_0 := F^{-1}(W)$ is an open subset of M containing p. Furthermore, let $\iota : H^n (\text{ or } \mathbb{R}^n_{\geq 0}) \hookrightarrow \mathbb{R}^n$ be the inclusion map (here $n = \dim(N)$). Then $\iota \circ \psi \circ F|_{U_0}$ is a C^k map from M to a C^k manifold without boundary. Also, as $\mathrm{d}\iota_a$ and $\mathrm{d}\psi_b$ are bijections for all $a \in H^n$ and $b \in W$, we can conclude that $\iota \circ \psi \circ F|_{U_0}$ is a C^k-immersion still. Thus, we can apply the earlier reasoning to this map to find an open set $U_1 \subseteq U_0$ containing p such that $(\iota \circ \psi \circ F|_{U_0})|_{U_1}$ is an injection. Equivalently, this implies $F|_{U_1}$ is an injection.

Now consider any $p \in M$ and let $U_1 \subseteq M$ be an open set containing p on which F is injective. Since M is an LCH space, we can find a precompact open set U such that $p \in U \subseteq \overline{U} \subseteq U_1$. But now $F|_{\overline{U}}$ is an injective continuous function from a compact set. Therefore, $F|_{\overline{U}}$ is a topological embedding. Next since restrictions of topological embeddings are still topological embeddings, we know $F|_U$ is a topological embedding. And since $F|_U$ is a C^k immersion, we can conclude that $F|_U : U \to N$ is a C^k embedding. ■

Suppose M and N are C^r manifolds. If $\pi : M \to N$ is a continuous map then a section for π is a continuous right inverse function $\sigma : N \to M$ (i.e. $\pi \circ \sigma = \text{Id}_N$). A local section for π is a continuous map $\sigma : U \to M$ where $U \subseteq N$ is an open set and $\pi \circ \sigma = \text{Id}_U$.

Theorem 4.26 (Local Section Theorem): Suppose M and N are C^r manifolds without boundaries, and $\pi : M \to N$ is a C^k map (where $k \leq r$). Then π is a C^k submersion if and only if every $p \in M$ is in the image of a C^k local section of π.

$(\Longleftarrow)$

Given any $p \in M$, let $\sigma : U \to M$ be a C^k local section such that $q \in U$ and $\sigma(q) = p$. In other words, $q = \pi(\sigma(q)) = \pi(p)$. But now since $\pi \circ \sigma = \text{Id}_U$, we can conclude that $\mathrm{d}\pi_p \circ \mathrm{d}\sigma_q = \text{Id}_{T_qN}$. In turn, we must have that $\mathrm{d}\pi_p$ is surjective. And since p was arbitrary, we can conclude π is a submersion.

$(\Longrightarrow)$

Suppose π is a C^k submersion. Then without loss of generality we can assume M and N are equipped with the unique C^k structures containing their original C^r structures. After all, this change won't effect whether π is a C^k submersion, or whether some local section of π is C^k or not. Also, this assumption lets us apply the rank theorem.

Pick any $p \in M$ and let $q = \pi(p)$. By the rank theorem, we can choose smooth charts (V_0, φ) on M containing p and (U_0, ψ) containing q such that $\varphi(p) = 0$, $\pi(V_0) \subseteq U_0$, and $\widehat{\pi} := \psi \circ \pi \circ \varphi^{-1}$ has the form $\widehat{\pi}(x_1, \ldots, x_n, x_{n+1}, \ldots, x_m) = (x_1, \ldots, x_n)$.

For a small enough $\varepsilon > 0$ we have that the coordinate cube
$$C_\varepsilon := \{(x_1, \ldots, x_m) : |x_i| < \varepsilon \text{ for all } i\}$$
is an open subset of $\varphi(V_0)$ containing $0 = \varphi(p)$. In turn we have that:
$$C'_\varepsilon := \{(y_1, \ldots, y_n) : |y_i| < \varepsilon \text{ for all } i\}$$
is an open subset of $\psi(U_0)$ containing $0 = \widehat{\pi}(0) = \widehat{\pi} \circ \varphi(p) = \psi(q)$ and satisfying that $\widehat{\pi}(C_\varepsilon) = C'_\varepsilon$. Let $U = \psi^{-1}(C'_\varepsilon)$. Then define $\iota : C'_\varepsilon \to C_\varepsilon$ by:
$$\iota(y_1, \ldots, y_n) = (y_1, \ldots, y_n, \underbrace{0, \ldots, 0}_{m-n \text{ zeros}}).$$

Finally, let $\sigma = \varphi^{-1} \circ \iota \circ \psi|_U$. Then if $q' \in U$ and $\varphi(q') = (y_1, \ldots, y_n)$ we have that:
$$\begin{aligned} \pi \circ \sigma(q') &= \psi^{-1} \circ \widehat{\pi} \circ \varphi \circ (\varphi^{-1} \circ \iota \circ \psi)(q') \\ &= \psi^{-1} \circ \widehat{\pi} \circ \iota(y_1, \ldots, y_n) = \psi^{-1} \circ \widehat{\pi}(y_1, \ldots, y_n, 0, \ldots, 0) \\ &= \psi^{-1}(y_1, \ldots, y_n) = q' \end{aligned}$$

So $\pi \circ \sigma = \mathrm{Id}_U$. Also σ is a C^k map and
$$\sigma(q) = \varphi^{-1} \circ \iota(0, \ldots, 0) = \varphi^{-1}(0, \ldots, 0) = p. \blacksquare$$

Based on theorems 4.25 and 4.26, we also get the following purely topological definitions. Suppose X and Y are topological spaces and $f : X \to Y$ is a continuous function.

- f is a topological immersion if every point $x \in X$ has a neighborhood $U \subseteq X$ such that $f|_U : U \to Y$ is a topological embedding.
- f is a topological submersion if every point $x \in X$ is in the image of a (continuous) local section for f.

The prior two theorems say that any C^k immersion / submersion is a topological immersion / submersion. That said, be aware that a topological immersion or submersion that happens to also be a C^k map is not necessarily a "C^k immersion" or "C^k submersion".

For example, let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(t) = t^3$. Then f is a C^∞ map which is also a homeomorphism. And clearly homeomorphisms are trivially topological immersions and submersions. That said, as $\mathrm{d}f_0$ is the trivial linear map as $f'(0) = 0$, we know f is not a C^∞ immersion or submersion.

Theorem 4.27: Let M and N be C^r manifolds without boundaries, and suppose $\pi : M \to N$ is a C^k submersion. Then π is an open map. Also consequently, if π is surjective then π is a quotient map.

Proof:
Suppose $W \subseteq M$ is an open set and $q \in \pi(W) \subseteq N$. We can pick $p \in W$ with $\pi(p) = q$. And then by the local section theorem, we know there is an open subset $U \subseteq N$ containing q as well as a C^k map $\sigma : U \to M$ such that $\sigma(q) = p$ and $\pi \circ \sigma = \mathrm{Id}_U$. Yet now note

that $\sigma^{-1}(W)$ is an open subset of U (since σ is continuous) and it contains q. Also note that if $y \in \sigma^{-1}(W)$ then $y = \pi(\sigma(y)) \in \pi(W)$ Therefore, we've shown that q is in the topological interior of $\pi(W)$. And since q was arbitrary, this proves that $\pi(W)$ is an open set. ■

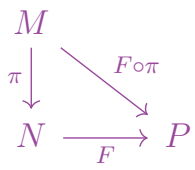
Exercise 4-6: Let M be a nonempty compact C^r manifold without a boundary, and let N be a connected non-compact C^r without a boundary. Then there is no C^k submersion $\pi : M \to N$ (where $k \leq r$).

Proof:
By the prior theorem we must have that such a function π is an open map. Thus in particular we can conclude that $\pi(M)$ is an open subset of N. That said, as M is compact and π is continuous, we must also have that $\pi(M)$ is a compact subset of N. Since N is Hausdorff, we can conclude $\pi(M)$ is a clopen set. And since N is connected, we thus must have that $\pi(M)$ is either the empty set or all of N.

$\pi(M)$ can't be the empty set however because $M \neq \emptyset$. Simultaneously, since N is not compact we can't have that $\pi(M) = N$. Thus we have a contradiction. ■

The next theorem is a generalization of exercise 4.10(b) on *pages 1061-1062*.

Theorem 4.29: Suppose M and N are C^r manifolds without boundaries and $\pi : M \to N$ is a surjective C^k submersion. For any C^r manifold P we have that any map $F : N \to P$ is C^k iff $F \circ \pi : M \to P$ is a C^k map.



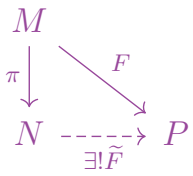
Proof:
If F is C^k then the composition $F \circ \pi$ will also be C^k. So, one direction of this theorem is trivial.

Conversely, suppose $F \circ \pi$ is C^k and let $q \in N$ be any point. Then as π is surjective we know that there exists $p \in M$ with $\pi(p) = q$. By the local section theorem, we know there exists an open set $U \subseteq N$ containing q as well as a C^k map $\sigma : U \to M$ such that $\sigma(q) = p$ and $\pi \circ \sigma = \mathrm{Id}_U$. And now $F|_U = F \circ \mathrm{Id}_U = F \circ (\pi \circ \sigma) = (F \circ \pi) \circ \sigma$ is a C^k map.

Since every point in N has a neighborhood on which F is a C^k map, we can conclude that F is globally a C^k map. ■

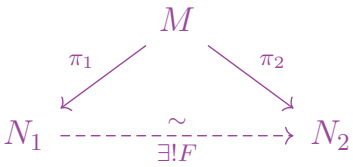
Corollary 4.30: Suppose M and N are C^r manifolds without boundaries and $\pi : M \to N$ is a surjective C^k submersion. If P is any C^r manifold and $F : M \to P$ is any C^k map that is constant on the fibers of π (meaning F is constant on each set $\pi^{-1}(q)$ for every $q \in N$),

then there exists a unique C^k map $\widetilde{F} : N \to P$ such that $F = \widetilde{F} \circ \pi$.

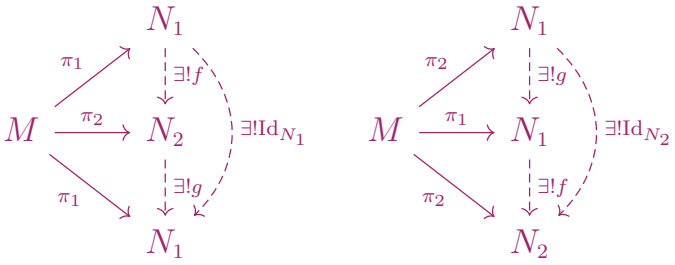


Proof:
Since π is a quotient map, we know from the universal property of quotient maps that there exists a unique continuous function $\widetilde{F} : N \to P$ such that $F = \widetilde{F} \circ \pi$. Then as F is C^k, the prior theorem tells us that $\widetilde{F}$ is also C^k. ■

Corollary 4.31: Suppose M, N_1, N_2 are C^r manifolds without boundaries and $\pi_1 : M \to N_1$ and $\pi_2 : M \to N_2$ are surjective C^k submersions that are constant on each other's fibers (i.e. $\pi_1(x) = \pi_1(y) \iff \pi_2(x) = \pi_2(y)$). Then there exists a unique C^k diffeomorphism $F : N_1 \to N_2$ such that $\pi_2 = F \circ \pi_1$.



Why?
Just consider the following two diagrams:



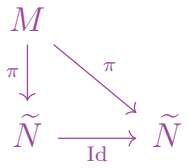
Something I want to quickly mention is that an anlogous corollary can be proven for quotient maps $p_1 : X \to Y_1$ and $p_2 : X \to Y_2$ (where X, Y_1, Y_2 are topological spaces). My original learning about quotient maps was fragmented so I never wrote that down before, but I think it's worth remembering in the future.

One last exercise I want to do before forcing myself to try to sleep is as follows:

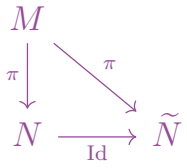
Exercise 4-7: Let M and N be C^r manifolds without boundaries, and let $\pi : M \to N$ be a surjective C^r submersion. Show there is no other manifold topology and C^r structure that we can equip N with such that for any C^r manifold P and map $F : N \to P$ we have that F is C^r if and only if $F \circ \pi$ is C^r.

Proof:
Let $\widetilde{N}$ be the same set as N except equipped with a potentially different manifold topology and C^r structure. Since the following diagram commutes and Id is a C^r map from $\widetilde{N}$ to $\widetilde{N}$,

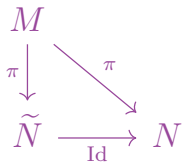
we can conclude that $\pi : M \to \widetilde{N}$ is a C^r map.



But then since $\pi : M \to \widetilde{N}$ is a C^r map we can also conclude from the following diagram that $\mathrm{Id} : N \to \widetilde{N}$ is a C^r map.



Meanwhile, as $\pi : M \to N$ is a C^r map we can conclude from the following diagram that $\mathrm{Id} : \widetilde{N} \to N$ is a C^r map.



So, we've proven that $\mathrm{Id} : N \to \widetilde{N}$ is a C^r-diffeomorphism. Since this also means Id is a homeomorphism, we can conclude that the topologies on N and $\widetilde{N}$ are identical. Also, $\mathrm{Id} : N \to \widetilde{N}$ being a C^r diffeomorphism easily lets us conclude that every smooth chart on N is C^r compatible with every smooth chart on $\widetilde{N}$ and vice versa.

Why?
Suppose (U, φ) is a smooth chart with respect to N and (V, ψ) is a smooth chart with respect to $\widetilde{N}$ such that $U \cap V \neq \emptyset$. Then as $\mathrm{Id} : N \to \widetilde{N}$ is a C^r-diffeomorphism, we know that $\psi \circ \varphi^{-1} = \psi \circ \mathrm{Id} \circ \varphi^{-1}$ and $\varphi \circ \psi^{-1} = \varphi \circ \mathrm{Id} \circ \psi^{-1}$ are C^r maps.

It follows that the C^r structures on N and $\widetilde{N}$ are also the same. ■

In particular, this shows that if M is a C^r manifold without a boundary and $\pi : M \to N$ is any surjective function then there is at most one manifold topology without boundary and C^r structure on N such that π is a C^r submersion.

Chapter 5 of John Lee's book will let us fully connect the abstract definition of a manifold with the more naive definition studied on *pages 77-99*.

Suppose M is a C^r manifold. A C^r embedded submanifold of M is a subset $S \subseteq M$ which is a topological manifold when equipped with the subspace topology and which has a C^r structure such that the inclusion map $\iota : S \to M$ is a C^r embedding.

Remarks:

- If $k \leq r$ then M has a unique C^k structure containing its C^r structure. Based on this structure on M, if S is a C^k manifold then we can ask if $\iota : S \to M$ is a C^k embedding. And if so we call S a C^k embedded submanifold of M.
- Some authors also refer to embedded submanifolds as regular submanifolds.

If M is a C^r manifold and $S \subseteq M$ is a C^k embedded submanifold of M then the codimension of S in M is defined to be the difference $\dim(M) - \dim(S)$. Here M is called the ambient manifold for S. If S has codimension 1 we say S is an embedded hypersurface.

Proposition 5.1: Suppose M is a C^r manifold without a boundary. Then the embedded C^k submanifolds of codimension 0 (without boundaries) in M are precisely the open submanifolds of M.

Proof:
If $U \subseteq M$ is an open submanfold then we know $\dim(U) = \dim(M)$. Furthermore, U is equipped with the subspace topology and we know from before that the inclusion map $\iota : U \hookrightarrow M$ is a C^k immersion and submersion. It follows that ι is a C^k embedding. And as M has no boundary, neither does U.

Conversely, suppose $U \subseteq M$ is an embedded C^k submanifold in M of codimension 0. By definition the inclusion map $\iota : U \hookrightarrow M$ is a C^k embedding. Yet as $\dim(M) = \dim(U)$ and ι is an immersion, we know from proposition 4.8(b) on *page 1061* that ι is a local C^k-diffeomorphism. In turn, by proposition 4.6(c) on *page 1060* we know ι is an open map. So, $\iota(U) = U$ is an open subset of M.

Finally, suppose (V, φ) is a smooth chart on M (equipped with its unique C^k structure). Then $U \cap V$ is an open subset of U and I claim that $(U \cap V, \varphi)$ is a smooth map on U. After all, suppose (W, ψ) is any other smooth chart on U as an embedded submanifold. Then as $\iota : U \hookrightarrow M$ is a C^k map we know that $\varphi \circ \iota \circ \psi^{-1} = \varphi \circ \psi^{-1}$ is C^k on $\psi((U \cap V) \cap W)$.

Meanwhile, as ι is a local C^k-diffeomorphism, we know for every $p \in (U \cap V) \cap W$ that there is some open set $\Omega \subseteq U$ containing p such that ι is a C^k-diffeomorphism from Ω (as an open submanifold of U) to Ω (as an open submanifold of M). In turn, $\psi \circ \varphi^{-1}$ is a C^k map on $\psi((U \cap V) \cap W \cap \Omega)$. So, $\psi \circ \varphi^{-1}$ is C^k at every $p \in (U \cap V) \cap W$. And with that, we've proven that the C^k structure on U as an embedded submanifold must be the same as the C^k structure on U as an open submanifold. ■

Proposition 5.2: Suppose M and N are C^r manifolds, and $F : N \to M$ is a C^r embedding. Let $S = F(N)$. With the subspace topology S is a topological manifold. Also S has a unique C^r structure making it into an embedded submanifold of M and satisfying that F is a C^r-diffeomorphism from N to S.

Proof:
S being a subspace of M and M being second countable and Hausdorff means that S is also second countable and Hausdorff. Also by assumption F is a homeomorphism from N to S. Therefore, N being locally Euclidean of dimension $\dim(N)$ means that S is as well.

Now we equip S with the C^r structure given by the charts $(F(U), \varphi \circ F^{-1})$ where (U, φ) is any smooth chart on N. Them being C^r compatible with each other is a result of the smooth charts on N being C^r compatible with each other. So, S is a C^r manifold. It's also easy to see now that F is a C^r-diffeomorphism. After all, for any $p \in S$ we can pick a smooth chart (U, φ) on N containing $F^{-1}(p)$. Then $(F(U), \varphi \circ F^{-1})$ is a smooth chart on S containing p, and:

$$\varphi \circ F \circ (\varphi \circ F^{-1})^{-1} = \mathrm{Id}_{\varphi(U)} \text{ is a } C^r \text{ map.}$$

So, $F^{-1} : S \to N$ is a C^r map.

Next, to show uniqueness just note that if F is to be a diffeomorphism from N to S then for any smooth chart (U, φ) on N we must have that $(F(U), \varphi \circ F^{-1})$ is a smooth chart on S. Similarly, for every smooth chart (V, ψ) on S we must have that $(F^{-1}(V), \psi \circ F)$ is a smooth chart on N. So the atlas described in the prior paragraph is maximal and F being a C^r-diffeomorphism from N to S automatically implies that S is equipped with that maximal structure.

Finally, I need to show that the map $\iota : S \to M$ is a C^r-embedding. Fortunately, consider that ι is given by the composition:

$$S \xrightarrow{F^{-1}} N \xrightarrow{F} M \qquad (\iota : S \to M)$$

Since F^{-1} is a C^r-diffeomorphism and F is a C^r-embedding, we can conclude ι is a C^r embedding. So, S is an embedded submanifold of M. ■

> As a side note, if $F : N \to M$ is a C^k embedding (where $k \leq r$) we can also now say that $S = F(N)$ has a unique C^k structure making it into an embedded submanifold of M such that F is a C^k-diffeomorphism from N to S. After all, we can just re-equip M and N with the unique C^k structures containing their original C^r structures and then everything works the same.

Proposition 5.3: Suppose M and N are C^r manifolds. For each $p \in N$ the subset $M \times \{p\} \subseteq M \times N$ (called a slice of the product manifold) is an embedded submanifold of $M \times N$ diffeomorphic to M.

Proof:
The map $F : M \to M \times N$ given by $F(m) := (m, p)$ is a C^r embedding. After all, it's a C^r map that is a topological embedding. Furthermore, recall from *page 1059* how we standardly identify $T_{(m,p)}(M \times N)$ with $T_m M \oplus T_p N$.

Lemma: Suppose $N, M_1, \ldots, M_k$ are C^r manifolds. Then let $M := \prod_{i=1}^k M_i$ and for each i let $\pi_i : M \to M_i$ be the projection map. Given any $(p_1, \ldots, p_k) \in M$ we identify $T_{(p_1,\ldots,p_k)} M$ with $\bigoplus_{i=1}^k T_{p_i} M_i$ via the standard identification. Now suppose for each $1 \leq i \leq k$ that $F_i : N \to M_i$ is a C^1 map. Then let $F : N \to M$ be defined by $F(q) = (F_1(q), \ldots, F_k(q))$. Now for any $q \in N$ we have that:

$$\mathrm{d}F_q(v) = (\mathrm{d}(F_1)_q(v), \ \ldots, \ \mathrm{d}(F_k)_q(v)).$$

Why?
Suppose $v \in T_qN$ and $f : M \to \mathbb{R}$ is a C^1 map. Then $\mathrm{d}F_q(v)f$ is identified with:
$$(\mathrm{d}(\pi_1)_{F(q)} \circ \mathrm{d}F_q(v), \ldots \mathrm{d}(\pi_k)_{F(q)} \circ \mathrm{d}F_q(v)) = (\mathrm{d}(\pi_1 \circ F)_q(v), \ldots, \mathrm{d}(\pi_k \circ F)_q(v))$$
$$= (\mathrm{d}(F_1)_q(v), \ \ldots, \ \mathrm{d}(F_k)_q(v))$$

Based on the prior lemma, we can see for any $v \in T_mM$ that:
$$\mathrm{d}F_m(v) = (v, 0) \in T_mM \oplus T_pN.$$

It follows that $\mathrm{d}F_m$ is injective for all $m \in M$. So, F is an immersion. ■

Proposition 5.4: Suppose M is a C^r m-dimensional manifold, N is a C^r n-dimensional manifold, $U \subseteq M$ is open, and $f : U \to N$ is a C^k map. Let $\Gamma(f) \subseteq M \times N$ denote the graph of f:
$$\Gamma(f) \coloneqq \{(m, f(M)) \in M \times N : m \in U\}$$
Then $\Gamma(f)$ is a C^k m-dimensional embedded submanifold of $M \times N$.

Proof:
Define a map $\gamma_f : U \to M \times N$ by $\gamma_f(m) = (m, f(m))$. This will be a C^k map because the inclusion map $\iota : U \to M$ is C^r and similarly the map $f : U \to N$ is C^k. Also let $\pi_M : M \times N \to M$ be the projection map. Then $\mathrm{Id}_U = \pi_M \circ \gamma_f$ implies that γ_f injective. Also for any $m \in U$ we have $\mathrm{d}(\pi_M)_{(m,f(m))} \circ \mathrm{d}(\gamma_f)_m = \mathrm{d}(\mathrm{Id}_U)_m = \mathrm{Id}_{T_mU}$. So, for every $m \in U$ we have that $\mathrm{d}(\gamma_f)_m$ is injective. In other words, γ_f is an immersion. Finally, γ_f is a topological embedding since its inverse function is given by $\pi_M|_{\Gamma(f)}$.

And now we've shown that $\Gamma(f)$ is the image of a C^k embedding γ_f from U. ■

An embedded submanifold $S \subseteq M$ is said to be properly embedded if the inclusion map $S \hookrightarrow M$ is a proper map.

Proposition 5.5: Suppose M is a C^r manifold and $S \subseteq M$ is an embedded submanifold. S is properly embedded if and only if S is a closed subset of M.

Proof:
Suppose the inclusion map $\iota : S \to M$ is a proper map. Then we know that it is a closed map (see *page 1104*). In turn, $S = \iota(S)$ must be a closed subset of M.

Conversely, note that ι being a topological embedding with a closed image implies that ι is proper.

This is because if $K \subseteq M$ is a compact set, then $S \cap K$ will also be compact since S is closed. And now as ι^{-1} is a continuous map, we know that $\iota^{-1}(K) = \iota^{-1}(S \cap K)$ is compact. So, ι is proper. ■

Corollary 5.6: Every compact embedded submanifold is properly embedded.

Proof:
Compact subsets of Haudorff spaces are closed. ■

Proposition 5.7: Suppose M and N are C^r manifolds, and $f : M \to N$ is a C^k map. Then after equipping $\Gamma(f) \coloneqq \{(m, f(M)) \in M \times N : m \in M\}$ with the smooth structure defined in proposition 5.4, we have that $\Gamma(f)$ is properly embedded.

Proof:
Note that $\gamma_f : M \to M \times N$ given by $\gamma_f(m) = (m, f(m))$ has the map $\pi_M : M \times N \to M$ given by $\pi_M(m, n) = m$ as a continuous left inverse. In other words, $\pi_M \circ \gamma_f = \mathrm{Id}_M$. As a result we can conclude that $\gamma_f : M \to M \times N$ is a proper map. To see why, I claim the following:

Lemma: If X, Y are topological spaces and $g : X \to Y$ and $h : Y \to X$ are continuous functions such that $h \circ g = \mathrm{Id}_X$, then g must be proper.

Why?
Suppose $K \subseteq Y$ is a compact set. Since h is continuous, we know that $h(K) \subseteq X$ is a compact set. But also let $F \coloneqq g^{-1}(K)$. Then:

$$\begin{aligned} F = \mathrm{Id}_X(F) = (h \circ g)(F) &= h(K \cap g(X)) \\ &= h(K) \cap (h \circ g)(X) \\ &= h(K) \cap \mathrm{Id}_X(X) = h(K) \cap X = h(K). \end{aligned}$$

So, we've shown that $g^{-1}(K)$ is compact.

Since γ_f is a continuous proper map though and $M \times N$ is an LCH space, we can conclude that γ_f is a closed map. So $\gamma_f(M) = \Gamma(f)$ is a closed subset of $M \times N$. And by proposition 5.5, we are done. ■

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If $U \subseteq \mathbb{R}^n$ is an open set and $k \in \{0, 1, \ldots, n\}$ then a k-slice of U is a set of the form

$$S \coloneqq \{(x_1, \ldots, x_k, x_{k+1}, \ldots, x_n) \in U : x_{k+1} = c_{k+1}, \ldots, x_n = c_n\}$$

where $c_{k+1}, \ldots, c_n \in \mathbb{R}$. In practice we'll usually have $c_{k+1}, \ldots, c_n = 0$.

More generally, if M is an n-dimensional C^r manifold without a boundary and (U, φ) is a smooth chart on M, then we say a set $S \subseteq U$ is a k-slice of U if $\varphi(S)$ is a k-slice of $\varphi(U) \subseteq \mathbb{R}^n$.

Given a subset S of a C^r n-dimensional manifold M and a nonnegative integer k, we say S satisfies the local k-slice condition if every $p \in S$ is contained in a smooth chart (U, φ) such that $S \cap U$ is a k-slice of U. Such a chart (U, φ) is then called a slice chart and the corresponding local coordinates are called slice coordinates.

Theorem 5.8 (Local Slice Criterion): Let M be a C^r n-dimensional manifold without a boundary.

- If $S \subseteq M$ is an embedded k-dimensional C^r submanifold without a boundary then S satisfies the local k-slice condition.

  Proof:
  Let $p \in S$. Since the inclusion map $\iota : S \hookrightarrow M$ is an immersion, we can use the rank theorem to conclude that there are smooth charts (U, φ) on S and (V, ψ) on M such that $p \in U$, $U \subseteq V \cap S$, $\varphi(p) = 0$, and:

$$\psi \circ \iota \circ \varphi^{-1}(x_1, \ldots, x_k) = (x_1, \ldots, x_k, \underbrace{0, \ldots, 0}_{n-k \text{ zeros}}).$$

Choose $\varepsilon > 0$ small enough so that U and V both contain coordinate cubes $(-\varepsilon, \varepsilon)^k$ and $(-\varepsilon, \varepsilon)^n$ respectively centered at p and denote these coordinate balls as U_0 and V_0. Then U_0 is a k-slice of V_0.

Finally, as S has the subspace topology, we know there is an open set $W \subseteq M$ such that $W \cap S = U_0$. Set $V_1 = V_0 \cap W$. Then $(V_1, \psi|_{V_1})$ is a smooth chart for M containing p such that $V_1 \cap S = U_0$ and U_0 is still a k-slice of V_1. Finally, as S has the subspace topology, we know there is an open set $W \subseteq M$ such that $W \cap S = U_0$. Set $V_1 = V_0 \cap W$. Then $(V_1, \psi|_{V_1})$ is a smooth chart for M containing p such that $V_1 \cap S = U_0$ and U_0 is still a k-slice of V_1.

As a side note if $\pi : \mathbb{R}^n \to \mathbb{R}^k$ is the projection map onto the first k coordinates, then $\varphi = \pi \circ \psi|_{U_0}$. So we've proven for all $p \in S$ can find a slice chart (Ω, ϕ) on M containing p such that $(\Omega \cap S, \pi \circ \phi)$ is a smooth chart on S containing p. This observation will be referenced later on *page 1122*.

- Conversely, if $S \subseteq M$ is a subset satisfying the local k-slice condition, then it is a k-dimensional topological manifold without a boundary when equipped with the subspace topology relative to M. Also, there is a C^r structure on S which makes S into a C^r embedded submanifold of M.

  Proof:
  If we give S the subspace topology, then we know S is second countable and Hausdorff because M is. Plus, we also have that the inclusion map $\iota : S \hookrightarrow M$ is a topological embedding. Meanwhile, the idea for constructing an atlas for S is that if $(x_1, \ldots, x_n)$ is the coordinates of a slice chart for S then we can define a chart with the coordinates $(x_1, \ldots, x_k)$.

  We let $\pi : \mathbb{R}^n \to \mathbb{R}^k$ be the projection onto the first k coordinates. Now suppose (U, φ) is any slice chart on M. We let:

  $$V \coloneqq U \cap S, \widehat{V} \coloneqq (\pi \circ \varphi)(V), \text{ and } \psi \coloneqq \pi \circ \varphi|_V.$$

  Note that as S is equipped with the subspace topology, V is an open subset of S. Also, by the definition of a slice chart, $\varphi(V) = A \cap \varphi(U)$ for some set:

  $$A = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_{k+1} = c_{k+1}, \ldots, x_n = c_n\}$$

  where $c_{k+1}, \ldots, c_n \in \mathbb{R}$ are constants. It follows that $\varphi(V)$ is an open subset of A. And since $\pi|_A : A \to \mathbb{R}^k$ is a C^∞ diffeomorphism from A to $\mathbb{R}^k$, we know that $\widehat{V} = \pi(\varphi(V))$ is an open subset of $\mathbb{R}^k$.

  Moreover note that $\psi : V \to \widehat{V}$ has a continuous inverse function $\varphi^{-1} \circ j|_{\widehat{V}}$ where $j : \mathbb{R}^k \to \mathbb{R}^n$ is the C^∞ function $(x_1, \ldots, x_k) \mapsto (x_1, \ldots, x_k, c_{k+1}, \ldots, c_n)$. So, (V, ψ) is a well-defined interior chart on S. Since $S \subseteq M$ has the local k-slice condition, we know every $p \in S$ is contained in some chart constructed above. So, S is a topological manifold without a boundary.

To equip S with a C^r structure, we prove that the charts on S constructed priorly are all C^r compatible with each other. So let (U, φ) and (U', φ') be two different slice charts for S in M and then let (V, ψ) and (V', ψ') be the corresponding charts for S constructed priorly. The transition map $\psi' \circ \psi^{-1}$ will be given by $\pi \circ \varphi \circ \varphi^{-1} \circ j$ where $j : \mathbb{R}^k \to \mathbb{R}^n$ is the map defined priorly. Thus, $\psi' \circ \psi^{-1}$ is a C^r map since it is a composition of four C^r maps. And since ψ and ψ' were arbitrary, we've proven that our priorly constructed atlas defines a C^r structure on S.

Finally, consider any $p \in S$ and let (U, φ) be a slice chart for S in M containing p. Then let (V, ψ) be the corresponding smooth chart on S. We know $p \in V$, $\iota(V) \subseteq U$ and also that there exists $c_{k+1}, \ldots, c_n \in \mathbb{R}$ such that $\varphi \circ \iota \circ \psi^{-1}$ is given by $(x_1, \ldots, x_k) \mapsto (x_1, \ldots, x_k, c_{k+1}, \ldots, c_n)$. It follows that ι is a C^r map. And by looking at the Jacobian of $(x_1, \ldots, x_k) \mapsto (x_1, \ldots, x_k, c_{k+1}, \ldots, c_n)$, we can also conclude that $\mathrm{d}\iota_p$ is full rank (and injective). ■

As a side note, the local k-slice condition is importantly a local property. One illustration of this fact is as follows:

Suppose M is a C^r n-dimensional manifold and $S \subseteq M$ is a set equipped with the subspace topology on M such that for every $p \in S$ there is an open set $U \subseteq M - \partial M$ containing p such that $S \cap U$ is a k-dimensional embedded C^r submanifold without a boundary. Then S is a k-dimensional topological manifold without a boundary that can be equipped with a C^r structure such that S is an embedded C^r submanifold of M.

This is because $S \cap U$ being an embedded submanifold of M without a boundary implies that we can find a slice chart (V, φ) for S in $U \subseteq M$ containing p. And since p was arbitrary, S satisfies the local k-slice condition.

I'll later show that if M is a C^r manifold without a boundary and $S \subseteq M$ is equipped with the subspace topology then the C^r structure making S into an embedded submanifold (without a boundary) is unique if it exists.

Theorem 5.11: Let M be a C^r n-dimensional manifold without corners (where $n \geq 2$) Then ∂M equipped with the subspace topology is an $(n-1)$-dimensional topological manifold without a boundary that can be equipped with a C^r structure such that ∂M is an embedded
submanifold of M.

Proof:
∂M is trivially Hausdorff and second countable when equipped with the subspace topology, and similarly the inclusion map $\iota : \partial M \hookrightarrow M$ is trivially a topological embedding when ∂M is equipped with the subspace topology.

To show that ∂M is locally Euclidean, we construct an atlas. Let $\pi : \mathbb{R}^n \to \mathbb{R}^{n-1}$ be given by $\pi(x_1, \ldots, x_n) = (x_1, \ldots, x_{n-1})$ and let $j : \mathbb{R}^{n-1} \to \mathbb{R}^n$ be given by:

$$j(x_1, \ldots, x_{n-1}) = (x_1, \ldots, x_{n-1}, 0).$$

Now given any smooth boundary chart (U, φ) for M, we define a chart (V, ψ) on ∂M by $V = U \cap \partial M$ and $\psi = \pi \circ \varphi|_V$.

By the invariance of the boundary, we know that:

$$\varphi(U \cap \partial M) = \{(x_1, \ldots, x_n) \in \varphi(U) : x_n = 0\} = \varphi(U) \cap \partial H^n.$$

It follows that $\varphi(U \cap \partial M)$ is an open subset of ∂H^n. And since π is a homeomorphism from ∂H^n to $\mathbb{R}^{n-1}$, we can conclude that $\psi(V)$ is an open subset of $\mathbb{R}^{n-1}$.

Furthermore, note that ψ has the continuous inverse $\varphi^{-1} \circ j|_{\psi(V)}$. So, ψ is a homeomorphism. And since every $p \in \partial M$ is contained in some boundary chart, we can conclude that the charts constructed here form an atlas for ∂M.

By identical reasoning as in the prior theorem, we can show that the charts constructed prior are C^r compatible with each other, and also that ∂M equipped with the C^r structure they generate is an embedded submanifold of M. ■

As a side note, if $n = 1$ then ∂M will be a discrete set. So, this theorem also holds when $n = 1$ if we allow ourselves to consider 0-dimensional manifolds.

There isn't really a way to adapt the k-slice condition to an ambient manifold M with a boundary and so usually the strategy people will instead use to study those manifolds is that they will embed an ambient manifold with a boundary into a larger ambient manifold without a boundary. I'll talk about that in the future.

Suppose M and N are C^k manifolds without boundaries, $\Phi : M \to N$ is any map, and $c \in N$ is any point. The set $\Phi^{-1}(\{c\})$ is called a level set of Φ. If $N = \mathbb{R}^n$ and $c = 0$, the level set $\Phi^{-1}(\{c\})$ is called the zero set of Φ.

In general, a level set or zero set of a C^k map is not going to be a submanifold.

Theorem 2.29: Let M be a C^k manifold and suppose $K \subseteq M$ is any closed set. Then there is a C^k nonnegative function $f : K \to \mathbb{R}$ such that $f^{-1}(\{0\}) = K$.

Proof:

We start by considering the case where $M = \mathbb{R}^n$ and $K \subseteq \mathbb{R}^n$ is a closed set. Then for every $x \in K^C$ let $r_x \in (0, 1)$ be such that $B_{r_x}(x) \subseteq K^C$. Thus $\{B_{r_x/2}(x) : x \in K^C\}$ is an open cover of K^C, and since K^C is second countable, we know there is a countable subcover. So, we've proven there are sequences $\{x_i\}_{i\in\mathbb{N}} \subseteq \mathbb{R}^n$ and $\{r_i\}_{i\in\mathbb{N}} \subseteq (0, 1)$ satisfying that:

$$K^C = \bigcup_{i=1}^{\infty} B_{r_i/2}(x_i) = \bigcup_{i=1}^{\infty} B_{r_i}(x_i).$$

Now let $h : \mathbb{R}^n \to [0, 1]$ be a C^∞ function such that $h \equiv 1$ on $B_{1/2}(0)$ and $\text{supp}(h) \subseteq B_1(0)$. Also for each $i \in \mathbb{Z}_{\geq 0}$ let $C_i \geq 1$ be some constant such that $\|\partial^\alpha h\|_u \leq C_i$ whenever $|\alpha| \leq i$. And finally, define $f : \mathbb{R}^n \to [0, 1]$ by:

$$f(x) = \sum_{i=1}^{\infty} \frac{(r_i)^i}{2^i C_i} h\left(\frac{x - x_i}{r_i}\right)$$

More generally, for any multi-index α let $g_\alpha(x) = \sum_{i=1}^{\infty} \frac{(r_i)^{i-|\alpha|}}{2^i C_i} \partial^\alpha h\left(\frac{x - x_i}{r_i}\right)$. Clearly $g_{(0,\ldots,0)} = f$. By the Weierstrass M-test we can see that each g_α is continuous. Also I claim $\partial^\alpha f = g_\alpha$ for every multi-index α. This specifically follows as an immediate consequence of the following lemma.

Lemma: Suppose $U \subseteq \mathbb{R}^n$ is an open set and let $e_1, \ldots, e_n$ be the standard basis for $\mathbb{R}^n$. Also denote $\partial_j = \frac{\partial}{\partial x_j}$. Next suppose $\{f_k\}_{k\in\mathbb{N}} \in C^1(U, \mathbb{R})$, and also that $g, h \in C(U, \mathbb{R})$ satisfy that $f_k \to g$ pointwise and $\partial_j f_k \to h$ uniformly on compact sets. Then $\partial_j g = h$.

Why?

Let $x \in U$ be any point. Then for small enough $t > 0$ we have that $B_t(x) \subseteq U$. And then by the fundamental theorem of calculus we have that:

$$f_k(x + te_j) - f_k(x) = \int_0^t \partial_j f_k(x + se_j)\mathrm{d}s$$

Since $[-t, t]$ is compact, $\partial_j f$ is continuous, and $\partial_j f_k \to h$ uniformly, we can conclude by the dominated convergence theorem that:

$$\lim_{k\to\infty} \int_0^t \partial_j f_k(x + se_j)\mathrm{d}s = \int_0^t h(x + se_j)\mathrm{d}s.$$

Also since $f_k \to g$ pointwise we know:

$$g(x + te_j) - g(x) = \int_0^t h(x + se_j)\mathrm{d}s$$

And now by the fundamental theorem of calculus, we can conclude that $\frac{\mathrm{d}}{\mathrm{d}t}(g(x + te_j)) = h(x + te_j)$ exists and equals h when $|t|$ is small enough. In other words, $\partial_j g = h$ at x.

By the prior lemma, we can conclude that if β and α are multi-indices such that $\partial^\alpha = \frac{\partial}{\partial x_j} \circ \partial^\beta$ then $\frac{\partial}{\partial x_j} g_\beta = g_\alpha$. Then the fact that $\partial^\alpha f = g_\alpha$ immediately follows by induction.

To finish, off, note that if $x \in K$ then $x \notin B_{r_i}(x_i)$ for any i and so $f(x_i) = 0$. Meanwhile, if $x \notin K$ then $x \in B_{r_i/2}(x_i)$ for some i and hence $h(\frac{x - x_i}{r_i}) = 1$. It follows that $f(x) > 0$. So, $K = f^{-1}(\{0\})$.

Now that this theorem has been shown in the case that $M = \mathbb{R}^n$, we can now consider the general case for a C^k manifold. Suppose $K \subseteq M$ is any closed set and let $\{(B_\alpha, \varphi_\alpha)\}_{\alpha\in A}$ be a covering of M with smooth coordinate balls, half balls, and corner balls. We can let $\{\psi_\alpha\}_{\alpha\in A}$ be a C^r subordinate partition of unity as constructed on *page 1032*.

Note that $B_\alpha \cap K$ is a closed subset relative to B_α for every $\alpha \in A$. In turn, we can see that $\varphi(B_\alpha \cap K)$ is closed relative to $\varphi(B_\alpha)$. Then there is some closed set $F_\alpha \subseteq \mathbb{R}^n$ such that $F_\alpha \cap \varphi(B_\alpha) = \varphi(B_\alpha \cap K)$. And now by the prior reasoning, we let $f_\alpha : \mathbb{R}^n \to [0, \infty)$ be a C^∞ map such that $f_\alpha^{-1}(\{0\}) = F_\alpha$.

But now $f = \sum_{\alpha\in A} \psi_\alpha \cdot (f_\alpha \circ \varphi_\alpha)$ defines a C^r map on M with $f^{-1}(\{0\}) = K$. ■

However, by making assumptions about the rank of a C^k map Φ, we can guarentee that the level sets of Φ are embedded submanifolds.

Theorem 5.12 (Constant-Rank Level Set Theorem): Let M and N be C^k manifolds without boundaries and let $\Phi : M \to N$ be a C^k map with constant rank r. Then each level set of Φ is a properly embedded submanifold of codimension r in M without a boundary.

Proof:

Write $m = \dim(M)$ and $n = \dim(N)$. Also let $c \in N$ be arbitrary and let $S \coloneqq \Phi^{-1}(\{c\}) \subseteq M$. Now for any $p \in S$, the rank theorem gives us smooth charts (U, φ) on M and (V, ψ) on N such that $\varphi(p) = 0$ and:

$$\psi \circ \Phi \circ \varphi^{-1}(x_1, \dots, x_r, x_{r+1}, \dots, x_m) = (x_1, \dots, x_r, \underbrace{0, \dots, 0}_{n-r \text{ zeros}}).$$

Yet note that $\Phi(p) = c$ implies that $\psi(c) = 0$. And now:

$$U \cap S = U \cap \Phi^{-1}(\{c\}) = \varphi^{-1}(\{(x_1, \dots, x_m) \in \varphi(U) : x_1 = \cdots = x_r = 0\}).$$

It follows that $U \cap S$ is a $(m-r)$-slice of (U, φ). Since $p \in S$ was arbitrary, we can conclude that S satisfies the local $(m-r)$-slice condition. So, S is an embedded $(m-r)$-dimensional C^k submanifold of M without a boundary.

Also as S is a closed set (since $\{c\}$ is a closed subset of N and Φ is continuous), we know S is properly embedded. ■

Corollary 5.13 (Submersion Level Set Theorem): If M and N are C^k manifolds without boundaries and $\Phi : M \to N$ is a C^k submersion, then each level set of Φ is a properly embedded submanifold of M without a boundary whose codimension is $\dim(N)$

Proof:
Apply the prior theorem and use the fact that $\operatorname{rank}(\Phi) = \dim(N)$. ■

Suppose M and N are C^k manifolds and $\Phi : M \to N$ is a C^k map. We say a point $p \in M$ is a regular point of Φ if $\mathrm{d}\Phi_p$ is surjective. Otherwise we say p is a critical point of Φ. Also we say a point $c \in N$ is a regular value of Φ if $\Phi^{-1}(\{c\})$ if every $p \in \Phi^{-1}(\{c\})$ is a regular point of Φ. Otherwise, we say c is a critical value of Φ. Finally, a level set $\Phi^{-1}(\{c\})$ is a regular level set if c is a regular value of Φ.

As a side note, the set of regular points in M is always an open subset of U. This is because of proposition 4.1 on *pages 1054-1056*. So, we can always restrict the domain of Φ to just an open-submanifold where Φ is a submersion.

Corollary 5.14 (Regular level set Theorem): Suppose M and N are C^k manifolds without boundaries and $\Phi : M \to N$ is a C^k map. Every regular level set of Φ is a properly embedded C^k submanifold without a boundary and with codimension equal to $\dim(N)$.

Proof:
Let $c \in N$ be a regular value of Φ. The set $U \subseteq M$ of regular points of Φ is open in M and contains $\Phi^{-1}(\{c\})$ since c is a regular value. So, $\Phi|_U$ is a C^k submersion and the prior corollary shows that $\Phi^{-1}(\{c\})$ is a C^k submanifold of U without a boundary and with codimension equal to $\dim(N)$. Since the composition of two C^k embeddings is a C^k embedding, and the inclusion $\Phi^{-1}(\{c\}) \hookrightarrow M$ is the composition of the inclusions $\Phi^{-1}(\{c\}) \hookrightarrow U$ and $U \hookrightarrow M$, we can also conclude $\Phi^{-1}(\{c\})$ is an embedded C^k submanifold of M.

Also once again, Φ being continuous and $\{c\}$ being a closed subset of N means that $\Phi^{-1}(\{c\})$ is a closed subset of M. So, $\Phi^{-1}(\{c\})$ is a properly embedded submanifold. ■

Proposition 5.16: Let S be a subset of a C^r m-dimensional manifold M without a boundary. Then S is a k-dimensional embedded submanifold of M without a boundary if and only if every $p \in S$ has an open neighborhood $U \subseteq M$ such that $U \cap S$ is a zero set of a C^r submersion $\Phi : U \to \mathbb{R}^{m-k}$.

$(\Longrightarrow)$

Consider any $p \in S$ and let (U, φ) be a slice chart for S in M containing p. By translating the codomain of φ, we can without loss of generality assume that:

$$\varphi(U \cap S) := \{(x_1, \ldots, x_m) \in \varphi(U) : x_{k+1} = 0, \ldots, x_m = 0\}.$$

Now let us write $\varphi(q) = (x_1, \ldots, x_m)$ for all $q \in U$. Next define $\pi : \mathbb{R}^m \to \mathbb{R}^{m-k}$ by $(x_1, \ldots, x_m) \mapsto (x_{k+1}, \ldots, x_m)$. And finally, define $\Phi := \pi \circ \varphi$. Since both π and φ are C^r submersions, we can conclude that $\Phi : U \to \mathbb{R}^{m-k}$ is also a C^r submersion. Also, $\Phi^{-1}(\{(0, \ldots, 0)\}) = U \cap S$.

$(\Longleftarrow)$

By the submersion level set theorem, we can conclude that for every $p \in S$ there is an open set $U \subseteq M$ containing p such that $S \cap U$ is a k-dimensional C^r embedded submanifold of M without a boundary. And now we just apply the comment on *page 1116*. ■

Up until now, we've worked with submanifolds equipped with the subspace topology. However, this can be limiting. So, we adopt the following terminology.

Let M be a C^r manifold. A C^r immersed submanifold of M is a subset $S \subseteq M$ endowed with some topology (not necessarily the subspace topology) turning it into a topological manifold, as well a C^r structure such that the inclusion map $\iota : S \hookrightarrow M$ is a C^r immersion.

Remarks:

- If $k \leq r$ then we can define immersed C^k submanifolds of M identically as on *page 1111*.
- Since all embedded submanifolds are also immersed submanifolds, we take the convention that a "C^r submanifold" of M is only understood by default to be immersed. If we want that submanifold to also be embedded then we will add that adjective.
- The codimension of a C^r submanifold S of M is still defined as $\dim(M) - \dim(S)$. And S is a C^r hypersurface if S has codimension 1.

Proposition 5.18: Suppose M and N are C^r manifolds, and $F : N \to M$ is an injective C^r immersion. Also let $S := F(N)$. Then S has a unique topology and C^r structure making S into a C^r submanifold of M such that F is a C^r-diffeomorphism from N to S.

Proof:

To start off, we define a topology on S by saying that $U \subseteq S$ is open if $F^{-1}(U)$ is an open subset of N.

Note that this is a well-defined topology. After all, $F^{-1}(S) = N$ and $F^{-1}(\emptyset) = \emptyset$ are both open in N and hence S and $\emptyset$ are open subsets of S. Also suppose $\{U_\alpha\}_{\alpha \in A}$ is a collection of subsets of S such that $F^{-1}(U_\alpha)$ is open in N for all α. Then $F^{-1}(\bigcup_{\alpha \in A} U_\alpha) = \bigcup_{\alpha \in A} F^{-1}(U_\alpha)$ is open in N. Similar reasoning shows the topology on S is closed under finite intersections.

Importantly, this topology on S is the *unique* topology S must be equipped with in order for F to be a homeomorphism from N to S (which is necessary if we want F to also be a C^r-diffeomorphism).

Now N being second countable and Hausdorff means that S is as well. Plus we can equip S with an atlas given by the charts $(F(U), \varphi \circ F^{-1})$ where (U, φ) is any chart on N. By identical reasoning as in the proof of proposition 5.2 on *pages 1111-1112*, this atlas defines a C^r structure on S such that S is a topological manifold and $F : N \to S$ is a C^r-diffeomorphism. (In fact this atlas is itself maximal and F being a C^r-diffeomorphism from N to S would automatically imply S is equipped with this maximal atlas).

Finally, the inclusion map $S \hookrightarrow M$ is the composition of the diffeomorphism $F^{-1} : S \to N$ and the immersion $F : N \to M$. So, $S \hookrightarrow M$ is an immersion. ■

Proposition 5.22: If M is a C^r manifold and $S \subseteq M$ is a C^r (immersed) submanifold without corners, then for every $p \in S$ there is an open set $U \subseteq S$ containing p such that U is a C^r embedded submanifold of M.

Proof:
By the local embedding theorem on *pages 1105-1106*, since the inclusion $\iota : S \hookrightarrow M$ is an immersion we know every $p \in S$ has an open neighborhood $U \subseteq S$ such that $\iota|_U$ is an embedding. And then the image of an embedding is always an embedded submanifold. ■

As a side note, the prior proposition does *not* say that we can find an open set U in the topology of M containing p such that $U \cap S$ is an embedded submanifold. It only says we can find an open set U in the topology of S containing p such that U is an embedded submanifold.

Before talking about parametrizations of submanifolds, I want to skip ahead to some results which I think are more interesting yet still important. Firstly, we will establish two results about restricting the domain or codomain of a C^r map.

Theorem 5.27: Suppose M and N are C^r manifolds, $F : M \to N$ is a C^k map, and $S \subseteq M$ is an (immersed) C^r submanifold. Then $F|_S : S \to N$ is a C^k map.

Proof:
$F|_S = F \circ \iota$ where $\iota : S \hookrightarrow M$ is the C^r inclusion map. ■

Theorem 5.29: Suppose M is a C^r manifold without a boundary, $S \subseteq M$ is an (immersed) C^r submanifold without a boundary, and N is another C^r manifold. Next suppose $F : N \to M$ is a C^k map such that $F(N) \subseteq S$. If F is continuous as a map from N to S then $F : N \to S$ is a C^k map.

Proof:
Let $p \in N$ be arbitrary and set $q = F(p) \in S$. By proposition 5.22, we know there is an open neighborhood $V \subseteq S$ of q such that the inclusion map $\iota|_V : V \hookrightarrow M$ is a C^r embedding. But now V (equipped with the open submanifold structure relative to S) is an embedded C^r submanifold of M without a boundary. So, we let $\pi : \mathbb{R}^{\dim(M)} \to \mathbb{R}^{\dim(S)}$ be the projection map onto the first $\dim(S)$ many coordinates. Then by the comment on *page 1115* we can find a smooth chart (W, ψ) on M containing q such that $(W \cap V, \pi \circ \psi|_V)$ is a smooth chart on V containing q. Since every smooth chart on an open submanifold is also a smooth chart on the original manifold, we can further conclude $(W \cap V, \pi \circ \psi|_V)$ is a smooth chart on S containing q.

Now since $F : N \to S$ is continuous, we know that $U := F^{-1}(W \cap V)$ is an open subset of N containing p. So, we let (U_0, φ) be any smooth chart on N such that $p \in U_0 \subseteq U$. And now $(\pi \circ \psi|_V) \circ F \circ \varphi^{-1} = \pi \circ (\psi \circ F \circ \varphi^{-1})$ will be a C^k map since $F : N \to M$ is a C^k map.

This finishes proving that $F : N \to S$ is a C^k map. ■

Corollary 5.30 (The Embedded Case): Suppose M is a C^r manifold without a boundary, $S \subseteq M$ is an embedded C^r submanifold, and N is another C^r manifold. If $F : N \to M$ is a C^k map with $F(N) \subseteq S$ then $F : N \to S$ is also a C^k map.

Proof:
Since S is equipped with the subspace topology, we automatically have that $F : N \to M$ being continuous means that $F : N \to S$ is continuous. So, we can apply the prior theorem. ■

Suppose M is a C^r manifold and $S \subseteq M$ is an immersed C^r submanifold of M. We say S is weakly embedded in M and also an initial submanifold of M if every C^k map $F : N \to M$ (where $k \leq r$) whose image lies in S is also a C^k map from N into S. The prior corollary says that every embedded C^r submanifold is also a weakly embedded C^r submanifold.

A particularly important consequence of the prior corollary can be stated as follows:

Theorem 5.31: Suppose M is a C^r manifold without a boundary and $S \subseteq M$ is a weakly embedded C^r submanifold without a boundary. Then there is no other topology and C^r structure that S can be equipped with such that S is an immersed C^r submanifold of M without a boundary.

Proof:
Suppose $\widetilde{S}$ is the set S equipped with some other topology and C^r structure such that $\widetilde{S}$ is an immersed C^r submanifold of M without a boundary. Also let ι denote the inclusion map $S \hookrightarrow M$ and $\widetilde{\iota}$ denote inclusion map $\widetilde{S} \hookrightarrow M$.

Because $\widetilde{\iota}(S)$ is contained in S which is weakly embedded, we know $\widetilde{\iota}$ is a C^r map from $\widetilde{S}$ into S. Yet then note for any $p \in S$ that $\mathrm{d}\widetilde{\iota}_p : T_p\widetilde{S} \to T_pM$ is both injective and the composition of the maps:

$$T_p\widetilde{S} \xrightarrow{\mathrm{d}\widetilde{\iota}_p} T_pS \xrightarrow{\mathrm{d}\iota_p} T_pM$$

Thus we must have that $\widetilde{\iota}$ is a C^r immersion from $\widetilde{S}$ into S. In other words, $\widetilde{\iota}$ has constant rank. And since $\widetilde{\iota}$ is a bijection between S and $\widetilde{S}$, we can then conclude by the corollary on *page 1066* that $\widetilde{\iota} : \widetilde{S} \to S$ is a C^r-diffeomorphism. This implies that the topology and C^r structure on S and $\widetilde{S}$ are the same. ■

In particular, the prior theorem tells us that there is only one C^r structure that an embedded submanifold S of M (without a boundary) can be equipped with. So, the C^r structure constructed in theorem 5.8 based on the local k-slice condition (see *pages 1114-1116*) is the unique structure on S.

For a general immersed submanifold of M though, we have the following weaker uniqueness result.

Theorem 5.32: Suppose M is a C^r manifold without a boundary and $S \subseteq M$ is an immersed C^r submanifold without a boundary. Then with the given topology on S there is no other C^r structure that S can be equipped with such that S is an immersed C^r submanifold of M without a boundary.

Proof:
Let $\widetilde{S}$ be the set S equipped with the same topology but just a different C^r structure. Then the proof of theorem 5.31 works entirely the same, except that instead of using corollary 5.30, we instead note that $\widetilde{\iota}$ is a continuous function from $\widetilde{S}$ into S and so theorem 5.29 applies. ■

8/31/2026

I could allow for more types of corner charts and then try to prove an immersion rank theorem for those manifolds but ehhh I don't think it would likely be the end of the world to just limit myself to having a rank theorem for boundary charts (non-corner charts) in the domain.

Let M be an n-dimensional C^r manifold without a boundary (where $n \geq 1$) and let (U, φ) be a smooth chart on M. Given an integer $k \in \{0, 1, \ldots, n\}$, we say a k-dimensional half slice of U is a set $V \subseteq U$ such that:

$$\varphi(V) = \{(x_1, \ldots, x_n) \in \varphi(U) : x_{k+1} = c_{k+1}, \ldots, x_n = c_n, \text{ and } x_k \geq 0\},$$

where $c_{k+1}, \ldots, c_n \in \mathbb{R}$.

We say a subset $S \subseteq M$ satisfies the local k-slice condition for a manifold with a boundary (what a mouthful) if every $p \in S$ is contained in a smooth chart (U, φ) such that $S \cap U$ is either a k-slice or a k-half slice of U. In the former case, (U, φ) is called an interior slice chart for S in M. Meanwhile, in the latter case (U, φ) is called a boundary slice chart for S in M.

By using the local immersion theorem for manifolds with boundaries (see *pages 1101-1102*) in place of the rank theorem, we can prove an analogous result to theorem 5.8 on *pages 1114-1116*. Then in turn, we can also prove an analogous result to theorem 5.29. I shall write those results right belows. There's no new ideas in proving them though.

Theorem 5.51: Let M be a C^r n-dimensional manifold without a boundary and let $S \subseteq M$ be a C^r embedded k-dimensional submanifold without corners. Then S satisfies the local k-slice condition for a manifold with a boundary.

In particular, let $\pi : \mathbb{R}^n \to \mathbb{R}^k$ be the projection map onto the first k-coordinates. Then for any $p \in S$ we can find a boundary or interior slice chart (U, φ) on M such that $(S \cap U, \pi \circ \varphi|_S)$ is a smooth chart on S containing p and:

$$S \cap U = \varphi^{-1}(\{(x_1, \ldots, x_n) : x_{k+1} = \cdots = x_n = 0 \text{ (and } x_k \geq 0 \text{ if } p \in \partial S)\}).$$

Conversely, if $S \subseteq M$ satisfies the local k-slice condition for a manifold with a boundary, then S equipped with the subspace topology is a topological manifold. Furthermore, it has a C^r structure making it into an embedded C^r submanifold in M without corners.

Specifically, we equip S with the structure generated by the charts $(U \cap S, \pi \circ \varphi|_S)$ where (U, φ) is any boundary or interior slice chart for S and $\pi : \mathbb{R}^n \to \mathbb{R}^k$ is the projection map onto the first k-coordinates.

Theorem 5.53(b): Suppose M and N are C^r manifolds such that M has no boundary, and also suppose $S \subseteq M$ is an (immersed) C^r submanifold without corners. Now let $F : N \to M$ be a C^k map such that $F(N) \subseteq S$. If F is continuous as a function from N to S then $F : N \to S$ is a C^k map.

As a side note, theorem 5.51 is enough to tell us that there is at most one C^r structure on a set $S \subseteq M$ equipped with the subspace topology turning it into an embedded C^r submanifold of M without corners.

To see why, suppose $S \subseteq M$ is any subset equipped with a C^r structure such that it becomes a C^r embedded submanifold of M. Next let $\widetilde{S}$ be the subset S equipped with the k-slice C^r structure constructed on the converse direction of theorem 5.51. By the red text, we can see that the C^r structures of S and $\widetilde{S}$ both contain a common atlas of smooth charts. So, both structures must be identical. And since the C^r structure of $\widetilde{S}$ is defined entirely independently from the C^r structure on S, we can conclude that the C^r structure on $\widetilde{S}$ is the unique C^r structure turning S into an embedded C^r submanifold of M.

Now the connection between abstract manifolds and the definition of manifolds given on *pages 77-99* is that Munkres works with embedded submanifolds of $\mathbb{R}^n$ (equipped with the standard structure).

To see one direction of this equivalence, consider the following:

Note that Munkres' definition of a coordinate patch on a C^r manifold $S \subseteq \mathbb{R}^n$ is equivalent to that of a C^r embedding $\alpha : U \to \mathbb{R}^n$ where $U \subseteq \mathbb{R}^k$ (or H^k) is an open set and $\alpha(U) = V$ is an open subset of S (equipped with the subspace topoloy). Since every point is in the image of a coordinate patch, we can conclude there is a covering for S consisting of open sets $\{W_\alpha\}_{\alpha \in A}$ in M such that $W_\alpha \cap S$ is an embedded submanifold (without corners because none of the coordinate patches are from $\mathbb{R}^k_{\geq 0}$) for all α. It follows that S satisfies the local k-slice condition for a manifold with a boundary. So, S is an embedded C^r submanifold of $\mathbb{R}^n$.

Going the other way, I want to introduce parametrizations more generally.

Suppose M is a C^r manifold and $S \subseteq M$ is an immersed k-dimensional C^r submanifold of M. Then a local parametrization of S is a continuous map $X : U \to M$ (where $U \subseteq \mathbb{R}^k$, H^k, or $\mathbb{R}^k_{\geq 0}$ is an open set) whose image is an open subset of S and which is a homeomorphism when considered as a function from U to $X(U) \subseteq S$. If X is also a C^r diffeomorphism from U to $X(U) \subseteq S$, it is called a C^r parametrization. Meanwhile, if $X(U) = S$ then we say X is a global parametrization.

Proposition 5.23: Suppose M is a C^r manifold, $S \subseteq M$ is an immersed k-dimensional C^r submanifold of M, $\iota : S \hookrightarrow M$ is the inclusion map, and $U \subseteq \mathbb{R}^k$ (or H^k or $\mathbb{R}^k_{\geq 0}$) is an open set. A map $X : U \to M$ is a C^r parametrization if and only if there is a smooth coordinate chart (V, φ) on S such that $U = \varphi(V)$ and $X = \iota \circ \varphi^{-1}$.

$(\Longleftarrow)$

Let (V, φ) be a smooth chart on S such that $U = \varphi(V)$ and $X = \iota \circ \varphi^{-1}$. Then X is a continuous map with a continuous inverse φ from $X(U) = V \subseteq S$ to U. Also X is a C^r diffeomorphism from U to V.

$(\Longrightarrow)$

Let $X : U \to M$ be a C^r paremtrization. Then let $V = X(U)$ and $\varphi = X^{-1}$ (considered as a C^r map from $V \subseteq S$ to U). Clearly $X = \iota \circ \varphi^{-1}$ and $U = \varphi(V)$. Also I claim that (V, φ) is a smooth chart on M. After all, let (W, ψ) be any other smooth chart on M.

Since φ^{-1} is a C^r map by assumption, we know that $\mathrm{Id}_U \circ \varphi \circ \psi^{-1} = \varphi \circ \psi^{-1}$ is a C^r map. Similarly, as X is a C^r map from U into V, we know $\psi \circ X \circ \mathrm{Id}_U = \psi \circ \varphi^{-1}$ is a C^r map. ■

In particular, this proposition shows that every point in an immersed submanifold is contained in some parametrization.

9/5/2026

Augh never mind I want to broaden the corner charts I'm allowed to use.

Suppose M is an n-dimensional C^r manifold with corners and $p \in M$ is a corner point. Then given any chart (U, φ) on M containing p we must have that $\varphi(U)$ is open in $\mathbb{R}^n_{\geq 0}$ and that $\varphi(p) := (x_1, \ldots, x_n)$ where two of the coordinates of $\varphi(p)$ are zero. In particular, suppose exactly k coordinates of $\varphi(p)$ are zero. By permuting those coordinates, we can

assume without loss of generality it is the first k coordinates of $\varphi(p)$ are zero. And then by translating the codomain of φ by the vector $(0,\ldots,0,x_{k+1},\ldots,x_n)$, we get a "chart" (U,φ') where $\varphi'(U)$ is an open subset of $[0,\infty)^k \times \mathbb{R}^{n-k}$ with $\varphi'(p)=0$. And this "chart" will be C^r compatible with the C^r structure on M.

Lemma: Suppose $k_1, k_2 < n$ with $k_1 \neq k_2$ and that $U \subseteq [0,\infty)^{k_1} \times \mathbb{R}^{n-k_1}$ and $V \subseteq [0,\infty)^{k_2} \times \mathbb{R}^{n-k_2}$ are open subsets of the respective subspace topologies which both contain 0. Then there is no C^1 diffeomorphism $\varphi : U \to V$ such that $\varphi(0)=0$.

Proof:
We basically proceed by identical reasoning as on *page 1041*. Without loss of generality we can assume $k_1 < k_2$. Then suppose for the sake of contradiction that such a map $\varphi : U \to V$ did exist.

Let $S := \{y=(y_1,\ldots,y_n) \in \mathbb{R}^n : y_1=0,\ldots,y_{k_1}=0\}$. Thus S is an $(n-k_1)$-dimensional vector subspace of $\mathbb{R}^n$. Next note that $\mathrm{D}\varphi(0)$ must be an invertible linear map (with the inverse map $\mathrm{D}\varphi^{-1}(0)$). Therefore, we can conclude that $T := (\mathrm{D}\varphi(0))(S)$ is also an $(n-k_1)$-dimensional vector subspace.

Yet then there must exist $v=(v_1,\ldots,v_n) \in T$ such that one of the $v_1,\ldots,v_{k_2}$ equals zero. Otherwise, T would be contained in an $(n-k_2)$-dimensional vector space. And since $k_1 < k_2$, this would contradict that $(n-k_1)$-dimensional vector subspace of $\mathbb{R}^n$.

By negating v if necessary we can assume $v_i < 0$ for some $1 \leq i \leq k_2$. Finally, let $\gamma : (-\varepsilon,\varepsilon) \to U \cap S$ be a C^1 curve such that $\gamma(0)=0$ and

$$(\varphi \circ \gamma)'(0) = (\mathrm{D}\varphi(0))\gamma'(0) = v.$$

Then for small enough $t \in (0,\varepsilon)$ we must have that $(\alpha \circ \gamma)(t) \cdot e_i < 0$. But this contradicts that $\varphi(U \cap S) \subseteq [0,\infty)^{k_2} \times \mathbb{R}^{n-k_2}$. ■

Given a smooth chart (W,ψ) about a corner point p in a C^r manifold M such that $\psi(W)$ is an open subset of $\mathbb{R}^n_{\geq 0}$, we say p has depth k if exactly k of the components of $\psi(p)$ are zero. The prior lemma lets us conclude that the depth of p is defined independently of which chart we choose.

That said, because I want to allow smooth charts (W,ψ) about a corner point $p \in M$ to be "centered" (i.e. satisfy that $\psi(p)=0$), I'm going to start letting corner charts map to $[0,\infty)^k \times \mathbb{R}^{n-k}$ instead of just $[0,\infty)^n$.

Importantly, once we allow these new charts then we can by identical reasoning prove that the local immersion theorem for manifolds with boundaries on *pages 1101-1102* holds for a C^r manifold M with corner points. Then in turn, the local embedding theorem on *pages 1105-1106* also holds if the C^r manifold M has corners.

9/6/2026

If M is a C^r manifold, we say $D \subseteq M$ is a regular domain in M if it is a properly embedded codimension 0 submanifold of M.

Proposition 5.46: Suppose M is a C^r manifold without a boundary and $D \subseteq M$ is a regular domain. Then the topological interior and boundary of D is equivalent to the manifold interior and boundary of D respectively.

Proof:
For the sake of notation, I will let ∂D refer to the manifold boundary of D and $\text{Int}(D)$ refer to the manifold interior of D.

Suppose $p \in \partial D$ with depth $k \geq 1$. Then by the local immersion theorem for manifolds with boundaries, there exists a smooth chart (U, φ) centered at p in D and a smooth chart (V, ψ) centered at p in M such that $\varphi(p) = 0$ and $\psi \circ \varphi(x_1, \ldots, x_n) = (x_1, \ldots, x_n)$. (Here $n = \dim(M) = \dim(D)$).

But now since D has the subspace topology, we know there is an open set $W \subseteq M$ such that $U = W \cap D$. In turn, $V_0 := V \cap W$ is an open set in M such that $V_0 \cap D = \psi^{-1}([0, \infty)^k \times \mathbb{R}^{n-k})$. And since $\psi(p) = (0, \ldots, 0)$ and $\psi(V_0)$ is an open subset of $\mathbb{R}^n$ containing $\psi(p)$, we can conclude that any open subset of p will contain points inside and outside D. So, p is in the topological boundary of D.

Meanwhile, suppose p is in the manifold interior of D. Importantly, the manifold interior is an open submanifold of M (see *page 1111*). So, p is contained in an open subset of M which is contained in D. Thus p is in the topological interior of D.

Since the topological interior of D is contained in D we can now conclude that the topological interior of D is precisely $\text{Int}(D)$. Meanwhile, because D is a closed subset of M (since it is properly embedded), we know that the topological boundary of D is also contained in D. So, it must be precisely ∂D. ■

Proposition 5.47: Suppose M is a C^r manifold without a boundary and $f \in C^r(M, \mathbb{R})$.

(a) For each regular value $b \in \mathbb{R}$ we have that $f^{-1}((-\infty, b])$ is a regular domain in M without corners.

Proof:
To start off, f being continuous implies that $D := f^{-1}((-\infty, b])$ will be a closed subset of M. Thus, once we show D is an embedded C^r submanifold of M, it will be obvious that D is properly embedded.

We proceed by showing that if $p \in D$ then there is an open set $U \subseteq M$ containing p such that $D \cap U$ is an embedded submanifold of codimension 0. Fortunately, by the continuity of f, if $f(p) \neq b$ then $f^{-1}((-\infty, b))$ is an open subset of M and of D containing p. Since open subsets of M are submanifolds of codimension 0 without boundaries, we are done in that case.

(In fact this would show that $f^{-1}((-\infty, b)) \subseteq \text{Int}(D)$.)

Meanwhile suppose $f(p) = b$. Since b is a regular value of f, we know that $\mathrm{d}f_p$ is surjective. In turn, by proposition 4.1 on *page 1054* we can find an open set $U \subseteq M$ containing p such that $\mathrm{d}f_q$ is surjective for all $q \in U$.

By the rank theorem, there's a smooth chart (V, φ) on U and a smooth chart (W, ψ) on $\mathbb{R}$ such that $\varphi(p) = 0$ and $\psi \circ f \circ \varphi^{-1}$ equals the projection map $\pi : \mathbb{R}^n \to \mathbb{R}$ onto the first coordinate. (Here $n = \dim(M)$.) Yet since $\psi(b) = 0$ and ψ must map connected components of $W - \{b\}$ to $\psi(W) - \{0\}$, we can conclude that $W \cap (-\infty, b]$ is precisely given by either $\psi^{-1}((-\infty, 0])$ or $\psi^{-1}([0, \infty))$. By negating components of φ and ψ if necessary, we can without loss of generality assume $W \cap (-\infty, b] = \psi^{-1}([0, \infty))$.

Now $q \in V \cap D$ iff $q \in V$ and $f(q) \leq b$. Yet $f(q) \leq b$ iff:

$$\psi(f(q)) = (\psi \circ f \circ \varphi^{-1})(\varphi(q)) \geq 0.$$

So $V \cap D = \varphi^{-1}([0, \infty) \times \mathbb{R}^{n-1})$. In other words, $V \cap D$ is an n-half slice of the smooth chart (V, φ) on M containing p. It follows that $D \cap V$ is an embedded submanifold of M of codimension 0.

(And in particular p is a boundary point. Thus $f^{-1}(\{b\}) = \partial D$).

(b) Given any two regular values $a, b \in \mathbb{R}$ with $a < b$, we have that $f^{-1}([a, b])$ is a regular domain in M without corners.

Proof:
The proof of part (a) translates basically verbatim. The only additional step is that you need to shrink the open set U containing $p \in f^{-1}(\{b\})$ by intersecting it with $f^{-1}(b - \varepsilon, b + \varepsilon)$ for some small enough $\varepsilon > 0$. And you'd do an analogous thing when showing the case for some $p \in f^{-1}(\{a\})$. ■

(In particular, if $D \coloneqq f^{-1}([a, b])$ then $f^{-1}((a, b)) = \mathrm{Int}(D)$ and $f^{-1}(\{a, b\}) = \partial D$).

9/10/2026

Lemma 5.34: Suppose M is a C^r manifold without a boundary, $S \subseteq M$ is an (immersed) C^r submanifold, and $f \in C^r(S, \mathbb{R})$ (meaning f is a C^r map according to the differentiable structure on S).

(a) If S is embedded and without a boundary, then there exists an open set $U \subseteq M$ containing S and a function $\widetilde{f} \in C^r(U, \mathbb{R})$ such that $f = \widetilde{f}|_S$.

Proof:
First I shall show that for every $p \in S$ there is an open set $U_p \subseteq M$ and a function $f_p \in C^r(U_p, \mathbb{R})$ such that $f = f_p$ on $U \cap S_p$.

Let $k = \dim(S)$, $n = \dim(M)$, and $\pi : \mathbb{R}^n \to \mathbb{R}^k$ be the projection onto the first k coordinates. Then we can find a slice chart (U_p, φ) for S in M containing p. In particular, $(S \cap U_p, \pi \circ \varphi)$ is a smooth chart on S with $S \cap U_p = \varphi^{-1}(\{(x_1, \ldots, x_n) : x_{k+1} = c_{k+1}, \ldots, x_n = c_n\})$. Next, let $\iota : \mathbb{R}^k \to \mathbb{R}^n$ be the function $(x_1, \ldots, x_k) \mapsto (x_1, \ldots, x_k, c_{k+1}, \ldots, c_n)$.

Given the local coordinates defined by (V, φ) we want $f_p(x_1, \ldots, x_n) = f(x_1, \ldots, x_k)$. So, let:

$$f_p = f \circ \varphi^{-1} \circ \iota \circ \pi \circ \varphi$$

Then $f_p \in C^r(U_p, \mathbb{R})$. Also for all $(x_1, \ldots, x_k, c_{k+1}, \ldots, c_n) \in S \cap U_p$ we know that:

$$f_p \circ \varphi^{-1}(x_1, \ldots, x_k, c_{k+1}, \ldots, c_n) = f \circ \varphi^{-1}(x_1, \ldots, x_k, c_{k+1}, \ldots, c_n)$$

So, $f = f_p$ on $S \cap U_p$.

Now for every $p \in S$ let $U_p \subseteq M$ be an open set containing p and $f_p \in C^r(U_p, \mathbb{R})$ be a function with $f = f_p$ on $U \cap S_p$. Next, let $\{\psi_p\}_{p \in S}$ be a C^r partition of unity on the open submanifold $U := \bigcup_{p \in S} U_p$ of M subordinate to the sets U_p. (In other words $\operatorname{supp}(\psi_p) \subseteq U_p$ for every p and $\sum_{p \in S} \psi_p \equiv 1$.)

Then set $\widetilde{f} = \sum_{p \in S} f_p \psi_p$. This will be a C^r map from the open set U containing S. After all, every $q \in U$ has a neighborhood in U where only finitely many ψ_p are nonzero. And then $\widetilde{f}$ is automatically a C^r function on that neighborhood. Plus, since $f_p = f$ on $S \cap \operatorname{supp}(\psi_p)$ for all $p \in S$, we have that $\widetilde{f} = f \cdot 1 = f$ on S.

(b) If S is properly embedded then we can take U to be all of M.

Proof:
Let $\{(U_p, \varphi_p)\}_{p \in S}$ be the open covering of S consisting of slice charts from part (a). Then also let $U = M - S$. Since S is closed, we know that U is an open set. And now, we let $\{\psi_p\}_{p \in S} \cup \{\psi\}$ be a partition of unity on M subordinate to those slice charts as well as U. In other words, $\operatorname{supp}(\psi_p) \subseteq U_p$ for all $p \in S$, $\operatorname{supp}(\psi) \subseteq U$, and $\psi + \sum_{p \in S} \psi_p \equiv 1$ on M.

Now set $\widetilde{f} = \sum_{p \in S} f_p \psi_p$. Then $\widetilde{f}$ will be a C^r map from M into $\mathbb{R}$ After all, every $q \in M$ has a neighborhood where only finitely many of the ψ_p and ψ are nonzero. Furthermore, since $\psi \equiv 0$ on S, we know that $\sum_{p \in S} \psi_p \equiv 1$ on S. So, $\widetilde{f} = f \cdot 1 = f$ on S. ■

I'd like to take a break from John Lee chapter 5 and introduce a different important topic in the theory of manifolds.

Let M be a topological space. A (real) vector bundle of rank k over M is a topological space E with a surjective continuous map $\pi : E \to M$ satisfying that:

(a) For each $p \in M$ the fiber $E_p := \pi^{-1}(\{p\})$ over p is endowed with a structure turning it into a k-dimensional real vector space.

(b) For each $p \in M$ there is an open neighborhood U of p in M and a homeomorphism $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ (called a local trivialization of E over U) satisfying that:

- $\pi_U \circ \Phi = \pi$ (where $\pi_U : U \times \mathbb{R}^k \to U$ is the projection);
- For each $q \in U$ the restriction of Φ to E_q is a vector space isomorphism from E_q to $\{q\} \times \mathbb{R}^k \cong \mathbb{R}^k$.

If E and M are C^r manifolds, $\pi : E \to M$ is a C^r map, and the local trivializations can be chosen to be C^r diffeomorphisms, then E is said to be a C^r vector bundle over M. Also, any local trivialization that is a C^r diffeomorphism is called a "smooth" local trivialization.

We call E the total space of the bundle, M the base space of the bundle, and $\pi : E \to M$ the projection of the bundle.

Exercise 10.1: Suppose $\pi : E \to M$ is a C^r vector bundle (of rank k). Then the projection map π is a C^r submersion.

Proof:
Let $q \in E$ be any point, and then set $p = \pi(q)$. We can find an open set $U \subseteq M$ containing p and a C^r diffeomorphism $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ that is a local trivialization. Because Φ is continuous, we know $V := \pi^{-1}(U)$ is an open subset of E containing q.

But now note that $\pi|_V = \pi_U \circ \Phi$ where $\pi_U : U \times \mathbb{R}^k \to U$ is the projection map. Therefore, $\mathrm{d}\pi_q = \mathrm{d}(\pi_U)_{\Phi(q)} \circ \mathrm{d}\Phi_q$. Yet we can also see that π_U is a submersion. After all, for any $x \in U \times \mathbb{R}^k$ we have that:
$T_x(U \times \mathbb{R}^k) \cong T_{\pi_U(x)}U \oplus T_{\pi_{\mathbb{R}^k}(x)}\mathbb{R}^k$ via the map sending v to $(\mathrm{d}(\pi_U)_x v,\ \mathrm{d}(\pi_{\mathbb{R}^k})_x v)$.

Thus, $\mathrm{d}(\pi_U)_x$ must be a surjection onto $T_{\pi_U(q)}U$. And since x was arbitrary, we can conclude π_U is a submersion.

More generally, if $M_1, \ldots, M_k$ are any manifolds, $1 \leq i \leq k$, and $\pi : M_1 \times \cdots \times M_k \to M_i$ is the projection map, then this reasoning shows that π is a submersion.

Also $\mathrm{d}\Phi_q$ must be an invertible map since Φ is a diffeomorphism. So, $\mathrm{d}\pi_q$ is the composition of two surjective linear maps, and in turn must be a surjective linear map itself. Since q was arbitrary, this proves that $\pi : E \to M$ is a submersion. ■

Lemma 1.35: Let M be a set, $r \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$ and suppose we are given a collection $\{U_\alpha\}_{\alpha \in A}$ of subsets of M together with maps $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$ such that:

(i) For each $\alpha \in A$, φ_α is a bijection between U_α and an open subset $\varphi_\alpha(U_\alpha)$ of $[0, \infty)^k \times \mathbb{R}^{n-k}$ (where $0 \leq k \leq n$);

(ii) For every $\alpha, \beta \in A$ we have that $\varphi_\alpha(U_\alpha \cap U_\beta)$ is an open subset of $\varphi_\alpha(U_\alpha)$;

(iii) Whenever $U_\alpha \cap U_\beta \neq \emptyset$ then the map $\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$ is a C^r map;

(iv) Countably many of the U_α cover M;

(v) For any $p, q \in M$ either there exists U_α with both $p, q \in U_\alpha$ or there exists disjoint U_α and U_β with $p \in U_\alpha$ and $q \in U_\beta$.

Then M has a unique topology such that each $(U_\alpha, \varphi_\alpha)$ is a chart. And if $r \geq 1$ then M has a unique C^r structure such that each $(U_\alpha, \varphi_\alpha)$ is a smooth chart.

To start off, we equip M with the topology generated by the collection:

$$\{\varphi_\alpha^{-1}(V) : \alpha \in A \text{ and } V \text{ is an open subset of } \varphi_\alpha(U_\alpha)\}.$$

To prove that the latter set is a basis, first note by (iv) that every $x \in M$ is contained in some U_α. Also suppose that $x \in \varphi_\alpha^{-1}(V) \cap \varphi_\beta^{-1}(W)$. Then (iii) implies that:

$$(\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W) \text{ is an open subset of } \varphi_\alpha(U_\alpha \cap U_\beta).$$

By (ii), we also know $\varphi_\alpha(U_\alpha \cap U_\beta)$ is an open subset of $\varphi_\alpha(U_\alpha)$. Thus, $(\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W)$ is also an open subset of $\varphi_\alpha(U_\alpha)$. And now, we can say that:

$$\varphi_\alpha^{-1}(V \cap (\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W)) = \varphi_\alpha^{-1}(V) \cap \varphi_\beta^{-1}(W)$$

is also a basis set. Equality here is due to (i) saying φ_α and φ_β are bijections.

Having equipped M with this topology, it's now clear that each φ_α is a homeomorphism between U_α and $\varphi_\alpha(U_\alpha)$.

To see why, first note by the definition of the topology that each φ_α is continuous. I also claim each φ_α is an open map. After all, suppose $V \subseteq U_\alpha$ is any open set and $x \in V$ is any point. Then we know there exists some open set $B \coloneqq \varphi_\beta^{-1}(W)$ where W is an open subset of $\varphi_\beta(U_\beta)$ and $x \in \varphi_\beta^{-1}(W) \subseteq V$. Yet also note that as $B \subseteq V \subseteq U_\alpha$, we can further say that W is an open subset of $\varphi_\beta(U_\beta \cap U_\alpha)$. In turn, by (ii) and (iii) we know that $(\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W) = \varphi_\alpha(B)$ is an open subset of $\varphi_\alpha(U_\alpha)$. And since $\varphi_\alpha(B) \subseteq \varphi_\alpha(V)$, we've proven that $\varphi_\alpha(x)$ is in the topological interior of $\varphi_\alpha(V)$. Yet x was arbitrary. So, $\varphi_\alpha(V)$ is an open subset of $\varphi_\alpha(U_\alpha)$.

It follows that M is locally Euclidean. Furthermore, (iv) guarentees that M is second countable. Also, (v) guarentees that M is Hausdorff. So, we've equipped M with a topology turning it into a topological n-dimensional manifold.

As a side note, this topology is the coarsest topology such that every φ_α is continuous. Also suppose $\mathcal{T}'$ is another topology on M turning it into a topological manifold with each $(U_\alpha, \varphi_\alpha)$ as a chart. Then suppose $V \in \mathcal{T}'$. Since each φ_α is a homeomorphism, we must have that $\varphi_\alpha(V \cap U_\alpha)$ is an open subset of $\varphi_\alpha(U_\alpha)$ for all α. Yet in turn $V \cap U_\alpha$ must be in our original topology. And since $V = V \cap M = V \cap \bigcup_{\alpha \in A} U_\alpha = \bigcup_{\alpha \in A}(V \cap U_\alpha)$, we can conclude that V is open in our original topology. So, our original topology is also the finest topology that works. And this proves the topology we equipped M with is unique.

Finally, if $r \geq 1$ then (iii) also tells us the $(U_\alpha, \varphi_\alpha)$ are C^r compatible. And since they cover all of M, we know that they define a unique C^r structure on M. ■

The prior lemma is useful for constructing C^r manifolds. Here's one other observation that's necessary for what I'm about to do.

Suppose M is a C^r manifold (where $r \geq 1$) and (U, φ) and (V, ψ) are smooth charts such that $U \cap V = \emptyset$. We'll write $\varphi(p) = (x_1(p), \ldots, x_n(p))$ and $\psi(p) = (\widetilde{x_1}(p), \ldots, \widetilde{x_n}(p))$ Then for any $p \in U \cap V$ we can write $v \in T_pM$ as either $\sum_{i=1}^n c_i \left.\frac{\partial}{\partial x_i}\right|_p$ or as $\sum_{i=1}^n \widetilde{c_i} \left.\frac{\partial}{\partial \widetilde{x_i}}\right|_p$ (where each representation is unique). Importantly though, by taking coordinate representations of the identity map Id_M, we can see that:

$$|\widetilde{c_1}, \ldots, \widetilde{c_n}\rangle = (\mathrm{D}(\psi \circ \varphi^{-1})(\varphi(p)))|c_1, \ldots, c_n\rangle.$$

The most important vector bundle is the tangent bundle TM of a C^r manifold M.

We'll define $TM := \coprod_{p \in M} T_pM = \{(p, v) : p \in M \text{ and } v \in T_pM\}$. Then we'll also define $\pi : TM \to M$ by $\pi(p, v) = p$.

Lemma 3.18: Suppose $r \geq 1$. For any C^r n-dimensional manifold M, TM has a natural topology (and C^{r-1} structure if $r \geq 2$) turning TM into a $2n$-dimensional C^{r-1} manifold. Also, $\pi : TM \to M$ is a C^{r-1} map.

Proof:
Let (U, φ) be any smooth chart and let $\varphi(p) = (x_1(p), \ldots, x_n(p))$ be the local coordinates on that chart. Then we define a map $\widetilde{\varphi} : \pi^{-1}(U) \to \varphi(U) \times \mathbb{R}^n$ by:

$$\widetilde{\varphi}(p, \sum_{i=1}^n c_i \left.\frac{\partial}{\partial x_i}\right|_p) = (x_1(p), \ldots, x_n(p), c_1, \ldots, c_n).$$

This is a bijection to the open set $\varphi(U) \times \mathbb{R}^n$. So, we let $(\pi^{-1}(U), \widetilde{\varphi})$ be a chart. By constructing such charts for all smooth charts on M, we get a collection of smooth charts for which condition (i) in the prior lemma is satisfied.

Next suppose (U, φ) and (V, ψ) are smooth charts on M. Then we can easily see that

- $\widetilde{\varphi}(\pi^{-1}(U) \cap \pi^{-1}(V)) = \varphi(U \cap V) \times \mathbb{R}^n$;
- $\widetilde{\psi}(\pi^{-1}(U) \cap \pi^{-1}(V)) = \psi(U \cap V) \times \mathbb{R}^n$.

The latter sets are all open so condition (ii) is satisfied.

Furthermore, $\widetilde{\psi} \circ \widetilde{\varphi}^{-1} : \widetilde{\varphi}(\pi^{-1}(U) \cap \pi^{-1}(V)) \to \widetilde{\psi}(\pi^{-1}(U) \cap \pi^{-1}(V))$ is given by the map:

$$((x_1, \ldots, x_n), (c_1, \ldots, c_n)) \mapsto (\psi \circ \varphi^{-1}(x_1, \ldots, x_n), (\mathrm{D}(\psi \circ \varphi^{-1})(\varphi(p)))|c_1, \ldots, c_n\rangle)$$

Since $\psi \circ \varphi^{-1}$ is a C^r map, $\widetilde{\psi} \circ \widetilde{\varphi}^{-1}$ is thus a C^{r-1} map. And this fulfills condition (iii).

To show condition (iv), note that a countable collection $\{(U_n, \varphi_n)\}_{n \in \mathbb{N}}$ of smooth charts covers M. In turn, we can see that $\{(\pi^{-1}(U_n), \widetilde{\varphi_n})\}$ is a countable collection of charts covering TM.

Finally, to show condition (v) just note that if $p, q \in TM$ are in the same fiber of π then there is a single chart $(\pi^{-1}(U), \widetilde{\varphi})$ containing p and q. Meanwhile, if $\pi(p) \neq \pi(q)$ then we can find disjoint charts (U, φ) and (V, ψ) on M such that $\pi(p) \in U$ and $\pi(q) \in V$. And then $\pi^{-1}(U)$ and $\pi^{-1}(V)$ are disjoint sets containing p and q respectively.

And now we just apply lemma 1.35 to get our topology and C^{r-1} structure on TM turning it into a $2n$-dimensional C^{r-1} manifold.

To see that $\pi : TM \to M$ is a C^{r-1} map, just note for any smooth chart (U, φ) on M that $\pi(\pi^{-1}(U)) = U$ (since π is surjective) and $\varphi \circ \pi \circ \widetilde{\varphi}^{-1} = \pi_U$ where π_U is the projection map $U \times \mathbb{R}^n \to U$. It follows that every $q \in TM$ has a neighborhood on which π is a C^{r-1} map. So, π is a C^{r-1} map.

Now I claim that $\pi : TM \to M$ (where TM is equipped with the topology described priorly) is a vector bundle of rank $n \coloneqq \dim(M)$ over M.

Proof:
For every $p \in M$ we have that $\pi^{-1}(\{p\}) \cong T_pM$. Thus every fiber is an n-dimensional real vector space.

Also for any $p \in M$ we can pick a smooth chart (U, φ) on M containing p. And then after writing $\varphi(q) = (x_1(q), \ldots, x_n(q))$, we define $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^n$ by:

$$\Phi(q, \textstyle\sum_{i=1}^n c_i \left.\frac{\partial}{\partial x_i}\right|_p) = (q, (c_1, \ldots, c_n)).$$

To see that Φ is a local trivialization, first note that $\pi_U \circ \Phi = \pi$ (where $\pi_U : U \times \mathbb{R}^n \to U$ is the projection map). Also note for each $q \in U$ that $\Phi|_{T_qM}$ is a linear map from T_qM to $\mathbb{R}^n$. ■

Furthermore, if M is a C^r manifold where $r \geq 2$, then TM is a C^{r-1} vector bundle over M.

Proof:
It suffices to show that Φ from before is a C^{r-1} diffeomorphism. Fortunately, note that $\varphi^{-1} \times \mathrm{Id}_{\mathbb{R}^n} : \varphi(U) \times \mathbb{R}^n \to U \times \mathbb{R}^n$ is a C^r diffeomorphism. Also, recall that $\widetilde{\varphi} : \pi^{-1}(U) \to \varphi(U) \times \mathbb{R}^n$ given by $\widetilde{\varphi}(p, \sum_{i=1}^n \left.\frac{\partial}{\partial x_i}\right|_p) = (\varphi(p), (c_1, \ldots, c_n))$ is a smooth chart on TM (and thus a C^{r-1} diffeomorphism). And since $\Phi = (\varphi^{-1} \times \mathrm{Id}_{\mathbb{R}^n}) \circ \widetilde{\varphi}$, we know that Φ is a C^{r-1} diffeomorphism. ■

Given a C^k map $F : M \to N$ between C^r manifolds, we define a map $\mathrm{d}F : TM \to TN$ (called the global differential) by $\mathrm{d}F(p, v) \coloneqq (F(p), \mathrm{d}F_p(v))$.

Proposition 3.21: Suppose $k \geq 1$. Then $\mathrm{d}F$ is a C^{k-1} map.

Proof:
We'll let $\pi_1 : TM \to M$ and $\pi_2 : TN \to N$ denote the projection maps of our vector bundles. Now suppose $(p, v) \in TM$. Then we know there exists smooth charts (U, φ) on

M containing p and (V, ψ) on N containing $F(U)$. In turn, we consider the smooth charts $(\pi^{-1}(U), \widetilde{\varphi})$ on TM and $(\pi^{-1}(V), \widetilde{\psi})$ on TN (where $\widetilde{\varphi}$, $\widetilde{\psi}$ were constructed in lemma 3.18). These charts will satisfy that $(p, v) \in \pi^{-1}(U)$ and $\mathrm{d}F(\pi^{-1}(U)) \subseteq \pi^{-1}(V)$.

But now let us write $\varphi(q) = (x_1(q), \ldots, x_m(q))$ and $\psi(q) = (y_1(q), \ldots, y_n(q))$. Also let us set $\widehat{F} \coloneqq \psi \circ F \circ \varphi^{-1}$. Then:

$$\widetilde{\psi} \circ \mathrm{d}F \circ \widetilde{\varphi}^{-1}((x_1, \ldots, x_m), (c_1, \ldots, c_m)) = (\widehat{F}(x_1, \ldots, x_m),\ (\mathrm{D}\widehat{F}(x_1, \ldots, x_m))|c_1, \ldots, c_m\rangle)$$

This proves that $\mathrm{d}F : TM \to TN$ is a C^{k-1} map. ■

As a side note, if $r \geq 2$ we can now define a functor $\mathbf{Diff}_c^r \to \mathbf{Diff}_c^{r-1}$ sending any manifold M to TM and any C^r map $F : M \to N$ to $\mathrm{d}F$. Meanwhile, if $r = 1$ then we can similarly define a functor $\mathbf{Diff}_c^1 \to \mathbf{Man}_b$. After all, because $\mathrm{d}(G \circ F)_p = \mathrm{d}G_{F(p)} \circ \mathrm{d}F_p$, we can easily conclude that $\mathrm{d}(G \circ F) = \mathrm{d}G \circ \mathrm{d}F$. Similarly, we can conclude $\mathrm{d}(\mathrm{Id}_M) = \mathrm{Id}_{TM}$. So, these functors are well-defined.

Typically, given the map $\pi : TM \to M$ we'll just identify T_pM with $\pi^{-1}(\{p\})$ for every $p \in M$. That way, we can formally write $\mathrm{d}F$ instead of $\mathrm{d}F_p$ if the point p is obvious from context.

Let $J \subseteq \mathbb{R}$ be some interval and suppose M is a C^r manifold. Given a C^1 map $\gamma : J \to M$ and any $t_0 \in J$ we define the velocity of γ at t_0 to be $\gamma'(t_0) \coloneqq \mathrm{d}\gamma(\frac{\mathrm{d}}{\mathrm{d}t}\big|_{t_0})$. Some other common notations for $\gamma'(t_0)$ are $\dot{\gamma}(t_0)$ and $\frac{\mathrm{d}\gamma}{\mathrm{d}t}(t_0)$.

Note that if $f \in C^1(U, \mathbb{R})$ where U is an open set containing $\gamma(t_0)$ then:

$$\gamma'(t_0)f = \mathrm{d}\gamma\left(\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t_0}\right) f = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t_0}(f \circ \gamma) = (f \circ \gamma)'(t_0)$$

Next let (U, φ) be a smooth chart on M containing $\gamma(t_0)$. Then by restricting to a small enough neighborhood of t_0 we can assume $\gamma(J) \subseteq U$. Thus, we can write $\varphi \circ \gamma(t) = (\gamma_1(t), \ldots, \gamma_n(t))$. And now in coordinates, we have that:

$$\gamma'(t_0) = \sum_{i=1}^{n} \frac{\mathrm{d}\gamma_i}{\mathrm{d}t}(t_0)\ \frac{\partial}{\partial x_i}\Big|_p$$

Proposition 3.23: Suppose M is a C^r manifold without corners and $p \in M$. Then every $v \in T_pM$ is the velocity of some C^r curve γ at 0.

Hopefully this is obvious from *pages 1046-1051* and *page 1057-1058*.

Now back to chapter 5.

Given an immersed C^r submanifold S of a C^r manifold M, we can view its tangent space T_pS at a point $p \in S$ as a subspace of the tangent space T_pM of M at p. To see how, recall by definition that the inclusion map $\iota : S \hookrightarrow M$ is an immersion. Thus at every $p \in S$ we have that $\mathrm{d}\iota_p : T_pS \to T_pM$ is an injection.

In particular, suppose $v \in T_pS$ and $\widetilde{v} = \mathrm{d}\iota_p v \in T_pM$. Then:

$$\widetilde{v}([f]) = \widetilde{v}([f, U]) = v([f \circ \iota, U \cap S]) = v([f|_S]).$$

Now we shall view the injection $\mathrm{d}\iota_p$ as an identification of T_pS as a subspace of T_pM.

Proposition 5.35: Suppose M is a C^r manifold, $S \subseteq M$ is an immersed C^r submanifold without corners, and $p \in S$. A vector $v \in T_pM$ is in T_pS if and only if there is a C^r curve $\gamma : J \to M$ such that:

- $J \subseteq \mathbb{R}$ is a compact interval containing 0,
- $\gamma(J) \subseteq S$,
- $\gamma(0) = p$,
- $\gamma'(0) = v$,
- γ is a C^r curve when viewed as a function from J into S.

$(\Longrightarrow)$
Suppose there exists $\widetilde{v} \in T_pS$ such that $v = \mathrm{d}\iota_p\widetilde{v}$. Then we know there is a C^r curve $\gamma : J \to S$ such that $J \subseteq \mathbb{R}$ is a compact interval containing 0, $\gamma(0) = p$, and $\gamma'(0) = \widetilde{v}$. But now $\iota \circ \gamma : J \to M$ is also a C^r curve (since it is a composition of two C^r functions). Also $(\iota \circ \gamma)'(0) = \mathrm{d}\iota_p \circ \mathrm{d}\gamma_0(\frac{\mathrm{d}}{\mathrm{d}t}\big|_0) = \mathrm{d}\iota_p(\gamma'(0)) = \mathrm{d}\iota_p(\widetilde{v}) = v$

$(\Longleftarrow)$
If we let $\widetilde{\gamma}$ be the curve γ with it's codomain restricted to S and then set $\widetilde{v} = \gamma'(0)$, we have that:

$$\mathrm{d}\iota_p(\widetilde{v}) = \mathrm{d}\iota_p \circ \mathrm{d}\widetilde{\gamma}_0(\tfrac{\mathrm{d}}{\mathrm{d}t}\big|_0) = \mathrm{d}(\iota \circ \widetilde{\gamma})_0(\tfrac{\mathrm{d}}{\mathrm{d}t}\big|_0) = (\iota \circ \widetilde{\gamma})'(0) = \gamma'(0) = v.$$

So, $v \in T_pM$ is also in T_pS. ■

Proposition 5.37: Suppose M is a C^r manifold without a boundary, $S \subseteq M$ is an embedded C^r submanifold (without corners), and $p \in S$. As a subspace of T_pM, we have that:

$$T_pS = \{v \in T_pM : v(f) = 0 \text{ if } f \in C^r(M, \mathbb{R}) \text{ with } f \equiv 0 \text{ on } S\}.$$

Proof:
First suppose $v \in T_pS \subseteq T_pM$. In other words, if $\iota : S \hookrightarrow M$ is the inclusion map then we know there exists $w \in T_pS$ such that $\mathrm{d}\iota_p(w) = v$. But now if $f \in C^r(M, \mathbb{R})$ is a function vanishing on S, we know that $f \circ \iota \equiv 0$. So:

$$v(f) = \mathrm{d}\iota_p(w)f = w(f \circ \iota) = w(0) = 0.$$

Conversely, suppose $v \in T_pM$ satisfies that $v(f) = 0$ for all $f \in C^r(M, \mathbb{R})$ with $f \equiv 0$ on S. Now we want to show that there is some $w \in T_pS$ with $v = \mathrm{d}\iota_p(w)$. Fortunately, we can pick some slice (or half slice) chart (U, φ) on M containing p such that:

$$U \cap S = \{\varphi^{-1}(x_1, \ldots, x_n) : (x_1, \ldots, x_n) \in \varphi(U) \text{ and } x_{k+1} = 0, \ldots, x_n = 0\}$$

If we let $\pi : \mathbb{R}^n \to \mathbb{R}^k$ be the projection onto the first k coordinates, we have that $(U \cap S, \pi \circ \varphi)$ is a smooth chart on S. But now with respect to this coordinate chart and the chart (U, φ) on M, we have that the coordinate representation of $\iota : S \hookrightarrow M$ is $\iota(x_1, \ldots, x_k) = (x_1, \ldots, x_k, 0, \ldots, 0)$. It then follows that if $\varphi(q) = (x_1(q), \ldots, x_n(q))$ then T_pS is the subspace of T_pM spanned by $\frac{\partial}{\partial x_1}\Big|_p, \ldots, \frac{\partial}{\partial x_k}\Big|_p$.

Now write $v = \sum_{i=1}^n c_i \frac{\partial}{\partial x_i}\Big|_p \in T_pM$. We just need to show that $c_j = 0$ for all $j \geq k+1$ in order to prove that $v \in T_pS$.

Fortunately, we can define a C^r function $f : M \to [0,1]$ such that $\operatorname{supp}(f) \subseteq U$ and $f \equiv 1$ on a neighborhood of p in M. Next given any given any $j \geq k+1$ we can define a C^r map $g : M \to \mathbb{R}$ by $g(q) = f(q) \cdot x_j(q)$. Then:

$$v(g) = v(q \mapsto x_j(q)) = c_j.$$

Yet we also know that $g \equiv 0$ on U^{C} and $g \equiv 0$ on $U \cap S$. So, $g \equiv 0$ on S and in turn $v(g) = 0$ by assumption. So, $c_j = 0$ for all $j \geq k+1$. ■

If $S \subseteq M$ is an embedded C^r submanifold of a manifold M, we say a C^r map $\Phi : M \to N$ such that S is a regular level set of Φ is a defining map for S. If $N = \mathbb{R}^{\dim(M)-\dim(S)}$, we also call Φ a defining function.

Also suppose $U \subseteq M$ is an open subset and $\Phi : U \to N$ is a C^r map such that $S \cap U$ is a regular level set of Φ. Then Φ is called a local defining map / function for S. Note that proposition 5.16 on *page 1120* guarentees that all embedded submanifolds without boundaries admit local defining functions on neighborhoods around all their points.

Proposition 5.38: Suppose M is a C^r manifold without a boundary and $S \subseteq M$ is an embedded submanifold without a boundary. If $\Phi : U \to N$ is a local defining map for S then $T_pS = \ker(\mathrm{d}\Phi_p)$ for all $p \in S \cap U$.

Proof:
Let $\iota : S \cap U \hookrightarrow M$ be the inclusion map and recall that T_pS is identified with the subspace $\operatorname{im}(\mathrm{d}\iota_p)$ of T_pM. But also note $\Phi \circ \iota$ is a constant map. Therefore, $\mathrm{d}\Phi_p \circ \mathrm{d}\iota_p$ will be the zero map. And this proves that $\operatorname{im}(\mathrm{d}\iota_p) \subseteq \ker(\mathrm{d}\Phi_p)$.

On the other hand, since $\mathrm{d}\Phi_p$ is surjective, the rank-nullity theorem tells us that $\dim(\ker(\mathrm{d}\Phi_p)) = \dim(T_pM) - \dim(T_{\Phi(p)}N) = \dim(T_pS) = \dim(\operatorname{im}(\mathrm{d}\iota_p))$. So, we can actually conclude that $\operatorname{im}(\mathrm{d}\iota_p) = \ker(\mathrm{d}\Phi_p)$. ■

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Problem 5-3: Suppose M is a C^r manifold and $S \subseteq M$ is an immersed C^r submanifold. If any of the following conditions holds then S is an embedded submanifold.

(a) S has no boundary and has codimension 0 in M.

(b) The inclusion map $\iota : S \hookrightarrow M$ is proper.

(c) S is compact.

Proof:
In all cases, we need to show that $\iota : S \hookrightarrow M$ is a topological embedding. After all, that is equivalent to saying that S is equipped with the subspace topology relative to M.

For condition (a), note that ι being full rank actually let's us conclude by the inverse function theorem and by exercise 4-2 on *pages 1056-1057* that $S \subseteq \text{Int}(M)$ and ι is a local C^r diffeomorphism. In turn, as ι is also injective we know that it is a topological embedding.

For condition (b), recall that any continuous proper map into an LCH space is a closed map. Therefore, ι is a closed map. And since any injective continuous closed map is a topological embedding, we can conclude ι is a topological embedding.

For condition (c), note that since S is compact and M is Hausdorff, we can conclude that $\iota : S \hookrightarrow M$ is a closed map. And since ι is a continuous injection, we can conclude it is a topological embedding. ■

Problem 5-7: Let $F : \mathbb{R}^2 \to \mathbb{R}$ be defined by $F(x, y) = x^3 + xy + y^3$. Which level sets of F are smooth embedded submanifolds of $\mathbb{R}^2$?

To start off, $DF(x, y) = [3x^2 + y, 3y^2 + x]$. Thus, $(x, y) \in \mathbb{R}^2$ is a critical point of F iff $3x^2 + y = 0 = 3y^2 + x$. By solving this simple system of equations, we can in turn see the only critical points in $\mathbb{R}^2$ are $(0, 0)$ and $(-\frac{1}{3}, -\frac{1}{3})$. Since $F(0, 0) = 0$ and $F(-\frac{1}{3}, -\frac{1}{3}) = \frac{1}{27}$, we can conclude that if $c \notin \{0, \frac{1}{27}\}$ then $F^{-1}(\{c\})$ is a regular level set and in turn an embedded C^∞ submanifold without a boundary.

This leaves us with the task of concluding whether $F^{-1}(\{0\})$ or $F^{-1}(\{\frac{1}{27}\})$ are embedded C^∞ submanifolds.

- I claim $F^{-1}(\{\frac{1}{27}\})$ is not an embedded C^∞ submanifold. To see why, note that:

$$x^3 + xy + y^3 - \tfrac{1}{27} = x^3 + y^3 + (-\tfrac{1}{3})^3 - 3(\tfrac{-1}{3}xy)$$

Lemma: $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - ac - bc)$.

Motivation:
Substitute in $a = -(b + c)$. Then:

$$a^3 + b^3 + c^3 - 3abc = -(b^3 + 3b^2c + 3bc^2 + c^3) + b^3 + c^3 + 3(b + c)bc = 0.$$

It follows that if $a + b + c = 0$ then $a^3 + b^3 + c^3 - 3abc = 0$. Or in other words, $(a + b + c)$ should be a factor of the polynomial $a^3 + b^3 + c^3 - 3abc$.

Next note that if $(a + b + c) \cdot P(a, b, c) = a^3 + b^3 + c^3 - 3abc$ then we can make the reasonable guess that P is a symmetric polynomial due to the fact that both other polynomials in the equation are symmetric. Also, since every term in $a + b + c$ has degree 1 and every term in $a^3 + b^3 + c^3 - 3abc$ has degree 3, we should have that every term of P has degree 2. So, we can assume $P(a, b, c)$ has the form:

$$P(a, b, c) = A(a^2 + b^2 + c^2) + B(ab + ac + bc)$$

By matching terms, we can see that $A = 1$ and $B = -1$. And now we just hope that this works.

Proof:

$$\begin{aligned}(a+b+c)(a^2+b^2+c^2-ab-ac-bc) &= a^3+ab^2+ac^2-a^2b-a^2c-abc\\ &\quad +a^2b+b^3+bc^2-ab^2-abc-b^2c\\ &\quad +a^2c+b^2c+c^3-abc-ac^2-bc^2\\ &= a^3+b^3+c^3-3abc.\end{aligned}$$

Based on the prior lemma, we can now say that:

$$x^3+y^3+(-\tfrac{1}{3})^3-3(\tfrac{-1}{3}xy)=(x+y-\tfrac{1}{3})(x^2+y^2+\tfrac{1}{9}-xy+\tfrac{1}{3}(x+y))$$

Our level set is thus the line drawn by $y=-x+\frac{1}{3}$ unioned with whatever the zero set is of the function $G(x,y)=x^2+y^2+\frac{1}{9}-xy+\frac{1}{3}(x+y)$. To calculate this set, note that:

$$\begin{aligned}2G(x,y) &= x^2+x^2+y^2+y^2-2xy+\tfrac{1}{9}+\tfrac{1}{9}+2\tfrac{1}{3}x+2\tfrac{1}{3}y\\ &= (x-y)^2+(x+\tfrac{1}{3})^2+(y+\tfrac{1}{3})^2.\end{aligned}$$

Since squares are always nonnegative and scaling a function shouldn't change its zero set, we can conclude that $G(x,y)=0$ iff $x=y$, $x=-\frac{1}{3}$, and $y=-\frac{1}{3}$. So, we've proven that:

$$F^{-1}(\{\tfrac{1}{27}\})=\{(-\tfrac{1}{3},-\tfrac{1}{3})\}\cup\{(x,y):y=-x+\tfrac{1}{3}\}$$

This is not an embedded submanifold because it is a union of a 0-dimensional embedded submanifold and 1-dimensional embedded submanifold which are both disjoint.

- Next I claim $F^{-1}(\{0\})$ is also not an embedded C^∞ submanifold. To see why, consider that we can parametrize $F^{-1}(\{0\})$ as follows.

  Suppose $x\neq 0$. Then for any $(x,y)\in F^{-1}(\{0\})$ there exists some scalar $t\in\mathbb{R}$ such that $y=tx$. And now $x^3+tx^2+t^3x^3=0$. Since $x\neq 0$, we can divide by it to get that $x(1+t^3)+t=0$. So, $x=\frac{-t}{1+t^3}$. Next note that t can't equal -1 since that would give a contradiction that $t=0$. So, we can safely rearrange our equation to get that $x=\frac{-t}{1+t^3}$. And in turn, we can solve that $y=\frac{-t^2}{1+t^3}$.

  Meanwhile, note that $F(0,y)=y^3=0$ iff $y=0$. But then $(0,0)=(\frac{-0}{1+0^3},\frac{-0^2}{1+0^3})$. So, we can conclude that:

  $$F^{-1}(\{0\})\subseteq\{(\tfrac{-t}{1+t^3},\tfrac{-t^2}{1+t^3}):t\in\mathbb{R}-\{-1\}\}$$

  Checking the other inclusion is also now easy. So, we have a parametrization for $F^{-1}(\{0\})$.

  But now to finish off, note that as $t\to 0$ or ∞ or $-\infty$, we have that our parametrization approaches the origin from different directions. This would imply $T_0(F^{-1}(\{0\}))=T_0\mathbb{R}^2$ and in turn $F^{-1}(\{0\})$ is a 2-dimensional embedded submanifold. Yet we know that every other point of $F^{-1}(\{0\})$ has a neighborhood on which $F^{-1}(\{0\})$ is a 1-dimensional embedded submanifold. So, we can't have that $F^{-1}(\{0\})$ is an embedded submanifold. ■

I'm going to switch to chapter 6 of John Lee now. Note that we consider any measurable subset of $\mathbb{R}^n$ (for any n) as being equipped with the complete Lebesgue measure (which I'll denote μ).

Lemma 6.3: Suppose $n \geq 2$, A is an open or closed subset of $\mathbb{R}^{n-1}$ (or $[0,\infty)^k \times \mathbb{R}^{n-k-1}$ or $[0,\infty)^{n-1}$), and $f : A \to \mathbb{R}$ is a continuous function. Then the graph:

$$\Gamma(f) := \{(x_1, \ldots, x_{n-1}, f(x_1, \ldots, x_n)) \in \mathbb{R}^n : (x_1, \ldots, x_{n-1} \in A)\}$$

is a Borel set of measure zero in $\mathbb{R}^n$.

Proof:

To start off $\Gamma(f)$ is a Borel measurable set. To see why, let $g : A \to A \times \mathbb{R}$ be defined by $g(x) = (x, f(x))$. This will be continuous function.

Next I claim that in the case that either A is open or closed, we can write A as a union of countably many compact sets $\{F_k\}_{k\in\mathbb{N}}$.

If A is closed then we merely intersect A with parallelotopes in a lattice to get our compact sets. Meanwhile, if A is open then we know A is an LCH space. So, every $x \in A$ has a compact neighborhood in A. And then since A is second countable, we can just take a countable subcover.

But now since g is continuous, we then know each $g(F_k)$ is a compact subset of $A \times \mathbb{R}$ (and in turn of $\mathbb{R}^n$). So, each $g(F_k)$ is a measurable set and $g(A) = \bigcup_{k\in\mathbb{N}} g(F_k)$. So, $\Gamma(f) = g(A)$ is a measurable set.

Now to show that $\Gamma(f)$ has measure zero, we just note by Fubini's theorem that:

$$\mu(\Gamma(f)) = \int_A \mu(\{f(x)\})\mathrm{d}x = \int_A 0\mathrm{d}x = 0. \blacksquare$$

Propositon 6.5: Suppose $U \subseteq \mathbb{R}^n$ is an open set and $f : U \to \mathbb{R}^n$ is a C^1 map. If $A \subseteq U$ is a set of Lebesgue measure zero then so is $f(A)$.

Proof:

By Corollary C.30 on *page 292*, we know that f is locally Lipschitz continuous. So every $x \in A$ has an open neighborhood N_x on which F is Lipschitz continuous. Yet because U is second countable, we then can pick countably many of those neighborhoods $\{N_{x_k}\}_{k\in\mathbb{N}}$ to cover A.

Next, write $A = \bigcup_{k\in\mathbb{N}}(A \cap N_{x_k})$. Since $A \cap N_{x_k} \subseteq A$ we know that it is Lebesgue measurable with measure zero. And since f is Lipschitz continuous on N_{x_k}, we can conclude by the lemma on *page 1009* that $f(A \cap N_{X_k})$ is also Lebesgue measurable with measure zero.

Finally, since $f(A) = \bigcup_{k\in\mathbb{N}} f(A_k)$ we can conclude that $f(A)$ is a Lebesgue measurable. Also $\mu(f(A)) \leq \sum_{k\in\mathbb{N}} \mu(f(A \cap N_{x_k})) = \sum_{k\in\mathbb{N}} 0 = 0$. So, $f(A)$ has measure zero. $\blacksquare$

As a side note, it is now also an easy corollary that if $U \subseteq [0,\infty)^k \times \mathbb{R}^{n-k}$ is an open set, $f : U \to \mathbb{R}^n$ is a C^1 map, and $A \subseteq U$ is a set of measure zero, then so is $f(A)$. After all, by definition f can be extended to a C^1 map to an open set $\widetilde{U} \subseteq \mathbb{R}^n$ containing U. And then we just apply the proposition from before.

Now for the time being, I don't have the background to define full on measures on abstract manifolds. However, it will still be useful to be able to classify if a subset of a manifold is negligable or not. Later on when I learn proper measure theory for manifolds, I can tie this content to that.

Let M be a C^r n-dimensional manifold (where $r \geq 1$ and $n \geq 1$). We can equip M with it's Borel σ-algebra $\mathcal{B}_M$.

We say $A \subseteq M$ has measure zero in M if there exists a Borel set $B \subseteq M$ such that $A \subseteq B$ and for every smooth chart (U, φ) on M we have that $\varphi(B \cap U)$ is a subset of $\mathbb{R}^n$ with Lebesgue measure zero.

As a side note, since $U \subseteq M$ is an open set and φ is a homeomorphism from U to a measurable subset of $\mathbb{R}^n$, we can guarentee that $\varphi(E \cap U)$ will be Borel measurable in $\mathbb{R}^n$ for every Borel measurable set $E \in \mathcal{B}_M$.

Remark:
It's easy to see that if $\mathcal{N}$ is the collection of sets of measure zero in M, then:

- $\emptyset \in \mathcal{N}$,
- $A \in \mathcal{N}$ and $A' \subseteq A \implies A' \in \mathcal{N}$,
- $\{A_k\}_{k \in \mathbb{N}} \subseteq \mathcal{N} \implies \bigcup_{k \in \mathbb{N}} A_k \in \mathcal{N}$.

The following two lemmas prove almost converse statements.

Lemma: If $A \subseteq M$ is a set of measure zero in M and (U, φ) is any smooth chart, then $\varphi(A \cap U)$ is Lebesgue measurable with measure zero.

Proof:
Since A has measure zero in M, we can find a Borel set $B \in \mathcal{B}_M$ such that $A \subseteq B$ and $\varphi(B \cap U) = 0$. Then as $\varphi(A \cap U) \subseteq \varphi(B \cap U)$, we know by the completeness of the Lebesgue measure that $\varphi(A \cap U)$ is Lebesgue measurable and a null set. ■

Lemma 6.6: Suppose $A \subseteq M$ and that $\{(U_i, \varphi_i)\}_{i \in I}$ is a collection of smooth charts covering A such that $\varphi_i(A \cap U_i)$ is Lebesgue measurable with measure zero for all $i \in I$. Then A has measure zero in M.

Proof:
Since M is second countable, we can without loss of generality assume our indexing set I equals $\mathbb{N}$ by taking a countable subcover of A. Then note that $A = \bigcup_{i \in \mathbb{N}} (A \cap U_i)$. Next, since $\varphi(A \cap U_i)$ is Lebesgue measurable with measure zero for every i by assumption, we know there is a Borel measurable set $E_i \subseteq \varphi(U_i)$ of measure zero containing $\varphi(A \cap U_i)$. And now we set $B_i := \varphi^{-1}(E_i)$. This will be a Borel subset of M (since φ is continuous.) Also, $A \cap U_i \subseteq B_i \subseteq U_i$ for each i.

Let $B := \bigcup_{i \in \mathbb{N}} B_i$. Then B is a Borel subset of M containing A. If we can justify for any smooth chart (U, φ) on M now that $\mu(\varphi(U \cap B)) = 0$ then we will be done.

Fortunately, note that:

$$\varphi(U \cap B) = \bigcup_{i \in \mathbb{N}} \varphi(U \cap U_i \cap B) = \bigcup_{i \in \mathbb{N}} (\varphi \circ \varphi_i^{-1})(\varphi_i(U \cap U_i \cap B)).$$

Then because $\varphi_i(U \cap U_i \cap B) \subseteq \varphi_i(U_i \cap B)$ and we know the latter is a Lebesgue null set, we know that $\varphi_i(U \cap U_i \cap B)$ is a Lebesgue null set. And by an application of proposition 6.5 from two pages ago plus the fact that the transition function $(\varphi \circ \varphi_i^{-1})$ is a C^r map, we can conclude that $(\varphi \circ \varphi_i^{-1})(\varphi_i(U \cap U_i \cap B))$ is also a Lebesgue null set. Since this works for all i, we can conclude that $\varphi(U \cap B)$ is a Lebesgue null set. ■

Proposition 6.8: If $A \subseteq M$ is a set of measure zero in M then $M - A$ is dense in M.

Proof:
If $M - A$ is not dense in M then we can find an open set U contained in A. But now let (V, ψ) be a smooth chart on M such that $V \cap U \neq \emptyset$. Then $\psi(V \cap U)$ is an open subset of $[0, \infty)^k \times \mathbb{R}^{n-k}$ (or $[0, \infty)^n$ or $\mathbb{R}^n$). It follows that $\psi(V \cap U)$ is measurable and will contain a coordinate ball, half ball, or corner ball. So, $\mu(\psi(V \cap U)) > 0$. And since $\mu(\psi(V \cap A)) \geq \mu(\psi(V \cap U))$ (assuming $\psi(V \cap A)$ is even measurable), we can conclude that A does not have measure zero in M. ■

Theorem 6.9: Suppose M and N are C^r n-dimensional manifolds and $F : M \to N$ is a C^1 map. If $A \subseteq M$ is a set of measure zero in M then $F(A)$ is a set of measure zero in N.

Proof:
Let $\{(U_k, \varphi_k)\}_{k \in \mathbb{N}}$ be a countable collection of smooth charts covering M. Then suppose (ψ, V) is any smooth chart on N. By lemma 6.6 it suffices to show that $\psi(V \cap F(A))$ is Lebesgue measurable with measure zero. Fortunately, note that:

$$\begin{aligned}\psi(V \cap F(A)) &= \psi(\textstyle\bigcup_{k \in \mathbb{N}}(V \cap F(A \cap U_k))) \\ &= \textstyle\bigcup_{k \in \mathbb{N}} \psi(V \cap F(A \cap U_k)) = \bigcup_{k \in \mathbb{N}}(\psi \circ F)(A \cap U_k \cap F^{-1}(V)) \\ &= \textstyle\bigcup_{k \in \mathbb{N}}(\psi \circ F \circ \varphi^{-1})(\varphi(A \cap U_k \cap F^{-1}(V))).\end{aligned}$$

Now because A has measure zero in M and $(U_k \cap F^{-1}(V), \varphi|_{F^{-1}(V)})$ is a smooth chart on M, we know from a prior lemma that $\varphi(A \cap U_k \cap F^{-1}(V))$ is Lebesgue measurable with measure zero for each k. In turn, because $\psi \circ F \circ \varphi^{-1}$ is a C^1 map, we can also say that $(\psi \circ F \circ \varphi^{-1})(\varphi(A \cap U_k \cap F^{-1}(V)))$ is Lebesgue measurable with measure zero for each k. And finally, since $\psi(V \cap F(A))$ is a countable union of these sets, we can conclude that $\psi(V \cap F(A))$ is Lebesgue measurable with measure zero. ■

9/16/2026

Today I want to cover Sard's theorem. This will be split into a few results. In particualar, I will first focus on the case of a function whose domain is an open subset of $\mathbb{R}^m$ and whose codomain is $\mathbb{R}^n$.

Lemma A: Suppose X is an LCH space, Y is a Hausdorff topological space, and $f : X \to Y$ is continuous.

- If X is second countable and $U \subseteq X$ is open then $f(U)$ is a Borel set.

  Proof:
  Note that if X is a second countable LCH space then U is a σ-compact set. As a result,

we can find a sequence of compact sets $\{K_\ell\}_{\ell\in\mathbb{N}}$ such that $U = \bigcup_{\ell\in\mathbb{N}} K_\ell$. But then $f(U) = \bigcup_{\ell\in\mathbb{N}} f(K_\ell)$. Since f is continuous and K_ℓ is compact, we know $f(K_\ell)$ is compact. In turn, as Y is Hausdorff we know that $f(K_\ell)$ is a closed set and thus Borel. Finally, $f(U)$ being a countable union of Borel sets means that $f(U)$ is a Borel set.

- If X is σ-compact and $C \subseteq X$ is closed then $f(C)$ is a Borel set.

  Proof:
  Since X is σ-compact, we can find a sequence of compact sets $\{K_\ell\}_{\ell\in\mathbb{N}}$ such that $X = \bigcup_{\ell\in\mathbb{N}} K_\ell$. Then $f(C) = \bigcup_{\ell\in\mathbb{N}} f(C \cap K_\ell)$. And by identical reasoning as before, we can conclude that $f(C \cap K_\ell)$ is a closed set for all ℓ. ■

Lemma B: Suppose $m < n$ are integers, $U \subseteq \mathbb{R}^m$ is an open set, and $f : U \to \mathbb{R}^n$ is a C^1 map. Then $f(U)$ is a Borel null set.

Proof:
By the prior lemma, we know that $f(U)$ is a Borel set. Also, we can define a C^1 map $g : U \times \mathbb{R}^{n-m} \to \mathbb{R}^n$ by $g(x_1, \ldots, x_n) = f(x_1, \ldots, x_k)$. But now $f(U) = g(U \times \{(0, \ldots, 0)\})$. Since $U \times \{(0, \ldots, 0)\}$ is a null set of $\mathbb{R}^n$, we can conclude by proposition 6.5 on *page 1139* that $g(U \times \{(0, \ldots, 0)\})$ has Lebesgue measure zero. ■

This is enough to to prove a trivial case of Sard's theorem.

Corollary (Sard Part 1): Suppose $m < n$, $U \subseteq \mathbb{R}^m$ is an open subset, and $f : U \to \mathbb{R}^n$ is a C^1 map. Then the set of critical values of f is a Lebesgue null set in $\mathbb{R}^n$.

Proof:
The set of critical values of f will be a subset of $f(U)$. But the prior lemma says that $f(U)$ is a Lebesgue null set. So, the critical values of f will also be a Lebesgue null set. ■

Next I need to cover the actual base case for the difficult version of Sard's Theorem. However, first I want to go on a tangent about vector valued integration.

In my math 140b notes I defined integration of an $\mathbb{R}^n$-valued function $f = (f_1, \ldots, f_n)$ by just taking the Riemann integral of f component wise. Meanwhile, on *pages 515-517* I defined a generalization of the Lebesgue integral for $\mathcal{Y}$-valued functions where $\mathcal{Y}$ is a separable Banach space. I'd like to try to merge those approaches.

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $\mathcal{H}$ be a separable Hilbert space with an orthonormal basis $\{u_k\}_{k\in\mathbb{N}}$. If f is $\mathcal{H}$-valued then we will let $\int f \mathrm{d}\mu$ refer to the Bochner integral from *pages 515-517*.

Lemma C: Suppose X is a second countable and regular (i.e. T_3) topological space. Then X is normal (i.e. T_4).

Proof:
Let A and B be disjoint closed subsets of X. For every $x \in A$ we can use regularity to find an open set U_x containing x that is disjoint from B. Next, we can use regularity to find two disjoint open sets U'_x and $\widetilde{U'_x}$ such that $x \in U'_x$ and $(U_x)^{\mathsf{C}} \subseteq \widetilde{U'_x}$. Finally, we can use

regularity one more time to find an open set $U_x'' \subseteq U_x'$ containing x. The result of doing this is that the closure of U_x'' is disjoint from B. After all, $B \subseteq (U_x)^C \subseteq \widetilde{U_x'}$. So, $(\widetilde{U_x'})^C$ is a closed set disjoint from B that contains U_x'. And since $U_x'' \subseteq U_x'$, we can conclude there is a closed set containing U_x'' that is disjoint from B. So, $\overline{U_x''} \cap B = \emptyset$.

By identical reasoning, for every $y \in B$ we can find an open set V_y'' containing y such that $\overline{V_y''} \cap A = \emptyset$.

But now since X is second countable, we know there are sequences $\{x_n\}_{n\in\mathbb{N}} \subseteq A$ and $\{y_n\}_{n\in\mathbb{N}} \subseteq B$ such that $A \subseteq \bigcup_{n\in\mathbb{N}} U_{x_n}''$ and $B \subseteq \bigcup_{n\in\mathbb{N}} V_{y_n}''$.

To finish off, let $\widehat{U}_n := U_{x_n} - \bigcup_{k=1}^n \overline{V_{y_k}''}$ and $\widehat{V}_n := V_{y_n} - \bigcup_{k=1}^n \overline{U_{x_k}''}$. Then for every n we have that $\widehat{U}_n$ and $\widehat{V}_n$ are open subsets of X. Furthermore, we still have that $A \subseteq \bigcup_{n\in\mathbb{N}} \widehat{U}_n$. After all, every $x \in A$ is contained in some U_{x_n} but is not contained in any of the $\overline{V_{y_n}}$. Similarly, we have that $B \subseteq \bigcup_{n\in\mathbb{N}} \widehat{V}_n$.

Let $U := \bigcup_{n\in\mathbb{N}} \widehat{U}_n$ and $V := \bigcup_{n\in\mathbb{N}} \widehat{V}_n$. Then U and V are open neighborhoods of A and B respectively. I also claim U and V are disjoint. After all, suppose $x \in U \cap V$. Then $x \in \widehat{U}_j \cap \widehat{V}_k$ for some j and k. By symmetry, we can assume without loss of generality that $j \leq k$. But now $x \in V_k'' - \bigcup_{j'=1}^k \overline{U_{j'}''}$ and $x \in U_j$. This is a contradiction. ■

Lemma D: Suppose X is a second countable and regular (i.e. T_3) topological space. Then given any closed subset $C \subseteq X$ we can find a decreasing sequence of open sets $\{V_k\}_{k\in\mathbb{N}}$ in X such that $\overline{V_{k+1}} \subseteq V_k$ for all $k \in \mathbb{N}$ and $C = \bigcap_{k\in\mathbb{N}} V_k$.

Proof:
If $C = X$ then this lemma is trivial. Just set $V_k = X$ for all k. So from now on we'll suppose C is a proper subset of X.

Consider any $x \in C$. By regularity we can find open sets U_x and N_x such that $C \subseteq U_x$ and $x \in N_x$, and $U_x \subseteq X - N_x$. But then because X is second countable, we can find a sequence $\{x_k\}_{k\in\mathbb{N}} \subseteq X - C$ such that $X - C = \bigcup_{k\in\mathbb{N}} N_{x_k}$. In turn, we have that $C = \bigcap_{k\in\mathbb{N}}(X - N_{x_k}) \supseteq \bigcap_{k\in\mathbb{N}}(U_{x_k}) \supseteq C$. And now we set $W_k := \bigcap_{j=1}^k U_{x_j}$ for all $k \in \mathbb{N}$. By doing this we get a decreasing sequence of open sets $\{W_k\}_{k\in\mathbb{N}}$ in X such that $C = \bigcap_{k\in\mathbb{N}} W_k$.

Next, we want to modify our new sets to get the desired closure property. Fortunately, by lemma C we know that X is a normal (i.e. T_4) topological space. As a result, given any open set U containing C we can easily find an open set W such that $C \subseteq W \subseteq \overline{W} \subseteq U$.
Namely, by Urysohn's lemma we can find a function $f : X \to [0, 1]$ such that $C \subseteq f^{-1}(\{0\})$ and $U^C \subseteq f^{-1}(\{1\})$ Then just set $W = f^{-1}([0, \frac{1}{2}))$.

Set $V_1 = W_1$. Then for every $k > 1$ we inductively define V_k as follows.
Let Ω_k be an open subset of X such that $C \subseteq \Omega_k \subseteq \overline{\Omega_k} \subseteq V_{k-1}$. Then set $V_k := \Omega_k \cap W_k$. Thus $\overline{V_k} \subseteq \overline{\Omega_k} \subseteq V_{k-1}$. Also $V_k \subseteq W_k$. And since C is contained in W_k and Ω_k, we have that $C \subseteq V_k$.

Because $C = \bigcap_{k\in\mathbb{N}} W_k$, and $C \subseteq V_k \subseteq W_k$ for all k, we can conclude that $C = \bigcap_{k\in\mathbb{N}} V_k$ still. ■

Lemma E: Suppose $(X, \mathcal{M})$ is a measurable space, Y is a regular (i.e T_3) second countable topological space equipped with it's Borel σ-algebra, and $\{f_n : X \to Y\}_{n\in\mathbb{N}}$ is a sequence of functions converging pointwise to $f : X \to Y$. If every f_n is $(\mathcal{M}, \mathcal{B}_Y)$-measurable then so is f.

Proof:
Since the closed subsets of Y generate the Borel σ-algebra, it suffices to show that $f^{-1}(C) \in \mathcal{M}$ for any arbitrary closed sets $C \subseteq Y$. Fortunately, by the prior lemma we can find a decreasing sequence of open sets $\{V_k\}_{k\in\mathbb{N}}$ such that $C = \bigcap_{k\in\mathbb{N}} V_k$ and $\overline{V_{k+1}} \subseteq V_k$ for all $k \in \mathbb{N}$.

Next note that if $f(x) \in C$ then we know by the definition of a pointwise limit that for every set V_k there exists an integer $N \in \mathbb{N}$ such that $f_n(x) \in V_k$ for all $n \geq N$. In other words:

$$f^{-1}(C) \subseteq \bigcap_{k\in\mathbb{N}} \bigcup_{N\in\mathbb{N}} \bigcap_{n\geq N} f_n^{-1}(V_k)$$

On the other hand, if $f(x) \notin C$ then because $C = \bigcap_{k\in\mathbb{N}} \overline{V_k}$ we know that $Y - C = \bigcup_{k\in\mathbb{N}}(Y - \overline{V_k})$ Hence, there exists some $k \in \mathbb{N}$ such that $f(x) \notin \overline{V_k}$. Next, because $Y - \overline{V_k}$ is an open set, we know by the definition of a limit that there exists an integer $N \in \mathbb{N}$ such that $f_n(x) \in X - \overline{V_k}$ for all $n \geq N$. Consequently, for all $N \in \mathbb{N}$ we have that $x \notin \bigcap_{n\geq N} f_n^{-1}(V_k)$. So, $x \notin \bigcup_{N\in\mathbb{N}}(\bigcap_{n\geq N} f_n^{-1}(V_k))$. And this proves that:

$$f^{-1}(C) \supseteq \bigcap_{k\in\mathbb{N}} \bigcup_{N\in\mathbb{N}} \bigcap_{n\geq N} f_n^{-1}(V_k)$$

So, now that we know that $f^{-1}(C) = \bigcap_{k\in\mathbb{N}}(\bigcup_{N\in\mathbb{N}}(\bigcap_{n\geq N} f_n^{-1}(V_k)))$, we can conclude that $f^{-1}(C) \in \mathcal{M}$. After all, f_n being measurable and V_k being open means that $f_n^{-1}(V_k)$ is open for all k and n. Plus, $\mathcal{M}$ is closed under countable unions and intersections. ■

In particular, if $\mathcal{H}$ is a separable Hilbert space, then $\mathcal{H}$ will be second countable and normal (since every metric space is normal and every separable metric space is second countable). So, lemma E applies to $\mathcal{H}$.

Proposition: Let $L_{\mathcal{H}}$ be the space of all $(\mathcal{M}, \mathcal{B}_{\mathcal{H}})$-measurable maps from X to $\mathcal{H}$ (where $\mathcal{B}_{\mathcal{H}}$ is the Borel σ-algebra on $\mathcal{H}$). Then $f : X \to \mathcal{H}$ is in $L_{\mathcal{H}}$ if and only if $f_k(x) \coloneqq \langle f(x), u_k\rangle$ is $(\mathcal{M}, \mathcal{B}_{\mathbb{C}})$-measurable for all $k \in \mathbb{N}$.

$(\Longrightarrow)$
Let $\pi_k : \mathcal{H} \to \mathbb{C}$ be the map $\pi_k(x) = \langle x, u_k\rangle$. This is a continuous linear functional on $\mathcal{H}$ and so it is trivially $(\mathcal{B}_{\mathcal{H}}, \mathcal{B}_{\mathbb{C}})$-measurable. Also note that $f_k = \pi_k \circ f$. So if f is $(\mathcal{M}, \mathcal{B}_{\mathcal{H}})$-measurable then we can conclude that f_k is $(\mathcal{M}, \mathcal{B}_{\mathbb{C}})$-measurable.

$(\Longleftarrow)$
Because $\{u_k\}_{k\in\mathbb{N}}$ is a (complete) orthonormal basis for $\mathcal{H}$, we know that:

$$f(x) = \textstyle\sum_{k\in\mathbb{N}} \langle f(x), u_k\rangle u_k = \sum_{k\in\mathbb{N}} f_k(x) u_k.$$

But now if $m : \mathbb{C} \times \mathcal{H} \to \mathcal{H}$ is the scalar multiplication function, we know that m is continuous and in turn m is $(\mathcal{B}_{\mathbb{C}\times\mathcal{H}}, \mathcal{B}_{\mathcal{H}})$-measurable. But by a propositon at the top of page 14 in my math 240a notes as well as the fact that $\mathbb{C}$ and $\mathcal{H}$ are separable metric spaces, we know that $\mathcal{B}_{\mathbb{C}\times\mathcal{H}} = \mathcal{B}_{\mathbb{C}} \otimes \mathcal{B}_{\mathcal{H}}$. Also, $g_k(x) = (f_k(x), u_k)$ is going to be a $(\mathcal{M}, \mathcal{B}_{\mathbb{C}} \otimes \mathcal{B}_{\mathcal{H}})$-measurable function for every k. So by composing functions, we get that $x \mapsto f_k(x)u_k$ is a $(\mathcal{M}, \mathcal{B}_{\mathcal{H}})$-measurable function for all k.

To finish off, since finite sums of measurable functions from X to $\mathcal{H}$ are always measurable, we can just apply lemma E to get that f is $(\mathcal{M}, \mathcal{B}_{\mathcal{H}})$-measurable. ■

Proposition: Suppose $f \in L^1_{\mathcal{H}} := \{g \in L_{\mathcal{H}} : \int \|g(x)\| \mathrm{d}\mu(x) < \infty\}$. Then $f_k(x) := \langle f(x), u_k \rangle \in L^1(X)$ and $\int f \mathrm{d}\mu = \sum_{k\in\mathbb{N}} (\int f_k \mathrm{d}\mu) u_k$.

Proof:

Because of the prior proposition we know that f_k is measurable for all k. Also note $|f_k(x)| \leq \|u_k\| \cdot \|f(x)\| = \|f(x)\|$. Therefore $\int |f_k| \mathrm{d}\mu \leq \int \|f\| \mathrm{d}\mu < \infty$ and this proves that $f_k \in L^1(X)$ for all $k \in \mathbb{N}$.

Next note that by reusing the reasoning in the prior proposition, we know that the map $x \mapsto f_k(x)u_k$ is a $(\mathcal{M}, \mathcal{B}_{\mathcal{H}})$-measurable map for all k. In turn the partial sums of $f(x) = \sum_{k=1}^{\infty} f_k(x)u_k$ are also measurable. Also note that:

$$\|\sum_{k=1}^{n} f_k(x)u_k\| = (\sum_{k=1}^{n} |f_k(x)|^2)^{1/2} \leq (\sum_{k=0}^{\infty} |f_k(x)|^2)^{1/2} = \|f(x)\|$$

So, $\int \|\sum_{k=1}^{n} f_k(x)u_k\| \mathrm{d}\mu \leq \int \|f(x)\| \mathrm{d}\mu < \infty$, meaning all our partial sums are in $L^1_{\mathcal{H}}$. And now we can use the dominated convergence theorem (with an upper bound of $g(x) = \|f(x)\|$) to say that:

$$\int f(x) \mathrm{d}\mu = \lim_{n\to\infty} \int \sum_{k=1}^{n} f_k(x)u_k \mathrm{d}\mu = \lim_{n\to\infty} \sum_{k=1}^{n} \int f_k(x)u_k \mathrm{d}\mu.$$

By considering the continuous linear map $T : \mathbb{C} \to \mathcal{H}$ given by $c_k \mapsto c_k u_k$, we know from *page 517* that $(\int f_k \mathrm{d}\mu)u_k = T(\int f_k \mathrm{d}\mu) = \int (T \circ f_k) \mathrm{d}\mu = \int f_k(x)u_k \mathrm{d}\mu$. So:

$$\int f(x) \mathrm{d}\mu = \lim_{n\to\infty} \sum_{k=1}^{n} \int f_k(x)u_k \mathrm{d}\mu = \lim_{n\to\infty} \sum_{k=1}^{n} (\int f_k \mathrm{d}\mu)u_k = \sum_{k=1}^{\infty} (\int f_k \mathrm{d}\mu)u_k. \blacksquare$$

For the next bit of my notes I'm learning from a different pdf on the internet (see *citation 13*).

Theorem (Sard Part 2): Suppose $U \subseteq \mathbb{R}^n$ is an open set and $f : U \to \mathbb{R}^n$ is a C^1 map. Then the set of critical values of f is a Borel null set in $\mathbb{R}^n$.

Proof:

Recall from proposition 4.1 on *Pages 1054-1056* that the set of regular points of f is open. It follows that the set C of critical points of f is closed (relative to U). And since $f(C)$ is the set of critical values of f, we can conclude from lemma A that $f(C)$ is a Borel subset of $\mathbb{R}^n$.

Next note that since U is second countable, we can cover U with countably many closed

coordinate cubes. In other words, for all $k \in \mathbb{N}$ we can find $x_k = (x_k^1, \ldots, x_k^n) \in U$ and $\varepsilon_k > 0$ such that $U = \bigcup_{k\in\mathbb{N}}(\prod_{i=1}^n [x_k^i - \varepsilon, x_k^i + \varepsilon])$. And now upon setting $E_k := \prod_{i=1}^n [x_k^i - \varepsilon, x_k^i + \varepsilon]$, it suffices to prove that $f(C \cap E_k)$ is a Borel null set for all k.

As a side note, $f(C \cap E_k)$ will be Borel measurable for identical reasons that $f(C)$ is Borel measurable.

Fortunately, note that by applying a sufficient affine transformation to the domain of f, we can without loss of generality assume $E_k = [0,1]^n =: I^n$. After all, an affine transformation is a C^∞ diffeomorphism and will thus not effect the critical values of f at all. So, we are now left with the slightly simpler task of proving that if $f : I^n \to \mathbb{R}^n$ is a C^1 map and $C \subseteq I^n$ is the set of critical points of f then $f(C)$ is a Borel null set. Two advantages of our simplified task though are as follows.

1. I^n a compact subset of $\mathbb{R}^n$.
2. By proposition C.29 on *page 291*, we know f is Lipschitz on I^n. So let $M \geq 0$ be some constant such that $|f(x) - f(y)| \leq M|x - y|$ for all $x, y \in I^n$.

Next note for any $x, c \in I^n$ that:

$$\begin{aligned} f(x) - f(c) - (\mathrm{D}f(c))(x-c) &= \textstyle\int_0^1 (\mathrm{D}f(c + t(x-c)))(x-c)\mathrm{d}t - (\mathrm{D}f(c))(x-c) \\ &= \textstyle\int_0^1 [\mathrm{D}f(c + t(x-c)) - \mathrm{D}f(c)](x-c)\mathrm{d}t. \end{aligned}$$

In turn we have that:

$$\begin{aligned} |f(x) - f(c) - (\mathrm{D}f(c))(x-c)| &\leq \left(\textstyle\int_0^1 \|\mathrm{D}f(c + t(x-c)) - \mathrm{D}f(c)\|_{\mathrm{op}}\mathrm{d}t\right)|x-c| \\ &\leq \left(\sup_{x' \in B_{|x-c|}(c)} \|\mathrm{D}f(x') - \mathrm{D}f(c)\|_{\mathrm{op}}\right) \cdot |x-c| \end{aligned}$$

Then as f is a C^1 map, we know that $x \mapsto \mathrm{D}f(x)$ is continuous as a map from I^n to $B(\mathbb{R}^n)$. Next, since I^n is compact we know that $x \mapsto \mathrm{D}f(x)$ is uniformly continuous as a map from I^n to $B(\mathbb{R}^n)$. So, for any $\varepsilon > 0$ there exists $\delta > 0$ such that $|x' - c| < \delta$ implies that $\|\mathrm{D}(x') - \mathrm{D}(c)\|_{\mathrm{op}} < \varepsilon$. So in particular, for all $\varepsilon > 0$ we there exists $\delta > 0$ such that for all $x, c \in I^n$ with $|x-c| < \delta$ we have that $|f(x) - f(c) - (\mathrm{D}f(c))(x-c)| \leq \varepsilon|x-c|$.

Yet now suppose $c \in C$. Then $\mathrm{im}(\mathrm{D}f(c))$ is contained in an $(n-1)$-dimensional hyperplane H. And if $|x - c| < \kappa < \delta$ then:

$$\inf_{x' \in H} |f(x) - f(x')| \leq |f(x) - f(c) - \mathrm{D}f(c)(x-c)| \leq \varepsilon\kappa.$$

Similarly, $|f(x) - f(c)| \leq M\kappa$. So, all in all we can say that there is some n-dimensional parallelotope with volume $(2M\kappa)^{n-1} \cdot (2\varepsilon\kappa) = 2^n \kappa^n M^{n-1} \cdot \varepsilon$ containing $f(B_\kappa(c) \cap I^n)$.

To finish off, we can subdivide I^n into h^n many subcubes of side length h^{-1} (where h is a sufficiently large integer such that $h^{-1}\sqrt{n} < \delta$). Note that the side-lengths of any subcube (say Q) are small enough to guarentee that if $c \in Q \cap C$ then $Q \subseteq B_{h^{-1}\sqrt{n}}(c)$. So in turn, by the prior reasoning we'd have that $f(Q) \subseteq f(B_{h^{-1}\sqrt{n}}(c) \cap I^n)$ where the latter is contained in a parallelotope with volume $2^n h^{-n} n^{n/2} M^{n-1} \cdot \varepsilon$. There are only h^n many subcubes though so at most h^n of them actually intersect C. Hence, by just taking the unions of the images of the subcubes that intersect C, we can guarentee that $f(C)$ is contained in a set of measure at most $h^n(2^n h^{-n} n^{n/2} M^{n-1} \cdot \varepsilon) = 2^n n^{n/2} M^{n-1} \cdot \varepsilon$.

But ε can be any positive real number. So, we just take $\varepsilon \to 0$ to get that $\mu(f(C)) = 0$. ■

Finally, note that if $m > n$, $U \subseteq \mathbb{R}^m$ is an open set, and $f : U \to \mathbb{R}^n$ is a C^1 map, then Sard's theorem as was priorly stated does not hold anymore. I don't have the background to explicitely construct a counter example, but here's some vague reasoning for why we wouldn't expect Sard's theorem to hold.

Because the Cantor set $C \subseteq \mathbb{R}$ is a null set, it wouldn't contradict the prior theorem that there exists a C^1 map $f : \mathbb{R} \to \mathbb{R}$ such that the set of critical values of f includes C. (As it turns out, such a function does exist although I don't know how to construct it.)

Next define $g : \mathbb{R}^2 \to \mathbb{R}$ by $g(x, y) = f(x) + f(y)$. Then g is a C^1 map. But also note that $C + C$ is a subset of the critical values of f. And we can fairly easily conclude that $C + C = [0, 2]$.

Why?
Since $C \subseteq [0, 1]$ we know that $C+C \subseteq [0, 2]$. On the other hand, suppose $x \in [0, 2]$ and let $y = \frac{x}{2}$. We write the base 3 representation of y as $\sum_{k=1}^{\infty} a_k \frac{1}{3^k}$. Now if $a_k = 0$ we write $b_k = c_k = 0$. If $a_k = 1$ we write $b_k = 1$ and $c_k = 0$. And if $a_k = 2$ we write $b_k = c_k = 1$. Then upon setting $z_1 = \sum_{k=1}^{\infty} b_k \frac{1}{3^k}$ and $z_2 = \sum_{k=1}^{\infty} c_k \frac{1}{3^k}$, we have that $y = z_1 + z_2$. In turn, $x = 2z_1 + 2z_2$. Also $2z_1, 2z_2 \in C$ since all their ternary-digits are 0 and 2. ■

So, the set of critical values of g would have positive measure.

As it turns out we can get Sard's theorem to still work though by requiring f to have more derivatives.

The standard proof on the internet assumes f is smooth. I would rather not make that assumption but unfortunately each call to the induction hypothesis of this proof would require f to have an unoptimal number derivatives. So I think what I'm going to do is I'm going to just assume $f : U \to \mathbb{R}^n$ (where $U \subseteq \mathbb{R}^m$ is an open set) is a smooth map. After I complete that proof, I will then do a proper accounting of the number of derivatives that were actually used by the proof.

Theorem (Sard Part 3): Suppose $m > n$, $U \subseteq \mathbb{R}^m$ is an open subset, and $f : U \to \mathbb{R}^n$ is a smooth map. Then the set of critical values of f is a Borel measurable set in $\mathbb{R}^n$ with Lebesgue measure 0.

Proof:
We'll do induction on m, noting that Sard parts 1 and 2 already proved the base case of when $m = 1$. Also, it suffices to assume that the domain of f is $\overline{I^m}$ where $I^m = (0, 1)^m$, and we only need to prove that $f(C \cap I^m)$ is a null set. This is because we can cover U with countably many open cubes whose closures are also contained in U. Then each closed cube can be diffeomorphically mapped to $[0, 1]^m$ sending the interior to $(0, 1)^m$. Also importantly, diffeomorphisms do not change which points are critical values.

Let $k := \lceil \frac{m}{n} \rceil$. Then given any $1 \leq j \leq k$ let:

$$C_j := \{x \in I^m : \partial^\alpha f(x) = 0 \text{ for all multi-indices } \alpha \text{ such that } |\alpha| \leq j\}.$$

Also let C be the set of critical points of f in I^m. We know C and all the C_j are closed subsets of points in I^m (either because the set of critical values is always closed or because the preimage under a continuous function of a closed set is closed). Also $C \supseteq C_1 \supseteq \cdots \supseteq C_k$.

Note that $C - C_1$ and $C_i - C_{i+1}$ are open subsets of C or C_i respectively where both of those are second countable (due to being subspaces of a second countable space) and locally compact Hausdorff (due to being closed subspace of a LCH space). So by lemma A on *pages 1141-1142* we can conclude that $f(C-C_1)$ and $f(C_i-C_{i+1})$ are Borel measurable for all $i < k$. Also because C_k is a closed subset of I^m, we can similarly conclude that $f(C_k)$ is Borel measurable.

Now clearly $f(C) = f(C_k) \cup (f(C) - f(C_1)) \cup \bigcup_{i=1}^{k-1}(f(C_i) - f(C_{i+1}))$. Our strategy will be to show that each individual set in this union is a null set.

- Part I: (Showing that $f(C_k)$ is a null set)
  Given any multidex α and any $x \in \mathbb{R}^m$, recall from my math 240c notes how ∂^α, x^α, $\alpha!$, and $|\alpha|$ are defined.

Lemma (Another Version of Taylor's Theorem): Suppose $U \subseteq \mathbb{R}^n$ is open and $g \in C^{k+1}(U, \mathbb{R})$. Then for any $c \in U$ and convex subset $W \subseteq U$ containing c, we have for all $x \in W$ that:

$$g(x) = \sum_{|\alpha| \leq k} \frac{\partial^\alpha g(c)}{\alpha!}(x-c)^\alpha + R_k(x)$$

where $R_k(x) = \sum_{|\alpha|=k+1} \frac{k+1}{\alpha!}(x-c)^\alpha \int_0^1 (1-t)^k \partial^\alpha g(c + t(x-c))\mathrm{d}t$.

Proof:
Let $S_\ell := \{1, \ldots, n\}^\ell$. Also let us write $x = (x_1, \ldots, x_n)$ and $c = (c_1, \ldots, c_n)$. Then I'm going to prove that $g(x) = \widetilde{P}_k(x) + \widetilde{R}_k(x)$ where:

- $\widetilde{P}_k(x) := g(c) + \sum_{\ell=1}^{k} \left(\frac{1}{\ell!} \sum_{(i_1,\ldots,i_\ell) \in S_\ell} \left(\prod_{j=1}^{\ell} (x_{i_j} - c_{i_j}) \right) \frac{\partial^\ell g}{\partial x_{i_1} \cdots \partial x_{i_\ell}}(c) \right)$,
- $\widetilde{R}_k(x) := \frac{1}{k!} \sum_{(i_1,\ldots,i_{k+1}) \in S_{k+1}} \left(\prod_{j=1}^{k+1} (x_{i_j} - c_{i_j}) \right) \int_0^1 (1-t)^k \frac{\partial^{k+1} g}{\partial x_{i_1} \cdots \partial x_{i_{k+1}}}(c + t(x-c))\mathrm{d}t$.

Note that if $|\alpha| = \ell$ then there are $\frac{\ell!}{\alpha!}$ many orders one can differentiate in to get $\partial^\alpha g$. So by collecting together like terms, we can see that $\widetilde{R}_k = R_k$ and $\widetilde{P}_k = P_k$.

To start off, note that if $k = 0$ then I've already shown this result on *page 1035*.
So now we proceed by induction on k, supposing we've already shown that $g(x) = \widetilde{P}_{k-1}(x) + \widetilde{R}_{k-1}(x)$

We get via integration by parts that:

$$\int_0^1 (1-t)^{k-1} \frac{\partial^k g}{\partial_{x_{i_1}} \cdots \partial_{x_{i_k}}}(c+t(x-c))\mathrm{d}t$$
$$= \left[\frac{-(1-t)^k}{k} \frac{\partial^k g}{\partial_{x_{i_1}} \cdots \partial_{x_{i_k}}}(c+t(x-c)) \right]_{t=0}^{t=1} + \int_0^1 \frac{(1-t)^k}{k} \cdot \frac{\mathrm{d}}{\mathrm{d}s}\left[\frac{\partial^k g}{\partial_{x_{i_1}} \cdots \partial_{x_{i_k}}}(c+s(x-c)) \right](t)\mathrm{d}t$$
$$= \frac{1}{k} \frac{\partial^k g}{\partial_{x_{i_1}} \cdots \partial_{x_{i_k}}}(c) + \sum_{j=1}^{n} \frac{(x_j - c_j)}{k} \int_0^1 (1-t)^k \frac{\partial^{k+1} g}{\partial_{x_{i_1}} \cdots \partial_{x_{i_k}} \partial_{x_j}}(c+t(x-c))\mathrm{d}t$$

And now we shift the term without the integral into being part of $\widetilde{P_k}$ and we let the term with the integral be in $\widetilde{R}_k$. ■

Corollary: Suppose $U \subseteq \mathbb{R}^n$ is open and $g \in C^{k+1}(U, \mathbb{R})$. Given any $c \in U$ let P_k and R_k be from the prior lemma. Then suppose $W \subseteq U$ is a convex subset such that $c \in W$ and all $(k+1)$th partial derivatives of g are bounded in absolute value on W by a constant M. Then for all $x \in W$ we have that:

$$|f(x) - P_k(x)| = |R_k(x)| \leq \frac{n^{k+1} M}{(k+1)!} |x-c|^{k+1}.$$

Proof:

Note that:

$$\frac{k+1}{\alpha!}(x-c)^\alpha \int_0^1 (1-t)^k \partial^\alpha g(c+t(x-c))\mathrm{d}t \leq \frac{M(k+1)}{\alpha!}|x-c|^{|\alpha|} \int_0^1 (1-t)^k \mathrm{d}t$$
$$\leq \frac{M}{\alpha!}|x-c|^{k+1}$$

So, we can say that $|R_k(x)| \leq M|x-c|^{k+1} \sum_{|\alpha|=k+1} \frac{1}{\alpha!}$.

Also note that: $\sum_{|\alpha|=k+1} \frac{1}{\alpha!} = \frac{1}{(k+1)!} \sum_{|\alpha|=k+1} \frac{(k+1)!}{\alpha!} = \frac{n^{k+1}}{(k+1)!}$. ■

Corollary 2: Suppose $U \subseteq \mathbb{R}^m$ is open and $g \in C^k(U, \mathbb{R}^n)$. Also suppose $K \subseteq U$ is a compact convex subset. Let $P_k(x,c) = \sum_{|\alpha| \leq k} \frac{(x-c)^\alpha}{\alpha!} \partial^\alpha g(c)$. Then for any $\varepsilon > 0$ we can find $\delta > 0$ such that $|x-c| < \delta \implies |g(x) - P(x,c)| \leq M\varepsilon |x-c|^k$. (Here M does not depend on ε and δ.)

Proof:

Write $g(x) = (g_1(x), \ldots, g_n(x))$. By the prior lemma, we know that:

$$g_i(x) - \sum_{|\alpha|<k} \frac{\partial^\alpha g_i(c)}{\alpha!}(x-c)^\alpha = \sum_{|\alpha|=k} \frac{k}{\alpha!}(x-c)^\alpha \int_0^1 (1-t)^{k-1} \partial^\alpha g_i(c+t(x-c))\mathrm{d}t$$

But now note:

$$\frac{k}{\alpha!}(x-c)^\alpha \int_0^1 (1-t)^{k-1} \partial^\alpha g_i(c+t(x-c))\mathrm{d}t - \frac{1}{\alpha!}(x-c)^\alpha \partial^\alpha g_i(c)$$
$$= \frac{k}{\alpha!}(x-c)^\alpha \int_0^1 (1-t)^{k-1} \partial^\alpha g_i(c+t(x-c))\mathrm{d}t - \frac{k}{\alpha!}(x-c)^\alpha \int_0^1 (1-t)^{k-1} \partial^\alpha g_i(c)\mathrm{d}t$$
$$= \frac{k}{\alpha!}(x-c)^\alpha \int_0^1 (1-t)^{k-1} \left[\partial^\alpha g_i(c+t(x-c)) - \partial^\alpha g_i(c) \right] \mathrm{d}t$$

So by subtracting all needed terms from both sides, we can get that:

$$g_i(x) - \textstyle\sum_{|\alpha| \leq k} \frac{(x-c)^\alpha}{\alpha!} \partial^\alpha g_i(c) = \sum_{|\alpha|=k} \frac{k}{\alpha!}(x-c)^\alpha \int_0^1 (1-t)^{k-1} \left[\partial^\alpha g_i(c+t(x-c)) - \partial^\alpha g_i(c) \right] \mathrm{d}t.$$

And by putting all the vector components together, we get that:

$$g(x) - P(x,c) = \sum_{|\alpha|=k} \frac{k}{\alpha!}(x-c)^\alpha \int_0^1 (1-t)^{k-1} \left[\partial^\alpha g(c+t(x-c)) - \partial^\alpha g(c) \right] \mathrm{d}t.$$

But now since $\partial^\alpha g$ is a continuous function on K for all $|\alpha| = k$, we can guarentee that each of them are uniformly continuous. So, given any $\varepsilon > 0$ we can find $0 < \delta < \varepsilon$ such that $|x - c| < \delta \implies |\partial^\alpha g(c + t(x-c)) - \partial^\alpha g(c)| < \varepsilon$ for all multi-indices α with $|\alpha| = k$ and all $t \in [0,1]$. And now:

$$\begin{aligned}|g(x) - P(x,c)| &\leq \sum_{|\alpha|=k} \tfrac{k}{\alpha!}|(x-c)^\alpha| \cdot \left|\int_0^1 (1-t)^{k-1}\left[\partial^\alpha g(c+t(x-c)) - \partial^\alpha g(c)\right] \mathrm{d}t\right| \\ &\leq \sum_{|\alpha|=k} \tfrac{k}{\alpha!}|x-c|^k \cdot \int_0^1 (1-t)^{k-1}\left|\partial^\alpha g(c+t(x-c)) - \partial^\alpha g(c)\right| \mathrm{d}t \\ &\leq \sum_{|\alpha|=k} \tfrac{k}{\alpha!}|x-c|^k \cdot \varepsilon \int_0^1 (1-t)^{k-1} \mathrm{d}t = \sum_{|\alpha|=k} \tfrac{\varepsilon}{\alpha!}|x-c|^k = \tfrac{m^k \varepsilon}{k!}|x-c|^k\end{aligned}$$

So upon setting $M = \frac{m^k}{k!}$ we have that $|g(x) - P(x,c)| \leq M\varepsilon|x-c|^k$.

By the prior corollary, for any $\varepsilon > 0$ we can find some $0 < \delta < \varepsilon$ such that for all $c \in C_k$ and $x \in \overline{I^m}$, $|x-c| < \delta \implies |f(x) - f(c)| \leq M\varepsilon|x-c|^k$ (where $M = \frac{m^k}{k!}$ is independent of ε).

But now subdivide $\overline{I^m}$ into h^m many subcubes with sidelength h^{-1} where h is an integer with $h^{-1}\sqrt{m} < \delta$. Then for any subcube Q, can guarentee that if $c \in Q \cap C$ then $Q \subseteq B_{h^{-1}\sqrt{m}}(c)$. So by the prior line, this in turn guarentees that $f(Q) \subseteq f(B_{h^{-1}\sqrt{m}}(c) \cap I^m) \subseteq B_{M\varepsilon h^{-k} m^{k/2}}(f(c))$. So $f(Q)$ is contained in a set of measure $(M\varepsilon h^{-k} m^{k/2})^n \cdot \mu(B_1(0))$.

Since there are h^m many possible cubes that can intercept C_k, we can guarentee that:

$$\mu(f(C_k)) \leq h^m (M\varepsilon h^{-k} m^{k/2})^n \cdot \mu(B_1(0)) = \varepsilon^n \cdot M m^{kn/2} \mu(B_1(0)) h^{m-kn}$$

But now as $k \geq \frac{m}{n}$, we know that $m - kn \leq 0$. And since $h \geq 1$, we can thus conclude that $\mu(f(C_k)) \leq \varepsilon^n \cdot M m^{kn/2} \mu(B_1(0))$. Finally, by taking $\varepsilon \to 0$ we can thus conclude that $\mu(f(C_k)) = 0$.

> As a side note, the paper I was following uses the first corollary instead of the second corollary I made in the prior aside in order to say that this reasoning works if $k > \frac{m}{n} - 1$. That said, using the first corollary requires f to have $k+1$ many derivatives and I'd rather use a result only requiring f to have $k = \lceil \frac{m}{n} \rceil$ many derivatives.

- Part II: (Showing that $f(C - C_1)$ is a null set)
  If $n = 1$ then $C = C_1$. So trivially $f(C - C_1)$ would be a null set. And thus without loss of generality we can assume $n \geq 1$.

  Note that $c \in I^m$ is in C iff $\mathrm{d}f_c$ is not surjective. Meanwhile, $c \in I^m$ is in C_1 iff $\mathrm{d}f_c$ is the zero map. We already know that these properties behave nicely with respect to diffeomorphisms of either the domain or codomain. After all, if ψ and φ are diffeomorphisms then $\mathrm{d}f_p$ is surjective / a zero map iff $\mathrm{d}\psi_{f(p)} \circ \mathrm{d}f_p \circ \mathrm{d}\varphi^{-1}_{\varphi(p)}$ is surjective / a zero map. Also diffeomorphisms preserve null sets. So $f(C - C_1)$ would be null iff $\psi(f(C - C_1))$ is null.

Now suppose $c \in C - C_1$ is any point. I claim there is a neighborhood $V \subseteq I^m$ of c such that $\mu(f((C - C_1) \cap V)) = 0$

To see why, note that one consequence of the first paragraph of this part is that by permuting the standard basis vectors of the domain and codomain of f, we can assume without loss of generality that $f = (f_1, \ldots, f_n)$ and that $\frac{\partial f_1}{\partial x_1}(c) \neq 0$.

Next, we define $\varphi : I^m \to \mathbb{R}^m$ by:

$$\varphi(x_1, \ldots, x_m) := (f_1(x_1, \ldots, x_m), x_2, \ldots, x_m).$$

We can easily see that $\mathrm{D}\varphi(c) = \begin{bmatrix} a & * \\ 0 & \mathbf{I}_{m-1} \end{bmatrix}$ where $a \neq 0$.

So, $\mathrm{D}\varphi(c)$ is invertible. And by the inverse function theorem, we can find an open set $V \subseteq I^m$ containing c such that φ is a C^k-diffeomorphism from V (which contains c) to some open set $\widehat{V} \subseteq \mathbb{R}^m$.

Let $\widetilde{f} = f \circ \varphi^{-1}$ and let $\widetilde{C}$ be the set of critical points of $\widetilde{f}$ in $\widehat{V}$. Importantly, we know that $\varphi((C - C_1) \cap V) = \varphi(C \cap V) = \widetilde{C}$. After all, $V \cap C_1 = \emptyset$ since $\nabla f_1 \neq 0$ must be nonzero for all $x \in V$. Otherwise, $\mathrm{D}\varphi$ would not be full rank for some $x \in V$ and that would contradict that φ is a diffeomorphism.

Now, we just need to show that $\widetilde{f}(\widetilde{C}) = f((C - C_1) \cap V)$ is a null set.

Fortunately, note that:

$$\widetilde{f}(t, x_2, \ldots, x_m) = (t, f_2 \circ \varphi^{-1}(t, x_2, \ldots, x_m), \ldots, f_2 \circ \varphi^{-1}(t, x_2, \ldots, x_m)).$$

In turn, the image of any slice $\widehat{V}_t := \{(x_1, \ldots, x_m) \in \widetilde{V} : x_1 = t\}$ is contained in the hyperplane $\{(y_1, \ldots, y_n) \in \mathbb{R}^n : y_1 = t\}$. So consider any fixed $t \in \mathbb{R}$ and let $\widetilde{f}_t$ be the function defined by:

$$\widetilde{f}_t(x_2, \ldots, x_m) = (f_2 \circ \varphi^{-1}(t, x_2, \ldots, x_m), \ldots, f_2 \circ \varphi^{-1}(t, x_2, \ldots, x_m)).$$

$\widetilde{f}_t$ is a C^k map from an open subset $\mathbb{R}^{m-1}$ to $\mathbb{R}^{n-1}$. Therefore, by induction we can conclude that the set of critical values of $\widetilde{f}_t$ is a null set. But also note that $(y_2, \ldots, y_n)$ is a critical value of $\widetilde{f}_t$ iff there exists a point $(x_2, \ldots, x_m)$ in the domain of $\widetilde{f}_t$ that is a critical point. In turn that happens iff $(t, x_2, \ldots, x_m)$ is a critical point of $\widetilde{f}$, and that happens iff $(t, y_2, \ldots, y_n)$ is a critical value of $\widetilde{f}$.

So, each slice of $\widetilde{f}(\widetilde{C})$ where $y_1 = t$ has $(n-1)$-dimensional measure zero. This is enough by Fubini's theorem to conclude that $\widetilde{f}(\widetilde{C})$ is Lebesgue measurable with measure zero.

By covering I^m with countably many open sets $\{V_j\}_{j \in \mathbb{N}}$ such that $f((C - C_1) \cap V_j)$ is a Lebesgue null set for each $j \in \mathbb{N}$ we can now conclude that $f(C - C_1)$ is a Lebesgue null set.

As a side note, if $n > 1$ then we can see that this reasoning required f to be as many times continuously differentiable as whatever number of derivatives are needed for Sard's theorem to apply to a function $f : \mathbb{R}^{m-1} \to \mathbb{R}^{n-1}$.

Meanwhile, if $n = 1$ then this step doesn't require any derivatives.

- Part III: (Showing that $f(C_i - C_{i+1})$ is a null set for each $1 \leq i < k$)
  Like in part II, it suffices to show that any $c \in C_i - C_{i+1}$ has a neighborhood $V \subseteq I^m$ such that $\mu(f((C_i - C_{i+1}) \cap V)) = 0$. After all, we can then cover I^m with countably many such neighborhoods in order to prove that $\mu(f(C_i - C_{i+1})) = 0$.

  Fortunately, for any $c \in C_i - C_{i-1}$ we can find an ith order derivative $h : I^m \to \mathbb{R}$ of some component function of f such that $h(c) = 0$ and $\nabla h(c) \neq 0$. By continuity, we can find an open set $W \subseteq I^m$ with $\nabla h \neq 0$ everywhere on W. And now it follows that $S := (h|_W)^{-1}(\{0\})$ is a regular level set of $h|_W$. So, S is an embedded $(m-1)$-dimensional C^∞ submanifold of W (and in turn of I^m).
  As a side note, if f were only a C^q map then S would be an embedded C^{q-i} submanifold.

  Next, we can find a smooth (or C^{q-i}) parametrization $X : \widehat{V} \to I^m$ such that $\widehat{V}$ is an open subset of $\mathbb{R}^{m-1}$ (since S has no boundary), $c \in X(\widehat{V})$ and $X(\widehat{V})$ is an open subset of S. Since S is an embedded submanifold of W, we in turn know there is an open set $V \subseteq W \subseteq I^m$ such that $X(\widehat{V}) = V \cap S$. And now we define $g := f \circ X$.

  Importantly, $C_i \cap V \subseteq C_i \cap W \subseteq S$. And similarly, $C_{i+1} \cap V \subseteq C_{i+1} \cap W = \emptyset$. So, we can conclude that $(C_i - C_{i+1}) \cap V = C_i \cap V \subseteq S$.

  Also let D be the set of critical points of g. We can see that $C \cap (V \cap S) \subseteq X(D)$. After all, if $q \in C \cap (V \cap S)$ then we know that there exists p such that $X(p) = q$. But now if $\mathrm{d}f_{X(p)}$ is not surjective then neither will $\mathrm{d}f_{(X(p))} \circ \mathrm{d}X_p = \mathrm{d}g_p$ be surjective. So, $p \in D$, meaning $q = X(p) \in X(D)$.

  Putting the prior two groups of text together:
  $$(C_i - C_{i+1}) \cap V = C_i \cap V = C_i \cap V \cap S \subseteq C \cap V \cap S \subseteq X(D).$$

  So, it now suffices to show that $f(X(D)) = g(D)$ is a null set. Fortunately, we can just apply our induction hypothesis to this case. ■
  In particular, this means that this step required f to be $k - 1 + S(m-1, n)$ times continuously differentiable where $S(m-1, n)$ is the number of times a function $g : \mathbb{R}^{m-1} \to \mathbb{R}^n$ needs to be continuous differentiable for Sard's theorem to apply.

9/23/2026

The number of derivatives of f that the prior proof uses is not optimal. That said, I will still give an accounting of how many derivatives were needed at each step. I think its a good exercise.

So let $S(m,n)$ denote the number of times continuously differentiable that we need f to be in order to prove Sard's theorem for a function from $f : U \subseteq \mathbb{R}^m \to \mathbb{R}^n$ using the prior reasoning. Importantly, we have the following recursive relations:

- $S(m,1) = \max(m-1+S(m-1,1), m)$,
- $S(m,n) = \max(S(m-1,n-1), \lceil \frac{m}{n} \rceil - 1 + S(m-1,1), \lceil \frac{m}{n} \rceil)$ if $m > n > 1$,
- $S(n,n) = 1$.

I'm going to start by solving for $s_m \coloneqq S(m,1)$ for all m.

To start, we can easily see that $S(m-1,1) \geq m-1 \geq 1$ for all $m \geq 2$. Hence, we can conclude that $s_m = s_{m-1} + m - 1$ for all m. Also since $s_1 = 1$, we can conclude $s_2 = 2$. And it will be convenient to let $s_0 = 1$. That way $s_1 = s_0 + 1 - 1$. And now note that:

$$\begin{aligned} s_m - s_{m-1} &= (s_{m-1} + m - 1) - (s_{m-2} + m - 2) \\ \implies s_m &= 2s_{m-1} - s_{m-2} + 1 \\ \implies s_m - s_{m-1} &= (2s_{m-1} - s_{m-2} + 1) - (2s_{m-2} - s_{m-3} + 1) \\ \implies s_m &= 3s_{m-1} - 3s_{m-2} + s_{m-3} \end{aligned}$$

Let $A(x) \coloneqq \sum_{m \geq 0} s_m x^m$ be a formal power series (see my math 188 notes). Then note that:

$$\begin{aligned} A(x) &= s_0 + s_1 x + s_2 x^2 + \sum_{m \geq 3} (3s_{m-1} - 3s_{m-2} + s_{m-3}) x^m \\ &= s_0 + s_1 x + s_2 x^2 + 3x(A(x) - s_1 x - s_0) - 3x^2(A(x) - s_0) + x^3 A(x) \end{aligned}$$

In other words, $(1 - 3x + 3x^2 - x^3)A(x) = s_0 + (s_1 - 3s_0)x + (s_2 - 3s_1 + 3s_0)x^2$. And when we fill in what s_0, s_1, and s_2 are, we get that:

$$A(x) = \frac{1-2x+2x^2}{1-3x+3x^2-x^3} = \frac{1-2x+2x^2}{(1-x)^3} = (1 - 2x + 2x^2) \sum_{m \geq 0} \binom{m+2}{m} x^m$$

It follows that if $m \geq 2$ then:

$$\begin{aligned} s_m &= \tbinom{m+2}{m} - 2\tbinom{m+1}{m-1} + 2\tbinom{m}{m-2}. \\ &= \tfrac{(m+2)(m+1)}{2} - 2\left(\tfrac{m(m+1)}{2}\right) + 2\left(\tfrac{m(m-1)}{2}\right) \\ &= \tfrac{1}{2}(m^2 + 3m + 2) - m^2 - m + m^2 - m = \tfrac{1}{2}m^2 - \tfrac{1}{2}m + 1. \end{aligned}$$

Also we can manually verify that this formula holds when $m = 1$.

With that we've proven that $S(m,1) = \frac{1}{2}m^2 - \frac{1}{2}m + 1$ for all $m \geq 1$.

Next note that since we only care here about the case where $m \geq n$, we will always have that $\lceil \frac{m}{n} \rceil \geq 1$. So, we can actually conclude that if $m > n > 1$ then:

$$S(m,n) = \max(S(m-1,n-1), \lceil \tfrac{m}{n} \rceil - 1 + S(m-1,n)).$$

I shall prove by induction though that $S(m-1,n-1) \geq \lceil \frac{m}{n} \rceil - 1 + S(m-1,n)$ and in turn $S(n+k,n) = S(1+k,1)$ for all $n \geq 1$ and $k \geq 0$.

I shall have two layers of induction. For the first layer, note that trivially if $n = 1$ then for any $k \geq 0$ we have that $S(n+k,n) = S(1+k,1)$. So, I shall now consider a fixed $n > 1$ and assume I've already proven that $S(n_0+k,n_0) = S(1+k,1)$ for all $k \geq 0$ and $1 \leq n_0 < n$.

To prove that $S(n+k,n) = S(1+k,1)$ for all $k \geq 0$, I shall do a second layer of induction on k. For my base case, just note that $S(n+0,n) = 1 = S(1+0,1)$. So, I can now proceed by induction on k.

In particular, by our two layers of induction we know that:

$$\begin{aligned} S(n+k,n) &= \max(S(n+k-1,n-1), \lceil \tfrac{n+k}{n} \rceil - 1 + S(n+k-1,n)) \\ &= \max(S(1+k,1), \lceil \tfrac{n+k}{n} \rceil - 1 + S(k,1)) \\ &= \max(\tfrac{1}{2}(k+1)^2 - \tfrac{1}{2}(k+1) + 1, \lceil \tfrac{n+k}{n} \rceil - 1 + \tfrac{1}{2}k^2 - \tfrac{1}{2}k + 1) \end{aligned}$$

Thus we are left now with the task of showing that:

$$\tfrac{1}{2}(k+1)^2 - \tfrac{1}{2}(k+1) + 1 \geq \lceil \tfrac{n+k}{n} \rceil - 1 + \tfrac{1}{2}k^2 - \tfrac{1}{2}k + 1$$

Fortunately, $\tfrac{1}{2}(k+1)^2 - \tfrac{1}{2}(k+1) + 1 = \tfrac{1}{2}k^2 + \tfrac{1}{2}k + 1$. Therefore, by rearranging terms we can equivalently say that we need to prove that:

$$k + 1 \geq \lceil \tfrac{n+k}{n} \rceil$$

Since $n \geq 1$ though, we know that $(1) + 1 = 2 = \lceil \tfrac{(1)+n}{n} \rceil$. Also note that $\lceil \tfrac{n+k}{n} \rceil \leq \tfrac{n+k}{n} + 1 = 2 + \tfrac{k}{n}$. Since $n > 1$, we can thus say that $\lceil \tfrac{n+k}{n} \rceil \leq 2 + \tfrac{1}{2}k$. And when $k \geq 2$ we have that $2 + \tfrac{1}{2}k \leq k + 1$. So, we are done.

In other words, in the prior theorem we managed to prove that if $f : U \to \mathbb{R}^n$ is a function where $U \subseteq \mathbb{R}^m$ is an open set, then f being a C^q map where:

$$q \geq \tfrac{1}{2}(m-n+1)^2 - \tfrac{1}{2}(m-n+1) + 1 = \tfrac{1}{2}(m-n)^2 + \tfrac{1}{2}(m-n) + 1$$

guarentees that the set of critical values of f has measure zero in $\mathbb{R}^n$.

As a side note, this is not close to being the optimal lower bound on the number of derivatives of f. In fact in the original proof that Sard came up with, he shows that the optimal low bound on q is actually that $q \geq m - n + 1$. His proof is longer though and not in most books since most books only consider the smooth case.

To finish off, this topic before school starts again, we can also generalize each part of Sard's theorem to applying to C^k maps $F : M \to N$ between C^r manifolds M and N. Because I don't want to rewrite this theorem after hypothetically proving the bound Sard proved, I'm just going to denote $\lambda(k)$ to be the number of times continuously differentiable we need a function $f : \mathbb{R}^{n+k} \to \mathbb{R}^n$ to be to prove that Sard's theorem holds for f.

For the time being, I'll suppose that $\lambda(k) = \begin{cases} \tfrac{1}{2}k^2 + \tfrac{1}{2}k + 1 & \text{if } k \geq 0 \\ 1 & \text{if } k \leq 0 \end{cases}$.

However, we can all keep in mind that $\lambda(k)$ should actually equal $\max(k+1, 1)$.

See *page* ____ for a proof of this better bound if I actually ever show it.

Sard's Theorem: Suppose M and N are C^r manifolds of dimension m and n respectively, and let $F : M \to N$ be a C^q map where $\lambda(m-n) \leq q \leq r$. Then the set of critical values of F is a set of measure zero in N.

Proof:
By the second countability of M and the smoothness of F, we can find a countable collection of smooth charts $\{(U_n, \varphi_n)\}_{n\in\mathbb{N}}$ on M and $\{(V_n, \psi_n)\}_{n\in\mathbb{N}}$ on N such that $M = \bigcup_{n\in\mathbb{N}} U_n$ and $F(U_n) \subseteq V_n$ for all n. And now if C is the set of critical points of F in M, it suffices to show that $F(C \cap U_n)$ has measure zero in N for all $n \in \mathbb{N}$.

Fortunately, note that $p \in U_n$ is contained in C iff $\mathrm{D}(\psi \circ F \circ \varphi_n^{-1})(\varphi_n(p))$ is not a surjective linear map. So if we let D be the set of critical points of $\widehat{F} := \psi_n \circ F \circ \varphi_n^{-1}$, then we have that $D = \varphi_n(C \cap U_n)$.

Next, note that $F(C \cap U_n) = \psi_n^{-1} \circ (\psi_n \circ F \circ \varphi_n^{-1})(\varphi_n(C \cap U_n)) = \psi_n^{-1} \circ \widehat{F}(D)$. Then because C^1 maps send null subsets of n-dimensional manifolds to null subsets of n-dimensional manifolds, we can conclude that if $\widehat{F}(D)$ is a null set then so is $F(C \cap U_n)$. Yet $\widehat{F}$ is a map that can be extended to being a C^q map from an open subset of $\mathbb{R}^m$ into $\mathbb{R}^n$. Therefore, we can apply Sard's theorem in the Euclidean case to $\widehat{F}$ to conclude that $\widehat{F}(D)$ is a Lebesgue null set. ■

I guess I'll also mention that if we allow ourselves to consider 0-dimensional manifolds (i.e. countable sets equipped with the discrete topology), then Sard's theorem can still be said to trivially apply. After all, suppose $F : M \to N$ is a C^1 map between manifolds. If M is 0-dimensional and N isn't, then $F(M)$ is a countable subset and in turn has measure zero. Meanwhile, if N is 0-dimensional and M isn't, then trivially $\mathrm{d}F_p$ is surjective for all p. (Here I'm considering T_qN to just be the zero vector space). Finally, if both M and N are zero dimensional then $\mathrm{d}F_p$ is trivially a surjection. So, F would have no critical points or values.

9/24/2026

I'm gonna study a bit more of John Lee smooth manifolds today but I will start to get distracted now that school has started again.

Corollary 6.11: Suppose M and N are C^r manifolds such that $\dim(M) < \dim(N)$. Then if $F : M \to N$ is a C^1 map we know that $F(M)$ is a set of measure zero in N.
Proof:
Every point in $F(M)$ is trivially a critical value because every point of M is trivially a critical point of F. So Sard's theorem tells us that $F(M)$ is a set of measure zero in N. ■

Corollary 6.12: Suppose M is a C^r manifold and $S \subseteq M$ is an immersed C^r submanifold such that $\dim(S) < \dim(M)$. Then S is a set of measure zero in M.
Proof:
Apply corollary 6.11 to the inclusion map $\iota : S \hookrightarrow M$. ■

Another application of Sard's theorem is that it lets us prove that we can embed any n-dimensional C^∞ manifold into $\mathbb{R}^{2n+1}$.

Recall from *pages 131-132* that $\mathbb{RP}^N$ is a topological manifold. In particular, let $\pi : \mathbb{R}^{N+1} - \{0\} \to \mathbb{RP}^N$ be the quotient map, and for every $1 \leq i \leq N+1$ let $U_i = \pi(\{(x_1, \ldots, x_{N+1}) : x_i \neq 0\})$. Also let $\varphi_i : U_i \to \mathbb{R}^N$ be the homeomorphism given by $\varphi_i([x_1, \ldots, x_{n+1}]) := (\frac{x_1}{x_i}, \ldots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \ldots, \frac{x_{N+1}}{x_i})$.

Lemma: All the φ_i are C^∞-compatible with each other. Also if $\mathbb{RP}^N$ is equipped with the C^∞ structure generated by the charts (U_i, φ_i), then $\pi : \mathbb{R}^{N+1} - \{0\} \to \mathbb{PR}^N$ is a C^∞ submersion.

To see why, note that if $(x_1, \ldots, x_{N+1}) \in \varphi_i(U_i \cap U_j)$ then we have that:

$$\begin{aligned}\varphi_j \circ \varphi_i^{-1}(x_1, \ldots, x_{N+1}) &= \varphi_j([x_1, \ldots, x_{i-1}, 1, x_{i+1}, \ldots, x_n]) \\ &= (\tfrac{x_1}{x_j}, \ldots, \tfrac{x_{i-1}}{x_j}, \tfrac{1}{x_j}, \tfrac{x_{i+1}}{x_j}, \ldots, \tfrac{x_{j-1}}{x_j}, \tfrac{x_{j+1}}{x_j}, \ldots, \tfrac{x_{N+1}}{x_j}).\end{aligned}$$

But now $[(x_1, \ldots, x_{N+1})] \in U_i \cap U_j$ iff $x_i \neq 0$ and $x_j \neq 0$. Therefore, $\varphi_j \circ \varphi_i^{-1}$ is a smooth map. And since i and j were arbitrary, this proves the first claim.

As for the second claim, note that $\mathbb{R}^{N+1} - \{0\}$ is covered by open sets of the form $\widetilde{U}_i := \{(x_1, \ldots, x_{N+1}) : x_i \neq 0\}$. Also, $\pi(\widetilde{U}_i) \subseteq U_i$. So, it suffices to say that $\varphi \circ \pi$ is a smooth submersion from $\widetilde{U}_i \subseteq \mathbb{R}^{N+1}$ onto $\mathbb{R}^n$. Fortunately,

$$\varphi \circ \pi(x_1, \ldots, x_{N+1}) = (\tfrac{x_1}{x_i}, \ldots, \tfrac{x_{i-1}}{x_i}, \tfrac{x_{i+1}}{x_i}, \ldots, \tfrac{x_{N+1}}{x_i}).$$

This is a smooth map on $\widetilde{U}_i$. Also, it's derivative matrix after removing the ith column will be the matrix $x_i^{-1}\mathbf{I}_N$, and so the derivative matrix is full rank. ■

Lemma 6.13: Suppose $n \leq N$ and $M \subseteq \mathbb{R}^N$ is a C^r n-dimensional (immersed) submanifold of M (where $r \geq 2$). We also will identify $\mathbb{R}^{N-1}$ with the subspace of $\mathbb{R}^N$ of vectors whose last coordinate is zero, and for any $v \in \mathbb{R}^N - \mathbb{R}^{N-1}$ we will let $\pi_v : \mathbb{R}^N \to \mathbb{R}^{N-1}$ be the projection linear map with kernel $\mathbb{R}v$.

In particular, π_v is defined by noting that since $v \notin \mathbb{R}^{N-1}$, any $x \in \mathbb{R}^N$ can be uniquely written as $x = x' + cv$ where $x' \in \mathbb{R}^{N-1}$ and $c \in \mathbb{R}$. And now we define $\pi_v(x) = x'$.

Now if $N > 2n+1$ then there is a dense set of vectors $v \in \mathbb{R}^N - \mathbb{R}^{N-1}$ for which $\pi_v|_M$ is an injective C^r immersion of M into $\mathbb{R}^{N-1}$.

Proof:
In order for $\pi_v|_M$ to be an injection, we need $p - q \notin \operatorname{span}\{v\}$ for all distinct $p, q \in M$. After all, if $p = \pi_v(p) + cv$ and $q = \pi_v(q) + c'v$ then $p - q = (\pi_v(p) - \pi_v(q)) + (c - c')v$. So $\pi_v(p) = \pi_v(q)$ iff $p - q \in \operatorname{span}\{v\}$.

Meanwhile, note that $\pi_v|_M$ is an immersion if and only if $T_pM \cap \ker(\mathrm{d}\pi_v) = \{0\}$. After all, if $\iota : M \hookrightarrow \mathbb{R}^N$ is the inclusion map then $\mathrm{d}(\pi_v|_M)_p = \mathrm{d}(\pi_v)_p \circ \mathrm{d}\iota_p$. So $\mathrm{d}(\pi_v|_M)_p$ is injective iff it has trivial kernel iff $\operatorname{im}(\mathrm{d}\iota_p) \cap \ker(\mathrm{d}(\pi_v)_p) = \{0\}$. Yet also note that $\ker(\mathrm{d}\pi_v) = \operatorname{span}\{v\}$. So, $\pi_v|_M$ is an immersion iff $v \notin T_pM$ for all p.

Let $\Delta_M \coloneqq \{(p,p) \in M \times M : p \in M\}$. By proposition 5.7 on *1113-1114*, we know that $\Delta(M)$ is a closed subset of $M \times M$. Also let $M_0 \coloneqq \{(p,v) \in TM : p \in M \text{ and } v = 0\}$. I claim M_0 is also closed.

To see why, let $f : M \to TM$ be defined by $f(m) = (m, 0)$. I claim this is a continuous map. After all, given any $p \in M$, we can pick a smooth chart (U, φ) on M containing p as well as a smooth chart $(\widetilde{U}, \widetilde{\varphi})$ on TM containing $(p,0)$ such that $F(U) \subseteq \widetilde{U}$ and $\widetilde{\varphi} \circ F \circ \varphi^{-1}(x_1, \ldots, x_n) = ((x_1, \ldots, x_n), 0)$. So, $\widetilde{\varphi} \circ F \circ \varphi^{-1}$ is a continuous map. And this proves that f is continuous on a neighborhood of p. Since p was arbitrary, f is continuous from all of M

Next note that $\widetilde{\pi} : TM \to M$ sending (p,v) to p is also a continuous map. And since $\mathrm{Id}_M = \pi \circ f$, we can conclude via the lemma on *1114* that f is a proper map. But a proper continuous map into an LCH space is also a closed map. So, $f(M) = M_0$ must be a closed subset of TM.

Now let $\kappa : (M \times M) - \Delta_M \to \mathbb{RP}^{N-1}$ be defined by $\kappa(p,q) \coloneqq [p-q]$. Meanwhile let $\tau : TM \to \mathbb{RP}^{N-1}$ be defined by $\tau(p,v) \coloneqq [v]$. These are C^r and C^{r-1} maps respectively.

To see why, let $\pi : \mathbb{R}^N - \{0\} \to \mathbb{RP}^{N-1}$ be the quotient map. This is a C^∞ map. Also let $\iota : M \to \mathbb{R}^n$ the inclusion map. This is a C^r map. Finally, note that $s : \mathbb{R}^N \times \mathbb{R}^N$ defined by $s(x,y) = x - y$ is a C^∞ map. And now κ is a C^r map because $\kappa = \pi \circ s \circ (\iota \times \iota)|_{(M\times M)-\Delta_M}$.

Meanwhile, let $\widetilde{\pi} : TM \to M$ be the map sending (p,v) to p. This is a C^{r-1} map. Thus τ is a C^{r-1} map since $\tau = \pi \circ \iota \circ \widetilde{\pi}|_{TM - M_0}$.

Yet note that $\dim(M \times M) = \dim(TM) = 2n$. Meanwhile $\dim(\mathbb{RP}^{N-1}) = N-1 > 2n$. So by corollary 6.12 from two pages ago, we can conclude that $\kappa((M \times M) - \Delta_M)$ and $\tau(TM - M_0)$ are null subsets of $\mathbb{RP}^{N-1}$. It follows that their union is also a null subset of $\mathbb{RP}^{N-1}$.

But now we've shown that the set of equivalence classes of vectors $[v] \in \mathbb{RP}^{N-1}$ such that v fulfills the needs of this lemma is dense in $\mathbb{RP}^{N-1}$. We can in turn conclude that the set of $v \in \mathbb{R}^N$ fulfilling the needs of this lemma is dense in $\mathbb{R}^N$. More generally:

Lemma: If $\pi : \mathbb{R}^N - \{0\} \to \mathbb{RP}^{N-1}$ is the quotient map then A being a dense subset of $\mathbb{RP}^{N-1}$ implies that $\pi^{-1}(A)$ is dense in $\mathbb{R}^N$.

Proof:
Suppose $v \in \mathbb{R}^N$. Then as $\mathbb{RP}^{N-1}$ is first countable and A is dense, we can find a sequence of elements $\{[w_n]\}_{n\in\mathbb{N}}$ converging to $[v]$ in $\mathbb{RP}^{N-1}$. Without loss of generality, we can assume $|w_n| = |v|$. And then $\{w_n\}_{N\in\mathbb{N}}$ will be a sequence in $\mathbb{R}^N$ with the limit points v and $-v$. We can take a subsequence to finish. ■

An easy corollary of the prior lemma is that if a C^r n-dimensional manifold (where $r \geq 2$) admits any C^r immersion into $\mathbb{R}^N$ for some N, then it admits an immersion into $\mathbb{R}^{2n+1}$.

9/25/2026

I need to cram some other topics quickly. So I'm going to make a hard departure from John Lee for now. I'll come back to the topic of embedding manifolds into $\mathbb{R}^{2n+1}$ on *page 1239*.

I need to learn about singular homology. So I'm going to read Haynes Miller's lectures on it. See *citation 14* for more info.

If $0 \leq n \leq m$ and $p_0, p_1, \ldots, p_n \in \mathbb{R}^m$ then we define $[p_0, p_1, \ldots, p_n]$ to be the convex hull $\text{conv}(\{p_0, p_1, \ldots, p_n\}) = \{\sum_{i=0}^n c_i p_i : \sum_{i=0}^n c_i = 1\}$.

If $e_0, e_1, \ldots, e_n$ denote the standard basis on $\mathbb{R}^{n+1}$, then we'll define:

$$\Delta^n := [e_0, e_1, \ldots, e_n].$$

Note that if $u_0, u_1, \ldots, u_{n-1}$ is the standard basis on $\mathbb{R}^n$, then for each $0 \leq i \leq n$ we can define a continuous map $d^{(i)} : \Delta^{n-1} \to \Delta^n$ by:

$$d^{(i)}(\textstyle\sum_{j=0}^{n-1} c_j u_j) = \left(\sum_{j=0}^{i-1} c_j e_j\right) + \left(\sum_{j=i}^{n} c_j e_{j+1}\right).$$

> Importantly the image of $d^{(i)}$ is the face of Δ^n not containing the point e_i as a corner. In other words, $d^{(i)}([u_0, u_1, \ldots, u_{n-1}]) = [e_0, e_1, \ldots, \widehat{e_i}, \ldots, e_n]$.
>
> (Here we add a circumflex above the vector in the list to denote that we removed it.)
>
> Also note for any $0 \leq j \leq n-1$ that $d^{(i)}(u_j) = e_k$ for some k. And importantly if $j < j'$ and $d^{(i)}(u_{j'}) = e_{k'}$ then $k < k'$.

Let X be any topological space. A singular n-simplex in X is a continuous map $\sigma : \Delta^n \to X$. We'll denote by $\text{Sin}_n(X)$ the set of all singular n-simplices on X. Next, we'll let $S_n(X)$ denote the free abelian group (i.e. $\mathbb{Z}$-module) generated by $\text{Sin}_n(X)$. An element of $S_n(X)$ (called a singular n-chain) will be a formal sum of singular n-simplices on X.

If $n < 0$, then we take $\text{Sin}_n(X) = \emptyset$ and $\text{Sin}_n(X) = \{0\}$.

And now for every $n \geq 1$ we can define a boundary operator / group homomorphism $\partial : S_n(X) \to S_{n-1}(X)$ by setting:

$$\partial(\sigma) := \textstyle\sum_{i=0}^n (-1)^i (\sigma \circ d^{(i)})$$

for all $\sigma \in \text{Sin}_n(X)$.

> Remark 1: By the universal property of free modules there exists a unique group homomorphism from $S_n(X)$ whose restriction to $\text{Sin}_n(X)$ is as defined here.
>
> Remark 2: I will be making a consistent abuse of notation where ∂ can denote the map $S_n(X) \to S_{n-1}(X)$ or $S_{n-1}(X) \to S_{n-2}(X)$ or etc. It will always be clear from context what map ∂ is.

We'll also define ∂ to just be the zero map if $n \leq 0$. And now I shall show that

$$\cdots \longrightarrow S_3 \xrightarrow{\partial} S_2 \xrightarrow{\partial} S_1 \xrightarrow{\partial} S_0 \xrightarrow{\partial} S_{-1} \longrightarrow \cdots$$

is a complex (see *page 614* for a definition).

Lemma: Suppose $0 \leq i \leq j \leq n-1$. Then $d^{(i)} \circ d^{(j)} = d^{(j+1)} \circ d^{(i)}$ as maps from Δ^{n-2} to Δ^n.

Proof:
Note that both maps send e_k to e_k if $0 \leq k < i$, e_k to e_{k+1} if $i \leq k < j$, and e_k to e_{k+2} if $j \leq k \leq n-2$. In other words, both maps send the corners of Δ^{n-2} to the same points in Δ^n. Furthermore, as both maps are compositions of affine maps, we know that both are affine maps themselves. In other words, they preserve convex linear combinations. And this is enough to now guarentee that both maps agree everywhere on Δ^{n-2}.

Proposition: $\partial^2 : S_n \to S_{n-2}$ is the zero map.

Proof:
Since $\mathrm{Sin}_n(X)$ generates the group S_n, it suffices to show that $\partial^2(\mathrm{Sin}_n(X)) = \{0\}$. When $n \leq 1$, this is obvious. So, we can assume that $n \geq 2$.

Let $\sigma : \Delta^n \to X$ be a singular n-simplex. Then note that:

$$\partial^2(\sigma) = \sum_{i=0}^{n}(-1)^i \partial(\sigma \circ d^{(i)}) = \sum_{i=0}^{n}\sum_{j=0}^{n-1}(-1)^{i+j} \cdot (\sigma \circ d^{(i)} \circ d^{(j)})$$

$$= \sum_{j=0}^{n-1}\sum_{i=0}^{j}(-1)^{i+j}(\sigma \circ d^{(i)} \circ d^{(j)}) + \sum_{j=0}^{n-1}\sum_{i=j+1}^{n}(-1)^{i+j}(\sigma \circ d^{(i)} \circ d^{(j)}).$$

But now by the prior lemma, we know that:

$$\sum_{j=0}^{n-1}\sum_{i=0}^{j}(-1)^{i+j}(\sigma \circ d^{(i)} \circ d^{(j)}) = \sum_{j=0}^{n-1}\sum_{i=0}^{j}(-1)^{i+j}(\sigma \circ d^{(j+1)} \circ d^{(i)}) \text{ (using lemma)}$$

$$= \sum_{i=0}^{n-1}\sum_{j=i}^{n-1}(-1)^{i+j}(\sigma \circ d^{(j+1)} \circ d^{(i)}) \text{ (changing order of indexing)}$$

$$= \sum_{j=0}^{n-1}\sum_{i=j}^{n-1}(-1)^{i+j}(\sigma \circ d^{(i+1)} \circ d^{(j)}) \text{ (relabeling indices)}$$

$$= \sum_{j=0}^{n-1}\sum_{i=j+1}^{n}(-1)^{i+j-1}(\sigma \circ d^{(i)} \circ d^{(j)})$$

Therefore:

$$\sum_{j=0}^{n-1}\sum_{i=0}^{j}(-1)^{i+j}(\sigma \circ d^{(i)} \circ d^{(j)}) + \sum_{j=0}^{n-1}\sum_{i=j+1}^{n}(-1)^{i+j}(\sigma \circ d^{(i)} \circ d^{(j)})$$

$$= \sum_{j=0}^{n-1}\sum_{i=j+1}^{n}(-1)^{i+j-1}(\sigma \circ d^{(i)} \circ d^{(j)}) + \sum_{j=0}^{n-1}\sum_{i=j+1}^{n}(-1)^{i+j}(\sigma \circ d^{(i)} \circ d^{(j)}) = 0. \blacksquare$$

We say an n-cycle in X is an element of $\ker(\partial : S_n(X) \to S_{n-1}(X))$. The set of n-cycles is denoted $Z_n(X)$.

Meanwhile, we say an n-boundary in X is an element of $\mathrm{im}(\partial : S_{n+1}(X) \to S_n(X))$. The set of n-boundaries is denoted $B_n(X)$, and by the prior proposition we know that $B_n(X) \subseteq Z_n(X)$.

And now we define the nth singular holomology group of X to be $H_n(X) := \frac{Z_n(X)}{B_n(X)}$.

As a side note, this definition of n-cycles, n-boundaries, and n-homologies works for any complex of modules over commutative rings.

Let $f : X \to Y$ be a continuous map. It induces a map $f_\sharp : \mathrm{Sin}_n(X) \to \mathrm{Sin}_n(Y)$ by $f_\sharp(\sigma) = f \circ \sigma$. And then by the universal property of free groups, this extends to a unique group homomorphism $f_\sharp : S_n(X) \to S_n(Y)$. Also, we can easily see that the following commuting diagram holds for all n.

$$\begin{array}{ccc} S_n(X) & \xrightarrow{\partial} & S_{n-1}(X) \\ {\scriptstyle f_\sharp}\downarrow & & \downarrow{\scriptstyle f_\sharp} \\ S_n(Y) & \xrightarrow{\partial} & S_{n-1}(Y) \end{array}$$

More importantly though, we want to show that $f_\sharp$ quotients out to a homomorphism $f_* : H_n(X) \to H_n(Y)$.

To start off, we shall prove that $f_\sharp(Z_n(X)) \subseteq Z_n(Y)$.

Suppose $c \in Z_n(X)$. Then we know that $\partial(c) = 0$. In turn:

$$\partial(f_\sharp(c)) = f_\sharp(\partial(c)) = f_\sharp(0) = 0.$$

This proves that $f_\sharp(c) \in Z_n(Y)$.

Next, we shall prove that $f_\sharp(B_n(X)) \subseteq B_n(Y)$.

Suppose $b \in B_n(X)$. Then we know there exists $c \in S_{n+1}(X)$ such that $\partial(c) = b$. Yet now $f_\sharp(b) = f_\sharp(\partial(c)) = \partial(f_\sharp(c))$. This proves $f_\sharp(b) \in B_n(Y)$.

It now follows that $f_*(c + B_n(X)) := f_\sharp(c) + B_n(Y)$ is a well-defined group homomorphism from $H_n(X)$ to $H_n(Y)$.

As a side note, we can also use similar reasoning in more general cases. Suppose we have the following chains of R-modules (where R is a commutative ring):

$$\cdots \longrightarrow A_3 \longrightarrow A_2 \longrightarrow A_1 \longrightarrow A_0 \longrightarrow A_{-1} \longrightarrow \cdots$$

$$\cdots \longrightarrow B_3 \longrightarrow B_2 \longrightarrow B_1 \longrightarrow B_0 \longrightarrow B_{-1} \longrightarrow \cdots$$

A chain homomorphism between those complexes would be a sequence of R-module homomorphisms $\{f_n\}_{n\in\mathbb{N}}$ such that the following diagram commutes for all n.

$$\begin{array}{ccc} A_n & \longrightarrow & A_{n-1} \\ {\scriptstyle f_n}\downarrow & & \downarrow{\scriptstyle f_{n-1}} \\ B_n & \longrightarrow & B_{n-1} \end{array}$$

The same reasoning from before shows that any chain homomorphism quotients out to a homomorphism between the nth homology modules of this chain.

Exercise 3.7: Let $\pi_0(X)$ denote the set of path components of a topological space X. Then construct an isomorphism from the free abelian group $\mathbb{Z}\pi_0(X)$ to $H_0(X)$.

To start off, a 0-simplex on X is just a map from a singleton set to a point in X. Therefore, we can identify $S_0(X)$ with $\mathbb{Z}X$ where the latter denotes the free abelian group on X. Next, we trivially have that every singular 0-chain is a 0-cycle (because $S_{-1}(X) = \{0\}$). Therefore, $Z_0(X) = S_0(X)$.

Now we can define a map $f : X \to \pi_0(X)$ mapping each $p \in X$ to the path component of X containing p. By the universal property of free groups, this then extends to a unique group homomorphism $f : Z_0(X) \to \mathbb{Z}\pi_0(X)$. Also, we trivially have that this group homomorphism is surjective because every generator of $\mathbb{Z}\pi_0(X)$ is in the image.

To finish off, it suffices to prove that $\ker(f) = B_0(X)$.

To start off, suppose $b \in B_0(X)$. In turn, we know there is a singular 1-chain $\sum_{i=1}^m n_i\gamma_i$ such that $\partial(\sum_{i=1}^m n_i\gamma_i) = b$. Yet we can then solve that:

$$b = \sum_{i=1}^m n_i[\gamma_i(1,0) - \gamma_i(0,1)]$$

In turn, $f(b) = \sum_{i=1}^m n_i[f(\gamma_i(1,0)) - f(\gamma_i(0,1))]$. Yet also note that a 1-simplex can be reparametrized into being a path. Therefore, $f(\gamma_i(1,0))$ and $f(\gamma_i(0,1))$ will be in the same path component of X. So, $f(b) = \sum_{i=1}^m n_i 0 = 0$ and this proves that $B_0(X) \subseteq \ker(f)$.

Conversely, suppose $c = \sum_{i=1}^m n_i p_i \in \ker(f)$. For ease of notation, by allowing the p_i to repeat themselves I can assume without loss of generality that:

$$c = \sum_{i=1}^{m_1} p_i - \sum_{j=1}^{m_2} q_j.$$

But now I claim that $m_1 = m_2$.

To see why, let $h : \mathbb{Z}\pi_0(X) \to \mathbb{Z}$ be defined by sending $\sum n_i E_i$ to $\sum n_i$. This a group homomorphism which importantly satisfies that $h(f(p)) = +1$ for all $p \in X$. Therefore, $h(f(c)) = m_1 - m_2$. At the same time though, we also have that $h(0) = 0$. So, we can conclude that $m_1 - m_2 = 0$.

So, we've shown that there is an integer $m \geq 0$ such that $c = \sum_{i=1}^m p_i - \sum_{i=1}^m q_i$. Next I shall prove that all these elements must cancel $\bmod B_0(X)$.

I shall proceed by induction on m. For our base case, note that if $m = 0$ then c is obviously in $B_0(X)$. Meanwhile, suppose $m \geq 1$.

Let E be the path component containing p_1. Then we can define $h_i : \mathbb{Z}\pi_0(X) \to \mathbb{Z}$ to be the group homomorphism projecting $\mathbb{Z}\pi_0(X)$ onto the Eth coordinate. Again we importantly have that $h_i(f(p)) = +1$ if $p \in E$ and $h_i(f(p)) = 0$ if $p \notin E$. Yet also note that $h_i(0) = 0$. Therefore:

$$0 = h_i(0) = h_i(f(c)) = (\#\text{ of } p_i \text{ in E}) - (\#\text{ of } q_i \text{ in E})$$

Since $(\#\text{ of } p_i \text{ in E}) \geq 1$ since $p_1 \in E$, we can conclude that there must exist some $q_{i'} \in E$ as well. But then we also can find a path (which via a reparametrization is a 1-simplex) from p_1 to $q_{i'}$. So, $p_1 + B_0(X) = q_{i'} + B_0(X)$. And after canceling out those terms, we get that:

$$c + B_0(X) = \sum_{i=2}^{m} p_i - \sum_{\substack{i=1 \\ i \neq i'}}^{m} q_i.$$

By induction, we can conclude that $c + B_0(X) = 0 + B_0(X)$. So, $\ker(f) \subseteq B_0(X)$.

With that we can now just apply the first isomorphism theorem to get our result. ■

Here is a writeup for @hchan. Although I'm getting all this from Munkres Topology. As part of this I'm also referencing an exercise on *pages 139-140* of my notes.

Lemma 1.4: Let M be a paracompact Hausdorff space and $\{U_\alpha\}_{\alpha\in A}$ *any* open cover. Then there exists a locally finite open cover $\{V_\alpha\}_{\alpha\in A}$ such that $\overline{V_\alpha} \subseteq U_\alpha$ for all α.

Proof:

Our first step is showing that paracompact and Hausdorff implies regular (i.e. T_3).

Let $A \subseteq M$ be a closed set and suppose $p \notin A$. Then since M is Hausdorff, for all $q \in A$ we can pick an open set $V_q \subseteq M$ such that $p \notin \overline{V_q}$. Next, note that $\{M - A\} \cup \{V_q\}_{q\in A}$ is an open cover of M. So by paracompactness, we can take a locally finite refinement $\{W_\beta\}_{\beta\in B}$ covering M.

Let B' be the collection of $\beta \in B$ such that $W_\beta \cap A \neq \emptyset$. Since we know that $M = \bigcup_{\beta\in B} W_\beta$, we can thus conclude that $A \subseteq \bigcup_{\beta\in B'} W_\beta$. At the same time though, because $\{W_\beta\}_{\beta\in B'}$ is locally finite, we can say that:

$\overline{\bigcup_{\beta\in B'} W_\beta} = \bigcup_{\beta\in B'} \overline{W_\beta} \subseteq \bigcup_{q\in A} \overline{V_q}$ where the latter set doesn't contain p.

So, the open sets $U := \bigcup_{\beta\in B'} W_\beta$ and $V := M - \overline{U}$ separate p and the closed set A.

Next, let $\mathcal{A}$ be the collection of open subsets of M whose closures are contained in a set U_α. Because M is regular, we know that $\mathcal{A}$ covers all of M. So by paracompactness, we can take a locally finite refinement $\{W_\beta\}_{\beta\in B}$ of $\mathcal{A}$ covering M.

Define a function $f : B \to A$ such that $\overline{W_\beta} \subseteq U_{f(\beta)}$ for all β. Then for every α define $V_\alpha := \bigcup_{\beta\in f^{-1}(\{\alpha\})} W_\beta$. Clearly $\bigcup_{\alpha\in A} V_\alpha = \bigcup_{\beta\in B} W_\beta = M$. Also note that because the W_β are locally finite:

$$\overline{V_\alpha} = \overline{\bigcup_{\beta\in f^{-1}(\{\alpha\})} W_\beta} = \bigcup_{\beta\in f^{-1}(\{\alpha\})} \overline{W_\beta} \subseteq U_\alpha. \blacksquare$$

Theorem 2.23: Let M be a paracompact Hausdorff space and let $\mathcal{U} = \{U_\alpha\}_{\alpha\in A}$ be *any* open cover. Then there is a continuous partition of unity $\{f_\alpha : M \to [0,1]\}_{\alpha\in A}$ subordinate to $\mathcal{U}$ such that $\operatorname{supp}(f_\alpha) \subseteq U_\alpha$ for all $\alpha \in A$.

Proof:

We know from before that paracompact and Hausorff implies regular. As it turns out, we can also use an identical argument (except that we replace p with a disjoint closed set B from A) in order to prove that paracompact and regular implies normal (i.e. T_4). This is useful because we can apply Urysohn's lemma to any normal space.

By lemma 1.4, we can find two locally finite open covers $\{V_\alpha\}_{\alpha\in A}$ and $\{W_\alpha\}_{\alpha\in A}$ such that $\overline{W_\alpha} \subseteq V_\alpha \subseteq \overline{V_\alpha} \subseteq U_\alpha$ for all α. And then by Urysohn's lemma we can for every $\alpha \in A$ find a function $g \in C(M, [0,1])$ such that $g \equiv 1$ on $\overline{W_\alpha}$ and $g \equiv 0$ on $M - V_\alpha$.

But now note that as $\operatorname{supp}(g_\alpha) \subseteq \overline{V_\alpha}$ and $\{\overline{V_\alpha}\}_{\alpha\in A}$ is locally finite, we know that every $p \in M$ has a neighborhood on which only finitely many g_α are nonzero. So, we can define a continuous function $h = \sum_{\alpha\in A} g_\alpha$. And then we define $f_\alpha = g_\alpha/h$ for all α. This satisfies what we want. $\blacksquare$

9/29/2026

The following terminology is taken from my math 290a class. Recall from *page 117* what a homotopy and path homotopy is. We can give the following slight generalization of what a path homotopy is.

> Suppose $f_0, f_1 : X \to Y$ are continuous maps that agree on a subset $A \subseteq X$. Then a homotopy $\operatorname{rel} A$ from f_0 to f_1 is a continuous map $F : X \times [0,1] \to Y$ such that $F(x,0) = f_0(x)$, $F(x,1) = f_1(x)$, and $F(a,t) = f_0(a,t) = f_1(a,t)$ for all $a \in A$.

Remarks:

1. A path homotopy is a just a homotopy between two maps $f_0, f_1 : [0,1] \to Y$ rel $\{0,1\}$.
2. If $f_0, f_1 : X \to Y$ satisfy that there is a homotopy rel $A \subseteq X$ between f_0 and f_1, we write $f_0 \simeq_A f_1$ and say f_0 is homotopic to $f_1 \operatorname{rel} A$. Also we still write $\simeq_p$ for path homotopies. And if $A = \emptyset$ we just write $\simeq$.
3. $\simeq_A$ is an equivalence relation via identical reasoning as in the proof of lemma 51.1 on *page 117*.

Here is some more terminology (given two topological spaces X and Y):

- We let $[X,Y]$ denote the set of homotopy classes of continuous maps from X to Y.
- We say X and Y are homotopy equivalent if there exists maps $f : X \to Y$ and $g : Y \to X$ such that $f \circ g \simeq \mathrm{Id}_Y$ and $g \circ f \simeq \mathrm{Id}_X$. In this case we say f and g are homotopy equivalences.

  Note that while being homeomorphic trivially implies being homotopy equivalent, it is also possible to be homotopy equivalent and *not* homeomorphic.

  Lemma: Given any topological space X and any convex subspace K of a topological vector space, we have that any two functions $f_0 : X \to K$ and $f_1 : X \to K$ are homotopic.

  Why?
  We can just use the linear homotopy $F(x,t) = (1-t)f_0(x) + tf_1(x)$.

  Based on this lemma, we can now for example see that $\mathbb{R}^n$ is homotopy equivalent to a singleton set $\{*\}$ for all $n \geq 1$.

  Why?
  We can pick any point $x_0 \in \mathbb{R}^n$. Then we let $f : \mathbb{R}^n \to \{*\}$ be the unique function that exists going from $\mathbb{R}^n$ to $\{*\}$, and we let $g : \{*\} \to \mathbb{R}^n$ be the map sending $*$ to x_0.

  $f \circ g = \mathrm{Id}_{\{*\}}$ so we have no issues there. Meanwhile both $g \circ f$ and $\mathrm{Id}_{\mathbb{R}^n}$ are functions from $\mathbb{R}^n$ to $\mathbb{R}^n$. So by the prior lemma we know that they are homotopic. And this finishes proving that $\mathbb{R}^n$ is homotopy equivalent to $\{*\}$.
- A space X that is homotopy equivalent to a singleton set is called contractible. We can easily see that X is contractible iff there is a homotopy sending Id_X to a constant map.

One more concept is that if X is a topological space, $A \subseteq X$, and $r : X \to A$ is a retraction (i.e. a continuous map such that $r|_A = \mathrm{Id}_A$), then we say r is a deformation retraction if $\iota \circ r \simeq \mathrm{Id}_X \text{ rel } A$. (Here $\iota : A \hookrightarrow X$ is the inclusion map.)

There's a category theory lense that I can give to all of the homology theory established so far.

We say a graded abelian group (or R-module where R is a commutative ring) is a sequence $\{A_n\}_{n\in\mathbb{Z}}$ of abelian groups (or R-modules). Typically, we just denote the graded abelian group by A_*.

A (chain) complex is then just a graded abelian group (or graded R-module) equipped with a sequence of homomorphisms $\{d : A_n \to A_{n-1}\}_{n\in\mathbb{Z}}$ such that $d^2 : A_n \to A_{n-2}$ is the zero map.

Also recall from *page 1160* what a homomorphism $\{f_n\}_{n\in\mathbb{Z}}$ between chains is. Typically, we just denote this homomorphism by $f_\sharp$. Also its clear that a composition of chain homomorphisms (given by $\{g_n\}_{n\in\mathbb{Z}} \circ \{f_n\}_{n\in\mathbb{Z}} = \{g_n \circ f_n\}_{n\in\mathbb{Z}}$) is another chain homomorphism. Plus, $\mathrm{Id}_{B_*} \circ f_\sharp = f_\sharp = f_\sharp \circ \mathrm{Id}_{A^*}$ for all chain homomorphisms $f_\sharp : A_* \to B_*$.

We let $\mathrm{ch}\mathbf{Ab}$ denote the category whose objects are all chain complexes of abelian groups and whose morphisms are all chain homomorphisms.

> Under this lens, we have a functor S_* from **Top** (the category of topological spaces and continuous maps) to $\mathrm{ch}\mathbf{Ab}$. After all, for any topological space we let:
> $$S_*(X) = \{S_n(X)\}_{n\in\mathbb{N}}$$
> where the latter is a graded abelian group that becomes a chain complex when equipped with the boundary operator ∂.
>
> Also, we also, we already noted for any continuous map $f : X \to Y$ that $S_*(f) := f_\sharp$ is a chain homomorphism from $S_*(X)$ to $S_*(Y)$.
>
> > Technically we still need to show that if $f : X \to Y$ and $g : Y \to Z$ are continuous maps then $(g \circ f)_\sharp = g_\sharp \circ f_\sharp$. This is easy though. If $\sigma : \Delta^n \to X$ is any singular n-simplex then:
> > $$(g \circ f)_\sharp(\sigma) = g \circ f \circ \sigma = g \circ (f \circ \sigma) = g_\sharp(f_\sharp(\sigma)).$$
> >
> > Similarly, we can easily show that Id_X induces the identity chain homomorphism.

Next, we can define a homology functor $H_n : \mathrm{ch}\mathbf{Ab}$ to **Ab** as follows.

> Suppose A_* is a chain complex equiped with the maps $d : A_m \to A_{m-1}$ for all m such that $d^2 = 0$. Then we define a functor H_n sending A_* to $\frac{\ker(d:A_n\to A_{n-1})}{\mathrm{im}(d:A_{n+1}\to A_n)}$.

Also by the note on *page 1160*, we know that any chain homomorphism $f_\sharp : A_* \to B_*$ can be restricted and quotiented to a well defined map $f_* : H_n(A_*) \to H_m(B_*)$. So, we define the nth homology functor to send $f_\sharp$ to the map $a + d(A_{n+1}) \mapsto f_\sharp(a) + d(B_{n+1})$. This similarly behaves well with composition of homomorphisms and identity homomorphisms.

For time being I'll let the nth singular homology functor H_n : **Top** $\to$ **Ab** be the functor defined by composing the prior two functors I've defined.

I'm sorry if its confusing that the same letters and names keep getting reused for different slightly adjacent functors / functions.

One more category I want to define is **HoTop**.

The objects of **HoTop** are the same as the objects of **Top**. Meanwhile, we let the morphisms of **HoTop** between any topological spaces X and Y be the homotopy equivalence classes in $[X, Y]$.

Lemma: Suppose X, Y, Z are topological spaces, $f_0, f_1 \in C(X, Y)$ with $f_0 \simeq f_1$, and $g_0, g_1 \in C(Y, Z)$ with $g_0 \simeq g_1$. Then $g_0 \circ f_0 \simeq g_1 \circ f_1$.

Proof:
Let $F : X \times [0, 1] \to Y$ be a homotopy from f_0 to f_1 and $G : Y \times [0, 1] \to Z$ be a homotopy from g_0 to g_1. Then $H : X \times [0, 1] \to Z$ defined by:

$$H(x, t) := G(F(x, t), t)$$

is a homotopy from $g_0 \circ f_0$ to $g_1 \circ f_1$. ■

Also note that trivially $\mathrm{Id}_X \circ f \simeq f$. So, **HTop** is a well-defined category.

We can also see that there is a well-defined functor Ho : **Top** $\to$ **HoTop** sending any topological space X to X and any continuous map $f : X \to Y$ to its equivalence class $[f]_\simeq$.

9/30/2026

I'm going to take a break from studying singular homology so that I can instead briefly come up with a basic formulation of divergence theorem. I'm doing this because I will need it tomorrow in my PDE class.

Firstly, here's some terminology from John Lee:

Suppose M is a C^r manifold without corners. If $p \in \partial M$ then $v \in T_pM - T_p(\partial M)$ is said to be inward pointing if for some $\varepsilon > 0$ there is a C^1 curve $\gamma : [0, \varepsilon) \to M$ such that $\gamma(0) = p$ and $\gamma'(0) = v$. Similarly, $v \in T_pM - T_p(\partial M)$ is said to be outward pointing if for some $\varepsilon > 0$ there is a C^1 curve $\gamma : (-\varepsilon, 0] \to M$ such that $\gamma(0) = p$ and $\gamma'(0) = v$.

Proposition 5.41: Suppose M is a C^r manifold, $p \in \partial M$, and (U, φ) is a smooth boundary chart on M. (In other words $\widehat{U} = \varphi(U)$ is an open subset of $H^n := \mathbb{R}^{n-1} \times [0, \infty)$.) We'll

let $\varphi(p) = (x_1(p), \ldots, x_n(p))$. Then any $v \in T_pM$ can be written uniquely as $\sum_{i=1}^n c_i \frac{\partial}{\partial x_i}$. v is inward pointing iff $c_n > 0$. v is outward pointing iff $c_n < 0$. And finally, $v \in T_p(\partial M)$ iff $c_n = 0$.

Proof:
If $v \in T_p(\partial M)$ then we know we can find a C^1 path $\gamma : J \to M$ such that $\gamma(J) \subseteq \partial M$, $\gamma(0) = p$, and $\gamma'(0) = v$. But then if we look at the local coordinates on (U, φ), we have that $\varphi(\partial M \cap U) = \{(x_1, \ldots, x_n) \in \widehat{U} : x_n = 0\}$. So, $(\varphi \circ \gamma)(t) \cdot e_n = 0$ for all t. And this proves that if $v = \gamma'(0) = \sum_{i=1}^n c_i \frac{\partial}{\partial x_i}$ then $c_n = 0$.

Conversely, if $v = \sum_{i=1}^{n-1} c_i \frac{\partial}{\partial x_i} \in T_pM$ then the straight line path $\gamma : J \to M$ in the direction of v (according to the local coordinates) satisfies that $\gamma(0) = p$, $\gamma'(0) = v$, and that the nth coordinate of $\gamma(t)$ is always 0. Yet this equivalently means that $\gamma(J) \subseteq \partial M$. And since ∂M has the subspace topology, we can conclude that γ is a C^1 path into ∂M. So, $v \in T_p(\partial M)$.

Next, suppose v is inward pointing and let $\gamma : [0, \varepsilon) \to M$ be such that $\gamma(0) = p$ and $\gamma'(0) = v$. By shrinking ε we can include that $\gamma([0, \varepsilon)) \subseteq U$. But now note that the nth component of $(\varphi \circ \gamma)'(0)$ must be positive. This is because the nth coordinate of $\gamma(t)$ is always nonnegative with it equaling zero at $t = 0$. So, if $(\varphi \circ \gamma)'(0) \cdot e_n < 0$ we'd have a contradiction. At the same time, since $v \notin T_p(\partial M)$ we know from before that $(\varphi \circ \gamma)'(0) \cdot e_n \neq 0$. So, upon writing $v = \gamma'(0) = \sum_{i=1}^n c_i \frac{\partial}{\partial x_i}$ we have that $c_n > 0$.

Conversely, if $v = \sum_{i=1}^n c_i \frac{\partial}{\partial x_i}$ then the straight line path (according to the local coordinates) will give an example path proving that v is inward pointing.

Finally, we can use analogous reasoning when v is outward pointing. ■

Corollary: Suppose M is a C^r manifold and $p \in \partial M$. Then $v \in T_pM$ is outward pointing iff $-v$ is inward pointing.

Now the next reasoning will be hyper-specific to any C^1 regular domain D without corners in $\mathbb{R}^N$ (equipped with the standard smooth structure). To start off, note that ∂D will be a C^1 $(N-1)$-dimensional embedded submanifold of $\mathbb{R}^N$ without a boundary.

One consequence of this is that from *pages 95-99* of my journal we have a standard way of defining a surface measure S on ∂D.

Namely, we let S be the unique measure on ∂D satisfying that if (U, φ) is a coordinate chart on ∂D and $E \subseteq U$ then:

$$S(E) = \int_{\varphi(E)} \sqrt{\det\left((\mathrm{D}\varphi^{-1})^T(\mathrm{D}\varphi^{-1})\right)}\mathrm{d}x$$

Also, because $T_p\mathbb{R}^N$ can be identified with $\mathbb{R}^N$ for all $p \in \mathbb{R}^N$ via a standard identification (see *proposition 3.13 on pages 1043-1044*), we can identify $T_p(\partial D)$ with a geometric hyperplane for all $p \in \partial D$. And in turn, it makes sense to think of the orthogonal complement to that tangent plane.

Thus, for all $p \in \partial D$ I'll let $v(p) \in \mathbb{R}^N$ be the unique outward pointing unit vector in the orthogonal complement of $T_p(\partial D)$. My next goal will be to show that it is always possible locally to find a nice parametrization of the boundary of ∂D that also makes it easy to calculate the surface measure and normal vector γ.

Theorem 5.48: If M is a C^r manifold without boundary and $D \subseteq M$ is a regular domain without corners, then there exists a C^r function $f : M \to \mathbb{R}$ such that $\partial D = f^{-1}(\{0\})$, $D = f^{-1}((-\infty, 0])$, and 0 is a regular value of f.

Proof:
Let D° be equipped with the constant function -1. Meanwhile, let D^C be equipped with the constant function $+1$. And finally, for every $p \in \partial D$ let (U_p, φ_p) be a half-slice chart on M containing p. Thus $\varphi_p(U_p \cap D) = \{(x_1, \ldots, x_n) \in \varphi(U_p) : x_n \geq 0\}$. We can define a function $g_p : U_p \to \mathbb{R}$ by just setting $g_p(p) = -x_n(p)$. And this will be a C^r map such that $g_p < 0$ on $D^\circ \cap U_p$, $g_p = 0$ on $\partial D \cap U_p$,and $g_p > 0$ on $U_p - D^\circ$.

Now take a C^r partition of unity $\{\phi_{D^\circ}, \phi_{D^\mathsf{C}}\} \cup \{\psi_p\}_{p \in \partial D}$ subordinate to the open sets D°, D^C, and all the U_p. Next, define:

$$f = \psi_{D^\mathsf{C}} - \psi_{D^\circ} + \textstyle\sum_{p \in \partial D} \psi_p g_p$$

If $q \notin D$ then $\psi_{D^\circ}(q) = 0$ and all the other summands of f are positive or zero (and some will be positive). So, $D^\mathsf{C} \subseteq f^{-1}((0, \infty))$. On the other hand, if $q \in D$ then we know that $\psi_{D^\mathsf{C}}(q) = 0$ and the other terms are all nonpositive. So $D \subseteq f^{-1}((-\infty, 0])$. Putting these observations together, we have that $D = f^{-1}((-\infty, 0])$.

Next I claim that 0 is a regular value of f. To see why, first note that $f^{-1}(\{0\}) = \partial D$.
Hopefully this is obvious to see.

So, it suffices to show for all $p \in \partial D$ that $\mathrm{d}f_p$ is not the zero map. Fortunately, given any fixed $p \in \partial D$ we can find a neighborhood V containing p that intercepts the support of only finitely many $\psi_{p'}$. Furthermore, as $p \notin D^\circ$ and $p \notin D^\mathsf{C}$, we can shrink V to assume that $\psi_{D^\circ} = 0$ and $\psi_{D^\mathsf{C}} = 0$ on V. So, there are $p_1, \ldots, p_M$ such that
$f(q) = \sum_{i=1}^M \psi_{p_i}(q) g_{p_i}(q)$ for all $q \in V$.

By shrinking V more, we can assume that $p \in \operatorname{supp}(\psi_{p_i})$ for each i. This is important because it also now lets us conclude that $p \in \bigcap_{i=1}^M U_{p_i}$. So by shrinking V one more time we can assume without loss of generality that $V \subseteq \bigcap_{i=1}^M U_{p_i}$.

One last note I'll make is that by reordering the p_i, I can assume without loss of generality that $\psi_{p_1}(p) > 0$.

Now to show that $\mathrm{d}f_p$ does not vanish it suffices to show that gradient of f at p with respect to the smooth coordinate chart (U_{p_1}, φ_{p_1}) does not vanish. So, let $\pi : \mathbb{R}^n \to \mathbb{R}$ be the projection onto the nth coordinate. Then for all $x \in \varphi_{p_1}(U_{p_1})$ we have that:

$$\begin{aligned}
&\nabla(f \circ \varphi_{p_1}^{-1})(x) \\
&\quad = -\sum_{i=1}^m \left[(\psi_{p_i} \circ \varphi_{p_1}^{-1})(x) \nabla(\pi \circ (\varphi_{p_i} \circ \varphi_{p_1}^{-1}))(x) + (\pi \circ (\varphi_{p_i} \circ \varphi_{p_1}^{-1}))(x) \nabla(\psi_{p_i} \circ \varphi_{p_1}^{-1})(x)\right]
\end{aligned}$$

In the special case that $x \in \varphi_p(\partial D)$ though, we know that $(\pi \circ (\varphi_{p_i} \circ \varphi_{p_1}^{-1}))(x) = 0$ for all i. Therefore, our expression simplies to:

$$\begin{aligned}\nabla(f \circ \varphi_{p_1}^{-1})(\varphi_{p_1}(p)) &= -\sum_{i=1}^{m}(\psi_{p_i} \circ \varphi_p^{-1})(\varphi_{p_1}(p))\nabla(\pi \circ (\varphi_{p_i} \circ \varphi_{p_1}^{-1}))(\varphi_{p_1}(p)) \\ &= -\sum_{i=1}^{m}\psi_{p_i}(p)\nabla(\pi \circ (\varphi_{p_i} \circ \varphi_{p_1}^{-1}))(\varphi_{p_1}(p)). \\ &= -\psi_{p_1}(p)(0,\ldots,0,1) - \sum_{i=2}^{m}\psi_{p_i}(p)\nabla(\pi \circ (\varphi_{p_i} \circ \varphi_{p_1}^{-1}))(\varphi_{p_1}(p)).\end{aligned}$$

To finish off, write $x = (x_1, \ldots, x_n)$. We know that $\pi \circ (\varphi_{p_i} \circ \varphi_{p_1}^{-1})(x) > 0$ if $x_n > 0$ and that $\pi \circ (\varphi_{p_i} \circ \varphi_{p_1}^{-1})(x) < 0$ if $x_n < 0$. (This is just because the transition map $\varphi_{p_i} \circ \varphi_{p_1}^{-1}$ has to send the upper half space to the upper half space and the lower half space to the lower half space). But now in turn, we must have that the nth partial derivative of $\pi \circ (\varphi_{p_i} \circ \varphi_{p_1}^{-1})$ is nonnegative when $x_n = 0$.

Putting everything together, we have that the nth component of $\nabla(f \circ \varphi_{p_1}^{-1})(\varphi_{p_1}(p))$ is equal to something strictly negative minus a bunch of nonnegative terms. So, the nth component of $\nabla(f \circ \varphi_{p_1}^{-1})(\varphi_{p_1}(p))$ doesn't vanish. ■

We can use the prior result to show that if D is a regular domain without corners of a C^r manifold M without a boundary, then $\partial(D^\circ) = \partial D$. In other words, D is a regular closed set (see *page 927* for what that means).

Proof:
Suppose $p \in \partial(D^\circ)$. Therefore, we know that $p \notin \partial(D^\circ)$ and yet we can find a sequence $\{p_k\}_{k\in\mathbb{N}}$ in D° converging to p. Next, let $f : M \to \mathbb{R}$ be a C^r function such that $D = f^{-1}((-\infty, 0])$ and 0 is a regular value of f. Because f is continuous, we know that $f(p_k) \to f(p)$ as $k \to \infty$. Yet $f(p_k) \leq 0$ for all k. So, we can also guarentee that $f(p) \leq 0$. So, $p \in D$. And since $p \notin \partial(D^\circ)$, we can guarentee that $p \in \partial D$.

With that we've shown that $\partial(D^\circ) \subseteq \partial D$. To get the other inclusion, suppose $p \in \partial D$. Then we know that $p \notin D^\circ$. That said, we can take a chart (U, φ) about the point p. And then using that chart we can easily get a sequence of interior points $\{p_k\}_{k\in\mathbb{N}}$ of D converging to p. Since the manifold interior of D corresponds to the topological interior, this proves that $p \in \partial(D^\circ)$. ■

Equivalently, this means that D° is a regular open set because the interior of any regular closed is a regular open set.

Proof:
Lemma: If X is a toplogical space, a closed set $F \subseteq X$ is regular iff $F = \overline{F^\circ}$.

$(\Longrightarrow)$
If F is regular we know that $\partial F = \partial(F^\circ)$. But now:

$$\overline{F^\circ} = F^\circ \cup \partial(F^\circ) = F^\circ \cup \partial F = F^\circ \cup (F - F^\circ) = F.$$

$(\Longleftarrow)$
Suppose $F = \overline{F^\circ}$. Then $\partial F = F - F^\circ = \overline{F^\circ} - F^\circ = \partial(F^\circ)$.

Based on the prior lemma, we can now conclude that:

$$\partial(\overline{D^\circ}) = \overline{D^\circ} - (\overline{D^\circ})^\circ = D - D^\circ = \partial D.$$

And since D is regular, we know that $\partial D = \partial(D^\circ)$. So, $\partial(\overline{D^\circ}) = \partial(D^\circ)$. And this proves that D° is a regular open set. ■

More generally, this shows that the interior of a regular closed set is a regular open set.

This is important because this would let us apply the lemma on *page 928*. Although, I'm also now realizing that we can also show the following lemma way more directly. Oops.

Lemma: Suppose M is a C^r manifold without boundary and $D \subseteq M$ is a regular domain without corners. Suppose $p \in \partial D$ and $U \subseteq M$ is an open set containing p. Then we can find an open set $V \subseteq U$ containing p such that $V - \partial D$ has two nonempty connected components: $V \cap D^\circ$ and $V \cap D^\mathsf{C}$. Furthermore, we can find a C^r map $f : V \to \mathbb{R}$ such that 0 is a regular value of f,

$$V \cap D^\circ = f^{-1}(-\infty, 0),\ V \cap \partial D = f^{-1}(\{0\}),\ \text{and } V \cap D^\mathsf{C} = f^{-1}((0, \infty))$$

Proof:
We can find a half-slice chart (W, φ) on M containing p such that:

- $\varphi(U \cap D) = \{(x_1, \ldots, x_n) \in \varphi(U) : x_n \geq 0\}$
- $\varphi(U \cap \partial D) = \{(x_1, \ldots, x_n) \in \varphi(U) : x_n = 0\}$.

By shrinking U, we can without loss of generality assume $U \subseteq W$. And now since $\varphi(U)$ is an open subset of $\varphi(W)$ which is an open subset of $\mathbb{R}^n$, there exists $r > 0$ such that $B_r(\varphi(p)) \subseteq \varphi(U)$. So, let $V \coloneqq \varphi^{-1}(B_r(\varphi(p)))$. Then we know $V \subseteq U$.

Now, $\varphi(V - \partial D) = B_r(\varphi(p)) - (\mathbb{R}^{n-1} \times \{0\})$ has two connected (even convex) disjoint components, which are the subset where $x_n < 0$ and the subset where $x_n > 0$. Yet note that the subset where $x_n < 0$ is precisely $\varphi(V \cap D^\mathsf{C})$ and the subset where $x_n > 0$ is precisely $\varphi(V \cap D^\circ)$. Since homeomorphisms map connected components to connected components, we can conclude that $V \cap D^\mathsf{C}$ and $V \cap D^\circ$ are the conected components of $V - \partial D$.

As for getting our function f, just apply theorem 5.48 from two pages ago and then restrict the function you get to the set V. ■

I realize at this rate I'm just letting myself distracted but I thought this was a cool thought of mine so I wanted to write it down.

Proposition: Suppose M is a C^r manifold without a boundary and $D \subseteq M$ is a regular domain without corners. Then $\overline{D^\mathsf{C}} = D^\mathsf{C} \cup \partial(D^\mathsf{C}) = D^\mathsf{C} \cup \partial D$ is also a regular domain without corners and its boundary is ∂D.

Proof:

By theorem 5.48 we can find a function $f : M \to \mathbb{R}$ such that 0 is a regular value of f, $D = f^{-1}((-\infty, 0])$ and $\partial D = f^{-1}(\{0\})$. But now consider the function $g = -f$. Then we have that 0 is a regular value of g, $\overline{D^C} = g^{-1}((-\infty, 0])$, and $\partial D = g^{-1}(\{0\})$. So by proposition 5.47 on *page 1127*, we know ghat $\overline{D^C}$ is a regular domain with boundary ∂D. ■

Now I get to the important result that I will need to get my basic formulation of divergence theorem. In particular, from here on out I will assume $M = \mathbb{R}^N$ and $D \subseteq M$ is a C^1 regular domain without corners.

Theorem: Suppose D is a C^r regular domain without corners of $\mathbb{R}^N$ and $p \in \partial D$. Then for some $1 \leq i \leq N$ there exists open sets $U \subseteq \mathbb{R}^N$ and $V \subseteq \mathbb{R}^{N-1}$, as well as a C^r function $g : V \to \mathbb{R}$ such that the following holds:

(a) if $\widehat{\pi}_i : \mathbb{R}^N \to \mathbb{R}^{N-1}$ is the projection onto all but the ith component, we have that $\widehat{\pi}_i(U) = V$. In other words, $(x_1, \ldots, x_n) \in U$ implies that $(x_1, \ldots, \widehat{x_i}, \ldots, x_N) \in V$.

(b) $U \cap \partial D = \{(x_1, \ldots, x_N) \in U : x_i = g(x_1, \ldots, \widehat{x_i}, \ldots, x_N)\}$.

(c) Also, we can guarentee that one of the two things holds.

- $U \cap D = \{(x_1, \ldots, x_N) \in U : x_i \leq g(x_1, \ldots, \widehat{x_i}, \ldots, x_N)\}$
- $U \cap D = \{(x_1, \ldots, x_N) \in U : x_i \geq g(x_1, \ldots, \widehat{x_i}, \ldots, x_N)\}$.

As a side note, the vector $(x_1, \ldots, \widehat{x_i}, \ldots, x_N) \in \mathbb{R}^N$ just denotes the result of removing the ith component from a vector $(x_1, \ldots, x_N) \in \mathbb{R}^N$.

Proof:
Let $f : \mathbb{R}^N \to \mathbb{R}$ be a C^r map such that $D = f^{-1}((-\infty, 0])$, $\partial D = f^{-1}(\{0\})$, and 0 is a regular value of f. Then suppose $p = (p_1, \ldots, p_N) \in \partial D$. By the implicit function theorem, we know for some $1 \leq i \leq N$ that there is an open set $U \subseteq \mathbb{R}^N$, an open set $V \subseteq \mathbb{R}^{N-1}$ and a C^r function $g : V \to \mathbb{R}$ satisfying that:

- $p \in U$ and $\widehat{p} = (p_1, \ldots, \widehat{p_i}, \ldots, p_N) \in V$,
- For all $\widehat{q} = (q_1, \ldots, \widehat{q_i}, \ldots, q_N) \in V$ we have that $q_i = g(\widehat{q}) \in \mathbb{R}$ is the unique real number such that $q = (q_1, \ldots, q_N) \in U$ with $f(q) = 0$.

Technically, in my math 140c notes I only proved g is C^1. But the same proof can show it is C^r.

Next, let $G : V \to U$ be the graph function given by:

$$G(x_1, \ldots, \widehat{x_i}, \ldots, x_n) = (x_1, \ldots, x_{i-1}, g(x_1, \ldots, \widehat{x_i}, \ldots, x_n), x_{i+1}, \ldots, x_N)$$

We know that G is a C^r map. Furthermore, if we let $\widehat{\pi}_i : \mathbb{R}^N \to \mathbb{R}^{N-1}$ be the projection onto all but the ith component, then we have that $G(V) = U \cap (\widehat{\pi}_i)^{-1}(V) \cap \partial D$. So, I'll firstly replace U with the open set $U' = U \cap (\widehat{\pi}_i)^{-1}(V)$. This is useful because it guarentees for every $\vec{y} \in U'$ that $\widehat{\pi}_i(\vec{y}) \in V$. Also, now $G(V) = U' \cap \partial D$.

In other words, we've shown that U', V', and $g|_{V'}$ satisfy properties (a) and (b) that we wanted.

Next let us define $H : U' \to \mathbb{R}$ by:

$$H(x_1, \ldots, x_N) = x_i - g(x_1, \ldots, \widehat{x_i}, \ldots, x_N).$$

A key observation is that H is C^r and that $H^{-1}(\{0\}) = U' \cap \partial D$.

Now by the prior lemma we know there is an open set $U'' \subseteq U'$ containing p such that $U'' - \partial D$ has two nonempty connected components: $U'' \cap D^\circ$ and $U'' - D$. Now granted, by restricting to U'', we might have lost properties (a) and (b). That said, connectedness is useful here because H must map $U'' \cap D^\circ$ and $U'' - D$ each to connected subsets (i.e. intervals) of $\mathbb{R}$. Yet we also know that 0 cannot be contained in the image of either set (since $H^{-1}(\{0\}) = U' \cap \partial D$). So, we know that either:

$$H(U'' \cap D^\circ) \subseteq (-\infty, 0) \text{ or } H(U'' \cap D^\circ) \subseteq (0, \infty).$$

And similarly we know that either:

$$H(U'' - D) \subseteq (-\infty, 0) \text{ or } H(U'' - D) \subseteq (0, \infty).$$

Yet because U'' is an open subset containing p, we also know that there must be some $\varepsilon > 0$ such that $p + te_i \in U''$ for all $|t| < \varepsilon$. (Here e_i is the ith standard basis vector.) So, we can guarentee that $H(U'')$ contains both positive and negative values. Hence, we can conclude that either:

$$U'' \cap D^\circ = H|_{U''}^{-1}((-\infty, 0)) \text{ and } U'' - D = H|_{U''}^{-1}((0, \infty))$$

or:

$$U'' \cap D^\circ = H|_{U''}^{-1}((0, \infty)) \text{ and } U'' - D = H|_{U''}^{-1}((-\infty, 0))$$

This is equivalent to property (c) that we wanted.

To finish off, we just need to regain properties (a) and (b). So let $V''' := G^{-1}(U'')$ and let $U''' := U'' \cap (\widehat{\pi}_i)^{-1}(V''')$. Then U''', V''', and $g|_{V'''}$ satisfies our theorem. ■

As a side note, I might have destroyed the connectedness of $U''' - \partial D$. That said, I don't care about it anymore.

In particular, the prior theorem show that the boundary of any C^1 regular domain D can be locally parametrized as the graph of a C^1 function $g : V \subseteq \mathbb{R}^{N-1} \to \mathbb{R}$.

To see why, let V, U, and g be from the original theorem. Then define $G : V \to \mathbb{R}^N$ by:

$$G(x_1, \ldots, \widehat{x_i}, \ldots, x_n) = (x_1, \ldots, x_{i-1}, g(x_1, \ldots, \widehat{x_i}, \ldots, x_n), x_{i+1}, \ldots, x_N)$$

Then G is an injective C^1 map of rank $N - 1 = \dim(\partial D)$ whose image is contained in ∂D. Since ∂D is an embedded submanifold, we thus know that G is a C^1 map from V to ∂D. Also, G must still have rank $N - 1$ as a map from V to ∂D. By the inverse function theorem, we can conclude that G is a local C^1 diffeomorphism from V to ∂D. And since G is injective, we can conclude that G is a C^1 diffeomorphism from V to an open set $G(V) \subseteq \partial D$.

It follows that $(G(V), G^{-1}) = (U \cap \partial D, G^{-1})$ is a smooth chart on ∂D by proposition 5.23 on *page 1125*.

Furthermore, this parametrization will satisfy that $D \cap U$ is the volume below or above the graph G (intercepted with U).

This is useful for us because the surface measure of a graph is incredibly easy to calculate.

Suppose $V \subseteq \mathbb{R}^{N-1}$, $g : V \to \mathbb{R}$ is a C^1 map, and $G : V \to \mathbb{R}^N$ is the map:

$$G(x_1, \ldots, x_{N-1}) = (x_1, \ldots, x_{N-1}, g(x_1, \ldots, x_{N-1})).$$

We know $\mathrm{D}G = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \\ \partial_{x_1} g & \cdots & \partial_{x_{N-1}} g \end{bmatrix}$.

Therefore, if we write the gradient of g as a column vector, we can calculate that:

$$(\mathrm{D}G)^T(\mathrm{D}G) = \mathbf{I}_{N-1} + (\nabla g)(\nabla g)^T.$$

Then we have the following result.

Lemma: Suppose $v = (v_1, \ldots, v_n) \in \mathbb{R}^n$ is interpretted as a column vector. Then:

$$\det(\mathbf{I}_n + vv^T) = 1 + |v|^2.$$

Proof:
Note that $(\mathbf{I}_n + vv^T)v = v + vv^Tv = (1 + |v|^2)v$. Therefore, we can see that $(1 + |v|^2)$ is an eigenvalue of $\mathbf{I}_n + vv^T$ with a corresponding eigenvector v.

Next let $u_2, \ldots, u_n$ be some more vectors such that $v, u_2, \ldots, u_n$ is an orthogonal basis for $\mathbb{R}^n$. Then for each i we have that $(\mathbf{I}_n + vv^T)u_i = u_i + 0v = u_i$. So, we have that 1 is repeated $n - 1$ times as an eigenvalue of $\mathbf{I}_n + vv^T$, and its corresponding eigenspace is the span of $u_2, \ldots, u_n$.

Since the determinant of a matrix is equal to the product of its eigenvalues, we can thus conclude that $\det(\mathbf{I}_n + vv^T) = 1 + |v|^2$. ■

It follows that $\sqrt{\det\left((\mathrm{D}G)^T(\mathrm{D}G)\right)} = \sqrt{1 + |\nabla g|^2}$.

As a side note, if $S : \mathbb{R}^N \to \mathbb{R}^N$ is an orthogonal map (which any permutation of coordinates in the codomain would be), then:

$$\begin{aligned}(\mathrm{D}(S \circ G))^T(\mathrm{D}(S \circ G)) = (S(\mathrm{D}G))^T(S(\mathrm{D}G)) &= (\mathrm{D}G)^T S^T S(\mathrm{D}G) \\ &= (\mathrm{D}G)^T(\mathrm{D}G)\end{aligned}$$

So, we can also now conclude that the prior reasoning also holds for any graph where the dependent variable is x_i even when $i \neq N$.

Calculating a normal vector to a graph is also really easy.

Proof:
Let $V \subseteq \mathbb{R}^{N-1}$, $g : V \to \mathbb{R}$, and $G : V \to \mathbb{R}^N$ be the function defined by:

$$G(x_1, \ldots, \widehat{x_i}, \ldots, x_N) = (x_1, \ldots, x_{i-1}, g(x_1, \ldots, \widehat{x_i}, \ldots, x_N), x_{i+1}, \ldots, x_N).$$

We need our normal vector v to be orthogonal to each column of $\mathrm{D}G$ (which I've written on the next page). As it turns out, a vector that works is:

$$(-\partial_{x_1} g, \ldots, -\partial_{x_{i-1}} g, 1, -\partial_{x_i} g, \ldots, -\partial_{x_{N-1}} g).$$

Also the norm of this vector is $\sqrt{1+|\nabla g|^2}$. So we can divide by this to normalize our normal vector. Also, depending on whether the regular domain D is below or above the graph G, we can negate or not negate our vector in order to make sure it points outwards away from D.

Note: the derivative matrix of G is:

$$\mathrm{D}G = \begin{bmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ \partial_{x_1} g & \cdots & \partial_{x_{i-1}} g & \partial_{x_i} g & \cdots & \partial_{x_{N-1}} g & \\ & & & 1 & & & \\ & & & & \ddots & & \\ & & & & & 1 & \end{bmatrix}.$$

As a side note, the prior theorem plus the prior calculation would let us prove that for a regular domain $D \subseteq \mathbb{R}^N$, the map $p \mapsto v(p)$ (sending $p \in \partial D$ to the unit outward pointing normal vector to $T_p(\partial D)$) is continuous on a neighborhood about any point in ∂D. As a result, the map $p \mapsto v(p)$ is globally continuous from ∂D to $\mathbb{R}^N$. Hence, it is also measurable.

Theorem: Let $D \subseteq \mathbb{R}^N$ be a bounded C^1 regular domain without corners (where $N \geq 2$). Then suppose there exists an open set $\Omega \subseteq \mathbb{R}^N$ containing D such that $f \in C^1(\Omega, \mathbb{R})$. For every $i \in \{1, \ldots, N\}$ we have that $\int_D \partial_{x_i} f(x) \mathrm{d}x = \int_{\partial D} f(p) v_i(p) \mathrm{d}S(p)$.

Here we write $v(p) = (v_1(p), \ldots, v_N(p))$ for all $p \in \partial D$.

Proof:

To start off, without loss of generality we can assume f is compactly supported. After all, as D is compact we can find precompact open sets E_1 and E_2 such that $D \subseteq E_1 \subseteq \overline{E_1} \subseteq E_2 \subseteq \overline{E_2} \subseteq \Omega$, as well as a C^∞ function $\phi : \mathbb{R}^N \to [0,1]$ such that $\phi \equiv 1$ on $\overline{E_1}$ and $\mathrm{supp}(\phi) \subseteq \overline{E_2} \subseteq \Omega$.

But now $\partial_{x_i} \phi \equiv 0$ on D and $\phi \equiv 1$ on D. Hence, we can calculate that:

- $\int_D \partial_{x_i}(f\phi)(x) \mathrm{d}x = \int_D \phi(x) \partial_{x_i} f(x) + 0 \mathrm{d}x = \int_D \partial_{x_i} f(x) \mathrm{d}x$;
- $\int_{\partial D} (f\phi)(p) v_i(p) \mathrm{d}S(p) = \int_{\partial D} f(p) v_i(p) \mathrm{d}S(p)$.

Thus, we might as well replace f with $f\phi$.

Next, for every $p \in \partial D$ we can pick open sets $U_p \subseteq \mathbb{R}^N$, $V \subseteq \mathbb{R}^{N-1}$, and a C^1 function $g_p : V \to \mathbb{R}$ with the properties described in the theorem on *pages 1170-1171*. But now because ∂D is compact, we know that it is covered by finitely many of the U_p. Hence, we can pick finitely many $p_1, \ldots, p_n$ such that $D \subseteq D^\circ \cup U_{p_1} \cup \cdots \cup U_{p_n}$. Then, we can pick a C^∞ partition of unity $\phi, \phi^{(1)}, \ldots, \phi^{(n)}$ subordinate to the sets $D^\circ, U_{p_1}, \ldots, U_{p_n}$ respectively such that $\phi + \sum_{j=1}^n \phi^{(j)} \equiv 1$ on D.

But now let $u = \phi f$ and $u^{(j)} = \phi^{(j)} f$. Then:

$$f = u + \sum_{j=1}^{n} u^{(j)} \text{ and } \partial_{x_i} f = \partial_{x_i} u + \sum_{j=1}^{n} \partial_{x_i} u^{(j)}.$$

Also, $\text{supp}(u) \subseteq D^\circ$ and $\text{supp}(u^{(j)}) \subseteq U_{p_j}$ for each j. In turn, we have that:

- $\int_D \partial_{x_i} f(x) \mathrm{d}x = \int u(x) \mathrm{d}x + \sum_{j=1}^{n} \int_{D \cap U_{p_j}} \partial_{x_i} u^{(j)}(x) \mathrm{d}x.$
- $\int_{\partial D} f(q) v_i(q) \mathrm{d}S(q) = \sum_{j=1}^{n} \int_{\partial D \cap U_{p_i}} u^{(j)}(q) v_i(q) \mathrm{d}S(q).$

Calculation 1: $\int_{D^\circ} \partial_{x_i} u(x) \mathrm{d}x = 0$

Why?

By Fubini's theorem and the fundamental theorem of calculus, we get that:

$$\begin{aligned} \int \partial_{x_i} u(x) \mathrm{d}x &= \int_{\mathbb{R}^{N-1}} \left(\int_{-\infty}^{\infty} \partial_{x_i} u(x) \mathrm{d}x_i \right) \mathrm{d}(x_1, \ldots, \widehat{x_i}, \ldots, x_N) \\ &= \int_{\mathbb{R}^{N-1}} 0 \mathrm{d}(x_1, \ldots, \widehat{x_i}, \ldots, x_N) = 0. \end{aligned}$$

Calculation 2: For each j we have that $\int_{D \cap U_{p_j}} \partial_{x_i} u^{(j)}(x) \mathrm{d}x = \int_{\partial D \cap U_{p_i}} u^{(j)}(q) v_i(q) \mathrm{d}S(q)$.

Why?

Recall that for some $k \in \{1, \ldots, N\}$ we have that $U^{p_j} \cap \partial D$ is parametrized by the function:

$$G(x_1, \ldots, \widehat{x_k}, \ldots, x_N) = (x_1, \ldots, x_{k-1}, g(x_1, \ldots, \widehat{x_k}, \ldots, x_N), x_{k+1}, \ldots, x_N)$$

Furthermore, we can guarentee that one of the two following things is true:

- $U_{p_j} \cap D = \{(x_1, \ldots, x_N) \in U_{p_j} : x_k \leq g_{p_j}(x_1, \ldots, \widehat{x_k}, \ldots, x_N)\}$
- $U_{p_j} \cap D = \{(x_1, \ldots, x_N) \in U_{p_j} : x_k \geq g_{p_j}(x_1, \ldots, \widehat{x_k}, \ldots, x_N)\}.$

Easy case: Suppose $i = k$. Then the first of two possibilities is that:

$$v_i(G(y)) = \frac{1}{\sqrt{1 + |\nabla g(y)|^2}},$$

and by fundamental theorem of calculus plus the fact that $u^{(j)}$ is compactly supported in U_{p_j}, we can conclude that:

$$\begin{aligned} \int_{D \cap U_{p_j}} \partial_{x_i} u^{(j)}(x) \mathrm{d}x &= \int_V \int_{-\infty}^{g(x_1, \ldots, \widehat{x_i}, \ldots, x_n)} \partial_{x_i} u^{(j)}(x_1, \ldots, x_n) \mathrm{d}x_i \mathrm{d}(x_1, \ldots, \widehat{x_i}, \ldots, x_n) \\ &= \int_V u^{(j)}(G(y)) - 0 \mathrm{d}y \\ &= \int_V u^{(j)}(G(y)) v_i(G(y)) \sqrt{1 + |\nabla g(y)|^2} \mathrm{d}y \\ &= \int_{\partial D \cap U_{p_j}} u^{(j)}(q) v_i(q) \mathrm{d}S(q) \end{aligned}$$

The other possibility is that $v_i(G(y)) = \frac{-1}{\sqrt{1 + |\nabla g(y)|^2}}$ and:

$$\int_{D \cap U_{p_j}} \partial_{x_i} u^{(j)}(x) \mathrm{d}x = \int_V \int_{g(x_1, \ldots, \widehat{x_i}, \ldots, x_n)}^{\infty} \partial_{x_i} u^{(j)}(x_1, \ldots, x_n) \mathrm{d}x_i \mathrm{d}(x_1, \ldots, \widehat{x_i}, \ldots, x_n)$$

But then our calculation just introduces two minus signs which cancel each other out.

Hard case 1: Suppose $i < k$. Then we have that $v_i(G(y)) = \mp \frac{\partial_{y_i} g(y)}{\sqrt{1 + |\nabla g(y)|^2}}$.

(Specifically, the sign is $-$ if D lies below the graph G and the sign is $+$ if D lies above the graph G.)

In order to keep notation cleaner, let $\theta : V \times \mathbb{R} \to \mathbb{R}^N$ be defined by:
$$\theta(y_1, \dots, y_{N-1}, s) = (y_1, \dots, y_{k-1}, s, y_{k+1}, \dots, y_N).$$

Then for all $y \in V$ and $s \in \mathbb{R}$ define $\eta(\theta(y,s)) := u^{(j)}(\theta(y, g(y) + s))$. In other words, we have that:
$$\eta(x_1, \dots, x_{k-1}, s, x_{k+1}, \dots, x_N) = \\ u^{(j)}(x_1, \dots, x_{k-1}, g(x_1, \dots, \widehat{x_k}, \dots, x_N) + s, x_{k+1}, \dots, x_N).$$

Importantly, η is continuously differentiable. Also, η is the composition of $u^{(j)}$ and the inverse map of a homeomorphism. So, $u^{(j)}$ being compactly supported implies that η is as well. By chain rule, we can calculate that:
$$(\partial_{x_i}\eta)(\theta(y,s)) = (\partial_{x_i}u^{(j)})(\theta(y, g(y)+s)) + (\partial_{x_k}u^{(j)})(\theta(y, g(y)+s)) \cdot \partial_{y_i} g(y)$$

In turn, in the case that D lies below the graph G we have that:
$$\begin{aligned}\int_{D \cap U_{p_j}} \partial_{x_i} u^{(j)}(x) \mathrm{d}x &= \int_V \int_{-\infty}^{g(y)} (\partial_{x_i} u^{(j)})(\theta(y, x_k)) \mathrm{d}x_k \mathrm{d}y \\ &= \int_V \int_{-\infty}^{0} (\partial_{x_i} u^{(j)})(\theta(y, g(y)+s)) \mathrm{d}s\mathrm{d}y \\ &= \int_V \int_{-\infty}^{0} (\partial_{x_i} \eta)(\theta(y,s)) \mathrm{d}s\mathrm{d}y \\ &\quad - \int_V \int_{-\infty}^{0} (\partial_{x_k} u^{(j)})(\theta(y, g(y)+s)) \cdot \partial_{y_i} g(y) \mathrm{d}s\mathrm{d}y.\end{aligned}$$

(As for the case that D lies above the graph G, we instead have that:
$$\begin{aligned}\int_{D \cap U_{p_j}} \partial_{x_i} u^{(j)}(x) \mathrm{d}x &= \int_V \int_{0}^{\infty} (\partial_{x_i} \eta)(\theta(y,s)) \mathrm{d}s\mathrm{d}y \\ &\quad - \int_V \int_{0}^{\infty} (\partial_{x_k} u^{(j)})(\theta(y, g(y)+s)) \cdot \partial_{y_i} g(y) \mathrm{d}s\mathrm{d}y.)\end{aligned}$$

But now I claim that $\int_V \int_{-\infty}^{0} (\partial_{x_i}\eta)(\theta(y,s)) \mathrm{d}s\mathrm{d}y = 0$. To see why, for all $(y_1, \dots, \widehat{y_i}, \dots, y_N) \in \mathbb{R}$ let $V_{(y_1, \dots, \widehat{y_i}, \dots, y_{N-1})} := \{y_i \in \mathbb{R} : (y_1, \dots, y_{N-1}) \in V\}$. Then by fubini's theorem, we have that:
$$\begin{aligned}&\int_V \int_{-\infty}^{0} (\partial_{x_i}\eta)(\theta(y,s)) \mathrm{d}s\mathrm{d}y \\ &= \int_{\mathbb{R}^{N-2}} \int_{V_{(y_1, \dots, \widehat{y_i}, \dots, y_{N-1})}} \int_{-\infty}^{0} (\partial_{x_i}\eta)(\theta((y_1, \dots, y_{N-1}), s)) \mathrm{d}s\mathrm{d}y_i \mathrm{d}(y_1, \dots, \widehat{y_i}, \dots, y_{N-1}) \\ &= \int_{\mathbb{R}^{N-2}} \int_{-\infty}^{0} \int_{V_{(y_1, \dots, \widehat{y_i}, \dots, y_{N-1})}} (\partial_{x_i}\eta)(\theta((y_1, \dots, y_{N-1}), s)) \mathrm{d}y_i \mathrm{d}s \mathrm{d}(y_1, \dots, \widehat{y_i}, \dots, y_{N-1})\end{aligned}$$

But now $V_{(y_1, \dots, \widehat{y_i}, \dots, y_{N-1})}$ is an open subset of $\mathbb{R}$. So, there exists countably many intervals $\{I_\ell\}_{\ell \in \mathbb{N}}$ such that:
$$\begin{aligned}&\int_{V_{(y_1, \dots, \widehat{y_i}, \dots, y_{N-1})}} (\partial_{x_i}\eta)(\theta((y_1, \dots, y_{N-1}), s)) \mathrm{d}y_i \\ &\qquad = \sum_{\ell \in \mathbb{N}} \int_{I_\ell} (\partial_{y_i}\eta)(\theta((y_1, \dots, y_{N-1}), s)) \mathrm{d}y_i\end{aligned}$$

Then since $* \mapsto \eta(\theta(y_1, \dots, y_{i-1}, *, y_{i+1}, \dots, y_{N-1}, s))$ is compactly supported in $\bigcup_{\ell \in \mathbb{N}} I_\ell$, we know by the fundamental theorem of calculus that:
$$\sum_{\ell \in \mathbb{N}} \int_{I_\ell} (\partial_{y_i}\eta)(\theta((y_1, \dots, y_{N-1}), s)) \mathrm{d}y_i = 0$$

So, $\int_V \int_{-\infty}^{0} (\partial_{x_i}\eta)(\theta(y,s)) \mathrm{d}s\mathrm{d}y = \int_{\mathbb{R}^{N-2}} \int_{-\infty}^{0} 0 \mathrm{d}s \mathrm{d}(y_1, \dots, \widehat{y_i}, \dots, y_{N-1}) = 0.$

(Or in the case that D lies above the graph G, we instead have that:
$\int_V \int_{0}^{\infty} (\partial_{x_i}\eta)(\theta(y,s)) \mathrm{d}s\mathrm{d}y = \int_{\mathbb{R}^{N-2}} \int_{0}^{\infty} 0 \mathrm{d}s \mathrm{d}(y_1, \dots, \widehat{y_i}, \dots, y_{N-1}) = 0.$)

And now in the case that D lies below the graph G, we have that:

$$\begin{aligned}\int_{D\cap U_{p_j}} \partial_{x_i} u^{(j)}(x)\mathrm{d}x &= \int_V \left(\int_{-\infty}^{0} (\partial_{x_k} u^{(j)})(\theta(y, g(y)+s))\mathrm{d}s\right)\cdot -\partial_{y_i} g(y)\mathrm{d}y \\ &= \int_V \left(\int_{-\infty}^{g(y)} (\partial_{x_k} u^{(j)})(\theta(y, x_k))\mathrm{d}x_k\right)\cdot -\partial_{y_i} g(y)\mathrm{d}y \\ &= \int_V (u^{(j)}(G(y)) - 0)\cdot -\partial_{y_i} g(y)\mathrm{d}y = \int_{U_{p_j}\cap \partial D} u^{(j)}(q) v_i(q)\mathrm{d}S(q)\end{aligned}$$

(Alternatively, if D lies below the graph G then:

$$\begin{aligned}\int_{D\cap U_{p_j}} \partial_{x_i} u^{(j)}(x)\mathrm{d}x &= \int_V \left(\int_{0}^{\infty} (\partial_{x_k} u^{(j)})(\theta(y, g(y)+s))\mathrm{d}s\right)\cdot -\partial_{y_i} g(y)\mathrm{d}y \\ &= \int_V \left(\int_{g(y)}^{\infty} (\partial_{x_k} u^{(j)})(\theta(y, x_k))\mathrm{d}x_k\right)\cdot -\partial_{y_i} g(y)\mathrm{d}y \\ &= \int_V (u^{(j)}(0 - G(y)))\cdot -\partial_{y_i} g(y)\mathrm{d}y = \int_{U_{p_j}\cap \partial D} u^{(j)}(q) v_i(q)\mathrm{d}S(q).)\end{aligned}$$

Hard Case 2: Suppose $k < i$. The only difference between this case and the prior case is that now the indexing is slightly fucked up. After all, we have that:

$$v_i(G(y)) = \mp \frac{\partial_{y_{i-1}} g(y)}{\sqrt{1+|\nabla g(y)|^2}}.$$

(Specifically, the sign is $-$ if D lies below the graph G and the sign is $+$ if D lies above the graph G.)

But also when we define η and θ as before, we now get that:

$$(\partial_{x_i}\eta)(\theta(y,s)) = (\partial_{x_i} u^{(j)})(\theta(y, g(y)+s)) + (\partial_{x_k} u^{(j)})(\theta(y, g(y)+s))\cdot \partial_{y_{i-1}} g(y)$$

And now the entire rest of the proof goes identically as in the prior case.

Putting the prior two calculations together, we can now conclude the desired theorem. ■

10/3/2026

A weak version of the divergence theorem is a direct consequence of the prior theorem. Recall from *pages 829-830* some notations for derivative operators.

Corollary (Divergence Theorem): Let $D \subseteq \mathbb{R}^N$ (where $N \geq 2$) be a bounded C^1 regular domain without corners. Then suppose there exists an open set $\Omega \subseteq \mathbb{R}^N$ containing D and a map $f \in C^1(\Omega, \mathbb{R}^N)$. If $v(p)$ denotes the outward pointing normal vector to ∂D at a point $p \in \partial D$, we have that:

$$\int_D \nabla \cdot F \mathrm{d}x = \int_{\partial D} F \cdot v \mathrm{d}S$$

Proof:
Write $F = (F_1, \ldots F_N)$. Then by the prior theorem:

$$\int_D \nabla \cdot F \mathrm{d}x = \sum_{i=1}^N \int_D \partial_{x_i} F_i \mathrm{d}x = \sum_{i=1}^N \int_\partial DF_i v_i \mathrm{d}S = \int_\partial DF \cdot v \mathrm{d}S.$$ ■

As a side note, if $N = 1$, $[a, b] \subseteq \mathbb{R}$, we let $v(b) := 1$ and $v(a) := -1$, and we equip $\partial([a, b]) = \{a, b\}$ with the counting measure, then the divergence theorem becomes just the fundamental theorem of calculus. We'll use these conventions when dealing with regular domains of $\mathbb{R}$.

Since we have a formulation of divergence theorem, we can also prove Green's identities:

Theorem: Let $D \subseteq \mathbb{R}^N$ be a C^1 regular domain without corners.

(a) (Green's 1st Identity): Let $f \in C^1(D)$ and $g \in C^2(D)$ be scalar-valued functions. Then $\int_{\partial D} f \frac{\partial g}{\partial v} \mathrm{d}S = \int_D \nabla f \cdot \nabla g \mathrm{d}x + \int_D f \triangle g \mathrm{d}x$.

(Here, a function h being in $C^k(D)$ is used to mean that there exists an open set $\Omega \subseteq \mathbb{R}^N$ containing D such that $f \in C^k(\Omega)$).

Proof:
Note that $\nabla \cdot (f \nabla g) = \sum_{i=1}^N (f_{x_i} g_{x_i} + f g_{x_i, x_i}) = (\nabla f \cdot \nabla g) + f \triangle g$. In turn, we can just integrate and apply divergence theorem.

(b) (Green's 2nd Identity): Let $f, g \in C^2(D)$ be scalar-valued fnctions. Then:

$$\int_{\partial D} f \frac{\partial g}{\partial v} - g \frac{\partial f}{\partial v} \mathrm{d}S = \int_D (f \triangle g - g \triangle f) \mathrm{d}x$$

Proof:
Apply the first identity to $\int_{\partial D} f \frac{\partial g}{\partial v} \mathrm{d}S$ and $\int_{\partial D} g \frac{\partial f}{\partial v} \mathrm{d}S$ separately and then take their difference. ■

For the sake of being able to do calculations, I do need to actually know how to compute the manifold surface measure of at least a sphere. Fortunately, I can show as follows that I already know how to do that.

In math 240a we showed there's a unique Borel measure σ on $S^{N-1} := \partial B_1(0) \subseteq \mathbb{R}^N$ with $\int_E \mathrm{d}x = \int_0^\infty \int_{S^{N-1}} \chi_E(rx') \mathrm{d}\sigma(x') r^{N-1} \mathrm{d}r$ for any measurable set $E \subseteq \mathbb{R}^N - \{0\}$.
Here χ_E denotes the characteristic function of the set E.

Lemma: The manifold surface measure S on S^{N-1} (where $N \geq 2$) is precisely equal to σ.

Proof:
To start off, I'll denote by S_ε the manifold surface measure of $\partial B_\varepsilon(0)$. My first claim is that $S_\varepsilon(\partial B_\varepsilon(0)) = \varepsilon^{N-1} \sigma(S^{N-1})$.

Why?
First note that if $x \in \partial B_\varepsilon(0)$ then $|x| = \varepsilon$ and the outward pointing unit normal vector is $\frac{x}{|x|} = \varepsilon^{-1} x$. Also note that $x \mapsto x$ is a C^∞ map from $\mathbb{R}^N$ to $\mathbb{R}^N$. Therefore, by the divergence theorem we have that:

$$\begin{aligned} \varepsilon S_\varepsilon(\partial B_\varepsilon(0)) &= \int_{\partial B_\varepsilon(0)} \varepsilon \mathrm{d}S_\varepsilon(x) \\ &= \int_{\partial B_\varepsilon(0)} x \cdot \tfrac{x}{|x|} \mathrm{d}S_\varepsilon(x) = \int_{B_\varepsilon(0)} \nabla \cdot x \mathrm{d}x \\ &= N \int_{B_\varepsilon(0)} \mathrm{d}x = N \sigma(S^{N-1}) \int_0^\varepsilon r^{N-1} \mathrm{d}r \\ &= \varepsilon^N \sigma(S^{N-1}). \end{aligned}$$

We can divide both sides by ε to get that $S_\varepsilon(\partial B_\varepsilon(0)) = \varepsilon^{N-1}\sigma(S^{N-1})$.

The prior result is useful because it now lets us show for any $f \in C^1(S^{N-1}, \mathbb{R})$ that:
$$\int_{S^{N-1}} f(x')\mathrm{d}S(x') = \int_{S^{N-1}} f\mathrm{d}\sigma(x').$$

Why?
To start off, note that given such an f, we can define a function $g \in C^1(\mathbb{R}^N - \{0\}, \mathbb{R})$ by $g(x) = f(\frac{x}{|x|})$. This map is C^1 because it is a composition of C^1 maps. Also importantly, we know it stays constant as one travels radially outwards or inwards. And when $x \in S^{N-1}$ we know $f(x) = g(x)$.

But now for all $x \in S^{N-1}$ we have that $|x| = 1$ and the outward pointing normal vector from x is x. Thus by the divergence theorem, we have for any $\varepsilon > 0$ that:
$$\begin{aligned}\int_{S^{N-1}} f(x')\mathrm{d}S(x') &= \int_{S^{N-1}} g(x')x' \cdot x'\mathrm{d}S(x') \\ &= \int_{\varepsilon \le |x| \le 1} \nabla \cdot (g(x)x)\mathrm{d}x + \int_{\partial B_\varepsilon(0)} (g(x')x') \cdot (\varepsilon^{-1}x')\mathrm{d}S_\varepsilon(x')\end{aligned}$$

> Beware that because g is not continuously extendable to 0, we can only apply the divergence theorem to a solid shell and not the entire unit ball.

But now note that $\nabla \cdot (g(x)x) = Ng(x) + x \cdot \nabla g(x)$. Since $g(x)$ is constant as one travels radially, we know that $x \cdot \nabla g(x) = 0$. Therefore, we can now calculate via polar coordinates that:
$$\begin{aligned}\int_{\varepsilon \le |x| \le 1} \nabla \cdot (g(x)x)\mathrm{d}x &= N\int_{\varepsilon \le |x| \le 1} g(x)\mathrm{d}x \\ &= N\int_\varepsilon^1 \int_{S^{N-1}} g(rx')\mathrm{d}\sigma(x')r^{N-1}\mathrm{d}r \\ &= N\int_\varepsilon^1 \int_{S^{N-1}} f(x')\mathrm{d}\sigma(x')r^{N-1}\mathrm{d}r \\ &= N\int_\varepsilon^1 r^{N-1}\mathrm{d}r \int_{S^{N-1}} f(x')\mathrm{d}\sigma(x') \\ &= (1-\varepsilon^N)\int_{S^{N-1}} f(x')\mathrm{d}\sigma(x')\end{aligned}$$

Meanwhile, note that since S^{N-1} is compact, we know that $|f|$ (and in turn $|g|$) is bounded from above by some constant $M \ge 0$. So,
$$\begin{aligned}\left|\int_{\partial B_\varepsilon(0)} (g(x)x) \cdot (\varepsilon^{-1}x)\mathrm{d}S_\varepsilon(x)\right| &\le \int_{\partial B_\varepsilon(0)} \varepsilon|g(x)|\mathrm{d}S_\varepsilon(x) \\ &\le \varepsilon M S_\varepsilon(\partial B_\varepsilon(0)) = \varepsilon^N M\sigma(S^{N-1})\end{aligned}$$

And now we can just take $\varepsilon \to 0$ in order to prove that:
$$\int_{S^{N-1}} f(x')\mathrm{d}S(x') = \int_{S^{N-1}} f(x')\mathrm{d}\omega(x').$$

Having proven the prior result, we are now basically done. After all, given any open set $U \subseteq S^{N-1}$ we can let $\{K_n\}_{n\in\mathbb{N}}$ be a sequence of compact sets whose union is U. And then we can inductively pick C^1 maps $f_n \in C^1(S^{N-1}, [0,1])$ such that $\mathrm{supp}(f_n) \subseteq U$ and $f_n \equiv 1$ on $(\bigcup_{m=1}^n K_m) \cup (\bigcup_{m=1}^{n-1} \mathrm{supp}(f_m))$. But now by the monotone convergence theorem, we have that:
$$S(U) = \lim_{n\to\infty} \int_{S^{N-1}} f_n(x')\mathrm{d}S(x') = \lim_{n\to\infty} \int_{S^{N-1}} f_n(x')\mathrm{d}\sigma(x') = \sigma(U).$$

Next since both S and σ are Radon measures (they are both finite on a second countable LCH space), them equaling on open sets implies they equal on all Borel sets. ■

As a side note, if we consider the surface measure on the 0-dimensional manifold $S^0 = \{-1, 1\}$ to be the counting measure, then this result also trivially holds. So, we shall make that a convention.

I also want to be able to apply affine transformations to my manifolds before integrating. So, I'll also prove the following results.

Lemma: Suppose M is a k-dimensional C^r embedded submanifold of $\mathbb{R}^N$ without corners and $\Phi : \mathbb{R}^N \to \mathbb{R}^N$ is a C^r diffeomorphism. Then $M' := \phi(M)$ is also a k-dimensional C^r embedded submanifold of $\mathbb{R}^N$.

Proof:
The fact that M' is an embedded submanifold of $\mathbb{R}^N$ is just a result of proposition 5.2 on *pages 1111-1112*. In particular, given a smooth chart (U, φ) on M containing a point p we have that $(\Phi(U), \varphi \circ \Phi^{-1})$ is a smooth chart on M' containing the point p'. Or in other words, if $X : V \subseteq \mathbb{R}^k \to \mathbb{R}^N$ is a C^r parametrization for M, then we have that $(X^{-1} \circ \Phi^{-1})^{-1} = \Phi \circ X : V \to \mathbb{R}^N$ is also a C^r parametrization of M' by proposition 5.23 on *page 1125*. ■

Next suppose $\Phi(x) := Ax + a$ where A is an orthogonal matrix and $a \in \mathbb{R}^N$. This is a C^∞ diffeomorphism and I claim the following result...

Lemma 2: Let M and M' be defined as in the prior lemma. Also let S_M be the surface measure on M and let $S_{M'}$ be the surface measure on M'. Then for any borel set $E \subseteq M$ we have that $S_{M'}(\Phi(E)) = S_M(E)$ (or equivalently $S_{M'}(E) = S_M(\Phi^{-1}(E))$ for all $E \in \mathcal{B}_{M'}$).

Proof:
By a result I showed on *pages 98-99*, it suffices to prove this lemma for any Borel set E contained in the image of some C^r parametrization $X : V \to \mathbb{R}^N$. But now note that:

$$S_M(E) = \int_{X^{-1}(E)} \sqrt{\det(\mathrm{D}X^T \mathrm{D}X)}\mathrm{d}x$$

Meanwhile:

$$\begin{aligned} S_{M'}(\Phi(E)) &= \int_{(\Phi \circ X)^{-1}(\Phi(E))} \sqrt{\det((\mathrm{D}(\Phi \circ X))^T \mathrm{D}(\Phi \circ X))}\mathrm{d}x \\ &= \int_{X^{-1}(E)} \sqrt{(\mathrm{D}X)^T A^T A \mathrm{D}X}. \end{aligned}$$

And because A is orthogonal, $A^T A$ cancels and we are also left with:

$$S_{M'}(\Phi(E)) = \int_{X^{-1}(E)} \sqrt{\det(\mathrm{D}X^T \mathrm{D}X)}\mathrm{d}x = S_M(E). \blacksquare$$

Similarly, we can also consider the map $\Phi(x) := cx$ where $c \in \mathbb{R}$. This is also a C^∞ diffeomorphism.

Lemma 3: Let M and M' be k-dimensional defined as in two lemmas ago. Also let S_M be the surface measure on M and let $S_{M'}$ be the surface measure on M'. Then for any borel set $E \subseteq M$ we have that $S_{M'}(\Phi(E)) = |c|^k S_M(E)$ (or equivalently $S_{M'}(E) = \frac{1}{|c|^k} S_M(\Phi^{-1}(E))$ for all $E \in \mathcal{B}_{M'}$).

The idea of this proof is identical to that in lemma 2. ■

Just for convenience sake, I also want to list a few results here from Folland that I proved in math 240a. Here, m will denote the Lebesgue measure on $\mathbb{R}^N$ and σ will denote the surface measure of S^{N-1}.

- $\sigma(S^{N-1}) = \frac{2\pi^{N/2}}{\Gamma(N/2)}$.
- $m(B_1(0)) = \frac{\pi^{N/2}}{\Gamma(1+\frac{N}{2})} = \frac{1}{N}\sigma(S^{N-1})$. (My 231 professor refers to this number as $\alpha(N)$.)
- If $N \geq 0$ then $\Gamma(N) = (N-1)!$.
- If $N \geq 0$ then $\Gamma(N+\frac{1}{2}) = (N-\frac{1}{2})(N-\frac{3}{2})\cdots(\frac{1}{2})\sqrt{\pi}$.

10/4/2026

I should be working on other classes right now but I want to finally take notes on my PDE class.

Math 231A Lecture Notes:
Like in the undergrad PDE class, in this class gradients, laplacians, divergences, and etc. will all be assumed to only be with respect to the spacial variables.

We first address the Laplace equation on $\mathbb{R}^n$. Namely we want to find a function $u \in C^2(\mathbb{R}^n)$ such that $-\triangle u \equiv 0$. To find such a u, we first consider that the Laplacian is invariant under rotations and reflections.

Proof:
Suppose T is a linear isometry. (Hence $|\det(T)| = 1$ and $T^{-1} = T^*$. Also here I'm using a "$*$" to denote taking a matrix transpose.) Then letting $\widehat{f}$ or $f^\wedge$ denote the Fourier transform of a function (which I am taking to be $\widehat{f}(\xi) := \int f(x)e^{-2\pi i\xi\cdot x}$), recall from *page 919* that for any Schwartz function $u : \mathbb{R}^n \to \mathbb{R}$ that:

$$(\triangle(u\circ T))^\wedge(\xi) = -4\pi^2|\xi|^2(u\circ T)^\wedge(\xi) = -4\pi^2|T(\xi)|^2\widehat{u}(T(\xi)) = (\triangle u)^\wedge \circ T(\xi).$$

But now by applying the inverse Fourier transform, we get that:

$$\triangle(u\circ T)(x) = ((\triangle u)^\wedge \circ T)^\vee(x) = ((\triangle u)^\wedge)^\vee \circ T(x) = (\triangle u)\circ T(x)$$

Thus in the special case of a Schwartz function u, we are done. The next logical step is to show that $\triangle(F\circ T) = (\triangle F)\circ T$ for any distribution $F \in \mathcal{D}'(\mathbb{R}^n)$.

Let $\phi \in C_c^\infty$ be a test function. Then recalling *pages 959-974* for a reminder on how we work with distributions, we know that:

$$(\triangle(F\circ T),\phi) = (F\circ T, \triangle\phi) = |\det(T)|^{-1}(F, (\triangle\phi)\circ T^{-1}).$$

Yet because ϕ is Schwartz, we then know that:

$$\begin{aligned}|\det(T)|^{-1}(F,(\triangle\phi)\circ T^{-1}) = |\det(T)|^{-1}(F,\triangle(\phi\circ T^{-1})) &= |\det(T)|^{-1}(\triangle F, \phi\circ T^{-1})\\ &= ((\triangle F)\circ T,\phi)\end{aligned}$$

This also proves that classically $\triangle$ is rotation / reflection invariant for every C^2 function u. After all, since $\triangle(u\circ T)$ and $(\triangle u)\circ T$ defining the same distributions means that they equal a.e. And since they are continuous, that means they are equal everywhere. ■

Based on the prior result, it would seem a good idea to look for radial solutions. In other words, we shall try to find a solution $u(x) = v(r)$ where $v : [0, \infty) \to \mathbb{R}$ is a function and $r = |x|$.

Fortunately though, note that $\frac{\partial r}{\partial x_i} = \frac{x_i}{r}$. In turn, $\partial_{x_i} u(x) = v'(r)\frac{x_i}{r}$. And then differentiating again we get that:

$$\partial^2_{x_i} y(x) = v''(r)\frac{x_i^2}{r^2} + v'(r)\left(\frac{1}{r} - \frac{x_i^2}{r^3}\right)$$

So, $0 = \triangle u = v''(r) + (\frac{n-1}{r})v'(r)$. Or in other words:

$$(\ln|v'|)'(r) = \frac{v''(r)}{v'(r)} = \frac{1-n}{r}$$

By taking antiderivatives we get that:

$$\ln|v'| = (1-n)\ln(r) + C = \ln(r^{1-n}) + C.$$

So, $|v'| = Cr^{1-n}$. By allowing C to be negative we can just say $v' = Cr^{1-n}$. Finally, we can take one more antiderivative to get that:

$$v(r) = \begin{cases} C_1 \ln(r) + C_2 & \text{if } n = 2 \\ \frac{C_1}{r^{n-2}} + C_2 & \text{if } n \geq 3. \end{cases}$$

Based on the prior reasoning, we define a function $\Phi : \mathbb{R}^n - \{0\} \to \mathbb{R}$ by:

$$\Phi(x) = \begin{cases} \frac{-1}{2\pi}\ln|x| & \text{if } n = 2 \\ \frac{1}{n(n-2)\alpha(n)} \cdot \frac{1}{|x|^{n-2}} & \text{if } n \geq 3. \end{cases}$$

This is called the fundamental solution of $-\triangle$.

The reason this is called the fundamental solution will become obvious when I pick up this subject again on *page 1184*. However, until then I will just note that Φ is locally integrable on $\mathbb{R}^n$.

Math 290A Lecture Notes:

Everything in this class will be building off notes I already took while studying Munkres' Topology on *pages 117-130*. That said, the class uses slightly different notation. I'll list that notation here:

- Given a topological space X and a point $a \in X$, $\pi_1(X, a)$ shall denote the first fundamental group of X based at a.
- Given any path $\gamma : [0, 1] \to X$ (where X is a topological space), I'll denote by $[\gamma]$ the path homotopy equivalence class of $\gamma : [0, 1] \to X$. The composition of paths γ (going from a to b) and σ (going from b to c) will be denoted $\gamma \cdot \sigma$. Also, the reversed parametrization of a path γ will be denoted $\overline{\gamma}$.
- If $\gamma(0) = a$ and $\gamma(1) = b$ then we define the map $\gamma_\sharp : \pi_1(X, a) \to \pi_1(X, b)$ by $[f] \mapsto [\overline{\gamma} \cdot f \cdot \gamma]$.

To start off, when I was introducing singular homology before getting distracted by vector calculus, I made a point of trying to give a category theoretic perspective to homology groups. I'd also like to give a category theoretic perspective to the 1st fundamental group.

We let $\mathbf{Top}_*$ be the category consisting of pairs (X, a) where X is a topological space and $a \in X$ is some element. Also, we say a morphism from (X, a) to (Y, b) is a continuous function $f : X \to Y$ such that $f(a) = b$. Checking that this is a category is easy.
Next note that we can define a functor $\pi_1 : \mathbf{Top}_* \to \mathbf{Grp}$ as follows:

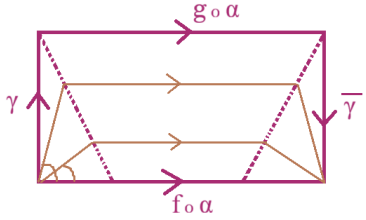
> Map any pair (X, a) to the 1st fundamental group $\pi_1(X, a)$. Also, associate to any morphism $f : (X, a) \to (Y, b)$ the map $f_* : \pi_1(X, a) \to \pi_2(Y, b)$ given by $f_*([\gamma]) := [f \circ \gamma]$. Theorem 52.4 on *page 120* shows that this is a well-defined functor.

It's obvious that for any homeomorphism $f : X \to Y$ we'll have that $f_* : \pi_1(X, a) \to \pi_1(Y, f(a))$ is a group isomorphism. I'd like to prove that this also holds if we weaken f to only being a homotopy equivalence.

Theorem: Suppose X and Y are topological spaces and $f, g : X \to Y$ are continuous maps that are homotopic by a continuous map $H : X \times [0, 1] \to Y$. Fix $a \in X$, and let $\gamma : [0, 1] \to Y$ be the function defined by $\gamma(t) = H(a, t)$. Then we have the following commuting diagram:

$$\begin{array}{ccc} & & \pi_1(Y, f(a)) \\ & \overset{f_*}{\nearrow} & \wr \downarrow \gamma_\sharp \\ \pi_1(X, a) & \xrightarrow[g_*]{} & \pi_1(Y, g(a)) \end{array}$$

> Proof:
> Suppose $[\alpha] \in \pi_1(X, a)$. Then we have that $(\gamma_\sharp \circ f_*)([\alpha]) = \gamma_\sharp([f \circ \alpha]) = [\overline{\gamma} \cdot (f \circ \alpha) \cdot \gamma]$.
> So to prove the theorem we need to show that $[\overline{\gamma} \cdot (f \circ \alpha) \cdot \gamma] = [g \circ \alpha]$.
> Equivalently, it suffices to show that:
> $$[f \circ \alpha] = [\gamma \cdot (g \circ \alpha) \cdot \overline{\gamma}]$$
>
> Fortunately, we can define a continuous map $H_0 : [0, 1] \times [0, 1] \to Y$ by:
> $$H_0(s, t) := H(\alpha(s), t).$$
>
> This will be a homotopy (but not a path homotopy necessarily) from $f \circ \alpha$ to $g \circ \alpha$ because we have that:
> $H_0(s, 0) = (f \circ \alpha)(s)$, $H_0(s, 1) = (g \circ \alpha)(s)$, and $H_0(0, t) = H_0(1, t) = \gamma(t)$.
>
> And now we can reparametrize the homotopy H_0 as shown below to get a path homotopy from $f \circ \alpha$ to $\gamma \cdot (g \circ \alpha) \cdot \overline{\gamma}$.



One way of doing this is to define a reparametrization $k : [0,1]^2 \to [0,1]^2$ by:

$$k(s,t) \coloneqq \begin{cases} (s - st, 3st) & \text{if } 0 \leq s \leq \frac{1}{3} \\ (s + 2st - t, t) & \text{if } \frac{1}{3} \leq s \leq \frac{2}{3} \\ (s + t - st, (3-3s)t) & \text{if } \frac{2}{3} \leq s \leq 1 \end{cases}$$

By pasting lemma we can see that k is a continuous map. Also, $\widetilde{H_0}(s,t) \coloneqq H_0(k(s,t))$ is a path homotopy from $f \circ \alpha$ to $\gamma \cdot (g \circ \alpha) \cdot \overline{\gamma}$. ■

Technically $\gamma \cdot (g \circ \alpha) \cdot \overline{\gamma}$ was formally defined as having a different parametrization from what $s \mapsto \widetilde{H_0}(s,1)$ would give. That said, reparametrizing a curve does not change its path homotopy equivalence class.

More specifically, given any path $\alpha : [0,1] \to X$ and any continuous map $k : [0,1] \to [0,1]$ such that $k(0) = 0$ and $k(1) = 1$, we have that $\alpha \simeq_p \alpha \circ k$ via the homotopy $H(s,t) \coloneqq \alpha((1-t)s + tk(s))$.

Theorem: If $f : X \to Y$ and $g : Y \to X$ are inverse homotopy equivalences and $x_0 \in X$, then $f_* : \pi_1(X, x_0) \to \pi_1(Y, f(x_0))$ is an isomorphism.

Proof:
Let $y_1 = f(x_0)$, let $x_2 = g(y_1)$, and let $y_3 = f(x_2)$. Then by the prior theorem we know there are group isomorphisms ϕ and ψ such that the following commuting diagrams holds:

$$\begin{array}{ccccc} & & & & \pi_1(Y,y_1) \\ & & \overset{(\mathrm{Id}_Y)_* = \mathrm{Id}_{\pi_1(Y,y_1)}}{\nearrow} & & \uparrow \psi \wr \\ \pi_1(Y,y_1) & \xrightarrow[g_*]{} & \pi_1(X,x_2) & \xrightarrow[f_*]{} & \pi_1(Y,y_3) \end{array}$$

$$\begin{array}{ccccc} \pi_1(X,x_0) & \xrightarrow{f_*} & \pi_1(Y,y_1) & \xrightarrow{g_*} & \pi_1(X,x_2) \\ & \underset{(\mathrm{Id}_X)_* = \mathrm{Id}_{\pi_1(X,x_0)}}{\searrow} & & & \downarrow \phi \wr \\ & & & & \pi_1(X,x_0) \end{array}$$

Yet note that top and bottom commuting diagrams show that $g_* : \pi_1(Y,y_1) \to \pi_1(X,x_2)$ must be injective and surjective respectively. Hence it is a bijection (and thus a group isomorphism). Then using that information, I claim that $(f_*)^{-1} = \phi \circ g_*$. After all, it was already clear that $(\phi \circ g_*) \circ f_* = \mathrm{Id}_{\pi_1(X,x_0)}$. In turn, we have that:

$f_*([\alpha]) = (\phi \circ g_*)^{-1}([\alpha])$ for all $[\alpha] \in \pi_1(X, x_0)$. ■

10/6/2026

Math 231A Lecture Notes:

I'm going to now return to discussing the fundamental solution of $-\triangle$. Scroll back three pages for a reminder on its definition.

The reason Φ is called the fundamental solution of $-\triangle$ is that we can calculate that in the weak sense, $-\triangle\Phi = \delta_0$ where δ_x denotes the dirac delta function at x.

Case when $n = 2$:
Recalling that $\Phi(x) = \frac{-1}{2\pi}\ln|x|$ define for any $\varepsilon \geq 0$ the function:

$$\Phi_\varepsilon(x) = \frac{-1}{4\pi}\ln(|x|^2 + \varepsilon^2)$$

Note that Φ_ε and Φ are both locally integrable. Also $\Phi_\varepsilon \to \Phi$ on $\mathbb{R}^2 - \{0\}$ as $\varepsilon \to 0$ (where specifically $|\Phi_\varepsilon|$ is increasing on $B_1(0)$ as $\varepsilon \to 0$ and decreasing on $B_1(0)^C$ as $\varepsilon \to 0$). So, we can show by exerciese 9.1(b) on *pages 963-964* that $\Phi_\varepsilon \to \Phi$ in $\mathcal{D}'$. And since partial derivatives are continuous on $\mathcal{D}'$, we can conclude that $-\triangle\Phi_\varepsilon \to -\triangle\Phi$.

Yet also after some manipulations, we can compute for all $\varepsilon > 0$ and $x \in \mathbb{R}^2 - \{0\}$ that:

$$-\triangle\Phi_\varepsilon(x) = \frac{1}{\pi}\left(\frac{\varepsilon}{|x|^2+\varepsilon^2}\right)^2 = \frac{1}{\pi}\left(\frac{\varepsilon}{\varepsilon^2|\varepsilon^{-1}x|^2+\varepsilon^2}\right)^2 = \frac{1}{\pi}\left(\frac{1}{\varepsilon|\varepsilon^{-1}x|^2+\varepsilon}\right)^2 = \frac{1}{\varepsilon^2}\cdot\frac{1}{\pi}\left(\frac{1}{|\varepsilon^{-1}x|^2+1}\right)^2.$$

In other words, we can let $g(x) = \frac{1}{\pi}\left(\frac{1}{|x|^2+1}\right)^2$. And then we have that:

$$-\triangle\Phi_\varepsilon(x) = \frac{1}{\varepsilon^2}g(\varepsilon^{-1}x).$$

Also note that $\int_{\mathbb{R}^2}\frac{1}{\pi}\left(\frac{1}{|x|^2+1}\right)^2 dx = \int_0^\infty\left(\frac{1}{r^2+1}\right)^2 2r\,dr = \int_1^\infty\frac{1}{u^2}du = 1$. Therefore, we can say by proposition 9.1 on *page 960* that $-\triangle\Phi_\varepsilon \to \delta_0$ in $\mathcal{D}'$.

Case when $n > 2$:
We can adapt the prior reasoning but with different functions Φ_ε and g. Or alternatively, I can show this through more a direct path (which we could have also taken in the $n = 2$ case by the way).

Let $\psi \in C_c^2(\mathbb{R}^n)$. We have to be careful because Φ is not differentiable at the origin. That said, since ψ is compactly supported, we can choose some $R > 0$ such that $\text{supp}(\psi) \subseteq B_{R/2}(0)$. And now for any $0 < \varepsilon < 1$ we have that:

$$\int_{\mathbb{R}^2}\Phi\cdot(-\triangle\psi)dx = \int_{B_\varepsilon(0)}\Phi\cdot(-\triangle\psi)dx + \int_{\{\varepsilon\leq|x|\leq R\}}\Phi\cdot(-\triangle\psi)dx$$

Now as $\triangle\psi \in C_c(\mathbb{R}^n)$, there is a constant $M \geq 0$ such that $|-\triangle\psi| \leq M$. And now we have that:

$$\left|\int_{B_\varepsilon(0)}\Phi\cdot(-\triangle\psi)dx\right| \leq M\int_{B_\varepsilon(0)}|\Phi|dx = M\int_0^\varepsilon\frac{1}{(n-2)}\cdot\frac{r^{n-1}}{r^{n-2}}dr = \frac{M\varepsilon^2}{2(n-2)}$$

So as $\varepsilon \to 0$ we have that $\int_{B_\varepsilon(0)}\Phi\cdot(-\triangle\psi)dx \to 0$.

Meanwhile, by Green's first identity we have that:

$$\int_{\{\varepsilon\leq|x|\leq R\}}\Phi\cdot(-\triangle\psi)dx = -\int_{\partial B_R(0)}\Phi(x')\nabla\psi(x')\cdot\frac{x'}{|x'|}dS(x') + \int_{\partial B_\varepsilon(0)}\Phi(x')\nabla\psi(x')\cdot\frac{x'}{|x'|}dS(x') + \int_{\{\varepsilon\leq|x|\leq R\}}\nabla\Phi\cdot\nabla\psi dx$$

Based on how we chose R, we know that $-\int_{\partial B_R(0)}\Phi(x')\nabla\psi(x')\cdot\frac{x'}{|x'|}dS(x') = 0$. So, our expression simplifies to:

$$\int_{\{\varepsilon\leq|x|\leq R\}}\Phi\cdot(-\triangle\psi)dx = \int_{\partial B_\varepsilon(0)}\Phi(x')\nabla\psi(x')\cdot\frac{x'}{|x'|}dS(x') + \int_{\{\varepsilon\leq|x|\leq R\}}\nabla\Phi\cdot\nabla\psi dx.$$

But now because $\nabla\psi(x')$ is continuous and nonzero only on a compact set, we can find some constant $M' \geq 0$ such that $|\nabla\psi| \leq M'$. And now we have that:

$$\left|\int_{\partial B_\varepsilon(0)} \Phi(x')\nabla\psi(x') \cdot \frac{x'}{|x'|} \mathrm{d}S(x')\right| \leq \frac{M'}{n(n-2)\alpha(n)\varepsilon^{n-2}} \cdot \varepsilon^{n-1} n\alpha(n) = \frac{M'\varepsilon}{n-2}$$

The latter goes to 0 as $\varepsilon \to 0$. Hence to recap, we have that:

$$\int_{\mathbb{R}^2} \Phi \cdot (-\triangle\psi)\mathrm{d}x = \lim_{\varepsilon\to 0} \int_{\{\varepsilon \leq x \leq R\}} \nabla\Phi \cdot \nabla\psi \mathrm{d}x$$

By another application of Green's first identity, we have that:

$$\int_{\{\varepsilon\leq|x|\leq R\}} \nabla\Phi \cdot \nabla\psi \mathrm{d}x = \int_{\partial B_R(0)} \psi(x')\nabla\Phi(x') \cdot \frac{x'}{|x'|}\mathrm{d}S(x') - \int_{\partial B_\varepsilon(0)} \psi(x')\nabla\Phi(x') \cdot \frac{x'}{|x'|}\mathrm{d}S(x') - \int_{\varepsilon\leq|x|\leq R} \psi(x')\psi\triangle\Phi \mathrm{d}x.$$

$\int_{\partial B_R(0)} \psi(x')\nabla\Phi(x') \cdot \frac{x'}{|x'|}\mathrm{d}S(x')$ cancels though because $\psi \equiv 0$ on $\partial B_R(0)$. Similarly, $\int_{\varepsilon\leq|x|\leq R} \psi(x')\psi\triangle\Phi\mathrm{d}x$ cancels because we can compute through a straightforward computation that $\triangle\Phi \equiv 0$ on $\mathbb{R}^n - \{0\}$. Hence, we are left with:

$$\int_{\mathbb{R}^2} \Phi \cdot (-\triangle\psi)\mathrm{d}x = -\lim_{\varepsilon\to 0} \int_{\partial B_\varepsilon(0)} \psi(x')\nabla\Phi(x') \cdot \frac{x'}{|x'|}\mathrm{d}S(x')$$

We can compute that $\nabla\Phi(x') = \frac{-1}{n\alpha(n)} \cdot \frac{x'}{|x'|^n}$. In turn, we get that:

$$-\int_{\partial B_\varepsilon(0)} \psi(x')\nabla\Phi(x') \cdot \frac{x'}{|x'|}\mathrm{d}S(x') = \frac{1}{n\alpha(n)\varepsilon^{n-1}} \int_{\partial B_\varepsilon(0)} \psi(x')\mathrm{d}x'$$

But now as ψ is continuous, we can say that:

$$\frac{1}{n\alpha(n)\varepsilon^{n-1}} \int_{\partial B_\varepsilon(0)} \psi(x')\mathrm{d}x' \to \psi(0) \text{ as } \varepsilon \to 0$$

Why?
We can write $\psi(x) = \psi(0) + u(x)$ where $u(x) \to 0$ as $x \to 0$. And then as $S(\partial B_\varepsilon(0)) = n\alpha(n)\varepsilon^{n-1}$, we have that:

$$\frac{1}{n\alpha(n)\varepsilon^{n-1}} \int_{\partial B_\varepsilon(0)} \psi(x')\mathrm{d}x' = \psi(0) + \frac{1}{n\alpha(n)\varepsilon^{n-1}} \int_{\partial B_\varepsilon(0)} u(x')\mathrm{d}x'$$

Meanwhile, $\left|\frac{1}{n\alpha(n)\varepsilon^{n-1}} \int_{\partial B_\varepsilon(0)} u(x')\mathrm{d}x'\right| \leq \max_{x\in B_\varepsilon(0)} |u(x)| \to 0$ as $\varepsilon \to 0$.

So, we've finished showing that $\int_{\mathbb{R}^2} \Phi \cdot (-\triangle\psi)\mathrm{d}x = \psi(0)$. ■

As mentioned already, I could have done both of the prior approaches in the opposite cases I actually did them in. The approach I took in the $n > 2$ case has the advantage that I only had to assume ψ is twice differentiable instead of infinitely differentiable. Although, I think it's computationally less neat.

Anyways, this then leads us to the following theorem:

Theorem: Assume $f \in C_c^2(\mathbb{R}^n)$ (where $n \geq 2$). Then the function $u = \Phi * f$ satisfies that $u \in C^2(\mathbb{R}^n)$ and $-\triangle u = f$ in $\mathbb{R}^n$.

Proof:
To see that $u \in C^2(\mathbb{R}^n)$, we just need to note that Φ is locally integrable and f is compactly supported and in C^2. So by Folland exercise 8.7 (which I did for homework in math 240c), we know $u \in C^2(\mathbb{R}^n)$.

Also by the prior reasoning, plus some Leibniz rule spam, we can conclude that:
$$-\triangle u = \int_{\mathbb{R}^n} \Phi(y) \cdot -\triangle f(x-y)\mathrm{d}x = f(x-0) = f(x). \blacksquare$$

Some properties of harmonic functions: (Also for all these results assume $n \geq 2$)

Mean Value Formula: Suppose $U \subseteq \mathbb{R}^n$ is an open set and $u \in C^2(U)$ satisfies that $\triangle u = 0$. Then for any $x \in U$ and $r > 0$ such that $\overline{B_r(x)} \subseteq U$ we have that:
$$u(x) = \frac{1}{nr^{n-1}\alpha(n)} \int_{\partial B_r(x)} u(y)\mathrm{d}S(y) = \frac{1}{r^n\alpha(n)} \int_{B_r(x)} u(y)\mathrm{d}y$$

Proof:
To start off, note for any r small enough so that $\overline{B_r(x)} \subseteq U$ as well as any measurable function $u : U \to \mathbb{R}$ we have that:
$$\begin{aligned}\frac{1}{nr^{n-1}\alpha(n)} \int_{\partial B_r(x)} u(y)\mathrm{d}S(y) &= \frac{1}{nr^{n-1}\alpha(n)} \int_{\partial B_r(0)} u(y+x)\mathrm{d}S(y) \\ &= \frac{1}{n\alpha(n)} \int_{\partial B_1(0)} u(ry+x)\mathrm{d}S(y).\end{aligned}$$

All of the prior equalities can be checked first for simple functions, then nonnegative functions, and then general complex or extended-real valued functions. Hence, we can now define:
$$\phi(r) \coloneqq \frac{1}{nr^{n-1}\alpha(n)} \int_{\partial B_r(x)} u(y)\mathrm{d}S(y).$$

By Leibniz's rules, we will get that:
$$\begin{aligned}\phi'(r) &= \frac{1}{n\alpha(n)} \int_{\partial B_1(0)} \nabla u(ry+x) \cdot y \mathrm{d}S(y) \\ &= \frac{1}{n\alpha(n)r^{n-1}} \int_{\partial B_r(x)} \nabla u(y) \cdot \tfrac{y-x}{r} \mathrm{d}S(y) \\ &= \frac{1}{n\alpha(n)r^{n-1}} \int_{B_r(x)} \triangle u(y)\mathrm{d}y = 0 \text{ (by divergence theorem)}\end{aligned}$$

Therefore ϕ is constant. And since $\phi(r) \to u(x)$ as $r \to 0$ because u is continuous, we can conclude that $\phi(r) = u(x)$ for all r.

With that we can also now conclude the other desired formula because:
$$\begin{aligned}\frac{1}{r^n\alpha(n)} \int_{B_r(x)} u(y)\mathrm{d}y &= \frac{1}{r^n\alpha(n)} \int_0^r \rho^{n-1} \int_{\partial B_1(0)} u(\rho y' + x)\mathrm{d}S(y')\mathrm{d}\rho \\ &= \frac{1}{r^n\alpha(n)} \int_0^r \int_{\partial B_\rho(x)} u(y')\mathrm{d}S(y')\mathrm{d}\rho \\ &= \frac{1}{r^n\alpha(n)} \int_0^r n\rho^{n-1}\alpha(n)u(x)\mathrm{d}\rho = u(x). \blacksquare\end{aligned}$$

A function $u : U \to \mathbb{R}$ satisfying the formula $u(x) = \frac{1}{nr^{n-1}\alpha(n)} \int_{\partial B_r(x)} u(y)\mathrm{d}S(y)$ whenever $\overline{B_r(x)} \subseteq U$ is said to have the mean value property. Note that the reasoning above generalizes the theorem on *page 804*.

Theorem: If $U \subseteq \mathbb{R}^n$ is an open set and $u \in C^2(U, \mathbb{R})$ satisfies the mean value property then $\triangle u = 0$ on U.

Proof:
Assume to the contrary that there exists $x_0 \in U$ such that $\triangle u(x_0) \neq 0$. Then define $\phi(r) \coloneqq \frac{1}{nr^{n-1}\alpha(n)} \int_{\partial B_r(x_0)} u(y)\mathrm{d}S(y)$ like in the proof of the prior result. Trivially, we now

have that $\phi'(r) = 0$ (since $\phi \equiv u(x_0)$ is a constant function). Yet we can also calculate identically to before that $\phi'(r) = \frac{1}{n\alpha(n)r^{n-1}} \int_{B_r(x_0)} \triangle u(y)\mathrm{d}y$. Since $\triangle u$ is a continuous function, we can shrink r so that eventually $u(y) \neq 0$ for any $y \in B_r(x_0)$ and $u(y)$ and $u(x_0)$ share a sign for all $y \in B_r(x_0)$. And now we have a contradiction. ■

(Regularity of Harmonic Functions): Suppose $U \subseteq \mathbb{R}^n$ is an open set and $u \in C(U, \mathbb{R})$ satisfies the mean value property. Then $u \in C^\infty(U, \mathbb{R})$.

Proof:
Let $\{V_j\}_{j\in\mathbb{N}}$ be an increasing sequence of precompact open sets in U whose union is all of U. If we can show that u restricted to each V_j is smooth, then we will be done.

Fortunately, for any fixed j we can find some constant $R > 0$ such that $|x - y| > R$ for all $x \in \overline{V_j}$ and $y \in U^{\mathsf{C}}$. And now we let $\eta \in C_c^\infty(B_R(0))$ be a radial nonnegative bump function with $\int \eta = 1$. In other words, there exists some smooth compactly supported function $\widetilde{\eta} : [0, R) \to [0, \infty)$ such that $\eta(x) = \widetilde{\eta}(|x|)$ and $\int_0^R r^{n-1}\widetilde{\eta}(r)\mathrm{d}r = 1$.

We can convolve u and η to get a well-defined smooth function $f : V_j \to \mathbb{R}$ defined by $f(x) = \int u(x - y)\eta(y)\mathrm{d}y$. Yet now note by an easy change of variables that:

$$\begin{aligned} f(x) = \int_{B_R(0)} u(x + y)\eta(-y)\mathrm{d}y &= \int_0^R \int_{\partial B_1(0)} u(x + rz)\widetilde{\eta}(r)r^{n-1}\mathrm{d}S(z)\mathrm{d}r \\ &= \int_0^R \widetilde{\eta}(r)r^{n-1} \cdot \int_{\partial B_r(x)} u(z)\mathrm{d}S(z)\mathrm{d}r \\ &= u(x) \int_0^R \widetilde{\eta}(r)r^{n-1}\mathrm{d}r = u(x) \cdot 1 \end{aligned}$$

Thus, $u = f$ on V_j and we've actually shown that u is a smooth function on V_j. ■

As a side note, because smoothness of harmonicity are local properties, the combination of the prior three results also shows that if the mean value property holds locally for a continuous function, then the function is smooth and harmonic globally. So we have the following theorem.

Theorem: Given an open set $U \subseteq \mathbb{R}^n$ and a continuous function $f : U \to \mathbb{R}$, the following are equivalent:

- f is C^2 and harmonic.
- f is C^∞ and harmonic.
- f locally has the mean value property.

By basically identical reasoning as on *page 804*, we can show that if $U \subseteq \mathbb{R}^n$ is a connected open set and $u : U \to \mathbb{R}$ is a continuous function satisfying the mean value property, then u satisfies the maximal principle. In other words, if there exists $a \in U$ with $u(a) \geq u(z)$ for all $z \in G$ then u is a constant function.

In a similar vein, we can then prove analogous results to three results starting at the *bottom of page 805* (using identical reasoning). I don't feel like proving these things since there are no new ideas.

Since we know harmonic functions are smooth, the next leap I want to make is that harmonic functions are (real) analytic.

Theorem: Assume $U \subseteq \mathbb{R}^n$ is an open set and $u : U \to \mathbb{R}$ is harmonic. Then for all $x_0 \in U$, $r > 0$ such that $\overline{B_r(x_0)} \subseteq U$, and all multi-indices α with $|\alpha| = k$, we have that:

$$|\partial^\alpha u(x_0)| \leq \frac{C_k}{r^{n+k}} \|u\|_{L^1(B_r(x_0))}$$

where $C_0 = \frac{1}{\alpha(n)}$ and $C_k = \frac{(2^{n+1}nk)^k}{\alpha(n)}$ for all $k \geq 1$.

Proof:

We proceed by induction. Firstly, when $k = 0$ then $u(x_0) = \frac{1}{r^n \alpha(n)} \int_{B_r(x_0)} u$ by the mean value theorem. Therefore:

$$|u(x_0)| \leq \frac{1}{r^n \alpha(n)} \|u\|_{L^1(B_r(x_0))} = \frac{C_0}{r^{n+k}} \|u\|_{L^1(B_r(x_0))}.$$

Next, since u is smooth in addition to being harmonic, partial derivative operators commute. Therefore, for any $1 \leq i \leq n$ we have that $\partial_{x_i} u$ is harmonic on U since:

$$0 = \partial_{x_i}(0) = \partial_{x_i} \triangle u = \triangle \partial_{x_i} u$$

And now upon writing $y = (y_1, \ldots, y_n)$ for any $y \in \mathbb{R}^n$, we can apply the mean value theorem in order to get that:

$$\begin{aligned}
|\partial_{x_i} u(x_0)| &= \left| \frac{2^n}{r^n \alpha(n)} \int_{B_{\frac{r}{2}}(x_0)} \partial_{x_i} u(y) \mathrm{d}y \right| \\
&= \frac{2^n}{r^n \alpha(n)} \left| \int_{\partial B_{\frac{r}{2}}(x_0)} u(y) \frac{2y_i}{r} \mathrm{d}S(y) \right| \text{ (See the theorem on } \textit{pages 1173-1176}\text{)} \\
&\leq \frac{2^n}{r^n \alpha(n)} \int_{\partial B_{\frac{r}{2}}(x_0)} |u(y)| \, \mathrm{d}S(y) \\
&\leq \frac{2^n}{r^n \alpha(n)} \cdot \frac{nr^{n-1}\alpha(n)}{k^{n-1}} \cdot \|u\|_{L^\infty(B_{\frac{r}{2}}(x_0))} = \frac{2n}{r} \|u\|_{L^\infty(B_{\frac{r}{2}}(x_0))}
\end{aligned}$$

But now by induction plus the fact that $\overline{B_{\frac{r}{2}}(y)} \subseteq U$ for all $y \in \overline{B_{\frac{r}{2}}(x_0)}$, we can conclude that:

$$\begin{aligned}
\frac{2n}{r} \|u\|_{L^\infty(B_{\frac{r}{2}}(x_0))} \leq \frac{2n}{r} \cdot \frac{2^n}{\alpha(n) r^n} \sup_{y \in B_{\frac{r}{2}}(x_0)} \|u\|_{L^1(B_{\frac{r}{2}}(y))} &\leq \frac{2^{n+1} n}{\alpha(n)} \cdot \frac{1}{r^{n+1}} \|u\|_{L^1(B_r(x_0))} \\
&\leq \frac{C_1}{r^{n+1}} \|u\|_{L^1(B_r(x_0))}
\end{aligned}$$

Finally, we get to the induction step. Suppose α is a multi-index of order $k - 1$ (where $k \geq 2$) and then let $1 \leq i \leq n$. By identical reasoning as before, we can conclude that $\partial_{x_i} \partial^\alpha u$ is harmonic. So, we can apply the mean value theorem identically to before in order to show that $|\partial_{x_i} \partial^\alpha u(x_0)| \leq \frac{nk}{r} \|\partial^\alpha u\|_{L^\infty(B_{\frac{r}{k}}(x_0))}$.

But now by induction plus the fact that $\overline{B_{\frac{r(k-1)}{k}}(y)} \subseteq U$ for all $y \in \partial B_{\frac{r}{k}}(x_0)$, we can conclude that:

$$\|\partial^\alpha u\|_{L^\infty(B_{\frac{r}{2}}(x_0))} \leq C_{k-1} \left(\frac{k}{r(k-1)} \right)^{n+k-1} \sup_{y \in B_{\frac{r}{k}}(x_0)} \|u\|_{L^1(B_{\frac{r(k-1)}{k}}(y))}$$

$$\leq \frac{(2^{n+1}n(k-1))^{k-1}}{\alpha(n)} \cdot \frac{k^{n+k-1}}{r^{n+k-1}(k-1)^{n+k-1}} \|u\|_{L^1(B_r(x_0))}$$
$$= \frac{(2^{n+1}nk)^{k-1}}{\alpha(n)} \cdot \frac{1}{r^{n+k-1}} \left(\frac{k}{k-1}\right)^n \|u\|_{L^1(B_r(x_0))}$$

So, we get that:

$$|\partial_{x_i}\partial^\alpha u(x_0)| \leq \frac{nk}{r} \cdot \frac{(2^{n+1}nk)^{k-1}}{\alpha(n)} \cdot \frac{1}{r^{n+k-1}} \left(\frac{k}{k-1}\right)^n \|u\|_{L^1(B_r(x_0))}$$
$$\frac{(2^{n+1}nk)^k}{\alpha(n)} \cdot \frac{1}{r^{n+k}} \cdot \|u\|_{L^1(B_r(x_0))} = \frac{C_k}{r^{n+k}} \|u\|_{L^1(B_r(x_0))}. \blacksquare$$

Corollary (Liouville's Theorem): Suppose $u : \mathbb{R}^n \to \mathbb{R}$ is harmonic and bounded on $\mathbb{R}^n$. Then u is constant.

Proof:
We shall show for every $1 \leq i \leq n$ that $\partial_{x_i} u \equiv 0$. So, let $C > 0$ be a constant such that $|u(x)| \leq C$ for all $x \in \mathbb{R}^n$. Then by the prior theorem, for any $x \in \mathbb{R}^n$ and $r > 0$ we have that:

$$|\partial_{x_i} u(x)| \leq \frac{2^{n+1}n}{\alpha(n)r^{n+1}} \|u\|_{L^1(B_r(x))} \leq \frac{2^{n+1}n}{\alpha(n)r^{n+1}} \cdot Cr^n\alpha(n) = \frac{2^{n+1}nC}{r}$$

And now we just take $r \to 0$ to get that $|\partial_{x_i} u(x)| = 0$. Since all the partial derivatives of u are identically zero on $\mathbb{R}^n$, this proves that u is constant. $\blacksquare$

Corollary 2: Let $f \in C_c^2(\mathbb{R}^n)$ (where $n \geq 3$). Then any bounded solution of $-\triangle u = f$ on $\mathbb{R}^n$ has the form $u = (\Phi * f) + C$ (where Φ is the fundamental solution of $-\triangle$ and C is a real constant).

Proof:
To start off, we can easily see that $v := \Phi * f$ is a bounded function. Plus, by the theorem on *page 1185* we know that it does solve $-\triangle v = f$.

Now supposing u is another bounded solution, we then know that $w = u - v$ is bounded and solves that $-\triangle w = 0$ on $\mathbb{R}^n$. So by the prior corollary, we know $w \equiv C$ for some constant. And then $u = v + C$. $\blacksquare$

I'm about finally ready to prove that harmonic functions are analytic. But first there are two things I need to go over.

Firstly, what do I mean when I say a function $f : U \to \mathbb{R}$ is analytic?

Suppose $f : U \to \mathbb{R}$ is a smooth function and $c \in U$. For any $k \geq 0$ we define the kth Taylor polynomial of f centered at c by the expression:

$$P_k(x) = \sum_{|\alpha| \leq k} \frac{\partial^\alpha f(c)}{\alpha!}(x-c)^\alpha$$

Notably, P_k will have all the same derivatives as f at c up to order k.

We shall say that $f \in C^\infty(\mathbb{R}^n, \mathbb{R})$ is analytic at c if there exists some $r > 0$ such that $P_k(x) \to f(x)$ for all $x \in B_r(c)$. Equivalently, this means that the remainder formula $R_k(x) := f(x) - P_{k-1}(x)$ must go to 0 on $B_r(c)$ as $k \to \infty$. If f is analytic at every $c \in U$ we say f is analytic. I will hopefully deal with more of the loose ends of this definition in the future as I learn about functions of multiple complex variables.

Secondly, I need to prove an approximation which will undoubtedly come up many more times in my life. I'm going to follow the steps layed out by an exercise in Folland Real Analysis. This is a proof from probability theory so I will be using probability notation.

Exercise 10.21: Let $\{X_n\}_{n\in\mathbb{N}}$ be a sequence of independent random variables, each having the Poisson distribution with mean 1.

In other words, the probability distribution on $\mathbb{R}$ induced by each X_i is the measure:

$$\lambda_1 := \frac{1}{e}\sum_{k=0}^{\infty}\frac{1}{k!}\delta_k.$$

Define $S_n := \sum_{i=1}^{n} X_i$ and $Y_n := \max\left(\frac{n-S_n}{\sqrt{n}}, 0\right)$.

(a) $E(Y_n) = e^{-n}\sum_{k=0}^{n-1}\left(\frac{n-k}{\sqrt{n}}\cdot\frac{n^k}{k!}\right) = \frac{n^{n+\frac{1}{2}}e^{-n}}{n!}$.

Proof:

To start off, note that the probability distribution of S_n is the Poisson distribution $\lambda_n := \frac{1}{e^n}\sum_{k=0}^{\infty}\frac{n^k}{k!}\delta_k$.

In fact more generally, if $X_1, \ldots, X_n$ are independent and have Poisson distributions with means $a_1, \ldots, a_n$ respectively then $X_1 + \ldots + X_n$ has a Poisson distribution with mean $a_1 + \ldots + a_n$. For a proof of this see my math 180a notes.

And now we can calculate that $E(Y_n) = \sum_{k=0}^{n-1}\frac{n-k}{\sqrt{n}}P(S_n = k) = e^{-n}\sum_{k=0}^{n-1}\left(\frac{n-k}{\sqrt{n}}\cdot\frac{n^k}{k!}\right)$.

To prove the other equation, it suffices to show that:

$$\sum_{k=0}^{n-1}\left(\frac{n-k}{\sqrt{n}}\cdot\frac{n^k}{k!}\right) = \frac{n^{n+\frac{1}{2}}}{n!}$$

Fortunately, obeserve that:

$$\begin{aligned}\sum_{k=0}^{n-1}\left(\frac{n-k}{\sqrt{n}}\cdot\frac{n^k}{k!}\right) = \sum_{k=0}^{n}\left(\frac{n-k}{\sqrt{n}}\cdot\frac{n^k}{k!}\right) &= \sum_{k=0}^{n}\frac{n^{k+1/2}}{k!} - \sum_{k=0}^{n}\frac{kn^{k-1/2}}{k!}\\ &= \sum_{k=0}^{n}\frac{n^{k+1/2}}{k!} - \sum_{k=1}^{n}\frac{n^{k-1/2}}{(k-1)!}\\ &= \sum_{k=0}^{n}\frac{n^{k+1/2}}{k!} - \sum_{k=0}^{n-1}\frac{n^{k+1/2}}{k!} = \frac{n^{n+1/2}}{n!}.\end{aligned}$$

(b) $P_{Y_n} \to \frac{1}{2}\delta_0 + \chi_{[0,\infty)}\nu_0^1$ vaguely (where ν_0^1 is the standard normal distribution $\frac{1}{\sqrt{2\pi}}e^{\frac{-t^2}{2}}\mathrm{d}t$).

Note that it makes sense to multiply $\chi_{[0,\infty)}$ and ν_0^1 since the measure ν_0^1 can be identified with a function in $L^1(\mathbb{R})$ (considered as a subspace of the set of complex measurs on $\mathbb{R}$).

Proof:

Let $F_n(x) := P_{Y_n}((-\infty, x])$ be the CDF function of Y_n for each n. Similarly, let $F(x) := (\frac{1}{2}\delta_0 + \chi_{[0,\infty)}\nu_0^1)((-\infty, x])$ be the CDF function of the probability distribution we claim the P_{Y_n} vaguely approach. Since all the measures we are dealing

with trivially have total measure 1, it suffices by *proposition 7.19 on pages 170-172* to show that $F_n(x) \to F(x)$ pointwise (Lebesgue)-a.e.

We can calculate that $F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{2} + \frac{1}{\sqrt{2\pi}} \int_0^x e^{\frac{-t^2}{2}} \mathrm{d}t & \text{if } x \geq 0. \end{cases}$

Meanwhile, note for each n that:

$$P_{Y_n} = \left(1 - e^{-n} \sum_{k=0}^{n-1} \frac{n^k}{k!}\right) \delta_0 + \sum_{k=0}^{n-1} (e^{-n} \frac{n^k}{k!}) \delta_{\frac{n-k}{\sqrt{n}}} = \left(e^{-n} \sum_{k=n}^{\infty} \frac{n^k}{k!}\right) \delta_0 + \sum_{k=0}^{n-1} (e^{-n} \frac{n^k}{k!}) \delta_{\frac{n-k}{\sqrt{n}}}$$

Its immediate that $F_n(x) = 0$ for all $x < 0$ and so trivially $F_n(x) \to F(x)$ if $x < 0$. Meanwhile observe that $\frac{n-k}{\sqrt{n}} \leq x$ iff $n - x\sqrt{n} \leq k$. Therefore, for any *fixed* $x \geq 0$ we have that:

$$F_n(x) = \left(e^{-n} \sum_{k=n}^{\infty} \frac{n^k}{k!}\right) + \sum_{k=\lceil n - x\sqrt{n} \rceil}^{n-1} e^{-n} \frac{n^k}{k!} = \left(e^{-n} \sum_{k=n+1}^{\infty} \frac{n^k}{k!}\right) + \sum_{k=\lceil n - x\sqrt{n} \rceil}^{n} e^{-n} \frac{n^k}{k!}.$$

To evaluate the limit of the expression as $n \to \infty$, we employ the following trick:

Note that $\frac{S_n - n}{\sqrt{n}} = \sum_{k=0}^{\infty} \left(e^{-n} \frac{n^k}{k!}\right) \delta_{\frac{k-n}{\sqrt{n}}} \to \nu_0^1$ vaguely by the central limit theorem.

Also note for any $a \in \mathbb{R}$ that $\frac{k-n}{\sqrt{n}} \leq a$ iff $k \leq n + a\sqrt{n}$. This has the consequence that:

- If $a = 0$ then $e^{-n} \sum_{k=0}^{n} \frac{n^k}{k!} = P(\frac{S_n - n}{\sqrt{n}} \leq 0) \to \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0} e^{\frac{-t^2}{2}} \mathrm{d}t = \frac{1}{2}$.
- If $a = -x$ then $e^{-n} \sum_{k=0}^{\lfloor n - x\sqrt{n} \rfloor} \frac{n^k}{k!} = P(\frac{S_n - n}{\sqrt{n}} \leq -x) \to \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{-x} e^{\frac{-t^2}{2}} \mathrm{d}t$.

In turn, as $e^{-n} \sum_{k=n+1}^{\infty} \frac{n^k}{k!} = 1 - e^{-n} \sum_{k=0}^{n} \frac{n^k}{k!}$ we can also conclude that:

$$e^{-n} \sum_{k=n+1}^{\infty} \frac{n^k}{k!} \to \frac{1}{2} \text{ as } n \to \infty$$

Similarly, we can conclude that:

$$\sum_{\lfloor n - x\sqrt{n} \rfloor + 1}^{n} e^{-n} \frac{n^k}{k!} \to \frac{1}{\sqrt{2\pi}} \int_{-x}^{0} e^{\frac{-t^2}{2}} \mathrm{d}t = \frac{1}{\sqrt{2\pi}} \int_{0}^{x} e^{\frac{-t^2}{2}} \mathrm{d}t \text{ as } n \to \infty$$

To finish off, note that $\lfloor n - x\sqrt{n} \rfloor + 1 = \lceil n - x\sqrt{n} \rceil$ iff $n - x\sqrt{n} \notin \mathbb{Z}$, and that happens iff $x\sqrt{n} \notin \mathbb{Z}$. Thus, so long as $x \notin \bigcup_{n \in \mathbb{N}} \frac{1}{\sqrt{n}} \mathbb{Z}$ (where the latter is a countable and thus null subset of $\mathbb{R}$), we have finished proving that $F_n(x) \to F(x)$ as $n \to \infty$.

(c) $\int t^2 \mathrm{d}P_{Y_n}(t) = E(Y_n^2) \leq 1$ for all n.

Proof:
To start off, note that $0 \leq Y_n^2 \leq (\frac{n - S_n}{\sqrt{n}})^2 = \frac{(S_n - n)^2}{n}$. As a result, we can say that

$E(Y_n^2) \leq \frac{1}{n}E((S_n - n)^2)$. Yet now note that since S_n has a Poisson distribution with mean n, we have that $E((S_n - n)^2)$ is equal to the variance of S_n (which will also equal n). So, $\frac{(S_n - n)^2}{n} = \frac{n}{n} = 1$.

(d) Conclude Stirling's approximation that $\lim_{n\to\infty} \frac{n^n\sqrt{2\pi n}}{n!e^n} \to 1$.

Proof:
Recall the moment convergence theorem on *pages 198-199*. By parts (b) and (c) of this problem in addition to that theorem, we are able to conclude that:

$$E(Y_n) \to E(\tfrac{1}{2}\delta_0 + \chi_{[0,\infty)}\nu_0^1) \text{ as } n \to \infty.$$

But now by part (a) we know that $E(Y_n) = \frac{n^n\sqrt{n}}{n!e^n}$. Meanwhile:

$$E(\tfrac{1}{2}\delta_0 + \chi_{[0,\infty)}\nu_0^1) = \int_0^\infty \frac{t}{\sqrt{2\pi}} e^{\frac{-t^2}{2}} \mathrm{d}t = \frac{1}{\sqrt{2\pi}} \int_0^\infty e^{-u} \mathrm{d}u = \frac{1}{\sqrt{2\pi}}$$

So, $\frac{n^n\sqrt{n}}{n!e^n} \to \frac{1}{\sqrt{2\pi}}$ as $n \to \infty$. ■

Finally, I'm ready for the big result I wanted to show.

Theorem: Suppose $U \subseteq \mathbb{R}^n$ is an open set and $u : U \to \mathbb{R}$ is a harmonic function. Then u is analytic on U.

Proof:
If $x_0 \in U$ then set $r = \frac{1}{4} \inf_{y \in U^c} |x_0 - y|$.

Importantly for all $x \in B_r(x_0)$ we have that $B_r(x) \subseteq B_{2r}(x_0)$ and also that the closure of $B_{2r}(x_0)$ is contained in U. Now let $M > 0$ be such that $|u(x)| \leq M$ for all $x \in \overline{B_{2r}(x_0)}$. By the theorem on *pages 1188-1189*, we know for any multi-index α that:

$$\|\partial^\alpha u\|_{L^\infty(B_r(x_0))} \leq \frac{(2^{n+1}n)^{|\alpha|}|\alpha|^{|\alpha|}}{\alpha(n)r^{n+|\alpha|}} \sup_{x \in B_r(x_0)} \|u\|_{L^1(B_r(x))} \leq M\frac{(2^{n+1}n)^{|\alpha|}|\alpha|^{|\alpha|}}{r^{|\alpha|}}$$

(Here when $|\alpha| = 0$ I'm interpretting $0^0 = 1$.)

Next, by Stirling's approximation we know there is a constant $C > 0$ such that $|\alpha|^{|\alpha|} \leq Ce^{|\alpha|}|\alpha|!$ for all multi-indices α.
(Actually fuck I'm realizing the professor is massively over complicating this. By looking at a series representation of $e^{|\alpha|}$ we can always see that $\frac{|\alpha|^{|\alpha|}}{|\alpha|!} \leq e^{|\alpha|}$. So, literally the constant $C = 1$ works.)

So, we have that $\|\partial^\alpha u\|_{L^\infty(B_r(x_0))} \leq M\left(\frac{2^{n+1}ne}{r}\right)^{|\alpha|}|\alpha|!$

For one more bound, note that $n^k = \sum_{|\alpha|=k} \frac{|\alpha|!}{\alpha!}$. So, we can conclude $|\alpha|! \leq \alpha! n^{|\alpha|}$ and in turn that:

$$\|\partial^\alpha u\|_{L^\infty(B_r(x_0))} \leq M\left(\frac{2^{n+1}n^2e}{r}\right)^{|\alpha|}\alpha!.$$

Based on that, I shall now only consider x such that $|x - x_0| < \frac{r}{2^{n+2}n^3e}$.

Consider the Nth remainder formula $R_N(x) = u(x) - \sum_{|\alpha| \leq N-1} \frac{\partial^\alpha u(x_0)}{\alpha!}(x - x_0)^\alpha$.

By using Taylor's theorem from math 140b, we can show for any fixed $x \in B_r(x_0)$ that there exists some $t \in (0, 1)$ such that:

$$R_N(x) = \sum_{|\alpha|=N} \frac{\partial^\alpha u(x_0 + t(x - x_0))}{\alpha!}(x - x_0)^\alpha$$

In turn, if $|x - x_0| < \frac{r}{2^{n+2}n^3e}$ then:

$$\begin{aligned} |R_N(x)| &\leq M \sum_{|\alpha|=N} \left(\frac{2^{n+1}n^2e}{r}\right)^N \left(\frac{r}{2^{n+2}n^3e}\right)^N \\ &= \frac{M}{(2n)^N} \sum_{|\alpha|=N} 1 = \frac{M}{2^N} \to 0 \text{ as } N \to \infty. \blacksquare \end{aligned}$$

One last comment I'd like to make is that in the $n = 1$ case every harmonic function has the form $u(x) = ax + b$. So harmonic functions on the real number line aren't that interesting and that's why I've been avoiding talking about them.

10/11/2026

Math 250A Homework:

Set 1 Problem 3: Firstly some notes on notation. Oprea denotes a "presheaf of sets" on a topological space X (i.e. a functor $\mathcal{F} : \mathbf{Op}(X)^{\text{op}} \to \mathbf{Set}$) by "$\mathcal{F} \to X$". Also Oprea denotes the restriction homomorphism $\mathcal{F}(U) \to \mathcal{F}(V)$ by r_{UV}.

(ii) Suppose M is a smooth manifold with at least 2 points, $\mathcal{F} \to M$ is the presheaf given by the assignment $\mathcal{F}(U) = \{$ constant functions on $U\}$ for all open sets U, and the restriction maps are given by domain restrictions. This $\mathcal{F}$ is called the presheaf of constant functions. Show that $\mathcal{F}$ is *not* a sheaf.

Since M is Hausdorff with two distinct points, we can find disjoint open sets U_1 and U_2. And now we let $U := U_1 \cup U_2$. Now if c_1 and c_2 are distinct constants and $f : U_1 \to \{c_1\}$ and $f_2 : U_2 \to \{c_2\}$ are functions, automatically we have that $f_1|_{U_1 \cap U_2} = f_2|_{U_1 \cap U_2}$ since $U_1 \cap U_2 = \emptyset$. Yet there is no constant function f on U such that $f|_{U_1} = f_1$ and $f|_{U_2} = f_2$ simultaneously.

(iii) Suppose X is a topological space and $\mathcal{F} \to X$ and $\mathcal{G} \to X$ are sheaves. A morphism $f : \mathcal{F} \to \mathcal{G}$ between sheaves is just a natural transformation between $\mathcal{F}$ and $\mathcal{G}$.

Prove that for any $p \in X$, the morphism $f : \mathcal{F} \to \mathcal{G}$ induces a morphism $f_p : \mathcal{F}_p \to \mathcal{G}_p$ between stalks.

Given any germ (Oprea denotes as (s, U)) in $\mathcal{F}_p$ I define $f_p((s, U)) = (f_U(s), U)$.I claim this is a well-defined function. After all, suppose $(s, U) = (t, V)$ as germs. Thus, there exists an open set $W \subseteq U \cap V$ such that $r_{UW}(s) = r_{VW}(t)$. And now we have that:

$$r_{UW}(f_U(s)) = f_W(r_{UW}(s)) = f_W(r_{VW}(t)) = r_{VW}(f_V(t)).$$

So, $(f_U(s), U) = (f_V(t), V)$.

Note: If $\mathcal{F}$ and $\mathcal{G}$ are abelian group valued or R-module valued or R-algebra valued, clearly f_p will preserve the group/ring/module structure of $\mathcal{F}_p$ and $\mathcal{G}_p$.

(iv) Now we say a sheaf of continuous functions on a topological space X is a sheaf $\mathcal{F} \to X$ such that $\mathcal{F}(U)$ is a subset of $C(U, \mathbb{R})$ for every open set U and $r_{UV}(f : U \to \mathbb{R})$ is given by restricting the domain of f to V.

A topological space X together with a sheaf of continuous functions is called a ringed space. A morphism of ringed spaces $f : (X, \mathcal{F}) \to (Y, \mathcal{G})$ is a continuous map $f : X \to Y$ such that if $V \subseteq Y$ is open and $\phi \in \mathcal{G}(V)$ is a section, then the $\phi \circ f \in \mathcal{F}(f^{-1}(V))$.

How would you define an isomorphism of ringed spaces?

This took me four hours to think about and understand the definitions, and I'm probably not going to finish my homework because of this :(. Oh well.

We can easily see that the identity map on X is always a morphism of ringed spaces. (In fact it is the identity morphism). Thus, the obvious starting point is that $f : (X, \mathcal{F}) \to (Y, \mathcal{G})$ will be an isomorphism iff there exists a morphism of ringed spaces $g : (Y, \mathcal{G}) \to (X, \mathcal{F})$ such that $f \circ g = \mathrm{Id}_Y$ and $g \circ f = \mathrm{Id}_X$. I'd like to get a better (actually testable) statement of a definition though.

Clearly f and g are inverse homeomorphisms. Also, I claim that if $\phi \in C(V)$ where $V \subseteq Y$ is open, then $\phi \in \mathcal{G}(V)$ iff $\phi \circ f \in \mathcal{F}(f^{-1}(V))$.

Why?
We know by the definition of a morphism of ringed spaces that if $\phi \in \mathcal{G}(V)$ then $\phi \circ f \in \mathcal{F}(f^{-1}(V))$. As for the other implication, note that if $\phi \circ f \in \mathcal{F}(f^{-1}(V))$ then we know by definition of a morphism of ringed spaces that $\phi = \phi \circ f \circ g$ is in $\mathcal{G}(V)$.

Based on the prior reasoning, we now have a necessary condition for f to be an isomorphism of ringed spaces. I'd also like to show this condition is sufficient.

Suppose $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ are ringed spaces and $f : X \to Y$ is a homeomorphism such that for any open set $V \subseteq Y$, $\phi \in \mathcal{G}(V)$ iff $\phi \circ f \in \mathcal{F}(f^{-1}(V))$. nmWe can see that if $U \subseteq X$ is open and $\phi \in \mathcal{F}(U)$, then $\phi \circ f^{-1} \in \mathcal{G}(f(U))$ because we know $(\phi \circ f^{-1}) \circ f \in \mathcal{F}(f^{-1}(f(U))) = \mathcal{F}(U)$. So, f^{-1} is a morphism of ringed spaces.

So all in all, we say an isomorphism of ringed spaces $f : (X, \mathcal{F}) \to (Y, \mathcal{G})$ is a homeomorphism $f : X \to Y$ such that for any open set $V \subseteq Y$, $\phi \in \mathcal{G}(V)$ iff $\phi \circ f \in \mathcal{F}(f^{-1}(V))$.

Note that if $(X, \mathcal{F})$ is a ringed space then U is also a ringed space when equipped with the restriction of sheaf $\mathcal{F}$ to just the open sets $V \subseteq U$.

(Hopefully this is obvious...)

Also note the inclusion map $\iota : U \hookrightarrow M$ is a morphism of ringed spaces.

Why?
If $V \subseteq X$ is an open set then $\iota^{-1}(V) = V \cap U$. But also note that if $\phi \in \mathcal{F}(V)$ then we know that $\phi \circ \iota|_{V \cap U} = \phi|_{V \cap U} \in \mathcal{F}(V \cap U)$.

Now observe that any C^r manifold M is a ringed space when the sheaf of C^r functions (denoted $\mathscr{C}^r_M$.)

Theorem: If M is an n-dimensional C^r manifold then the ringed space $(M, \mathscr{C}^r_M)$ is locally isomorphic to an open subset of $\mathbb{R}^n$ or $[0,\infty)^k \times \mathbb{R}^{n-k}$ (where $k < n$) equipped with the sheaf of C^r functions.

In other words, for every $p \in M$ there is an open set $U \subseteq M$ containing p and an open set $V \subseteq \mathbb{R}^n$ or $[0,\infty)^k \times \mathbb{R}^{n-k}$ (where $k < n$) such that $(U, \mathscr{C}^r_U)$ and $(V, \mathscr{C}^r_V)$ are isomorphic as ringed spaces.

Proof:
Given any $p \in M$ we can pick a smooth chart (U, φ) on M containing p. Also we let $V = \varphi(U)$. I claim that φ is a ringed space isomorphism between $(U, \mathscr{C}^r_U)$ and $(V, \mathscr{C}^r_V)$. After all, clearly $\varphi : U \to V$ is a homeomorphism. Also suppose $W \subseteq V$ is an open set and $f \in C(W, \mathbb{R})$

If $f \in C^r(W, \mathbb{R})$ then $f \circ \varphi \in C^r(\varphi^{-1}(W), \mathbb{R})$ since $(f \circ \varphi) \circ \varphi^{-1} = f$ is a C^r map from $W = \varphi(\varphi^{-1}(W))$ to $\mathbb{R}$.

Conversely, suppose $f \circ \varphi \in C^r(\varphi^{-1}(W), \mathbb{R})$. Then by definition we know that $f = (f \circ \varphi) \circ \varphi^{-1}$ is a C^r map from $W = \varphi(\varphi^{-1}(W))$ to $\mathbb{R}$.

Conversely, given any ringed space $(M, \mathcal{S})$ that is Hausdorff, second countable, and locally isomorphic to an open subset of $\mathbb{R}^n$ or $[0,\infty)^k \times \mathbb{R}^{n-k}$ (where $k \leq n$) equipped with the sheaf of C^r functions, we have that there is a unique C^r structure $\mathcal{A}$ on M turning M into a C^r manifold with $\mathcal{S} = \mathscr{C}^r_M$ when M is equipped with $\mathcal{A}$.

Proof:
Hopefully it is obvious that M is a topological n-dimensional manifold. Also given any $p \in M$ let (U_p, φ_p) be a coordinate chart containing p with $\varphi_p : (U_p, \mathcal{S}) \to (V_p, \mathscr{C}^r_{V_p})$ a ringed space isomorphism.

Firstly I claim the collection $\mathcal{U} := \{(U_p, \varphi_p)\}_{p \in M}$ are all C^r compatible with each other.

To see why, let $p, q \in M$ be distinct points such that $U_p \cap U_q \neq \emptyset$. We'll write $\varphi_p(m) = (x_1(m), \ldots, x_n(m))$ for all $m \in U_p$. And now in order to prove that $\varphi_p \circ \varphi_q^{-1}|_{\varphi_q(U_p \cap U_q)}$ is a C^r map it suffices to show to show that each component function $x_i \circ \varphi_q^{-1}|_{\varphi_q(U_p \cap U_q)}$ is a C^r map.

Note that $\mathrm{Id}_{V_p} \in C^r(V_p, \mathbb{R}^n)$ implies that $\mathrm{Proj}_i \circ \varphi_p = x_i \in \mathcal{S}(\varphi_p^{-1}(V_p)) = \mathcal{S}(U_p)$. And then by applying a restriction map, we get that $x_i|_{U_p \cap U_q} \in \mathcal{S}(U_p \cap U_q)$. Next, because φ_q^{-1} is a ringed isomorphism from $(V_q, \mathscr{C}^r_{V_q})$ to $(U_q, \mathcal{S})$, we must have that:

$$x_i|_{U_p \cap U_q} \circ \varphi_q^{-1}|_{\varphi_q(U_p \cap U_q)} = x_i \circ \varphi_q^{-1}|_{\varphi_q(U_p \cap U_q)} \text{ is a } C^r \text{ map.}$$

Secondly, I claim that if M is equipped with any C^r structure $\mathcal{A}$ turning M into a C^r manifold with $\mathcal{S} = \mathscr{C}^r_M$ then we must have that $\mathcal{U} \subseteq \mathcal{A}$. (This will prove uniqueness since there is only one maximal atlas containing $\mathcal{U}$.)

To see why, let (W, ψ) be any chart in $\mathcal{A}$ and then consider any chart (U_p, φ_p) in $\mathcal{U}$. We need to show that these two charts are C^r compatible. After all, since (W, ψ) is arbitrary this will tell us that $(U_p, \varphi_p) \in \mathcal{A}$.

Fortunately, by the reasoning in the proof of the first part of this theorem (see the prior page), we know that ψ is a ringed space isomorphism between $(W, \mathscr{C}^r_W = \mathcal{S})$ and $(\psi(W), \mathscr{C}^r_{\psi(W)})$. Hence, we can now apply similar reasoning as in the proof that $\mathcal{U}$ is a C^r atlas to prove that (W, ψ) and (U_p, φ_p) are C^r compatible.

Thirdly, I claim that if M is equipped with the C^r structure generated by $\mathcal{U}$ then $\mathcal{S} = \mathscr{C}^r_M$.
To see why, let $W \subseteq M$ be any open set and first suppose $f \in \mathcal{S}(W)$.
For any $p \in W$ there is a smooth chart $(U_p, \varphi_p) \in \mathcal{U}$ containing p. In turn, $(W \cap U_p, \varphi_p|_{W \cap U_p})$ is also a smooth chart. But also note $f|_{W \cap U_p} \in \mathcal{S}(W \cap U_p)$. In turn, as φ_p is a ringed space isomorphism we know that:

$$f \circ \varphi_p^{-1}|_{\varphi_p(W \cap U_p)} \in C^r(\varphi_p(W \cap U_p), \mathbb{R}).$$

Since every $p \in W$ has a smooth chart (U', φ') satisfying that $p \in U' \subseteq W$ and $f \circ (\varphi')^{-1}$ is a C^r map, we can thus conclude that $f \in \mathscr{C}^r_M(W)$.

Conversely, suppose $f \in \mathscr{C}^r_M(W)$. Then as $(W \cap U_p, \varphi_p|_{W \cap U_p})$ is a smooth chart for every p, we know that $f \circ \varphi_p^{-1} \in C^r(\varphi(W \cap U_p), \mathbb{R})$ for every $p \in W$. Then in turn, as φ is a ringed space isomorphism we can conclude that $f|_{W \cap U_p} \in \mathcal{S}(W \cap U_p)$ for every $p \in W$. Finally, as $\mathcal{S}$ is a sheaf and $\bigcup_{p \in W}(W \cap U_p) = W$, we can conclude that $f \in \mathcal{S}(W)$. ■

The T.L.D.R. to the prior theorem is that we can equivalently consider a C^r manifold to be a ringed space. This approach has advantages because it generalizes to weirder spaces. (In fact Wells in *Differential Analysis on Complex Manifolds* [see *citation 9*] actually defines the C^r structure of a manifold to be a sheaf of a ringed space as opposed to a maximal atlas.)

Theorem: Suppose M and N are both C^r manifolds equipped with the sheafs $\mathscr{C}^r_M$ and $\mathscr{C}^r_N$ respectively. Then $F : M \to N$ is a ringed space isomorphism from $(M, \mathscr{C}^r_M)$ to $(N, \mathscr{C}^r_N)$ if and only if f is a C^r map.

$(\Longrightarrow)$
Let $p \in M$ be any point and let (V, ψ) be a chart on N containing $F(p)$. We'll write $\psi(q) = (y_1(q), \ldots, y_n(q))$ for all $q \in V$. Next, because F is continuous, we know that $F^{-1}(V)$ is an open set containing p. So, we can find a smooth chart (U, φ) on M such that $p \in U \subseteq F^{-1}(V)$. and now as F and φ^{-1} are both ringed homomorphisms, we know that $y_i \in C^r(V, \mathbb{R})$ implies that $y_i \circ F \circ \varphi^{-1} \in C^r(\varphi(U), \mathbb{R})$ for each i. In turn, $\psi \circ F \circ \varphi^{-1}$ is a C^r map.

$(\Longleftarrow)$
Suppose $V \subseteq N$ is any open set and $f \in C^r(V, \mathbb{R})$. Then $f \circ F \in C^r(F^{-1}(V), \mathbb{R})$. ■

Before doing more homework, I want to take some notes. These will be a mix of lecture notes (by Oprea) and notes from John Lee *Smooth Manifolds*.

Firstly, I want to study more about vector bundles. For a reminder on them, look at *pages 1129-1133*.

Note that $\mathrm{GL}_k(\mathbb{R})$ is an open submanifold of $\mathbb{R}^{k^2}$ and hence it makes sense to talk about C^r maps from a C^r manifold M to $\mathrm{GL}_k(\mathbb{R})$.

Proposition 10.5: Let $\pi : E \to M$ be a C^r vector bundle of rank k over M (where $r \geq 0$). Suppose $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ and $\Psi : \pi^{-1}(V) \to V \times \mathbb{R}^k$ are two C^r local trivializations of E with $U \cap V \neq \emptyset$. Then there exists a C^r map $\tau : U \cap V \to \mathrm{GL}_k(\mathbb{R})$ such that the composition $\Phi \circ \Psi^{-1} : (U \cap V) \times \mathbb{R}^k \to (U \cap V) \times \mathbb{R}^k$ has the form:

$$\Phi \circ \Psi^{-1}(p, v) = (p, \tau(p)v).$$

Proof:
Let $\pi_1 : (U \cap V) \times \mathbb{R}^k \to (U \cap V)$ be the projection map. Then we have the following commuting diagram:

$$\begin{array}{ccccc} (U \cap V) \times \mathbb{R}^k & \xleftarrow[\sim]{\Psi} & \pi^{-1}(U \cap V) & \xrightarrow[\sim]{\Phi} & (U \cap V) \times \mathbb{R}^k \\ & {\scriptstyle \pi_1} \searrow & \downarrow {\scriptstyle \pi} & \swarrow {\scriptstyle \pi_1} & \\ & & U \cap V & & \end{array}$$

In other words, we can see that $\pi_1 \circ (\Phi \circ \Psi^{-1}) = \pi_1$. And this is enough to guarentee that $(\Phi \circ \Psi^{-1})(p, v) = (p, \sigma(v))$ for some C^r function $\sigma : (U \cap V) \times \mathbb{R}^k \to \mathbb{R}^k$.

Next because Φ, Ψ are vector space isomorphisms from $\pi^{-1}(\{p\})$ to $\{p\} \times \mathbb{R}^k$ for every $p \in U \cap V$, we know for every $p \in U \cap V$ that there is a matrix $\tau(p) \in \mathrm{GL}_k(\mathbb{R})$ such that $\sigma(p, v) = \tau(p)v$.

Finally, to prove that the map $\tau : U \cap V \to \mathrm{GL}_k(\mathbb{R})$ is a C^r map, let $e_1, \ldots, e_k$ be the standard basis for $\mathbb{R}^k$. Also let us write $\tau(p) = |c_1(p), \ldots, c_k(p)\rangle$ where each $c_i(p)$ is the ith column of the matrix $\tau(p)$. To show that τ is a C^r map we equivalently need to show that each $c_i : U \cap V \to \mathbb{R}^k$ is a C^r map. Fortunately, this is as easy as noting that $c_i(p) = \sigma(p, e_i)$ is a C^r map. ■

Remark: The function $\tau : U \cap V \to \mathrm{GL}_k(\mathbb{R})$ is called a transition function between local trivializations Φ and Ψ.

Problem 10-5: Let $\pi : E \to M$ be a C^r vector bundle of rank k over M (where $r \geq 0$). Also suppose $\Phi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{R}^k$, $\Phi_\beta : \pi^{-1}(U_\beta) \to U_\beta \times \mathbb{R}^k$, and $\Phi_\gamma : \pi^{-1}(U_\gamma) \to U_\gamma \times \mathbb{R}^k$ are three C^r local trivializations of E with $U_\alpha \cap U_\beta \cap U_\gamma \neq \emptyset$. Let $\tau_{\alpha,\beta}$, $\tau_{\beta,\gamma}$ and $\tau_{\alpha,\gamma}$ be transition functions from the former trivialization in the subscript to the latter. Then $\tau_{\beta,\gamma} \circ \tau_{\alpha,\beta} = \tau_{\alpha,\gamma}$.

Proof:
Let $W := U_\alpha \cap U_\beta \cap U_\gamma$. Then we have the following commuting diagram:

$$\begin{array}{ccccc} (p,v) & \longmapsto & (p,\tau_{\alpha,\beta}(p)v) & \longmapsto & (p,\tau_{\beta,\gamma}(p)\tau_{\alpha,\beta}(p)v) \\ W \times \mathbb{R}^k & \xrightarrow{\Phi_\beta \circ \Phi_\alpha^{-1}} & W \times \mathbb{R}^k & \xrightarrow{\Phi_\gamma \circ \Phi_\beta^{-1}} & W \times \mathbb{R}^k \\ & & \underset{\Phi_\gamma \circ \Phi_\alpha^{-1}}{\searrow} & & \\ (p,v) & & \longmapsto & & (p,\tau_{\alpha,\gamma}(p)v) \end{array}$$

Thus for all $p \in W$ and $v \in \mathbb{R}^k$, we have that $\tau_{\beta,\gamma}(p)\tau_{\alpha,\beta}(p)v = \tau_{\alpha,\gamma}(p)v$. Since matrices are uniquely determined by where they send all $v \in \mathbb{R}^k$, we can conclude that $\tau_{\beta,\gamma} \circ \tau_{\alpha,\beta}(p) = \tau_{\alpha,\gamma}(p)$ for all $p \in W$. ■

Now it would be particularly helpful to have a general theorem for being to able construct vector bundles, especially as there are vector bundles other than the tangent bundle of a manifold that we care about.

Lemma 10.6: Let M be a C^r manifold and suppose for each $p \in M$ we are given a real k-dimensional vector space E_p. Let $E = \coprod_{p \in M} E_p$ and let $\pi : E \to M$ be the map that takes each element of E_p to the point p. Furthermore, suppose that we given:

(i) an open cover $\{U_\alpha\}_{\alpha \in A}$ of M;

(ii) for each $\alpha \in A$ a bijective map $\Phi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{R}^k$ whose restriction to each E_p is a vector space isomorphism from E_p to $\{p\} \times \mathbb{R}^k \cong \mathbb{R}^k$;

(iii) For every $\alpha, \beta \in A$ with $U_\alpha \cap U_\beta \neq \emptyset$ a C^r map $\tau_{\alpha,\beta} : U_\alpha \cap U_\beta \to \mathrm{GL}_k(\mathbb{R})$ such that the map $\Phi_\beta \circ \Phi_\alpha^{-1}$ from $(U_\alpha \cap U_\beta) \times \mathbb{R}^k$ to itself has the form:

$$\Phi_\beta \circ \Phi_\alpha^{-1}(p, v) = (p, \tau_{\alpha,\beta}(p)v).$$

Then E has a unique topology and C^r structure making it into a C^r manifold and a C^r rank k vector bundle over M with π as it's projection and $\{(U_\alpha, \Phi_\alpha)\}_{\alpha \in A}$ all as C^r local trivializations.

Proof:
Let $\{(V_\beta, \varphi_\beta)\}_{\beta \in B}$ be the collection of smooth charts on M such that V_β is contained in some open set U_α. Let $\widehat{V_\beta} = \varphi_\beta(V_\beta)$. Then for each β we can define a map $\widetilde{\varphi_\beta} : \pi^{-1}(V_\beta) \to \widehat{V_\beta} \times \mathbb{R}^k$ by $\widetilde{\varphi_\beta} = (\varphi_\beta \times \mathrm{Id}_{\mathbb{R}^k}) \circ \Phi_\alpha$ (where α is some fixed element such that $V_\beta \subseteq U_\alpha$).

We shall show that the charts $\{(\pi^{-1}(V_\beta), \widetilde{\varphi_\beta})\}_{\beta \in B}$ satisfy the conditions of lemma 1.35 on *pages 1130-1132*. Hence, E has a unique topology and C^r structure such that E is a C^r manifold and all those charts are smooth charts on E.

Requirement (i) holds since each $\widetilde{\varphi_\beta}$ is a composition of two bijections and whose image $\widehat{V_\beta} \times \mathbb{R}^k$ is an open subset of $[0, \infty)^\ell \times \mathbb{R}^{n-\ell+k}$ for some $0 \leq \ell \leq n$.

Similarly, to see requirement (ii) note for every $\beta_1, \beta_2 \in B$ that:
$\widetilde{\varphi_{\beta_1}}(\pi^{-1}(V_{\beta_1}) \cap \pi^{-1}(V_{\beta_2})) = \varphi_p(V_{\beta_1} \cap V_{\beta_2}) \times \mathbb{R}^k$ is an open subset of $\widehat{V_{\beta_1}} \times \mathbb{R}^k$.

To see requirement (iii), note for any $\beta_1, \beta_2 \in B$ with $V_{\beta_1} \cap V_{\beta_2} \neq \emptyset$ that there exists $\alpha_1, \alpha_2 \in A$ with:

$$\widetilde{\varphi_{\beta_2}} \circ \widetilde{\varphi_{\beta_1}}^{-1} = (\varphi_{\beta_2} \times \mathrm{Id}_{\mathbb{R}^k}) \circ (\Phi_{\alpha_2} \circ \Phi_{\alpha_1}^{-1}) \circ (\varphi_{\beta_1}^{-1} \times \mathrm{Id}_{\mathbb{R}^k})$$

The functions being composed on the right side though are C^r diffeomorphisms. So, $\widetilde{\varphi_{\beta_2}} \circ \widetilde{\varphi_{\beta_1}}^{-1} : \varphi_{\beta_1}(V_{\beta_1} \cap V_{\beta_2}) \times \mathbb{R}^k \to \varphi_{\beta_2}(V_{\beta_1} \cap V_{\beta_2}) \times \mathbb{R}^k$ is a C^r diffeomorphism.

Requirement (iv) follows from the fact that we can cover M with countably many of the sets V_β (since M is second countable).

Finally, to see requirement (v) first note that if $\xi, \zeta \in E$ both lie in some common E_p then they are obviously contained in a single common chart. Meanwhile if $\xi \in E_p$ and $\zeta \in E_q$ where p, q are distinct, then we can find disjoint smooth charts $(V_{\beta_p}, \varphi_{\beta_p})$ and $(V_{\beta_q}, \varphi_{\beta_q})$ containing p and q respectively.

Next, I claim that E is now equipped with the unique topology and C^r structure such that each of the $\Phi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{R}^k$ are C^r diffeomorphisms.

Firstly, note that when E is equipped with this topology then $\pi^{-1}(U_\alpha)$ is an open subset of E for every α. This because if (V, φ) is a smooth chart on M such that $V \subseteq U_\alpha$ then we know by definition that $\pi^{-1}(V)$ is open in E. And since $\pi^{-1}(U_\alpha)$ is a union of such subsets, we know $\pi^{-1}(U_\alpha)$ is open in E.

If E is equipped with the topology and C^r structure we gave it, then for any $p \in U_\alpha$ we can find a smooth chart (V, φ) on M with $p \in V \subseteq U_\alpha$. With respect to the smooth chart $(\pi^{-1}(V), \widetilde{\varphi} := (\varphi \times \mathrm{Id}_{\mathbb{R}^k}) \circ \Phi_\alpha)$ on E and $(V \times \mathbb{R}^k, \varphi \times \mathrm{Id}_{\mathbb{R}^k})$ on $U \times \mathbb{R}^k$ we have that:

$$(\varphi \times \mathrm{Id}_{\mathbb{R}^k}) \circ \Phi_\alpha \circ \widetilde{\varphi}^{-1} = \mathrm{Id}_{\varphi(V) \times \mathbb{R}^k}.$$

This proves that each Φ_α is a diffeomorphism when E is equipped with the topology and smooth structure we gave it.

Conversely, suppose E is equipped with another topology $\mathcal{T}$ and C^r structure $\mathcal{A}$ still satisfying that Φ_α is a C^r diffeomorphism from an open set $\pi^{-1}(U_\alpha) \subseteq E$ to the set $U_\alpha \times \mathbb{R}^k$. Then $\mathrm{Id}_{U_\alpha} = \Phi_\alpha^{-1} \circ \Phi_\alpha$ will be a C^r diffeomorphism from one topology and C^r structure on $\pi^{-1}(U_\alpha)$ to the other. It follows that the topology and C^r structure on E restricted to any $\pi^{-1}(U_\alpha)$ is unique.

But now as the $\pi^{-1}(U_\alpha)$ cover E, any set $U \in \mathcal{T}$ can be written as a union of sets $U \cap U_\alpha$ which are all open in our original topology. Hence, $\mathcal{T}$ must be our original topology. Secondly, since smooth charts in each of the open submanifolds $\pi^{-1}(U_\alpha)$ which are in $\mathcal{A}$ and our original structure cover all of M, we know they determine a unique maximal C^r atlas on E. Hence, $\mathcal{A}$ is our original atlas.

Thirdly, I claim $\pi : E \to M$ is a C^r map.

This is as simple as noting for any smooth chart (V, φ) on M and smooth chart $(\pi^{-1}(V), (\varphi \times \mathrm{Id}_{\mathbb{R}^k}) \circ \Phi_\alpha)$ on E we have that the representation of π with respect to those local coordinates is just the projection map $(x, v) \mapsto x$.

Finally, the fact that $\pi : E \to M$ is a vector bundle is just a result of the latter assumptions given in (ii). ■

Note that if $r = 0$ then this proposition still gives a unique topology. Just no smooth structure.

It will also be helpful to establish some terminology about frames and sections of vector bundles. Firstly, if M is a topological space, and $\pi : E \to M$ is a vector bundle, we say $\sigma : M \to E$ is a section / local section of E if it is a section / local section for the map π.

(See *page 1106* for a reminder on the definition of a section or local section.)

If we want to drop the continuity requirement of σ, then we'll say the section is rough.

Also if $\pi : E \to M$ is a C^r vector bundle then we shall say σ is a "C^r" section if it is a C^r map.

Next, suppose $\pi : E \to M$ is a vector bundle. If $U \subseteq M$ is an open set, a $(\sigma_1, \ldots, \sigma_k)$ is a k-tuple of local sections of E over U is said to be linearly independent if for every $p \in U$ we have that $(\sigma_1(p), \ldots, \sigma_k(p))$ forms a linearly independent k-tuple of vectors in E_p. Also $(\sigma_1, \ldots, \sigma_k)$ is said to span E if $\sigma_1(p), \ldots, \sigma_k(p)$ span E_p for all $p \in U$.

A local frame for E over U is an ordered k-tuple of local sections $(\sigma_1, \ldots, \sigma_k)$ of linearly independent sections over U that span E. It is called a global frame if $U = M$. Also, if every σ_i is a C^r map we say that it is a C^r frame.

Often, we denote a local frame by $(\sigma_i)_{1 \leq i \leq k}$.

A key result is that there is a one-to-one correspondance between C^r local frames and C^r local trivializations.

Example 10.18: Suppose $\pi : E \to M$ is a C^r vector bundle of rank k. We let $e_1, \ldots, e_k$ denote the standard basis on $\mathbb{R}^k$. If $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ is a C^r local trivialization of E then we can define maps $\sigma_1, \ldots, \sigma_k : U \to E$ by $\sigma_i(p) = \Phi^{-1}(p, e_i) = \Phi^{-1} \circ \widetilde{e_i}(p)$ (where $\widetilde{e_i} : U \to U \times \mathbb{R}^k$ is the C^r map sending p to (p, e_i)). I claim $(\sigma_1, \ldots, \sigma_k)$ is a C^r local frame for E over U.

Why?

Each σ_i is a C^r map since it is a composition of C^r maps. Also, note that if $\pi_1 : U \times \mathbb{R}^k \to U$ is the projection map, then:

$$\pi \circ \sigma_i(p) = \pi \circ \Phi^{-1}(p, e_i) = \pi_1(p, e_i) = p.$$

So, σ_i is a local section of E.

To finish off, to show that $(\sigma_i)_{1 \leq i \leq k}$ is linearly independent and spans E, we can note that Φ restricts to a vector space isomorphism from E_p to $\{p\} \times \mathbb{R}^k$ for all $p \in U$. Also, $\Phi(\sigma_i(p)) = (p, e_i)$. So, Φ takes the vectors $\sigma_1(p), \ldots, \sigma_k(p)$ to the standard basis on $\mathbb{R}^k$.

Remark: The local frame for E over U constructed in this example is called the local frame associated with Φ.

Proposition 10.19: Let $\pi : E \to M$ be a C^r vector bundle. Every C^r local frame $(\sigma_i)_{1 \leq i \leq k}$ for E over an open set $U \subseteq M$ is associated with a C^r local trivialization $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$.

Proof:
Define a map $\Psi : U \times \mathbb{R}^k \to \pi^{-1}(U)$ by $\Psi(p, (v_1, \ldots, v_k)) = \sum_{i=1}^k v_i \sigma_i(p)$.
(Here the right side is an element of the vector space E_p.)

Trivially, for every $p \in U$ the map Ψ restricts to a linear map between $\{p\} \times \mathbb{R}^k$ and E_p. Also, the $\sigma_i(p)$ forming a basis for E_p at every $p \in U$ guarentees that the linear map that Ψ restricts to is a vector space isomorphism for all p. Thirdly, if $e_1, \ldots, e_k$ is the standard basis for $\mathbb{R}^k$, and if for each i we define $\widetilde{e_i} : U \to U \times \mathbb{R}^k$ by $\widetilde{e_i}(p) = (p, e_i)$, then we can see that $\sigma_i = \Psi \circ \widetilde{e_i}$ for each i. Hence once we show that Ψ is a C^r diffeomorphism from $U \times \mathbb{R}^k$ to $\pi^{-1}(U)$ then we will be able to conclude that $\Psi^{-1} : \pi^{-1}(U) \to U \times \mathbb{R}^k$ is a local trivialization whose associated local frame is our original frame $(\sigma_i)_{1 \le i \le k}$.

Fortunately, because Ψ is a bijection, to prove that Ψ is a C^r diffeomorphism it suffices to prove that Ψ is a local C^r diffeomorphism. In particular, we know to show that there is an open cover $\{V_\alpha\}_{\alpha \in A}$ of U such that $\Psi|_{V_\alpha \times \mathbb{R}^k}$ is a C^r diffeomorphism for every $\alpha \in A$.

Consider any $q \in U$. There exists an open set $V \subseteq M$ containing q over which there is a C^r local trivialization $\Phi : \pi^{-1}(V) \to V \times \mathbb{R}^k$. By shrinking V, we can without loss of generality assume $V \subseteq U$. And now if $\pi_1 : V \times \mathbb{R}^k \to V$ is the projection map, we have the following commuting diagram:

$$\begin{array}{ccccc} V \times \mathbb{R}^k & \xrightarrow{\Psi|_{V \times \mathbb{R}^k}} & \pi^{-1}(V) & \xrightarrow[\sim]{\Phi} & V \times \mathbb{R}^k \\ & \searrow^{\pi_1} & \downarrow{\scriptstyle \pi} & \swarrow_{\pi_1} & \\ & & V & & \end{array}$$

Note that $\Phi \circ \Psi|_{V \times \mathbb{R}^k}$ is a bijection from $V \times \mathbb{R}^k$ to itself. Since Φ is a C^r diffeomorphism, in order to prove that $\Psi|_{V \times \mathbb{R}^k}$ is a C^r diffeomorphism it suffices to prove that $\Phi \circ \Psi|_{V \times \mathbb{R}^k}$ is a C^r diffeomorphism.

To do this, note that $\Phi \circ \sigma_i|_V : V \to V \times \mathbb{R}^k$ is a C^r map for each i. In turn, we know for every $1 \le i, j \le k$ that there is a C^r map $\sigma_i^{(j)} : V \to \mathbb{R}$ such that:

$$\Phi \circ \sigma_i|_V(p) = \Big(p, (\sigma_i^{(1)}(p), \ldots, \sigma_i^{(k)}(p))\Big).$$

But now as $\Phi : E_p \to \{p\} \times \mathbb{R}^k$ is a vector space isomorphism, we know for all $p \in V$ and $(v_1, \ldots, v_k) \in \mathbb{R}^k$ that:

$$\Phi \circ \Psi(p, (v_1, \ldots, v_k)) = \Phi(\textstyle\sum_{i=1}^k v_i \sigma_i(p)) = \sum_{i=1}^k v_i \Phi \circ \sigma_i(p)$$

We can define $\tau : V \to \mathrm{GL}_k(\mathbb{R})$ by $\tau(p) := \begin{bmatrix} \sigma_1^{(1)} & \cdots & \sigma_k^{(1)} \\ \vdots & \ddots & \vdots \\ \sigma_1^{(k)} & \cdots & \sigma_k^{(k)} \end{bmatrix}$.

Note that τ is a C^r map into the space of kxk matrices because each component function is a C^r map. Furthermore, by the prior reasoning we know that $\Phi \circ \Psi(p,v) = (p, \tau(p)v)$. And finally since $\Phi \circ \Psi|_{\{p\}\times\mathbb{R}^k}$ is the composition of two vector space isomorphisms, we know $\tau(p) \in \mathrm{GL}_k(\mathbb{R})$ for all $p \in V$.

Since matrix inversion is a C^∞ diffeomorphism on $\mathrm{GL}_k(\mathbb{R})$, we can conclude that $\Phi \circ \Psi|_{V\times\mathbb{R}^k}$ is a C^r map with the C^r inverse:

$$(\phi \circ \Psi)^{-1}(p,v) = (p, (\tau(p))^{-1}v). \blacksquare$$

Remark: We call $\Psi^{-1} : \pi^{-1}(U) \to U \times \mathbb{R}^k$ the local trivialization associated with $(\sigma_i)_{1\le i\le k}$.

Corollary 10.21: Let $\pi : E \to M$ be a C^r vector bundle of rank k, let $(V, \varphi = (x_1, \ldots, x_n))$ be a smooth chart on M, and suppose there exists a C^r local frame $(\sigma_i)_{1\le i\le k}$ for E over V. Define $\widetilde{\varphi} : \pi^{-1}(V) \to \varphi(V) \times \mathbb{R}^k$ by:

$$\widetilde{\varphi}(\textstyle\sum_{i=1}^k v_i\sigma_i(p)) = ((x_1(p), \ldots, x_n(p)), (v_1, \ldots, v_k)).$$

Then $(\pi^{-1}(V), \widetilde{\varphi})$ is a smooth chart on E.

Proof:
Note that $\widetilde{\varphi} = (\varphi \times \mathrm{Id}_{\mathbb{R}^k}) \circ \Phi$ where $\varphi \times \mathrm{Id}_{\mathbb{R}^k}$ is a C^r diffeomorphim and Φ (which is the local trivialization associated with $(\sigma_i)_{1\le i\le k}$) is also a C^r diffeomorphism. So, $\widetilde{\varphi}$ is a C^r diffeomorphism. $\blacksquare$

Using the prior results, we can also now give a neat criteria for showing if a rough section of a vector bundle $\pi : E \to M$ is continuous and C^r.

Let $\pi : E \to M$ be a C^r vector bundle of rank k. Also suppose $(\sigma_i)_{1\le i\le k}$ is a C^r local frame for E over some open subset $U \subseteq M$. Given any rough local section $\tau : V \to E$ we can for every $p \in V \cap U$ write $\tau(p) = \sum_{i=1}^k \tau_i(p)\sigma_i(p)$ for some uniquely determined numbers $\tau_1(p), \ldots, \tau_k(p)$. This gives us functions $\tau_1, \ldots, \tau_k$ from $V \cap U$ to $\mathbb{R}$ which we call the component functions of τ with respect to the frame $(\sigma_i)_{1\le i\le k}$.

Theorem 10.22: Let $\pi : E \to M$ be a C^r vector bundle, let $\tau : V \to E$ be a rough section, and let $(\sigma_i)_{1\le i\le k}$ be a C^r local frame for E over some open subset $U \subseteq M$. Then $\tau|_{U\cap V}$ is a C^r section iff its components functions with respect to the frame $(\sigma_i)_{1\le i\le k}$ are C^r maps.

Proof:
Let $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ be the local trivialization associated to the frame $(\sigma_i)_{1\le i\le k}$. Because $\Phi|_{\pi^{-1}(V\cap U)}$ is a C^r diffeomorphism to $(V \cap U) \times \mathbb{R}^k$, we have that τ is a C^r map iff $\Phi \circ \tau|_{V\cap U}$ is a C^r map. Yet we can calculate that $\Phi \circ \tau(p) = (p, (\tau_1(p), \ldots, \tau_k(p)))$. So, $\Phi \circ \tau|_{V\cap U}$ is a C^r map iff $\tau_1|_{U\cap V}, \ldots, \tau_k|_{U\cap V}$ are all C^r maps. $\blacksquare$

As a side note, the prior four results and proofs all still hold when $r = 0$, provided that we substitute in "homeomorphism" for "C^0 diffeomorphism".

As I mentioned when first defining vector bundles, the most important vector bundle is the tangent bundle TM of a C^r manifold M (where $r \geq 1$).

In the special case of the tangent bundle $\pi : TM \to M$, we call a section of TM a vector field. We can also add the adjectives "rough", "C^r", and "local" as we did for sections of more general vector bundles.

For an alternate lens, suppose $U \subseteq M$ is an open subset of a C^r manifold M and $X : U \to TM$ is a rough local vector field. Given any open set $W \subseteq U$ we can define:

$$\widetilde{X}_W : C^r(W, \mathbb{R}) \to \mathbb{R}^W \text{ by } (\widetilde{X}_W(f))(p) \coloneqq (X(p))(f)$$

for all $p \in U$ and $f \in C^r(W, \mathbb{R})$. This is clearly a linear map. Also, for convenience we denote $\widetilde{X}_U$ as just $\widetilde{X}$.

Propositon 8.14: Let $0 \leq k \leq r-1$. Then the following are equivalent:

(a) X is a C^k local vector field over U,

(b) the codomain of $\widetilde{X}$ is contained in $C^k(U, \mathbb{R})$,

(c) for any open set $W \subseteq U$ the codomain of $\widetilde{X}_W$ is contained in $C^k(W, \mathbb{R})$.

$(a \implies b)$

Let $f \in C^r(U, \mathbb{R})$. Then for any $p \in U$ let (V, φ) be a smooth chart on M such that $p \in V \subseteq U$. Also denote the local coordinate functions of φ by $x_1, \ldots, x_n$. Then, we can define a C^{r-1} local trivialization $(\pi^{-1}(V), \Phi)$ on TM by:

$$\Phi(p, \textstyle\sum_{i=1}^n c_i \left.\frac{\partial}{\partial x_i}\right|_p) = (p, (c_1, \ldots, c_n)).$$

Yet now for each $1 \leq i \leq n$ let $\sigma_i : V \to TM$ be the C^{r-1} vector field (i.e. section) defined by $\sigma_i(p) = \left.\frac{\partial}{\partial x_i}\right|_p$. Then $(\sigma_i)_{1 \leq i \leq n}$ is the frame associated with Φ.

From theorem 10.22 on the prior page applied to the C^k vector field X we thus know there are C^k component functions $X_1, \ldots, X_n : V \to \mathbb{R}$ such that:

$$X(p) = (p, \textstyle\sum_{i=1}^n X_i(p) \left.\frac{\partial}{\partial x_i}\right|_p).$$

In turn, we can calculate for every $p \in V$ that:

$$\left(\widetilde{X}(f)\right)(p) = \textstyle\sum_{i=1}^n X_i(p) \left.\frac{\partial}{\partial x_i}\right|_p f$$

Switching to local coordinates and setting $\widehat{p} = \varphi(p)$ for all $p \in V$, we then get:

$$\widetilde{X}(f) \circ \varphi^{-1}(\widehat{p}) = \textstyle\sum_{i=1}^n (X_i \circ \varphi^{-1})(\widehat{p}) \frac{\partial}{\partial x_i}(f \circ \varphi^{-1})(\widehat{p}).$$

Every summand on the right hand side is a C^k map. So, we are done.

$(b \implies c)$

Suppose $f \in C^r(W, \mathbb{R})$. It suffices to show locally in W that $\widetilde{X}_W(f)$ is a C^k map. Fortunately, for any $p \in W$ we can find a C^r bump function $\phi : U \to [0, 1]$ such that

$\phi \equiv 1$ on an open neighborhood Ω of p in W and $\operatorname{supp}(\phi) \subseteq W$. Then importantly, for all $q \in \Omega$ we have that $(\widetilde{X}(f))(q) = (X(q))(f) = (X(q))(\phi f) = (\widetilde{X}(\phi f))(q)$. By part (b) though, we know that $\widetilde{X}(\phi f)$ is a C^k map on U though. So, this proves f is a C^k map on Ω.

$(c \Longrightarrow a)$
Let (V, φ) be any smooth chart on M such that $V \subseteq U$ and write the coordinates on V as $\varphi = (y_1, \ldots, y_n)$. Like before, we can define a C^{r-1} local trivialization $(\pi^{-1}(V), \Phi)$ on TM by:

$$\Phi(p, \textstyle\sum_{i=1}^n c_i \left.\frac{\partial}{\partial y_i}\right|_p) = (p, (c_1, \ldots, c_n)).$$

And now for each $1 \leq i \leq n$ we let $\sigma_i : V \to TM$ be the C^{r-1} vector field defined by $\sigma_i(p) = \left.\frac{\partial}{\partial y_i}\right|_p$. Then $(\sigma_i)_{1 \leq i \leq n}$ is the frame associated with Φ.

If we consider the component functions $X_1, \ldots, X_n : V \to \mathbb{R}$ such that:

$$X(p) = (p, \textstyle\sum_{i=1}^n X_i(p) \left.\frac{\partial}{\partial y_i}\right|_p),$$

we can now say that X is a C^k vector field iff each X_i is a C^k map.

Yet also note that $y_j : V \to \mathbb{R}$ is a map in $C^r(V, \mathbb{R})$ for all $1 \leq j \leq n$. In turn, by assumption (c) we thus know that $\widetilde{X}(y_j) \in C^k(W, \mathbb{R})$. Yet also note that:

$$(\widetilde{X}(y_j))(p) = (X(p))(y_j) = \textstyle\sum_{i=1}^n X_i(p) \left.\frac{\partial}{\partial y_i}\right|_p y_j = X_j(p)$$

So, each X_i is a C^k map. ■

Also observe the following about the function $\widetilde{X} : C^r(U, \mathbb{R}) \to \mathbb{R}^U$.

Lemma: Suppose $f \in C^r(U, \mathbb{R})$ and $p \in U$ satisfies that $\mathrm{d}f_p = 0$. Then $(\widetilde{X}(f))(p) = 0$.
Why?
Note that $(\widetilde{X}(f))(p) = (X(p))(f) = (X(p))(\mathrm{Id}_{\mathbb{R}} \circ f) = \mathrm{d}f_p(X(p))\mathrm{Id}_{\mathbb{R}} = 0$.

Interestingly, I just realized I can also prove the following general result.

Proposition: Suppose M is a C^r manifold (with $r \geq 1$), $U \subseteq M$ is an open subset, and $T : C^r(U, \mathbb{R}) \to \mathbb{R}^U$ is a linear map.

- Suppose T satisfies that if $p \in U$ and $f \in C^r(U, \mathbb{R})$ has $\mathrm{d}f_p = 0$ then $(T(f))(p) = 0$. Then we have that $T(fg) = fT(g) + gT(f)$ for all $f, g \in C^r(U, \mathbb{R})$.

  Proof:
  Given a fixed p define $h = fg - g(p)f - f(p)g$. Then we can easily verify that $\mathrm{d}h_p = 0$. In turn, by linearity and our starting assumption we have that:

  $$0 = (T(h))(p) = (T(fg))(p) - g(p)(T(f))(p) - f(p)(T(g))(p).$$

  Or in other words, $(T(fg))(p) = g(p)(T(f))(p) + f(p)(T(g))(p)$. Since p was arbitrary, this proves that $T(fg) = gT(f) + fT(g)$.

- The converse claim holds if $r = \infty$.

Proof:
First I shall prove the easiest claim that if $f \equiv C$ on U is a constant function then $T(f) \equiv 0$ on U.

To see why, note that if $C = 0$ then $T(f) = 0T(f) \equiv 0$. Meanwhile, if $C \neq 0$ then

$$T(f) = \tfrac{1}{C}T(f^2) = \tfrac{1}{C}(2fT(f)) = 2T(f)$$

And now subtracting $T(f)$ from both sides we get that $T(f) = 0$.

Secondly, I shall prove that if $p \in U$ and $f \in C^r(U, \mathbb{R})$ satisfies that f is constant on a neighborhood V of p then $(T(f))(p) = 0$.

To start off, without loss of generality we can assume $f(p) = 0$. After all, if not then by the previous claim we have that:

$$T(f) = T(f - f(p)) + T(f(p)) = T(f - f(p)) + 0.$$

Next, we can find a C^r bump function $\omega : U \to [0, 1]$ such that $\omega \equiv 1$ on $\operatorname{supp}(f)$ and $\operatorname{supp}(\omega) \subseteq U - \{p\}$. Then $f = \omega f$ and in turn:

$$(T(f))(p) = (T(\omega f))(p) = \omega(p)(T(f))(p) + f(p)(T(\omega))(p) = 0 + 0 = 0$$

Now I'm ready for proving the actual proposition's claim. Suppose $f \in C^r(U, \mathbb{R})$ and $p \in U$ is a critical point of f. On a small enough smooth chart $(V, \varphi = (x_1, \ldots, x_n))$ on M containing p we can conclude that there are C^∞ functions $h_i : V \to \mathbb{R}$ with $f|_V = f(p) + (x_i - x_i(p))h_i$ and $h_i(p) = 0$ for each i.

Next let $W \subseteq V$ be an open set containing p and $\omega : U \to [0, 1]$ be a C^r bump function such that $\omega \equiv 1$ on W and $\operatorname{supp}(\omega) \subseteq V$. Notably, we also have that ω^2 is a C^r bump function with $\omega \equiv 1$ on W and $\operatorname{supp}(\omega) \subseteq V$.

But now ω^2 being constant on a neighborhood of p means that:

$$T(\omega^2 f) = f(p)(T(\omega^2))(p) + \omega^2(p)(T(f))(p) = 0 + 1(T(f))(p).$$

Yet also note that $\omega^2 f = f(p)\omega^2 + \sum_{i=1}^n [\omega(x_i - x_i(p)) \cdot \omega h_i]$.

Thus $T(\omega^2 f) = 0 + \sum_{i=1}^n [0T(\omega h_i) + 0T(\omega(x_i - x_i(p)))] = 0$. So,

$$T(f) = T(\omega^2 f) = 0. \blacksquare$$

Based on the lemma before, we can now ask if we can go in reverse from what we've been doing. In other words…

Theorem: Let M be a C^r manifold, let $U \subseteq M$ be an open set, and let $0 \leq k \leq r-1$. Suppose we are given a linear map $\widetilde{Y} : C^r(U,\mathbb{R}) \to C^k(U,\mathbb{R})$ satisfying that if $f \in C^r(U,\mathbb{R})$ and $p \in U$ is such that $\mathrm{d}f_p = 0$ then $(\widetilde{Y}(f))(p) = 0$. Then there is a unique rough local vector field $Y : U \to TM$ such that:

$$(Y(p))(f) = (\widetilde{Y}(f))(p) \text{ for all } p \in U \text{ and } f \in C^r(U,\mathbb{R}).$$

Furthermore, Y is a C^k vector field.

Proof:

To start off, we do have to exercise some caution for how we define Y. After all, for every $p \in U$ we want $Y(p)$ to be a linear map from $C^1(U,\mathbb{R})$. That said, if we were to simply define $(Y(p))(f) = (\widetilde{Y}(f))(p)$ then we would only be defining $Y(p)$ on the domain $C^r(U,\mathbb{R})$. So, we need to be more subtle.

I'll first show for any $p \in U$ that there exists an open set $W \subseteq U$ containing p as well as a C^k vector field $Y^{(W)} : W \to TM$ such that:

$$(\widetilde{Y}(f))(p) = (Y^{(W)}(p))(f) \text{ for all } f \in C^r(U,\mathbb{R}) \text{ and } p \in W.$$

To start off, let (V,φ) be a smooth chart on M such that $p \in V \subseteq U$, and for all $q \in V$ write $\varphi(q) = (x_1(q),\ldots,x_n(q))$. Next, let $W \subseteq V$ be an open set containing p, and let $\omega : U \to [0,1]$ be a C^r bump function with $\omega \equiv 1$ on W and $\mathrm{supp}(\omega) \subseteq V$.

Importantly, $\omega x_i : U \to \mathbb{R}$ is a well-defined C^r map. Therefore, we can define a C^k map $Y_i^{(W)} := \widetilde{Y}(\omega x_i)$. And now we define the vector field $Y^{(W)} : W \to TM$ by:

$$Y^{(W)}(p) = \left(p, \textstyle\sum_{i=1}^n Y_i^{(W)}(p) \left.\frac{\partial}{\partial x_i}\right|_p\right)$$

All the component functions $Y_i^{(W)}$ being C^k implies that $Y^{(W)}$ is a C^k vector field. Also I claim that if $f \in C^r(U,\mathbb{R})$ and $p \in W$ then $(\widetilde{Y}(f))(p) = (Y(p))(f)$.

To see why, for every i let $c_i = \left.\frac{\partial}{\partial x_i}\right|_p f$.

We know that $g := f - \sum_{i=1}^n c_i \omega x_i$ is a C^r map on U satisfying that $\mathrm{d}g_p = 0$. Hence:

$$0 = (\widetilde{Y}(g))(p) = (\widetilde{Y}(f))(p) - \textstyle\sum_{i=1}^n c_i Y_i^{(W)}(p)$$

In other words:

$$(\widetilde{Y}(f))(p) = \textstyle\sum_{i=1}^n c_i Y_i^{(W)}(p) = Y_i^{(W)}(p) \left.\frac{\partial}{\partial x_i}\right|_p f = (Y^{(W)}(p))(f)$$

Secondly, I shall show that if W and Ω are both open subsets of U with $W \cap \Omega \neq \emptyset$, and if $Y : W \to TM$ and $X : \Omega \to TM$ are both rough vector fields satisfying for all $p \in W$, $q \in \Omega$, and $f \in C^r(U,\mathbb{R})$ that:

1. $(\widetilde{Y}(f))(p) = (Y(p))(f)$ for all $p \in W$,
2. $(\widetilde{Y}(f))(q) = (X(q))(f)$ for all $q \in \Omega$,

then $Y = X$ on $W \cap \Omega$.

Again let $p \in W \cap \Omega$. Then we can pick a small enough smooth coordinate chart (V,φ) on M such that $p \in V \subseteq W \cap \Omega$. Also, we say $\varphi = (x_1,\ldots,x_n)$.

Next, we can find an open set V' such that $p \in V' \subseteq V$ as well a C^r bump function $\omega : U \to [0,1]$ with $\omega \equiv 1$ on V' and $\text{supp}(\omega) \subseteq V$. Then, we write:

$$Y(p) = \sum_{i=1}^n c_i \left.\frac{\partial}{\partial x_i}\right|_p \text{ and } X(p) = \sum_{i=1}^n d_i \left.\frac{\partial}{\partial x_i}\right|_p.$$

For any $1 \leq j \leq n$ we have that:

$$\begin{aligned} c_j = (X(p))(x_j) &= (X(p))(\omega x_j) \\ &= (\widetilde{Y}(\omega x_j))(p) = (Y(p))(\omega x_j) = (Y(p))(x_j) = d_j \end{aligned}$$

So, $X(p) = Y(p)$.

Hopefully it is clear how the second claim proves the uniqueness part of the theorem statement. Also importantly, the second claim let's us use the gluing lemma on a covering of C^k vector fields given by the first claim in order to get a single C^k vector field $Y : U \to TM$ doing exactly what we wanted. ■

Before moving on to the final topic I want to go over before starting the 250 homework, I'd like to say in what way the standard topology and C^{r-1} structure on TM (where M is a C^r manifold) is "natural" like I claimed without justification on *page 1132*.

Note that if (U, φ) is a smooth chart on a C^r manifold M and we write $\varphi = (x_1, \ldots, x_n)$, then for each $1 \leq i \leq n$ we have that $p \mapsto \left.\frac{\partial}{\partial x_i}\right|_p$ defines a rough vector field over U. This vector field is called a coordinate vector field.

Proposition 10.24: Let M be a C^r manifold. The topology and C^{r-1} structure that TM is equipped with are the unique such objects satisfying that $\pi : TM \to M$ is a C^{r-1} vector bundle and every coordinate vector field is a C^{r-1} vector field.

Proof:
Let (U, φ) be a smooth chart on M and write $\varphi = (x_1, \ldots, x_n)$. Then define $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ by $\Phi(p, \sum_{i=1}^n c_i \left.\frac{\partial}{\partial x_i}\right|_p) = (p, (c_1, \ldots, c_n))$. Also for every $1 \leq i \leq n$ let $\sigma_i : U \to TM$ be defined by $\sigma_i(p) = \left.\frac{\partial}{\partial x_i}\right|_p$.

In the standard topology on TM, we know that Φ is a C^{r-1} map and in turn $(\sigma_i)_{1 \leq i \leq n}$ is the local frame associated with Φ. So, each σ_i is a C^{r-1} vector field like we wanted.

Meanwhile, suppose TM is equipped with some alternate topology $\mathcal{T}$ and C^{r-1} structure $\mathcal{A}$ such that $\pi : TM \to M$ is a C^{r-1} vector bundle and every coordinate vector field is a C^{r-1} vector field.

Then $(\sigma_i)_{1 \leq i \leq n}$ is a C^r local frame for $(TM, \mathcal{T}, \mathcal{A})$ over U. Hence by corollary 10.21 on *page 1202*, we know that $(\pi^{-1}(U), \widetilde{\varphi})$ where $\widetilde{\varphi}(p, \sum_{i=1}^n c_i \left.\frac{\partial}{\partial x_i}\right|_p) := (\varphi(p), (c_1, \ldots, c_n))$ is a smooth chart on $(TM, \mathcal{T}, \mathcal{A})$. Yet $(\pi^{-1}(U), \widetilde{\varphi})$ is also a smooth chart in the original topology and C^{r-1} structure on TM.

Since such charts cover all of TM we can now conclude by standard arguments (such as those in the middle of *page 1199*) to conclude our new topology and C^{r-1} structure are identical to our old objects. ■

Given a C^r manifold M, another C^{r-1} vector bundle I want to define is M's cotangent bundle T^*M.

Let $p \in M$, let $U \subseteq M$ be an open set containing p, and let $f : U \to \mathbb{R}$ be a C^1 map. We can define a map in $T_p^*M := (T_pM)^*$ by $v \mapsto := v(f)$. Usually this map is denoted $\mathrm{d}f_p$ (and also called the differential of f). I'll briefly refer to this map though as $\widetilde{\mathrm{d}f_p}$ just to distinguish it from the typical differential.

Next, given any $a \in \mathbb{R}$ we can identify $T_a\mathbb{R}$ with $\mathbb{R}$ by the linear map sending $\frac{\mathrm{d}}{\mathrm{d}t}\big|_a$ to 1. And this gives us a more direct way of considering the differential $\mathrm{d}f_p : T_pM \to T_{f(p)}\mathbb{R}$ as a dual vector.

Propositon: $\mathrm{d}f_p = \widetilde{\mathrm{d}f_p}$.

Proof:

$$\widetilde{\mathrm{d}f_p}(v) = v(f) = v(\mathrm{Id}_{\mathbb{R}} \circ f) = \mathrm{d}f_p(v)\ \tfrac{\mathrm{d}}{\mathrm{d}t}\big|_{f(p)}\ \mathrm{Id}_{\mathbb{R}} = \mathrm{d}f_p(v) \cdot 1 \ ■$$

A big reason I wanted to show the prior equivalence is just because it shows that we already know how to calculate a matrix representation of $\mathrm{d}f_p \in T_p^*M$. Namely, we can just compute the gradient of f with respect to some local coordinates.

Furthermore, we have the following corollary:

Corollary: Let M be a C^r manifold and suppose $f : U \to \mathbb{R}$ is a C^1 map. Then f has a critical point at $p \in U$ iff $\mathrm{d}f_p = 0$ in T_p^*M.

Given any C^r manifold M and $p \in M$, we can consider a smooth chart (U, φ) on M containing p. In particular, we write $\varphi(q) = (x_1(q), \ldots, x_n(q))$ for all $q \in U$. Then recall that $\frac{\partial}{\partial x_1}\Big|_p, \ldots, \frac{\partial}{\partial x_n}\Big|_p$ forms a basis for T_pU.

In turn we can consider the dual basis $\frac{\partial}{\partial x_1}\Big|_p^*, \ldots, \frac{\partial}{\partial x_n}\Big|_p^*$ for T_p^*M.

Also note that each $x_i : U \to \mathbb{R}$ is a C^r map. So, we can consider the dual vector $\mathrm{d}(x_i)_p \in T_p^*M$.

Claim: $\mathrm{d}(x_i)_p = \left.\frac{\partial}{\partial x_i}\right|_p^*$.

Proof:
Let $v = \sum_{j=1}^n v_j \left.\frac{\partial}{\partial x_j}\right|_p$ be any vector in T_pM. Then:

$$\mathrm{d}(x_i)_p\left(\sum_{j=1}^n v_j \left.\frac{\partial}{\partial x_j}\right|_p\right) = \sum_{j=1}^n v_j \left(\left.\frac{\partial}{\partial x_j}\right|_p x_i\right) = v_i = \left(\left.\frac{\partial}{\partial x_i}\right|_p^*\right)\left(\sum_{j=1}^n v_j \left.\frac{\partial}{\partial x_j}\right|_p\right). \blacksquare$$

Because of the prior claim, we just refer to the dual basis for T_p^*M by $\mathrm{d}(x_1)_p, \ldots, \mathrm{d}(x_n)_p$.

Next, suppose (V, ψ) is another smooth chart on M containing p and write $\psi(q) = (y_1(q), \ldots, y_n(q))$ for all $q \in V$. What is the matrix representing the change of coordinates from $\mathrm{d}(x_1)_p, \ldots, \mathrm{d}(x_n)_p$ to $\mathrm{d}(y_1)_p, \ldots, \mathrm{d}(y_n)_p$?

The answer is incredibly straightforward. We already know the matrix representaion of the change from the coordinates $\left.\frac{\partial}{\partial x_i}\right|_p$ to the coordinates $\left.\frac{\partial}{\partial y_i}\right|_p$ in the tangent space T_pM is simply given by:

$$\mathrm{D}(\psi \circ \varphi^{-1})(\varphi(p)).$$

In turn, it is a well-known fact that the matrix representation of the change from the coordinates $\mathrm{d}(y_i)_p$ to the coordinates $\mathrm{d}(x_i)_p$ is given by the matrix transpose $\mathrm{D}(\psi \circ \varphi^{-1})(\varphi(p))^*$. And finally, we can invert that to get the change of coordinates going in the direction we want.

Now let $T^*M = \coprod_{p \in M} T_p^*M$ and for any $(p, \omega) \in T^*M$ (i.e. $\omega \in T_p^*M$) define $\pi(p, \omega) = p$. Note that given any open $U \subseteq M$, and C^1 map $f : U \to \mathbb{R}$, we have that $\mathrm{d}f : U \to T^*M$ defines a "rough" local section for $\pi : T^*M \to M$.

The adjective "rough" here is just because we haven't equipped T^*M with any topology for f to be continuous with respect to yet.

Proposition 11.9: Let M be a C^r manifold (where $r \geq 1$). Then there is a unique topology (and C^{r-1} structure if $r \geq 2$) on T^*M such that $\pi : T^*M \to M$ is a C^{r-1} vector bundle of rank $n = \dim(M)$ and for any local coordinates $(U, \varphi = (x_1, \ldots, x_n))$ on M we have that each $\mathrm{d}x_i$ is a C^{r-1} local section of π (called a coordinate covector field).

Proof:
Given any smooth chart $(U, \varphi = (x_1, \ldots, x_n))$ on M define the local trivialization $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^n$ by $\Phi(p, \sum_{i=1}^n \omega_i \mathrm{d}x_i) = (p, (\omega_1, \ldots, \omega_n))$.

Also given another smooth chart $(V, \psi = (y_1, \ldots, y_n))$ on M we can analogously define the local trivialization $\Psi : \pi^{-1}(V) \to V \times \mathbb{R}^n$. Then we can see that:

$$\Psi \circ \Phi^{-1}(p, \omega) = (p, (((\mathrm{D}(\psi \circ \varphi^{-1}))(\varphi(p)))^*)^{-1}\,\omega).$$

Importantly, $(((\mathrm{D}(\psi \circ \varphi^{-1}))(\varphi(p)))^*)^{-1}$ is a C^{r-1} map from U to $\mathrm{GL}_n(\mathbb{R})$. So, we can conclude by lemma 10.6 on page 1198 that there is a unique topology and C^{r-1} structure on T^*M such that all the Φ constructed priorly are C^{r-1} local trivializations and $\pi : T^*M \to M$ is a C^{r-1} vector bundle.

The uniqueness of the topology and C^{r-1} structure on T^*M follows identically as in the proof of uniqueness in proposition 10.24 of the topology and C^{r-1} structure on TM. $\blacksquare$

I'll briefly mention how to compute a covector field (i.e. section of T^*M) defined by a C^r map $f : M \to \mathbb{R}$.

Let (U, φ) be a smooth chart on M and write $\varphi = (x_1, \ldots, x_n)$ Then we know there are functions $\tau_1, \ldots, \tau_n : U \to \mathbb{R}$ such that $\mathrm{d}f = \sum_{i=1}^n \tau_i \mathrm{d}x_i$. But now for any fixed j we have that:

$$\frac{\partial f}{\partial x_j}(p) := \frac{\partial}{\partial x_j}\Big|_p f = \mathrm{d}f\left(\frac{\partial}{\partial x_j}\Big|_p\right) = \sum_{i=1}^n \tau_i \mathrm{d}x_i\left(\frac{\partial}{\partial x_j}\Big|_p\right) = \tau_j(p)$$

So, for each $1 \leq i \leq n$ we have that:

$$\tau_i(p) = \frac{\partial f}{\partial x_i}(p) = \frac{\partial (f \circ \varphi^{-1})}{\partial x_i}(\varphi(p)) \text{ for all } p \in U.$$

In other words, we have that $\mathrm{d}f(p) = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(p)\mathrm{d}x_i$ for all $p \in U$.

The prior computation also now lets us conclude that if f is a C^r map then $\mathrm{d}f$ is a C^{r-1} covector field since each component function with respect to the frame $(\mathrm{d}x_i)_{1 \leq i \leq n}$ is a C^{r-1} map.

One other note is that a covector field is also called a 1 form.

Set 2 Problem 2: Let M be a C^2 manifold, let $f : M \to \mathbb{R}$ be a C^2 map, and suppose $p \in M$ is a critical point of f. The Hessian at p is the "symmetric bilinear form (i.e. symmetric 2-tensor)" on T_pM given with respect to any local coordinates $(U, \varphi = (x_1, \ldots, x_n))$ by:

$$H_p(u, v) = \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2 f}{\partial x_i \partial x_j}(p) u_i v_j \text{ where } u = \sum_{i=1}^n u_i \frac{\partial}{\partial x_i}\Big|_p \text{ and } v = \sum_{j=1}^n v_j \frac{\partial}{\partial x_j}\Big|_p.$$

(Here $\frac{\partial^2 f}{\partial x_i \partial x_j}(p) := \frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j}(\varphi(p))$.) Show that this is well-defined independently of the local coordinates chosen.

The hard part is just showing that this is well-defined irrespective of which local coordinates are chosen. So, I'll let (U, φ) and (V, ψ) both be smooth charts on M containing the critical point p, and also I'll write $\varphi = (x_1, \ldots, x_n)$ and $\psi = (y_1, \ldots, y_n)$.

Next I'll let u and v be vectors in T_pM. We can uniquely write:

- $u = \sum_{i=1}^n u_i \frac{\partial}{\partial x_i}\Big|_p = \sum_{i=1}^n \overline{u_i} \frac{\partial}{\partial x_i}\Big|_p$
- $v = \sum_{i=1}^n v_i \frac{\partial}{\partial x_i}\Big|_p = \sum_{i=1}^n \overline{v_i} \frac{\partial}{\partial y_i}\Big|_p$.

And now, our task is to prove that:

$$\begin{bmatrix} u_1 & \cdots & u_n \end{bmatrix} \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(p) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(p) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(p) \end{bmatrix} \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$$

$$= \begin{bmatrix} \overline{u_1} & \cdots & \overline{u_n} \end{bmatrix} \begin{bmatrix} \frac{\partial^2 f}{\partial y_1^2}(p) & \cdots & \frac{\partial^2 f}{\partial y_1 \partial y_n}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial y_n \partial y_1}(p) & \cdots & \frac{\partial^2 f}{\partial y_n^2}(p) \end{bmatrix} \begin{bmatrix} \overline{v_1} \\ \vdots \\ \overline{v_n} \end{bmatrix}$$

But now note that if $\sum_{i=1}^n c_i \left.\frac{\partial}{\partial x_i}\right|_p = \sum_{i=1}^n \overline{c_i} \left.\frac{\partial}{\partial y_i}\right|_p$ then:

$$\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} \frac{\partial x_1}{\partial y_1}(p) & \cdots & \frac{\partial x_1}{\partial y_n}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial y_1}(p) & \cdots & \frac{\partial x_n}{\partial y_n}(p) \end{bmatrix} \begin{bmatrix} \overline{c_1} \\ \vdots \\ \overline{c_n} \end{bmatrix}$$

Therefore, it suffices to prove that:

$$\begin{bmatrix} \frac{\partial x_1}{\partial y_1}(p) & \cdots & \frac{\partial x_1}{\partial y_n}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial y_1}(p) & \cdots & \frac{\partial x_n}{\partial y_n}(p) \end{bmatrix}^{\mathsf{T}} \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(p) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(p) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(p) \end{bmatrix} \begin{bmatrix} \frac{\partial x_1}{\partial y_1}(p) & \cdots & \frac{\partial x_1}{\partial y_n}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial y_1}(p) & \cdots & \frac{\partial x_n}{\partial y_n}(p) \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\partial^2 f}{\partial y_1^2}(p) & \cdots & \frac{\partial^2 f}{\partial y_1 \partial y_n}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial y_n \partial y_1}(p) & \cdots & \frac{\partial^2 f}{\partial y_n^2}(p) \end{bmatrix}$$

Or going component by component, we need to show for every $1 \leq i \leq n$ and $1 \leq j \leq n$ that:

$$\frac{\partial^2 (f \circ \psi^{-1})}{\partial y_i \partial y_j}(\psi(p)) = \frac{\partial^2 f}{\partial y_i \partial y_j}(p) = \sum_{k=1}^n \sum_{\ell=1}^n \frac{\partial^2 f}{\partial x_k \partial x_\ell}(p) \frac{\partial x_k}{\partial y_i}(p) \frac{\partial x_\ell}{\partial y_j}(p)$$

$$= \sum_{k=1}^n \sum_{\ell=1}^n \frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_k \partial x_\ell}(\varphi(p)) \frac{\partial (x_k \circ \psi^{-1})}{\partial y_i}(\psi(p)) \frac{\partial (x_\ell \circ \psi^{-1})}{\partial y_j}(\psi(p))$$

But now note by chain rule that:

$$\frac{\partial (f \circ \psi^{-1})}{\partial y_j}(\psi(p)) = \frac{\partial ((f \circ \varphi^{-1}) \circ (\varphi \circ \psi^{-1}))}{\partial y_j}(\psi(p))$$

$$= \sum_{\ell=1}^n \frac{\partial (f \circ \varphi^{-1})}{\partial x_\ell}(\varphi(p)) \frac{\partial (x_\ell \circ \psi^{-1})}{\partial y_j}(\psi(p))$$

$$= \sum_{\ell=1}^n \left(\frac{\partial (f \circ \varphi^{-1})}{\partial x_\ell} \circ (\varphi \circ \psi^{-1}) \right)(\psi(p)) \cdot \frac{\partial (x_\ell \circ \psi^{-1})}{\partial y_j}(\psi(p)).$$

Next, we can apply both the chain and product rules again to get that:

$$\frac{\partial^2 (f \circ \psi^{-1})}{\partial y_i \partial y_j}(\psi(p)) = \sum_{\ell=1}^n \frac{\partial}{\partial y_i} \left[\left(\frac{\partial (f \circ \varphi^{-1})}{\partial x_\ell} \circ (\varphi \circ \psi^{-1}) \right)(\psi(p)) \cdot \frac{\partial (x_\ell \circ \psi^{-1})}{\partial y_j}(\psi(p)) \right]$$

$$= \sum_{\ell=1}^n \sum_{k=1}^n \frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_k \partial x_\ell}(\varphi(p)) \frac{\partial (x_k \circ \psi^{-1})}{\partial y_i}(\psi(p)) \frac{\partial (x_\ell \circ \psi^{-1})}{\partial y_j}(\psi(p))$$

$$+ \sum_{\ell=1}^n \frac{\partial (f \circ \varphi^{-1})}{\partial x_\ell}(\varphi(p)) \frac{\partial^2 (x_\ell \circ \psi^{-1})}{\partial y_i \partial y_j}(\psi(p))$$

And finally, because f has a critical point at p, we know that:

$$\frac{\partial (f \circ \varphi^{-1})}{\partial x_\ell}(\varphi(p)) = \left.\frac{\partial}{\partial x_\ell}\right| f = 0 \text{ for all } \ell.$$

So the terms we don't want all cancel and finished proving coordinate-independence.

The fact that H_p is a symmetric map (i.e. $H_p(u, v) = H_p(v, u)$) is because f is C^2 and so $\frac{\partial^2 f}{\partial x_i \partial x_j}(p) = \frac{\partial^2 f}{\partial x_j \partial x_i}(p)$. Also, H_p is clearly a bilinear map. ■

I'm short on time so I don't want to get into defining what a tensor product of bundles is / should be. However, the next question is about proving the existence of a Riemannian metric on a C^r manifold. So, I ought to at least give some notes.

If M is a C^r manifold, a Riemannian metric on M is a "C^{r-1} varying" choice of inner product $g_p : T_pM \times T_pM \to \mathbb{R}$ for every $p \in M$.

As for what that definition means, I first need to give some observations about inner products. So suppose V is an n-dimensional real vector space and let $v_1, \ldots, v_n$ be a basis for V. Thus, we can uniquely write any vector $v \in V$ as $(c_1, \ldots, c_n) \coloneqq \sum_{i=1}^n c_i v_i$.

I'll write out this next reasoning in a way that generalizes to complex vector spaces. If $\langle \cdot, \cdot \rangle$ is an inner product and we interpret $\vec{c} = (c_1, \ldots, c_n)$ as a column vector, we have that $\langle \vec{c}, \vec{d} \rangle = \vec{d}^* \mathbf{A} \vec{c}$ where $\mathbf{A}$ is the matrix:

$$\begin{bmatrix} \langle v_1, v_1 \rangle & \cdots & \langle v_n, v_1 \rangle \\ \vdots & \ddots & \vdots \\ \langle v_1, v_n \rangle & \cdots & \langle v_n, v_n \rangle \end{bmatrix}.$$

Note that $\mathbf{A}$ is a self-adjoint matrix, $\vec{c}^* \mathbf{A} \vec{c} \geq 0$ for all $\vec{c}$, and $\vec{c}^* \mathbf{A} \vec{c} = 0$ iff $\vec{c} = 0$. In other words, $\mathbf{A}$ is a positive definite matrix.

Conversely, suppose $\mathbf{B}$ is a self-adjoint positive definite matrix and define:

$$\langle \vec{c}, \vec{d} \rangle_{\text{new}} = \vec{d}^* \mathbf{B} \vec{c}.$$

Note that:

$$\langle \vec{d}, \vec{c} \rangle_{\text{new}} = \vec{c}^* B \vec{d} = \overline{\vec{d}^* B^* \vec{c}} = \overline{\vec{d}^* B \vec{c}} = \overline{\langle \vec{c}, \vec{d} \rangle_{\text{new}}}.$$

Also it's easy to see that $\langle \vec{c}, \vec{d} \rangle_{\text{new}}$ is linear with respect to the first argument $\vec{c}$. And by positive definiteness we know that the last requirement for $\langle \cdot, \cdot \rangle_{\text{new}}$ to be a well-defined inner product holds.

As a side note, in the real vector space and inner product case, all the prior reasoning holds just without any complex conjugates.

Focusing now on the case of a C^r manifold M, let (U, φ) be any smooth chart on M and write $\varphi = (x_1, \ldots, x_n)$. Then by defining $\sigma_i : U \to TM$ by $\sigma_i(p) = \frac{\partial}{\partial x_i}\Big|_p$, we get that $(\sigma_i)_{1 \leq i \leq n}$ forms a C^{r-1} frame for TM over U.

It follows that if $\{g_p\}_{p \in M}$ is a Riemannian metric then there are coordinate functions $A_{i,j} : U \to \mathbb{R}$ such that for all $p \in U$, if $u = \sum_{i=1}^n u_i \sigma_i(p)$ and $v = \sum_{i=1}^n v_i \sigma_i(p)$ are elements of T_pM then:

$$g_p(u, v) = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix} \begin{bmatrix} A_{1,1}(p) & \cdots & A_{1,n}(p) \\ \vdots & \ddots & \vdots \\ A_{n,1}(p) & \cdots & A_{n,n}(p) \end{bmatrix} \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$$

In particular, $A_{i,j}(p) = g_p(\sigma_j(p), \sigma_i(p)) = g_p(\frac{\partial}{\partial x_j}\Big|_p, \frac{\partial}{\partial x_i}\Big|_p)$.

When we say that g_p varies C^{r-1}-ly with p, what we mean is that there is an atlas of smooth charts $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ covering M and satisfying that the coordinate functions $A_{i,j}$ induced by each smooth chart in our atlas are C^{r-1} maps on U. Once I define what tensor bundles on M are, it will become clear that this is equivalent to us having a C^{r-1} "tensor field" (i.e. section) of a certain tensor bundle.

Observation: Suppose $\{g_p\}_{p \in M}$ is a Riemannian metric over a C^r manifold M, and let (U, φ) and (V, ψ) be overlapping smooth charts on M (i.e. $U \cap V \neq \emptyset$). Write $\varphi = (x_1, \ldots, x_n)$ and $\psi = (y_1, \ldots, y_n)$. Then for each $1 \leq i, j \leq n$ define:

- $A_{i,j} : U \to \mathbb{R}$ by $A_{i,j}(p) = g_p(\frac{\partial}{\partial x_j}\Big|_p, \frac{\partial}{\partial x_i}\Big|_p)$,
- $B_{i,j} : V \to \mathbb{R}$ by $B_{i,j}(p) = g_p(\frac{\partial}{\partial y_j}\Big|_p, \frac{\partial}{\partial y_i}\Big|_p)$.

If all the $A_{i,j}$ are C^{r-1} maps then all the $B_{i,j}$ are C^{r-1} maps on $U \cap V$.

To see why, let $\mathbf{A}(p)$ and $\mathbf{B}(p)$ be the matrices with entries $A_{i,j}(p)$ and $B_{i,j}(p)$ respectively. Also let $\tau : \psi(U \cap V) \to \varphi(U \cap V)$ be the C^r chart transition function. Then by doing a change of basis like in set 2 problem 2 two pages ago, we can get for all $p \in U \cap V$ that:

$$\mathbf{B}(p) = (\mathrm{D}\tau(\psi(p)))^{\mathsf{T}} \mathbf{A}(p) (\mathrm{D}\tau(\psi(p))).$$

Yet going component by component, this is equivalent to saying that:

$$B_{i,j}(p) = \sum_{k=1}^{n} \sum_{\ell=1}^{n} \frac{\partial x_k}{\partial y_i}(p) \frac{\partial x_\ell}{\partial y_j}(p) A_{k,\ell}(p).$$

So, each $B_{i,j}$ is a C^{r-1} map on $U \cap V$ since the right hand side of the above equation is C^{r-1}. ■

A Riemannian manifold is a manifold equipped with a Riemannian metric. These manifolds are very important and in Math 250b during Winter quarter we will go way more in depth about them. The T.L.D.R. though for now is that a Riemannian metric allows one to talk about curvature or lengths or areas/volumes. Hopefully that is enough background to make the next question make sense.

Set 2 Problem 3: Show that if M is a C^r manifold (where $r \geq 1$) then M admits a Riemannian metric.

Lemma: Suppose (U, φ) is a smooth chart on M. Then there is a well-defined Riemannian metric on U.

Why?

Write $\varphi = (x_1, \ldots, x_n)$. Then for all $p \in U$ define:

$$g_p(u, v) = \sum_{i=1}^{n} \mathrm{d}x_i(u) \mathrm{d}x_i(v).$$

Checking that this is an inner product is easy. Also all the coordinate functions with respect to the obvious frame are either the constant 0 or constant 1 functions.

Now let $\{(U_\alpha, \varphi)\}_{\alpha \in A}$ be an open cover of M with smooth charts. For each $\alpha \in A$ we can by the prior lemma find a Riemmanian metric $\{g_p^{(\alpha)}\}_{p \in U_\alpha}$ over U_α. Next, we can find a C^r partition of unity $\{\phi_\alpha : M \to [0,1]\}_{\alpha \in A}$ such that:

- $\sum_{\alpha \in A} \phi_\alpha = 1$ on M;
- $\text{supp}(\phi_\alpha) \subseteq U_\alpha$ for all $\alpha \in A$;
- Every $p \in M$ has an open neighborhod $V \subseteq M$ such that only finitely many ϕ_α are nonzero anywhere V.

Now set $h_p = \sum_{\alpha \in A} \phi_\alpha(p) g_p^{(\alpha)}$ for all $p \in M$. Note that this sum is well-defined because at the points p where any $g_p^{(\alpha)}$ is not defined, we know $\phi_\alpha(p) = 0$ and so we can just ignore that term.

I claim that $\{h_p\}_{p \in M}$ is a Riemmanian metric.

To see why $h_p : T_pM \times T_pM$ is a well-defined inner product, just note that finite summing of inner product functions and scaling of inner product functions by positive real numbers gives us back a well-defined inner product function.

Meanwhile, to prove that h^p varies C^{r-1}-ly over M, consider any $p \in M$ and let $V \subseteq M$ be an open neighborhood on which only finitely many ϕ_α are nonzero. (In particular, say that the only nonzero ϕ_α are $\phi_{\alpha_1}, \ldots, \phi_{\alpha_k}$). By shrinking V, we can further assume that $p \in \bigcap_{j=1}^k \text{supp}(\phi_{\alpha_j})$. This has the added consequence that $p \in \bigcap_{j=1}^k U_{\alpha_j}$. So by shrinking V again we can assume that $V \subseteq \bigcap_{j=1}^k U_{\alpha_j}$. And this is useful because it lets us say that $g_p^{(\alpha_j)}$ is well defined for all $p \in V$ and $1 \leq j \leq k$.

Now let $y_1, \ldots, y_n$ be smooth coordinates on V. Note that we have at least k choices of such smooth coordinates to pick from. So, we don't need to restrict V further to get those coordinates.

We denote the (k,ℓ)th coordinate function for $\{g_p^{(\alpha_j)}\}_{p \in V}$ as:

$$A_{k,\ell}^{(\alpha_j)}(p) \coloneqq g_p^{(\alpha_j)}\left(\frac{\partial}{\partial y_\ell}\Big|_p, \frac{\partial}{\partial y_k}\Big|_p\right).$$

And now the (k,ℓ)th coordinate function for $\{h_p\}_{p \in V}$ is:

$$B_{k,\ell}(p) = \sum_{j=1}^k \phi_{\alpha_j}(p) A_{k,\ell}^{(\alpha_j)}(p)$$

Thus the coordinate functions for $\{h_p\}_{p \in M}$ are C^{r-1} maps. ■

Set 2 Problem 7: Let α be an irrational number.

(i) Show that for all integers $N > 0$ there exists integers m, n with $1 \leq n \leq N$ such that $|n\alpha - m| < 1/N$.

Proof:
Let $\text{frac}(x)$ denote the non-integer part of x. In particular, we will make it so that $\text{frac}(x) \in [0,1)$ by defining:

$$\text{frac}(x) = \begin{cases} x - \lfloor x \rfloor & \text{if } x \geq 0 \\ \lceil x \rceil - x & \text{if } x < 0 \end{cases}$$

It suffices to show that there exists $n \in \{1, \ldots, N\}$ such that:

$$\text{frac}(n\alpha) \in [0, \tfrac{1}{N}) \cup (\tfrac{N-1}{N}, 1).$$

Yet note by partitioning $[0, 1)$ into N intervals of length $1/N$ and then considering the $N + 1$ many numbers $0, 1, \ldots, N$, we can conclude by the pigeonhole principle that there exists some $0 \leq k < n$ as well as distinct integers $i, j \in \{0, 1, \ldots, N\}$ with:

$$\tfrac{k}{N} \leq \text{frac}(i\alpha), \text{frac}(j\alpha) < \tfrac{k+1}{N}.$$

Without loss of generality suppose $i < j$. Then $n \coloneqq j - i \in \{1, \ldots, N\}$. Also, we will get that $\text{frac}(n\alpha) \in [0, \frac{1}{N}) \cup (\frac{N-1}{N}, 1)$ as we wanted.

(ii) Let $\mathbb{T} = S^1 \times S^1$ denote the torus.

Note that $\mathbb{T} = \Phi^{-1}((1, 1))$ where $\Phi : \mathbb{R}^4 \to \mathbb{R}^2$ is given by $\Phi(x, y, z, w) = (x^2 + y^2, z^2 + w^2)$. Also, Φ has full rank so long $(x, y) \neq 0$ and $(z, w) \neq 0$. It follows that $\mathbb{T}$ is a regular level set of Φ. So, $\mathbb{T}$ is a 2-dimensional embedded submanifold of $\mathbb{R}^4$ without a boundary.

We identify $\mathbb{R}^4 = \mathbb{R}^2 \times \mathbb{R}^2 \cong \mathbb{C} \times \mathbb{C}$. Now let $\gamma : \mathbb{R} \to \mathbb{T}$ be given by $\gamma(t) = (e^{2\pi it}, e^{2\pi i\alpha t})$. Then show γ is an injective immersion.

Injectivity is the hardest to show. So suppose $\gamma(t_1) = \gamma(t_2)$. Then we must have that there exists integers n, m such that:

$$2\pi t_1 = 2\pi t_2 + 2n\pi \text{ and } 2\pi\alpha t_1 = 2\pi\alpha t_2 + 2m\pi.$$

Dividing by 2π we equivalently need:

$$t_1 = t_2 + n \text{ and } \alpha t_1 = \alpha t_2 + m.$$

But then $\alpha t_2 + \alpha n = \alpha t_2 + m$. In other words, $\alpha n = m$. And finally, because α is irrational, the only integer sollutions to this are if $m = 0$ and $n = 0$. So, $t_1 = t_2 + 0$ and we've proven injectivity.

To show γ is an immersion into $\mathbb{T}$, it suffices to show that γ is an immersion into the ambient manifold $\mathbb{R}^4 \cong \mathbb{C} \times \mathbb{C}$. This is easy though. Just observe that $\gamma'(t) = (2\pi i e^{2\pi it}, 2\pi i\alpha e^{2\pi i\alpha t})$. Yet neither complex coordinate of $\gamma'(t)$ ever equals 0. So, $\gamma'(t) \neq 0$ for all $t \in \mathbb{R}$.

(iii) Show for all $\varepsilon > 0$ there exists an integer $n > 0$ such that $|\gamma(n) - \gamma(0)| < \varepsilon$.

Here, we set $|(z, w) - (z', w')| \coloneqq |z - z'| + |w - w'|$ for all $z, z', w, w' \in \mathbb{C}$. This metric generates the product topology on $\mathbb{C} \times \mathbb{C}$ which is identical to the Euclidean topology of $\mathbb{R}^4$. So, this metric induces the topology on $\mathbb{T}$.

We know the map $t \mapsto e^{2\pi it}$ is a uniformly continuous function from $\mathbb{R}$ to S^1. So, for any $\varepsilon > 0$ we can find $\delta > 0$ such that $|t_1 - t_2| < \delta$ implies that $|e^{2\pi it_1} - e^{2\pi it_2}| < \varepsilon$.

Next, pick a positive integer N such that $1/N < \delta$. Then from part (i) we know that there exist integers m, n with $1 \leq n \leq N$ such that $|n\alpha - m| < 1/N < \delta$. In turn, we have that:

$$|\gamma(n) - \gamma(0)| = |1 - 1| + |e^{2\pi i\alpha n} - 1| = |e^{2\pi i\alpha n} - e^{2\pi i m}| < \varepsilon.$$

(iv) Show that γ is not a homeomorphism onto its image. So, $\gamma(\mathbb{R})$ is not an embedded submanifold of $\mathbb{T}$.

If γ were a homeomorphism then there would exist a continuous inverse function $\gamma^{-1} : \gamma(\mathbb{R}) \to \mathbb{R}$ (where $\gamma(\mathbb{R})$ is equipped with the subspace topology). And now if $(z_k, w_k) \to (1, 1) \in \gamma(\mathbb{R})$, we ought to have $\gamma^{-1}(z_k, w_k) \to \gamma^{-1}(1, 1) = 0$ in $\mathbb{R}$.

Yet by (iii) we know there is a sequence $\{n_k\}_{k\in\mathbb{N}} \subseteq \mathbb{Z}_{>0}$ with $\gamma(n_k) \to \gamma(0) = (1, 1)$. And now we have a contradiction as $n_k = \gamma^{-1}(\gamma(n_k))$ does not converge to $0 = \gamma^{-1}(\gamma(0))$ as $k \to \infty$.

(v) Show that $\gamma(\mathbb{R})$ is a dense subset of $\mathbb{T}$.

Given any point $(z, w) \in \mathbb{T}$, we can write it as $(e^{2\pi iA}, e^{2\pi iB})$ for some real numbers A and B.

But now note for any integer n that:

$$\gamma(A + n) = (e^{2\pi iA}, e^{2\pi i\alpha(A+n)}).$$

Thus, to show $(z, w) \in \overline{\gamma(\mathbb{R})}$ it suffices to show there exists a sequence of integers $\{n_k\}_{k\in\mathbb{N}}$ such that $e^{2\pi i\alpha A}e^{2\pi i\alpha n_k} \to e^{2\pi iB}$. Or equivalently, upon letting $C = B - \alpha A$, we need to find a sequence of integers $\{n_k\}_{k\in\mathbb{N}}$ such that:

$$e^{2\pi i(\alpha n - C)} \to 1$$

By the continuity of the map $t \mapsto e^{2\pi it}$, in order to prove such a sequence exists it suffices to prove for any $\varepsilon > 0$ that there exists integers n and m such that $|\alpha n - C - m| < \varepsilon$. One more simplifying assumption is that if $C = 0$ then we are already done by part (i). Otherwise, by moving integers over onto m, we can assume without loss of generality that $C \in (0, 1)$.

Pick $N > 0$ with $\frac{1}{N} < \varepsilon$. Then we know from part (i) that there exists integers n and m with $|n\alpha - m| < \frac{1}{N}$. By negating n and m potentially, we can guarentee that:

$$0 < n\alpha - m < \tfrac{1}{N} < \varepsilon.$$

Let K be the unique integer such that $(K-1)(n\alpha-m) \leq C < K(n\alpha-m)$. Then we know that $0 < Kn\alpha - C - Km \leq n\alpha - m < \varepsilon$. ■

10/17/2026

Math 231A Lecture Notes:

I have a few more notes about harmonic functions to take and then I will move on to talking about the heat equation. To start off, the following is useful for bounding harmonic functions. (Also once again I'll just be assuming $n \geq 2$ always.)

Harnack's Inequality: Suppose $U \subseteq \mathbb{R}^n$ is an open set and K is a connected compact set contained in U. Then there exists a constant $C > 0$ such that:

$$\max_{x \in K}(u) \leq C \min_{x \in K}(u)$$

for all harmonic functions $u : U \to \mathbb{R}^n$ with $u \geq 0$ on U.

Note that C does not depend on the function u at all.

Proof:
Set $r = \frac{1}{4}d(K, \partial U) := \frac{1}{4}\inf\{|x - y| : x \in K, y \in \partial U\}$. Since K and ∂U are disjoint closed sets with K compact, we know $r > 0$.

Now for any $x \in K$ and $y \in \overline{B_r(x)}$ we know by the mean value theorem plus the fact that u is nonnegative that:

$$u(x) = \frac{1}{(2r)^n \alpha(n)} \int_{B_{2r}(x)} u \geq \frac{1}{(2r)^n \alpha(n)} \int_{B_r(y)} u = \frac{r^n \alpha(n)}{(2r)^n \alpha(n)} u(y) = \frac{1}{2^n} u(y)$$

But now observe that because K is compact, we can cover cover K with a finite number N of balls: $B^{(1)}, \ldots, B^{(N)}$, each with radius $r/2$ and centered at a point in K. Next, I claim that because K is connected, we can for any $x, y \in K$ always find $z_1, z_2, \ldots, z_\ell \in K$ (where $\ell \leq 2N - 2$) such that $|x - z_1| \leq r$, $|z_i - z_{i+1}| \leq r$ for all i, and $|z_\ell - y| \leq r$. In fact, I'll even prove a slightly more general lemma.

Lemma: Suppose (X, d) is any metric space, $A \subseteq X$ is a connected set, and there exists finitely many sets $E_1, \ldots, E_N$ of diameter at most r covering A. Given any $x, y \in A$ we can always find points $z_1, \ldots, z_\ell \in A$ (where $\ell \leq 2N - 2$) such that $d(x, z_1) \leq r$, $d(z_i, z_{i+1}) \leq r$ for each i, and $d(z_\ell, y) \leq r$.

Proof:
It suffices to fix $x \in A$. Now to start off, I shall disregard getting an upper bound for ℓ and just show that for any $y \in A$ there even exists points $z_1, \ldots, z_\ell \in A$ such that $d(x, z_1) \leq r$, $d(z_i, z_{i+1}) \leq r$ for each i, and $d(z_\ell, y) < r$. To do this, I shall let $C \subseteq A$ be the set of all $y \in A$ such that this is possible. Note that C is nonempty because trivially $x \in C$. Also, we can easily verify both C and $A - C$ are open subsets of A. Since A is connected, this thus proves that $C = A$.

Now fixing $y \in A$, we've shown that there are $z_1, \ldots, z_\ell \in A$ with $d(x, z_1) \leq r$, $d(z_i, z_{i+1}) \leq r$ for each i, and $d(z_\ell, y) \leq r$. For ease of notation, I'll let $z_0 := x$ and $z_{\ell+1} := y$.

If $\ell > 2N-2$ then we know by the pigeonhole principle that there are integers $0 \leq i_1 < i_2 < i_3 \leq \ell + 1$ with $z_{i_1}, z_{i_2}, z_{i_3}$ all in the same set E_j. Then as E_j has diameter at most r, we know $d(z_{i_1}, z_{i_3}) \leq r$. So, we can disregard each of the z_j with $i_1 < j < i_3$ to get another collection of points satisfying our lemma. In particular, this collection has at least one less element because we disregarded z_{i_2}.

By repeating the prior reasoning finitely many times, we will eventually get that $\ell \leq 2N-2$.

Based on the prior lemma, we can now see for any nonnegative harmonic function $u : U \to \mathbb{R}$ and any $x, y \in K$ that:

$$u(x) \geq \left(\tfrac{1}{2^n}\right)^{2N-1} u(y). \blacksquare$$

I also want to motivate why we'd care about Green's functions.

Firstly, I want to briefly go over how the textbook (see *citation 16*) that my 231 class is loosely based on defines what it means for an open set to have a C^r boundary (where $r \geq 1$).

Let $U \subseteq \mathbb{R}^n$ be an open set and let $k \geq 1$. Evans (the author of that textbook) says that U has a C^k boundary if for each $x^{(0)} \in \partial U$ there exists $r > 0$ and a C^k function $\gamma : \mathbb{R}^{n-1} \to \mathbb{R}$ such that upon relabeling and reorienting the coordinate axes if necessary, we have that:

$$U \cap B(x^{(0)}, r) = \{x \in B(x^{(0)}, r) : x_n < \gamma(x_1, \ldots, x_{n-1})\}$$

I claim that this happens if and only if U is the interior of a regular domain D without corners and whose boundary is ∂U.

$(\Longleftarrow)$

Hopefully it is obvious how this direction is an easy result of the theorem on *pages 1170-1171*.

$(\Longrightarrow)$

We know $D = U \cup \partial U$ is a closed subset of $\mathbb{R}^n$. So if it were a submanifold it would be properly embedded. Also it's clear that U is a C^r submanifold of codimension 0. At the same time, if $p \in \partial U$ and we let γ and $B(x^{(0)}, r)$ be as described in the above definition, then we can apply a change of coordinates:

$$(x_1, \ldots, x_n) \mapsto (x_1, \ldots, x_{n-1}, \gamma(x_1, \ldots, x_{n-1}) - x_n)$$

in order to get an n-dimensional half slice chart for D centered at p.

Now suppose $U \subseteq \mathbb{R}^n$ is a bounded open set with a C^1 boundary. At any point $p \in \partial U$ we let $v(p)$ denote the outward pointing unit normal vector. Also let $\Phi : \mathbb{R}^n - \{0\} \to \mathbb{R}$ be the fundamental solution of $-\triangle$. (See *page 1181*). And let $\tau_x f(y) \coloneqq f(y - x)$.

Observation: Let $u \in C^2(\overline{U})$ be an arbitrary function. Then for any $x \in U$:

$$u(x) = \int_{\partial U} \Phi(y-x)\frac{\partial u}{\partial v}(y) - u(y)\frac{\partial(\tau_x\Phi)}{\partial v}(y)\mathrm{d}S(y) - \int_U \Phi(y-x)\triangle u(y)\mathrm{d}y$$

Proof:
Fix any $x \in U$ and let $\varepsilon > 0$ be small enough so that $\overline{B_\varepsilon(x)} \subseteq U$. Then let $V_\varepsilon = U - B_\varepsilon$.

We have by Green's 2nd identity that:

$$\begin{aligned}\int_{V_\varepsilon} \Phi(y-x)\triangle u(y)\mathrm{d}y &= \int_{\partial V_\varepsilon} \Phi(y-x)\tfrac{\partial u}{\partial v}(y) - u(y)\tfrac{\partial(\tau_x\Phi)}{\partial v}(y)\mathrm{d}S(y)\\ &= \int_{\partial U} \Phi(y-x)\tfrac{\partial u}{\partial v}(y) - u(y)\tfrac{\partial(\tau_x\Phi)}{\partial v}(y)\mathrm{d}S(y)\\ &\quad - \int_{\partial B_\varepsilon(x)} \Phi(y-x)\tfrac{\partial u}{\partial v}(y) - u(y)\tfrac{\partial(\tau_x\Phi)}{\partial v}(y)\mathrm{d}S(y)\end{aligned}$$

But now $\left|\int_{\partial B_\varepsilon(x)} \Phi(y-x)\frac{\partial u}{\partial v}(y)\mathrm{d}S(y)\right| = \int_{\partial B_\varepsilon(x)} |\Phi(y-x)| \cdot |\nabla u(y)|\mathrm{d}S(y)$. For small enough ε we can guarentee that there is some $C > 0$ with $|\nabla u(y)| \leq C$ for all $y \in B_\varepsilon(x)$. And thus:

$$\left|\int_{\partial B_\varepsilon(x)} \Phi(y-x)\tfrac{\partial u}{\partial v}(y)\mathrm{d}S(y)\right| \leq C\varepsilon^{n-1}n\alpha(n)\max_{y\in\partial B_\varepsilon(0)}|\Phi(y)|$$

If $n = 2$ then $\varepsilon\ln(\varepsilon) \to 0$ as $\varepsilon \to 0$. In other words $\ln(\varepsilon) = o(\varepsilon^{-1})$ as $\varepsilon \to 0$. Similarly, if $n \geq 3$ then $\varepsilon^{2-n} = o(\varepsilon^{1-n})$ as $\varepsilon \to 0$. So in general, we can conclude:

$$C\varepsilon^{n-1}n\alpha(n)\max_{y\in\partial B_\varepsilon(0)}|\Phi(y)| = \varepsilon^{n-1}o(1-n) = o(1) \text{ as } \varepsilon \to 0.$$

Meanwhile, note that if $n = 2$ and $y \neq 0$ then $\nabla\Phi(y)\cdot\frac{y}{|y|} = \frac{-1}{2\pi|y|}$. Similarly, if $n \geq 3$ then $\nabla\Phi(y)\cdot\frac{y}{|y|} = \frac{-1}{n\alpha(n)|y|^{n-1}}$. So, we can say that:

$$\int_{\partial B_\varepsilon(x)} u(y)\tfrac{\partial(\tau_x\Phi)}{\partial v}(y)\mathrm{d}S(y) = \tfrac{-1}{n\alpha(n)\varepsilon^{n-1}}\int_{\partial B_\varepsilon(x)} u(y)\mathrm{d}S(y) \to -u(x) \text{ as } \varepsilon \to 0.$$

And thus we've proven that by taking $\varepsilon \to 0$, we have that:

$$\begin{aligned}\int_{V_\varepsilon} \Phi(y-x)\triangle u(y)\mathrm{d}y &= \int_{\partial V_\varepsilon} \Phi(y-x)\tfrac{\partial u}{\partial v}(y) - u(y)\tfrac{\partial(\tau_x\Phi)}{\partial v}(y)\mathrm{d}S(y)\\ &= \int_{\partial U} \Phi(y-x)\tfrac{\partial u}{\partial v}(y) - u(y)\tfrac{\partial(\tau_x\Phi)}{\partial v}(y)\mathrm{d}S(y) - u(x). \blacksquare\end{aligned}$$

Based on the prior observation, we can thus say that u is uniquely determined on U by u and $\frac{\partial u}{\partial v}$ on ∂U as well as $\triangle u$ on U. That said though, our PDE is actually over-determined if we give boundary conditions for both u and $\frac{\partial u}{\partial v}$ on ∂U.

We focus on the Dirichlet problem (i.e. where we don't control for $\frac{\partial u}{\partial v}$ at all). So, we need to cancel out the integral $\int_{\partial U}\Phi(y-x)\frac{\partial u}{\partial v}(y)\mathrm{d}S(y)$. To do this, we shall assume there exists a corrector function $\phi : U \times \overline{U} \to \mathbb{R}$ such that:

- $\triangle_y\phi(x,y) = 0$ for all $x, y \in U$
- $\phi(x,y) = \Phi(y-x)$ for all $x \in U$ and $y \in \partial U$.
- $\phi^x(y) \coloneqq \phi(x,y)$ satisfies that $\phi^x \in C^2(\overline{U})$.

  (Don't worry for now about if ϕ exists. It will be our goal later to prove it exists and calculate it in the cases we are interested in.)

By Green's 2nd identity we have that:

$$-\int_U \phi^x(y)\triangle u(y)\mathrm{d}y = \int_{\partial U} u(y)\frac{\partial \phi^x}{\partial v}(y) - \Phi(y-x)\frac{\partial u}{\partial v}(y)\mathrm{d}S(y).$$

And by combining that with our prior observation we get that:

$$u(x) = \int_{\partial U} u(y)\left[\frac{\partial \phi^x}{\partial v}(y) - \frac{\partial(\tau_x\Phi)}{\partial v}(y)\right]\mathrm{d}S(y) + \int_U \left[\phi^x(y) - \Phi(y-x)\right]\triangle u(y)\mathrm{d}y$$

Notably the prior expression writes u purely in terms of $u|_{\partial U}$ and $\triangle u$ on U.

We define Green's function associated to the domain $\overline{U}$ to be the map:

$$G(x,y) \coloneqq \Phi(y-x) - \phi^x(y).$$

And by the prior reasoning, we can conclude the following theorem:

Theorem: Suppose $u \in C^2(\overline{U})$ satisfies that $-\triangle u = f$ on U and $u = g$ on ∂U (where f and g are continuous). Then:

$$u(x) = -\int_{\partial U} u(y)\frac{\partial G}{\partial v}(x,y)\mathrm{d}S(y) + \int_U f(y)G(x,y)\mathrm{d}y.$$

(Here $\frac{\partial G}{\partial v}(x,y) = \nabla_y G(x,y)\cdot v$.)

Given an open set $U \subseteq \mathbb{R}^n$ with a C^1 boundary, another way we can interpret Green's function associated to $\overline{U}$ is that it is the map satisfying that $-\triangle_y G(x,y) = \delta_x$ for all $x, y \in U$ and $G(x,y) = 0$ for all $x \in U$ and $y \in \partial U$.

Theorem: For all $x, y \in U$ with $x \neq y$ we have that $G(x,y) = G(y,x)$.

Proof:

Fix $x, y \in U$ and define $w_1(z) \coloneqq G(x,z)$ and $w_2(z) \coloneqq G(y,z)$. Then $\triangle w_1(z) = 0$ for all $z \in U - \{x\}$, $\triangle w_2(z) = 0$ for all $z \in U - \{y\}$, and $w_1(z) = w_2(z) = 0$ if $z \in \partial U$. Now let $\varepsilon > 0$ be such that $B_{2\varepsilon}(x) \cap B_{2\varepsilon}(y) = \emptyset$ and $B_{2\varepsilon}(x) \cup B_{2\varepsilon}(y) \subseteq U$.

By Green's 2nd Identity applied to the domain $V = U - (B_\varepsilon(x) \cup B_\varepsilon(y))$ we have that:

$$0 = -\int_{\partial B_\varepsilon(x)\cup\partial B_\varepsilon(y)} w_1\frac{\partial w_2}{\partial v} - w_2\frac{\partial w_1}{\partial v}\mathrm{d}S.$$

Or after some rearranging, we can equivalently say that:

$$\int_{\partial B_\varepsilon(y)} w_1\frac{\partial w_2}{\partial v}\mathrm{d}S - \int_{\partial B_\varepsilon(y)} w_2\frac{\partial w_1}{\partial v}\mathrm{d}S = \int_{\partial B_\varepsilon(x)} w_2\frac{\partial w_1}{\partial v}\mathrm{d}S - \int_{\partial B_\varepsilon(x)} w_1\frac{\partial w_2}{\partial v}\mathrm{d}S$$

But now observe the following:

w_1 is a smooth function on $B_\varepsilon(y)$ (since it is harmonic on $U - \{x\}$). So, we can find a constant $C > 0$ with $|\nabla w_1| \leq C$ on $\overline{B_\varepsilon(y)}$. And then:

$$\left|\int_{\partial B_\varepsilon(y)} w_2\frac{\partial w_1}{\partial v}\mathrm{d}S\right| \leq C\int_{\partial B_\varepsilon(y)} |w_2|\mathrm{d}S \leq C\varepsilon^{n-1}n\alpha(n)\max_{z\in\partial B_\varepsilon(y)}|w_2(z)|$$

Next, $w_2(z) = \Phi(z-y) - \phi^y(z)$. But ϕ^y is a smooth (since it is harmonic) function around y. So there exists a constant $D > 0$ with $|\phi^y(z)| \leq D$ for all $z \in \overline{B_\varepsilon(y)}$.

And now:

$$C\varepsilon^{n-1}n\alpha(n)\max_{z\in\partial B_\varepsilon(y)}|w_2(z)|\leq CD\varepsilon^{n-1}n\alpha(n)+o(1)\text{ as }\varepsilon\to 0$$

This finishes proving that $\int_{\partial B_\varepsilon(y)} w_2\frac{\partial w_1}{\partial v}\mathrm{d}S\to 0$ as $\varepsilon\to 0$.

Analogous reasoning as in the prior observation shows that $\int_{\partial B_\varepsilon(x)} w_1\frac{\partial w_2}{\partial v}\mathrm{d}S\to 0$ as $\varepsilon\to 0$. And so, we can conclude that:

$$\lim_{\varepsilon\to 0^+}\int_{\partial B_\varepsilon(y)} w_1\frac{\partial w_2}{\partial v}\mathrm{d}S=\lim_{\varepsilon\to 0^+}\int_{\partial B_\varepsilon(x)} w_2\frac{\partial w_1}{\partial v}\mathrm{d}S$$

Finally, we just have to compute these two limits.

Note that:

$$\int_{\partial B_\varepsilon(y)} w_1\frac{\partial w_2}{\partial v}\mathrm{d}S=-\frac{1}{n\alpha(n)\varepsilon^{n-1}}\int_{\partial B_\varepsilon(y)} w_1\mathrm{d}S+\int_{\partial B_\varepsilon(y)} w_1\frac{\partial \phi^y}{\partial v}\mathrm{d}S.$$

As $\varepsilon\to 0$ we have that $-\frac{1}{n\alpha(n)\varepsilon^{n-1}}\int_{\partial B_\varepsilon(y)} w_1\mathrm{d}S\to -w_1(y)$.

Meanwhile, because w_1 and ϕ^y are smooth on $\overline{B_\varepsilon(y)}$ (since they are harmonic), we can find a constant C such that $|w_1\nabla\phi^y|\leq C$ on $\overline{B_\varepsilon(y)}$. And now:

$$\left|\int_{\partial B_\varepsilon(y)} w_1\frac{\partial \phi^y}{\partial v}\mathrm{d}S\right|\leq C\varepsilon^{n-1}\alpha(n)n\to 0\text{ as }\varepsilon\to 0.$$

This finishes showing that $\lim_{\varepsilon\to 0^+}\int_{\partial B_\varepsilon(y)} w_1\frac{\partial w_2}{\partial v}\mathrm{d}S=-w_1(y)$.

Analogous computations show $\lim_{\varepsilon\to 0^+}\int_{\partial B_\varepsilon(x)} w_2\frac{\partial w_1}{\partial v}\mathrm{d}S=-w_2(x)$

So, we've shown that $-G(x,y)=-w_1(y)=-w_2(x)=-G(y,x)$. ■

Right now I don't have the tools to prove when a Green's function does or does not exist. However, I'll at least construct Green's function explicitely for the domain $\overline{B_1(0)}$.

First I'll find the correction function ϕ^x when $x\neq 0$.

Given any $x\in\mathbb{R}^n-\{0\}$ let us denote $\widetilde{x}=\frac{x}{|x|^2}$. Importantly, if $x\in B_1(0)$ then $|\widetilde{x}|>1$. In turn, we know that $|x|(y-\widetilde{x})\neq 0$ for all $y\in\overline{B}_1(0)$. And in turn, upon letting $\phi^x(y)=\Phi(|x|(y-\widetilde{x}))$, we have that:

$$\triangle\phi^x(y)=|x|^2\triangle\Phi(|x|(y-\widetilde{x}))=|x|^2\cdot 0=0\text{ for all }y\in B_1(0).$$

Also note that if $|y|=1$ then:

$$|x|^2|y-\widetilde{x}|^2=|y|^2|x|^2-2x\cdot y+1=1|x|^2-2x\cdot y+|y|^2=|y-x|^2$$

So $|x||y-\widetilde{x}|=|y-x|$ and this proves that $\phi^x(y)=\Phi(|x|(y-\widetilde{x}))=\Phi(y-x)$ for all $y\in\partial B_1(0)$.

Meanwhile, if $x=0$ then just set the correction function equal to:

$$\phi^0\equiv\begin{cases}0 & \text{if } n=2\\ \frac{1}{n(n-2)\alpha(n)} & \text{if } n=3.\end{cases}$$

Constant functions are trivially harmonic and we know $\phi^0(y) = \Phi(y)$ trivially when $|y| = 1$.

With that, we are basically done. In the case that $n \geq 3$ we have that:

If $x \neq 0$ then:

$$G(x,y) = \Phi(y-x) - \Phi(|x|(y-\widetilde{x})) = \frac{1}{n(n-2)\alpha(n)}\left(\frac{1}{|y-x|^{n-2}} - \frac{1}{|x|^{n-2}|y-\widetilde{x}|^{n-2}}\right)$$

Meanwhile, $G(0,y) = \Phi(y) - \frac{1}{n(n-2)\alpha(n)} = \frac{1}{n(n-2)\alpha(n)}\left(\frac{1}{|y|^{n-2}} - 1\right)$.

As for the case $n = 2$, we have that:

If $x \neq 0$ then:

$$G(x,y) = \Phi(y-x) - \Phi(|x|(y-\widetilde{x})) = \frac{-1}{2\pi}\left(\ln|y-x| - \ln(|x|\cdot|y-\widetilde{x}|)\right)$$

Meanwhile, $G(0,y) = \Phi(y) - 0 = \frac{-1}{2\pi}\ln|y|$.

I'd like to use this computation to find an analog of the Poisson integral formula on *page 808*.

For all $n \geq 2$ and $1 \leq i \leq n$ we can calculate that $\frac{\partial}{\partial y_i}\left(\Phi(y-x)\right) = \frac{x_i - y_i}{n\alpha(n)|x-y|^n}$.

In a similar vein, if $x \neq 0$ and $n \geq 2$ then:

$$\frac{\partial}{\partial y_i}\left(\Phi(|x|(y-\widetilde{x}))\right) = \frac{1}{n\alpha(n)|x|^{n-1}|y-\widetilde{x}|^{n-1}} \cdot \frac{|x|}{2|y-\widetilde{x}|} \cdot 2(y_i - \frac{x_i}{|x|^2}) = \frac{1}{n\alpha(n)} \cdot \frac{x_i - y_i|x|^2}{|x|^n|y-\widetilde{x}|^n}$$

In particular, if $|y| = 1$ then $|x|^n|y-\widetilde{x}|^n = |y-x|^n$.

Therefore, we get for all $n \geq 2$, $1 \leq i \leq n$, $x \in B_1(0) - \{0\}$, and $y \in \partial B_1(0)$ that:

$$\frac{\partial}{\partial y_i}G(x,y) = \frac{-1}{n\alpha(n)} \cdot \frac{1-|x|^2}{|x-y|^n} \cdot y_i$$

Also, the prior result extends to when $x = 0$. After all, because the correction term is constant we know that:

$$\frac{\partial}{\partial y_i}G(x,y) = \frac{\partial \Phi}{\partial y_i}(y) = \frac{-1}{n\alpha(n)} \cdot \frac{1}{|y|^n} \cdot y_i = \frac{-1}{n\alpha(n)} \cdot \frac{1-|x|^2}{|x-y|^n} \cdot y_i$$

And finally, we can now compute that $\frac{\partial G}{\partial v}(x,y) = \frac{-1}{n\alpha(n)} \cdot \frac{1-|x|^2}{|x-y|^n} \cdot |y|^2 = \frac{-1}{n\alpha(n)} \cdot \frac{1-|x|^2}{|x-y|^n}$.

With that, we can now conclude the following corollary:

Theorem: Suppose $u \in C^2(\overline{B_1(0)})$ satisfies that $\triangle u = 0$ on $B_1(0)$ and $u|_{\partial B_1(0)} = g$. Then for all $x \in B_1(0)$ we have that:

$$u(x) = \frac{(1-|x|^2)}{n\alpha(n)} \int_{\partial B_1(0)} \frac{g(y)}{|x-y|^n} \mathrm{d}S(y)$$

I need to move onto other topics. So, its time to talk about the heat equation.

All exercises for this class are from the Evan's PDE book (see *citation 16*). Also as a reminder, $\triangle$ and ∇ and $\partial/\partial v$ are always assumed to be derivative operators on only the spatial variables.

Exercise 2.10: Suppose u is smooth and satisfies $u_t - \triangle u = 0$ on $\mathbb{R}^n \times \mathbb{R}_{>0}$.

(a) Show that $w(x,t) := u(\lambda x, \lambda^2 t)$ also satisfies that $w_t - \triangle w = 0$ on $\mathbb{R}^n \times \mathbb{R}_{>0}$ for every $\lambda \in \mathbb{R} - \{0\}$.

Proof:
So long as $\lambda \neq 0$ we know that $\lambda^2 t \in \mathbb{R}_{>0}$ for all $t \in \mathbb{R}_{>0}$. Hence, w is well-defined for all $\lambda \in \mathbb{R} - \{0\}$.

Next, note for any $1 \leq i \leq n$ that $\frac{\partial w}{\partial x_i}(x,t) = \lambda \frac{\partial u}{\partial x_i}(\lambda x, \lambda^2 t)$. In turn:

$$\frac{\partial^2 w}{\partial x_i^2}(x,t) = \lambda^2 \frac{\partial^2 u}{\partial^2 x_i}(\lambda x, \lambda^2 t) \text{ and } \triangle w(x,t) = \lambda^2 \triangle u(\lambda x, \lambda^2 t).$$

Similarly, we get that $w_t(x,t) = \lambda^2 u_t(\lambda x, \lambda^2 t)$. And now we can compute that:

$$w_t(x,t) - \triangle w(x,t) = \lambda^2 (u_t(\lambda x, \lambda^2 t) - \triangle u(\lambda x, \lambda^2 t)) = \lambda^2 \cdot 0 = 0.$$

(b) Show that $w(x,t) := x \cdot \nabla u(x,t) + 2t u_t(x,t)$ satisfies that $w_t - \triangle w = 0$ on $\mathbb{R}^n \times \mathbb{R}_{>0}$.

Proof:
Note that due to the product rule plus the fact that derivatives commute with each other, we know for any $1 \leq i \leq n$ that:

$$\begin{aligned} \partial_{x_i} w(x,t) &= e_i \cdot \nabla u(x,t) + x \cdot \tfrac{\partial}{\partial x_i} \nabla u(x,t) + 2t \partial_{x_i} u_t \\ &= u_{x_i}(x,t) + x \cdot \nabla u_{x_i}(x,t) + 2t (u_{x_i})_t(x,t). \end{aligned}$$

We can similarly take another partial derivative and get that:

$$\begin{aligned} \partial_{x_i}^2 w(x,t) &= u_{x_i,x_i}(x,t) + e_i \cdot \nabla u_{x_i}(x,t) + x \cdot \tfrac{\partial}{\partial x_i} \nabla u_{x_i}(x,t) + 2t \partial_{x_i} (u_{x_i})_t(x,t) \\ &= 2u_{x_i,x_i}(x,t) + x \cdot \nabla u_{x_i,x_i}(x,t) + 2t (u_{x_i,x_i})_t(x,t). \end{aligned}$$

And now we can conclude that:

$$\triangle w(x,t) = 2 \triangle u(x,t) + x \cdot \nabla \triangle u(x,t) + 2t \tfrac{\partial}{\partial t} \triangle u(x,t)$$

Similarly, we can calculate that:

$$w_t(x,t) = x \cdot \nabla u_t(x,t) + 2u_t(x,t) + 2t \tfrac{\partial}{\partial t} u_t(x,t)$$

And now we get that:

$$\begin{aligned} &w_t(x,t) - \triangle w(x,t) \\ &\quad = 2(u_t(x,t) - \triangle u(x,t)) + x \cdot \nabla (u_t - \triangle u)(x,t) + 2t \tfrac{\partial}{\partial t}(u_t - \triangle u)(x,t) \\ &\quad = 2(0) + x \cdot 0 + 2t(0) = 0. \blacksquare \end{aligned}$$

Now recall from *page 921* what the fundamental solution to the equation $u_t - \triangle u = 0$ on $\mathbb{R}^n \times \mathbb{R}_{>0}$ is. In math 148 we denoted it $S^{(n)}$. In math 231 we will denote it Φ.

The last thing I will probably get to take notes on before returning to differential geometry is finally addressing the incomplete proof on *pages 924-925*. For the sake of completeness I will not assume a normalized constant of heat transport when I'm showing the following.

Proposition: Let $\Phi(x,t) = \frac{1}{(2\pi\kappa t)^{n/2}} e^{\frac{-|x|^2}{4\kappa t}}$. Also suppose $f \in C^{2,1}(\mathbb{R}^n \times \mathbb{R}_{\geq 0})$ with f and all its derivatives bounded by a constant. Then:

$$u(x,t) \coloneqq \int_0^t \int_{\mathbb{R}^n} f(y,s)\Phi(x-y,t-s)\mathrm{d}y\mathrm{d}s$$

satisfies that:

$$\lim_{\substack{t\to 0 \\ x'\to x}} u(x',t) = 0, \text{ that } u \in C^{2,1}(\mathbb{R}^n \times \mathbb{R}_{>0}), \text{ and } u_t - \kappa\triangle u = f.$$

Proof:
See the work on *pages 924-925* for why $\lim_{\substack{t\to 0 \\ x'\to x}} u(x',t) = 0$.

Now we just need to show the other claims. To do this, I will first change variables and write $u(x,t) = \int_0^t \int_{\mathbb{R}^n} f(x-y,t-s)\Phi(y,s)\mathrm{d}y\mathrm{d}s$.

We'll in particular write $G(x,t,s) \coloneqq \int_{\mathbb{R}^n} f(x-y,t-s)\Phi(y,s)\mathrm{d}y$.

Note that G_t, G_{x_i}, and G_{x_i,x_j} all exist, equal what they should equal, and are continuous on $\{(x,t,s) : x \in \mathbb{R}^n, t > 0, \text{ and } 0 < s < t\}$.

I'll only show this for the tth partial derivative. Note that if $0 < s_0 < s < t < s_1$ then:

$|f_t(x-y,t-s)\Phi(y,s)| \leq \|f_t\|_u \frac{1}{\sqrt{4\pi\kappa s_0}} e^{\frac{-|y|^2}{4\kappa s_1}}$ where the latter is integrable.

Thus by Leibniz' rule applied to a properly chosen neighborhood containing any (x,t,s) we are interested in, we can conclude that:

$$G_t(x,t,s) = \int_{\mathbb{R}^n} f_t(x-y,t-s)\Phi(y,s)\mathrm{d}y.$$

Also as $f_t(x-y,t-s)\Phi(y,s)$ is continuous, the same bound from before shows that G_t is continuous.

I'll furthermore point out that f can be uniquely extended continuously to $\mathbb{R}^n \times \mathbb{R}_{\geq 0}$ via a result I am about to prove in a tangent, and then nonuniquely extended to a continuous bounded function on all of $\mathbb{R}^n \times \mathbb{R}$ via the Tietz extension theorem. We shall choose such an extension and it doesn't matter which we pick since for the computations of u_t and $\triangle u$ we will only access the value of f on $\mathbb{R}^n \times \mathbb{R}_{\geq 0}$.

I'll actually prove a slightly stronger statement in this tangent because why not?

Proposition: Suppose $D \subseteq \mathbb{R}^n$ is an embedded C^1 submanifold of codimension 0 with a nonempty boundary. If $f : D^\circ \to \mathbb{R}$ is a C^1 map with $|\nabla f|$ bounded on D°, then for all $x \in \partial D$ we have that $\lim_{x'\to x} f(x')$ exists.

Proof:
Let (V,φ) be a smooth chart on D containing x. Then let $\widehat{V} = \varphi(V)$ and $\widehat{x} = \varphi(x)$. Note

that (V, φ) must be a boundary chart as $x \in \partial D$. Hence, without loss of generality we can assume that $\widehat{x} \in \partial \mathbb{R}^n_{\geq 0}$ and $f \circ \varphi^{-1}$ is defined on $\widehat{V} \cap (\mathbb{R}^n_{>0})$. Furthermore, because φ is a homeomorphism, we know that $\lim_{x' \to x} f(x')$ exists iff $\lim_{\widehat{x'} \to \widehat{x}} f \circ \varphi^{-1}(\widehat{x'})$ exists. And finally, if we restrict V to the interior of a compact subneighborhood of x, we can guarentee that $\|Df\varphi^{-1}\|_{\text{op}}$ is bounded on $\widehat{V}$. In turn, we will get that $|\nabla(f \circ \varphi^{-1})|$ is also bounded on D°.

Based on the prior reasoning, it thus suffices to assume D is an open subset of $[0, \infty)^n$ and $f \in C^1(D \cap (0, \infty)^n, \mathbb{R})$ with $|\nabla f|$ bounded by some constant. Also, upon writing $x = (x_1, \ldots, x_n)$, we know at least one x_i is zero. By permuting our coordinates if necessary we can always make it so that specifically $x_n = 0$.

Pick $R > 0$ such that $B_R(x) \cap [0, \infty)^n \subseteq D$. Then make the key observation that $B_R(x) \cap (0, \infty)^n$ is convex. As a result, if $|\nabla f(y)| \leq C$ for all $y \in D$, we have that:

$$|f(y_1) - f(y_2)| \leq C|y_2 - y_1| \text{ for all } y_1, y_2 \in B_R(x) \cap (0, \infty)^n.$$

This lets us conclude that if $\{x_k\}_{k \in \mathbb{N}}$ is *any* sequence in D converging to x, then $\{f(x_k)\}_{k \in \mathbb{N}}$ is a Cauchy sequence (and thus converges).

> Why?
> For any $\varepsilon > 0$ there exists N with $|x_m - x_n| < \varepsilon / C$ for all $m, n \geq N$. In turn, $|f(x_m) - f(x_n)| < \varepsilon$ for all $m, n \geq N$.

Also, if $\{x_k\}_{k \in \mathbb{N}}$ and $\{y_k\}_{k \in \mathbb{N}}$ are differing sequences in D converging to x then $\{f(x_k)\}_{k \in \mathbb{N}}$ and $\{f(y_k)\}_{k \in \mathbb{N}}$ must converge to the same thing.

> Why?
> Suppose to the contrary that they don't and then construct a new sequence by $z_{2k-1} = x_k$ and $z_{2k} = y_k$ for all k. Then $\{f(z_k)\}_{k \in \mathbb{N}}$ will have two distinct subsequential limits (and thus not converge). Yet we also know $\{z_k\}_{k \in \mathbb{N}}$ converges to x in D. This contradicts our prior observation. ■

Continuing where I left off, if $x \in \mathbb{R}^n$, $t \in \mathbb{R}$, and $0 < s_0 < s < s_1$ then:

$$|f(x - y, t - s)\Phi(y, s)| \leq \|f\|_u \frac{1}{\sqrt{4\pi\kappa s_0}} e^{\frac{-|y|^2}{4\kappa s_1}}$$

So by Leibniz' rule, we know that G is a well-defined continuous function on the set $\{(x, t, s) \in \mathbb{R}^n \times \mathbb{R} \times (0, \infty)\}$.

Now, we're ready to define $H(x, t, a) \coloneqq \int_0^a G(x, t, s) \mathrm{d}s$ on $\mathbb{R}^n \times \mathbb{R} \times (0, \infty)$. Importantly, $\|G\|_u \leq \|f\|_u$ for all $(x, t, s) \in \mathbb{R}^n \times \mathbb{R} \times (0, \infty)$. So, H is well-defined and continuous. Plus, by the fundamental theorem of calculus we have that:

$$\frac{\partial H}{\partial a}(x, t, a) = G(x, t, a) \text{ for all } (x, t, a) \in \mathbb{R}^n \times \mathbb{R} \times (0, \infty).$$

Next note that $|G_t(x, t, s)| \leq \|f_t\|_u$ for all:

$$(x, t, s) \in \{(x, t, s) : x \in \mathbb{R}^n, t > 0, \text{ and } 0 < s < t\}.$$

This is enough to show by Leibniz's rule that for any $x \in \mathbb{R}^n$, $t > 0$, and $0 < a \leq t$ that $\frac{\partial H}{\partial t}(x, t, a)$ exists and equals $\int_0^a G_t(x, t, s) \mathrm{d}s$.

Now, we can apply chain rule to get that if $x \in \mathbb{R}^n$ and $t > 0$ then:

$$\begin{aligned} u_t(x,t) = \tfrac{\partial H(x,t,t)}{\partial t} &= \int_0^t G_t(x,t,s)\mathrm{d}s + G(x,t,t) \\ &= \int_0^t \int_{\mathbb{R}^n} f_t(x-y,t-s)\Phi(y,s)\mathrm{d}y\mathrm{d}s + \int_{\mathbb{R}^n} f(x-y,0)\Phi(y,t) \end{aligned}$$

Meanwhile, because $|G_{x_i}(x,t,s) \leq \|f_{x_i}\|_u$ and $|G_{x_i,x_j}(x,t,s)| \leq \|f_{x_i,x_j}\|_u$ for all:

$$(x,t,s) \in \{(x,t,s) : x \in \mathbb{R}^n, t > 0, \text{ and } 0 < s < t\},$$

we can analogously apply Leibniz's rule to get that:

$$\begin{aligned} u_{x_i,x_i}(x,t) = \tfrac{\partial^2 H}{\partial x_i^2}(x,t,t) &= \int_0^t G_{x_i,x_i}(x,t,s)\mathrm{d}s \\ &= \int_0^t \int_{\mathbb{R}^n} f_{x_i,x_i}(x-y,t-s)\Phi(y,s)\mathrm{d}y\mathrm{d}s. \end{aligned}$$

And with that the hard part is now done because we have successfully moved the derivative operators inside our integral expression and proven that:

$$\begin{aligned} &(\partial_t - \triangle)u(x,t) \\ &\quad = \int_0^t \int_{\mathbb{R}^n} (\partial_t - \triangle) f(x-y,t-s)\Phi(y,s)\mathrm{d}y\mathrm{d}s + \int_{\mathbb{R}^n} f(x-y,0)\Phi(y,t)\mathrm{d}y. \end{aligned}$$

Now we still need to actually show for any fixed $(x,t) \in \mathbb{R}^n \times \mathbb{R}_{>0}$ that our awful integral expression evaluates to $f(x,t)$. To do this, we consider any fixed $0 < \varepsilon < t$ and then we set:

- $I_\varepsilon := \int_\varepsilon^t \int_{\mathbb{R}^n} (\partial_t - \triangle) f(x-y,t-s)\Phi(y,s)\mathrm{d}y\mathrm{d}s$
- $J_\varepsilon := \int_0^\varepsilon \int_{\mathbb{R}^n} (\partial_t - \triangle) f(x-y,t-s)\Phi(y,s)\mathrm{d}y\mathrm{d}s$
- $K := \int_{\mathbb{R}^n} f(x-y,0)\Phi(y,t)\mathrm{d}y.$

Note that $(\partial_t - \triangle)u(x,t) = I_\varepsilon + J_\varepsilon + K.$

Claim 1: $J_\varepsilon \to 0$ as $\varepsilon \to 0^+$.

Why?

We know from our assumptions on f that $\|f_t - \triangle f\|_u < \infty$. In turn, we have that:

$$|J_\varepsilon| \leq \varepsilon \|f_t - \triangle f\|_u \to 0 \text{ as } \varepsilon \to 0$$

Claim 2: $I_\varepsilon \to f(x,t) - K$ as $\varepsilon \to 0^+$.

Why?

Observe that:

$$\begin{aligned} &\int_\varepsilon^t \int_{\mathbb{R}^n} f_t(x-y,t-s)\Phi(y,s)\mathrm{d}y\mathrm{d}s \\ &\quad = \int_{\mathbb{R}^n} \int_\varepsilon^t f_t(x-y,t-s)\Phi(y,s)\mathrm{d}s\mathrm{d}y \text{ (by Fubini)} \\ &\quad = -\int_{\mathbb{R}^n} \int_\varepsilon^t \tfrac{\partial f(x-y,t-s)}{\partial s}\Phi(y,s)\mathrm{d}s\mathrm{d}y \\ &\quad = -\int_{\mathbb{R}^n} f(x-y,0)\Phi(y,t)\mathrm{d}y + \int_{\mathbb{R}^n} f(x-y,t-\varepsilon)\Phi(y,\varepsilon)\mathrm{d}y \\ &\qquad\qquad + \int_{\mathbb{R}^n} \int_\varepsilon^t f(x-y,t-s)\Phi_t(y,s)\mathrm{d}s\mathrm{d}y \\ &\qquad\qquad \text{(by integration by parts)} \\ &\quad = -K + \int_{\mathbb{R}^n} f(x-y,t-\varepsilon)\Phi(y,\varepsilon)\mathrm{d}y \\ &\qquad\qquad + \int_\varepsilon^t \int_{\mathbb{R}^n} f(x-y,t-s)\Phi_t(y,s)\mathrm{d}y\mathrm{d}s \text{ (by Fubini).} \end{aligned}$$

Meanwhile, because Φ and all its derivatives vanish at infinity, we can conclude via integration by parts that for any $1 \leq i \leq n$ and fixed $\varepsilon \leq s \leq t$ that:

$$\begin{aligned}\int_{\mathbb{R}^n} f_{x_i,x_i}(x-y,t-s)\Phi(y,s)\mathrm{d}y &= -\int_{\mathbb{R}^n} \frac{\partial(f_{x_i}(x-y,t-s))}{\partial y_i}\Phi(y,s)\mathrm{d}y \\ &= \int_{\mathbb{R}^n} f_{x_i}(x-y,t-s)\Phi_{y_i}(y,s)\mathrm{d}y \\ &= -\int_{\mathbb{R}^n} \frac{\partial f(x-y,t-s)}{\partial y_i}\Phi_{y_i}(y,s)\mathrm{d}y \\ &= \int_{\mathbb{R}^n} f(x-y,t-s)\Phi_{y_i,y_i}(y,s)\mathrm{d}y.\end{aligned}$$

So in conclusion, we get that:

$$I_\varepsilon = -K + \int_{\mathbb{R}^n} f(x-y,t-\varepsilon)\Phi(y,\varepsilon)\mathrm{d}y + \int_\varepsilon^t \int_{\mathbb{R}^n} f(x-y,t-s)(\Phi_t - \triangle\Phi)(y,s)\mathrm{d}y\mathrm{d}s$$

But now Φ satisfies the (homogeneous) heat equation. So the last integral cancels and we are left with:

$$I_\varepsilon = \int_{\mathbb{R}^n} f(x-y,t-\varepsilon)\Phi(y,\varepsilon)\mathrm{d}y - K.$$

Now note that:

$$\begin{aligned}&\int_{\mathbb{R}^n} f(x-y,t-\varepsilon)\Phi(y,\varepsilon)\mathrm{d}y \\ &\qquad = \int_{\mathbb{R}^n} \left[f(x-y,t-\varepsilon) - f(x-y,t)\right]\Phi(y,\varepsilon)\mathrm{d}y + \int_{\mathbb{R}^n} f(x-y,t)\Phi(y,\varepsilon)\mathrm{d}y\end{aligned}$$

I claim that $\int_{\mathbb{R}^n} \left[f(x-y,t-\varepsilon) - f(x-y,t)\right]\Phi(y,\varepsilon)\mathrm{d}y \to 0$ as $\varepsilon \to 0^+$.

> After all, since f is continuous (and thus in particular uniformly continuous on a compact ball containing (x,t)) we can for any $\kappa > 0$ pick some $\delta > 0$ such that $|f(x-y,t-\varepsilon) - f(x-y,t)| < \kappa$ for all $y \in \overline{B_\delta}(x)$ and $\varepsilon < \delta$. In turn for $\varepsilon < \delta$ we have that:
>
> $$\begin{aligned}&\left|\int_{\overline{B_\delta(x)}} \left[f(x-y,t-\varepsilon) - f(x-y,t)\right]\Phi(y,\varepsilon)\mathrm{d}y\right| \\ &\qquad \leq \kappa \int_{\overline{B_\delta(x)}} \Phi(y,\varepsilon)\mathrm{d}y \leq \kappa \int_{\mathbb{R}^n} \Phi(y,\varepsilon)\mathrm{d}y = \kappa.\end{aligned}$$
>
> Yet simultaneously, because $|f(x-y,t-\varepsilon) - f(x-y,t)|$ is bounded by $2\|f\|_u$, it is an easy to establish fact that:
>
> $$\lim_{\varepsilon\to 0^+} \int_{B_\delta(x)^\mathsf{C}} \left[f(x-y,t-\varepsilon) - f(x-y,t)\right]\Phi(y,\varepsilon)\mathrm{d}y \to 0 \text{ as } \varepsilon \to 0$$
>
> Therefore:
>
> $$\begin{aligned}&\limsup_{\varepsilon\to 0^+} \left|\int_{\mathbb{R}^n} \left[f(x-y,t-\varepsilon) - f(x-y,t)\right]\Phi(y,\varepsilon)\mathrm{d}y\right| \\ &\qquad = \limsup_{\varepsilon\to 0^+} \left|\int_{\overline{B_\delta(x)}} \left[f(x-y,t-\varepsilon) - f(x-y,t)\right]\Phi(y,\varepsilon)\mathrm{d}y\right| \leq \kappa\end{aligned}$$
>
> And since κ was arbitrary, we can take $\kappa \to 0$ to finish showing this claim.

Meanwhile, note that $w(x,\varepsilon) := \int_{\mathbb{R}^n} f(x-y,t)\Phi(y,\varepsilon)\mathrm{d}y$ is just the solution to the (homogeneous) heat equation $w_t - \triangle w = 0$ with an initial value of $w(x,0) = f(x,t)$. So we can already identify from the proof on *pages 921-922* that $\int_{\mathbb{R}^n} f(x-y,t)\Phi(y,\varepsilon)\mathrm{d}y \to f(x,t)$ as $\varepsilon \to 0^+$.

This finishes showing that $I_\varepsilon \to f(x,t) - K$ as $\varepsilon \to 0$.

And now adding together the limits we just proved finishes showing that:

$$u_t(x,t) - \triangle u(x,t) = f(x,t). \blacksquare$$

10/23/2026

Math 250A Lecture Notes:
Today and maybe tomorrow I need to take lecture notes for this class and also do homework. Once I've dealt with that, I can go back to taking PDE notes.

To start off, I want to give an actual atlas for $S^n \subseteq \mathbb{R}^{n+1}$. Note that this is in fact an embedded submanifold as it is the regular zero set of $\Phi : \mathbb{R}^{n+1} \to \mathbb{R}$ given by $\Phi(x) = |x|^2 - 1$. Note that all the following reasoning is heavily inspired by work on *page 241* showing that $\mathbb{C}_\infty$ is homeomorphic to S^2.

Let $U_1 = S^n - \{(0, \dots, 0, 1)\}$ and let $\widehat{U}_1 = \mathbb{R}^n$. We define:

- $\varphi_1 : \mathbb{R}^{n+1} - \{(0, \dots, 0, 1)\} \to \widehat{U}_1$ by:
$$\varphi_1(x_1, \dots, x_n, x_{n+1}) := \left(\frac{x_1}{1-x_{n+1}}, \dots, \frac{x_n}{1-x_{n+1}}\right)$$
- $\psi_1 : \widehat{U}_1 \to \mathbb{R}^{n+1}$ by $\psi_1(x_1, \dots, x_n) := \left(\frac{2x_1}{1+|x|^2}, \dots, \frac{2x_n}{1+|x|^2}, \frac{|x|^2-1}{1+|x|^2}\right)$.

Note that φ_1 is a smooth map. It follows that it's restriction to U_1 will still be a smooth map with respect to the smooth structure on the open submanifold $U_1 \subseteq S^n \subseteq \mathbb{R}^{n+1}$.

Meanwhile, note that ψ_1 is a smooth map. Furthermore:
$$|\psi_1(x)|^2 = \frac{1}{(1+|x|^2)^2}(4|x|^2 + (|x|^2 - 1)^2) = \frac{1}{(1+|x|^2)^2}(|x|^4 + 2|x|^2 + 1) = 1.$$

So, ψ_1 maps into S^n.

We can further note that $(0, \dots, 0, 1) \notin \psi_1(\widehat{U}_1)$. And since $U_1 \subseteq S^n \subseteq \mathbb{R}^{n+1}$ is an *embedded* submanifold, we can thus conclude that restricting the codomain of ψ_1 does not change that ψ_1 is a smooth map.

With that, we've proven that $\varphi_1 : U_1 \to \widehat{U}_1$ and $\psi_1 : \widehat{U}_1 \to U_1$ are both smooth maps. Finally, I claim that $\psi_1 \circ \varphi_1 = \mathrm{Id}_{U_1}$ and $\varphi_1 \circ \psi_1 = \mathrm{Id}_{\widehat{U}_1}$.

You do this computation yourself. There's no fancy ideas.

And with that, we've proven that (U_1, φ_1) is a smooth chart on S^n.

In a similar vein, if $U_2 = S^n - \{(0, \dots, 0, -1)\}$ then we can define another smooth chart (U_2, φ_2) on S^n by:

- $\varphi_2(x_1, \dots, x_{n+1}) = \left(\frac{x_1}{1+x_{n+1}}, \dots, \frac{x_n}{1+x_{n+1}}\right)$,
- $\varphi_2^{-1}(x_1, \dots, x_n) = \left(\frac{2x_1}{1+|x|^2}, \dots, \frac{2x_n}{1+|x|^2}, \frac{1-|x|^2}{1+|x|^2}\right)$.

The prior two charts (U_1, φ_1) and (U_2, φ_2) are called the stereographic projections and they let you cover S^n with the fewest possible charts.

(We can't cover S^n with just a single chart (smooth or not) because that would imply the compact set S^n is homeomorphic to an open subset of $\mathbb{R}^n$ (since S^n has no boundary). But that's impossible.)

Sometimes, we want to use a smooth graph chart instead though. I'll construct that here.

Let $U = \{(x_1, \ldots, x_{n+1}) \in S^n : x_{n+1} > 0\}$. Next set $\widehat{U} = B_1(0) \subseteq \mathbb{R}^n$. And finally, define the maps:

- $\varphi : \mathbb{R}^{n+1} \to \mathbb{R}^n$ by $\varphi(x_1, \ldots, x_n, x_{n+1}) \coloneqq (x_1, \ldots, x_n)$,
- $\psi : \widehat{U} \to \mathbb{R}^{n+1}$ by $\psi(x_1, \ldots, x_n) = (x_1, \ldots, x_n, \sqrt{1 - x_1^2 - \ldots - x_n^2})$.

By identical reasoning as before, these become smooth maps from U to $\widehat{U}$ and $\widehat{U} \to U$ respectively. And identically to before, we can prove that $\psi \circ \varphi = \mathrm{Id}_U$ and $\varphi \circ \psi = \mathrm{Id}_{\widehat{U}}$.

This finishes showing that $(U, \varphi|_U)$ is a smooth chart on S^n.

In the prior construction, we can also obviously make the dependent variable in our graph any x_i instead of just x_{n+1}. Also, instead of considering the open set $\{(x_1, \ldots, x_{n+1}) \in S^n : x_{n+1} > 0\}$ we can insetad consider the open set $\{(x_1, \ldots, x_{n+1}) \in S^n : x_{n+1} < 0\}$. And then we just have to negate $\sqrt{1 - x_1^2 - \ldots - x_n^2}$ in our expression for the inverse map.

Set 3 Problem 1: Note that Oprea writes $\mathrm{span}((x_1, \ldots, x_{n+1})) \in \mathbb{RP}^n$ as $[x_1 : \cdots : x_{n+1}]$.

(i) Show that the map $q : S^2 \to \mathbb{RP}^2$ given by $q(x, y, z) = [x : y : z]$ is a smooth local diffeomorphism.

Given any $(x_0, y_0, z_0) \in S^2$ we know at least one coordinate is nonzero. Without loss of generality, I will assume $z_0 > 0$. After all, the argument remains basically identical if $z_0 < 0$. And if $z_0 = 0$ then we can just rename our coordinates and do the exact same proof on our renamed coordinates.

Let $U = \{(x, y, z) \in S^2 : z > 0\}$. Then we have a smooth parametrization $X : B_1(0) \subseteq \mathbb{R}^2 \to U$ given by $X(x, y) = (x, y, \sqrt{1 - x^2 - y^2})$. In other words, (U, X^{-1}) is a smooth chart on S^2 containing (x_0, y_0, z_0).

Meanwhile, recall that $V = \{[x : y : z] \in \mathbb{RP}^2 : z \neq 0\}$ and $\psi : V \to \mathbb{R}^2$ given by $\psi([x : y : z]) = (\frac{x}{z}, \frac{y}{z})$ define a smooth chart on $\mathbb{RP}^2$. In particular, we have that $q(U) \subseteq V$.

To check that q is smooth and a local diffeomorphism, it suffices to show that q is smooth with respect to the local coordinates (U, X^{-1}) and (V, ψ), and also that $\mathrm{D}(\psi \circ q \circ X)$ is invertible on $B_1(0)$. Fortunately, this is easy.

Note that $\widehat{q} \coloneqq \psi \circ q \circ X : B_1(0) \to \mathbb{R}^2$ is given by:

$$\widehat{q}(x, y) = (\tfrac{x}{\sqrt{1-x^2-y^2}}, \tfrac{y}{\sqrt{1-x^2-y^2}}).$$

This is a smooth map on $B_1(0)$. Also:

$$\mathrm{D}\widehat{q}(x, y) = \begin{bmatrix} \frac{1-y^2}{(1-x^2-y^2)^{3/2}} & \frac{xy}{(1-x^2-y^2)^{3/2}} \\ \frac{xy}{(1-x^2-y^2)^{3/2}} & \frac{1-x^2}{(1-x^2-y^2)^{3/2}} \end{bmatrix} = \frac{1}{(1 - x^2 - y^2)^{3/2}} \begin{bmatrix} 1 - y^2 & xy \\ xy & 1 - x^2 \end{bmatrix}$$

Since $\frac{1}{(1-x^2-y^2)^{3/2}} \neq 0$ ever, we can thus conclude that $\mathrm{D}\widehat{q}(x,y)$ is singular iff:

$$\begin{bmatrix} 1-y^2 & xy \\ xy & 1-x^2 \end{bmatrix} \text{ is singular.}$$

This happens iff $0 = (1-y^2)(1-x^2) - x^2y^2 = 1 - x^2 - y^2$. Yet the latter never happens for all $(x,y) \in B_1(0)$.

(ii) Let $f : S^2 \to \mathbb{R}^4$ be given by $f(x,y,z) = (x^2 - y^2, xy, xz, yz)$. Show that f is a smooth immersion.

To start off, let $\iota : S^2 \hookrightarrow \mathbb{R}^3$ be the embedding map. Also let $F : \mathbb{R}^3 \to \mathbb{R}^4$ be defined by $F(x,y,z) = (x^2 - y^2, xy, xz, yz)$. Then we have that $f = F \circ \iota$. Note that because F an ι are both smooth, we trivially have that f is smooth. The harder part is showing that f is an immersion.

Now $\mathrm{D}F(x,y,z) = \begin{bmatrix} 2x & -2y & 0 \\ y & x & 0 \\ z & 0 & x \\ 0 & z & y \end{bmatrix}$.

Based on the above expression, we can now conclude that $\mathrm{D}F$ is full rank at most points of S^2. After all:

- If $x \neq 0$, $y \neq 0$, and $z \neq 0$, then the determinant of the bottom three rows is $-2xyz \neq 0$.
- If $z = 0$ then our matrix simplifies and we can show that $\mathrm{D}F(x,y,z)$ has full rank if and only if either $x \neq 0$ or $y \neq 0$.
- If $x = 0$ and $y \neq 0$ and $z \neq 0$ then we can also simplify our matrix and show that $\mathrm{D}F(x,y,z)$ has full rank.
- And if $y = 0$ and $x \neq 0$ and $z \neq 0$, then we similarly can show that $\mathrm{D}F(x,y,z)$ has full rank.

In other words, the only points of S^2 where $\mathrm{D}F$ is not full rank are $(0,0,1)$ and $(0,0,-1)$. Since a composition of injective linear maps is injective, we can thus conclude that $\mathrm{d}f_p$ is injective for all $p \in S^2 - \{(0,0,1),(0,0,-1)\}$.

As for showing that $\mathrm{d}f_{(0,0,\pm 1)}$ is injective, we need to show that:

$$T_{(0,0,\pm1)}S^2 \cap \ker(\mathrm{d}F_{(0,0,\pm1)}) = \{0\}.$$

Fortunately, S^2 is regular zero set of $\Phi : \mathbb{R}^3 \to \mathbb{R}$ given by $\Phi(x) = |x|^2 - 1$. Therefore:

$$T_{(0,0,\pm1)}S^2 = \ker(\mathrm{D}\Phi(0,0,\pm 1)) = \ker([0 \quad 0 \quad \pm 2]) = \mathbb{R}^2 \times \{0\}$$

Meanwhile, $\ker(\mathrm{d}F_{(0,0,\pm1)}) = \ker\left(\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \pm 1 & 0 & 0 \\ 0 & \pm 1 & 0 \end{bmatrix}\right) = \{(0,0)\} \times \mathbb{R}$.

(c) Deduce there exists an embedding $g : \mathbb{RP}^2 \to \mathbb{R}^4$ such that $g \circ q = f$.

From part (a) we know that q is a smooth submersion from S^2 to $\mathbb{RP}^2$. Its pretty trivial that q is also surjective. Thirdly, note that the fibers of q are precisely the sets $\{\vec{x}, -\vec{x}\}$ for any $\vec{x} \in S^2$. This is notable because we in turn can see that f is constant on each of the fibers of q.

By corollary 4.30 on *pages 1108-1109*, we thus know there is a unique smooth map $g : \mathbb{RP}^2 \to \mathbb{R}^4$ satisfying that $g \circ q = f$.

Now to show that g is an immersion, note for any $[x] \in \mathbb{RP}^2$ that we can pick $x \in S^2$ with $q(x) = [x]$. Then $\mathrm{d}g_{[x]} \circ \mathrm{d}q_x = \mathrm{d}f_x$. Yet because q is a local diffeomorphism, we know $\mathrm{d}q_x$ is invertible. So, $\mathrm{d}g_{[x]} = \mathrm{d}f_x \circ (\mathrm{d}q_x)^{-1}$ is the composition of two injective linear maps and is thus injective.

Next, to prove that g is an injection it suffices to show that if $f(x_1, y_1, z_1) = f(x_2, y_2, z_2)$ then (x_1, y_1, z_1) and (x_2, y_2, z_2) are in the same fiber of p.

> In other words, we need to show that $(x_2, y_2, z_2) = \pm(x_1, y_1, z_1)$.
>
> To do this, recall from math 220 that the map $s(x, y) \coloneqq (x^2 - y^2, 2xy)$ acts homeomorphically (and in fact biholomorphically) from:
>
> - $\mathbb{R} \times (0, \infty) \to \mathbb{R}^2 - \{(x, 0) : x \geq 0\}$,
> - $\mathbb{R} \times (-\infty, 0) \to \mathbb{R}^2 - \{(x, 0) : x \geq 0\}$.
>
> Scaling the y axis afterwards doesn't modify the codomain. So, the fiber of any $(x', y') \in \mathbb{R}^2 - \{(x, 0) : x \geq 0\}$ intersects $\mathbb{R}^2 - \{\mathbb{R} \times \{0\}\}$ at exactly two points.
>
> Meanwhile, if $y = 0$ then we know $(x^2 - y^2, xy) = (x^2, 0) \in (0, \infty) \times \{0\}$. So, none of the $(x, 0)$ contribute any points to the fibers in the prior paragraph. Also, the fiber of every $(x', 0) \in \mathbb{R}^2 - \{(x, 0) : x > 0\}$ has exactly two points. Meanwhile, the fiber of $(0, 0)$ is just $\{(0, 0)\}$.
>
> With that, we've shown that if $h : \mathbb{R}^2 \to \mathbb{R}^2$ is given by $h(x, y) = (x^2 - y^2, xy)$ then $h^{-1}(\{(x', y')\})$ has exactly two points unless $(x', y') = (0, 0)$.
>
> Finally, note that $h(x, y) = h(-x, -y)$ for all $(x, y) \in \mathbb{R}^2$. Hence, we've proven that if $h(x_1, y_1) = a = h(x_2, y_2)$ then $(x_1, y_1) = \pm(x_2, y_2)$.
>
> ---
>
> Now the prior reasoning was relevant because if $f(x_1, y_1, z_1) = f(x_2, y_2, z_2)$ then automatically $h(x_1, y_1) = h(x_2, y_2)$. So, there exists $\kappa \in \{-1, +1\}$ with $x_2 = \kappa x_1$ and $y_2 = \kappa y_1$.
>
> Now suppose $x_1 \neq 0$. Then $x_1 z_1 = x_2 z_2 = \kappa x_1 z_2 \implies z_1 = \kappa z_2$. But $\kappa = \kappa^{-1}$. So, we can conclude $z_2 = \kappa z_1$ as well.
>
> Similarly, if $y_1 \neq 0$ then we can conclude that $z_2 = \kappa z_1$.

Finally, suppose that $x_1 = y_1 = 0$. Then we know $z_1, z_2 \in \{-1, +1\}$ since otherwise $(x_i, y_i, z_i) \notin S^2$. And thus, the claim becomes trivial that $(x_1, y_1, z_1) = \pm(x_2, y_2, z_2)$.

Finally, since g is a smooth injective immersion from the manifold $\mathbb{RP}^2$ which happens to be compact, we can conclude by *problem 5-3* that g is a smooth embedding. ■

The next exercise will be important to proving a theorem in class.

Set 3 Problem 5: Let M be a C^r manifold (where $r \geq 1$) and let $\delta : M \to (0, \infty)$ be a continuous function. Suppose $f : M \to \mathbb{R}^k$ is another continuous function. Then there is a C^r map $F : M \to \mathbb{R}^k$ such that $|F(x) - f(x)| < \delta(x)$ for all $x \in M$.

Proof:

Firstly, I shall find a sequence of points $\{a_n\}_{n\in\mathbb{N}}$ and open sets $\{U_n\}_{n\in\mathbb{N}}$ such that: $\bigcup_{n\in\mathbb{N}} U_n = M$, $a_n \in U_n$ for all n, and $|f(x) - f(a_n)| < \delta(x)$ for all $x \in U_n$.

Note that if $a \in M$ then we know that $\delta(a) > 0$. And now as δ is continuous, for any $0 < r < \delta(a)$ we can find an open set V_a containing a with $\delta(x) > r$ for all $x \in V_a$. Next, because f is continuous at a, we can find an open set $U_a \subseteq V_a$ containing a such that $|f(x) - f(a)| < r$ for all $x \in U_a$. In turn, $|f(x) - f(a)| < \delta(x)$ for all $x \in U_a$.

The prior paragraph shows us how to give an open covering of M consisting of open sets $\{U_a\}_{a\in M}$ satisfying that $a \in U_a$ for all a and $|f(x) - f(a)| < \delta(x)$ for all $x \in U_a$. Now we just use the second countability of M in order to take a countable subcover.

Now that I have the prior sequence, I'm going to construct a C^r partition of unity $\{\phi_n\}_{n\in\mathbb{N}}$ subordinate to the open sets U_n. Then, I'm going to define $F : M \to \mathbb{R}^k$ by:

$$F(x) = \sum_{n=0}^{\infty} \phi_n(x) f(a_n).$$

Note that since any $x \in M$ has a neighborhood on which only finitely many ϕ_n are nonzero, we can say that locally F is a linear combination of C^r maps. Hence, F is also a C^r map.

Also for any fixed $x \in X$, let $n_1, \ldots, n_k$ be the finitely many integers such that $\phi_{n_i}(x) \neq 0$. Then we have that:

$$\begin{aligned} |F(x) - f(x)| &= \left|\sum_{i=1}^{k} \phi_{n_i}(x) \cdot [f(a_n) - f(x)]\right| \\ &\leq \sum_{i=1}^{k} \phi_{n_i}(x)|f(x) - f(a_n)| < \sum_{i=1}^{k} \phi_{n_i}(x)\delta(x) = \delta(x). \end{aligned}$$ ■

The final exercises on the homework all are about introducing Lie groups. Since 251 and not 250 is the Lie groups class, I don't think these will come up again in my differential geometry class. I'll still take notes on this though.

Let G be a group (written multiplicatively). We let $m : G \times G \to G$ denote the multiplication map and $i : G \to G$ denote the inverse map. Also let e denote the identity of G. If G is also a smooth manifold and m and i are C^∞ functions, we say G is a Lie group.

Note in particular that all Lie groups are topological groups.

A special property of a Lie group G is that the group of C^∞ diffeo-automorphisms on G acts transitively on G. After all, given any $g \in G$ we define $L_g : G \to G$ and $R_g : G \to G$ by:

$$L_g(h) \coloneqq gh \text{ and } R_g(h) \coloneqq hg.$$

Note that L_g is the composition of the smooth map $h \mapsto (g, h)$ and the smooth map m. So, L_g is a smooth map. We similarly can show that R_g is a smooth map. Also, both L_g and R_g are diffeomorphisms since they have smooth inverses given by $L_{g^{-1}}$ and $R_{g^{-1}}$.

Now if we consider the obvious action of diffeo-automorphisms of G on G, we can see for any $g \in G$ that $L_g(e) = g$ for all g. This proves that the orbit of e is all of G.

Corollary: Lie groups have no boundary.

Proof:
Let g be an interior point of G and then suppose h is any other point in G. We know there is a diffeomorphism $F : G \to G$ satisfying that $f(h) = g$. Yet diffeomorphisms must map interior points to interior points (see exercise 4-2 on page 1057). So, we know h is an interior point of G. ■

A homomorphism $f : G \to H$ of Lie groups is a group homomorphism that is also a smooth map. An isomorphism of Lie groups is similarly a group isomorphism which is also a smooth diffeomorphism.

A Lie subgroup H of G is a subgroup of G endowed with a smooth structure making H into a Lie group and an immersed submanifold of G.

Set 3 Problem 2: Let G be a Lie group.

(i) Show that if H is a weakly embedded smooth submanifold of G that is also a subgroup of G, then H is a Lie subgroup of G.

Lemma: Suppose M_1, M_2, N_1, N_2 are C^r manifolds. If $F : M_1 \to N_1$ and $G : M_2 \to N_2$ are both C^r immersions, then $F \times G : M_1 \times M_2 \to N_1 \times N_2$ is also a C^r immersion.

Proof:
Let $(p, q) \in M_1 \times M_2$ and recall that there are standard identifications:

- $T_{(p,q)}(M_1 \times M_2) \cong T_pM_1 \oplus T_qM_2$,
- $T_{(F(p),G(q))}(N_1 \times N_2) \cong T_{F(p)}N_1 \oplus T_{G(q)}N_2$.

For each i let $\pi_i : M_1 \times M_2 \to M_i$ and $\widetilde{\pi}_i : N_1 \times N_2 \to N_i$ be the projection maps. Then importantly we have that $F \circ \pi_1 = \widetilde{\pi}_1 \circ (F \times G)$ and $G \circ \pi_2 = \widetilde{\pi}_2 \circ (F \times G)$. As a result for any $v \in T_{(p,q)}(M_1 \times M_2)$ we have that:

$$\begin{aligned}
\mathrm{d}(F \times G)_{(p,q)}(v) &= (\mathrm{d}(\widetilde{\pi}_1)_{(F(p),G(p))} \circ \mathrm{d}(F \times G)_{(p,q)}(v),\ \ \mathrm{d}(\widetilde{\pi}_2)_{(F(p),G(p))} \circ \mathrm{d}(F \times G)_{(p,q)}(v)) \\
&= (\mathrm{d}(\widetilde{\pi}_1 \circ (F \times G))_{(p,q)}(v),\ \ \mathrm{d}(\widetilde{\pi}_2 \circ (F \times G))_{(p,q)}(v)) \\
&= (\mathrm{d}(F \circ \pi_1)_{(p,q)}(v),\ \ \mathrm{d}(G \circ \pi_2)_{(p,q)}(v)) \\
&= (\mathrm{d}F_p \circ \mathrm{d}(\pi_1)_{(p,q)}(v),\ \ \mathrm{d}G_q \circ \mathrm{d}(\pi_2)_{(p,q)}(v)).
\end{aligned}$$

In other words, according to the standard identifications we have that:

$$\mathrm{d}(F \times G)_{(p,q)}(v_1, v_2) = (\mathrm{d}F_p(v_1), \mathrm{d}G_q(v_2)).$$

Even more succinctly, this means $\mathrm{d}(F \times G)_{(p,q)} = \mathrm{d}F_p \times \mathrm{d}G_q$ under this identication. Finally, note that:

$$\ker(\mathrm{d}F_p \times \mathrm{d}G_q) = \ker(\mathrm{d}F_p) \oplus \ker(\mathrm{d}G_q) = \{0\} \oplus \{0\} = \{0\}.$$

So, we've finished showing that $\mathrm{d}(F \times G)_{(p,q)}$ is injective.

I went to the effort of showing the prior lemma because we can now use it to conclude that $H \times H$ is an immersed submanifold of $G \times G$. In turn, if $m : G \times G \to G$ is a smooth map then so is $m|_{H \times H} : H \times H \to G$.

Next, as H is a subgroup of G, we know that the image of $m|_{H \times H}$ is contained in H. Therefore, as H is weakly embedded we know that we can restrict the codomain of $m|_{H \times H}$ in order to get that $m|_{H \times H} : H \times H \to H$ is a smooth map.

We can use analogous reasoning to show that the inversion map i is smooth from H into H.

(b) Give an example of a dense immersed Lie subgroup of the torus $\mathbb{T} := S^1 \times S^1$.

Let $\alpha \in \mathbb{R} - \mathbb{Q}$ and then define $\gamma : \mathbb{R} \to \mathbb{T}$ by $\gamma(t) = (e^{2\pi i t}, e^{2\pi i \alpha t})$. We know from set 2 problem 7 on *pages 1214-1216* that γ is an injective immersion whose image is dense in $\mathbb{T}$. Therefore, $H := \gamma(\mathbb{R})$ is an immersed submanifold of $\mathbb{T}$ that is dense in $S^1 \times S^1$.

Next, note that $\mathbb{T}$ is a Lie group under pointwise complex multiplication.

I don't feel like proving it since it seems obvious to me.

Furthermore, note that $\mathbb{R}$ is a Lie group under addition. Plus, $\gamma : \mathbb{R} \to \mathbb{T}$ is a group homomorphism. Thus, H is automatically a subgroup of $\mathbb{T}$. And by definition we have that γ is a smooth diffeomorphism from $\mathbb{R}$ to H.

It follows that multiplication and inversion are continuous with respect to the topology on H because addition and negation is smooth with respect to the topology on $\mathbb{R}$.

(c) If G is a Lie group, show that the connected component G^0 of the identity element e is a Lie subgroup. Furthermore, every other connected component of G is diffeomorphic to G^0.

To start off, because G is locally connected (because it is a manifold), we know that every connected component of G is an open set. Thus, G^0 is an open embedded submanifold of G^0.

Next, I claim that if $g \in G^0$ then so is g^{-1}. To see why, note that $i : G \to G$ is in particular a homeomorphism. So, it must permute the connected components of G. Yet also note that $e \in G^0$ and $i(e) = e$. So i must map G^0 back to G^0.

In a similar vein, I claim that if $g, h \in G^0$ then $gh \in G^0$ as well. After all, consider any fixed $g \in G^0$ and let $L_g : G \to G$ be the map $L_g(h) = gh$. Then L_g is in particular a homeomorphism. So, it also must permute the connected components of G. Yet $e \in G^0$ and $L_g(e) = g \in G^0$. So, L_g must map G^0 back to G^0. So, in particular, if $h \in G^0$ then $gh \in G^0$.

With that, we can now conclude by part (a) that G^0 is a Lie subgroup of G.

Finally, suppose G^1 is another connected component of G and let $g' \in G^1$. Then define $L_{g'} : G \to G$ by $L_{g'}(h) = g'h$. Because $L_{g'}$ is a homeomorphism, it must permute the connected components of G. Yet $e \in G^0$ and $L_{g'}(e) = g' \in G^1$. So, $L_{g'}(G^0) = G^1$. And now by restricting the domain and codomain of $L_{g'}$, we get that $L_{g'}$ is a smooth diffeomorphism from G^0 to G^1. ■

Let G be a Lie group and X a C^r manifold. We say G acts on X (on the left) if we have a C^r map $G \times X \to X$ written as $(g, x) \mapsto gx$ satisfying the standard group action axioms.

If X, Y are two C^r manifolds being acted on by a Lie group G and $f : X \to Y$ is any map, we say f is G-equivariant if $f(gx) = gf(x)$ for all $x \in X$ and $g \in G$.

Set 3 Problem 3:

(i) Suppose G is a Lie group acting on a C^r manifold X. Show that the maps $\mu_g : X \to X$ giveny by $\mu_g(x) = gx$ are all C^r-diffeomorphisms.

To see that μ_g is C^r for any fixed g, first note that $\iota_g : X \to G \times X$ given by $x \mapsto (g, x)$ is a C^r map. Then if $a : G \times X \to X$ is the group action, we know that $\mu_g = a \circ \iota_g$ is a composition of two C^r functions. So, μ_g is itself C^r.

Also note that μ_g is a diffeomorphism because it has the inverse function $\mu_{g^{-1}}$ which is a C^r map by identical reasoning.

(ii) As an example, show that the Lie group G acts on itself via the conjugation action $\mu_g(h) = ghg^{-1}$.

We define the $a : G \times G \to G$ by $(g, h) \mapsto \mu_g(h) = ghg^{-1}$. Then, we just need to show that the map a is smooth.

Fortunately, we know that the multiplication map $m : G \times G \to G$ given by $m(g, h) = gh$ is a smooth map. Meanwhile, if we let $\pi_1 : G \times G \to G$ be the map $\pi_1(g, h) = g$, then we know that function is smooth. Then by composing π_1 with the inverting function $i : G \to G$ which is also smooth, we get that $i \circ \pi_1 : G \times G \to G$ is a smooth map.

By the universal property of product manifolds, we now know that:

$$(g, h) \mapsto (m(g, h), i \circ \pi_1(g, h)) = (gh, g^{-1})$$

is a smooth map from $G \times G$ to $G \times G$. And now we just compose that with the multiplication map m one more time to get that $a(g, h) = ghg^{-1}$ is a smooth map.

(iii) Show that if $f : G \to H$ is a Lie group homomorphism then f has constant rank.

Note for any fixed $g \in G$ that:

$$f(h) = f(g)f(g^{-1}h) = (L' \circ f \circ L)(h).$$

where $L' : H \to H$ is the diffeomorphism given by the left multiplication of $f(g)$ and $L : G \to G$ is the diffeomorphism given by the left multiplication of g^{-1}.

But now in particular, if $h = g$ then we can see that:

$$\mathrm{d}f_g = \mathrm{d}(L' \circ f \circ L)_g = \mathrm{d}L'_e \circ \mathrm{d}f_e \circ \mathrm{d}L_g.$$

Since both $\mathrm{d}L'_e$ and $\mathrm{d}L_g$ are bijective linear maps, we can thus say that $\mathrm{d}f_g$ has the same rank as $\mathrm{d}f_e$. And since g was arbitrary, we've proven that f has constant rank equal to whatever the rank of $\mathrm{d}f_e$ is.

(iv) Suppose X, Y are two C^r manifolds both being acted on by a Lie group and $f : X \to Y$ is a G-equivariant C^1 map. If the action of G on X is transitive then f has constant rank.

Proof:
Fix any $x_0 \in X$. Then consider any other $x_1 \in X$. Because G acts on X transitively, we know there exists $g \in G$ such that $gx_0 = x_1$. And now we let $\mu_g : X \to X$ and $\widetilde{\mu}_g Y \to Y$ respectively be the maps $x \mapsto gx$ and $y \mapsto gy$.

Because μ_g is a diffeomorphism, we can say that:

$$\mathrm{d}f_{x_1} = \mathrm{d}f_{x_1} \circ \mathrm{Id}_{T_{x_1}X} = \mathrm{d}f_{x_1} \circ \mathrm{d}(\mu_g)_{x_0} \circ (\mathrm{d}(\mu_g)_{x_0})^{-1} = \mathrm{d}(f \circ \mu_g)_{x_0} \circ (\mathrm{d}(\mu_g)_{x_0})^{-1}$$

Yet then because f is G-equivariant, we know $f \circ \mu_g = \widetilde{\mu}_g \circ f$. In turn, we know that:

$$\mathrm{d}(f \circ \mu_g)_{x_0} = \mathrm{d}(\widetilde{\mu}_g \circ f)_{x_0} = \mathrm{d}(\widetilde{\mu}_g)_{f(x_0)} \circ \mathrm{d}f_{x_0}$$

So to summarize, $\mathrm{d}f_{x_1} = \mathrm{d}(\widetilde{\mu}_g)_{f(x_0)} \circ \mathrm{d}f_{x_0} \circ (\mathrm{d}(\mu_g)_{x_0})^{-1}$. Since $\mathrm{d}(\widetilde{\mu}_g)_{f(x_0)}$ and $(\mathrm{d}(\mu_g)_{x_0})^{-1}$ are both bijective linear maps, we can conclude that the rank of $\mathrm{d}f_{x_1}$ equals the rank of $\mathrm{d}f_{x_0}$. And since x_1 was arbitrary, we are done. ■

The next bit of stuff is a tangent I don't have time to do but I want to do anyways. My goal is just to illustrate that Lie groups have stronger topological properties than mere topological groups.

To start off, we have the following corollary to part (iii) of the prior problem.

Corollary: Suppose $f : G \to H$ is a Lie group homomorphism. Then f is a Lie group isomorphism if and only if f is a bijection.

Proof:
The forwards implication is trivial. As for the backwards implication, note by part (iii) of the prior problem that f has constant rank. Therefore, if f is also bijective then we know from the corollary on *page 1066* that f is a smooth diffeomorphism. And hence, f is a Lie group isomorphism. ■

Note though that we don't have an analogous result for topological groups. After all, we can give two topological groups G and H as a well as a bijective topological group homomorphism $f : G \to H$ that is not a topological group isomorphism (i.e. a group homomorphism as well as a homeomorphism).

Let $\mathbb{R}_d$ denote the real numbers with the discrete topology and let $\mathbb{R}$ denote the real numbers with the standard Euclidean topology. Both $\mathbb{R}$ and $\mathbb{R}_d$ are topological groups under addition.

Consider the map $f : \mathbb{R}_d \to \mathbb{R}$ given by $f(x) = x$. This is a bijective group homomrphism. Furthermore, f is trivially continuous. That said, f^{-1} is not continuous.

Another interesting result concerns when we know a group equipped with a topology and perhaps a smooth structure is a topological group or Lie group. But first we need a smaller result.

Lemma: Suppose G is a group equipped with a topology. Let $m : G \times G \to G$ and $i : G \to G$ denote the multiplication and inversion operations. Then G is a topological group if and only if the map $s : G \times G \to G$ given by $s(g, h) = gh^{-1}$ is continuous.

Proof:
For $i = 1, 2$ we let $\pi_i : G \times G \to G$ be the projection $(g_1, g_2) \mapsto g_i$. Both of these are continuous maps by the definition of the product topology.

$(\Longrightarrow)$
If G is a topological group then $s = m \circ (\pi_1, i \circ \pi_2)$ is a composition of continuous maps and is thus continuous.

$(\Longleftarrow)$
To start off, note that the map $\iota_e : G \to G \times G$ given by $\iota_e(g) = (e, g)$ is continuous. Also, $i = s \circ \iota_e$. So, i is a continuous map. Next, note that $m = s \circ (\pi_1, i \circ \pi_2)$ is a composition of two continuous functions and is thus continuous.

Furthermore, if G is a smoth manifold then G is a Lie group if and only if the map $s : G \times G \to G$ given by $s(g, h) = gh^{-1}$ is smooth.

The proof is literally identical to before. Just swap out the word "continuous" with the word "smooth". ■

I also thought I needed the next result (which is an exercise from John Lee).

Problem 7-1: Suppose G is a group that is also a smooth manifold. Also suppose the group map $m : G \times G \to G$ given by $m(g, h) = gh$ is smooth. Then m is a submersion.

Proof:
To start off, we only need the smoothness of m to prove that the map $R_g(h) = gh$ is a C^∞ diffeo-automorphisms for all $g \in G$. And thus by identical reasoning as on *page 1233*, we can conclude that G will have no boundary. This also in turn means that $G \times G$ has no boundary. And now in order to prove that m is a submersion it is equivalent due the local section theorem on *pages 1106-1107* to show that for any fixed $(g_1, g_2) \in G \times G$ there exists an open set $U \subseteq G$ containing $g_1 g_2$ and a smooth map $\sigma : U \to G \times G$ such that $\sigma(g_1 g_2) = (g_1, g_2)$.

This is easy though and the σ we will construct is even a *global* section. Just define $\sigma : G \to G \times G$ by $\sigma(g) = (gg_2^{-1}, g_2)$ Then σ is smooth, $\sigma(g_1 g_2) = (g_1, g_2)$, and $m \circ \sigma(g) = g$ for all $g \in G$. ■

Now we're ready for the actual interesting result (which is also an exercise from John Lee).

Problem 7-3: Suppose that G is a group that is also a smooth manifold. Also, let $m : G \times G \to G$ and $s : G \times G \to G$ be the maps $m(g, h) = gh$ and $s(g, h) = gh^{-1}$. If m is a smooth map then so is s. Then consequently by our prior lemma, we know G is a Lie group.

Proof:
Define $F : G \times G \to G \times G$ by $F(g, h) = (g, hg)$. Also for $i = 1, 2$ let $\pi_i : G \times G \to G$ be the map $(g_1, g_2) \mapsto g_i$.

My first obervation is that F is a smooth map.

After all, F can be written as $F = (\pi_1, \ \ m \circ (\pi_2, \pi_1))$.

Secondly, I claim that F is a local diffeomorphism.

Since G and thus $G \times G$ has no boundary (see my proof of the last problem), it suffices by the inverse function theorem to prove that $\mathrm{d}F_(g, h)$ has full rank at all $(g, h) \in G \times G$. The easiest way to go about this is to prove that $\mathrm{d}F_{(g,h)}$ is injective for all $(g, h) \in G \times G$.

We start by making the standard identification $T_{(g,h)}(G \times G) \cong T_g G \oplus T_h G$. Then note for any $(v, w) \in T_g G \oplus T_h G$ that:

$$\begin{aligned} \mathrm{d}F_{(g,h)}(v, w) &= (\mathrm{d}(\pi_1)_{(g,h)}(v, w), \ \ \mathrm{d}(m \circ (\pi_2, \pi_1))_{(g,h)}(v, w)) \\ &= (v, \ \ \mathrm{d}m_{(h,g)}\left(\mathrm{d}(\pi_2)_{(g,h)}(v, w), \ \mathrm{d}(\pi_1)_{(g,h)}(v, w)\right)) \\ &= (v, \ \ \mathrm{d}m_{(h,g)}\left(w, v\right)). \end{aligned}$$

It easily follows that $(v, w) \in \ker(\mathrm{d}m_{(g,h)})$ if and only if $v = 0$ and $(w, 0) \in \ker(\mathrm{d}m_{(h,g)})$

Next let $R_g : G \to G$ be the map $R_g(h) = hg$. We know that R_g is a diffeomorphism (see the prior problem). Also note that $R_g = m \circ (\mathrm{Id}_G, C_g)$ where $C_g : G \to G$ is the constant map sending everything to g. As a result, for any $w \in T_h G$ we have that:

$$\mathrm{d}(R_g)_h w = \mathrm{d}m_{(h,g)}(\mathrm{d}(\mathrm{Id}_G)_h w, \ \mathrm{d}(C_g)_g w) = \mathrm{d}m_{(h,g)}(w, 0).$$

But because R_g is a diffeomorphism, we know that $\mathrm{d}(R_g)_h w = 0$ if and only if $w = 0$. So, $(w, 0) \in \ker(\mathrm{d}m_{(h,g)})$ implies that $w = 0$. And with that, we've finished proving that the kernel of $\mathrm{d}F_{(g,h)}$ is trivial.

Thirdly I observe that F is a bijection whose inverse function is $F^{-1}(g, h) = (g, hg^{-1})$.

The three prior facts combined let us conclude that F is a smooth diffeomorphism from $G \times G$ to $G \times G$. In other words, we now know that F^{-1} is a smooth function.

Finally, we can now conclude that s is smooth since:

$$s = \pi_2 \circ F^{-1} \circ (\pi_2, \pi_1). \ ■$$

There is not an analogous result when we are only working with topological groups.

For example, let $\mathbb{R}_l$ be the set of real numbers equipped with the lower limit topology

(i.e. the topology generated by the basis of all half intervals $[a,b)$). We can consider $\mathbb{R}_l$ a group under addition.

I shall first show that the map $a:\mathbb{R}_l\times\mathbb{R}_l\to\mathbb{R}_l$ given by $a(x,y):=x+y$ is continuous.
It suffices to show for any $(x,y)\in\mathbb{R}_l\to\mathbb{R}_l$ and open set $U\subseteq\mathbb{R}_l$ cotaining $x+y$ we have that there is an open set $V\subseteq\mathbb{R}_l\times\mathbb{R}_l$ containing (x,y) with $a(V)\subseteq U$. Fortunately, by shrinking U we can always without loss of generality assume $U=[x+y,x+y+2\varepsilon)$ for some $\varepsilon>0$. And now we just set:
$$V=[x,x+\varepsilon)\times[y,y+\varepsilon).$$

That said, also observe for any sequence $\{x_k\}_{k\in\mathbb{N}}\subseteq\mathbb{R}_l$ that $x_k\to x$ in $\mathbb{R}_l$ iff $x_k\to x$ in the standard Euclidean topology and there exists $K\in\mathbb{N}$ such that for all $k\geq K$ we have that $x_k\geq x$. Using this fact, we can easily show that the negation map $n:\mathbb{R}_l\to\mathbb{R}_l$ given by $n(x)=-x$ is *not* continuous.
To see why, note that $\frac{1}{k}\to 0$ in $\mathbb{R}_\ell$ by $n(\frac{1}{k})=\frac{-1}{k}\not\to 0$ in $\mathbb{R}_\ell$.

Therefore, $\mathbb{R}_l$ is *not* a topological group even though its addition operation is continuous.

The next notes finally return to what I was doing on *page 1157*.

My end goal before getting distracted by singular homology was to show that most C^r manifolds we would care about can be embedded into Euclidean space if they weren't at first, and in fact if M is n-dimensional then it can be embedded into $\mathbb{R}^{2n+1}$. To accomplish this goal, it suffices to show that:

1. We can embed M into $\mathbb{R}^N$ for just some N.
2. Given an n dimensional embedded submanifold $M\subseteq\mathbb{R}^N$ where $N>2n+1$, there is an embedding of M into $\mathbb{R}^{N-1}$.

Lemma 6.13 on *pages 1156-1157* takes a step towards addressing the second task, albeit the lemma only gives us an immersion and not an embedding of M. At least in the case though that M is compact, then proposition 4.22 on *page 1104* tells us that our injective immersion is automatically an embedding. So for compact manifolds we are half done.

Lemma: Suppose M is a compact n-dimensional C^r manifold (where $r\geq 1$). Then there exists $N\in\mathbb{N}$ and a C^r embedding $f:M\hookrightarrow\mathbb{R}^N$.
Proof:
For every $p\in M$ we can pick a smooth chart (U_p,φ_p) such that $p\in U_p$. And then we can pick two more open sets W_p and V_p such that $p\in W_p\subseteq\overline{W_p}\subseteq V_p\subseteq\overline{V_p}\subseteq U_p$. But now as M is compact, we know there are finitely many $p_1,\ldots,p_\ell$ such that $M\subseteq\bigcup_{i=1}^{\ell}W_{p_i}$.

For each i let $b_i:M\to[0,1]$ be a C^r bump function with $b_i\equiv 1$ on $\overline{W_{p_i}}$ and $b_i\equiv 0$ on $M-V_{p_i}$. Then by pasting lemma, we can guarentee that the product $b_i\varphi_{p_i}$ defines a C^r map from M to $\mathbb{R}^n$ for each i (where the product equals 0 at all points outside the

domain of the chart (U_{p_i}, φ_{p_i})). So, we now define the map $f : M \to \mathbb{R}^{(n+1)\ell}$ by:

$$f(p) \coloneqq (b_1(p), \ldots, b_\ell(p), (b_1\varphi_1)(p), \ldots, (b_\ell\varphi_\ell)(p))$$

Because M is compact, in order to prove that f is a C^r embedding we just need to explain why f is a C^r injective immersion.

Fortunately, because every component function of f is a C^r map, we automatically have that f is a C^r map as well.

Also suppose $f(p) = f(q)$. Since $M \subseteq \bigcup_{i=1}^{\ell} W_{p_i}$ we know that $p \in W_{p_i}$ for some i. In turn $b_i(p) = 1$, and since $f(p) = f(q)$ this also guarentees that $b_i(q) = 1$. So, both p, q are covered by the chart (U_{p_i}, φ_{p_i}) and $\varphi_i(p) = (b_i\varphi_i)(p) = (b_i\varphi_i)(q) = \varphi_i(q)$. Finally, since φ_{p_i} is injective on U_{p_i} and $p, q \in U_{p_i}$, we can conclude that $p = q$. So, we have shown that f is injective.

Finally, to prove that f is an immersion, consider any $p \in M$ and then pick i with $p \in W_{p_i}$. Importantly, $b_i \equiv 1$ on a neighborhood of p. And now upon identifying:

$$T_{f(p)}\mathbb{R}^{(n+1)\ell} \cong \left(\bigoplus_{i=1}^{\ell} T_{b_i(p)}\mathbb{R}\right) \oplus \left(\bigoplus_{i=1}^{\ell} T_{(b_i\varphi_i)(p)}\mathbb{R}^n\right)$$

we have that:

$$\mathrm{d}f_p = (\mathrm{d}b_1, \ldots, \mathrm{d}b_\ell, \mathrm{d}(b_1\varphi_{p_1})_p, \ldots, \mathrm{d}(b_{i-1}\varphi_{p_{i-1}})_p, \mathrm{d}(\varphi_i)_p, \mathrm{d}(b_{i+1}\varphi_{p_{i+1}})_p, \ldots, (b_\ell\varphi_{p_\ell})_p).$$

Yet $\mathrm{d}(\varphi_i)_p : T_pM \to T_{\varphi_i(p)}\mathbb{R}^n$ is a linear bijection (and thus has rank $n = \dim(M)$). It then follows that $\mathrm{d}(f)_p$ must also have rank at least $n = \dim(M)$. So, $\mathrm{d}f_p$ is full rank and thus injective. And since p was arbitrary, we've finished proving that f is an immersion. ■

Corollary (Whitney embedding theorem for compact manifolds): Suppose M is a compact n-dimensional C^r manifold (where $r \geq 1$). Then there exists $N \in \mathbb{N}$ and a C^r embedding $f : M \hookrightarrow \mathbb{R}^N$. If in addition we assume $r \geq 2$, then we can specifically choose N to be equal to $2n + 1$.

The case of noncompact C^r manifolds is more delicate and in fact Oprea skips going over this case in lecture. I don't want to skip this though.

To start off, suppose X is a topological space. We say a function $f : X \to \mathbb{R}$ is an exhaustion function if for every $c \in \mathbb{R}$ the set $f^{-1}((-\infty, c])$ is compact.

Theorem 2.28: Suppose M is a C^r manifold (where $r \geq 1$). Then M admits a postive C^r exhaustion function $f : M \to (0, \infty)$.

Proof:

If M is already compact, then we can just pick f to be a constant function. So without loss of generality we will now assume M is not compact.

Using the fact that M is second countable as well as an LCH space, we can find a sequence $\{U_k\}_{k\in\mathbb{N}}$ of precompact open sets on M such that $M \subseteq \bigcup_{k=1}^{\infty} U_k$. Next, we can pick C^r partition of unity $\{\phi_k\}_{k\in\mathbb{N}}$ subordinate to the sets U_k. In other words, ϕ_k is positive and C^r for all k, $\text{supp}(\phi_k) \subseteq U_k$ for all k, $\sum_{k=1}^{\infty} \phi_k = 1$, and every $p \in M$ has a neighborhood on which only finitely many ϕ_k are ever nonzero.

Finally, we define $f \coloneqq \sum_{k=1}^{\infty} k\phi_k$. Note that f is a C^r map because locally around any $p \in M$ we have that f is just a linear combination of finitely many of the ϕ_K. Also note that $f(p) > 0$ for all $p \in M$. After all, $\phi_{k'}(p) > 0$ for some k'. And then since all the summands are nonnegative, we can conclude that $f(p) > k'\phi_{k'}(p) > 0$.

As for showing that f is an exhaustion function, consider any $c \in \mathbb{R}$. We can pick a positive integer $N > c$. And now if $p \notin \bigcup_{i=1}^{N} U_{p_i}$ we know that:

$$\phi_1(p) = \phi_2(p) = \cdots = \phi_N(p) = 0.$$

As a result: $f(p) = \sum_{k=N+1}^{\infty} k\phi_k(p) > N \sum_{k=N+1}^{\infty} \phi_k(p) = N \cdot 1.$

This proves that $f^{-1}((-\infty, c]) \subseteq f^{-1}((-\infty, N]) \subseteq \bigcup_{i=1}^{N} U_{p_i} \subseteq \bigcup_{i=1}^{N} \overline{U_{p_i}}$. Since the right most set is a union of compact sets and thus compact itself, and also because $f^{-1}((-\infty, c])$ is a closed subset of M due to f being continuous, we can thus conclude that $f^{-1}((-\infty, c])$ is compact. ■

Let $\lambda(k)$ denote the number of time continuously differentiable we need a function $f : \mathbb{R}^{n+k} \to \mathbb{R}^n$ to be to prove that Sard's theorem holds for f.

See *page 1154* to recall what's up with $\lambda(k)$. (Namely, $\lambda(k) = k + 1$ for all $k \geq 0$. That said, the lower bound which *I* managed to prove on $\lambda(k)$ is weaker.)

Recall from two pages ago the two tasks we need to accomplish. We'll accomplish the first task for a noncompact manifold by using a C^r exhaustion function, Sard's theorem, and the Whitney embedding theorem for compact manifolds that we already proved.

Theorem 6.15: Suppose M is a noncompact n-dimensional C^r manifold and that $r \geq \max(2, \lambda(n-1))$ (which equals $\max(2, n)$). Then there is a C^r embedding $F : M \hookrightarrow \mathbb{R}^{4n+3}$.

Proof:
Let $f : M \to (0, \infty)$ be a C^r exhaustion function. Because $r \geq \lambda(n-1)$, we know that the set of critical values of f in $(0, \infty)$ has measure zero. Hence, for every nonnegative integer i we can pick real numbers $i < a_i < b_i < i + 1$ such that a_i and b_i are regular values of f.

Now define:

- $D_0 \coloneqq f^{-1}((-\infty, 1])$ and $E_0 \coloneqq f^{-1}((-\infty, a_1])$,
- $D_i \coloneqq f^{-1}([i, i+1])$, and $E_i \coloneqq f^{-1}([b_{i-1}, a_{i+1}])$ for all $i \geq 1$.

By proposition 5.47 on *pages 1127-1128*, we can conclude that each E_i is a compact regular domain of M for all $i \geq 0$. Furthermore, each D_i will be a compact subset of M contained within the interior of each E_i.

For all $i \geq 0$ we let $\phi_i : M \to [0,1]$ be a C^r bump function such that $\phi_i \equiv 1$ on a neighborhood of D_i and $\operatorname{supp}(\phi_i) \subseteq E_i^\circ$. Also, for all $i \geq 0$ we let $g_i : E_i \hookrightarrow \mathbb{R}^{2n+1}$ be a C^r embedding.

We know such an embedding exists due to Whitney's embedding theorem for compact manifolds as well as the fact that $r \geq 2$.

Importantly, the product $\phi_i g_i$ then defines a C^r map on all of M (where $(\phi_i g_i)(p) = 0$ if $p \notin E_i$.) Also observe that every point $p \in M$ has a neighborhood U intersecting at most two of the E_i. In fact, we can even even say that U intersects E_i and E_{i+1} for some i.

To see, why note that $f^{-1}((-\infty, a_1))$ intersects only E_0 and E_1. Also, for any $i \geq 0$ we have that $f^{-1}((a_i, a_{i+1}))$ and $f^{-1}((b_i, b_{i+1}))$ intersect only E_i and E_{i+1} on a neighborhood of p.

Now we define $F : M \to \mathbb{R}^{4n+3} \cong \mathbb{R}^{2n+1} \times \mathbb{R}^{2n+1} \times \mathbb{R}$ by:

$$F(p) \coloneqq \left(\sum_{i \text{ even}} (\phi_i g_i)(p), \quad \sum_{i \text{ odd}} (\phi_i g_i)(p), \quad f(p) \right)$$

Note that F is a C^r map because every $p \in M$ has a neighborhood on which only one summand from each of the sums in the definition of F is nonzero. In a similar vein, to see that F is an immersion, note for any $p \in M$ that there exists i with $p \in D_i$. In turn, $(\phi_i g_i) = g_i$ on a neighborhood of p. Then combining the fact that $\mathrm{d}(g_i)_p$ is injective with the observation of what the formula for F locally looks like will prove $\mathrm{d}F_p$ is injective.

As for showing that F is injective, note that if $F(p) = F(q)$ then we know that $f(p) = f(q)$. In particular, this means that if $p \in D_i$ then $q \in D_i$ also. So, $\phi_i(p) = 1 = \phi_i(q)$. Next note that $F(p) = F(q)$ implies that:

$$g_i(p) = (\phi_i g_i)(p) = (\phi_i g_i)(q) = g_i(q).$$

Yet $p, q \in D_i$ implies that $p, q \in E_i$ as well. And since g_i is injective on E_i, we can conclude that $p = q$.

Finally, I claim that F is a proper map. To see why, suppose $K \subseteq \mathbb{R}^{4n+3}$ is any compact set. Then in particular, there must exist $A, B \in \mathbb{R}$ such that $K \subseteq \mathbb{R}^{2n+2} \times [A, B]$. Hence, $F^{-1}(K) \subseteq f^{-1}([A, B])$. Yet now note that $F^{-1}(K)$ is a closed set. Meanwhile, f being an exhaustion function means that $f^{-1}([A, B])$ is a compact set. It follows that $F^{-1}(K)$ is compact.

By proposition 4.22 on *page 1104*, we can conclude that F is a C^r embedding. ■

As for accomplishing the second task we first define for any 1-dimensional linear subspace $S \subseteq \mathbb{R}^N$ and any $R > 0$ the tube with axis S and radius R given by:

$$T_R(S) \coloneqq \{x \in \mathbb{R}^N : |x - y| < R \text{ for some } y \in S\}.$$

Note that if $S = \operatorname{span}(v)$ then $\min_{y \in S} |x - y| = \left| x - \frac{x \cdot v}{|v|^2} v \right|$.

Importantly, the map $x \mapsto \left| x - \frac{x \cdot v}{|v|^2} v \right|$ is a continuous function and so we can conclude that $T_R(S)$ is an open set.

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